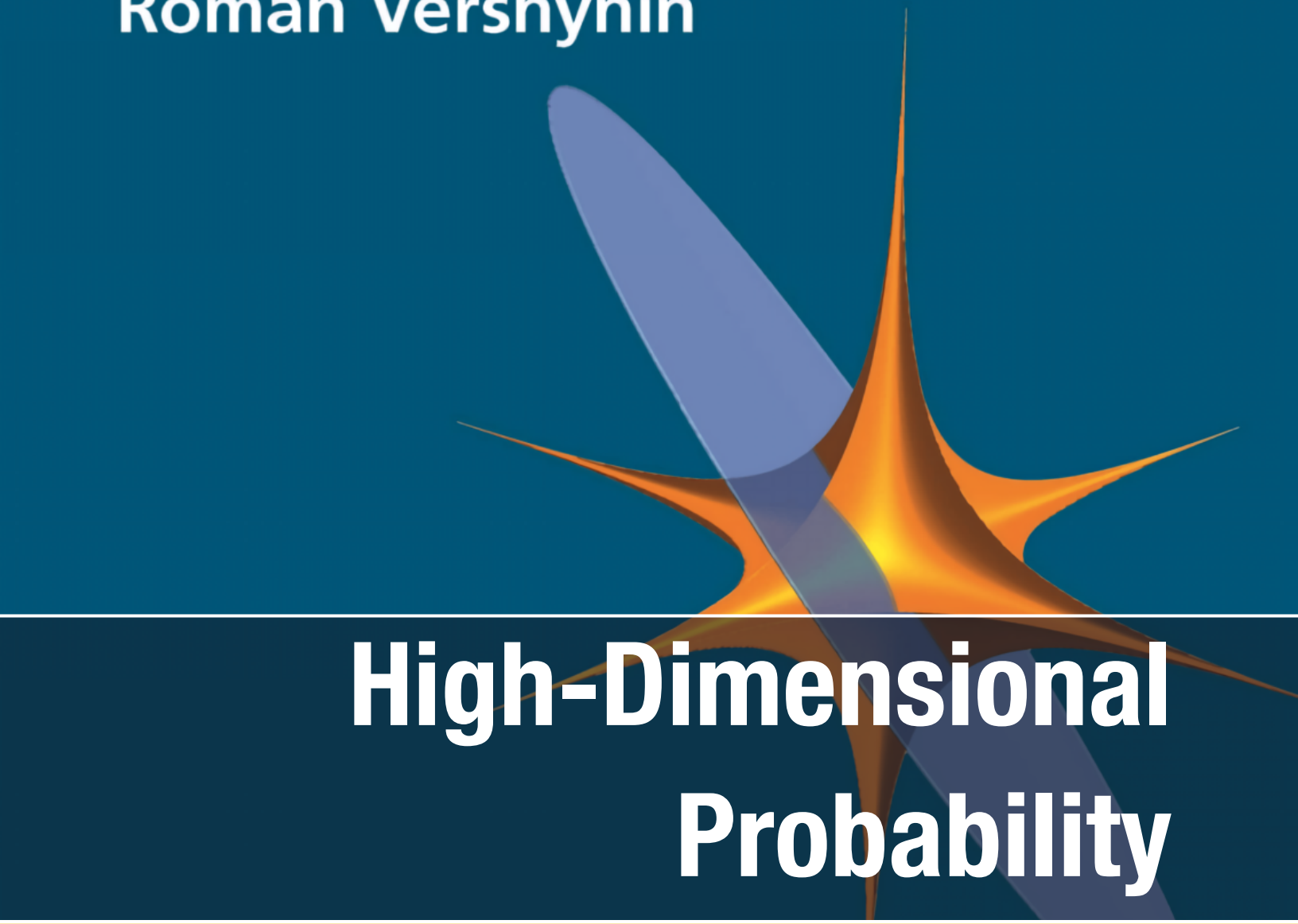
Second Edition

v1.0

Roman Vershynin



High-Dimensional Probability

Problems, Solutions and Proofs

WorseFive

2026/8/29

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Preface

Roman Vershynin's *High-Dimensional Probability: An Introduction with Applications in Data Science* is a classic work in high-dimensional probability theory. Unlike traditional textbooks grounded in the axiomatics of probability spaces and asymptotic limit theorems, this book directly confronts random phenomena in high-dimensional spaces, systematically expounding concentration inequalities, sub-Gaussian properties, random matrices, empirical processes, and random geometry together with their concrete realizations in data science. The second edition has been thoroughly revised and expanded on the basis of the first. The author reorganized the logical threads of some core chapters, while also introducing a large amount of new material and supplementary exercises.

(1) What Is High-Dimensional Probability Theory?

Classical probability theory mainly studies the statistical properties of finite-dimensional random variables, while high-dimensional probability theory investigates why, when the dimension of a system is extremely large, complex randomness comes to exhibit a highly deterministic structure.

For instance, although a high-dimensional random vector has many independent random coordinates, its Euclidean norm is often concentrated near some fixed value; although a high-dimensional random matrix is full of random entries, its spectral norm, singular values, and other characteristic quantities still admit very fine non-asymptotic estimates; and random point sets scattered in high-dimensional space may very well naturally present regular geometric shapes.

It can be seen that high dimension does not mean chaos; on the contrary, at an appropriate scale of measurement, high-dimensional random systems possess strong regularity and predictability.

(2) For Beginners

Although this book is aimed at doctoral students, advanced master's students, and beginning researchers in related fields, if you have already learned elementary probability theory and functional analysis in your sophomore year, you can work through this book without pain, and it will also be your first contact with technical proofs.

For readers encountering high-dimensional probability theory for the first time, the real difficulty is often not memorizing a particular theorem, but figuring out why these theorems hold, when they should be used, and the relationships among the different tools.

High-dimensional probability theory involves many new terms and new methods; at first glance one easily gets the feeling that "there are many formulas and many techniques." But as learning deepens, you will gradually discover that the truly recurring core tools are actually not that numerous. Many proofs can ultimately be reduced to a few basic questions:

How fast is the tail of a random variable decaying?

Will a certain function of a random vector exhibit concentration?

How do we discretize a problem involving infinitely many directions?

How do we cover an entire sphere with finitely many points?

How do we lift a pointwise estimate to a uniform estimate?

And how do we decompose a complicated random object into several easily controllable parts?

...

As readers become increasingly familiar with these theories, many of the technical proofs of high-dimensional probability theory gradually become an internalized framework of thought. Exercises are precisely the best way to move from "knowing what" to "knowing why."

Compared with the first edition, Vershynin's second edition adds a large number of exercises, a considerable part of which do not mechanically repeat the computations in the text, but serve to supplement the text, train proof techniques, and even build the entire theoretical framework. Some exercises actually give alternative proofs of certain conclusions in the text; some require the reader to combine the tools obtained in the previous chapters; and some appear to be only small technical problems, yet in fact reveal very important ideas in high-dimensional probability theory.

(3) Organization of the Solutions

These solutions are organized according to the chapter order of Vershynin's second edition, keeping correspondence with the original book so that readers can consult it at any time. In the solutions, I have not entirely copied the notation and terminology of the original textbook, and where necessary I supplement the intermediate steps of the proofs. At the same time, in order to make the proof process clearer, I also add appropriate verbal explanations or figures according to the specific problem, to help readers understand the motivation of the proof; for some places that are important or easily misunderstood, I also give additional remarks.

9 [**The Carl–Pajor Theorem**] Improve Theorem T0.1.4 (*Volume of a polytope*) by replacing N with N/n.
Let $P \subset \mathbb{B}_2^n$ be a polytope generated by $N \geqslant n$ vertices. Denote $B = \mathbb{B}_2^n$. Prove:

$$\left(\frac{\mathrm{Vol}(P)}{\mathrm{Vol}(B)}\right)^{1/n} \leqslant C\sqrt{\frac{\log(\mathrm{e}N/n)}{n}}. \qquad (C>0)$$

Prf. Let $P = \mathrm{conv}\{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N\}$, where $\|\boldsymbol{x}_i\|_2 \leqslant 1$.

Solution 1 Maurey's empirical method

- Covering-number estimate

  By Theorem T0.1.2 (*Approximate Caratheodory theorem*), for every $\boldsymbol{x} \in P$, there exist $\boldsymbol{x}_{i_1}, \cdots, \boldsymbol{x}_{i_k}$ such that

$$\left\| \boldsymbol{x} - \frac{1}{k}\sum^{k} \boldsymbol{x}_{\cdot} \right\| \leqslant \frac{1}{=}.$$

Multiple Solution

Solution 2 Entropy-number method

Construct the linear operator $T : \ell_1^N \to \ell_2^n$ by

$$T(\boldsymbol{a}) = \sum_{i=1}^{N} a_i \boldsymbol{x}_i, \qquad (\boldsymbol{a} = (a_1, \cdots, a_N) \in \ell_1^N)$$

This satisfies $\|T\| = \max_{1\leqslant i\leqslant N} \|\boldsymbol{x}_i\|_2 \leqslant 1$, so $T(\mathbb{B}_1^N) = \mathrm{conv}\{\pm\boldsymbol{x}_1, \cdots, \pm\boldsymbol{x}_N\}$, and hence $P \subset T(\mathbb{B}_1^N)$.
Then let $e_k(T)$ be the k-th dyadic entropy number of T (*the smallest radius with which* 2^{k-1} *balls cover* $T(\mathbb{B}_1^N)$), i.e.,

$$e_k(T) = \inf\{\varepsilon > 0 : \mathcal{N}(T(\mathbb{B}_1^N), \varepsilon\mathbb{B}_2^n \leqslant 2^{k-1})\}.$$

State the following lemmas:

Lemma 1 (*Carl's inequality, 1981*) Let T satisfy $\mathrm{rank}(T) = n$, and let $e_n(T)$ be its n-th dyadic entropy number. Then

$$\left(\frac{\mathrm{Vol}(T(\mathbb{B}_1^N))}{\mathrm{Vol}(B)}\right)^{1/n} \leqslant c_1 e_n(T).$$

43 **[Upper Bounds for Sub-Gaussian Matrices: Relaxing the Independence Assumption]**
(1) Matrix-level sub-Gaussianity: if $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ is viewed as a K-sub-Gaussian random vector in the space $\mathbb{R}^{m\times n}$, with $K = \|\boldsymbol{A}\|_{\psi_2}$, prove: even without any independence assumption, the conclusion of Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*) still holds.
(2) Row (or column)-level sub-Gaussianity: prove: if the rows $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_m \in \mathbb{R}^n$ of $\boldsymbol{A}$ are independent zero-mean K-sub-Gaussian random vectors, then Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*) still holds. (*The same conclusion holds for columns.*)

Explain Motivation

Rk. Recall that the ε-net proof of Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*) proceeds in three steps:
1. Approximation: choose a deterministic ε-net $\mathcal{N} \subset \mathbb{S}^{n-1}, \mathcal{M} \subset \mathbb{S}^{m-1}$, reducing the norm to $\sup_{\boldsymbol{x}\in\mathcal{N},\boldsymbol{y}\in\mathcal{M}}\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$.
2. Concentration: prove that for fixed $\boldsymbol{x} \in \mathcal{N}, \boldsymbol{y} \in \mathcal{M}$, the bilinear form $Z = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$ satisfies the sub-Gaussian tail bound $\mathbb{P}\{|Z| \geqslant u\} \leqslant 2\exp(-cu^2/K^2)$.
3. Union bound: take the union over all net points.

Since steps 1 and 3 are deterministic geometric operations, they are independent of the randomness assumptions on the matrix entries. Hence we only need to prove that under the new assumptions of (1)(2), the sub-Gaussian norm bound $\|Z\|_{\psi_2} \leqslant C'K$ for $Z = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$ in step 2 still holds.

Prf. (1) By the definition of a sub-Gaussian random vector,

$$\|\boldsymbol{A}\|_{\psi_2} = \sup \|\langle \boldsymbol{A}, \boldsymbol{W}\rangle\|_{\psi_2} = K,$$

37 **[Controlling Norm Deviation on an ε-Net]** Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, and let $\mathcal{N}$ be an ε-net of $\mathbb{S}^{n-1}$ with $\varepsilon \in [0, 1/2)$. For $\mu \in \mathbb{R}$, define the global deviation and the maximal deviation over the ε-net respectively by

$$E = \sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu|, \quad \delta = \sup_{\boldsymbol{x}\in\mathcal{N}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu|.$$

Prove: the global norm deviation over $\mathbb{S}^{n-1}$ can be controlled by the local deviation over $\mathcal{N}$, i.e.

$$E \leqslant \frac{C}{1-2\varepsilon}\delta. \qquad (C > 0)$$

Rk. The following is an erroneous attempt:

An incorrect solution with some heuristic value

By the definition of the ε-net, for every $\boldsymbol{x} \in \mathbb{S}^{n-1}$ there exists $\boldsymbol{x}_0 \in \mathcal{N}$ with $\|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \leqslant \varepsilon$; hence

$$\begin{aligned} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu| &\leqslant |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \|\boldsymbol{A}\boldsymbol{x}_0\|_2| + |\|\boldsymbol{A}\boldsymbol{x}_0\|_2 - \mu| \\ &\leqslant \|\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0)\|_2 + (\|\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0)\|_2 + \delta). \end{aligned}$$

Let $\boldsymbol{h} = \boldsymbol{x} - \boldsymbol{x}_0$, so that $\|\boldsymbol{h}\|_2 \leqslant \varepsilon$.
If $\boldsymbol{h} = \boldsymbol{0}$, the conclusion is trivial; we only need to consider $\boldsymbol{h} \neq \boldsymbol{0}$.

Let $\boldsymbol{u} = \bar{\boldsymbol{h}} \in \mathbb{S}^{n-1}$; then

$$\|\boldsymbol{A}\boldsymbol{h}\|_2 = \|\boldsymbol{h}\|_2 \cdot \|\boldsymbol{A}\boldsymbol{u}\|_2 \leqslant \varepsilon\|\boldsymbol{A}\boldsymbol{u}\|_2,$$

where

$$\|\boldsymbol{A}\boldsymbol{u}\|_2 \leqslant |\|\boldsymbol{A}\boldsymbol{u}\|_2 - \mu| + |\mu| \leqslant E + |\mu|,$$

hence

$$|\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu| \leqslant \varepsilon(E + |\mu|) + \delta,$$

and taking the supremum on both sides,

$$E \leqslant \varepsilon E + \varepsilon|\mu| + \delta \implies E \leqslant \frac{\varepsilon|\mu| + \delta}{1-\varepsilon}.$$

One can see that bounding the absolute values directly inevitably introduces a cross term involving μ, leaving the proof stuck.

Prf. First consider $\mu \leqslant 0$. Let $M = \|\boldsymbol{A}\|$ and $a_0 = \sup_{\boldsymbol{y}\in\mathcal{N}} \|\boldsymbol{A}\boldsymbol{y}\|_2$. By Exercise Q4.35 (*Computing Vector Norms on an ε-Net*), $M \leqslant a_0/(1-\varepsilon)$; since $-\mu \geqslant 0$,

$$E = M - \mu \leqslant \frac{a_0 - \mu}{1-\varepsilon} \leqslant \frac{\delta}{1-\varepsilon}.$$

37 [**Controlling Norm Deviation on an ε-Net**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, and let $\mathcal{N}$ be an ε-net of $\mathbb{S}^{n-1}$ with $\varepsilon \in [0, 1/2)$. For $\mu \in \mathbb{R}$, define the global deviation and the maximal deviation over the ε-net respectively by

$$E = \sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu|, \quad \delta = \sup_{\boldsymbol{x}\in\mathcal{N}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu|.$$

Prove: the global norm deviation over $\mathbb{S}^{n-1}$ can be controlled by the local deviation over $\mathcal{N}$, i.e.

$$E \leqslant \frac{C}{1-2\varepsilon}\delta. \qquad (C > 0)$$

Rk. The following is an erroneous attempt: **An incorrect solution with some heuristic value**

By the definition of the ε-net, for every $\boldsymbol{x} \in \mathbb{S}^{n-1}$ there exists $\boldsymbol{x}_0 \in \mathcal{N}$ with $\|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \leqslant \varepsilon$; hence

$$\begin{aligned} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu| &\leqslant |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \|\boldsymbol{A}\boldsymbol{x}_0\|_2| + |\|\boldsymbol{A}\boldsymbol{x}_0\|_2 - \mu| \\ &\leqslant \|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{x}_0)\|_2 + (\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{x}_0)\|_2 + \delta). \end{aligned}$$

Let $\boldsymbol{h} = \boldsymbol{x} - \boldsymbol{x}_0$, so that $\|\boldsymbol{h}\|_2 \leqslant \varepsilon$.

If $\boldsymbol{h} = \boldsymbol{0}$, the conclusion is trivial; we only need to consider $\boldsymbol{h} \neq \boldsymbol{0}$.

Let $\boldsymbol{u} = \bar{\boldsymbol{h}} \in \mathbb{S}^{n-1}$; then

$$\|\boldsymbol{A}\boldsymbol{h}\|_2 = \|\boldsymbol{h}\|_2 \cdot \|\boldsymbol{A}\boldsymbol{u}\|_2 \leqslant \varepsilon\|\boldsymbol{A}\boldsymbol{u}\|_2,$$

where

$$\|\boldsymbol{A}\boldsymbol{u}\|_2 \leqslant |\|\boldsymbol{A}\boldsymbol{u}\|_2 - \mu| + |\mu| \leqslant E + |\mu|,$$

hence

$$|\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu| \leqslant \varepsilon(E + |\mu|) + \delta,$$

and taking the supremum on both sides,

$$E \leqslant \varepsilon E + \varepsilon|\mu| + \delta \implies E \leqslant \frac{\varepsilon|\mu| + \delta}{1-\varepsilon}.$$

One can see that bounding the absolute values directly inevitably introduces a cross term involving μ, leaving the proof stuck.

Prf. First consider $\mu \leqslant 0$. Let $M = \|\boldsymbol{A}\|$ and $a_0 = \sup_{\boldsymbol{y}\in\mathcal{N}} \|\boldsymbol{A}\boldsymbol{y}\|_2$. By Exercise Q4.35 (*Computing Vector Norms on an ε-Net*), $M \leqslant a_0/(1-\varepsilon)$; since $-\mu \geqslant 0$,

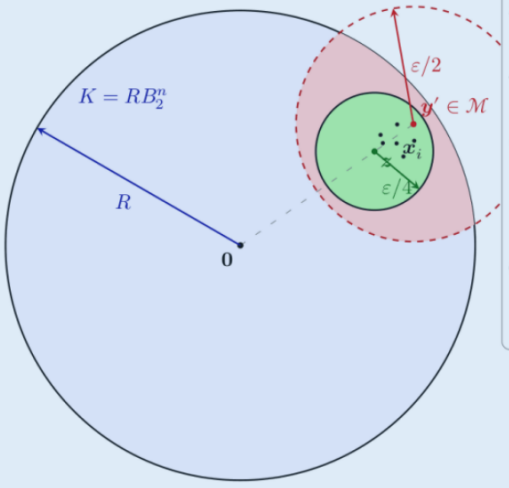
$$E = M - \mu \leqslant \frac{a_0 - \mu}{1-\varepsilon} \leqslant \frac{\delta}{1-\varepsilon}.$$

39 [**Random Point Sets Form ε-Nets**] Let $\boldsymbol{g}_1, \cdots, \boldsymbol{g}_N$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. For any radius $R > 0$ and accuracy $\varepsilon > 0$, prove:

when $N \geqslant \mathrm{e}^{C_{R,\varepsilon} n}$, the random set

$$\mathcal{N}_X = \left\{\frac{\boldsymbol{g}_1}{\sqrt{n}}, \cdots, \frac{\boldsymbol{g}_N}{\sqrt{n}}\right\} \cap R\mathbb{B}_2^n$$

with high probability forms an ε-net of the centered ball $K = R\mathbb{B}_2^n$, where $c > 0$ (w.p. $1 - \mathrm{e}^{-cn}$). **Additional Image Explain**



Geometric logic of the constructive proof:

1. The light blue large ball is the target space $K = RB_2^n$.
2. $\boldsymbol{y}'$ is a point of the predetermined reference $\varepsilon/2$-net $\mathcal{M}$.
3. The red dashed circle is $B(\boldsymbol{y}', \varepsilon/2)$; the red shaded region is the intersection $V_{\boldsymbol{y}'} = B(\boldsymbol{y}', \varepsilon/2) \cap K$.
4. To compute the probability lower bound, we look inside $V_{\boldsymbol{y}'}$ for the green small ball $B(\boldsymbol{z}, \varepsilon/4)$ fully contained in it.
5. Shrinking $\boldsymbol{y}'$ towards the origin by distance $\frac{\varepsilon}{4R}\|\boldsymbol{y}'\|_2$ yields the center $\boldsymbol{z}$, which guarantees the green small ball does not cross the boundary of the blue large ball or the red dashed boundary.
6. If a random sample point $\boldsymbol{x}_i$ falls in the green small ball, it certainly falls in the red intersection region, thus successfully covering $\boldsymbol{y}'$.

54 **[Max-Cut Approximation Algorithm: Deterministic Output and Randomized Running Time]** Theorem T3.6.4 (0.878-*approximation algorithm for maximum cut*) uses SDP relaxation and rounding (random hyperplane rounding) techniques to find a cut whose expected weight is at least 0.878 times the maximum cut.

For any $\varepsilon > 0$, design a $(0.878 - \varepsilon)$-approximation algorithm for the max-cut problem such that the final output cut always satisfies the approximation ratio requirement (*i.e., deterministic output*), but the running time of the algorithm is randomized. Then find an upper bound on the expected running time of the algorithm.

Prf. It remains to convert a Monte Carlo-type algorithm (*fixed running time, result guaranteed probabilistically*) into a Las Vegas-type algorithm (*deterministic result, randomized running time*).

Let $G = (V, E)$ with edge weights $w_{ij} \geqslant 0$, and let W_{opt} be the maximum cut weight of G.

Present the following algorithm:

Additional Algorithm Explained

Algorithm 1: Guaranteed-Max-Cut-Approximation(G, ε)

Input : Graph $G = (V, E)$, edge weights $\{w_{ij}\}_{i,j \in V} \geqslant 0$, approximation parameter $\varepsilon > 0$.

Output: A cut $S \subset V$ whose weight satisfies $W(S) \geqslant (\alpha_{GW} - \varepsilon) W_{\text{opt}}$.

Part 1: Solve the SDP relaxation problem

Solve the following SDP relaxation problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i<j} w_{ij} \frac{1 - \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle}{2}, \\ \text{subject to} \quad & \boldsymbol{v}_i \in \mathbb{S}^{n-1}, \end{aligned}$$

Let the optimal value be W_{SDP} and the optimal vector solution be $\{\boldsymbol{v}_i\}_{i=1}^n$.

Part 2: Iterative randomized rounding (Las Vegas process)

Set $\alpha \leftarrow \alpha_{GW} \approx 0.878$;

while True do

  Independently draw a standard normal random vector $\boldsymbol{g}$ from $N(\boldsymbol{0}, \boldsymbol{I}_n)$;

  Construct candidate cut: $S \leftarrow \{i \in V : \langle \boldsymbol{v}_i, \boldsymbol{g} \rangle \geqslant 0\}$;

  Compute weight: $W \leftarrow \sum_{i \in S, j \notin S} w_{ij}$;

  if $W \geqslant (\alpha - \varepsilon) W_{\text{SDP}}$ **then**

    return S;

  end

end

(4) Finally

Due to the tight schedule of writing, the book inevitably contains errors and omissions. Although I have tried my best to proofread, I still fear that there may be oversights. If you find mistakes while reading, or have questions about the solution ideas, I sincerely look forward to your criticism and correction.

AI Disclaimer. Although contemporary AI can solve these problems with ease, this work was not independently generated by AI: I used AI only for proofreading after completing the manuscript. In addition, the author is Chinese and wrote the first draft in Chinese; consequently, the English version was machine-translated from the Chinese version.

WorseFive

August 22, 2026

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Appetizer: Using Probability to Cover a Set

0

Keypoints

Section 0.1 Appetizer: Covering a Set Q0.1 ∼ Q0.9

Theorem 0.1.1 (*Caratheodory theorem*)
Every point in the convex hull of a set $T \subset \mathbb{R}^n$ can be expressed as a convex combination of at most $n+1$ points from T.

Theorem 0.1.2 (*Approximate Caratheodory theorem*)
Consider a set $T \subset \mathbb{B}_2^n$. Then, for every $\boldsymbol{x} \in \operatorname{conv}(T)$, there exist $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_k \in T$ such that

$$\left\| \boldsymbol{x} - \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{x}_j \right\|_2 \leqslant \frac{1}{\sqrt{k}}.$$

Corollary 0.1.3 (*Covering polytopes by balls*)
Let P be a polytope in $\mathbb{B}_2^n$ with N vertices. Then P can be covered by at most N^k Euclidean balls of radii $1/\sqrt{k}$.

Theorem 0.1.4 (*Volume of a polytope*)
Let $P \subset \mathbb{B}_2^n = B$ be a polytope with N vertices. Then

$$\frac{\operatorname{Vol}(P)}{\operatorname{Vol}(B)} \leqslant \left(3\sqrt{\frac{\log N}{n}} \right)^n.$$

Exercises

1 [**Two Identities Regarding Variance**]

(1) Given the moment expansion form of variance

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Prove the multidimensional version of the above identity.

Let $\boldsymbol{Z} \in \mathbb{R}^n$ be a random vector. Prove:

$$\mathbb{E}\|\boldsymbol{Z} - \mathbb{E}[\boldsymbol{Z}]\|_2^2 = \mathbb{E}\|\boldsymbol{Z}\|_2^2 - \|\mathbb{E}[\boldsymbol{Z}]\|_2^2.$$

(2) Given the pairwise difference form of variance

$$\mathrm{Var}(Z) = \mathbb{E}[(Z - \mathbb{E}[Z])^2] = \frac{1}{2}\mathbb{E}[(Z - Z')^2].$$

Prove the multidimensional version of the above identity.

Let $\boldsymbol{Z}, \boldsymbol{Z}' \in \mathbb{R}^n$ be i.i.d. random vectors. Prove:

$$\mathbb{E}\|\boldsymbol{Z} - \mathbb{E}[\boldsymbol{Z}]\|_2^2 = \frac{1}{2}\mathbb{E}\|\boldsymbol{Z} - \boldsymbol{Z}'\|_2^2.$$

Prf. Let $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{Z}]$.

(1) It is easy to see that

$$\begin{aligned}\|\boldsymbol{Z} - \boldsymbol{\mu}\|_2^2 &= \langle \boldsymbol{Z} - \boldsymbol{\mu}, \boldsymbol{Z} - \boldsymbol{\mu}\rangle \\ &= \langle \boldsymbol{Z}, \boldsymbol{Z}\rangle - \langle \boldsymbol{Z}, \boldsymbol{\mu}\rangle - \langle \boldsymbol{\mu}, \boldsymbol{Z}\rangle + \langle \boldsymbol{\mu}, \boldsymbol{\mu}\rangle \\ &= \|\boldsymbol{Z}\|_2^2 - 2\langle \boldsymbol{Z}, \boldsymbol{\mu}\rangle + \|\boldsymbol{\mu}\|_2^2,\end{aligned}$$

Taking expectation on both sides yields

$$\begin{aligned}\|\boldsymbol{Z} - \boldsymbol{\mu}\|_2^2 &= \langle \boldsymbol{Z} - \boldsymbol{\mu}, \boldsymbol{Z} - \boldsymbol{\mu}\rangle \\ &= \|\boldsymbol{Z}\|_2^2 - 2\langle \boldsymbol{Z}, \boldsymbol{\mu}\rangle + \|\boldsymbol{\mu}\|_2^2,\end{aligned}$$

where

$$\mathbb{E}[\langle \boldsymbol{Z}, \boldsymbol{\mu}\rangle] = \langle \mathbb{E}[\boldsymbol{Z}], \boldsymbol{\mu}\rangle = \langle \boldsymbol{\mu}, \boldsymbol{\mu}\rangle = \|\boldsymbol{\mu}\|_2^2,$$

hence

$$\begin{aligned}\mathbb{E}\|\boldsymbol{Z} - \boldsymbol{\mu}\|_2^2 &= \mathbb{E}\|\boldsymbol{Z}\|_2^2 - 2\|\boldsymbol{\mu}\|_2^2 + \|\boldsymbol{\mu}\|_2^2 \\ &= \mathbb{E}\|\boldsymbol{Z}\|_2^2 - \|\boldsymbol{\mu}\|_2^2.\end{aligned}$$

(2) It is easy to see that

$$\begin{aligned}\|\boldsymbol{Z} - \boldsymbol{Z}'\|_2^2 &= \langle \boldsymbol{Z} - \boldsymbol{Z}', \boldsymbol{Z} - \boldsymbol{Z}'\rangle \\ &= \|\boldsymbol{Z}\|_2^2 - 2\langle \boldsymbol{Z}, \boldsymbol{Z}'\rangle + \|\boldsymbol{Z}'\|_2^2,\end{aligned}$$

Taking expectation on both sides yields

$$\begin{aligned}\mathbb{E}\|\boldsymbol{Z} - \boldsymbol{Z}'\|_2^2 &= \mathbb{E}\|\boldsymbol{Z}\|_2^2 - 2\mathbb{E}[\langle \boldsymbol{Z}, \boldsymbol{Z}'\rangle] + \mathbb{E}\|\boldsymbol{Z}'\|_2^2 \\ &= 2\mathbb{E}\|\boldsymbol{Z}\|_2^2 - 2\mathbb{E}[\langle \boldsymbol{Z}, \boldsymbol{Z}'\rangle].\end{aligned}$$

where

$$\mathbb{E}[\langle \boldsymbol{Z}, \boldsymbol{Z}'\rangle] = \langle \mathbb{E}[\boldsymbol{Z}], \mathbb{E}[\boldsymbol{Z}']\rangle = \langle \boldsymbol{\mu}, \boldsymbol{\mu}\rangle = \|\boldsymbol{\mu}\|_2^2,$$

hence

$$\mathbb{E}\|\boldsymbol{Z} - \boldsymbol{Z}'\|_2^2 = 2\mathbb{E}\|\boldsymbol{Z} - \mathbb{E}[\boldsymbol{Z}]\|_2^2,$$

i.e.,

$$\mathbb{E}\|\boldsymbol{Z} - \mathbb{E}[\boldsymbol{Z}]\|_2^2 = \frac{1}{2}\mathbb{E}\|\boldsymbol{Z} - \boldsymbol{Z}'\|_2^2.$$

2 [**Expectation Minimizes Mean Squared Error**] It is known that the extremal property of variance is

$$\mathrm{Var}(X) = \min_{a \in \mathbb{R}} \mathbb{E}[(X - a)^2].$$

Prove the multidimensional version of the above property.

Let $\boldsymbol{Z} \in \mathbb{R}^n$ be a random vector. Prove:

$$\mathbb{E}\|\boldsymbol{Z} - \mathbb{E}[\boldsymbol{Z}]\|_2^2 = \min_{\boldsymbol{a}\in\mathbb{R}^n} \mathbb{E}\|\boldsymbol{Z} - \boldsymbol{a}\|_2^2.$$

Prf. Let $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{Z}]$, then

$$\begin{aligned}\|\boldsymbol{Z} - \boldsymbol{a}\|_2^2 &= \|(\boldsymbol{Z} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \boldsymbol{a})\|_2^2 \\ &= \langle(\boldsymbol{Z} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \boldsymbol{a}), (\boldsymbol{Z} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \boldsymbol{a})\rangle \\ &= \|\boldsymbol{Z} - \boldsymbol{\mu}\|_2^2 + \|\boldsymbol{\mu} - \boldsymbol{a}\|_2^2 + 2\langle\boldsymbol{Z} - \boldsymbol{\mu}, \boldsymbol{\mu} - \boldsymbol{a}\rangle,\end{aligned}$$

Taking expectation on both sides yields

$$\mathbb{E}\|\boldsymbol{Z} - \boldsymbol{a}\|_2^2 = \mathbb{E}\|\boldsymbol{Z} - \boldsymbol{\mu}\|_2^2 + \|\boldsymbol{\mu} - \boldsymbol{a}\|_2^2 + 2\mathbb{E}[\langle\boldsymbol{Z} - \boldsymbol{\mu}, \boldsymbol{\mu} - \boldsymbol{a}\rangle],$$

where

$$\mathbb{E}[\langle\boldsymbol{Z} - \boldsymbol{\mu}, \boldsymbol{\mu} - \boldsymbol{a}\rangle] = \langle\mathbb{E}[\boldsymbol{Z} - \boldsymbol{\mu}], \boldsymbol{\mu} - \boldsymbol{a}\rangle = \langle\boldsymbol{0}, \boldsymbol{\mu} - \boldsymbol{a}\rangle = 0,$$

hence

$$\mathbb{E}\|\boldsymbol{Z} - \boldsymbol{a}\|_2^2 = \mathbb{E}\|\boldsymbol{Z} - \boldsymbol{\mu}\|_2^2 + \|\boldsymbol{\mu} - \boldsymbol{a}\|_2^2.$$

When $\boldsymbol{a} = \boldsymbol{\mu}$, we have $\|\boldsymbol{\mu} - \boldsymbol{a}\|_2^2 = 0$, hence

$$\min_{\boldsymbol{a}\in\mathbb{R}^n} \mathbb{E}\|\boldsymbol{Z} - \boldsymbol{a}\|_2^2 = \mathbb{E}\|\boldsymbol{Z} - \mathbb{E}[\boldsymbol{Z}]\|_2^2.$$

3 [**Variance of a Sum**] It is known that if $Z_1, \cdots, Z_n$ are independent and satisfy $\mathbb{E}[Z_j] = 0$ for every j, then the variance of their sum is

$$\mathbb{E}\left[\left(\sum_{j=1}^k Z_j\right)^2\right] = \sum_{j=1}^k \mathbb{E}[Z_j^2].$$

Prove the multidimensional version of the above identity (*i.e., the Parseval identity or Pythagorean theorem in the space of random vectors*).

Let $\boldsymbol{Z}_1, \cdots, \boldsymbol{Z}_k \in \mathbb{R}^n$ be independent mean-zero random vectors. Prove:

$$\mathbb{E}\left[\left\|\sum_{j=1}^k \boldsymbol{Z}_j\right\|_2^2\right] = \sum_{j=1}^k \mathbb{E}\|\boldsymbol{Z}_j\|_2^2.$$

Prf. It is easy to see that

$$\begin{aligned}\left\|\sum_{j=1}^k \boldsymbol{Z}_j\right\|_2^2 &= \left\langle\sum_{i=1}^k \boldsymbol{Z}_i, \sum_{j=1}^k \boldsymbol{Z}_j\right\rangle \\ &= \sum_{i=1}^k \sum_{j=1}^k \langle\boldsymbol{Z}_i, \boldsymbol{Z}_j\rangle,\end{aligned}$$

Taking expectation on both sides yields

$$\begin{aligned}\mathbb{E}\left[\left\|\sum_{j=1}^k \boldsymbol{Z}_j\right\|_2^2\right] &= \mathbb{E}\left[\sum_{i=1}^k \sum_{j=1}^k \langle\boldsymbol{Z}_i, \boldsymbol{Z}_j\rangle\right] \\ &= \sum_{i=1}^k \sum_{j=1}^k \mathbb{E}[\langle\boldsymbol{Z}_i, \boldsymbol{Z}_j\rangle] \\ &= \sum_{i=1}^k \mathbb{E}[\langle\boldsymbol{Z}_i, \boldsymbol{Z}_i\rangle] + \sum_{i\neq j} \mathbb{E}[\langle\boldsymbol{Z}_i, \boldsymbol{Z}_j\rangle],\end{aligned}$$

where for $i \neq j$,

$$\mathbb{E}[\langle\boldsymbol{Z}_i, \boldsymbol{Z}_j\rangle] = \langle\mathbb{E}[\boldsymbol{Z}_i], \mathbb{E}[\boldsymbol{Z}_j]\rangle = \langle\boldsymbol{0}, \boldsymbol{0}\rangle = 0.$$

For $i = j$,

$$\mathbb{E}[\langle \boldsymbol{Z}_i, \boldsymbol{Z}_i \rangle] = \mathbb{E}\|\boldsymbol{Z}_i\|_2^2,$$

hence

$$\mathbb{E}\left[\left\|\sum_{j=1}^{k} \boldsymbol{Z}_j\right\|_2^2\right] = \sum_{i=1}^{k} \mathbb{E}\|\boldsymbol{Z}_i\|_2^2 + 0 = \sum_{j=1}^{k} \mathbb{E}\|\boldsymbol{Z}_j\|_2^2.$$

4 [**Vector Balancing Problem**] Let $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n \in \mathbb{R}^n$ satisfy $\|\boldsymbol{x}_i\|_2 \leqslant 1$.

(1) Prove: there exists a way to assign signs $\pm$ to each vector such that $\pm\boldsymbol{x}_1 \pm \boldsymbol{x}_2 \pm \cdots \pm \boldsymbol{x}_n \in \sqrt{n}\mathbb{B}_2^n$, i.e., there exist $\varepsilon_1, \cdots, \varepsilon_n \in \{-1, 1\}$ such that

$$\left\|\sum_{i=1}^{n} \varepsilon_i \boldsymbol{x}_i\right\|_2 \leqslant \sqrt{n}.$$

(2) Show that the above upper bound is optimal.

Rk. In fact, one can prove

$$\left\|\sum_{i=1}^{n} \varepsilon_i \boldsymbol{x}_i\right\|_\infty \leqslant \sqrt{n \log n}.$$

In 1985, Joel Spencer proposed "Six Standard Deviations Suffice," improving the upper bound of the above inequality to

$$\left\|\sum_{i=1}^{n} \varepsilon_i \boldsymbol{x}_i\right\|_\infty \leqslant 6\sqrt{n}.$$

In 1998, Wojciech Banaszczyk generalized the Spencer-type inequality: let $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n \in \mathbb{R}^n$ satisfy $\|\boldsymbol{x}_i\|_2 \leqslant 1$. For any convex body K with Gaussian measure $\gamma_n(K) \geqslant 1/2$, there exist signs $\varepsilon_i \in \{-1, 1\}$ such that

$$\sum_{k=1}^{n} \varepsilon_k \boldsymbol{x}_k \in C \cdot K,$$

i.e.,

$$\left\|\sum_{k=1}^{n} \varepsilon_k \boldsymbol{x}_k\right\|_\infty \leqslant C\sqrt{\log n}.$$

In 2010, Nikhil Bansal gave the first polynomial-time algorithm for finding such a coloring, using SDP and Brownian motion simulation. In 2012, Lovett and Meka proposed the Gram-Schmidt Walk algorithm, which achieves Spencer's bound constructively via random walks inside a polytope.

Prf. (1) Let $\varepsilon_1, \cdots, \varepsilon_n$ i.i.d. $\sim$ Rademacher(1/2), i.e.,

$$\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = \frac{1}{2},$$

then $\mathbb{E}[\varepsilon_i] = 0$, $\mathrm{Var}(\varepsilon_i) = 1$.

Let $\boldsymbol{S} = \sum_{i=1}^{n} \varepsilon_i \boldsymbol{x}_i$, then

$$\begin{aligned}
\mathbb{E}\|\boldsymbol{S}\|_2^2 &= \mathbb{E}\left[\left\|\sum_{i=1}^{n} \varepsilon_i \boldsymbol{x}_i\right\|_2^2\right] \\
&= \mathbb{E}\left[\left\langle \sum_{i=1}^{n} \varepsilon_i \boldsymbol{x}_i, \sum_{j=1}^{n} \varepsilon_j \boldsymbol{x}_j \right\rangle\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{n} \varepsilon_i \varepsilon_j \langle \boldsymbol{x}_i, \boldsymbol{x}_j \rangle\right]
\end{aligned}$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n}\langle \boldsymbol{x}_i, \boldsymbol{x}_j\rangle \mathbb{E}[\varepsilon_i\varepsilon_j],$$

where

$$\mathbb{E}[\varepsilon_i\varepsilon_j] = \begin{cases} 0, & i \neq j, \\ 1, & i = j \end{cases},$$

hence

$$\mathbb{E}\|\boldsymbol{S}\|_2^2 = \sum_{i=1}^{n}\langle \boldsymbol{x}_i, \boldsymbol{x}_i\rangle \cdot 1 = \sum_{i=1}^{n}\|\boldsymbol{x}_i\|_2^2 \leqslant \sum_{i=1}^{n} 1 = n,$$

Thus, by the averaging principle (*probabilistic method*), there exist $\varepsilon_1, \cdots, \varepsilon_n \in \{-1, 1\}$ such that

$$\left\|\sum_{i=1}^{n}\varepsilon_i\boldsymbol{x}_i\right\|_2^2 \leqslant n,$$

Taking square roots yields

$$\left\|\sum_{i=1}^{n}\varepsilon_i\boldsymbol{x}_i\right\|_2 \leqslant \sqrt{n}.$$

(2) Let $(\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n) = (\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n)$ be the standard orthonormal basis, satisfying $\|\boldsymbol{x}_i\|_2 = 1 \leqslant 1$, then

$$\boldsymbol{S} = \sum_{i=1}^{n}\varepsilon_i\boldsymbol{e}_i = (\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n)^\top,$$

hence

$$\|\boldsymbol{S}\|_2 = \sqrt{\sum_{i=1}^{n}\varepsilon_i^2} = \sqrt{\sum_{i=1}^{n} 1} = \sqrt{n},$$

so regardless of how the signs are assigned, the norm of the vector sum is identically $\sqrt{n}$. Thus the upper bound constructed above cannot be improved.

5 [**Asymptotic Tightness of the Approximate Carathéodory Theorem**] Construct a counterexample showing that the bound in Theorem T0.1.2 (*Approximate Caratheodory theorem*) is nearly tight.

Construct $T \subset \mathbb{R}^n$, let $\boldsymbol{x} \in \operatorname{conv}(T)$, such that for any convex combination $\sum_{j=1}^{k}\lambda_j\boldsymbol{x}_j$ of $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_k \in T$,

$$\left\|\boldsymbol{x} - \sum_{j=1}^{k}\lambda_j\boldsymbol{x}_j\right\|_2 \geqslant \sqrt{\frac{1}{k} - \frac{1}{n}}.$$

Then show: as $n \to \infty$, Theorem T0.1.2 (*Approximate Caratheodory theorem*) is asymptotically tight.

Prf. Construct $T = \{\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n\}$, and let

$$\boldsymbol{x} = \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{e}_i = \left(\frac{1}{n}, \frac{1}{n}, \cdots, \frac{1}{n}\right)^\top$$

be the barycenter of T.

(*Merge repeated vertices. If the support has size $r < k$, the calculation below with r in place of k yields a stronger lower bound; hence it suffices to consider $r = k$.*)

After relabeling, let $(\boldsymbol{x}_1, \cdots, \boldsymbol{x}_k) = (\boldsymbol{e}_1, \cdots, \boldsymbol{e}_k)$. Then

$$\boldsymbol{y} = \sum_{j=1}^{k}\lambda_j\boldsymbol{e}_j = (\lambda_1, \lambda_2, \cdots, \lambda_k, 0, \cdots, 0)^\top,$$

hence

$$\|\boldsymbol{x}-\boldsymbol{y}\|_2^2=\sum_{j=1}^{k}\left(\frac{1}{n}-\lambda_j\right)^2+\sum_{j=k+1}^{n}\left(\frac{1}{n}\right)^2=\sum_{j=1}^{k}\left(\frac{1}{n}-\lambda_j\right)^2+\frac{1}{n}-\frac{k}{n^2},$$

The problem reduces to a constrained optimization over $\lambda_1,\cdots,\lambda_j$.

For $\sum_{j=1}^{k}\lambda_j=1$, by symmetry (*or by the Cauchy–Schwarz inequality*), we obtain that when $\lambda_j=1/k$,

$$\min_{\sum_{j=1}^{k}\lambda_j=1}\sum_{j=1}^{k}\left(\frac{1}{n}-\lambda_j\right)^2=\sum_{j=1}^{k}\left(\frac{1}{n}-\frac{1}{k}\right)^2=k\left(\frac{1}{n}-\frac{1}{k}\right)^2=\frac{1}{k}-\frac{2}{n}+\frac{k}{n^2},$$

hence

$$\min\|\boldsymbol{x}-\boldsymbol{y}\|_2^2=\left(\frac{1}{k}-\frac{2}{n}+\frac{k}{n^2}\right)+\left(\frac{1}{n}-\frac{k}{n^2}\right)=\frac{1}{k}-\frac{1}{n},$$

thus

$$\left\|\boldsymbol{x}-\sum_{j=1}^{k}\lambda_j\boldsymbol{x}_j\right\|_2\geqslant\sqrt{\frac{1}{k}-\frac{1}{n}}\to\frac{1}{\sqrt{k}},$$

so the upper bound constructed above cannot be improved.

6 [**Bounds on Binomial Coefficients**] Prove:

$$\left(\frac{n}{k}\right)^k\leqslant\binom{n}{k}\leqslant\sum_{j=0}^{k}\binom{n}{j}\leqslant\left(\frac{\mathrm{e}n}{k}\right)^k.\qquad(1\leqslant k\leqslant n)$$

Prf. For $(n/k)^k\leqslant\binom{n}{k}$,

$$\binom{n}{k}=\frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots1}=\prod_{i=0}^{k-1}\frac{n-i}{k-i}\geqslant\prod_{i=0}^{k-1}\frac{n}{k}=\left(\frac{n}{k}\right)^k.$$

For $\binom{n}{k}\leqslant\sum_{j=0}^{k}\binom{n}{j}$,

$$\sum_{j=0}^{k}\binom{n}{j}=\binom{n}{k}+\binom{n}{k-1}+\cdots+\binom{n}{0}\geqslant\binom{n}{k}.$$

For $\sum_{j=0}^{k}\binom{n}{j}\leqslant(\mathrm{e}n/k)^k$, the binomial theorem gives

$$(1+x)^n=\sum_{j=0}^{n}\binom{n}{j}x^j\geqslant\sum_{j=0}^{k}\binom{n}{j}x^j.$$

Dividing both sides by x^k gives

$$\frac{(1+x)^n}{x^k}\geqslant\sum_{j=0}^{k}\binom{n}{j}x^{j-k}.$$

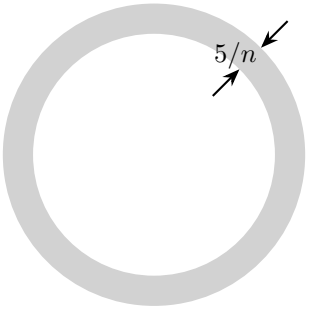
For $x\in(0,1]$, $x^{j-k}\geqslant1$, so

$$\sum_{j=0}^{k}\binom{n}{j}\leqslant\sum_{j=0}^{k}\binom{n}{j}x^{j-k}\leqslant\frac{(1+x)^n}{x^k}.$$

Set $x=k/n$. Then

$$\sum_{j=0}^{k}\binom{n}{j}\leqslant\frac{(1+k/n)^n}{(k/n)^k}=\left(\frac{n}{k}\right)^k\left(1+\frac{k}{n}\right)^n\leqslant\left(\frac{n}{k}\right)^k\left(\mathrm{e}^{\frac{k}{n}}\right)^n=\left(\frac{\mathrm{e}n}{k}\right)^k.$$

7 [**The Thin Shell Phenomenon**] For $n\geqslant6$, as shown in the figure, consider the set of points in $\mathbb{B}_2^n$ whose distance from the surface is at most $5/n$. Prove that the volume of this set accounts for more than 99% of the total volume of the ball.



Prf. Let $V_n(R) = \mathrm{Vol}(B_n(R))$ be the volume of the n-dimensional ball. Then $V_n(R) \propto R^n$, namely,

$$V_n(R) = \mathrm{Vol}(B_n(R)) = C_n R^n, \qquad \left(C_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)}\right)$$

Therefore

$$V_{\text{shell}} = V_n(1) - V_n\left(1 - \frac{5}{n}\right),$$

It remains to prove

$$\frac{V_n(1-5/n)}{V_n(1)} < 0.01.$$

For $n > 5$,

$$\frac{V_n(1-5/n)}{V_n(1)} = \frac{C_n(1-5/n)^n}{C_n(1)^n} = \left(1 - \frac{5}{n}\right)^n \leqslant (\mathrm{e}^{-5/n})^n = \mathrm{e}^{-5} = 0.00673795.$$

8 [**The Thin Shell Phenomenon (continued)**] Let $\boldsymbol{X} \sim U(\mathbb{B}_2^n)$. Prove:

$$\mathbb{E}\|\boldsymbol{X}\|_2 = \frac{n}{n+1}.$$

Prf. It is easy to see that

$$F_R(r) = \mathbb{P}\{\|\boldsymbol{X}\|_2 \leqslant r\} = \frac{\mathrm{Vol}(B_n(r))}{\mathrm{Vol}(B_n(1))} = \frac{C_n r^n}{C_n 1^n} = r^n, \qquad (0 < r < 1)$$

hence

$$f_R(r) = \frac{\mathrm{d}}{\mathrm{d}r} F_R(r) = n r^{n-1}, \qquad (0 < r < 1)$$

thus

$$\mathbb{E}[R] = \int_{-\infty}^{+\infty} r f_R(r) \mathrm{d}r = \int_0^1 n r^n \mathrm{d}r = \frac{n}{n+1}.$$

9 [**The Carl–Pajor Theorem**] Improve Theorem T0.1.4 (*Volume of a polytope*) by replacing N with N/n. Let $P \subset \mathbb{B}_2^n$ be a polytope generated by $N \geqslant n$ vertices. Denote $B = \mathbb{B}_2^n$. Prove:

$$\left(\frac{\mathrm{Vol}(P)}{\mathrm{Vol}(B)}\right)^{1/n} \leqslant C\sqrt{\frac{\log(\mathrm{e}N/n)}{n}}. \qquad (C > 0)$$

Prf. Let $P = \mathrm{conv}\{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N\}$, where $\|\boldsymbol{x}_i\|_2 \leqslant 1$.

Solution 1 Maurey's empirical method

- Covering-number estimate

  By Theorem T0.1.2 (*Approximate Caratheodory theorem*), for every $\boldsymbol{x} \in P$, there exist $\boldsymbol{x}_{i_1}, \cdots, \boldsymbol{x}_{i_k}$ such that

$$\left\| \boldsymbol{x} - \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{x}_{i_j} \right\|_2 \leqslant \frac{1}{\sqrt{k}}.$$

Construct the set of all averages of k vertices,

$$\mathcal{N}_k = \left\{ \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{x}_{i_j} : \ \boldsymbol{x}_{i_1}, \cdots, \boldsymbol{x}_{i_k} \in \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N\} \right\}.$$

Since an average does not depend on the order in which vertices are selected, the stars-and-bars argument gives

$$|\mathcal{N}_k| \leqslant \binom{N+k-1}{k} \leqslant \left(\frac{\mathrm{e}(N+k)}{k} \right)^k.$$

(*The last step follows from Exercise Q0.6 (Bounds on Binomial Coefficients).*)

Thus P is covered by at most $|\mathcal{N}_k|$ Euclidean balls with centers in $\mathcal{N}_k$ and radius $1/\sqrt{k}$, namely,

$$P \subset \bigcup_{c \in \mathcal{N}_k} \left(c + \frac{1}{\sqrt{k}} B \right).$$

By the subadditivity of Lebesgue measure,

$$\mathrm{Vol}(P) \leqslant |\mathcal{N}_k| \cdot k^{-n/2} \, \mathrm{Vol}(B).$$

Hence

$$\left(\frac{\mathrm{Vol}(P)}{\mathrm{Vol}(B)} \right)^{1/n} \leqslant |\mathcal{N}_k|^{1/n} \cdot \frac{1}{\sqrt{k}} \leqslant \underbrace{\left(\frac{\mathrm{e}(N+k)}{k} \right)^{k/n} \frac{1}{\sqrt{k}}}_{F(k)}.$$

(*It remains to optimize the upper bound.*)

- Choosing the optimal coefficient k

  (*To obtain the scale $\sqrt{\log(\mathrm{e}N/n)/n}$ for $F(k)$, $k^{-1/2}$ must have scale $\sqrt{L/n}$, where $L = \log(\mathrm{e}N/n)$.*

  If k has scale n/L, then k/n has scale $1/L$; if the logarithmic factor has scale L, the first factor in $F(k)$ is an absolute constant.)

  Consider the possible values of L:

  a. If $L \geqslant n/4$, then $P \subset B$, and

$$\left(\frac{\mathrm{Vol}(P)}{\mathrm{Vol}(B)} \right)^{1/n} \leqslant 1 \leqslant 2\sqrt{\frac{L}{n}}.$$

  b. For $L < n/4$, let $k = \lfloor n/2L \rfloor$. Then

$$\frac{\mathrm{e}(N+k)}{k} \leqslant \frac{2\mathrm{e}N}{k} \leqslant 8\mathrm{e}\left(\frac{N}{n}\right) L,$$

  and

$$\begin{aligned} \frac{k}{n} \log\left(\frac{\mathrm{e}(N+k)}{k} \right) &\leqslant \frac{1}{2L} \log\left[8\mathrm{e}\left(\frac{N}{n}\right) L \right] \\ &= \frac{\log(8\mathrm{e}) + \log(N/n) + \log L}{2L} \\ &\leqslant \frac{4L + \log(N/n) + \log L}{2L} \leqslant 3, \end{aligned}$$

  Thus

$$\left(\frac{\mathrm{Vol}(P)}{\mathrm{Vol}(B)} \right)^{1/n} \leqslant \mathrm{e}^3 \cdot 2\sqrt{\frac{L}{n}} =: C\sqrt{\frac{\log(\mathrm{e}N/n)}{n}}. \qquad (C = 2\mathrm{e}^3)$$

Solution 2 Entropy-number method

Construct the linear operator $T : \ell_1^N \to \ell_2^n$ by

$$T(\boldsymbol{a}) = \sum_{i=1}^{N} a_i \boldsymbol{x}_i, \qquad (\boldsymbol{a} = (a_1, \cdots, a_N) \in \ell_1^N)$$

This satisfies $\|T\| = \max_{1 \leqslant i \leqslant N} \|\boldsymbol{x}_i\|_2 \leqslant 1$, so $T(\mathbb{B}_1^N) = \mathrm{conv}\{\pm\boldsymbol{x}_1, \cdots, \pm\boldsymbol{x}_N\}$, and hence $P \subset T(\mathbb{B}_1^N)$.

Then let $e_k(T)$ be the k-th dyadic entropy number of T (*the smallest radius with which* 2^{k-1} *balls cover* $T(\mathbb{B}_1^N)$), i.e.,

$$e_k(T) = \inf\{\varepsilon > 0 : \mathcal{N}(T(\mathbb{B}_1^N), \varepsilon\mathbb{B}_2^n \leqslant 2^{k-1})\}.$$

State the following lemmas:

Lemma 1 (*Carl's inequality, 1981*) Let T satisfy $\operatorname{rank}(T) = n$, and let $e_n(T)$ be its n-th dyadic entropy number. Then

$$\left(\frac{\operatorname{Vol}(T(\mathbb{B}_1^N))}{\operatorname{Vol}(B)}\right)^{1/n} \leqslant c_1 e_n(T).$$

Lemma 2 (*Maurey's lemma, 1988*) Let H be a Hilbert space and $T : \ell_1^N \to H$ a linear operator, then

$$e_k(T) \leqslant c_2\|T\|\sqrt{\frac{\log(1+N/k)}{k}}. \qquad (1 \leqslant k \leqslant n)$$

(*The proofs appear below.*)

If $\operatorname{rank}(T) < n$, then $T(\mathbb{B}_1^N)$, and hence P, is contained in a proper subspace, so $\operatorname{Vol}(P) = 0$ and the conclusion is immediate. Henceforth assume $\operatorname{rank}(T) = n$, so Lemma 1 applies. Since $P \subset T(\mathbb{B}_1^N)$, Lemma 1 gives

$$\left(\frac{\operatorname{Vol}(P)}{\operatorname{Vol}(B)}\right)^{1/n} \leqslant c_1 e_n(T),$$

By Lemma 2, set $k = n$ to obtain

$$\begin{aligned} e_n(T) &\leqslant c_2\|T\|\sqrt{\frac{\log(1+N/n)}{n}} \\ &\leqslant C'\sqrt{\frac{\log(\mathrm{e}N/n)}{n}}. \end{aligned}$$

(*For* $x \geqslant 1$, $\log(1+x) \leqslant \log 2x = \log 2 + \log x \leqslant c' \log \mathrm{e}x$.) Thus,

$$\begin{aligned} \left(\frac{\operatorname{Vol}(P)}{\operatorname{Vol}(B)}\right)^{1/n} &\leqslant c_1 e_n(T) \\ &\leqslant c_1 \cdot C'\sqrt{\frac{\log(\mathrm{e}N/n)}{n}} \\ &=: C\sqrt{\frac{\log(\mathrm{e}N/n)}{n}}. \qquad (C = c_1 C') \end{aligned}$$

Rk. Proof of Lemma 1 (*Carl's inequality*):

By the definition of the dyadic entropy number $e_n(T)$,

$$e_n(T) = \inf\left\{\varepsilon > 0 : T(\mathbb{B}_1^N) \subset \bigcup_{j=1}^{2^{n-1}} B_2(\boldsymbol{x}_j, \varepsilon); \boldsymbol{x}_1, \cdots, \boldsymbol{x}_{2^{n-1}} \in \mathbb{R}^n\right\},$$

for every $\varepsilon > e_n(T)$, there exist $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_{2^{n-1}} \in \mathbb{R}^n$ such that $T(\mathbb{B}_1^N)$ is completely covered by these 2^{n-1} Euclidean balls of radius ε,

$$T(\mathbb{B}_1^N) \subset \bigcup_{j=1}^{2^{n-1}} (\boldsymbol{x}_j + \varepsilon B).$$

Hence, by the subadditivity of Lebesgue measure,

$$\operatorname{Vol}(T(\mathbb{B}_1^N)) \leqslant \operatorname{Vol}(\bigcup_{j=1}^{2^{n-1}} (\boldsymbol{x}_j + \varepsilon B)) \leqslant \sum_{j=1}^{2^{n-1}} \operatorname{Vol}(\boldsymbol{x}_j + \varepsilon B) = 2^{n-1} \cdot \varepsilon^n \operatorname{Vol}(B).$$

Dividing both sides by $\operatorname{Vol}(B)$ and taking n-th roots gives

$$\left(\frac{\operatorname{Vol}(T(\mathbb{B}_1^N))}{\operatorname{Vol}(B)}\right)^{1/n} \leqslant (2^{n-1}\varepsilon^n)^{1/n} = 2^{\frac{n-1}{n}}\varepsilon < 2\varepsilon.$$

Letting $\varepsilon \to e_n(T)^+$ yields

$$\left(\frac{\mathrm{Vol}(T(\mathbb{B}_1^N))}{\mathrm{Vol}(B)}\right)^{1/n} \leqslant 2^{\frac{n-1}{n}} e_n(T) \leqslant 2e_n(T). \qquad \left(c_1 = 2^{\frac{n-1}{n}} \leqslant 2\right)$$

Rk. Proof of Lemma 2 (*Maurey's lemma*)

Without loss of generality, suppose that $\|T\| = 1$ (*if* $\|T\| \neq 0$, *it suffices to consider* $T/\|T\|$).

Let $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_N$ be the standard basis of ℓ_1^N, and write $\boldsymbol{v}_i = T(\boldsymbol{e}_i) \in H$. Then the definition of the operator norm gives

$$\|T\| = \sup_{\|\boldsymbol{a}\|_1 \leqslant 1} \|T(\boldsymbol{a})\|_H = \max_{1 \leqslant i \leqslant N} \|\boldsymbol{v}_i\|_H = 1.$$

Since the closed unit ball of ℓ_1^N is $\mathbb{B}_1^N = \mathrm{conv}\{\pm\boldsymbol{e}_1, \cdots, \pm\boldsymbol{e}_N\}$, its image under T is the symmetric convex hull

$$T(\mathbb{B}_1^N) = \mathrm{conv}\{\pm\boldsymbol{v}_1, \cdots, \pm\boldsymbol{v}_N\},$$

and denote the set of extreme points by $V = \{\pm\boldsymbol{v}_1, \cdots, \pm\boldsymbol{v}_N\} \subset H$.

- Maurey's empirical approximation (*variance reduction*)

  Let $\boldsymbol{x} \in T(\mathbb{B}_1^N) = \mathrm{conv}(V)$. Then there exist nonnegative weights $\lambda_1, \cdots, \lambda_{2N} \geqslant 0$ (*satisfying* $\sum_{j=1}^{2N} \lambda_j = 1$) and $\boldsymbol{w}_1, \cdots, \boldsymbol{w}_{2N} \in V$ such that

  $$\boldsymbol{x} = \sum_{j=1}^{2N} \lambda_j \boldsymbol{w}_j.$$

  Define the discrete random vector $Z \in H$ by

  $$\mathbb{P}\{Z = \boldsymbol{w}_j\} = \lambda_j, \qquad (j = 1, \cdots, 2N)$$

  Its numerical characteristics are

  $$\mathbb{E}[Z] = \sum_{j=1}^{2N} \lambda_j \boldsymbol{w}_j = \boldsymbol{x}, \mathbb{E}[\|Z\|_H^2] = \sum_{j=1}^{2N} \lambda_j \|\boldsymbol{w}_j\|_H^2 \leqslant \sum_{j=1}^{2N} \lambda_j \cdot 1^2 = 1,$$

  so, for independent copies $Z_1, \cdots, Z_m$ i.i.d. $\sim Z$, the empirical mean $\bar{Z}_m = \frac{1}{m}\sum_{l=1}^m Z_l$ has mean-square error in the Hilbert space equal to

  $$\mathbb{E}\left[\left\|\boldsymbol{x} - \frac{1}{m}\sum_{l=1}^m Z_l\right\|_H^2\right] = \mathbb{E}\left[\left\|\frac{1}{m}\sum_{l=1}^m (Z_l - \mathbb{E}[Z_l])\right\|_H^2\right] = \frac{1}{m^2}\sum_{l=1}^m\sum_{r=1}^m \mathbb{E}[\langle Z_l - \mathbb{E}[Z_l], Z_r - \mathbb{E}[Z_r]\rangle_H],$$

  where the cross term $\mathbb{E}[\langle Z_l - \mathbb{E}[Z_l], Z_r - \mathbb{E}[Z_r]\rangle_H] = 0$ when $l \neq r$. Therefore,

  $$\mathbb{E}[\|\boldsymbol{x} - \bar{Z}_m\|_H^2] = \frac{1}{m^2}\sum_{l=1}^m \mathbb{E}[\|Z_l - \mathbb{E}[Z_l]\|_H^2] = \frac{1}{m}(-\|\boldsymbol{x}\|_H^2) \leqslant \frac{1}{m},$$

  so the probabilistic method gives $(\boldsymbol{z}_1, \cdots, \boldsymbol{z}_m) \in V^m$ such that

  $$\left\|\boldsymbol{x} - \frac{1}{m}\sum_{l=1}^m \boldsymbol{z}_l\right\|_H \leqslant \frac{1}{\sqrt{m}}.$$

  This shows that $T(\mathbb{B}_1^N)$ is completely covered by Euclidean balls centered at the averages $\frac{1}{m}\sum_{l=1}^m \boldsymbol{z}_l$ and with radius $\varepsilon = 1/\sqrt{m}$.

- Counting cover centers and stars-and-bars

  By stars-and-bars, the number of centers (*the number of unordered selections with replacement of* m *elements from the set* V *of size* $2N$) is

  $$\mathcal{N}\left(T(\mathbb{B}_1^N), \frac{1}{\sqrt{m}}\mathbb{B}_H\right) = \binom{2N+m-1}{m} \leqslant \left(\frac{\mathrm{e}(2N+m)}{m}\right)^m \leqslant \left(1 + \frac{2\mathrm{e}N}{m}\right)^m, \tag{I}$$

  For every continuous accuracy $\varepsilon \in (0, 1]$, let $m = \lceil \varepsilon^{-2} \rceil \leqslant 2\varepsilon^{-2}$. Substituting it into (I) and taking base-2 logarithms

on both sides gives

$$\log_2 \mathcal{N}(T(\mathbb{B}_1^N), \varepsilon\mathbb{B}_H) \leqslant \frac{C_1}{\varepsilon^2} \log_2(1 + N\varepsilon^2).$$

By the definition of the dyadic entropy number,

$$e_k(T) = \inf\{\varepsilon > 0 : \log_2 \mathcal{N}(T(\mathbb{B}_1^N), \varepsilon\mathbb{B}_H) \leqslant k - 1\},$$

so to prove $e_k(T) \leqslant \varepsilon$, it suffices to find $\varepsilon > 0$ satisfying the following implicit inequality,

$$\frac{1}{\varepsilon^2} \log_2(1 + N\varepsilon^2) \leqslant c_0 k. \tag{II}$$

- Inverting a monotone function

  Introduce $\delta = \varepsilon^{-2} \geqslant 1$. Then (II) is equivalent to

  $$\Phi(\delta) := \delta \log_2\left(1 + \frac{N}{\delta}\right) \leqslant c_0 k,$$

  where Φ is increasing on $[1, +\infty)$.

  Substituting the trial solution $\delta^* := \frac{c_1 k}{\log_2(1+N/k)}$ gives

  $$\begin{aligned}
  \Phi(\delta^*) &= \frac{c_1 k}{\log_2(1 + N/k)} \log_2\left(1 + \frac{N \log_2(1 + N/k)}{c_1 k}\right) \\
  &\leqslant \frac{c_1 k}{\log_2(1 + N/k)} \log_2\left[\left(1 + \frac{N}{k}\right)^2\right] \qquad \left(\text{choose } c_1 \text{ sufficiently small so that } \frac{\log_2(1+u)}{c_1} \leqslant (1+u)\right) \\
  &= \frac{c_1 k}{\log_2(1 + N/k)} \cdot 2\log_2\left(1 + \frac{N}{k}\right) \\
  &= 2c_1 k,
  \end{aligned}$$

  and hence

  $$e_k(T) \leqslant \frac{1}{\sqrt{\delta^*}} = \frac{1}{\sqrt{c_1}} \sqrt{\frac{\log_2(1 + N/k)}{k}},$$

  Restoring the operator-norm factor $\|T\|$ and changing the base of the logarithm gives

  $$e_k(T) \leqslant c_2 \|T\| \sqrt{\frac{\log(1 + N/k)}{k}}.$$

A Quick Refresher on Analysis and Probability

1

Keypoints

Section 1.1 Convex sets and functions Q1.1 ∼ Q1.6

Section 1.2 Norms and inner products

Section 1.3 Random variables and random vectors

Section 1.4 Union bound Q1.7 ∼ Q1.10

Lemma 1.4.1 (*Union bound*)

For any events $E_1, \cdots, E_n$, we have

$$\mathbb{P}\{\bigcup_{i=1}^{n} E_i\} \leqslant \sum_{i=1}^{n} \mathbb{P}(E_i).$$

Example 1.4.2 (*Dense random graphs have no isolated vertices*)

Suppose n freshmen arrive on campus, with each pair of students becoming friends with probability p independently of all other pairs. If $p \geqslant 4\log(n)/n$, then with high probability there are no friendless students (w.p. $1 - 1/n$).

Section 1.5 Conditioning

Example 1.5.1 (*Probability of a perfect cancelation*)

Let $a_1, \cdots, a_n$ be real numbers that are not all zero, let $X_1, \cdots, X_n$ i.i.d. ∼ Rademacher, and write $S_n = \sum_{i=1}^{n} a_i X_i$. Then

$$\mathbb{P}\{S_n = 0\} \leqslant \frac{1}{2}.$$

Section 1.6 Probabilistic inequalities Q1.15 ∼ Q1.16

Lemma 1.6.1 (*Integrated tail formula*)

Let X be a nonnegative random variable. Then

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}\{X > t\}\mathrm{d}t,$$

The two sides are either both finite or both infinite.

Proposition 1.6.2 (*Markov inequality*)

Let X be a nonnegative random variable. Then

$$\mathbb{P}\{X \geqslant t\} \leqslant \frac{\mathbb{E}[X]}{t}. \qquad (t > 0)$$

Corollary 1.6.3 (*Chebyshev inequality*)

Let X be a random variable with mean μ and variance σ^2. Then, for any $t > 0$, we have

$$\mathbb{P}\{|X - \mu| \geqslant t\} \leqslant \frac{\sigma^2}{t^2}.$$

Section 1.7 Limit theorems Q1.17 ~ Q1.19

Theorem 1.7.1 (*Strong law of large numbers*)

Let $\{X_N\}$ be a sequence of i.i.d. random variables with mean μ, and write $S_N = \sum_{i=1}^N X_i$. Then

$$\frac{S_N}{N} \to \mu \text{ (a.s.)}.$$

Definition 1.7.2 (*Normal distribution*)

A random variable X is called a standard normal random variable, denoted by $X \sim N(0,1)$, if it has density

$$f(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2},$$

in which case X has mean zero and variance 1.

More generally, $X \sim N(\mu, \sigma^2)$ if

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)},$$

in which case X can be written as $X = \mu + \sigma Z$, where $Z \sim N(0,1)$.

Theorem 1.7.3 (*Lindeberg-Levy central limit theorem*)

Let $\{X_N\}$ be a sequence of i.i.d. random variables with mean μ and variance σ^2, write $S_N = \sum_{i=1}^N a_i X_i$, and normalize it to a random variable with zero mean and unit variance,

$$Z_N = \frac{S_N - \mathbb{E}[S_N]}{\sqrt{\mathrm{Var}(S_N)}} = \frac{1}{\sigma\sqrt{N}}\sum_{i=1}^N (X_i - \mu),$$

then

$$Z_N \xrightarrow{d} N(0,1).$$

Definition 1.7.5 (*Poisson distribution*)

A random variable Z has Poisson distribution with parameter $\lambda > 0$, denoted by $Z \sim P(\lambda)$, if $\mathrm{supp}(Z) = \mathbb{Z}_0^+$ and

$$\mathbb{P}\{Z = k\} = \mathrm{e}^{-\lambda}\frac{\lambda^k}{k!}. \qquad (k \in \mathbb{Z}_0^+)$$

Theorem 1.7.6 (*Poisson limit theorem*)

Consider independent random variables $X_{N,i} \sim \text{Bernoulli}(p_{N,i})$, where $1 \leqslant i \leqslant N$, and write $S_N = \sum_{i=1}^{N} X_{N,i}$. If, as $N \to \infty$,

$$\max_{i \leqslant N} p_{N,i} \to 0, \mathbb{E}[S_N] = \sum_{i=1}^{N} p_{N,i} \to \lambda < \infty,$$

then, as $N \to \infty$,

$$S_N \xrightarrow{d} P(\lambda).$$

Lemma 1.7.7 (*Stirling approximation*)

As $n \to \infty$,

$$n! = \sqrt{2\pi n}\left(\frac{n}{\mathrm{e}}\right)^n (1 + o(1)).$$

Lemma 1.7.8 (*Bounds on the factorial*)

$$\left(\frac{n}{\mathrm{e}}\right)^n \leqslant n! \leqslant \mathrm{e}n\left(\frac{n}{\mathrm{e}}\right)^n.$$

Exercises

1 Let $T \subset \mathbb{R}^n$. Prove: $\text{conv}(T)$ is a convex set.

Prf. It is easy to see that

$$\text{conv}(T) = \left\{ \sum_{i=1}^{k} \lambda_i \boldsymbol{x}_i : k \in \mathbb{N}^+, \boldsymbol{x}_i \in T, \lambda_i \geqslant 0, \sum_{i=1}^{k} \lambda_i = 1 \right\}.$$

To prove that $\text{conv}(T)$ is convex, it suffices to show that for any $\boldsymbol{u}, \boldsymbol{v} \in \text{conv}(T)$, we have $\alpha\boldsymbol{u} + (1-\alpha)\boldsymbol{v} \in \text{conv}(T)$ for all $\alpha \in [0,1]$.

Let

$$\begin{aligned}
\boldsymbol{u} &= \sum_{i=1}^{p} \lambda_i \boldsymbol{x}_i, \\
\boldsymbol{v} &= \sum_{j=1}^{q} \mu_j \boldsymbol{y}_j, \\
\boldsymbol{w} &= \alpha\boldsymbol{u} + (1-\alpha)\boldsymbol{v} \\
&= \sum_{i=1}^{p} (\alpha\lambda_i)\boldsymbol{x}_i + \sum_{j=1}^{q} ((1-\alpha)\mu_j)\boldsymbol{y}_j,
\end{aligned}$$

then $\boldsymbol{w}$ satisfies:

1. Non-negativity:

Since $\alpha \in [0,1]$ and $\lambda_i, \mu_j \geqslant 0$, we have $\alpha\lambda_i \geqslant 0$, $(1-\alpha)\mu_j \geqslant 0$.

2. Normalization:

$$\sum_{i=1}^{p} \alpha\lambda_i + \sum_{j=1}^{q} (1-\alpha)\mu_j = \alpha \underbrace{\sum_{i=1}^{p} \lambda_i}_{1} + (1-\alpha) \underbrace{\sum_{j=1}^{q} \mu_j}_{1} = \alpha + (1-\alpha) = 1.$$

Hence $\boldsymbol{w} \in \operatorname{conv}(T)$, i.e., $\operatorname{conv}(T)$ is convex.

2 Let $f_1, \cdots, f_m$ be finitely many convex functions defined on a convex set $D \subset \mathbb{R}^n$, and $f(\boldsymbol{x}) = \max\{f_1(\boldsymbol{x}), \cdots, f_m(\boldsymbol{x})\}$. Prove: f is convex.

Prf. To prove that f is convex, it suffices to show that for any $\boldsymbol{x}, \boldsymbol{y} \in D$,

$$f(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}) \leqslant \alpha f(\boldsymbol{x}) + (1-\alpha) f(\boldsymbol{y}). \qquad (\alpha \in [0,1])$$

Let $f(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}) = f_k(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y})$. Then by the convexity of f_k,

$$\begin{aligned} f(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}) &= f_k(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}) \\ &\leqslant \alpha f_k(\boldsymbol{x}) + (1-\alpha) f_k(\boldsymbol{y}) \\ &\leqslant \alpha f(\boldsymbol{x}) + (1-\alpha) f(\boldsymbol{y}), \end{aligned}$$

hence f is convex.

3 [**Jensen's Inequality**] (1) It is known that the definition of a convex function is

$$f(\lambda x + (1-\lambda)y) \leqslant \lambda f(x) + (1-\lambda) f(y).$$

Prove the multidimensional version of the above definition.

Let $K \subset \mathbb{R}^n$ be a convex subset. Prove: $f : K \to \mathbb{R}$ is convex if and only if for any $\sum_{i=1}^{m} \lambda_i = 1$ with $\lambda_i \geqslant 0$,

$$f\left(\sum_{i=1}^{m} \lambda_i \boldsymbol{x}_i\right) \leqslant \sum_{i=1}^{m} \lambda_i f(\boldsymbol{x}_i).$$

(2) Let $\boldsymbol{X}$ be a random vector with finite $\operatorname{supp}(\boldsymbol{X})$ and $f : \mathbb{R}^n \to \mathbb{R}$ a convex function. Using the conclusion of (1), deduce Jensen's inequality

$$f(\mathbb{E}[\boldsymbol{X}]) \leqslant \mathbb{E}[f(\boldsymbol{X})].$$

Prf. (1) • ($\Longleftarrow$)

Setting $m = 2$ recovers the two-point definition of convexity, hence f is convex.

• ($\Longrightarrow$)

Suppose f is convex.

For $m = 1$: $f(\boldsymbol{x}_1) \leqslant f(\boldsymbol{x}_1)$.

For $m = 2$: $f(\lambda_1\boldsymbol{x}_1 + \lambda_2\boldsymbol{x}_2) \leqslant \lambda_1 f(\boldsymbol{x}_1) + \lambda_2 f(\boldsymbol{x}_2)$.

Assume the inequality holds for $m = k$. For $m = k+1$ with $\sum_{i=1}^{k+1} \lambda_i = 1$:

a. If $\lambda_{k+1} = 1$, then $\lambda_i = 0$ for all $i \neq k+1$, and the conclusion is trivial.

b. If $\lambda_{k+1} < 1$, let $L = \sum_{i=1}^{k} \lambda_i = 1 - \lambda_{k+1} > 0$, then

$$\sum_{i=1}^{k+1} \lambda_i \boldsymbol{x}_i = L\left(\sum_{i=1}^{k} \frac{\lambda_i}{L} \boldsymbol{x}_i\right) + \lambda_{k+1}\boldsymbol{x}_{k+1}.$$

By the two-point definition of convexity,

$$\begin{aligned} f\left(\sum_{i=1}^{k+1} \lambda_i \boldsymbol{x}_i\right) &\leqslant L f\left(\sum_{i=1}^{k} \frac{\lambda_i}{L} \boldsymbol{x}_i\right) + \lambda_{k+1} f(\boldsymbol{x}_{k+1}) \\ &= L \sum_{i=1}^{k} \frac{\lambda_i}{L} f(\boldsymbol{x}_i) + \lambda_{k+1} f(\boldsymbol{x}_{k+1}) \qquad \left(\sum_{i=1}^{k} \frac{\lambda_i}{L} = 1\right) \end{aligned}$$

$$= \sum_{i=1}^{k+1} \lambda_i f(\boldsymbol{x}_i).$$

(2) Let $\operatorname{supp}(\boldsymbol{X}) = \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_m\}$ with probabilities $p_i = \mathbb{P}\{\boldsymbol{X} = \boldsymbol{x}_i\}$, satisfying $p_i \geqslant 0$, $\sum_{i=1}^m p_i = 1$. Then

$$\mathbb{E}[\boldsymbol{X}] = \sum_{i=1}^m p_i \boldsymbol{x}_i, \mathbb{E}[f(\boldsymbol{X})] = \sum_{i=1}^m p_i f(\boldsymbol{x}_i).$$

Applying the conclusion of (1) with $\lambda_i = p_i$ yields

$$f\left(\sum_{i=1}^m p_i \boldsymbol{x}_i\right) \leqslant \sum_{i=1}^m p_i f(\boldsymbol{x}_i),$$

i.e., $f(\mathbb{E}[\boldsymbol{X}]) \leqslant \mathbb{E}[f(\boldsymbol{X})]$.

4 [**Maximum Principle**] Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Prove: for any $T \subset \mathbb{R}^n$,

$$\sup_{\boldsymbol{x} \in \operatorname{conv}(T)} f(\boldsymbol{x}) = \sup_{\boldsymbol{x} \in T} f(\boldsymbol{x}).$$

Prf. Since $T \subset \operatorname{conv}(T)$,

$$\sup_{\boldsymbol{x} \in T} f(\boldsymbol{x}) \leqslant \sup_{\boldsymbol{x} \in \operatorname{conv}(T)} f(\boldsymbol{x}).$$

Let $\boldsymbol{x} \in \operatorname{conv}(T)$. Then there exist $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m \in T$ such that

$$\boldsymbol{x} = \sum_{i=1}^m \lambda_i \boldsymbol{x}_i, \qquad \left(\textstyle\sum_{i=1}^m \lambda_i = 1\right)$$

By Jensen's inequality,

$$f(\boldsymbol{x}) = f\left(\sum_{i=1}^m \lambda_i \boldsymbol{x}_i\right) \leqslant \sum_{i=1}^m \lambda_i f(\boldsymbol{x}_i).$$

where

$$f(\boldsymbol{x}_i) \leqslant \sup_{\boldsymbol{y} \in T} f(\boldsymbol{y}),$$

thus

$$f(\boldsymbol{x}) \leqslant \sum_{i=1}^m \lambda_i \sup_{\boldsymbol{y} \in T} f(\boldsymbol{y}) = \sup_{\boldsymbol{y} \in T} f(\boldsymbol{y}),$$

i.e.,

$$\sup_{\boldsymbol{x} \in \operatorname{conv}(T)} f(\boldsymbol{x}) \leqslant \sup_{\boldsymbol{y} \in T} f(\boldsymbol{y}).$$

Thus,

$$\sup_{\boldsymbol{x} \in \operatorname{conv}(T)} f(\boldsymbol{x}) = \sup_{\boldsymbol{x} \in T} f(\boldsymbol{x}).$$

5 [**Representing the Unit Cube as the Convex Hull of Its Vertices**] Prove: $[-1, 1]^n = \operatorname{conv}(\{-1, 1\}^n)$.

Prf. • Prove $\operatorname{conv}(\{-1, 1\}^n) \subset [-1, 1]^n$

Notice that $[-1, 1]^n$ can be written as the intersection of $2n$ closed half-spaces,

$$[-1, 1]^n = \bigcap_{i=1}^n \{x \in \mathbb{R}^n : x_i \leqslant 1\} \cap \{x \in \mathbb{R}^n : x_i \geqslant -1\}.$$

Since the half-spaces are convex, $[-1, 1]^n$ is convex. Thus the minimality of the convex hull gives $\operatorname{conv}(\{-1, 1\}^n) \subset [-1, 1]^n$.

• Prove $[-1, 1]^n \subset \operatorname{conv}(\{-1, 1\}^n)$

It suffices to prove that for any $\boldsymbol{x} = (x_1, x_2, \cdots, x_n) \in [-1, 1]^n$, there exists $\{\lambda_{\boldsymbol{v}}\}_{\boldsymbol{v} \in \{-1,1\}^n}$ with $\lambda_{\boldsymbol{v}} \geqslant 0$ and

$\sum_{\boldsymbol{v}\in\{-1,1\}^n} \lambda_{\boldsymbol{v}} = 1$ such that

$$\boldsymbol{x} = \sum_{\boldsymbol{v}\in\{-1,1\}^n} \lambda_{\boldsymbol{v}} \boldsymbol{v}.$$

For any $\boldsymbol{v} \in \{-1,1\}^n$, define

$$\lambda_{\boldsymbol{v}} = \prod_{i=1}^{n} \Big(\frac{1+v_i x_i}{2} \Big).$$

(*The construction is explained in the note below.*)

Then:

1. Non-negativity: Omitted.

2. Normalization:

$$\begin{aligned}
\sum_{\boldsymbol{v}\in\{-1,1\}^n} \lambda_{\boldsymbol{v}} &= \sum_{v_1\in\{-1,1\}} \cdots \sum_{v_n\in\{-1,1\}} \prod_{i=1}^{n} \Big(\frac{1+v_i x_i}{2} \Big) \\
&= \prod_{i=1}^{n} \Bigg(\sum_{v_i\in\{-1,1\}} \frac{1+v_i x_i}{2} \Bigg) \\
&= \prod_{i=1}^{n} \Big(\frac{1+x_i+1-x_i}{2} \Big) = \prod_{i=1}^{n} 1 = 1.
\end{aligned}$$

3. Coordinate reconstruction: for the jth coordinate of $\sum_{\boldsymbol{v}\in\{-1,1\}^n} \lambda_{\boldsymbol{v}} \boldsymbol{v}$,

$$\begin{aligned}
\Bigg(\sum_{\boldsymbol{v}\in\{-1,1\}^n} \lambda_{\boldsymbol{v}} \boldsymbol{v} \Bigg)_j &= \sum_{\boldsymbol{v}\in\{-1,1\}^n} \lambda_{\boldsymbol{v}} v_j = \sum_{\boldsymbol{v}\in\{-1,1\}^n} v_j \prod_{i=1}^{n} \Big(\frac{1+v_i x_i}{2} \Big) \\
&= \Bigg(\sum_{v_j\in\{-1,1\}} v_j \frac{1+v_j x_j}{2} \Bigg) \prod_{i\neq j} \Bigg(\sum_{v_i\in\{-1,1\}} \frac{1+v_i x_i}{2} \Bigg) \\
&= \Bigg((+1)\cdot \frac{1+x_j}{2} + (-1)\cdot \frac{1-x_j}{2} \Bigg) \cdot \prod_{i\neq j} 1 \\
&= \Bigg(\frac{1+x_j-1+x_j}{2} \Bigg) \cdot 1 = x_j,
\end{aligned}$$

whence

$$\sum_{\boldsymbol{v}\in\{-1,1\}^n} \lambda_{\boldsymbol{v}} \boldsymbol{v} = \boldsymbol{x}.$$

Thus $[-1,1]^n \subset \operatorname{conv}(\{-1,1\}^n)$.

Rk. Explanation of the construction:

The n-dimensional unit cube is the Cartesian product of n independent one-dimensional line segments, so it suffices to consider the one-dimensional case.

Writing x as a convex combination of $\{-1,1\}$ gives

$$x = p\cdot(+1) + (1-p)\cdot(-1) \implies p = \frac{1+x}{2}.$$

Thus the endpoint weights are

$$(v=1) \quad \lambda_{+1} = p = \frac{1+x}{2},$$

$$(v=-1) \quad \lambda_{-1} = 1-p = \frac{1-x}{2},$$

and the weight is the affine interpolation $\lambda_v = (1+vx)/2$.

6 [**Representing the Cross-Polytope as the Convex Hull of Its Vertices**] Let $\mathbb{B}_1^n = \{\boldsymbol{x} \in \mathbb{R}^n : \sum_{i=1}^n |x_i| \leqslant 1\}$. Prove: $\mathbb{B}_1^n$ is the convex hull of the set consisting of $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n$ and their opposites, i.e.,

$$\mathbb{B}_1^n = \operatorname{conv}(\{\pm\boldsymbol{e}_1, \cdots, \pm\boldsymbol{e}_n\}),$$

and then find an explicit expression for the convex combination of $\boldsymbol{x} \in \mathbb{B}_1^n$ in terms of $\pm\boldsymbol{e}_1, \cdots, \pm\boldsymbol{e}_n$.

Prf. It is easy to see that $\operatorname{conv}(\{\pm\boldsymbol{e}_i\}) \subset \mathbb{B}_1^n$. (*It remains to prove* $\mathbb{B}_1^n \subset \operatorname{conv}(\{\pm\boldsymbol{e}_i\})$.)

For any $\boldsymbol{x} = (x_1, x_2, \cdots, x_n) \in \mathbb{B}_1^n$, let $L = \|\boldsymbol{x}\|_1 = \sum_{i=1}^n |x_i| \leqslant 1$ denote its ℓ^1-norm. Define the positive and negative parts of each component as

$$x_i^+ = \max(0, x_i), x_i^- = \max(0, -x_i),$$

Then

$$\boldsymbol{x} = \sum_{i=1}^n x_i \boldsymbol{e}_i = \sum_{i=1}^n (x_i^+ \boldsymbol{e}_i + x_i^- (-\boldsymbol{e}_i)).$$

(*It is easy to see that* $\sum_{i=1}^n (x_i^+ + x_i^-) = L \leqslant 1$. *If* $L < 1$, *we need to allocate the excess weight* $1 - L$. *Note that*

$$\mathbf{0} = \frac{1}{2}\boldsymbol{e}_1 + \frac{1}{2}(-\boldsymbol{e}_1),$$

so the remaining weight can be assigned to any pair of opposite vertices to cancel their contribution.)

Construct $\lambda_{i,\sigma} \geqslant 0$ for $i \in \{1, \cdots, n\}, \sigma \in \{+, -\}$ with $\sum_{i=1}^n (\lambda_{i,+} + \lambda_{i,-}) = 1$ such that

$$\boldsymbol{x} = \sum_{i=1}^n (\lambda_{i,+} \boldsymbol{e}_i + \lambda_{i,-}(-\boldsymbol{e}_i)).$$

Then define

$$\lambda_{i,+} = \begin{cases} x_1^+ + \frac{1}{2}(1 - L), & i = 1, \\ x_i^+, & i \geqslant 2 \end{cases}, \lambda_{i,-} = \begin{cases} x_1^- + \frac{1}{2}(1 - L), & i = 1, \\ x_i^-, & i \geqslant 2 \end{cases},$$

then:

1. Non-negativity: Since $x_i^{\pm} \geqslant 0$ and $L \leqslant 1$, we have $\lambda_{i,\sigma} \geqslant 0$.
2. Normalization:

$$\sum_{i=1}^n (\lambda_{i,+} + \lambda_{i,-}) = [x_1^+ + x_1^- + (1 - L)] + \sum_{i=2}^n (x_i^+ + x_i^-) = \sum_{i=1}^n |x_i| + 1 - L = L + 1 - L = 1.$$

3. Equivalence:

$$\sum_{i=1}^n (\lambda_{i,+} \boldsymbol{e}_i + \lambda_{i,-}(-\boldsymbol{e}_i)) = (x_1^+ - x_1^-)\boldsymbol{e}_1 + \frac{1 - L}{2}(\boldsymbol{e}_1 - \boldsymbol{e}_1) + \sum_{i=2}^n (x_i^+ - x_i^-)\boldsymbol{e}_i = \sum_{i=1}^n x_i \boldsymbol{e}_i = \boldsymbol{x}.$$

Hence every point in $\mathbb{B}_1^n$ can be expressed as a convex combination of $\{\pm\boldsymbol{e}_1, \cdots, \pm\boldsymbol{e}_n\}$.

7 For Example E1.4.2 (*Dense random graphs have no isolated vertices*), suppose the number N of incoming freshmen is no longer constant but satisfies $N \sim P(\lambda)$. Prove: if $p \geqslant 2\log\lambda/\lambda$, then the probability that there are no friendless students (*i.e., no isolated vertices*) is at least $1 - 1/\lambda$.

Prf. Let $X_i = \mathbf{1}_{A_i}$, where $A_i = \{\text{student } i \text{ is an isolated vertex}\}$.

(*These indicators are generally not independent; the argument below uses only linearity of expectation.*)

Let $S_n = \sum_{i=1}^n X_i$ be the number of isolated vertices. To prove $\mathbb{P}\{S_n = 0\} \geqslant 1 - 1/\lambda$, it suffices to prove $\mathbb{P}\{S_n \geqslant 1\} \leqslant 1/\lambda$. Conditional on $N = n$, the conditional expectation of S_n is

$$\begin{aligned} \mathbb{E}[S_n \mid N = n] &= \sum_{i=1}^n \mathbb{E}[X_i \mid N = n] \\ &= \sum_{i=1}^n \mathbb{P}\{A_i \mid N = n\} \end{aligned}$$

$$= \sum_{i=1}^{n} (1-p)^{n-1} = n(1-p)^{n-1},$$

Thus the expectation of S_n is

$$\begin{aligned}
\mathbb{E}[S_n] = \mathbb{E}[\mathbb{E}[S_n \mid N]] &= \sum_{n=1}^{\infty} \mathbb{E}[S_n \mid N = n] \cdot \mathbb{P}\{N = n\} \\
&= \sum_{n=1}^{\infty} n(1-p)^{n-1} \cdot \frac{\mathrm{e}^{-\lambda}\lambda^n}{n!} \\
&= \lambda \mathrm{e}^{-\lambda} \sum_{n=1}^{\infty} \frac{\lambda^{n-1}(1-p)^{n-1}}{(n-1)!} \\
&= \lambda \mathrm{e}^{-\lambda} \sum_{k=0}^{\infty} \frac{[\lambda(1-p)]^k}{k!} \\
&= \lambda \mathrm{e}^{-\lambda} \cdot \mathrm{e}^{\lambda(1-p)} = \lambda \mathrm{e}^{-\lambda p} \leqslant \lambda \mathrm{e}^{-\lambda \frac{2\log\lambda}{\lambda}} = \frac{1}{\lambda}.
\end{aligned}$$

By Markov's inequality,

$$\mathbb{P}\{S_n \geqslant 1\} \leqslant \mathbb{E}[S_n] \leqslant \frac{1}{\lambda},$$

hence

$$\mathbb{P}\{S_n = 0\} = 1 - \mathbb{P}\{S_n \geqslant 1\} \geqslant 1 - \frac{1}{\lambda}.$$

8 [**Independent Sets in Random Graphs**] Suppose $n \geqslant 7$ students are enrolled in a probability course. Each pair of students independently become friends with probability $1/2$ (*i.e., an edge exists with probability* $1/2$). Prove: the probability that there is no independent set of size greater than $2\log_2 n$ is at least $1 - 1/n$.

Prf. For any subset $S \subset \{1, 2, \cdots, n\}$ of size $k = \lfloor 2\log_2 n \rfloor + 1$, let X_S be the indicator that S is an independent set.

Let $X = \sum_{|S|=k} X_S$ be the number of independent sets of size k. It suffices to prove $\mathbb{P}\{X \geqslant 1\} \leqslant 1/n$, which is equivalent to $\mathbb{P}\{X = 0\} \geqslant 1 - 1/n$.

Since S contains $\binom{k}{2}$ potential edges that are mutually independent,

$$\mathbb{P}\{X_S = 1\} = \left(\frac{1}{2}\right)^{\binom{k}{2}} = 2^{-\frac{k(k-1)}{2}},$$

thus

$$\begin{aligned}
\mathbb{E}[X] = \sum_{|S|=k} \mathbb{E}[X_S] &= \binom{n}{k} 2^{-\frac{k(k-1)}{2}} \\
&\leqslant \frac{n^k}{k!} 2^{-\frac{k(k-1)}{2}} = \frac{1}{k!}\left(n \cdot 2^{-\frac{k-1}{2}}\right)^k && \left(\binom{n}{k} \leqslant \frac{n^k}{k!}\right) \\
&< \frac{1}{k!}\left(n \cdot 2^{-\frac{2\log_2 n - 1}{2}}\right)^k = \frac{1}{k!}\left(\frac{\sqrt{2}}{n}\right)^k. && (k > 2\log_2 n)
\end{aligned}$$

For $n \geqslant 7$, we have $k \geqslant 6$ and $(\sqrt{2}/n)^k/k! < 1/n$; hence Markov's inequality gives

$$\mathbb{P}\{X \geqslant 1\} \leqslant \mathbb{E}[X] < \frac{1}{n}.$$

9 [**Isolated Vertices in Dense Random Graphs (First Moment Method)**] Let $G(n, p_n)$ be a random graph, where n is the number of students (*vertices*) and p_n is the probability that any two students become friends (*an edge exists*). Prove: if for every $\varepsilon > 0$,

$$p_n > \frac{(1+\varepsilon)\log n}{n}.$$

then as $n \to \infty$, the probability that there are no friendless students (*i.e., no isolated vertices*) converges to 1.

Prf. Let $X_i = \mathbf{1}_{A_i}$, where $A_i = \{\text{student } i \text{ is an isolated vertex}\}$.

(*These indicators are generally not independent; the calculation below uses only the first-moment method.*)

Let $S_n = \sum_{i=1}^{n} X_i$ be the number of isolated vertices. To prove $\mathbb{P}\{S_n = 0\} \to 1$, it suffices to prove $\mathbb{P}\{S_n \geqslant 1\} \to 0$.

By Markov's inequality,

$$\mathbb{P}\{S_n \geqslant 1\} \leqslant \mathbb{E}[S_n],$$

so it suffices to prove $\mathbb{E}[S_n] \to 0$.

It is easy to see that

$$\begin{aligned}\mathbb{E}[S_n] &= \sum_{i=1}^{n} \mathbb{E}[X_i] = n\mathbb{P}\{X_1 = 1\} = n(1-p_n)^{n-1} \\ &\leqslant n\mathrm{e}^{-p_n(n-1)} \leqslant n\mathrm{e}^{-\frac{(1+\varepsilon)\log n}{n}(n-1)} \longrightarrow 0.\end{aligned}$$

10 [**Isolated Vertices in Sparse Random Graphs (Converse of the Isolated Vertex Property)**] Let $G(n, p_n)$ be a random graph, where n is the number of students (*vertices*) and p_n is the probability that any two students become friends (*an edge exists*). Prove the converse of Exercise Q1.9 (*Isolated Vertices in Dense Random Graphs (First Moment Method)*): if for every $\varepsilon > 0$,

$$p_n < \frac{(1-\varepsilon)\log n}{n}.$$

then as $n \to \infty$, the probability that there is at least one friendless student (*i.e., there exists an isolated vertex*) converges to 1.

Prove this result by the second-moment method, following these steps:

(1) Let $S_n = \sum_{i=1}^{n} X_i$, where X_i is the indicator that student i is an isolated vertex. Prove that $\mu_n = \mathbb{E}[S_n] \to \infty$.

(2) Expand the square and compute $\mathbb{E}[S_n^2]$. (*Then prove that* $\mathrm{Var}(S_n)/\mu_n^2 \to 0$.)

(3) Complete the proof using Chebyshev's inequality.

Prf. To prove that the probability of an isolated vertex tends to 1, it suffices to prove $\mathbb{P}\{S_n = 0\} \to 0$.

(1) Proof that $\mu_n = \mathbb{E}[S_n] \to \infty$

Since the edges are mutually independent,

$$\mathbb{P}\{X_i = 1\} = (1-p_n)^{n-1},$$

thus

$$\begin{aligned}\mu_n = \mathbb{E}[S_n] &= \sum_{i=1}^{n} \mathbb{E}[X_i] &&= n(1-p_n)^{n-1} \\ &\geqslant n\exp\left(-\frac{(n-1)p_n}{1-p_n} + \log(1-p_n)\right) \quad (\text{since } 1-p_n \geqslant 1/2 \text{ for large } n) \\ &\geqslant n\exp\left(-\frac{(n-1)p_n}{1-p_n} - \frac{p_n}{1-p_n}\right), \text{ and } \frac{1}{1-p_n} \leqslant 1 + \frac{\varepsilon}{2} \text{ for all sufficiently large } n \\ &= n\exp\left(-\frac{np_n}{1-p_n}\right) \geqslant n\exp\left(-(1+\varepsilon)p_n\right) \\ &\geqslant n\exp\left(-(1+\varepsilon)\frac{(1-\varepsilon)\log n}{n}\right) = n^{\varepsilon^2} \to \infty.\end{aligned}$$

(2) Proof that $\mathrm{Var}(S_n)/\mu_n^2 \to 0$

$$\mathbb{E}[S_n^2] = \sum_{i=1}^{n} \mathbb{E}[X_i^2] + \sum_{i \neq j} \mathbb{E}[X_i X_j] = \mu_n + \sum_{i \neq j} \mathbb{P}\{X_i = 1, X_j = 1\}.$$

(*It remains only to consider the cross terms.*)

The event $\{X_i = 1, X_j = 1\}$ means that both i and j are isolated, i.e., "(i, j) is absent, i has no edges to the remaining $n-2$ vertices, and j has no edges to the remaining $n-2$ vertices." The number of independent edges involved is $1 + (n-2) + (n-2) = 2n-3$, hence

$$\mathbb{E}[X_i X_j] = (1-p_n)^{2n-3},$$

Substitution gives

$$\mathbb{E}[S_n^2] = \mu_n + n(n-1)(1-p_n)^{2n-3},$$

thus

$$\frac{\mathbb{E}[S_n^2]}{\mu_n^2} = \frac{\mu_n}{\mu_n^2} + \frac{n(n-1)(1-p_n)^{2n-3}}{n^2(1-p_n)^{2n-2}} = \frac{1}{\mu_n} + \frac{n-1}{n}\cdot\frac{1}{1-p_n} \to 1,$$

thus

$$\frac{\mathrm{Var}(S_n)}{\mu_n^2} = \frac{\mathbb{E}[S_n^2]}{\mu_n^2} - 1 \to 0.$$

(3) By Chebyshev's inequality,

$$\mathbb{P}\{S_n = 0\} \leqslant \mathbb{P}\{|S_n - \mu_n| \geqslant \mu_n\} \leqslant \frac{\mathrm{Var}(S_n)}{\mu_n^2} \to 0.$$

11 [**Monotonicity of L^p Norms**] (1) Let X be a random variable on $(\Omega, \mathscr{F}, \mathbb{P})$. Prove: $\|X\|_{L^p} = (\mathbb{E}[|X|^p])^{1/p}$ is a monotonically increasing function of p, i.e., for any $0 < p \leqslant q \leqslant \infty$,

$$\|X\|_{L^p} \leqslant \|X\|_{L^q}.$$

(2) Show that the inequality in (1) is generally not reversible, i.e., construct a counterexample X such that for any $0 < p < q \leqslant \infty$, $\|X\|_{L^p} < \infty$ while $\|X\|_{L^q} = \infty$.

Prf. (1) We consider the possible values of q:

a. If $q < \infty$: Using Jensen's inequality $\phi(\mathbb{E}[Y]) \leqslant \mathbb{E}[\phi(Y)]$ with $Y = |X|^p$ and $\phi(x) = x^\alpha$ for $\alpha \geqslant 1$, we obtain

$$(\mathbb{E}[|X|^p])^{q/p} \leqslant \mathbb{E}[(|X|^p)^{q/p}] = \mathbb{E}[|X|^q],$$

Taking the $1/q$-th power on both sides,

$$(\mathbb{E}[|X|^p])^{\frac{1}{p}} \leqslant (\mathbb{E}[|X|^q])^{\frac{1}{q}},$$

i.e., $\|X\|_{L^p} \leqslant \|X\|_{L^q}$.

b. If $q = \infty$: Since $|X| \leqslant \|X\|_{L^\infty}$ a.s.,

$$|X|^p \leqslant \|X\|_{L^\infty}^p \text{ (a.s.)},$$

Taking expectation on both sides,

$$\mathbb{E}[|X|^p] \leqslant \mathbb{E}\|X\|_{L^\infty}^p = \|X\|_{L^\infty}^p,$$

Taking the $1/p$-th power on both sides,

$$\|X\|_{L^p} \leqslant \|X\|_{L^\infty}.$$

(2) a. If $q < \infty$:

(*Consider constructing a random variable with a heavy-tailed distribution.*)

Let $\Omega = [1, \infty)$ and $X \sim$ Pareto, i.e.,

$$f(x) = \frac{q}{x^{q+1}}\mathbf{1}_{\{x \geqslant 1\}},$$

then

$$\|X\|_{L^q}^q = \mathbb{E}[|X|^q] = \int_1^\infty x^q \cdot \frac{q}{x^{q+1}}\mathrm{d}x = \int_1^\infty \frac{q}{x}\mathrm{d}x = q[\log x]_1^\infty = \infty,$$

$$\|X\|_{L^p}^p = \mathbb{E}[|X|^p] = \int_1^\infty x^p \cdot \frac{q}{x^{q+1}}\mathrm{d}x = q\int_1^\infty x^{p-q-1}\mathrm{d}x = \frac{q}{q-p} < \infty.$$

b. If $q = \infty$:

(*Consider constructing an unbounded random variable whose p-th moment is finite.*)

Let $X \sim E(1)$, then

$$\|X\|_{L^\infty} = \infty,$$

$$\|X\|_{L^p}^p = \mathbb{E}[|X|^p] = \int_0^\infty x^p \mathrm{e}^{-x} \mathrm{d}x = \Gamma(p+1) < \infty.$$

12 [**Interpolation Inequality between L^1 and L^∞**] It is known that $\|X\|_{L^p}$ has the upper bound

$$\|X\|_{L^p} \leqslant \|X\|_{L^\infty}.$$

Prove: when the L^1-norm of X is small, there exists a tighter upper bound

$$\|X\|_{L^p} \leqslant \|X\|_{L^1}^{\frac{1}{p}} \|X\|_{L^\infty}^{1-\frac{1}{p}}.$$

Prf. By the definition of the L^∞-norm (*essential supremum*),

$$|X(\omega)| \leqslant \|X\|_{L^\infty} \text{ (a.s.)},$$

hence

$$\begin{aligned} |X|^p &= |X| \cdot |X|^{p-1} \\ &\leqslant |X| \cdot (\|X\|_{L^\infty})^{p-1} \text{ (a.s.)}, \end{aligned}$$

Taking expectation on both sides,

$$\begin{aligned} \mathbb{E}[|X|^p] &\leqslant \mathbb{E}[|X| \cdot (\|X\|_{L^\infty})^{p-1}] \\ &= (\|X\|_{L^\infty})^{p-1} \cdot \mathbb{E}[|X|] \\ &= (\|X\|_{L^\infty})^{p-1} \cdot \|X\|_{L^1}, \end{aligned}$$

Taking the $1/p$-th power on both sides,

$$\begin{aligned} \|X\|_{L^p} = (\mathbb{E}[|X|^p])^{\frac{1}{p}} &\leqslant (\|X\|_{L^1} \cdot \|X\|_{L^\infty}^{p-1})^{\frac{1}{p}} \\ &= \|X\|_{L^1}^{\frac{1}{p}} \cdot \|X\|_{L^\infty}^{\frac{p-1}{p}} \\ &= \|X\|_{L^1}^{\frac{1}{p}} \cdot \|X\|_{L^\infty}^{1-\frac{1}{p}}. \end{aligned}$$

13 [**Expectation of the Maximum**] Let $X_1, \cdots, X_n$ be nonnegative random variables.

(1) Prove:

$$\max_{1 \leqslant i \leqslant n} \mathbb{E}[X_i] \leqslant \mathbb{E}\left[\max_{1 \leqslant i \leqslant n} X_i\right] \leqslant n \cdot \max_{1 \leqslant i \leqslant n} \mathbb{E}[X_i].$$

(2) Prove: both inequalities in (1) are tight, i.e., construct $\{X_n; Y_n\}$ such that

$$\max_{1 \leqslant i \leqslant n} \mathbb{E}[X_i] = \mathbb{E}\left[\max_{1 \leqslant i \leqslant n} X_i\right] > 0, \mathbb{E}\left[\max_{1 \leqslant i \leqslant n} Y_i\right] = n \cdot \max_{1 \leqslant i \leqslant n} \mathbb{E}[Y_i] > 0.$$

(3) Prove: even for independent random variables, the upper bound in (1) cannot be improved in order of magnitude, i.e., construct a counterexample $X_1, \cdots, X_n$ such that

$$\mathbb{E}\left[\max_{1 \leqslant i \leqslant n} X_i\right] > cn \cdot \max_{1 \leqslant i \leqslant n} \mathbb{E}[X_i].$$

Prf. (1) For the left inequality, note that $\max_i X_i \geqslant X_k$; taking expectation on both sides gives

$$\mathbb{E}\left[\max_{1 \leqslant i \leqslant n} X_i\right] \geqslant \mathbb{E}[X_k],$$

Therefore

$$\mathbb{E}\left[\max_{1 \leqslant i \leqslant n} X_i\right] \geqslant \max_{1 \leqslant k \leqslant n} \mathbb{E}[X_k].$$

For the right inequality, note that $\max_i X_i \leqslant \sum_{i=1}^n X_i$; taking expectation on both sides gives

$$\mathbb{E}\left[\max_{1 \leqslant i \leqslant n} X_i\right] \leqslant \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i],$$

where

$$\sum_{i=1}^n \mathbb{E}[X_i] \leqslant \sum_{i=1}^n \left(\max_{1 \leqslant j \leqslant n} \mathbb{E}[X_j]\right) = n \cdot \max_{1 \leqslant j \leqslant n} \mathbb{E}[X_j],$$

Substitution gives

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} X_i\right] \leqslant n \cdot \max_{1\leqslant i\leqslant n} \mathbb{E}[X_i].$$

(2) (*For the left inequality, construct* $\{X_n\}$ *such that* $\mathbb{E}[\max_i X_i] = \max_i \mathbb{E}[X_i]$.)

Let $X_1 = \cdots = X_n \equiv 1$, then

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} X_i\right] = \max_{1\leqslant i\leqslant n} \mathbb{E}[X_i] = 1.$$

(*For the right inequality, construct* $\{Y_n\}$ *such that* $\mathbb{E}[\max Y_i] = n \cdot \max \mathbb{E}[Y_i]$.)

Partition Ω into n pairwise disjoint equal-probability events $A_1, \cdots, A_n$, namely

$$\Omega = \bigsqcup_{i=1}^{n} A_i, \mathbb{P}(A_i) = \frac{1}{n}.$$

Let $Y_i = \mathbf{1}_{A_i}$, then $\max_i Y_i = 1$ (*exactly one of* $Y_1(\omega), \cdots, Y_n(\omega)$ *takes the value* 1), hence

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} Y_i\right] = n \cdot \max_{1\leqslant i\leqslant n} \mathbb{E}[Y_i] = 1.$$

(3) Let $X_1, \cdots, X_n$ i.i.d. $\sim$ Bernoulli$(1/n)$ and $M_n = \max_i X_i$. Then

$$\begin{aligned}\mathbb{P}\{M_n = 1\} &= 1 - \mathbb{P}\{M_n = 0\} \\ &= 1 - \mathbb{P}\{X_1 = 0, \cdots, X_n = 0\} \\ &= 1 - \prod_{i=1}^{n} \mathbb{P}\{X_n = 0\} = 1 - \left(1 - \frac{1}{n}\right)^n.\end{aligned}$$

Hence

$$\mathbb{E}[M_n] = 1 \cdot \mathbb{P}\{M_n = 1\} = 1 - \left(1 - \frac{1}{n}\right)^n \to 1 - \frac{1}{\mathrm{e}}.$$

Thus there exists $c > 0$ such that

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} X_i\right] > c \cdot \left(n \cdot \max_{1\leqslant i\leqslant n} \mathbb{E}[X_i]\right) = c,$$

i.e.,

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} X_i\right] \gtrsim \left(n \cdot \max_{1\leqslant i\leqslant n} \mathbb{E}[X_i]\right).$$

14 [**Integral Form of Minkowski's Inequality**] Let $\{X_n\}$ be nonnegative. Prove: for any $1 \leqslant p < \infty$,

$$\left(\sum_{i=1}^{n} (\mathbb{E}[X_i])^p\right)^{1/p} \leqslant \mathbb{E}\left[\left(\sum_{i=1}^{n} X_i^p\right)^{1/p}\right] \leqslant \left(\sum_{i=1}^{n} \mathbb{E}[X_i^p]\right)^{1/p}.$$

Prf. For the right inequality, using Jensen's inequality $\phi(\mathbb{E}[Y]) \leqslant \mathbb{E}[\phi(Y)]$ with $Y = \sum_{i=1}^n X_i^p$ and $\phi(y) = y^{1/p}$,

$$\mathbb{E}\left[\left(\sum_{i=1}^{n} X_i^p\right)^{1/p}\right] \leqslant ()^{1/p} = \left(\sum_{i=1}^{n} \mathbb{E}[X_i^p]\right)^{1/p}.$$

For the left inequality, define $g : \mathbb{R}^n \to \mathbb{R}$ as (*i.e., the* ℓ^p*-norm*)

$$g(\boldsymbol{x}) = \|\boldsymbol{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p},$$

Then by the multidimensional Jensen inequality, $g(\mathbb{E}[\boldsymbol{X}]) \leqslant \mathbb{E}[g(\boldsymbol{X})]$, i.e.,

$$\left(\sum_{i=1}^{n} |\mathbb{E}[X_i]|^p\right)^{1/p} \leqslant \mathbb{E}\left[\left(\sum_{i=1}^{n} |X_i|^p\right)^{1/p}\right].$$

i.e.,

$$\left(\sum_{i=1}^{n}(\mathbb{E}[X_i])^p\right)^{1/p} \leqslant \mathbb{E}\left[\left(\sum_{i=1}^{n} X_i^p\right)^{1/p}\right].$$

15 [**Integral Tail Formula**] Prove the general version of Lemma L1.6.1 (*Integrated tail formula*).

(1) Let X be an arbitrary random variable (*not necessarily nonnegative*). Prove:

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}\{X > t\}\mathrm{d}t - \int_{-\infty}^0 \mathbb{P}\{X < t\}\mathrm{d}t.$$

(2) Let X be a nonnegative random variable and $f : \mathbb{R} \to \mathbb{R}$ a monotonically increasing differentiable function with $f(0) = 0$. Prove:

$$\mathbb{E}[f(X)] = \int_0^\infty \mathbb{P}\{X > t\}f'(t)\mathrm{d}t.$$

(3) Let X be an arbitrary random variable (*not necessarily nonnegative*). Prove: for any $0 < p < \infty$,

$$\mathbb{E}[|X|^p] = \int_0^\infty \mathbb{P}\{|X| > t\}pt^{p-1}\mathrm{d}t.$$

Prf. (1) Decompose X as $X = X^+ - X^-$, so that $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$.

Since

$$X^+ = \int_0^\infty \mathbf{1}_{\{X^+>t\}}\mathrm{d}t = \int_0^\infty \mathbf{1}_{\{X>t\}}\mathrm{d}t,$$

we have

$$\mathbb{E}[X^+] = \mathbb{E}\left[\int_0^\infty \mathbf{1}_{\{X>t\}}\mathrm{d}t\right] = \int_0^\infty \mathbb{E}[\mathbf{1}_{\{X>t\}}]\mathrm{d}t = \int_0^\infty \mathbb{P}\{X > t\}\mathrm{d}t; \qquad (t \geqslant 0)$$

Similarly,

$$\begin{aligned}\mathbb{E}[X^-] &= \int_0^\infty \mathbb{P}\{X^- > u\}\mathrm{d}u = \int_0^\infty \mathbb{P}\{X < -u\}\mathrm{d}u && (u \geqslant 0)\\ &= \int_{-\infty}^0 \mathbb{P}\{X < t\}\mathrm{d}t, && (t = -u)\end{aligned}$$

thus

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-] = \int_0^\infty \mathbb{P}\{X > t\}\mathrm{d}t - \int_{-\infty}^0 \mathbb{P}\{X < t\}\mathrm{d}t.$$

(2) Since $f(0) = 0$,

$$f(X) = \int_0^X f'(t)\mathrm{d}t = \int_0^\infty f'(t)\mathbf{1}_{\{t<X\}}\mathrm{d}t = \int_0^\infty f'(t)\mathbf{1}_{\{X>t\}}\mathrm{d}t,$$

Taking expectation on both sides and applying Fubini's theorem,

$$\begin{aligned}\mathbb{E}[f(X)] &= \mathbb{E}\left[\int_0^\infty f'(t)\mathbf{1}_{\{X>t\}}\mathrm{d}t\right]\\ &= \int_0^\infty \mathbb{E}[f'(t)\mathbf{1}_{\{X>t\}}]\mathrm{d}t\\ &= \int_0^\infty f'(t)\mathbb{E}[\mathbf{1}_{\{X>t\}}]\mathrm{d}t\\ &= \int_0^\infty f'(t)\mathbb{P}\{X > t\}\mathrm{d}t,\end{aligned}$$

(3) Using the conclusion of (2), take $Y = |X|$ and $f(y) = y^p$, then

$$\mathbb{E}[f(Y)] = \mathbb{E}[|X|^p] = \int_0^\infty \mathbb{P}\{Y > t\} f'(t) \mathrm{d}t,$$

i.e.,

$$\mathbb{E}[|X|^p] = \int_0^\infty \mathbb{P}\{|X| > t\} p t^{p-1} \mathrm{d}t.$$

16 [**Paley–Zygmund Inequality**] Markov's inequality tells us that the probability of a random variable being much larger than its expectation is small. Conversely, is the probability of a nonnegative random variable being much smaller than its expectation also small? The answer is generally no; however, if the second moment of the random variable is not too large, an affirmative answer can be given.

Let X be a nonnegative random variable with $\mathbb{E}[X^2] < \infty$. Prove: for any $\varepsilon \in [0, 1]$,

$$\mathbb{P}\{X > \varepsilon \mathbb{E}[X]\} \geqslant (1 - \varepsilon)^2 \frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]}.$$

Prf. Let $A = \{X > \varepsilon \mathbb{E}[X]\}$ and decompose X with respect to A:

$$\mathbb{E}[X] = \mathbb{E}[X \mathbf{1}_A] + \mathbb{E}[X \mathbf{1}_{A^c}],$$

where

$$\mathbb{E}[X \mathbf{1}_{A^c}] \leqslant \mathbb{E}[\varepsilon \mathbb{E}[X] \cdot \mathbf{1}_{A^c}] = \varepsilon \mathbb{E}[X] \mathbb{P}\{A^c\} \leqslant \varepsilon \mathbb{E}[X],$$

hence

$$\mathbb{E}[X] \leqslant \mathbb{E}[X \mathbf{1}_A] + \varepsilon \mathbb{E}[X],$$

i.e.,

$$\mathbb{E}[X \mathbf{1}_A] \geqslant (1 - \varepsilon) \mathbb{E}[X].$$

By Cauchy–Schwarz,

$$(\mathbb{E}[X \mathbf{1}_A])^2 \leqslant \mathbb{E}[X^2] \mathbb{E}[\mathbf{1}_A^2] = \mathbb{E}[X^2] \mathbb{E}[\mathbf{1}_A] = \mathbb{E}[X^2] \mathbb{P}(A),$$

thus

$$(1 - \varepsilon)^2 (\mathbb{E}[X])^2 \leqslant \mathbb{E}[X^2] \mathbb{P}(A),$$

i.e.,

$$\mathbb{P}(A) \geqslant (1 - \varepsilon)^2 \frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]}.$$

(*If* $\mathbb{E}[X^2] = 0$, *then* $X = 0$ *a.s., and both sides are* 0, *which is trivial.*)

17 [**Comparison of ℓ^p Norms**] Let $\boldsymbol{x} = (x_1, \cdots, x_n) \in \mathbb{R}^n$. For $p \in \bar{\mathbb{R}}^+$, define the ℓ^p norm as

$$\|\boldsymbol{x}\|_p = \begin{cases} \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, & 0 < p < \infty, \\ \max\limits_{1 \leqslant i \leqslant n} |x_i|, & p = \infty \end{cases},$$

(1) Prove: for any $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|\boldsymbol{x}\|_q \leqslant \|\boldsymbol{x}\|_p \leqslant n^{\frac{1}{p} - \frac{1}{q}} \|\boldsymbol{x}\|_q,$$

where when $q = \infty$, we adopt the convention $1/q = 0$.

(2) Prove: both inequalities in (1) are tight, i.e., construct $\boldsymbol{x}, \boldsymbol{y} \neq \boldsymbol{0}$ such that $\|\boldsymbol{x}\|_p = \|\boldsymbol{x}\|_q$ and

$$\|\boldsymbol{y}\|_p = n^{\frac{1}{p} - \frac{1}{q}} \|\boldsymbol{y}\|_q.$$

Prf. (1) For the left inequality:

a. If $q = \infty$: Since

$$|x_j|^p \leqslant \sum_{i=1}^{n} |x_i|^p,$$

taking the p-th root on both sides,

$$|x_j| \leqslant \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} = \|\boldsymbol{x}\|_p,$$

hence

$$\|\boldsymbol{x}\|_\infty = \max_{1 \leqslant j \leqslant n} |x_j| \leqslant \|\boldsymbol{x}\|_p.$$

b. If $q < \infty$: We may assume $\|\boldsymbol{x}\|_p = 1$ (*if* $\boldsymbol{x} = \boldsymbol{0}$*, the statement is trivial; otherwise consider the normalized vector* $\bar{\boldsymbol{x}} = \boldsymbol{x}/\|\boldsymbol{x}\|_p$), so $\sum_{i=1}^{n} |x_i|^p = 1$ and $|x_i| \in [0, 1]$.
Since $p \leqslant q$, we have $|x_i|^q \leqslant |x_i|^p$. Summing over i,

$$\sum_{i=1}^{n} |x_i|^q \leqslant \sum_{i=1}^{n} |x_i|^p = 1,$$

taking the $1/q$-th power,

$$\|\boldsymbol{x}\|_q = \left(\sum_{i=1}^{n} |x_i|^q \right)^{1/q} \leqslant 1^{1/q} = 1 = \|\boldsymbol{x}\|_p.$$

For the right inequality:

a. If $q = \infty$:

$$\|\boldsymbol{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} \leqslant \left(\sum_{i=1}^{n} (\|\boldsymbol{x}\|_\infty)^p \right)^{1/p} = (n\|\boldsymbol{x}\|_\infty^p)^{1/p} = n^{1/p}\|\boldsymbol{x}\|_\infty.$$

b. If $q < \infty$: Using Hölder's inequality with $\alpha = q/p$, $\beta = q/(q-p)$,

$$\sum_{i=1}^{n} |x_i|^p \cdot 1 \leqslant \left(\sum_{i=1}^{n} (|x_i|^p)^\alpha \right)^{1/\alpha} \left(\sum_{i=1}^{n} 1^\beta \right)^{1/\beta},$$

i.e.,

$$\|\boldsymbol{x}\|_p^p \leqslant \left(\sum_{i=1}^{n} |x_i|^q \right)^{p/q} \cdot n^{\frac{q-p}{q}} = \|\boldsymbol{x}\|_q^p \cdot n^{1-p/q},$$

taking the $1/p$-th power,

$$\|\boldsymbol{x}\|_p \leqslant \|\boldsymbol{x}\|_q \cdot n^{\frac{1}{p}-\frac{1}{q}}.$$

(2) For the left inequality, take $\boldsymbol{x} = \boldsymbol{e}_1 = (1, 0, \cdots, 0)^\top$ (*a sparse vector*), then

$$\|\boldsymbol{x}\|_p = \|\boldsymbol{x}\|_q = 1.$$

For the right inequality, take $\boldsymbol{y} = (1, 1, \cdots, 1)^\top$, then

$$\|\boldsymbol{y}\|_p = \left(\sum_{i=1}^{n} 1^p \right)^{\frac{1}{p}} = n^{\frac{1}{p}}, \|\boldsymbol{y}\|_q = \left(\sum_{i=1}^{n} 1^q \right)^{\frac{1}{q}} = n^{\frac{1}{q}},$$

Hence

$$n^{\frac{1}{p}-\frac{1}{q}}\|\boldsymbol{y}\|_q = n^{\frac{1}{p}} \cdot n^{-\frac{1}{q}} \cdot n^{\frac{1}{q}} = n^{\frac{1}{p}} = \|\boldsymbol{y}\|_p.$$

(*If* $q = \infty$*, then* $\|\boldsymbol{y}\|_\infty = 1$*, and* $n^{1/p-0} \cdot 1 = n^{1/p}$.)

18 [**The ℓ^∞ Norm as the Limit of ℓ^p Norms**] Let $\boldsymbol{x} \in \mathbb{R}^n$.

(1) Prove: as $p \to \infty$, $\|\boldsymbol{x}\|_p \to \|\boldsymbol{x}\|_\infty$.

(2) Indeed, p need not be very large for the ℓ^p norm to be sufficiently close to the ℓ^∞ norm. Prove: if $p \geqslant \log n$, then

$$\|\boldsymbol{x}\|_\infty \leqslant \|\boldsymbol{x}\|_p \leqslant \mathrm{e}\|\boldsymbol{x}\|_\infty.$$

Prf. If $\boldsymbol{x} = \boldsymbol{0}$, every norm is 0 and the conclusion is trivial; we only need to consider $\boldsymbol{x} \neq \boldsymbol{0}$.

(1) For the lower bound, by the definition of $\|\boldsymbol{x}\|_\infty$, there exists an index k such that $|x_k| = \|\boldsymbol{x}\|_\infty$. Then

$$\|\boldsymbol{x}\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} \geqslant (|x_k|^p)^{1/p} = |x_k| = \|\boldsymbol{x}\|_\infty.$$

For the upper bound, since $|x_i| \leqslant \|\boldsymbol{x}\|_\infty$,

$$\|\boldsymbol{x}\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p} \leqslant \left(\sum_{i=1}^n \|\boldsymbol{x}\|_\infty^p\right)^{1/p} = (n \cdot \|\boldsymbol{x}\|_\infty^p)^{1/p} = n^{1/p}\|\boldsymbol{x}\|_\infty.$$

Thus,

$$\|\boldsymbol{x}\|_\infty \leqslant \|\boldsymbol{x}\|_p \leqslant n^{1/p}\|\boldsymbol{x}\|_\infty,$$

hence by the squeeze theorem,

$$\lim_{p\to\infty} \|\boldsymbol{x}\|_p = \|\boldsymbol{x}\|_\infty.$$

(2) When $p \geqslant \log n$, we have $n^{1/p} \leqslant n^{1/\log n} = \mathrm{e}$, hence

$$\|\boldsymbol{x}\|_\infty \leqslant \|\boldsymbol{x}\|_p \leqslant \mathrm{e}\|\boldsymbol{x}\|_\infty.$$

19 [**Duality of ℓ_p Norms**] Let $p, p' \in [1, \infty]$ be conjugate exponents (*i.e.*, $1/p + 1/p' = 1$).

(1) Prove that Hölder's inequality is tight, i.e., for any $\boldsymbol{x} \in \mathbb{R}^n$, there exists $\boldsymbol{y} \neq \boldsymbol{0}$ such that

$$\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \|\boldsymbol{x}\|_p \|\boldsymbol{y}\|_{p'}.$$

(2) Using the conclusion of (1), prove: for any $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|\boldsymbol{x}\|_p = \max\{\langle \boldsymbol{x}, \boldsymbol{y}\rangle : \boldsymbol{y} \in \mathbb{B}_{p'}^n\},$$

where $\mathbb{B}_{p'}^n = \{\boldsymbol{y} \in \mathbb{R}^n : \|\boldsymbol{y}\|_{p'} \leqslant 1\}$ is the unit ball of the $\ell_{p'}$ norm.

Prf. If $\boldsymbol{x} = \boldsymbol{0}$, both sides are 0 and the conclusion is trivial; we only need to consider $\boldsymbol{x} \neq \boldsymbol{0}$.

(1) (*It remains to construct $\boldsymbol{y}$ according to the value of p.*)

a. If $1 < p' < \infty$: Let $y_i = \mathrm{sgn}(x_i)|x_i|^{p-1}$. Then

$$\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \sum_{i=1}^n x_i \cdot \mathrm{sgn}(x_i)|x_i|^{p-1} = \sum_{i=1}^n |x_i| \cdot |x_i|^{p-1} = \sum_{i=1}^n |x_i|^p = \|\boldsymbol{x}\|_p^p,$$

$$\|\boldsymbol{y}\|_{p'} = \left(\sum_{i=1}^n |y_i|^{p'}\right)^{1/p'} = \left(\sum_{i=1}^n (|x_i|^{p-1})^{p'}\right)^{1/p'} = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p'} = \|\boldsymbol{x}\|_p^{p/p'},$$

hence

$$\|\boldsymbol{x}\|_p \|\boldsymbol{y}\|_{p'} = \|\boldsymbol{x}\|_p \|\boldsymbol{x}\|_p^{p/p'} = \|\boldsymbol{x}\|_p^{1+p/p'} = \|\boldsymbol{x}\|_p^p.$$

b. If $p = 1, p' = \infty$: Let $y_i = \mathrm{sgn}(x_i)$. Then

$$\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \sum_{i=1}^n x_i \mathrm{sgn}(x_i) = \sum_{i=1}^n |x_i| = \|\boldsymbol{x}\|_1,$$

$$\|\boldsymbol{y}\|_\infty = \max_{1\leqslant i\leqslant n} |\mathrm{sgn}(x_i)| = 1,$$

giving

$$\|\boldsymbol{x}\|_1 \|\boldsymbol{y}\|_\infty = \|\boldsymbol{x}\|_1 \cdot 1 = \langle \boldsymbol{x}, \boldsymbol{y}\rangle.$$

c. If $p=\infty, p'=1$: Let the index k satisfy $|x_k|=\|\boldsymbol{x}\|_\infty$, and construct

$$\boldsymbol{y}=(0,\cdots,0,\underbrace{\operatorname{sgn}(x_k)}_{y_k},0,\cdots,0),$$

Then

$$\begin{aligned}\langle \boldsymbol{x},\boldsymbol{y}\rangle &= x_k\operatorname{sgn}(x_k)=|x_k|=\|\boldsymbol{x}\|_\infty,\\ \|\boldsymbol{y}\|_1 &= |\operatorname{sgn}(x_k)|=1,\end{aligned}$$

hence

$$\|\boldsymbol{x}\|_\infty\|\boldsymbol{y}\|_1=\|\boldsymbol{x}\|_\infty\cdot 1=\langle \boldsymbol{x},\boldsymbol{y}\rangle.$$

(2) To prove $\|\boldsymbol{x}\|_p=\max\{\langle \boldsymbol{x},\boldsymbol{y}\rangle : \boldsymbol{y}\in\mathbb{B}_{p'}^n\}$, it suffices to prove

$$\|\boldsymbol{x}\|_p=\sup\{\langle \boldsymbol{x},\boldsymbol{y}\rangle : \|\boldsymbol{y}\|_{p'}\leqslant 1\}.$$

(*First prove* $\|\boldsymbol{x}\|_p\leqslant\max\{\langle \boldsymbol{x},\boldsymbol{y}\rangle : \|\boldsymbol{y}\|_{p'}\leqslant 1\}$.)

For any $\boldsymbol{y}\in\mathbb{B}_{p'}^n$, by Hölder's inequality,

$$\langle \boldsymbol{x},\boldsymbol{y}\rangle\leqslant|\langle \boldsymbol{x},\boldsymbol{y}\rangle|\leqslant\|\boldsymbol{x}\|_p\|\boldsymbol{y}\|_{p'}\leqslant\|\boldsymbol{x}\|_p\cdot 1=\|\boldsymbol{x}\|_p.$$

(*Next prove* $\|\boldsymbol{x}\|_p\geqslant\max\{\langle \boldsymbol{x},\boldsymbol{y}\rangle : \|\boldsymbol{y}\|_{p'}\leqslant 1\}$ *(attainability).*)

Consider the $\boldsymbol{y}$ constructed in (1), which satisfies $\langle \boldsymbol{x},\boldsymbol{y}\rangle=\|\boldsymbol{x}\|_p\|\boldsymbol{y}\|_{p'}$. Let $\bar{\boldsymbol{y}}\in\mathbb{B}_{p'}^n$; then

$$\langle \boldsymbol{x},\bar{\boldsymbol{y}}\rangle=\left\langle \boldsymbol{x},\frac{\boldsymbol{y}}{\|\boldsymbol{y}\|_{p'}}\right\rangle=\frac{1}{\|\boldsymbol{y}\|_{p'}}\langle \boldsymbol{x},\boldsymbol{y}\rangle=\frac{1}{\|\boldsymbol{y}\|_{p'}}(\|\boldsymbol{x}\|_p\|\boldsymbol{y}\|_{p'})=\|\boldsymbol{x}\|_p,$$

Concentration of Sums of Independent Random Variables

2

Keypoints

Section 2.1 Why concentration inequalities? Q2.1 ~ Q2.4

Proposition 2.1.2 (*Gaussian tails*)

Let $g \sim N(0,1)$. Then for any $t > 0$, we have

$$\frac{t}{t^2+1} \cdot \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-t^2/2} \leqslant \mathbb{P}\{g \geqslant t\} \leqslant \frac{1}{t} \cdot \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-t^2/2}.$$

In particular, for $t \geqslant 1$ the tail is bounded by the density,

$$\mathbb{P}\{g \geqslant t\} \leqslant \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-t^2/2}.$$

Theorem 2.1.4 (*Berry-Esseen central limit theorem*)

In the setting of Theorem T1.7.3 (*Lindeberg-Levy central limit theorem*),

$$|\mathbb{P}\{Z_N \geqslant t\} - \mathbb{P}\{g \geqslant t\}| \leqslant \frac{\rho}{\sqrt{N}},$$

where $g \sim N(0,1)$ and $\rho = \mathbb{E}[|X_1 - \mu|^3]/\sigma^3$.

Section 2.2 Hoeffding inequality Q2.5 ~ Q2.10

Theorem 2.2.1 (*Hoeffding inequality*)

Let $X_1, \cdots, X_N$ i.i.d. $\sim$ Rademacher, and let $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$. Then, for any $t \geqslant 0$, we have

$$\mathbb{P}\left\{\sum_{i=1}^N a_i X_i \geqslant t\right\} \leqslant \exp\left(-\frac{t^2}{2\|\boldsymbol{a}\|_2^2}\right).$$

Theorem 2.2.5 (*Hoeffding inequality, two-sided*)

Let $X_1, \cdots, X_N$ i.i.d. $\sim$ Rademacher, and let $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$. Then, for any $t > 0$, we have

$$\mathbb{P}\left\{\left|\sum_{i=1}^N a_i X_i\right| \geqslant t\right\} \leqslant 2\exp\left(-\frac{t^2}{2\|\boldsymbol{a}\|_2^2}\right).$$

Theorem 2.2.6 (*Hoeffding inequality for bounded random variables*)

Let $X_1, \cdots, X_N$ be mutually independent random variables satisfying $\operatorname{supp}(X_i) \subset [m_i, M_i]$. Then, for any $t > 0$, we have

$$\mathbb{P}\left\{\sum_{i=1}^N (X_i - \mathbb{E}[X_i]) \geqslant t\right\} \leqslant \exp\left(-\frac{2t^2}{\sum_{i=1}^N (M_i - m_i)^2}\right).$$

Section 2.3 Chernoff inequality Q2.11 ∼ Q2.15

Theorem 2.3.1 (*Chernoff inequality*)

Let $X_1, \cdots, X_N \sim \text{Bernoulli}(p)$, $S_N = \sum_{i=1}^N X_i$, and $\mu = \mathbb{E}[S_N]$. Then, for any $t > \mu$, we have

$$\mathbb{P}\{S_N \geqslant t\} \leqslant \exp(-\mu)\left(\frac{e\mu}{t}\right)^t.$$

Remark 2.3.2 (*Chernoff inequality: left tails*)

By slightly modifying the proof of Chernoff inequality, we can also obtain a bound on the left tail: in the setting of Theorem T2.3.1 (*Chernoff inequality*), for every $0 < t \leqslant \mu$,

$$\mathbb{P}\{S_N \leqslant t\} \leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{t}\right)^t.$$

Corollary 2.3.4 (*Chernoff inequality: small deviations*)

In the setting of Theorem T2.3.1 (*Chernoff inequality*), for any $\delta \in (0, 1]$ we have

$$\mathbb{P}\{|S_N - \mu| \geqslant \delta\mu\} \leqslant 2\exp(-c\mu\delta^2).$$

Section 2.4 Median-of-means estimator Q2.16 ∼ Q2.17

Theorem 2.4.1 (*Median-of-means estimator*)

Let X be a random variable with mean μ and variance σ^2. Partition the data into k blocks of equal size. Let $\hat{\mu}_{MOM}$ be the median of the k block means. Then, for any $\varepsilon \in (0, 1)$ we have

$$\mathbb{P}\{|\hat{\mu}_{MOM} - \mu| \geqslant \varepsilon\sigma\} \leqslant \mathrm{e}^{-ck\varepsilon^2}. \qquad (c > 0)$$

Section 2.5 Degrees of random graphs Q2.18 ∼ Q2.21

Proposition 2.5.1 (*Dense graphs are almost regular*)

Let $G \sim G(n, p)$ be an Erdos–Renyi random graph with $p \geqslant \log n/n$. Then all vertices of G have degrees $(1 \pm o(1))np$ with high probability.

Section 2.6 Subgaussian distributions Q2.22 ∼ Q2.28

Proposition 2.6.1 (*Subgaussian properties*)

Let X be a random variable. The following properties are equivalent up to absolute constant factors:

(1) For any $t \geqslant 0$, $\mathbb{P}\{|X| \geqslant t\} \leqslant 2\exp(-t^2/K_1^2)$.

(2) For any $p \geqslant 1$, $(\mathbb{E}|X|^p)^{1/p} \leqslant K_2\sqrt{p}$.
(3) For any $|\lambda| \leqslant 1/K_3$, $\mathbb{E}[\exp(\lambda^2 X^2)] \leqslant \exp(K_3^2\lambda^2)$.
(4) If $\mathbb{E}[X] = 0$, then $\mathbb{E}[\exp(\lambda X)] \leqslant \exp(K_4^2\lambda^2)$.

Definition 2.6.4 (*Subgaussian distributions*)

The subgaussian norm of a random variable X is defined as

$$\|X\|_{\psi_2} = \inf\{t > 0 : \mathbb{E}[\exp(X^2/t^2)] \leqslant 2\}.$$

A random variable X is called subgaussian if $\|X\|_{\psi_2} < \infty$.

Proposition 2.6.6 (*Subgaussian bounds*)

Let X be subgaussian. Then:
(1) For any $t \geqslant 0$, $\mathbb{P}\{|X| \geqslant t\} \leqslant 2\exp(-ct^2/\|X\|_{\psi_2}^2)$.
(2) For any $p \geqslant 1$, $\|X\|_{L_p} \leqslant C\|X\|_{\psi_2}\sqrt{p}$.
(3) If $\mathbb{E}[X] = 0$, then $\mathbb{E}[\exp(\lambda X)] \leqslant \exp(C\lambda^2\|X\|_{\psi_2}^2)$.

Section 2.7 Subgaussian Hoeffding and Khintchine inequalities Q2.29 ∼ Q2.40

Proposition 2.7.1 (*Subgaussian norm of a sum*)

Let $X_1, \cdots, X_N$ be independent, mean-zero subgaussian random variables. Then

$$\left\|\sum_{i=1}^N X_i\right\|_{\psi_2}^2 \leqslant C\sum_{i=1}^N \|X_i\|_{\psi_2}^2.$$

Theorem 2.7.3 (*Subgaussian Hoeffding inequality*)

Let $X_1, \cdots, X_N$ be mutually independent, mean-zero, K-subgaussian random variables (abbreviated below as K-subgaussian), and let $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$. Then, for any $t \geqslant 0$, we have

$$\mathbb{P}\left\{\sum_{i=1}^N a_i X_i \geqslant t\right\} \leqslant \exp\left(-\frac{ct^2}{K^2\|\boldsymbol{a}\|_2^2}\right).$$

Theorem 2.7.5 (*Khintchine inequality*)

Let $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$ and $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher. Then for any $p \geqslant 1$ we have

$$\left(\mathbb{E}\left|\sum_{i=1}^N a_i\varepsilon_i\right|^p\right)^{1/p} \leqslant C\sqrt{p}\|\boldsymbol{a}\|_2.$$

The same conclusion holds if ε_i i.i.d. $\sim$ Rademacher are replaced by g_i i.i.d. $\sim N(0,1)$.

Proposition 2.7.6 (*Maximum of subgaussians*)

Let $X_1, \cdots, X_N$ be K-subgaussian random variables (not necessarily independent). Then

$$\mathbb{E}\left[\max_{i\leqslant N}|X_i|\right] \leqslant CK\sqrt{\log N}.$$

Lemma 2.7.8 (*Centering*)

For any random variable X, we have

$$\|X - \mathbb{E}[X]\|_{\psi_2} \leqslant C\|X\|_{\psi_2}.$$

Section 2.8 Subexponential distributions Q2.41 ∼ Q2.44

Proposition 2.8.1 (*Subexponential properties*)

Let X be a random variable. The following properties are equivalent up to absolute constant factors:

(1) For any $t \geqslant 0$, $\mathbb{P}\{|X| \geqslant t\} \leqslant 2\exp(-t/K_1)$.

(2) For any $p \geqslant 1$, $(\mathbb{E}|X|^p)^{1/p} \leqslant K_2 p$.

(3) For any $|\lambda| \leqslant 1/K_3$, $\mathbb{E}[\exp(\lambda X)] \leqslant \exp(K_3^2\lambda^2)$.

Definition 2.8.4 (*Subexponential distributions*)

The subexponential norm of a random variable X is defined as

$$\|X\|_{\psi_1} = \inf\{t > 0 : \mathbb{E}[\exp(|X|/t)] \leqslant 2\}.$$

A random variable X is called subexponential if $\|X\|_{\psi_1} < \infty$.

Lemma 2.8.5 (*Subexponential is subgaussian squared*)

A random variable X is subgaussian if and only if X^2 is subexponential, in which case

$$\|X^2\|_{\psi_1} \asymp \|X\|_{\psi_2}^2.$$

Lemma 2.8.6 (*Subgaussian $\times$ subgaussian = subexponential*)

Let X, Y be subgaussian random variables. Then XY is subexponential and satisfies

$$\|XY\|_{\psi_1} \leqslant \|X\|_{\psi_2}\|Y\|_{\psi_2}.$$

Section 2.9 Bernstein inequality Q2.45 ~ Q2.48

Theorem 2.9.1 (*Subexponential Bernstein inequality*)

Let $X_1, \cdots, X_N$ be mutually independent, mean-zero, K-subexponential random variables, and let $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$. Then, for any $t \geqslant 0$, we have

$$\mathbb{P}\left\{\sum_{i=1}^N a_i X_i \geqslant t\right\} \leqslant 2\exp\left[-c\min\left(\frac{t^2}{K^2\|\boldsymbol{a}\|_2^2}, \frac{t}{K\|\boldsymbol{a}\|_\infty}\right)\right],$$

where $K = \max_i \|X_i\|_{\psi_1}$.

Corollary 2.9.2 (*Subexponential Bernstein inequality, simplified*)

In the setting of Theorem T2.9.1 (*Subexponential Bernstein inequality*), for any $t \geqslant 0$, we have

$$\mathbb{P}\left\{\left|\sum_{i=1}^N a_i X_i\right| \geqslant t\right\} \leqslant 2\exp\left[-c\min\left(\frac{t^2}{K^2\|\boldsymbol{a}\|_2^2}, \frac{t}{K\|\boldsymbol{a}\|_\infty}\right)\right].$$

Theorem 2.9.5 (*Bernstein inequality for bounded distributions*)

Let $X_1, \cdots, X_N$ be mutually independent, mean-zero random variables satisfying $|X_i| \leqslant M$ (a.s.). Then, for any $t \geqslant 0$, we have

$$\mathbb{P}\left\{\sum_{i=1}^N X_i \geqslant t\right\} \leqslant \exp\left(-\frac{t^2/2}{\sum_{i=1}^N \mathbb{E}[X_i^2] + Mt/3}\right).$$

Exercises

1 [**Non-Concentration of Products of I.I.D. Random Variables**] Typically, the average of a large number of independent random variables concentrates around its expectation. However, for products, the situation can be quite different. Here is an example of "non-concentration," where the probability that the product exceeds its expectation is exponentially small.

Let $X_1, \cdots, X_n$ i.i.d. $\sim U(0,1)$, $Y_n = \prod_{i=1}^n X_i$. Prove that

$$(0.5)^n \leqslant \mathbb{P}\{Y_n \geqslant \mathbb{E}[Y_n]\} \leqslant (0.95)^n.$$

Prf. For the lower bound, since

$$\mathbb{E}[Y_n] = \mathbb{E}\left[\prod_{i=1}^n X_i\right] = \prod_{i=1}^n \mathbb{E}[X_i] = \left(\frac{1}{2}\right)^n = 2^{-n},$$

therefore

$$\mathbb{P}\{Y_n \geqslant 2^{-n}\} \geqslant \mathbb{P}\{X_i \geqslant 1/2 \text{ for every } i\} = \prod_{i=1}^n \left(1 - \frac{1}{2}\right) = (0.5)^n.$$

For the upper bound, consider the Chernoff bound. Let $Z_n = -\log X_n$ i.i.d. $\sim E(1)$, and write $S_n = \sum_{i=1}^n Z_i$. Then

$$\begin{aligned}
\{Y_n \geqslant 2^{-n}\} &= \{\log Y_n \geqslant -n \log 2\} \\
&= \left\{\sum_{i=1}^n \log X_i \geqslant -n \log 2\right\} \\
&= \left\{\sum_{i=1}^n Z_i \leqslant n \log 2\right\} \\
&= \{S_n \leqslant \lambda n\}. \qquad (\lambda = \log 2)
\end{aligned}$$

(*Note that $\lambda < 1$, so the above problem is a small-deviation problem (deviation to the left of the mean).*)

By the Chernoff bound, we obtain that for any $t < 0$,

$$\mathbb{P}\{S_n \leqslant n\lambda\} = \mathbb{P}\{\mathrm{e}^{tS_n} \leqslant \mathrm{e}^{tn\lambda}\} \leqslant \mathrm{e}^{-tn\lambda}\mathbb{E}[\mathrm{e}^{tS_n}] = \mathrm{e}^{-tn\lambda}(\mathbb{E}[\mathrm{e}^{tZ_1}])^n = (\mathrm{e}^{-t\lambda}\mathbb{E}[\mathrm{e}^{tZ_1}])^n,$$

where $\mathbb{E}[\mathrm{e}^{tZ_1}] = 1/(1-t)$. Substituting this gives

$$\mathbb{P}\{S_n \leqslant n\lambda\} \leqslant \left(\frac{\mathrm{e}^{-t\lambda}}{1-t}\right)^n.$$

(*Let the bound on the right be $g(t)$. Differentiating $\log g(t) = -t\lambda - \log(1-t)$ and setting it to 0 gives*

$$t_0 = 1 - \frac{1}{\lambda} < 0,$$

and the derivative calculation will be omitted below in favor of stating its value directly.)

The optimal parameter on the right-hand side is $t_0 = 1 - 1/\lambda$. Substituting it gives

$$\min_{t<0} g(t) = g(t_0) = \frac{\mathrm{e}^{-(1-1/\lambda)\lambda}}{1/\lambda} = \lambda \mathrm{e}^{1-\lambda} = \frac{\mathrm{e}\log 2}{2},$$

hence

$$\mathbb{P}\{Y_n \geqslant \mathbb{E}[Y_n]\} \leqslant \left(\frac{\mathrm{e}\log 2}{2}\right)^n \leqslant (0.95)^n.$$

2 [**Lower Bound for Gaussian Tails**] The lower-bound part of Proposition P2.1.2 (*Gaussian tails*) remains unproved; we now prove it.

Let $g \sim N(0,1)$. Prove that for any $t > 0$,

$$\mathbb{P}\{g \geqslant t\} \geqslant \frac{t}{t^2+1} \cdot \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-t^2/2}.$$

Prf. Construct an auxiliary function

$$f(t) = \mathbb{P}\{g \geqslant t\} - \frac{t}{t^2+1}\phi(t) = \int_t^\infty \phi(x)\mathrm{d}x - \frac{t}{t^2+1}\phi(t),$$

Then

$$f'(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\int_t^\infty \phi(x)\mathrm{d}x\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{t}{t^2+1}\phi(t)\right),$$

where

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_t^\infty \phi(x)\mathrm{d}x\right) &= -\phi(t), \\
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{t}{t^2+1}\phi(t)\right) &= \left(\frac{t}{t^2+1}\right)'\phi(t) + \frac{t}{t^2+1}\phi'(t) \\
&= \frac{1\cdot(t^2+1) - t\cdot(2t)}{(t^2+1)^2}\phi(t) + \frac{t}{t^2+1}(-t)\phi(t) && (\phi'(t) = -t\phi(t)) \\
&= \frac{1-2t^2-t^4}{(t^2+1)^2}\phi(t),
\end{aligned}$$

Substituting it gives

$$f'(t) = -\phi(t)\frac{2}{(t^2+1)^2} < 0,$$

Therefore $f(t)$ is decreasing on $\mathbb{R}^+$. Noting that $\lim_{t\to\infty} f(t) = 0$, we have $f(t) > 0$, i.e.,

$$\mathbb{P}\{g \geqslant t\} \geqslant \frac{t}{t^2+1}\cdot\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-t^2/2}.$$

3 [**Asymptotic Bounds for the Mills Ratio**] Let $g \sim N(0,1)$. Prove that for any $t > 0$, the ratio of the tail probability to the density (Mills Ratio) satisfies

$$\frac{1}{t} - \frac{1}{t^3} < \frac{\mathbb{P}\{g > t\}}{\phi(t)} < \frac{1}{t} - \frac{1}{t^3} + \frac{3}{t^5}.$$

Prove it by following these steps:

(1) Prove that

$$\phi'(x) + x\phi(x) = 0.$$

(2) Using (1), repeatedly integrate by parts

$$\mathbb{P}\{g > t\} = \int_t^\infty \phi(x)\mathrm{d}x$$

to complete the proof.

Prf. (1) Proof omitted.

(2) Let $I(t) = \mathbb{P}\{g > t\}$. For the first integration by parts,

$$\begin{aligned}
I(t) &= \int_t^\infty \phi(x)\mathrm{d}x \\
&= \int_t^\infty -\frac{1}{x}\phi'(x)\mathrm{d}x && (\phi'(x) = -x\phi(x)) \\
&= \left[-\frac{1}{x}\phi(x)\right]_t^\infty - \int_t^\infty (-\phi(x))\left(-\frac{1}{x^2}\right)\mathrm{d}x && \left(u = \frac{1}{x}, \mathrm{d}v = -\phi'(x)\mathrm{d}x\right) \\
&= \frac{\phi(t)}{t} - \int_t^\infty \frac{1}{x^2}\phi(x)\mathrm{d}x,
\end{aligned}$$

Therefore

$$\mathbb{P}\{g>t\}<\frac{\phi(t)}{t} \implies \frac{\mathbb{P}\{g>t\}}{\phi(t)}<\frac{1}{t}.$$

For the remainder term, the second integration by parts gives

$$\begin{aligned}\int_t^\infty \frac{1}{x^2}\phi(x)\mathrm{d}x &= \int_t^\infty \frac{1}{x^3}(-\phi'(x))\mathrm{d}x && (\phi'(x)=-x\phi(x))\\ &= \left[-\frac{\phi(x)}{x^3}\right]_t^\infty - \int_t^\infty (-\phi(x))\left(-\frac{3}{x^4}\right)\mathrm{d}x && \left(u=\frac{1}{x^3}, \mathrm{d}v=-\phi'(x)\mathrm{d}x\right)\\ &= \frac{\phi(t)}{t^3} - \int_t^\infty \frac{3}{x^4}\phi(x)\mathrm{d}x,\end{aligned}$$

Substituting it gives

$$\begin{aligned}I(t) &= \frac{\phi(t)}{t} - \left(\frac{\phi(t)}{t^3} - \int_t^\infty \frac{3}{x^4}\phi(x)\mathrm{d}x\right)\\ &= \phi(t)\left(\frac{1}{t}-\frac{1}{t^3}\right) + \int_t^\infty \frac{3}{x^4}\phi(x)\mathrm{d}x,\end{aligned}$$

Therefore

$$\mathbb{P}\{g>t\}>\phi(t)\left(\frac{1}{t}-\frac{1}{t^3}\right) \implies \frac{\mathbb{P}\{g>t\}}{\phi(t)}>\frac{1}{t}-\frac{1}{t^3}.$$

For the remainder term, the third integration by parts gives

$$\begin{aligned}\int_t^\infty \frac{3}{x^4}\phi(x)\mathrm{d}x &= \int_t^\infty \frac{3}{x^5}(-\phi'(x))\mathrm{d}x && (\phi'(x)=-x\phi(x))\\ &= \left[-\frac{3\phi(x)}{x^5}\right]_t^\infty - \int_t^\infty (-\phi(x))\left(-\frac{15}{x^6}\right)\mathrm{d}x && \left(u=\frac{3}{x^5}, \mathrm{d}v=-\phi'(x)\mathrm{d}x\right)\\ &= \frac{3\phi(t)}{t^5} - \int_t^\infty \frac{15}{x^6}\phi(x)\mathrm{d}x,\end{aligned}$$

therefore

$$\begin{aligned}I(t) &= \phi(t)\left(\frac{1}{t}-\frac{1}{t^3}\right) + \left(\frac{3\phi(t)}{t^5} - \int_t^\infty \frac{15}{x^6}\phi(x)\mathrm{d}x\right)\\ &= \phi(t)\left(\frac{1}{t}-\frac{1}{t^3}+\frac{3}{t^5}\right) - \int_t^\infty \frac{15}{x^6}\phi(x)\mathrm{d}x,\end{aligned}$$

Therefore

$$\mathbb{P}\{g>t\}<\phi(t)\left(\frac{1}{t}-\frac{1}{t^3}+\frac{3}{t^5}\right) \implies \frac{\mathbb{P}\{g>t\}}{\phi(t)}<\frac{1}{t}-\frac{1}{t^3}+\frac{3}{t^5}.$$

Thus,

$$\frac{1}{t}-\frac{1}{t^3}<\frac{\mathbb{P}\{g>t\}}{\phi(t)}<\frac{1}{t}-\frac{1}{t^3}+\frac{3}{t^5}.$$

4 [**Moments of the Truncated Gaussian Distribution**] Let $g\sim N(0,1)$. Prove:

(1) (*First-order truncated moment*) For any $t>0$,

$$\mathbb{E}[g\mathbf{1}_{\{g>t\}}]=\phi(t).$$

(2) (*Second-order truncated moment*) For any $t>0$,

$$\mathbb{E}[g^2\mathbf{1}_{\{g>t\}}]\leqslant\left(t+\frac{1}{t}\right)\phi(t).$$

Prf. (1) Writing the expectation as an integral and then integrating by parts gives

$$\mathbb{E}[g\mathbf{1}_{\{g>t\}}]=\int_{-\infty}^{\infty} x\mathbf{1}_{\{x>t\}}\phi(x)\mathrm{d}x$$

$$
\begin{aligned}
&= \int_t^\infty x\phi(x)\mathrm{d}x \\
&= \int_t^\infty -\phi'(x)\mathrm{d}x && (\phi'(x) = -x\phi(x)) \\
&= [-\phi(x)]_t^\infty = \phi(t).
\end{aligned}
$$

(2) Similarly to the conclusion of (1), writing the expectation as an integral and then integrating by parts gives

$$
\begin{aligned}
\mathbb{E}[g^2\mathbf{1}_{\{g>t\}}] &= \int_{-\infty}^\infty x^2\mathbf{1}_{\{x>t\}}\phi(x)\mathrm{d}x \\
&= \int_t^\infty x^2\phi(x)\mathrm{d}x \\
&= \int_t^\infty x\cdot(-\phi'(x))\mathrm{d}x && (\phi'(x) = -x\phi(x)) \\
&= [-x\phi(x)]_t^\infty - \int_t^\infty (-\phi(x))\mathrm{d}x && (u = x, \mathrm{d}v = x\phi(x)\mathrm{d}x) \\
&= t\phi(t) + \mathbb{P}\{g > t\}.
\end{aligned}
$$

By the Mills Ratio (Q2.3 (*Asymptotic Bounds for the Mills Ratio*)), we obtain

$$\mathbb{P}\{g > t\} < \frac{\phi(t)}{t},$$

(*or by a simple bound,*

$$\int_t^\infty \phi(x)\mathrm{d}x = \int_t^\infty \frac{x}{x}\phi(x)\mathrm{d}x < \frac{1}{t}\int_t^\infty x\phi(x)\mathrm{d}x = \frac{\phi(t)}{t}.)$$

Substituting it gives

$$\mathbb{E}[g^2\mathbf{1}_{\{g>t\}}] \leqslant t\phi(t) + \frac{1}{t}\phi(t) = \left(t + \frac{1}{t}\right)\phi(t).$$

5 [**Completing the Proof of Hoeffding's Inequality**] In the proof of Hoeffding's inequality, a key numerical bound is needed.

Prove that

$$\cosh(x) \leqslant \mathrm{e}^{x^2/2}.$$

Prf. **Method 1 Taylor series expansion**

It is easy to see that

$$
\begin{aligned}
\cosh(x) &= \sum_{n=0}^\infty \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots, \\
\mathrm{e}^{x^2/2} &= \sum_{n=0}^\infty \frac{(x^2/2)^n}{n!} = \sum_{n=0}^\infty \frac{x^{2n}}{2^n n!} = 1 + \frac{x^2}{2} + \frac{x^4}{8} + \cdots,
\end{aligned}
$$

Therefore, to prove $\cosh(x) \leqslant \mathrm{e}^{x^2/2}$, it suffices to prove

$$\frac{1}{(2n)!} \leqslant \frac{1}{2^n n!},$$

i.e., $(2n)! \geqslant 2^n n!$.

It is easy to see that

$$(2n)! = (2n)!!\cdot(2n-1)!! = (2^n n!)\cdot(2n-1)!! \geqslant 2^n n!.$$

Method 2 Property of the derivative

Construct an auxiliary function $f(x) = x^2/2 - \log(\cosh(x))$. It is even and satisfies $f(0) = 0$. Differentiating, for $x > 0$

we have $f'(x) = x - \tanh(x) > 0$, and hence $f(x) \geqslant 0$, i.e.,

$$\frac{x^2}{2} \geqslant \log(\cosh(x)).$$

Consequently,

$$\mathrm{e}^{x^2/2} \geqslant \cosh(x).$$

6 [**Bounding Gaussian Tails via the Exponential Moment Method**] Let $g \sim N(0,1)$. Using the Chernoff bound, prove that for any $t \geqslant 0$,

$$\mathbb{P}\{g \geqslant t\} \leqslant \mathrm{e}^{-t^2/2}.$$

(Remark: Although this upper bound is slightly looser than the exact Mills Ratio, it is already sufficiently effective in some applications.)

Prf. By the Chernoff bound, we obtain

$$\mathbb{P}\{g \geqslant t\} = \mathbb{P}\{\mathrm{e}^{\lambda g} \geqslant \mathrm{e}^{\lambda t}\} \leqslant \mathrm{e}^{-\lambda t}\mathbb{E}[\mathrm{e}^{\lambda g}],$$

where

$$\begin{aligned}
\mathbb{E}[\mathrm{e}^{\lambda g}] &= \int_{-\infty}^{\infty} \mathrm{e}^{\lambda x} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-x^2/2} \mathrm{d}x \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-(x^2-2\lambda x)/2} \mathrm{d}x \\
&= \mathrm{e}^{\lambda^2/2} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{(x-\lambda)^2}{2}} \mathrm{d}x}_{1} \\
&= \mathrm{e}^{\lambda^2/2},
\end{aligned}$$

Substituting this in, we obtain

$$\mathbb{P}\{g \geqslant t\} \leqslant \mathrm{e}^{-\lambda t} \cdot \mathrm{e}^{\lambda^2/2},$$

The optimal parameter on the right-hand side is $\lambda_0 = t$. Substituting it gives

$$\mathbb{P}\{g \geqslant t\} \leqslant \mathrm{e}^{-t^2/2}.$$

7 [**Small Ball Probabilities**] Let $X_1, \cdots, X_N$ be independent nonnegative random variables with $X_i \sim f_{X_i}$, and write $S_N = \sum_{i=1}^N X_i$. Prove that if the density functions are uniformly bounded, namely

$$\|f_{X_i}\|_\infty = \operatorname*{ess\,sup}_{x \in \mathbb{R}} f_{X_i}(x) \leqslant K,$$

then for any $\varepsilon > 0$,

$$\mathbb{P}\{S_N \leqslant \varepsilon N\} \leqslant (\mathrm{e}K\varepsilon)^N.$$

Rk. The event $\{S_N \leqslant \varepsilon N\}$ is equivalent to the empirical average lying in the small ball around the origin, $\{\frac{1}{N}\sum_{i=1}^N X_i \leqslant \varepsilon\}$; hence the name small ball probability.

Prf. By the Chernoff bound, we obtain

$$\begin{aligned}
\mathbb{P}\{S_N \leqslant \varepsilon N\} &= \mathbb{P}\{\mathrm{e}^{-tS_N} \geqslant \mathrm{e}^{-t\varepsilon N}\} \\
&\leqslant \mathrm{e}^{t\varepsilon N}\mathbb{E}[\mathrm{e}^{-tS_N}] \\
&= \mathrm{e}^{t\varepsilon N} \prod_{i=1}^N \mathbb{E}[\mathrm{e}^{-tX_i}],
\end{aligned}$$

where

$$\mathbb{E}[\mathrm{e}^{-tX_i}] = \int_0^\infty \mathrm{e}^{-tx} f_i(x) \mathrm{d}x \leqslant \int_0^\infty \mathrm{e}^{-tx} K \mathrm{d}x = K\left[-\frac{1}{t}\mathrm{e}^{-tx}\right]_0^\infty = \frac{K}{t},$$

Substituting it gives

$$\mathbb{P}\{S_N \leqslant \varepsilon N\} \leqslant \mathrm{e}^{t\varepsilon N}\left(\frac{K}{t}\right)^N,$$

The optimal parameter on the right-hand side is $t_0 = 1/\varepsilon$. Substituting it gives

$$\mathbb{P}\{S_N \leqslant \varepsilon N\} \leqslant (\mathrm{e}K\varepsilon)^N.$$

8 [**MGF Comparison Inequality**] Let X, Y satisfy $\mathbb{E}[X] = \mathbb{E}[Y] = \mu$, $\operatorname{supp}(X) = [a, b]$, $\operatorname{supp}(Y) = \{a, b\}$. Prove that for any $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\mathrm{e}^{\lambda X}] \leqslant \mathbb{E}[\mathrm{e}^{\lambda Y}].$$

Prf. For Y, which is a two-point distribution, let $p = \mathbb{P}\{Y = b\}$. Then

$$\mathbb{E}[Y] = \mu = a(1-p) + bp \implies p = \frac{\mu - a}{b - a},$$

therefore

$$\mathbb{E}[\mathrm{e}^{\lambda Y}] = (1-p)\mathrm{e}^{\lambda a} + p\mathrm{e}^{\lambda b} = \frac{b-\mu}{b-a}\mathrm{e}^{\lambda a} + \frac{\mu - a}{b-a}\mathrm{e}^{\lambda b}.$$

For X, write $X = x \in [a, b]$ as a convex combination,

$$x = (1-\alpha)a + \alpha b,$$

Since $f(x) = \mathrm{e}^{\lambda x}$ is convex,

$$f(x) \leqslant (1-\alpha)f(a) + \alpha f(b),$$

i.e.,

$$\mathrm{e}^{\lambda x} \leqslant \frac{b-x}{b-a}\mathrm{e}^{\lambda a} + \frac{x-a}{b-a}\mathrm{e}^{\lambda b},$$

i.e.,

$$\mathrm{e}^{\lambda X} \leqslant \frac{b-X}{b-a}\mathrm{e}^{\lambda a} + \frac{X-a}{b-a}\mathrm{e}^{\lambda b},$$

Taking expectations on both sides,

$$\begin{aligned}\mathbb{E}[\mathrm{e}^{\lambda X}] &\leqslant \mathbb{E}\left[\frac{b-X}{b-a}\mathrm{e}^{\lambda a} + \frac{X-a}{b-a}\mathrm{e}^{\lambda b}\right] \\ &= \frac{b-\mathbb{E}[X]}{b-a}\mathrm{e}^{\lambda a} + \frac{\mathbb{E}[X]-a}{b-a}\mathrm{e}^{\lambda b} \\ &= \frac{b-\mu}{b-a}\mathrm{e}^{\lambda a} + \frac{\mu-a}{b-a}\mathrm{e}^{\lambda b} = \mathbb{E}[\mathrm{e}^{\lambda Y}].\end{aligned}$$

9 [**Hoeffding's Lemma**] Let $\operatorname{supp}(X) = [a, b]$. Prove that for any $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\mathrm{e}^{\lambda(X - \mathbb{E}[X])}] \leqslant \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Prove it by following these steps:

(1) Show that the assumptions may be simplified: without loss of generality, suppose that X is mean-zero, $b - a = 1$, and X takes values in $\{a, b\}$.

(2) Compute $K(\lambda) = \log \mathbb{E}[\mathrm{e}^{\lambda X}]$, where $K(0) = K'(0) = 0$ and $K''(\lambda) \leqslant 1/4$. Then derive $K(\lambda) \leqslant \lambda^2/8$.

Prf. (1) Let $Y = X - \mathbb{E}[X]$. Then $\mathbb{E}[Y] = 0$, $\operatorname{supp}(Y) = [a - \mathbb{E}[X], b - \mathbb{E}[X]]$, and the objective becomes

$$\mathbb{E}[\mathrm{e}^{\lambda Y}] \leqslant \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Thus we only need to consider the case $\mu = 0$.

By Exercise Q2.8 (*MGF Comparison Inequality*), we only need to consider the case $\operatorname{supp}(X) = \{a, b\}$.

Let $W = X/(b-a)$, then $\operatorname{supp}(W) = [a/(b-a), b/(b-a)]$, and the interval length is 1, so the objective becomes

$$\mathbb{E}[\mathrm{e}^{\mu W}] \leqslant \exp\left(\frac{\mu^2}{8}\right).$$

Thus the problem reduces to: Let X satisfy $\operatorname{supp}(X) = \{a, b\}$ $(b-a=1)$, with $\mathbb{E}[X] = 0$. Prove:

$$\log \mathbb{E}[\mathrm{e}^{\lambda X}] \leqslant \frac{\lambda^2}{8}.$$

(2) Let $\mathbb{P}\{X = b\} = p$, then

$$\begin{cases} b - a = 1, \\ \mathbb{E}[X] = bp + a(1-p) = 0 \end{cases},$$

solving gives $(a, b) = (-p, 1-p)$.

Let

$$K(\lambda) = \log \mathbb{E}[\mathrm{e}^{\lambda X}] = \log((1-p)\mathrm{e}^{-p\lambda} + p\mathrm{e}^{(1-p)\lambda}),$$

By Taylor expansion,

$$K(\lambda) = K(0) + \lambda K'(0) + \frac{\lambda^2}{2} K''(\xi), \qquad (\xi \text{ between } 0 \text{ and } \lambda)$$

It remains to calculate $K(0)$ and $K'(0)$ and bound $K''(\xi)$.

Differentiating,

$$K'(\lambda) = \frac{(1-p)(-p)\mathrm{e}^{-p\lambda} + p(1-p)\mathrm{e}^{(1-p)\lambda}}{(1-p)\mathrm{e}^{-p\lambda} + p\mathrm{e}^{(1-p)\lambda}},$$

so $K(0) = K'(0) = 0$ (*in fact,* $K'(0) = \mathbb{E}[X]$).

Differentiating once more,

$$\begin{aligned} K''(\lambda) &= \frac{p\mathrm{e}^{\lambda}(1-p+p\mathrm{e}^{\lambda}) - p\mathrm{e}^{\lambda}(p\mathrm{e}^{\lambda})}{(1-p+p\mathrm{e}^{\lambda})^2} \\ &= \frac{p\mathrm{e}^{\lambda}(1-p)}{(1-p+p\mathrm{e}^{\lambda})^2}, \end{aligned}$$

Note that

$$K''(\lambda) = \frac{p\mathrm{e}^{\lambda}}{1-p+p\mathrm{e}^{\lambda}} \cdot \frac{1-p}{1-p+p\mathrm{e}^{\lambda}} = \theta(1-\theta), \qquad \left(\theta = \frac{p\mathrm{e}^{\lambda}}{1-p+p\mathrm{e}^{\lambda}} \in [0,1]\right)$$

therefore $K''(\lambda) \leqslant 1/4$.

Thus,

$$\begin{aligned} K(\lambda) &= K(0) + \lambda K'(0) + \frac{\lambda^2}{2} K''(\xi) \\ &\leqslant 0 + 0 + \frac{\lambda^2}{2} \cdot \frac{1}{4} = \frac{\lambda^2}{8}, \end{aligned}$$

We obtain

$$\log \mathbb{E}[\mathrm{e}^{\lambda X}] \leqslant \frac{\lambda^2}{8}.$$

10 [**Hoeffding's Inequality for Bounded Random Variables**] Prove Theorem T2.2.6 (*Hoeffding inequality for bounded random variables*).

Let $X_1, \cdots, X_N$ be independent random variables with $\operatorname{supp}(X_i) = [a_i, b_i]$. Using Exercise Q2.9 (*Hoeffding's Lemma*), prove that for any $t > 0$,

$$\mathbb{P}\left\{\sum_{i=1}^{N}(X_i - \mathbb{E}[X_i]) \geqslant t\right\} \leqslant \exp\left(-\frac{2t^2}{\sum_{i=1}^{N}(b_i - a_i)^2}\right).$$

Prf. Let $S_N = \sum_{i=1}^N (X_i - \mathbb{E}[X_i])$, then

$$\begin{aligned}\mathbb{E}[\mathrm{e}^{\lambda S_N}] &= \mathbb{E}\left[\exp\left(\lambda \sum_{i=1}^N (X_i - \mathbb{E}[X_i])\right)\right] \\ &= \prod_{i=1}^N \mathbb{E}[\mathrm{e}^{\lambda(X_i - \mathbb{E}[X_i])}],\end{aligned}$$

By Hoeffding's lemma (Q2.9 (*Hoeffding's Lemma*)), we obtain

$$\begin{aligned}\mathbb{E}[\mathrm{e}^{\lambda S_N}] &\leqslant \prod_{i=1}^N \exp\left(\frac{\lambda^2 (b_i - a_i)^2}{8}\right) \\ &= \exp\left(\sum_{i=1}^N \frac{\lambda^2 (b_i - a_i)^2}{8}\right) \\ &= \exp\left(\frac{\lambda^2}{8} \sum_{i=1}^N (b_i - a_i)^2\right).\end{aligned}$$

By the Chernoff bound, we obtain

$$\mathbb{P}\{S_N \geqslant t\} = \mathbb{P}\{\mathrm{e}^{\lambda S_N} \geqslant \mathrm{e}^{\lambda t}\} \leqslant \mathrm{e}^{-\lambda t}\mathbb{E}[\mathrm{e}^{\lambda S_N}] \leqslant \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^N (b_i - a_i)^2\right),$$

The optimal parameter on the right-hand side is $\lambda_0 = 4t / \sum_{i=1}^N (b_i - a_i)^2$. Substituting it gives

$$\mathbb{P}\left\{\sum_{i=1}^N (X_i - \mathbb{E}[X_i]) \geqslant t\right\} \leqslant \exp\left(-\frac{2t^2}{\sum_{i=1}^N (b_i - a_i)^2}\right).$$

11 [**Chernoff Inequality: Left-Tail Bound**] Let $X_1, \cdots, X_N$ be independent random variables with $X_i \sim \text{Bernoulli}(p_i)$, and let $\mu = \mathbb{E}[S_N] = \sum_{i=1}^N p_i$.

By Remark R2.3.2 (*Chernoff inequality: left tails*), for any $0 < t < \mu$,

$$\mathbb{P}\{S_N \leqslant t\} \leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{t}\right)^t.$$

Prove that for any $0 < \delta < 1$, the following left-tail deviation inequality holds:

$$\mathbb{P}\{S_N \leqslant (1-\delta)\mu\} \leqslant \exp\left(-\frac{\mu\delta^2}{2}\right).$$

(*Remark: This conclusion is tighter than the right-tail deviation bound* $\exp(-\mu\delta^2/3)$.)

Prf. By Remark R2.3.2 (*Chernoff inequality: left tails*), let $t = (1-\delta)\mu$, we obtain

$$\begin{aligned}\mathbb{P}\{S_N \leqslant (1-\delta)\mu\} &\leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{(1-\delta)\mu}\right)^{(1-\delta)\mu} \\ &= \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}}{1-\delta}\right)^{(1-\delta)\mu} \\ &= \mathrm{e}^{-\delta\mu} \cdot (1-\delta)^{-(1-\delta)\mu} \\ &= \exp[-\mu(\delta + (1-\delta)\log(1-\delta))],\end{aligned}$$

Therefore it suffices to prove

$$\delta + (1-\delta)\log(1-\delta) \geqslant \frac{\delta^2}{2}.$$

It is easy to see that

$$\delta + (1-\delta)\log(1-\delta) = \delta + (1-\delta)\left(-\delta - \frac{\delta^2}{2} - \frac{\delta^3}{3} - \cdots\right)$$

$$
\begin{aligned}
&= \delta - \delta - \frac{\delta^2}{2} - \frac{\delta^3}{3} - \cdots + \delta^2 + \frac{\delta^3}{2} + \frac{\delta^4}{3} + \cdots \\
&= (\delta - \delta) + \left(\delta^2 - \frac{\delta^2}{2}\right) + \left(\frac{\delta^3}{2} - \frac{\delta^3}{3}\right) + \cdots \\
&= \frac{\delta^2}{2} + \frac{\delta^3}{6} + \cdots \\
&\geqslant \frac{\delta^2}{2}.
\end{aligned}
$$

12 [**Inverse Chernoff Inequality**] Let $S_N \sim B(N, \mu/N)$. Prove that for any integer t satisfying $\mu \leqslant t \leqslant N$,

$$
\mathbb{P}\{S_N = t\} \geqslant \mathrm{e}^{-\mu}\left(\frac{\mu}{t}\right)^t.
$$

Prf. It is easy to see that

$$
\begin{aligned}
\mathbb{P}\{S_N = t\} &= \binom{N}{t}\frac{\mu^t}{N^t}\left(1 - \frac{\mu}{N}\right)^{N-t} \\
&= \left[\binom{N}{t}\frac{t^t}{N^t}\right]\left(\frac{\mu}{t}\right)^t\left(1 - \frac{\mu}{N}\right)^{N-t},
\end{aligned}
$$

Therefore, to prove $\mathbb{P}\{S_N = t\} \geqslant \mathrm{e}^{-\mu}(\mu/t)^t$, it suffices to prove

$$
\left[\binom{N}{t}\left(\frac{t}{N}\right)^t\right] \cdot \left(1 - \frac{\mu}{N}\right)^{N-t} \geqslant \mathrm{e}^{-\mu}.
$$

We state the following lemmas:

Lemma 1 For $1 \leqslant t \leqslant N$,

$$
\binom{N}{t}\left(\frac{t}{N}\right)^t \geqslant 1.
$$

Lemma 2 For $\mu \leqslant t \leqslant N$,

$$
\left(1 - \frac{\mu}{N}\right)^{N-t} \geqslant \mathrm{e}^{-\mu}.
$$

Rk. Proof of Lemma 1:

It is easy to see that

$$
\begin{aligned}
\binom{N}{t}\left(\frac{t}{N}\right)^t &= \frac{N(N-1)\cdots(N-t+1)}{t!} \cdot \frac{t^t}{N^t} \\
&= \frac{N(N-1)\cdots(N-t+1)}{N^t} \cdot \frac{t^t}{t!} \\
&= \left(\frac{N}{N} \cdot \frac{N-1}{N} \cdots \frac{N-t+1}{N}\right) \cdot \left(\frac{t}{1} \cdot \frac{t}{2} \cdots \frac{t}{t}\right) \\
&= \prod_{k=0}^{t-1}\left(\frac{N-k}{N} \cdot \frac{t}{t-k}\right) \\
&= \prod_{k=0}^{t-1} \frac{Nt - kt}{Nt - Nk} \geqslant 1.
\end{aligned}
$$

By Lemmas 1 and 2, we obtain

$$
\begin{aligned}
\mathbb{P}\{S_N = t\} &= \left[\binom{N}{t}\left(\frac{t}{N}\right)^t\right] \cdot \left(1 - \frac{\mu}{N}\right)^{N-t} \cdot \left(\frac{\mu}{t}\right)^t \\
&\geqslant 1 \cdot \mathrm{e}^{-\mu} \cdot \left(\frac{\mu}{t}\right)^t
\end{aligned}
$$

$$= \mathrm{e}^{-\mu}\left(\frac{\mu}{t}\right)^t.$$

Rk. Proof of Lemma 2:

It is easy to prove that

$$\left(1-\frac{\mu}{N}\right)^{N-t} \geqslant \left(1-\frac{\mu}{N}\right)^{N-\mu},$$

Let $x = \mu/N \in (0,1)$ (*if $\mu = N$, the conclusion is trivial*), then it suffices to prove

$$(1-x)^{N(1-x)} \geqslant \mathrm{e}^{-Nx},$$

Taking the N-th root on both sides and then taking logarithms, this is equivalent to proving

$$(1-x)\log(1-x) \geqslant -x.$$

Construct an auxiliary function $f(x) = (1-x)\log(1-x) + x$, satisfying $f(0) = 0$, then

$$f'(x) = -1 \cdot \log(1-x) + (1-x)\cdot\frac{-1}{1-x} + 1 = -\log(1-x) > 0,$$

therefore $f(x) > 0$, i.e., $(1-x)\log(1-x) > -x$.

13 [**Tail Bounds for the Poisson Distribution**] Thanks to Theorem T1.7.6 (*Poisson limit theorem*), it is reasonable to expect that the Chernoff inequality also applies to Poisson-distributed random variables. Indeed, this is the case. Let $X \sim P(\mu)$. Using the exponential moment method, prove:

(1) (Version of Theorem T2.3.1 (*Chernoff inequality*)) For any $t \geqslant \mu$,

$$\mathbb{P}\{X \geqslant t\} \leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{t}\right)^t.$$

(2) (Version of Remark R2.3.2 (*Chernoff inequality: left tails*)) For any $0 < t \leqslant \mu$,

$$\mathbb{P}\{X \leqslant t\} \leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{t}\right)^t.$$

(3) (Version of Corollary C2.3.4 (*Chernoff inequality: small deviations*)) For any $0 \leqslant \delta \leqslant 1$,

$$\mathbb{P}\{|X-\mu| \geqslant \delta\mu\} \leqslant 2\exp\left(-\frac{\delta^2\mu}{3}\right).$$

(4) (Version of Exercise Q2.12 (*Inverse Chernoff Inequality*): point probability lower bound / inverse Chernoff) For any integer $t > 0$,

$$\mathbb{P}\{X = t\} \geqslant \mathrm{e}^{-\mu}\left(\frac{\mu}{t}\right)^t.$$

Prf. (1) By the Chernoff bound, we obtain that for any $\lambda > 0$,

$$\mathbb{P}\{X \geqslant t\} = \mathbb{P}\{\mathrm{e}^{\lambda X} \geqslant \mathrm{e}^{\lambda t}\} \leqslant \mathrm{e}^{-\lambda t}\mathbb{E}[\mathrm{e}^{\lambda X}] = \exp(-\lambda t + \mu(\mathrm{e}^{\lambda}-1)),$$

The optimal parameter on the right-hand side is $\lambda_0 = \log(t/\mu)$. Substituting it gives

$$\mathbb{P}\{X \geqslant t\} \leqslant \exp\left(\log\left(\left(\frac{\mu}{t}\right)^t\right) + t - \mu\right)$$

$$= \left(\frac{\mu}{t}\right)^t \mathrm{e}^t\mathrm{e}^{-\mu}$$

$$= \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{t}\right)^t.$$

(2) Similarly to the conclusion of (1), we obtain that for any $\lambda < 0$,

$$\mathbb{P}\{X \leqslant t\} = \mathbb{P}\{\mathrm{e}^{\lambda X} \geqslant \mathrm{e}^{\lambda t}\} \leqslant \mathrm{e}^{-\lambda t}\mathbb{E}[\mathrm{e}^{\lambda X}] = \exp(-\lambda t + \mu(\mathrm{e}^{\lambda}-1)),$$

The optimal parameter on the right-hand side is $\lambda_0 = \log(t/\mu)$. Substituting it gives

$$\mathbb{P}\{X \geqslant t\} = \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{t}\right)^t.$$

(3) Decompose $\mathbb{P}\{|X-\mu| \geqslant \delta\mu\}$,

$$\mathbb{P}\{|X-\mu| \geqslant \delta\mu\} = \mathbb{P}\{X \geqslant (1+\delta)\mu\} + \mathbb{P}\{X \leqslant (1-\delta)\mu\}.$$

For the right tail, using the conclusion of (1), let $t = (1+\delta)\mu$, we obtain

$$\begin{aligned}
\mathbb{P}\{X \geqslant (1+\delta)\mu\} &\leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{(1+\delta)\mu}\right)^{(1+\delta)\mu} \\
&= \mathrm{e}^{-\mu}\cdot \mathrm{e}^{(1+\delta)\mu}\cdot(1+\delta)^{-(1+\delta)\mu} \\
&= \exp(-\mu + (1+\delta)\mu - (1+\delta)\mu\log(1+\delta)) \\
&= \exp[-\mu((1+\delta)\log(1+\delta)-\delta)] \\
&\leqslant \exp\left(-\frac{\delta^2\mu}{3}\right),
\end{aligned}$$

For the left tail, using the conclusion of (2), let $t = (1-\delta)\mu$, we obtain

$$\begin{aligned}
\mathbb{P}\{X \geqslant (1-\delta)\mu\} &\leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{(1-\delta)\mu}\right)^{(1-\delta)\mu} \\
&= \mathrm{e}^{-\mu}\cdot \mathrm{e}^{(1+\delta)\mu}\cdot(1-\delta)^{-(1-\delta)\mu} \\
&= \exp(-\mu + (1-\delta)\mu - (1-\delta)\mu\log(1-\delta)) \\
&= \exp[-\mu((1-\delta)\log(1-\delta)+\delta)] \\
&\leqslant \exp\left(-\frac{\delta^2\mu}{2}\right),
\end{aligned}$$

therefore

$$\begin{aligned}
\mathbb{P}\{|X-\mu| \geqslant \delta\mu\} &= \mathbb{P}\{X \geqslant (1+\delta)\mu\} + \mathbb{P}\{X \leqslant (1-\delta)\mu\} \\
&\leqslant \exp\left(-\frac{\delta^2\mu}{2}\right) + \exp\left(-\frac{\delta^2\mu}{3}\right) \\
&\leqslant 2\exp\left(-\frac{\delta^2\mu}{3}\right).
\end{aligned}$$

(4) It is easy to see that

$$\mathbb{P}\{X=t\} = \frac{\mu^t\mathrm{e}^{-\mu}}{t!} \geqslant \frac{\mu^t\mathrm{e}^{-\mu}}{t^t} = \mathrm{e}^{-\mu}\left(\frac{\mu}{t}\right)^t.$$

14 [**Chernoff Inequality: Small Deviation Generalization**] Generalize Corollary C2.3.4 (*Chernoff inequality: small deviations*).

Let $\{X_N\}$ be independent with $X_i \sim$ Bernoulli(p_i), and let $\mu = \mathbb{E}[S_N] = \sum_{i=1}^N p_i$. Prove that for any $\delta \geqslant 0$, the following two-sided deviation inequality holds:

$$\mathbb{P}\{|S_N-\mu| \geqslant \delta\mu\} \leqslant 2\exp\left(-\frac{\delta^2\mu}{2+\delta}\right).$$

Prf. Decompose $\mathbb{P}\{|S_N-\mu| \geqslant \delta\mu\}$,

$$\mathbb{P}\{|S_N-\mu| \geqslant \delta\mu\} = \mathbb{P}\{S_N \geqslant (1+\delta)\mu\} + \mathbb{P}\{S_N \leqslant (1-\delta)\mu\}.$$

For the right tail, by the Chernoff bound, we obtain that for any $\delta > 0$,

$$\mathbb{P}\{S_N \geqslant (1+\delta)\mu\} \leqslant \left(\frac{\mathrm{e}^\delta}{(1+\delta)^{1+\delta}}\right)^\mu = \exp(-\mu((1+\delta)\log(1+\delta)-\delta)),$$

Then it suffices to prove

$$(1+\delta)\log(1+\delta) - \delta \geqslant \frac{\delta^2}{2+\delta}.$$

Construct an auxiliary function $f(\delta) = (1+\delta)\log(1+\delta) - \delta - \delta^2/(2+\delta)$, then

$$\begin{aligned} f'(\delta) &= \left[1 \cdot \log(1+\delta) + (1+\delta)\frac{1}{1+\delta}\right] - 1 - \frac{2\delta(2+\delta) - \delta^2 \cdot 1}{(2+\delta)^2} \\ &= \log(1+\delta) - \frac{\delta(\delta+4)}{(\delta+2)^2}, \\ f''(\delta) &= \frac{1}{1+\delta} - \frac{(\delta+2)^2(2\delta+4) - (\delta^2+4\delta)\cdot 2(\delta+2)}{(\delta+2)^4} \\ &= \frac{1}{1+\delta} - \frac{8}{(\delta+2)^3} \geqslant 0, \end{aligned}$$

satisfying $f(0) = f'(0) = 0$, therefore $f'(\delta) > 0$, hence $f(\delta) > 0$.

For the left tail:

a. When $\delta > 1$, $(1-\delta)\mu < 0$, and $\mathbb{P}\{S_N < 0\} = 0$, which is trivial.

b. When $0 \leqslant \delta \leqslant 1$, by the Chernoff bound, we obtain that for any $\delta > 0$,

$$\mathbb{P}\{S_N \geqslant (1+\delta)\mu\} \leqslant \left(\frac{\mathrm{e}^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{\mu} = \exp(-\mu((1-\delta)\log(1-\delta) + \delta)),$$

Then it suffices to prove

$$(1-\delta)\log(1-\delta) + \delta \geqslant \frac{\delta^2}{2+\delta}.$$

It is easy to see that

$$(1-\delta)\log(1-\delta) + \delta \geqslant \frac{\delta^2}{2} \geqslant \frac{\delta^2}{2+\delta}.$$

Thus,

$$\begin{aligned} \mathbb{P}\{|S_N - \mu| \geqslant \delta\mu\} &= \mathbb{P}\{S_N \geqslant (1+\delta)\mu\} + \mathbb{P}\{S_N \leqslant (1-\delta)\mu\} \\ &\leqslant \exp\left(-\frac{\delta^2\mu}{2+\delta}\right) + \exp\left(-\frac{\delta^2\mu}{2+\delta}\right) \\ &= 2\exp\left(-\frac{\delta^2\mu}{2+\delta}\right). \end{aligned}$$

15 [**Generalization of Chernoff's Inequality for Bounded Random Variables**] Show that all versions of Theorem T2.3.1 (*Chernoff inequality*), Remark R2.3.2 (*Chernoff inequality: left tails*), Corollary C2.3.4 (*Chernoff inequality: small deviations*) hold not only for Bernoulli random variables, but also for any independent random variables X_i taking values in $[0,1]$. Specifically, let $\{X_N\}$ be independent with $\mathrm{supp}(X_i) = [0,1]$, $\mathbb{E}[X_i] = p_i$, and let $\mu = \mathbb{E}[S_N] = \sum_{i=1}^N p_i$. Prove:

(1) By Theorem T2.3.1 (*Chernoff inequality*), for any $t \geqslant \mu$,

$$\mathbb{P}\{S_N \geqslant t\} \leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{t}\right)^t.$$

(2) By Remark R2.3.2 (*Chernoff inequality: left tails*), for any $0 < t \leqslant \mu$,

$$\mathbb{P}\{S_N \leqslant t\} \leqslant \mathrm{e}^{-\mu}\left(\frac{\mathrm{e}\mu}{t}\right)^t.$$

(3) By Corollary C2.3.4 (*Chernoff inequality: small deviations*), for any $0 \leqslant \delta \leqslant 1$,

$$\mathbb{P}\{|S_N - \mu| \geqslant \delta\mu\} \leqslant 2\exp\left(-\frac{\delta^2\mu}{3}\right).$$

Prf. By Exercise Q2.8 (*MGF Comparison Inequality*), we obtain

$$\mathbb{E}[\mathrm{e}^{\lambda X}] \leqslant \mathbb{E}[\mathrm{e}^{\lambda Y}] = 1 - p + p\mathrm{e}^{\lambda}.$$

where $Y \sim \text{Bernoulli}(p)$.

Since

$$\begin{aligned}\mathbb{E}[\mathrm{e}^{\lambda S_N}] = \prod_{i=1}^{N} \mathbb{E}[\mathrm{e}^{\lambda X_i}] &\leqslant \prod_{i=1}^{N} (1 - p_i + p_i \mathrm{e}^{\lambda}) \\ &= \prod_{i=1}^{N} (1 + p_i(\mathrm{e}^{\lambda} - 1)) \\ &\leqslant \prod_{i=1}^{N} \exp(p_i(\mathrm{e}^{\lambda} - 1)) \\ &= \exp\left((\mathrm{e}^{\lambda} - 1) \sum_{i=1}^{N} p_i\right) \\ &= \exp(\mu(\mathrm{e}^{\lambda} - 1)),\end{aligned}$$

This is the same MGF upper bound as $P(\mu), B(N, \mu/N)$, so all subsequent Chernoff bound derivation steps depend only on the MGF upper bound and are independent of the specific distribution of X_i, hence the proofs of (1)(2)(3) are identical to Q13.

16 [**Mean-Median Method: Correcting the Proof**] In the proof of Theorem T2.4.1 (*Median-of-means estimator*), there is a slight flaw: the original proof, for convenience, divided the sample into $B = t^2/4$ groups, and then assumed that each group size $L = N/B$ is an integer, i.e., $B \mid N$. However, this does not hold in general.

Correct the above proof so that it still holds when $B \nmid N$.

By Theorem T2.4.1 (*Median-of-means estimator*), let X satisfy $\mu = \mathbb{E}[X]$, $\sigma^2 = \mathrm{Var}(X)$, and let $X_1, \cdots, X_N$ be independent copies of X. Prove that for any $0 \leqslant t \leqslant \sqrt{N}$, there exists an estimator $\hat{\mu} = \hat{\mu}(X_1, \cdots, X_N)$ such that

$$\mathbb{P}\left\{|\hat{\mu} - \mu| \geqslant \frac{t\sigma}{\sqrt{N}}\right\} \leqslant 2\mathrm{e}^{-ct^2}. \qquad (c > 0)$$

Prf. • Defining the grouping parameters

a. When $t < 4$, let $c = \log 2/16$. Then $2\mathrm{e}^{-ct^2} \geqslant 1$, so the claim is trivial.

b. When $t \geqslant 4$, let $B = \lfloor t^2/8 \rfloor$ be the number of groups, and $G_1, \cdots, G_B$ be B disjoint index sets, each of size $L = \lfloor N/B \rfloor$. (*i.e., we only use the first $B \cdot L$ samples and discard the remaining $N - BL$ samples, at most $B - 1$ samples. Since $B \leqslant N/8$, we have $N/B \geqslant 8$ and hence $L \geqslant 1$.*)

• Analyzing the properties of a single group mean

For any group $b \in \{1, \cdots, B\}$, construct the sample mean of the b-th group as

$$\mu_b = \frac{1}{L} \sum_{i \in G_b} X_i,$$

satisfying

$$\mathbb{E}[\mu_b] = \mu, \mathrm{Var}(\mu_b) = \frac{\sigma^2}{L},$$

Then by Chebyshev's inequality, we obtain

$$p_b = \mathbb{P}\left\{|\mu_b - \mu| \geqslant \frac{t\sigma}{\sqrt{N}}\right\} \leqslant \frac{\mathrm{Var}(\mu_b)}{(t\sigma/\sqrt{N})^2} = \frac{\sigma^2/L}{t^2\sigma^2/N} = \frac{N}{Lt^2},$$

where

$$\frac{N}{Lt^2} \leqslant \frac{N}{(N/2B)t^2} = \frac{2B}{t^2} \leqslant \frac{1}{4}, \qquad \left(L = \left\lfloor \frac{N}{B} \right\rfloor \geqslant \frac{N}{2B}, \quad B = \left\lfloor \frac{t^2}{8} \right\rfloor \leqslant \frac{t^2}{8}\right)$$

(*When $x \geqslant 1$, $\lfloor x \rfloor \geqslant x/2$, so we need to ensure $N/B \geqslant 1$, which explains why we needed to discuss the $B < 1$ case earlier.*)

Therefore

$$\mathbb{P}\left\{|\mu_b - \mu| \geqslant \frac{t\sigma}{\sqrt{N}}\right\} \leqslant \frac{1}{4},$$

i.e., the probability of a single group estimate failing satisfies $p_b \leqslant 1/4$.

- Using the median property and Hoeffding's inequality

  Define $Z_b = \mathbf{1}_{\{|\mu_b - \mu| \geqslant \varepsilon\}}$, where $\varepsilon = t\sigma/\sqrt{N}$, then $\{Z_B\}$ are independent, satisfying $\mathbb{P}\{Z_b = 1\} = \mathbb{E}[Z_b] \leqslant 1/4$.

  Define the median of the B group means as

$$\hat{\mu} = \mathrm{Med}(\mu_1, \cdots, \mu_B),$$

  then

$$\{|\hat{\mu} - \mu| \geqslant \varepsilon\} \subset \left\{\sum_{b=1}^{B} Z_b \geqslant \frac{B}{2}\right\},$$

  (*If $\hat{\mu}$ satisfies $|\hat{\mu} - \mu| \geqslant \varepsilon$, then at least half of the group means μ_b deviate from μ by more than ε.*)

  Therefore

$$\begin{aligned}
\mathbb{P}\left\{\sum_{b=1}^{B} Z_b \geqslant \frac{B}{2}\right\} &= \mathbb{P}\left\{\frac{1}{B}\sum_{b=1}^{B} Z_b \geqslant \frac{1}{2}\right\} \\
&= \mathbb{P}\left\{\frac{1}{B}\sum_{b=1}^{B}(Z_b - \mathbb{E}[Z_b]) \geqslant \frac{1}{2} - \mathbb{E}[Z_b]\right\} \\
&\leqslant \mathbb{P}\left\{\frac{1}{B}\sum_{b=1}^{B}(Z_b - \mathbb{E}[Z_b]) \geqslant \frac{1}{4}\right\},
\end{aligned}$$

  Hence by Hoeffding's inequality, we obtain

$$\begin{aligned}
\mathbb{P}\left\{\frac{1}{B}\sum_{b=1}^{B} Z_b \geqslant \frac{1}{2}\right\} &\leqslant \mathbb{P}\left\{\frac{1}{B}\sum_{b=1}^{B}(Z_b - \mathbb{E}[Z_b]) \geqslant \frac{1}{4}\right\} \\
&\leqslant \exp\left(-2B\left(\frac{1}{4}\right)^2\right) \\
&= \exp\left(-\frac{B}{8}\right) \\
&\sim \exp\left(-\frac{t^2}{64}\right).
\end{aligned}$$

17 [**No Mean Estimator Is Sub-Gaussian Everywhere**] We consider whether Theorem T2.4.1 (*Median-of-means estimator*) holds for all $t > 0$. The answer is negative. In fact, no mean estimator can provide sub-Gaussian tail bounds for all confidence levels.

Using a variant of Le Cam's two-point method, prove that the following assertion is false:

False assertion Let $\{X_N\}$ be i.i.d. with $\mu = \mathbb{E}[X_1]$, $\sigma^2 = \mathrm{Var}(X_1)$. For any $N \in \mathbb{Z}^+$ and any $t \geqslant 0$, there exists $\hat{\mu} : \mathbb{R}^N \to \mathbb{R}$ such that

$$\mathbb{P}\left\{|\hat{\mu}(X_1, \cdots, X_N) - \mu| \geqslant \frac{t\sigma}{\sqrt{N}}\right\} \leqslant 2\mathrm{e}^{-ct^2}. \qquad (c > 0)$$

Prove as follows:

(1) Consider two random vectors $\boldsymbol{X} = (X_1, \cdots, X_N)$ and $\boldsymbol{Y} = (Y_1, \cdots, Y_N)$ with independent coordinates: $X_i \sim \mathrm{Laplace}(0, 1)$ and $Y_i \sim \mathrm{Laplace}(\mu, 1)$ for some $\mu > 0$. Show that $\mathbb{P}\{\boldsymbol{X} \in B\} \leqslant \mathrm{e}^{\mu}\mathbb{P}\{\boldsymbol{Y} \in B\}$ for any measurable subset $B \subset \mathbb{R}^N$.

(2) Assume for contradiction that the assertion holds. Apply it to bound the probabilities of the events $|\hat{\mu}(\boldsymbol{X})| \geqslant \mu/2$ and $|\hat{\mu}(\boldsymbol{Y}) - \mu| \geqslant \mu/2$.

(3) Choose μ large enough and replace $\boldsymbol{Y}$ with $\boldsymbol{X}$ using (1), so that both $|\hat{\mu}(\boldsymbol{X})| < \mu/2$ and $|\hat{\mu}(\boldsymbol{X}) - \mu| < \mu/2$ are likely. Use

the triangle inequality to arrive at a contradiction; thus, the assertion is false.

Prf. It suffices to prove that when $N = 1$, the above assertion leads to a contradiction. In this case, the observation is a single random variable, and $\hat{\mu}$ is a univariate function $\hat{\mu}(x)$.

- Proving the measure change inequality

  Define $\mathbb{P}_0, \mathbb{P}_\mu$ as measures induced on $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$ satisfying:

  1. $\mathbb{P}_0$: $X \sim \text{Laplace}(0, 1)$, satisfying

  $$\mu_0 = 0, \sigma^2 = 2, f_0(x) = \frac{1}{2}\mathrm{e}^{-|x|};$$

  2. $\mathbb{P}_\mu$: $Y \sim \text{Laplace}(\mu, 1)$, satisfying

  $$\mu_Y = \mu, \sigma^2 = 2, f_\mu(x) = \frac{1}{2}\mathrm{e}^{-|x-\mu|}, \qquad (\mu > 0)$$

  then

  $$\frac{\mathrm{d}\mathbb{P}_0}{\mathrm{d}\mathbb{P}_\mu}(x) = \frac{\mathrm{d}\mathbb{P}_0/\mathrm{d}\lambda}{\mathrm{d}\mathbb{P}_\mu/\mathrm{d}\lambda}(x) = \frac{f_0(x)}{f_\mu(x)} = \frac{\mathrm{e}^{-|x|}/2}{\mathrm{e}^{-|x-\mu|}/2} = \mathrm{e}^{|x-\mu|-|x|} \leqslant \mathrm{e}^{\mu},$$

  therefore

  $$\mathbb{P}_0(B) = \int_B \mathrm{d}\mathbb{P}_0 = \int_B \frac{\mathrm{d}\mathbb{P}_0}{\mathrm{d}\mathbb{P}_\mu}\mathrm{d}\mathbb{P}_\mu \leqslant \int_B \mathrm{e}^{\mu}\mathrm{d}\mathbb{P}_\mu = \mathrm{e}^{\mu}\mathbb{P}_\mu(B).$$

- Proving the assertion is false

  Suppose the assertion holds. Let

  $$\frac{t\sqrt{2}}{\sqrt{1}} = \frac{\mu}{2} \implies t = \frac{\mu}{2\sqrt{2}},$$

  then

  $$2\mathrm{e}^{-ct^2} = 2\exp\left(-c\left(\frac{\mu}{2\sqrt{2}}\right)^2\right) = \underbrace{2\exp\left(-\frac{c\mu^2}{8}\right)}_{\delta(\mu)},$$

  Therefore, for the same estimator $\hat{\mu}$,

  $$\mathbb{P}_0\left\{|\hat{\mu}(X)| \geqslant \frac{\mu}{2}\right\} \leqslant \delta(\mu), \mathbb{P}_\mu\left\{|\hat{\mu}(Y) - \mu| \geqslant \frac{\mu}{2}\right\} \leqslant \delta(\mu).$$

  Define $B = \{x \in \mathbb{R} : |\hat{\mu}(x)| < \mu/2\}$, then:

  a. If $x \in B$, then $|\hat{\mu}(x)| < \mu/2$.

  b. If $x \in B^c$, then $|\hat{\mu}(x)| \geqslant \mu/2$.

  For $\mathbb{P}_0$,

  $$\mathbb{P}_0(B^c) = \mathbb{P}\left\{|\hat{\mu}(X)| \geqslant \frac{\mu}{2}\right\} \leqslant \delta(\mu),$$

  therefore

  $$\mathbb{P}_0(B) = 1 - \mathbb{P}\{X \in B^c\} \geqslant 1 - \delta(\mu).$$

  For $\mathbb{P}_\mu$, by the triangle inequality, we obtain

  $$|\hat{\mu}(y) - \mu| \geqslant |\mu| - |\hat{\mu}(y)| > \mu - \frac{\mu}{2} = \frac{\mu}{2},$$

  therefore $B \subset \{y : |\hat{\mu}(y) - \mu| \geqslant \mu/2\}$, hence

  $$\mathbb{P}\{Y \in B\} \leqslant \mathbb{P}\left\{|\hat{\mu}(Y) - \mu| \geqslant \frac{\mu}{2}\right\} \leqslant \delta(\mu).$$

Thus,

$$\begin{cases} \mathbb{P}_0(B) \geqslant 1-\delta(\mu), \\ \mathbb{P}_\mu(B) \leqslant \delta(\mu), \\ \mathbb{P}_0(B) \leqslant \mathrm{e}^{\mu}\mathbb{P}_\mu(B) \end{cases},$$

hence

$$1-\delta(\mu) \leqslant \mathbb{P}_0(B) \leqslant \mathrm{e}^{\mu}\delta(\mu),$$

i.e.,

$$1 \leqslant (\mathrm{e}^{\mu}+1)\delta(\mu) = (\mathrm{e}^{\mu}+1)\cdot 2\exp\left(-\frac{c\mu^2}{8}\right),$$

where

$$\lim_{\mu\to\infty}(\mathrm{e}^{\mu}+1)2\mathrm{e}^{-\frac{c\mu^2}{8}} = 0.$$

A contradiction.

18 [**Most Vertices in a Sparse Random Graph Have Normal Degrees**] In the previous exercise Q1.10 (*Isolated Vertices in Sparse Random Graphs (Converse of the Isolated Vertex Property)*), we saw that a typical sparse random graph $G \sim G(n,p)$ often contains isolated vertices, which leads to a minimum degree of 0. However, despite this, the degrees of the vast majority of vertices still concentrate around the expected degree $d = (n-1)p$.

Prove that there exists $C > 0$ such that if the expected degree $d \geqslant C$, then with high probability at least 99% of the vertices have degrees lying in $[0.9d, 1.1d]$ (w.p. 0.99).

Prf. Let the vertex set of G be $V = \{1, \cdots, n\}$, and D_v be the degree of vertex v, then $D_v \sim B(n-1,p)$, so $\mathbb{E}[D_v] = (n-1)p = d$. If the degree of v deviates from its expectation by more than 10%, we call it a bad vertex and define the corresponding event $B_v = \{|D_v - d| > 0.1d\}$. Let $N_{\text{bad}} = \sum_{v=1}^{n} \mathbf{1}_{B_v}$, then the objective becomes $\mathbb{P}\{N_{\text{bad}} \leqslant 0.01n\} \geqslant 0.99$, i.e., $\mathbb{P}\{N_{\text{bad}} > 0.01n\} \leqslant 0.01$.

By the Chernoff inequality, we obtain that for any $\delta \in (0,1)$,

$$\mathbb{P}\{|D_v - d| > \delta d\} \leqslant 2\mathrm{e}^{-\frac{\delta^2 d}{3}},$$

Let $\delta = 0.1$, then

$$p_{\text{bad}} = \mathbb{P}(B_v) \leqslant 2\mathrm{e}^{-\frac{(0.1)^2 d}{3}} = 2\mathrm{e}^{-\frac{0.01d}{3}} = 2\mathrm{e}^{-\frac{d}{300}},$$

therefore

$$\mathbb{E}[N_{\text{bad}}] = \mathbb{E}\left[\sum_{v=1}^{n} \mathbf{1}_{B_v}\right] = \sum_{v=1}^{n} \mathbb{P}(B_v) = n \cdot p_{\text{bad}} = 2n\mathrm{e}^{-\frac{d}{300}},$$

Hence by Markov's inequality, we obtain

$$\mathbb{P}\{N_{\text{bad}} > 0.01n\} \leqslant \frac{\mathbb{E}[N_{\text{bad}}]}{0.01n} = \frac{n \cdot p_{\text{bad}}}{0.01n} = 200\mathrm{e}^{-\frac{d}{300}}.$$

Let

$$200\mathrm{e}^{-\frac{d}{300}} \leqslant \frac{1}{100} \implies d \geqslant 300\log(20000) = 2970,$$

then there exists $C = 3000$ such that when $d \geqslant C$,

$$\mathbb{P}\{N_{\text{bad}} > 0.01n\} \leqslant 0.01.$$

19 [**Maximum Degree of a Sparse Random Graph**] In Exercise Q1.10 (*Isolated Vertices in Sparse Random Graphs (Converse of the Isolated Vertex Property)*), we observed that the minimum degree of a typical sparse random graph $G \sim G(n,p)$ is often zero.

Now we focus on the maximum degree $\Delta(G)$ (i.e., the degree of the student with the most friends). Prove that if the expected degree $d = (n-1)p$ satisfies the sparsity condition $d \leqslant c(\log n)^{0.99}$, then there exist $c, c_1, c_2 > 0$ such that when $n \geqslant 3$, the

following inequality holds with high probability,

$$c_1\frac{\log n}{\log\log n}\leqslant\Delta(G)\leqslant c_2\frac{\log n}{\log\log n}\ \text{(w.p. 0.99)}.$$

Prf. Let the vertex set of G be $V=\{1,2,\cdots,n\}$, and D_v be the degree of vertex v, then $D_v\sim B(n-1,p)\xrightarrow{\mathrm{d}}P(d)$.

- Proof of the upper bound

  By the Chernoff bound, we obtain that for any $k>d$,

  $$\mathbb{P}\{D_v\geqslant k\}\leqslant\left(\frac{\mathrm{e}d}{k}\right)^k,$$

  i.e.,

  $$\log\mathbb{P}\{D_v\geqslant k\}\leqslant k(\log(\mathrm{e}d)-\log k),$$

  Let $k=c_2K$, where $K=\log n/\log\log n$, then

  $$\log k=\log c_2+\log\log n-\log\log\log n\sim\log\log n,$$
  $$\log(\mathrm{e}d)=1+\log c+0.99\log\log n\sim0.99\log\log n,$$

  i.e.,

  $$\log(\mathrm{e}d)-\log k\sim(0.99-1)\log\log n=-0.01\log\log n,$$

  therefore

  $$\log\mathbb{P}\{D_v\geqslant k\}\lesssim c_2\frac{\log n}{\log\log n}(-0.01\log\log n)=-0.01c_2\log n,$$

  i.e.,

  $$\mathbb{P}\{D_v\geqslant k\}\leqslant n^{-0.01c_2+o(1)},$$

  hence

  $$\mathbb{P}\{\Delta(G)\geqslant k\}=\mathbb{P}\{\bigcup_{v=1}^n\{D_v\geqslant k\}\}\leqslant\sum_{v=1}^n\mathbb{P}\{D_v\geqslant k\}=n\mathbb{P}\{D_1\geqslant k\}$$
  $$\leqslant n\cdot n^{-0.01c_2}=n^{1-0.01c_2},$$

  (*To ensure* $\mathbb{P}\{\Delta(G)\geqslant k\}\leqslant0.005$) (*it then suffices to let* $c_2>100$.)

- Proof of the lower bound

  (*It suffices to prove that there exists at least one vertex with degree at least* $k'=c_1K$ (w.h.p.).)

  Let $X_v=\mathbf{1}_{\{D_v\geqslant k'\}}$, $Y=\sum_{v=1}^nX_v$ be the number of vertices with degree at least k'. It suffices to prove $\mathbb{P}\{Y>0\}\geqslant0.99$.

  When $k'\gg d$,

  $$p_{k'}=\mathbb{P}\{D_v=k'\}\sim\mathrm{e}^{-d}\frac{d^{k'}}{k'!}\geqslant\mathrm{e}^{-d}\left(\frac{d}{k'}\right)^{k'},$$

  Since

  $$\log p_{k'}\sim k'(\log d-\log k')$$
  $$\sim c_1\frac{\log n}{\log\log n}(O(\log\log n)-\log\log n)$$
  $$\sim-c_1\log n,$$

  therefore $p_{k'}\sim n^{-c_1}$, hence

  $$\mathbb{E}[Y]=np_{k'}\sim n\cdot n^{-c_1}=n^{1-c_1},$$

  hence let $c_1<1$, then $\mathbb{E}[Y]\sim n^{1-c_1}\to\infty$.

  We make the following inference:

  Inference $\mathrm{Var}(Y)=o(\mathbb{E}[Y]^2)$, i.e., $\mathrm{Var}(Y)\sim\mathbb{E}[Y]$.

It is easy to see that

$$\begin{aligned}\mathrm{Var}(Y) &= \sum_{v=1}^{n}\mathrm{Var}(X_v) + \sum_{u\neq v}\mathrm{Cov}(X_u, X_v)\\ &\leqslant \sum_{v=1}^{n}\mathbb{E}[X_v] + \sum_{u\neq v}\mathrm{Cov}(X_u, X_v)\\ &= \mathbb{E}[Y] + \sum_{u\neq v}\mathrm{Cov}(X_u, X_v).\end{aligned}$$

Let I_{uv} be the indicator of the existence of edge (u,v), satisfying $I_{uv}\sim$ Bernoulli(p), and D'_u, D'_v i.i.d. $\sim B(n-2,p)$, *then*

$$\begin{aligned}\mathbb{E}[X_uX_v] &= \mathbb{E}[\mathbf{1}_{\{D'_u+I_{uv}\geqslant k\}}\cdot\mathbf{1}_{\{D'_v+I_{uv}\geqslant k\}}]\\ &= p\cdot\mathbb{P}\{D'_u+1\geqslant k\}\mathbb{P}\{D'_v+1\geqslant k\} + (1-p)\cdot\mathbb{P}\{D'_u\geqslant k\}\mathbb{P}\{D'_v\geqslant k\}\\ &= pA_{k-1}^2 + (1-p)A_k^2, \qquad (A_k = \mathbb{P}\{D'_u\geqslant k\})\\ \mathbb{E}[X_u] &= \mathbb{E}[\mathbf{1}_{\{D'_u+I_{uv}\geqslant k\}}] = pA_{k-1} + (1-p)A_k,\\ \mathbb{E}[X_u]\mathbb{E}[X_v] &= (pA_{k-1}+(1-p)A_k)^2,\end{aligned}$$

therefore

$$\begin{aligned}\mathrm{Cov}(X_u,X_v) &= (pA_{k-1}^2+(1-p)A_k^2) - (pA_{k-1}+(1-p)A_k)^2\\ &= p(1-p)(A_{k-1}-A_k)^2,\end{aligned}$$

where

$$A_{k-1}-A_k = \mathbb{P}\{D'_u\geqslant k-1\} - \mathbb{P}\{D'_u\geqslant k\} = \mathbb{P}\{D'_u = k-1\},$$

hence

$$\mathrm{Cov}(X_u,X_v) = p(1-p)()^2 \leqslant p()^2,$$

hence

$$\begin{aligned}\frac{\mathrm{Var}(Y)}{(\mathbb{E}[Y])^2} &\leqslant \frac{\mathbb{E}[Y]+\sum_{u\neq v}\mathrm{Cov}(X_u,X_v)}{(\mathbb{E}[Y])^2} = \frac{1}{\mathbb{E}[Y]} + \frac{n(n-1)\,\mathrm{Cov}(X_1,X_2)}{n^2(\mathbb{E}[X_1])^2}\\ &\sim \frac{\mathrm{Cov}(X_1,X_2)}{(\mathbb{E}[X_1])^2} = \frac{p()^2}{()^2}.\end{aligned}$$

Since $D_1\sim D'_1$, $k\gg d$, the tail probability is dominated by the leading term, i.e.,

$$\mathbb{P}\{D_1\geqslant k\}\sim\mathbb{P}\{D_1=k\} = \frac{d}{k}\frac{k!}{d^k}\frac{d^{k-1}}{(k-1)!}\sim\frac{d}{k}\mathbb{P}\{D_1=k-1\},$$

hence

$$\begin{aligned}\frac{\mathrm{Var}(Y)}{(\mathbb{E}[Y])^2} &\sim \frac{p()^2}{(d/k)^2()^2} = p\left(\frac{k}{d}\right)^2\\ &\sim \frac{d}{n}\frac{k^2}{d^2} = \frac{k^2}{nd}\sim\frac{(\log n)^2}{nd}\to 0. \qquad (p=d/(n-1)\sim d/n, k\sim\log n)\end{aligned}$$

By Chebyshev's inequality, from the inference, we obtain

$$\mathbb{P}\{Y=0\}\leqslant\mathbb{P}\{|Y-\mathbb{E}[Y]|\geqslant\mathbb{E}[Y]\}\leqslant\frac{\mathrm{Var}(Y)}{(\mathbb{E}[Y])^2}\sim\frac{1}{\mathbb{E}[Y]}\to 0.$$

In summary, there exist $c_1,c_2>0$ such that $\Delta(G)$ with high probability lies in $[c_1K, c_2K]$ (w.p. 0.99).

20 [**Expansion Properties of Random Graphs**] Random graphs $G(n,p)$ exhibit a remarkable property: for any $S,T\subset V$, as long as they are not too small, the number of edges $e(S,T)$ connecting them is proportional to the product of the sizes of the above sets.

Prove that there exists $C > 0$ such that (w.p. $1 - 2^{-n}$)[1], the following event holds: Let $S, T \subset V$ satisfy $S \cap T = \emptyset$. Then whenever $|S||T| \geqslant Cn/p$,

$$0.9p \leqslant \frac{e(S,T)}{|S||T|} \leqslant 1.1p.$$

Prf. Let $|S| = s, |T| = t$, then $e(S,T) \sim B(st, p)$, so $\mu_{S,T} = \mathbb{E}[e(S,T)] = stp$, and the objective becomes

$$0.9stp \leqslant e(S,T) \leqslant 1.1stp,$$

i.e.,

$$|e(S,T) - \mu_{S,T}| \leqslant 0.1\mu_{S,T},$$

By the Chernoff bound, we obtain that for any $\delta \in (0,1)$,

$$\mathbb{P}\{|X - \mu| \geqslant \delta\mu\} \leqslant 2\exp\left(-\frac{\delta^2\mu}{3}\right),$$

Let $\delta = 0.1$, $\mu = stp$, then for a fixed (S,T), the probability that the inequality fails (*i.e., a large deviation occurs*) is

$$p_{\text{fail}}(S,T) = \mathbb{P}\{|e(S,T) - stp| > 0.1stp\} \leqslant 2\exp\left(-\frac{0.01stp}{3}\right) \leqslant 2\exp\left(-\frac{Cn}{300}\right). \qquad \left(st \geqslant \frac{Cn}{p}\right)$$

Let $\mathcal{F}$ be the collection of all disjoint subset pairs (S,T), then the total failure probability is

$$\begin{aligned} P_{\text{total fail}} &= \mathbb{P}(\text{there exists } (S,T) \text{ satisfying } stp \geqslant Cn,\ |e(S,T) - stp| > 0.1stp) \\ &\leqslant \sum_{\substack{(S,T):S\cap T=\emptyset \\ stp\geqslant Cn}} p_{\text{fail}}(S,T), \end{aligned}$$

When (S,T) are disjoint, either $v \in S$, or $v \in T$, or $v \notin S \cup T$, therefore

$$\begin{aligned} P_{\text{total fail}} &= \sum_{(S,T):S\cap T=\emptyset} 2\exp\left(-\frac{Cn}{300}\right) \\ &= 3^n \cdot 2\exp\left(-\frac{Cn}{300}\right) \\ &= 2\exp\left(n\log 3 - \frac{Cn}{300}\right). \end{aligned}$$

To ensure $P_{\text{total fail}} \leqslant 2^{-n} = \exp(-n\log 2)$, it suffices to satisfy

$$n\log 3 - \frac{Cn}{300} \leqslant -n\log 2 - \log 2 \implies C \geqslant 300\left(\log 6 + \frac{\log 2}{n}\right),$$

therefore let $C \geqslant 600$.

21 [**Boosting Randomized Algorithms**] Consider a randomized algorithm $\mathcal{A}$ for solving a decision problem (e.g., primality testing) whose accuracy is slightly better than random guessing, i.e., for any input, the probability that the algorithm gives the correct answer is $p = 1/2 + \varepsilon$, where $\varepsilon > 0$.

To improve accuracy, run $\mathcal{A}$ independently N times and use the majority vote rule as the final output. Prove that for any given error tolerance $\delta \in (0,1)$, if

$$N \geqslant \frac{1}{2\varepsilon^2}\log\left(\frac{1}{\delta}\right)$$

then the probability that the final output is correct is at least $1 - \delta$.

Prf. Let X_i be the indicator variable for the i-th run of the algorithm, then $\{X_N\}$ i.i.d. $\sim$ Bernoulli(p), and $\mu = \mathbb{E}[S_N] = Np = N/2 + N\varepsilon$, then

$$P_{\text{fail}} = \mathbb{P}\left\{S_N \leqslant \frac{N}{2}\right\} = \mathbb{P}\{S_N - \mu \leqslant -N\varepsilon\},$$

[1] The original text was $1 - 2^n$, which has been corrected here.

Therefore by Hoeffding's inequality, letting $a_i = 0, b_i = 1, t = N\varepsilon$, we obtain

$$P_{\text{fail}} = \mathbb{P}\{S_N - \mu \leqslant -N\varepsilon\} \leqslant \exp\left(-\frac{2(N\varepsilon)^2}{N}\right) = \exp(-2N\varepsilon^2).$$

To ensure $P_{\text{fail}} \leqslant \delta$, it suffices to satisfy

$$\exp(-2N\varepsilon^2) \leqslant \delta \implies N \geqslant \frac{1}{2\varepsilon^2}\log\left(\frac{1}{\delta}\right).$$

22 [**Absolute Moments of the Normal Distribution**] Let $g \sim N(0,1)$.

(1) Prove that for any $p \geqslant 1$,

$$\mathbb{E}[|g|^p] = \frac{2^{p/2}}{\sqrt{\pi}}\Gamma\left(\frac{p+1}{2}\right).$$

(2) Prove that as $p \to \infty$,

$$\|g\|_{L^p} = ()^{1/p} = \sqrt{\frac{p}{\mathrm{e}}}(1 + o(1)).$$

Prf. (1) It is easy to see that

$$\begin{aligned}\mathbb{E}[|g|^p] &= \int_{-\infty}^{\infty} |x|^p \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2}\mathrm{d}x \\ &= \sqrt{\frac{2}{\pi}}\int_0^{\infty} x^p \mathrm{e}^{-x^2/2}\mathrm{d}x,\end{aligned}$$

where

$$\begin{aligned}\int_0^{\infty} x^p \mathrm{e}^{-x^2/2}\mathrm{d}x &= \int_0^{\infty} (2^{1/2}u^{1/2})^p \mathrm{e}^{-u}\frac{1}{2^{1/2}u^{1/2}}\mathrm{d}u \qquad \left(u = \frac{x^2}{2}\right) \\ &= 2^{(p-1)/2}\int_0^{\infty} u^{\frac{p+1}{2}-1}\mathrm{e}^{-u}\mathrm{d}u \\ &= 2^{(p-1)/2}\Gamma\left(\frac{p+1}{2}\right),\end{aligned}$$

Substituting this in, we obtain

$$\mathbb{E}[|g|^p] = \frac{2^{p/2}}{\sqrt{\pi}}\Gamma\left(\frac{p+1}{2}\right).$$

(2) From the conclusion of (1), we obtain

$$\|g\|_{L^p} = \left(\frac{2^{p/2}}{\sqrt{\pi}}\Gamma\left(\frac{p+1}{2}\right)\right)^{1/p} = \frac{\sqrt{2}}{(\sqrt{\pi})^{1/p}}\left[\Gamma\left(\frac{p+1}{2}\right)\right]^{1/p},$$

Using Stirling's formula $\Gamma(z) \sim \sqrt{2\pi}z^{z-1/2}\mathrm{e}^{-z}$, we obtain

$$\begin{aligned}[\Gamma(z)]^{1/p} &\sim [\sqrt{2\pi}z^{z-1/2}\mathrm{e}^{-z}]^{1/p} \\ &= (2\pi)^{\frac{1}{2p}} \cdot z^{\frac{z-1/2}{p}} \cdot \mathrm{e}^{-\frac{z}{p}},\end{aligned}$$

where

$$\mathrm{e}^{-\frac{z}{p}} = \mathrm{e}^{-\frac{p+1}{2p}} \to \frac{1}{\sqrt{\mathrm{e}}},$$

$$z^{\frac{z-1/2}{p}} = \left(\frac{p+1}{2}\right)^{\frac{p}{2p}} = \left(\frac{p(1+1/p)}{2}\right)^{\frac{1}{2}} = \left(\frac{p}{2}\right)^{1/2}\left(1+\frac{1}{p}\right)^{1/2} \sim \sqrt{\frac{p}{2}},$$

Substituting this in, we obtain

$$\|g\|_{L^p} \sim \sqrt{2} \cdot 1 \cdot \sqrt{\frac{p}{2\mathrm{e}}} = \sqrt{\frac{p}{\mathrm{e}}},$$

i.e.,

$$\|g\|_{L^p} = \sqrt{\frac{p}{\mathrm{e}}}(1 + o(1)).$$

23 [**The Sub-Gaussian MGF Property Implies Zero Mean**] In the definition of sub-Gaussian random variables (e.g., property (iv) of Proposition P2.6.1 (*Subgaussian properties*)), the condition of zero mean is usually assumed or included. Prove that this is necessary.

Specifically, let X be a random variable. Prove that if for any $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\mathrm{e}^{\lambda X}] \leqslant \mathrm{e}^{C\lambda^2}, \qquad (C > 0)$$

then $\mathbb{E}[X] = 0$.

Prf. **Method 1 Jensen's inequality**

By Jensen's inequality, let $\phi(x) = \mathrm{e}^x$, $Y = \lambda X$, then $\mathrm{e}^{\mathbb{E}[\lambda X]} \leqslant \mathbb{E}[\mathrm{e}^{\lambda X}]$, i.e.,

$$\mathrm{e}^{\lambda \mathbb{E}[X]} \leqslant C\lambda^2.$$

a. If $\lambda > 0$, then $\mathbb{E}[X] \leqslant C\lambda$. Letting $\lambda \to 0^+$, we obtain $\mathbb{E}[X] \leqslant 0$.

b. If $\lambda < 0$, then $\mathbb{E}[X] \geqslant C\lambda$. Letting $\lambda \to 0^-$, we obtain $\mathbb{E}[X] \geqslant 0$.

Thus, $\mathbb{E}[X] = 0$.

Method 2 Derivative property

Construct the auxiliary function

$$h(\lambda) = M(\lambda) - g(\lambda) = \mathbb{E}[\mathrm{e}^{\lambda X}] - \mathrm{e}^{C\lambda^2},$$

where $M(0) = g(0) = 1$, $M(\lambda) \leqslant g(\lambda)$, then $h(0) = 0$, $h(\lambda) \geqslant 0$, so $\lambda = 0$ is the global minimum of $h(\lambda)$, hence

$$h'(0) = M'(0) - g'(0) = 0,$$

substituting $g'(0) = 0$, $M'(0) = \mathbb{E}[X]$, we obtain $\mathbb{E}[X] = 0$.

24 [**Examples of Sub-Gaussian Distributions**] Prove the sub-Gaussian norm $\|\cdot\|_{\psi_2}$ of the following random variables.

(1) (Constant) If $X = c$ (a.s.), then

$$\|X\|_{\psi_2} = \frac{|c|}{\sqrt{\log 2}}.$$

(2) (Bounded) If $\|X\|_\infty < \infty$, then

$$\|X\|_{\psi_2} \leqslant \frac{\|X\|_\infty}{\sqrt{\log 2}}.$$

(3) (Rademacher) If $X \sim$ Rademacher, then

$$\|X\|_{\psi_2} = \frac{1}{\sqrt{\log 2}}.$$

(4) (Normal) If $X \sim N(0, \sigma^2)$, then

$$\|X\|_{\psi_2} = \sqrt{\frac{8}{3}}\sigma.$$

(5) (Bernoulli) If $X \sim$ Bernoulli(p), then

$$\|X\|_{\psi_2} = \frac{1}{\sqrt{\log(1 + 1/p)}}.$$

Prf. (1) Let $X = c$ (a.s.), then

$$\mathbb{E}\left[\exp\left(\frac{X^2}{K^2}\right)\right] = \exp\left(\frac{c^2}{K^2}\right) \leqslant 2 \implies K \geqslant \frac{|c|}{\sqrt{\log 2}},$$

therefore $\|X\|_{\psi_2} = |c|/\sqrt{\log 2}$.

(2) Let $|X| \leqslant \|X\|_\infty = M$ (a.s.), then

$$\mathbb{E}\left[\exp\left(\frac{X^2}{K^2}\right)\right] \leqslant \exp\left(\frac{M^2}{K^2}\right) \leqslant 2 \implies K \geqslant \frac{M}{\sqrt{\log 2}},$$

therefore $\|X\|_{\psi_2} \leqslant \|X\|_\infty / \sqrt{\log 2}$.

(3) Note that $X^2 = 1$, which is the same as the $c = 1$ case of (1), we obtain

$$\mathbb{E}\left[\exp\left(\frac{X^2}{K^2}\right)\right] = \exp\left(\frac{1}{K^2}\right) \leqslant 2 \implies K \geqslant \frac{1}{\sqrt{\log 2}},$$

hence $\|X\|_{\psi_2} = 1/\sqrt{\log 2}$.

(4) It is easy to see that

$$\begin{aligned}\mathbb{E}\left[\exp\left(\frac{X^2}{K^2}\right)\right] &= \int_{-\infty}^{\infty} \mathrm{e}^{x^2/K^2} \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-x^2/(2\sigma^2)} \mathrm{d}x \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{2}\left(\frac{1}{\sigma^2} - \frac{2}{K^2}\right)\right) \mathrm{d}x,\end{aligned}$$

If the above integral converges, then $1/\sigma^2 - 2/K^2 > 0$, i.e., $K^2 > 2\sigma^2$, at this point

$$\mathbb{E}\left[\exp\left(\frac{X^2}{K^2}\right)\right] = \frac{1}{\sqrt{2\pi}\sigma} \cdot \sqrt{\frac{2\pi}{1/\sigma^2 - 2/K^2}} = \frac{1}{\sqrt{1 - 2\sigma^2/K^2}} \leqslant 2 \implies K^2 \geqslant \frac{8}{3}\sigma^2,$$

therefore $\|X\|_{\psi_2} = \sqrt{8/3}\sigma$.

(5) It is easy to see that

$$\mathbb{E}\left[\exp\left(\frac{X^2}{K^2}\right)\right] = (1-p)\mathrm{e}^{0^2/K^2} + p\mathrm{e}^{1^2/K^2} = 1 - p + p\mathrm{e}^{1/K^2} \leqslant 2 \implies K \geqslant \frac{1}{\sqrt{\log(1 + 1/p)}},$$

therefore $\|X\|_{\psi_2} = 1/\sqrt{\log(1 + 1/p)}$.

25 [**Counterexamples of Non-Sub-Gaussian Distributions**] Sub-Gaussian distributions require that the decay rate of their tail probabilities be at least as fast as that of the Gaussian distribution (*i.e.,* $\exp(-ct^2)$). Explain why the following common probability distributions are not sub-Gaussian.

(1) Sub-exponential class: exponential distribution, Gamma distribution, chi-squared distribution, geometric distribution, Poisson distribution.

(2) Heavy-tailed class (polynomial decay): Cauchy distribution, Pareto distribution.

Sol. (1) For $X \sim E(\lambda)$, $\mathbb{P}\{X \geqslant t\} = \mathrm{e}^{-\lambda t}$.

For $X \sim G(p)$, $\mathbb{P}\{X \geqslant k\} = (1-p)^k = \mathrm{e}^{-k\log(1-p)}$.

For $X \sim \Gamma(k, \theta)$, $f(x) \propto x^{k-1}\mathrm{e}^{-k/\theta}$.

For $X \sim P(\lambda)$, by the Chernoff bound, we obtain

$$\mathbb{P}\{X \geqslant t\} \leqslant \mathrm{e}^{-st}\mathbb{E}[\mathrm{e}^{sX}] = \exp(-st + \lambda(\mathrm{e}^s - 1)),$$

The optimal parameter on the right-hand side is $s_0 = \log(t/\lambda)$. Substituting it gives

$$\mathbb{P}\{X \geqslant t\} \leqslant \exp(-t(\log t - \log\lambda) + t - \lambda) \sim \exp(-t\log t).$$

(*When t is close to λ, $P(\lambda)$ is locally sub-Gaussian.*)

(2) For $X \sim \mathrm{Pareto}(\alpha, x_m)$, $\mathbb{P}\{X > t\} = (x_m/t)^\alpha$.

For $X \sim \mathrm{Cauchy}(0, 1)$, it does not even have a first moment, so it is not sub-Gaussian (*a necessary condition for a sub-Gaussian distribution is that all moments exist and are finite*).

In summary,

Distribution type	Tail decay form	Conclusion
Gaussian, sub-Gaussian	$\sim \exp(-t^2)$	Baseline
Exponential, Gamma, Chi-Squared	$\sim \exp(-t)$	No
Poisson	$\sim \exp(-t\log t)$	No
Cauchy, Pareto	$\sim t^{-\alpha}$	No

26 [**Equivalent Characterization of Sub-Gaussian: MGF of X^2**] In Proposition P2.6.1 (*Subgaussian properties*), we have already seen several equivalent ways to characterize sub-Gaussian distributions (e.g., tail probability decay, moment growth rate, etc.). Here we provide another characterization.

Prove that X is a sub-Gaussian random variable if and only if there exists $K > 0$ such that whenever $|\lambda| \leqslant 1/K$,

$$\mathbb{E}[\exp(\lambda^2 X^2)] \leqslant \exp(\lambda^2 K^2).$$

Prf. ($\Longrightarrow$)

If X is a K-sub-Gaussian random variable, then $\mathbb{E}[\exp(X^2/K^2)] \leqslant 2$.

By Jensen's inequality, let $\phi(y) = y^\alpha$, where $\alpha = \lambda^2 K^2 \in [0,1]$, we obtain

$$\mathbb{E}[\exp(\lambda^2 X^2)] = \mathbb{E}\left[\left(\exp\left(\frac{X^2}{K^2}\right)\right)^\alpha\right] \leqslant ()^\alpha \leqslant 2^\alpha < \mathrm{e}^\alpha = \exp(\lambda^2 K^2).$$

($\Longleftarrow$)

To prove that X is a K-sub-Gaussian random variable, it suffices to prove that there exists $t > 0$ such that $\mathbb{E}[\exp(X^2/t^2)] \leqslant 2$.

Let $\lambda = 1/t$, then t satisfies:

1. Domain: if $|\lambda| \leqslant 1/K$, then $t \geqslant K$;
2. The upper bound does not exceed 2: if $\exp(\lambda^2 K^2) = \exp(K^2/t^2) \leqslant 2$, then $t \geqslant K/\sqrt{\log 2}$;

Hence let $t = 2K$, we obtain

$$\mathbb{E}[\exp(\lambda^2 K^2)] = \mathbb{E}\left[\exp\left(\frac{X^2}{4K^2}\right)\right] \leqslant \exp\left(\frac{K^2}{4K^2}\right) = \mathrm{e}^{1/4} < 2,$$

i.e., $\|X\|_{\psi_2} < 2K$.

27 [**Characterization of Sub-Gaussian Distributions: Almost Stochastic Dominance**] Let X, Y be random variables. If $\mathbb{P}\{X \geqslant t\} \leqslant 2\mathbb{P}\{Y \geqslant t\}$, we write $X \preceq Y$.

(1) Prove that X is sub-Gaussian if and only if there exists $K > 0$ such that

$$|X| \preceq K|g|, \qquad (g \sim N(0,1))$$

that is,

$$K \leqslant C\|X\|_{\psi_2}. \qquad (C > 0)$$

(2) Give an example showing that if the factor 2 in the definition of $X \preceq Y$ is replaced by 1, then the conclusion of (1) does not necessarily hold.

Prf. (1) ($\Longleftarrow$)

If $X \preceq K|g|$, then

$$\mathbb{P}\{|X| \geqslant t\} \leqslant 2\mathbb{P}\{K|g| \geqslant t\} = 2\mathbb{P}\left\{|g| \geqslant \frac{t}{K}\right\}.$$

Using the Gaussian tail upper bound $\mathbb{P}\{|g| \geqslant x\} \leqslant 2\exp(-x^2/2)$, let $x = t/K$, we obtain

$$\mathbb{P}\{|X| \geqslant t\} \leqslant 2 \cdot 2\exp\left(-\frac{t^2}{2K^2}\right) = 4\exp\left(-\frac{t^2}{2K^2}\right),$$

therefore X is a sub-Gaussian random variable.

($\Longrightarrow$)

If X is a sub-Gaussian random variable, then there exists $c > 0$ such that

$$\mathbb{P}\{|X| \geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{\sigma^2}\right), \qquad (\sigma = \|X\|_{\psi_2})$$

($\Longrightarrow$)

therefore it suffices to prove that

$$\mathbb{P}\{|X| \geqslant t\} \leqslant 2\mathbb{P}\left\{|g| \geqslant \frac{t}{K}\right\},$$

Simplifying the above objective, let $y = t/K$, we obtain

$$\mathbb{P}\{|X| \geqslant Ky\} \leqslant 2\mathbb{P}\{|g| \geqslant y\}.$$

Let M be the median of the absolute value of g, i.e., $\mathbb{P}\{|g| \geqslant M\} = 1/2$. We discuss the values of y:

a. If $0 \leqslant y \leqslant M$, then $\mathbb{P}\{|g| \geqslant y\} \geqslant \mathbb{P}\{|g| \geqslant M\} = 1/2$, therefore

$$2\mathbb{P}\{|g| \geqslant y\} \geqslant 2\left(\frac{1}{2}\right) = 1,$$

hence $\mathbb{P}\{|X| \geqslant Ky\} \leqslant 2\mathbb{P}\{|g| \geqslant y\}$.

b. If $y > M$, then it suffices to prove that there exists K such that

$$\mathbb{P}\{|X| \geqslant Ky\} \leqslant 2\exp\left(-\frac{K^2y^2}{\sigma^2}\right) \leqslant 2\mathbb{P}\{|g| \geqslant y\},$$

i.e.,

$$\frac{K^2}{\sigma^2} \geqslant -\frac{\log \mathbb{P}\{|g| \geqslant y\}}{y^2}.$$

(*Using the Mills Ratio, we obtain* $\log \mathbb{P}\{|g| \geqslant y\} \sim -y^2/2$*, hence the right-hand side is bounded.*)

($\Longrightarrow$)

Let $f(y) = -\log \mathbb{P}\{|g| \geqslant y\}/y^2$, substituting the Mills Ratio

$$\log \mathbb{P}\{|g| \geqslant y\} \sim -\frac{y^2}{2},$$

we obtain $f(y) \sim -1/2$, hence there exists $C_{\sup} = \sup_{y \geqslant M} g(y)$ such that

$$\frac{K^2}{\sigma^2} \geqslant C_{\sup} \Longrightarrow K \geqslant \sqrt{C_{\sup}}\sigma.$$

(2) We only need to refute the following conclusion:

$$\mathbb{P}\{|X| \geqslant t\} \leqslant \mathbb{P}\{K|g| \geqslant t\}.$$

Suppose the above conclusion holds. Let $X \sim$ Rademacher, then according to Exercise Q2.24 (*Examples of Sub-Gaussian Distributions*), X is a sub-Gaussian random variable, at this point

$$\mathbb{P}\{1 \geqslant t\} \leqslant \mathbb{P}\left\{|g| \geqslant \frac{t}{K}\right\}.$$

But when $t \in (0, 1]$,

$$1 \leqslant \mathbb{P}\left\{|g| \geqslant \frac{t}{K}\right\} < 1,$$

A contradiction.

28 [**Characterization of Sub-Gaussian Distributions: Convex Dominance Definition**] Let X, Y be random variables. If for any convex increasing function $\phi : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}[\phi(X)] \leqslant \mathbb{E}[\phi(Y)],$$

we write $X \precsim Y$. Prove that X is a sub-Gaussian random variable if and only if there exists $K > 0$ such that

$$|X| \precsim K|g|. \qquad (g \sim N(0,1))$$

Rk. In the theory of stochastic ordering, there are the following three concepts:

1. First-order Stochastic Dominance (FSD)

   If

   $$F_X(t) \leqslant F_Y(t),$$

   or for any non-decreasing function $\phi : \mathbb{R} \to \mathbb{R}$, (if the expectations exist)

   $$\mathbb{E}[\phi(X)] \geqslant \mathbb{E}[\phi(Y)],$$

   then X is said to first-order stochastically dominate Y, denoted $X \geqslant_{\text{st}} Y$ or $X \geqslant_{\text{fsd}} Y$.
2. Second-order Stochastic Dominance (SSD)

   If

   $$\int_{-\infty}^{x} F_X(t)\mathrm{d}t \leqslant \int_{-\infty}^{x} F_Y(t)\mathrm{d}t,$$

   or for any non-decreasing concave function $u : \mathbb{R} \to \mathbb{R}$ (*utility function*),

   $$\mathbb{E}[u(X)] \geqslant \mathbb{E}[u(Y)],$$

   then X is said to second-order stochastically dominate Y, denoted $X \geqslant_{\text{ssd}} Y$.
3. Increasing Convex Order (ICX)

   If

   $$\int_{x}^{\infty} \bar{F}_X(t)\mathrm{d}t \leqslant \int_{x}^{\infty} \bar{F}_Y(t)\mathrm{d}t,$$

   or for any non-decreasing convex function $\phi : \mathbb{R} \to \mathbb{R}$,

   $$\mathbb{E}[\phi(X)] \leqslant \mathbb{E}[\phi(Y)],$$

   then X is said to be less than or equal to Y in the increasing convex order, denoted $X \leqslant_{\text{icx}} Y$.

Intuitively:

1. FSD: means X tends to be numerically larger than Y (*distribution shifted to the right*).
2. SSD: means X has a larger mean and lower risk (*dispersion*), and is the choice of risk-averse agents (*concave utility*).
3. ICX: means Y has a heavier right-tail risk. In sub-Gaussian studies, it is often used to characterize that the tail of $|X|$ is "covered" by the tail of the Gaussian distribution $|Kg|$.

Prf. If $X \preceq Y$, then for any $t \geqslant 0$,

$$\int_{x}^{\infty} \mathbb{P}\{X \geqslant t\}\mathrm{d}t \leqslant \int_{x}^{\infty} \mathbb{P}\{Y \geqslant t\}\mathrm{d}t.$$

- ($\Longrightarrow$)

  If X is a sub-Gaussian random variable, then for any $t \geqslant 0$,

  $$\mathbb{P}\{|X| \geqslant t\} \leqslant 2\mathrm{e}^{-ct^2/L^2},$$

  Then it suffices to prove

  $$\underbrace{\int_{x}^{\infty} \mathbb{P}\{|X| \geqslant t\}\mathrm{d}t}_{f(x)} \leqslant \underbrace{\int_{x}^{\infty} \mathbb{P}\{K|g| \geqslant t\}\mathrm{d}t}_{h(x,K)}.$$

  Since

  $$\begin{aligned} f(x) &= \int_{x}^{\infty} 2\mathrm{e}^{-ct^2/L^2}\mathrm{d}t \\ &= \frac{L}{\sqrt{2c}} \int_{x\sqrt{2c}/L}^{\infty} 2\mathrm{e}^{-u^2/2}\mathrm{d}u \qquad \left(u = \frac{t\sqrt{2c}}{L}\right) \end{aligned}$$

$$= \frac{L\sqrt{2\pi}}{\sqrt{c}}\mathbb{P}\left\{|g| \geqslant \frac{x\sqrt{2c}}{L}\right\}$$
$$\sim \frac{L^2}{cx}\mathrm{e}^{-cx^2/L^2}, \qquad \left(\mathbb{P}\{|g| \geqslant x\} \sim \frac{2}{x\sqrt{2\pi}}\mathrm{e}^{-x^2/2}\right)$$
$$h(x,K) = \int_x^\infty \mathbb{P}\{K|g| \geqslant t\}\mathrm{d}t$$
$$= 2K\int_{x/K}^\infty (1-\Phi(u))\mathrm{d}u$$
$$\sim \frac{2K^3}{x^2\sqrt{2\pi}}\mathrm{e}^{-x^2/(2K^2)}, \qquad \left(\int_y^\infty (1-\Phi(u))\mathrm{d}u \sim \frac{1}{y^2\sqrt{2\pi}}\mathrm{e}^{-y^2/2}\right)$$

therefore

$$h(x,K) \geqslant f(x) \Longrightarrow \frac{1}{2K^2} < \frac{c}{L^2},$$

hence let $K > L/\sqrt{2C}$, where C is sufficiently large so that $h(0,K) = \mathbb{E}[K|g|] \geqslant \mathbb{E}[|X|]$, $h(x,K) \geqslant f(x)$.

- ($\Longleftarrow$)

  Construct $\Psi(t) = \mathrm{e}^{t^2/a^2}$, satisfying

$$\Psi'(t) = \frac{2t}{a^2}\mathrm{e}^{t^2/a^2} \geqslant 0,$$
$$\Psi''(t) = \left(\frac{2}{a^2} + \frac{4t^2}{a^4}\right)\mathrm{e}^{t^2/a^2} > 0,$$

  then Ψ is convex and increasing on $\mathbb{R}^+$.

  If $|X| \preceq K|g|$, then

$$\mathbb{E}[\mathrm{e}^{X^2/a^2}] = \mathbb{E}[\Psi(|X|)] \leqslant \mathbb{E}[\Psi(K|g|)] = \mathbb{E}[\mathrm{e}^{K^2g^2/a^2}] = \left(1 - \frac{2K^2}{a^2}\right)^{-1/2},$$

  (*Note that the MGF of* $g^2 \sim \chi^2(1)$ *is* $\mathbb{E}[\mathrm{e}^{\lambda g^2}] = (1-2\lambda)^{-1/2}$ $(\lambda < 1/2)$*, so letting* $\lambda = K^2/a^2$ *gives the last equality.*)

  Therefore

$$\mathbb{E}[\mathrm{e}^{X^2/a^2}] \leqslant 2 \Longleftarrow \left(1 - \frac{2K^2}{a^2}\right)^{-1/2} \leqslant 2$$
$$\Longleftarrow a > \sqrt{\frac{8}{3}}K,$$

  i.e., $\|X\|_{\psi_2} \leqslant \sqrt{8/3}K$.

29 [**Hoeffding's Inequality for Bounded Random Variables**] Using Theorem T2.7.3 (*Subgaussian Hoeffding inequality*), derive the Hoeffding inequality for bounded random variables. Note that the absolute constant in the exponent of the derived result differs from the original theorem.

Theorem T2.2.6: Let $\{X_N\}$ be independent with $\mathrm{supp}(X_i) = [a_i, b_i]$. Then for any $t > 0$,

$$\mathbb{P}\left\{\sum_{i=1}^N (X_i - \mathbb{E}[X_i]) \geqslant t\right\} \leqslant \exp\left(-\frac{2t^2}{\sum_{i=1}^N (b_i - a_i)^2}\right).$$

Theorem T2.7.3: Let $\{X_N\}$ be independent sub-Gaussian random variables with $\mathbb{E}[X_i] = 0$. Then for any $t \geqslant 0$, there exists $c > 0$ such that

$$\mathbb{P}\left\{\left|\sum_{i=1}^N X_i\right| \geqslant t\right\} \leqslant 2\exp\left(-\frac{ct^2}{\sum_{i=1}^N \|X_i\|_{\psi_2}^2}\right).$$

Prf. Center X_i by setting $Y_i = X_i - \mathbb{E}[X_i]$. Then $\mathbb{E}[Y_i] = 0$, and the length of $\mathrm{supp}(Y_i)$ is $b_i - a_i$, so by Hoeffding's lemma

(Q2.9 (*Hoeffding's Lemma*)), we obtain

$$\mathbb{E}[\exp(\lambda Y_i)] \leqslant \exp\left(\frac{\lambda^2(b_i-a_i)^2}{8}\right).$$

By the ψ_2 norm condition for sub-Gaussian distributions $\|Y_i\|_{\psi_2} \leqslant CK$, let

$$K^2 = \frac{(b_i-a_i)^2}{8},$$

we obtain

$$\|Y_i\|_{\psi_2} \leqslant C \cdot \frac{b_i-a_i}{2\sqrt{2}} = C_1(b_i-a_i). \qquad \left(C_1 = \frac{C}{2\sqrt{2}}\right)$$

Therefore $\{Y_N\}$ are sub-Gaussian.

By Theorem T2.7.3 (*Subgaussian Hoeffding inequality*), we obtain

$$\mathbb{P}\left\{\left|\sum_{i=1}^N Y_i\right| \geqslant t\right\} \leqslant 2\exp\left(-\frac{ct^2}{\sum_{i=1}^N \|Y_i\|_{\psi_2}^2}\right),$$

where

$$\sum_{i=1}^N \|Y_i\|_{\psi_2}^2 \leqslant \sum_{i=1}^N C_1^2(b_i-a_i)^2 = C_1^2\sum_{i=1}^N (b_i-a_i)^2,$$

therefore

$$\mathbb{P}\left\{\left|\sum_{i=1}^N (X_i-\mathbb{E}[X_i])\right| \geqslant t\right\} \leqslant 2\exp\left(-\frac{ct^2}{C_1^2\sum_{i=1}^N (b_i-a_i)^2}\right).$$

where

$$\mathbb{P}\left\{\sum_{i=1}^N (X_i-\mathbb{E}[X_i]) \geqslant t\right\} \leqslant \mathbb{P}\left\{\left|\sum_{i=1}^N (X_i-\mathbb{E}[X_i])\right| \geqslant t\right\},$$

hence

$$\mathbb{P}\left\{\sum_{i=1}^N (X_i-\mathbb{E}[X_i]) \geqslant t\right\} \leqslant 2\exp\left(-\frac{C't^2}{\sum_{i=1}^N (b_i-a_i)^2}\right). \qquad \left(C' = \tfrac{c}{C_1^2}\right)$$

30 [**Inverse Hoeffding Inequality**] Let $\{X_N\}$ be independent centered K-sub-Gaussian random variables with $\operatorname{Var}(X_i) = 1$. Prove that for any $\boldsymbol{a} \in \mathbb{R}^N$,

$$\mathbb{P}\left\{\left|\sum_{i=1}^N a_iX_i\right| \geqslant \frac{1}{2}\left(\sum_{i=1}^N a_i^2\right)^{1/2}\right\} \geqslant \frac{c}{K^4}. \qquad (c>0)$$

(Remark: This result is a general quantitative version of Example E1.5.1 (*Probability of a perfect cancelation*), although it does not achieve the exact probability value of 1/2.)

Prf. Let $S = \sum_{i=1}^N a_iX_i$, $\sigma^2 = \sum_{i=1}^N a_i^2$, then

$$\mathbb{E}[S^2] = \operatorname{Var}(S) = \sum_{i=1}^N a_i^2 \operatorname{Var}(X_i) = \sum_{i=1}^N a_i^2 = \sigma^2,$$

Therefore the objective becomes

$$\mathbb{P}\left\{|S| \geqslant \frac{1}{2}\sigma\right\} \geqslant \frac{c}{K^4}.$$

(*To use the Paley-Zygmund inequality, we first need to control* $\mathbb{E}[S^4]$.)

By Proposition P2.7.1 (*Subgaussian norm of a sum*), we obtain

$$\begin{aligned}\|S\|_{\psi_2}^2 &\leqslant C_1 \sum_{i=1}^N a_i^2 \|X_i\|_{\psi_2}^2 \\ &\leqslant C_1 \sum_{i=1}^N a_i^2 K^2 = C_1 K^2 \sigma^2, \qquad (C_1 > 0)\end{aligned}$$

Taking square roots on both sides,

$$\|S\|_{\psi_2} \leqslant \sqrt{C_1} K \sigma.$$

By the moment property of sub-Gaussian random variables $(\mathbb{E}|Z|^p)^{1/p} \leqslant C\sqrt{p}\|Z\|_{\psi_2}$, let $p = 4$, we obtain

$$(\mathbb{E}[S^4])^{1/4} \leqslant C_2 \sqrt{4} \|S\|_{\psi_2} = 2C_2 \|S\|_{\psi_2} \leqslant 2C_2(\sqrt{C_1} K \sigma), \qquad (C_2 > 0)$$

Raising both sides to the 4th power,

$$\mathbb{E}[S^4] \leqslant C_3 K^4 \sigma^4. \qquad (C_3 > 0)$$

By the Paley-Zygmund inequality

$$\mathbb{P}\{Z > \theta \mathbb{E}[Z]\} \geqslant (1-\theta)^2 \frac{(\mathbb{E}[Z])^2}{\mathbb{E}[Z^2]},$$

let $Z = S^2$, then $\mathbb{E}[Z] = \mathbb{E}[S^2] = \sigma^2$, $\mathbb{E}[Z^2] = \mathbb{E}[S^4]$, $\theta = 1/4$, we obtain

$$\begin{aligned}\mathbb{P}\left\{|S| \geqslant \frac{1}{2}\sigma\right\} = \mathbb{P}\left\{S^2 \geqslant \frac{1}{4}\sigma^2\right\} & \\ \geqslant \left(1 - \frac{1}{4}\right)^2 \frac{(\mathbb{E}[S^2])^2}{\mathbb{E}[S^4]} & \\ = \frac{9}{16} \frac{\sigma^4}{C_3 K^4 \sigma^4} = \frac{c}{K^4}. & \left(c = \frac{9}{16C_3}\right)\end{aligned}$$

31 [**Hoeffding's Inequality Requires Zero Mean**] Let $\{X_N\}$ be i.i.d. such that for any $\boldsymbol{a} \in \mathbb{R}^N$,

$$\mathbb{P}\left\{\left|\sum_{i=1}^N a_i X_i\right| \geqslant t\right\} \leqslant 2\exp\left(-\frac{ct^2}{\|\boldsymbol{a}\|_2^2}\right). \qquad (c > 0)$$

Prove that $\mathbb{E}[X_i] = 0$.

Prf. Suppose $\mu = \mathbb{E}[X_1] \neq 0$. Let $\boldsymbol{a} = (1, 1, \cdots, 1)$, then $S_N = \sum_{i=1}^N X_i$, $\|\boldsymbol{a}\|_2^2 = \sum_{i=1}^N 1^2 = N$, satisfying

$$\mathbb{P}\{|S_N| \geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{N}\right). \qquad (t \geqslant 0)$$

By the WLLN, $S_N/N \xrightarrow{\mathbb{P}} \mu$, i.e., for any $\varepsilon > 0$,

$$\mathbb{P}\left\{\left|\frac{S_N}{N} - \mu\right| < \varepsilon\right\} \to 1,$$

Let $\varepsilon = |\mu|/2$, we obtain

$$\mathbb{P}\left\{|S_N| > \frac{N|\mu|}{2}\right\} \to 1.$$

But

$$\begin{aligned}\mathbb{P}\left\{|S_N| \geqslant \frac{N|\mu|}{2}\right\} &\leqslant 2\exp\left(-\frac{c(N|\mu|/2)^2}{N}\right) \\ &= 2\exp\left(-\frac{cN^2\mu^2}{4N}\right) \\ &= 2\exp\left(-\frac{c\mu^2}{4}N\right) \to 0,\end{aligned}$$

i.e.,

$$\mathbb{P}\left\{|S_N| \geqslant \frac{N|\mu|}{2}\right\} \to 0,$$

A contradiction. Thus, $\mu = 0$.

32 [**Sub-Gaussian Norm of a Sum**] Let X, Y be independent sub-Gaussian random variables with $\mathbb{E}[X] = \mathbb{E}[Y] = 0$. Prove that

$$\|X+Y\|_{\psi_2} \asymp \|X\|_{\psi_2} + \|Y\|_{\psi_2},$$

where the symbol "$\asymp$" means equivalence up to absolute constant factors, i.e., there exist $c, C > 0$ such that

$$c(\|X\|_{\psi_2} + \|Y\|_{\psi_2}) \leqslant \|X+Y\|_{\psi_2} \leqslant C(\|X\|_{\psi_2} + \|Y\|_{\psi_2}).$$

Prf. • ($\lesssim$)

Since $\|\cdot\|_{\psi_2}$ is a norm in Orlicz space, by the triangle inequality, we obtain

$$\|X+Y\|_{\psi_2} \leqslant \|X\|_{\psi_2} + \|Y\|_{\psi_2},$$

where $C = 1$.

• ($\gtrsim$)

Note that

$$\mathbb{E}[X+Y \mid X] = X + \mathbb{E}[Y] = X,$$

Therefore by Jensen's inequality, let $\phi(u) = \mathrm{e}^{u^2}$, we obtain

$$\exp\left(\frac{X^2}{t^2}\right) = \exp\left(\frac{(\mathbb{E}[X+Y \mid X])^2}{t^2}\right) \leqslant \mathbb{E}\left[\exp\left(\frac{(X+Y)^2}{t^2}\right)\middle| X\right].$$

Taking expectations on both sides,

$$\mathbb{E}\left[\exp\left(\frac{X^2}{t^2}\right)\right] \leqslant \mathbb{E}\left[\mathbb{E}\left[\exp\left(\frac{(X+Y)^2}{t^2}\right)\middle| X\right]\right] = \mathbb{E}\left[\exp\left(\frac{(X+Y)^2}{t^2}\right)\right].$$

Let $t = \|X+Y\|_{\psi_2}$, we obtain

$$\mathbb{E}\left[\exp\left(\frac{X^2}{\|X+Y\|_{\psi_2}^2}\right)\right] \leqslant \mathbb{E}\left[\exp\left(\frac{(X+Y)^2}{\|X+Y\|_{\psi_2}^2}\right)\right] \leqslant 2,$$

therefore

$$\|X\|_{\psi_2} \leqslant \|X+Y\|_{\psi_2},$$

(*The definition of* $\|X\|_{\psi_2}$ *is the smallest qualifying parameter, so* $\|X\|_{\psi_2} \leqslant t = \|X+Y\|_{\psi_2}$.)

By symmetry, we obtain

$$\|Y\|_{\psi_2} \leqslant \|X+Y\|_{\psi_2},$$

thus

$$\|X\|_{\psi_2} + \|Y\|_{\psi_2} \leqslant 2\|X+Y\|_{\psi_2}.$$

In summary,

$$\frac{1}{2}(\|X\|_{\psi_2} + \|Y\|_{\psi_2}) \leqslant \|X+Y\|_{\psi_2} \leqslant 1 \cdot (\|X\|_{\psi_2} + \|Y\|_{\psi_2}),$$

i.e., $\|X+Y\|_{\psi_2} \asymp \|X\|_{\psi_2} + \|Y\|_{\psi_2}$.

33 [**Sub-Gaussian Norm of an I.I.D. Sum**] In view of the Pythagorean identity (Eq. 2.19, which is the sum-of-squares property of the L^2 norm), it is natural to ask whether the inequality in Proposition P2.7.1 (*Subgaussian norm of a sum*) can be reversed.

In this and the next exercise, you will show that: if $\{X_N\}$ are identically distributed, then the answer is positive; but for the general case, the answer is negative.

(1) Let $X, X_1, \cdots, X_N$ be i.i.d. sub-Gaussian random variables with zero mean. Prove that

$$\left\| \sum_{i=1}^{N} X_i \right\|_{\psi_2} \asymp \sqrt{N} \|X\|_{\psi_2}$$

(2) Let $S_N \sim B(N, p)$. Prove that

$$\|S_N - Np\|_{\psi_2} \asymp \sqrt{\frac{N}{\log(2/p)}}.$$

Prf. ***Sol.*** ($\lesssim$)

By Proposition P2.7.1 (*Subgaussian norm of a sum*), we obtain

$$\left\| \sum_{i=1}^{N} X_i \right\|_{\psi_2}^2 \leqslant C \sum_{i=1}^{N} \|X_i\|_{\psi_2}^2 = CN \|X\|_{\psi_2}^2, \qquad (C > 0)$$

Taking square roots on both sides,

$$\left\| \sum_{i=1}^{N} X_i \right\|_{\psi_2} \leqslant C' \sqrt{N} \|X\|_{\psi_2}.$$

- ($\gtrsim$)

  Let $L = \|S_N\|_{\psi_2}$, then

  $$\mathbb{E}[\exp(\lambda S_N)] \leqslant \exp(C_1 \lambda^2 L^2),$$

  where

  $$\begin{aligned} \mathbb{E}\left[\exp\left(\lambda \sum_{i=1}^{N} X_i \right) \right] &= \mathbb{E}\left[\prod_{i=1}^{N} \exp(\lambda X_i) \right] \\ &= \prod_{i=1}^{N} \mathbb{E}[\exp(\lambda X_i)] \\ &= ()^N, \end{aligned}$$

  therefore

  $$()^N \leqslant \exp(C_1 \lambda^2 L^2),$$

  Taking the N-th root on both sides,

  $$\mathbb{E}[\exp(\lambda X)] \leqslant \exp\left(\frac{C_1 \lambda^2 L^2}{N} \right) = \exp(\lambda^2 \sigma^2), \qquad \left(\sigma^2 = \frac{C_1 L^2}{N} \right)$$

  thus

  $$\|X\|_{\psi_2} \leqslant c_2 \cdot \sigma = c_2 \sqrt{\frac{C_1}{N}} L = c_2 \sqrt{\frac{C_1}{N}} \left\| \sum_{i=1}^{N} X_i \right\|_{\psi_2},$$

  i.e.,

  $$\left\| \sum_{i=1}^{N} X_i \right\|_{\psi_2} \geqslant c_2 \sqrt{\frac{C_1}{N}} \|X\|_{\psi_2}.$$

Sol. Let $S_N = \sum_{i=1}^{N} Y_i$, where $Y_i \sim \text{Bernoulli}(p)$, then

$$S_N - Np = \sum_{i=1}^{N} (Y_i - p) = \sum_{i=1}^{N} X_i,$$

where $X = Y - p$, satisfying

$$\mathbb{P}\{X = 1 - p\} = p, \ \mathbb{P}\{X = -p\} = 1 - p.$$

Therefore by the conclusion of (1), we obtain

$$\|S_N - Np\|_{\psi_2} = \left\|\sum_{i=1}^N X_i\right\|_{\psi_2} \asymp \sqrt{N}\|X_1\|_{\psi_2},$$

so it suffices to prove

$$\|X_1\|_{\psi_2} \asymp \frac{1}{\sqrt{\log(2/p)}}.$$

Thus it suffices to prove, for

$$f(t) = \mathbb{E}\left[\exp\left(\frac{X^2}{t^2}\right)\right] = \underbrace{p\cdot\exp\left(\frac{(1-p)^2}{t^2}\right)}_{I_1} + \underbrace{(1-p)\cdot\exp\left(\frac{p^2}{t^2}\right)}_{I_2},$$

there exists t^* such that $f(t^*) \asymp 1$.

(*Let* $S'_N = N - S_N \sim B(N, 1-p)$, *then*

$$S_N - Np = (N - S'_N) - Np = -(S'_N - N(1-p)),$$

noting that the sub-Gaussian norm satisfies $\|-Z\|_{\psi_2} = \|Z\|_{\psi_2}$, *therefore*

$$\|S_N - Np\|_{\psi_2} = \|S'_N - N(1-p)\|_{\psi_2},$$

thus $p \leftrightarrow 1-p$, *and we only need to consider the case* $p \leqslant 1/2$.)

Without loss of generality assume $p \leqslant 1/2$, let $t = 1/\sqrt{\log(2/p)}$, i.e., $1/t^2 = \log(2/p)$, then:

Sol. For I_1,

$$I_1 = p\cdot\exp((1-p)^2\log(2/p)) = p\cdot\left(\frac{2}{p}\right)^{(1-p)^2} = 2^{(1-p)^2}\cdot p^{1-(1-p)^2},$$

where when $p \in (0, 1/2]$, $2^{(1-p)^2} \in [\sqrt{2}, 2]$; as $p \to 0$, $p^{2p-p^2} = \exp((2p-p^2)\log p) \to \exp(0) = 1$, we obtain $I_1 \asymp 1$.

Sol. For I_2,

$$I_2 = (1-p)\cdot\exp(p^2\log(2/p)) = (1-p)\cdot\left(\frac{2}{p}\right)^{p^2} = (1-p)\cdot 2^{p^2}\cdot p^{-p^2},$$

where when $p \in (0, 1/2]$, $(1-p)2^{p^2} \in [1/2, \sqrt{2}]$; as $p \to 0$, $p^{-p^2} = \exp(-p^2\log p) \to \exp(0) = 1$, we obtain $I_2 \asymp 1$.

Therefore $f(t) \asymp 1$.

($f(t) = I_1 + I_2$ *is a constant bounded above and below independently of* p, *so by adjusting the absolute constant factor of* t, *we can precisely make* $f(t)$ *satisfy* $f(t) \leqslant 2$ *or* $f(t) \geqslant 2$.)

Rk. For (2), in fact, by symmetry of $p, 1-p$, a more rigorous proof target is

$$\|S_N - Np\|_{\psi_2} \asymp \sqrt{\frac{N}{\log(2/\min(p, 1-p))}}.$$

34 **[Sub-Gaussian Norm of a Non-I.I.D. Sum]**

(1) Construct independent sub-Gaussian random variables $\{X_N\}$ with $\mathbb{E}[X_i] = 0$, such that for any $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$,

$$\left\|\sum_{i=1}^N a_i X_i\right\|_{\psi_2} \asymp \|\boldsymbol{a}\|_\infty.$$

(2) Using the conclusion of (1), show that the inequality in Proposition P2.7.1 (*Subgaussian norm of a sum*) (the Hoeffding-type inequality for sums of sub-Gaussian random variables) is generally not reversible.

Prf. ***Sol.*** Without loss of generality, assume $\|\boldsymbol{a}\|_\infty = 1$.

(*Construct sparse random variables.*)

Construct $\{X_N\}$ as independent random variables with distribution

$$\mathbb{P}\{X_i=\sqrt{i}\}=\mathbb{P}\{X_i=-\sqrt{i}\}=\frac{1}{2}\mathrm{e}^{-i},\mathbb{P}\{X_i=0\}=1-\mathrm{e}^{-i}.$$

Then $\{X_N\}$ are all sub-Gaussian distributed (*bounded random variables are necessarily sub-Gaussian*), with $\mathbb{E}[X_i]=0$, $\mathbb{E}[|X_i|^p]=i^{p/2}\mathrm{e}^{-i}$, therefore

$$\|X_i\|_{L_p}=(i^{p/2}\mathrm{e}^{-i})^{1/p}=\sqrt{i}\mathrm{e}^{-i/p},$$

thus

$$\|X_i\|_{\psi_2}\asymp\sup_{p\geqslant 1}\frac{\sqrt{i}\mathrm{e}^{-i/p}}{\sqrt{p}}=\sup_{p\geqslant 1}\sqrt{\frac{i}{p}}\mathrm{e}^{-i/p},$$

The supremum on the right-hand side is attained at $u=i/p=1/2$, where it equals $1/\sqrt{2\mathrm{e}}$. Therefore $\|X_i\|_{\psi_2}\asymp 1$. Let $S=\sum_{i=1}^N a_iX_i$, then:

Sol. Lower bound: By Jensen's inequality, we obtain

$$\|S\|_{\psi_2}\geqslant c\max_{1\leqslant i\leqslant N}\|a_iX_i\|_{\psi_2}=c\|\boldsymbol{a}\|_\infty\max_{1\leqslant i\leqslant N}\|X_i\|_{\psi_2}\asymp\|\boldsymbol{a}\|_\infty.$$

Sol. Upper bound: By Rosenthal's inequality for sums of independent random variables (*or the generalized Marcinkiewicz–Zygmund inequality*), we obtain for any $p\geqslant 2$,

$$\|S\|_{L_p}\leqslant C\left[\underbrace{\sqrt{p}\left(\sum_{i=1}^N\mathbb{E}[a_i^2X_i^2]\right)^{1/2}}_{I_1}+\underbrace{\left(\sum_{i=1}^N\mathbb{E}[|a_iX_i|^p]\right)^{1/p}}_{I_2}\right],$$

where

$$\begin{aligned}
I_1&=\sqrt{p}\left(\sum_{i=1}^N a_i^2i\mathrm{e}^{-i}\right)^{1/2}\leqslant\sqrt{p}\left(\sum_{i=1}^\infty i\mathrm{e}^{-i}\right)^{1/2}\leqslant C_1\sqrt{p},\\
I_2&=\left(\sum_{i=1}^N|a_i|^pi^{p/2}\mathrm{e}^{-i}\right)^{1/p}\leqslant\left(\sum_{i=1}^\infty i^{p/2}\mathrm{e}^{-i}\right)^{1/p}\\
&\sim\left(\int_0^\infty t^{p/2}\mathrm{e}^{-t}\mathrm{d}t\right)^{1/p}=\left(\Gamma\left(\frac{p}{2}+1\right)\right)^{1/p}\\
&\sim\left(\sqrt{2\pi\frac{p}{2}}\left(\frac{p}{2\mathrm{e}}\right)^{p/2}\right)^{1/p}\asymp\sqrt{p},
\end{aligned}$$

therefore $\|S\|_{L_p}\leqslant C\sqrt{p}$.

Therefore

$$\|S\|_{\psi_2}\asymp\sup_{p\geqslant 2}\frac{\|S\|_{L_p}}{\sqrt{p}}\leqslant C=C\|\boldsymbol{a}\|_\infty,$$

i.e.,

$$\left\|\sum_{i=1}^N a_iX_i\right\|_{\psi_2}\asymp\|\boldsymbol{a}\|_\infty.$$

Sol. Proposition P2.7.1 (*Subgaussian norm of a sum*) gives an upper bound based on the l_2 norm of the coefficient vector,

$$\left\|\sum_{i=1}^N a_iX_i\right\|_{\psi_2}\leqslant C\sqrt{\sum_{i=1}^N a_i^2\|X_i\|_{\psi_2}^2}\asymp\sqrt{\sum a_i^2}=\|\boldsymbol{a}\|_2,$$

Let $\boldsymbol{a}=(1,1,\cdots,1)$, then the upper bound order from Proposition P2.7.1 (*Subgaussian norm of a sum*) is $\|\boldsymbol{a}\|_2=$

$\sqrt{N}$, while the true sub-Gaussian norm order is $\|\boldsymbol{a}\|_\infty = 1$, so this is not reversible.

35 [**Interpolation Between L^1 and L^∞**] In Exercise Q1.12 (*Interpolation Inequality between L^1 and L^∞*), we showed how to bound the L^p norm of a random variable X using upper bounds on its L^1 and L^∞ norms.

Now, let us prove a similar result for the sub-Gaussian norm.

(1) Prove: If $\|X\|_{L^1} = a$, $\|X\|_{L^\infty} = b$, then

$$\|X\|_{\psi_2} \leqslant \frac{Cb}{\sqrt{\log(2b/a)}}. \qquad (C > 0)$$

(2) Prove: For any $a \leqslant b$, the upper bound in (1) is optimal up to absolute constants.

Prf. ***Sol.*** Since $\mathbb{E}[|X|] = a$, $|X| \leqslant b$ (a.s.), we have

$$\mathbb{E}[|X|^p] = \mathbb{E}[|X| \cdot |X|^{p-1}] \leqslant \mathbb{E}[|X|] \cdot \|X\|_{L^\infty}^{p-1} = a \cdot b^{p-1},$$

therefore

$$\|X\|_{L^p} \leqslant \mathbb{E}[|X|^p]^{1/p} = (a \cdot b^{p-1})^{1/p} = b\left(\frac{a}{b}\right)^{1/p},$$

thus

$$\|X\|_{\psi_2} \leqslant C_0 \sup_{p \geqslant 1} \frac{\|X\|_{L^p}}{\sqrt{p}} \leqslant C_0 \sup_{p \geqslant 1} \frac{b(a/b)^{1/p}}{\sqrt{p}} = C_0 \sup_{p \geqslant 1} \frac{br^{1/p}}{\sqrt{p}}. \qquad \left(r = \frac{a}{b} \in (0,1)\right)$$

The optimal parameter on the right-hand side is $p^* = 2\log(1/r) = 2\log(b/a)$. Substituting it gives

a. If $p^* \geqslant 1$, then

$$\max_{p \geqslant 1} \frac{br^{1/p}}{\sqrt{p}} = \frac{b\mathrm{e}^{-1/2}}{\sqrt{2\log(b/a)}} \asymp \frac{1}{\sqrt{\log(b/a)}}.$$

b. If $p^* < 1$, then

$$\max_{p \geqslant 1} \frac{br^{1/p}}{\sqrt{p}} = a,$$

Thus,

$$\|X\|_{\psi_2} \asymp \frac{b}{\sqrt{\log(2b/a)}}.$$

Sol. For any $0 < a \leqslant b$, construct X with distribution

$$\mathbb{P}\{X = b\} = \frac{a}{b}, \mathbb{P}\{X = 0\} = 1 - \frac{a}{b},$$

then $\|X\|_{L^1} = b \cdot (a/b) + 0 \cdot (1 - a/b) = a$, $\|X\|_{L^\infty} = b$.

Let $K = \|X\|_{\psi_2}$, then

$$\mathbb{E}[\mathrm{e}^{X^2/K^2}] = (1 - p_0)\mathrm{e}^0 + p_0\mathrm{e}^{b^2/K^2} = 1 - \frac{a}{b} + \frac{a}{b}\mathrm{e}^{b^2/K^2} \leqslant 2 \implies K \geqslant \frac{b}{\sqrt{\log(2b/a)}}.$$

36 [**Khintchine Inequality for $p < 2$**] In Theorem T2.7.5 (*Khintchine inequality*), we proved the Khintchine inequality for $p \in [2, \infty)$.

Now extend the result to $p \in [1, 2]$.

(1) (Extrapolation property) Prove: For any random variable Z with finite L^3 norm,

$$\|Z\|_{L^2} \leqslant \|Z\|_{L^1}^{1/4} \|Z\|_{L^3}^{3/4}.$$

(2) Let $\{X_N\}$ be independent centered K-sub-Gaussian random variables with $\mathrm{Var}(X_i) = 1$. Prove: For any $\boldsymbol{a} \in \mathbb{R}^N$,

$$cK^{-3}\left(\sum_{i=1}^N a_i^2\right)^{1/2} \leqslant \mathbb{E}\left[\left|\sum_{i=1}^N a_i X_i\right|\right] \leqslant \left(\sum_{i=1}^N a_i^2\right)^{1/2}. \qquad (c > 0)$$

(3) Using the conclusion of (2), prove: For any $p \in [1, 2]$, the same inequality for the L^p norm of the sum holds.

Prf. ***Sol.*** Using the interpolation property of L^p spaces (*Hölder's inequality or Lyapunov's inequality*), for $1 \leqslant p < q < r$,

$$\|Z\|_{L^q} \leqslant \|Z\|_{L^p}^{\theta} \|Z\|_{L^r}^{1-\theta}, \qquad \left(\frac{1}{q} = \frac{\theta}{p} + \frac{1-\theta}{r}\right)$$

Let $p = 1, q = 2, r = 3$, then $\theta = 1/4$, we obtain

$$\|Z\|_{L^2} \leqslant \|Z\|_{L^1}^{1/4} \|Z\|_{L^3}^{3/4},$$

i.e.,

$$\mathbb{E}[Z^2]^{1/2} \leqslant (\mathbb{E}|Z|)^{1/4} (\mathbb{E}|Z|^3)^{1/4}.$$

Sol. Let $S = \sum_{i=1}^N a_i X_i$, then

$$\|S\|_{L^2} = (\mathbb{E}[S^2])^{1/2} = \left(\sum_{i=1}^N a_i^2 \operatorname{Var}(X_i)\right)^{1/2} = \left(\sum_{i=1}^N a_i^2\right)^{1/2}.$$

For the right-hand side, by Jensen's inequality, we obtain

$$\mathbb{E}|S| \leqslant (\mathbb{E}[S^2])^{1/2} = \|S\|_{L^2} \leqslant \left(\sum_{i=1}^N a_i^2\right)^{1/2}.$$

For the left-hand side, by the conclusion of (1), we obtain $\|S\|_{L^2} \leqslant \|S\|_{L^1}^{1/4} \|S\|_{L^3}^{3/4}$, i.e.,

$$\|S\|_{L^1} \geqslant \frac{\|S\|_{L^2}^4}{\|S\|_{L^3}^3},$$

and it suffices to bound $\|S\|_{L^3}$ from above.

Since X_i are sub-Gaussian with $\|X_i\|_{\psi_2} \leqslant K$, by Proposition P2.7.1 (*Subgaussian norm of a sum*), S is sub-Gaussian with

$$\|S\|_{\psi_2} \leqslant C_1 \left(\sum_{i=1}^N a_i^2 \|X_i\|_{\psi_2}^2\right)^{1/2} \leqslant C_1 K \left(\sum_{i=1}^N a_i^2\right)^{1/2},$$

thus setting $p = 3$, we obtain

$$\|S\|_{L^3} \leqslant C_2 \sqrt{3} \|S\|_{\psi_2} \leqslant C_3 K \left(\sum_{i=1}^N a_i^2\right)^{1/2} = C_3 K \|S\|_{L^2},$$

therefore

$$\|S\|_{L^1} \geqslant \frac{\|S\|_{L^2}^4}{(C_3 K \|S\|_{L^2})^3} = \frac{1}{C_3^3 K^3} \|S\|_{L^2} = cK^{-3} \left(\sum_{i=1}^N a_i^2\right)^{1/2}.$$

Sol. For any $p \in [1, 2]$, by monotonicity of L^p norms,

$$\|S\|_{L^1} \leqslant \|S\|_{L^p} \leqslant \|S\|_{L^2}.$$

By the conclusion of (2), for the left-hand side,

$$\|S\|_{L^p} \geqslant \|S\|_{L^1} \geqslant cK^{-3} \|S\|_{L^2} = cK^{-3} \left(\sum_{i=1}^N a_i^2\right)^{1/2},$$

for the right-hand side,

$$\|S\|_{L^p} \leqslant \|S\|_{L^2} = \left(\sum_{i=1}^N a_i^2\right)^{1/2},$$

therefore for any $p \in [1,2]$,

$$cK^{-3}\left(\sum_{i=1}^N a_i^2\right)^{1/2} \leqslant \left(\mathbb{E}\left|\sum_{i=1}^N a_i X_i\right|^p\right)^{1/p} \leqslant \left(\sum_{i=1}^N a_i^2\right)^{1/2}.$$

37 [**Maximal Inequality**] Prove a strengthened form of Proposition P2.7.6 (*Maximum of subgaussians*). Let $\{X_N\}$ be sub-Gaussian random variables (not necessarily independent). Prove:

$$\left\|\sup_N \frac{X_N}{\sqrt{\log(2N)}}\right\|_{\psi_2} \leqslant C \sup_N \|X_N\|_{\psi_2}. \qquad (C>0)$$

Prf. Without loss of generality, let $K = \max_i \|X_i\|_{\psi_2} = 1$; it suffices to prove

$$\|Y\| := \left\|\sup_{i\geqslant 1} \frac{\tilde{X}_i}{\sqrt{\log(2i)}}\right\|_{\psi_2} \leqslant C.$$

- First consider the positive part

  For Y_+, for any $t \geqslant 0$,

  $$\mathbb{P}\{Y_+ \geqslant t\} = \mathbb{P}\left\{\exists i : \frac{X_i}{\sqrt{\log(2i)}} \geqslant t\right\} \leqslant \sum_{i=1}^\infty \mathbb{P}\{X_i \geqslant t\sqrt{\log(2i)}\},$$

  where

  $$\mathbb{P}\{X_i \geqslant t\sqrt{\log(2i)}\} \leqslant 2\mathrm{e}^{-(t\sqrt{\log(2i)})^2/1} = 2\mathrm{e}^{-t^2\log(2i)} = 2(2i)^{-t^2},$$

  substituting it gives

  $$\mathbb{P}\{Y_+ \geqslant t\} \leqslant \sum_{i=1}^\infty 2(2i)^{-t^2} = 2^{1-t^2}\sum_{i=1}^\infty i^{-t^2} = 2^{1-t^2}\zeta(t^2).$$

  Discuss the value of t^2:

  a. If $t^2 < 2$, i.e., $0 \leqslant t < \sqrt{2}$, then $\mathbb{P}\{Y_+ \geqslant t\} \leqslant 1$, which is trivial.

  b. If $t^2 \geqslant 2$, i.e., $t \geqslant \sqrt{2}$, then $\zeta(t^2) \leqslant \zeta(2) = \pi^2/6$, so

  $$\mathbb{P}\{Y_+ \geqslant t\} \leqslant 2^{1-t^2}\cdot\frac{\pi^2}{6} = \frac{\pi^2}{3}\mathrm{e}^{-t^2\log 2}.$$

  Therefore

  $$\mathbb{P}\{Y_+ \geqslant t\} \leqslant A\mathrm{e}^{-c_0 t^2}, \qquad (A, c_0 > 0)$$

  thus $\|Y_+\|_{\psi_2} \leqslant C_1$.

- Next consider the negative part

  For Y_-, since $Y \geqslant X_1/\sqrt{\log 2}$, we have $Y_- \leqslant X_1/\sqrt{\log 2}$, so

  $$\|Y_-\|_{\psi_2} \leqslant \frac{1}{\sqrt{\log 2}}\|X_1\|_{\psi_2} \leqslant \frac{1}{\sqrt{\log 2}},$$

  thus by the triangle inequality,

  $$\|Y\|_{\psi_2} = \|Y_+ - Y_-\|_{\psi_2} \leqslant \|Y_+\|_{\psi_2} + \|Y_-\|_{\psi_2} \leqslant C_1 + \frac{1}{\sqrt{\log 2}} = C.$$

In summary, $\|Y\|_{\psi_2} \leqslant C$, i.e.,

$$\left\|\sup_{N\geqslant 1} \frac{X_N}{\sqrt{\log(2N)}}\right\|_{\psi_2} \leqslant CK = C\sup_{N\geqslant 1}\|X_N\|_{\psi_2}.$$

38 [**Maximum of Gaussian Random Variables**] Let us prove an asymptotically sharp version of the maximal inequality (2.22) for standard normal random variables.

(1) Let $g_1, \cdots, g_N \sim N(0,1)$ (not necessarily independent). Prove:

$$\mathbb{E}\left[\max_{i\leqslant N} g_i\right] \leqslant \sqrt{2\log N}, \mathbb{E}\left[\max_{i\leqslant N} |g_i|\right] \leqslant \sqrt{2\log(2N)}.$$

(2) Let $g_1, \cdots, g_N$ i.i.d. $\sim N(0,1)$. Prove:

$$\mathbb{E}\left[\max_{i\leqslant N} g_i\right] = \sqrt{2\log N}(1+o(1)), \mathbb{E}\left[\max_{i\leqslant N} |g_i|\right] = \sqrt{2\log N}(1+o(1)).$$

Prf. ***Sol.*** • By Jensen's inequality, for any $\lambda > 0$,

$$\exp(\lambda) \leqslant \mathbb{E}\left[\exp\left(\lambda \max_{i\leqslant N} g_i\right)\right] = \mathbb{E}\left[\max_{i\leqslant N} \mathrm{e}^{\lambda g_i}\right]$$
$$\leqslant \sum_{i=1}^{N} \mathbb{E}[\mathrm{e}^{\lambda g_i}] = N\mathrm{e}^{\lambda^2/2},$$

taking logarithms on both sides,

$$\lambda\mathbb{E}\left[\max_{i\leqslant N} g_i\right] \leqslant \log N + \frac{\lambda^2}{2} \implies \mathbb{E}\left[\max_{i\leqslant N} g_i\right] \leqslant \frac{\log N}{\lambda} + \frac{\lambda}{2} \leqslant 2\sqrt{\frac{\log N}{\lambda}\frac{\lambda}{2}} = \sqrt{2\log N}.$$

• Note that $\max_{i\leqslant N} |g_i| = \max(g_1, \cdots, g_N, -g_1, \cdots, -g_N)$ is the maximum of $2N$ standard normal random variables, so replacing N by $2N$, we obtain

$$\mathbb{E}\left[\max_{i\leqslant N} |g_i|\right] \leqslant \sqrt{2\log(2N)}.$$

Sol. • Let $M_N = \max_{i\leqslant N} g_i$. For any $\varepsilon > 0$, set the threshold $T_N = (1-\varepsilon)\sqrt{2\log N}$, then

$$\mathbb{P}\{M_N \leqslant T_N\} = \prod_{i=1}^{N} \mathbb{P}\{g_i \leqslant T_N\} = (1 - \mathbb{P}\{g_1 > T_N\})^N \leqslant \exp(-N\mathbb{P}\{g_1 > T_N\}),$$

where

$$\mathbb{P}\{g_1 > T_N\} \sim \frac{1}{(1-\varepsilon)\sqrt{2\log N}\sqrt{2\pi}} \mathrm{e}^{-(1-\varepsilon)^2\log N} \qquad \left(\mathbb{P}\{g_1 > t\} \sim \frac{1}{t\sqrt{2\pi}}\mathrm{e}^{-t^2/2}\right)$$
$$= C_\varepsilon \frac{1}{\sqrt{\log N} N^{(1-\varepsilon)^2}},$$

substituting it gives

$$\mathbb{P}\{M_N \leqslant T_N\} \leqslant \exp\left(-C_\varepsilon \frac{N^{1-(1-\varepsilon)^2}}{\sqrt{\log N}}\right) \to 0,$$

therefore $M_N > (1-\varepsilon)\sqrt{2\log N}$ (w.h.p.).

Since M_N is uniformly integrable (*its sub-Gaussian norm is bounded*), we have

$$\liminf_{N\to\infty} \frac{\mathbb{E}[M_N]}{\sqrt{2\log N}} \geqslant 1-\varepsilon,$$

thus by the conclusion of (1)

$$\frac{\mathbb{E}[M_N]}{\sqrt{2\log N}} \leqslant 1,$$

we obtain

$$\mathbb{E}\left[\max_{i\leqslant N} g_i\right] = \sqrt{2\log N}(1+o(1)).$$

• Since $\max_{i\leqslant N} |g_i| \geqslant \max_{i\leqslant N} g_i$, we have

$$\mathbb{E}\left[\max_{i\leqslant N} |g_i|\right] \geqslant \mathbb{E}\left[\max_{i\leqslant N} g_i\right] = \sqrt{2\log N}(1+o(1)),$$

thus by the conclusion of (1)

$$\mathbb{E}\left[\max_{i\leqslant N} |g_i|\right] \leqslant \sqrt{2\log(2N)} = \sqrt{2\log N + 2\log 2} = \sqrt{2\log N}(1+o(1)),$$

we obtain

$$\mathbb{E}\left[\max_{i\leqslant N}|g_i|\right]=\sqrt{2\log N}(1+o(1)).$$

Rk. In fact, according to the Gumbel distribution, we obtain

$$\mathbb{E}\left[\max_{i} X_i\right]=\sqrt{2\log n}-\frac{\log\log n+\log(4\pi)}{2\sqrt{2\log n}}\sim\sqrt{2\log n},$$

39 [**Characterization of Sub-Gaussian Distributions: Maximal Inequality**] Prove: X is a sub-Gaussian random variable if and only if

$$\mathbb{E}\left[\max_{i\leqslant N}|X_i|\right]\leqslant K\sqrt{\log N}. \qquad (K>0)$$

where $\{X_N\}_{N\geqslant 2}$ are independent copies of X.

Proposition P2.7.6 (*Maximum of subgaussians*) has already proved the necessity, i.e., if X is sub-Gaussian, then there exists K with $K\leqslant C\|X\|_{\psi_2}$ such that (2.35) holds. Now prove the sufficiency, i.e., if (2.35) holds, then X is sub-Gaussian with $\|X\|_{\psi_2}\leqslant CK$.

Prf. To prove that X is a sub-Gaussian random variable, it suffices to prove that its tail probability satisfies sub-Gaussian decay, i.e., to prove

$$\mathbb{P}\{|X|\geqslant t\}\leqslant 2\exp\left(-c\frac{t^2}{K^2}\right)\quad(\forall t\geqslant 0). \qquad (c>0)$$

Set $p_t=\mathbb{P}\{|X|\geqslant t\}$.

a. If $p_t\leqslant 1/2$, then $\frac{1}{p_t}\geqslant 2$. Choose a positive integer $N=\left\lfloor\frac{1}{p_t}\right\rfloor\geqslant 2$, which clearly satisfies $\frac{1}{2p_t}\leqslant N\leqslant\frac{1}{p_t}$.
Using the truncated lower bound for the expectation of a nonnegative random variable, we obtain

$$\mathbb{E}\left[\max_{i\leqslant N}|X_i|\right]\geqslant t\cdot\mathbb{P}\left\{\max_{i\leqslant N}|X_i|\geqslant t\right\}=t\Big(1-(1-p_t)^N\Big),$$

since $(1-p_t)^N\leqslant \mathrm{e}^{-Np_t}\leqslant \mathrm{e}^{-1/2}$ (*by* $N\geqslant\frac{1}{2p_t}$), so

$$1-(1-p_t)^N\geqslant 1-\mathrm{e}^{-1/2}=c_0>0,$$

combining with the assumed upper bound, we obtain

$$c_0 t\leqslant\mathbb{E}\left[\max_{i\leqslant N}|X_i|\right]\leqslant K\sqrt{\log N}\leqslant K\sqrt{\log\left(\frac{1}{p_t}\right)},$$

squaring both sides and rearranging, we obtain

$$\log\left(\frac{1}{p_t}\right)\geqslant\frac{c_0^2t^2}{K^2}\implies p_t\leqslant\exp\left(-\frac{c_0^2t^2}{K^2}\right).$$

b. If $p_t>1/2$, then t is necessarily small, and clearly $p_t\leqslant 1\leqslant 2\exp(-\frac{c_0^2t^2}{K^2})$ (*by multiplying the coefficient by* 2 *to absorb the constant*).

In summary, for all $t\geqslant 0$,

$$\mathbb{P}\{|X|\geqslant t\}\leqslant 2\exp\left(-c\frac{t^2}{K^2}\right).$$

By the equivalent characterization of sub-Gaussian distributions, this proves that X is a sub-Gaussian random variable with $\|X\|_{\psi_2}\leqslant CK$.

40 [**A Precise View: Exact Sub-Gaussian Norm**] You may have noticed that most results involving the sub-Gaussian norm hold only up to absolute constant factors, rather than exact equality.

To refine these results, we can redefine the sub-Gaussian norm using the equivalent property (iv) of Proposition P2.6.1 (*Sub-gaussian properties*).

As background, recall that the MGF of $X\sim N(\mu,\sigma^2)$ satisfies $\mathbb{E}[\mathrm{e}^{\lambda(X-\mu)}]=\mathrm{e}^{\sigma^2\lambda^2/2}$. Inspired by this, define the sub-Gaussian

variance of X as

$$\mathrm{Var}_G(X) = \inf\{\sigma^2 : \mathbb{E}[\mathrm{e}^{\lambda(X-\mathbb{E}[X])}] \leqslant \mathrm{e}^{\sigma^2\lambda^2/2}\}.$$

Recall that the L^2 norm of a random variable is $\|X\|_{L^2}^2 = \mathbb{E}[X^2] = \mathrm{Var}(X) + (\mathbb{E}[X])^2$. Inspired by this, define the exact sub-Gaussian norm of X as

$$\|X\|_G^2 = \mathrm{Var}_G(X) + (\mathbb{E}[X])^2.$$

Prove:

(1) $\|X\|_G$ is a norm on the space of sub-Gaussian random variables.

(2) $\|X\|_{\psi_2} \asymp \|X\|_G$, i.e., the exact sub-Gaussian norm is equivalent to the standard sub-Gaussian norm.

(3) $\mathrm{Var}(X) \leqslant \mathrm{Var}_G(X)$, $\|X\|_{L^2} \leqslant \|X\|_G$, with equality if and only if X is normally distributed.

(4) If $\{X_N\}$ are independent, then $\mathrm{Var}_G(\sum_{i=1}^N X_i) \leqslant \sum_{i=1}^N \mathrm{Var}_G(X_i)$; if $\mathbb{E}[X_i] = 0$, then $\left\|\sum_{i=1}^N X_i\right\|_G^2 \leqslant \sum_{i=1}^N \|X_i\|_G^2$, which is the exact version of Proposition P2.7.1 (*Subgaussian norm of a sum*).

(5) $\|X - \mathbb{E}[X]\|_G \leqslant \|X\|_G$, which is the exact version of Lemma L2.7.8 (*Centering*).

Prf. ***Sol.*** To prove $\|X\|_G$ is a norm, it suffices to prove:

Sol. Non-negativity and homogeneity:

Since $\mathrm{Var}_G(X) \geqslant 0$, $(\mathbb{E}[X])^2 \geqslant 0$, we have $\|X\|_G \geqslant 0$.

Since $\mathrm{Var}_G(\alpha X) = \alpha^2 \mathrm{Var}_G(X)$ (*obtained via the substitution* $\lambda \to \alpha\lambda$), $\mathbb{E}[\alpha X]^2 = \alpha^2(\mathbb{E}[X])^2$, we have $\|\alpha X\|_G = |\alpha|\|X\|_G$.

Sol. Triangle inequality:

We make the following deduction:

Claim For

$$\sigma_G(X) = \sqrt{\mathrm{Var}_G(X)} = \inf\{\sigma : \mathbb{E}[\mathrm{e}^{\lambda(X-\mathbb{E}[X])}] \leqslant \mathrm{e}^{\sigma^2\lambda^2/2}\},$$

we have $\sigma_G(X+Y) \leqslant \sigma_G(X) + \sigma_G(Y)$.

Using Hölder's inequality, let $p, q > 1$ *with* $1/p + 1/q = 1$*, we obtain*

$$\begin{aligned}
\mathbb{E}[\mathrm{e}^{\lambda(X+Y-\mathbb{E}[X+Y])}] &= \mathbb{E}[\mathrm{e}^{\lambda(X-\mathbb{E}[X])} \cdot \mathrm{e}^{\lambda(Y-\mathbb{E}[Y])}] \\
&\leqslant (\mathbb{E}[\mathrm{e}^{p\lambda(X-\mathbb{E}[X])}])^{1/p} (\mathbb{E}[\mathrm{e}^{q\lambda(Y-\mathbb{E}[Y])}])^{1/q} \\
&\leqslant (\mathrm{e}^{\sigma_G(X)^2(p\lambda)^2/2})^{1/p} (\mathrm{e}^{\sigma_G(Y)^2(q\lambda)^2/2})^{1/q} \\
&= \mathrm{e}^{(p\sigma_G(X)^2 + q\sigma_G(Y)^2)\lambda^2/2},
\end{aligned}$$

where

$$\min_{\substack{p,q>1 \\ 1/p+1/q=1}} (pA^2 + qB^2) = \frac{A+B}{A}A^2 + \frac{A+B}{B}B^2 = (A+B)^2, \qquad (A = \sigma_G(X), B = \sigma_G(Y))$$

therefore

$$\mathbb{E}[\mathrm{e}^{\lambda(X+Y-\mathbb{E}[X+Y])}] \leqslant \mathrm{e}^{(\sigma_G(X)+\sigma_G(Y))^2\lambda^2/2},$$

thus by the minimality definition of σ_G*, we obtain* $\sigma_G(X+Y) \leqslant \sigma_G(X) + \sigma_G(Y)$.

By Minkowski's inequality,

$$\begin{aligned}
\|X+Y\|_G &= \sqrt{\sigma_G(X+Y)^2 + (\mathbb{E}[X+Y])^2} \\
&\leqslant \sqrt{(\sigma_G(X) + \sigma_G(Y))^2 + (\mathbb{E}[X] + \mathbb{E}[Y])^2} \\
&\leqslant \sqrt{\sigma_G(X)^2 + (\mathbb{E}[X])^2} + \sqrt{\sigma_G(Y)^2 + (\mathbb{E}[Y])^2} \\
&= \|X\|_G + \|Y\|_G.
\end{aligned}$$

Sol. It suffices to prove $c_1\|X\|_{\psi_2} \leqslant \|X\|_G \leqslant c_2\|X\|_{\psi_2}$.

- Proof of $\|X\|_G \leqslant c_2\|X\|_{\psi_2}$

  It suffices to prove that each part of $\|X\|_G^2 = \mathrm{Var}_G(X) + (\mathbb{E}[X])^2$ is controlled by $\|X\|_{\psi_2}^2$.

For $\mathrm{Var}_G(X)$, by Proposition P2.6.1 (*Subgaussian properties*), we obtain

$$\mathbb{E}[\mathrm{e}^{\lambda(X-\mathbb{E}[X])}] \leqslant \mathrm{e}^{C\lambda^2\|X\|_{\psi_2}^2},$$

therefore by the definition of $\mathrm{Var}_G(X)$,

$$\frac{1}{2}\mathrm{Var}_G(X) \leqslant C\|X\|_{\psi_2}^2 \implies \mathrm{Var}_G(X) \leqslant 2C\|X\|_{\psi_2}^2.$$

For $(\mathbb{E}[X])^2$, by properties of norms, (*the absolute expectation is bounded by the second moment, and the second moment is bounded by the sub-Gaussian norm*)

$$|\mathbb{E}[X]| \leqslant \|X\|_{L^2} \leqslant C'\|X\|_{\psi_2} \implies (\mathbb{E}[X])^2 \leqslant (C')^2\|X\|_{\psi_2}^2.$$

Thus, $\|X\|_G^2 \leqslant (2C + (C')^2)\|X\|_{\psi_2}^2$, i.e., $\|X\|_G \leqslant c_2\|X\|_{\psi_2}$.

- Proof of $c_1\|X\|_{\psi_2} \leqslant \|X\|_G$

  By the definition of $\mathrm{Var}_G(X)$, we have $\mathbb{E}[\mathrm{e}^{\lambda(X-\mathbb{E}[X])}] \leqslant \mathrm{e}^{\mathrm{Var}_G(X)\lambda^2/2}$, thus by the equivalent properties of sub-Gaussian distributions (*or using Chernoff's inequality*), we obtain

  $$\mathbb{P}\{|X - \mathbb{E}[X]| \geqslant t\} \leqslant 2\mathrm{e}^{-t^2/(2\,\mathrm{Var}_G(X))},$$

  therefore

  $$\|X - \mathbb{E}[X]\|_{\psi_2} \leqslant C''\sqrt{\mathrm{Var}_G(X)},$$

  thus by the triangle inequality,

  $$\begin{aligned}\|X\|_{\psi_2} &= \|(X - \mathbb{E}[X]) + \mathbb{E}[X]\|_{\psi_2} \\ &\leqslant \|X - \mathbb{E}[X]\|_{\psi_2} + \|\mathbb{E}[X]\|_{\psi_2} \\ &\leqslant C''\sqrt{\mathrm{Var}_G(X)} + C'''|\mathbb{E}[X]|.\end{aligned}$$

  (*For the last inequality, let $c = \mathbb{E}[X]$, then*

  $$\|c\|_{\psi_2} = \inf\{t > 0 : \mathbb{E}[\mathrm{e}^{c^2/t^2}] \leqslant 2\},$$

  therefore

  $$\mathrm{e}^{c^2/t^2} \leqslant 2 \implies t \geqslant \frac{|c|}{\sqrt{\log 2}},$$

  thus

  $$\|\mathbb{E}[X]\|_{\psi_2} = \frac{1}{\sqrt{\log 2}}|\mathbb{E}[X]|,$$

  and we set $C''' = 1/\sqrt{\log 2}$.)

Sol. Expand $\mathrm{e}^{\lambda(X-\mathbb{E}[X])}$ in a Taylor series around $\lambda = 0$,

$$\begin{aligned}\mathbb{E}[\mathrm{e}^{\lambda(X-\mathbb{E}[X])}] &= 1 + \lambda\mathbb{E}[X - \mathbb{E}[X]] + \frac{\lambda^2}{2}\mathbb{E}[(X - \mathbb{E}[X])^2] + o(\lambda^2) \\ &= 1 + \frac{\lambda^2}{2}\mathrm{Var}(X) + o(\lambda^2),\end{aligned}$$

where by the definition of $\mathrm{Var}_G(X)$,

$$1 + \frac{\lambda^2}{2}\mathrm{Var}(X) + o(\lambda^2) \leqslant \mathrm{e}^{\mathrm{Var}_G(X)\lambda^2/2} = 1 + \frac{\lambda^2}{2}\mathrm{Var}_G(X) + o(\lambda^2),$$

therefore $\mathrm{Var}(X) \leqslant \mathrm{Var}_G(X)$, i.e., $\|X\|_{L^2}^2 \leqslant \|X\|_G^2$.

Sol. Let $S = \sum_{i=1}^N X_i$, we obtain

$$\mathbb{E}[\mathrm{e}^{\lambda S}] = \prod_{i=1}^N \mathbb{E}[\mathrm{e}^{\lambda X_i}] \leqslant \prod_{i=1}^N \mathrm{e}^{\mathrm{Var}_G(X_i)\lambda^2/2} = \mathrm{e}^{(\sum \mathrm{Var}_G(X_i))\lambda^2/2},$$

therefore by the minimality definition of $\mathrm{Var}_G(S)$, we obtain $\mathrm{Var}_G(S) \leqslant \sum_{i=1}^N \mathrm{Var}_G(X_i)$.

Since $\mathbb{E}[X_i]=0$, we have $\|S\|_G^2=\mathrm{Var}_G(S)$, $\|X_i\|_G^2=\mathrm{Var}_G(X_i)$, thus $\|S\|_G^2\leqslant\sum_{i=1}^N\|X_i\|_G^2$.

Sol. By definition,

$$\|X-\mathbb{E}[X]\|_G^2=\mathrm{Var}_G(X-\mathbb{E}[X])+(\mathbb{E}[X-\mathbb{E}[X]])^2=\mathrm{Var}_G(X)+0=\mathrm{Var}_G(X)\leqslant\|X\|_G^2,$$

therefore $\|X-\mathbb{E}[X]\|_G\leqslant\|X\|_G$, which avoids the possible increase of constant factors due to centering in the standard sub-Gaussian norm.

41 [**Properties of Sub-Gaussian Distributions**] By modifying the proof of Proposition P2.6.1 (*Subgaussian properties*), prove the equivalence of properties (i)–(iii) of Proposition P2.8.1 (*Subexponential properties*).

By Proposition P2.8.1 (*Subexponential properties*), let X be a random variable. Then the following properties are equivalent, with parameters $K_i>0$ differing by at most an absolute constant factor.

(1) (Tail probability) There exists $K_1>0$ such that for all $t\geqslant 0$, $\mathbb{P}\{|X|\geqslant t\}\leqslant 2\exp(-t/K_1)$.

(2) (Moments) There exists $K_2>0$ such that for all $p\geqslant 1$, $\|X\|_{L^p}=(\mathbb{E}[|X|^p])^{1/p}\leqslant K_2p$.

(3) (MGF of $|X|$) There exists $K_3>0$ such that $\mathbb{E}[\exp(|X|/K_3)]\leqslant 2$.

Prf. We prove the cyclic equivalence in the order (i) $\Longrightarrow$ (ii) $\Longrightarrow$ (iii) $\Longrightarrow$ (i).

- (i) $\Longrightarrow$ (ii)

  If $\mathbb{P}\{|X|\geqslant t\}\leqslant 2\mathrm{e}^{-t/K_1}$, then

$$\begin{aligned}\mathbb{E}[|X|^p]&=\int_0^\infty pt^{p-1}\mathbb{P}\{|X|\geqslant t\}\mathrm{d}t\\&\leqslant\int_0^\infty pt^{p-1}\cdot 2\mathrm{e}^{-t/K_1}\mathrm{d}t\\&\leqslant 2pK_1^p\int_0^\infty v^{p-1}\mathrm{e}^{-v}\mathrm{d}v&\left(v=\frac{t}{K_1}\right)\\&=2pK_1^p\Gamma(p)=2K_1^p\Gamma(p+1).\end{aligned}$$

  By Stirling's formula $\Gamma(p+1)\sim\sqrt{2\pi p}(p/\mathrm{e})^p$, there exists $C>0$ such that $\Gamma(p+1)^{1/p}\leqslant Cp$, therefore

$$\|X\|_{L^p}\leqslant(2\Gamma(p+1))^{1/p}K_1\leqslant K_2p,\qquad(K_2\leqslant CK_1)$$

- (ii) $\Longrightarrow$ (iii)

  If $\|X\|_{L^p}\leqslant K_2p$, then $\mathbb{E}[|X|^p]\leqslant(K_2p)^p$.

  Consider the Taylor expansion of $\exp(|X|/K)$,

$$\begin{aligned}\mathbb{E}\left[\exp\left(\frac{|X|}{K}\right)\right]&=\mathbb{E}\left[\sum_{p=0}^\infty\frac{(|X|/K)^p}{p!}\right]\\&=1+\sum_{p=1}^\infty\frac{\mathbb{E}[|X|^p]}{K^pp!}\\&\leqslant 1+\sum_{p=1}^\infty\frac{(K_2p)^p}{K^pp!}\\&\leqslant 1+\sum_{p=1}^\infty\left(\frac{\mathrm{e}K_2}{K}\right)^p&\left(p!\geqslant\left(\frac{p}{\mathrm{e}}\right)^p\right)\end{aligned}$$

  Therefore set $K_3=2\mathrm{e}K_2$. When $K\geqslant K_3$,

$$\mathbb{E}\left[\exp\left(\frac{|X|}{K_3}\right)\right]\leqslant 1+\sum_{p=1}^\infty\left(\frac{1}{2}\right)^p=1+1=2.$$

- (3) $\Longrightarrow$ (1)

If $\mathbb{E}[\exp(|X|/K_3)] \leqslant 2$, then by Chernoff's bound,

$$\mathbb{P}\{|X| \geqslant t\} = \mathbb{P}\left\{\exp\left(\frac{|X|}{K_3}\right) \geqslant \exp\left(\frac{t}{K_3}\right)\right\} \leqslant \frac{\mathbb{E}[\exp(|X|/K_3)]}{\exp(t/K_3)} \leqslant 2\mathrm{e}^{-t/K_3},$$

therefore set $K_1 = K_3$.

42 [**A Bird's Eye View: Orlicz Norms**] This is a general framework that encompasses most of the norms seen so far. Let $\psi : \mathbb{R}_0^+ \to \mathbb{R}_0^+$ be a non-decreasing convex function with $\psi(0) = 0$. Define the Orlicz norm of a random variable X as

$$\|X\|_\psi = \inf\left\{K > 0 : \mathbb{E}\left[\psi\left(\frac{|X|}{K}\right)\right] \leqslant 1\right\}.$$

(1) Prove: On the same probability space, the set of all random variables satisfying $\|X\|_\psi < \infty$ forms a normed linear space under this definition.

(2) Prove: The L^p $(1 \leqslant p < \infty)$ norm, the sub-Gaussian norm $\|X\|_{\psi_2}$, and the sub-Exponential norm $\|X\|_{\psi_1}$ are all special cases of the Orlicz norm.

Prf. ***Sol.*** Let $\mathcal{L}_\psi$ be the space of all random variables with $\|X\|_\psi < \infty$.

Sol. Non-negativity and definiteness:

It is easy to see that $\|X\|_\psi \geqslant 0$.

If $\|X\|_\psi = 0$, then for any $K > 0$, $\mathbb{E}[\psi(|X|/K)] \leqslant 1$.

Suppose $X \neq 0$ (a.s.), then there exists $\varepsilon > 0$ such that $\mathbb{P}\{|X| > \varepsilon\} > 0$, therefore

$$\lim_{K\to 0} \psi\left(\frac{\varepsilon}{K}\right) \to \infty,$$

thus

$$\lim_{K\to 0} \mathbb{E}\left[\psi\left(\frac{\varepsilon}{K}\right)\right] \to \infty,$$

contradicting $\mathbb{E}[\psi(|X|/K)] \leqslant 1$; hence $X = 0$ (a.s.).

Sol. Homogeneity:

It suffices to prove

$$\|\alpha X\|_\psi = \inf\left\{K > 0 : \mathbb{E}\left[\psi\left(\frac{|\alpha X|}{K}\right)\right] \leqslant 1\right\}.$$

If $\alpha = 0$, the conclusion is trivial; we only need to consider $\alpha \neq 0$.

Let $K' = K/|\alpha|$, i.e., $K = |\alpha|K'$, then

$$\inf\left\{|\alpha|K' > 0 : \mathbb{E}\left[\psi\left(\frac{|X|}{K'}\right)\right] \leqslant 1\right\} = |\alpha| \inf\left\{K' > 0 : \mathbb{E}\left[\psi\left(\frac{|X|}{K'}\right)\right] \leqslant 1\right\} = |\alpha|\|X\|_\psi.$$

Sol. Triangle inequality:

Let $\|X\|_\psi = K_X$, $\|Y\|_\psi = K_Y$, then $\mathbb{E}[\psi(|X|/K_X)], \mathbb{E}[\psi(|Y|/K_Y)] \leqslant 1$.

Using the convexity of ψ, we obtain

$$\begin{aligned}\psi\left(\frac{|X+Y|}{K_X+K_Y}\right) &\leqslant \psi\left(\frac{|X|+|Y|}{K_X+K_Y}\right) \\ &= \psi\left(\frac{K_X}{K_X+K_Y}\cdot\frac{|X|}{K_X} + \frac{K_Y}{K_X+K_Y}\cdot\frac{|Y|}{K_Y}\right),\end{aligned}$$

Taking expectations on both sides,

$$\begin{aligned}\mathbb{E}\left[\psi\left(\frac{|X+Y|}{K_X+K_Y}\right)\right] &\leqslant \frac{K_X}{K_X+K_Y}\mathbb{E}\left[\psi\left(\frac{|X|}{K_X}\right)\right] + \frac{K_Y}{K_X+K_Y}\mathbb{E}\left[\psi\left(\frac{|Y|}{K_Y}\right)\right] \\ &\leqslant \frac{K_X}{K_X+K_Y}\cdot 1 + \frac{K_Y}{K_X+K_Y}\cdot 1 = 1,\end{aligned}$$

therefore by the definition of infimum, $\|X+Y\|_\psi \leqslant \|X\|_\psi + \|Y\|_\psi$.

Sol. • L^p norm $\|X\|_{L^p}$

Let $\psi(x) = x^p$ $(p \geqslant 1)$, which is non-decreasing and convex on $\mathbb{R}_0^+$ with $\psi(0) = 0$, then

$$\|X\|_\psi = \inf\{K > 0 : \mathbb{E}[|X|^p/K^p] \leqslant 1\} = \inf\{K > 0 : \mathbb{E}[|X|^p] \leqslant K^p\} = (\mathbb{E}[|X|^p])^{1/p} = \|X\|_{L^p}.$$

• Sub-Gaussian norm $\|X\|_{\psi_2}$

Let $\psi_2(x) = \mathrm{e}^{x^2} - 1$, which is non-decreasing and convex on $\mathbb{R}_0^+$ with $\psi_2(0) = 0$, then

$$\|X\|_{\psi_2} = \inf\{K > 0 : \mathbb{E}[\mathrm{e}^{X^2/K^2} - 1] \leqslant 1\} = \inf\{K > 0 : \mathbb{E}[\mathrm{e}^{X^2/K^2}] \leqslant 2\}.$$

• Sub-Exponential norm $\|X\|_{\psi_1}$:

Let $\psi_1(x) = \mathrm{e}^x - 1$, which is non-decreasing and convex on $\mathbb{R}_0^+$ with $\psi_1(0) = 0$, then

$$\|X\|_{\psi_1} = \inf\{K > 0 : \mathbb{E}[\mathrm{e}^{|X|/K} - 1] \leqslant 1\} = \inf\{K > 0 : \mathbb{E}[\mathrm{e}^{|X|/K}] \leqslant 2\}.$$

43 [ψ_α **Distributions**] Consider the family of distributions whose tail probabilities decay at rate $\exp(-ct^\alpha)$ or faster. When $\alpha = 2$, these are sub-Gaussian distributions; when $\alpha = 1$, these are sub-Exponential distributions. For such distributions, give a generalized version of Proposition P2.8.1 (*Subexponential properties*) (i)–(iii), define the corresponding ψ_α norm, and prove the equivalence of the above properties.

(1) (Tail probability) There exists $K_1 > 0$ such that for all $t \geqslant 0$, $\mathbb{P}\{|X| \geqslant t\} \leqslant 2\exp(-(t/K_1)^\alpha)$.

(2) (Moments) There exists $K_2 > 0$ such that for all $p \geqslant 1$, $\|X\|_{L^p} = (\mathbb{E}[|X|^p])^{1/p} \leqslant K_2 p^{1/\alpha}$.

(3) (MGF) There exists $K_3 > 0$ such that $\mathbb{E}[\exp(|X|^\alpha/K_3^\alpha)] \leqslant 2$.

Prf. We prove the cyclic equivalence in the order (i) $\Longrightarrow$ (ii) $\Longrightarrow$ (iii) $\Longrightarrow$ (i).

• (i) $\Longrightarrow$ (ii)

It is easy to see that

$$\begin{aligned}
\mathbb{E}[|X|^p] &= \int_0^\infty pt^{p-1}\mathbb{P}\{|X| \geqslant t\}\mathrm{d}t \\
&\leqslant \int_0^\infty pt^{p-1} \cdot 2\mathrm{e}^{-(t/K_1)^\alpha}\mathrm{d}t \\
&\leqslant \frac{2pK_1^p}{\alpha}\int_0^\infty v^{p/\alpha-1}\mathrm{e}^{-v}\mathrm{d}v && \left(v = \left(\frac{t}{K_1}\right)^\alpha\right) \\
&= \frac{2pK_1^p}{\alpha}\Gamma\left(\frac{p}{\alpha}\right) = 2K_1^p\Gamma\left(\frac{p}{\alpha}+1\right).
\end{aligned}$$

By Stirling's formula, there exists $C > 0$ such that $\Gamma(p/\alpha+1)^{1/p} \leqslant Cp^{1/\alpha}$, therefore

$$\|X\|_{L^p} \leqslant (2\Gamma(p/\alpha+1))^{1/p}K_1 \leqslant K_2 p^{1/\alpha}, \qquad (K_2 \leqslant CK_1)$$

• (ii) $\Longrightarrow$ (iii)

If $\|X\|_{L^p} \leqslant K_2 p^{1/\alpha}$, then $\mathbb{E}[|X|^{\alpha k}] \leqslant (K_2(\alpha k)^{1/\alpha})^{\alpha k} = K_2^{\alpha k}\alpha^k k^k$.

Consider the Taylor expansion of $\exp(|X|^\alpha/K^\alpha)$,

$$\begin{aligned}
\mathbb{E}\left[\exp\left(\frac{|X|^\alpha}{K^\alpha}\right)\right] &= 1 + \sum_{k=1}^\infty \frac{\mathbb{E}[|X|^{\alpha k}]}{K^{\alpha k}k!} \\
&\leqslant 1 + \sum_{k=1}^\infty \frac{K_2^{\alpha k}\alpha^k k^k}{K^{\alpha k}k!} \\
&\leqslant 1 + \sum_{k=1}^\infty \left(\frac{\mathrm{e}\alpha K_2^\alpha}{K^\alpha}\right)^k && (k! \geqslant (k/\mathrm{e})^k)
\end{aligned}$$

Therefore set $K_3^\alpha = 2\mathrm{e}\alpha K_2^\alpha$. When $K \geqslant K_3$,

$$\mathbb{E}\left[\exp\left(\frac{|X|^\alpha}{K_3^\alpha}\right)\right] \leqslant 2.$$

- (3) $\Longrightarrow$ (1)

  By Chernoff's bound,

$$\mathbb{P}\{|X| \geqslant t\} = \mathbb{P}\left\{\exp\left(\frac{|X|^\alpha}{K_3^\alpha}\right) \geqslant \exp\left(\frac{t^\alpha}{K_3^\alpha}\right)\right\} \leqslant 2\mathrm{e}^{-(t/K_3)^\alpha},$$

  therefore set $K_1 = K_3$.

44 [**Generalization of Sub-Exponential Properties**] Many properties of sub-Gaussian random variables, after appropriate modification, can be carried over in parallel to sub-Exponential random variables.

Define the sub-Exponential norm of X as

$$\|X\|_{\psi_1} = \inf\{t > 0 : \mathbb{E}[\exp(|X|/t)] \leqslant 2\},$$

Prove the following properties:

(1) Centering (see Lemma L2.7.8 (*Centering*)): If X is a sub-Exponential random variable, then

$$\|X - \mathbb{E}[X]\|_{\psi_1} \leqslant C\|X\|_{\psi_1}. \qquad (C > 0)$$

(2) Maximal Inequality (see Exercise Q2.39 (*Characterization of Sub-Gaussian Distributions: Maximal Inequality*)): X is a sub-Exponential random variable if and only if

$$\mathbb{E}\left[\max_{i \leqslant N} |X_i|\right] \leqslant K \log N, \qquad (K > 0)$$

where $\{X_N\}$ are independent copies of X.

(3) Convex Dominance (see Exercise Q2.28 (*Characterization of Sub-Gaussian Distributions: Convex Dominance Definition*)): X is a sub-Exponential random variable if and only if

$$|X| \preceq KE, \qquad (K > 0)$$

where $E \sim E(1)$.

Prf. ***Sol.*** Since $\|\cdot\|_{\psi_1}$ is an Orlicz norm (see Exercise Q2.45 (*Restatement of Bernstein's Inequality*)), by the triangle inequality,

$$\|X - \mathbb{E}[X]\|_{\psi_1} \leqslant \|X\|_{\psi_1} + \|\mathbb{E}[X]\|_{\psi_1}.$$

For $\|\mathbb{E}[X]\|_{\psi_1}$, by Proposition P2.8.1 (*Subexponential properties*),

$$\|X\|_{L^1} \leqslant C_1\|X\|_{\psi_1} = \frac{C_1}{\log 2}|X|, \qquad \left(\|a\|_{\psi_1} = \frac{|a|}{\log 2}\right)$$

therefore

$$\|X - \mathbb{E}[X]\|_{\psi_1} \leqslant \|X\|_{\psi_1} + \frac{C_1}{\log 2}\|X\|_{\psi_1} = \left(1 + \frac{C_1}{\log 2}\right)\|X\|_{\psi_1},$$

so set $C = 1 + C_1/\log 2$.

Sol. ($\Longrightarrow$)

If X is sub-Exponential, let $L = \|X\|_{\psi_1}$, then

$$\mathbb{P}\{|X| \geqslant t\} \leqslant 2\mathrm{e}^{-ct/L}. \qquad (c > 0)$$

Let $M_N = \max_{i \leqslant N} |X_i|$, then

$$\mathbb{P}\{M_N \geqslant t\} \leqslant N\mathbb{P}\{|X| \geqslant t\} \leqslant 2N\mathrm{e}^{-ct/L}.$$

Split $\mathbb{E}[M_N]$ at t_0,

$$\begin{aligned}
\mathbb{E}[M_N] &= \int_0^\infty \mathbb{P}\{M_N \geqslant t\}\mathrm{d}t \\
&= \int_0^{t_0} \mathbb{P}\{M_N \geqslant t\}\mathrm{d}t + \int_{t_0}^\infty \mathbb{P}\{M_N \geqslant t\}\mathrm{d}t \\
&\leqslant \int_0^{t_0} 1\mathrm{d}t + \int_{t_0}^\infty 2N\mathrm{e}^{-ct/\|X\|_{\psi_1}}\mathrm{d}t
\end{aligned}$$

$$= t_0 + \left[2N\left(-\frac{L}{c}\right)\mathrm{e}^{-ct/L}\right]_{t_0}^{\infty}$$
$$= t_0 + \frac{L}{c}(2N\mathrm{e}^{-ct_0/L}),$$

Set $2N\mathrm{e}^{-ct_0/L} = 1$, i.e., $t_0 = L\log(2N)/c$, then

$$\mathbb{E}[M_N] \leqslant \frac{L}{c}\log(2N) + \frac{L}{c} \lesssim \log N.$$

($\Longleftarrow$)

If $\mathbb{E}[M_N] \leqslant K\log N$, let $p = \mathbb{P}\{|X| \geqslant t\}$, then

$$\mathbb{E}[M_N] \geqslant t \cdot \mathbb{P}\{M_N \geqslant t\}$$
$$= t[1-(1-p)^N],$$

Let $N = \lfloor 1/p \rfloor$, then

$$t\left[1-\left(1-\frac{1}{N}\right)^N\right] \to t(1-\mathrm{e}^{-1}) =: c_0 t, \qquad (c_0 = 1-\mathrm{e}^{-1})$$

therefore

$$K\log N \geqslant t\cdot c_0 \implies \log N \geqslant \frac{c_0}{K}t$$
$$\implies \log\left(\frac{1}{p}\right) \geqslant \frac{c_0}{K}t$$
$$\implies p = \mathbb{P}\{|X| \geqslant t\} \leqslant \mathrm{e}^{-c_0 t/K}.$$

Sol. ($\Longrightarrow$)

To prove $|X| \preceq KE$, it suffices to verify the second-order stochastic dominance condition, i.e., for any $x \geqslant 0$,

$$\int_x^{\infty} \mathbb{P}\{|X| \geqslant t\}\mathrm{d}t \leqslant \int_x^{\infty} \mathbb{P}\{KE \geqslant t\}\mathrm{d}t.$$

For the left-hand side, since $\mathbb{P}\{|X| \geqslant t\} \leqslant 2\mathrm{e}^{-ct/L}$ where $L = \|X\|_{\psi_1}$,

$$\int_x^{\infty} \mathbb{P}\{|X| \geqslant t\}\mathrm{d}t = \int_x^{\infty} 2\mathrm{e}^{-ct/L}\mathrm{d}t = \frac{2L}{c}\mathrm{e}^{-cx/L};$$

For the right-hand side, since $\mathbb{P}\{KE \geqslant t\} = \mathrm{e}^{-t/K}$,

$$\int_x^{\infty} \mathbb{P}\{KE \geqslant t\}\mathrm{d}t = \int_x^{\infty} \mathrm{e}^{-t/K}\mathrm{d}t = K\mathrm{e}^{-x/K},$$

Therefore if $|X| \preceq KE$, then $K\mathrm{e}^{-x/K} \geqslant \frac{2L}{c}\mathrm{e}^{-cx/L}$, so set K such that

$$K \geqslant \frac{2L}{c}, \frac{1}{K} \leqslant \frac{c}{L} \implies K = \max\left(\frac{2L}{c}, \frac{L}{c}\right) = \frac{2L}{c}.$$

($\Longleftarrow$)

If $|X| \preceq KE$, it suffices to prove that for any monotone increasing convex function $\Phi : \mathbb{R} \to \mathbb{R}$, $\mathbb{E}[\Phi(|X|)] \leqslant \mathbb{E}[\Phi(KE)]$.

Let $\Phi(u) = \mathrm{e}^{u/T}$ $(T > 0)$, then

$$\mathbb{E}[\mathrm{e}^{|X|/T}] \leqslant \mathbb{E}[\mathrm{e}^{(KE)/T}],$$

where

$$\mathbb{E}[\mathrm{e}^{(KE)/T}] = \mathbb{E}[\mathrm{e}^{(K/T)E}] = \int_0^{\infty} \mathrm{e}^{(K/T)t}\cdot\mathrm{e}^{-t}\mathrm{d}t = \int_0^{\infty} \mathrm{e}^{(K/T-1)t}\mathrm{d}t,$$

When $K/T < 1$, i.e., $T > K$, the above integral converges, and

$$\mathbb{E}[\mathrm{e}^{(KE)/T}] = \frac{1}{1-K/T},$$

therefore set

$$\frac{1}{1-K/T} \leqslant 2 \implies T \geqslant 2K,$$

we obtain $\mathbb{E}[\mathrm{e}^{|X|/(2K)}] \leqslant 2$, so $\|X\|_{\psi_1} \leqslant 2K < \infty$.

45 [**Restatement of Bernstein's Inequality**] Theorem T2.9.1 (*Subexponential Bernstein inequality*) gives tail probability bounds for sums of random variables.

In probability theory and high-dimensional statistics, there is a more popular and intuitive way to express such results: by specifying the confidence level (i.e., the tail probability level u) to invert the error bound for the sum of random variables.

Let $\{X_N\}$ be independent K-sub-exponential random variables with $\mathbb{E}[X_i] = 0$. Prove: For any $u \geqslant 0$,

$$\mathbb{P}\left\{\left|\sum_{i=1}^{N} X_i\right| \geqslant C(\sigma\sqrt{u} + Ku)\right\} \leqslant 2\mathrm{e}^{-u}. \qquad (C > 0)$$

where $\sigma^2 = \sum_{i=1}^{N} \|X_i\|_{\psi_1}^2$.

Prf. By Theorem T2.9.1 (*Subexponential Bernstein inequality*), for any $t \geqslant 0$,

$$\mathbb{P}\{|S_N| \geqslant t\} \leqslant 2\exp\left[-c\min\left(\frac{t^2}{\sigma^2}, \frac{t}{K}\right)\right],$$

It suffices to prove that there exists t such that

$$c\min\left(\frac{t^2}{\sigma^2}, \frac{t}{K}\right) = u \implies \min\left(\frac{t^2}{\sigma^2}, \frac{t}{K}\right) = \frac{u}{c}.$$

a. When $t^2/\sigma^2 \leqslant t/K$, i.e., $t \leqslant \sigma^2/K$, the inequality exhibits sub-Gaussian behavior, with

$$\frac{t^2}{\sigma^2} = \frac{u}{c} \implies t = \frac{\sigma\sqrt{u}}{\sqrt{c}},$$

therefore

$$u \leqslant \frac{c\sigma^2}{K^2}.$$

b. When $t^2/\sigma^2 > t/K$, i.e., $t > \sigma^2/K$, the inequality exhibits sub-Exponential behavior, with

$$\frac{t}{K} = \frac{u}{c} \implies t = \frac{Ku}{c},$$

therefore

$$u > \frac{c\sigma^2}{K^2}.$$

Thus, the critical value is

$$t = \max\left(\frac{\sigma\sqrt{u}}{\sqrt{c}}, \frac{Ku}{c}\right),$$

Using the inequality $\max(A, B) \leqslant A + B$, we obtain

$$t \leqslant \frac{1}{\sqrt{c}}\sigma\sqrt{u} + \frac{1}{c}Ku = C(\sigma\sqrt{u} + Ku). \qquad \left(C = \max\left(\frac{1}{\sqrt{c}}, \frac{1}{c}\right)\right)$$

Rk. This restatement reveals the mixed-tail nature of Bernstein's inequality:

1. Fluctuation-dominated term: $\sigma\sqrt{u}$ reflects the normal fluctuation similar to CLT, related to variance.
2. Extreme-value-dominated term: Ku reflects the Poisson-type fluctuation of the worst single component (the thickest sub-Exponential tail).
3. Phase transition: When u is small (low confidence requirement), $\sqrt{u}$ dominates and the system behaves like a Gaussian distribution; when u is large (very small probability events), u dominates and the system is limited by the maximum thick tail of individual components.

46 [**Sub-Exponential Khintchine Inequality**] Prove the sub-Exponential version of Theorem T2.7.5 (*Khintchine inequality*).

Let $\{X_N\}$ be independent mean-zero K-sub-Exponential random variables. Prove: For any $\boldsymbol{a} \in \mathbb{R}^N$, when $p \in [2, \infty)$,

$$\left\| \sum_{i=1}^{N} a_i X_i \right\|_{L^p} \leqslant CK(\sqrt{p}\|\boldsymbol{a}\|_2 + p\|\boldsymbol{a}\|_\infty). \qquad (C > 0)$$

Prf. Let $S_N = \sum_{i=1}^N a_i X_i$. By Theorem T2.9.1 (*Subexponential Bernstein inequality*), for any $t \geqslant 0$,

$$\mathbb{P}\{|S_N| \geqslant t\} \leqslant 2\exp\left[-c\min\left(\frac{t^2}{\sigma^2}, \frac{t}{L}\right)\right], \qquad (c > 0)$$

where

$$\sigma^2 = \sum_{i=1}^{N} a_i^2 \|X_i\|_{\psi_1}^2 \leqslant K^2 \sum_{i=1}^{N} a_i^2 = K^2\|\boldsymbol{a}\|_2^2,$$

$$L = \max_{i \leqslant N} |a_i| \|X_i\|_{\psi_1} \leqslant K \max_{i \leqslant N} |a_i| = K\|\boldsymbol{a}\|_\infty,$$

Let $A = K\|\boldsymbol{a}\|_2$, $B = K\|\boldsymbol{a}\|_\infty$, then

$$\mathbb{P}\{|S_N| \geqslant t\} \leqslant 2\exp\left[-c\min\left(\frac{t^2}{A^2}, \frac{t}{B}\right)\right].$$

Substituting the tail bound into the moment identity, we obtain

$$\begin{aligned}
\|S_N\|_{L^p}^p = \mathbb{E}[|S_N|^p] &= \int_0^\infty pt^{p-1}\mathbb{P}\{|S_N| \geqslant t\}\mathrm{d}t \\
&\leqslant \int_0^\infty 2pt^{p-1}\exp\left(-c\frac{t^2}{A^2}\right)\mathrm{d}t + \int_0^\infty 2pt^{p-1}\exp\left(-c\frac{t}{B}\right)\mathrm{d}t \\
&\leqslant (C_1 A\sqrt{p})^p + (C_2 Bp)^p \\
&\leqslant (C_1 A\sqrt{p} + C_2 Bp)^p,
\end{aligned}$$

therefore

$$\|S_N\|_{L^p} \leqslant C_1 K\sqrt{p}\|\boldsymbol{a}\|_2 + C_2 Kp\|\boldsymbol{a}\|_\infty \leqslant CK(\sqrt{p}\|\boldsymbol{a}\|_2 + p\|\boldsymbol{a}\|_\infty).$$

47 **[Bernstein's Inequality for Bounded Distributions]**

(1) Let X satisfy $\mathbb{E}[X] = 0$, $|X| \leqslant K$ (a.s.). Prove: When $|\lambda| < 3/K$,

$$\mathbb{E}[\mathrm{e}^{\lambda X}] \leqslant \exp(g(\lambda)\mathbb{E}[X^2]),$$

where

$$g(\lambda) = \frac{\lambda^2/2}{1 - |\lambda|K/3}.$$

(2) Using the exponential moment method, derive Theorem T2.9.5 (*Bernstein inequality for bounded distributions*) (Bernstein's concentration inequality for sums of independent bounded random variables).

Prf. ***Sol.*** It is easy to see that $|X^k| = |X|^{k-2} \cdot X^2 \leqslant K^{k-2}X^2$, therefore

$$\mathbb{E}[X^k] \leqslant K^{k-2}\mathbb{E}[X^2].$$

Consider the Taylor expansion of $\mathrm{e}^{\lambda X}$,

$$\begin{aligned}
\mathbb{E}[\mathrm{e}^{\lambda X}] &= 1 + \lambda\mathbb{E}[X] + \sum_{k=2}^{\infty} \frac{\lambda^k \mathbb{E}[X^k]}{k!} \\
&\leqslant 1 + \sum_{k=2}^{\infty} \frac{|\lambda|^k K^{k-2}\mathbb{E}[X^2]}{k!} \\
&= 1 + \mathbb{E}[X^2]\lambda^2 \sum_{k=2}^{\infty} \frac{|\lambda|^{k-2}K^{k-2}}{k!},
\end{aligned}$$

where when $|\lambda| < 3/K$, substituting $k! \geqslant 2 \cdot 3^{k-2}$, we obtain

$$\sum_{k=2}^{\infty} \frac{|\lambda|^{k-2} K^{k-2}}{k!} \leqslant \frac{1}{2} \sum_{j=0}^{\infty} \left(\frac{|\lambda| K}{3} \right)^j = \frac{1/2}{1 - |\lambda| K/3},$$

substituting it, we obtain

$$\mathbb{E}[\mathrm{e}^{\lambda X}] \leqslant 1 + \mathbb{E}[X^2] \cdot \frac{\lambda^2/2}{1 - |\lambda| K/3} \leqslant \exp(g(\lambda) \mathbb{E}[X^2]). \tag{I}$$

Sol. Let $S_N = \sum_{i=1}^N X_i$, and let the variance sum be $\Sigma^2 = \sum_{i=1}^N \mathbb{E}[X_i^2]$.
By Chernoff's bound,

$$\mathbb{P}\{S_N \geqslant t\} = \mathbb{P}\{\mathrm{e}^{\lambda S_N} \geqslant \mathrm{e}^{\lambda t}\} \leqslant \mathrm{e}^{-\lambda t} \mathbb{E}[\mathrm{e}^{\lambda S_N}] = \mathrm{e}^{-\lambda t} \prod_{i=1}^N \mathbb{E}[\mathrm{e}^{\lambda X_i}], \tag{II}$$

substituting (II) into (I), we obtain

$$\mathbb{P}\{S_N \geqslant t\} \leqslant \exp\left(-\lambda t + \frac{\lambda^2 \Sigma^2/2}{1 - \lambda K/3} \right).$$

The optimal parameter on the right-hand side is $\lambda = t/(\Sigma^2 + Kt/3) < 3/K$. Substituting it, we obtain

$$\mathbb{P}\{S_N \geqslant t\} \leqslant \exp\left(-\frac{t^2}{2\Sigma^2 + 2Kt/3} \right),$$

taking the union bound on both sides,

$$\mathbb{P}\{|S_N| \geqslant t\} \leqslant 2 \exp\left(-\frac{t^2}{2\Sigma^2 + 2Kt/3} \right).$$

48 [**Bennett's Inequality**] Let us prove the following strengthening of Theorem T2.9.5 (*Bernstein inequality for bounded distributions*).
Let $\{X_N\}$ be independent with $\mathbb{E}[X_i] = 0$, $|X_i| \leqslant K$. Prove: For any $t \geqslant 0$,

$$\mathbb{P}\left\{ \left| \sum_{i=1}^N X_i \right| \geqslant t \right\} \leqslant 2 \exp\left[-\frac{\sigma^2}{K^2} h\left(\frac{Kt}{\sigma^2} \right) \right].$$

where $\sigma^2 = \sum_{i=1}^N \mathbb{E}[X_i^2]$, $h(u) = (1+u)\log(1+u) - u$. Prove this result as follows:
(1) Let X be a mean-zero random variable satisfying $|X| \leqslant K$ (a.s.). Show that for any $\lambda > 0$,

$$\mathbb{E}[\mathrm{e}^{\lambda X}] \leqslant \exp\left(\frac{\mathbb{E}[X^2]}{K^2} (\mathrm{e}^{\lambda K} - 1 - \lambda K) \right).$$

(2) Use the exponential moment method to prove the Bennett inequality above.
(3) (*Small deviations: sub-Gaussian tail bound*) Show that when $u = Kt/\sigma^2 \to 0$, we have $h(u) \sim u^2/2$, so Bennett's inequality reduces to the Gaussian tail bound $\exp(-t^2/\sigma^2)$.
(4) (*Large deviations: Poisson tail bound*) Show that when u is sufficiently large, we have $h(u) \geqslant u \log u/2$, so Bennett's inequality reduces to the Poisson tail bound $2(\sigma^2/Kt)^{t/(2K)}$, similar to Exercise Q2.13 (*Tail Bounds for the Poisson Distribution*).
Prf. (1) Let $f(x) = (\mathrm{e}^{\lambda x} - 1 - \lambda x)/x^2$, and define $f(0) = \lambda^2/2$. Then

$$f(x) = \sum_{k=2}^{\infty} \frac{\lambda^k x^{k-2}}{k!} = \frac{\lambda^2}{2!} + \frac{\lambda^3 x}{3!} + \frac{\lambda^4 x^2}{4!} + \cdots,$$

differentiating both sides with respect to x,

$$f'(x) = \sum_{k=3}^{\infty} \frac{\lambda^k (k-2) x^{k-3}}{k!} = \frac{\lambda^3}{3!} + \frac{2\lambda^4 x}{4!} + \cdots = \frac{\lambda^3 g(\lambda x)}{(\lambda x)^3},$$

where $g(y) = y\mathrm{e}^y - 2\mathrm{e}^y + 2 + y$.

Since $g(0) = g'(0) = 0$ and $g''(y) = ye^y$, we have $f'(x) > 0$ $(x \neq 0)$, so $f(x)$ is increasing; hence

$$f(X) \leqslant f(K) = \frac{e^{\lambda K} - 1 - \lambda K}{K^2},$$

Multiplying both sides by X^2,

$$e^{\lambda X} - 1 - \lambda X = X^2 f(X) \leqslant X^2 f(K) = X^2 \cdot \frac{e^{\lambda K} - 1 - \lambda K}{K^2},$$

taking expectations on both sides and substituting $\mathbb{E}[X] = 0$, we obtain

$$\mathbb{E}[e^{\lambda X}] - 1 \leqslant \mathbb{E}[X^2] \cdot \frac{e^{\lambda K} - 1 - \lambda K}{K^2}.$$

That is,

$$\begin{aligned}\mathbb{E}[e^{\lambda X}] &\leqslant 1 + \frac{\mathbb{E}[X^2]}{K^2}(e^{\lambda K} - 1 - \lambda K) \\ &\leqslant \exp\left(\frac{\mathbb{E}[X^2]}{K^2}(e^{\lambda K} - 1 - \lambda K)\right).\end{aligned}$$

(2) Let $S_N = \sum_{i=1}^N X_i$. Then by the result of (1),

$$\mathbb{E}[e^{\lambda S_N}] = \prod_{i=1}^N \mathbb{E}[e^{\lambda X_i}] \leqslant \prod_{i=1}^N \exp\left(\frac{\mathbb{E}[X_i^2]}{K^2}(e^{\lambda K} - 1 - \lambda K)\right) = \exp\left(\frac{\sigma^2}{K^2}(e^{\lambda K} - 1 - \lambda K)\right),$$

hence, by Chernoff's bound, for any $\lambda > 0$,

$$\mathbb{P}\{S_N \geqslant t\} \leqslant e^{-\lambda t}\mathbb{E}[e^{\lambda S_N}] \leqslant \exp\left(-\lambda t + \frac{\sigma^2}{K^2}(e^{\lambda K} - 1 - \lambda K)\right),$$

The optimal parameter on the right-hand side is $\lambda_0 = \log(1 + u)/K$. Substituting it gives

$$\begin{aligned}\mathbb{P}\{S_N \geqslant t\} &\leqslant \exp\left(-\frac{\sigma^2}{K^2}[(1 + u)\log(1 + u) - u]\right) \\ &\leqslant \exp\left[-\frac{\sigma^2}{K^2}h\left(\frac{Kt}{\sigma^2}\right)\right]. \qquad \text{(I)}\end{aligned}$$

Similarly, replacing X_i by $-X_i$ and taking the union bound,

$$\mathbb{P}\{S_N \leqslant -t\} \leqslant \exp\left[-\frac{\sigma^2}{K^2}h\left(\frac{Kt}{\sigma^2}\right)\right]. \qquad \text{(II)}$$

Combining (I) and (II), we obtain

$$\mathbb{P}\left\{\left|\sum_{i=1}^N X_i\right| \geqslant t\right\} \leqslant 2\exp\left[-\frac{\sigma^2}{K^2}h\left(\frac{Kt}{\sigma^2}\right)\right].$$

(3) When $u = Kt/\sigma^2 \to 0$,

$$h(u) = (1 + u)\left(u - \frac{u^2}{2} + O(u^3)\right) - u = u - \frac{u^2}{2} + u^2 + O(u^3) - u = \frac{u^2}{2} + O(u^3) \sim \frac{u^2}{2},$$

substituting it, the exponent becomes

$$-\frac{\sigma^2}{K^2}h(u) \sim -\frac{\sigma^2}{K^2} \cdot \frac{1}{2}\left(\frac{Kt}{\sigma^2}\right)^2 = -\frac{t^2}{2\sigma^2},$$

therefore

$$\mathbb{P}\left\{\left|\sum_{i=1}^N X_i\right| \geqslant t\right\} \lesssim 2\exp\left(-\frac{t^2}{2\sigma^2}\right),$$

(4) When u is sufficiently large, we may assume without loss of generality that $u \geqslant e^2$,

$$h(u) = (1 + u)\log(1 + u) - u \geqslant u\log u - u \geqslant \frac{1}{2}u\log u.$$

Substituting it, the exponent becomes

$$-\frac{\sigma^2}{K^2}h(u) \leqslant -\frac{\sigma^2}{K^2}\left(\frac{1}{2}u\log u\right) = -\frac{t}{2K}\log\left(\frac{Kt}{\sigma^2}\right) = \log\left[\left(\frac{\sigma^2}{Kt}\right)^{t/(2K)}\right],$$

therefore

$$\mathbb{P}\left\{\left|\sum_{i=1}^{N} X_i\right| \geqslant t\right\} \leqslant 2\exp\left[-\frac{\sigma^2}{K^2}h\left(\frac{Kt}{\sigma^2}\right)\right] \leqslant 2\left(\frac{\sigma^2}{Kt}\right)^{t/(2K)}.$$

High-Dimensional Random Vectors

3

Keypoints

Section 3.1 Concentration of the norm — Q3.1 ~ Q3.3

Theorem 3.1.1 (*Concentration of the norm*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector with independent, mean-zero, subgaussian coordinates X_i. Then

$$\mathbb{P}\{|\|\boldsymbol{X}\|_2 - \sqrt{n}| \geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{K^4}\right).$$

for all $t \geqslant 0$.

Section 3.2 Covariance matrices and PCA — Q3.4

Proposition 3.2.1 (*Covariance matrix helps compute interesting quantities*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector with mean zero, and let its covariance matrix be $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]$. Then:

(1) For any $\boldsymbol{v} \in \mathbb{S}^{n-1}$,

$$\mathrm{Var}(\langle \boldsymbol{X}, \boldsymbol{v}\rangle) = \boldsymbol{v}^\top \boldsymbol{\Sigma} \boldsymbol{v}.$$

(2) For any $S \subset \{1, \cdots, n\}$, the sub-vector $\boldsymbol{X}_S := (X_i)_{i\in S}$ has covariance matrix $\boldsymbol{\Sigma}_S$, the principal submatrix of $\boldsymbol{\Sigma}$ restricted to S.

Proposition 3.2.2 (*Optimization-based characterization of eigenvalues*)

For a symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, the largest eigenvalue $\lambda_{\max}(\boldsymbol{A})$ satisfies

$$\lambda_{\max}(\boldsymbol{A}) = \max_{\|\boldsymbol{x}\|_2=1} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x},$$

and the smallest eigenvalue $\lambda_{\min}(\boldsymbol{A})$ satisfies

$$\lambda_{\min}(\boldsymbol{A}) = \min_{\|\boldsymbol{x}\|_2=1} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}.$$

Corollary 3.2.3 (*Principal component analysis*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a mean-zero random vector with covariance matrix $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]$. Then the first principal direction $\boldsymbol{v}_1$, which captures the maximal variance $\mathrm{Var}(\langle \boldsymbol{X}, \boldsymbol{v}\rangle)$ over all unit vectors $\boldsymbol{v}$, is the eigenvector corresponding to $\lambda_{\max}(\boldsymbol{\Sigma})$; the

variance in that direction equals $\lambda_{\max}(\boldsymbol{\Sigma})$.

Definition 3.2.5 (*Isotropic random vectors*)

A random vector $\boldsymbol{X} \in \mathbb{R}^n$ is called isotropic if its covariance matrix is the identity: $\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \boldsymbol{I}_n$. Equivalently, $\mathbb{E}[\langle \boldsymbol{X}, \boldsymbol{v}\rangle^2] = \|\boldsymbol{v}\|_2^2$ for all $\boldsymbol{v} \in \mathbb{R}^n$.

Section 3.3 Examples of high-dimensional distributions Q3.5 ∼ Q3.31

Proposition 3.3.1 (*Rotation invariance*)

The standard normal distribution $N(\boldsymbol{0}, \boldsymbol{I}_n)$ is rotation invariant: if $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ and $\boldsymbol{Q} \in \mathbb{R}^{n\times n}$ is an orthogonal matrix, then $\boldsymbol{Q}\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

Corollary 3.3.2 (*1D marginals of the standard normal distribution*)

Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Then for any fixed unit vector $\boldsymbol{v} \in \mathbb{R}^n$, we have

$$\langle \boldsymbol{g}, \boldsymbol{v}\rangle \sim N(0, 1).$$

Corollary 3.3.3 (*The sum of independent normals is normal*)

Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ be independent. Then

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

Definition 3.3.4 (*General normal distribution*)

A random vector $\boldsymbol{X} \in \mathbb{R}^n$ has a normal distribution, denoted $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, if there exists a matrix $\boldsymbol{B}$ such that $\boldsymbol{X}$ can be expressed as $\boldsymbol{X} = \boldsymbol{\mu} + \boldsymbol{B}\boldsymbol{g}$ where $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_k)$, and $\boldsymbol{\Sigma} = \boldsymbol{B}\boldsymbol{B}^\top$.

Proposition 3.3.5 (*Uniqueness*)

If $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are uniquely determined by the distribution of $\boldsymbol{X}$; indeed, $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{X}]$ and $\boldsymbol{\Sigma} = \mathrm{Cov}(\boldsymbol{X})$.

Proposition 3.3.6 (*General normal density*)

If $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ invertible, then $\boldsymbol{X}$ has density

$$f(\boldsymbol{x}) = \frac{1}{(2\pi)^{n/2}\sqrt{\det(\boldsymbol{\Sigma})}} \exp\Big(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\Big), \quad \boldsymbol{x} \in \mathbb{R}^n.$$

Corollary 3.3.7 (*Jointly normal random variables*)

Random variables $X_1, \cdots, X_n$ are jointly normal (i.e., the vector $(X_1, \cdots, X_n)$ has a multivariate normal distribution) if and only if every linear combination $\sum_{i=1}^n a_i X_i$ is a normal random variable.

Proposition 3.3.8 (*A sphere is isotropic*)

A random vector $\boldsymbol{\theta}$ uniformly distributed on the unit Euclidean sphere $\mathbb{S}^{n-1}$ is isotropic:

$$\mathbb{E}[\boldsymbol{\theta}\boldsymbol{\theta}^\top] = \frac{1}{n}\boldsymbol{I}_n.$$

Theorem 3.3.9 (*Projective central limit theorem*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be an isotropic random vector with independent coordinates. Then, for most unit vectors $\boldsymbol{v} \in \mathbb{S}^{n-1}$, the marginal $\langle \boldsymbol{X}, \boldsymbol{v}\rangle$ is approximately normal $N(0, 1)$.

Proposition 3.3.11 (*Parseval frames*)

A set of vectors $\{\boldsymbol{u}_i\}_{i=1}^m$ in $\mathbb{R}^n$ is called a tight frame if $\sum_{i=1}^m \langle \boldsymbol{x}, \boldsymbol{u}_i\rangle^2 = A\|\boldsymbol{x}\|_2^2$ for all $\boldsymbol{x} \in \mathbb{R}^n$ (Parseval identity) for some constant $A > 0$ (the frame bound).

Section 3.4 Subgaussian distributions in higher dimensions Q3.32 ∼ Q3.46

Definition 3.4.1 (*Subgaussian random vectors*)

A random vector $\boldsymbol{X} \in \mathbb{R}^n$ is subgaussian if the one-dimensional marginals $\langle \boldsymbol{X}, \boldsymbol{v} \rangle$ are subgaussian random variables for all $\boldsymbol{v} \in \mathbb{R}^n$. The subgaussian norm of $\boldsymbol{X}$ is $\|\boldsymbol{X}\|_{\psi_2} = \sup_{\|\boldsymbol{v}\|_2=1} \|\langle \boldsymbol{X}, \boldsymbol{v} \rangle\|_{\psi_2}$.

Lemma 3.4.2 (*Distributions with independent subgaussian coordinates*)

If $\boldsymbol{X} = (X_1, \cdots, X_n)$ has independent, mean-zero, subgaussian coordinates, then $\boldsymbol{X}$ is a subgaussian random vector with

$$\|\boldsymbol{X}\|_{\psi_2} \leqslant C \max_i \|X_i\|_{\psi_2}.$$

Theorem 3.4.5 (*The uniform distribution on the sphere is subgaussian*)

A random vector $\boldsymbol{\theta}$ uniformly distributed on the Euclidean sphere of radius $\sqrt{n}$ in $\mathbb{R}^n$ is subgaussian, and

$$\|\boldsymbol{\theta}\|_{\psi_2} \leqslant C,$$

where C is an absolute constant.

Section 3.5 Grothendieck inequality and SDP Q3.47 ∼ Q3.52

Theorem 3.5.1 (*Grothendieck inequality*)

There exists an absolute constant $K_G \leqslant 1.783$ such that the following holds. For any $m \times n$ real matrix $\boldsymbol{A} = (a_{ij})$,

$$\max_{u_i, v_j \in \{-1,1\}} \sum_{i,j} a_{ij} u_i v_j \leqslant \max_{\boldsymbol{u}_i, \boldsymbol{v}_j \in \mathbb{S}^{m+n-1}} \sum_{i,j} a_{ij} \langle \boldsymbol{u}_i, \boldsymbol{v}_j \rangle \leqslant K_G \cdot \max_{u_i, v_j \in \{-1,1\}} \sum_{i,j} a_{ij} u_i v_j,$$

where the middle maximum is over unit vectors $\boldsymbol{u}_i, \boldsymbol{v}_j$ in a Hilbert space.

Remark 3.5.3 (*Quadratic Grothendieck: $x_i = y_i$*)

The assumption of the Grothendieck inequality can often be relaxed by assuming $x_i = y_i$, letting us bound a quadratic instead of a bilinear form. Let $\boldsymbol{A} = (a_{ij}) \in \mathbb{R}^{n \times n}$ be a matrix, either symmetric positive-semidefinite or diagonal-free. Assume that

$$\left| \sum_{i,j} a_{ij} x_i x_j \right| \leqslant 1$$

for any numbers $x_i \in \{-1, 1\}$. Then, for any Hilbert space $\mathcal{H}$, we have:

$$\left| \sum_{i,j} a_{ij} \langle \boldsymbol{u}_i, \boldsymbol{v}_j \rangle \right| \leqslant 2K_G$$

for any unit vectors $\boldsymbol{u}_i, \boldsymbol{v}_j \in \mathcal{H}$. Here K_G is the Grothendieck constant. You will check this in Exercises Q3.49 (*Quadratic Grothendieck Inequality for Positive Semidefinite Matrices*) and Q3.50 (*Quadratic Grothendieck Inequality for Matrices without Diagonal Entries*).

Definition 3.5.4 (*Semidefinite program*)

A semidefinite program (SDP) is an optimization problem of the form: maximize $\langle \boldsymbol{C}, \boldsymbol{X} \rangle_F$ subject to $\langle \boldsymbol{A}_i, \boldsymbol{X} \rangle_F = b_i$ for $i = 1, \cdots, m$, and $\boldsymbol{X} \succeq 0$, where the inner product of matrices is $\langle \boldsymbol{A}, \boldsymbol{B} \rangle_F = \operatorname{tr}(\boldsymbol{A}^\top \boldsymbol{B})$, and $\boldsymbol{X} \succeq 0$ means that $\boldsymbol{X}$ is symmetric and positive semidefinite.

Proposition 3.5.6 (*The relaxation is an SDP*)

The semidefinite relaxation of Grothendieck's problem is an SDP in the matrix variable $\boldsymbol{Z} = (z_{ij})$ with $z_{ij} = \langle \boldsymbol{u}_i, \boldsymbol{v}_j \rangle$, namely: maximize $\sum_{i,j} a_{ij} z_{ij}$ subject to $\boldsymbol{Z} \succeq 0$ and $z_{ii} = 1$ for all i.

Theorem 3.5.7 (*The guarantee of relaxation*)

For any matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, let $\mathrm{SDP}(\boldsymbol{A})$ denote the optimal value of the semidefinite relaxation. Then

$$\mathrm{SDP}(\boldsymbol{A}) \leqslant K_G \cdot \mathrm{OPT}(\boldsymbol{A}),$$

where $\mathrm{OPT}(\boldsymbol{A}) = \max_{u_i, v_j \in \{-1,1\}} \sum_{i,j} a_{ij} u_i v_j$ and K_G is the Grothendieck constant.

Section 3.6 Maximum cut for graphs

Q3.53 ~ Q3.54

Definition 3.6.1 (*Maximum cut*)

For a graph $G = (V, E)$ with edge weights $w_{ij} \geqslant 0$, the maximum cut problem (MAX-CUT) asks to find a partition of the vertices into two sets that maximizes the total weight of edges crossing the partition.

Definition 3.6.2 (*Weighted adjacency matrix*)

The weighted adjacency matrix of a weighted graph $G = (V, E)$ is defined as $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, satisfying the diagonal entries $a_{ii} = 0$, and the off-diagonal entries are

$$a_{ij} = \begin{cases} w_{ij}, & (i,j) \in E, \\ 0, & (i,j) \notin E \end{cases},$$

The total edge weight is denoted by $W = \sum_{i<j} w_{ij} = \frac{1}{2} \sum_{i,j} a_{ij}$.

Proposition 3.6.3 (0.5-*approximation algorithm for maximum cut*)

There is a simple randomized polynomial-time algorithm (*i.e., assign each vertex independently to S or S^c with probability* $1/2$) such that for any weighted graph G with nonnegative weights, the expected weight of the output cut satisfies

$$\mathbb{E}[\mathrm{CUT}(\boldsymbol{x})] = \frac{1}{2} W \geqslant 0.5 \cdot \text{MAX-CUT}(G).$$

Equivalently, let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric matrix satisfying $a_{ii} = 0$. Then there exists $\boldsymbol{u} \in \{-1, 1\}^n$ such that

$$\sum_{i,j} a_{ij} u_i u_j \leqslant 0.$$

Theorem 3.6.4 (0.878-*approximation algorithm for maximum cut*)

There is a polynomial-time algorithm based on SDP relaxation and random hyperplane rounding (*Goemans-Williamson algorithm*) such that for any weighted graph G with nonnegative weights, the expected weight of the output cut satisfies

$$\mathbb{E}[\mathrm{CUT}(\boldsymbol{x})] \geqslant \alpha_{\mathrm{GW}} \cdot \text{MAX-CUT}(G) \geqslant 0.878 \cdot \text{MAX-CUT}(G). \qquad \left(\alpha_{\mathrm{GW}} = \min_{0\leqslant\theta\leqslant\pi} \frac{2}{\pi} \frac{\theta}{1-\cos\theta}\right)$$

Lemma 3.6.5 (*Grothendieck identity*)

Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Then for any $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{S}^{n-1}$,

$$\mathbb{E}[\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}\rangle) \cdot \mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{v}\rangle)] = \frac{2}{\pi} \arcsin(\langle \boldsymbol{u}, \boldsymbol{v}\rangle).$$

Section 3.7 Kernel trick

Q3.55 ~ Q3.58

Definition 3.7.1 (*Tensors*)

A d-tensor $T = (T_{i_1 \cdots i_d})$ is a d-dimensional array of real numbers indexed by $i_1, \cdots, i_d \in \{1, \cdots, n\}$.

Lemma 3.7.4 (*Powers*)

Applying the kernel trick to the power function: $\langle \boldsymbol{x}, \boldsymbol{y}\rangle^d = \langle \boldsymbol{x}^{\otimes d}, \boldsymbol{y}^{\otimes d}\rangle$ where $\boldsymbol{x}^{\otimes d}$ denotes the d-fold tensor product of $\boldsymbol{x}$ with itself, appropriately flattened into a vector.

Lemma 3.7.7 (*Real analytic functions*)

Any real analytic function $f : (-1, 1) \to \mathbb{R}$ with nonnegative Taylor coefficients gives rise to a kernel $K(\boldsymbol{x}, \boldsymbol{y}) = f(\langle \boldsymbol{x}, \boldsymbol{y} \rangle)$ that is positive definite on the unit sphere.

Exercises

1 [**The Thin Shell Phenomenon**] Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector whose entries are independent standardized K-sub-Gaussian random variables. By Theorem T3.1.1 (*Concentration of the norm*), prove that

$$\operatorname{Var}(\|\boldsymbol{X}\|_2) \leqslant CK^4. \qquad (C > 0)$$

Prf. Let $Z = \|\boldsymbol{X}\|_2 - \sqrt{n}$. Using the minimality property of the variance $\operatorname{Var}(Y) \leqslant \mathbb{E}[(Y - a)^2]$, we have

$$\operatorname{Var}(\|\boldsymbol{X}\|_2) \leqslant \mathbb{E}[(\|\boldsymbol{X}\|_2 - \sqrt{n})^2] = \mathbb{E}[Z^2]. \qquad \text{(I)}$$

(We only need to control $\mathbb{E}[Z^2]$.)

By Theorem T3.1.1 (*Concentration of the norm*), for any $t \geqslant 0$,

$$\mathbb{P}\{|Z| \geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{K^4}\right), \qquad (c > 0)$$

Using the tail integral formula for nonnegative variables, we have

$$\mathbb{E}[Z^2] = \int_0^\infty 2t\mathbb{P}\{|Z| \geqslant t\}\mathrm{d}t \leqslant 4\int_0^\infty t\exp\left(-\frac{ct^2}{K^4}\right)\mathrm{d}t = \frac{2}{c}K^4 =: CK^4, \qquad \text{(II)}$$

Substituting (II) into (I), we have

$$\operatorname{Var}(\|\boldsymbol{X}\|_2) \leqslant CK^4.$$

2 [**The Generalized Thin Shell Phenomenon**] Relax the sub-Gaussian assumption from Exercise Q3.1 (*The Thin Shell Phenomenon*) and show that the corresponding variance bound still holds for any distribution with bounded fourth moment. Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector whose entries are independent, satisfying $\mathbb{E}[X_i^2] = 1$, $\mathbb{E}[X_i^4] \leqslant K^4$. Prove that

$$\operatorname{Var}(\|\boldsymbol{X}\|_2) \leqslant K^4, \sqrt{n} - \frac{K^4}{\sqrt{n}} \leqslant \mathbb{E}\|\boldsymbol{X}\|_2 \leqslant \sqrt{n}.$$

Prf. Let $Y = \|\boldsymbol{X}\|_2$, then $\mathbb{E}[Y^2] = \mathbb{E}\left[\sum_{i=1}^n X_i^2\right] = \sum_{i=1}^n 1 = n$.

(1) Using the minimality property of the variance $\operatorname{Var}(Y) \leqslant \mathbb{E}[(Y - a)^2]$, we have

$$\operatorname{Var}(Y) \leqslant \mathbb{E}[(Y - \sqrt{n})^2] = \mathbb{E}\left[\frac{(Y^2 - n)^2}{(Y + \sqrt{n})^2}\right] \leqslant \frac{1}{n}\mathbb{E}[(Y^2 - n)^2] = \frac{1}{n}\operatorname{Var}(Y^2),$$

where

$$\operatorname{Var}(Y^2) = \sum_{i=1}^n \operatorname{Var}(X_i^2) = \sum_{i=1}^n (\mathbb{E}[X_i^4] - (\mathbb{E}[X_i^2])^2) \leqslant \sum_{i=1}^n (K^4 - 1) \leqslant nK^4,$$

substituting it back, we have

$$\operatorname{Var}(Y) \leqslant \mathbb{E}[(Y - \sqrt{n})^2] \leqslant \operatorname{Var}(Y^2) \leqslant \frac{1}{n} \cdot nK^4 = K^4.$$

(2) For the upper bound, by Jensen's inequality, let $\phi(y) = \sqrt{y}$. Then

$$\mathbb{E}[Y] = \mathbb{E}[\sqrt{Y^2}] \leqslant \sqrt{\mathbb{E}[Y^2]} = \sqrt{n}.$$

For the lower bound, by the definition of variance,

$$\operatorname{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = n - (\mathbb{E}[Y])^2 = (\sqrt{n} - \mathbb{E}[Y])(\sqrt{n} + \mathbb{E}[Y]),$$

hence, using the conclusion of (1),

$$(\sqrt{n}-\mathbb{E}[Y])(\sqrt{n}+\mathbb{E}[Y])\leqslant K^4 \implies \sqrt{n}-\mathbb{E}[Y]\leqslant \frac{K^4}{\sqrt{n}+\mathbb{E}[Y]}\leqslant \frac{K^4}{\sqrt{n}}$$
$$\implies \mathbb{E}[Y]\geqslant \sqrt{n}-\frac{K^4}{\sqrt{n}}.$$

3 [**The Reverse Thin Shell Phenomenon**] Prove a reverse estimate of the conclusion of Exercise Q3.2 (*The Generalized Thin Shell Phenomenon*).

Let $\boldsymbol{X}\in\mathbb{R}^n$ be a random vector whose entries are independent, satisfying $\mathbb{E}[X_i^2]=1$, $\mathrm{Var}(X_i^2)>\alpha$ and $\mathbb{E}[X_i^6]\leqslant\beta$.

(1) Prove that when n is sufficiently large (*depending on* α *and* β),

$$\mathrm{Var}(\|\boldsymbol{X}\|_2)\geqslant c\alpha, \mathbb{E}\|\boldsymbol{X}\|_2\leqslant\sqrt{n}-\frac{c\alpha}{\sqrt{n}}. \qquad (c>0)$$

(2) Explain why the lower bound on the variance is an essential assumption.

Solution (1) Let $S=\|\boldsymbol{X}\|_2^2=\sum_{i=1}^n X_i^2$, then

$$\mathbb{E}[S]=\sum_{i=1}^n\mathbb{E}[X_i^2]=n, \mathrm{Var}(S)=\sum_{i=1}^n\mathrm{Var}(X_i^2)>n\alpha;$$

Since $\mathbb{E}[X_i^6]\leqslant\beta$, the higher moments of X_i^2 are bounded, i.e., there exists a constant C_β such that $\mathbb{E}[|X_i^2-1|^3]\leqslant C_\beta$.

- Prove $\mathbb{E}\|\boldsymbol{X}\|_2\leqslant\sqrt{n}-c\alpha/\sqrt{n}$

  Using the Delta method, consider the second-order Taylor expansion of $f(s)=\sqrt{s}$ at $s=n$,

  $$\sqrt{s}=\sqrt{n}+\frac{s-n}{2\sqrt{n}}-\frac{(s-n)^2}{8n^{3/2}}+R_2(s,n),$$

  where

  $$R(s,n)=\frac{1}{16\xi^{5/2}}(s-n)^3, \qquad (\xi \text{ lies between } s \text{ and } n)$$

  Let $s=S$. Taking expectations on both sides,

  $$\mathbb{E}\|\boldsymbol{X}\|_2=\mathbb{E}[\sqrt{S}]=\sqrt{n}+\frac{\mathbb{E}[S-n]}{2\sqrt{n}}-\frac{\mathbb{E}[(S-n)^2]}{8n^{3/2}}+\mathbb{E}[R_2(S,n)]$$
  $$=\sqrt{n}-\frac{\mathrm{Var}(S)}{8n^{3/2}}+\mathbb{E}[R_2(S,n)]$$
  $$>\sqrt{n}-\frac{\alpha}{8\sqrt{n}}+\mathbb{E}[R_2(S,n)],$$

  (It remains to estimate $\mathbb{E}[R(S,n)]$*. Intuitively,*

  $$\mathbb{E}[R(s,n)]\approx\frac{1}{6}f'''(n)\mathbb{E}[(S-n)^3]\sim n^{-5/2}\cdot O(n)=O(n^{-3/2}),$$

  assuming here that $f'''(\xi)\approx f'''(n)$*, i.e.,* s *and* n *are close, but when* s *is close to* 0*,* $f'''(\xi)$ *may blow up to infinity, which is not rigorous.)*

  We make the following claim:

  Claim $\mathbb{E}[R(S,n)]=O(n^{-1})$.

  (The detailed proof is given in the remark.)

  By the claim,

  $$\mathbb{E}\|\boldsymbol{X}\|_2\leqslant\sqrt{n}-\frac{\alpha}{8\sqrt{n}}+o(n^{-1/2})\leqslant\sqrt{n}-\frac{c\alpha}{\sqrt{n}}. \qquad (c>0)$$

- Prove $\mathrm{Var}(\|\boldsymbol{X}\|_2)\geqslant c\alpha$

It is easy to see that

$$\begin{aligned}\mathrm{Var}(\|\boldsymbol{X}\|_2) &= \mathbb{E}\|\boldsymbol{X}\|_2^2 - (\mathbb{E}\|\boldsymbol{X}\|_2)^2 \\ &= n - \left(\sqrt{n} - \frac{\alpha}{8\sqrt{n}} + O(n^{-1})\right)^2 \\ &= n - \left(n - \frac{\alpha}{4} + O(n^{-1})\right) \\ &= \frac{\alpha}{4} + O(n^{-1}) \geqslant c\alpha.\end{aligned}$$

(2) If the condition $\mathrm{Var}(X_i^2) > \alpha$ is omitted, it may happen that X_i^2 is almost surely 1.

Consider the following counterexample: let $X_1, \cdots, X_n$ i.i.d. $\sim$ Rademacher, in which case $\|\boldsymbol{X}\|_2 = \sqrt{\sum_{i=1}^n 1} = \sqrt{n}$, so $\mathrm{Var}(\|\boldsymbol{X}\|_2) = 0$, $\mathbb{E}\|\boldsymbol{X}\|_2 = \sqrt{n}$, i.e., the thickness of the "shell" collapses completely to 0.

Thus, the lower bound α on the variance measures the degree of fluctuation of the coordinate magnitudes. The reverse thin shell bound holds under the premise that the "lengths" of the individual coordinates must exhibit random fluctuations α that do not vanish as the dimension increases.

Rk. Detailed proof that $\mathbb{E}[R(s,n)] = O(n^{-1})$:

Define the "good set" $A = \{|S-n| \leqslant n/2\}$, so that $\xi \in [n/2, 3n/2]$, in which case

$$|f'''(\xi)| \leqslant \frac{3}{8(n/2)^{5/2}} = Cn^{-5/2},$$

hence

$$\mathbb{E}[|R(S,n)|\mathbf{1}_A] \leqslant Cn^{-5/2}\mathbb{E}[|S-n|^3].$$

- On A, note that $S-n$ is a sum of independent zero-mean variables, so by Rosenthal's inequality, let $p=3$. Then

$$\mathbb{E}[|S-n|^3] \leqslant C_3\left(\sum_{i=1}^n \mathbb{E}[|W_i|^3] + \left(\sum_{i=1}^n \mathbb{E}[W_i^2]\right)^{3/2}\right) \leqslant C'(n + n^{3/2}) \leqslant C''n^{3/2};$$

  hence $\mathbb{E}[|R(S,n)|\mathbf{1}_A] \leqslant O(n^{-5/2} \cdot n^{3/2}) = O(n^{-1})$.

- On A^c, by Chebyshev's inequality,

$$\mathbb{P}\{A^c\} = \mathbb{P}\left\{|S-n| > \frac{n}{2}\right\} \leqslant \frac{\mathbb{E}[|S-n|^3]}{(n/2)^3} \leqslant \frac{C''n^{3/2}}{n^3/8} = O(n^{-3/2}),$$

  It remains to verify that the contribution of each term of

$$R(S,n) = \sqrt{S} - \left(\sqrt{n} + \frac{S-n}{2\sqrt{n}} - \frac{(S-n)^2}{8n^{3/2}}\right)$$

  integrated over A^c is $O(n^{-1})$.

  For the second term,

$$\mathbb{E}[\sqrt{n}\mathbf{1}_{A^c}] = \sqrt{n}\mathbb{P}\{A^c\} \leqslant \sqrt{n} \cdot O(n^{-3/2}) = O(n^{-1});$$

  For the third term, using $\mathbb{E}[|W|\mathbf{1}_{\{|W|>\varepsilon\}}] \leqslant \mathbb{E}[|W|^3]/\varepsilon^2$, let $W = S-n, \varepsilon = n/2$. Then

$$\mathbb{E}[|S-n|\mathbf{1}_{A^c}] \leqslant \frac{\mathbb{E}[|S-n|^3]}{(n/2)^2} \sim \frac{n^{3/2}}{n^2} = n^{-1/2},$$

  hence

$$\mathbb{E}\left[\frac{|S-n|}{2\sqrt{n}}\mathbf{1}_{A^c}\right] = \frac{1}{\sqrt{n}} \cdot O(n^{-1/2}) = O(n^{-1});$$

  For the fourth term,

$$\mathbb{E}\left[\frac{(S-n)^2}{8n^{3/2}}\mathbf{1}_{A^c}\right] \leqslant \frac{1}{n^{3/2}}\frac{\mathbb{E}[|S-n|^3]}{n/2} \sim \frac{n^{3/2}}{n^{5/2}} = O(n^{-1});$$

For the first term, one cannot directly use the Taylor expansion (S may be close to 0). Split A^c further into $\{S < n/2\} \sqcup \{S > 3n/2\}$:

a. When $S < n/2$,

$$\mathbb{E}[\sqrt{S}\mathbf{1}_{\{S<n/2\}}] \leqslant \sqrt{n}\mathbb{P}\{A^c\} = O(n^{-1}).$$

b. When $S > 3n/2$, using the linear upper bound (i.e., the tangent line of $f(s) = \sqrt{s}$ at $s = n$)

$$\sqrt{s} \leqslant \frac{s+n}{2\sqrt{n}},$$

we obtain

$$\mathbb{E}[\sqrt{S}\mathbf{1}_{\{S>3n/2\}}] \leqslant \mathbb{E}\left[\frac{S+n}{2\sqrt{n}}\mathbf{1}_{\{S>3n/2\}}\right] = \frac{1}{2\sqrt{n}}(+n),$$

where

$$\mathbb{E}[S\mathbf{1}_{A^c}] = \mathbb{E}[(n+S-n)\mathbf{1}_{A^c}] = O(n\cdot n^{-3/2} + n^{-1/2}) = O(n^{-1/2}),$$

hence

$$\mathbb{E}[\sqrt{S}\mathbf{1}_{\{S>3n/2\}}] = \frac{1}{\sqrt{n}}\cdot O(n^{-1/2}) = O(n^{-1}).$$

(Supplementary remark on Cauchy's inequality: if higher-order moments exist (e.g., $\mathbb{E}[X_i^8]$, i.e., $\mathbb{E}[S^4]$), one can directly use Cauchy's inequality,

$$\mathbb{E}[\sqrt{S}\mathbf{1}_{A^c}] \leqslant \sqrt{\mathbb{E}[S]}\sqrt{\mathbb{P}\{A^c\}} = \sqrt{n}\cdot\sqrt{\mathbb{P}\{A^c\}},$$

but if only the third moment exists, $\sqrt{\mathbb{P}\{A^c\}} \sim \sqrt{n^{-3/2}} = n^{-3/4}$, in which case $\mathbb{E}[\sqrt{S}\mathbf{1}_{A^c}] = O(n^{-1/4})$, which is insufficient as a remainder term.)

4 [**PCA Maximizes Explained Variance**] Let $\boldsymbol{X} \in \mathbb{R}^n$ be a mean-zero random vector, and denote by $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_n$ the eigenvalues of $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]$, with corresponding orthonormal eigenvectors $\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n$.

(1) Prove that the orthogonal projection operator $\boldsymbol{P}_k$ onto the subspace $\operatorname{span}\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_k\}$ satisfies

$$\mathbb{E}\|\boldsymbol{P}_k\boldsymbol{X}\|_2^2 = \sum_{i=1}^k \operatorname{Var}(\langle \boldsymbol{X}, \boldsymbol{v}_i\rangle) = \sum_{i=1}^k \lambda_i.$$

(2) Prove that for every $\boldsymbol{P} \in \mathbb{R}^n$ with $\operatorname{rank}(\boldsymbol{P}) = k$,

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{X}\|_2^2 \leqslant \mathbb{E}\|\boldsymbol{P}_k\boldsymbol{X}\|_2^2.$$

Consequently, PCA maximizes the explained variance of the data, with the proportion of explained variance given by s_k/s_n, where $s_k = \sum_{i=1}^k \lambda_i$.

Prf. (1) Since $\boldsymbol{v}_1, \cdots, \boldsymbol{v}_k$ form an orthonormal basis,

$$\boldsymbol{P}_k\boldsymbol{X} = \sum_{i=1}^k \langle \boldsymbol{X}, \boldsymbol{v}_i\rangle \boldsymbol{v}_i,$$

by the orthogonal decomposition property of the norm, we have

$$\|\boldsymbol{P}_k\boldsymbol{X}\|_2^2 = \sum_{i=1}^k \langle \boldsymbol{X}, \boldsymbol{v}_i\rangle^2.$$

Taking expectations on both sides,

$$\mathbb{E}\|\boldsymbol{P}_k\boldsymbol{X}\|_2^2 = \sum_{i=1}^k \mathbb{E}[\langle \boldsymbol{X}, \boldsymbol{v}_i\rangle^2].$$

Since $\mathbb{E}[\boldsymbol{X}] = \boldsymbol{0}$, we have

$$\mathbb{E}[\langle \boldsymbol{X}, \boldsymbol{v}_i\rangle] = \langle \mathbb{E}[\boldsymbol{X}], \boldsymbol{v}_i\rangle = 0,$$

hence

$$\begin{aligned}\mathbb{E}[\langle \boldsymbol{X}, \boldsymbol{v}_i\rangle^2] &= \operatorname{Var}(\langle \boldsymbol{X}, \boldsymbol{v}_i\rangle)\\ &= \boldsymbol{v}_i^\top \boldsymbol{\Sigma} \boldsymbol{v}_i = \boldsymbol{v}_i^\top(\lambda_i \boldsymbol{v}_i) = \lambda_i \|\boldsymbol{v}_i\|_2^2 = \lambda_i,\end{aligned}$$

and therefore

$$\mathbb{E}\|\boldsymbol{P}_k \boldsymbol{X}\|_2^2 = \sum_{i=1}^k \operatorname{Var}(\langle \boldsymbol{X}, \boldsymbol{v}_i\rangle) = \sum_{i=1}^k \lambda_i.$$

(2) Let $\boldsymbol{P} \in \mathbb{R}^n$ be an orthogonal projection operator with $\operatorname{rank}(\boldsymbol{P}) = k$. Then there exist orthonormal vectors $\boldsymbol{u}_1, \cdots, \boldsymbol{u}_k$ such that

$$\boldsymbol{P}\boldsymbol{X} = \sum_{j=1}^k \langle \boldsymbol{X}, \boldsymbol{u}_j\rangle \boldsymbol{u}_j,$$

hence, similarly to the conclusion of (1), the expected squared norm of the projection is

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{X}\|_2^2 = \sum_{j=1}^k \boldsymbol{u}_j^\top \boldsymbol{\Sigma} \boldsymbol{u}_j,$$

i.e.,

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{X}\|_2^2 = \operatorname{tr}(\boldsymbol{U}^\top \boldsymbol{\Sigma} \boldsymbol{U}).$$

By the variational characterization of eigenvalues (*i.e., the Poincaré separation theorem or Ky Fan trace theorem*), for a symmetric positive semidefinite matrix $\boldsymbol{\Sigma}$ the sum of its k largest eigenvalues satisfies

$$\sum_{i=1}^k \lambda_i = \max_{\boldsymbol{U}^\top \boldsymbol{U} = \boldsymbol{I}_k} \operatorname{tr}(\boldsymbol{U}^\top \boldsymbol{\Sigma} \boldsymbol{U}),$$

hence

$$\operatorname{tr}(\boldsymbol{U}^\top \boldsymbol{\Sigma} \boldsymbol{U}) \leqslant \sum_{i=1}^k \lambda_i,$$

and by the conclusion of (1),

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{X}\|_2^2 \leqslant \mathbb{E}\|\boldsymbol{P}_k \boldsymbol{X}\|_2^2.$$

This shows that the projection space chosen by PCA (*spanned by the first k principal eigenvectors*) retains the maximum possible variance of the data (*i.e., explained variance*).

5 [**Expected ℓ^p Norm of a Random Vector**] Let $p \in [1, \infty]$. Let $\boldsymbol{X} = (X_1, \cdots, X_n) \in \mathbb{R}^n$ be a random vector whose coordinates X_i are independent sub-Gaussian random variables. Prove that

$$\mathbb{E}\|\boldsymbol{X}\|_p \leqslant \begin{cases} CK\sqrt{p} n^{1/p}, & p \leqslant \log n, \\ CK\sqrt{\log n}, & p \geqslant \log n \end{cases}.$$

Prf. Since $\|X_i\|_{\psi_2} \leqslant K$, for any $q \geqslant 1$,

$$\|X_i\|_{L_q} = (\mathbb{E}[|X_i|^q])^{1/q} \leqslant C_1 K \sqrt{q},$$

hence $\mathbb{E}[|X_i|^q] \leqslant (C_1 K\sqrt{q})^q$.

We discuss the values of p:

a. If $p \leqslant \log n$, then by Jensen's inequality, let $\phi(x) = x^{1/p}$. Then

$$\begin{aligned}\mathbb{E}\|\boldsymbol{X}\|_p = \mathbb{E}\left[\left(\sum_{i=1}^n |X_i|^p\right)^{1/p}\right] &\leqslant ()^{1/p} = \left(\sum_{i=1}^n \mathbb{E}[|X_i|^p]\right)^{1/p}\\ &\leqslant \left(\sum_{i=1}^n (C_1 K\sqrt{p})^p\right)^{1/p} = (n(C_1 K\sqrt{p})^p)^{1/p} = C_1 K\sqrt{p} n^{1/p}.\end{aligned}$$

b. If $p \geqslant \log n$, then

$$\begin{aligned}\mathbb{E}\|\boldsymbol{X}\|_p &\leqslant \mathbb{E}\|\boldsymbol{X}\|_{\log n} \\ &\leqslant C_1 K\sqrt{\log n} n^{1/\log n} = \mathrm{e}C_1 K\sqrt{\log n} = CK\sqrt{\log n}. \qquad (C, C_1 > 0, C = \mathrm{e}C_1)\end{aligned}$$

6 [**Lower Bound for the Expected ℓ^p Norm of a Gaussian Vector**] Show that the bound in Exercise Q3.5 (*Expected ℓ^p Norm of a Random Vector*) is reversible for standard Gaussian random vectors.

Let $\boldsymbol{X} = (X_1, \cdots, X_n) \in \mathbb{R}^n$ be a random vector where $X_1, \cdots, X_n$ i.i.d. $\sim N(0,1)$. Prove that for any $p \in [1, \infty]$,

$$\mathbb{E}\|\boldsymbol{X}\|_p \geqslant \begin{cases} c\sqrt{p} n^{1/p}, & p \leqslant \log n, \\ c\sqrt{\log n}, & p \geqslant \log n \end{cases}.$$

Rk. Recall Exercise Q2.22 (*Absolute Moments of the Normal Distribution*). For $X \sim N(0,1)$,

$$\|X\|_{L_p} = (\mathbb{E}[|X|^p])^{1/p} = \left(\frac{2^{p/2}\Gamma((p+1)/2)}{\sqrt{\pi}}\right)^{1/p} \sim (c_1\sqrt{p})^p. \qquad (c_1 > 0)$$

Prf. We discuss the values of p:

a. If $p \leqslant \log n$, let $Z = \|\boldsymbol{X}\|_p^p = \sum_{i=1}^n |X_i|^p$, then

$$\mathbb{E}[Z] = \mathbb{E}\left[\sum_{i=1}^n |X_i|^p\right] = \sum_{i=1}^n \mathbb{E}[|X_i|^p] = n\mathbb{E}[|X|^p],$$

$$(\mathbb{E}[Z])^{1/p} = (n\mathbb{E}[|X|^p])^{1/p} = n^{1/p} \cdot (\mathbb{E}[|X|^p])^{1/p} = n^{1/p}\|X\|_{L_p} \geqslant c_1\sqrt{p} n^{1/p},$$

hence it suffices to prove $\mathbb{E}[Z^{1/p}] \geqslant c(\mathbb{E}[Z])^{1/p}$.

(*Jensen's inequality only gives an upper bound for $\mathbb{E}[Z^{1/p}]$; to reverse Jensen's inequality for the concave function, one must prove that Z satisfies anti-concentration, i.e., Z does not collapse near 0.*

The most common method is indicator truncation,

$$\mathbb{E}[Z^{1/p}] \geqslant \mathbb{E}[Z^{1/p} \cdot \mathbf{1}_{\{Z \geqslant \theta\mathbb{E}[Z]\}}] \geqslant (\theta\mathbb{E}[Z])^{1/p} \cdot \mathbb{P}\{Z \geqslant \theta\mathbb{E}[Z]\},$$

where $(\mathbb{E}[Z])^{1/p}$ is known and $\mathbb{P}\{Z \geqslant \theta\mathbb{E}[Z]\}$ is unknown; it remains to estimate it using the Paley-Zygmund inequality (the second moment method).)

The second moment ratio of Z is

$$\frac{\mathbb{E}[Z^2]}{(\mathbb{E}[Z])^2} = 1 + \frac{1}{n}\frac{\mathrm{Var}(|X|^p)}{(\mathbb{E}[|X|^p])^2} = 1 + \frac{1}{n}\frac{n(\mathbb{E}[|X|^{2p}] - (\mathbb{E}[|X|^p])^2)}{(\mathbb{E}[|X|^p])^2} \leqslant 1 + \frac{1}{n}\frac{\mathbb{E}[|X|^{2p}]}{(\mathbb{E}[|X|^p])^2},$$

where

$$\frac{\mathbb{E}[|X|^{2p}]}{(\mathbb{E}[|X|^p])^2} \sim \frac{\sqrt{2}(2p/\mathrm{e})^p}{(\sqrt{2}(p/\mathrm{e})^{p/2})^2} = \frac{1}{\sqrt{2}}2^p \leqslant \frac{1}{\sqrt{2}}2^{\log n} \leqslant \frac{1}{\sqrt{2}}n^{\log 2},$$

Substituting this in, we obtain

$$\frac{\mathbb{E}[Z^2]}{(\mathbb{E}[Z])^2}\sqrt{2}n^{\log 2 - 1} \leqslant 1 + \sqrt{2} =: C_0 \implies \mathbb{E}[Z^2] \leqslant C_0(\mathbb{E}[Z])^2,$$

hence, by the Paley-Zygmund inequality, let $\theta = 1/2$. Then

$$\mathbb{P}\left\{Z > \frac{1}{2}\mathbb{E}[Z]\right\} \geqslant \frac{(1-1/2)^2}{C_0} = \frac{1}{4C_0} =: \delta,$$

hence

$$\begin{aligned}\mathbb{E}[Z^{1/p}] &\geqslant \mathbb{E}\left[Z^{1/p} \cdot \mathbf{1}_{\{Z > \frac{1}{2}\mathbb{E}[Z]\}}\right] \geqslant \left(\frac{1}{2}\mathbb{E}[Z]\right)^{1/p} \cdot \mathbb{P}\left\{Z > \frac{1}{2}\mathbb{E}[Z]\right\} \\ &\geqslant \frac{1}{2^{1/p}} \cdot \delta \cdot (\mathbb{E}[Z])^{1/p} \geqslant \frac{1}{2} \cdot \delta \cdot (n(c_1\sqrt{p})^p)^{1/p} \\ &= \frac{\delta c_1}{2}\sqrt{p} n^{1/p} = C\sqrt{p} n^{1/p}. \qquad \left(C = \frac{\delta c_1}{2}\right)\end{aligned}$$

b. If $p \geqslant \log n$, then

$$\|\boldsymbol{X}\|_p \geqslant \|\boldsymbol{X}\|_\infty = \max_{1\leqslant i\leqslant n} |X_i|,$$

Taking expectations on both sides,

$$\mathbb{E}\|\boldsymbol{X}\|_p \geqslant \mathbb{E}\left[\max_{1\leqslant i\leqslant n} |X_i|\right],$$

where, by Exercise Q2.38 (*Maximum of Gaussian Random Variables*), $\mathbb{E}[\max_i X_i] \sim \sqrt{2\log n}$. Substituting this in,

$$\mathbb{E}\|\boldsymbol{X}\|_p \geqslant c_2\sqrt{\log n}. \qquad (c_2 > 0)$$

7 [**Small Ball Probabilities**] Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector whose entries follow independent continuous distributions, satisfying $\sup f_i(x) \leqslant K$.

(1) Prove that $\boldsymbol{X}$ falls into a ball of radius $o(\sqrt{n})$ with extremely low probability. Specifically, for any $\boldsymbol{a} \in \mathbb{R}^n$,

$$\mathbb{P}\{\|\boldsymbol{X}-\boldsymbol{a}\|_2 \leqslant \varepsilon\sqrt{n}\} \leqslant (\sqrt{2\pi\mathrm{e}}K\varepsilon)^n.$$

(2) Using the conclusion of (1), prove that when $n \geqslant 2$,

$$\mathbb{E}\left[\frac{1}{\|\boldsymbol{X}\|_2}\right] \leqslant \frac{CK}{\sqrt{n}}.$$

Then explain why the above expectation may diverge to infinity when $n = 1$.

Prf. (1) It is easy to see that

$$f(\boldsymbol{x}) = \prod_{i=1}^n f_i(x_i) \leqslant \prod_{i=1}^n K = K^n.$$

The probability that $\boldsymbol{X}$ falls inside $B(\boldsymbol{a}, r)$ equals the integral of the density function over the ball, i.e.,

$$\mathbb{P}\{\|\boldsymbol{X}-\boldsymbol{a}\|_2 \leqslant r\} = \int_{B(\boldsymbol{a},r)} f(\boldsymbol{x})\mathrm{d}\boldsymbol{x} \leqslant \int_{B(\boldsymbol{a},r)} K^n \mathrm{d}\boldsymbol{x} = K^n \cdot \mathrm{Vol}(B(\boldsymbol{a},r)),$$

where

$$\mathrm{Vol}(B(\boldsymbol{a},r)) = \frac{\pi^{n/2}r^n}{\Gamma(n/2+1)} \leqslant \frac{\pi^{n/2}(\varepsilon\sqrt{n})^n}{(n/2\mathrm{e})^{n/2}} = \frac{\pi^{n/2}\varepsilon^n n^{n/2}}{n^{n/2}(2\mathrm{e})^{-n/2}} = (2\pi\mathrm{e})^{n/2}\varepsilon^n = (\sqrt{2\pi\mathrm{e}}\varepsilon)^n,$$

Substituting this in, we obtain

$$\mathbb{P}\{\|\boldsymbol{X}-\boldsymbol{a}\|_2 \leqslant \varepsilon\sqrt{n}\} \leqslant K^n(\sqrt{2\pi\mathrm{e}}\varepsilon)^n = (\sqrt{2\pi\mathrm{e}}K\varepsilon)^n.$$

(2) Let $t_0 = \sqrt{2\pi\mathrm{e}}K/\sqrt{n}$. Splitting the integral at t_0,

$$\begin{aligned}\mathbb{E}\left[\frac{1}{\|\boldsymbol{X}\|_2}\right] &= \int_0^\infty \mathbb{P}\left\{\|\boldsymbol{X}\|_2 \leqslant \frac{1}{t}\right\}\mathrm{d}t \\ &= \underbrace{\int_0^{t_0} \mathbb{P}\left\{\|\boldsymbol{X}\|_2 \leqslant \frac{1}{t}\right\}\mathrm{d}t}_{I_1} + \underbrace{\int_{t_0}^\infty \mathbb{P}\left\{\|\boldsymbol{X}\|_2 \leqslant \frac{1}{t}\right\}\mathrm{d}t}_{I_2},\end{aligned}$$

For I_1,

$$I_1 \leqslant \int_0^{t_0} 1\mathrm{d}t = t_0 = \frac{\sqrt{2\pi\mathrm{e}}K}{\sqrt{n}};$$

For I_2, let $1/t = \varepsilon\sqrt{n}$, i.e., $\varepsilon = 1/t\sqrt{n}$. Then by the conclusion of (1), let $\boldsymbol{a} = \boldsymbol{0}$. Then

$$\mathbb{P}\left\{\|\boldsymbol{X}\|_2 \leqslant \frac{1}{t}\right\} \leqslant \left(\sqrt{2\pi\mathrm{e}}K \cdot \frac{1}{t\sqrt{n}}\right)^n = \left(\frac{t_0}{t}\right)^n,$$

Substituting this in, we obtain

$$I_2 \leqslant \int_{t_0}^\infty \left(\frac{t_0}{t}\right)^n \mathrm{d}t = t_0^n\left[\frac{t^{-n+1}}{-n+1}\right]_{t_0}^\infty = \frac{t_0^n \cdot t_0^{-n+1}}{n-1} = \frac{t_0}{n-1} \leqslant t_0. \qquad (n \geqslant 2)$$

Thus,

$$\mathbb{E}\left[\frac{1}{\|\boldsymbol{X}\|_2}\right] \leqslant t_0 + \frac{t_0}{n-1} \leqslant 2t_0 = \frac{2\sqrt{2\pi\mathrm{e}}K}{\sqrt{n}} = \frac{CK}{\sqrt{n}}. \qquad (C = 2\sqrt{2\pi\mathrm{e}})$$

When $n = 1$, $\|\boldsymbol{X}\|_2 = |X_1|$, in which case

$$\mathbb{E}\left[\frac{1}{|X_1|}\right] = \int_{-\infty}^{\infty} \frac{1}{|x|} f_1(x)\mathrm{d}x,$$

If $f_1(x)$ is nonzero near the origin, the integrand behaves like $1/|x|$, and the above integral will diverge.

8 [**Expectation of an Isotropic Random Vector**]

(1) Prove that if $\boldsymbol{X} \in \mathbb{R}^n$ is isotropic, then $\|\mathbb{E}[\boldsymbol{X}]\|_2 \leqslant 1$.

(2) Prove that the above upper bound is sharp in general, i.e., there exists an isotropic random vector $\boldsymbol{X} \in \mathbb{R}^n$ such that $\|\mathbb{E}[\boldsymbol{X}]\|_2 = 1$.

Prf. (1) It is easy to see that

$$\mathrm{Cov}(\boldsymbol{X}) = \mathbb{E}[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^\top] = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] - \mathbb{E}[\boldsymbol{X}]\mathbb{E}[\boldsymbol{X}]^\top = \boldsymbol{I}_n - \boldsymbol{\mu}\boldsymbol{\mu}^\top,$$

therefore for any $\boldsymbol{u} \in \mathbb{R}^n$,

$$\boldsymbol{u}^\top \mathrm{Cov}(\boldsymbol{X})\boldsymbol{u} = \boldsymbol{u}^\top(\boldsymbol{I}_n - \boldsymbol{\mu}\boldsymbol{\mu}^\top)\boldsymbol{u} = \boldsymbol{u}^\top\boldsymbol{u} - (\boldsymbol{u}^\top\boldsymbol{\mu})(\boldsymbol{\mu}^\top\boldsymbol{u}) = \|\boldsymbol{u}\|_2^2 - \langle\boldsymbol{u},\boldsymbol{\mu}\rangle^2 \geqslant 0.$$

When $\boldsymbol{\mu} \neq \boldsymbol{0}$, let $\boldsymbol{u} = \boldsymbol{\mu}$. Then

$$\begin{aligned}\|\boldsymbol{\mu}\|_2^2 - \langle\boldsymbol{\mu},\boldsymbol{\mu}\rangle^2 = \|\boldsymbol{\mu}\|_2^2 - \|\boldsymbol{\mu}\|_2^4 \geqslant 0 &\implies \|\boldsymbol{\mu}\|_2^2(1-\|\boldsymbol{\mu}\|_2^2) \geqslant 0\\ &\implies \|\boldsymbol{\mu}\|_2^2 \leqslant 1,\end{aligned}$$

Taking square roots on both sides yields $\|\mathbb{E}[\boldsymbol{X}]\|_2 \leqslant 1$.

(2) Let $\boldsymbol{X} = (X_1, \cdots, X_n)$, where $X_1 = 1$ (a.s.), $X_2, \cdots, X_n$ i.i.d. $\sim N(0,1)$ (*or let* $X_2, \cdots, X_n$ i.i.d. $\sim$ Rademacher). Then

$$\mathbb{E}[\boldsymbol{X}] = (1, 0, \cdots, 0)^\top, \|\mathbb{E}[\boldsymbol{X}]\|_2 = \sqrt{1^2 + 0 + \cdots + 0} = 1,$$

It remains to verify the isotropy condition $\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \boldsymbol{I}_n$.

a. For the diagonal entries, when $i = 1$, $\mathbb{E}[X_1^2] = 1^2 = 1$; when $i \geqslant 2$, $\mathbb{E}[X_i^2] = \mathrm{Var}(X_i) + (\mathbb{E}[X_i])^2 = 1 + 0^2 = 1$.

b. For the off-diagonal entries, when $i \geqslant 2$, $\mathbb{E}[X_1X_i] = \mathbb{E}[1 \cdot X_i] = \mathbb{E}[X_i] = 0$; when $i, j \geqslant 2$ and $i \neq j$, $\mathbb{E}[X_iX_j] = \mathbb{E}[X_i]\mathbb{E}[X_j] = 0 \cdot 0 = 0$.

Thus, $\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \boldsymbol{I}_n$.

9 [**One-Dimensional Marginals of an Isotropic Distribution**] Let $\boldsymbol{X} \in \mathbb{R}^n$ be an isotropic random vector. For any $\boldsymbol{u} \in \mathbb{R}^n$, consider the random variable $X_{\boldsymbol{u}} = \langle\boldsymbol{X},\boldsymbol{u}\rangle$. Prove that

$$\mathbb{E}[X_{\boldsymbol{u}}X_{\boldsymbol{v}}] = \langle\boldsymbol{u},\boldsymbol{v}\rangle, \|X_{\boldsymbol{u}} - X_{\boldsymbol{v}}\|_{L^2} = \|\boldsymbol{u}-\boldsymbol{v}\|_2.$$

Prf. (1) It is easy to see that

$$X_{\boldsymbol{u}}X_{\boldsymbol{v}} = (\boldsymbol{u}^\top\boldsymbol{X})(\boldsymbol{v}^\top\boldsymbol{X}) = \boldsymbol{u}^\top\boldsymbol{X}\boldsymbol{X}^\top\boldsymbol{v},$$

Taking expectations on both sides,

$$\mathbb{E}[X_{\boldsymbol{u}}X_{\boldsymbol{v}}] = \mathbb{E}[\boldsymbol{u}^\top\boldsymbol{X}\boldsymbol{X}^\top\boldsymbol{v}] = \boldsymbol{u}^\top\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]\boldsymbol{v} = \boldsymbol{u}^\top\boldsymbol{v} = \langle\boldsymbol{u},\boldsymbol{v}\rangle.$$

(2) It is easy to see that

$$X_{\boldsymbol{u}} - X_{\boldsymbol{v}} = \langle\boldsymbol{X},\boldsymbol{u}\rangle - \langle\boldsymbol{X},\boldsymbol{v}\rangle = \langle\boldsymbol{X},\boldsymbol{u}-\boldsymbol{v}\rangle = X_{\boldsymbol{u}-\boldsymbol{v}} = X_{\boldsymbol{w}}, \qquad (\boldsymbol{w} = \boldsymbol{u}-\boldsymbol{v})$$

Using the conclusion of (1), let $\boldsymbol{u} = \boldsymbol{v} = \boldsymbol{w}$. Then

$$\mathbb{E}[X_{\boldsymbol{w}}^2] = \mathbb{E}[X_{\boldsymbol{w}}X_{\boldsymbol{w}}] = \langle\boldsymbol{w},\boldsymbol{w}\rangle = \|\boldsymbol{w}\|_2^2,$$

therefore

$$\|X_{\boldsymbol{u}} - X_{\boldsymbol{v}}\|_{L^2}^2 = \mathbb{E}[(X_{\boldsymbol{u}} - X_{\boldsymbol{v}})^2] = \mathbb{E}[X_{\boldsymbol{u}-\boldsymbol{v}}^2] = \|\boldsymbol{u}-\boldsymbol{v}\|_2^2,$$

Taking square roots on both sides yields $\|X_{\boldsymbol{u}} - X_{\boldsymbol{v}}\|_{L^2} = \|\boldsymbol{u} - \boldsymbol{v}\|_2$.

10 [**Standardizing a Random Vector**]
(1) Let $\boldsymbol{\mu} \in \mathbb{R}^n$, $\boldsymbol{\Sigma} \in \mathbb{R}^{n\times n}$ be a positive semidefinite matrix, and $\boldsymbol{Z} \in \mathbb{R}^n$ be an isotropic standardized random vector. Prove that $\boldsymbol{X} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\boldsymbol{Z}$ satisfies

$$\mathbb{E}[\boldsymbol{X}] = \boldsymbol{\mu}, \mathrm{Cov}(\boldsymbol{X}) = \boldsymbol{\Sigma}.$$

(2) Conversely, let $\boldsymbol{X}$ satisfy $\mathbb{E}[\boldsymbol{X}] = \boldsymbol{\mu}$, $\mathrm{Cov}(\boldsymbol{X}) = \boldsymbol{\Sigma}$. Construct an isotropic random vector $\boldsymbol{Z}$ satisfying $\mathbb{E}[\boldsymbol{Z}] = 0$ such that $\boldsymbol{X} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\boldsymbol{Z}$ (a.s.).

Prf. (1) For the expectation,

$$\mathbb{E}[\boldsymbol{X}] = \mathbb{E}[\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\boldsymbol{Z}] = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbb{E}[\boldsymbol{Z}] = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\cdot\boldsymbol{0} = \boldsymbol{\mu}.$$

For the covariance matrix,

$$\begin{aligned}\mathrm{Cov}(\boldsymbol{X}) &= \mathbb{E}[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^\top] = \mathbb{E}[(\boldsymbol{\Sigma}^{1/2}\boldsymbol{Z})(\boldsymbol{\Sigma}^{1/2}\boldsymbol{Z})^\top] = \mathbb{E}[\boldsymbol{\Sigma}^{1/2}\boldsymbol{Z}\boldsymbol{Z}^\top(\boldsymbol{\Sigma}^{1/2})^\top] \\ &= \boldsymbol{\Sigma}^{1/2}\mathbb{E}[\boldsymbol{Z}\boldsymbol{Z}^\top]\boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{I}\boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}.\end{aligned}$$

(2) Let $\mathrm{rank}(\boldsymbol{\Sigma}) = r$. The eigendecomposition of $\boldsymbol{\Sigma}$ is

$$\boldsymbol{\Sigma} = \sum_{i=1}^n \lambda_i \boldsymbol{u}_i\boldsymbol{u}_i^\top,$$

where $\lambda_1 \geqslant \cdots \geqslant \lambda_r > \lambda_{r+1} = \cdots = \lambda_n = 0$, and $\{\boldsymbol{u}_i\}_{i=1}^n$ are orthonormal eigenvectors. Then

$$\boldsymbol{\Sigma}^{1/2} = \sum_{i=1}^r \sqrt{\lambda_i}\boldsymbol{u}_i\boldsymbol{u}_i^\top.$$

Note that $\boldsymbol{X} - \boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$. We perform the following construction:

i. First construct the main component in $\mathcal{C}(\boldsymbol{\Sigma})$,

$$\boldsymbol{Z}_0 = \sum_{i=1}^r \frac{1}{\sqrt{\lambda_i}}\langle \boldsymbol{X}-\boldsymbol{\mu}, \boldsymbol{u}_i\rangle \boldsymbol{u}_i,$$

then $\mathbb{E}[\boldsymbol{Z}_0] = \boldsymbol{0}$, $\mathrm{Cov}(\boldsymbol{Z}_0) = \sum_{i=1}^r \boldsymbol{u}_i\boldsymbol{u}_i^\top$ (*i.e., the projection matrix onto the subspace*).

ii. Then add independent isotropic white noise in $\ker(\boldsymbol{\Sigma})$. Introduce a standard random vector $\boldsymbol{W} = (W_{r+1}, \cdots, W_n)^\top$ independent of $\boldsymbol{X}$, satisfying $\mathbb{E}[\boldsymbol{W}] = \boldsymbol{0}$, $\mathrm{Cov}(\boldsymbol{W}) = \boldsymbol{I}_{n-r}$ (*e.g., independent standard normal*), and let

$$\boldsymbol{Z} = \boldsymbol{Z}_0 + \sum_{i=r+1}^n W_i\boldsymbol{u}_i,$$

then

$$\mathbb{E}[\boldsymbol{Z}] = \boldsymbol{0}, \mathrm{Cov}(\boldsymbol{Z}) = \sum_{i=1}^r \boldsymbol{u}_i\boldsymbol{u}_i^\top + \sum_{i=r+1}^n \boldsymbol{u}_i\boldsymbol{u}_i^\top = \boldsymbol{I}_n,$$

hence $\boldsymbol{Z}$ is isotropic.

Since $\boldsymbol{u}_i \in \ker(\boldsymbol{\Sigma}^{1/2})$ for $i > r$,

$$\boldsymbol{\Sigma}^{1/2}\boldsymbol{Z} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{Z}_0 + \boldsymbol{0} = \sum_{j=1}^r \sqrt{\lambda_j}\boldsymbol{u}_j\boldsymbol{u}_j^\top \left(\sum_{i=1}^r \frac{1}{\sqrt{\lambda_i}}\langle \boldsymbol{X}-\boldsymbol{\mu}, \boldsymbol{u}_i\rangle \boldsymbol{u}_i\right) = \sum_{i=1}^r \langle \boldsymbol{X}-\boldsymbol{\mu}, \boldsymbol{u}_i\rangle \boldsymbol{u}_i = \boldsymbol{X}-\boldsymbol{\mu} \text{ (a.s.)},$$

i.e., $\boldsymbol{X} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\boldsymbol{Z}$ (a.s.).

11 [**Random Permutation Vector**] Let $\boldsymbol{x} \in \mathbb{R}^n$ satisfy $\sum_{i=1}^n x_i = 0$, $\sum_{i=1}^n x_i^2 = 1$, and let $\boldsymbol{X}$ be the random vector obtained by randomly permuting the coordinates of $\boldsymbol{x}$, i.e., $\boldsymbol{X} = (X_1, \cdots, X_n) = (x_{\sigma(1)}, \cdots, x_{\sigma(n)})$, where σ is a random permutation of $\{1, \cdots, n\}$.

(1) Find $\mathbb{E}[\boldsymbol{X}], \mathrm{Cov}(\boldsymbol{X})$.

(2) Explain why the coordinates of $\boldsymbol{X}$ are negatively correlated.

Sol. (1) The permutation σ is chosen uniformly at random, i.e., $\mathbb{P}\{\sigma(i)=j\}=1/n$. Therefore

$$\mathbb{E}[X_i]=\sum_{j=1}^{n}\mathbb{P}\{\sigma(i)=j\}x_j=\frac{1}{n}\sum_{j=1}^{n}x_j=0,$$

i.e., $\mathbb{E}[\boldsymbol{X}]=\mathbf{0}$, hence

$$\mathrm{Var}(X_i)=\mathbb{E}[X_i^2]=\sum_{j=1}^{n}\mathbb{P}\{\sigma(i)=j\}x_j^2=\sum_{j=1}^{n}\frac{1}{n}x_j^2=\frac{1}{n}\sum_{j=1}^{n}x_j^2=\frac{1}{n}.$$

Note that $\boldsymbol{X}$ is a permutation of $\boldsymbol{x}$, satisfying $\sum_{k=1}^{n}X_k=\sum_{k=1}^{n}x_k=0$, therefore

$$0=\mathrm{Var}\left(\sum_{k=1}^{n}X_k\right)=\sum_{k=1}^{n}\mathrm{Var}(X_k)+\sum_{k\neq l}\mathrm{Cov}(X_k,X_l),$$

By symmetry, all off-diagonal covariances are equal, denote $\mathrm{Cov}(X_i,X_j)=\rho$. Substituting this in, we obtain

$$0=n\cdot\frac{1}{n}+n(n-1)\rho\implies\rho=-\frac{1}{n(n-1)},$$

i.e.,

$$\mathrm{Cov}(\boldsymbol{X})=\frac{1}{n-1}\left(\boldsymbol{I}_n-\frac{1}{n}\mathbf{1}\mathbf{1}^\top\right).$$

(2) Subject to the constraint $\sum_{i=1}^{n}X_i=0$, if one of the coordinates X_i takes a large positive value, the remaining coordinates $\{X_k\}_{k\neq i}$ must jointly bear the corresponding negative value, manifesting as negative correlation. In high-dimensional space, this constraint means that the random point is restricted to the hyperplane perpendicular to the all-ones vector, causing the coordinates to be unable to fluctuate completely independently.

12 [**Distance Between Independent Isotropic Random Vectors**] Let $\boldsymbol{X},\boldsymbol{Y}\in\mathbb{R}^n$ be independent isotropic random vectors satisfying $\mathbb{E}[\boldsymbol{X}]=\mathbb{E}[\boldsymbol{Y}]=\mathbf{0}$. Prove that

$$\mathbb{E}\|\boldsymbol{X}-\boldsymbol{Y}\|_2^2=2n.$$

Prf. Expanding the Euclidean norm,

$$\|\boldsymbol{X}-\boldsymbol{Y}\|_2^2=\|\boldsymbol{X}\|_2^2+\|\boldsymbol{Y}\|_2^2-2\langle\boldsymbol{X},\boldsymbol{Y}\rangle,$$

Taking expectations on both sides,

$$\mathbb{E}\|\boldsymbol{X}-\boldsymbol{Y}\|_2^2=\mathbb{E}\|\boldsymbol{X}\|_2^2+\mathbb{E}\|\boldsymbol{Y}\|_2^2-2\mathbb{E}[\langle\boldsymbol{X},\boldsymbol{Y}\rangle],$$

where

$$\begin{aligned}\mathbb{E}\|\boldsymbol{X}\|_2^2&=\mathbb{E}[\mathrm{tr}(\boldsymbol{X}\boldsymbol{X}^\top)]=\mathrm{tr}(\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top])=\mathrm{tr}(\boldsymbol{I}_n)=n,\\ \mathbb{E}\|\boldsymbol{Y}\|_2^2&=\mathbb{E}[\mathrm{tr}(\boldsymbol{Y}\boldsymbol{Y}^\top)]=\mathrm{tr}(\mathbb{E}[\boldsymbol{Y}\boldsymbol{Y}^\top])=\mathrm{tr}(\boldsymbol{I}_n)=n,\\ \mathbb{E}[\langle\boldsymbol{X},\boldsymbol{Y}\rangle]&=\sum_{i=1}^{n}\mathbb{E}[X_iY_i]=\sum_{i=1}^{n}\mathbb{E}[X_i]\mathbb{E}[Y_i]=\sum_{i=1}^{n}0\cdot0=0,\end{aligned}$$

Substituting this in, we obtain

$$\mathbb{E}\|\boldsymbol{X}-\boldsymbol{Y}\|_2^2=n+n-2\cdot0=2n.$$

13 [**Maximum Norm of a Random Vector**] Prove the high-dimensional version of Proposition P2.7.6 (*Maximum of sub-gaussians*).

(1) Let $\boldsymbol{X}_1,\cdots,\boldsymbol{X}_N\in\mathbb{R}^n$ be random vectors (*not necessarily independent*) whose inner entries are independent, centered K-sub-Gaussian random variables. Prove that

$$\mathbb{E}\left[\max_{i\leqslant N}\|\boldsymbol{X}_i\|_2\right]\leqslant\sqrt{n}+CK^2\sqrt{\log N}.\qquad(C>0)$$

(2) Show that the bound in (1) is asymptotically tight for i.i.d. standard normal random vectors. Specifically, if $\boldsymbol{X}_1,\cdots,\boldsymbol{X}_N$ i.i.d. $\sim$

$N(\mathbf{0}, \boldsymbol{I}_n)$, prove that

$$\mathbb{E}\left[\max_{i\leqslant N}\|\boldsymbol{X}_i\|_2\right] \geqslant c(\sqrt{n}+\sqrt{\log N}). \qquad (c>0)$$

Prf. (1) To control the maximum norm of a random vector, decompose it into the deterministic geometric radius $\sqrt{n}$ and a random residual term,

$$\max_{1\leqslant i\leqslant N}\|\boldsymbol{X}_i\|_2 = \sqrt{n} + \max_{1\leqslant i\leqslant N}(\|\boldsymbol{X}_i\|_2 - \sqrt{n}),$$

Let $Z_i := \|\boldsymbol{X}_i\|_2 - \sqrt{n}$. Then by Theorem T3.1.1 (*Concentration of the norm*), for any $t \geqslant 0$,

$$\mathbb{P}\{|\|\boldsymbol{X}_i\|_2 - \sqrt{n}| \geqslant t\} \leqslant 2\exp\left(-\frac{c_0 t^2}{K^4}\right) \implies \|Z_i\|_{\psi_2} \leqslant C_1 K^2,$$

hence, by Proposition P2.7.6 (*Maximum of subgaussians*),

$$\mathbb{E}\left[\max_{i\leqslant N} Z_i\right] \leqslant C_2\sqrt{\log N}\max_{i\leqslant N}\|Z_i\|_{\psi_2} \leqslant C_2\sqrt{\log N}(C_1K^2) =: CK^2\sqrt{\log N}, \qquad (C=C_1C_2)$$

hence

$$\mathbb{E}\left[\max_{i\leqslant N}\|\boldsymbol{X}_i\|_2\right] = \mathbb{E}\left[\max_{i\leqslant N}(\sqrt{n}+Z_i)\right] = \sqrt{n} + \mathbb{E}\left[\max_{i\leqslant N} Z_i\right] \leqslant \sqrt{n} + CK^2\sqrt{\log N}.$$

(2) If $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ i.i.d. $\sim N(\mathbf{0}, \boldsymbol{I}_n)$, then $K = O(1)$.

Similarly to the conclusion of (1), let $\mathbb{E}\|\boldsymbol{X}_i\|_2 = \mu_n \asymp \sqrt{n}$, $Z_i = \|\boldsymbol{X}_i\|_2 - \mu_n$. Then

$$\begin{aligned}\mathbb{E}\left[\max_{i\leqslant N}\|\boldsymbol{X}_i\|_2\right] &= \mathbb{E}\left[\max_{i\leqslant N}(\mu_n + Z_i)\right] = \mu_n + \mathbb{E}\left[\max_{i\leqslant N} Z_i\right] \\ &\geqslant c_1'\sqrt{n} + c_2\sqrt{\log N} \geqslant c(\sqrt{n}+\sqrt{\log N}). \qquad (c=\min(c_1,c_2))\end{aligned}$$

14 [**One-Dimensional Marginals of the Normal Distribution**] Prove that if $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then for any $\boldsymbol{v} \in \mathbb{R}^n$,

$$\langle \boldsymbol{X}, \boldsymbol{v}\rangle \sim N(\langle \boldsymbol{\mu}, \boldsymbol{v}\rangle, \boldsymbol{v}^\top \boldsymbol{\Sigma}\boldsymbol{v}).$$

Prf. For $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, its characteristic function is

$$\varphi_{\boldsymbol{X}}(\boldsymbol{t}) = \mathbb{E}[\exp(\mathrm{i}\boldsymbol{t}^\top \boldsymbol{X})] = \exp\left(\mathrm{i}\boldsymbol{t}^\top\boldsymbol{\mu} - \frac{1}{2}\boldsymbol{t}^\top\boldsymbol{\Sigma}\boldsymbol{t}\right),$$

For $Y = \langle \boldsymbol{X}, \boldsymbol{v}\rangle = \boldsymbol{v}^\top\boldsymbol{X}$, its characteristic function is

$$\varphi_Y(t) = \mathbb{E}[\exp(\mathrm{i}tY)] = \mathbb{E}[\exp(\mathrm{i}t(\boldsymbol{v}^\top\boldsymbol{X}))] = \mathbb{E}[\exp(\mathrm{i}(t\boldsymbol{v})^\top\boldsymbol{X})],$$

therefore

$$\begin{aligned}\varphi_Y(t) = \varphi_{\boldsymbol{X}}(t\boldsymbol{v}) &= \exp\left(\mathrm{i}(t\boldsymbol{v})^\top\boldsymbol{\mu} - \frac{1}{2}(t\boldsymbol{v})^\top\boldsymbol{\Sigma}(t\boldsymbol{v})\right) \\ &= \exp\left(\mathrm{i}t(\boldsymbol{v}^\top\boldsymbol{\mu}) - \frac{1}{2}t^2(\boldsymbol{v}^\top\boldsymbol{\Sigma}\boldsymbol{v})\right) \\ &:= \exp\left(\mathrm{i}t\mu_Y - \frac{1}{2}t^2\sigma_Y^2\right), \qquad (\mu_Y = \langle\boldsymbol{\mu},\boldsymbol{v}\rangle, \sigma_Y^2 = \boldsymbol{v}^\top\boldsymbol{\Sigma}\boldsymbol{v})\end{aligned}$$

Hence, by the uniqueness theorem for characteristic functions,

$$\langle \boldsymbol{X}, \boldsymbol{v}\rangle \sim N(\langle \boldsymbol{\mu}, \boldsymbol{v}\rangle, \boldsymbol{v}^\top \boldsymbol{\Sigma}\boldsymbol{v}).$$

15 [**General Normal Density Function**] Prove Proposition P3.3.6 (*General normal density*).

Prf. For any $B \subset \mathscr{B}(\mathbb{R}^n)$,

$$\mathbb{P}\{\boldsymbol{X} \in B\} = \int_B f_{\boldsymbol{X}}(\boldsymbol{x})\mathrm{d}\boldsymbol{x}.$$

Let $\boldsymbol{Y} = \boldsymbol{X} - \boldsymbol{\mu} \sim N(\mathbf{0}, \boldsymbol{\Sigma})$. Then $f_{\boldsymbol{X}}(\boldsymbol{x}) = f_{\boldsymbol{Y}}(\boldsymbol{x}-\boldsymbol{\mu})$, so it suffices to derive the density function in the zero-mean case.

Let $\boldsymbol{Z} \sim N(\mathbf{0}, \boldsymbol{I}_n)$, whose density function is

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \frac{1}{(2\pi)^{n/2}}\mathrm{e}^{-\frac{1}{2}\|\boldsymbol{z}\|_2^2},$$

By the linear transformation property of the normal distribution, $\boldsymbol{\Sigma}^{1/2}\boldsymbol{Z} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}^{1/2}\boldsymbol{I}_n(\boldsymbol{\Sigma}^{1/2})^\top) = N(\boldsymbol{0}, \boldsymbol{\Sigma})$. Therefore, let $\boldsymbol{X} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{Z}$. Then for any $B \subset \mathscr{B}(\mathbb{R}^n)$,

$$\mathbb{P}\{\boldsymbol{X} \in B\} = \mathbb{P}\{\boldsymbol{\Sigma}^{1/2}\boldsymbol{Z} \in B\} = \mathbb{P}\{\boldsymbol{Z} \in \boldsymbol{\Sigma}^{-1/2}B\} = \int_{\boldsymbol{\Sigma}^{-1/2}B} \frac{1}{(2\pi)^{n/2}} \mathrm{e}^{-\frac{1}{2}\boldsymbol{z}^\top \boldsymbol{z}} \mathrm{d}\boldsymbol{z},$$

Let $\boldsymbol{x} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{z}$. Then $\boldsymbol{z} = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{x}$, in which case

$$\boldsymbol{z}^\top \boldsymbol{z} = (\boldsymbol{\Sigma}^{-1/2}\boldsymbol{x})^\top (\boldsymbol{\Sigma}^{-1/2}\boldsymbol{x}) = \boldsymbol{x}^\top (\boldsymbol{\Sigma}^{-1/2})^\top \boldsymbol{\Sigma}^{-1/2}\boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{x},$$

Its Jacobian determinant is

$$\left|\det\left(\frac{\mathrm{d}\boldsymbol{z}}{\mathrm{d}\boldsymbol{x}}\right)\right| = |\det(\boldsymbol{\Sigma}^{-1/2})| = \frac{1}{\sqrt{\det \boldsymbol{\Sigma}}},$$

Substituting this in, we obtain

$$\mathbb{P}\{\boldsymbol{X} \in B\} = \int_B \frac{1}{(2\pi)^{n/2}} \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{x}} \frac{1}{\sqrt{\det \boldsymbol{\Sigma}}} \mathrm{d}\boldsymbol{x} = \int_B \frac{1}{(2\pi)^{n/2}\sqrt{\det \boldsymbol{\Sigma}}} \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{x}} \mathrm{d}\boldsymbol{x}.$$

Thus, letting $\boldsymbol{x} \leftarrow \boldsymbol{x} - \boldsymbol{\mu}$, we obtain the density function in the general case as

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{n/2}\sqrt{\det \boldsymbol{\Sigma}}} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}.$$

16 [**Characterization of Joint Normality**]

(1) Prove that $\boldsymbol{X} \in \mathbb{R}^n$ is normally distributed if and only if for every $\boldsymbol{u} \in \mathbb{R}^n$, its one-dimensional marginal $\langle \boldsymbol{X}, \boldsymbol{u} \rangle$ is normally distributed.

(2) Using the conclusion of (1), prove that $X_1, \cdots, X_n$ are jointly normal if and only if every linear combination $a_1X_1 + \cdots + a_nX_n$ is normally distributed.

Prf. (1) ($\Longrightarrow$)

If $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then for any $\boldsymbol{u} \in \mathbb{R}^n$, $\langle \boldsymbol{X}, \boldsymbol{u} \rangle \sim N(\boldsymbol{u}^\top \boldsymbol{\mu}, \boldsymbol{u}^\top \boldsymbol{\Sigma} \boldsymbol{u})$.

($\Longleftarrow$)

Let $Y_{\boldsymbol{u}} = \langle \boldsymbol{X}, \boldsymbol{u} \rangle$. Then

$$\mathbb{E}[Y_{\boldsymbol{u}}] = \mathbb{E}[\boldsymbol{u}^\top \boldsymbol{X}] = \boldsymbol{u}^\top \boldsymbol{\mu}, \mathrm{Var}(Y_{\boldsymbol{u}}) = \mathrm{Var}(\boldsymbol{u}^\top \boldsymbol{X}) = \boldsymbol{u}^\top \boldsymbol{\Sigma} \boldsymbol{u},$$

Its characteristic function is

$$\varphi_{Y_{\boldsymbol{u}}}(t) = \mathbb{E}[\mathrm{e}^{\mathrm{i}tY_{\boldsymbol{u}}}] = \mathrm{e}^{\mathrm{i}t(\boldsymbol{u}^\top \boldsymbol{\mu}) - \frac{1}{2}t^2(\boldsymbol{u}^\top \boldsymbol{\Sigma} \boldsymbol{u})},$$

Therefore $\varphi_{\boldsymbol{X}}(\boldsymbol{u}) = \varphi_{Y_{\boldsymbol{u}}}(1)$. Letting $t = 1$, we obtain

$$\varphi_{\boldsymbol{X}}(\boldsymbol{u}) = \mathrm{e}^{\mathrm{i}\boldsymbol{u}^\top \boldsymbol{\mu} - \frac{1}{2}\boldsymbol{u}^\top \boldsymbol{\Sigma} \boldsymbol{u}},$$

which is precisely the characteristic function of $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Hence, by the uniqueness theorem for characteristic functions, $\boldsymbol{X}$ must be multivariate normal.

(2) It is easy to see that

$$a_1X_1 + \cdots + a_nX_n = \langle \boldsymbol{X}, \boldsymbol{a} \rangle,$$

Therefore, by the conclusion of (1), $\boldsymbol{X}$ is normally distributed (i.e., $X_1, \cdots, X_N$ are jointly normal) if and only if for every $\boldsymbol{a} \in \mathbb{R}^n$, $\langle \boldsymbol{X}, \boldsymbol{a} \rangle$ is normally distributed.

17 Construct normally distributed random variables X, Y that are uncorrelated but not independent.

Sol. Let $X \sim N(0,1)$, $S \sim$ Rademacher, $Y = SX$. Then

$$\begin{aligned}
\mathbb{P}\{Y \leqslant y\} &= \mathbb{P}\{SX \leqslant y\} = \mathbb{P}\{SX \leqslant y \mid S = 1\}\mathbb{P}\{S = 1\} + \mathbb{P}\{SX \leqslant y \mid S = -1\}\mathbb{P}\{S = -1\} \\
&= \frac{1}{2}\mathbb{P}\{X \leqslant y\} + \frac{1}{2}\mathbb{P}\{-X \leqslant y\} \\
&= \frac{1}{2}\Phi(y) + \frac{1}{2}\Phi(y) = \Phi(y),
\end{aligned}$$

hence $Y \sim N(0,1)$.

Since $\mathbb{E}[XY] = \mathbb{E}[SX^2] = \mathbb{E}[S]\mathbb{E}[X^2] = 0$, therefore

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - 0 \cdot 0 = 0.$$

Since $Y^2 = X^2$, X^2 and Y^2 are not independent, hence X and Y are not independent.

18 [**Ginibre Random Matrix**] A fundamental model of random matrices is the Ginibre ensemble. $\boldsymbol{G} \in \mathbb{R}^{n\times n}$ is called a Ginibre random matrix if its entries satisfy g_{ij} i.i.d. $\sim N(0,1)$. Prove that the Ginibre distribution is invariant under multiplication by orthogonal matrices, i.e., for any orthogonal matrix $\boldsymbol{U} \in \mathbb{R}^{n\times n}$, $\boldsymbol{UG} \overset{d}{=} \boldsymbol{GU} \overset{d}{=} \boldsymbol{G}$.

Prf. Let $\boldsymbol{G} = (\boldsymbol{Z}_1, \cdots, \boldsymbol{Z}_n)$ with $\boldsymbol{Z}_j$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

- For left multiplication, consider $\boldsymbol{G}' = \boldsymbol{UG}$, whose column vectors are $\boldsymbol{Z}'_j = \boldsymbol{UZ}_j$. Then

$$\mathbb{E}[\boldsymbol{Z}'_j] = \mathbb{E}[\boldsymbol{UZ}_j] = \boldsymbol{U}\mathbb{E}[\boldsymbol{Z}_j] = \boldsymbol{0},$$
$$\mathrm{Cov}(\boldsymbol{Z}'_j) = \mathrm{Cov}(\boldsymbol{UZ}_j) = \boldsymbol{U}\,\mathrm{Cov}(\boldsymbol{Z}_j)\boldsymbol{U}^\top = \boldsymbol{UI}_n\boldsymbol{U}^\top = \boldsymbol{UU}^\top,$$

  Therefore $\boldsymbol{Z}'_j$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_n)$, hence $\boldsymbol{G}'$ follows the Ginibre distribution.

- For right multiplication, consider $\boldsymbol{G}'' = \boldsymbol{GU}$. Note that

$$\boldsymbol{G}'' = (\boldsymbol{U}^\top \boldsymbol{G}^\top)^\top,$$

  Since $\boldsymbol{U}^\top$ is orthogonal and $\boldsymbol{G}^\top$ is a Ginibre random matrix (*the entries of* $\boldsymbol{G}$ *are i.i.d.* $N(0,1)$*, so the entries of* $\boldsymbol{G}^\top$ *are i.i.d.* $N(0,1)$), $\boldsymbol{G}^\top$ is a Ginibre random matrix. Hence, by the conclusion of (1), $\boldsymbol{U}^\top\boldsymbol{G}^\top \overset{d}{=} \boldsymbol{G}^\top$ follows the Ginibre distribution, so $\boldsymbol{G}'' = (\boldsymbol{U}^\top\boldsymbol{G}^\top)^\top$ follows the Ginibre distribution.

19 [**GOE Random Matrix**] A fundamental model of symmetric random matrices is the Gaussian Orthogonal Ensemble. A symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is called a GOE random matrix if its diagonal and upper-triangular entries are independent normal random variables satisfying

$$A_{ij} \sim \begin{cases} N(0,2), & i = j, \\ N(0,1), & i < j \end{cases}.$$

Prove that:

(1) $\boldsymbol{A}$ has the same distribution as $\frac{1}{\sqrt{2}}(\boldsymbol{G} + \boldsymbol{G}^\top)$, where $\boldsymbol{G}$ is a Ginibre random matrix from Exercise Q3.18 (*Ginibre Random Matrix*); this explains the different scaling of the diagonal and off-diagonal entries of $\boldsymbol{A}$ in the definition.

(2) the GOE distribution is invariant under orthogonal similarity transformations, i.e., for any orthogonal matrix $\boldsymbol{U} \in \mathbb{R}^{n\times n}$,

$$\boldsymbol{U}^\top\boldsymbol{AU} \overset{d}{=} \boldsymbol{A},$$

which explains the name "Gaussian Orthogonal (Invariant) Ensemble."

Prf. **Solution 1**

Let $\boldsymbol{G} \in \mathbb{R}^{n\times n}$ be a Ginibre random matrix. Construct

$$\boldsymbol{M} = \frac{1}{\sqrt{2}}(\boldsymbol{G} + \boldsymbol{G}^\top),$$

Then:

1. Normality: $M_{ii} \sim N(0,2)$, $M_{ij} \sim N(0,1)$.

2. Independence: $\{G_{ii}\}$ are independent, and $\{(g_{ij}, G_{ji})\}$ $(i < j)$ are independent.

Therefore $\{M_{ii}, M_{ij}\}$ are independent, hence $\boldsymbol{A} \overset{d}{=} \boldsymbol{M}$. Thus

$$\boldsymbol{U}^\top\boldsymbol{AU} \overset{d}{=} \frac{1}{\sqrt{2}}(\boldsymbol{U}^\top(\boldsymbol{G} + \boldsymbol{G}^\top)\boldsymbol{U}) = \frac{1}{\sqrt{2}}(\boldsymbol{U}^\top\boldsymbol{GU} + \boldsymbol{U}^\top\boldsymbol{G}^\top\boldsymbol{U}) = \frac{1}{\sqrt{2}}(\boldsymbol{U}^\top\boldsymbol{GU} + (\boldsymbol{U}^\top\boldsymbol{GU})^\top).$$

By the conclusion of Exercise Q3.18 (*Ginibre Random Matrix*), $\boldsymbol{U}^\top\boldsymbol{GU} \overset{d}{=} \boldsymbol{G}$. Therefore

$$\boldsymbol{U}^\top\boldsymbol{AU} \overset{d}{=} \frac{1}{\sqrt{2}}(\boldsymbol{G} + \boldsymbol{G}^\top) \overset{d}{=} \boldsymbol{A}.$$

Solution 2 Density Function Method

The joint density function of a GOE matrix is related to the sum of squares of its entries. Its density satisfies

$$f(\boldsymbol{A}) \propto \exp\left(-\frac{1}{4}\sum_i A_{ii}^2 - \frac{1}{2}\sum_{i<j} A_{ij}^2\right) = \exp\left(-\frac{1}{4}\operatorname{tr}(\boldsymbol{A}^2)\right).$$

Since $\operatorname{tr}((\boldsymbol{U}^\top \boldsymbol{A}\boldsymbol{U})^2) = \operatorname{tr}(\boldsymbol{U}^\top \boldsymbol{A}^2\boldsymbol{U}) = \operatorname{tr}(\boldsymbol{A}^2)$, the form of the density function remains unchanged after the transformation, which is the invariance of the distribution.

20 [**Gaussian Random Matrices Make Orthogonal Vectors Independent**] Let $\boldsymbol{G} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix. Let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ satisfy $\|\boldsymbol{u}\| = \|\boldsymbol{v}\| = 1$, $\boldsymbol{u}^\top \boldsymbol{v} = 0$. Then $\boldsymbol{G}\boldsymbol{u}, \boldsymbol{G}\boldsymbol{v}$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_m)$.

Prf. Let $Y_i = \boldsymbol{Z}_i^\top \boldsymbol{u}$. Then

$$\mathbb{E}[Y_i] = \mathbb{E}[\boldsymbol{Z}_i^\top \boldsymbol{u}] = \mathbb{E}[\boldsymbol{Z}_i^\top]\boldsymbol{u} = 0,$$
$$\operatorname{Var}(Y_i) = \boldsymbol{u}^\top \operatorname{Var}(\boldsymbol{Z}_i)\boldsymbol{u} = \boldsymbol{u}^\top \boldsymbol{I}_n \boldsymbol{u} = \|\boldsymbol{u}\|^2 = 1,$$

and $Y_1, \cdots, Y_m$ are independent (*since* $\boldsymbol{Z}_1, \cdots, \boldsymbol{Z}_m$ *are independent*), therefore $\boldsymbol{G}\boldsymbol{u} \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$; similarly, $\boldsymbol{G}\boldsymbol{v} \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$.

Let $\boldsymbol{W} = \begin{pmatrix} \boldsymbol{G}\boldsymbol{u} \\ \boldsymbol{G}\boldsymbol{v} \end{pmatrix}$. Then $\boldsymbol{W}$ is multivariate normal, so to prove that $\boldsymbol{G}\boldsymbol{u}$ and $\boldsymbol{G}\boldsymbol{v}$ are independent, it suffices to prove $\operatorname{Cov}(\boldsymbol{G}\boldsymbol{u}, \boldsymbol{G}\boldsymbol{v}) = \boldsymbol{O}_{m\times m}$.

Consider $\operatorname{Cov}(\boldsymbol{Z}_i^\top \boldsymbol{u}, \boldsymbol{Z}_j^\top \boldsymbol{v})$:

a. When $i \neq j$, $\boldsymbol{Z}_i$ and $\boldsymbol{Z}_j$ are independent, so $\operatorname{Cov}(\boldsymbol{Z}_i^\top \boldsymbol{u}, \boldsymbol{Z}_j^\top \boldsymbol{v}) = 0$.

b. When $i = j$,

$$\begin{aligned}\operatorname{Cov}(\boldsymbol{Z}_i^\top \boldsymbol{u}, \boldsymbol{Z}_i^\top \boldsymbol{v}) &= \mathbb{E}[(\boldsymbol{Z}_i^\top \boldsymbol{u})(\boldsymbol{Z}_i^\top \boldsymbol{v})] - \mathbb{E}[\boldsymbol{Z}_i^\top \boldsymbol{u}]\mathbb{E}[\boldsymbol{Z}_i^\top \boldsymbol{v}] \\ &= \mathbb{E}[\boldsymbol{u}^\top \boldsymbol{Z}_i \boldsymbol{Z}_i^\top \boldsymbol{v}] = \boldsymbol{u}^\top \mathbb{E}[\boldsymbol{Z}_i \boldsymbol{Z}_i^\top]\boldsymbol{v} = \boldsymbol{u}^\top \boldsymbol{I}_n \boldsymbol{v} = \boldsymbol{u}^\top \boldsymbol{v} = 0.\end{aligned}$$

Therefore $\operatorname{Cov}(\boldsymbol{G}\boldsymbol{u}, \boldsymbol{G}\boldsymbol{v}) = \boldsymbol{O}_{m\times m}$.

21 [**Orthogonal Invariance of Gaussian Random Vectors**] Let $\boldsymbol{X}, \boldsymbol{Y}$ i.i.d. $\sim (0, \boldsymbol{I}_n)$. Define

$$\boldsymbol{U} = \frac{\boldsymbol{X}+\boldsymbol{Y}}{\sqrt{2}}, \boldsymbol{V} = \frac{\boldsymbol{X}-\boldsymbol{Y}}{\sqrt{2}},$$

Prove that $\boldsymbol{U}, \boldsymbol{V}$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

Prf. It is easy to see that

$$\begin{pmatrix} \boldsymbol{U} \\ \boldsymbol{V} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \boldsymbol{I}_n & \boldsymbol{I}_n \\ \boldsymbol{I}_n & -\boldsymbol{I}_n \end{pmatrix} \begin{pmatrix} \boldsymbol{X} \\ \boldsymbol{Y} \end{pmatrix},$$

Therefore $\boldsymbol{U}, \boldsymbol{V}$ are jointly normal, where

$$\mathbb{E}[\boldsymbol{U}] = \frac{1}{\sqrt{2}}(\mathbb{E}[\boldsymbol{X}] + \mathbb{E}[\boldsymbol{Y}]) = \boldsymbol{0},$$
$$\mathbb{E}[\boldsymbol{V}] = \frac{1}{\sqrt{2}}(\mathbb{E}[\boldsymbol{X}] - \mathbb{E}[\boldsymbol{Y}]) = \boldsymbol{0},$$
$$\operatorname{Var}(\boldsymbol{U}) = \frac{1}{2}(\operatorname{Var}(\boldsymbol{X}) + \operatorname{Var}(\boldsymbol{Y})) = \frac{1}{2}(\boldsymbol{I}_n + \boldsymbol{I}_n) = \boldsymbol{I}_n,$$
$$\operatorname{Var}(\boldsymbol{V}) = \frac{1}{2}(\operatorname{Var}(\boldsymbol{X}) + \operatorname{Var}(\boldsymbol{Y})) = \frac{1}{2}(\boldsymbol{I}_n + \boldsymbol{I}_n) = \boldsymbol{I}_n,$$

hence $\boldsymbol{U}, \boldsymbol{V} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

Since

$$\operatorname{Cov}(\boldsymbol{U}, \boldsymbol{V}) = \mathbb{E}[\boldsymbol{U}\boldsymbol{V}^\top] - \mathbb{E}[\boldsymbol{U}]\mathbb{E}[\boldsymbol{V}]^\top = \mathbb{E}[\boldsymbol{U}\boldsymbol{V}^\top]$$

$$= \mathbb{E}\left[\frac{\boldsymbol{X}+\boldsymbol{Y}}{\sqrt{2}}\left(\frac{\boldsymbol{X}-\boldsymbol{Y}}{\sqrt{2}}\right)^{\top}\right] = \frac{1}{2}\mathbb{E}[(\boldsymbol{X}+\boldsymbol{Y})(\boldsymbol{X}^{\top}-\boldsymbol{Y}^{\top})]$$
$$= \frac{1}{2}\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^{\top}-\boldsymbol{X}\boldsymbol{Y}^{\top}+\boldsymbol{Y}\boldsymbol{X}^{\top}-\boldsymbol{Y}\boldsymbol{Y}^{\top}]$$
$$= \frac{1}{2}(\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^{\top}]-\mathbb{E}[\boldsymbol{Y}\boldsymbol{Y}^{\top}])$$
$$= \frac{1}{2}(\boldsymbol{I}_n-\boldsymbol{I}_n) = \boldsymbol{O},$$

therefore $\boldsymbol{U}$ and $\boldsymbol{V}$ are independent.

22 [**Norm and Direction of a Normal Random Vector**] Let $\boldsymbol{Z}\sim N(\boldsymbol{0},\boldsymbol{I}_n)$. Define

$$R = \|\boldsymbol{Z}\|_2, \boldsymbol{U} = \bar{\boldsymbol{Z}}.$$

Prove that $\|\boldsymbol{Z}\|_2$ and $\bar{\boldsymbol{Z}}$ are independent.

Rk. Note that

$$f_{\boldsymbol{Z}}(\boldsymbol{x}) = \prod_{k=1}^{n}\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{1}{2}x_k^2} = \frac{1}{(2\pi)^{n/2}}\mathrm{e}^{-\frac{1}{2}\|\boldsymbol{x}\|_2^2},$$

which depends only on $\|\boldsymbol{x}\|_2$, geometrically representing the rotational invariance of the distribution.

Prf. Consider the mapping $\mathbb{R}^n\to(r,\boldsymbol{\theta})$, where $r\geqslant 0$, $\boldsymbol{\theta}\in\mathbb{S}^{n-1}$. Then

$$\mathrm{d}\boldsymbol{x} = r^{n-1}\mathrm{d}r\mathrm{d}\sigma(\boldsymbol{\theta}),$$

where σ is the surface area measure on the unit sphere $\mathbb{S}^{n-1}$. Therefore, for any measurable sets $A\subset[0,\infty), B\subset\mathbb{S}^{n-1}$,

$$\begin{aligned}\mathbb{P}\{R\in A,\boldsymbol{U}\in B\} &= \int_{\mathbb{R}^n}\mathbf{1}_{\{\|\boldsymbol{x}\|_2\in A,\bar{\boldsymbol{x}}\in B\}}f_{\boldsymbol{Z}}(\boldsymbol{x})\mathrm{d}\boldsymbol{x}\\ &= \int_{r\in A}\int_{\boldsymbol{\theta}\in B}\frac{1}{(2\pi)^{n/2}}\mathrm{e}^{-\frac{1}{2}r^2}\cdot r^{n-1}\mathrm{d}r\mathrm{d}\sigma(\boldsymbol{\theta})\\ &= \underbrace{\left(\int_A\frac{r^{n-1}\mathrm{e}^{-\frac{1}{2}r^2}}{(2\pi)^{n/2}}\mathrm{d}r\right)}_{I_1(A)}\cdot\underbrace{\left(\int_B\mathrm{d}\sigma(\boldsymbol{\theta})\right)}_{I_2(B)}.\end{aligned}$$

For the marginal distribution of R, let $B=\mathbb{S}^{n-1}$. Then

$$\mathbb{P}\{R\in A\} = \mathbb{P}\{R\in A,\boldsymbol{U}\in\mathbb{S}^{n-1}\} = I_1(A)\cdot I_2(\mathbb{S}^{n-1}),$$

Denote $C_{\boldsymbol{U}} = I_2(\mathbb{S}^{n-1}) = \int_{\mathbb{S}^{n-1}}\mathrm{d}\sigma(\boldsymbol{\theta})$ as the total surface area of the unit sphere. Then

$$I_1(A) = \frac{\mathbb{P}\{R\in A\}}{C_{\boldsymbol{U}}};$$

For the marginal distribution of $\boldsymbol{U}$, let $A=[0,\infty)$. Then

$$\mathbb{P}\{\boldsymbol{U}\in B\} = \mathbb{P}\{R\in[0,\infty),\boldsymbol{U}\in B\} = I_1([0,\infty))\cdot I_2(B),$$

Denote

$$C_R = I_1([0,\infty)) = \int_0^{\infty}\frac{r^{n-1}\mathrm{e}^{-\frac{1}{2}r^2}}{(2\pi)^{n/2}}\mathrm{d}r.$$

Then

$$I_2(B) = \frac{\mathbb{P}\{\boldsymbol{U}\in B\}}{C_R},$$

Therefore, by the normalization condition of total probability, taking A and B simultaneously to be the full spaces,

$$1 = \mathbb{P}\{R\in[0,\infty),\boldsymbol{U}\in\mathbb{S}^{n-1}\} = C_R\cdot C_{\boldsymbol{U}},$$

hence

$$\begin{aligned} I_1(A)\cdot I_2(B) &= \left(\frac{\mathbb{P}\{R\in A\}}{C_{\boldsymbol{U}}}\right)\cdot\left(\frac{\mathbb{P}\{\boldsymbol{U}\in B\}}{C_R}\right) \\ &= \frac{1}{C_R\cdot C_{\boldsymbol{U}}}\cdot\mathbb{P}\{R\in A\}\cdot\mathbb{P}\{\boldsymbol{U}\in B\} \\ &= \mathbb{P}\{R\in A\}\cdot\mathbb{P}\{\boldsymbol{U}\in B\}, \end{aligned}$$

i.e., $\mathbb{P}\{R\in A,\boldsymbol{U}\in B\}=\mathbb{P}\{R\in A\}\cdot\mathbb{P}\{\boldsymbol{U}\in B\}$, so R and $\boldsymbol{U}$ are independent.

23 [**High-Dimensional Random Points Are in Convex Position**] A finite set of points in $\mathbb{R}^n$ is said to be in convex position if no point lies in the convex hull of the remaining points.

Let $N\leqslant \mathrm{e}^{cn}$, where $c>0$ is a sufficiently small constant. Consider $\boldsymbol{g}_1,\cdots,\boldsymbol{g}_N$ i.i.d. $\sim N(\boldsymbol{0},\boldsymbol{I}_n)$. Following the steps below, prove that the above N random points are in convex position with high probability (w.p. $1-\mathrm{e}^{-cn}$):

(1) Show that $\boldsymbol{g}_1$ is linearly separable from the other points, i.e. there exists a normal vector $\boldsymbol{x}\in\mathbb{R}^n$ (*possibly random*) such that $\langle\boldsymbol{g}_1,\boldsymbol{x}\rangle>\max_{j=2,\cdots,N}\langle\boldsymbol{g}_j,\boldsymbol{x}\rangle$ (w.h.p.).

(2) Deduce that $\boldsymbol{g}_1$ does not lie in the convex hull $\mathrm{conv}(\{\boldsymbol{g}_j\}_{j=2}^N)$ of the other points (w.h.p.).

(3) Conclude by the union bound that all N points are in convex position.

Prf. (1) Choose the separating normal vector $\boldsymbol{x}=\boldsymbol{g}_1$. We need to prove that the linear separability event $\mathcal{B}=\{\|\boldsymbol{g}_1\|_2^2>\max_{2\leqslant j\leqslant N}\langle\boldsymbol{g}_j,\boldsymbol{g}$ holds (w.h.p.). Let $\mathcal{A}=\{\frac{n}{2}\leqslant\|\boldsymbol{g}_1\|_2^2\leqslant\frac{3n}{2}\}$ be the benign event of norm concentration. Since $\|\boldsymbol{g}_1\|_2^2=\sum_{k=1}^n g_{1k}^2$ follows the chi-square distribution $\chi^2(n)$ with n degrees of freedom, by Theorem T3.1.1 (*Concentration of the norm*),

$$\mathbb{P}\{\mathcal{A}^c\}=\mathbb{P}\left\{\left|\|\boldsymbol{g}_1\|_2^2-n\right|>\frac{n}{2}\right\}\leqslant 2\mathrm{e}^{-c_1 n},\qquad (c_1>0)$$

Fix $\boldsymbol{g}_1=\boldsymbol{x}$, where $\boldsymbol{x}\in\mathcal{A}$, satisfying $\|\boldsymbol{x}\|_2^2\geqslant n/2$. For every $j\in\{2,\cdots,N\}$, let $Z_j=\langle\boldsymbol{g}_j,\boldsymbol{g}_1\rangle=\sum_{k=1}^n g_{jk}x_k$ be the one-dimensional inner-product projection variable. Since $\boldsymbol{g}_j$ is independent of $\boldsymbol{g}_1$, and the coordinates g_{jk} i.i.d. $\sim N(0,1)$, $Z_j\mid\{\boldsymbol{g}_1=\boldsymbol{x}\}$ strictly follows a one-dimensional Gaussian distribution, with conditional expectation, conditional variance

$$\mathbb{E}[Z_j\mid\boldsymbol{g}_1=\boldsymbol{x}]=\sum_{k=1}^n x_k\mathbb{E}[g_{jk}]=0,$$

$$\mathrm{Var}(Z_j\mid\boldsymbol{g}_1=\boldsymbol{x})=\sum_{k=1}^n x_k^2\,\mathrm{Var}(g_{jk})=\sum_{k=1}^n x_k^2=\|\boldsymbol{x}\|_2^2.$$

Denote $\sigma^2=\|\boldsymbol{x}\|_2^2$; then $Z_j\mid\{\boldsymbol{g}_1=\boldsymbol{x}\}\sim N(0,\sigma^2)$. By Proposition P2.1.2 (*Gaussian tails*) (*Gaussian tail*), $\mathbb{P}\{Z\geqslant t\}\leqslant\mathrm{e}^{-t^2/2\sigma^2}$. Taking the threshold $t=\|\boldsymbol{x}\|_2^2=\sigma^2$,

$$\mathbb{P}\{Z_j\geqslant\|\boldsymbol{x}\|_2^2\mid\boldsymbol{g}_1=\boldsymbol{x}\}\leqslant\mathrm{e}^{-\frac{(\sigma^2)^2}{2\sigma^2}}=\mathrm{e}^{-\frac{\sigma^2}{2}}=\mathrm{e}^{-\frac{1}{2}\|\boldsymbol{x}\|_2^2}.$$

Applying the union bound over the remaining $N-1$ points, together with the condition $\|\boldsymbol{x}\|_2^2\geqslant n/2$, $N\leqslant\mathrm{e}^{cn}$,

$$\begin{aligned}\mathbb{P}\left\{\max_{2\leqslant j\leqslant N}Z_j\geqslant\|\boldsymbol{x}\|_2^2\mid\boldsymbol{g}_1=\boldsymbol{x}\right\}&\leqslant\sum_{j=2}^N\mathbb{P}\{Z_j\geqslant\|\boldsymbol{x}\|_2^2\mid\boldsymbol{g}_1=\boldsymbol{x}\}\\&\leqslant N\mathrm{e}^{-\frac{1}{2}\|\boldsymbol{x}\|_2^2}\leqslant\mathrm{e}^{cn}\cdot\mathrm{e}^{-\frac{n}{4}}=\mathrm{e}^{-(\frac{1}{4}-c)n}.\end{aligned}$$

By the law of total probability, letting $c<1/4$,

$$\begin{aligned}\mathbb{P}\{\mathcal{B}^c\}&\leqslant\mathbb{P}\{\mathcal{B}^c\mid\mathcal{A}\}\mathbb{P}(\mathcal{A})+\mathbb{P}\{\mathcal{A}^c\}\\&\leqslant\mathrm{e}^{-(\frac{1}{4}-c)n}+2\mathrm{e}^{-c_1 n}\leqslant\mathrm{e}^{-c_2 n},\end{aligned}\qquad (c_2>0)$$

(2) We prove by contradiction that the event $\mathcal{B}$ implies $\boldsymbol{g}_1\notin\mathrm{conv}(\{\boldsymbol{g}_j\}_{j=2}^N)$. Suppose the contrary holds, i.e. $\boldsymbol{g}_1\in\mathrm{conv}(\{\boldsymbol{g}_j\}_{j=2}^N)$; then there must exist a set of nonnegative convex combination coefficients $\lambda_2,\cdots,\lambda_N\geqslant 0$ satisfying

$\sum_{j=2}^{N}\lambda_j = 1$ such that

$$\boldsymbol{g}_1 = \sum_{j=2}^{N}\lambda_j \boldsymbol{g}_j.$$

Taking the inner product of both sides with $\boldsymbol{g}_1$,

$$\begin{aligned}\|\boldsymbol{g}_1\|_2^2 &= \langle \boldsymbol{g}_1, \boldsymbol{g}_1\rangle \\ &= \left\langle \sum_{j=2}^{N}\lambda_j \boldsymbol{g}_j, \boldsymbol{g}_1 \right\rangle = \sum_{j=2}^{N}\lambda_j \langle \boldsymbol{g}_j, \boldsymbol{g}_1\rangle \\ &\leqslant \sum_{j=2}^{N}\lambda_j \left(\max_{2\leqslant k\leqslant N}\langle \boldsymbol{g}_k, \boldsymbol{g}_1\rangle\right) \\ &= \left(\sum_{j=2}^{N}\lambda_j\right)\max_{2\leqslant k\leqslant N}\langle \boldsymbol{g}_k, \boldsymbol{g}_1\rangle = \max_{2\leqslant k\leqslant N}\langle \boldsymbol{g}_k, \boldsymbol{g}_1\rangle,\end{aligned}$$

which directly contradicts the strict inequality $\|\boldsymbol{g}_1\|_2^2 > \max_{2\leqslant k\leqslant N}\langle \boldsymbol{g}_k, \boldsymbol{g}_1\rangle$ defined in the event $\mathcal{B}$; thus, the occurrence of the event $\mathcal{B}$ must guarantee $\boldsymbol{g}_1 \notin \operatorname{conv}(\{\boldsymbol{g}_j\}_{j=2}^{N})$. By the inclusion relation of complementary events,

$$\mathbb{P}\{\boldsymbol{g}_1 \in \operatorname{conv}(\{\boldsymbol{g}_j\}_{j=2}^{N})\} \leqslant \mathbb{P}\{\mathcal{B}^c\} \leqslant \mathrm{e}^{-c_2 n}.$$

(3) Let $E = \bigcup_{i=1}^{N}\{\boldsymbol{g}_i \in \operatorname{conv}(\{\boldsymbol{g}_j\}_{j\neq i})\}$ be the global failure event (*i.e. there exists at least one point lying in the convex hull of the remaining points*). By the identical-distribution symmetry of the points $\boldsymbol{g}_1, \cdots, \boldsymbol{g}_N$, applying the union bound over all N points,

$$\begin{aligned}\mathbb{P}(E) &\leqslant \sum_{i=1}^{N}\mathbb{P}\{\boldsymbol{g}_i \in \operatorname{conv}(\{\boldsymbol{g}_j\}_{j\neq i})\} \\ &= N\cdot\mathbb{P}\{\boldsymbol{g}_1 \in \operatorname{conv}(\{\boldsymbol{g}_j\}_{j=2}^{N})\} \leqslant N\mathrm{e}^{-c_2 n} \leqslant \mathrm{e}^{cn}\cdot\mathrm{e}^{-c_2 n} = \mathrm{e}^{-(c_2-c)n}.\end{aligned}$$

Letting $c < \min\{1/4, c_2/2\}$, we have $c_2 - c \geqslant c$; hence

$$\mathbb{P}(E) \leqslant \mathrm{e}^{-cn}.$$

Taking the complementary event, the probability that all N random points are simultaneously in convex position satisfies

$$\mathbb{P}\{E^c\} = 1 - \mathbb{P}(E) \geqslant 1 - \mathrm{e}^{-cn}.$$

24 [**Polar Decomposition of Uniform Distribution on the Unit Ball**] Let $\boldsymbol{X} \sim U(\mathbb{B}_2^n)$, decompose it as

$$\boldsymbol{X} = R\boldsymbol{Z},$$

where $R = \|\boldsymbol{X}\|_2$ is the random norm, $\boldsymbol{Z} = \bar{\boldsymbol{X}}$ is the random direction vector. Prove that:

(1) $\boldsymbol{Z} \sim U(\mathbb{S}^{n-1})$.

(2) $R \sim f_R(r) = nr^{n-1}\mathbf{1}_{r\in[0,1]}$.

(3) R and $\boldsymbol{Z}$ are independent.

Prf. Let $V_n = \operatorname{Vol}(\mathbb{B}_2^n)$, $A_{n-1} = \operatorname{Area}(\mathbb{S}^{n-1})$, then $A_{n-1} = nV_n$.

It is easy to see that

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \begin{cases} 1/V_n, & \|\boldsymbol{x}\|_2 \leqslant 1, \\ 0, & \text{otherwise}\end{cases}$$

Consider the polar coordinate transformation $\boldsymbol{x} \to (r, \boldsymbol{\theta})$, where $r = \|\boldsymbol{x}\|_2 \in [0,1]$, $\boldsymbol{\theta} = \bar{\boldsymbol{x}} \in \mathbb{S}^{n-1}$. Then

$$\mathrm{d}\boldsymbol{x} = r^{n-1}\mathrm{d}r\mathrm{d}\sigma(\boldsymbol{\theta}),$$

where σ is the surface area measure on $\mathbb{S}^{n-1}$. Therefore, for any measurable sets $A \subset [0,1], B \subset \mathbb{S}^{n-1}$,

$$\begin{aligned}\mathbb{P}\{R \in A, \boldsymbol{Z} \in B\} &= \int_{\mathbb{R}^n} \mathbf{1}_{\{\|\boldsymbol{x}\|_2 \in A, \tilde{\boldsymbol{x}} \in B\}} f_{\boldsymbol{X}}(\boldsymbol{x}) \mathrm{d}\boldsymbol{x} \\ &= \int_{r \in A} \int_{\boldsymbol{\theta} \in B} \frac{1}{V_n} r^{n-1} \mathrm{d}r \mathrm{d}\sigma(\boldsymbol{\theta}) \\ &= \left(\int_A \frac{1}{V_n} r^{n-1} \mathrm{d}r \right) \cdot \left(\int_B \mathrm{d}\sigma(\boldsymbol{\theta}) \right),\end{aligned}$$

hence R and $\boldsymbol{Z}$ are independent.

Let $A = [0,1]$. Then

$$\mathbb{P}\{\boldsymbol{Z} \in B\} = \left(\int_0^1 \frac{r^{n-1}}{V_n} \mathrm{d}r \right) \cdot \int_B \mathrm{d}\sigma(\boldsymbol{\theta}) = \frac{1}{nV_n} \int_B \mathrm{d}\sigma(\boldsymbol{\theta}) = \frac{\sigma(B)}{A_{n-1}},$$

hence $\boldsymbol{Z}$ is proportional to the surface area on the sphere, i.e., $\boldsymbol{Z} \sim U(\mathbb{S}^{n-1})$.

Let $B = \mathbb{S}^{n-1}$. Then

$$\mathbb{P}\{R \in A\} = \int_A \frac{r^{n-1}}{V_n} \cdot \sigma(\mathbb{S}^{n-1}) \mathrm{d}r = \int_A \frac{A_{n-1}}{V_n} r^{n-1} \mathrm{d}r = \int_A n r^{n-1} \mathrm{d}r,$$

hence $f_R(r) = n r^{n-1}$.

25 [**Isotropy of the Euclidean Ball Distribution**] Let $\boldsymbol{Y} \sim U(\mathbb{B}_2^n(0, \sqrt{n+2}))$. Prove that $\boldsymbol{Y}$ is isotropic.

Prf. Since $\boldsymbol{Y}$ is uniformly distributed on the ball centered at the origin:

1. By symmetry, $f_{\boldsymbol{Y}}(\boldsymbol{y}) = f_{\boldsymbol{Y}}(-\boldsymbol{y})$, hence
$$\mathbb{E}[\boldsymbol{Y}] = \int_{\mathbb{B}_2^n(0,R)} \boldsymbol{y} \cdot \frac{1}{\mathrm{Vol}(\mathbb{B}_2^n(0,R))} \mathrm{d}\boldsymbol{y} = \boldsymbol{0}.$$
2. By rotational invariance, $\mathbb{E}[\boldsymbol{Y}\boldsymbol{Y}^\top]$ is a scalar multiple of the identity matrix, denote $\mathbb{E}[\boldsymbol{Y}\boldsymbol{Y}^\top] = \lambda \boldsymbol{I}_n$. Taking the trace on both sides,
$$\begin{aligned}\mathrm{LHS} &= \mathrm{tr}(\mathbb{E}[\boldsymbol{Y}\boldsymbol{Y}^\top]) = \mathbb{E}[\mathrm{tr}(\boldsymbol{Y}\boldsymbol{Y}^\top)] = \mathbb{E}\|\boldsymbol{Y}\|_2^2, \\ \mathrm{RHS} &= \mathrm{tr}(\lambda \boldsymbol{I}_n) = n\lambda,\end{aligned}$$
hence $\lambda = n / \mathbb{E}\|\boldsymbol{Y}\|_2^2$. It remains to compute $\mathbb{E}\|\boldsymbol{Y}\|_2^2$.
Let $L = \|\boldsymbol{Y}\|_2$. Using the conclusion of Exercise Q3.24 (*Polar Decomposition of Uniform Distribution on the Unit Ball*), we have
$$f_L(r) = \frac{n r^{n-1}}{R^n} \mathbf{1}_{[0,R]},$$
therefore
$$\mathbb{E}\|\boldsymbol{Y}\|_2^2 = \int_0^R r^2 \cdot \frac{n r^{n-1}}{R^n} \mathrm{d}r = \frac{n}{R^n} \int_0^R r^{n+1} \mathrm{d}r = \frac{nR^2}{n+2} = \frac{n \cdot (\sqrt{n+2})^2}{n+2} = n,$$
hence $\lambda = 1$, i.e.,
$$\mathbb{E}[\boldsymbol{Y}\boldsymbol{Y}^\top] = 1 \cdot \boldsymbol{I}_n = \boldsymbol{I}_n.$$
Thus, $\boldsymbol{Y}$ is isotropic.

26 [**Delocalization of Random Vectors on the Unit Sphere**] If $\boldsymbol{x} \in \mathbb{R}^n$ satisfies that all coordinates are equal in magnitude, i.e., for every i, $|x_i| = 1/\sqrt{n}$, then it is completely spread out or delocalized.
In fact, random vectors on the unit sphere are almost delocalized. Prove that if $\boldsymbol{X} \sim U(\mathbb{S}^{n-1})$, then

$$\mathbb{E}\|\boldsymbol{X}\|_\infty \asymp \sqrt{\frac{\log n}{n}},$$

where the symbol "$\asymp$" denotes equality up to absolute constant factors.

Prf. Let $\boldsymbol{G} = (G_1, \cdots, G_n)^\top \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. By the construction of the uniform distribution on the sphere, we have

$$\boldsymbol{X} \overset{d}{=} \bar{\boldsymbol{G}}.$$

Using the rotational invariance of Gaussian random vectors (Q3.22 (*Norm and Direction of a Normal Random Vector*)), $\|\boldsymbol{G}\|_2$ and $\bar{\boldsymbol{G}}$ are independent, hence

$$\mathbb{E}\|\boldsymbol{X}\|_\infty = \mathbb{E}\left[\frac{\max_{1\leqslant i\leqslant n}|G_i|}{\|\boldsymbol{G}\|_2}\right] = \mathbb{E}\left[\max_{1\leqslant i\leqslant n}|G_i|\right]\cdot\mathbb{E}\left[\frac{1}{\|\boldsymbol{G}\|_2}\right].$$

(*It remains to estimate the two integrals on the right-hand side.*)

- Prove $\mathbb{E}[\max_{1\leqslant i\leqslant n}|G_i|] \asymp \sqrt{\log n}$
  - For the upper bound, by Jensen's inequality,

$$\mathrm{e}^{\lambda\mathbb{E}[\max|G_i|]} \leqslant \mathbb{E}\left[\mathrm{e}^{\lambda\max|G_i|}\right] = \mathbb{E}\left[\max_{1\leqslant i\leqslant n}|G_i|\right] \leqslant \sum_{i=1}^n \mathbb{E}[\mathrm{e}^{\lambda|G_i|}],$$

    where $\mathrm{e}^{\lambda|G_i|} \leqslant \mathrm{e}^{\lambda G_i} + \mathrm{e}^{-\lambda G_i}$, with $\mathbb{E}[\mathrm{e}^{\lambda G_i}] = \mathrm{e}^{\lambda^2/2}$, hence

$$\mathrm{e}^{\lambda\mathbb{E}[\max_{1\leqslant i\leqslant n}|G_i|]} \leqslant 2n\mathrm{e}^{\lambda^2/2},$$

    Taking logarithms on both sides,

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n}|G_i|\right] \leqslant \frac{\log(2n)}{\lambda} + \frac{\lambda}{2},$$

    The optimal parameter on the right-hand side is $\lambda_0 = \sqrt{2\log(2n)}$. Substituting it gives

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n}|G_i|\right] \leqslant \sqrt{2\log(2n)}.$$

  - For the lower bound, let $M = \max_{1\leqslant i\leqslant n}|G_i|$. Then

$$\begin{aligned}\mathbb{E}[M] &= \int_0^\infty \mathbb{P}\{M > t\}\mathrm{d}t \\ &\geqslant \int_0^{t_0}\mathbb{P}\{M>t\}\mathrm{d}t \geqslant \int_0^{t_0}\mathbb{P}\{M>t_0\}\mathrm{d}t \qquad (t_0>0)\\ &= t_0\mathbb{P}\{M>t_0\}.\end{aligned}$$

    It is easy to see that

$$\mathbb{P}\{M>t_0\} = 1-\mathbb{P}\{M\leqslant t_0\} = 1-[1-\mathbb{P}\{|G_1|>t_0\}]^n,$$

    Using Mill's ratio, we have

$$\mathbb{P}\{|G_1|>t\} = 2\mathbb{P}\{G>t\} \geqslant 2\cdot\frac{1}{\sqrt{2\pi}}\cdot\frac{t}{t^2+1}\mathrm{e}^{-t^2/2} \geqslant \frac{\mathrm{e}^{-t^2/2}}{t},$$

    hence, letting $t_0 = \sqrt{\log n}$, we obtain

$$\mathbb{P}\{|G_1|>\sqrt{\log n}\} \geqslant \frac{\mathrm{e}^{-\frac{1}{2}\log n}}{\sqrt{\log n}} = \frac{n^{-1/2}}{\sqrt{\log n}} = \frac{1}{\sqrt{n\log n}},$$

    hence

$$\mathbb{P}\{M\leqslant\sqrt{\log n}\} = [1-\mathbb{P}\{|G_1|>\sqrt{\log n}\}]^n \leqslant \left(1-\frac{1}{\sqrt{n\log n}}\right)^n \leqslant \mathrm{e}^{-\frac{n}{\sqrt{n\log n}}} = \mathrm{e}^{-\sqrt{\frac{n}{\log n}}} \to 0,$$

    hence, when n is sufficiently large,

$$\mathbb{P}\{M>\sqrt{\log n}\} = 1-\mathbb{P}\{M\leqslant\sqrt{\log n}\} \geqslant 1-\mathrm{e}^{-\sqrt{\frac{n}{\log n}}} > \frac{1}{2},$$

    hence

$$\mathbb{E}[M] \geqslant t_0\cdot\mathbb{P}\{M>t_0\} \geqslant \sqrt{\log n}\cdot\frac{1}{2},$$

i.e., there exists $c > 0$ such that $\mathbb{E}[M] \geqslant c\sqrt{\log n}$.

- Prove $\mathbb{E}[1/\|\boldsymbol{G}\|_2] \asymp 1/\sqrt{n}$

  Let $R = \|\boldsymbol{G}\|_2 \sim \chi^2(n)$, whose density function is

$$f_R(r) = \frac{2^{1-n/2}}{\Gamma(n/2)} r^{n-1} \mathrm{e}^{-r^2/2} \mathbf{1}_{(0,\infty)},$$

  Then

$$\begin{aligned}
\mathbb{E}[R^{-1}] &= \int_0^\infty \frac{1}{r} f_R(r) \mathrm{d}r = \frac{2^{1-n/2}}{\Gamma(n/2)} \int_0^\infty r^{n-2} \mathrm{e}^{-r^2/2} \mathrm{d}r \\
&= \frac{2^{1-n/2}}{\Gamma(n/2)} \int_0^\infty (2u)^{\frac{n-2}{2}} \mathrm{e}^{-u} (2u)^{-1/2} \mathrm{d}u = \frac{1}{\sqrt{2}} \frac{\Gamma((n-1)/2)}{\Gamma(n/2)} && \left(u = \frac{r^2}{2}\right) \\
&\sim \frac{1}{\sqrt{2}} \frac{1}{\sqrt{n/2}} \asymp \frac{1}{\sqrt{n}}. && \left(\frac{\Gamma(z+1/2)}{\Gamma(z)} \sim \sqrt{z}\right)
\end{aligned}$$

  Thus,

$$\mathbb{E}\|\boldsymbol{X}\|_\infty \asymp \sqrt{\log n} \cdot \frac{1}{\sqrt{n}} = \sqrt{\frac{\log n}{n}}.$$

27 [**One-Dimensional Marginals of the Sphere and Ball**] Let $\boldsymbol{X} \sim U(\mathbb{S}^{n-1})$, $\boldsymbol{Y} \sim U(\mathbb{B}_2^n)$.

(1) Prove that the density function of any one-dimensional marginal (*projection distribution*) of $\boldsymbol{Y}$ is proportional to $(1-x^2)^{\frac{n-1}{2}}$, where $x \in [-1, 1]$.

(2) Prove that the density function of any one-dimensional marginal of $\boldsymbol{X}$ is proportional to $(1-x^2)^{\frac{n-3}{2}}$, where $x \in [-1, 1]$.

Rk. From (2) it follows that the one-dimensional marginal of the 3-dimensional sphere is $U(-1, 1)$, known as Archimedes' Hat-Box Theorem

Prf. (1) Since $\boldsymbol{Y} \sim U(\mathbb{B}_2^n)$, $f_{\boldsymbol{Y}}(\boldsymbol{y}) \propto \mathbf{1}_{\{\|\boldsymbol{y}\|_2 \leqslant 1\}}$, hence

$$f_{Y_1}(x) \propto \int \cdots \int_{\sum_{i=2}^n y_i^2 \leqslant 1-x^2} \mathrm{d}y_2 \cdots \mathrm{d}y_n = \int_{B(0, \sqrt{1-x^2})} \mathrm{d}y_2 \cdots \mathrm{d}y_n = V_{n-1}(\sqrt{1-x^2}),$$

where $V_k(r) = V_k(1) \cdot r^k$, hence

$$f_{Y_1}(x) \propto V_{n-1}(1) \cdot (\sqrt{1-x^2})^{n-1} \propto (1-x^2)^{\frac{n-1}{2}}.$$

(2) (*For $\boldsymbol{X} \sim U(\mathbb{S}^{n-1})$, the density $f_{X_1}(x)$ of its one-dimensional marginal is proportional to the surface area of the "annular band" at height x on the sphere.*)

Consider the region $X_1 \in [x, x + \mathrm{d}x]$, which is an $(n-2)$-dimensional sphere of radius $r = \sqrt{1-x^2}$. The surface area $\mathrm{d}A$ of the annular band is determined by two factors:

1. The surface area measure $\mathrm{d}\sigma$ of the cross-section (*the $(n-2)$-dimensional sphere*), satisfying

$$\mathrm{d}\sigma \propto r^{n-2} = (1-x^2)^{\frac{n-2}{2}}.$$

2. The arc length $\mathrm{d}s$ along the meridian direction of the sphere, satisfying

$$\mathrm{d}s = \frac{\mathrm{d}x}{\sin\theta} = \frac{\mathrm{d}x}{\sqrt{1-x^2}} = (1-x^2)^{-1/2} \mathrm{d}x,$$

where θ is the angle between the tangent vector and the vertical axis.

Therefore

$$f_{X_1}(x) \propto (1-x^2)^{\frac{n-2}{2}} \cdot (1-x^2)^{-1/2} = (1-x^2)^{\frac{n-3}{2}}.$$

Rk. In particular, when $n = 3$, $f_{X_1}(x) \propto (1-x^2)^0 = 1$, i.e., $X_1 \sim U(-1, 1)$; when $n = 2$, $f_{X_1}(x) \propto (1-x^2)^{-1/2}$, which is the Arcsine distribution.

28 [**The Big Cube Probability Paradox**] Let $\boldsymbol{g} \sim N(\mathbf{0}, \boldsymbol{I}_n)$. We will prove that: $\boldsymbol{g}$ is close with high probability in Euclidean distance to a unit cube of fixed side length, but simultaneously extremely unlikely to fall inside a cube even with side length a

hundred times larger.

(1) Let $\pi_a : \mathbb{R}^n \to [-a,a]^n$ be the metric projection onto the cube $[-a,a]^n$, i.e.,

$$\pi_a(\boldsymbol{x}) = \arg\min_{\boldsymbol{y}\in[-a,a]^n} \|\boldsymbol{x}-\boldsymbol{y}\|_2.$$

Prove that $\boldsymbol{y}$ can be obtained by truncating the coordinates, i.e., for every i,

$$y_i = \begin{cases} x_i, & |x_i| \leqslant a, \\ a\cdot \operatorname{sgn}(x_i), & |x_i| > a \end{cases}.$$

(2) Prove that there exist a, n_0 such that when $n > n_0$, $\boldsymbol{g} \sim N(0, \boldsymbol{I}_n)$ simultaneously satisfies with high probability,

$$\operatorname{dist}(\boldsymbol{g}, [-a,a]^n) < 0.01\|\boldsymbol{g}\|_2, \boldsymbol{g} \notin [-100a, 100a]^n \text{ (w.p. 0.99)}.$$

Prf. (1) It is easy to see that π_a is the solution to the optimization problem

$$\min_{\boldsymbol{y}} \sum_{i=1}^n (x_i - y_i)^2 \quad \text{s.t.} \quad |y_i| \leqslant a.$$

Since the objective function is separable (*the pairs (x_i, y_i) are uncoupled*) and the constraint set is a hypercube, the above problem is equivalent to

$$\min_{|y_i|\leqslant a} (x_i - y_i)^2.$$

a. If $|x_i| \leqslant a$, the minimum is attained at $y_i = x_i$, with distance 0.

b. If $x_i > a$, the minimum is attained at the boundary $y_i = a$.

c. If $x_i < -a$, the minimum is attained at the boundary $y_i = -a$.

Thus, $y_i = \operatorname{clip}(x_i, -a, a)$.

(2) Let $\mathcal{E}_1 = \{\operatorname{dist}(\boldsymbol{g}, [-a,a]^n) < 0.01\|\boldsymbol{g}\|_2\}$, $\mathcal{E}_2 = \{\boldsymbol{g} \notin [-100a, 100a]^n\}$.

(*It then suffices to prove $\mathbb{P}\{\mathcal{E}_1 \cap \mathcal{E}_2\} \geqslant 0.99$, i.e., $\mathbb{P}(\mathcal{E}_1), \mathbb{P}(\mathcal{E}_2) \geqslant 0.995$.*)

- Prove $\mathbb{P}(\mathcal{E}_1) \geqslant 0.995$

  Let $D_n^2 = \operatorname{dist}(\boldsymbol{g}, [-a,a]^n)^2$, $V_n^2 = \|\boldsymbol{g}\|_2^2 = \sum_{i=1}^n G_i^2$.

  For V_n^2, by the WLLN,

$$\frac{1}{n} V_n^2 \xrightarrow{\mathbb{P}} \mathbb{E}[G_1^2] = 1.$$

  For D_n^2, since

$$D_n^2 = \operatorname{dist}(\boldsymbol{g}, [-a,a]^n)^2 = \sum_{i=1}^n (|g_i| - a)_+^2,$$

  where $g_1, \cdots, g_n$ i.i.d. $\sim N(0,1)$. Hence, by the WLLN, letting $h(s) = (|s| - a)_+^2$, we have

$$\frac{1}{n} D_n^2 = \frac{1}{n}\sum_{i=1}^n h(G_i) \xrightarrow{\mathbb{P}} \mathbb{E}[h(G_1)].$$

  Define

$$f(a) = \mathbb{E}[(|g_1| - a)_+^2] = 2\int_a^\infty (x-a)^2 \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-x^2/2} \mathrm{d}x,$$

  By the Lebesgue Dominated Convergence Theorem, $\lim_{a\to\infty} f(a) = 0$. Hence there exists $a_0 > 0$ such that when $a > a_0$,

$$f(a) < \frac{(0.01)^2}{2},$$

  hence

$$\frac{D_n^2}{V_n^2} = \frac{D_n^2/n}{V_n^2/n} \xrightarrow{\mathbb{P}} \frac{f(a)}{1} < \frac{(0.01)^2}{2},$$

Hence there exists n_1 such that when $n \geqslant n_1$, $\mathbb{P}\{D_n/V_n < 0.01\} \geqslant 0.995$, i.e., $\mathbb{P}(\mathcal{E}_1) \geqslant 0.995$.

- Prove $\mathbb{P}(\mathcal{E}_2) \geqslant 0.995$

  For $\mathcal{E}_2$, consider $\mathcal{E}_2^c = \{\boldsymbol{g} \in [-100a, 100a]^n\}$. Then

$$\mathbb{P}(\mathcal{E}_2^c) = \mathbb{P}\left\{\max_{1\leqslant i\leqslant n} |G_i| \leqslant 100a\right\} = \prod_{i=1}^{n} \mathbb{P}\{|G_i| \leqslant 100a\} = [\mathbb{P}\{|G_1| \leqslant 100a\}]^n \to 0,$$

  Hence there exists n_2 such that when $n \geqslant n_2$, $\mathbb{P}(\mathcal{E}_2^c) \leqslant 0.005$, i.e., $\mathbb{P}(\mathcal{E}_2) \geqslant 0.995$.

Thus, let $n_0 = \max\{n_1, n_2\}$. Then when $n > n_0$,

$$\mathbb{P}\{\mathcal{E}_1 \cap \mathcal{E}_2\} \geqslant \mathbb{P}(\mathcal{E}_1) + \mathbb{P}(\mathcal{E}_2) - 1 \geqslant 0.995 + 0.995 - 1 = 0.99.$$

29 [**Parseval Frames and Orthogonal Matrices**] Let $\boldsymbol{U} \in \mathbb{R}^{n\times N}$. Prove that the column vectors of $\boldsymbol{U}$ form a Parseval frame for $\mathbb{R}^n$ if and only if the row vectors of $\boldsymbol{U}$ are orthonormal.

Prf. Denote the column vectors and row vectors of $\boldsymbol{U}$ by $\{\boldsymbol{u}_i\}_{i=1}^N$ and $\{\boldsymbol{r}_i\}_{i=1}^n$, respectively.

For the column vectors, if $\{\boldsymbol{u}_i\}_{i=1}^N$ forms a Parseval frame for $\mathbb{R}^n$, then First consider the column vectors. By the Parseval frame property, we have

$$\boldsymbol{U}\boldsymbol{U}^\top = \sum_{i=1}^{N} \boldsymbol{u}_i \boldsymbol{u}_i^\top = \boldsymbol{I}_n.$$

Now consider the row vectors. Since $(\boldsymbol{U}\boldsymbol{U}^\top)_{jk} = \langle \boldsymbol{r}_j, \boldsymbol{r}_k \rangle$, we have $\boldsymbol{U}\boldsymbol{U}^\top = \boldsymbol{I}_n$ if and only if

$$\langle \boldsymbol{r}_j, \boldsymbol{r}_k \rangle = \delta_{jk} = \begin{cases} 1, j = k, \\ 0, j \neq k \end{cases},$$

i.e., $\{\boldsymbol{r}_j\}_{j=1}^n$ is an orthonormal set of vectors in $\mathbb{R}^N$.

30 [**The Mercedes-Benz Frame**] Consider N equally spaced points lying on $\mathbb{S}^2(0, \sqrt{2/N})$ (*when $N = 3$, the configuration resembles the Mercedes-Benz logo*).

Prove that the set of vectors $\{\boldsymbol{x}_k\}_{k=1}^N$ formed by the above N points is a Parseval frame for $\mathbb{R}^2$ (*also known as a 1-tight frame*).

Prf. (*When $N = 2$, the two points are collinear and cannot span $\mathbb{R}^2$, so we only need to consider $N \geqslant 3$.*)

Let $R = \sqrt{2/N}$, and set

$$\boldsymbol{x}_k = R\begin{pmatrix} \cos\theta_k \\ \sin\theta_k \end{pmatrix}, \theta_k = \frac{2\pi(k-1)}{N},$$

Then

$$\begin{aligned} \boldsymbol{x}_k \boldsymbol{x}_k^\top &= R^2 \begin{pmatrix} \cos^2\theta_k & \cos\theta_k \sin\theta_k \\ \cos\theta_k \sin\theta_k & \sin^2\theta_k \end{pmatrix} \\ &= \frac{R^2}{2} \begin{pmatrix} 1+\cos 2\theta_k & \sin 2\theta_k \\ \sin 2\theta_k & 1-\cos 2\theta_k \end{pmatrix}, \end{aligned}$$

When $N \geqslant 3$ (*ensuring that the spacing of $2\theta_k$ is not a multiple of π*), by the balancing property of roots of unity,

$$\sum_{k=1}^{N} \cos 2\theta_k = \sum_{k=1}^{N} \sin 2\theta_k = 0,$$

hence

$$S_{11} = \frac{R^2}{2} \sum_{k=1}^{N} (1 + \cos 2\theta_k) = \frac{R^2}{2}(N+0) = \frac{NR^2}{2},$$

$$S_{22} = \frac{R^2}{2} \sum_{k=1}^{N} (1 - \cos 2\theta_k) = \frac{R^2}{2}(N-0) = \frac{NR^2}{2},$$

$$S_{12} = S_{21} = \frac{R^2}{2}\sum_{k=1}^{N}\sin 2\theta_k = 0,$$

i.e.,

$$\boldsymbol{S} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \boldsymbol{I}_2.$$

31 [**Conversion Between Isotropic Distributions and Parseval Frames**] Let $\boldsymbol{X} \in \mathbb{R}^n$ be an isotropic random variable taking values in the finite set $\{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N\} \subset \mathbb{R}^n$, with probabilities $\mathbb{P}\{\boldsymbol{X} = \boldsymbol{x}_k\} = p_k$ (*satisfying* $\sum_{k=1}^{N} p_k = 1$). Prove that the probability-weighted set of vectors $\{\sqrt{p_1}\boldsymbol{x}_1, \cdots, \sqrt{p_N}\boldsymbol{x}_N\}$ forms a Parseval frame for $\mathbb{R}^n$.

Prf. Let $\boldsymbol{u}_k = \sqrt{p_k}\boldsymbol{x}_k$. Then

$$\boldsymbol{I}_n = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \sum_{k=1}^{N} p_k(\boldsymbol{x}_k\boldsymbol{x}_k^\top) = \sum_{k=1}^{N} \boldsymbol{u}_k\boldsymbol{u}_k^\top,$$

Therefore, for any $\boldsymbol{v} \in \mathbb{R}^n$,

$$\sum_{k=1}^{N}\langle \boldsymbol{v}, \boldsymbol{u}_k\rangle^2 = \sum_{k=1}^{N}\boldsymbol{v}^\top(\boldsymbol{u}_k\boldsymbol{u}_k^\top)\boldsymbol{v} = \boldsymbol{v}^\top\left(\sum_{k=1}^{N}\boldsymbol{u}_k\boldsymbol{u}_k^\top\right)\boldsymbol{v} = \boldsymbol{v}^\top\boldsymbol{I}_n\boldsymbol{v} = \|\boldsymbol{v}\|^2,$$

hence $\{\sqrt{p_k}\boldsymbol{x}_k\}_{k=1}^{N}$ forms a Parseval frame for $\mathbb{R}^n$.

32 [**Sub-Gaussian Norm of One-Dimensional Marginals**] Let $\boldsymbol{X} \in \mathbb{R}^n$ be a sub-Gaussian random vector. Prove that for any $\boldsymbol{v} \in \mathbb{R}^n$, the sub-Gaussian norm of its one-dimensional marginal (*i.e., the inner product* $\langle \boldsymbol{X}, \boldsymbol{v}\rangle$) satisfies

$$\|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} \leqslant \|\boldsymbol{X}\|_{\psi_2}\|\boldsymbol{v}\|_2,$$

Rk. The sub-Gaussian norm of Y is defined as

$$\|Y\|_{\psi_2} = \inf\{t > 0 : \mathbb{E}[\mathrm{e}^{Y^2/t^2}] \leqslant 2\},$$

The sub-Gaussian norm of $\boldsymbol{X}$ is defined as the maximum sub-Gaussian norm of its projections onto the unit sphere, i.e.,

$$\|\boldsymbol{X}\|_{\psi_2} = \sup_{\boldsymbol{u}\in\mathbb{S}^{n-1}}\|\langle \boldsymbol{X}, \boldsymbol{u}\rangle\|_{\psi_2}.$$

Prf. If $\boldsymbol{v} = \boldsymbol{0}$, then $\langle \boldsymbol{X}, \boldsymbol{0}\rangle = 0$, and the conclusion is trivial. We only need to consider $\boldsymbol{v} \neq \boldsymbol{0}$.

Decompose $\boldsymbol{v}$ as $\boldsymbol{v} = \|\boldsymbol{v}\|_2 \cdot \bar{\boldsymbol{v}}$, where $\bar{\boldsymbol{v}} \in \mathbb{S}^{n-1}$. Then

$$\|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} = \|\langle \boldsymbol{X}, \|\boldsymbol{v}\|_2\bar{\boldsymbol{v}}\rangle\|_{\psi_2} = \|\boldsymbol{v}\|_2\|\langle \boldsymbol{X}, \bar{\boldsymbol{v}}\rangle\|_{\psi_2},$$

where

$$\|\langle \boldsymbol{X}, \bar{\boldsymbol{v}}\rangle\|_{\psi_2} \leqslant \sup_{\boldsymbol{z}\in\mathbb{S}^{n-1}}\|\langle \boldsymbol{X}, \boldsymbol{z}\rangle\|_{\psi_2} = \|\boldsymbol{X}\|_{\psi_2},$$

substituting it back, we have

$$\|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} \leqslant \|\boldsymbol{v}\|_2 \cdot \|\boldsymbol{X}\|_{\psi_2}.$$

33 [**Rotational Invariance of the Sub-Gaussian Norm**] Proposition P3.3.1 (*Rotation invariance*) showed that the standard normal distribution is rotation invariant. Sub-Gaussian distributions need not in general satisfy rotational invariance, but the sub-Gaussian norm does.

Let $\boldsymbol{U} \in \mathbb{R}^{n\times n}$ be an orthogonal matrix and $\boldsymbol{X} \in \mathbb{R}^n$ a sub-Gaussian random vector. Prove that $\boldsymbol{U}\boldsymbol{X}$ is a sub-Gaussian random vector satisfying

$$\|\boldsymbol{U}\boldsymbol{X}\|_{\psi_2} = \|\boldsymbol{X}\|_{\psi_2}.$$

Prf. For

$$\|\boldsymbol{U}\boldsymbol{X}\|_{\psi_2} = \sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}}\|\langle \boldsymbol{U}\boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} = \sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}}\|\langle \boldsymbol{X}, \boldsymbol{U}^\top\boldsymbol{v}\rangle\|_{\psi_2},$$

let $\boldsymbol{w} = \boldsymbol{U}^\top \boldsymbol{v}$. Then

$$\|\boldsymbol{w}\|_2 = \|\boldsymbol{U}^\top \boldsymbol{v}\|_2 = \|\boldsymbol{v}\|_2,$$

hence $\boldsymbol{v} \mapsto \boldsymbol{U}^\top \boldsymbol{v}$ is a bijection from $\mathbb{S}^{n-1}$ onto itself, so

$$\|\boldsymbol{U}\boldsymbol{X}\|_{\psi_2} = \sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}} \|\langle \boldsymbol{X}, \boldsymbol{U}^\top \boldsymbol{v}\rangle\|_{\psi_2} = \sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}} \|\langle \boldsymbol{X}, \boldsymbol{w}\rangle\|_{\psi_2} = \|\boldsymbol{X}\|_{\psi_2}.$$

i.e., $\boldsymbol{U}\boldsymbol{X}$ is a sub-Gaussian random vector.

34 [**Sub-Gaussian Bilinear Forms**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix whose row vectors $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_m \in \mathbb{R}^n$ are independent zero-mean K-sub-Gaussian random vectors.

(1) Prove that for any $\boldsymbol{u} \in \mathbb{R}^m, \boldsymbol{v} \in \mathbb{R}^n$, $\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v}$ is a sub-Gaussian random vector satisfying

$$\|\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v}\|_{\psi_2} \leqslant C\|\boldsymbol{u}\|_2\|\boldsymbol{v}\|_2 \max_{1\leqslant i\leqslant m} \|\boldsymbol{A}_i\|_{\psi_2}. \qquad (C>0)$$

(2) Prove that the conclusion still holds if the condition of row-wise independent sub-Gaussian random vectors is replaced by column-wise independent sub-Gaussian random vectors.

Prf. (1) It is easy to see that

$$\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v} = \sum_{i=1}^m u_i(\boldsymbol{A}_i\boldsymbol{v}) = \sum_{i=1}^m u_i\langle \boldsymbol{A}_i, \boldsymbol{v}\rangle,$$

Let $Z_i = \langle \boldsymbol{A}_i, \boldsymbol{v}\rangle$. Using the conclusion of Exercise Q3.32 (*Sub-Gaussian Norm of One-Dimensional Marginals*), we have

$$\|Z_i\|_{\psi_2} \leqslant K\|\boldsymbol{v}\|_2 = K\|\boldsymbol{v}\|_2,$$

Therefore, by Proposition P2.7.1 (*Subgaussian norm of a sum*),

$$\begin{aligned}\|\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v}\|_{\psi_2}^2 = \left\|\sum_{i=1}^m u_i Z_i\right\|_{\psi_2}^2 &\leqslant C'\sum_{i=1}^m \|u_i Z_i\|_{\psi_2}^2 = C'\sum_{i=1}^m u_i^2\|Z_i\|_{\psi_2}^2 \qquad (C'>0)\\ &\leqslant C'\sum_{i=1}^m u_i^2(K\|\boldsymbol{v}\|_2)^2 = C'K^2\|\boldsymbol{v}\|_2^2\sum_{i=1}^m u_i^2 = C'K^2\|\boldsymbol{v}\|_2^2\|\boldsymbol{u}\|_2^2,\end{aligned}$$

Taking square roots on both sides,

$$\|\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v}\|_{\psi_2} \leqslant C\|\boldsymbol{u}\|_2\|\boldsymbol{v}\|_2 \max_i \|\boldsymbol{A}_i\|_{\psi_2}. \qquad (C=\sqrt{C'}>0)$$

(2) If the column vectors $\boldsymbol{C}_1, \cdots, \boldsymbol{C}_n \in \mathbb{R}^m$ of $\boldsymbol{A}$ are independent sub-Gaussian random vectors, then the row vectors $\boldsymbol{C}_1^\top, \cdots, \boldsymbol{C}_n^\top \in \mathbb{R}^m$ of $\boldsymbol{A}^\top$ are independent sub-Gaussian random vectors.

Note that the bilinear form $\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v}$ is a scalar, satisfying

$$\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v} = (\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v})^\top = \boldsymbol{v}^\top \boldsymbol{A}^\top \boldsymbol{u},$$

Hence, using the conclusion of (1), we have

$$\|\boldsymbol{v}^\top \boldsymbol{A}^\top \boldsymbol{u}\|_{\psi_2} \leqslant C\|\boldsymbol{v}\|_2\|\boldsymbol{u}\|_2 \max_{1\leqslant j\leqslant n} \|\boldsymbol{C}_j^\top\|_{\psi_2},$$

i.e.,

$$\|\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v}\|_{\psi_2} \leqslant C\|\boldsymbol{u}\|_2\|\boldsymbol{v}\|_2 \max_{1\leqslant j\leqslant n} \|\boldsymbol{C}_j\|_{\psi_2}.$$

35 [**Sub-Gaussian Norm of a Sum of Random Vectors**] The property for one-dimensional random variables can be generalized to high-dimensional spaces, namely Proposition P2.7.1 (*Subgaussian norm of a sum*).

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N \in \mathbb{R}^n$ be independent zero-mean sub-Gaussian random vectors. Prove that

$$\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_{\psi_2}^2 \leqslant C\sum_{i=1}^N \|\boldsymbol{X}_i\|_{\psi_2}^2. \qquad (C>0)$$

Prf. Let $\boldsymbol{S}=\sum_{i=1}^N \boldsymbol{X}_i$. Then for any $\boldsymbol{v}\in\mathbb{S}^{n-1}$,

$$\langle \boldsymbol{S},\boldsymbol{v}\rangle=\left\langle \sum_{i=1}^N \boldsymbol{X}_i,\boldsymbol{v}\right\rangle=\sum_{i=1}^N\langle \boldsymbol{X}_i,\boldsymbol{v}\rangle,$$

Let $Y_i=\langle \boldsymbol{X}_i,\boldsymbol{v}\rangle$. Then $Y_1,\cdots,Y_N$ are independent zero-mean random variables. Hence, by Proposition P2.7.1 (*Subgaussian norm of a sum*),

$$\left\|\sum_{i=1}^N Y_i\right\|_{\psi_2}^2\leqslant C\sum_{i=1}^N\|Y_i\|_{\psi_2}^2, \qquad (C>0)$$

i.e.,

$$\|\langle \boldsymbol{S},\boldsymbol{v}\rangle\|_{\psi_2}^2\leqslant C\sum_{i=1}^N\|\langle \boldsymbol{X}_i,\boldsymbol{v}\rangle\|_{\psi_2}^2,$$

where

$$\|\langle \boldsymbol{X}_i,\boldsymbol{v}\rangle\|_{\psi_2}\leqslant\sup_{\boldsymbol{u}\in\mathbb{S}^{n-1}}\|\langle \boldsymbol{X}_i,\boldsymbol{u}\rangle\|_{\psi_2}=\|\boldsymbol{X}_i\|_{\psi_2},$$

substituting it back, we have

$$\|\langle \boldsymbol{S},\boldsymbol{v}\rangle\|_{\psi_2}^2\leqslant C\sum_{i=1}^N\|\boldsymbol{X}_i\|_{\psi_2}^2,$$

Taking the supremum over $\boldsymbol{v}$ on both sides,

$$\|\boldsymbol{S}\|_{\psi_2}^2=\left(\sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}}\|\langle \boldsymbol{S},\boldsymbol{v}\rangle\|_{\psi_2}\right)^2=\sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}}\|\langle \boldsymbol{S},\boldsymbol{v}\rangle\|_{\psi_2}^2,$$

i.e.,

$$\|\boldsymbol{S}\|_{\psi_2}^2\leqslant C\sum_{i=1}^N\|\boldsymbol{X}_i\|_{\psi_2}^2.$$

36 [**Random Vectors with Sub-Gaussian Coordinates**] In Lemma L3.4.2 (*Distributions with independent subgaussian coordinates*), if the coordinates are mutually independent, the high-dimensional sub-Gaussian norm satisfies the dimension-free bound $\|\boldsymbol{X}\|_{\psi_2}\leqslant C\max_{1\leqslant i\leqslant n}\|X_i\|_{\psi_2}$. This exercise studies how the norm bound changes after dropping the independence assumption.

Let $\boldsymbol{X}\in\mathbb{R}^n$ be a random vector whose coordinates $X_1,\cdots,X_n$ are sub-Gaussian random variables (*not necessarily independent*).

(1) Prove that $\boldsymbol{X}$ is a sub-Gaussian random vector satisfying

$$\max_{i\leqslant n}\|X_i\|_{\psi_2}\leqslant\|\boldsymbol{X}\|_{\psi_2}\leqslant\sqrt{n}\max_{i\leqslant n}\|X_i\|_{\psi_2}.$$

(2) Show that the bound in (1) is optimal.

Prf. (1) By Definition D3.4.1 (*Subgaussian random vectors*), we have

$$\|\boldsymbol{X}\|_{\psi_2}=\sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}}\|\langle \boldsymbol{X},\boldsymbol{v}\rangle\|_{\psi_2}.$$

For the lower bound, let $\boldsymbol{v}=\boldsymbol{e}_i=(0,\cdots,1,\cdots,0)^\top\in\mathbb{S}^{n-1}$. Then

$$\|\boldsymbol{X}\|_{\psi_2}\geqslant\|\langle \boldsymbol{X},\boldsymbol{e}_i\rangle\|_{\psi_2}=\|X_i\|_{\psi_2}.$$

For the upper bound, let $\boldsymbol{v}=(v_1,\cdots,v_n)^\top\in\mathbb{S}^{n-1}$ satisfy $\sum_{i=1}^n v_i^2=1$. Then

$$\|\langle \boldsymbol{X},\boldsymbol{v}\rangle\|_{\psi_2}=\left\|\sum_{i=1}^n v_iX_i\right\|_{\psi_2}\leqslant\sum_{i=1}^n|v_i|\|X_i\|_{\psi_2}\leqslant K\sum_{i=1}^n|v_i|, \qquad \left(K=\max_{1\leqslant i\leqslant n}\|X_i\|_{\psi_2}\right)$$

By Cauchy's inequality,

$$\sum_{i=1}^n |v_i| \cdot 1 \leqslant \sqrt{\sum_{i=1}^n v_i^2}\sqrt{\sum_{i=1}^n 1^2} = 1 \cdot \sqrt{n} = \sqrt{n},$$

substituting it back,

$$\|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} \leqslant \sqrt{n}K,$$

Taking the supremum on both sides,

$$\|\boldsymbol{X}\|_{\psi_2} \leqslant \sqrt{n}\max_{i\leqslant n}\|X_i\|_{\psi_2}.$$

(2) For the lower bound, let $X_1, \cdots, X_n$ i.i.d. $\sim N(0,1)$. Then for any $\boldsymbol{v} \in \mathbb{S}^{n-1}$,

$$\langle \boldsymbol{X}, \boldsymbol{v}\rangle = \sum_{i=1}^n v_i X_i = N\left(0, \sum_{i=1}^n v_i^2\right) = N(0,1),$$

hence $\|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} = \|g\|_{\psi_2}$.

For the upper bound, let $X_1 = \cdots = X_n = X \sim N(0,1)$. Then

$$\|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} = \left\|\sum_{i=1}^n v_i X\right\|_{\psi_2} \leqslant \|X\|_{\psi_2}\sum_{i=1}^n |v_i|,$$

Let $\boldsymbol{v} = (1/\sqrt{n}, \cdots, 1/\sqrt{n})^\top$. Then

$$\|\boldsymbol{X}\|_{\psi_2} = \sqrt{n}\|X\|_{\psi_2},$$

i.e.,

$$\|\boldsymbol{X}\|_{\psi_2} = \sqrt{n}\max_{1\leqslant i\leqslant n}\|X_i\|_{\psi_2}.$$

37 [**Counterexample: Sub-Gaussian Vector Norm May Not Concentrate**] Theorem T3.1.1 (*Concentration of the norm*) states that if the coordinates X_i of a random vector $\boldsymbol{X} = (X_1, \cdots, X_n)$ are independent and standardized sub-Gaussian random variables, then its Euclidean norm $\|\boldsymbol{X}\|_2$ is highly concentrated around $\sqrt{n}$.

In fact, the concentration of the Euclidean norm relies heavily on the independence of the coordinates. Even if a sub-Gaussian random vector is isotropic, the norm may fail to concentrate at all around $\sqrt{n}$ if there is strong coupling among the coordinates. Construct a counterexample that violates the above conclusion when the coordinate independence assumption is dropped.

Sol. Let $\boldsymbol{G} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$, $\eta \sim$ Bernoulli(1/2), and define $\boldsymbol{X} = \sqrt{2}\eta\boldsymbol{G}$. Then:

1. Isotropy:

$$\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \mathbb{E}[(\sqrt{2}\eta\boldsymbol{G})(\sqrt{2}\eta\boldsymbol{G})^\top] = 2\mathbb{E}[\eta^2]\mathbb{E}[\boldsymbol{G}\boldsymbol{G}^\top] = 2\mathbb{E}[\eta]\mathbb{E}[\boldsymbol{G}\boldsymbol{G}^\top] = 2 \cdot \frac{1}{2} \cdot \boldsymbol{I}_n = \boldsymbol{I}_n.$$

2. Sub-Gaussian property:

For any $\boldsymbol{v} \in \mathbb{S}^{n-1}$, consider the projection $\langle \boldsymbol{X}, \boldsymbol{v}\rangle = \sqrt{2}\eta\langle \boldsymbol{G}, \boldsymbol{v}\rangle$, where $Z = \langle \boldsymbol{G}, \boldsymbol{v}\rangle \sim N(0,1)$. Hence

$$\|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} = \sqrt{2}\|\eta Z\|_{\psi_2} \leqslant \sqrt{2}\|Z\|_{\psi_2} = \sqrt{2}K_0,$$

Taking the supremum over $\boldsymbol{v}$,

$$\|\boldsymbol{X}\|_{\psi_2} \leqslant \sqrt{2}K_0.$$

Hence $\boldsymbol{X}$ is an isotropic sub-Gaussian vector.

However, for $\|\boldsymbol{X}\|_2 = \sqrt{2}\eta\|\boldsymbol{G}\|_2$, we have

$$\mathbb{P}\{\|\boldsymbol{X}\|_2 = 0\} = \mathbb{P}\{\|\boldsymbol{X}\|_2 = \sqrt{2}\|\boldsymbol{G}\|_2\} = \frac{1}{2},$$

hence

$$\mathbb{P}\{|\|\boldsymbol{X}\|_2 - \sqrt{n}| \geqslant 0.1\sqrt{n}\} = \frac{1}{2} + \frac{1}{2} = 1.$$

38 [**Sub-Gaussian Norm of Gaussian Random Vectors**] In Exercise Q2.24 (*Examples of Sub-Gaussian Distributions*) (3), we proved that the sub-Gaussian norm of a one-dimensional normal random variable $Y \sim N(0, \sigma^2)$ satisfies $\|Y\|_{\psi_2} = \sqrt{8/3}\sigma$. Generalize this conclusion to the high-dimensional setting: Let $\boldsymbol{X} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma})$. Prove that

$$\|\boldsymbol{X}\|_{\psi_2} = \sqrt{\frac{8}{3}}\|\boldsymbol{\Sigma}\|^{1/2},$$

where $\|\boldsymbol{\Sigma}\| = \lambda_{\max}(\boldsymbol{\Sigma})$.

Prf. The sub-Gaussian norm of $\boldsymbol{X}$ is

$$\|\boldsymbol{X}\|_{\psi_2} = \sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}} \|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2},$$

Let $Y_{\boldsymbol{v}} = \langle \boldsymbol{X}, \boldsymbol{v}\rangle \sim N(0, \boldsymbol{v}^\top \boldsymbol{\Sigma} \boldsymbol{v})$. Then

$$\|\boldsymbol{X}\|_{\psi_2} = \sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}} \sqrt{\frac{8}{3}}(\boldsymbol{v}^\top \boldsymbol{\Sigma} \boldsymbol{v})^{1/2} = \sqrt{\frac{8}{3}}\left(\sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}} \boldsymbol{v}^\top \boldsymbol{\Sigma} \boldsymbol{v}\right)^{1/2},$$

hence

$$\|Y_{\boldsymbol{v}}\|_{\psi_2} = \sqrt{\frac{8}{3}}\sigma_{\boldsymbol{v}} = \sqrt{\frac{8}{3}}(\boldsymbol{v}^\top \boldsymbol{\Sigma} \boldsymbol{v})^{1/2} = \sqrt{\frac{8}{3}}\lambda_{\max}(\boldsymbol{\Sigma}) = \sqrt{\frac{8}{3}}\|\boldsymbol{\Sigma}\|.$$

39 [**Properties of the Euclidean Norm of Sub-Gaussian Random Vectors**]
(1) Let $X_1, \cdots, X_n$ be sub-Gaussian random variables (*not necessarily independent*). Prove that

$$\left\| \left(\sum_{i=1}^n X_i^2 \right)^{1/2} \right\|_{\psi_2} \leqslant \left(\sum_{i=1}^n \|X_i\|_{\psi_2}^2 \right)^{1/2}.$$

(2) Let $\boldsymbol{X} \in \mathbb{R}^n$ be a sub-Gaussian random vector. Prove that $\|\boldsymbol{X}\|_2$ is a sub-Gaussian random variable satisfying

$$\|\|\boldsymbol{X}\|_2\|_{\psi_2} \leqslant \sqrt{n}\|\boldsymbol{X}\|_{\psi_2}.$$

Then explain why $\sqrt{n}$ is the best scaling factor in general. In Proposition P6.2.1 (*The norm of a subgaussian random vector*), we will prove a more informative bound.

Prf. (1) Let

$$K_i = \|X_i\|_{\psi_2}, \qquad K = \left(\sum_{i=1}^n K_i^2 \right)^{1/2} = \left(\sum_{i=1}^n \|X_i\|_{\psi_2}^2 \right)^{1/2}.$$

To prove that $Y := (\sum_{i=1}^n X_i^2)^{1/2}$ satisfies $\|Y\|_{\psi_2} \leqslant K$, it suffices to prove

$$\mathbb{E}\left[\exp\left(\frac{Y^2}{K^2}\right)\right] = \mathbb{E}\left[\exp\left(\frac{\sum_{i=1}^n X_i^2}{K^2}\right)\right] \leqslant 2.$$

Construct the nonnegative convex-combination weights $w_i = K_i^2/K^2$, satisfying $w_i \geqslant 0$, $\sum_{i=1}^n w_i = 1$. Rewrite the exponent as a convex combination based on the weights w_i,

$$\frac{\sum_{i=1}^n X_i^2}{K^2} = \sum_{i=1}^n \left(\frac{K_i^2}{\sum_{j=1}^n K_j^2} \right) \frac{X_i^2}{K_i^2} = \sum_{i=1}^n w_i \frac{X_i^2}{K_i^2},$$

Hence, by Jensen's inequality,

$$\exp\left(\sum_{i=1}^n w_i \frac{X_i^2}{K_i^2} \right) \leqslant \sum_{i=1}^n w_i \exp\left(\frac{X_i^2}{K_i^2} \right),$$

Taking expectations on both sides,

$$\mathbb{E}\left[\exp\left(\frac{\sum_{i=1}^n X_i^2}{K^2} \right)\right] \leqslant \sum_{i=1}^n w_i \mathbb{E}\left[\exp\left(\frac{X_i^2}{K_i^2} \right)\right] \leqslant \sum_{i=1}^n w_i \cdot 2 = 2.$$

(2) Using the conclusion of (1), we have

$$\|\|\boldsymbol{X}\|_2\|_{\psi_2} \leqslant \left(\sum_{i=1}^n \|X_i\|_{\psi_2}^2\right)^{1/2},$$

where

$$\|X_i\|_{\psi_2} = \|\langle \boldsymbol{X}, \boldsymbol{e}_i\rangle\|_{\psi_2} \leqslant \sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}} \|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2} = \|\boldsymbol{X}\|_{\psi_2},$$

substituting it back, we have

$$\|\|\boldsymbol{X}\|_2\|_{\psi_2} \leqslant \left(\sum_{i=1}^n \|\boldsymbol{X}\|_{\psi_2}^2\right)^{1/2} = \sqrt{n}\|\boldsymbol{X}\|_{\psi_2}.$$

40 [**Almost Orthogonality of High-Dimensional Random Vectors**] Two random vectors drawn independently from the uniform distribution on $\mathbb{S}^{n-1}$ are "almost orthogonal" with high probability. This exercise aims to generalize this geometric phenomenon to general coordinate-wise independent sub-Gaussian random vectors.

(1) Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector whose entries are mutually independent standardized K-sub-Gaussian random variables. Define the direction vector

$$\boldsymbol{\theta} = \begin{cases} \bar{\boldsymbol{X}}, & \boldsymbol{X} \neq \boldsymbol{0}, \\ \boldsymbol{0}, & \boldsymbol{X} = \boldsymbol{0} \end{cases},$$

and let $\boldsymbol{\eta}$ be an independent copy of $\boldsymbol{\theta}$. Prove:

$$|\langle \boldsymbol{\theta}, \boldsymbol{\eta}\rangle| \leqslant \frac{C_K}{\sqrt{n}} \text{ (w.p. 0.99).} \qquad (C_K > 0)$$

(2) It remains to explain why the coordinate independence assumption in (1) is crucial for the almost orthogonality of random vectors. Construct independent, identically distributed, isotropic K-sub-Gaussian random vectors $\boldsymbol{X}, \boldsymbol{Y}$, such that

$$\mathbb{P}\{\boldsymbol{X} = \boldsymbol{Y}\} \geqslant 0.99,$$

This shows that if the coordinate independence assumption is dropped, $\langle \boldsymbol{\theta}, \boldsymbol{\eta}\rangle$ cannot decay at the rate $O(1/\sqrt{n})$.

Prf. (1) Let $\boldsymbol{Y}$ be an independent copy of $\boldsymbol{X}$, and let $\boldsymbol{\eta} = \bar{\boldsymbol{Y}}$.

- Concentration estimate for the Euclidean norm

  By Theorem T3.1.1 (*Concentration of the norm*), we have

  $$\mathbb{P}\{|\|\boldsymbol{X}\|_2 - \sqrt{n}| \geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{K^4}\right). \qquad (c > 0)$$

  Let $t = \sqrt{n}/2$. Define the benign event that $\boldsymbol{X}$ concentrates near its expectation as $\mathcal{E}_X = \{\sqrt{n}/2 \leqslant \|\boldsymbol{X}\|_2 \leqslant 3\sqrt{n}/2\}$. Then

  $$\mathbb{P}(\mathcal{E}_X^c) \leqslant \mathbb{P}\left\{|\|\boldsymbol{X}\|_2 - \sqrt{n}| \geqslant \frac{1}{2}\sqrt{n}\right\} \leqslant 2\exp\left(-\frac{cn}{4K^4}\right), \qquad (c > 0)$$

  (*There is no need to control the upper bound of $\|\boldsymbol{Y}\|_2$ by defining $\mathcal{E}_Y$, since the direction vector $\boldsymbol{\eta}$ is itself a unit vector; we may simply condition on $\boldsymbol{\eta}$.*)

  Hence when n is sufficiently large,

  $$\mathbb{P}(\mathcal{E}_X^c) \leqslant 0.005. \qquad (\mathrm{I})$$

- Sub-Gaussian property of the conditional inner product

  Given $\boldsymbol{\eta} = \boldsymbol{v}$, the conditional inner product is

  $$\langle \boldsymbol{X}, \boldsymbol{v}\rangle \mid \boldsymbol{\eta} = \boldsymbol{v} = \sum_{i=1}^n v_i X_i$$

  which is a weighted sum of mutually independent sub-Gaussian random variables. Hence, by Proposition P2.7.1

(*Subgaussian norm of a sum*), we have

$$\|\langle \boldsymbol{X}, \boldsymbol{v}\rangle\|_{\psi_2}^2 \leqslant C_1 \sum_{i=1}^n \|v_i X_i\|_{\psi_2}^2 \leqslant C_1 \sum_{i=1}^n v_i^2 K^2 = C_1 K^2 \|\boldsymbol{v}\|_2^2 = C_1 K^2, \qquad (C_1 > 0)$$

hence

$$\mathbb{P}\{|\langle \boldsymbol{X}, \boldsymbol{v}\rangle| \geqslant t \mid \boldsymbol{\eta} = \boldsymbol{v}\} \leqslant 2\exp\left(-\frac{c_1 t^2}{C_1 K^2}\right). \qquad (c_1 > 0)$$

- Estimate the inner product by the law of total probability

  Consider the inner product of the direction vectors,

  $$|\langle \boldsymbol{\theta}, \boldsymbol{\eta}\rangle| = \frac{|\langle \boldsymbol{X}, \boldsymbol{\eta}\rangle|}{\|\boldsymbol{X}\|_2},$$

  Decompose it on $\mathcal{E}_X^c$,

  $$\mathbb{P}\left\{|\langle \boldsymbol{\theta}, \boldsymbol{\eta}\rangle| \geqslant \frac{C_K}{\sqrt{n}}\right\} \leqslant \mathbb{P}(\mathcal{E}_X^c) + \mathbb{P}\left\{\frac{|\langle \boldsymbol{X}, \boldsymbol{\eta}\rangle|}{\|\boldsymbol{X}\|_2} \geqslant \frac{C_K}{\sqrt{n}} \cap \mathcal{E}_X\right\}.$$

  (*It then suffices to estimate the second term.*)

  When $\mathcal{E}_X$ occurs, $\|\boldsymbol{X}\|_2 \geqslant \sqrt{n}/2$. Hence

  $$\mathbb{P}\left\{\frac{|\langle \boldsymbol{X}, \boldsymbol{\eta}\rangle|}{\|\boldsymbol{X}\|_2} \geqslant \frac{C_K}{\sqrt{n}} \cap \mathcal{E}_X\right\} \leqslant \mathbb{P}\left\{|\langle \boldsymbol{X}, \boldsymbol{\eta}\rangle| \geqslant \frac{C_K}{2}\right\} = \mathbb{E}\left[\mathbb{P}\left\{|\langle \boldsymbol{X}, \boldsymbol{\eta}\rangle| \geqslant \frac{C_K}{2} \;\middle|\; \boldsymbol{\eta}\right\}\right] \leqslant 2\exp\left(-\frac{c_2 (C_K/2)^2}{C_1 K^2}\right),$$

  hence when C_K is sufficiently large,

  $$\mathbb{P}\left\{\frac{|\langle \boldsymbol{X}, \boldsymbol{\eta}\rangle|}{\|\boldsymbol{X}\|_2} \geqslant \frac{C_K}{\sqrt{n}} \cap \mathcal{E}_X\right\} \leqslant 0.005, \qquad \text{(II)}$$

  Combining (I) and (II), we obtain

  $$\mathbb{P}\left\{|\langle \boldsymbol{\theta}, \boldsymbol{\eta}\rangle| \geqslant \frac{C_K}{\sqrt{n}}\right\} \leqslant 0.005 + 0.005 = 0.01 \implies \mathbb{P}\left\{|\langle \boldsymbol{\theta}, \boldsymbol{\eta}\rangle| \leqslant \frac{C_K}{\sqrt{n}}\right\} \geqslant 0.99.$$

(2) **Counterexample 1 One-dimensional counterexample**

Define $\boldsymbol{X} = 1$ (a.s.). Then:

1. Isotropy: $\mathbb{E}[\boldsymbol{X}^2] = 1^2 \cdot 1 = 1$.
2. Sub-Gaussian property: $\boldsymbol{X}$ is constant, satisfying $\|\boldsymbol{X}\|_{\psi_2} = 1/\sqrt{\log 2}$.
3. Independent copies: Let $\boldsymbol{Y}$ be an independent copy of $\boldsymbol{X}$. Then $\mathbb{P}\{\boldsymbol{X} = \boldsymbol{Y}\} = 1 \geqslant 0.99$.

However, $\boldsymbol{\theta} = \boldsymbol{X}/|\boldsymbol{X}| = 1$, $\boldsymbol{\eta} = \boldsymbol{Y}/|\boldsymbol{Y}| = 1$, so $|\langle \boldsymbol{\theta}, \boldsymbol{\eta}\rangle| = 1 \gg n^{-1/2}$, a contradiction.

Counterexample 2 High-dimensional counterexample

Let $\boldsymbol{v} \in \mathbb{S}^{n-1}$ be a unit vector, and let $\varepsilon = 0.005$ (*so that* $(1-\varepsilon)^2 \approx 0.99$). Define the distribution of the random vector $\boldsymbol{X}$ as follows:

$$\boldsymbol{X} = \begin{cases} \xi \cdot \frac{1}{\sqrt{1-\varepsilon}} \boldsymbol{v}, & \text{w.p. } 1-\varepsilon, \\ \boldsymbol{Z}, & \text{w.p. } \varepsilon, \end{cases}$$

where $\xi \in \{-1, +1\}$ is an equiprobable Rademacher random variable (*i.e., $\boldsymbol{X}$ takes $\pm\frac{\boldsymbol{v}}{\sqrt{1-\varepsilon}}$ each with probability $\frac{1-\varepsilon}{2}$*), and $\boldsymbol{Z} \sim N(\boldsymbol{0}, \frac{1}{\varepsilon}(\boldsymbol{I}_n - \boldsymbol{v}\boldsymbol{v}^\top))$ (*i.e., a degenerate zero-mean Gaussian random vector supported on $\boldsymbol{v}^\perp$*).

1. Zero mean: $\mathbb{E}[\boldsymbol{X}] = (1-\varepsilon)\mathbb{E}[\xi]\frac{\boldsymbol{v}}{\sqrt{1-\varepsilon}} + \varepsilon\mathbb{E}[\boldsymbol{Z}] = \boldsymbol{0}$.
2. Isotropy: $\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = (1-\varepsilon) \cdot \frac{1}{1-\varepsilon}\boldsymbol{v}\boldsymbol{v}^\top + \varepsilon \cdot \frac{1}{\varepsilon}(\boldsymbol{I}_n - \boldsymbol{v}\boldsymbol{v}^\top) = \boldsymbol{v}\boldsymbol{v}^\top + (\boldsymbol{I}_n - \boldsymbol{v}\boldsymbol{v}^\top) = \boldsymbol{I}_n$.
3. Bounded sub-Gaussian norm: For every unit vector $\boldsymbol{u} \in \mathbb{S}^{n-1}$, decompose $\boldsymbol{u} = a\boldsymbol{v} + b\boldsymbol{w}$ (*where $\boldsymbol{w} \perp \boldsymbol{v}$ and $a^2 + b^2 = 1$*). Then:

   a. On the branch of probability $1-\varepsilon$,

   $$|\langle \boldsymbol{X}, \boldsymbol{u}\rangle| = \frac{|a|}{\sqrt{1-\varepsilon}} \leqslant \frac{1}{\sqrt{1-\varepsilon}},$$

which is bounded.

b. On the branch of probability ε,

$$\langle \boldsymbol{X}, \boldsymbol{u} \rangle = b\langle \boldsymbol{Z}, \boldsymbol{w} \rangle \sim N\left(0, \frac{b^2}{\varepsilon}\right),$$

whose sub-Gaussian norm is bounded by $C/\sqrt{\varepsilon}$.

Therefore

$$\|\boldsymbol{X}\|_{\psi_2} \leqslant \max\left\{\frac{1}{\sqrt{1-\varepsilon}}, \frac{C}{\sqrt{\varepsilon}}\right\} \leqslant K.$$

Let $\boldsymbol{Y}$ be an independent copy of $\boldsymbol{X}$, with direction vectors $\boldsymbol{\theta}, \boldsymbol{\eta}$. Then

$$\mathbb{P}\{\boldsymbol{\theta} \in \{\pm \boldsymbol{v}\}, \boldsymbol{\eta} \in \{\pm \boldsymbol{v}\}\} = (1-\varepsilon)^2 = (0.995)^2 \geqslant 0.99,$$

When the event above occurs,

$$|\langle \boldsymbol{\theta}, \boldsymbol{\eta} \rangle| = |\langle \pm \boldsymbol{v}, \pm \boldsymbol{v} \rangle| = 1 \gg O\left(\frac{1}{\sqrt{n}}\right) \text{ (w.p. 0.99)}.$$

41 [**Exponentially Many Almost Orthogonal Vectors**] In linear algebra, we know that the number of mutually orthogonal vectors in $\mathbb{R}^n$ does not exceed n. Surprisingly, however, $\mathbb{R}^n$ can accommodate an exponentially large number of "almost orthogonal" vectors.

Follow the steps below to prove that for any integer $N \leqslant \mathrm{e}^{cn}$, there exist $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N \in \mathbb{S}^{n-1}$ satisfying

$$|\langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle| \leqslant 0.01. \qquad (i \neq j)$$

(1) Argue that independent random vectors $\boldsymbol{X}, \boldsymbol{Y} \sim U(\mathbb{S}^{n-1})$ satisfy $|\langle \boldsymbol{X}, \boldsymbol{Y} \rangle| \leqslant 0.01$ with probability at least $1 - 4\mathrm{e}^{-c_0 n}$, where c_0 is an absolute constant.

(2) Take a union bound over all pairs (i, j) with $i \neq j$, and conclude that the event above holds with positive probability.

Prf. • Estimate the almost orthogonality of two random vectors

Let $\boldsymbol{X}, \boldsymbol{Y} \sim U(\mathbb{S}^{n-1})$. By rotational invariance,

$$\langle \boldsymbol{X}, \boldsymbol{Y} \rangle \overset{d}{=} \langle \boldsymbol{X}, \boldsymbol{e}_1 \rangle = X_1.$$

We make the following claim:

Claim Let $\boldsymbol{X} \in \mathbb{S}^{n-1}$. Then

$$\|X_1\|_{\psi_2} \leqslant \frac{C}{\sqrt{n}}.$$

Method 1

Using the conclusion of Exercise Q3.27 (One-Dimensional Marginals of the Sphere and Ball) (2),

$$f_{X_1}(x) = \frac{1}{C_n}(1-x^2)^{\frac{n-3}{2}} \mathbf{1}_{[-1,1]},$$

where

$$(1-x^2)^{\frac{n-3}{2}} \leqslant (\mathrm{e}^{-x^2})^{\frac{n-3}{2}} = \mathrm{e}^{-\frac{n-3}{2}x^2},$$

hence

$$f_{X_1}(x) \leqslant \frac{1}{C_n}\mathrm{e}^{-\frac{(n-3)}{2}x^2},$$

i.e., $f_{X_1}(x)$ is dominated by a Gaussian kernel with mean 0 and variance $\sigma^2 = 1/(n-3)$. Hence

$$\mathbb{P}\{|X_1| \geqslant t\} = 2\int_t^1 f_{X_1}(x)\mathrm{d}x \leqslant \frac{2}{C_n}\int_t^\infty \mathrm{e}^{-\frac{n-3}{2}x^2}\mathrm{d}x \leqslant 2\mathrm{e}^{-\frac{n-3}{2}t^2},$$

Comparing with the definition of sub-Gaussian random variables

$$\mathbb{P}\{|X_1| \geqslant t\} \leqslant 2\mathrm{e}^{-\frac{n-3}{2}t^2},$$

let $K = 1/\sqrt{n-3}$. *Then*

$$\|X_1\|_{\psi_2} \leqslant \frac{C}{\sqrt{n-3}} \leqslant \frac{C'}{\sqrt{n}}.$$

Method 2 Gaussian Representation

Let $\boldsymbol{G} = (G_1, \cdots, G_n)^\top \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. *Then*

$$\boldsymbol{X} \overset{d}{=} \bar{\boldsymbol{G}}, X_1 = \frac{G_1}{\sqrt{G_1^2 + \cdots + G_n^2}},$$

Hence, by the SLLN,

$$X_1 \sim \frac{G_1}{\sqrt{n}} = O\left(\frac{1}{\sqrt{n}}\right).$$

By the claim,

$$\|X_1\|_{\psi_2} \leqslant \frac{C}{\sqrt{n}}, \qquad (C > 0)$$

hence

$$\mathbb{P}\{|X_1| \geqslant t\} \leqslant 2\mathrm{e}^{-c't^2/\|X_1\|_{\psi_2}^2} \leqslant 2\mathrm{e}^{-c''t^2 n}, \qquad (c', c'' > 0, c'' = C^2 c')$$

Let $t = 0.01$. Then

$$\mathbb{P}\{|\langle \boldsymbol{X}, \boldsymbol{Y}\rangle| > 0.01\} \leqslant 2\mathrm{e}^{-c_0 n}. \qquad (c_0 = c''(0.01)^2)$$

i.e., $\mathbb{P}\{|\langle \boldsymbol{X}, \boldsymbol{Y}\rangle| \leqslant 0.01\} \geqslant 1 - 2\mathrm{e}^{-c_0 n}$.

- Use the union bound to prove existence

  By the probabilistic method, let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ i.i.d. $\sim U(\mathbb{S}^{n-1})$ and construct

  $$\mathcal{E} = \bigcup_{1 \leqslant i < j \leqslant N} \{|\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle| > 0.01\},$$

  Then

  $$\mathbb{P}(\mathcal{E}) \leqslant \sum_{1 \leqslant i < j \leqslant N} \mathbb{P}\{|\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle| > 0.01\} \leqslant \binom{N}{2} \cdot 2\mathrm{e}^{-c_0 n},$$

  Hence when $N \leqslant \mathrm{e}^{cn}$,

  $$\mathbb{P}(\mathcal{E}) \leqslant \sum_{1 \leqslant i < j \leqslant N} \mathbb{P}\{|\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle| > 0.01\} < N^2 \cdot 2\mathrm{e}^{-c_0 n} \leqslant (\mathrm{e}^{cn})^2 \cdot \mathrm{e}^{-c_0 n} = \mathrm{e}^{(2c - c_0)n},$$

  Now let $2c < c_0$. Then $\mathbb{P}(\mathcal{E}) \to 0$, hence $\mathbb{P}\{\mathcal{E}^c\} > 0$.

42 [**Uniform Distribution on the Unit Ball is Sub-Gaussian**] Let $\boldsymbol{Y} \sim U(\mathbb{B}_2^n)$. Prove that $\boldsymbol{Y}$ is a sub-Gaussian random vector satisfying the conclusion of Theorem T3.4.5 (*The uniform distribution on the sphere is subgaussian*).

Prf. **Method 1**

By Exercise Q3.24 (*Polar Decomposition of Uniform Distribution on the Unit Ball*), perform the polar decomposition $\boldsymbol{Y} = R\boldsymbol{X}$, where $\boldsymbol{X} = \bar{\boldsymbol{Y}} \sim U(\mathbb{S}^{n-1})$, $R = \|\boldsymbol{Y}\|_2 \in [0, 1]$. Then for any $\boldsymbol{v} \in \mathbb{S}^{n-1}$,

$$|\langle \boldsymbol{Y}, \boldsymbol{v}\rangle| = R|\langle \boldsymbol{X}, \boldsymbol{v}\rangle| \leqslant |\langle \boldsymbol{X}, \boldsymbol{v}\rangle|,$$

hence

$$\mathbb{P}\{\langle \boldsymbol{Y}, \boldsymbol{v}\rangle \geqslant t\} \leqslant \mathbb{P}\{\langle \boldsymbol{X}, \boldsymbol{v}\rangle \geqslant t\} \leqslant 2\exp\left(-\frac{t^2 n}{2}\right),$$

Thus $\langle \boldsymbol{Y}, \boldsymbol{v}\rangle$ is a sub-Gaussian random variable satisfying

$$\mathbb{P}\{|\langle \boldsymbol{Y}, \boldsymbol{v}\rangle| \geqslant t\} \leqslant 4\exp\left(-\frac{t^2 n}{2}\right) \implies |\langle \boldsymbol{Y}, \boldsymbol{v}\rangle|_{\psi_2} \leqslant C_0\sqrt{\frac{2}{n}} = \frac{C_1}{\sqrt{n}}, \qquad (C_0, C_1 > 0)$$

Hence, by the equivalent definition of sub-Gaussian $\|\boldsymbol{Y}\|_{\psi_2} = \sup_{\boldsymbol{v}\in\mathbb{S}^{n-1}} \|\langle \boldsymbol{Y}, \boldsymbol{v}\rangle\|_{\psi_2}$,

$$\|\boldsymbol{Y}\|_{\psi_2} \leqslant \frac{C}{\sqrt{n}}. \qquad (C>0)$$

Method 2

By the rotational invariance of $\boldsymbol{Y} \sim U(\mathbb{B}_2^n)$, for any $\boldsymbol{v} \in \mathbb{S}^{n-1}$, $\langle \boldsymbol{Y}, \boldsymbol{v}\rangle \overset{d}{=} \langle \boldsymbol{Y}, \boldsymbol{e}_1\rangle = Y_1$. (*It then suffices to consider the sub-Gaussian property of* Y_1.)

- Compute $f_{Y_1}(x)$

  Using the conclusion of Exercise Q3.27 (*One-Dimensional Marginals of the Sphere and Ball*) (1),

  $$f_{Y_1}(x) = \frac{1}{C_n}(1-x^2)^{\frac{n-1}{2}}\mathbf{1}_{[-1,1]},$$

  where

  $$(1-x^2)^{\frac{n-1}{2}} \leqslant (\mathrm{e}^{-x^2})^{\frac{n-1}{2}} = \mathrm{e}^{-\frac{n-1}{2}x^2},$$

  hence

  $$f_{Y_1}(x) \leqslant \frac{1}{C_n}\mathrm{e}^{-\frac{(n-1)}{2}x^2},$$

  Using Stirling's formula, the normalization constant C_n satisfies

  $$C_n = \int_{-1}^{1}(1-x^2)^{\frac{n-1}{2}}\mathrm{d}x = \beta\Big(\frac{1}{2}, \frac{n+1}{2}\Big) = \frac{\sqrt{\pi}\Gamma\big(\frac{n+1}{2}\big)}{\Gamma(\frac{n}{2}+1)} \asymp \frac{1}{\sqrt{n}},$$

  hence

  $$f_{Y_1}(x) \leqslant c_1\sqrt{n}\mathrm{e}^{-\frac{n-1}{2}x^2}, \qquad (c_1>0)$$

- Compute $\mathbb{P}\{|Y_1| \geqslant t\}$

  For any $t \in (0,1]$,

  $$\mathbb{P}\{|Y_1| \geqslant t\} = 2\int_t^1 f_{Y_1}(x)\mathrm{d}x \leqslant 2c_1\sqrt{n}\int_t^\infty \mathrm{e}^{-\frac{n-1}{2}x^2}\mathrm{d}x.$$

  This is the tail integral of $N(0,\sigma^2)$ with $\sigma^2 = 1/(n-1)$. Using Mill's Ratio,

  $$\int_t^\infty \mathrm{e}^{-x^2/2\sigma^2}\mathrm{d}x \leqslant \frac{\sigma^2}{t}\mathrm{e}^{-t^2/2\sigma^2},$$

  hence

  $$\mathbb{P}\{|Y_1| \geqslant t\} \leqslant 2c_1\sqrt{n}\cdot\frac{1}{(n-1)t}\mathrm{e}^{-\frac{n-1}{2}t^2} = \frac{2c_1\sqrt{n}}{(n-1)t}\mathrm{e}^{-\frac{n-1}{2}t^2}.$$

  (*Note that the denominator contains* t*; when* t *is very small, the coefficient above tends to infinity, so a case distinction is necessary.*)

  a. When $t \geqslant 1/\sqrt{n}$,

  $$\frac{\sqrt{n}}{(n-1)t} \leqslant \frac{n}{(n-1)} \leqslant 2, \qquad (n \geqslant 2)$$

  hence

  $$\mathbb{P}\{|Y_1| \geqslant t\} \leqslant 2\mathrm{e}^{-c_2 n t^2}.$$

  b. When $t < 1/\sqrt{n}$, $\mathrm{e}^{-c_2 n t^2} \geqslant 1$, and

  $$\mathbb{P}\{|Y_1| \geqslant t\} \leqslant 1 \leqslant 2\mathrm{e}^{-c_2 n t^2},$$

  which is trivial.

  Thus,

  $$\mathbb{P}\{|Y_1| \geqslant t\} \leqslant 2\mathrm{e}^{-c_2 n t^2}.$$

Hence, by the definition of sub-Gaussian (if $\mathbb{P}\{|Y_1| \geqslant t\} \leqslant 2\mathrm{e}^{-t^2/K^2}$, then $\|Y_1\|_{\psi_2} \leqslant C'K$), letting $1/K^2 = c_2 n$, we have

$$\|Y_1\|_{\psi_2} \leqslant \frac{C}{\sqrt{n}},$$

The above estimate holds for any projection direction, hence

$$\|\boldsymbol{Y}\|_{\psi_2} \leqslant \frac{C}{\sqrt{n}}.$$

Rk. By Exercise Q3.25 (*Isotropy of the Euclidean Ball Distribution*), $\boldsymbol{Y} \sim U(\mathbb{B}_2^n)$ is not isotropic; its covariance matrix is

$$\boldsymbol{\Sigma} = \frac{1}{n+2}\boldsymbol{I}_n,$$

Hence, construct the isotropic random vector $\tilde{\boldsymbol{Y}} = \sqrt{n+2}\boldsymbol{Y}$. By the homogeneity of the sub-Gaussian norm,

$$\|\tilde{\boldsymbol{Y}}\|_{\psi_2} = \sqrt{n+2}\|\boldsymbol{Y}\|_{\psi_2} \leqslant \sqrt{n+2} \cdot \frac{C}{\sqrt{n}} = C \cdot O(1).$$

43 [**Sub-Gaussian Norm of the Standard Basis Distribution**] Let $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n$ be the standard orthonormal basis of $\mathbb{R}^n$ ($n \geqslant 2$), and let $\boldsymbol{X} \sim U\{\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n\}$. Prove that

$$\|\boldsymbol{X}\|_{\psi_2} \asymp \frac{1}{\sqrt{\log n}}.$$

Prf. Let $\boldsymbol{v} = (v_1, v_2, \cdots, v_n)^\top \in \mathbb{S}^{n-1}$ satisfy $\sum_{k=1}^n v_k^2 = 1$, and let $Y_{\boldsymbol{v}} = \langle \boldsymbol{X}, \boldsymbol{v} \rangle \sim U\{v_1, \cdots, v_n\}$ be the sub-Gaussian random variable of the projection onto the unit direction. By the equivalent definition of sub-Gaussian,

$$\|\boldsymbol{X}\|_{\psi_2} = \sup_{\boldsymbol{v} \in \mathbb{S}^{n-1}} \|Y_{\boldsymbol{v}}\|_{\psi_2}.$$

Here $\|Y_{\boldsymbol{v}}\|_{\psi_2}$ is the order of the smallest positive number t such that

$$G(\boldsymbol{v}, t) = \mathbb{E}\left[\exp\left(\frac{Y_{\boldsymbol{v}}^2}{t^2}\right)\right] = \frac{1}{n}\sum_{k=1}^n \exp\left(\frac{v_k^2}{t^2}\right) \leqslant 2.$$

Let $u_i = v_i^2$ satisfy $\sum_{i=1}^n u_i = 1$, $u_i \geqslant 0$, and construct

$$\phi(u) = \exp\left(\frac{u_i}{t^2}\right), \Phi(u_1, \cdots, u_n) = \sum_{i=1}^n \phi(u_i) = \sum_{i=1}^n \exp\left(\frac{u_i}{t^2}\right), \qquad (t > 0)$$

Then $\phi(u)$ is strictly convex for $u \geqslant 0$, and its maximum is attained at extreme points (*the extreme points of the probability simplex are* $\boldsymbol{u} \in \{\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n\}$). Hence, when $\boldsymbol{v}$ is a standard basis vector (*say* $\boldsymbol{e}_1$), $\sup_{\boldsymbol{v} \in \mathbb{S}^{n-1}} G(\boldsymbol{v}, t)$ attains its maximum. In this case,

$$\begin{aligned} G(\boldsymbol{e}_1, t) = 2 &\Longrightarrow \frac{1}{n}\left(\exp\left(\frac{1}{t^2}\right) + (n-1)\exp(0)\right) = 2 \\ &\Longrightarrow t = \frac{1}{\sqrt{\log(n+1)}}, \end{aligned}$$

hence

$$\|\boldsymbol{X}\|_{\psi_2} = \frac{1}{\sqrt{\log(n+1)}} \asymp \frac{1}{\sqrt{\log n}}.$$

44 [**Sub-Gaussian Analysis of the ℓ^1 Ball**] Consider the n-dimensional ℓ^1 cross-polytope in $\mathbb{R}^n$

$$K = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_1 \leqslant r\},$$

Let $\boldsymbol{X} \sim U(K)$.

(1) Find $f_{X_1}(x)$.

(2) Prove that there exists $r \asymp n$ such that $\boldsymbol{X}$ is isotropic.

(3) For the choice of r in (2), prove that $\|\boldsymbol{X}\|_{\psi_2} \asymp \sqrt{n}$.

Sol. (1) It is easy to see that

$$V_n(r) = V(K) = \frac{2^n r^n}{n!},$$

hence

$$\begin{aligned} f_{X_1}(x) &= \int \cdots \int_{\sum_{i=2}^n |x_i| \leqslant r-|x|} \frac{1}{V_n(r)} \mathrm{d}x_2 \cdots \mathrm{d}x_n \\ &= \frac{V_{n-1}(r-|x|)}{V_n(r)} = \frac{\frac{2^{n-1}(r-|x|)^{n-1}}{(n-1)!}}{\frac{2^n r^n}{n!}} = \frac{n}{2r}\left(1-\frac{|x|}{r}\right)^{n-1}. \end{aligned}$$

(2) Since K is symmetric about the coordinate axes, $\mathbb{E}[\boldsymbol{X}] = \mathbf{0}$, and $\mathrm{Cov}(\boldsymbol{X})$ is diagonal. It then suffices to prove $\mathbb{E}[X_1^2] = 1$.

It is easy to see that

$$\begin{aligned} \mathbb{E}[X_1^2] &= \int_{-r}^{r} x^2 \frac{n}{2r}\left(1-\frac{|x|}{r}\right)^{n-1} \mathrm{d}x = \frac{n}{r}\int_0^r x^2\left(1-\frac{x}{r}\right)^{n-1}\mathrm{d}x \\ &= nr^2 \int_0^1 u^2(1-u)^{n-1}\mathrm{d}u \qquad \left(u = \frac{x}{r}\right) \\ &= nr^2 \frac{\Gamma(3)\Gamma(n)}{\Gamma(n+3)} = nr^2 \frac{2\cdot(n-1)!}{(n+2)!} = \frac{2r^2}{(n+1)(n+2)}, \end{aligned}$$

Set $\mathbb{E}[X_1^2] = 1$,

$$\frac{2r^2}{(n+1)(n+2)} = 1 \implies r \asymp n.$$

(3) By property (2) of Proposition P2.6.1 (*Subgaussian properties*) (*moment characterization*),

$$\|\boldsymbol{X}\|_{\psi_2} \geqslant \|X_1\|_{\psi_2} \asymp \sup_{p\geqslant 1} \frac{(\mathbb{E}|X_1|^p)^{1/p}}{\sqrt{p}},$$

It then suffices to estimate $\mathbb{E}[|X_1|^p]$.

It is easy to see that

$$\mathbb{E}[|X_1|^p] = 2\int_0^r x^p \frac{n}{2r}\left(1-\frac{x}{r}\right)^{n-1}\mathrm{d}x = nr^p\beta(p+1,n) = \frac{nr^p\Gamma(p+1)\Gamma(n)}{\Gamma(n+p+1)},$$

Let $p = n$. Using Stirling's formula,

$$\begin{aligned} \mathbb{E}[|X_1|^n] &= \frac{nr^n n!(n-1)!}{(2n)!} = \frac{r^n(n!)^2}{(2n)!} \\ &\asymp \frac{r^n(2\pi n)(n/\mathrm{e})^{2n}}{\sqrt{4\pi n}(2n/\mathrm{e})^{2n}} \asymp \sqrt{n}\left(\frac{r}{4}\right)^n, \end{aligned}$$

hence

$$\|X_1\|_n = (\mathbb{E}|X_1|^n)^{1/n} \asymp (\sqrt{n})^{1/n}\frac{r}{4} \asymp r,$$

Using the conclusion of (2), let $r \asymp n$. Then

$$\|X_1\|_{\psi_2} \geqslant C\frac{\|X_1\|_n}{\sqrt{n}} \asymp \frac{n}{\sqrt{n}} = \sqrt{n}.$$

45 [**Probability Mass Function Bound for Sub-Gaussian Random Vectors**] Let $\boldsymbol{X} \in \mathbb{R}^n$ be a K-sub-Gaussian random vector. Prove that for any $\boldsymbol{x} \in \mathbb{R}^n$,

$$\mathbb{P}\{\boldsymbol{X} = \boldsymbol{x}\} \leqslant 2\exp\left(-\frac{\|\boldsymbol{x}\|_2^2}{K^2}\right).$$

Prf. If $\boldsymbol{x} = \mathbf{0}$, then the right-hand side is $2\exp(0) = 2$ and the left-hand side is $\mathbb{P}\{\boldsymbol{X} = \mathbf{0}\} \leqslant 1$, which is trivial. We only need

to consider $\boldsymbol{x} \neq \boldsymbol{0}$.

When $\boldsymbol{X} = \boldsymbol{x}$ is given,

$$\langle \boldsymbol{X}, \bar{\boldsymbol{x}} \rangle = \langle \boldsymbol{x}, \bar{\boldsymbol{x}} \rangle = \left\langle \boldsymbol{x}, \frac{\boldsymbol{x}}{\|\boldsymbol{x}\|_2} \right\rangle = \frac{\|\boldsymbol{x}\|_2^2}{\|\boldsymbol{x}\|_2} = \|\boldsymbol{x}\|_2,$$

hence $\{\boldsymbol{X} = \boldsymbol{x}\} \subset \{\langle \boldsymbol{X}, \bar{\boldsymbol{x}} \rangle = \|\boldsymbol{x}\|_2\}$, thus

$$\mathbb{P}\{\boldsymbol{X} = \boldsymbol{x}\} \leqslant \mathbb{P}\{\langle \boldsymbol{X}, \bar{\boldsymbol{x}} \rangle = \|\boldsymbol{x}\|_2\} \leqslant \mathbb{P}\{|\langle \boldsymbol{X}, \bar{\boldsymbol{x}} \rangle| \geqslant \|\boldsymbol{x}\|_2\}.$$

Let $Y_{\bar{\boldsymbol{x}}} = \langle \boldsymbol{X}, \bar{\boldsymbol{x}} \rangle$. By the equivalent definition of sub-Gaussian,

$$\|Y_{\bar{\boldsymbol{x}}}\|_{\psi_2} \leqslant \|\boldsymbol{X}\|_{\psi_2} \leqslant K \implies \mathbb{P}\{|Y_{\bar{\boldsymbol{x}}}| \geqslant t\} \leqslant 2\exp\left(-\frac{t^2}{\|Y_{\bar{\boldsymbol{x}}}\|_{\psi_2}^2}\right),$$

hence

$$\mathbb{P}\{|\langle \boldsymbol{X}, \bar{\boldsymbol{x}} \rangle| \geqslant \|\boldsymbol{x}\|_2\} \leqslant 2\exp\left(-\frac{\|\boldsymbol{x}\|_2^2}{\|Y_{\bar{\boldsymbol{x}}}\|_{\psi_2}^2}\right) \leqslant 2\exp\left(-\frac{\|\boldsymbol{x}\|_2^2}{K^2}\right),$$

i.e.,

$$\mathbb{P}\{\boldsymbol{X} = \boldsymbol{x}\} \leqslant 2\exp\left(-\frac{\|\boldsymbol{x}\|_2^2}{K^2}\right).$$

46 [**Isotropic Sub-Gaussian Distributions Have High Entropy**] Let a discrete random vector $\boldsymbol{X}$ take values $\boldsymbol{x}_i$ with probabilities p_i. Define its entropy as

$$H(\boldsymbol{X}) = -\sum_i p_i \log p_i,$$

Intuitively, $H(\boldsymbol{X})$ is proportional to the number of binary digits needed to encode $\boldsymbol{X}$.

(1) (Extremely spread distributions have low entropy) Prove that for any random vector $\boldsymbol{X}$ with N possible values (*i.e.,* $|\operatorname{supp}(\boldsymbol{X})| = N$), $H(\boldsymbol{X}) \leqslant \log N$.

(2) (Isotropic sub-Gaussian distributions have high entropy) If $\boldsymbol{X} \in \mathbb{R}^n$ is a discrete, isotropic, K-sub-Gaussian random vector, prove that

$$H(\boldsymbol{X}) \geqslant \frac{n}{K^2} - \log 2.$$

(3) (Isotropic sub-Gaussian distributions have huge supp) Prove that a random vector $\boldsymbol{X}$ satisfying condition (2) has at least $\mathrm{e}^{K^{-2}n}/2$ possible values.

Prf. (1) Let $\operatorname{supp}(P) = \{p_1, \cdots, p_N\}$. Construct the convex function $g(x) = \log x$. By Jensen's inequality,

$$H(\boldsymbol{X}) = \sum_{i=1}^N p_i \log \frac{1}{p_i} \leqslant \log\left(\sum_{i=1}^N p_i \cdot \frac{1}{p_i}\right) = \log\left(\sum_{i=1}^N 1\right) = \log N,$$

with equality attained if and only if $p_i = 1/N$ (*the uniform distribution*).

(2) Since $\boldsymbol{X}$ is isotropic,

$$\mathbb{E}[\|\boldsymbol{X}\|_2^2] = \mathbb{E}[\operatorname{tr}(\boldsymbol{X}\boldsymbol{X}^\top)] = \operatorname{tr}(\boldsymbol{I}_n) = n.$$

By Exercise Q3.45 (*Probability Mass Function Bound for Sub-Gaussian Random Vectors*),

$$p_i = \mathbb{P}\{\boldsymbol{X} = \boldsymbol{x}_i\} \leqslant 2\mathrm{e}^{-\|\boldsymbol{x}_i\|_2^2/K^2},$$

Taking logarithms on both sides,

$$\log p_i \leqslant \log 2 - \frac{\|\boldsymbol{x}_i\|_2^2}{K^2},$$

hence

$$H(\boldsymbol{X}) = \sum_i p_i(-\log p_i) = \mathbb{E}[-\log p_{\boldsymbol{X}}]$$

$$\geqslant \mathbb{E}\left[\frac{\|\boldsymbol{X}\|_2^2}{K^2} - \log 2\right] = \frac{1}{K^2}\mathbb{E}\|\boldsymbol{X}\|_2^2 - \log 2 = \frac{n}{K^2} - \log 2.$$

(3) Using the conclusions of (1) and (2),

$$\log N \geqslant H(\boldsymbol{X}) \geqslant \frac{n}{K^2} - \log 2,$$

Exponentiating both sides,

$$N \geqslant \mathrm{e}^{n/K^2} \cdot \mathrm{e}^{-\log 2} = \frac{1}{2}\mathrm{e}^{n/K^2} = \frac{1}{2}\exp(K^{-2}n).$$

47 [**Rewriting the Assumptions of Grothendieck's Inequality**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$. Prove that the following four properties are equivalent:

(1) For any $\boldsymbol{x} \in \{-1,1\}^m, \boldsymbol{y} \in \{-1,1\}^n$, $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} \leqslant 1$.

(2) For any $\boldsymbol{x} \in [-1,1]^m, \boldsymbol{y} \in [-1,1]^n$, $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} \leqslant 1$.

(3) For any $\boldsymbol{x} \in \mathbb{R}^m, \boldsymbol{y} \in \mathbb{R}^n$, $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} \leqslant \|\boldsymbol{x}\|_\infty \|\boldsymbol{y}\|_\infty$.

(4) Replacing $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}$ by its absolute value $|\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}|$ in any of the above properties, the statement still holds.

Prf. (*First prove* (1) $\iff$ (2) $\iff$ (3).)

For (3) $\iff$ (2), if $\boldsymbol{x} \in [-1,1]^m$, $\boldsymbol{y} \in [-1,1]^n$, then $\|\boldsymbol{x}\|_\infty, \|\boldsymbol{y}\|_\infty \leqslant 1$, hence $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} \leqslant 1 \cdot 1 = 1$.

For (2) $\iff$ (3), if $\boldsymbol{x} = \boldsymbol{0}$ or $\boldsymbol{y} = \boldsymbol{0}$, the conclusion is trivial. We only need to consider $\boldsymbol{x}, \boldsymbol{y} \neq \boldsymbol{0}$.

Let $\boldsymbol{x}' = \boldsymbol{x}/\|\boldsymbol{x}\|_\infty \in [-1,1]^m$, $\boldsymbol{y}' = \boldsymbol{y}/\|\boldsymbol{y}\|_\infty \in [-1,1]^n$. Then

$$(\boldsymbol{x}')^\top \boldsymbol{A}\boldsymbol{y}' \leqslant 1 \implies \frac{\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}}{\|\boldsymbol{x}\|_\infty \|\boldsymbol{y}\|_\infty} \leqslant 1$$
$$\implies \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} \leqslant \|\boldsymbol{x}\|_\infty \|\boldsymbol{y}\|_\infty.$$

For (2) $\iff$ (1), since $\{-1,1\}^m \subset [-1,1]^m$ and $\{-1,1\}^n \subset [-1,1]^n$, the conclusion is immediate.

For (1) $\iff$ (2), let $f(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}$. A linear function on a convex compact set ($[-1,1]^m$) attains its maximum at an extreme point ($\{-1,1\}^m$). Hence

$$\max_{\boldsymbol{x}\in[-1,1]^m} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} = \max_{\boldsymbol{x}\in\{-1,1\}^m} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} \leqslant 1, \max_{\boldsymbol{y}\in[-1,1]^n} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} = \max_{\boldsymbol{y}\in\{-1,1\}^n} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} \leqslant 1,$$

thus $f(\boldsymbol{x},\boldsymbol{y}) \leqslant 1$.

(*Next prove* (1)(2)(3) $\iff$ (4).)

It suffices to prove (1) $\iff$ (1′), where (1') is $|\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}| \leqslant 1$.

For (1′) $\implies$ (1), the conclusion is immediate.

For (1) $\implies$ (1′), if $\boldsymbol{x} \in \{-1,1\}^m$, then $-\boldsymbol{x} \in \{-1,1\}^m$, hence

$$(-\boldsymbol{x})^\top \boldsymbol{A}\boldsymbol{y} \leqslant 1 \implies \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} \geqslant -1,$$

i.e., $|\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}| \leqslant 1$.

48 [**Refinement of the Probabilistic Proof of Grothendieck's Inequality**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ satisfy $\left|\sum_{i,j} a_{ij}x_iy_j\right| \leqslant 1$ for all $x_i, y_j \in \{-1,1\}$. Theorem T3.5.1 (*Grothendieck inequality*) states that for any unit vectors $\boldsymbol{u}_i, \boldsymbol{v}_j \in \mathcal{H}$,

$$\left|\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle\right| \leqslant K_G. \qquad (K_G > 0)$$

In the preliminary probabilistic proof in the main text, the crude upper bound $4/R^2$ was used to handle the Taylor series remainder. By retaining the exact form of the series and numerically optimizing the scaling parameter R, prove that the tighter bound $K_G \leqslant 14.1$ can be obtained.

Prf. • Introduce the probabilistic construction and Grothendieck's identity

Let $\mathcal{H} = \mathbb{R}^d$, $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_d)$, and define

$$X_i = \operatorname{sgn}(\langle \boldsymbol{u}_i, \boldsymbol{g} \rangle), Y_j = \operatorname{sgn}(\langle \boldsymbol{v}_j, \boldsymbol{g} \rangle),$$

By Grothendieck's identity (Q3.53 (*Grothendieck's Identity*)), for any unit vectors $\boldsymbol{u}, \boldsymbol{v}$,

$$\mathbb{E}[X_i Y_j] = \frac{2}{\pi} \arcsin(\langle \boldsymbol{u}_i, \boldsymbol{v}_j \rangle).$$

Let $\rho_{ij} = \langle \boldsymbol{u}_i, \boldsymbol{v}_j \rangle$. Note that $X_i, Y_j \in \{-1, 1\}$, hence

$$\left| \sum_{i,j} a_{ij} \mathbb{E}[X_i Y_j] \right| = \left| \mathbb{E}\left[\sum_{i,j} a_{ij} X_i Y_j \right] \right| \leqslant \mathbb{E}\left| \sum_{i,j} a_{ij} X_i Y_j \right| \leqslant 1,$$

hence

$$\left| \sum_{i,j} a_{ij} \frac{2}{\pi} \arcsin(\rho_{ij}) \right| \leqslant 1.$$

Introduce a scaling parameter $R > 1$. In an enlargement of $\mathcal{H}$, choose orthonormal systems $\{\boldsymbol{e}_i\}_{i=1}^m$ and $\{\boldsymbol{f}_j\}_{j=1}^n$ which are mutually orthogonal and orthogonal to $\mathcal{H}$, and set

$$\boldsymbol{u}_i' = R^{-1} \boldsymbol{u}_i + \sqrt{1 - R^{-2}}\, \boldsymbol{e}_i, \qquad \boldsymbol{v}_j' = R^{-1} \boldsymbol{v}_j + \sqrt{1 - R^{-2}}\, \boldsymbol{f}_j.$$

These are unit vectors and satisfy $\langle \boldsymbol{u}_i', \boldsymbol{v}_j' \rangle = \rho_{ij} / R^2$. Applying the preceding random-sign argument to these vectors gives

$$\left| \sum_{i,j} a_{ij} \frac{2}{\pi} \arcsin\left(\frac{\rho_{ij}}{R^2} \right) \right| \leqslant 1,$$

where

$$\frac{2}{\pi} \arcsin(z) = \frac{2}{\pi} z + \frac{2}{\pi} \sum_{k=1}^{\infty} c_{2k+1} z^{2k+1}, \qquad \left(c_{2k+1} = \frac{(2k)!}{4^k (k!)^2 (2k+1)} \right)$$

hence

$$\left| \frac{2}{\pi R^2} \sum_{i,j} a_{ij} \rho_{ij} + \sum_{k=1}^{\infty} \frac{2 c_{2k+1}}{\pi R^{2(2k+1)}} \sum_{i,j} a_{ij} \rho_{ij}^{2k+1} \right| \leqslant 1.$$

- Use tensor product properties to bound the higher-order terms

  Note that

  $$\rho_{ij}^{2k+1} = \langle \boldsymbol{u}_i, \boldsymbol{v}_j \rangle^{2k+1} = \langle \boldsymbol{u}_i^{\otimes(2k+1)}, \boldsymbol{v}_j^{\otimes(2k+1)} \rangle,$$

  where $\boldsymbol{u}_i^{\otimes(2k+1)}, \boldsymbol{v}_j^{\otimes(2k+1)}$ are still unit vectors in some higher-dimensional Hilbert space. Hence, when K_G is the optimal constant,

  $$\left| \sum_{i,j} a_{ij} \rho_{ij}^{2k+1} \right| \leqslant K_G,$$

  Thus, by the triangle inequality,

  $$\underbrace{\frac{2}{\pi R^2} \left| \sum_{i,j} a_{ij} \rho_{ij} \right|}_{\leqslant K_G \cdot \frac{2}{\pi R^2}} \leqslant 1 + K_G \cdot \sum_{k=1}^{\infty} \frac{2 c_{2k+1}}{\pi R^{2(2k+1)}}.$$

  Define the series remainder on the right-hand side as $S(R) = \sum_{k=1}^{\infty} c_{2k+1} (R^{-2})^{2k}$. Note that $\arcsin(z) - z = \sum_{k=1}^{\infty} c_{2k+1} z^{2k+1}$, hence

  $$S(R) = R^2 (\arcsin(R^{-2}) - R^{-2}) = R^2 \arcsin(R^{-2}) - 1,$$

  Thus

  $$K_G \frac{2}{\pi R^2} \leqslant 1 + K_G \cdot \frac{2}{\pi R^2} S(R) \implies K_G \leqslant \frac{\pi R^2}{2(1 - S(R))} = \frac{\pi R^2}{2(2 - R^2 \arcsin(R^{-2}))}$$

$$\Longrightarrow K_G \leqslant \underbrace{\frac{\pi}{4z - 2\arcsin z}}_{h(z)}, \qquad (z = R^{-2} \in (0,1))$$

where

$$\min_{z\in(0,1)} h(z) = h\left(\frac{\sqrt{3}}{2}\right) \approx 2.29,$$

hence $K_G \leqslant 2.29$, which is far better than the required upper bound 14.1.

Rk. In fact, we need not directly project the original vectors $\boldsymbol{u}_i$; we can first "preprocess" the vectors via a nonlinear operator into new vectors $\tilde{\boldsymbol{u}}_i$, then perform random projection, ultimately obtaining the optimal upper bound $K_G = \pi/(2\log(1+\sqrt{2})) \approx 1.783$.

49 [**Quadratic Grothendieck Inequality for Positive Semidefinite Matrices**] In this and the next exercise, we relax the assumption of Grothendieck's inequality by assuming $x_i = y_i$, as announced in Remark R3.5.3 (*Quadratic Grothendieck:* $x_i = y_i$).

(1) (Polarization identity) Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, and set $\bar{\boldsymbol{x}} = (\boldsymbol{x}+\boldsymbol{y})/2$, $\bar{\boldsymbol{y}} = (\boldsymbol{x}-\boldsymbol{y})/2$. Prove that for any symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$,

$$\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} = \bar{\boldsymbol{x}}^\top \boldsymbol{A}\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}^\top \boldsymbol{A}\bar{\boldsymbol{y}}.$$

(2) (Quadratic Grothendieck inequality) Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric positive semidefinite matrix satisfying $\sum_{i,j} a_{ij}x_ix_j \leqslant 1$ for all $x_i \in \{-1,1\}$. Prove that for any unit vectors $\boldsymbol{u}_i, \boldsymbol{v}_j \in \mathcal{H}$,

$$\left|\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle\right| \leqslant 2K_G,$$

where $K_G \leqslant 1.783$ is the Grothendieck constant.

(3) Prove that if the positive semidefinite assumption is dropped, (2) may fail (*i.e., the left-hand side may not be controlled by a constant*).

Prf. (1) Since $\boldsymbol{A}$ is symmetric,

$$\begin{aligned}
\bar{\boldsymbol{x}}^\top \boldsymbol{A}\bar{\boldsymbol{x}} &= \left(\frac{\boldsymbol{x}+\boldsymbol{y}}{2}\right)^\top \boldsymbol{A}\left(\frac{\boldsymbol{x}+\boldsymbol{y}}{2}\right) \\
&= \frac{1}{4}(\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} + \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{y}) \\
&= \frac{1}{4}(\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{y} + 2\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}), \\
\bar{\boldsymbol{y}}^\top \boldsymbol{A}\bar{\boldsymbol{y}} &= \frac{1}{4}(\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{y} - 2\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}),
\end{aligned}$$

hence

$$\bar{\boldsymbol{x}}^\top \boldsymbol{A}\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}^\top \boldsymbol{A}\bar{\boldsymbol{y}} = \frac{1}{4}(4\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}.$$

(2) Construct the convex function $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}$. When $\boldsymbol{x} \in \{-1,1\}^n$, $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \leqslant 1$. By the property of convex functions (*a convex function on a compact convex set* $[-1,1]^n$ *attains its maximum at an extreme point* $\{-1,1\}^n$), for $\boldsymbol{z} \in [-1,1]^n$, $\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} \leqslant 1$.

Let $\boldsymbol{x}, \boldsymbol{y} \in \{-1,1\}^n$. Using the conclusion of (1),

$$|\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}| = |\bar{\boldsymbol{x}}^\top \boldsymbol{A}\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}^\top \boldsymbol{A}\bar{\boldsymbol{y}}| \leqslant |\bar{\boldsymbol{x}}^\top \boldsymbol{A}\bar{\boldsymbol{x}}| + |\bar{\boldsymbol{y}}^\top \boldsymbol{A}\bar{\boldsymbol{y}}|,$$

where $\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}} \in [-1,1]^n$, hence

$$|\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}| \leqslant 1 + 1 = 2,$$

Thus, by Exercise Q3.27 (*One-Dimensional Marginals of the Sphere and Ball*), using Theorem T3.5.1 (*Grothendieck*

inequality) (*Grothendieck inequality*), for any unit vectors $\boldsymbol{u}_i, \boldsymbol{v}_j \in \mathcal{H}$,

$$\left|\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle\right| \leqslant K_G \cdot \max_{\boldsymbol{x},\boldsymbol{y}\in\{-1,1\}^n} |\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}| \leqslant 2K_G.$$

(3) If $\boldsymbol{A}$ is not positive semidefinite, then $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}$ need not be convex.

Construct the following counterexample: let

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

Then for $x_i \in \{-1, 1\}$, $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = x_1^2 - x_2^2 = 1 - 1 = 0 \leqslant 1$.

Take the vertices $\boldsymbol{x} = (1, 1)^\top, \boldsymbol{y} = (1, -1)^\top$. Then $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} = (1)(1) + (1)(-1)(-1) = 1 + 1 = 2$. In this case, there is no fixed constant ratio between the quadratic form and the bilinear form.

(*Similarly, for the high-dimensional case, construct* $\boldsymbol{A}$ *with all diagonal entries* $a_{ii} = 1$*, then pair up the off-diagonal entries so that the quadratic form is always* 0*, but the bilinear form grows with* n.)

50 [**Quadratic Grothendieck Inequality for Matrices without Diagonal Entries**]

(1) (Componentwise convexity) A function $f(x_1, \cdots, x_n)$ is called componentwise convex if it is convex in each variable x_i (*when the other variables are fixed*).

i. Give an example of a function that is componentwise convex but not globally convex.

ii. Prove that a componentwise convex function f defined on the hypercube $[-1, 1]^n$ must attain its maximum at the vertex set $\{-1, 1\}^n$.

(2) (Quadratic Grothendieck inequality without diagonal entries) Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric real matrix with zero diagonal entries (*i.e.,* $\boldsymbol{A} = \boldsymbol{A}^\top$ *and* $a_{ii} = 0$). If $\left|\sum_{i,j} a_{ij}x_i x_j\right| \leqslant 1$ for all $x_i \in \{-1, 1\}$, prove that for any unit vectors $\boldsymbol{u}_i, \boldsymbol{v}_j \in \mathcal{H}$,

$$\left|\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle\right| \leqslant 2K_G,$$

where K_G is the Grothendieck constant.

Prf. (1) • Example

Construct the componentwise convex function $f(x, y) = xy$. Its Hessian matrix is

$$\boldsymbol{H} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

with eigenvalues $1, -1$ (*neither positive definite nor negative definite, i.e., a saddle point*), so f is not globally convex.

• Proof that the maximum is attained at vertices

By mathematical induction:

a. When $n = 1$, $f(x)$ is convex on $[-1, 1]$, so its maximum must be attained at $\{-1, 1\}$.

b. Assume the conclusion holds for dimension $n - 1$. For dimension n, fix the first $n - 1$ variables $(x_1, \cdots, x_{n-1})$. Then $g(x_n) = f(x_1, \cdots, x_n)$ is convex in x_n, so its maximum is attained at $x_n \in \{-1, 1\}$. Hence

$$\max_{\boldsymbol{x}\in[-1,1]^n} f(\boldsymbol{x}) = \max_{x_1,\cdots,x_{n-1}\in[-1,1]} \left(\max_{x_n\in\{-1,1\}} f(x_1, \cdots, x_n)\right),$$

Repeating this process for the remaining $n - 1$ variables,

$$\max_{\boldsymbol{x}\in[-1,1]^n} f(\boldsymbol{x}) = \max_{x_1\in\{-1,1\}} \left(\cdots \left(\max_{x_n\in\{-1,1\}} f(x_1, \cdots, x_n)\right)\right),$$

i.e., the maximum of $f(x_1, \cdots, x_n)$ is attained at $x_i \in \{-1, 1\}$.

(2) Let $f(\boldsymbol{z}) = \boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} = \sum_{i,j} a_{ij} z_i z_j$. When $\boldsymbol{x} \in \{-1,1\}^n$, $|f(\boldsymbol{x})| \leqslant 1$, i.e.,

$$\max_{\boldsymbol{z}\in[-1,1]^n} |\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z}| \leqslant 1.$$

Since $a_{ii} = 0$, the terms z_i^2 do not appear in the above sum, hence

$$\frac{\partial^2 f}{\partial z_i^2} = 2a_{ii} = 0.$$

Thus f is linear in z_i, so using the conclusion of (1), the extremum of f on the cube $[-1,1]^n$ must be attained at the vertices $\{-1,1\}^n$.

- First consider the case where $\boldsymbol{A}$ is symmetric

  Using the conclusion of Exercise Q3.49 (*Quadratic Grothendieck Inequality for Positive Semidefinite Matrices*) (1), for a symmetric matrix $\boldsymbol{A}$,

  $$\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} = \bar{\boldsymbol{x}}^\top \boldsymbol{A}\bar{\boldsymbol{x}} - \bar{\boldsymbol{y}}^\top \boldsymbol{A}\bar{\boldsymbol{y}},$$

  where $\bar{\boldsymbol{x}} = (\boldsymbol{x}+\boldsymbol{y})/2, \bar{\boldsymbol{y}} = (\boldsymbol{x}-\boldsymbol{y})/2$. When $\boldsymbol{x},\boldsymbol{y} \in \{-1,1\}^n$, $\bar{\boldsymbol{x}},\bar{\boldsymbol{y}} \in [-1,1]^n$, hence

  $$|\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}| \leqslant |\bar{\boldsymbol{x}}^\top \boldsymbol{A}\bar{\boldsymbol{x}}| + |\bar{\boldsymbol{y}}^\top \boldsymbol{A}\bar{\boldsymbol{y}}| \leqslant 1+1 = 2,$$

  Thus, using the equivalent form of Theorem T3.5.1 (*Grothendieck inequality*) (*Grothendieck inequality*) (Q3.47 (*Rewriting the Assumptions of Grothendieck's Inequality*)),

  $$\left|\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle\right| \leqslant K_G \cdot \sup_{\boldsymbol{x},\boldsymbol{y}\in\{-1,1\}^n} |\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}| \leqslant K_G \cdot 2 = 2K_G.$$

- Now consider the case where $\boldsymbol{A}$ is a general matrix

  Let $\boldsymbol{B} = (\boldsymbol{A}+\boldsymbol{A}^\top)/2$, satisfying $\boldsymbol{x}^\top \boldsymbol{B}\boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}$, $\boldsymbol{B}_{ii} = (a_{ii}+a_{ii})/2 = 0$. Then, similarly to the previous proof,

  $$\left|\sum_{i,j} b_{ij}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle\right| \leqslant K_G \cdot \sup_{\boldsymbol{x},\boldsymbol{y}\in\{-1,1\}^n} |\boldsymbol{x}^\top \boldsymbol{B}\boldsymbol{y}| \leqslant K_G \cdot 2 = 2K_G,$$

  where

  $$\sum_{i,j} b_{ij}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle = \sum_{i,j} \frac{a_{ij}+a_{ji}}{2}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle,$$

  hence

  $$\left|\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{v}_j\rangle\right| \leqslant 2K_G.$$

51 [**Properties and Characterization of Gram Matrices**] The Gram matrix of vectors $\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n$ is the $n \times n$ matrix $g_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j\rangle$. Prove the following.

(1) The Gram matrix of any set of vectors is symmetric and positive semidefinite.

(2) Conversely, any symmetric and positive semidefinite matrix is a Gram matrix of some set of vectors.

Prf. ($\Longrightarrow$). Symmetry:

$$g_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j\rangle = \langle \boldsymbol{v}_j, \boldsymbol{v}_i\rangle = g_{ji}.$$

Positive semidefiniteness: For any $\boldsymbol{c} = (c_1, \cdots, c_n)^\top \in \mathbb{R}^n$,

$$\begin{aligned}\boldsymbol{c}^\top \boldsymbol{G}\boldsymbol{c} &= \sum_{i=1}^n \sum_{j=1}^n c_i c_j g_{ij} = \sum_{i=1}^n \sum_{j=1}^n c_i c_j \langle \boldsymbol{v}_i, \boldsymbol{v}_j\rangle \\ &= \left\langle \sum_{i=1}^n c_i \boldsymbol{v}_i, \sum_{j=1}^n c_j \boldsymbol{v}_j \right\rangle = \left\| \sum_{i=1}^n c_i \boldsymbol{v}_i \right\|^2 \geqslant 0.\end{aligned}$$

($\Longleftarrow$).

Let $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$, where $\boldsymbol{Q}$ is an orthogonal matrix and $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \cdots, \lambda_n)$ is a diagonal matrix. Set $\boldsymbol{B} = \boldsymbol{Q}\boldsymbol{\Lambda}^{1/2} = (\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n)$. Then

$$\boldsymbol{B}\boldsymbol{B}^\top = (\boldsymbol{Q}\boldsymbol{\Lambda}^{1/2})(\boldsymbol{Q}\boldsymbol{\Lambda}^{1/2})^\top = \boldsymbol{Q}\boldsymbol{\Lambda}^{1/2}(\boldsymbol{\Lambda}^{1/2})^\top\boldsymbol{Q}^\top = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top = \boldsymbol{A},$$

satisfying

$$A_{ij} = \sum_{k=1}^{n} B_{ik}B_{jk} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle,$$

hence $\boldsymbol{A}$ is exactly the Gram matrix of $\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n\} \subset \mathbb{R}^n$.

52 [**SDP Relaxation of Boolean Bilinear Programming**] Consider the following Boolean bilinear programming:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}x_i y_j, \\ \text{subject to} \quad & x_i, y_j \in \{-1, 1\}, \end{aligned} \tag{I}$$

where $\boldsymbol{A} \in \mathbb{R}^{m\times n}$. Relax the above problem to the following vector-form optimization problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}\langle \boldsymbol{X}_i, \boldsymbol{Y}_j \rangle, \\ \text{subject to} \quad & \|\boldsymbol{X}_i\|_2 = \|\boldsymbol{Y}_j\|_2 = 1, \end{aligned} \tag{II}$$

whose maximum is taken over all unit vectors $\boldsymbol{X}_i, \boldsymbol{Y}_j \in \mathbb{R}^k$ for some fixed dimension $k \in \mathbb{N}$.

(1) Rewrite (II) in SDP form.

(2) Prove that (I) and (II) are approximable up to an absolute constant factor.

Prf. (1) By Exercise Q3.51 (*Properties and Characterization of Gram Matrices*), Gram matrices are symmetric positive semidefinite. Construct a symmetric matrix $\boldsymbol{M} \in \mathbb{R}^{(m+n)\times(m+n)}$ whose entries are the inner products of $\{\boldsymbol{X}_1, \cdots, \boldsymbol{X}_m, \boldsymbol{Y}_1, \cdots, \boldsymbol{Y}_n\}$,

$$\boldsymbol{M} = \begin{pmatrix} \langle \boldsymbol{X}_i, \boldsymbol{X}_{i'} \rangle & \langle \boldsymbol{X}_i, \boldsymbol{Y}_j \rangle \\ \langle \boldsymbol{Y}_j, \boldsymbol{X}_i \rangle & \langle \boldsymbol{Y}_j, \boldsymbol{Y}_{j'} \rangle \end{pmatrix},$$

then by the properties of Gram matrices, $\boldsymbol{M} \succcurlyeq 0$; by the unit vector constraint $\|\boldsymbol{X}_i\|_2 = \|\boldsymbol{Y}_j\|_2 = 1$, $M_{ii} = 1$.

Define the block matrix $\tilde{\boldsymbol{A}}$ as

$$\tilde{\boldsymbol{A}} = \frac{1}{2}\begin{pmatrix} \boldsymbol{O} & \boldsymbol{A} \\ \boldsymbol{A}^\top & \boldsymbol{O} \end{pmatrix},$$

then

$$\sum_{i,j} A_{ij}\langle \boldsymbol{X}_i, \boldsymbol{Y}_j \rangle = \operatorname{tr}(\tilde{\boldsymbol{A}}\boldsymbol{M}) = \langle \tilde{\boldsymbol{A}}, \boldsymbol{M} \rangle,$$

hence the SDP form of (II) is

$$\begin{aligned} \text{maximize} \quad & \langle \tilde{\boldsymbol{A}}, \boldsymbol{M} \rangle, \\ \text{subject to} \quad & \boldsymbol{M} \succcurlyeq 0, \\ & M_{ii} = 1. \end{aligned}$$

(2) Let $V_{\int}$ be the optimal value of the original problem (I), and V_{SDP} be the optimal value of the relaxed problem (II).

- Lower bound (trivial case)

  Note that any scalars $x_i, y_j \in \{-1, 1\}$ can be expressed as unit vectors in $\mathbb{R}^1$, so the feasible region of (I) is a subset of the feasible region of (II), hence $V_{\int} \leqslant V_{\text{SDP}}$.

- Upper bound (using Grothendieck's inequality)

  By Theorem T3.5.1 (*Grothendieck inequality*) (*Grothendieck inequality*), for any unit vectors $\boldsymbol{X}_i, \boldsymbol{Y}_j \in \mathcal{H}$, there

exists a Grothendieck constant $K_G \geqslant 1.67$ such that

$$\left|\sum_{i,j} A_{ij}\langle \boldsymbol{X}_i, \boldsymbol{Y}_j\rangle\right| \leqslant K_G \cdot \max_{x_i,y_j\in\{-1,1\}}\left|\sum_{i,j} A_{ij}x_i y_j\right|,$$

i.e., $V_{\text{SDP}} \leqslant K_G \cdot V_{\int}$.

Thus, $V_{\int} \leqslant V_{\text{SDP}} \leqslant K_G \cdot V_{\int}$, i.e., $V_{\int} \asymp V_{\text{SDP}}$.

53 [**Grothendieck's Identity**] Prove Lemma L3.6.5 (*Grothendieck identity*).

Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Prove: For any $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{S}^{n-1}$,

$$\mathbb{E}[\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}\rangle)\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{v}\rangle)] = \frac{2}{\pi}\arcsin\langle \boldsymbol{u}, \boldsymbol{v}\rangle.$$

Prf. Let $X = \langle \boldsymbol{g}, \boldsymbol{u}\rangle$ and $Y = \langle \boldsymbol{g}, \boldsymbol{v}\rangle$. Then (X, Y) follows a bivariate normal distribution, satisfying

$$\begin{aligned}\mathbb{E}[X] = \mathbb{E}[Y] &= \langle \mathbb{E}[\boldsymbol{g}], \boldsymbol{u}\rangle = 0,\\ \operatorname{Var}(X) = \operatorname{Var}(Y) &= \|\boldsymbol{u}\|_2^2 = 1,\\ \rho = \operatorname{Cov}(X, Y) &= \mathbb{E}[XY] = \boldsymbol{u}^\top \mathbb{E}[\boldsymbol{g}\boldsymbol{g}^\top]\boldsymbol{v} = \boldsymbol{u}^\top \boldsymbol{I}_n \boldsymbol{v} = \langle \boldsymbol{u}, \boldsymbol{v}\rangle,\end{aligned}$$

after which we only need to compute $I = \mathbb{E}[\operatorname{sgn}(X)\operatorname{sgn}(Y)]$.

It is easy to see that

$$\begin{aligned}1 &= \mathbb{P}(X, Y \text{ have the same sign}) + \mathbb{P}(X, Y \text{ have opposite signs}),\\ I &= \mathbb{P}(X, Y \text{ have the same sign}) - \mathbb{P}(X, Y \text{ have opposite signs}),\end{aligned}$$

hence $I = 2\mathbb{P}\{X, Y \text{ have the same sign}\} - 1$. (*It then remains to compute the probability that X and Y have the same sign.*)

In the support plane of the Gaussian distribution, the value of $\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}\rangle)$ depends on the direction of $\boldsymbol{g}$ relative to the hyperplane defined by $\boldsymbol{u}$. Then $\{\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}\rangle) = 1\}$ and $\{\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{v}\rangle) = 1\}$ correspond to two half-circle regions respectively, and their intersection (*i.e., the region where both are positive*) has angle $\pi - \alpha$, where $\alpha \in [0, \pi]$ is the angle between the unit vectors $\boldsymbol{u}$ and $\boldsymbol{v}$, satisfying $\rho = \cos\alpha$; similarly, the region where both are negative has angle $\pi - \alpha$. Hence

$$\mathbb{P}(X, Y \text{ have the same sign}) = \frac{2(\pi - \alpha)}{2\pi} = 1 - \frac{\alpha}{\pi},$$

thus

$$\begin{aligned}I &= 2\left(1 - \frac{\alpha}{\pi}\right) - 1 = 1 - \frac{2\alpha}{\pi}\\ &= 1 - \frac{2}{\pi}\left(\frac{\pi}{2} - \arcsin\rho\right) = \frac{2}{\pi}\arcsin\rho, \qquad (\alpha = \arccos\rho)\end{aligned}$$

i.e.,

$$\mathbb{E}[\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}\rangle)\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{v}\rangle)] = \frac{2}{\pi}\arcsin\langle \boldsymbol{u}, \boldsymbol{v}\rangle.$$

54 [**Max-Cut Approximation Algorithm: Deterministic Output and Randomized Running Time**] Theorem T3.6.4 (0.878-*approximation algorithm for maximum cut*) uses SDP relaxation and rounding (random hyperplane rounding) techniques to find a cut whose expected weight is at least 0.878 times the maximum cut.

For any $\varepsilon > 0$, design a $(0.878 - \varepsilon)$-approximation algorithm for the max-cut problem such that the final output cut always satisfies the approximation ratio requirement (*i.e., deterministic output*), but the running time of the algorithm is randomized. Then find an upper bound on the expected running time of the algorithm.

Prf. It remains to convert a Monte Carlo-type algorithm (*fixed running time, result guaranteed probabilistically*) into a Las Vegas-type algorithm (*deterministic result, randomized running time*).

Let $G = (V, E)$ with edge weights $w_{ij} \geqslant 0$, and let W_{opt} be the maximum cut weight of G.

Present the following algorithm:

Since $W_{\text{SDP}} \geqslant W_{\text{opt}}$, when the decision condition is satisfied, $W \geqslant (0.878 - \varepsilon)W_{\text{opt}}$.

Algorithm 1: Guaranteed-Max-Cut-Approximation(G, ε)

Input : Graph $G = (V, E)$, edge weights $\{w_{ij}\}_{i,j\in V} \geqslant 0$, approximation parameter $\varepsilon > 0$.

Output: A cut $S \subset V$ whose weight satisfies $W(S) \geqslant (\alpha_{GW} - \varepsilon)W_{\text{opt}}$.

Part 1: Solve the SDP relaxation problem

Solve the following SDP relaxation problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i<j} w_{ij} \frac{1 - \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle}{2}, \\ \text{subject to} \quad & \boldsymbol{v}_i \in \mathbb{S}^{n-1}, \end{aligned}$$

Let the optimal value be W_{SDP} and the optimal vector solution be $\{\boldsymbol{v}_i\}_{i=1}^n$.

Part 2: Iterative randomized rounding (Las Vegas process)

Set $\alpha \leftarrow \alpha_{GW} \approx 0.878$;

while True do

  Independently draw a standard normal random vector $\boldsymbol{g}$ from $N(\boldsymbol{0}, \boldsymbol{I}_n)$;

  Construct candidate cut: $S \leftarrow \{i \in V : \langle \boldsymbol{v}_i, \boldsymbol{g} \rangle \geqslant 0\}$;

  Compute weight: $W \leftarrow \sum_{i\in S, j\notin S} w_{ij}$;

  if $W \geqslant (\alpha - \varepsilon)W_{\text{SDP}}$ **then**

    return S;

  end

end

Let W be the cut weight produced by a single iteration. Then by the Goemans–Williamson theorem,

$$\mathbb{E}[W] \geqslant \alpha_{GW} W_{\text{SDP}},$$

where $0 \leqslant W \leqslant W_{\text{SDP}}$.

Consider $A = \{W < (\alpha_{GW} - \varepsilon)W_{\text{SDP}}\}$ (iteration failure), and let the deficit random variable $Z = W_{\text{SDP}} - W \geqslant 0$, satisfying

$$\mathbb{E}[Z] = W_{\text{SDP}} - \mathbb{E}[W] \leqslant (1 - \alpha_{GW})W_{\text{SDP}},$$

Then using Proposition P1.6.2 (*Markov inequality*) (*Markov's inequality*),

$$\mathbb{P}(A) = \mathbb{P}\{Z > (1 - \alpha_{GW} + \varepsilon)W_{\text{SDP}}\} \leqslant \frac{\mathbb{E}[Z]}{(1 - \alpha_{GW} + \varepsilon)W_{\text{SDP}}} \leqslant \frac{(1 - \alpha_{GW})W_{\text{SDP}}}{(1 - \alpha_{GW} + \varepsilon)W_{\text{SDP}}} = \frac{1 - \alpha_{GW}}{1 - \alpha_{GW} + \varepsilon},$$

Hence the probability that a single iteration succeeds is

$$p = \mathbb{P}\{A^c\} \geqslant 1 - \frac{1 - \alpha_{GW}}{1 - \alpha_{GW} + \varepsilon} = \frac{\varepsilon}{1 - \alpha_{GW} + \varepsilon}.$$

Let $K \sim G(p)$ be the number of iterations, satisfying

$$\mathbb{E}[K] = \frac{1}{p} = \frac{1 - \alpha_{GW}}{\varepsilon} + 1 \sim \frac{0.122}{\varepsilon} + 1,$$

Let T_{SDP} be the time for a single SDP solve and T_{round} be the time for a single random projection. Then the total expected running time is

$$\mathbb{E}[T_{total}] = T_{SDP} + \left(\frac{1 - \alpha_{GW} + \varepsilon}{\varepsilon}\right) T_{round},$$

which is linear in $1/\varepsilon$. Thus for fixed ε, the above algorithm is polynomial-time in expectation.

55 [**Kernel Trick for Real Analytic Functions**] To prove Lemma L3.7.7 (*Real analytic functions*), it remains only to introduce the infinite-dimensional version of $\mathbb{R}^n$, namely the square-summable sequence space ℓ^2, defined as the set of all sequences $\boldsymbol{x} = (x_1, x_2, \cdots)$ satisfying $\sum_{i=1}^{\infty} x_i^2 < \infty$, endowed with the following norm and inner product:

$$\|\boldsymbol{x}\|_2 = \left(\sum_{i=1}^{\infty} x_i^2\right)^{1/2}, \langle \boldsymbol{x}, \boldsymbol{y}\rangle = \sum_{i=1}^{\infty} x_i y_i.$$

Let $f(z) = \sum_{k=0}^{\infty} a_k z^k$ be a real analytic function, satisfying that $f(z)$ converges absolutely on $\mathbb{R}$. Prove that there exists a Hilbert space $\mathcal{H}$ (*taken here as the sequence space* ℓ^2) and transformations $\Phi, \Psi : \mathbb{R}^n \to \mathcal{H}$ such that:
(1) For any $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$,

$$\langle \Phi(\boldsymbol{u}), \Psi(\boldsymbol{v})\rangle = f(\langle \boldsymbol{u}, \boldsymbol{v}\rangle).$$

(2) For any $\boldsymbol{u} \in \mathbb{S}^{n-1}$, the norms of their images satisfy

$$\|\Phi(\boldsymbol{u})\|_2^2 = \|\Psi(\boldsymbol{u})\|_2^2 = \sum_{k=0}^{\infty} |a_k|.$$

Prf. (1) By the properties of inner products,

$$\langle \boldsymbol{u}^{\otimes k}, \boldsymbol{v}^{\otimes k}\rangle = (\langle \boldsymbol{u}, \boldsymbol{v}\rangle)^k.$$

Define Φ by weighting with the square root of the absolute values of the coefficients,

$$\Phi(\boldsymbol{u}) = (\sqrt{|a_0|}, \sqrt{|a_1|}\boldsymbol{u}^{\otimes 1}, \sqrt{|a_2|}\boldsymbol{u}^{\otimes 2}, \cdots, \sqrt{|a_k|}\boldsymbol{u}^{\otimes k}, \cdots),$$

and Ψ by introducing the signs s_k of the original coefficients on top of the weighting,

$$\Psi(\boldsymbol{u}) = (s_0\sqrt{|a_0|}, s_1\sqrt{|a_1|}\boldsymbol{u}^{\otimes 1}, s_2\sqrt{|a_2|}\boldsymbol{u}^{\otimes 2}, \cdots, s_k\sqrt{|a_k|}\boldsymbol{u}^{\otimes k}, \cdots),$$

then

$$\begin{aligned}\langle \Phi(\boldsymbol{u}), \Psi(\boldsymbol{v})\rangle &= \sum_{k=0}^{\infty} \langle \sqrt{|a_k|}\boldsymbol{u}^{\otimes k}, s_k\sqrt{|a_k|}\boldsymbol{v}^{\otimes k}\rangle \\ &= \sum_{k=0}^{\infty} s_k|a_k|\langle \boldsymbol{u}^{\otimes k}, \boldsymbol{v}^{\otimes k}\rangle = \sum_{k=0}^{\infty} a_k(\langle \boldsymbol{u}, \boldsymbol{v}\rangle)^k = f(\langle \boldsymbol{u}, \boldsymbol{v}\rangle). \qquad (a_k = s_k|a_k|)\end{aligned}$$

(2) Note that $\|\boldsymbol{u}^{\otimes k}\|_2 = \|\boldsymbol{u}\|_2^k = 1$ and $s_k^2 \in \{0, 1\}$. Hence

$$\begin{aligned}\|\Phi(\boldsymbol{u})\|_2^2 &= \sum_{k=0}^{\infty} \|\sqrt{|a_k|}\boldsymbol{u}^{\otimes k}\|_2^2 = \sum_{k=0}^{\infty} |a_k| \cdot \|\boldsymbol{u}\|_2^{2k} = \sum_{k=0}^{\infty} |a_k|, \\ \|\Psi(\boldsymbol{u})\|_2^2 &= \sum_{k=0}^{\infty} (s_k\sqrt{|a_k|})^2 \cdot \|\boldsymbol{u}\|_2^{2k} = \sum_{k\in\{k:a_k\neq 0\}} |a_k| = \sum_{k=0}^{\infty} |a_k|,\end{aligned}$$

where $\sum_{k=0}^{\infty} |a_k| < \infty$, so $\|\Phi(\boldsymbol{u})\|_2^2, \|\Psi(\boldsymbol{u})\|_2^2 < \infty$, i.e., the mapping is well-defined in ℓ^2.

56 [**Linearization of Grothendieck's Identity**] Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Let's get rid of the nonlinearity "arcsin" in Lemma L3.6.5 (*Grothendieck identity*) by adding a correction term.
Prove the following for any $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{S}^{n-1}$:
(1)

$$\mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{u}\rangle \mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{v}\rangle)] = \sqrt{\frac{2}{\pi}}\langle \boldsymbol{u}, \boldsymbol{v}\rangle.$$

(2)

$$\mathbb{E}[\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}\rangle)\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{v}\rangle)] = \frac{2}{\pi}\langle \boldsymbol{u}, \boldsymbol{v}\rangle + \mathbb{E}[Z_{\boldsymbol{u}} Z_{\boldsymbol{v}}].$$

where the correction random variables are defined as

$$Z_{\boldsymbol{w}} = \sqrt{\frac{2}{\pi}}\langle \boldsymbol{g}, \boldsymbol{w}\rangle - \mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{w}\rangle). \qquad (\boldsymbol{w} \in \{\boldsymbol{u}, \boldsymbol{v}\})$$

Prf. (1) Let $X = \langle \boldsymbol{g}, \boldsymbol{u} \rangle$ and $Y = \langle \boldsymbol{g}, \boldsymbol{v} \rangle$, satisfying

$$\begin{aligned} \mathbb{E}[X] = \mathbb{E}[Y] &= 0, \\ \mathrm{Var}(X) = \mathrm{Var}(Y) &= 1, \\ \rho = \mathbb{E}[XY] &= \langle \boldsymbol{u}, \boldsymbol{v} \rangle, \end{aligned}$$

Then $(X, Y) \sim N(0, 0, 1, 1, \rho)$.

Decompose $X = \rho Y + \sqrt{1-\rho^2} W$, where $Y \perp W \sim N(0,1)$. Then

$$\begin{aligned} \mathbb{E}[X\mathrm{sgn}(Y)] &= \mathbb{E}[(\rho Y + \sqrt{1-\rho^2} W)\mathrm{sgn}(Y)] \\ &= \rho \mathbb{E}[Y\mathrm{sgn}(Y)] + \sqrt{1-\rho^2}\mathbb{E}[W]\mathbb{E}[\mathrm{sgn}(Y)] \\ &= \rho \mathbb{E}[|Y|], \end{aligned}$$

where

$$\mathbb{E}[|Y|] = \int_{-\infty}^{\infty} |y| \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-y^2/2} \mathrm{d}y = 2 \int_0^{\infty} y \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-y^2/2} \mathrm{d}y = \sqrt{\frac{2}{\pi}},$$

hence

$$\mathbb{E}[X\mathrm{sgn}(Y)] = \rho \cdot \sqrt{\frac{2}{\pi}} = \sqrt{\frac{2}{\pi}} \langle \boldsymbol{u}, \boldsymbol{v} \rangle.$$

(2) Let

$$L_{\boldsymbol{w}} = \sqrt{\frac{2}{\pi}} \langle \boldsymbol{g}, \boldsymbol{w} \rangle,$$

and $S_{\boldsymbol{w}} = \mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{w} \rangle)$. Then

$$\begin{aligned} \mathbb{E}[Z_{\boldsymbol{u}} Z_{\boldsymbol{v}}] &= \mathbb{E}[(L_{\boldsymbol{u}} - S_{\boldsymbol{u}})(L_{\boldsymbol{v}} - S_{\boldsymbol{v}})] \\ &= \mathbb{E}[L_{\boldsymbol{u}} L_{\boldsymbol{v}}] - \mathbb{E}[L_{\boldsymbol{u}} S_{\boldsymbol{v}}] - \mathbb{E}[S_{\boldsymbol{u}} L_{\boldsymbol{v}}] + \mathbb{E}[S_{\boldsymbol{u}} S_{\boldsymbol{v}}], \end{aligned}$$

where

$$\begin{aligned} \mathbb{E}[L_{\boldsymbol{u}} L_{\boldsymbol{v}}] &= \frac{2}{\pi} \mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{u} \rangle \langle \boldsymbol{g}, \boldsymbol{v} \rangle] = \frac{2}{\pi} \langle \boldsymbol{u}, \boldsymbol{v} \rangle, \\ \mathbb{E}[L_{\boldsymbol{u}} S_{\boldsymbol{v}}] &= \sqrt{\frac{2}{\pi}} \mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{u} \rangle \mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{v} \rangle)] = \sqrt{\frac{2}{\pi}} \cdot \sqrt{\frac{2}{\pi}} \langle \boldsymbol{u}, \boldsymbol{v} \rangle = \frac{2}{\pi} \langle \boldsymbol{u}, \boldsymbol{v} \rangle, \\ \mathbb{E}[S_{\boldsymbol{u}} L_{\boldsymbol{v}}] &= \frac{2}{\pi} \langle \boldsymbol{u}, \boldsymbol{v} \rangle, \end{aligned}$$

hence

$$\begin{aligned} \mathbb{E}[Z_{\boldsymbol{u}} Z_{\boldsymbol{v}}] &= \frac{2}{\pi} \langle \boldsymbol{u}, \boldsymbol{v} \rangle - \frac{2}{\pi} \langle \boldsymbol{u}, \boldsymbol{v} \rangle - \frac{2}{\pi} \langle \boldsymbol{u}, \boldsymbol{v} \rangle + \mathbb{E}[S_{\boldsymbol{u}} S_{\boldsymbol{v}}] \\ &= -\frac{2}{\pi} \langle \boldsymbol{u}, \boldsymbol{v} \rangle + \mathbb{E}[\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{u} \rangle)\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{v} \rangle)], \end{aligned}$$

i.e.,

$$\mathbb{E}[\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{u} \rangle)\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{v} \rangle)] = \frac{2}{\pi} \langle \boldsymbol{u}, \boldsymbol{v} \rangle + \mathbb{E}[Z_{\boldsymbol{u}} Z_{\boldsymbol{v}}].$$

57 [**Refined Grothendieck Inequality for Positive Semidefinite Matrices**] For positive semidefinite matrices, Theorem T3.5.1 (*Grothendieck inequality*) holds with a better constant $\pi/2 \approx 1.571$, with a simpler proof. Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be a symmetric positive semidefinite matrix satisfying $\sum_{i,j} a_{ij} x_i x_j \leqslant 1$ for all $\boldsymbol{x} \in \{-1, 1\}^n$. Use Exercise Q3.56 (*Linearization of Grothendieck's Identity*) to show that for any unit vectors $\boldsymbol{u}_i \in \mathcal{H}$,

$$\sum_{i,j=1}^{n} a_{ij} \langle \boldsymbol{u}_i, \boldsymbol{u}_j \rangle \leqslant \frac{\pi}{2}.$$

Prf. By Exercise Q3.56 (*Linearization of Grothendieck's Identity*), for any unit vectors $\boldsymbol{u}_i, \boldsymbol{u}_j \in \mathcal{H}$,

$$\mathbb{E}[\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}_i \rangle)\mathrm{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}_j \rangle)] = \frac{2}{\pi} \langle \boldsymbol{u}_i, \boldsymbol{u}_j \rangle + \mathbb{E}[Z_i Z_j],$$

where the correction random variables are defined as

$$Z_k = \sqrt{\frac{2}{\pi}}\langle \boldsymbol{g}, \boldsymbol{u}_k\rangle - \operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}_k\rangle),$$

Multiply both sides by a_{ij} and sum,

$$\sum_{i,j} a_{ij}\mathbb{E}[\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}_i\rangle)\operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}_j\rangle)] = \frac{2}{\pi}\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{u}_j\rangle + \sum_{i,j} a_{ij}\mathbb{E}[Z_iZ_j],$$

i.e.,

$$\frac{2}{\pi}\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{u}_j\rangle = \mathbb{E}\left[\sum_{i,j} a_{ij}X_iX_j\right] - \mathbb{E}\left[\sum_{i,j} a_{ij}Z_iZ_j\right],$$

where $X_k = \operatorname{sgn}(\langle \boldsymbol{g}, \boldsymbol{u}_k\rangle) \in \{-1, 1\}$ (*since* $\boldsymbol{u}_k$ *is a unit vector*).

It is easy to see that

$$\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \leqslant 1 \implies \sum_{i,j} a_{ij}X_iX_j \leqslant 1 \implies \mathbb{E}\left[\sum_{i,j} a_{ij}X_iX_j\right] \leqslant 1,$$

Since $\boldsymbol{A}$ is positive semidefinite, for any $\boldsymbol{z} \in \mathbb{R}^n$,

$$\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} \geqslant 0 \implies \sum_{i,j} a_{ij}Z_iZ_j \geqslant 0 \implies \mathbb{E}\left[\sum_{i,j} a_{ij}Z_iZ_j\right] \geqslant 0,$$

hence

$$\frac{2}{\pi}\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{u}_j\rangle \leqslant 1 - 0 = 1,$$

i.e.,

$$\sum_{i,j} a_{ij}\langle \boldsymbol{u}_i, \boldsymbol{u}_j\rangle \leqslant \frac{\pi}{2}.$$

58 [**SDP Relaxation Analysis Based on Grothendieck's Inequality**] Let's use the solution of Exercise Q3.57 (*Refined Grothendieck Inequality for Positive Semidefinite Matrices*) to improve Theorem T3.5.7 (*The guarantee of relaxation*). Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric positive semidefinite matrix. Define the optimal value of the corresponding Boolean quadratic programming as

$$\operatorname{int}(\boldsymbol{A}) = \max\left\{\sum_{i,j=1}^{n} A_{ij}x_ix_j : x_i \in \{-1, 1\}\right\},$$

and the optimal value of its SDP relaxation as

$$\operatorname{sdp}(\boldsymbol{A}) = \max\left\{\sum_{i,j=1}^{n} A_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle : \boldsymbol{X}_i \in \mathbb{S}^{n-1}\right\}.$$

(1) Prove that

$$\operatorname{int}(\boldsymbol{A}) \leqslant \operatorname{sdp}(\boldsymbol{A}) \leqslant \frac{\pi}{2}\cdot \operatorname{int}(\boldsymbol{A}).$$

(2) Design a randomized algorithm that maps the SDP solution $\{\boldsymbol{X}_i\}$ to integer solutions $x_i = \pm 1$, and then prove that the above solution approximates the original problem in expectation.

Prf. (1) • Left-hand inequality

Define the feasible set of the integer program as

$$\mathcal{F}_{\text{int}} = \{\boldsymbol{x} \in \mathbb{R}^n : x_i \in \{-1, 1\}\},$$

and the feasible set of the SDP relaxation problem as

$$\mathcal{F}_{\text{sdp}} = \{(\boldsymbol{X}_1, \cdots, \boldsymbol{X}_n) : \boldsymbol{X}_i \in \mathbb{S}^{n-1}\},$$

(*Then consider constructing the embedding map* $\phi : \mathcal{F}_{int} \to \mathcal{F}_{sdp}$.)

For any $\boldsymbol{x} \in \mathcal{F}_{\text{int}}$, set $\boldsymbol{X}_i = x_i \boldsymbol{e}_1 \in \mathbb{R}^n$, satisfying

$$\|\boldsymbol{X}_i\|_2 = |x_i| \cdot \|\boldsymbol{e}_1\|_2 = 1,$$

Then $(\boldsymbol{X}_1, \cdots, \boldsymbol{X}_n) \in \mathcal{F}_{\text{sdp}}$, and

$$\langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle = \langle x_i \boldsymbol{e}_1, x_j \boldsymbol{e}_1 \rangle = x_i x_j \langle \boldsymbol{e}_1, \boldsymbol{e}_1 \rangle = x_i x_j,$$

Hence the SDP objective function simplifies to

$$\sum_{i,j=1}^{n} A_{ij} \langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle = \sum_{i,j=1}^{n} A_{ij} x_i x_j.$$

Let $f(\boldsymbol{X}_1, \cdots, \boldsymbol{X}_n) = \sum_{i,j} A_{ij} \langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle$. Then all possible values of the integer program are contained in the set of values of the SDP objective function, i.e.,

$$\left\{ \sum_{i,j} A_{ij} x_i x_j : \boldsymbol{x} \in \mathcal{F}_{\text{int}} \right\} \subset \{ f(\boldsymbol{X}_1, \cdots, \boldsymbol{X}_n) : (\boldsymbol{X}_1, \cdots, \boldsymbol{X}_n) \in \mathcal{F}_{\text{sdp}} \},$$

Taking the supremum on both sides,

$$\max_{\mathcal{F}_{\text{int}}} \sum_{i,j} A_{ij} x_i x_j \leqslant \max_{\mathcal{F}_{\text{sdp}}} f(\boldsymbol{X}_1, \cdots, \boldsymbol{X}_n),$$

i.e., $\text{int}(\boldsymbol{A}) \leqslant \text{sdp}(\boldsymbol{A})$.

- Right-hand inequality

For

$$\text{int}(\boldsymbol{A}) = \max_{\boldsymbol{x} \in \{-1,1\}^n} \sum_{i,j=1}^{n} a_{ij} x_i x_j, \text{sdp}(\boldsymbol{A}) = \max_{\boldsymbol{X}_i \in \mathbb{S}^{n-1}} \sum_{i,j=1}^{n} a_{ij} \langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle,$$

If $\boldsymbol{A} = \boldsymbol{O}$, the conclusion is trivial; we only need to consider $\boldsymbol{A} \neq \boldsymbol{O}$ afterwards.

(*We consider constructing a normalized matrix.*)

Construct a new positive semidefinite matrix $\tilde{\boldsymbol{A}} = (\tilde{a}_{ij})$ as

$$\tilde{a}_{ij} = \frac{a_{ij}}{\text{int}(\boldsymbol{A})},$$

Then for any $\boldsymbol{x} \in \{-1, 1\}^n$,

$$\sum_{i,j=1}^{n} \tilde{a}_{ij} x_i x_j = \frac{1}{\text{int}(\boldsymbol{A})} \sum_{i,j=1}^{n} a_{ij} x_i x_j \leqslant \frac{\text{int}(\boldsymbol{A})}{\text{int}(\boldsymbol{A})} = 1,$$

Hence by Exercise Q3.57 (*Refined Grothendieck Inequality for Positive Semidefinite Matrices*),

$$\sum_{i,j=1}^{n} \tilde{a}_{ij} \langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle \leqslant \frac{\pi}{2},$$

i.e.,

$$\frac{1}{\text{int}(\boldsymbol{A})} \sum_{i,j=1}^{n} a_{ij} \langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle \leqslant \frac{\pi}{2},$$

where $\text{int}(\boldsymbol{A}) > 0$ (*since* $\boldsymbol{A}$ *is a positive semidefinite matrix*), thus

$$\sum_{i,j=1}^{n} a_{ij} \langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle \leqslant \frac{\pi}{2} \cdot \text{int}(\boldsymbol{A}).$$

Thus,

$$\text{sdp}(\boldsymbol{A}) = \max_{\boldsymbol{X}_i \in \mathbb{S}^{n-1}} \sum_{i,j=1}^{n} a_{ij} \langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle \leqslant \frac{\pi}{2} \text{int}(\boldsymbol{A}).$$

(2) Present the following algorithm:

Let $V_{round} = \sum_{i,j} A_{ij} x_i x_j$ be the random value output by the algorithm. Then

Algorithm 2: Random Hyperplane Rounding Algorithm

```
Solve the SDP problem to obtain the optimal vector set {X_1, ..., X_n} ⊂ S^{n-1};
Independently draw a standard normal random vector g from N(0, I_n);
for i = 1, ..., n do
    x_i ← sgn(⟨X_i, g⟩);
end
return {x_1, ..., x_n};
```

$$\mathbb{E}[V_{round}] = \sum_{i,j} A_{ij} \mathbb{E}[x_i x_j],$$

Hence using Grothendieck's identity (Exercise Q3.53 (*Grothendieck's Identity*)),

$$\mathbb{E}[\operatorname{sgn}(\langle \boldsymbol{X}_i, \boldsymbol{g} \rangle) \operatorname{sgn}(\langle \boldsymbol{X}_j, \boldsymbol{g} \rangle)] = \frac{2}{\pi} \arcsin(\langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle),$$

Let $\rho_{ij} = \langle \boldsymbol{X}_i, \boldsymbol{X}_j \rangle$. Then

$$\mathbb{E}[V_{round}] = \frac{2}{\pi} \sum_{i,j} A_{ij} \arcsin(\rho_{ij}),$$

where $\arcsin \rho = \sum_{k=0}^{\infty} c_{2k+1} \rho^{2k+1}$. Thus

$$\mathbb{E}[V_{round}] = \frac{2}{\pi} \sum_{i,j} A_{ij} \rho_{ij} + \frac{2}{\pi} \sum_{k=1}^{\infty} c_{2k+1} \left(\sum_{i,j} A_{ij} \rho_{ij}^{2k+1} \right),$$

Since $\boldsymbol{A}$ is positive semidefinite and ρ_{ij}^{2k+1} is the Gram matrix of the corresponding tensor product vectors (*also positive semidefinite*), we have $\operatorname{tr}(\boldsymbol{A}\boldsymbol{G}^{(2k+1)}) \geqslant 0$. Hence

$$\mathbb{E}[V_{round}] \geqslant \frac{2}{\pi} \sum_{i,j} A_{ij} \rho_{ij} = \frac{2}{\pi} \operatorname{sdp}(\boldsymbol{A}) \geqslant \frac{2}{\pi} \operatorname{int}(\boldsymbol{A}),$$

i.e., the above algorithm achieves an approximation ratio of at least $2/\pi \approx 0.637$ in expectation.

4 Random Matrices

Keypoints

Section 4.1 Linear algebra review — Q4.1 ∼ Q4.17

Theorem 4.1.1 (*SVD decomposition*)

Any $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ can be written as

$$\boldsymbol{A} = \sum_{i=1}^{r} s_i \boldsymbol{u}_i \boldsymbol{v}_i^\top, \qquad (r = \min(m,n))$$

In this case, we call $s_i \geqslant 0$ the singular values, $\boldsymbol{u}_i \in \mathbb{R}^m$ the orthonormal left singular vectors, and $\boldsymbol{v}_i \in \mathbb{R}^n$ the orthonormal right singular vectors.

Theorem 4.1.6 (*Minimax theorem for eigenvalues*)

For a symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, the k-th principal eigenvalue satisfies

$$\lambda_k(\boldsymbol{A}) = \max_{\dim E = k} \min_{\boldsymbol{x}\in\mathbb{S}(E)} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = \min_{\dim E = n-k+1} \max_{\boldsymbol{x}\in\mathbb{S}(E)} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x},$$

where $\mathbb{S}(E)$ is the unit sphere of the subspace E.

Corollary 4.1.7 (*Minimax theorem for singular values*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ have singular values $s_1 \geqslant s_2 \geqslant \cdots$. Then

$$s_k(\boldsymbol{A}) = \max_{\dim E = k} \min_{\boldsymbol{x}\in\mathbb{S}(E)} \|\boldsymbol{A}\boldsymbol{x}\|_2 = \min_{\dim E = n-k+1} \max_{\boldsymbol{x}\in\mathbb{S}(E)} \|\boldsymbol{A}\boldsymbol{x}\|_2.$$

Definition 4.1.8 (*Operator norm*)

$$\|\boldsymbol{A}\| = \max_{\boldsymbol{x}\neq 0} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2} = \max_{\|\boldsymbol{x}\|_2=1} \|\boldsymbol{A}\boldsymbol{x}\|_2 = \max_{\|\boldsymbol{x}\|_2=\|\boldsymbol{y}\|_2=1} |\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}|.$$

Lemma 4.1.10 (*Orthogonal invariance*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, $\boldsymbol{U} \in \mathbb{R}^{m\times m}$, $\boldsymbol{V} \in \mathbb{R}^{n\times n}$ be orthogonal matrices. Then

$$\|\boldsymbol{U}\boldsymbol{A}\boldsymbol{V}\|_F = \|\boldsymbol{A}\|_F, \|\boldsymbol{U}\boldsymbol{A}\boldsymbol{V}\| = \|\boldsymbol{A}\|.$$

Lemma 4.1.11 (*Matrix norms via singular values*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ have singular values $s_1 \geqslant s_2 \geqslant \cdots$. Then

$$\|\boldsymbol{A}\|_F = \left(\sum_i s_i(\boldsymbol{A})^2\right)^{1/2}, \|\boldsymbol{A}\| = s_1(\boldsymbol{A}).$$

Theorem 4.1.13 (*Eckart–Young–Mirsky theorem*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ have SVD $\boldsymbol{A} = \sum_i s_i \boldsymbol{u}_i \boldsymbol{v}_i^\top$. Then for any $1 \leqslant k < r$, among the matrices of rank at most k, the best approximation error of $\boldsymbol{A}$ in the operator norm is

$$\min_{\operatorname{rank}(\boldsymbol{B})\leqslant k} \|\boldsymbol{A}-\boldsymbol{B}\| = s_{k+1},$$

The minimum is attained at the truncated singular value matrix $\boldsymbol{A}_k = \sum_{i=1}^k s_i \boldsymbol{u}_i \boldsymbol{v}_i^\top$.

Lemma 4.1.14 (*Weyl's inequality*)

For any symmetric matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n\times n}$,

$$|\lambda_i(\boldsymbol{A}+\boldsymbol{B}) - \lambda_i(\boldsymbol{A})| \leqslant \|\boldsymbol{B}\|.$$

Theorem 4.1.15 (*Davis–Kahan inequality*)

Let $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n\times n}$ be symmetric matrices with unit eigenvectors $\boldsymbol{v}_i(\boldsymbol{A})$, $\boldsymbol{v}_i(\boldsymbol{B})$. Then

$$\|\boldsymbol{v}_i(\boldsymbol{A}) - \boldsymbol{v}_i(\boldsymbol{B})\|_2 \leqslant \frac{\|\boldsymbol{A}-\boldsymbol{B}\|}{\min_{j\neq i}|\lambda_i(\boldsymbol{A}) - \lambda_j(\boldsymbol{A})|}.$$

Lemma 4.1.16 (*Davis–Kahan inequality for spectral projections*)

Let $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n\times n}$ be symmetric matrices, and let $\boldsymbol{P}_A$, $\boldsymbol{P}_B$ be orthogonal projections onto the span of eigenvectors corresponding to $I \subset \mathbb{R}$. Then

$$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \frac{\|\boldsymbol{A}-\boldsymbol{B}\|}{\operatorname{dist}(I, \sigma(\boldsymbol{A}) \setminus I)},$$

Lemma 4.1.17 (*Approximate isometries*)

If a matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ satisfies $\|\boldsymbol{A}^\top \boldsymbol{A} - \boldsymbol{I}_n\| \leqslant \varepsilon$ ($\varepsilon \in (0,1)$), then for any $\boldsymbol{x} \in \mathbb{R}^n$,

$$(1-\varepsilon)\|\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{A}\boldsymbol{x}\|_2 \leqslant (1+\varepsilon)\|\boldsymbol{x}\|_2.$$

Section 4.2 Nets, covering, and packing Q4.18 ∼ Q4.32

Definition 4.2.1 (ε-*net*)

Let $\mathbb{T} \subset \mathbb{R}^n$ and $\mathcal{N} \subset \mathbb{T}$. If for every $\boldsymbol{x} \in \mathbb{T}$, there exists $\boldsymbol{y} \in \mathcal{N}$ such that $\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \varepsilon$, then $\mathcal{N}$ is called an ε-net of $\mathbb{T}$ (*with respect to the Euclidean metric*).

(*If a set is unbounded, then the covering number is* $+\infty$*, which is trivial; in the sequel,* $\mathbb{T}$ *refers to a bounded subset.*)

Definition 4.2.2 (*Covering number*)

The covering number of $\mathbb{T} \subset \mathbb{R}^n$, denoted $\mathcal{N}(\mathbb{T}, \varepsilon) = \mathcal{N}(\mathbb{T}, \|\cdot\|_2, \varepsilon)$, is defined as the minimal cardinality of an ε-net of $\mathbb{T}$,

$$\mathcal{N}(T, \varepsilon) := \inf\{|\mathcal{N}| : \mathcal{N} \subset \mathbb{T} \text{ is an } \varepsilon\text{-net of } \mathbb{T}\}.$$

Definition 4.2.4 (*Packing number*)
The packing number of $\mathbb{T} \subset \mathbb{R}^n$, denoted $\mathcal{P}(\mathbb{T}, \varepsilon)$, is defined as the maximal cardinality of an ε-separated subset of $\mathbb{T}$,

$$\mathcal{P}(\mathbb{T}, \varepsilon) := \sup\{|\mathcal{P}| : \mathcal{P} \subset \mathbb{T} \text{ is an } \varepsilon\text{-separated set of } \mathbb{T}\}.$$

Remark 4.2.5 (*Packing balls into K*)
If $\mathcal{N}$ is ε-separated, the closed $\varepsilon/2$-balls centered at points in $\mathcal{N}$ are disjoint by the triangle inequality. So we can always pack at least $\mathcal{P}(K, d, \varepsilon)$ disjoint $\varepsilon/2$-balls into K。

Lemma 4.2.6 (*A net from a separated set*)
Any maximal ε-separated subset of $\mathbb{T}$ is an ε-net of $\mathbb{T}$.

Remark 4.2.7 (*Constructing a net*)
Lemma L4.2.6 (*A net from a separated set*) gives an iterative algorithm to construct an ε-net of a given set K: pick a point $\boldsymbol{x}_1 \in K$ arbitrarily, then pick $\boldsymbol{x}_2 \in K$ that is farther than ε from $\boldsymbol{x}_1$, then pick $\boldsymbol{x}_3$ that is farther than ε from both $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, and so on. If K is compact, this process stops in finite time (why?) and gives an ε-net of K。

Lemma 4.2.8 (*Equivalence of covering and packing numbers*)

$$\mathcal{P}(\mathbb{T}, 2\varepsilon) \leqslant \mathcal{N}(\mathbb{T}, \varepsilon) \leqslant \mathcal{P}(\mathbb{T}, \varepsilon).$$

Definition 4.2.9 (*Minkowski sum*)
$A + B = \{a + b : a \in A, b \in B\}$.

Proposition 4.2.10 (*Covering and volume*)
Let $\mathbb{T} \subset \mathbb{R}^n$. Then

$$\mathcal{N}(\mathbb{T}, \varepsilon) \leqslant \frac{\mathrm{Vol}(\mathbb{T} + (\varepsilon/2)\mathbb{B}_2^n)}{\mathrm{Vol}((\varepsilon/2)\mathbb{B}_2^n)}.$$

Corollary 4.2.11 (*Covering the Euclidean ball*)
For $\varepsilon \in (0, 1)$,

$$\mathcal{N}(\mathbb{B}_2^n, \varepsilon) \leqslant (3/\varepsilon)^n.$$

Definition 4.2.14 (*Hamming distance*)
For $x, y \in \{0,1\}^n$, $d_H(x, y) = |\{i : x_i \neq y_i\}|$, the number of coordinates where x and y differ.

Proposition 4.2.15 (*Covering and packing the Hamming cube*)
For the Hamming cube $\{0,1\}^n$ with normalized Hamming distance $d(x, y) = \frac{1}{n} d_H(x, y)$:

$$\exp(nh(\varepsilon/2)) \lesssim \mathcal{N}(\{0,1\}^n, \varepsilon) \lesssim \exp(nh(\varepsilon)),$$

where $h(p) = -p \log p - (1-p)\log(1-p)$ is the binary entropy.

Section 4.3 Error-correcting codes Q4.33

Proposition 4.3.1 (*Metric entropy and encoding*)
The minimal number of binary digits needed to encode a point $\boldsymbol{x} \in \mathbb{T}$ with precision ε is at least $\log_2 \mathcal{N}(\mathbb{T}, \varepsilon)$.

Definition 4.3.3 (*Error-correcting codes*)
An (n, M, d)-code is a set of M codewords in $\{0,1\}^n$ with Hamming distance at least d.

Lemma 4.3.4 (*Error correction and packing*)

An (n, M, d)-code can correct up to $\lfloor (d-1)/2 \rfloor$ errors.

Theorem 4.3.5 (*Guarantee of error-correcting codes*)

For any $\delta \in (0, 1/2)$, there exists an (n, M, d)-code whose rate $R = \log_2 M/n$ and relative distance d/n satisfy:

$$R = \frac{\log_2 M}{n} \geqslant 1 - h(\delta), \frac{d}{n} \geqslant \delta,$$

where h is the binary entropy function.

Section 4.4 Upper bounds for sub-Gaussian random matrices Q4.34 ∼ Q4.44

Lemma 4.4.1 (*Estimating the operator norm on a discrete net*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, let $\mathcal{N}$ be an ε-net of $\mathbb{S}^{n-1}$, and let $\mathcal{M}$ be an ε-net of $\mathbb{S}^{m-1}$, where $\varepsilon < 1/2$. Then

$$\|\boldsymbol{A}\| \leqslant \frac{1}{1-2\varepsilon} \max_{\boldsymbol{x}\in\mathcal{N},\, \boldsymbol{y}\in\mathcal{M}} \langle \boldsymbol{y}, \boldsymbol{A}\boldsymbol{x}\rangle.$$

Lemma 4.4.2 (*Estimating the quadratic form of a symmetric matrix on a discrete net*)

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric matrix and $\mathcal{N}$ an ε-net of $\mathbb{S}^{n-1}$, where $\varepsilon < 1/2$. Then

$$\|\boldsymbol{A}\| \leqslant \frac{1}{1-2\varepsilon} \max_{\boldsymbol{x}\in\mathcal{N}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{x}\rangle|.$$

Theorem 4.4.3 (*Norm bounds for independent sub-Gaussian matrices*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix whose entries A_{ij} are independent, zero-mean K-sub-Gaussian random variables. Then:

(1)

$$\mathbb{E}\|\boldsymbol{A}\| \leqslant CK(\sqrt{m} + \sqrt{n}),$$

(2) for any $t \geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{A}\| \geqslant CK(\sqrt{m} + \sqrt{n} + t)\} \leqslant 2\exp(-ct^2).$$

Corollary 4.4.7 (*Norm bounds for symmetric sub-Gaussian matrices*)

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric random matrix whose entries on and above the diagonal are independent, zero-mean K-sub-Gaussian random variables. Then:

(1)

$$\mathbb{E}\|\boldsymbol{A}\| \leqslant CK\sqrt{n},$$

(2) for any $t \geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{A}\| \geqslant CK(\sqrt{n} + t)\} \leqslant 2\exp(-ct^2).$$

Section 4.5 Community detection in networks Q4.45

Definition 4.5.1 (*Stochastic block model*)

The stochastic block model $\mathrm{SBM}(n, p, q)$ with two equal-size communities generates a random graph on n vertices as follows:

1. Label the first $n/2$ vertices as community $+1$ and the remaining $n/2$ vertices as community -1.

2. Vertices within the same community are connected independently with probability p, and vertices in different communities are connected independently with probability q.

Theorem 4.5.2 (*Spectral clustering for the stochastic block model*)

Consider the SBM with $p = a\log n/n$ and $q = b\log n/n$. If $|\sqrt{a}-\sqrt{b}| > \sqrt{2}$, then spectral clustering applied to the adjacency matrix recovers the community partition of all vertices with high probability.

Section 4.6 Two-sided bounds for sub-Gaussian matrices Q4.46 ∼ Q4.47

Theorem 4.6.1 (*Two-sided bounds for sub-Gaussian matrices*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix whose entries are independent, centered K-sub-Gaussian random variables. Then, for any $t \geqslant 0$,

$$\sqrt{m} - CK\sqrt{n} - t \leqslant s_{\min}(\boldsymbol{A}) \leqslant s_{\max}(\boldsymbol{A}) \leqslant \sqrt{m} + CK\sqrt{n} + t \text{ (w.p.1} - 2\exp(-ct^2)),$$

where $s_{\min}(\boldsymbol{A}), s_{\max}(\boldsymbol{A})$ denote the smallest and largest singular values of $\frac{1}{\sqrt{n}}\boldsymbol{A}$, respectively.

Section 4.7 Covariance estimation and clustering Q4.48 ∼ Q4.51

Theorem 4.7.1 (*Covariance estimation*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a sub-Gaussian random vector with covariance $\boldsymbol{\Sigma}$, and let $\{\boldsymbol{X}_N\}$ be independent copies of $\boldsymbol{X}$. Define the sample covariance

$$\boldsymbol{\Sigma}_N = \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{X}_i \boldsymbol{X}_i^\top,$$

If $N \geqslant Cn/\varepsilon^2$, then

$$\|\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}\| \leqslant \varepsilon\|\boldsymbol{\Sigma}\| \text{ (w.h.p.)}.$$

Remark 4.7.3 (*High-probability bound*)

In fact, the proof of Theorem T4.7.1 (*Covariance estimation*) gives the following non-asymptotic high-probability bound: for any $u \geqslant 0$,

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant CK^2\left(\sqrt{\frac{n+u}{m}} + \frac{n+u}{m}\right)\|\boldsymbol{\Sigma}\| \text{ (w.p. } 1 - 2\mathrm{e}^{-u}).$$

Definition 4.7.4 (*Gaussian mixture model*)

The two-component Gaussian mixture model (GMM) is a probability distribution on $\mathbb{R}^n$ defined as drawing a sample $X \sim N(\boldsymbol{\mu}, \boldsymbol{I}_n)$ or $X \sim N(-\boldsymbol{\mu}, \boldsymbol{I}_n)$ with equal probability $1/2$, where $\boldsymbol{\mu} \in \mathbb{R}^n$ is unknown.

Theorem 4.7.5 (*Spectral clustering for Gaussian mixtures*)

If $\|\boldsymbol{\mu}\|_2 \geqslant C\sqrt{n}$ and the sample size satisfies $N \geqslant Cn$, then spectral clustering using the largest eigenvector of the sample covariance matrix recovers all cluster members exactly with high probability.

Exercises

1 [**SVD of the Inverse Matrix**] Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ have the SVD

$$\boldsymbol{A} = \sum_{i=1}^{n} s_i \boldsymbol{u}_i \boldsymbol{v}_i^\top,$$

where $s_1 \geqslant \cdots \geqslant s_n \geqslant 0$ are its singular values and $\{\boldsymbol{u}_i\}_{i=1}^n$, $\{\boldsymbol{v}_i\}_{i=1}^n$ are the corresponding left and right singular vector bases (both orthonormal bases of $\mathbb{R}^n$).

(1) Prove that $\boldsymbol{A}$ is invertible if and only if $s_i \neq 0$.

(2) Prove that if $\boldsymbol{A}$ is invertible, then the SVD of $\boldsymbol{A}^{-1}$ is

$$\boldsymbol{A}^{-1} = \sum_{i=1}^n s_i^{-1} \boldsymbol{v}_i \boldsymbol{u}_i^\top.$$

Prf. (1) ($\Longrightarrow$)

If $\boldsymbol{A}$ is invertible, then

$$0 \neq \det(\boldsymbol{A}) = \det(\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top) = \det(\boldsymbol{U})\det(\boldsymbol{\Sigma})\det(\boldsymbol{V}^\top),$$

where $\boldsymbol{U}, \boldsymbol{V}$ are orthogonal matrices with $|\det(\boldsymbol{U})| = |\det(\boldsymbol{V})| = 1$, so

$$0 \neq |\det(\boldsymbol{A})| = |\det(\boldsymbol{\Sigma})| = \prod_{i=1}^n s_i,$$

and hence $s_i \neq 0$.

($\Longleftarrow$)

If $s_i > 0$, then $\boldsymbol{\Sigma}$ is invertible with $\boldsymbol{\Sigma}^{-1} = \operatorname{diag}(s_1^{-1}, \cdots, s_n^{-1})$, so $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$ is invertible.

(2) Let $\boldsymbol{B} = \sum_{j=1}^n s_j^{-1} \boldsymbol{v}_j \boldsymbol{u}_j^\top$. Then

$$\begin{aligned}
\boldsymbol{A}\boldsymbol{B} &= \left(\sum_{i=1}^n s_i \boldsymbol{u}_i \boldsymbol{v}_i^\top\right)\left(\sum_{j=1}^n s_j^{-1} \boldsymbol{v}_j \boldsymbol{u}_j^\top\right) \\
&= \sum_{i=1}^n \sum_{j=1}^n s_i s_j^{-1} \boldsymbol{u}_i (\boldsymbol{v}_i^\top \boldsymbol{v}_j) \boldsymbol{u}_j^\top \\
&= \sum_{i=1}^n \sum_{j=1}^n s_i s_j^{-1} \boldsymbol{u}_i \delta_{ij} \boldsymbol{u}_j^\top \\
&= \sum_{i=1}^n s_i s_i^{-1} \boldsymbol{u}_i \boldsymbol{u}_i^\top \\
&= \sum_{i=1}^n \boldsymbol{u}_i \boldsymbol{u}_i^\top = \boldsymbol{I}_n;
\end{aligned}$$

similarly, $\boldsymbol{B}\boldsymbol{A} = \boldsymbol{I}_n$, so $\boldsymbol{B}$ is the inverse matrix, with singular values $s_n^{-1} \geqslant \cdots \geqslant s_1^{-1}$ and left/right singular vectors $\{\boldsymbol{v}_i\}_{i=1}^n$, $\{\boldsymbol{u}_i\}_{i=1}^n$, respectively.

2 [**Basic Properties of the Operator Norm**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$. Define the operator norm $\|\boldsymbol{A}\|$ of $\boldsymbol{A}$ as

$$\|\boldsymbol{A}\| = \sup\{\|\boldsymbol{A}\boldsymbol{x}\|_2 : \boldsymbol{x} \in \mathbb{R}^n, \|\boldsymbol{x}\|_2 = 1\}.$$

(1) Prove that the operator norm defined above is a norm on $\mathbb{R}^{m\times n}$.

(2) Prove that $\|\boldsymbol{A}^\top\| = \|\boldsymbol{A}\|$.

(3) Prove (submultiplicativity): for any matrices $\boldsymbol{A}, \boldsymbol{B}$ of compatible dimensions, $\|\boldsymbol{A}\boldsymbol{B}\| \leqslant \|\boldsymbol{A}\| \cdot \|\boldsymbol{B}\|$. Then give an example where the left-hand side is 0 while the right-hand side is not.

(4) Prove that the operator norm of any submatrix of a matrix does not exceed the operator norm of the original matrix.

Prf. (1) 1. Nonnegativity and definiteness:

Since $\|\boldsymbol{A}\boldsymbol{x}\|_2 \geqslant 0$, we have $\|\boldsymbol{A}\| \geqslant 0$.

If $\|\boldsymbol{A}\| = 0$, then for every $\boldsymbol{x} \in \mathbb{S}^{n-1}$, $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$, so $\boldsymbol{A} = \boldsymbol{O}$.

2. Homogeneity:

$$\|c\boldsymbol{A}\| = \sup_{\|\boldsymbol{x}\|_2=1} \|c\boldsymbol{A}\boldsymbol{x}\|_2 = \sup_{\|\boldsymbol{x}\|_2=1} |c| \cdot \|\boldsymbol{A}\boldsymbol{x}\|_2 = |c| \cdot \|\boldsymbol{A}\|.$$

3. Triangle inequality: for any $\boldsymbol{x} \in \mathbb{S}^{n-1}$,

$$\|(\boldsymbol{A}+\boldsymbol{B})\boldsymbol{x}\|_2 = \|\boldsymbol{A}\boldsymbol{x}+\boldsymbol{B}\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{A}\boldsymbol{x}\|_2 + \|\boldsymbol{B}\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{A}\| + \|\boldsymbol{B}\|,$$

and taking the supremum on both sides yields $\|\boldsymbol{A}+\boldsymbol{B}\| \leqslant \|\boldsymbol{A}\| + \|\boldsymbol{B}\|$.

(2) By the variational characterization 4.1.8 of the operator norm,

$$\|\boldsymbol{A}\| = \sup_{\|\boldsymbol{x}\|_2=1, \|\boldsymbol{y}\|_2=1} \langle \boldsymbol{y}, \boldsymbol{A}\boldsymbol{x}\rangle,$$

where $\langle \boldsymbol{y}, \boldsymbol{A}\boldsymbol{x}\rangle = \boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x} = (\boldsymbol{A}^\top \boldsymbol{y})^\top \boldsymbol{x} = \langle \boldsymbol{A}^\top \boldsymbol{y}, \boldsymbol{x}\rangle$, so

$$\|\boldsymbol{A}\| = \sup_{\|\boldsymbol{x}\|_2=1, \|\boldsymbol{y}\|_2=1} \langle \boldsymbol{A}^\top \boldsymbol{y}, \boldsymbol{x}\rangle = \sup_{\|\boldsymbol{y}\|_2=1} \|\boldsymbol{A}^\top \boldsymbol{y}\|_2 = \|\boldsymbol{A}^\top\|.$$

(*For a fixed $\boldsymbol{y}$, the supremum $\sup_{\|\boldsymbol{x}\|_2=1}\langle \boldsymbol{A}^\top \boldsymbol{y}, \boldsymbol{x}\rangle$ is attained when $\boldsymbol{x}$ points in the same direction as $\boldsymbol{A}^\top \boldsymbol{y}$, and its value is $\|\boldsymbol{A}^\top \boldsymbol{y}\|_2$.*)

(3) For any $\boldsymbol{x} \in \mathbb{S}^{n-1}$,

$$\|\boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{A}\| \cdot \|\boldsymbol{B}\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{A}\| \cdot \|\boldsymbol{B}\| \cdot \|\boldsymbol{x}\|_2 = \|\boldsymbol{A}\| \cdot \|\boldsymbol{B}\|,$$

and taking the supremum on both sides yields $\|\boldsymbol{A}\boldsymbol{B}\| \leqslant \|\boldsymbol{A}\| \cdot \|\boldsymbol{B}\|$.

Here is a counterexample: let

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \boldsymbol{B} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

then $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{O}$, so $\|\boldsymbol{A}\boldsymbol{B}\| = 0$, but $\|\boldsymbol{A}\| = 1, \|\boldsymbol{B}\| = 1$, and $\|\boldsymbol{A}\| \cdot \|\boldsymbol{B}\| = 1 \neq 0$.

(4) Let $\boldsymbol{A}_0 \subset \boldsymbol{A}$; without loss of generality, assume $\boldsymbol{A}_0 = (\boldsymbol{A})_{1:k;1:l}$.

For any $\boldsymbol{x}_0 \in \mathbb{S}^{l-1}$, construct $\boldsymbol{x} = (\boldsymbol{x}_0^\top, 0, \cdots, 0)^\top$ with $\|\boldsymbol{x}\|_2 = \|\boldsymbol{x}_0\|_2 = 1$ and $(\boldsymbol{A}\boldsymbol{x})_{1:k} = \boldsymbol{A}_0\boldsymbol{x}_0$. Then

$$\|\boldsymbol{A}_0\boldsymbol{x}_0\|_2^2 \leqslant \|\boldsymbol{A}\boldsymbol{x}\|_2^2 \leqslant \|\boldsymbol{A}\|^2,$$

and taking the supremum over $\boldsymbol{x}_0$ on both sides yields $\|\boldsymbol{A}_0\| \leqslant \|\boldsymbol{A}\|$.

3 **[Special Cases of the Operator Norm]**

(1) Rank-one matrices: let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$. Prove that the rank-one matrix $\boldsymbol{u}\boldsymbol{v}^\top$ has equal operator and Frobenius norms, both equal to the product of the norms of the factors:

$$\|\boldsymbol{u}\boldsymbol{v}^\top\| = \|\boldsymbol{u}\boldsymbol{v}^\top\|_F = \|\boldsymbol{u}\|_2 \|\boldsymbol{v}\|_2.$$

(2) Diagonal matrices: let $\boldsymbol{A} = \operatorname{diag}(a_1, \cdots, a_n) \in \mathbb{R}^{n\times n}$. Prove that its operator norm equals the maximum absolute value of the diagonal entries:

$$\|\boldsymbol{A}\| = \max_{1\leqslant i\leqslant n} |a_i|.$$

Prf. (1) • Operator norm part

It is easy to see that

$$\|\boldsymbol{u}\boldsymbol{v}^\top\| = \sup_{\|\boldsymbol{x}\|_2=1} \|\boldsymbol{u}\boldsymbol{v}^\top \boldsymbol{x}\|_2,$$

where $\boldsymbol{v}^\top \boldsymbol{x}$ is a scalar satisfying

$$\|\boldsymbol{u}\boldsymbol{v}^\top \boldsymbol{x}\|_2 = \|(\boldsymbol{v}^\top \boldsymbol{x})\boldsymbol{u}\|_2 = |\boldsymbol{v}^\top \boldsymbol{x}| \cdot \|\boldsymbol{u}\|_2,$$

so

$$\|\boldsymbol{u}\boldsymbol{v}^\top\| = \|\boldsymbol{u}\|_2 \cdot \sup_{\|\boldsymbol{x}\|_2=1} |\boldsymbol{v}^\top \boldsymbol{x}|,$$

where, by Cauchy's inequality, for any $\boldsymbol{x} \in \mathbb{S}^{n-1}$,

$$|\boldsymbol{v}^\top \boldsymbol{x}| \leqslant \|\boldsymbol{v}\|_2 \|\boldsymbol{x}\|_2 = \|\boldsymbol{v}\|_2,$$

with equality if and only if $\boldsymbol{x} = \bar{\boldsymbol{v}}$, so

$$\|\boldsymbol{u}\boldsymbol{v}^\top\| = \|\boldsymbol{u}\|_2\|\boldsymbol{v}\|_2.$$

- Frobenius norm part

  It is easy to see that

$$\|\boldsymbol{u}\boldsymbol{v}^\top\|_F^2 = \sum_{i=1}^m\sum_{j=1}^n (u_iv_j)^2 = \sum_{i=1}^m u_i^2\left(\sum_{j=1}^n v_j^2\right) = \|\boldsymbol{u}\|_2^2\|\boldsymbol{v}\|_2^2,$$

  and taking the square root on both sides,

$$\|\boldsymbol{u}\boldsymbol{v}^\top\|_F = \|\boldsymbol{u}\|_2\|\boldsymbol{v}\|_2.$$

(2) For any $\boldsymbol{x} \in \mathbb{S}^{n-1}$,

$$\|\boldsymbol{A}\boldsymbol{x}\|_2^2 = \sum_{i=1}^n a_i^2x_i^2,$$

and letting $M = \max_{1\leqslant i\leqslant n}|a_i|$, so that $a_i^2 \leqslant M^2$, we get

$$\|\boldsymbol{A}\boldsymbol{x}\|_2^2 \leqslant \sum_{i=1}^n M^2x_i^2 = M^2\sum_{i=1}^n x_i^2 = M^2,$$

i.e. $\|\boldsymbol{A}\boldsymbol{x}\|_2 \leqslant M$.

Next we show that the upper bound is attainable: let k satisfy $|a_k| = M$ and set $\boldsymbol{x} = \boldsymbol{e}_k$. Then

$$\|\boldsymbol{A}\boldsymbol{e}_k\|_2 = \sqrt{a_k^2} = |a_k| = M,$$

and therefore

$$\|\boldsymbol{A}\| = \sup_{\|\boldsymbol{x}\|_2=1}\|\boldsymbol{A}\boldsymbol{x}\|_2 = M = \max_{1\leqslant i\leqslant n}|a_i|.$$

4 [**Operator and Frobenius Norms: Relations and Properties**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$.

(1) Let $\operatorname{rank}(\boldsymbol{A}) = r$. Prove:

$$\|\boldsymbol{A}\| \leqslant \|\boldsymbol{A}\|_F \leqslant \sqrt{r}\|\boldsymbol{A}\|.$$

Then show that for any $1 \leqslant r \leqslant \min(m, n)$, both bounds are attainable.

(2) Let $\boldsymbol{Z} \in \mathbb{R}^n$ be an isotropic random vector. Prove:

$$\mathbb{E}\|\boldsymbol{A}\boldsymbol{Z}\|_2^2 = \|\boldsymbol{A}\|_F^2.$$

(3) Let $\boldsymbol{B} \in \mathbb{R}^{k\times m}$. Prove:

$$\|\boldsymbol{B}\boldsymbol{A}\|_F \leqslant \|\boldsymbol{B}\| \cdot \|\boldsymbol{A}\|_F.$$

(4) If $\boldsymbol{A}$ is a diagonal matrix, prove that the bound in (3) can be improved to

$$\|\boldsymbol{B}\boldsymbol{A}\|_F \leqslant \|\boldsymbol{B}\|_{1\to 2}\|\boldsymbol{A}\|_F,$$

where $\|\boldsymbol{B}\|_{1\to 2} = \max_{1\leqslant j\leqslant m}\|\boldsymbol{b}_j\|_2$ is the maximum Euclidean norm of the columns of $\boldsymbol{B}$.

Prf. (1) Let the singular values of $\boldsymbol{A}$ be $s_1 \geqslant s_2 \geqslant \cdots \geqslant s_r > 0$, so that

$$\|\boldsymbol{A}\| = s_1, \quad \|\boldsymbol{A}\|_F = \left(\sum_{i=1}^r s_i^2\right)^{1/2}.$$

For the left-hand side, it is clear that

$$\|\boldsymbol{A}\|_F = \left(\sum_{i=1}^r s_i^2\right)^{1/2} \geqslant (s_1^2)^{1/2} = s_1 = \|\boldsymbol{A}\|.$$

For the right-hand side,

$$\|\boldsymbol{A}\|_F^2 = \sum_{i=1}^{r} s_i^2 \leqslant \sum_{i=1}^{r} s_1^2 = r s_1^2 = r\|\boldsymbol{A}\|^2,$$

i.e. $\|\boldsymbol{A}\|_F \leqslant \sqrt{r}\|\boldsymbol{A}\|$.

(2) By the cyclic invariance $\operatorname{tr}(\boldsymbol{XY}) = \operatorname{tr}(\boldsymbol{YX})$,

$$\begin{aligned}\mathbb{E}\|\boldsymbol{AZ}\|_2^2 &= \mathbb{E}[(\boldsymbol{AZ})^\top(\boldsymbol{AZ})] = \mathbb{E}[\operatorname{tr}(\boldsymbol{Z}^\top\boldsymbol{A}^\top\boldsymbol{AZ})] \\ &= \mathbb{E}[\operatorname{tr}(\boldsymbol{A}^\top\boldsymbol{AZZ}^\top)] = \operatorname{tr}(\boldsymbol{A}^\top\boldsymbol{A}\mathbb{E}[\boldsymbol{ZZ}^\top]) \\ &= \operatorname{tr}(\boldsymbol{A}^\top\boldsymbol{A}) = \|\boldsymbol{A}\|_F^2. \qquad (\mathbb{E}[\boldsymbol{ZZ}^\top] = \boldsymbol{I}_n)\end{aligned}$$

(3) It is easy to see that

$$\begin{aligned}\|\boldsymbol{BA}\|_F^2 &= \sum_{j=1}^{n} \|\boldsymbol{Ba}_j\|_2^2 \\ &\leqslant \sum_{j=1}^{n} \|\boldsymbol{B}\|^2\|\boldsymbol{a}_j\|_2^2 = \|\boldsymbol{B}\|^2\left(\sum_{j=1}^{n}\|\boldsymbol{a}_j\|_2^2\right) = \|\boldsymbol{B}\|^2\|\boldsymbol{A}\|_F^2,\end{aligned}$$

and taking the square root on both sides yields $\|\boldsymbol{BA}\|_F \leqslant \|\boldsymbol{B}\| \cdot \|\boldsymbol{A}\|_F$.

(4) Let $\boldsymbol{A} = \operatorname{diag}(d_1, \cdots, d_n)$ and $\boldsymbol{B} = (\boldsymbol{b}_1, \cdots, \boldsymbol{b}_n)$. Then $\boldsymbol{BA} = (d_1\boldsymbol{b}_1, \cdots, d_n\boldsymbol{b}_n)$, so

$$\|\boldsymbol{BA}\|_F^2 = \sum_{j=1}^{n} \|d_j\boldsymbol{b}_j\|_2^2 = \sum_{j=1}^{n} d_j^2\|\boldsymbol{b}_j\|_2^2,$$

and letting $M = \|\boldsymbol{B}\|_{1\to 2}^2 = \max_j \|\boldsymbol{b}_j\|_2^2$, we get

$$\|\boldsymbol{BA}\|_F^2 \leqslant \sum_{j=1}^{n} d_j^2 M = M\sum_{j=1}^{n} d_j^2 = M \cdot \|\boldsymbol{A}\|_F^2,$$

and taking the square root on both sides yields $\|\boldsymbol{BA}\|_F \leqslant \|\boldsymbol{B}\|_{1\to 2}\|\boldsymbol{A}\|_F$.

5 [**Frobenius Norm Bound for Singular Values**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ have singular values $s_1(\boldsymbol{A}) \geqslant \cdots \geqslant s_r(\boldsymbol{A}) > 0$, where $r = \operatorname{rank}(\boldsymbol{A})$. Prove that for any $k \in \{1, \cdots, r\}$,

$$s_k(\boldsymbol{A}) \leqslant \frac{\|\boldsymbol{A}\|_F}{\sqrt{k}}.$$

Prf. It is easy to see that

$$\|\boldsymbol{A}\|_F^2 = \sum_{i=1}^{r} s_i(\boldsymbol{A})^2 \geqslant \sum_{i=1}^{k} s_i(\boldsymbol{A})^2 = k \cdot s_k(\boldsymbol{A})^2,$$

and taking the square root on both sides yields

$$s_k(\boldsymbol{A}) \leqslant \frac{\|\boldsymbol{A}\|_F}{\sqrt{k}}.$$

6 [**Estimating the Operator Norm by the Power Method**] The power method is a computationally friendly way to estimate the operator norm of a matrix without performing a full spectral decomposition.

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric matrix (*if it is not symmetric, the limit converges to the spectral radius* $\rho(\boldsymbol{A})$) with initial vector $\boldsymbol{x} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Prove:

$$\sqrt[k]{\|\boldsymbol{A}^k\boldsymbol{x}\|_2} \to \|\boldsymbol{A}\| \text{ (a.s.)}.$$

Prf. Since $\boldsymbol{A}$ is symmetric, there exists an orthonormal basis $\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n\}$ of eigenvectors with corresponding eigenvalues $\lambda_1, \cdots, \lambda_n$; order them by absolute value so that $|\lambda_1| = \cdots = |\lambda_r| > |\lambda_{r+1}| \geqslant \cdots \geqslant |\lambda_n| \geqslant 0$ for some $r \geqslant 1$. Then

$$\|\boldsymbol{A}\| = |\lambda_1|, \quad \boldsymbol{x} = \sum_{i=1}^{n} \xi_i\boldsymbol{v}_i,$$

where $\xi_i = \langle \boldsymbol{x}, \boldsymbol{v}_i \rangle$ i.i.d. $\sim N(0,1)$ are the projection coefficients of $\boldsymbol{x}$ onto the eigendirections.

For continuous random variables, $\mathbb{P}(\xi_i = 0) = 0$, so

$$\mathbb{P}\{\bigcup_{i=1}^{n} \{\xi_i = 0\}\} = 0,$$

i.e. all eigendirection components are nonzero (w.p. 1).

It is easy to see that

$$\boldsymbol{A}^k \boldsymbol{x} = \sum_{i=1}^{n} \xi_i \lambda_i^k \boldsymbol{v}_i,$$

hence

$$\|\boldsymbol{A}^k \boldsymbol{x}\|_2^2 = \sum_{i=1}^{n} \xi_i^2 \lambda_i^{2k} = |\lambda_1|^{2k} \left(\sum_{i=1}^{r} \xi_i^2 + \sum_{i=r+1}^{n} \xi_i^2 \left(\frac{\lambda_i}{\lambda_1} \right)^{2k} \right),$$

For $i > r$, $|\lambda_i / \lambda_1| < 1$, and hence

$$\sum_{i=1}^{r} \xi_i^2 \leqslant \sum_{i=1}^{r} \xi_i^2 + \sum_{i=r+1}^{n} \xi_i^2 \left(\frac{\lambda_i}{\lambda_1} \right)^{2k} \leqslant \sum_{i=1}^{n} \xi_i^2 = \|\boldsymbol{x}\|_2^2.$$

Taking the $2k$-th root gives

$$|\lambda_1| \cdot \left(\sum_{i=1}^{r} \xi_i^2 \right)^{\frac{1}{2k}} \leqslant \sqrt[k]{\|\boldsymbol{A}^k \boldsymbol{x}\|_2} \leqslant |\lambda_1| \cdot (\|\boldsymbol{x}\|_2^2)^{\frac{1}{2k}}.$$

Since $\sum_{i=1}^{r} \xi_i^2, \|\boldsymbol{x}\|_2^2 \neq 0$ almost surely, the squeeze theorem gives

$$\lim_{k \to \infty} \sqrt[k]{\|\boldsymbol{A}^k \boldsymbol{x}\|_2} = |\lambda_1| = \|\boldsymbol{A}\| \text{ (a.s.)}.$$

7 [**Operator Norm and Column Norms**] The operator norm of a matrix is usually hard to express explicitly in terms of its entries, but it can be bounded efficiently in terms of its columns.

Let $\boldsymbol{A} = (\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n) \in \mathbb{R}^{m \times n}$.

(1) Prove:

$$\|\boldsymbol{A}\| \geqslant \max_{1 \leqslant i \leqslant n} \|\boldsymbol{a}_i\|_2,$$

with equality if and only if $\{\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n\}$ are mutually orthogonal.

(2) Prove: if there exists an index i such that $\|\boldsymbol{A}\| = \|\boldsymbol{a}_i\|_2$, then $\boldsymbol{a}_i$ is necessarily orthogonal to the other columns.

(3) Prove that the above conclusions also hold for the rows of the matrix.

Prf. (1) By the single-vector definition 4.1.8 of the operator norm, we obtain

$$\|\boldsymbol{A}\| = \sup_{\boldsymbol{x} \in \mathbb{S}^{n-1}} \|\boldsymbol{A}\boldsymbol{x}\|_2.$$

Taking $\boldsymbol{x} = \boldsymbol{e}_i$ gives

$$\|\boldsymbol{A}\| \geqslant \|\boldsymbol{A}\boldsymbol{e}_i\|_2 = \|\boldsymbol{a}_i\|_2,$$

and taking the maximum on both sides,

$$\|\boldsymbol{A}\| \geqslant \max_{1 \leqslant i \leqslant n} \|\boldsymbol{a}_i\|_2.$$

When $\langle \boldsymbol{a}_i, \boldsymbol{a}_j \rangle = 0$ $(i \neq j)$, consider the symmetric matrix $\boldsymbol{M} = \boldsymbol{A}^\top \boldsymbol{A} = \operatorname{diag}(\|\boldsymbol{a}_1\|_2^2, \cdots, \|\boldsymbol{a}_n\|_2^2)$. Then

$$\|\boldsymbol{A}\|^2 = \lambda_{\max}(\boldsymbol{A}^\top \boldsymbol{A}) = \max_{1 \leqslant i \leqslant n} \|\boldsymbol{a}_i\|_2^2,$$

and taking the square root on both sides,

$$\|\boldsymbol{A}\| = \max_{1 \leqslant i \leqslant n} \|\boldsymbol{a}_i\|_2.$$

(2) As in (1), let k be the index of the largest eigenvalue of $\boldsymbol{M}$, so that

$$M_{kk} = \langle \boldsymbol{a}_k, \boldsymbol{a}_k \rangle = \|\boldsymbol{a}_k\|_2^2 = \lambda_{\max}(\boldsymbol{M}).$$

By the Rayleigh–Ritz variational principle, for any $\boldsymbol{x} \in \mathbb{S}^{n-1}$, $\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x} \leqslant \lambda_{\max}(\boldsymbol{M})$; note that for $\boldsymbol{x} = \boldsymbol{e}_k$,

$$\boldsymbol{e}_k^\top \boldsymbol{M}\boldsymbol{e}_k = M_{kk} = \lambda_{\max}(\boldsymbol{M}),$$

so $\boldsymbol{e}_k$ is an eigenvector of $\boldsymbol{M}$ for the largest eigenvalue, satisfying

$$\boldsymbol{m}_k = \boldsymbol{M}\boldsymbol{e}_k = \lambda_{\max}(\boldsymbol{M})\boldsymbol{e}_k = \begin{pmatrix} 0 & \cdots & \lambda_{\max} & \cdots & 0 \end{pmatrix}^\top,$$

hence $M_{ik} = 0$ $(i \neq k)$, where $M_{ik} = \langle \boldsymbol{a}_i, \boldsymbol{a}_k \rangle$; thus $\boldsymbol{a}_k$ is orthogonal to the other columns.

(3) Using the conclusion of Exercise Q4.2 (*Basic Properties of the Operator Norm*) (2), we obtain $\|\boldsymbol{A}\| = \|\boldsymbol{A}^\top\|$, so using the conclusion of (1)(2):

1. $\|\boldsymbol{A}\| = \|\boldsymbol{A}^\top\| \geqslant \max_j \|\mathrm{row}_j(\boldsymbol{A})\|_2$, with equality if and only if the rows are mutually orthogonal.
2. If $\|\boldsymbol{A}\|$ equals the norm of some row, then that row is orthogonal to the others.

8 [**Schur's Bound**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, and write $\boldsymbol{r}_i^\top = \boldsymbol{A}_{i*}$, $\boldsymbol{a}_j = \boldsymbol{A}_{*j}$. Prove:

$$\|\boldsymbol{A}\| \leqslant \sqrt{\left(\max_{1\leqslant i\leqslant m} \|\boldsymbol{r}_i\|_1\right) \cdot \left(\max_{1\leqslant j\leqslant n} \|\boldsymbol{a}_j\|_1\right)}.$$

Prf. By the bilinear form definition 4.1.8 of the operator norm, we obtain

$$\|\boldsymbol{A}\| = \sup_{\|\boldsymbol{x}\|_2=1, \|\boldsymbol{y}\|_2=1} \boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}.$$

Consider

$$\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} y_i x_j,$$

and using the triangle inequality and Cauchy's inequality,

$$\begin{aligned} |\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}| &\leqslant \sum_{i=1}^m \sum_{j=1}^n |a_{ij}| \cdot |y_i| \cdot |x_j| \\ &= \sum_{i,j} (\sqrt{|a_{ij}|}|y_i|) \cdot (\sqrt{|a_{ij}|}|x_j|) \\ &\leqslant \left(\sum_{i,j} |a_{ij}| y_i^2\right)^{1/2} \cdot \left(\sum_{i,j} |a_{ij}| x_j^2\right)^{1/2}, \end{aligned}$$

where

$$\begin{aligned} \sum_{i=1}^m \sum_{j=1}^n |a_{ij}| y_i^2 &= \sum_{i=1}^m y_i^2 \left(\sum_{j=1}^n |a_{ij}|\right) = \sum_{i=1}^m y_i^2 \|\boldsymbol{r}_i\|_1 \\ &\leqslant \left(\max_{1\leqslant i\leqslant m} \|\boldsymbol{r}_i\|_1\right) \sum_{i=1}^m y_i^2 = \max_{1\leqslant i\leqslant m} \|\boldsymbol{r}_i\|_1, \\ \sum_{j=1}^n \sum_{i=1}^m |a_{ij}| x_j^2 &= \sum_{j=1}^n x_j^2 \left(\sum_{i=1}^m |a_{ij}|\right) = \sum_{j=1}^n x_j^2 \|\boldsymbol{a}_j\|_1 \\ &\leqslant \left(\max_{1\leqslant j\leqslant n} \|\boldsymbol{a}_j\|_1\right) \sum_{j=1}^n x_j^2 = \max_{1\leqslant j\leqslant n} \|\boldsymbol{a}_j\|_1, \end{aligned}$$

hence

$$|\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}| \leqslant \sqrt{\max_i \|\boldsymbol{r}_i\|_1} \cdot \sqrt{\max_j \|\boldsymbol{a}_j\|_1} = \sqrt{\max_i \|\boldsymbol{r}_i\|_1 \cdot \max_j \|\boldsymbol{a}_j\|_1},$$

and taking the supremum on both sides,

$$\|\boldsymbol{A}\| \leqslant \sqrt{\max_i \|\boldsymbol{r}_i\|_1 \cdot \max_j \|\boldsymbol{a}_j\|_1}.$$

9 [**Construction and Orthogonality of Walsh Matrices**] To construct an orthogonal matrix whose entries all have the same absolute value (the most "spread out"), we define the sequence $\{\boldsymbol{W}_k\}$ recursively.
First define the 2×2 matrix

$$\boldsymbol{W}_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and then iterate with block matrices:

$$\boldsymbol{W}_{k+1} = \begin{pmatrix} \boldsymbol{W}_k & \boldsymbol{W}_k \\ \boldsymbol{W}_k & -\boldsymbol{W}_k \end{pmatrix},$$

Clearly, $\boldsymbol{W}_k$ has order $n = 2^k$ and all its entries are ± 1. Prove that $\boldsymbol{W}_k/\sqrt{n}$ are orthogonal matrices.

Prf. To prove

$$\left(\frac{1}{\sqrt{n}}\boldsymbol{W}_k\right)\left(\frac{1}{\sqrt{n}}\boldsymbol{W}_k\right)^\top = \boldsymbol{I}_n,$$

it suffices to prove $\boldsymbol{W}_k\boldsymbol{W}_k^\top = n\boldsymbol{I}_n$.

Use mathematical induction:

a. For $k = 1$,

$$\boldsymbol{W}_1\boldsymbol{W}_1^\top = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2\boldsymbol{I}_2.$$

b. Assume the claim holds for $k = m$, i.e. for order $n_m = 2^m$,

$$\boldsymbol{W}_m\boldsymbol{W}_m^\top = n_m\boldsymbol{I}_{n_m},$$

then for $k = m+1$, with order $n_{m+1} = 2^{m+1} = 2n_m$,

$$\begin{aligned}
\boldsymbol{W}_{m+1}\boldsymbol{W}_{m+1}^\top &= \begin{pmatrix} \boldsymbol{W}_m & \boldsymbol{W}_m \\ \boldsymbol{W}_m & -\boldsymbol{W}_m \end{pmatrix}\begin{pmatrix} \boldsymbol{W}_m^\top & \boldsymbol{W}_m^\top \\ \boldsymbol{W}_m^\top & -\boldsymbol{W}_m^\top \end{pmatrix} \\
&= \begin{pmatrix} \boldsymbol{W}_m\boldsymbol{W}_m^\top + \boldsymbol{W}_m\boldsymbol{W}_m^\top & \boldsymbol{W}_m\boldsymbol{W}_m^\top - \boldsymbol{W}_m\boldsymbol{W}_m^\top \\ \boldsymbol{W}_m\boldsymbol{W}_m^\top - \boldsymbol{W}_m\boldsymbol{W}_m^\top & \boldsymbol{W}_m\boldsymbol{W}_m^\top + \boldsymbol{W}_m\boldsymbol{W}_m^\top \end{pmatrix} \\
&= \begin{pmatrix} 2n_m\boldsymbol{I}_{n_m} & \boldsymbol{O} \\ \boldsymbol{O} & 2n_m\boldsymbol{I}_{n_m} \end{pmatrix} = 2n_m\begin{pmatrix} \boldsymbol{I}_{n_m} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{I}_{n_m} \end{pmatrix} = 2^{m+1}\boldsymbol{I}_{n_{m+1}}.
\end{aligned}$$

10 [**Norm Bounds for ± 1 Matrices**] Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ satisfy $a_{ij} \in \{-1, 1\}$. Prove:

$$\sqrt{n} \leqslant \|\boldsymbol{A}\| \leqslant n,$$

and then show that both bounds are attainable for infinitely many n. We will explore the $1 \to 2$ norm and general $p \to q$ norms in Exercises Q4.18 (*$p \to q$ Operator Norms*) and Q4.19 (*Explicit Formulas for the $1 \to \infty$ and $1 \to 2$ Operator Norms*).

Prf. It is easy to see that

$$\|\boldsymbol{A}\|_F = \left(\sum_{i=1}^n\sum_{j=1}^n a_{ij}^2\right)^{1/2} = \left(\sum_{i=1}^n\sum_{j=1}^n 1\right)^{1/2} = n.$$

- Proving the upper bound $\|\boldsymbol{A}\| \leqslant n$

  By Lemma L4.4.1 (*Estimating the operator norm on a discrete net*),

  $$\|\boldsymbol{A}\| \leqslant \|\boldsymbol{A}\|_F = n.$$

For the equality case, let $\boldsymbol{A}$ be the all-ones matrix (*or any rank-one matrix with* ± 1 *entries, e.g.* $\boldsymbol{x}\boldsymbol{y}^\top$ *with* $x_i, y_j \in \{-1, 1\}$), and take the unit vector $\boldsymbol{u} = (1, 1, \cdots, 1)^\top / \sqrt{n}$. Then

$$\boldsymbol{A}\boldsymbol{u} = \frac{1}{\sqrt{n}}(n, n, \cdots, n)^\top = \sqrt{n}(1, 1, \cdots, 1)^\top,$$

so for any $n \geqslant 1$,

$$\|\boldsymbol{A}\boldsymbol{u}\|_2 = \sqrt{n \cdot n} = n.$$

- Proving the lower bound $\|\boldsymbol{A}\| \geqslant \sqrt{n}$

  Let the singular values of $\boldsymbol{A}$ be $s_1 \geqslant \cdots \geqslant s_n \geqslant 0$; then $\|\boldsymbol{A}\| = s_1$, so

$$n^2 = \|\boldsymbol{A}\|_F^2 = \sum_{i=1}^{n} s_i^2 \leqslant n s_1^2 = n\|\boldsymbol{A}\|^2 \implies \|\boldsymbol{A}\| \geqslant \sqrt{n}.$$

  For the equality case, let $s_1^2 = \cdots = s_n^2 = n$; then $\boldsymbol{A}^\top \boldsymbol{A} = n\boldsymbol{I}_n$ (*i.e. the columns of* $\boldsymbol{A}$ *are mutually orthogonal*), so $\boldsymbol{A}$ is a Hadamard matrix.

  By Exercise Q4.9 (*Construction and Orthogonality of Walsh Matrices*),

$$\boldsymbol{W}_k^\top \boldsymbol{W}_k = n\boldsymbol{I}_n, \qquad (n = 2^k)$$

  hence for any $n = 2^k$, take $\boldsymbol{A} = \boldsymbol{W}_k$.

Rk. The upper bound corresponds to the "most non-uniform" case of highly correlated entries (*rank one*), while the lower bound corresponds to the "most uniform" case of perfectly orthogonal entries (*Hadamard matrices*).

11 [**Norm of the Difference of Orthogonal Projections**] Let $\boldsymbol{P}, \boldsymbol{Q} \in \mathbb{R}^{n\times n}$ be orthogonal projection matrices. Prove that the operator norm (*spectral norm*) of their difference satisfies

$$\|\boldsymbol{P} - \boldsymbol{Q}\| \leqslant 1.$$

Rk. By the Loewner partial order,

$$\boldsymbol{O} \preccurlyeq \boldsymbol{P}, \boldsymbol{Q} \preccurlyeq \boldsymbol{I}_n.$$

Therefore

$$-\boldsymbol{I}_n \preccurlyeq \boldsymbol{P} - \boldsymbol{Q} \preccurlyeq \boldsymbol{I}_n.$$

Since $\boldsymbol{P} - \boldsymbol{Q}$ is symmetric, the Loewner bounds above show that its eigenvalues satisfy $\mu \in [-1, 1]$, hence

$$\|\boldsymbol{P} - \boldsymbol{Q}\| = \max_i |\mu_i| \leqslant 1.$$

12 [**Norms of Differences and Products of Orthogonal Projections**] Let $\boldsymbol{P}, \boldsymbol{Q} \in \mathbb{R}^{n\times n}$ be orthogonal projection matrices onto subspaces U, V, respectively. Write $\boldsymbol{P}_\perp = \boldsymbol{I}_n - \boldsymbol{P}$ for the orthogonal projection onto $U^\perp$ and $\boldsymbol{Q}_\perp = \boldsymbol{I}_n - \boldsymbol{Q}$ for the orthogonal projection onto $V^\perp$. Prove:

(1) (Equivalence of norm maxima) $\|\boldsymbol{P} - \boldsymbol{Q}\| = \max(\|\boldsymbol{P}_\perp \boldsymbol{Q}\|, \|\boldsymbol{P}\boldsymbol{Q}_\perp\|)$.

(2) (Norm equals 1 for unequal ranks) If $\operatorname{rank}(\boldsymbol{P}) \neq \operatorname{rank}(\boldsymbol{Q})$, then $\|\boldsymbol{P} - \boldsymbol{Q}\| = 1$.

(3) (Equivalence of the three for equal ranks) If $\operatorname{rank}(\boldsymbol{P}) = \operatorname{rank}(\boldsymbol{Q})$, then $\|\boldsymbol{P} - \boldsymbol{Q}\| = \|\boldsymbol{P}_\perp \boldsymbol{Q}\| = \|\boldsymbol{P}\boldsymbol{Q}_\perp\|$.

Prf. (1) Note that $\boldsymbol{I}_n = \boldsymbol{P} + \boldsymbol{P}_\perp = \boldsymbol{Q} + \boldsymbol{Q}_\perp$, so

$$\begin{aligned}\boldsymbol{P} - \boldsymbol{Q} &= \boldsymbol{P}(\boldsymbol{Q} + \boldsymbol{Q}_\perp) - (\boldsymbol{P} + \boldsymbol{P}_\perp)\boldsymbol{Q} \\ &= \boldsymbol{P}\boldsymbol{Q} + \boldsymbol{P}\boldsymbol{Q}_\perp - \boldsymbol{P}\boldsymbol{Q} - \boldsymbol{P}_\perp \boldsymbol{Q} \\ &= \boldsymbol{P}\boldsymbol{Q}_\perp - \boldsymbol{P}_\perp \boldsymbol{Q},\end{aligned}$$

and letting $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{Q}_\perp$, $\boldsymbol{B} = -\boldsymbol{P}_\perp \boldsymbol{Q}$, we get

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{P}(\boldsymbol{Q}_\perp \boldsymbol{x}) \in \operatorname{Range}(\boldsymbol{P}), \quad \boldsymbol{B}\boldsymbol{x} = -\boldsymbol{P}_\perp(\boldsymbol{Q}\boldsymbol{x}) \in \operatorname{Range}(\boldsymbol{P}_\perp).$$

Since $U \perp U^\perp$, for any $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|(\boldsymbol{P} - \boldsymbol{Q})\boldsymbol{x}\|_2^2 = \|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{x}\|_2^2 = \|\boldsymbol{A}\boldsymbol{x}\|_2^2 + \|\boldsymbol{B}\boldsymbol{x}\|_2^2.$$

Next consider $\ker(\boldsymbol{A})$ and $\ker(\boldsymbol{B})$:

1. If $\boldsymbol{x} \in \text{Range}(\boldsymbol{Q})$, then $\boldsymbol{Q}_\perp \boldsymbol{x} = \boldsymbol{0}$, so $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$.
2. If $\boldsymbol{x} \in \text{Range}(\boldsymbol{Q}_\perp)$, then $\boldsymbol{Q}\boldsymbol{x} = \boldsymbol{0}$, so $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{0}$.

For any $\boldsymbol{x} \in \mathbb{S}^{n-1}$, note that $\mathbb{R}^n = \text{Range}(\boldsymbol{Q}) \oplus \text{Range}(\boldsymbol{Q}_\perp)$; decompose it uniquely as

$$\boldsymbol{x} = \boldsymbol{x}_Q + \boldsymbol{x}_{Q_\perp},$$

where $\boldsymbol{x}_Q \in \text{Range}(\boldsymbol{Q}), \boldsymbol{x}_{Q_\perp} \in \text{Range}(\boldsymbol{Q}_\perp)$ satisfy $\|\boldsymbol{x}\|_2^2 = \|\boldsymbol{x}_Q\|_2^2 + \|\boldsymbol{x}_{Q_\perp}\|_2^2 = 1$. Then

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}\boldsymbol{x}_{Q_\perp}, \quad \boldsymbol{B}\boldsymbol{x} = \boldsymbol{B}\boldsymbol{x}_Q,$$

so

$$\|(\boldsymbol{P} - \boldsymbol{Q})\boldsymbol{x}\|_2^2 = \|\boldsymbol{A}\boldsymbol{x}_{Q_\perp}\|_2^2 + \|\boldsymbol{B}\boldsymbol{x}_Q\|_2^2,$$

i.e.

$$\|\boldsymbol{P} - \boldsymbol{Q}\|^2 = \sup_{\|\boldsymbol{x}\|_2 = 1} (\|\boldsymbol{A}\boldsymbol{x}_{Q_\perp}\|_2^2 + \|\boldsymbol{B}\boldsymbol{x}_Q\|_2^2).$$

Let $a = \|\boldsymbol{A}\| = \|\boldsymbol{P}\boldsymbol{Q}_\perp\|$ and $b = \|\boldsymbol{B}\| = \|\boldsymbol{P}_\perp \boldsymbol{Q}\|$. Then

$$\|\boldsymbol{A}\boldsymbol{x}_{Q_\perp}\|_2^2 + \|\boldsymbol{B}\boldsymbol{x}_Q\|_2^2 \leqslant a^2\|\boldsymbol{x}_{Q_\perp}\|_2^2 + b^2\|\boldsymbol{x}_Q\|_2^2 \leqslant \max(a^2, b^2) \cdot (\|\boldsymbol{x}_{Q_\perp}\|_2^2 + \|\boldsymbol{x}_Q\|_2^2) = \max(a^2, b^2),$$

i.e.

$$\|\boldsymbol{P} - \boldsymbol{Q}\| \leqslant \max(\|\boldsymbol{A}\|, \|\boldsymbol{B}\|).$$

Conversely, take a unit vector $\boldsymbol{x} = \boldsymbol{x}_{Q_\perp}$ at which $\boldsymbol{A}$ attains its norm; then $\|\boldsymbol{x}_Q\|_2 = 0$, so

$$\|\boldsymbol{A}\boldsymbol{x}_{Q_\perp}\|_2^2 + \|\boldsymbol{B}\boldsymbol{x}_Q\|_2^2 \leqslant a^2;$$

similarly,

$$\|\boldsymbol{A}\boldsymbol{x}_{Q_\perp}\|_2^2 + \|\boldsymbol{B}\boldsymbol{x}_Q\|_2^2 \leqslant b^2,$$

so

$$\|\boldsymbol{P} - \boldsymbol{Q}\|^2 = \max(a^2, b^2),$$

and taking the square root on both sides,

$$\|\boldsymbol{P} - \boldsymbol{Q}\| = \max(\|\boldsymbol{P}\boldsymbol{Q}_\perp\|, \|\boldsymbol{P}_\perp \boldsymbol{Q}\|).$$

(2) Let $\text{rank}(\boldsymbol{P}) = \dim(U) = d_1$ and $\text{rank}(\boldsymbol{Q}) = \dim(V) = d_2$; without loss of generality, assume $d_1 > d_2$. Then

$$\dim(U \cap V^\perp) \geqslant \dim(U) + \dim(V^\perp) - n = d_1 + (n - d_2) - n = d_1 - d_2 > 0,$$

so there exists a unit vector $\boldsymbol{u} \in U \cap V^\perp$ with $\boldsymbol{P}\boldsymbol{u} = \boldsymbol{u}$ and $\boldsymbol{Q}\boldsymbol{u} = \boldsymbol{0}$, hence

$$(\boldsymbol{P} - \boldsymbol{Q})\boldsymbol{u} = \boldsymbol{P}\boldsymbol{u} - \boldsymbol{Q}\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{0} = \boldsymbol{u},$$

so

$$\|\boldsymbol{P} - \boldsymbol{Q}\| \geqslant \|\boldsymbol{u}\|_2 = 1.$$

By Exercise Q4.11 (*Norm of the Difference of Orthogonal Projections*), $\|\boldsymbol{P} - \boldsymbol{Q}\| \leqslant 1$, so $\|\boldsymbol{P} - \boldsymbol{Q}\| = 1$.

(3) When $\text{rank}(\boldsymbol{P}) = \text{rank}(\boldsymbol{Q})$, we have $\dim(U) = \dim(V)$; consider the operator $\boldsymbol{P}_\perp \boldsymbol{Q}$. **Version 1 Variational geometry method**

- Setting the motivation and goal

  By the definition of the operator norm 4.1.8, we get

  $$\|\boldsymbol{P}_\perp \boldsymbol{Q}\| = \sup_{\boldsymbol{v} \in V, \|\boldsymbol{v}\|_2 = 1} \|\boldsymbol{P}_\perp \boldsymbol{v}\|_2.$$

  Since $\boldsymbol{Q}$ is the orthogonal projection onto the subspace V, for every $\boldsymbol{x} \in \mathbb{S}^{n-1}$, $\boldsymbol{Q}\boldsymbol{x}$ is a vector in V. Computing $\|\boldsymbol{P}_\perp \boldsymbol{Q}\|$ is equivalent to finding, among all unit vectors in V, the one whose projection under $\boldsymbol{P}_\perp$ has the largest norm.

- Converting the orthogonal complement into the main projection via the Pythagorean theorem

  For every unit vector $\boldsymbol{v}$ in V (*with* $\|\boldsymbol{v}\|_2 = 1$), the vector $\boldsymbol{v}$ admits the orthogonal decomposition into the projection $\boldsymbol{P}\boldsymbol{v}$ onto U and the projection $\boldsymbol{P}_\perp\boldsymbol{v}$ onto $U^\perp$. By the Pythagorean theorem,

  $$\|\boldsymbol{P}_\perp\boldsymbol{v}\|_2^2 = \|\boldsymbol{v}\|_2^2 - \|\boldsymbol{P}\boldsymbol{v}\|_2^2 = 1 - \|\boldsymbol{P}\boldsymbol{v}\|_2^2,$$

  so maximizing $\|\boldsymbol{P}_\perp\boldsymbol{v}\|_2$ is equivalent to minimizing $\|\boldsymbol{P}\boldsymbol{v}\|_2$.
- Connecting with the principal angles

  By the geometric definition of the principal angles $\theta_1 \leqslant \cdots \leqslant \theta_d = \theta_{\max}$ between U and V, the norm of the projection of a unit vector in V onto U ranges exactly over $[\cos\theta_{\max}, \cos\theta_1]$, where

  $$\min_{\boldsymbol{v}\in V, \|\boldsymbol{v}\|_2=1} \|\boldsymbol{P}\boldsymbol{v}\|_2 = \cos\theta_{\max}.$$

  Substituting this into the Pythagorean relation,

  $$\|\boldsymbol{P}_\perp\boldsymbol{Q}\| = \sup_{\boldsymbol{v}\in V, \|\boldsymbol{v}\|_2=1} \sqrt{1 - \|\boldsymbol{P}\boldsymbol{v}\|_2^2} = \sqrt{1 - \left(\min_{\boldsymbol{v}\in V, \|\boldsymbol{v}\|_2=1} \|\boldsymbol{P}\boldsymbol{v}\|_2\right)^2} = \sqrt{1 - \cos^2\theta_{\max}} = \sin\theta_{\max}.$$

Version 2 Principal orthonormal bases / CS decomposition

- Principal orthonormal bases and the paired projection relations

  By the theory of principal angles between subspaces, there exist orthonormal bases $\{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_d\}$, $\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_d\}$ of U and V, respectively, satisfying the paired projection relations

  $$\boldsymbol{P}\boldsymbol{v}_i = \cos\theta_i\boldsymbol{u}_i, \quad i = 1, \cdots, d.$$
- Geometric effect of the operator $\boldsymbol{P}_\perp\boldsymbol{Q}$ on the basis vectors

  Consider the geometric effect of the operator $\boldsymbol{P}_\perp\boldsymbol{Q}$ on the basis vector $\boldsymbol{v}_i$:

  $$\boldsymbol{P}_\perp\boldsymbol{Q}\boldsymbol{v}_i = \boldsymbol{P}_\perp\boldsymbol{v}_i = \boldsymbol{v}_i - \boldsymbol{P}\boldsymbol{v}_i = \boldsymbol{v}_i - \cos\theta_i\boldsymbol{u}_i.$$

  Computing the squared norm of this residual vector (*since* $\langle\boldsymbol{v}_i, \boldsymbol{u}_i\rangle = \cos\theta_i$),

  $$\|\boldsymbol{P}_\perp\boldsymbol{v}_i\|_2^2 = \|\boldsymbol{v}_i\|_2^2 - 2\cos\theta_i\langle\boldsymbol{v}_i, \boldsymbol{u}_i\rangle + \cos^2\theta_i\|\boldsymbol{u}_i\|_2^2 = 1 - 2\cos^2\theta_i + \cos^2\theta_i = 1 - \cos^2\theta_i = \sin^2\theta_i.$$
- Nonzero singular values and the operator norm

  Since the principal directions are mutually orthogonal, the operator $\boldsymbol{P}_\perp\boldsymbol{Q}$ maps the basis $\{\boldsymbol{v}_i\}$ to an orthogonal system of vectors with lengths $\sin\theta_i$. Hence the set of nonzero singular values of $\boldsymbol{P}_\perp\boldsymbol{Q}$ is exactly $\{\sin\theta_1, \cdots, \sin\theta_d\}$, and its operator norm is determined by the largest principal angle:

  $$\|\boldsymbol{P}_\perp\boldsymbol{Q}\| = \max_{1\leqslant i\leqslant d} \sin\theta_i = \sin\theta_{\max}.$$
- Duality symmetry and synthesis of the conclusion

  By the transpose invariance of the operator norm, we get

  $$\|\boldsymbol{P}\boldsymbol{Q}_\perp\| = \|(\boldsymbol{P}\boldsymbol{Q}_\perp)^\top\| = \|\boldsymbol{Q}_\perp\boldsymbol{P}\|.$$

  Since $\dim(U) = \dim(V)$, exchanging the roles of U and V leaves the principal angles unchanged, and similarly we get

  $$\|\boldsymbol{Q}_\perp\boldsymbol{P}\| = \sin\theta_{\max}.$$

  Then, combining $\|\boldsymbol{P}_\perp\boldsymbol{Q}\| = \sin\theta_{\max}$, we get

  $$\|\boldsymbol{P}_\perp\boldsymbol{Q}\| = \|\boldsymbol{P}\boldsymbol{Q}_\perp\| = \sin\theta_{\max}.$$

  Combining with the conclusion of (1), $\|\boldsymbol{P} - \boldsymbol{Q}\| = \max(\|\boldsymbol{P}\boldsymbol{Q}_\perp\|, \|\boldsymbol{P}_\perp\boldsymbol{Q}\|)$, we get

  $$\|\boldsymbol{P} - \boldsymbol{Q}\| = \|\boldsymbol{P}_\perp\boldsymbol{Q}\| = \|\boldsymbol{P}\boldsymbol{Q}_\perp\| = \sin\theta_{\max}.$$

13 [**Davis-Kahan Theorem for Spectral Projections**] This problem aims to prove a variant of Theorem T4.1.15 (*Davis–*

Kahan inequality), namely a perturbation bound for the subspace spanned by the first k eigenvectors.

Let $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n\times n}$ be symmetric matrices with spectral decompositions

$$\boldsymbol{A} = \sum_{i=1}^{n} \lambda_i \boldsymbol{u}_i \boldsymbol{u}_i^\top, \quad \boldsymbol{B} = \sum_{i=1}^{n} \mu_i \boldsymbol{v}_i \boldsymbol{v}_i^\top,$$

where $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ and $\mu_1 \geqslant \cdots \geqslant \mu_n$, and take $1 \leqslant k < n$.

Define the spectral projections onto the first k eigenvalues respectively as

$$\boldsymbol{P}_A = \sum_{i=1}^{k} \boldsymbol{u}_i \boldsymbol{u}_i^\top, \quad \boldsymbol{P}_B = \sum_{i=1}^{k} \boldsymbol{v}_i \boldsymbol{v}_i^\top.$$

Prove: if $\lambda_k > \lambda_{k+1}$, then the operator norm of the difference of the projections satisfies

$$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \frac{2\|\boldsymbol{A} - \boldsymbol{B}\|}{\lambda_k - \lambda_{k+1}}.$$

Prf. **Solution 1 Weyl perturbation method**

- Establishing the Sylvester equation for the residual operator

  Let the perturbation matrix $\boldsymbol{E} = \boldsymbol{A} - \boldsymbol{B}$; let $\boldsymbol{P}_A^\perp = \boldsymbol{I}_n - \boldsymbol{P}_A$ be the projection onto the remaining eigenspace of $\boldsymbol{A}$. Consider the operator product $\boldsymbol{P}_A^\perp \boldsymbol{E} \boldsymbol{P}_B$,

  $$\boldsymbol{P}_A^\perp \boldsymbol{E} \boldsymbol{P}_B = \boldsymbol{P}_A^\perp (\boldsymbol{A} - \boldsymbol{B}) \boldsymbol{P}_B = \boldsymbol{P}_A^\perp \boldsymbol{A} \boldsymbol{P}_B - \boldsymbol{P}_A^\perp \boldsymbol{B} \boldsymbol{P}_B.$$

  Since $\boldsymbol{A}$ commutes with its principal spectral projection $\boldsymbol{P}_A$, we have $\boldsymbol{P}_A^\perp \boldsymbol{A} = \boldsymbol{P}_A^\perp \boldsymbol{A} \boldsymbol{P}_A^\perp$; since $\boldsymbol{B}$ commutes with its principal spectral projection $\boldsymbol{P}_B$, we have $\boldsymbol{B}\boldsymbol{P}_B = \boldsymbol{P}_B \boldsymbol{B} \boldsymbol{P}_B$.

  Substituting the above, setting $\boldsymbol{A}_2 = \boldsymbol{P}_A^\perp \boldsymbol{A} \boldsymbol{P}_A^\perp$, $\boldsymbol{B}_1 = \boldsymbol{P}_B \boldsymbol{B} \boldsymbol{P}_B$, $\boldsymbol{Y} = \boldsymbol{P}_A^\perp \boldsymbol{P}_B$, we obtain the Sylvester equation

  $$\boldsymbol{A}_2 \boldsymbol{Y} - \boldsymbol{Y} \boldsymbol{B}_1 = \boldsymbol{P}_A^\perp \boldsymbol{E} \boldsymbol{P}_B.$$

- Estimating the operator norm via the spectral gap lower bound

  Consider the spectral ranges of the two matrices: the set of nonzero eigenvalues of $\boldsymbol{A}_2$ is $\{\lambda_{k+1}, \cdots, \lambda_n\}$, whose largest eigenvalue is λ_{k+1}, so $\boldsymbol{A}_2 \preccurlyeq \lambda_{k+1} \boldsymbol{I}$; the set of nonzero eigenvalues of $\boldsymbol{B}_1$ is $\{\mu_1, \cdots, \mu_k\}$, whose smallest eigenvalue is μ_k, so $\boldsymbol{B}_1 \succcurlyeq \mu_k \boldsymbol{I}$.

  If $\mu_k > \lambda_{k+1}$, then the spectral gap of the two matrices is $\mu_k - \lambda_{k+1} > 0$. By the inverse estimate for the Sylvester operator (*Bhatia-Davis lemma*), we get

  $$\|\boldsymbol{A}_2 \boldsymbol{Y} - \boldsymbol{Y} \boldsymbol{B}_1\| \geqslant (\mu_k - \lambda_{k+1}) \|\boldsymbol{Y}\|.$$

  Combining with the bound on the right-hand side,

  $$(\mu_k - \lambda_{k+1}) \|\boldsymbol{Y}\| \leqslant \|\boldsymbol{P}_A^\perp \boldsymbol{E} \boldsymbol{P}_B\| \leqslant \|\boldsymbol{P}_A^\perp\| \cdot \|\boldsymbol{E}\| \cdot \|\boldsymbol{P}_B\| \leqslant \|\boldsymbol{A} - \boldsymbol{B}\|,$$

  since $\operatorname{rank}(\boldsymbol{P}_A) = \operatorname{rank}(\boldsymbol{P}_B) = k$, using the conclusion of Exercise Q4.12 (*Norms of Differences and Products of Orthogonal Projections*) (3),

  $$\|\boldsymbol{Y}\| = \|\boldsymbol{P}_A^\perp \boldsymbol{P}_B\| = \|\boldsymbol{P}_A - \boldsymbol{P}_B\|,$$

  hence we get the conditional perturbation bound (*when* $\mu_k > \lambda_{k+1}$)

  $$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \frac{\|\boldsymbol{A} - \boldsymbol{B}\|}{\mu_k - \lambda_{k+1}}.$$

- Closing the argument case by case with Weyl's eigenvalue inequality

  Let $\delta = \lambda_k - \lambda_{k+1}$. By Weyl's eigenvalue perturbation theorem, the k-th eigenvalue of $\boldsymbol{B}$ satisfies $|\mu_k - \lambda_k| \leqslant \|\boldsymbol{A} - \boldsymbol{B}\|$, hence

  $$\mu_k \geqslant \lambda_k - \|\boldsymbol{A} - \boldsymbol{B}\|,$$

  subtracting λ_{k+1} on both sides,

  $$\mu_k - \lambda_{k+1} \geqslant (\lambda_k - \lambda_{k+1}) - \|\boldsymbol{A} - \boldsymbol{B}\| = \delta - \|\boldsymbol{A} - \boldsymbol{B}\|.$$

a. Small perturbation ($\|\boldsymbol{A}-\boldsymbol{B}\| \leqslant \frac{1}{2}\delta$): in this case the denominator satisfies

$$\mu_k - \lambda_{k+1} \geqslant \delta - \frac{1}{2}\delta = \frac{1}{2}\delta > 0,$$

substituting this into the conditional perturbation bound, we get

$$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \frac{\|\boldsymbol{A}-\boldsymbol{B}\|}{\frac{1}{2}\delta} = \frac{2\|\boldsymbol{A}-\boldsymbol{B}\|}{\lambda_k - \lambda_{k+1}}.$$

b. Large perturbation ($\|\boldsymbol{A}-\boldsymbol{B}\| > \frac{1}{2}\delta$): in this case the right-hand side is already greater than 1,

$$\frac{2\|\boldsymbol{A}-\boldsymbol{B}\|}{\lambda_k - \lambda_{k+1}} > \frac{2\cdot\left(\frac{1}{2}\delta\right)}{\delta} = 1,$$

while the operator norm of the difference of any two orthogonal projection operators always satisfies the trivial upper bound

$$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \max(\|\boldsymbol{P}_A\|, \|\boldsymbol{P}_B\|) = 1,$$

hence the inequality holds

$$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant 1 < \frac{2\|\boldsymbol{A}-\boldsymbol{B}\|}{\lambda_k - \lambda_{k+1}}.$$

Thus, for symmetric matrices $\boldsymbol{A}, \boldsymbol{B}$ satisfying $\lambda_k > \lambda_{k+1}$, we get

$$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \frac{2\|\boldsymbol{A}-\boldsymbol{B}\|}{\lambda_k - \lambda_{k+1}}.$$

Supplement: commutator estimate

- Constructing and decomposing the commutator

  Define the commutator $\boldsymbol{R} = \boldsymbol{A}\boldsymbol{P}_B - \boldsymbol{P}_B\boldsymbol{A}$ and expand it:

$$\boldsymbol{R} = \boldsymbol{A}\boldsymbol{P}_B - \boldsymbol{P}_B\boldsymbol{B} + \boldsymbol{P}_B\boldsymbol{B} - \boldsymbol{P}_B\boldsymbol{A} = (\boldsymbol{A}-\boldsymbol{B})\boldsymbol{P}_B + \boldsymbol{P}_B(\boldsymbol{B}-\boldsymbol{A}),$$

  hence by the triangle inequality (*note that* $\|\boldsymbol{P}_B\| = 1$),

$$\|\boldsymbol{R}\| \leqslant \|(\boldsymbol{A}-\boldsymbol{B})\boldsymbol{P}_B\| + \|\boldsymbol{P}_B(\boldsymbol{B}-\boldsymbol{A})\| \leqslant 2\|\boldsymbol{A}-\boldsymbol{B}\|. \tag{I}$$

  Let $\boldsymbol{P}_A^{\perp} = \boldsymbol{I}_n - \boldsymbol{P}_A$ be the projection onto the remaining eigenspace of $\boldsymbol{A}$ (*the eigenvectors* $k+1$ *through* n). Restricting $\boldsymbol{R}$ between Range($\boldsymbol{P}_A$) and Range($\boldsymbol{P}_A^{\perp}$),

$$\begin{aligned}\boldsymbol{P}_A\boldsymbol{R}\boldsymbol{P}_A^{\perp} &= \boldsymbol{P}_A(\boldsymbol{A}\boldsymbol{P}_B - \boldsymbol{P}_B\boldsymbol{A})\boldsymbol{P}_A^{\perp} \\ &= (\boldsymbol{P}_A\boldsymbol{A}\boldsymbol{P}_A)(\boldsymbol{P}_A\boldsymbol{P}_B\boldsymbol{P}_A^{\perp}) - (\boldsymbol{P}_A\boldsymbol{P}_B\boldsymbol{P}_A^{\perp})(\boldsymbol{P}_A^{\perp}\boldsymbol{A}\boldsymbol{P}_A^{\perp}),\end{aligned}$$

  and setting $\boldsymbol{A}_1 = \boldsymbol{P}_A\boldsymbol{A}\boldsymbol{P}_A$ (*its eigenvalues are* $\lambda_1, \cdots, \lambda_k$), $\boldsymbol{A}_2 = \boldsymbol{P}_A^{\perp}\boldsymbol{A}\boldsymbol{P}_A^{\perp}$ (*its eigenvalues are* $\lambda_{k+1}, \cdots, \lambda_n$), $\boldsymbol{X} = \boldsymbol{P}_A\boldsymbol{P}_B\boldsymbol{P}_A^{\perp}$, we obtain

$$\boldsymbol{A}_1\boldsymbol{X} - \boldsymbol{X}\boldsymbol{A}_2 = \boldsymbol{P}_A\boldsymbol{R}\boldsymbol{P}_A^{\perp}, \tag{II}$$

  which is a Sylvester equation.

- Estimating the commutator block

  Consider the linear operator $T(\boldsymbol{X}) = \boldsymbol{A}_1\boldsymbol{X} - \boldsymbol{X}\boldsymbol{A}_2$; its eigenvalues are $\{\lambda_i - \lambda_j\}$ ($i \in \{1, \cdots, k\}$, $j \in \{k+1, \cdots, n\}$). Since the eigenvalues are in decreasing order,

$$\min_{i \leqslant k, j > k}(\lambda_i - \lambda_j) = \lambda_k - \lambda_{k+1},$$

  hence

$$\|\boldsymbol{P}_A\boldsymbol{R}\boldsymbol{P}_A^{\perp}\| = \|\boldsymbol{A}_1\boldsymbol{X} - \boldsymbol{X}\boldsymbol{A}_2\| \geqslant (\lambda_k - \lambda_{k+1})\|\boldsymbol{X}\|. \tag{III}$$

  Since

$$\boldsymbol{X} = -\boldsymbol{P}_A\boldsymbol{P}_B^{\perp}\boldsymbol{P}_A^{\perp},$$

Exercise Q4.12 (*Norms of Differences and Products of Orthogonal Projections*) (3) gives

$$\begin{aligned}\|\boldsymbol{X}\| &\leqslant \|\boldsymbol{P}_A \boldsymbol{P}_B^{\perp}\| \\ &= \|\boldsymbol{P}_A - \boldsymbol{P}_B\|.\end{aligned}$$

Combining (I) and (III), we obtain

$$(\lambda_k - \lambda_{k+1})\|\boldsymbol{X}\| \leqslant 2\|\boldsymbol{A} - \boldsymbol{B}\|.$$

14 [**Spectral Properties of the Hermitian Expansion**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$. To study $\boldsymbol{A}$ with the tools for symmetric matrices, a common approach is to construct its Hermitian expansion.

Define the block symmetric matrix $\boldsymbol{H} \in \mathbb{R}^{(m+n)\times(m+n)}$

$$\boldsymbol{H} = \begin{pmatrix} \boldsymbol{O} & \boldsymbol{A} \\ \boldsymbol{A}^{\top} & \boldsymbol{O} \end{pmatrix},$$

and let the SVD of $\boldsymbol{A}$ be $\boldsymbol{A} = \sum_i s_i \boldsymbol{u}_i \boldsymbol{v}_i^{\top}$, where $s_i > 0$ are the singular values and $\boldsymbol{u}_i \in \mathbb{R}^m$, $\boldsymbol{v}_i \in \mathbb{R}^n$ are the corresponding left and right singular vectors.

Prove: all nonzero eigenvalues of $\boldsymbol{H}$ are exactly $\pm s_i$, with corresponding eigenvectors

$$\boldsymbol{w}_i^{(\pm)} = \begin{pmatrix} \boldsymbol{u}_i \\ \pm \boldsymbol{v}_i \end{pmatrix}.$$

Prf. It is easy to see that

$$\boldsymbol{A}\boldsymbol{v}_i = s_i \boldsymbol{u}_i, \quad \boldsymbol{A}^{\top}\boldsymbol{u}_i = s_i \boldsymbol{v}_i,$$

and it remains to verify $\boldsymbol{H}\boldsymbol{w} = \lambda \boldsymbol{w}$.

For the positive and negative eigenvalues $\pm s_i$, consider respectively

$$\boldsymbol{w}_i^{(+)} = \begin{pmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{pmatrix}, \quad \boldsymbol{w}_i^{(-)} = \begin{pmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{pmatrix}.$$

Since

$$\begin{aligned}\boldsymbol{H}\boldsymbol{w}_i^{(+)} &= \begin{pmatrix} \boldsymbol{O} & \boldsymbol{A} \\ \boldsymbol{A}^{\top} & \boldsymbol{O} \end{pmatrix} \begin{pmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{pmatrix} = \begin{pmatrix} \boldsymbol{A}\boldsymbol{v}_i \\ \boldsymbol{A}^{\top}\boldsymbol{u}_i \end{pmatrix} \\ &= \begin{pmatrix} s_i \boldsymbol{u}_i \\ s_i \boldsymbol{v}_i \end{pmatrix} = s_i \begin{pmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{pmatrix} = s_i \boldsymbol{w}_i^{(+)}, \\ \boldsymbol{H}\boldsymbol{w}_i^{(-)} &= \begin{pmatrix} \boldsymbol{O} & \boldsymbol{A} \\ \boldsymbol{A}^{\top} & \boldsymbol{O} \end{pmatrix} \begin{pmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{pmatrix} = \begin{pmatrix} \boldsymbol{A}(-\boldsymbol{v}_i) \\ \boldsymbol{A}^{\top}\boldsymbol{u}_i \end{pmatrix} \\ &= \begin{pmatrix} -s_i \boldsymbol{u}_i \\ s_i \boldsymbol{v}_i \end{pmatrix} = -s_i \begin{pmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{pmatrix} = -s_i \boldsymbol{w}_i^{(-)},\end{aligned}$$

so s_i and $-s_i$ are eigenvalues of $\boldsymbol{H}$.

Thus, if $\operatorname{rank}(\boldsymbol{A}) = r$, the construction above yields $2r$ eigenvalues of $\boldsymbol{H}$ $(\pm s_1, \cdots, \pm s_r)$.

Since $\boldsymbol{H}$ is symmetric, its eigenvectors span the whole $\mathbb{R}^{m+n}$ space, so the remaining $(m+n)-2r$ dimensions correspond to the zero eigenvalue of $\boldsymbol{H}$, whose eigenspace is spanned by block combinations of the left nullspace of $\boldsymbol{A}$ ($\boldsymbol{u} \perp \operatorname{Range}(\boldsymbol{A})$) and the right nullspace ($\boldsymbol{v} \in \ker(\boldsymbol{A})$).

15 [**Wedin's Theorem: Stability of the SVD**] Theorem T4.1.15 (*Davis–Kahan inequality*) applies to symmetric matrices. This problem aims to prove its version for general rectangular matrices.

Let $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m\times n}$, with SVDs

$$\boldsymbol{A} = \sum_i s_i \boldsymbol{u}_i \boldsymbol{v}_i^\top, \quad \boldsymbol{B} = \sum_i t_i \boldsymbol{w}_i \boldsymbol{z}_i^\top,$$

Let $r = \min(m, n)$; the singular values satisfy $s_1 \geqslant \cdots \geqslant s_r \geqslant 0$ and $t_1 \geqslant \cdots \geqslant t_r \geqslant 0$.
Define the projections onto the first k left singular vectors as

$$\boldsymbol{P}_A = \sum_{i=1}^k \boldsymbol{u}_i \boldsymbol{u}_i^\top, \quad \boldsymbol{P}_B = \sum_{i=1}^k \boldsymbol{w}_i \boldsymbol{w}_i^\top.$$

(1) Prove: if $1 \leqslant k < r$ and the spectral gap $s_k - s_{k+1} > 0$, then the left singular projections satisfy

$$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \frac{2\|\boldsymbol{A} - \boldsymbol{B}\|}{s_k - s_{k+1}}.$$

(2) For the projections onto the first k right singular vectors

$$\boldsymbol{Q}_A = \sum_{i=1}^k \boldsymbol{v}_i \boldsymbol{v}_i^\top, \quad \boldsymbol{Q}_B = \sum_{i=1}^k \boldsymbol{z}_i \boldsymbol{z}_i^\top,$$

give a similar bound.
(3) Prove a Davis-Kahan type inequality for the k-th singular vector (*let* $1 \leqslant k < r$, $s_{k-1} > s_k > s_{k+1}$, *and set* $s_0 = +\infty$).

Rk. The following is an erroneous attempt: let

$$\boldsymbol{W}_A = \sum_{i=1}^k \zeta_i \zeta_i^\top = \frac{1}{2}\sum_{i=1}^k \begin{pmatrix} \boldsymbol{u}_i \boldsymbol{u}_i^\top & \boldsymbol{u}_i \boldsymbol{v}_i^\top \\ \boldsymbol{v}_i \boldsymbol{u}_i^\top & \boldsymbol{v}_i \boldsymbol{v}_i^\top \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \boldsymbol{P}_A & \boldsymbol{X}_A \\ \boldsymbol{X}_A^\top & \boldsymbol{Q}_A \end{pmatrix}$$

be the spectral projection onto the first k positive eigenvalues; then by Exercise Q4.13 (*Davis-Kahan Theorem for Spectral Projections*),

$$\|\boldsymbol{W}_A - \boldsymbol{W}_B\| \leqslant \frac{2\|\boldsymbol{H}_A - \boldsymbol{H}_B\|}{s_k - s_{k+1}} = \frac{2\|\boldsymbol{A} - \boldsymbol{B}\|}{s_k - s_{k+1}}.$$

Note that $(\boldsymbol{P}_A - \boldsymbol{P}_B)/2$ is the top-left block of $\boldsymbol{W}_A - \boldsymbol{W}_B$; hence using the conclusion of Exercise Q4.2 (*Basic Properties of the Operator Norm*) (4),

$$\frac{1}{2}\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \|\boldsymbol{W}_A - \boldsymbol{W}_B\|,$$

and therefore

$$\|\boldsymbol{P}_A - \boldsymbol{P}_B\| \leqslant \frac{4\|\boldsymbol{A} - \boldsymbol{B}\|}{s_k - s_{k+1}}.$$

One can see that this is not the desired coefficient 2; the reason is that we did not take into account the symmetric distribution of the positive and negative singular values.

Prf. (1) By Exercise Q4.14 (*Spectral Properties of the Hermitian Expansion*), construct the symmetric matrices $\boldsymbol{H}_A, \boldsymbol{H}_B \in \mathbb{R}^{(m+n)\times(m+n)}$

$$\boldsymbol{H}_A = \begin{pmatrix} \boldsymbol{O} & \boldsymbol{A} \\ \boldsymbol{A}^\top & \boldsymbol{O} \end{pmatrix}, \quad \boldsymbol{H}_B = \begin{pmatrix} \boldsymbol{O} & \boldsymbol{B} \\ \boldsymbol{B}^\top & \boldsymbol{O} \end{pmatrix},$$

where the eigenvalues of $\boldsymbol{H}_A$ are $\{\pm s_i\}$, and the orthonormal eigenvectors corresponding to the first k positive eigenvalues $s_1, \cdots, s_k$ are

$$\zeta_i = \frac{1}{\sqrt{2}}\begin{pmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{pmatrix}.$$

Let the unit eigenvectors corresponding to $\pm s_i$ be respectively

$$\zeta_i = \frac{1}{\sqrt{2}}\begin{pmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{pmatrix}, \quad \boldsymbol{\chi}_i = \frac{1}{\sqrt{2}}\begin{pmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{pmatrix},$$

then the sum of their projections is

$$\boldsymbol{\zeta}_i\boldsymbol{\zeta}_i^\top+\boldsymbol{\chi}_i\boldsymbol{\chi}_i^\top=\frac{1}{2}\begin{pmatrix}\boldsymbol{u}_i\boldsymbol{u}_i^\top & \boldsymbol{u}_i\boldsymbol{v}_i^\top\\ \boldsymbol{v}_i\boldsymbol{u}_i^\top & \boldsymbol{v}_i\boldsymbol{v}_i^\top\end{pmatrix}+\frac{1}{2}\begin{pmatrix}\boldsymbol{u}_i\boldsymbol{u}_i^\top & -\boldsymbol{u}_i\boldsymbol{v}_i^\top\\ -\boldsymbol{v}_i\boldsymbol{u}_i^\top & \boldsymbol{v}_i\boldsymbol{v}_i^\top\end{pmatrix}=\begin{pmatrix}\boldsymbol{u}_i\boldsymbol{u}_i^\top & \boldsymbol{O}\\ \boldsymbol{O} & \boldsymbol{v}_i\boldsymbol{v}_i^\top\end{pmatrix},$$

hence the total projection matrix is

$$\boldsymbol{\Pi}_A=\sum_{i=1}^{k}(\boldsymbol{\zeta}_i\boldsymbol{\zeta}_i^\top+\boldsymbol{\chi}_i\boldsymbol{\chi}_i^\top)=\begin{pmatrix}\sum_{i=1}^{k}\boldsymbol{u}_i\boldsymbol{u}_i^\top & \boldsymbol{O}\\ \boldsymbol{O} & \sum_{i=1}^{k}\boldsymbol{v}_i\boldsymbol{v}_i^\top\end{pmatrix}=\begin{pmatrix}\boldsymbol{P}_A & \boldsymbol{O}\\ \boldsymbol{O} & \boldsymbol{Q}_A\end{pmatrix},$$

and similarly $\boldsymbol{\Pi}_B=\operatorname{diag}(\boldsymbol{P}_B,\boldsymbol{Q}_B)$.

Let $\delta=s_k-s_{k+1}$ and $e=\|\boldsymbol{A}-\boldsymbol{B}\|=\|\boldsymbol{H}_A-\boldsymbol{H}_B\|$.

If $e<\delta/2$, Weyl's eigenvalue inequality gives

$$\begin{aligned}t_k-t_{k+1}&\geqslant(s_k-e)-(s_{k+1}+e)\\&=\delta-2e>0.\end{aligned}$$

Thus $\boldsymbol{\Pi}_B$ is the spectral projection of $\boldsymbol{H}_B$ for the corresponding two-sided spectral cluster $\{\pm t_1,\cdots,\pm t_k\}$. The distance between the spectral cluster $\{\pm s_1,\cdots,\pm s_k\}$ of $\boldsymbol{H}_A$ and its remaining spectrum is δ. By the spectral-projection form of Lemma L4.1.16 (*Davis–Kahan inequality for spectral projections*),

$$\|\boldsymbol{\Pi}_A-\boldsymbol{\Pi}_B\|\leqslant\frac{e}{\delta-e}\leqslant\frac{2e}{\delta}.$$

If $e\geqslant\delta/2$, then $\boldsymbol{\Pi}_A,\boldsymbol{\Pi}_B$ have equal rank, so Exercise Q4.12 (*Norms of Differences and Products of Orthogonal Projections*) (3) gives

$$\|\boldsymbol{\Pi}_A-\boldsymbol{\Pi}_B\|\leqslant 1\leqslant\frac{2e}{\delta}.$$

Thus

$$\begin{aligned}\max\{\|\boldsymbol{P}_A-\boldsymbol{P}_B\|,\|\boldsymbol{Q}_A-\boldsymbol{Q}_B\|\}&=\|\boldsymbol{\Pi}_A-\boldsymbol{\Pi}_B\|\\&\leqslant\frac{2\|\boldsymbol{A}-\boldsymbol{B}\|}{s_k-s_{k+1}}.\end{aligned}$$

(2) Let s_k be a simple singular value with $1\leqslant k<r$ (*set* $s_0=+\infty$). Its spectral gap is $\delta_k=\min\{s_{k-1}-s_k,s_k-s_{k+1}\}$. Let $\boldsymbol{u}_k,\boldsymbol{w}_k$ be the k-th left singular vectors of $\boldsymbol{A},\boldsymbol{B}$ respectively, and let $\boldsymbol{v}_k,\boldsymbol{z}_k$ be the corresponding right singular vectors. The one-vector spectral-projection form of the Davis–Kahan inequality gives

$$\max\{\sin\Theta(\boldsymbol{u}_k,\boldsymbol{w}_k),\sin\Theta(\boldsymbol{v}_k,\boldsymbol{z}_k)\}\leqslant\frac{2\|\boldsymbol{A}-\boldsymbol{B}\|}{\delta_k}.$$

Thus the stability of the left and right singular vectors depends on the perturbation size and the separation between this singular value and its neighbors.

16 [**Angles between Vectors and Euclidean Distance**] Theorem T4.1.15 (*Davis–Kahan inequality*) bounds the deviation of spectral projections.

To simplify the conclusion, we examine the geometric relation between unit vectors.

Let $\boldsymbol{u},\boldsymbol{v}\in\mathbb{R}^n$ be unit vectors. Let $\alpha\in[0,\pi/2]$ be the angle between the lines spanned by $\boldsymbol{u},\boldsymbol{v}$ (*i.e. the principal angle*), satisfying:

$$\sin\alpha\leqslant\varepsilon.\qquad(\varepsilon>0)$$

Prove: $\boldsymbol{u},\boldsymbol{v}$ are close up to a sign, i.e. there exists $\theta\in\{-1,1\}$ such that

$$\|\boldsymbol{u}-\theta\boldsymbol{v}\|_2\leqslant\sqrt{2}\varepsilon.$$

Prf. Let the angle between $\boldsymbol{u},\boldsymbol{v}$ be $\phi\in[0,\pi]$, satisfying $\langle\boldsymbol{u},\boldsymbol{v}\rangle=\cos\phi$; then

$$\|\boldsymbol{u}-\theta\boldsymbol{v}\|_2^2=\|\boldsymbol{u}\|_2^2+\|\theta\boldsymbol{v}\|_2^2-2\langle\boldsymbol{u},\theta\boldsymbol{v}\rangle=1+1-2\theta\langle\boldsymbol{u},\boldsymbol{v}\rangle=2(1-\theta\cos\phi).$$

To minimize the distance above:

a. If $\cos\phi \geqslant 0$, choose $\theta = 1$; then the angle between $\boldsymbol{u}, \boldsymbol{v}$ is $\alpha = \phi \in [0, \pi/2]$.

b. If $\cos\phi < 0$, choose $\theta = -1$; then the angle between $\boldsymbol{u}, -\boldsymbol{v}$ is $\alpha = \pi - \phi \in [0, \pi/2]$.

Hence $\cos\alpha = |\cos\phi| \geqslant 0$, satisfying

$$\|\boldsymbol{u} - \theta\boldsymbol{v}\|_2^2 = 2(1 - \cos\alpha) = 4\sin^2\left(\frac{\alpha}{2}\right),$$

where

$$\sin\alpha = 2\sin\left(\frac{\alpha}{2}\right)\cos\left(\frac{\alpha}{2}\right) \geqslant 2\sin\left(\frac{\alpha}{2}\right)\cos\left(\frac{\pi}{4}\right) \geqslant 2\sin\left(\frac{\alpha}{2}\right)\cdot\frac{1}{\sqrt{2}} = \sqrt{2}\sin\left(\frac{\alpha}{2}\right),$$

and therefore

$$\|\boldsymbol{u} - \theta\boldsymbol{v}\|_2 = 2\sin\left(\frac{\alpha}{2}\right) \leqslant \sqrt{2}\sin\alpha \leqslant \sqrt{2}\varepsilon.$$

17 [**Approximate Projections and Approximate Isometries**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ $(m \geqslant n)$, and let $\varepsilon \geqslant 0$ be an accuracy constant. Recall Lemma L4.1.17 (*Approximate isometries*):

$\boldsymbol{A}$ is called an ε-approximate isometry if it satisfies one of the following equivalent properties:

(1) $\|\boldsymbol{A}^\top\boldsymbol{A} - \boldsymbol{I}_n\| \leqslant \varepsilon$;

(2) for every $\boldsymbol{x} \in \mathbb{R}^n$, one has $(1-\varepsilon)\|\boldsymbol{x}\|_2^2 \leqslant \|\boldsymbol{A}\boldsymbol{x}\|_2^2 \leqslant (1+\varepsilon)\|\boldsymbol{x}\|_2^2$;

(3) all singular values $s_i(\boldsymbol{A})$ satisfy $1 - \varepsilon \leqslant s_i(\boldsymbol{A})^2 \leqslant 1 + \varepsilon$.

Prove:

(1) $\boldsymbol{A}$ is an isometry (*i.e.* $\boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{I}_n$) if and only if $\boldsymbol{A}\boldsymbol{A}^\top \in \mathbb{R}^{m\times m}$ is an orthogonal projection matrix of rank n.

(2) More generally, $\boldsymbol{A}$ is an ε-approximate isometry if and only if there exists an orthogonal projection matrix $\boldsymbol{P} \in \mathbb{R}^{m\times m}$ of rank n such that

$$\|\boldsymbol{A}\boldsymbol{A}^\top - \boldsymbol{P}\| \leqslant \varepsilon.$$

Prf. (1) $(\Longrightarrow)$

If $\boldsymbol{A}$ is an isometry, then $\boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{I}_n$.

Let $\boldsymbol{Q} = \boldsymbol{A}\boldsymbol{A}^\top$; then:

1. Symmetry: $\boldsymbol{Q}^\top = (\boldsymbol{A}\boldsymbol{A}^\top)^\top = \boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{Q}$.
2. Idempotence: $\boldsymbol{Q}^2 = (\boldsymbol{A}\boldsymbol{A}^\top)(\boldsymbol{A}\boldsymbol{A}^\top) = \boldsymbol{A}(\boldsymbol{A}^\top\boldsymbol{A})\boldsymbol{A}^\top = \boldsymbol{A}\boldsymbol{I}_n\boldsymbol{A}^\top = \boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{Q}$.
3. Rank property: since $\boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{I}_n$, $\boldsymbol{A}$ has full column rank, hence $\operatorname{rank}(\boldsymbol{Q}) = \operatorname{rank}(\boldsymbol{A}\boldsymbol{A}^\top) = \operatorname{rank}(\boldsymbol{A}) = n$.

$(\Longleftarrow)$

If $\boldsymbol{Q} = \boldsymbol{A}\boldsymbol{A}^\top$ is an orthogonal projection matrix of rank n, let the SVD of $\boldsymbol{A}$ be $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, where $\boldsymbol{U} \in \mathbb{R}^{m\times m}$, $\boldsymbol{V} \in \mathbb{R}^{n\times n}$ are orthogonal matrices and $\boldsymbol{\Sigma} \in \mathbb{R}^{m\times n}$ contains the singular values $s_1, \cdots, s_n$; then

$$\boldsymbol{Q} = \boldsymbol{U}(\boldsymbol{\Sigma}\boldsymbol{\Sigma}^\top)\boldsymbol{U}^\top.$$

Since $\boldsymbol{Q}$ is a projection matrix, its eigenvalues can only be 1 and 0.

Note that the nonzero eigenvalues of $\boldsymbol{Q}$ are exactly the nonzero diagonal entries s_i^2 of $\boldsymbol{\Sigma}\boldsymbol{\Sigma}^\top$, and $\operatorname{rank}(\boldsymbol{Q}) = n$; hence

$$s_i^2 = 1 \implies s_i = 1,$$

so $\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma} = \boldsymbol{I}_n$, and therefore $\boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{V}(\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma})\boldsymbol{V}^\top = \boldsymbol{V}\boldsymbol{I}_n\boldsymbol{V}^\top = \boldsymbol{I}_n$.

(2) Let the singular values of $\boldsymbol{A}$ be $s_1, \cdots, s_n$; then by Lemma L4.1.17 (*Approximate isometries*), $\boldsymbol{A}$ is an ε-approximate isometry if and only if

$$\max_i |s_i^2 - 1| \leqslant \varepsilon.$$

$(\Longleftarrow)$

If there exists an orthogonal projection matrix $\boldsymbol{P}$ of rank n with $\|\boldsymbol{A}\boldsymbol{A}^\top - \boldsymbol{P}\| \leqslant \varepsilon$, consider the eigenvalue perturbation. We use the following lemma:

Lemma (Weyl's inequality) If the norm of the difference of two symmetric matrices does not exceed ε, then the

difference of their sorted eigenvalues also does not exceed ε.

It is easy to see that the eigenvalues of $\boldsymbol{P}$ are n copies of 1 and $m-n$ copies of 0, while the eigenvalues of $\boldsymbol{A}\boldsymbol{A}^\top$ are $s_1^2, \cdots, s_n^2$ and $m-n$ copies of 0; hence by the lemma, for the first n eigenvalues,

$$|s_i^2 - 1| \leqslant \|\boldsymbol{A}\boldsymbol{A}^\top - \boldsymbol{P}\| \leqslant \varepsilon,$$

so $\boldsymbol{A}$ is an ε-approximate isometry.

($\Longrightarrow$)

If $\boldsymbol{A}$ is an ε-approximate isometry, i.e. $|s_i^2 - 1| \leqslant \varepsilon$, let the SVD of $\boldsymbol{A}$ be $\boldsymbol{A} = \sum_{i=1}^n s_i \boldsymbol{u}_i \boldsymbol{v}_i^\top$ and construct the projection matrix

$$\boldsymbol{P} = \sum_{i=1}^n \boldsymbol{u}_i \boldsymbol{u}_i^\top$$

onto the range space of $\boldsymbol{A}$ (*i.e. the space spanned by the first n left singular vectors*); then

$$\boldsymbol{A}\boldsymbol{A}^\top - \boldsymbol{P} = \left(\sum_{i=1}^n s_i^2 \boldsymbol{u}_i \boldsymbol{u}_i^\top\right) - \left(\sum_{i=1}^n \boldsymbol{u}_i \boldsymbol{u}_i^\top\right) = \sum_{i=1}^n (s_i^2 - 1)\boldsymbol{u}_i \boldsymbol{u}_i^\top$$

is symmetric, hence

$$\|\boldsymbol{A}\boldsymbol{A}^\top - \boldsymbol{P}\| = \max_{1 \leqslant i \leqslant n} |s_i^2 - 1| \leqslant \varepsilon.$$

18 [$p \to q$ **Operator Norms**] The operator norm of $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ measures the ability of the matrix to "stretch" vectors. To quantify this stretching, we may use any ℓ^p and ℓ^q norms with $p, q \in [1, \infty]$.

Define the $p \to q$ operator norm of $\boldsymbol{A}$ as

$$\|\boldsymbol{A}\|_{p \to q} = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_q}{\|\boldsymbol{x}\|_p}.$$

(1) Prove that the definition above is equivalent to the following bilinear form:

$$\|\boldsymbol{A}\|_{p \to q} = \max_{\|\boldsymbol{x}\|_p = 1, \|\boldsymbol{y}\|_{q'} = 1} |\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}|,$$

where q' is the conjugate exponent of q.

(2) Prove that $\|\cdot\|_{p \to q}$ defines a norm on the matrix space $\mathbb{R}^{m \times n}$.

(3) (Duality) Prove that $\|\boldsymbol{A}^\top\|_{p \to q} = \|\boldsymbol{A}\|_{q' \to p'}$, where p', q' are the conjugate exponents of p, q respectively.

Prf. (1) We use the following lemma:

Lemma (Dual representation of the ℓ^q norm / a corollary of Hölder's inequality) For any $\boldsymbol{z} \in \mathbb{R}^m$,

$$\|\boldsymbol{z}\|_q = \max_{\|\boldsymbol{y}\|_{q'} = 1} |\boldsymbol{y}^\top \boldsymbol{z}|.$$

By definition,

$$\|\boldsymbol{A}\|_{p \to q} = \max_{\|\boldsymbol{x}\|_p = 1} \|\boldsymbol{A}\boldsymbol{x}\|_q,$$

hence by the lemma, taking $\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$,

$$\|\boldsymbol{A}\|_{p \to q} = \max_{\|\boldsymbol{x}\|_p = 1} \left(\max_{\|\boldsymbol{y}\|_{q'} = 1} |\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}| \right) = \max_{\|\boldsymbol{x}\|_p = 1, \|\boldsymbol{y}\|_{q'} = 1} |\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}|.$$

(*Since the objective function is continuous on a compact set, the supremum is attained as a maximum.*)

(2) 1. Nonnegativity and definiteness:

It is easy to see that $\|\boldsymbol{A}\boldsymbol{x}\|_q \geqslant 0$.

If $\|\boldsymbol{A}\|_{p \to q} = 0$, then for every $\boldsymbol{x} \in \mathbb{R}^n$ one has $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$, hence $\boldsymbol{A} = \boldsymbol{O}$.

2. Absolute homogeneity:

For any $c \in \mathbb{R}$,

$$\|c\boldsymbol{A}\|_{p\to q} = \max_{\|\boldsymbol{x}\|_p=1} \|c\boldsymbol{A}\boldsymbol{x}\|_q = \max_{\|\boldsymbol{x}\|_p=1} |c| \cdot \|\boldsymbol{A}\boldsymbol{x}\|_q = |c| \cdot \|\boldsymbol{A}\|_{p\to q}.$$

3. Triangle inequality:

For any $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m\times n}$,

$$\|(\boldsymbol{A}+\boldsymbol{B})\boldsymbol{x}\|_q = \|\boldsymbol{A}\boldsymbol{x}+\boldsymbol{B}\boldsymbol{x}\|_q \leqslant \|\boldsymbol{A}\boldsymbol{x}\|_q + \|\boldsymbol{B}\boldsymbol{x}\|_q \leqslant (\|\boldsymbol{A}\|_{p\to q} + \|\boldsymbol{B}\|_{p\to q})\|\boldsymbol{x}\|_p,$$

and maximizing both sides over $\|\boldsymbol{x}\|_p = 1$,

$$\|\boldsymbol{A}+\boldsymbol{B}\|_{p\to q} \leqslant \|\boldsymbol{A}\|_{p\to q} + \|\boldsymbol{B}\|_{p\to q}.$$

(3) Using the conclusion of (1),

$$\|\boldsymbol{A}\|_{q'\to p'} = \max_{\|\boldsymbol{x}\|_{q'}=1, \|\boldsymbol{y}\|_{(p')'}=1} |\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}| = \max_{\|\boldsymbol{x}\|_{q'}=1, \|\boldsymbol{y}\|_p=1} |\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}|,$$

$$\|\boldsymbol{A}^\top\|_{p\to q} = \max_{\|\boldsymbol{y}\|_p=1, \|\boldsymbol{x}\|_{q'}=1} |\boldsymbol{x}^\top \boldsymbol{A}^\top \boldsymbol{y}| = \max_{\|\boldsymbol{y}\|_p=1, \|\boldsymbol{x}\|_{q'}=1} |\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}|,$$

hence $\|\boldsymbol{A}^\top\|_{p\to q} = \|\boldsymbol{A}\|_{q'\to p'}$.

19 [**Explicit Formulas for the $1\to\infty$ and $1\to 2$ Operator Norms**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$; denote its i-th row by $\boldsymbol{r}_i^\top$ and its j-th column by $\boldsymbol{a}_j$. Recall the definition of the $p\to q$ operator norm:

$$\|\boldsymbol{A}\|_{p\to q} = \sup_{\|\boldsymbol{x}\|_p=1} \|\boldsymbol{A}\boldsymbol{x}\|_q.$$

(1) Prove: the $1\to\infty$ norm equals the maximum absolute value of the matrix entries, i.e.

$$\|\boldsymbol{A}\|_{1\to\infty} = \max_{1\leqslant i\leqslant m, 1\leqslant j\leqslant n} |a_{ij}|.$$

(2) Prove: the $1\to 2$ norm equals the maximum Euclidean norm of the columns, and the $2\to\infty$ norm equals the maximum Euclidean norm of the rows, i.e.

$$\|\boldsymbol{A}\|_{1\to 2} = \max_{1\leqslant j\leqslant n} \|\boldsymbol{a}_j\|_2, \quad \|\boldsymbol{A}\|_{2\to\infty} = \max_{1\leqslant i\leqslant m} \|\boldsymbol{r}_i\|_2.$$

Prf. (1) By definition,

$$\|\boldsymbol{A}\|_{1\to\infty} = \sup_{\|\boldsymbol{x}\|_1=1} \max_{1\leqslant i\leqslant m} |(\boldsymbol{A}\boldsymbol{x})_i|.$$

- Upper bound

For any $\boldsymbol{x}$ with $\|\boldsymbol{x}\|_1 = \sum_{j=1}^n |x_j| = 1$, consider its i-th component:

$$\begin{aligned} |(\boldsymbol{A}\boldsymbol{x})_i| &= \left|\sum_{j=1}^n a_{ij}x_j\right| \leqslant \sum_{j=1}^n |a_{ij}| \cdot |x_j| \\ &\leqslant \left(\max_{1\leqslant j\leqslant n} |a_{ij}|\right) \sum_{j=1}^n |x_j| = \max_j |a_{ij}|, \end{aligned}$$

and maximizing over all rows i,

$$\|\boldsymbol{A}\boldsymbol{x}\|_\infty \leqslant \max_{i,j} |a_{ij}|,$$

i.e. $\|\boldsymbol{A}\|_{1\to\infty} \leqslant \max_{i,j} |a_{ij}|$.

- Lower bound

Let the maximum absolute value be attained at (i^*, j^*), i.e. $|A_{i^*j^*}| = \max_{i,j} |a_{ij}|$; take $\boldsymbol{x} = \boldsymbol{e}_{j^*}$; then $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{a}_{j^*}$, and

$$\|\boldsymbol{A}\boldsymbol{x}\|_\infty = \max_i |A_{ij^*}| \geqslant |A_{i^*j^*}| = \max_{i,j} |a_{ij}|,$$

hence $\|\boldsymbol{A}\|_{1\to\infty} \geqslant \max_{i,j} |a_{ij}|$.

(2) - Proof of $\|\boldsymbol{A}\|_{1\to 2} = \max_j \|\boldsymbol{a}_j\|_2$

For the upper bound, let $\|\boldsymbol{x}\|_1 = 1$; then $\boldsymbol{A}\boldsymbol{x} = \sum_{j=1}^n x_j \boldsymbol{a}_j$, so by the triangle inequality,

$$\|\boldsymbol{A}\boldsymbol{x}\|_2 = \left\|\sum_{j=1}^n x_j \boldsymbol{a}_j\right\|_2 \leqslant \sum_{j=1}^n |x_j| \cdot \|\boldsymbol{a}_j\|_2 \leqslant \max_j \|\boldsymbol{a}_j\|_2 \left(\sum_{j=1}^n |x_j|\right) = \max_j \|\boldsymbol{a}_j\|_2.$$

For the lower bound, take $\boldsymbol{x} = \boldsymbol{e}_{j^*}$, where j^* is an index achieving the maximal column norm; then

$$\|\boldsymbol{A}\boldsymbol{e}_{j^*}\|_2 = \|\boldsymbol{a}_{j^*}\|_2 = \max_j \|\boldsymbol{a}_j\|_2,$$

i.e. $\|\boldsymbol{A}\|_{1\to 2} = \max_j \|\boldsymbol{a}_j\|_2$.

- Proof of $\|\boldsymbol{A}\|_{2\to\infty} = \max_i \|\boldsymbol{r}_i\|_2$

  For the upper bound, by definition,

$$\|\boldsymbol{A}\|_{2\to\infty} = \sup_{\|\boldsymbol{x}\|_2=1} \max_i |\langle \boldsymbol{r}_i, \boldsymbol{x}\rangle|,$$

  and by the Cauchy inequality,

$$|\langle \boldsymbol{r}_i, \boldsymbol{x}\rangle| \leqslant \|\boldsymbol{r}_i\|_2 \cdot \|\boldsymbol{x}\|_2 = \|\boldsymbol{r}_i\|_2,$$

  i.e. $\max_i |\langle \boldsymbol{r}_i, \boldsymbol{x}\rangle| \leqslant \max_i \|\boldsymbol{r}_i\|_2$.

  For the lower bound, take $\boldsymbol{x} = \boldsymbol{r}_{i^*}/\|\boldsymbol{r}_{i^*}\|_2$, where i^* is an index achieving the maximal row norm; then

$$\|\boldsymbol{A}\boldsymbol{x}\|_\infty \geqslant |\langle \boldsymbol{r}_{i^*}, \boldsymbol{x}\rangle| = \frac{\|\boldsymbol{r}_{i^*}\|_2^2}{\|\boldsymbol{r}_{i^*}\|_2} = \|\boldsymbol{r}_{i^*}\|_2 = \max_i \|\boldsymbol{r}_i\|_2,$$

  i.e. $\|\boldsymbol{A}\|_{2\to\infty} = \max_i \|\boldsymbol{r}_i\|_2$.

20 [**Properties and Duality of the $\infty \to 1$ Operator Norm**] We have seen this norm before! It was implicit in the assumption of Theorem T3.5.1 (*Grothendieck inequality*).

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$. Recall the definition of the operator norm

$$\|\boldsymbol{A}\|_{p\to q} = \sup_{\|\boldsymbol{y}\|_p \leqslant 1} \|\boldsymbol{A}\boldsymbol{y}\|_q.$$

(1) Prove: the $\infty \to 1$ norm can be expressed as a maximization over the discrete vertex set $\{-1, 1\}$,

$$\|\boldsymbol{A}\|_{\infty\to 1} = \max_{\boldsymbol{x}\in\{-1,1\}^m, \boldsymbol{y}\in\{-1,1\}^n} |\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}|.$$

(2) (Duality) Prove: the $\infty \to 1$ norm is equivalent to a search over rank-one matrices,

$$\|\boldsymbol{A}\|_{\infty\to 1} = \sup\{|\langle \boldsymbol{A}, \boldsymbol{Z}\rangle| : \|\boldsymbol{Z}\|_\infty \leqslant 1, \operatorname{rank}(\boldsymbol{Z}) = 1\},$$

where $\|\boldsymbol{Z}\|_\infty = \max_{i,j} |Z_{ij}|$ is the entrywise maximum norm and the matrix inner product is defined by $\langle \boldsymbol{A}, \boldsymbol{Z}\rangle = \operatorname{tr}(\boldsymbol{A}^\top \boldsymbol{Z})$.

(3) (Quadratic forms do not generalize) Prove that (2) cannot be directly generalized to quadratic forms. Construct a symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ such that

$$\max_{\boldsymbol{x}\in[-1,1]^n} |\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}| \neq \max_{\boldsymbol{x}\in\{-1,1\}^n} |\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}|.$$

Prf. (1) By definition,

$$\|\boldsymbol{A}\|_{\infty\to 1} = \sup_{\|\boldsymbol{y}\|_\infty \leqslant 1} \|\boldsymbol{A}\boldsymbol{y}\|_1.$$

Using the duality of the ℓ^1 norm $\|\boldsymbol{z}\|_1 = \sup_{\|\boldsymbol{x}\|_\infty \leqslant 1} \langle \boldsymbol{x}, \boldsymbol{z}\rangle$, we obtain

$$\|\boldsymbol{z}\|_1 = \sup_{\|\boldsymbol{x}\|_\infty \leqslant 1} \langle \boldsymbol{x}, \boldsymbol{z}\rangle,$$

where $f(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}$ is a bilinear function and $K = [-1, 1]^k$ is a convex polytope, so the supremum is attained on the discrete set:

$$\|\boldsymbol{A}\|_{\infty\to 1} = \max_{\boldsymbol{x}\in\{-1,1\}^m, \boldsymbol{y}\in\{-1,1\}^n} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y},$$

i.e.

$$\|\boldsymbol{A}\|_{\infty\to 1} = \max_{\boldsymbol{x}\in\{-1,1\}^m, \boldsymbol{y}\in\{-1,1\}^n} |\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y}|.$$

(2) Let $\boldsymbol{Z} = \boldsymbol{x}\boldsymbol{y}^\top$; then $\operatorname{rank}(\boldsymbol{Z}) = 1$ and $\|\boldsymbol{Z}\|_\infty \leqslant 1$ (*since* $|x_i|, |y_j| \leqslant 1$ *implies* $|Z_{ij}| \leqslant 1$); then

$$\langle \boldsymbol{A}, \boldsymbol{Z}\rangle = \operatorname{tr}(\boldsymbol{A}^\top \boldsymbol{x}\boldsymbol{y}^\top) = \operatorname{tr}(\boldsymbol{y}^\top \boldsymbol{A}^\top \boldsymbol{x}) = \boldsymbol{y}^\top \boldsymbol{A}^\top \boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y},$$

hence, using the conclusion of (1),

$$\begin{aligned}\|\boldsymbol{A}\|_{\infty\to 1} &= \sup\{\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{y} : \boldsymbol{x} \in [-1,1]^m, \boldsymbol{y} \in [-1,1]^n\} \\ &= \sup\{|\langle \boldsymbol{A}, \boldsymbol{Z}\rangle| : \|\boldsymbol{Z}\|_\infty \leqslant 1, \operatorname{rank}(\boldsymbol{Z}) = 1\}.\end{aligned}$$

(3) Construct

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and consider its quadratic form $g(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = x_1^2 - x_2^2$; then

$$\max_{\boldsymbol{x}\in\{-1,1\}^2} |x_1^2 - x_2^2| = |1-1| = 0, \qquad \max_{\boldsymbol{x}\in[-1,1]^2} |g(\boldsymbol{x})| = 1.$$

(*The maximum of a quadratic form is not necessarily attained at a vertex, since* $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}$ *is neither linear nor positive semidefinite.*)

21 [**The Cut Norm and Its Equivalence with the** $\infty \to 1$ **Norm**] In theoretical computer science and graph theory, the cut norm is an important measure of matrix properties.

For $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, the cut norm is defined as the maximum absolute sum of entries over all submatrices,

$$\|\boldsymbol{A}\|_{\text{cut}} = \max_{I,J} \left| \sum_{i\in I, j\in J} a_{ij} \right|,$$

where the maximum is taken over all subsets $I \subset \{1, \cdots, m\}$ and $J \subset \{1, \cdots, n\}$ of the index sets.

Prove: the cut norm is equivalent to the $\infty \to 1$ operator norm up to an absolute constant factor, i.e.

$$\|\boldsymbol{A}\|_{\text{cut}} \leqslant \|\boldsymbol{A}\|_{\infty\to 1} \leqslant 4\|\boldsymbol{A}\|_{\text{cut}}.$$

Prf. • The left inequality $\|\boldsymbol{A}\|_{\text{cut}} \leqslant \|\boldsymbol{A}\|_{\infty\to 1}$

By Exercise Q4.20 (*Properties and Duality of the* $\infty \to 1$ *Operator Norm*), the $\infty \to 1$ norm is equivalent to maximizing the bilinear form over the hypercube:

$$\|\boldsymbol{A}\|_{\infty\to 1} = \max_{\boldsymbol{u}\in[-1,1]^m, \boldsymbol{v}\in[-1,1]^n} \left|\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v}\right|.$$

For any given subsets I, J, define the corresponding indicator vectors $\boldsymbol{u}_I = \mathbf{1}_I \in \{0,1\}^m$ and $\boldsymbol{v}_J = \mathbf{1}_J \in \{0,1\}^n$, with $\boldsymbol{u}_I, \boldsymbol{v}_J \in [-1,1]^m \times [-1,1]^n$; then

$$\sum_{i\in I, j\in J} a_{ij} = \boldsymbol{u}_I^\top \boldsymbol{A}\boldsymbol{v}_J,$$

hence

$$\left| \sum_{i\in I, j\in J} a_{ij} \right| = \left|\boldsymbol{u}_I^\top \boldsymbol{A}\boldsymbol{v}_J\right| \leqslant \max_{\boldsymbol{u}\in[-1,1]^m, \boldsymbol{v}\in[-1,1]^n} |\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{v}| = \|\boldsymbol{A}\|_{\infty\to 1},$$

and maximizing both sides,

$$\|\boldsymbol{A}\|_{\text{cut}} \leqslant \|\boldsymbol{A}\|_{\infty\to 1}.$$

• The right inequality $\|\boldsymbol{A}\|_{\infty\to 1} \leqslant 4\|\boldsymbol{A}\|_{\text{cut}}$

Let $\boldsymbol{x} \in \{-1,1\}^n, \boldsymbol{y} \in \{-1,1\}^m$ be vectors attaining the maximum of $\|\boldsymbol{A}\|_{\infty\to 1}$; then

$$\|\boldsymbol{A}\|_{\infty\to 1} = \left| \sum_{i=1}^m \sum_{j=1}^n a_{ij} y_i x_j \right|.$$

Partition the index sets according to the signs of the components:

$$I^+ = \{i : y_i = 1\}, \quad I^- = \{i : y_i = -1\},$$
$$J^+ = \{j : x_j = 1\}, \quad J^- = \{j : x_j = -1\},$$

and expand the double sum:

$$\begin{aligned}\sum_{i,j} a_{ij} y_i x_j &= \sum_{i\in I^+, j\in J^+} a_{ij}(1)(1) + \sum_{i\in I^-, j\in J^-} a_{ij}(-1)(-1) + \sum_{i\in I^+, j\in J^-} a_{ij}(1)(-1) + \sum_{i\in I^-, j\in J^+} a_{ij}(-1)(1) \\ &= \sum_{i\in I^+, j\in J^+} a_{ij} + \sum_{i\in I^-, j\in J^-} a_{ij} - \sum_{i\in I^+, j\in J^-} a_{ij} - \sum_{i\in I^-, j\in J^+} a_{ij},\end{aligned}$$

hence by the triangle inequality,

$$\|\boldsymbol{A}\|_{\infty\to 1} \leqslant \left|\sum_{i\in I^+, j\in J^+} a_{ij}\right| + \left|\sum_{i\in I^-, j\in J^-} a_{ij}\right| + \left|\sum_{i\in I^+, j\in J^-} a_{ij}\right| + \left|\sum_{i\in I^-, j\in J^+} a_{ij}\right|,$$

i.e.

$$\|\boldsymbol{A}\|_{\infty\to 1} \leqslant \|\boldsymbol{A}\|_{\text{cut}} + \|\boldsymbol{A}\|_{\text{cut}} + \|\boldsymbol{A}\|_{\text{cut}} + \|\boldsymbol{A}\|_{\text{cut}} = 4\|\boldsymbol{A}\|_{\text{cut}}.$$

22 [**SDP Relaxation Approximation of Operator Norms**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$. Prove that $\|\boldsymbol{A}\|_{\infty\to 1}$, the cut norm $\|\boldsymbol{A}\|_{\text{cut}}$ (see Exercise Q4.21 (*The Cut Norm and Its Equivalence with the* $\infty \to 1$ *Norm*)), and $\|\boldsymbol{A}\|_{\infty\to 2}$ can all be computed approximately, up to an absolute constant factor, by solving an SDP problem.

Prf. By Exercise Q4.20 (*Properties and Duality of the* $\infty \to 1$ *Operator Norm*), the $\infty \to 1$ norm is defined by the following integer programming problem:

$$\|\boldsymbol{A}\|_{\infty\to 1} = \max_{x_j, y_i \in \{-1,1\}} \sum_{i,j} a_{ij} x_j y_i.$$

Construct its SDP relaxation by replacing the scalars x_j, y_i with unit vectors $\boldsymbol{u}_j, \boldsymbol{v}_i \in \mathbb{S}^{k-1}$ in a Hilbert space:

$$\text{sdp}(\boldsymbol{A}) = \max_{\|\boldsymbol{u}_j\|_2 = 1, \|\boldsymbol{v}_i\|_2 = 1} \sum_{i,j} a_{ij} \langle \boldsymbol{u}_j, \boldsymbol{v}_i \rangle,$$

hence by Theorem T3.5.1 (*Grothendieck inequality*),

$$\|\boldsymbol{A}\|_{\infty\to 1} \leqslant \text{sdp}(\boldsymbol{A}) \leqslant K_G \|\boldsymbol{A}\|_{\infty\to 1}, \qquad (K_G > 0)$$

i.e. $\|\boldsymbol{A}\|_{\infty\to 1} \asymp \text{sdp}(\boldsymbol{A})$.

By Exercise Q4.21 (*The Cut Norm and Its Equivalence with the* $\infty \to 1$ *Norm*),

$$\|\boldsymbol{A}\|_{\text{cut}} \asymp \|\boldsymbol{A}\|_{\infty\to 1},$$

and using the conclusion of (1), $\|\boldsymbol{A}\|_{\infty\to 1}$ can be computed approximately by solving an SDP; hence so can $\|\boldsymbol{A}\|_{\text{cut}}$.

Let $\boldsymbol{B} = \boldsymbol{A}^\top \boldsymbol{A}$; then

$$\|\boldsymbol{A}\|_{\infty\to 2}^2 = \left(\max_{\|\boldsymbol{x}\|_\infty \leqslant 1} \|\boldsymbol{A}\boldsymbol{x}\|_2\right)^2 = \max_{\boldsymbol{x}\in\{-1,1\}^n} \boldsymbol{x}^\top (\boldsymbol{A}^\top \boldsymbol{A}) \boldsymbol{x} = \max_{\boldsymbol{x}\in\{-1,1\}^n} \boldsymbol{x}^\top \boldsymbol{B} \boldsymbol{x},$$

which is a Boolean quadratic program over a PSD matrix.

By Exercise Q3.58 (*SDP Relaxation Analysis Based on Grothendieck's Inequality*),

$$\text{int}(\boldsymbol{B}) \leqslant \text{sdp}(\boldsymbol{B}) \leqslant \frac{\pi}{2} \text{int}(\boldsymbol{B}),$$

i.e. $\|\boldsymbol{A}\|_{\infty\to 2}^2 \asymp \text{sdp}(\boldsymbol{A}^\top \boldsymbol{A})$.

23 [**Transitivity of ε-Nets**] Let (K, d) be a metric space. Suppose $\mathcal{M}, \mathcal{N} \subset K$ satisfy:

1. $\mathcal{M}$ is a δ-net of K, i.e. for every $\boldsymbol{x} \in K$ there exists $\boldsymbol{m} \in \mathcal{M}$ with $d(\boldsymbol{x}, \boldsymbol{m}) \leqslant \delta$.
2. $\mathcal{N}$ is an ε-net of $\mathcal{M}$, i.e. for every $\boldsymbol{m} \in \mathcal{M}$ there exists $\boldsymbol{n} \in \mathcal{N}$ with $d(\boldsymbol{m}, \boldsymbol{n}) \leqslant \varepsilon$.

Prove: $\mathcal{N}$ is an $(\varepsilon + \delta)$-net of K.

Prf. By the triangle inequality,

$$d(\boldsymbol{x},\boldsymbol{n}) \leqslant d(\boldsymbol{x},\boldsymbol{m}) + d(\boldsymbol{m},\boldsymbol{n}) \leqslant \delta + \varepsilon.$$

24 [**Ball Packing and ε-Separated Sets**] Let $(\mathbb{T}, d)$ be a metric space. A set $\mathcal{N} \subset \mathbb{T}$ is called ε-separated if $d(x, y) > \varepsilon$ for all $x, y \in \mathcal{N}$. By the triangle inequality, $\{B(x, \varepsilon/2)\}_{x\in\mathcal{N}}$ are pairwise disjoint.

(1) Prove: the converse of the statement above fails in general metric spaces. That is, even if $\{B(x, \varepsilon/2)\}_{x\in\mathcal{N}}$ are pairwise disjoint, $\mathcal{N}$ need not be ε-separated.

(2) Prove: if $(\mathbb{T}, d)$ is a normed vector space (*i.e.* $d(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|$), then the converse holds.

Prf. (1) (*Construct a metric space* $(\mathbb{T}, d)$ *with* $B(x, \varepsilon/2) \cap B(y, \varepsilon/2) = \emptyset$ *but* $d(x, y) \leqslant \varepsilon$.)

Take $\mathbb{T} = \{0, 1\}$, $d(x, y) = |x - y|$, $\varepsilon = 1$; then $B(0, 0.5) \cap B(1, 0.5) = \emptyset$.

If $\mathcal{N} = \{0, 1\}$ were ε-separated, we would need $d(0, 1) > \varepsilon$; but $d(0, 1) = 1 \not> \varepsilon = 1$.

(2) Argue by contradiction: suppose there exist $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{N}$ that are not ε-separated, i.e. $\|\boldsymbol{x} - \boldsymbol{y}\| \leqslant \varepsilon$.

Let $\boldsymbol{m} = (\boldsymbol{x} + \boldsymbol{y})/2$; then

$$\|\boldsymbol{x} - \boldsymbol{m}\| = \|\boldsymbol{y} - \boldsymbol{m}\| = \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{y}\| \leqslant \frac{\varepsilon}{2},$$

so $\boldsymbol{m} \in B(\boldsymbol{x}, \varepsilon/2) \cap B(\boldsymbol{y}, \varepsilon/2)$, contradicting the disjointness of the balls.

25 [**External Covering Numbers and Their Equivalence**] Let $(\mathbb{T}, d)$ be a metric space and $K \subset \mathbb{T}$. In general, the covering number $\mathcal{N}(K, d, \varepsilon)$ requires the centers x_i to lie in K (i.e. internal covering). If this restriction is relaxed, allowing centers outside K (i.e. $x_i \in \mathbb{T}$), the minimal number of balls is called the external covering number, denoted by $\overline{\mathcal{N}}(K, d, \varepsilon)$.

Prove: the internal and external covering numbers are essentially equivalent, i.e.

$$\overline{\mathcal{N}}(K, d, \varepsilon) \leqslant \mathcal{N}(K, d, \varepsilon) \leqslant \overline{\mathcal{N}}(K, d, \varepsilon/2).$$

Prf. By definition, the feasible set for the external covering number is at least as large as that for the internal one, so

$$\overline{\mathcal{N}}(K, d, \varepsilon) \leqslant \mathcal{N}(K, d, \varepsilon),$$

and it remains to prove the right inequality.

Let $\{y_1, y_2, \cdots, y_m\}$ be an optimal external $(\varepsilon/2)$-net of K, with $m = \overline{\mathcal{N}}(K, d, \varepsilon/2)$. (*Then* $y_j \in \mathbb{T}$ *and* $K \subset \bigcup_{j=1}^{m} B(y_j, \varepsilon/2)$.)

a. If $B(y_j, \varepsilon/2) \cap K = \emptyset$, delete it.

b. If $B(y_j, \varepsilon/2) \cap K \neq \emptyset$, choose $x_j \in B(y_j, \varepsilon/2) \cap K$.

We claim:

Claim $\{x_1, x_2, \cdots, x_m\}$ is an internal ε-net of K.

Indeed, for any $z \in K$ *there exists* j *such that*

$$d(z, y_j) \leqslant \frac{\varepsilon}{2};$$

similarly,

$$d(x_j, y_j) \leqslant \frac{\varepsilon}{2},$$

hence

$$d(z, x_j) \leqslant d(z, y_j) + d(y_j, x_j) \leqslant \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

By the claim, $\mathcal{N}(K, d, \varepsilon) \leqslant m = \overline{\mathcal{N}}(K, d, \varepsilon/2)$.

26 [**Monotonicity Analysis of Covering Numbers**] Let $(\mathbb{T}, d)$ be a metric space and $K, L \subset \mathbb{T}$. Denote by $\mathcal{N}(K, d, \varepsilon)$ the internal covering number of K.

It is known that the covering number is monotone in the radius ε: if $\varepsilon_1 \leqslant \varepsilon_2$, then $\mathcal{N}(K, d, \varepsilon_1) \geqslant \mathcal{N}(K, d, \varepsilon_2)$ (think: why?). However, it is not directly monotone in K.

(1) Construct a counterexample showing that $L \subset K$ does not necessarily imply $\mathcal{N}(L,d,\varepsilon) \leqslant \mathcal{N}(K,d,\varepsilon)$.

(2) Prove: the covering number satisfies an "approximate monotonicity", i.e.

$$L \subset K \implies \mathcal{N}(L,d,\varepsilon) \leqslant \mathcal{N}(K,d,\varepsilon/2).$$

Rk. Proof of the monotonicity of the covering number in the radius ε:

Let $\mathcal{N}_1 = \{x_1, \cdots, x_M\}$ be an optimal internal ε_1-net of K, with $M = \mathcal{N}(K,d,\varepsilon_1)$. (*Then* $x_i \in K$ *and* $K \subset \bigcup_{i=1}^M B(x_i,\varepsilon_1)$.) Since

$$\varepsilon_1 \leqslant \varepsilon_2 \implies B(x,\varepsilon_1) \subset B(x,\varepsilon_2),$$

we have

$$K \subset \bigcup_{i=1}^M B(x_i,\varepsilon_1) \subset \bigcup_{i=1}^M B(x_i,\varepsilon_2),$$

so $\mathcal{N}_1$ is an internal ε_2-net of K, and therefore

$$\mathcal{N}(K,d,\varepsilon_2) \leqslant \operatorname{card}(\mathcal{N}_1) = \mathcal{N}(K,d,\varepsilon_1).$$

Prf. (1) Construct the following counterexample: take $(\mathbb{T},d)$ to be the standard metric space, $\varepsilon = 1$, $L = \{-1,1\}$, $K = \{-1,0,1\}$; then $\mathcal{N}(K,d,1) = 1$ while $\mathcal{N}(L,d,1) = 2$, so monotonicity with respect to the set fails.

(2) By Exercise Q4.25 (*External Covering Numbers and Their Equivalence*),

$$\overline{\mathcal{N}}(K,d,\varepsilon) \leqslant \mathcal{N}(K,d,\varepsilon) \leqslant \overline{\mathcal{N}}(K,d,\varepsilon/2); \tag{I}$$

by the absolute monotonicity of the external covering number with respect to the set,

$$\overline{\mathcal{N}}(L,d,\varepsilon) \leqslant \overline{\mathcal{N}}(K,d,\varepsilon). \tag{II}$$

Using (I),

$$\mathcal{N}(L,d,\varepsilon) \leqslant \overline{\mathcal{N}}(L,d,\varepsilon/2), \quad \overline{\mathcal{N}}(K,d,\varepsilon/2) \leqslant \mathcal{N}(K,d,\varepsilon/2);$$

using (II),

$$\overline{\mathcal{N}}(L,d,\varepsilon/2) \leqslant \overline{\mathcal{N}}(K,d,\varepsilon/2),$$

hence

$$\mathcal{N}(L,d,\varepsilon) \leqslant \overline{\mathcal{N}}(L,d,\varepsilon/2) \leqslant \overline{\mathcal{N}}(K,d,\varepsilon/2) \leqslant \mathcal{N}(K,d,\varepsilon/2).$$

27 [**Volume of the Euclidean Ball: Geometric Method**] This problem and the following ones compute the volume of $\mathbb{B}_2^n$ by three different methods (geometric, probabilistic, analytic).

First, we proceed from the geometric point of view.

(1) Standard simplex: define the standard simplex Δ_n in $\mathbb{R}^n$ as the set of points with nonnegative coordinates whose sum does not exceed 1:

$$\Delta_n = \left\{ \boldsymbol{x} \in \mathbb{R}^n : x_i \geqslant 0, \sum_{i=1}^n x_i \leqslant 1 \right\}.$$

Prove: $\operatorname{Vol}(\Delta_n) = 1/n!$.

(2) ℓ^1 unit ball: define the ℓ^1 unit ball in $\mathbb{R}^n$ as

$$\mathbb{B}_1^n = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_1 \leqslant 1\}.$$

Using the conclusion of (1), prove:

$$\operatorname{Vol}(\mathbb{B}_1^n) = \frac{2^n}{n!} \leqslant \left(\frac{2\mathrm{e}}{n}\right)^n.$$

(3) Sandwich bound: squeeze $\mathbb{B}_2^n = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_2 \leqslant 1\}$ between appropriately scaled cubes and ℓ^1 balls, and prove:

$$\left(\frac{2}{\sqrt{n}}\right)^n \leqslant \operatorname{Vol}(\mathbb{B}_2^n) \leqslant \left(\frac{2\mathrm{e}}{\sqrt{n}}\right)^n.$$

This shows that in high dimensions, the volume of the unit ball decreases exponentially in the dimension n.

Prf. (1) Use mathematical induction.

a. For $n=1$, $\Delta_1=\{x_1\in\mathbb{R}:x_1\geqslant 0,x_1\leqslant 1\}$, whose measure is clearly 1.

b. Assume the conclusion holds in dimension $n-1$. For dimension n, slice-integrate over x_n: on the slice $\sum_{i=1}^{n-1}x_i\leqslant 1-x_n$, the cross-section is an $(n-1)$-dimensional simplex with linear scaling factor $r=1-x_n$.

By the homogeneity of the Lebesgue measure,

$$\mathrm{Vol}_{n-1}(\Delta_{n-1}\cdot r)=r^{n-1}\,\mathrm{Vol}_{n-1}(\Delta_{n-1}),$$

hence

$$\begin{aligned}\mathrm{Vol}(\Delta_n)&=\int_0^1\mathrm{Vol}_{n-1}(\Delta_{n-1}(1-x_n))\mathrm{d}x_n=\int_0^1\frac{(1-x_n)^{n-1}}{(n-1)!}\mathrm{d}x_n\\&=\frac{1}{(n-1)!}\int_0^1u^{n-1}\mathrm{d}u=\frac{1}{(n-1)!}\cdot\frac{1}{n}=\frac{1}{n!}.\end{aligned}\qquad(u=1-x_n)$$

(2) $\mathbb{B}_1^n$ can be partitioned into 2^n congruent orthant pieces, each exactly the standard simplex Δ_n; hence

$$\mathrm{Vol}(\mathbb{B}_1^n)=2^n\cdot\mathrm{Vol}(\Delta_n)=\frac{2^n}{n!}\leqslant 2^n\left(\frac{\mathrm{e}}{n}\right)^n=\left(\frac{2\mathrm{e}}{n}\right)^n.$$

(3) For the lower bound, construct the hypercube C inscribed in $\mathbb{B}_2^n$:

$$\|\boldsymbol{x}\|_2^2=\sum_{i=1}^n x_i^2\leqslant\sum_{i=1}^n\frac{1}{n}=1,$$

so $C=[-1/\sqrt{n},1/\sqrt{n}]^n$, whose volume is

$$\mathrm{Vol}(\mathbb{B}_2^n)\geqslant\mathrm{Vol}(C)=\left(\frac{2}{\sqrt{n}}\right)^n.$$

For the upper bound, by the Cauchy inequality,

$$\|\boldsymbol{x}\|_1\leqslant\sqrt{n}\|\boldsymbol{x}\|_2\leqslant\sqrt{n},$$

hence $\mathbb{B}_2^n\subset\sqrt{n}\mathbb{B}_1^n$, and using the conclusion of (2),

$$\begin{aligned}\mathrm{Vol}(\mathbb{B}_2^n)&\leqslant\mathrm{Vol}(\sqrt{n}\mathbb{B}_1^n)=(\sqrt{n})^n\,\mathrm{Vol}(\mathbb{B}_1^n)\\&\leqslant n^{n/2}\cdot\left(\frac{2\mathrm{e}}{n}\right)^n=\frac{(2\mathrm{e})^n}{n^{n/2}}=\left(\frac{2\mathrm{e}}{\sqrt{n}}\right)^n.\end{aligned}$$

28 [**Volume of the Euclidean Ball: Probabilistic Method**] Let $\mathbb{B}_2^n$ be the unit Euclidean ball in $\mathbb{R}^n$. This problem aims to tighten the upper bound of Exercise Q4.27 (*Volume of the Euclidean Ball: Geometric Method*) using probabilistic tools. Using the small ball probability estimate (Q3.7 (*Small Ball Probabilities*)), prove:

$$\mathrm{Vol}(\mathbb{B}_2^n)\leqslant\left(\sqrt{\frac{2\pi\mathrm{e}}{n}}\right)^n,$$

and this bound is asymptotically tight as $n\to\infty$.

Prf. Let $\boldsymbol{g}\sim N(\boldsymbol{0},\boldsymbol{I}_n)$, with probability density function

$$f_{\boldsymbol{g}}(\boldsymbol{x})=\frac{1}{(2\pi)^{n/2}}\mathrm{e}^{-\|\boldsymbol{x}\|_2^2/2},$$

hence

$$\begin{aligned}\mathbb{P}\{\|\boldsymbol{g}\|_2\leqslant\varepsilon\}&=\int_{\varepsilon\mathbb{B}_2^n}\frac{1}{(2\pi)^{n/2}}\mathrm{e}^{-\|\boldsymbol{x}\|_2^2/2}\mathrm{d}\boldsymbol{x}\\&\geqslant\frac{\mathrm{e}^{-\varepsilon^2/2}}{(2\pi)^{n/2}}\int_{\varepsilon\mathbb{B}_2^n}\mathrm{d}\boldsymbol{x}=\frac{\mathrm{e}^{-\varepsilon^2/2}}{(2\pi)^{n/2}}\,\mathrm{Vol}(\varepsilon\mathbb{B}_2^n)\end{aligned}$$

$$= \frac{\mathrm{e}^{-\varepsilon^2/2}}{(2\pi)^{n/2}} \varepsilon^n \operatorname{Vol}(\mathbb{B}_2^n),$$

i.e.

$$\operatorname{Vol}(\mathbb{B}_2^n) \leqslant \frac{(2\pi)^{n/2}\mathrm{e}^{\varepsilon^2/2}}{\varepsilon^n} \mathbb{P}\{\|\boldsymbol{g}\|_2 \leqslant \varepsilon\}.$$

By Exercise Q3.7 (*Small Ball Probabilities*), for every $\varepsilon > 0$,

$$\mathbb{P}\{\|\boldsymbol{g}\|_2 \leqslant \varepsilon\} \leqslant \left(\frac{\varepsilon}{\sqrt{n}} \mathrm{e}^{\frac{n-\varepsilon^2}{2n}}\right)^n,$$

hence

$$\operatorname{Vol}(\mathbb{B}_2^n) \leqslant \frac{(2\pi)^{n/2}\mathrm{e}^{\varepsilon^2/2}}{\varepsilon^n} \cdot \left[\frac{\varepsilon}{\sqrt{n}} \exp\left(\frac{1}{2} - \frac{\varepsilon^2}{2n}\right)\right]^n = \left(\sqrt{\frac{2\pi\mathrm{e}}{n}}\right)^n.$$

29 [**Volume of the Euclidean Ball: Analytic Method**] This problem computes the volume of $\mathbb{B}_2^n$ exactly, using analytic methods.

(1) Radial integration identity: let $\|\cdot\|$ be a norm on $\mathbb{R}^n$, and let $B = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\| \leqslant 1\}$ be the unit ball of this norm. Let $f : [0,\infty) \to [0,\infty)$ be a monotone decreasing differentiable function with $\lim_{t\to\infty} f(t) = 0$ and $\lim_{t\to\infty} t^n f(t) = 0$, and assume that the integral below is finite. Prove:

$$\int_{\mathbb{R}^n} f(\|\boldsymbol{x}\|)\mathrm{d}\boldsymbol{x} = -\operatorname{Vol}(B) \int_0^\infty t^n f'(t)\mathrm{d}t.$$

(2) Exact volume formula: in Euclidean space, take $f(t) = \mathrm{e}^{-t^2/2}$; using the conclusion of (1), prove

$$\operatorname{Vol}(\mathbb{B}_2^n) = \frac{\pi^{n/2}}{\Gamma(n/2+1)}.$$

(3) Radius of the unit-volume ball: let R_n be the radius of the n-dimensional Euclidean ball of volume 1. Prove:

$$R_n = \sqrt{\frac{n}{2\pi\mathrm{e}}}(1 + o(1)).$$

This shows that R_n grows with the dimension, reflecting the rapid decay of the unit ball volume in high dimensions.

Prf. (1) Let $V(t) = \operatorname{Vol}(tB)$ be the volume of the ball of radius t; then $V(t) = t^n \operatorname{Vol}(B)$.

Integrating by parts,

$$\begin{aligned}
\int_{\mathbb{R}^n} f(\|\boldsymbol{x}\|)\mathrm{d}\boldsymbol{x} &= \int_0^\infty f(t)\mathrm{d}V(t) \\
&= [f(t)V(t)]_0^\infty - \int_0^\infty V(t)f'(t)\mathrm{d}t \\
&= -\int_0^\infty \operatorname{Vol}(B) t^n f'(t)\mathrm{d}t \qquad (V(0) = 0, \lim_{t\to\infty} t^n f(t) = 0) \\
&= -\operatorname{Vol}(B) \int_0^\infty t^n f'(t)\mathrm{d}t,
\end{aligned}$$

(2) Take $f(t) = \mathrm{e}^{-t^2/2}$; then

$$\int_{\mathbb{R}^n} \mathrm{e}^{-\|\boldsymbol{x}\|_2^2/2}\mathrm{d}\boldsymbol{x} = \prod_{j=1}^n \int_{-\infty}^\infty \mathrm{e}^{-x_j^2/2}\mathrm{d}x_j = (2\pi)^{n/2},$$

and using the conclusion of (1),

$$(2\pi)^{n/2} = -\operatorname{Vol}(\mathbb{B}_2^n) \int_0^\infty t^n(-t\mathrm{e}^{-t^2/2})\mathrm{d}t = \operatorname{Vol}(\mathbb{B}_2^n) \int_0^\infty t^{n+1}\mathrm{e}^{-t^2/2}\mathrm{d}t,$$

where

$$\int_0^\infty t^{n+1}\mathrm{e}^{-t^2/2}\mathrm{d}t = \int_0^\infty (2u)^{n/2}\mathrm{e}^{-u}\mathrm{d}u = 2^{n/2}\int_0^\infty u^{n/2}\mathrm{e}^{-u}\mathrm{d}u = 2^{n/2}\Gamma\Big(\frac{n}{2}+1\Big), \qquad \Big(u=\frac{t^2}{2}\Big)$$

hence

$$(2\pi)^{n/2} = \mathrm{Vol}(\mathbb{B}_2^n)\cdot 2^{n/2}\Gamma\Big(\frac{n}{2}+1\Big),$$

i.e.

$$\mathrm{Vol}(\mathbb{B}_2^n) = \frac{\pi^{n/2}}{\Gamma(n/2+1)}.$$

(3) It is easy to see that

$$\mathrm{Vol}(R_n\mathbb{B}_2^n) = R_n^n\,\mathrm{Vol}(\mathbb{B}_2^n) = 1 \implies R_n^n = \frac{1}{\mathrm{Vol}(\mathbb{B}_2^n)} = \frac{\Gamma(n/2+1)}{\pi^{n/2}},$$

and by Stirling's formula $\Gamma(k+1)\sim\sqrt{2\pi k}(k/\mathrm{e})^k$ with $k=n/2$,

$$\Gamma(n/2+1)\sim\sqrt{\pi n}\Big(\frac{n}{2\mathrm{e}}\Big)^{n/2},$$

hence

$$R_n^n \sim \frac{\sqrt{\pi n}(\frac{n}{2\mathrm{e}})^{n/2}}{\pi^{n/2}} = \sqrt{\pi n}\Big(\frac{n}{2\pi\mathrm{e}}\Big)^{n/2},$$

i.e.

$$R_n = \Big(\frac{n}{2\pi\mathrm{e}}\Big)^{1/2}(1+o(1)).$$

30 [**Volume Analysis of the ℓ^p Ball in $\mathbb{R}^n$**] Let $p\in[1,\infty]$, and define the unit ℓ^p ball in $\mathbb{R}^n$ as

$$\mathbb{B}_p^n = \{\boldsymbol{x}\in\mathbb{R}^n : \|\boldsymbol{x}\|_p \leqslant 1\},$$

where $\|\boldsymbol{x}\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ (for $p=\infty$, $\|\boldsymbol{x}\|_p = \max|x_i|$).

(1) Geometric sandwich bound: following the geometric argument of Exercise Q4.27 (*Volume of the Euclidean Ball: Geometric Method*), prove that the volume of $\mathbb{B}_p^n$ satisfies

$$\Big(\frac{2}{n^{1/p}}\Big)^n \leqslant \mathrm{Vol}(\mathbb{B}_p^n) \leqslant \Big(\frac{2\mathrm{e}}{n^{1/p}}\Big)^n.$$

(2) Exact analytic computation: following the analytic identity of Exercise Q4.29 (*Volume of the Euclidean Ball: Analytic Method*), prove the exact volume formula

$$\mathrm{Vol}(\mathbb{B}_p^n) = \frac{(2\Gamma(1/p+1))^n}{\Gamma(n/p+1)}.$$

Prf. (1) For the lower bound, construct the hypercube $C=[-1/n^{1/p},1/n^{1/p}]^n$ inscribed in $\mathbb{B}_p^n$; then for every $\boldsymbol{x}\in C$,

$$\|\boldsymbol{x}\|_p^p = \sum_{i=1}^n |x_i|^p \leqslant \sum_{i=1}^n \Big(\frac{1}{n^{1/p}}\Big)^p = \sum_{i=1}^n \frac{1}{n} = 1,$$

hence $C\subset\mathbb{B}_p^n$, and therefore

$$\mathrm{Vol}(\mathbb{B}_p^n) \geqslant \mathrm{Vol}(C) = \Big(\frac{2}{n^{1/p}}\Big)^n.$$

For the upper bound, by Hölder's inequality $\|\boldsymbol{x}\|_1 \leqslant n^{1-1/p}\|\boldsymbol{x}\|_p$, so $\mathbb{B}_p^n \subset n^{1-1/p}\mathbb{B}_1^n$, and therefore

$$\mathrm{Vol}(\mathbb{B}_p^n) \leqslant (n^{1-1/p})^n\,\mathrm{Vol}(\mathbb{B}_1^n).$$

Using the conclusion of Exercise Q4.27 (*Volume of the Euclidean Ball: Geometric Method*) (2),

$$\mathrm{Vol}(\mathbb{B}_p^n) \leqslant n^{n-n/p}\cdot\Big(\frac{2\mathrm{e}}{n}\Big)^n = \frac{n^n}{n^{n/p}}\cdot\frac{(2\mathrm{e})^n}{n^n} = \Big(\frac{2\mathrm{e}}{n^{1/p}}\Big)^n.$$

(2) Using the conclusion of Exercise Q4.29 (*Volume of the Euclidean Ball: Analytic Method*) (1),

$$\int_{\mathbb{R}^n} f(\|\boldsymbol{x}\|_p)\mathrm{d}\boldsymbol{x} = -\operatorname{Vol}(\mathbb{B}_p^n)\int_0^\infty t^n f'(t)\mathrm{d}t,$$

and taking $f(t)=\mathrm{e}^{-t^p}$,

$$\begin{aligned}\int_{\mathbb{R}^n} f(\|\boldsymbol{x}\|_p)\mathrm{d}\boldsymbol{x} &= \int_{\mathbb{R}^n}\mathrm{e}^{-\|\boldsymbol{x}\|_p^p}\mathrm{d}\boldsymbol{x} = \left(\int_{-\infty}^{\infty}\mathrm{e}^{-|x|^p}\mathrm{d}x\right)^n = \left(2\int_0^\infty \mathrm{e}^{-|x|^p}\mathrm{d}x\right)^n\\ &= \left(\frac{2}{p}\int_0^\infty u^{1/p-1}\mathrm{e}^{-u}\mathrm{d}u\right)^n = \left(\frac{2}{p}\Gamma\left(\frac{1}{p}\right)\right)^n = \left(2\Gamma\left(\frac{1}{p}+1\right)\right)^n, \qquad (u=x^p)\\ \int_0^\infty t^n f'(t)\mathrm{d}t &= -p\int_0^\infty t^{n+p-1}\mathrm{e}^{-t^p}\mathrm{d}t\\ &= -p\int_0^\infty t^{n+p-1}\mathrm{e}^{-t^p}\mathrm{d}t = -\int_0^\infty u^{n/p}\mathrm{e}^{-u}\mathrm{d}u = -\Gamma\left(\frac{n}{p}+1\right), \qquad (u=t^p)\end{aligned}$$

hence

$$\left(2\Gamma\left(\frac{1}{p}+1\right)\right)^n = \operatorname{Vol}(\mathbb{B}_p^n)\cdot\Gamma\left(\frac{n}{p}+1\right),$$

i.e.

$$\operatorname{Vol}(\mathbb{B}_p^n) = \frac{(2\Gamma(1/p+1))^n}{\Gamma(n/p+1)}.$$

(3) For $p=\infty$, $\mathbb{B}_\infty^n=[-1,1]^n$, so $\operatorname{Vol}(\mathbb{B}_\infty^n)=2^n$; this is also the limit of the displayed formula as $p\to\infty$.

31 [**Lattice-Based Construction of ε-Nets and Cardinality Estimates**] In metric spaces, the greedy iterative construction of ε-nets usually lacks structure and is difficult to store.

Consider instead a more algorithmically efficient construction: let

$$\mathcal{N} = \left(\frac{\varepsilon}{\sqrt{n}}\mathbb{Z}^n\right)\cap\mathbb{B}_2^n.$$

Prove:

(1) $\mathcal{N}$ is an ε-net of $\mathbb{B}_2^n$ in the Euclidean metric.

(2) The cardinality of this lattice net satisfies

$$|\mathcal{N}| \leqslant \mathrm{e}^n\left(\frac{2}{\varepsilon}+1\right)^n.$$

(3) Describe how to quickly map any vector of the unit ball to an approximate vector in $\mathcal{N}$.

Prf. (1) It suffices to prove that for every $\boldsymbol{x}\in\mathbb{B}_2^n$ there exists $\boldsymbol{u}\in\mathcal{N}$ with $\|\boldsymbol{x}-\boldsymbol{u}\|_2\leqslant\varepsilon$.

Let the lattice step be $h=\varepsilon/\sqrt{n}$ and take $\boldsymbol{x}\in\mathbb{B}_2^n$.

For each coordinate x_i, choose the lattice point u_i by rounding towards the origin:

$$u_i = \begin{cases} h\cdot\lfloor x_i/h\rfloor, & x_i\geqslant 0,\\ h\cdot\lceil x_i/h\rceil, & x_i<0\end{cases},$$

then $u_i\in h\mathbb{Z}$ and

$$\|\boldsymbol{u}\|_2^2 = \sum_{i=1}^n u_i^2 \leqslant \sum_{i=1}^n x_i^2 = \|\boldsymbol{x}\|_2^2 \implies \|\boldsymbol{u}\|_2\leqslant\|\boldsymbol{x}\|_2 = 1,$$

hence

$$\boldsymbol{u}\in\left(\frac{\varepsilon}{\sqrt{n}}\mathbb{Z}^n\right)\cap\mathbb{B}_2^n = \mathcal{N}.$$

Since $|x_i-u_i|\leqslant h$,

$$\|\boldsymbol{x}-\boldsymbol{u}\|_2^2 = \sum_{i=1}^n (x_i-u_i)^2 \leqslant \sum_{i=1}^n h^2 = nh^2 = \varepsilon^2,$$

i.e. $\|\boldsymbol{x}-\boldsymbol{u}\|_2 \leqslant \varepsilon$, so $\mathcal{N}$ is an ε-net of $\mathbb{B}_2^n$ in the Euclidean metric.

(2) For each lattice point $\boldsymbol{x} \in \mathcal{N}$, construct the hypercube $Q_{\boldsymbol{x}}$ centered at it with side length h; its volume is $h^n = (\varepsilon/\sqrt{n})^n$. Since $\boldsymbol{x} \in \mathbb{B}_2^n$, the maximum distance from any point of the hypercube to its center $\boldsymbol{x}$ is $\frac{h\sqrt{n}}{2} = \varepsilon/2$; hence all cubes are contained in the ball of radius $1+\varepsilon/2$:

$$\bigcup_{\boldsymbol{x}\in\mathcal{N}} Q_{\boldsymbol{x}} \subset \left(1+\frac{\varepsilon}{2}\right)\mathbb{B}_2^n,$$

and therefore

$$|\mathcal{N}| \cdot \left(\frac{\varepsilon}{\sqrt{n}}\right)^n \leqslant \mathrm{Vol}\left(\left(1+\frac{\varepsilon}{2}\right)\mathbb{B}_2^n\right) = \left(1+\frac{\varepsilon}{2}\right)^n \mathrm{Vol}(\mathbb{B}_2^n),$$

and by the upper bound $\mathrm{Vol}(\mathbb{B}_2^n) \leqslant (\sqrt{2\pi \mathrm{e}/n})^n$ of Exercise Q4.28 (*Volume of the Euclidean Ball: Probabilistic Method*),

$$\begin{aligned} |\mathcal{N}| &\leqslant \left(\frac{\sqrt{n}(1+\varepsilon/2)}{\varepsilon}\right)^n \left(\sqrt{\frac{2\pi\mathrm{e}}{n}}\right)^n = \left(\frac{\sqrt{2\pi\mathrm{e}}(1+\varepsilon/2)}{\varepsilon}\right)^n \\ &\leqslant (2\mathrm{e})^n \left(\frac{1+\varepsilon/2}{\varepsilon}\right)^n = \mathrm{e}^n \left(\frac{2}{\varepsilon}+1\right)^n. \end{aligned}$$

(3) Consider the following algorithm:

It is easy to see that the algorithm above runs in time $O(n)$.

Algorithm 3: Fast-Lattice-Quantization$(\boldsymbol{x}, \varepsilon)$

Input : $\boldsymbol{x} \in \mathbb{B}_2^n$, accuracy parameter $\varepsilon > 0$.

Output: A lattice vector

$$\boldsymbol{u} \in \mathcal{N} = \left(\frac{\varepsilon}{\sqrt{n}}\mathbb{Z}^n\right) \cap \mathbb{B}_2^n,$$

with $\|\boldsymbol{x}-\boldsymbol{u}\|_2 \leqslant \varepsilon$.

Part 1: Parameter setup and space scaling
Compute the lattice step: $h \leftarrow \frac{\varepsilon}{\sqrt{n}}$;
Map the input vector to the integer lattice frame: $\tilde{\boldsymbol{x}} \leftarrow \frac{\boldsymbol{x}}{h}$;
Part 2: Coordinate-wise Quantization
for $i = 1$ **to** n **do**
    // Round towards the origin to ensure $\|\boldsymbol{u}\|_2 \leqslant \|\boldsymbol{x}\|_2 \leqslant 1$
    if $\tilde{x}_i \geqslant 0$ **then**
        $\hat{x}_i \leftarrow \lfloor \tilde{x}_i \rfloor$;
    end
    else
        $\hat{x}_i \leftarrow \lceil \tilde{x}_i \rceil$;
    end
end
Part 3: Inverse mapping and lattice recovery
Construct the quantized lattice vector: $\boldsymbol{u} \leftarrow h \cdot (\hat{x}_1, \hat{x}_2, \cdots, \hat{x}_n)^\top$;
return $\boldsymbol{u}$;

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32 [**Covering and Packing Numbers of the Hamming Cube**] Let $K = \{0,1\}^n$ be the n-dimensional Hamming cube,

equipped with the Hamming metric $d_H(\boldsymbol{x},\boldsymbol{y}) = \sum_{i=1}^{n} \mathbf{1}_{\{x_i \neq y_i\}}$. For $m \in [\![0,n]\!]$, let $\mathcal{N}(K,d_H,m)$ be the m-covering number of K and $\mathcal{P}(K,d_H,m)$ its m-packing number.

By improving the volume method (*replacing the Lebesgue measure by the cardinality of sets*), prove the conclusions of Proposition P4.2.15 (*Covering and packing the Hamming cube*):

$$\frac{2^n}{\sum_{k=0}^{m}\binom{n}{k}} \leqslant \mathcal{N}(K,d_H,m) \leqslant \mathcal{P}(K,d_H,m) \leqslant \frac{2^n}{\sum_{k=0}^{\lfloor m/2\rfloor}\binom{n}{k}}.$$

Rk. In Hamming space, the closed ball centered at $\boldsymbol{x} \in K$ with integer radius r is

$$B(\boldsymbol{x},r) = \{\boldsymbol{y} \in K : d_H(\boldsymbol{x},\boldsymbol{y}) \leqslant r\},$$

whose cardinality is $|B(\boldsymbol{x},r)| = \sum_{k=0}^{r}\binom{n}{k}$, independent of the center. Here one uses cardinality rather than Euclidean volume.

Solution • Proving the left inequality

Let $\mathcal{C} = \{\boldsymbol{x}_1,\cdots,\boldsymbol{x}_N\}$ be a minimal m-covering of K, with $N = \mathcal{N}(K,d_H,m)$; then

$$K \subset \bigcup_{i=1}^{N} B(\boldsymbol{x}_i,m),$$

hence

$$|K| \leqslant \sum_{i=1}^{N} |B(\boldsymbol{x}_i,m)| = N \cdot \sum_{k=0}^{m}\binom{n}{k},$$

where $|K| = 2^n$; hence

$$N \geqslant \frac{2^n}{\sum_{k=0}^{m}\binom{n}{k}}.$$

• Proving the middle inequality

Let $\mathcal{S} = \{\boldsymbol{y}_1,\cdots,\boldsymbol{y}_P\}$ be a maximal m-packing of K; then $d_H(\boldsymbol{y}_i,\boldsymbol{y}_j) > m$.

Since the packing is maximal, for every $\boldsymbol{z} \in K \setminus \mathcal{S}$, the set $\mathcal{S} \cup \{\boldsymbol{z}\}$ is no longer an m-packing, so there exists $\boldsymbol{x} \in \mathcal{S}$ with $d_H(\boldsymbol{x},\boldsymbol{z}) \leqslant m$; hence $\mathcal{S}$ is an m-covering of K, satisfying

$$\mathcal{N}(K,d_H,m) \leqslant \operatorname{card}(\mathcal{S}) = \mathcal{P}(K,d_H,m).$$

• Let $\mathcal{S} = \{\boldsymbol{y}_1,\cdots,\boldsymbol{y}_P\}$ be an m-packing of K, with $P = \mathcal{P}(K,d_H,m)$.

Suppose there exists $\boldsymbol{z} \in B(\boldsymbol{y}_i,r) \cap B(\boldsymbol{y}_j,r)$ with $i \neq j$; then by the triangle inequality,

$$d_H(\boldsymbol{y}_i,\boldsymbol{y}_j) \leqslant d_H(\boldsymbol{y}_i,\boldsymbol{z}) + d_H(\boldsymbol{z},\boldsymbol{y}_j) \leqslant r + r = 2\lfloor m/2\rfloor \leqslant m,$$

contradicting the packing condition $d_H(\boldsymbol{y}_i,\boldsymbol{y}_j) > m$; hence $\{B(\boldsymbol{y}_i,\lfloor m/2\rfloor)\}_{i=1}^{P}$ are pairwise disjoint, and

$$\left|\bigcup_{i=1}^{P} B(\boldsymbol{y}_i,r)\right| = \sum_{i=1}^{P} |B(\boldsymbol{y}_i,r)| = P \cdot \sum_{k=0}^{r}\binom{n}{k} \leqslant 2^n,$$

which gives

$$P \leqslant \frac{2^n}{\sum_{k=0}^{\lfloor m/2\rfloor}\binom{n}{k}}.$$

33 [**Limitations of Error-Correcting Codes: Converse of Theorem T4.3.5**] Consider an error-correcting code that encodes k-bit information strings into n-bit codewords (i.e. code rate $R = k/n$). If $1 \leqslant r \leqslant n$ and this coding scheme can correct up to r errors, prove that its redundancy $n-k$ satisfies

$$n - k \geqslant r\log_2\left(\frac{n}{r}\right).$$

Prf. Let $K = \{0,1\}^n$ be the Hamming cube of all possible n-bit strings, and let the code be $\mathcal{C} \subset K$; then $|K| = 2^n$ and $|\mathcal{C}| = 2^k$.

If the code can correct r errors, then for any $\boldsymbol{c}_1, \boldsymbol{c}_2 \in \mathcal{C}$,

$$d_H(\boldsymbol{c}_1, \boldsymbol{c}_2) \geqslant 2r+1,$$

so the balls $B_H(\boldsymbol{c}, r) = \{\boldsymbol{x} \in \{0,1\}^n : d_H(\boldsymbol{c}, \boldsymbol{x}) \leqslant r\}$ are disjoint for distinct codewords.

Since

$$|B_H(\boldsymbol{c}, r)| = \sum_{i=0}^{r} \binom{n}{i} \geqslant \binom{n}{r},$$

by the Hamming bound,

$$\sum_{\boldsymbol{c}\in\mathcal{C}} |B_H(\boldsymbol{c}, r)| \leqslant |K|, \quad 2^k \cdot \binom{n}{r} \leqslant 2^n,$$

which gives

$$n - k \geqslant \log_2 \binom{n}{r} \geqslant \log_2 \left(\frac{n}{r}\right)^r = r \log_2 \left(\frac{n}{r}\right). \qquad \left(\binom{n}{r} \geqslant \left(\frac{n}{r}\right)^r\right)$$

34 [**Series Expansion over ε-Nets and Applications**] Let $\mathcal{N}$ be an ε-net of the unit sphere $\mathbb{S}^{n-1} = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_2 = 1\}$ (where $0 < \varepsilon < 1$).

(1) Series representation: prove that every $\boldsymbol{x} \in \mathbb{S}^{n-1}$ can be represented as

$$\boldsymbol{x} = \sum_{k=0}^{\infty} \lambda_k \boldsymbol{x}_k,$$

where $0 \leqslant \lambda_k \leqslant \varepsilon^k$ and $\boldsymbol{x}_k \in \mathcal{N}$.

(2) Operator norm bound: using the series expansion of (1), give an alternative proof of Lemma L4.4.1 (*Estimating the operator norm on a discrete net*).

Rk. Lemma L4.4.1 (*Estimating the operator norm on a discrete net*): let $\boldsymbol{A}$ be a symmetric matrix; then

$$\|\boldsymbol{A}\| \leqslant \frac{1}{1-2\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{x}\rangle|.$$

For a general matrix $\boldsymbol{A}$,

$$\|\boldsymbol{A}\| \leqslant \frac{1}{1-\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N}} \|\boldsymbol{A}\boldsymbol{x}\|_2.$$

Prf. Use the iterative approximation method.

1. For $k = 0$: since $\mathcal{N}$ is an ε-net of $\mathbb{S}^{n-1}$, for every $\boldsymbol{x} \in \mathbb{S}^{n-1}$ there exists $\boldsymbol{x}_0 \in \mathcal{N}$ with

$$\|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \leqslant \varepsilon.$$

Let the residual vector be $\boldsymbol{r}_1 = \boldsymbol{x} - \boldsymbol{x}_0$ and set $\lambda_0 = 1$ (*satisfying* $\lambda_0 \leqslant \varepsilon^0$).

2. Suppose the first m terms have been obtained, with $\boldsymbol{x} = \sum_{k=0}^{m-1} \lambda_k \boldsymbol{x}_k + \boldsymbol{r}_m$ and $\|\boldsymbol{r}_m\|_2 \leqslant \varepsilon^m$.

   a. If $\boldsymbol{r}_m = \boldsymbol{0}$, set all remaining terms to 0.

   b. If $\boldsymbol{r}_m \neq \boldsymbol{0}$, let $\bar{\boldsymbol{r}}_m \in \mathbb{S}^{n-1}$; then there exists $\boldsymbol{x}_m \in \mathcal{N}$ with $\|\bar{\boldsymbol{r}}_m - \boldsymbol{x}_m\|_2 \leqslant \varepsilon$.

   Set $\lambda_m = \|\boldsymbol{r}_m\|_2$, satisfying $\lambda_m \leqslant \varepsilon^m$, and update the residual

$$\boldsymbol{r}_{m+1} = \boldsymbol{r}_m - \lambda_m \boldsymbol{x}_m = \lambda_m(\bar{\boldsymbol{r}}_m - \boldsymbol{x}_m),$$

   with $\|\boldsymbol{r}_{m+1}\|_2 = \lambda_m \|\bar{\boldsymbol{r}}_m - \boldsymbol{x}_m\|_2 \leqslant \varepsilon^m \cdot \varepsilon = \varepsilon^{m+1}$.

   Repeating this process, $\|\boldsymbol{r}_m\|_2 \leqslant \varepsilon^m \to 0$, so the series converges, with

$$\boldsymbol{x} = \sum_{k=0}^{\infty} \lambda_k \boldsymbol{x}_k, \quad 0 \leqslant \lambda_k \leqslant \varepsilon^k.$$

3. Using the conclusion of (1), for every $\boldsymbol{x} \in \mathbb{S}^{n-1}$,

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}\left(\sum_{k=0}^{\infty} \lambda_k \boldsymbol{x}_k\right) = \sum_{k=0}^{\infty} \lambda_k \boldsymbol{A}\boldsymbol{x}_k,$$

and letting $M = \sup_{\boldsymbol{u}\in\mathcal{N}} \|\boldsymbol{A}\boldsymbol{u}\|_2$,

$$\|\boldsymbol{A}\boldsymbol{x}\|_2 \leqslant \sum_{k=0}^{\infty} \lambda_k \|\boldsymbol{A}\boldsymbol{x}_k\|_2 \leqslant \sum_{k=0}^{\infty} \varepsilon^k M = M \cdot \frac{1}{1-\varepsilon},$$

and taking the supremum on both sides,

$$\|\boldsymbol{A}\| \leqslant \frac{1}{1-\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N}} \|\boldsymbol{A}\boldsymbol{x}\|_2.$$

In this way, the complicated supremum problem is reduced to a finite search over the ε-net times a simple geometric series constant.

35 [**Computing Vector Norms on an ε-Net**] Let $\boldsymbol{x} \in \mathbb{R}^n$, and let $\mathcal{N}$ be an ε-net of $\mathbb{S}^{n-1}$ with $\varepsilon \in (0,1)$. Prove:

$$\sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{x}, \boldsymbol{y}\rangle \leqslant \|\boldsymbol{x}\|_2 \leqslant \frac{1}{1-\varepsilon} \sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{x}, \boldsymbol{y}\rangle.$$

Prf. Proving the lower bound.

It is easy to see that

$$\sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{x}, \boldsymbol{y}\rangle \leqslant \sup_{\boldsymbol{u}\in\mathbb{S}^{n-1}} \langle \boldsymbol{x}, \boldsymbol{u}\rangle = \|\boldsymbol{x}\|_2.$$

Proving the upper bound.

If $\boldsymbol{x} = \boldsymbol{0}$, the conclusion is trivial; we only need to consider $\boldsymbol{x} \neq \boldsymbol{0}$.

Let $\bar{\boldsymbol{x}} \in \mathbb{S}^{n-1}$; then $\|\boldsymbol{x}\|_2 = \langle \boldsymbol{x}, \bar{\boldsymbol{x}}\rangle$.

Since $\mathcal{N}$ is an ε-net of $\mathbb{S}^{n-1}$, there exists $\boldsymbol{y}_0 \in \mathcal{N}$ with $\|\bar{\boldsymbol{x}} - \boldsymbol{y}_0\|_2 \leqslant \varepsilon$.

Expanding the inner product expression of $\|\boldsymbol{x}\|_2$ at the net point $\boldsymbol{y}_0$,

$$\begin{aligned}\|\boldsymbol{x}\|_2 = \langle \boldsymbol{x}, \bar{\boldsymbol{x}}\rangle &= \langle \boldsymbol{x}, \boldsymbol{y}_0\rangle + \langle \boldsymbol{x}, \bar{\boldsymbol{x}} - \boldsymbol{y}_0\rangle \\ &\leqslant \langle \boldsymbol{x}, \boldsymbol{y}_0\rangle + \|\boldsymbol{x}\|_2 \|\bar{\boldsymbol{x}} - \boldsymbol{y}_0\|_2 \\ &\leqslant \langle \boldsymbol{x}, \boldsymbol{y}_0\rangle + \varepsilon \|\boldsymbol{x}\|_2,\end{aligned}$$

which gives

$$\|\boldsymbol{x}\|_2 \leqslant \frac{1}{1-\varepsilon} \sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{x}, \boldsymbol{y}\rangle.$$

Hence

$$\sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{x}, \boldsymbol{y}\rangle \leqslant \|\boldsymbol{x}\|_2 \leqslant \frac{1}{1-\varepsilon} \sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{x}, \boldsymbol{y}\rangle.$$

36 [**Maximizing Bilinear and Quadratic Forms over ε-Nets**] Prove Lemma L4.4.2 (*Estimating the quadratic form of a symmetric matrix on a discrete net*) by modifying the proof of Lemma L4.4.1 (*Estimating the operator norm on a discrete net*). Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, and let $\varepsilon \in [0, 1/2)$. Let $\mathcal{N}, \mathcal{M}$ be ε-nets of $\mathbb{S}^{n-1}, \mathbb{S}^{m-1}$ respectively.

(1) Prove:

$$\sup_{\boldsymbol{x}\in\mathcal{N},\boldsymbol{y}\in\mathcal{M}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle| \leqslant \|\boldsymbol{A}\| \leqslant \frac{1}{1-2\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N},\boldsymbol{y}\in\mathcal{M}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle|.$$

(2) In particular, if $m = n$ and $\boldsymbol{A}$ is symmetric, prove: when $\mathcal{N} = \mathcal{M}$, the operator norm can be controlled by the quadratic form,

$$\|\boldsymbol{A}\| \leqslant \frac{1}{1-2\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{x}\rangle|.$$

Prf. (1) • Proving the lower bound

Since $\mathcal{N} \subset \mathbb{S}^{n-1}$ and $\mathcal{M} \subset \mathbb{S}^{m-1}$, by the variational definition 4.1.8 of the operator norm, we obtain

$$\|\boldsymbol{A}\| = \sup_{\|\boldsymbol{u}\|_2=1, \|\boldsymbol{v}\|_2=1} |\langle \boldsymbol{A}\boldsymbol{u}, \boldsymbol{v}\rangle|.$$

Hence

$$\sup_{\boldsymbol{x}\in\mathcal{N}, \boldsymbol{y}\in\mathcal{M}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle| \leqslant \|\boldsymbol{A}\|.$$

• Proving the upper bound

By the definition 4.1.8 of the operator norm, there exist $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{S}^{n-1}$ such that $\|\boldsymbol{A}\| = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$.

By the definition of the ε-net, there exist $\boldsymbol{x}_0 \in \mathcal{N}$, $\boldsymbol{y}_0 \in \mathcal{M}$ with

$$\|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \leqslant \varepsilon, \quad \|\boldsymbol{y} - \boldsymbol{y}_0\|_2 \leqslant \varepsilon.$$

Expanding the bilinear form,

$$\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle = \langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{y}_0\rangle + \langle \boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0), \boldsymbol{y}\rangle + \langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{y} - \boldsymbol{y}_0\rangle,$$

and taking absolute values on both sides,

$$\begin{aligned} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle| &\leqslant |\langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{y}_0\rangle| + \|\boldsymbol{A}\| \cdot \|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \cdot \|\boldsymbol{y}\|_2 + \|\boldsymbol{A}\| \cdot \|\boldsymbol{x}_0\|_2 \cdot \|\boldsymbol{y} - \boldsymbol{y}_0\|_2 \\ &\leqslant \sup_{\boldsymbol{x}_0\in\mathcal{N}, \boldsymbol{y}_0\in\mathcal{M}} |\langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{y}_0\rangle| + 2\varepsilon\|\boldsymbol{A}\|, \end{aligned}$$

i.e.

$$\|\boldsymbol{A}\| \leqslant \frac{1}{1-2\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N}, \boldsymbol{y}\in\mathcal{M}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle|.$$

(2) If $\boldsymbol{A}$ is symmetric, then $\|\boldsymbol{A}\| = \sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{x}\rangle|$.

Let $\boldsymbol{x} \in \mathbb{S}^{n-1}$ be such that $\|\boldsymbol{A}\| = |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{x}\rangle|$, and let $\boldsymbol{x}_0 \in \mathcal{N}$ satisfy $\|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \leqslant \varepsilon$.

Consider the deviation of the quadratic form:

$$|\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{x}\rangle - \langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{x}_0\rangle| = |\langle \boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0), \boldsymbol{x}\rangle + \langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{x} - \boldsymbol{x}_0\rangle|,$$

hence

$$\begin{aligned} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{x}\rangle| &\leqslant |\langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{x}_0\rangle| + \|\boldsymbol{A}\| \cdot \|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \cdot \|\boldsymbol{x}\|_2 + \|\boldsymbol{A}\| \cdot \|\boldsymbol{x}_0\|_2 \cdot \|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \\ &\leqslant \sup_{\boldsymbol{x}_0\in\mathcal{N}} |\langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{x}_0\rangle| + 2\varepsilon\|\boldsymbol{A}\|, \end{aligned}$$

i.e.

$$\|\boldsymbol{A}\| \leqslant \frac{1}{1-2\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{x}\rangle|.$$

37 [**Controlling Norm Deviation on an ε-Net**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, and let $\mathcal{N}$ be an ε-net of $\mathbb{S}^{n-1}$ with $\varepsilon \in [0, 1/2)$. For $\mu \in \mathbb{R}$, define the global deviation and the maximal deviation over the ε-net respectively by

$$E = \sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu|, \quad \delta = \sup_{\boldsymbol{x}\in\mathcal{N}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu|.$$

Prove: the global norm deviation over $\mathbb{S}^{n-1}$ can be controlled by the local deviation over $\mathcal{N}$, i.e.

$$E \leqslant \frac{C}{1-2\varepsilon}\delta. \qquad (C > 0)$$

Rk. The following is an erroneous attempt:

By the definition of the ε-net, for every $\boldsymbol{x} \in \mathbb{S}^{n-1}$ there exists $\boldsymbol{x}_0 \in \mathcal{N}$ with $\|\boldsymbol{x} - \boldsymbol{x}_0\|_2 \leqslant \varepsilon$; hence

$$\begin{aligned} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu| &\leqslant |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \|\boldsymbol{A}\boldsymbol{x}_0\|_2| + |\|\boldsymbol{A}\boldsymbol{x}_0\|_2 - \mu| \\ &\leqslant \|\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0)\|_2 + (\|\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0)\|_2 + \delta). \end{aligned}$$

Let $\boldsymbol{h} = \boldsymbol{x} - \boldsymbol{x}_0$, so that $\|\boldsymbol{h}\|_2 \leqslant \varepsilon$.

If $\boldsymbol{h} = \boldsymbol{0}$, the conclusion is trivial; we only need to consider $\boldsymbol{h} \neq \boldsymbol{0}$.

Let $\boldsymbol{u} = \bar{\boldsymbol{h}} \in \mathbb{S}^{n-1}$; then

$$\|\boldsymbol{A}\boldsymbol{h}\|_2 = \|\boldsymbol{h}\|_2 \cdot \|\boldsymbol{A}\boldsymbol{u}\|_2 \leqslant \varepsilon \|\boldsymbol{A}\boldsymbol{u}\|_2,$$

where

$$\|\boldsymbol{A}\boldsymbol{u}\|_2 \leqslant |\|\boldsymbol{A}\boldsymbol{u}\|_2 - \mu| + |\mu| \leqslant E + |\mu|,$$

hence

$$|\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu| \leqslant \varepsilon(E + |\mu|) + \delta,$$

and taking the supremum on both sides,

$$E \leqslant \varepsilon E + \varepsilon|\mu| + \delta \implies E \leqslant \frac{\varepsilon|\mu| + \delta}{1 - \varepsilon}.$$

One can see that bounding the absolute values directly inevitably introduces a cross term involving μ, leaving the proof stuck.

Prf. First consider $\mu \leqslant 0$. Let $M = \|\boldsymbol{A}\|$ and $a_0 = \sup_{\boldsymbol{y}\in\mathcal{N}} \|\boldsymbol{A}\boldsymbol{y}\|_2$. By Exercise Q4.35 (*Computing Vector Norms on an ε-Net*), $M \leqslant a_0/(1-\varepsilon)$; since $-\mu \geqslant 0$,

$$E = M - \mu \leqslant \frac{a_0 - \mu}{1 - \varepsilon} \leqslant \frac{\delta}{1 - \varepsilon}.$$

Now assume $\mu \geqslant 0$. Construct the symmetric matrix $\boldsymbol{M} = \boldsymbol{A}^\top\boldsymbol{A} - \mu^2\boldsymbol{I}_n$; then

$$|\boldsymbol{x}^\top\boldsymbol{M}\boldsymbol{x}| = |\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - \mu^2|.$$

By Lemma L4.4.2 (*Estimating the quadratic form of a symmetric matrix on a discrete net*) (maximizing the quadratic form of a symmetric matrix over an ε-net),

$$\sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - \mu^2| \leqslant \frac{1}{1 - 2\varepsilon} \sup_{\boldsymbol{y}\in\mathcal{N}} |\|\boldsymbol{A}\boldsymbol{y}\|_2^2 - \mu^2|.$$

For the left-hand side, note that

$$|\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - \mu^2| = |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu| \cdot (\|\boldsymbol{A}\boldsymbol{x}\|_2 + \mu),$$

and since $\mathbb{S}^{n-1}$ is compact, there exists $\boldsymbol{x}^* \in \mathbb{S}^{n-1}$ with $E = |\|\boldsymbol{A}\boldsymbol{x}^*\|_2 - \mu|$; hence

$$\sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - \mu^2| \geqslant |\|\boldsymbol{A}\boldsymbol{x}^*\|_2^2 - \mu^2| = E \cdot (\|\boldsymbol{A}\boldsymbol{x}^*\|_2 + \mu).$$

For the right-hand side,

$$\begin{aligned}
\sup_{\boldsymbol{y}\in\mathcal{N}} |\|\boldsymbol{A}\boldsymbol{y}\|_2^2 - \mu^2| &= \sup_{\boldsymbol{y}\in\mathcal{N}} [|\|\boldsymbol{A}\boldsymbol{y}\|_2 - \mu| \cdot (\|\boldsymbol{A}\boldsymbol{y}\|_2 + \mu)] \\
&= \left(\sup_{\boldsymbol{y}\in\mathcal{N}} |\|\boldsymbol{A}\boldsymbol{y}\|_2 - \mu|\right) \cdot \left(\sup_{\boldsymbol{y}\in\mathcal{N}} (\|\boldsymbol{A}\boldsymbol{y}\|_2 + \mu)\right) \\
&\leqslant \delta \cdot (\|\boldsymbol{A}\| + \mu).
\end{aligned}$$

Hence

$$E \cdot (\|\boldsymbol{A}\boldsymbol{x}^*\|_2 + \mu) \leqslant \frac{1}{1 - 2\varepsilon}\delta \cdot (\|\boldsymbol{A}\| + \mu),$$

i.e.

$$E \leqslant \frac{1}{1 - 2\varepsilon}\delta \underbrace{\frac{\|\boldsymbol{A}\| + \mu}{\|\boldsymbol{A}\boldsymbol{x}^*\|_2 + \mu}}_{R}.$$

Define $g(\boldsymbol{x}) = \|\boldsymbol{A}\boldsymbol{x}\|_2$; its maximum over $\mathbb{S}^{n-1}$ is $M = \|\boldsymbol{A}\|$ and its minimum is $m \geqslant 0$. Since $E = \max_{\boldsymbol{x}} |g(\boldsymbol{x}) - \mu|$ can only be attained at an extreme point, i.e. $\|\boldsymbol{A}\boldsymbol{x}^*\|_2 = M$ or m, we consider the following cases:

a. If $\|\boldsymbol{A}\boldsymbol{x}^*\|_2 = M = \|\boldsymbol{A}\|$, then

$$R = \frac{\|\boldsymbol{A}\| + \mu}{\|\boldsymbol{A}\| + \mu} = 1, \text{and hence } E \leqslant \frac{1}{1 - 2\varepsilon}\delta.$$

b. If $\|\boldsymbol{A}\boldsymbol{x}^*\|_2 = m$, then the lower deviation exceeds the upper one, i.e.

$$\mu - m \geqslant M - \mu, \text{and hence } 2\mu \geqslant M + m \geqslant M, \text{so } \mu \geqslant \frac{1}{2}M = \frac{1}{2}\|\boldsymbol{A}\|,$$

and then

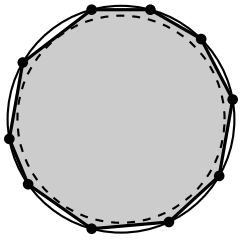
$$R = \frac{M+\mu}{m+\mu} \leqslant \frac{\mu/2+\mu}{0+\mu} = 3, \text{hence } E \leqslant \frac{3}{1-2\varepsilon}\delta.$$

In conclusion, taking $C = 3$,

$$\sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu| \leqslant \frac{3}{1-2\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mu|.$$

38 [**Convex Hull Containment Property of ε-Nets**] Let $\mathcal{N}$ be an ε-net of $\mathbb{S}^{n-1}$ with $\varepsilon \in (0,1)$. Prove: the convex hull of $\mathcal{N}$ must contain the concentric closed ball of radius $1-\varepsilon$, i.e.

$$(1-\varepsilon)\mathbb{B}_2^n \subset \operatorname{conv}(\mathcal{N}).$$



Prf. Argue by contradiction: suppose the conclusion fails, i.e. there exists $\boldsymbol{x} \in (1-\varepsilon)\mathbb{B}_2^n$ with $\boldsymbol{x} \notin \operatorname{conv}(\mathcal{N})$.

Since $\mathbb{S}^{n-1}$ is compact, $\mathcal{N}$ can be taken to be finite, in which case $\operatorname{conv}(\mathcal{N})$ is a compact convex set.

By the strict convex separation theorem [*a point and a closed convex set not containing it can be strictly separated by a hyperplane*], there exists $\boldsymbol{\theta} \in \mathbb{S}^{n-1}$ such that

$$\sup_{\boldsymbol{y}\in\operatorname{conv}(\mathcal{N})} \langle \boldsymbol{y}, \boldsymbol{\theta}\rangle < \langle \boldsymbol{x}, \boldsymbol{\theta}\rangle.$$

Since the supremum of a linear functional over a convex hull equals its supremum over the generating set,

$$\sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{y}, \boldsymbol{\theta}\rangle < \langle \boldsymbol{x}, \boldsymbol{\theta}\rangle,$$

where

$$\langle \boldsymbol{x}, \boldsymbol{\theta}\rangle \leqslant \|\boldsymbol{x}\|_2\|\boldsymbol{\theta}\|_2 \leqslant (1-\varepsilon)\cdot 1 = 1-\varepsilon,$$

hence

$$\sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{y}, \boldsymbol{\theta}\rangle < 1-\varepsilon.$$

By the definition of the ε-net, there exists $\boldsymbol{y}_0 \in \mathcal{N}$ with $\|\boldsymbol{\theta} - \boldsymbol{y}_0\|_2 \leqslant \varepsilon$; hence

$$\begin{aligned}\langle \boldsymbol{y}_0, \boldsymbol{\theta}\rangle &= \langle \boldsymbol{\theta} - (\boldsymbol{\theta} - \boldsymbol{y}_0), \boldsymbol{\theta}\rangle \\ &= \|\boldsymbol{\theta}\|_2^2 - \langle \boldsymbol{\theta} - \boldsymbol{y}_0, \boldsymbol{\theta}\rangle = 1 - \langle \boldsymbol{\theta} - \boldsymbol{y}_0, \boldsymbol{\theta}\rangle \\ &\geqslant 1 - \|\boldsymbol{\theta} - \boldsymbol{y}_0\|_2\|\boldsymbol{\theta}\|_2 \geqslant (1-\varepsilon)\cdot 1 = 1-\varepsilon,\end{aligned}$$

i.e.

$$\sup_{\boldsymbol{y}\in\mathcal{N}} \langle \boldsymbol{y}, \boldsymbol{\theta}\rangle \geqslant 1-\varepsilon,$$

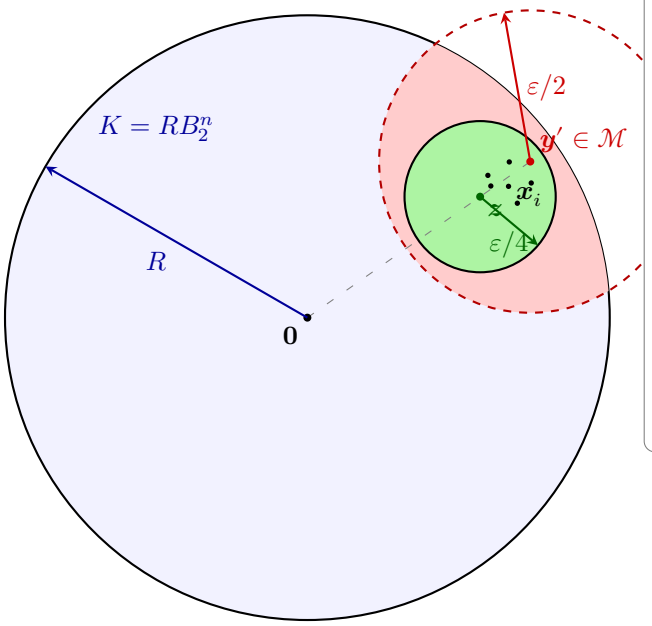
a contradiction; hence $(1-\varepsilon)\mathbb{B}_2^n \subset \operatorname{conv}(\mathcal{N})$.

39 [**Random Point Sets Form ε-Nets**] Let $\boldsymbol{g}_1, \cdots, \boldsymbol{g}_N$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. For any radius $R > 0$ and accuracy $\varepsilon > 0$, prove:

when $N \geqslant \mathrm{e}^{C_{R,\varepsilon} n}$, the random set

$$\mathcal{N}_X = \left\{ \frac{\boldsymbol{g}_1}{\sqrt{n}}, \cdots, \frac{\boldsymbol{g}_N}{\sqrt{n}} \right\} \cap R\mathbb{B}_2^n$$

with high probability forms an ε-net of the centered ball $K = R\mathbb{B}_2^n$, where $c > 0$ (w.p. $1 - \mathrm{e}^{-cn}$).



Geometric logic of the constructive proof:

1. The light blue large ball is the target space $K = R\mathbb{B}_2^n$.
2. $\boldsymbol{y}'$ is a point of the predetermined reference $\varepsilon/2$-net $\mathcal{M}$.
3. The red dashed circle is $B(\boldsymbol{y}', \varepsilon/2)$; the red shaded region is the intersection $V_{\boldsymbol{y}'} = B(\boldsymbol{y}', \varepsilon/2) \cap K$.
4. To compute the probability lower bound, we look inside $V_{\boldsymbol{y}'}$ for the green small ball $B(\boldsymbol{z}, \varepsilon/4)$ fully contained in it.
5. Shrinking $\boldsymbol{y}'$ towards the origin by distance $\frac{\varepsilon}{4R}\|\boldsymbol{y}'\|_2$ yields the center $\boldsymbol{z}$, which guarantees the green small ball does not cross the boundary of the blue large ball or the red dashed boundary.
6. If a random sample point $\boldsymbol{x}_i$ falls in the green small ball, it certainly falls in the red intersection region, thus successfully covering $\boldsymbol{y}'$.

Prf. If $\varepsilon > 2R$, then the origin itself is a trivial ε-net, so the conclusion is trivial; we only need to consider $\varepsilon \leqslant 2R$.

- Building a discrete reference net

  By Corollary C4.2.11 (*Covering the Euclidean ball*), there exists an $(\varepsilon/2)$-net $\mathcal{M}$ of $K = R\mathbb{B}_2^n$ with

  $$|\mathcal{M}| \leqslant \left(1 + \frac{2R}{\varepsilon/2}\right)^n = \left(1 + \frac{4R}{\varepsilon}\right)^n.$$

  If $\mathcal{N}_X$ is an ε-net of K, then for every $\boldsymbol{y}' \in \mathcal{M}$,

  $$\mathcal{N}_X \cap B(\boldsymbol{y}', \varepsilon/2) \neq \emptyset.$$

  Indeed, for every $\boldsymbol{y} \in K$, there exists $\boldsymbol{y}' \in \mathcal{M}$ with $\|\boldsymbol{y} - \boldsymbol{y}'\|_2 \leqslant \varepsilon/2$, and $\boldsymbol{x}_i \in \mathcal{N}_X$ with $\|\boldsymbol{x}_i - \boldsymbol{y}'\|_2 \leqslant \varepsilon/2$; hence

  $$\|\boldsymbol{x}_i - \boldsymbol{y}\|_2 \leqslant \|\boldsymbol{x}_i - \boldsymbol{y}'\|_2 + \|\boldsymbol{y} - \boldsymbol{y}'\|_2 \leqslant \varepsilon.$$

- Constructing an inner small ball to lower-bound the probability

  Let $\boldsymbol{y}' \in \mathcal{M}$ and $V_{\boldsymbol{y}'} = B(\boldsymbol{y}', \varepsilon/2) \cap K$.

  Consider $B(\boldsymbol{z}, \varepsilon/4)$, where

  $$\boldsymbol{z} = \left(1 - \frac{\varepsilon}{4R}\right)\boldsymbol{y}';$$

  then for every $\boldsymbol{x} \in B(\boldsymbol{z}, \varepsilon/4)$,

  $$\|\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{z}\|_2 + \frac{\varepsilon}{4} = \left(1 - \frac{\varepsilon}{4R}\right)\|\boldsymbol{y}'\|_2 + \frac{\varepsilon}{4} \leqslant \left(1 - \frac{\varepsilon}{4R}\right)R + \frac{\varepsilon}{4} = R \implies \boldsymbol{x} \in K,$$

  $$\|\boldsymbol{x} - \boldsymbol{y}'\|_2 \leqslant \|\boldsymbol{x} - \boldsymbol{z}\|_2 + \|\boldsymbol{z} - \boldsymbol{y}'\|_2 \leqslant \frac{\varepsilon}{4} + \frac{\varepsilon}{4R}\|\boldsymbol{y}'\|_2 \leqslant \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2} \implies \boldsymbol{x} \in B\left(\boldsymbol{y}', \frac{\varepsilon}{2}\right),$$

  hence $B(\boldsymbol{z}, \varepsilon/4) \subset V_{\boldsymbol{y}'}$.

- Lower-bounding the probability that a single point falls in the small ball

  Let $\boldsymbol{x}_1 = \boldsymbol{g}_1/\sqrt{n}$ be a single random sample, and write $p_0 = \mathbb{P}\{\boldsymbol{x}_1 \in V_{\boldsymbol{y}'}\}$; then

  $$p_0 = \mathbb{P}\{\boldsymbol{x}_1 \in V_{\boldsymbol{y}'}\} \geqslant \mathbb{P}\{\boldsymbol{x}_1 \in B(\boldsymbol{z}, \varepsilon/4)\}.$$

  Since $\boldsymbol{x} \in K$, we have $\|\boldsymbol{x}\|_2 \leqslant R$, hence

  $$f(\boldsymbol{x}) = \left(\frac{n}{2\pi}\right)^{n/2} \mathrm{e}^{-n\|\boldsymbol{x}\|_2^2/2} \geqslant \left(\frac{n}{2\pi}\right)^{n/2} \mathrm{e}^{-nR^2/2}.$$

  Using the conclusion of Exercise Q4.27 (*Volume of the Euclidean Ball: Geometric Method*) (3) ($\mathrm{Vol}(\mathbb{B}_2^n) \geqslant (2/\sqrt{n})^n$),

  $$\mathrm{Vol}\left(B\left(\boldsymbol{z}, \frac{\varepsilon}{4}\right)\right) = \left(\frac{\varepsilon}{4}\right)^n \mathrm{Vol}(\mathbb{B}_2^n) \geqslant \left(\frac{\varepsilon}{4}\right)^n \left(\frac{2}{\sqrt{n}}\right)^n = \left(\frac{\varepsilon}{2\sqrt{n}}\right)^n,$$

  hence

  $$p_0 \geqslant \left(\frac{\varepsilon}{2\sqrt{n}}\right)^n \left(\frac{n}{2\pi}\right)^{n/2} \mathrm{e}^{-nR^2/2} = \left(\frac{\varepsilon}{2\sqrt{2\pi}}\right)^n \mathrm{e}^{-nR^2/2} = \mathrm{e}^{-c_1 n}. \qquad \left(c_1 = \frac{R^2}{2} + \log\left(\frac{2\sqrt{2\pi}}{\varepsilon}\right)\right)$$

- Applying the union bound to obtain the global bound

  For a fixed $\boldsymbol{y}' \in \mathcal{M}$, the probability that none of the N points falls in its $\varepsilon/2$-neighborhood is $(1-p_0)^N \leqslant \mathrm{e}^{-p_0 N}$.

  Applying the union bound over all points of $\mathcal{M}$, the probability of failure of the net construction is

  $$\mathbb{P}(\text{Fail}) \leqslant |\mathcal{M}|\mathrm{e}^{-p_0 N} \leqslant \left(1 + \frac{4R}{\varepsilon}\right)^n \exp(-N\mathrm{e}^{-c_1 n}).$$

  Requiring $\mathbb{P}(\text{Fail}) \leqslant \mathrm{e}^{-cn}$ with $c = 1$, we get

  $$\exp\left(n \log\left(1 + \frac{4R}{\varepsilon}\right) - N\mathrm{e}^{-c_1 n}\right) \leqslant \mathrm{e}^{-n},$$

  which gives

  $$N \geqslant \mathrm{e}^{c_1 n} \cdot n\left(\log\left(1 + \frac{4R}{\varepsilon}\right) + 1\right),$$

  i.e.

  $$N \geqslant \mathrm{e}^{c_1 n} \cdot \mathrm{e}^{n}\left(\log\left(1 + \frac{4R}{\varepsilon}\right) + 1\right).$$

  Thus, defining

  $$C_{R,\varepsilon} = c_1 + 1 + \log\left(\log\left(1 + \frac{4R}{\varepsilon}\right) + 1\right),$$

  whenever $N \geqslant \mathrm{e}^{C_{R,\varepsilon} n}$ (w.p. $1 - \mathrm{e}^{-n}$), the random set forms an ε-net of K.

40 [**Too Many Random Points Are Not in Convex Position**] In Exercise Q3.23 (*High-Dimensional Random Points Are in Convex Position*), we know that if $N \leqslant \mathrm{e}^{cn}$, then $\boldsymbol{g}_1, \cdots, \boldsymbol{g}_N$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ are in convex position (w.h.p.).

Prove the converse: there exist $C, c > 0$ such that when $N \geqslant \mathrm{e}^{Cn}$, the points $\boldsymbol{g}_1, \cdots, \boldsymbol{g}_N$ are with high probability not in convex position (w.p. $1 - \mathrm{e}^{-cn}$).

Prf. By Exercise Q4.38 (*Convex Hull Containment Property of ε-Nets*),

$$(R - \varepsilon)\mathbb{B}_2^n \subset \mathrm{conv}(\mathcal{N}).$$

Take $R = 2$ and $\varepsilon = 0.5$; split the point set into the target point $\boldsymbol{g}_N$ and the remaining set $\mathcal{P} = \{\boldsymbol{g}_1, \cdots, \boldsymbol{g}_{N-1}\}$, then rescale the space by $1/\sqrt{n}$, write $\tilde{\boldsymbol{g}}_i = \boldsymbol{g}_i/\sqrt{n}$, and let the rescaled set be

$$\tilde{\mathcal{P}} = \left\{\frac{\boldsymbol{g}_1}{\sqrt{n}}, \cdots, \frac{\boldsymbol{g}_{N-1}}{\sqrt{n}}\right\}.$$

For $\tilde{\mathcal{P}}$, by Exercise Q4.39 (*Random Point Sets Form ε-Nets*), let

$$\mathcal{E}_1 = \{\tilde{\mathcal{P}} \cap 2\mathbb{B}_2^n \text{ forms a } 0.5\text{-net of } 2\mathbb{B}_2^n\},$$

so there exists $C_0 = C_{2,0.5}$ such that when $N - 1 \geqslant \mathrm{e}^{C_0 n}$, we have $\mathbb{P}(\mathcal{E}_1) \geqslant 1 - \mathrm{e}^{-c_1 n}$ for some $c_1 > 0$.

When $\mathcal{E}_1$ occurs,

$$(2-0.5)\mathbb{B}_2^n \subset \operatorname{conv}(\tilde{\mathcal{P}} \cap 2\mathbb{B}_2^n) \subset \operatorname{conv}(\tilde{\mathcal{P}}),$$

i.e. $1.5\mathbb{B}_2^n \subset \operatorname{conv}(\tilde{\boldsymbol{g}}_1, \cdots, \tilde{\boldsymbol{g}}_{N-1})$.

For $\boldsymbol{g}_N$, let

$$\mathcal{E}_2 = \{\|\boldsymbol{g}_N\|_2 \leqslant 1.5\sqrt{n}\},$$

and by Theorem T3.1.1 (*Concentration of the norm*) [*concentration inequality for the norm of a Gaussian vector*], there exists $c_2 > 0$ such that

$$\mathbb{P}\{|\|\boldsymbol{g}_N\|_2 - \sqrt{n}| \geqslant 0.5\sqrt{n}\} \leqslant 2\mathrm{e}^{-c_2 n},$$

i.e. $\mathbb{P}(\mathcal{E}_2) \geqslant 1 - 2\mathrm{e}^{-c_2 n}$.

When $\mathcal{E}_2$ occurs,

$$\tilde{\boldsymbol{g}}_N = \frac{\boldsymbol{g}_N}{\sqrt{n}} \in 1.5\mathbb{B}_2^n.$$

Hence, with $C = C_0 + 1$, when $N \geqslant \mathrm{e}^{Cn}$,

$$\mathbb{P}\{\mathcal{E}_1 \cap \mathcal{E}_2\} = 1 - \mathbb{P}\{\mathcal{E}_1^c \cup \mathcal{E}_2^c\} \geqslant 1 - \mathbb{P}(\mathcal{E}_1^c) - \mathbb{P}(\mathcal{E}_2^c) \geqslant 1 - \mathrm{e}^{-c_1 n} - 2\mathrm{e}^{-c_2 n},$$

and taking $c = \min\{c_1, c_2\}/2$, we get $\mathbb{P}\{\mathcal{E}_1 \cap \mathcal{E}_2\} \geqslant 1 - \mathrm{e}^{-cn}$.

When $\mathcal{E}_1 \cap \mathcal{E}_2$ occurs, $\tilde{\boldsymbol{g}}_N \in 1.5\mathbb{B}_2^n$ and $1.5\mathbb{B}_2^n \subset \operatorname{conv}(\tilde{\boldsymbol{g}}_1, \cdots, \tilde{\boldsymbol{g}}_{N-1})$; hence

$$\tilde{\boldsymbol{g}}_N \in \operatorname{conv}(\tilde{\boldsymbol{g}}_1, \cdots, \tilde{\boldsymbol{g}}_{N-1}),$$

i.e.

$$\boldsymbol{g}_N \in \operatorname{conv}(\boldsymbol{g}_1, \cdots, \boldsymbol{g}_{N-1}).$$

41 [**Expected Operator Norm and Singular Value Bounds for Random Matrices**] Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, where $m \geqslant n$, have rows $\boldsymbol{A}_i$ that are independent zero-mean sub-Gaussian random vectors with $\max_i \|\boldsymbol{A}_i\|_{\psi_2} \leqslant K$. Assume $\boldsymbol{A}$ satisfies the unit covariance normalization condition

$$\frac{1}{m}\mathbb{E}[\boldsymbol{A}^\top \boldsymbol{A}] = \boldsymbol{I}_n.$$

(1) Using Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*), prove:

$$\mathbb{E}\|\boldsymbol{A}\| \leqslant CK(\sqrt{m} + \sqrt{n}).$$

(2) Using Theorem T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*), prove:

$$\mathbb{E}\left\|\frac{1}{m}\boldsymbol{A}^\top \boldsymbol{A} - \boldsymbol{I}_n\right\| \leqslant CK^2\left(\sqrt{\frac{n}{m}} + \frac{n}{m}\right).$$

Then derive the expected bounds for the singular values:

$$\sqrt{m} - CK^2\sqrt{n} \leqslant \mathbb{E}[s_n(\boldsymbol{A})] \leqslant \mathbb{E}[s_1(\boldsymbol{A})] \leqslant \sqrt{m} + CK^2\sqrt{n}. \qquad (C > 0)$$

Prf. (1) By Theorem T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*),

$$\begin{aligned}\mathbb{E}\|\boldsymbol{A}\| &\leqslant C\left(\sqrt{\sum_{i=1}^m \|\boldsymbol{A}_i\|_{\psi_2}^2} + \max_i \|\boldsymbol{A}_i\|_{\psi_2} \cdot \sqrt{n}\right) \\ &\leqslant C(\sqrt{mK^2} + K\sqrt{n}) = CK(\sqrt{m} + \sqrt{n}).\end{aligned}$$

(2) Let the random covariance matrix be

$$\boldsymbol{S} = \frac{1}{m}\boldsymbol{A}^\top \boldsymbol{A},$$

with $\mathbb{E}[\boldsymbol{S}] = \boldsymbol{I}_n$; then by Theorem T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*), when $m \geqslant n$,

$$\mathbb{E}\|\boldsymbol{S} - \boldsymbol{I}_n\| \leqslant \delta_n = CK^2\left(\sqrt{\frac{n}{m}} + \frac{n}{m}\right).$$

Let $\sigma_i(\boldsymbol{S})$ be the eigenvalues of $\boldsymbol{S}$; then

$$\sigma_i(\boldsymbol{S}) = s_i\left(\frac{1}{\sqrt{m}}\boldsymbol{A}\right)^2 = \frac{s_i(\boldsymbol{A})^2}{m},$$

and by Lemma L4.1.14 (*Weyl's inequality*) (*Weyl's inequality*),

$$|\sigma_i(\boldsymbol{S}) - \sigma_i(\boldsymbol{I}_n)| \leqslant \|\boldsymbol{S} - \boldsymbol{I}_n\|,$$

i.e.

$$\left|\frac{s_i(\boldsymbol{A})^2}{m} - 1\right| \leqslant \|\boldsymbol{S} - \boldsymbol{I}_n\|.$$

Taking expectations on both sides,

$$\mathbb{E}\left[\left|\frac{s_i(\boldsymbol{A})^2}{m} - 1\right|\right] \leqslant \mathbb{E}\|\boldsymbol{S} - \boldsymbol{I}_n\| \leqslant \delta_n.$$

Set

$$a = \frac{s_i(\boldsymbol{A})^2}{m}.$$

Since $a \geqslant 0$, we obtain

$$|\sqrt{a} - 1| = \frac{|a-1|}{\sqrt{a}+1} \leqslant |a-1|.$$

That is,

$$1 - \left|\frac{s_i(\boldsymbol{A})^2}{m} - 1\right| \leqslant \frac{s_i(\boldsymbol{A})}{\sqrt{m}} \leqslant 1 + \left|\frac{s_i(\boldsymbol{A})^2}{m} - 1\right|.$$

Taking expectations gives

$$\mathbb{E}\left[\frac{s_i(\boldsymbol{A})}{\sqrt{m}}\right] \geqslant 1 - \mathbb{E}\left[\left|\frac{s_i(\boldsymbol{A})^2}{m} - 1\right|\right] \geqslant 1 - \delta_n,$$
$$\mathbb{E}\left[\frac{s_i(\boldsymbol{A})}{\sqrt{m}}\right] \leqslant 1 + \mathbb{E}\left[\left|\frac{s_i(\boldsymbol{A})^2}{m} - 1\right|\right] \leqslant 1 + \delta_n.$$

Since $m \geqslant n$, we obtain

$$\sqrt{m}\delta_n \leqslant CK^2\left(\sqrt{n} + \frac{n}{\sqrt{m}}\right) \leqslant 2CK^2\sqrt{n}.$$

i.e.

$$\sqrt{m} - C'K^2\sqrt{n} \leqslant \mathbb{E}[s_n(\boldsymbol{A})] \leqslant \mathbb{E}[s_1(\boldsymbol{A})] \leqslant \sqrt{m} + C'K^2\sqrt{n}.$$

42 [**Lower Bound for the Operator Norm of Random Matrices**] Let's prove a matching lower bound for Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*). Let $\boldsymbol{A} = (A_{ij}) \in \mathbb{R}^{m\times n}$, where the A_{ij} are independent standardized K-sub-Gaussian random variables (*i.e.* $\mathbb{E}[A_{ij}] = 0, \mathbb{E}[A_{ij}^2] = 1, \|A_{ij}\|_{\psi_2} \leqslant K$). Prove: for every $t > 0$,

$$\mathbb{P}\left\{\|\boldsymbol{A}\| \geqslant \frac{1}{2}(\sqrt{m} + \sqrt{n} - t)\right\} \geqslant 1 - 2\exp\left(-\frac{ct^2}{K^4}\right). \qquad (c > 0)$$

Rk. The following is an erroneous attempt:

Let $\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^m$ be independent standard normal random vectors, and define $\bar{\boldsymbol{x}} \in \mathbb{S}^{n-1}$, $\bar{\boldsymbol{y}} \in \mathbb{S}^{m-1}$; then by the variational definition 4.1.8 of the operator norm,

$$\|\boldsymbol{A}\| = \sup_{\bar{\boldsymbol{x}}\in\mathbb{S}^{n-1}, \bar{\boldsymbol{y}}\in\mathbb{S}^{m-1}} \langle \boldsymbol{A}\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}}\rangle \geqslant \frac{\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}}{\|\boldsymbol{x}\|_2\|\boldsymbol{y}\|_2}.$$

Let $L = \boldsymbol{y}^\top\boldsymbol{A}\boldsymbol{x} = \sum_{i,j} a_{ij}x_jy_i$; then $\boldsymbol{y}^\top\boldsymbol{A}\boldsymbol{x}$ is a K-sub-Gaussian random variable with

$$\mathbb{E}[L] = \sum_{i,j} A_{ij}\mathbb{E}[x_j]\mathbb{E}[y_i] = 0,$$

$$\mathbb{E}[L^2] = \sum_{i,j} \mathbb{E}[A_{ij}^2]\mathbb{E}[x_j^2]\mathbb{E}[y_i^2] = \sum_{i,j} 1 = mn,$$

but trying to lower-bound via the bilinear form of two random vectors introduces complicated coupling issues.

Prf. By the definition 4.1.8 of the operator norm $\|\boldsymbol{A}\| = \sup_{\|\boldsymbol{x}\|_2=1} \|\boldsymbol{A}\boldsymbol{x}\|_2$, taking $\boldsymbol{x} = \boldsymbol{e}_1 \in \mathbb{R}^n$,

$$\|\boldsymbol{A}\| \geqslant \|\boldsymbol{A}\boldsymbol{e}_1\|_2 = \|\boldsymbol{A}_{\cdot 1}\|_2;$$

noting that $\|\boldsymbol{A}\| = \|\boldsymbol{A}^\top\|$ and taking $\boldsymbol{x} = \tilde{\boldsymbol{e}}_1 \in \mathbb{R}^m$,

$$\|\boldsymbol{A}\| = \|\boldsymbol{A}^\top\| \geqslant \|\boldsymbol{A}^\top \tilde{\boldsymbol{e}}_1\|_2 = \|\boldsymbol{A}_{1\cdot}^\top\|_2,$$

hence a robust lower bound for $\|\boldsymbol{A}\|$ is

$$\|\boldsymbol{A}\| \geqslant \frac{1}{2}(\|\boldsymbol{A}_{\cdot 1}\|_2 + \|\boldsymbol{A}_{1\cdot}^\top\|_2).$$

By Theorem T3.1.1 (*Concentration of the norm*),

$$\mathbb{P}\{\|\boldsymbol{A}_{\cdot 1}\|_2 \leqslant \sqrt{m} - s\} \leqslant \exp\left(-\frac{cs^2}{K^4}\right).$$

Taking $s = t/2$, the probability of the event $\mathcal{E}_1 = \{\|\boldsymbol{A}\| \geqslant \sqrt{m} - t/2\}$ satisfies

$$\mathbb{P}(\mathcal{E}_1) \geqslant \mathbb{P}\{\|\boldsymbol{A}_{\cdot 1}\|_2 > \sqrt{m} - t/2\} \geqslant 1 - \exp\left(-\frac{c(t/2)^2}{K^4}\right) = 1 - \exp\left(-\frac{c't^2}{K^4}\right);$$

similarly, the probability of $\mathcal{E}_2 = \{\|\boldsymbol{A}\| \geqslant \sqrt{n} - t/2\}$ satisfies

$$\mathbb{P}(\mathcal{E}_2) \geqslant \mathbb{P}\{\|\boldsymbol{A}_{1\cdot}^\top\|_2 > \sqrt{n} - t/2\} \geqslant 1 - \exp\left(-\frac{c't^2}{K^4}\right).$$

Hence

$$\mathbb{P}\{\mathcal{E}_1 \cap \mathcal{E}_2\} = 1 - \mathbb{P}\{\mathcal{E}_1^c \cup \mathcal{E}_2^c\} \geqslant 1 - \mathbb{P}(\mathcal{E}_1^c) - \mathbb{P}(\mathcal{E}_2^c) \geqslant 1 - 2\exp\left(-\frac{c't^2}{K^4}\right).$$

When $\mathcal{E}_1 \cap \mathcal{E}_2$ occurs,

$$\|\boldsymbol{A}\| \geqslant \frac{1}{2}\left(\sqrt{m} - \frac{t}{2} + \sqrt{n} - \frac{t}{2}\right) = \frac{1}{2}(\sqrt{m} + \sqrt{n} - t),$$

and therefore

$$\mathbb{P}\left\{\|\boldsymbol{A}\| \geqslant \frac{1}{2}(\sqrt{m} + \sqrt{n} - t)\right\} \geqslant 1 - 2\exp\left(-\frac{c't^2}{K^4}\right).$$

43 [**Upper Bounds for Sub-Gaussian Matrices: Relaxing the Independence Assumption**]

(1) Matrix-level sub-Gaussianity: if $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ is viewed as a K-sub-Gaussian random vector in the space $\mathbb{R}^{m\times n}$[1], prove: even without any independence assumption, the conclusion of Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*) still holds.

(2) Row (or column)-level sub-Gaussianity: prove: if the rows $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_m \in \mathbb{R}^n$ of $\boldsymbol{A}$ are independent zero-mean K-sub-Gaussian random vectors, then Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*) still holds. (*The same conclusion holds for columns.*)

Rk. Recall that the ε-net proof of Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*) proceeds in three steps:

1. Approximation: choose a deterministic ε-net $\mathcal{N} \subset \mathbb{S}^{n-1}, \mathcal{M} \subset \mathbb{S}^{m-1}$, reducing the norm to $\sup_{\boldsymbol{x}\in\mathcal{N},\boldsymbol{y}\in\mathcal{M}} \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$.
2. Concentration: prove that for fixed $\boldsymbol{x} \in \mathcal{N}, \boldsymbol{y} \in \mathcal{M}$, the bilinear form $Z = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$ satisfies the sub-Gaussian tail bound $\mathbb{P}\{|Z| \geqslant u\} \leqslant 2\exp(-cu^2/K^2)$.
3. Union bound: take the union over all net points.

Since steps 1 and 3 are deterministic geometric operations, they are independent of the randomness assumptions on the

[1]The original text repeated $K = \|\boldsymbol{A}\|_{\psi_2}$ after defining the vector as K-sub-Gaussian. This repeated assignment is removed because K is already determined by the preceding definition; retaining it is semantically redundant.

matrix entries. Hence we only need to prove that under the new assumptions of (1)(2), the sub-Gaussian norm bound $\|Z\|_{\psi_2} \leqslant C'K$ for $Z = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$ in step 2 still holds.

Prf. (1) By the definition of a sub-Gaussian random vector,

$$\|\boldsymbol{A}\|_{\psi_2} = \sup_{\|\boldsymbol{W}\|_F=1} \|\langle \boldsymbol{A}, \boldsymbol{W}\rangle\|_{\psi_2} = K,$$

where $\boldsymbol{W} \in \mathbb{R}^{m\times n}$ and the matrix inner product is defined by $\langle \boldsymbol{A}, \boldsymbol{W}\rangle = \operatorname{tr}(\boldsymbol{W}^\top \boldsymbol{A})$.

For any $\boldsymbol{x} \in \mathbb{S}^{n-1}$, $\boldsymbol{y} \in \mathbb{S}^{m-1}$, consider

$$\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle = \boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x} = \operatorname{tr}(\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}) = \operatorname{tr}(\boldsymbol{x}\boldsymbol{y}^\top \boldsymbol{A}),$$

and let the rank-one matrix be $\boldsymbol{W} = \boldsymbol{y}\boldsymbol{x}^\top$; then using the conclusion of Exercise Q4.3 (*Special Cases of the Operator Norm*) (1),

$$\|\boldsymbol{W}\|_F = \|\boldsymbol{y}\boldsymbol{x}^\top\|_F = \|\boldsymbol{y}\|_2 \|\boldsymbol{x}\|_2 = 1 \cdot 1 = 1,$$

hence

$$\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle = \langle \boldsymbol{A}, \boldsymbol{y}\boldsymbol{x}^\top\rangle = \langle \boldsymbol{A}, \boldsymbol{W}\rangle,$$

and therefore

$$\|\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle\|_{\psi_2} = \|\langle \boldsymbol{A}, \boldsymbol{W}\rangle\|_{\psi_2} \leqslant \sup_{\|\boldsymbol{M}\|_F=1} \|\langle \boldsymbol{A}, \boldsymbol{M}\rangle\|_{\psi_2} = K,$$

i.e. the bilinear form $Z = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$ is K-sub-Gaussian, and the rest of the union bound proof is identical to the original theorem.

(2) Expanding the bilinear form $Z = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$ by rows,

$$\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle = \sum_{i=1}^m y_i(\boldsymbol{A}_i^\top \boldsymbol{x}) = \sum_{i=1}^m y_i \langle \boldsymbol{A}_i, \boldsymbol{x}\rangle,$$

and letting $Z_i = \langle \boldsymbol{A}_i, \boldsymbol{x}\rangle$,

$$\|Z_i\|_{\psi_2} = \|\langle \boldsymbol{A}_i, \boldsymbol{x}\rangle\|_{\psi_2} \leqslant \|\boldsymbol{A}_i\|_{\psi_2} \leqslant K,$$

so $Z_1, \cdots, Z_m$ are independent zero-mean sub-Gaussian random variables; hence by Proposition P2.7.1 (*Subgaussian norm of a sum*),

$$\|\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle\|_{\psi_2}^2 = \left\|\sum_{i=1}^m y_i Z_i\right\|_{\psi_2}^2 \leqslant C \sum_{i=1}^m \|y_i Z_i\|_{\psi_2}^2 = C \sum_{i=1}^m y_i^2 \|Z_i\|_{\psi_2}^2 \leqslant CK^2 \sum_{i=1}^m y_i^2 = CK^2,$$

i.e.

$$\|Z\|_{\psi_2} \leqslant \sqrt{C}K,$$

so the bilinear form $Z = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle$ is K-sub-Gaussian, and the rest of the union bound proof is identical to the original theorem.

By the row-column symmetry, if the columns are independent, it suffices to apply the same argument to $\boldsymbol{A}^\top$.

44 [**Other $p \to q$ Operator Norms of Random Matrices**] In Theorem T4.4.3 (*Norm bounds for independent sub-Gaussian matrices*), we studied the $2 \to 2$ operator norm (i.e. the spectral norm) of random matrices.

This problem explores other $p \to q$ operator norms (defined in Exercise Q4.18 ($p \to q$ *Operator Norms*), with special cases in Exercise Q4.19 (*Explicit Formulas for the* $1 \to \infty$ *and* $1 \to 2$ *Operator Norms*) and Exercise Q4.20 (*Properties and Duality of the* $\infty \to 1$ *Operator Norm*)).

Let $\boldsymbol{A}$ be an $m \times n$ matrix in $\mathbb{R}^{m\times n}$ whose entries A_{ij} are independent standardized K-sub-Gaussian random variables.

(1) Prove:

$$\|\boldsymbol{A}\|_{1\to\infty} \leqslant CK(\sqrt{\log m} + \sqrt{\log n}). \qquad (C > 0)$$

(2) Prove:

$$\|\boldsymbol{A}\|_{1\to 2} = \|\boldsymbol{A}^\top\|_{2\to\infty} \leqslant \sqrt{m} + CK^2\sqrt{\log n}. \qquad (C>0)$$

(*Note: with more advanced tools one can handle all $p\to q$ norms; see Exercise Q8.41 ($p\to q$ Norm of a Random Matrix) below.*)

(3) Prove: the bounds in (1)(2) are essentially optimal. Specifically, let the entries of $\boldsymbol{A}$ be independent standard normal random variables, and prove:

$$\mathbb{E}\|\boldsymbol{A}\|_{1\to\infty} \geqslant c(\sqrt{\log m} + \sqrt{\log n}), \quad \mathbb{E}\|\boldsymbol{A}\|_{1\to 2} \geqslant c(\sqrt{m} + \sqrt{\log n}). \qquad (c>0)$$

Prf. (1) Using the conclusion of Exercise Q4.19 (*Explicit Formulas for the $1\to\infty$ and $1\to 2$ Operator Norms*) (1),

$$\|\boldsymbol{A}\|_{1\to\infty} = \max_{1\leqslant i\leqslant m, 1\leqslant j\leqslant n} |A_{ij}|,$$

and since the A_{ij} are independent K-sub-Gaussian random variables,

$$\mathbb{P}\{|A_{ij}|\geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{K^2}\right),$$

and taking the union bound on both sides,

$$\mathbb{P}\left\{\max_{i,j}|A_{ij}|\geqslant t\right\} = \mathbb{P}\{\bigcup_{i,j}\{|A_{ij}|\geqslant t\}\} \leqslant \sum_{i,j}\mathbb{P}\{|A_{ij}|\geqslant t\} \leqslant 2mn\exp\left(-\frac{ct^2}{K^2}\right),$$

and taking $t = C_1K\sqrt{\log(mn)}$,

$$\mathbb{P}\left\{\max_{i,j}|A_{ij}|\geqslant t\right\} \leqslant 2mn\exp(-cC_1^2\log(mn)) = 2(mn)^{1-cC_1^2}.$$

When C_1 is large enough that $cC_1^2\geqslant 2$,

$$\|\boldsymbol{A}\|_{1\to\infty} \leqslant C_1K\sqrt{\log(mn)} = C_1K\sqrt{\log m + \log n} \leqslant C_1K(\sqrt{\log m} + \sqrt{\log n}).$$

(2) For the equality, using the conclusion of Exercise Q4.18 ($p\to q$ *Operator Norms*) (3),

$$\|\boldsymbol{A}^\top\|_{p\to q} = \|\boldsymbol{A}\|_{q'\to p'},$$

and taking $p=2, q=\infty$, the conjugate exponents are $q'=1, p'=2$; then

$$\|\boldsymbol{A}^\top\|_{2\to\infty} = \|\boldsymbol{A}\|_{1\to 2}.$$

For the upper bound, using the conclusion of Exercise Q4.19 (*Explicit Formulas for the $1\to\infty$ and $1\to 2$ Operator Norms*) (2),

$$\|\boldsymbol{A}\|_{1\to 2} = \max_{1\leqslant j\leqslant n}\|\boldsymbol{A}_j\|_2,$$

and by Theorem T3.1.1 (*Concentration of the norm*), for every $u\geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{A}_j\|_2\geqslant\sqrt{m}+u\} \leqslant 2\exp\left(-\frac{c_2u^2}{K^4}\right), \qquad (c_2>0)$$

and taking the union bound,

$$\mathbb{P}\left\{\max_{1\leqslant j\leqslant n}\|\boldsymbol{A}_j\|_2\geqslant\sqrt{m}+u\right\} \leqslant 2n\exp\left(-\frac{c_2u^2}{K^4}\right),$$

and taking $u = CK^2\sqrt{\log n}$,

$$\mathbb{P}\left\{\max_{1\leqslant j\leqslant n}\|\boldsymbol{A}_j\|_2\geqslant\sqrt{m}+u\right\} \leqslant 2n\exp(-c_2C^2\log n) = 2n^{1-c_2C^2}.$$

When C is large enough that $c_2C^2\geqslant 2$,

$$\|\boldsymbol{A}\|_{1\to 2} \leqslant \sqrt{m} + CK^2\sqrt{\log n}.$$

(3) Assume WLOG that $A_{ij}\sim N(0,1)$.

- Proving the lower bound for the $1\to\infty$ norm

  For $\|\boldsymbol{A}\|_{1\to\infty} = \max_{i,j}|A_{ij}|$, by Exercise Q2.38 (*Maximum of Gaussian Random Variables*) and Exercise Q2.39

(*Characterization of Sub-Gaussian Distributions: Maximal Inequality*), taking $N = mn$,

$$\mathbb{E}\|\boldsymbol{A}\|_{1\to\infty} \asymp \sqrt{\log(mn)} = \sqrt{\log m + \log n} \geqslant \frac{1}{\sqrt{2}}(\sqrt{\log m} + \sqrt{\log n}),$$

with $c = 1/\sqrt{2}$.

- Proving the lower bound for the $1 \to 2$ norm

  On one hand,

$$\mathbb{E}\left[\max_j \|\boldsymbol{A}_j\|_2\right] \geqslant \mathbb{E}\|\boldsymbol{A}_1\|_2,$$

  where $\|\boldsymbol{A}_1\|_2^2 \sim \chi^2(m)$, so by Theorem T3.1.1 (*Concentration of the norm*),

$$\mathbb{E}\|\boldsymbol{A}_1\|_2 \asymp \sqrt{m}.$$

  On the other hand,

$$\mathbb{E}\left[\max_j \|\boldsymbol{A}_j\|_2\right] \geqslant \mathbb{E}\left[\max_j |A_{1j}|\right],$$

  where A_{1j} i.i.d. $\sim N(0,1)$; hence by Theorem T3.1.1 (*Concentration of the norm*),

$$\mathbb{E}\left[\max_j |A_{1j}|\right] \asymp \sqrt{\log n}.$$

  Hence

$$\mathbb{E}\|\boldsymbol{A}\|_{1\to 2} \geqslant \max(c_1\sqrt{m}, c_2\sqrt{\log n}) \geqslant \frac{1}{2}(c_1\sqrt{m} + c_2\sqrt{\log n}) \geqslant c(\sqrt{m} + \sqrt{\log n}),$$

  showing that the growth of the orders $\sqrt{m}, \sqrt{\log n}$ is unavoidable, i.e. the bounds are optimal.

45 [**Community Detection: An Improved Spectral Clustering Algorithm**] In community detection under the stochastic block model (SBM), let n be even and split the n vertices of G equally into two hidden communities. Vertices in the same community are connected with probability p, and across communities with probability q, where $0 \leqslant q < p \leqslant 1$.

Theorem T4.5.2 (*Spectral clustering for the stochastic block model*) bounds the number of misclassified vertices under the assumption that $p - q$ is large enough, and also that q is not too small.

Design an improved spectral clustering algorithm that removes the lower bound assumption on q. Assume that $n(p-q) \geqslant 4p$ and $\sqrt{n}(p-q) \geqslant C_0$, where $C_0 > 0$ is a sufficiently large absolute constant, and prove that with high probability, the number of misclassified vertices in the output of this algorithm is at most

$$|M| \leqslant \frac{C}{(p-q)^2}. \qquad (C > 0)$$

Prf. Let $\boldsymbol{A} \in \{0,1\}^{n\times n}$ be the adjacency matrix of G, and let the true community label vector be $\boldsymbol{x}^* \in \{-1,1\}^n$ (*with* $+1$ *for community 1 and* -1 *for community 2*); then $\boldsymbol{x}^* \perp \mathbf{1}$.

Set $\operatorname{sgn}(0) = 1$, and construct the following improved algorithm:

- Spectral analysis

  Let $\boldsymbol{D} = \mathbb{E}[\boldsymbol{A}]$; its diagonal entries are 0, and the off-diagonal entries take p or q according to whether the labels agree:

$$\boldsymbol{D} = \frac{p+q}{2}\mathbf{1}\mathbf{1}^\top + \frac{p-q}{2}\boldsymbol{x}^*\boldsymbol{x}^{*\top} - p\boldsymbol{I}_n.$$

  Let the centered expectation matrix be $\boldsymbol{D}' = \mathbb{E}[\boldsymbol{A}'] = \boldsymbol{P}\mathbb{E}[\boldsymbol{A}]\boldsymbol{P}$; then (*note that* $\boldsymbol{P}(\mathbf{1}\mathbf{1}^\top)\boldsymbol{P} = \boldsymbol{O}$ *and* $\boldsymbol{P}(\boldsymbol{x}^*\boldsymbol{x}^{*\top})\boldsymbol{P} = \boldsymbol{x}^*\boldsymbol{x}^{*\top}$)

$$\boldsymbol{D}' = \frac{p-q}{2}\boldsymbol{x}^*\boldsymbol{x}^{*\top} - p\boldsymbol{P}.$$

  Hence its eigenvalues are

$$\boldsymbol{v}_1 = \frac{1}{\sqrt{n}}\boldsymbol{x}^* \implies \boldsymbol{D}'\boldsymbol{v}_1 = \left(\frac{n(p-q)}{2} - p\right)\boldsymbol{v}_1 \implies \lambda_1 = \frac{n(p-q)}{2} - p,$$

Algorithm 4: Decentered Spectral Clustering

Input : The adjacency matrix $\boldsymbol{A} \in \{0,1\}^{n\times n}$ of the graph G.
Output: The estimated community label vector $\hat{\boldsymbol{x}} \in \{-1,1\}^n$.

Step 1: Construct the centering projection matrix
Build the all-ones vector $\mathbf{1} = (1,1,\cdots,1)^\top \in \mathbb{R}^n$;
$\boldsymbol{P} \leftarrow \boldsymbol{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$; `// Orthogonal projection onto the subspace` $\mathbf{1}^\perp$
Step 2: Decenter the matrix
$\boldsymbol{A}' \leftarrow \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$; `// Remove the effect of the degree distribution (the all-connected component)`
Step 3: Compute the leading eigenvector
Compute the unit eigenvector $\boldsymbol{u} \in \mathbb{R}^n$ of $\boldsymbol{A}'$ corresponding to its largest eigenvalue;
Step 4: Sign clustering
for $i = 1$ **to** n **do**
  $\hat{x}_i \leftarrow \operatorname{sgn}(u_i)$; `// Partition communities by the sign of the eigenvector entries`
end
return $\hat{\boldsymbol{x}} = (\hat{x}_1, \hat{x}_2, \cdots, \hat{x}_n)^\top$;

$$\boldsymbol{v}_2 = \frac{1}{\sqrt{n}}\mathbf{1} \implies \boldsymbol{D}'\mathbf{1} = \mathbf{0} \implies \lambda_2 = 0.$$

$$\boldsymbol{w} \perp \operatorname{span}(\mathbf{1}, \boldsymbol{x}^*) \implies \boldsymbol{D}'\boldsymbol{w} = -p\boldsymbol{w} \implies \lambda_3 = -p,$$

i.e. the largest eigenvalue of $\boldsymbol{D}'$ is λ_1 and the remaining eigenvalues are $0, -p$. By the assumption $n(p-q) \geqslant 4p$, we have $\lambda_1 \geqslant n(p-q)/4 > 0$, so the leading eigenvector is $\boldsymbol{v}$ and its spectral gap from the rest of the spectrum is

$$\delta = \lambda_1 - 0 = \frac{n(p-q)}{2} - p.$$

Therefore

$$\delta \geqslant c_0 n(p-q), \qquad (c_0 > 0)$$

and this lower bound on the spectral gap no longer depends on q.

- Applying the Davis-Kahan theorem to bound the eigenvector error

  Consider the perturbation matrix $\boldsymbol{A}' - \boldsymbol{D}' = \boldsymbol{P}(\boldsymbol{A} - \boldsymbol{D})\boldsymbol{P}$; it is easy to see that

  $$\|\boldsymbol{A}' - \boldsymbol{D}'\| \leqslant \|\boldsymbol{P}\|^2 \|\boldsymbol{A} - \boldsymbol{D}\| = \|\boldsymbol{A} - \boldsymbol{D}\|,$$

  where $\boldsymbol{A} - \boldsymbol{D}$ is a symmetric random matrix whose upper triangular entries are independent zero-mean $O(1)$-sub-Gaussian random variables (*i.e. bounded in absolute value by* 1); hence by Corollary C4.4.7 (*Norm bounds for symmetric sub-Gaussian matrices*),

  $$\|\boldsymbol{A}' - \boldsymbol{D}'\| \leqslant \|\boldsymbol{A} - \boldsymbol{D}\| \leqslant C_1\sqrt{n} \text{ (w.h.p.).} \qquad (C_1 > 0)$$

  Since the leading eigenvector of $\boldsymbol{A}'$ is $\boldsymbol{u}$ and that of $\boldsymbol{D}'$ is $\boldsymbol{v} = \boldsymbol{x}^*/\sqrt{n}$, by the Davis-Kahan theorem (or the eigenvector version of the $\sin\Theta$ theorem),

  $$\min_{s\in\{-1,1\}} \|\boldsymbol{u} - s\boldsymbol{v}\|_2 \leqslant \frac{2\sqrt{2}\|\boldsymbol{A}' - \boldsymbol{D}'\|}{\delta},$$

  i.e.

  $$\min_{s\in\{-1,1\}} \|\boldsymbol{u} - s\boldsymbol{v}\|_2 \leqslant \frac{2\sqrt{2}C_1\sqrt{n}}{c_0 n(p-q)} = \frac{C_2}{\sqrt{n}(p-q)}, \qquad \left(C_2 = \frac{2\sqrt{2}C_1}{c_0} > 0\right)$$

  and WLOG let the global sign minimizing the distance be $s = 1$; then

  $$\|\boldsymbol{u} - \boldsymbol{v}\|_2 \leqslant \frac{C_2}{\sqrt{n}(p-q)}.$$

- Bounding the number of misclassified vertices

  Let $M \subset \{1, \cdots, n\}$ be the set of vertices misclassified by the algorithm; then for every $i \in M$, the sign $\operatorname{sgn}(u_i)$ given by the algorithm is opposite to the true sign $\operatorname{sgn}(v_i)$.

  Since the true eigenvector $\boldsymbol{v}$ satisfies $|v_i| = 1/\sqrt{n}$, if u_i and v_i have opposite signs, their distance is at least the distance from v_i to the origin, i.e.

$$|u_i - v_i| \geqslant |v_i| = \frac{1}{\sqrt{n}},$$

  hence

$$\|\boldsymbol{u} - \boldsymbol{v}\|_2^2 = \sum_{i=1}^{n} (u_i - v_i)^2 \geqslant \sum_{i \in M} (u_i - v_i)^2 \geqslant \sum_{i \in M} \left(\frac{1}{\sqrt{n}}\right)^2 = \frac{|M|}{n},$$

  i.e.

$$\frac{|M|}{n} \leqslant \|\boldsymbol{u} - \boldsymbol{v}\|_2^2 \leqslant \left(\frac{C_2}{\sqrt{n}(p-q)}\right)^2 = \frac{C_2^2}{n(p-q)^2},$$

  i.e.

$$|M| \leqslant \frac{C}{(p-q)^2}. \qquad (C = C_2^2 > 0)$$

46 [**Alternative Proof of Two-Sided Bounds for Random Matrices**] Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ have rows $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_m \in \mathbb{R}^n$ that are independent zero-mean isotropic sub-Gaussian random vectors with $\max_i \|\boldsymbol{A}_i\|_{\psi_2} \leqslant K$.

Using Theorem T3.1.1 (*Concentration of the norm*) and Exercise Q4.37 (*Controlling Norm Deviation on an ε-Net*), give a simpler alternative proof of Theorem T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*): for every $t \geqslant 0$,

$$\sqrt{m} - CK^2(\sqrt{n} + t) \leqslant s_n(\boldsymbol{A}) \leqslant s_1(\boldsymbol{A}) \leqslant \sqrt{m} + CK^2(\sqrt{n} + t) \text{ (w.p. } 1 - 2\exp(-t^2)).$$

Prf. • Applying the union bound over an ε-net

Let $\boldsymbol{x} \in \mathbb{S}^{n-1}$, $\boldsymbol{Z} = \boldsymbol{A}\boldsymbol{x} \in \mathbb{R}^m$, and set $Z_i = \langle \boldsymbol{A}_i, \boldsymbol{x} \rangle$; then $Z_1, \cdots, Z_m$ are independent and satisfy:

1. Unit variance: since the $\boldsymbol{A}_i$ are isotropic,

$$\mathbb{E}[Z_i^2] = \mathbb{E}[\boldsymbol{x}^\top \boldsymbol{A}_i \boldsymbol{A}_i^\top \boldsymbol{x}] = \boldsymbol{x}^\top \mathbb{E}[\boldsymbol{A}_i \boldsymbol{A}_i^\top] \boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{I}_n \boldsymbol{x} = \|\boldsymbol{x}\|_2^2 = 1.$$

2. sub-Gaussianity:

$$\|Z_i\|_{\psi_2} = \|\langle \boldsymbol{A}_i, \boldsymbol{x} \rangle\|_{\psi_2} \leqslant \|\boldsymbol{A}_i\|_{\psi_2} \|\boldsymbol{x}\|_2 \leqslant K.$$

Hence by Theorem T3.1.1 (*Concentration of the norm*), for every $s \geqslant 0$,

$$\mathbb{P}\{|\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| \geqslant s\} \leqslant 2\exp\left(-\frac{c_1 s^2}{K^4}\right). \qquad (c_1 > 0)$$

Take $\varepsilon = 1/4$; by Corollary C4.2.11 (*Covering the Euclidean ball*), there exists a $1/4$-net $\mathcal{N} \subset \mathbb{S}^{n-1}$ with $|\mathcal{N}| \leqslant 9^n$.

Applying the union bound over all points of $\mathcal{N}$,

$$\mathbb{P}(\text{Fail}) = \mathbb{P}\left\{\sup_{\boldsymbol{x} \in \mathcal{N}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| \geqslant s\right\} \leqslant |\mathcal{N}| \cdot 2\exp\left(-\frac{c_1 s^2}{K^4}\right) \leqslant 2\exp\left(n \log 9 - \frac{c_1 s^2}{K^4}\right),$$

and requiring $\mathbb{P}(\text{Fail}) \leqslant 2\exp(-t^2)$ gives

$$n \log 9 - \frac{c_1 s^2}{K^4} \leqslant -t^2 \implies s^2 \geqslant \frac{K^4}{c_1}(n \log 9 + t^2),$$

hence with $s = C_2 K^2(\sqrt{n} + t)$,

$$\mathbb{P}\left\{\underbrace{\sup_{\boldsymbol{x} \in \mathcal{N}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| \leqslant C_2 K^2(\sqrt{n} + t)}_{\mathcal{E}}\right\} \geqslant 1 - 2\exp(-t^2).$$

- Extending the discrete net to the whole sphere

By Exercise Q4.37 (*Controlling Norm Deviation on an ε-Net*), taking $\mu = \sqrt{m}$,

$$\sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| \leqslant \frac{C_3}{1-2\varepsilon} \sup_{\boldsymbol{x}\in\mathcal{N}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}|, \qquad (C_3 > 0)$$

and with $\varepsilon = 1/4$, whenever $\mathcal{E}$ occurs,

$$\sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| \leqslant 2C_3 \cdot C_2K^2(\sqrt{n}+t) =: CK^2(\sqrt{n}+t), \qquad (C = 2C_3C_2 > 0)$$

hence

$$\sqrt{m} - CK^2(\sqrt{n}+t) \leqslant \|\boldsymbol{A}\boldsymbol{x}\|_2 \leqslant \sqrt{m} + CK^2(\sqrt{n}+t).$$

Since

$$s_1(\boldsymbol{A}) = \sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} \|\boldsymbol{A}\boldsymbol{x}\|_2, \quad s_n(\boldsymbol{A}) = \inf_{\boldsymbol{x}\in\mathbb{S}^{n-1}} \|\boldsymbol{A}\boldsymbol{x}\|_2,$$

it follows that

$$\sqrt{m} - CK^2(\sqrt{n}+t) \leqslant s_n(\boldsymbol{A}) \leqslant s_1(\boldsymbol{A}) \leqslant \sqrt{m} + CK^2(\sqrt{n}+t).$$

47 [**Lower Bound for the Intermediate Singular Values of Random Matrices**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ have rows $\boldsymbol{A}_i$ that are independent zero-mean isotropic K-sub-Gaussian random vectors.

Using Theorem T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*), prove: for every $t \geqslant 0$, $s_k(\boldsymbol{A})$ satisfies

$$s_k(\boldsymbol{A}) \geqslant \sqrt{m} - CK^2(\sqrt{k}+t) \text{ (w.p. } 1-2\exp(-t^2)). \qquad (C > 0)$$

Prf. We use the following lemma:

Lemma (Singular value interlacing / Cauchy interlacing theorem) $s_k(\boldsymbol{A}) \geqslant s_k(\boldsymbol{A}_k)$.

Indeed, by the min-max theorem for singular values,

$$s_k(\boldsymbol{A}) = \max_{\dim(E)=k} \min_{\boldsymbol{x}\in E, \|\boldsymbol{x}\|_2=1} \|\boldsymbol{A}\boldsymbol{x}\|_2,$$

where $E \subset \mathbb{R}^n$ ranges over k-dimensional subspaces; hence for any fixed $E_0 \subset \mathbb{R}^n$,

$$s_k(\boldsymbol{A}) \geqslant \min_{\boldsymbol{x}\in E_0, \|\boldsymbol{x}\|_2=1} \|\boldsymbol{A}\boldsymbol{x}\|_2.$$

Taking $E_0 = \operatorname{span}(\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_k)$, for every $\boldsymbol{x} \in E_0$ the last $n-k$ coordinates are zero, i.e.

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}_k\tilde{\boldsymbol{x}},$$

where $\boldsymbol{A}_k \in \mathbb{R}^{m\times k}$ consists of the first k columns of $\boldsymbol{A}$, and $\tilde{\boldsymbol{x}} \in \mathbb{R}^k$ is the representation of $\boldsymbol{x}$ in the nonzero coordinates of E_0, with $\|\tilde{\boldsymbol{x}}\|_2 = 1$; hence

$$s_k(\boldsymbol{A}) \geqslant \min_{\tilde{\boldsymbol{x}}\in\mathbb{R}^k, \|\tilde{\boldsymbol{x}}\|_2=1} \|\boldsymbol{A}_k\tilde{\boldsymbol{x}}\|_2 = s_{\min}(\boldsymbol{A}_k) = s_k(\boldsymbol{A}_k).$$

Let the rows of $\boldsymbol{A}_k$ be $\tilde{\boldsymbol{A}}_i \in \mathbb{R}^k$; then the $\tilde{\boldsymbol{A}}_i$ are independent and zero-mean, and satisfy:

1. Isotropy:

Let $\boldsymbol{P} = (\boldsymbol{I}_k \mid \boldsymbol{O}_{k\times(n-k)})$ be the truncation matrix and $\tilde{\boldsymbol{A}}_i = \boldsymbol{P}\boldsymbol{A}_i$; then

$$\begin{aligned}\mathbb{E}[\tilde{\boldsymbol{A}}_i\tilde{\boldsymbol{A}}_i^\top] &= \mathbb{E}[(\boldsymbol{P}\boldsymbol{A}_i)(\boldsymbol{P}\boldsymbol{A}_i)^\top] = \mathbb{E}[\boldsymbol{P}\boldsymbol{A}_i\boldsymbol{A}_i^\top\boldsymbol{P}^\top] \\ &= \boldsymbol{P}\mathbb{E}[\boldsymbol{A}_i\boldsymbol{A}_i^\top]\boldsymbol{P}^\top = \boldsymbol{P}\boldsymbol{P}^\top \\ &= (\boldsymbol{I}_k \mid \boldsymbol{O})\begin{pmatrix}\boldsymbol{I}_k \\ \boldsymbol{O}\end{pmatrix} = \boldsymbol{I}_k\boldsymbol{I}_k + \boldsymbol{O}\boldsymbol{O} = \boldsymbol{I}_k.\end{aligned}$$

2. For every $\boldsymbol{v} \in \mathbb{S}^{k-1}$, extend it by zeros to $\boldsymbol{v}' \in \mathbb{S}^{n-1}$; then

$$\|\langle\tilde{\boldsymbol{A}}_i, \boldsymbol{v}\rangle\|_{\psi_2} = \|\langle\boldsymbol{A}_i, \boldsymbol{v}'\rangle\|_{\psi_2} \leqslant \|\boldsymbol{A}_i\|_{\psi_2} \leqslant K,$$

i.e. $\max_i \|\tilde{\boldsymbol{A}}_i\|_{\psi_2} \leqslant K$.

Hence $\boldsymbol{A}_k \in \mathbb{R}^{m\times k}$ satisfies the assumptions of Theorem T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*); by Theorem

T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*),

$$\mathbb{P}\{s_k(\boldsymbol{A}_k) \geqslant \sqrt{m} - CK^2(\sqrt{k}+t)\} \geqslant 1 - 2\exp(-t^2),$$

and since

$$\{s_k(\boldsymbol{A}_k) \geqslant \sqrt{m} - CK^2(\sqrt{k}+t)\} \subset \{s_k(\boldsymbol{A}) \geqslant \sqrt{m} - CK^2(\sqrt{k}+t)\},$$

we obtain

$$\mathbb{P}\{s_k(\boldsymbol{A}) \geqslant \sqrt{m} - CK^2(\sqrt{k}+t)\} \geqslant 1 - 2\exp(-t^2).$$

48 [**Covariance Estimation with Relative Error**] Let's give a more sensitive version of Theorem T4.7.1 (*Covariance estimation*) with relative, rather than absolute, error.

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_m \in \mathbb{R}^n$ be independent and identically distributed, with population covariance matrix $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]$, assumed invertible.

Let $\boldsymbol{X}$ be a sub-Gaussian random vector satisfying, for every $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|\langle \boldsymbol{X}, \boldsymbol{x}\rangle\|_{\psi_2} \leqslant K\|\langle \boldsymbol{X}, \boldsymbol{x}\rangle\|_{L^2}, \qquad (K \geqslant 1)$$

and let

$$\boldsymbol{\Sigma}_m = \frac{1}{m}\sum_{i=1}^m \boldsymbol{X}_i \boldsymbol{X}_i^\top,$$

be the sample covariance matrix based on m samples. Prove: for the relative error of the covariance,

$$\mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{R}^n\setminus\{\boldsymbol{0}\}}\left|\frac{\boldsymbol{v}^\top\boldsymbol{\Sigma}_m\boldsymbol{v}}{\boldsymbol{v}^\top\boldsymbol{\Sigma}\boldsymbol{v}} - 1\right|\right] \leqslant CK^2\left(\sqrt{\frac{n}{m}} + \frac{n}{m}\right).$$

(*In other words, we obtain a uniform relative approximation of the variance of $\boldsymbol{X}$ in all one-dimensional projection directions.*)

Prf. Let $\boldsymbol{u} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{v}$; then

$$\text{(true variance)} \quad \boldsymbol{v}^\top\boldsymbol{\Sigma}\boldsymbol{v} = \boldsymbol{v}^\top\boldsymbol{\Sigma}^{1/2}\boldsymbol{\Sigma}^{1/2}\boldsymbol{v} = \boldsymbol{u}^\top\boldsymbol{u} = \|\boldsymbol{u}\|_2^2,$$

$$\text{(difference between sample and true variance)} \quad \boldsymbol{v}^\top\boldsymbol{\Sigma}_m\boldsymbol{v} - \boldsymbol{v}^\top\boldsymbol{\Sigma}\boldsymbol{v} = \boldsymbol{u}^\top\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}_m\boldsymbol{\Sigma}^{-1/2}\boldsymbol{u} - \boldsymbol{u}^\top\boldsymbol{u} = \boldsymbol{u}^\top(\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}_m\boldsymbol{\Sigma}^{-1/2} - \boldsymbol{I}_n)\boldsymbol{u},$$

hence by the definition of the Rayleigh quotient,

$$\sup_{\boldsymbol{v}\neq\boldsymbol{0}}\left|\frac{\boldsymbol{v}^\top\boldsymbol{\Sigma}_m\boldsymbol{v}}{\boldsymbol{v}^\top\boldsymbol{\Sigma}\boldsymbol{v}} - 1\right| = \sup_{\boldsymbol{u}\neq\boldsymbol{0}}\frac{\left|\boldsymbol{u}^\top(\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}_m\boldsymbol{\Sigma}^{-1/2} - \boldsymbol{I}_n)\boldsymbol{u}\right|}{\|\boldsymbol{u}\|_2^2} = \left\|\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}_m\boldsymbol{\Sigma}^{-1/2} - \boldsymbol{I}_n\right\|,$$

so it remains to compute the expectation on the right.

Let the whitened random vectors be $\boldsymbol{Z}_i = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{X}_i$:

$$\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}_m\boldsymbol{\Sigma}^{-1/2} = \frac{1}{m}\sum_{i=1}^m(\boldsymbol{\Sigma}^{-1/2}\boldsymbol{X}_i)(\boldsymbol{\Sigma}^{-1/2}\boldsymbol{X}_i)^\top = \frac{1}{m}\sum_{i=1}^m \boldsymbol{Z}_i\boldsymbol{Z}_i^\top.$$

Then $\boldsymbol{Z}_1, \cdots, \boldsymbol{Z}_m$ are independent and satisfy

1. Isotropy:

$$\mathbb{E}[\boldsymbol{Z}_i\boldsymbol{Z}_i^\top] = \boldsymbol{\Sigma}^{-1/2}\mathbb{E}[\boldsymbol{X}_i\boldsymbol{X}_i^\top]\boldsymbol{\Sigma}^{-1/2} = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1/2} = \boldsymbol{I}_n.$$

2. Sub-Gaussian norm control:

   For every $\boldsymbol{w} \in \mathbb{S}^{n-1}$,

$$\langle \boldsymbol{Z}_i, \boldsymbol{w}\rangle = \langle \boldsymbol{\Sigma}^{-1/2}\boldsymbol{X}_i, \boldsymbol{w}\rangle = \langle \boldsymbol{X}_i, \boldsymbol{\Sigma}^{-1/2}\boldsymbol{w}\rangle,$$

   and letting $\boldsymbol{x} = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{w}$,

$$\|\langle \boldsymbol{X}_i, \boldsymbol{x}\rangle\|_{\psi_2} \leqslant K\|\langle \boldsymbol{X}_i, \boldsymbol{x}\rangle\|_{L^2},$$

   where

$$\|\langle \boldsymbol{X}_i, \boldsymbol{x}\rangle\|_{L^2}^2 = \mathbb{E}[\langle \boldsymbol{X}_i, \boldsymbol{x}\rangle^2] = \boldsymbol{x}^\top\boldsymbol{\Sigma}\boldsymbol{x} = \boldsymbol{w}^\top\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1/2}\boldsymbol{w} = \boldsymbol{w}^\top\boldsymbol{w} = 1,$$

   hence $\|\langle \boldsymbol{Z}_i, \boldsymbol{w}\rangle\|_{\psi_2} \leqslant K \cdot 1 = K$.

Let

$$\boldsymbol{R}_m = \frac{1}{m}\sum_{i=1}^m \boldsymbol{Z}_i\boldsymbol{Z}_i^\top - \boldsymbol{I}_n;$$

then

$$\mathbb{E}\left[\sup_{\boldsymbol{v}\neq\boldsymbol{0}}\left|\frac{\boldsymbol{v}^\top\boldsymbol{\Sigma}_m\boldsymbol{v}}{\boldsymbol{v}^\top\boldsymbol{\Sigma}\boldsymbol{v}}-1\right|\right] = \mathbb{E}\|\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}_m\boldsymbol{\Sigma}^{-1/2}-\boldsymbol{I}_n\| = \mathbb{E}\|\boldsymbol{R}_m\|,$$

and by Theorem T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*),

$$\mathbb{E}\|\boldsymbol{R}_m\| \leqslant CK^2\left(\sqrt{\frac{n}{m}}+\frac{n}{m}\right),$$

i.e.

$$\mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{R}^n}\left|\frac{\boldsymbol{v}^\top\boldsymbol{\Sigma}_m\boldsymbol{v}}{\boldsymbol{v}^\top\boldsymbol{\Sigma}\boldsymbol{v}}-1\right|\right] \leqslant CK^2\left(\sqrt{\frac{n}{m}}+\frac{n}{m}\right).$$

49 [**High-Probability Covariance Estimation**] Let $\boldsymbol{X}\in\mathbb{R}^n$ be a sub-Gaussian random vector with covariance matrix $\boldsymbol{\Sigma}=\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]$, satisfying, for every $\boldsymbol{x}\in\mathbb{R}^n$,

$$\|\langle\boldsymbol{X},\boldsymbol{x}\rangle\|_{\psi_2} \leqslant K\|\langle\boldsymbol{X},\boldsymbol{x}\rangle\|_{L^2}. \qquad (K\geqslant 1)$$

Let $\boldsymbol{X}_1,\cdots,\boldsymbol{X}_m$ be independent copies of $\boldsymbol{X}$, with sample covariance matrix

$$\boldsymbol{\Sigma}_m = \frac{1}{m}\sum_{i=1}^m \boldsymbol{X}_i\boldsymbol{X}_i^\top.$$

Prove: for every $u\geqslant 0$, with high probability,

$$\|\boldsymbol{\Sigma}_m-\boldsymbol{\Sigma}\| \leqslant CK^2\left(\sqrt{\frac{n+u}{m}}+\frac{n+u}{m}\right)\|\boldsymbol{\Sigma}\| \text{ (w.p. } 1-2\mathrm{e}^{-u}). \qquad (C>0)$$

Prf. WLOG assume $\boldsymbol{\Sigma}$ is invertible; let the whitened random vectors be $\boldsymbol{Z}_i=\boldsymbol{\Sigma}^{-1/2}\boldsymbol{X}_i$, satisfying $\|\langle\boldsymbol{Z}_i,\boldsymbol{x}\rangle\|_{\psi_2}\leqslant K\|\boldsymbol{x}\|_2$; then by Theorem T4.7.1 (*Covariance estimation*), the $\boldsymbol{Z}_i$ are independent zero-mean isotropic random vectors, hence

$$\|\boldsymbol{\Sigma}_m-\boldsymbol{\Sigma}\| = \|\boldsymbol{\Sigma}^{1/2}\boldsymbol{R}_m\boldsymbol{\Sigma}^{1/2}\| \leqslant \|\boldsymbol{\Sigma}\|\cdot\|\boldsymbol{R}_m\|,$$

where

$$\boldsymbol{R}_m = \frac{1}{m}\sum_{i=1}^m \boldsymbol{Z}_i\boldsymbol{Z}_i^\top - \boldsymbol{I}_n.$$

By Lemma L4.4.2 (*Estimating the quadratic form of a symmetric matrix on a discrete net*), taking $\varepsilon=1/4$, there exists a $1/4$-net $\mathcal{N}$ with $|\mathcal{N}|\leqslant 9^n$ such that

$$\|\boldsymbol{R}_m\| \leqslant 2\max_{\boldsymbol{x}\in\mathcal{N}}|\langle\boldsymbol{R}_m\boldsymbol{x},\boldsymbol{x}\rangle| = 2\max_{\boldsymbol{x}\in\mathcal{N}}\left|\frac{1}{m}\sum_{i=1}^m\langle\boldsymbol{Z}_i,\boldsymbol{x}\rangle^2-1\right|.$$

For $\boldsymbol{x}\in\mathcal{N}$, let $Y_i=\langle\boldsymbol{Z}_i,\boldsymbol{x}\rangle^2-1$; then:

1. Zero mean:

   Since the $\boldsymbol{Z}_i$ are isotropic, $\mathbb{E}[\langle\boldsymbol{Z}_i,\boldsymbol{x}\rangle^2]=\|\boldsymbol{x}\|_2^2=1$, i.e. $\mathbb{E}[Y_i]=0$.

2. Sub-exponentiality:

   Since $\langle\boldsymbol{Z}_i,\boldsymbol{x}\rangle$ is a K-sub-Gaussian random variable, $\langle\boldsymbol{Z}_i,\boldsymbol{x}\rangle^2$ is sub-exponential, with $\|Y_i\|_{\psi_1}\leqslant c_1K^2$.

Hence by Proposition P2.8.1 (*Subexponential properties*), for every $\delta\geqslant 0$,

$$\mathbb{P}\left\{\left|\frac{1}{m}\sum_{i=1}^m Y_i\right|\geqslant\delta\right\} \leqslant 2\exp\left(-c_2 m\min\left(\frac{\delta^2}{K^4},\frac{\delta}{K^2}\right)\right). \qquad (c_2>0)$$

Taking the union bound,

$$\mathbb{P}(\text{Fail}) = \mathbb{P}\left\{\max_{\boldsymbol{x}\in\mathcal{N}}\left|\frac{1}{m}\sum_{i=1}^{m} Y_i\right| \geqslant \delta\right\} \leqslant |\mathcal{N}|\cdot 2\exp\left(-c_2 m\min\left(\frac{\delta^2}{K^4},\frac{\delta}{K^2}\right)\right)$$
$$\leqslant 2\exp\left(n\log 9 - c_2 m\min\left(\frac{\delta^2}{K^4},\frac{\delta}{K^2}\right)\right).$$

Requiring $\mathbb{P}(\text{Fail}) \leqslant 2\mathrm{e}^{-u}$ gives

$$c_2 m\min\left(\frac{\delta^2}{K^4},\frac{\delta}{K^2}\right) - n\log 9 = u$$

i.e.

$$\min\left(\frac{\delta^2}{K^4},\frac{\delta}{K^2}\right) = \frac{n\log 9 + u}{c_2 m} \leqslant c_3\frac{n+u}{m}. \qquad (c_3 > 0)$$

Let $\alpha = c_3\frac{n+u}{m}$ and $x = \frac{\delta}{K^2}$; then

$$\min(x^2, x) = \alpha,$$

a. If $\alpha \leqslant 1$, then $x^2 \leqslant x$, so $x^2 = \alpha$, i.e. $x = \sqrt{\alpha}$.

b. If $\alpha > 1$, then $x < x^2$, so $x = \alpha$.

Hence

$$x \leqslant \sqrt{\alpha} + \alpha,$$

i.e.

$$\delta \leqslant K^2\left(\sqrt{c_3\frac{n+u}{m}} + c_3\frac{n+u}{m}\right) \leqslant C'K^2\left(\sqrt{\frac{n+u}{m}} + \frac{n+u}{m}\right). \qquad (C' > 0)$$

In conclusion,

$$\|\boldsymbol{R}_m\| \leqslant 2\delta \leqslant 2C'K^2\left(\sqrt{\frac{n+u}{m}} + \frac{n+u}{m}\right),$$

i.e.

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant CK^2\left(\sqrt{\frac{n+u}{m}} + \frac{n+u}{m}\right)\|\boldsymbol{\Sigma}\| \text{ (w.p. } 1-2\mathrm{e}^{-u}) \qquad (C = 2C')$$

50 [**Covering Numbers of Low-Rank Matrices**] A rank-r $m\times n$ matrix can be described by parameters via the SVD: the left singular vectors contribute mr degrees of freedom, the right singular vectors nr, and the singular values r.

Since covering numbers usually grow exponentially in the dimension (see Corollary C4.2.11 (*Covering the Euclidean ball*)), we guess that the covering number of the set

$$M_{m,n,r} = \{\boldsymbol{A}\in\mathbb{R}^{m\times n} : \operatorname{rank}(\boldsymbol{A}) \leqslant r, \|\boldsymbol{A}\|_F \leqslant 1\}$$

depends exponentially on $(m+n+1)r$. Indeed, this is the case.

(1) Let $O_{m,r}$ be the set of all $m\times r$ matrices with orthonormal columns. Prove: for every $\varepsilon > 0$, in the $\|\cdot\|_{1\to 2}$ norm (i.e. the maximum ℓ^2 norm of the columns),

$$\mathcal{N}(O_{m,r}, \|\cdot\|_{1\to 2}, \varepsilon) \leqslant \left(\frac{C}{\varepsilon}\right)^{mr}. \qquad (C > 0)$$

(2) Prove: for every $\varepsilon > 0$, in the Frobenius norm, the upper bound satisfies

$$\mathcal{N}(M_{m,n,r}, \|\cdot\|_F, \varepsilon) \leqslant \left(\frac{C}{\varepsilon}\right)^{(m+n+1)r}. \qquad (C > 0)$$

(3) Conversely, prove: the lower bound satisfies

$$\mathcal{N}(M_{m,n,r}, \|\cdot\|_F, \varepsilon) \geqslant \left(\frac{c}{\varepsilon}\right)^{\frac{1}{2}(m+n)r}. \qquad (c > 0)$$

Prf. (1) Let $\boldsymbol{U}=(\boldsymbol{u}_1,\boldsymbol{u}_2,\cdots,\boldsymbol{u}_r)\in O_{m,r}$; then the $\boldsymbol{u}_i\in\mathbb{S}^{m-1}$ are orthonormal, so $O_{m,r}$ embeds into the Cartesian product of r unit spheres:

$$O_{m,r}\subset\mathbb{S}^{m-1}\times\mathbb{S}^{m-1}\times\cdots\times\mathbb{S}^{m-1}.$$

By Corollary C4.2.11 (*Covering the Euclidean ball*), there exists an ε-net $\mathcal{N}_0\subset\mathbb{S}^{m-1}$ with $|\mathcal{N}_0|\leqslant(3/\varepsilon)^m$.

Construct the Cartesian product net $\mathcal{N}=\mathcal{N}_0\times\cdots\times\mathcal{N}_0$; then

$$|\mathcal{N}|=|\mathcal{N}_0|^r\leqslant\left(\frac{3}{\varepsilon}\right)^{mr}.$$

For each $\boldsymbol{u}_i$, find the nearest $\boldsymbol{x}_i\in\mathcal{N}_0$ with $\|\boldsymbol{u}_i-\boldsymbol{x}_i\|_2\leqslant\varepsilon$, and set $\boldsymbol{X}=(\boldsymbol{x}_1,\cdots,\boldsymbol{x}_r)\in\mathcal{N}$; then

$$\|\boldsymbol{U}-\boldsymbol{X}\|_{1\to2}=\max_{1\leqslant i\leqslant r}\|\boldsymbol{u}_i-\boldsymbol{x}_i\|_2\leqslant\varepsilon,$$

so $\mathcal{N}$ is an ε-net of $O_{m,r}$ in the $\|\cdot\|_{1\to2}$ norm, with $C=3$.

(2) Let the SVD of $\boldsymbol{A}\in M_{m,n,r}$ be $\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, where $\boldsymbol{U}\in O_{m,r}$, $\boldsymbol{V}\in O_{n,r}$, $\boldsymbol{\Sigma}=\operatorname{diag}(s_1,\cdots,s_r)$.

Since $\|\boldsymbol{A}\|_F^2=\sum s_i^2\leqslant1$, we have $\boldsymbol{s}=(s_1,\cdots,s_r)^\top\in\mathbb{B}_2^r$.

Let $\varepsilon_0=\varepsilon/3$ and construct internal nets separately:

1. For $O_{m,r}$, take an ε_0-net $\mathcal{N}_U$ in the $\|\cdot\|_{1\to2}$ norm, with $|\mathcal{N}_U|\leqslant(C/\varepsilon_0)^{mr}$.
2. For $O_{n,r}$, take an ε_0-net $\mathcal{N}_V$ in the $\|\cdot\|_{1\to2}$ norm, with $|\mathcal{N}_V|\leqslant(C/\varepsilon_0)^{nr}$.
3. For $\mathbb{B}_2^r$, take an ε_0-net $\mathcal{N}_\Sigma$ in the Euclidean norm, with $|\mathcal{N}_\Sigma|\leqslant(3/\varepsilon_0)^r$.

Construct the Cartesian product net

$$\mathcal{W}=\{\tilde{\boldsymbol{U}}\tilde{\boldsymbol{\Sigma}}\tilde{\boldsymbol{V}}^\top:\tilde{\boldsymbol{U}}\in\mathcal{N}_U,\tilde{\boldsymbol{\Sigma}}\in\mathcal{N}_\Sigma,\tilde{\boldsymbol{V}}\in\mathcal{N}_V\},$$

with $|\mathcal{W}|\leqslant(C'/\varepsilon)^{(m+n+1)r}$; then for any $\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, by the triangle inequality,

$$\|\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top-\tilde{\boldsymbol{U}}\tilde{\boldsymbol{\Sigma}}\tilde{\boldsymbol{V}}^\top\|_F\leqslant\|(\boldsymbol{U}-\tilde{\boldsymbol{U}})\boldsymbol{\Sigma}\boldsymbol{V}^\top\|_F+\|\tilde{\boldsymbol{U}}(\boldsymbol{\Sigma}-\tilde{\boldsymbol{\Sigma}})\boldsymbol{V}^\top\|_F+\|\tilde{\boldsymbol{U}}\tilde{\boldsymbol{\Sigma}}(\boldsymbol{V}-\tilde{\boldsymbol{V}})^\top\|_F.$$

For the first term, since the $\boldsymbol{v}_i$ are orthonormal,

$$\|(\boldsymbol{U}-\tilde{\boldsymbol{U}})\boldsymbol{\Sigma}\boldsymbol{V}^\top\|_F=\|(\boldsymbol{U}-\tilde{\boldsymbol{U}})\boldsymbol{\Sigma}\|_F=\sqrt{\sum_{i=1}^r s_i^2\|\boldsymbol{u}_i-\tilde{\boldsymbol{u}}_i\|_2^2}\leqslant\varepsilon_0\sqrt{\sum_{i=1}^r s_i^2}\leqslant\varepsilon_0.$$

For the second term, since $\tilde{\boldsymbol{U}}$ and the $\boldsymbol{v}_i$ are orthonormal,

$$\|\tilde{\boldsymbol{U}}(\boldsymbol{\Sigma}-\tilde{\boldsymbol{\Sigma}})\boldsymbol{V}^\top\|_F=\|\boldsymbol{\Sigma}-\tilde{\boldsymbol{\Sigma}}\|_F=\|\boldsymbol{s}-\tilde{\boldsymbol{s}}\|_2\leqslant\varepsilon_0.$$

For the third term, similarly,

$$\|\tilde{\boldsymbol{U}}\tilde{\boldsymbol{\Sigma}}(\boldsymbol{V}-\tilde{\boldsymbol{V}})^\top\|_F=\|(\boldsymbol{V}-\tilde{\boldsymbol{V}})\tilde{\boldsymbol{\Sigma}}\|_F\leqslant\varepsilon_0\|\tilde{\boldsymbol{s}}\|_2\leqslant\varepsilon_0.$$

Hence

$$\|(\boldsymbol{U}-\tilde{\boldsymbol{U}})\boldsymbol{\Sigma}\boldsymbol{V}^\top\|_F\leqslant3\varepsilon_0=\varepsilon,$$

so $\mathcal{W}$ is an ε-net.

(3) (*There is no need to compute the volume of the whole complicated manifold; it suffices to embed simple Euclidean subspaces into $M_{m,n,r}$.*)

- Constructing the subspace E_1

  Consider matrices with only the first r columns nonzero, $\boldsymbol{A}=(\boldsymbol{X},\boldsymbol{O})$, where $\boldsymbol{X}\in\mathbb{R}^{m\times r}$ with $\|\boldsymbol{X}\|_F\leqslant1$; then $\operatorname{rank}(\boldsymbol{A})\leqslant r$ and $\|\boldsymbol{A}\|_F=\|\boldsymbol{X}\|_F\leqslant1$, so $E_1\subset M_{m,n,r}(\mathbb{R})\cong\mathbb{B}_2^{mr}$; hence

$$\mathcal{N}(M_{m,n,r},\|\cdot\|_F,\varepsilon)\geqslant\mathcal{N}(E_1,\|\cdot\|_F,\varepsilon)\geqslant\left(\frac{c}{\varepsilon}\right)^{mr}.$$

- Constructing the subspace E_2

  By symmetry, consider matrices with only the first r rows nonzero, $\boldsymbol{A}=(\boldsymbol{Y},\boldsymbol{O})^\top$; then $E_2\subset M_{m,n,r}(\mathbb{R})\cong\mathbb{B}_2^{nr}$; hence

$$\mathcal{N}(M_{m,n,r},\|\cdot\|_F,\varepsilon)\geqslant\mathcal{N}(E_2,\|\cdot\|_F,\varepsilon)\geqslant\left(\frac{c}{\varepsilon}\right)^{nr}.$$

Both lower bounds must hold simultaneously; hence

$$\mathcal{N} \geqslant \max\left(\left(\frac{c}{\varepsilon}\right)^{mr}, \left(\frac{c}{\varepsilon}\right)^{nr}\right) = \left(\frac{c}{\varepsilon}\right)^{\max(m,n)r} \geqslant \left(\frac{c}{\varepsilon}\right)^{\frac{1}{2}(m+n)r}.$$

51 [**Spectral Clustering Guarantees for Mixtures of Gaussians**] Prove Theorem T4.7.5 (*Spectral clustering for Gaussian mixtures*), which guarantees that spectral clustering successfully learns mixtures of Gaussians.

Theorem T4.7.5 (*Spectral clustering for Gaussian mixtures*): Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_m \in \mathbb{R}^n$ be independent samples from the uniform mixture of the Gaussian components $N(\boldsymbol{\mu}, \boldsymbol{I}_n)$ and $N(-\boldsymbol{\mu}, \boldsymbol{I}_n)$; this mixture satisfies $\mathbb{E}[\boldsymbol{X}] = \boldsymbol{0}$. If the sample size satisfies $m \geqslant Cn$ and the separation $\|\boldsymbol{\mu}\|_2 \geqslant C$, then spectral clustering with high probability correctly identifies the communities (w.p. 0.99), misclassifying at most 1% of the points. Here is how to prove it:

(1) Compute the covariance matrix $\boldsymbol{\Sigma}$ of $\boldsymbol{X}$, and note that the top eigenvector $\boldsymbol{u}$ is collinear with $\boldsymbol{\mu}$.

(2) Conclude that the signs of $\langle \boldsymbol{X}_i, \boldsymbol{u} \rangle$ classify most points $\boldsymbol{X}_i$ correctly.

(3) Use results about covariance estimation to show that the sample covariance matrix $\boldsymbol{\Sigma}_m$ approximates $\boldsymbol{\Sigma}$.

(4) Use Theorem T4.1.15 (*Davis–Kahan inequality*) to deduce that the top eigenvector $\boldsymbol{v} = \boldsymbol{v}_1(\boldsymbol{\Sigma}_m)$ approximates $\boldsymbol{u}$.

(5) Using (2), conclude that the signs of $\langle \boldsymbol{X}_i, \boldsymbol{v} \rangle$ classify most points $\boldsymbol{X}_i$ correctly.

Prf. • Analyzing the structure of the population covariance

Let the true latent labels be y_i i.i.d. $\sim$ Rademacher; then

$$\boldsymbol{X}_i = y_i \boldsymbol{\mu} + \boldsymbol{g}_i, \qquad (\boldsymbol{g}_i \sim N(\boldsymbol{0}, \boldsymbol{I}_n))$$

and the population covariance matrix $\boldsymbol{\Sigma}$ is

$$\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \mathbb{E}[(y\boldsymbol{\mu} + \boldsymbol{g})(y\boldsymbol{\mu} + \boldsymbol{g})^\top] = \mathbb{E}[y^2]\boldsymbol{\mu}\boldsymbol{\mu}^\top + \mathbb{E}[\boldsymbol{g}\boldsymbol{g}^\top] = \boldsymbol{\mu}\boldsymbol{\mu}^\top + \boldsymbol{I}_n.$$

Spectral analysis of $\boldsymbol{\Sigma}$:

$$\boldsymbol{\Sigma}\boldsymbol{\mu} = (\boldsymbol{\mu}\boldsymbol{\mu}^\top + \boldsymbol{I}_n)\boldsymbol{\mu} = (\|\boldsymbol{\mu}\|_2^2 + 1)\boldsymbol{\mu} \implies \lambda_1 = \|\boldsymbol{\mu}\|_2^2 + 1,$$
$$\boldsymbol{w} \perp \boldsymbol{\mu} \implies \boldsymbol{\Sigma}\boldsymbol{w} = \boldsymbol{w} \implies \lambda_2 = 1,$$

so the leading eigenvector is $\boldsymbol{u} = \bar{\boldsymbol{\mu}}$, collinear with $\boldsymbol{\mu}$.

• Classification performance of the ideal eigenvector $\boldsymbol{u}$

Consider

$$\langle \boldsymbol{X}_i, \boldsymbol{u} \rangle = y_i \langle \boldsymbol{\mu}, \boldsymbol{u} \rangle + \langle \boldsymbol{g}_i, \boldsymbol{u} \rangle = y_i \|\boldsymbol{\mu}\|_2 + Z_i,$$

where $Z_i = \langle \boldsymbol{g}_i, \boldsymbol{u} \rangle \sim N(0, 1)$.

The algorithm misclassifies i iff $\operatorname{sgn}(\langle \boldsymbol{X}_i, \boldsymbol{u} \rangle) \neq y_i$, equivalently $y_i Z_i < -\|\boldsymbol{\mu}\|_2$. Since $y_i Z_i \sim N(0, 1)$,

$$\begin{aligned} p_{\text{err}} &= \mathbb{P}\{y_i Z_i < -\|\boldsymbol{\mu}\|_2\} \\ &= \mathbb{P}\{Z_i > \|\boldsymbol{\mu}\|_2\} \\ &\leqslant \exp\left(-\frac{\|\boldsymbol{\mu}\|_2^2}{2}\right), \end{aligned}$$

and since $\|\boldsymbol{\mu}\|_2 \geqslant C$, for C large enough $p_{\text{err}} \leqslant 0.005$; hence by Theorem T2.3.1 (*Chernoff inequality*) (*Chernoff's inequality*), the ideal classifier $\boldsymbol{u}$ misclassifies at most 0.5% of the points, i.e.

$$\frac{1}{m} \sum_{i=1}^{m} \mathbf{1}_{\{\operatorname{sgn}(\langle \boldsymbol{X}_i, \boldsymbol{u} \rangle) \neq y_i\}} \leqslant 0.005.$$

• Approximation by the sample covariance

Let the sample covariance matrix be

$$\boldsymbol{\Sigma}_m = \frac{1}{m} \sum_{i=1}^{m} \boldsymbol{X}_i \boldsymbol{X}_i^\top;$$

then $\boldsymbol{\Sigma}_m$ is a sub-Gaussian random matrix, so by Theorem T4.7.1 (*Covariance estimation*), we have

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant C_1\sqrt{\frac{n}{m}}\|\boldsymbol{\Sigma}\| = C_1\sqrt{\frac{n}{m}}(1+\|\boldsymbol{\mu}\|_2^2) \text{ (w.p. 0.995).} \qquad (C_1 > 0)$$

Since $m \geqslant Cn$, for C large enough,

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant \frac{1}{10}\|\boldsymbol{\mu}\|_2^2.$$

(*Since $\|\boldsymbol{\mu}\|_2 \geqslant C \geqslant 1$, we have $1+\|\boldsymbol{\mu}\|_2^2 \leqslant 2\|\boldsymbol{\mu}\|_2^2$, so the condition above is easily satisfied.*)

- Davis-Kahan bounds the eigenvector perturbation

  Let $\boldsymbol{v}$ be the leading eigenvector of $\boldsymbol{\Sigma}_m$, with spectral gap $\delta = \lambda_1 - \lambda_2 = \|\boldsymbol{\mu}\|_2^2$; then by the Davis-Kahan theorem (in the $\sin\Theta$ form, see Exercise Q4.16 (*Angles between Vectors and Euclidean Distance*)),

$$\min_{s\in\{-1,1\}} \|\boldsymbol{v} - s\boldsymbol{u}\|_2 \leqslant \frac{2\sqrt{2}\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\|}{\delta} \leqslant \frac{2\sqrt{2}\cdot\frac{1}{10}\|\boldsymbol{\mu}\|_2^2}{\|\boldsymbol{\mu}\|_2^2} \leqslant \frac{1}{3}.$$

  WLOG assume the signs agree; then $\boldsymbol{h} = \boldsymbol{v} - \boldsymbol{u}$ satisfies $\|\boldsymbol{h}\|_2 \leqslant 1/3$.

  Substituting the sample covariance estimate above gives

$$\begin{aligned}\|\boldsymbol{h}\|_2 &\leqslant 2\sqrt{2}C_1\sqrt{\frac{n}{m}}\frac{1+\|\boldsymbol{\mu}\|_2^2}{\|\boldsymbol{\mu}\|_2^2}\\ &\leqslant 4\sqrt{2}C_1\sqrt{\frac{n}{m}}.\end{aligned}$$

  Set $\varepsilon_0 = 4\sqrt{2}C_1\sqrt{n/m}$. Enlarging the absolute constant in the assumption gives $\varepsilon_0 \leqslant 0.01$. This estimate is conditional on the sample covariance event and will be used in the misclassification count below.

- Bounding the classification error under perturbation

  For the actual test statistic $\langle \boldsymbol{X}_i, \boldsymbol{v}\rangle = \langle \boldsymbol{X}_i, \boldsymbol{u}\rangle + \langle \boldsymbol{X}_i, \boldsymbol{h}\rangle$: if the actual classifier errs but the ideal one is correct, the perturbation term must have flipped the sign, i.e.

$$|\langle \boldsymbol{X}_i, \boldsymbol{h}\rangle| \geqslant |\langle \boldsymbol{X}_i, \boldsymbol{u}\rangle|.$$

  Define the set of samples whose projections lie "near the decision boundary" as

$$S_1 = \left\{i \in \{1,\cdots,m\} : |\langle \boldsymbol{X}_i, \boldsymbol{u}\rangle| < \frac{1}{2}\|\boldsymbol{\mu}\|_2\right\},$$

  so for $i \in S_1$, $|Z_i| > \|\boldsymbol{\mu}\|_2/2$.

  Since $Z_i \sim N(0,1)$, by Theorem T2.2.1 (*Hoeffding inequality*) (*Hoeffding inequality*) (*or another concentration inequality for Gaussian tails*), for $\|\boldsymbol{\mu}\|_2 \geqslant C$,

$$|S_1| \leqslant 0.001m \text{ (w.h.p.).}$$

  For the remaining samples, consider those whose signs are flipped by the eigenvector perturbation $\boldsymbol{h} = \boldsymbol{v} - \boldsymbol{u}$; define the index set of such samples as

$$S_2 = \{i \notin S_1 : \operatorname{sgn}(\langle \boldsymbol{X}_i, \boldsymbol{v}\rangle) \neq \operatorname{sgn}(\langle \boldsymbol{X}_i, \boldsymbol{u}\rangle)\},$$

  and for every $i \in S_2$, the actual projection $\langle \boldsymbol{X}_i, \boldsymbol{v}\rangle = \langle \boldsymbol{X}_i, \boldsymbol{u}\rangle + \langle \boldsymbol{X}_i, \boldsymbol{h}\rangle$ has flipped sign; hence the absolute value of the perturbation term must be at least that of the ideal projection:

$$|\langle \boldsymbol{X}_i, \boldsymbol{h}\rangle| \geqslant |\langle \boldsymbol{X}_i, \boldsymbol{u}\rangle| \geqslant \frac{1}{2}\|\boldsymbol{\mu}\|_2 \implies \langle \boldsymbol{X}_i, \boldsymbol{h}\rangle^2 \geqslant \frac{1}{4}\|\boldsymbol{\mu}\|_2^2,$$

  and summing the squared perturbation terms over all samples in S_2,

$$\sum_{i\in S_2}\langle \boldsymbol{X}_i, \boldsymbol{h}\rangle^2 \leqslant \sum_{i=1}^m \langle \boldsymbol{X}_i, \boldsymbol{h}\rangle^2 = \boldsymbol{h}^\top\left(\sum_{i=1}^m \boldsymbol{X}_i\boldsymbol{X}_i^\top\right)\boldsymbol{h} = m\cdot\boldsymbol{h}^\top\boldsymbol{\Sigma}_m\boldsymbol{h} \leqslant m\cdot\|\boldsymbol{\Sigma}_m\|\|\boldsymbol{h}\|_2^2,$$

where $\|\boldsymbol{\Sigma}_m\| \leqslant 2\|\boldsymbol{\Sigma}\| = 2(1 + \|\boldsymbol{\mu}\|_2^2)$ and $\|\boldsymbol{h}\|_2 \leqslant \varepsilon_0$; hence

$$\sum_{i \in S_2} \langle \boldsymbol{X}_i, \boldsymbol{h} \rangle^2 \leqslant 2m(1 + \|\boldsymbol{\mu}\|_2^2)\varepsilon_0^2,$$

and therefore

$$|S_2| \cdot \frac{1}{4}\|\boldsymbol{\mu}\|_2^2 \leqslant 2m\varepsilon_0^2(1 + \|\boldsymbol{\mu}\|_2^2),$$

i.e.

$$|S_2| \leqslant 8m\varepsilon_0^2 \frac{1 + \|\boldsymbol{\mu}\|_2^2}{\|\boldsymbol{\mu}\|_2^2} \leqslant 16m\varepsilon_0^2.$$

Since $\varepsilon_0 \propto \sqrt{n/m}$, for C large enough $\varepsilon_0 \leqslant 0.01$; then

$$|S_2| \leqslant 16m \cdot 10^{-4} = 0.0016m \leqslant 0.004m.$$

- The three error sets give the final classification bound

  Let M_0 be the misclassification set of the ideal classifier $\boldsymbol{u}$, and let M be that of the actual classifier $\boldsymbol{v}$. The preceding part gives

  $$|M_0| \leqslant 0.005m.$$

  For the boundary set S_1 and the sign-flip set S_2, the preceding estimates give

  $$|S_1| \leqslant 0.001m,$$
  $$|S_2| \leqslant 16m\varepsilon_0^2 \leqslant 0.004m.$$

  If $i \in M \setminus M_0$, the ideal classifier is correct at i and the actual classifier has the opposite sign. If $i \in S_1$, the point belongs to the boundary set; if $i \notin S_1$, then $i \in S_2$. Thus

  $$M \subset M_0 \cup S_1 \cup S_2.$$

  Applying the union bound to the three counting events and enlarging the absolute constant in the assumption gives

  $$\begin{aligned} |M| &\leqslant |M_0| + |S_1| + |S_2| \\ &\leqslant (0.005 + 0.001 + 0.004)m \\ &= 0.01m \text{ (w.h.p.)}. \end{aligned}$$

Concentration Without Independence

5

Keypoints

Section 5.1 Lipschitz functions on the sphere — Q5.1 ∼ Q5.7

Definition 5.1.1 (*Lipschitz functions*)

A function $f : \mathbb{R}^n \to \mathbb{R}$ is called L-Lipschitz with respect to the Euclidean norm if $|f(\boldsymbol{x}) - f(\boldsymbol{y})| \leqslant L\|\boldsymbol{x} - \boldsymbol{y}\|_2$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. The smallest such L is denoted $\|f\|_{\mathrm{Lip}}$.

Example 5.1.2 (*Vectors, matrices and norms define natural Lipschitz functions*)

(a) For a fixed vector $\boldsymbol{a} \in \mathbb{R}^n$, the linear functional $f(\boldsymbol{x}) = \langle \boldsymbol{x}, \boldsymbol{a} \rangle$ has Lipschitz norm $\|f\|_{\mathrm{Lip}} = \|\boldsymbol{a}\|_2$. (b) More generally, for any matrix $\boldsymbol{A}$, the linear operator $f(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$ has Lipschitz norm $\|f\|_{\mathrm{Lip}} = \|\boldsymbol{A}\|$. (c) For any norm $\|\cdot\|$ on $\mathbb{R}^n$, the function $f(\boldsymbol{x}) = \|\boldsymbol{x}\|$ has Lipschitz norm equal to the smallest L such that $\|\boldsymbol{x}\| \leqslant L\|\boldsymbol{x}\|_2$ for all $\boldsymbol{x} \in \mathbb{R}^n$.

Theorem 5.1.3 (*Concentration of Lipschitz functions on the sphere*)

Let $\boldsymbol{X}$ be uniformly distributed on the Euclidean sphere of radius $\sqrt{n}$ in $\mathbb{R}^n$. Then for any 1-Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$, we have

$$\mathbb{P}\{|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]| \geqslant t\} \leqslant 2\exp(-ct^2)$$

for all $t \geqslant 0$.

Theorem 5.1.4 (*Isoperimetric inequality on* $\mathbb{R}^n$)

Among all sets $A \subset \mathbb{R}^n$ with fixed Gaussian measure $\gamma_n(A)$, the half-spaces minimize the Gaussian perimeter (Minkowski content in the Gaussian metric).

Theorem 5.1.5 (*Isoperimetric inequality on the sphere*)

Among all sets $A \subset \mathbb{S}^{n-1}$ with fixed spherical measure $\sigma(A)$, the spherical caps minimize the spherical perimeter.

Lemma 5.1.6 (*Blow-up*)

Let A be a subset of the sphere $\mathbb{S}^{n-1}$ with $\sigma(A) \geqslant 1/2$. Then the t-neighborhood A_t satisfies

$$\sigma(A_t) \geqslant 1 - 2\exp(-cnt^2).$$

Section 5.2 Concentration on other metric measure spaces Q5.8 ∼ Q5.13

Theorem 5.2.2 (*Gaussian isoperimetric inequality*)

Let $\varepsilon > 0$. Then, among all sets $A \subset \mathbb{R}^n$ with fixed Gaussian measure $\gamma_n(A)$, the half-spaces minimize $\gamma_n(A_\varepsilon)$, where $A_\varepsilon = \{\boldsymbol{x} : \operatorname{dist}(\boldsymbol{x}, A) \leqslant \varepsilon\}$.

Theorem 5.2.3 (*Gaussian concentration*)

Consider a random vector $\boldsymbol{X} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ and an L-Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$. Then

$$\|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2} \leqslant CL.$$

Theorem 5.2.5 (*Concentration on the Hamming cube*)

Let $f : \{-1, 1\}^n \to \mathbb{R}$ be a function such that

$$|f(x) - f(y)| \leqslant \frac{1}{\sqrt{n}} \sum_{i=1}^{n} |x_i - y_i|.$$

If $\boldsymbol{X}$ is uniformly distributed on $\{-1, 1\}^n$, then

$$\mathbb{P}\{|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]| \geqslant t\} \leqslant 2\exp(-cnt^2).$$

Theorem 5.2.6 (*Concentration on the symmetric group*)

Let $f : S_n \to \mathbb{R}$ be a function on the symmetric group with $\|f\|_{\text{Lip}} \leqslant 1$ with respect to the normalized Hamming metric. If π is uniformly distributed on S_n, then

$$\mathbb{P}\{|f(\pi) - \mathbb{E}[f(\pi)]| \geqslant t\} \leqslant 2\exp(-cnt^2).$$

Theorem 5.2.7 (*Concentration on the special orthogonal group*)

Let $f : SO(n) \to \mathbb{R}$ be a 1-Lipschitz function with respect to the Hilbert-Schmidt metric, and let U be uniformly distributed on $SO(n)$. Then

$$\mathbb{P}\{|f(U) - \mathbb{E}[f(U)]| \geqslant t\} \leqslant 2\exp(-cnt^2).$$

Theorem 5.2.9 (*Concentration on the Grassmannian*)

Let f be a 1-Lipschitz function on the Grassmannian $G_{n,k}$ with respect to the projection metric, and let E be uniformly distributed. Then

$$\mathbb{P}\{|f(E) - \mathbb{E}[f(E)]| \geqslant t\} \leqslant 2\exp(-cnt^2).$$

Theorem 5.2.10 (*Concentration on the continuous cube and ball*)

Let $\boldsymbol{X}$ be uniformly distributed on $[0, 1]^n$ or $\mathbb{B}_2^n$. Then for any 1-Lipschitz function f,

$$\mathbb{P}\{|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]| \geqslant t\} \leqslant 2\exp(-cnt^2).$$

Theorem 5.2.12 (*Talagrand concentration inequality*)

Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ have independent coordinates and f be a 1-Lipschitz convex function on $\mathbb{R}^n$. Then, for every $t \geqslant 0$,

$$\mathbb{P}\{|f(\boldsymbol{X}) - M| \geqslant t\} \leqslant 4\exp(-ct^2),$$

where M is a median of $f(\boldsymbol{X})$.

Section 5.3 Johnson-Lindenstrauss lemma Q5.14 ∼ Q5.15

Theorem 5.3.1 (*Johnson-Lindenstrauss Lemma*)

For any set $\mathbb{T} \subset \mathbb{R}^n$ of N points and any $\varepsilon \in (0,1)$, there exists a linear map $\boldsymbol{Q} : \mathbb{R}^n \to \mathbb{R}^m$ with $m \leqslant C\varepsilon^{-2} \log N$ such that

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \|\boldsymbol{Q}\boldsymbol{x}-\boldsymbol{Q}\boldsymbol{y}\|_2 \leqslant (1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2$$

for every $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{T}$.

Lemma 5.3.2 (*Random projection*)

Let $\boldsymbol{Q} = \frac{1}{\sqrt{m}}\boldsymbol{A}$, where $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ is a standard Gaussian random matrix. Then, for any fixed $\boldsymbol{x} \in \mathbb{R}^n$, we have

$$\mathbb{P}\{(1-\varepsilon)\|\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{Q}\boldsymbol{x}\|_2 \leqslant (1+\varepsilon)\|\boldsymbol{x}\|_2\} \geqslant 1 - 2\exp(-cm\varepsilon^2).$$

Section 5.4 Matrix Bernstein inequality Q5.16 ∼ Q5.24

Theorem 5.4.1 (*Matrix Bernstein inequality*)

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ be independent, mean-zero, n-by-n symmetric random matrices such that $\|\boldsymbol{X}_i\| \leqslant M$ almost surely. Then, for any $t \geqslant 0$, we have

$$\mathbb{P}\left\{\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\| \geqslant t\right\} \leqslant 2n \cdot \exp\left(-\frac{t^2/2}{\sigma^2 + Mt/3}\right),$$

where $\sigma^2 = \left\|\sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i^2]\right\|$.

Definition 5.4.2 (*Functions of matrices*)

For a symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ with eigendecomposition $\boldsymbol{A} = \sum_i \lambda_i \boldsymbol{u}_i \boldsymbol{u}_i^\top$ and $f : \mathbb{R} \to \mathbb{R}$, define $f(\boldsymbol{A}) = \sum_i f(\lambda_i)\boldsymbol{u}_i\boldsymbol{u}_i^\top$.

Definition 5.4.3 (*Loewner order*)

For symmetric matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n\times n}$, we write $\boldsymbol{A} \preceq \boldsymbol{B}$ if $\boldsymbol{B} - \boldsymbol{A}$ is positive semidefinite.

Proposition 5.4.4 (*Simple properties of Loewner order*)

(i) $\boldsymbol{A} \preceq \boldsymbol{B}$ implies $\operatorname{tr}(\boldsymbol{A}) \leqslant \operatorname{tr}(\boldsymbol{B})$. (ii) If $0 \preceq \boldsymbol{A} \preceq \boldsymbol{B}$, then $\|\boldsymbol{A}\| \leqslant \|\boldsymbol{B}\|$. (iii) If $0 \prec \boldsymbol{A} \preceq \boldsymbol{B}$, then $\log \boldsymbol{A} \preceq \log \boldsymbol{B}$ (operator monotonicity of the log); if $\boldsymbol{I}_n \preceq \boldsymbol{A}$, then $0 \preceq \log \boldsymbol{A}$.

Theorem 5.4.7 (*Golden-Thompson inequality*)

For any symmetric matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n\times n}$, we have

$$\operatorname{tr}(\mathrm{e}^{\boldsymbol{A}+\boldsymbol{B}}) \leqslant \operatorname{tr}(\mathrm{e}^{\boldsymbol{A}}\mathrm{e}^{\boldsymbol{B}}).$$

Theorem 5.4.8 (*Lieb inequality*)

For any fixed symmetric matrix $\boldsymbol{H} \in \mathbb{R}^{n\times n}$, the function $\boldsymbol{A} \mapsto \operatorname{tr}[\exp(\boldsymbol{H} + \log \boldsymbol{A})]$ is concave on the set of positive definite matrices $\boldsymbol{A}$.

Lemma 5.4.9 (*Lieb inequality for random matrices*)

Let $\boldsymbol{H} \in \mathbb{R}^{n\times n}$ be a fixed symmetric matrix. Then

$$\mathbb{E}\left[\operatorname{tr}\exp(\boldsymbol{H} + \sum_{i=1}^N \boldsymbol{X}_i)\right] \leqslant \operatorname{tr}[\exp\left(\boldsymbol{H} + \sum_{i=1}^N \log \mathbb{E}[\exp \boldsymbol{X}_i]\right)],$$

where $\boldsymbol{X}_i$ are independent random symmetric matrices.

Lemma 5.4.10 (*Moment generating function*)

For a symmetric random matrix $\boldsymbol{X} \in \mathbb{R}^{n\times n}$, whenever $\mathbb{E}[\boldsymbol{X}] = 0$ and $\|\mathbb{E}[\boldsymbol{X}^2]\| \leqslant \sigma^2$, for $|\lambda| \leqslant 1/M$ with $\|\boldsymbol{X}\| \leqslant M$ a.s., we have

$$\mathbb{E}[\operatorname{tr}\exp(\lambda \boldsymbol{X})] \leqslant n \cdot \exp(\lambda^2\sigma^2/2).$$

Theorem 5.4.13 (*Matrix Hoeffding inequality*)

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N \in \mathbb{R}^{n\times n}$ be independent, mean-zero symmetric random matrices such that $\|\boldsymbol{X}_i\| \leqslant M_i$ almost surely. Then, for any $t \geqslant 0$,

$$\mathbb{P}\left\{\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\| \geqslant t\right\} \leqslant 2n \cdot \exp\left(-\frac{t^2}{2\sum_{i=1}^N M_i^2}\right).$$

Theorem 5.4.14 (*Matrix Khintchine inequality*)

Let $\varepsilon_1, \cdots, \varepsilon_N$ be independent Rademacher random variables, and let $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_N$ be fixed n-by-n symmetric matrices. Then, for any $p \geqslant 1$,

$$\left(\mathbb{E}\left\|\sum_i \varepsilon_i \boldsymbol{A}_i\right\|^p\right)^{1/p} \leqslant C\sqrt{p+\log n} \cdot \left\|\sum_i \boldsymbol{A}_i^2\right\|^{1/2}.$$

Section 5.5 Community detection in sparse networks Q5.25

Theorem 5.5.1 (*Spectral clustering for sparse stochastic block model*)

Consider the SBM with n vertices and edge probabilities a/n and b/n (sparse regime). If $(a-b)^2 > C(a+b)$, then spectral clustering applied to the adjacency matrix partially recovers the communities (w.h.p.) (the fraction of misclassified vertices is $o(1)$).

Section 5.6 Covariance estimation for general distributions Q5.26 ∼ Q5.32

Theorem 5.6.1 (*General covariance estimation*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a centered[1]random vector with covariance matrix $\boldsymbol{\Sigma}$, so $\mathbb{E}[\boldsymbol{X}] = \boldsymbol{0}$. Assume that for any $\boldsymbol{v} \in \mathbb{S}^{n-1}$, the marginal $\langle \boldsymbol{X}, \boldsymbol{v}\rangle$ has subexponential norm at most K. Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ be independent copies and

$$\boldsymbol{\Sigma}_N = \frac{1}{N}\sum_{i=1}^N \boldsymbol{X}_i \boldsymbol{X}_i^\top,$$

be the sample covariance. Then, for any $t \geqslant 0$, with high probability, we have

$$\mathbb{P}\left\{\|\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}\| \leqslant CK^2 \max\left(\sqrt{\frac{n+t}{N}}, \frac{n+t}{N}\right)\|\boldsymbol{\Sigma}\|\right\} \geqslant 1 - 2\exp(-ct).$$

Remark 5.6.5 (*High-probability guarantees*)

The argument actually gives a more informative high-probability guarantee: under the conditions of Theorem T5.6.1 (*General covariance estimation*), for any $u \geqslant 0$,

$$\mathbb{P}\left\{\|\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}\| \leqslant CK^2 r\frac{\log n + u}{N}\|\boldsymbol{\Sigma}\|\right\} \geqslant 1 - 2\mathrm{e}^{-u},$$

where $r = \operatorname{tr}(\boldsymbol{\Sigma})/\|\boldsymbol{\Sigma}\|$ is the effective rank.

Section 5.7 Bounded differences inequality Q5.1 ∼ Q5.32

Theorem 5.7.1 (*Bounded differences inequality*)

Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ have independent coordinates, and let $f : \mathbb{R}^n \to \mathbb{R}$ satisfy the bounded differences condition: $|f(\boldsymbol{x}) - f(\boldsymbol{x}')| \leqslant c_i$ whenever $\boldsymbol{x}$ and $\boldsymbol{x}'$ differ only in the i-th coordinate. Then

$$\mathbb{P}\{f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})] \geqslant t\} \leqslant \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right)$$

for every $t \geqslant 0$.

Exercises

1 **[Continuity, Differentiability, and Lipschitz Functions]**

(1) (Lipschitz implies uniform continuity) Prove: any Lipschitz continuous function is necessarily uniformly continuous.

(2) (Gradient controls the Lipschitz norm) Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable. Prove: f satisfies the Lipschitz condition (provided the right-hand side is bounded), and its Lipschitz constant $\|f\|_{\mathrm{Lip}}$ is controlled by the supremum of the Euclidean norm of its gradient,

$$\|f\|_{\mathrm{Lip}} \leqslant \sup_{\boldsymbol{x} \in \mathbb{R}^n} \|\nabla f(\boldsymbol{x})\|_2.$$

(3) (Counterexample 1) Construct a function f defined on $[-1, 1]$ that is uniformly continuous but not Lipschitz continuous.

(4) (Counterexample 2) Construct a function f defined on $[-1, 1]$ that is Lipschitz continuous but not everywhere differentiable.

Prf. (1) Let $f : D \to \mathbb{R}^m$ be Lipschitz continuous, i.e., there exists $L \geqslant 0$ such that for any $\boldsymbol{x}, \boldsymbol{y} \in D$,

$$\|f(\boldsymbol{x}) - f(\boldsymbol{y})\| \leqslant L\|\boldsymbol{x} - \boldsymbol{y}\|.$$

(*Note that the definition of uniform continuity is: for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $\boldsymbol{x}, \boldsymbol{y} \in D$ satisfying $\|\boldsymbol{x} - \boldsymbol{y}\| < \delta$, we have $\|f(\boldsymbol{x}) - f(\boldsymbol{y})\| < \varepsilon$.*)

a. If $L = 0$, then f is a constant function, which is trivial.

b. If $L > 0$, then for any $\varepsilon > 0$, letting $\delta = \varepsilon / L$, we obtain

$$\|f(\boldsymbol{x}) - f(\boldsymbol{y})\| \leqslant L\|\boldsymbol{x} - \boldsymbol{y}\| < L \cdot \frac{\varepsilon}{L} = \varepsilon,$$

hence f is uniformly continuous.

(2) Construct the auxiliary function

$$g(t) = f(\boldsymbol{x} + t(\boldsymbol{y} - \boldsymbol{x})), \qquad (t \in (0, 1))$$

then

$$g'(t) = \nabla f(\boldsymbol{x} + t(\boldsymbol{y} - \boldsymbol{x}))^\top (\boldsymbol{y} - \boldsymbol{x}).$$

By the Lagrange mean value theorem, there exists $\theta \in (0, 1)$ such that

$$g(1) - g(0) = g'(\theta) \cdot (1 - 0),$$

Denote $\boldsymbol{z} = \boldsymbol{x} + \theta(\boldsymbol{y} - \boldsymbol{x})$, i.e.,

$$f(\boldsymbol{y}) - f(\boldsymbol{x}) = \nabla f(\boldsymbol{z})^\top (\boldsymbol{y} - \boldsymbol{x}),$$

Taking absolute values on both sides and using Cauchy's inequality, we obtain

$$|f(\boldsymbol{y}) - f(\boldsymbol{x})| = |\langle \nabla f(\boldsymbol{z}), \boldsymbol{y} - \boldsymbol{x} \rangle| \leqslant \|\nabla f(\boldsymbol{z})\|_2 \|\boldsymbol{y} - \boldsymbol{x}\|_2 \leqslant \left(\sup_{\boldsymbol{w} \in \mathbb{R}^n} \|\nabla f(\boldsymbol{w})\|_2\right) \|\boldsymbol{y} - \boldsymbol{x}\|_2,$$

hence

$$\|f\|_{\mathrm{Lip}} = \sup_{\boldsymbol{x} \neq \boldsymbol{y}} \frac{|f(\boldsymbol{x}) - f(\boldsymbol{y})|}{\|\boldsymbol{x} - \boldsymbol{y}\|_2} \leqslant \sup_{\boldsymbol{x} \in \mathbb{R}^n} \|\nabla f(\boldsymbol{x})\|_2.$$

(*If the supremum of the gradient on the right-hand side is infinite, the conclusion is trivial, but the function is not globally Lipschitz. When the gradient is bounded, the function is necessarily Lipschitz continuous.*)

(3) Consider $f(x)=\sqrt{|x|}$, then $f\in UC[-1,1]$, but

$$\frac{|f(x)-f(0)|}{|x-0|}=\frac{\sqrt{x}-0}{x}=\frac{1}{\sqrt{x}}\to\infty,$$

i.e., f is not Lipschitz continuous.

(4) Consider $f(x)=|x|$, then for any $x,y\in[-1,1]$,

$$|f(x)-f(y)|=||x|-|y||\leqslant|x-y|,$$

i.e., f is Lipschitz continuous, but

$$\lim_{h\to0^+}\frac{|0+h|-|0|}{h}=\lim_{h\to0^+}\frac{h}{h}=1,$$
$$\lim_{h\to0^-}\frac{|0+h|-|0|}{h}=\lim_{h\to0^-}\frac{-h}{h}=-1,$$

i.e., $f\notin D\{0\}$.

(*Lipschitz continuity (controlling the maximum absolute value of the slope) does not mean the curve is necessarily smooth (there may be corners).*)

2 [**Verifying Properties of Natural Lipschitz Functions**] Verify all assertions in Example E5.1.2 (*Vectors, matrices and norms define natural Lipschitz functions*). Specifically, find the Lipschitz norm $\|f\|_{\text{Lip}}$ of the following functions.

(1) (Linear functional) For any $\boldsymbol{\theta}\in\mathbb{R}^n$, define the linear functional $f(\boldsymbol{x})=\langle\boldsymbol{x},\boldsymbol{\theta}\rangle$. Prove: $\|f\|_{\text{Lip}}=\|\boldsymbol{\theta}\|_2$.

(2) (Linear operator) For any $\boldsymbol{A}\in\mathbb{R}^{m\times n}$, define the linear operator $f(\boldsymbol{x})=\boldsymbol{A}\boldsymbol{x}$. Prove: $\|f\|_{\text{Lip}}=\|\boldsymbol{A}\|$.

(3) (General norm) For any norm $\|\cdot\|$ on $\mathbb{R}^n$, define $f(\boldsymbol{x})=\|\boldsymbol{x}\|$. Prove: $\|f\|_{\text{Lip}}$ is the smallest constant L satisfying $\|\boldsymbol{x}\|\leqslant L\|\boldsymbol{x}\|_2$.

Prf. The Lipschitz norm is defined as

$$\|f\|_{\text{Lip}}=\sup_{\boldsymbol{x}\neq\boldsymbol{y}}\frac{\|f(\boldsymbol{x})-f(\boldsymbol{y})\|}{\|\boldsymbol{x}-\boldsymbol{y}\|_2},$$

(*When the codomain is $\mathbb{R}$, the norm in the numerator is simply the absolute value $|\cdot|$.*)

The operator norm is defined as

$$\|\boldsymbol{A}\|=\sup_{\boldsymbol{v}\neq\boldsymbol{0}}\frac{\|\boldsymbol{A}\boldsymbol{v}\|_2}{\|\boldsymbol{v}\|_2},$$

$L_{\min}$ is defined as

$$L_{\min}=\sup_{\boldsymbol{v}\neq\boldsymbol{0}}\frac{\|\boldsymbol{v}\|}{\|\boldsymbol{v}\|_2}.$$

(1) For the upper bound, let $\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n$, then using Cauchy's inequality, we obtain

$$|f(\boldsymbol{x})-f(\boldsymbol{y})|=|\langle\boldsymbol{x},\boldsymbol{\theta}\rangle-\langle\boldsymbol{y},\boldsymbol{\theta}\rangle|=|\langle\boldsymbol{x}-\boldsymbol{y},\boldsymbol{\theta}\rangle|\leqslant\|\boldsymbol{x}-\boldsymbol{y}\|_2\|\boldsymbol{\theta}\|_2,$$

hence $\|f\|_{\text{Lip}}\leqslant\|\boldsymbol{\theta}\|_2$.

For the lower bound, if $\boldsymbol{\theta}=\boldsymbol{0}$, the conclusion is trivial; if $\boldsymbol{\theta}\neq\boldsymbol{0}$, let $\boldsymbol{x}=\boldsymbol{\theta}$, $\boldsymbol{y}=\boldsymbol{0}$, then

$$\frac{|f(\boldsymbol{\theta})-f(\boldsymbol{0})|}{\|\boldsymbol{\theta}-\boldsymbol{0}\|_2}=\frac{|\langle\boldsymbol{\theta},\boldsymbol{\theta}\rangle|}{\|\boldsymbol{\theta}\|_2}=\frac{\|\boldsymbol{\theta}\|_2^2}{\|\boldsymbol{\theta}\|_2}=\|\boldsymbol{\theta}\|_2,$$

hence $\|f\|_{\text{Lip}}\geqslant\|\boldsymbol{\theta}\|_2$.

Thus, $\|f\|_{\text{Lip}}=\|\boldsymbol{\theta}\|_2$.

(2) For the upper bound, let $\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n$, then

$$\|f(\boldsymbol{x})-f(\boldsymbol{y})\|_2=\|\boldsymbol{A}\boldsymbol{x}-\boldsymbol{A}\boldsymbol{y}\|_2=\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2\leqslant\|\boldsymbol{A}\|\|\boldsymbol{x}-\boldsymbol{y}\|_2,$$

hence $\|f\|_{\text{Lip}}\leqslant\|\boldsymbol{A}\|$.

For the lower bound, let $\boldsymbol{y}=\boldsymbol{0}$, then for any $\boldsymbol{x}\neq\boldsymbol{0}$,

$$\frac{\|f(\boldsymbol{x})-f(\boldsymbol{0})\|_2}{\|\boldsymbol{x}-\boldsymbol{0}\|_2}=\frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2},$$

Taking the supremum on both sides,

$$\|f\|_{\mathrm{Lip}}=\sup_{\boldsymbol{x}\neq\boldsymbol{y}}\frac{\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2}{\|\boldsymbol{x}-\boldsymbol{y}\|_2}\geqslant\sup_{\boldsymbol{x}\neq\boldsymbol{0}}\frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2}=\|\boldsymbol{A}\|,$$

hence $\|f\|_{\mathrm{Lip}}\geqslant\|\boldsymbol{A}\|$.

Thus, $\|f\|_{\mathrm{Lip}}=\|\boldsymbol{A}\|$.

(3) For the upper bound, let $\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n$, then

$$|f(\boldsymbol{x})-f(\boldsymbol{y})|=|\|\boldsymbol{x}\|-\|\boldsymbol{y}\||\leqslant\|\boldsymbol{x}-\boldsymbol{y}\|,$$

hence

$$|f(\boldsymbol{x})-f(\boldsymbol{y})|\leqslant L_{\min}\|\boldsymbol{x}-\boldsymbol{y}\|_2,$$

i.e., $\|f\|_{\mathrm{Lip}}\leqslant L_{\min}$.

For the lower bound, similarly to the conclusion of (2), let $\boldsymbol{y}=\boldsymbol{0}$, then for any $\boldsymbol{x}\neq\boldsymbol{0}$,

$$\frac{|f(\boldsymbol{x})-f(\boldsymbol{0})|}{\|\boldsymbol{x}-\boldsymbol{0}\|_2}=\frac{|\|\boldsymbol{x}\|-0|}{\|\boldsymbol{x}\|_2}=\frac{\|\boldsymbol{x}\|}{\|\boldsymbol{x}\|_2},$$

Taking the supremum on both sides,

$$\|f\|_{\mathrm{Lip}}\geqslant\sup_{\boldsymbol{x}\neq\boldsymbol{0}}\frac{\|\boldsymbol{x}\|}{\|\boldsymbol{x}\|_2}=L_{\min}.$$

Thus, $\|f\|_{\mathrm{Lip}}=L_{\min}$.

3 [**Expansion of Exponentially Small Sets**] Let σ be the normalized uniform probability measure on $\sqrt{n}\mathbb{S}^{n-1}$. For any $X\subset\sqrt{n}\mathbb{S}^{n-1}$, define its r-neighborhood as $X_r=\{\boldsymbol{x}\in\sqrt{n}\mathbb{S}^{n-1}:\operatorname{dist}(\boldsymbol{x},X)\leqslant r\}$. Let $A\subset\sqrt{n}\mathbb{S}^{n-1}$ whose measure satisfies

$$\sigma(A)>2\exp(-cs^2).\qquad(c,s>0)$$

(1) Prove: after s-expansion, the measure of A crosses the half-way mark, i.e., $\sigma(A_s)>1/2$.

(2) Deduce: for any $t\geqslant s$, after $2t$-expansion, A will occupy almost all the volume, satisfying

$$\sigma(A_{2t})\geqslant1-2\exp(-ct^2).$$

Prf. (1) Call the following conclusion the separation lemma. Let $X,Y\subset\sqrt{n}\mathbb{S}^{n-1}$ satisfy $\sigma(Y)\geqslant1/2$. If $d(X,Y)=\inf_{\boldsymbol{x}\in X,\boldsymbol{y}\in Y}\|\boldsymbol{x}-\boldsymbol{y}\|_2\geqslant r$, then

$$\sigma(X)\leqslant2\exp(-cr^2).$$

By Theorem T5.1.5 (Isoperimetric inequality on the sphere), the measure of the closed t-neighborhood $Y_t=\{\boldsymbol{z}:d(\boldsymbol{z},Y)\leqslant t\}$ satisfies

$$\sigma(Y_t)\geqslant1-2\exp(-ct^2).$$

Equivalently, considering its complement $(Y_t)^c=\{\boldsymbol{z}:d(\boldsymbol{z},Y)>t\}$, we have

$$\sigma(\{\boldsymbol{z}:d(\boldsymbol{z},Y)>t\})\leqslant2\exp(-ct^2).$$

Let $\boldsymbol{x}\in X$, then $d(\boldsymbol{x},Y)\geqslant r$, i.e., $d(\boldsymbol{x},Y)>r-\varepsilon$ where $\varepsilon\in(0,r)$, hence

$$X\subset\{\boldsymbol{z}:d(\boldsymbol{z},Y)>r-\varepsilon\}.$$

Therefore

$$\sigma(X)\leqslant\sigma(\{\boldsymbol{z}:d(\boldsymbol{z},Y)>r-\varepsilon\})\leqslant2\exp(-c(r-\varepsilon)^2),$$

Letting $\varepsilon\to0^+$, we obtain $\sigma(X)\leqslant2\exp(-cr^2)$.

Let $Y=(A_s)^c$, then $d(A,Y)\geqslant s$.

Assume, for contradiction, that $\sigma(A_s) \leqslant 1/2$, then the measure of its complement satisfies

$$\sigma(Y) = 1 - \sigma(A_s) \geqslant \frac{1}{2},$$

Hence by the lemma, we obtain $\sigma(A) \leqslant 2\exp(-cs^2)$, which contradicts $\sigma(A) > 2\exp(-cs^2)$. Thus, $\sigma(A_s) > 1/2$.

(2) Expand A_s by radius t, then its measure satisfies

$$\sigma((A_s)_t) \geqslant 1 - 2\exp(-ct^2).$$

By the triangle inequality for the metric, we obtain $(A_s)_t \subset A_{s+t}$. Note that $t \geqslant s$, hence $A_{s+t} \subset A_{2t}$, i.e.,

$$(A_s)_t \subset A_{2t},$$

therefore

$$\sigma(A_{2t}) \geqslant \sigma((A_s)_t) \geqslant 1 - 2\exp(-ct^2).$$

4 [**Measure Concentration in the Geodesic Metric**] In Theorem T5.1.4 (*Isoperimetric inequality on* $\mathbb{R}^n$) (*Isoperimetric inequality on* $\mathbb{R}^n$), we gave a measure concentration inequality for Lipschitz continuous functions on the sphere with respect to the Euclidean metric.

Prove: the same concentration result also holds for functions that are Lipschitz continuous with respect to the geodesic metric.

Rk. The Euclidean metric is the chord length between two points, denoted $\rho(x, y) = \|x - y\|_2$, and the geodesic metric is the arc length between two points, denoted $d(x, y) = \theta$, where $\theta = \arccos(\langle x, y\rangle)$.

Prf. Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{S}^{n-1}$ with geodesic distance $\theta = d(\boldsymbol{x}, \boldsymbol{y}) \in [0, \pi]$, then

$$\|\boldsymbol{x} - \boldsymbol{y}\|_2^2 = \|\boldsymbol{x}\|_2^2 + \|\boldsymbol{y}\|_2^2 - 2\langle \boldsymbol{x}, \boldsymbol{y}\rangle = 1 + 1 - 2\cos\theta = 2(1 - \cos\theta) = 4\sin^2\frac{\theta}{2},$$

Taking square roots on both sides,

$$\|\boldsymbol{x} - \boldsymbol{y}\|_2 = 2\sin\frac{\theta}{2} = 2\sin\frac{d(\boldsymbol{x}, \boldsymbol{y})}{2} \geqslant 2\left(\frac{2}{\pi}\frac{d(\boldsymbol{x}, \boldsymbol{y})}{2}\right) = \frac{2d(\boldsymbol{x}, \boldsymbol{y})}{\pi},$$

i.e.,

$$d(\boldsymbol{x}, \boldsymbol{y}) \leqslant \frac{\pi}{2}\|\boldsymbol{x} - \boldsymbol{y}\|_2.$$

Let $f : \mathbb{S}^{n-1} \to \mathbb{R}$ be L-Lipschitz with respect to the geodesic metric, then for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{S}^{n-1}$,

$$|f(\boldsymbol{x}) - f(\boldsymbol{y})| \leqslant L \cdot d(\boldsymbol{x}, \boldsymbol{y}) \leqslant L\frac{\pi}{2}\|\boldsymbol{x} - \boldsymbol{y}\|_2,$$

hence f is $L\pi/2$-Lipschitz with respect to the Euclidean metric. Denote $L' = L\pi/2$, then by Theorem T5.1.5 (*Isoperimetric inequality on the sphere*), we obtain

$$\mathbb{P}\{|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]| \geqslant t\} \leqslant 2\exp\left(-cn\frac{t^2}{(L')^2}\right). \qquad (c, C > 0)$$

5 [**Measure Concentration on the Unit Sphere**] It is known that Theorem T5.1.4 (*Isoperimetric inequality on* $\mathbb{R}^n$) (*Isoperimetric inequality on* $\mathbb{R}^n$) gives a measure concentration inequality for Lipschitz functions on the scaled sphere $\sqrt{n}\mathbb{S}^{n-1}$. Let $\boldsymbol{X} \sim U(\mathbb{S}^{n-1})$, and $f : \mathbb{S}^{n-1} \to \mathbb{R}$ be a Lipschitz continuous function. Prove: the fluctuation of $f(\boldsymbol{X})$ can be bounded by its sub-Gaussian norm,

$$\|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2} \leqslant \frac{C\|f\|_{\mathrm{Lip}}}{\sqrt{n}},$$

This conclusion is equivalent to: for any $t \geqslant 0$, the following Gaussian-type tail probability bound holds,

$$\mathbb{P}\{|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]| \geqslant t\} \leqslant 2\exp\left(-\frac{cnt^2}{\|f\|_{\mathrm{Lip}}^2}\right).$$

Prf. (*To apply Theorem T5.1.4 (Isoperimetric inequality on* $\mathbb{R}^n$*) (Isoperimetric inequality on* $\mathbb{R}^n$*), we need to map the unit*

sphere $\mathbb{S}^{n-1}$ to the scaled sphere $\sqrt{n}\mathbb{S}^{n-1}$.)

Let $\boldsymbol{Y} = \sqrt{n}\boldsymbol{X} \sim U(\sqrt{n}\mathbb{S}^{n-1})$, and construct the auxiliary function $g : \sqrt{n}\mathbb{S}^{n-1} \to \mathbb{R}$ as

$$g(\boldsymbol{y}) = f\left(\frac{\boldsymbol{y}}{\sqrt{n}}\right),$$

then

$$g(\boldsymbol{Y}) = f\left(\frac{\sqrt{n}\boldsymbol{X}}{\sqrt{n}}\right) = f(\boldsymbol{X}).$$

For any $\boldsymbol{y}_1, \boldsymbol{y}_2 \in \sqrt{n}\mathbb{S}^{n-1}$,

$$\begin{aligned} |g(\boldsymbol{y}_1) - g(\boldsymbol{y}_2)| &= \left| f\left(\frac{\boldsymbol{y}_1}{\sqrt{n}}\right) - f\left(\frac{\boldsymbol{y}_2}{\sqrt{n}}\right) \right| \\ &\leqslant \|f\|_{\mathrm{Lip}} \left\| \frac{\boldsymbol{y}_1}{\sqrt{n}} - \frac{\boldsymbol{y}_2}{\sqrt{n}} \right\|_2 \\ &\leqslant \frac{\|f\|_{\mathrm{Lip}}}{\sqrt{n}} \|\boldsymbol{y}_1 - \boldsymbol{y}_2\|_2, \end{aligned}$$

hence by Theorem T5.1.4 (*Isoperimetric inequality on $\mathbb{R}^n$*) (*Isoperimetric inequality on $\mathbb{R}^n$*), we obtain

$$\|g(\boldsymbol{Y}) - \mathbb{E}[g(\boldsymbol{Y})]\|_{\psi_2} \leqslant C_0 \|g\|_{\mathrm{Lip}} \leqslant \frac{C_0 \|f\|_{\mathrm{Lip}}}{\sqrt{n}},$$

Now set $C = C_0$.

By Proposition P2.6.1 (*Subgaussian properties*), letting $Z = f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]$, we obtain

$$\begin{aligned} \mathbb{P}\{|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]| \geqslant t\} &\leqslant 2\exp\left(-\frac{ct^2}{(C\|f\|_{\mathrm{Lip}}/\sqrt{n})^2}\right) && \left(K = \frac{C\|f\|_{\mathrm{Lip}}}{\sqrt{n}}, c > 0\right) \\ &= 2\exp\left(-\frac{c' n t^2}{\|f\|_{\mathrm{Lip}}^2}\right). && \left(c' = \frac{c}{C}\right) \end{aligned}$$

6 [**Equivalence of Mean and Median Concentration**] Let Z be a sub-Gaussian random variable with median M. Prove:

$$\|Z - M\|_{\psi_2} \asymp \|Z - \mathbb{E}[Z]\|_{\psi_2}.$$

(*This conclusion shows that in all concentration inequalities, we can freely replace the mean with the median, and vice versa, without changing the order of the tail decay.*)

Prf. It suffices to prove

$$c\|Z - \mathbb{E}[Z]\|_{\psi_2} \leqslant \|Z - M\|_{\psi_2} \leqslant C\|Z - \mathbb{E}[Z]\|_{\psi_2}.$$

- Proof of the left inequality $c\|Z - \mathbb{E}[Z]\|_{\psi_2} \leqslant \|Z - M\|_{\psi_2}$

  Let $Y = Z - M$, then $\mathbb{E}[Y] = \mathbb{E}[Z] - M$. By the definition of the ψ_2 norm $\mathbb{E}[\exp(a^2/K^2)] \leqslant 2$, we obtain

  $$\|\mathbb{E}[Y]\|_{\psi_2} = \frac{|\mathbb{E}[Y]|}{\sqrt{\log 2}}.$$

  Consider

  $$Z - \mathbb{E}[Z] = (Z - M) - (\mathbb{E}[Z] - M) = Y - \mathbb{E}[Y],$$

  Taking the ψ_2 norm on both sides,

  $$\|Z - \mathbb{E}[Z]\|_{\psi_2} = \|Y - \mathbb{E}[Y]\|_{\psi_2} \leqslant \|Y\|_{\psi_2} + \|\mathbb{E}[Y]\|_{\psi_2} \leqslant \|Y\|_{\psi_2} + \frac{1}{\sqrt{\log 2}} |\mathbb{E}[Y]|,$$

  where

  $$|\mathbb{E}[Y]| \leqslant \mathbb{E}[|Y|] \leqslant C_1 \|Y\|_{\psi_2}, \qquad (C_1 > 0)$$

hence

$$\|Z-\mathbb{E}[Z]\|_{\psi_2} \leqslant \left(1+\frac{C_1}{\sqrt{\log 2}}\right)\|Y\|_{\psi_2} = C_2\|Z-M\|_{\psi_2}, \qquad \left(C_2 = 1+\frac{C_1}{\sqrt{\log 2}}\right)$$

i.e.,

$$c\|Z-\mathbb{E}[Z]\|_{\psi_2} \leqslant \|Z-M\|_{\psi_2}. \qquad (c = 1/C_2)$$

- Proof of the right inequality $\|Z-M\|_{\psi_2} \leqslant C\|Z-\mathbb{E}[Z]\|_{\psi_2}$

  (*To prove the right-hand upper bound, we need to control the absolute deviation between the mean and the median* $|\mathbb{E}[Z]-M|$.)

  State the following lemma:

  Lemma $|\mathbb{E}[Z]-M| \leqslant C_3\|Z-\mathbb{E}[Z]\|_{\psi_2}$.

  Let $X = Z-\mathbb{E}[Z]$, *and denote* $K = \|X\|_{\psi_2}$. *If* $K=0$, *then* $\mathbb{E}[Z]=M$, *which is trivial; we only need to consider the case* $K>0$.

  Since X *is a zero-mean* K*-sub-Gaussian random variable, for any* $t \geqslant 0$,

$$\mathbb{P}\{|X| \geqslant t\} \leqslant 2\exp\left(-\frac{c_0 t^2}{K^2}\right), \qquad (c_0>0)$$

  Denote the median of X *as* $M(X) = M-\mathbb{E}[Z]$, *satisfying*

$$\mathbb{P}\{X \geqslant M(X)\} \geqslant \frac{1}{2},$$
$$\mathbb{P}\{X \leqslant M(X)\} \geqslant \frac{1}{2},$$

  Consider the sign of $M(X)$*:*

  a. If $M(X)>0$, *then* X *falls in the right tail, satisfying*

$$\frac{1}{2} \leqslant \mathbb{P}\{X \geqslant M(X)\} \leqslant \mathbb{P}\{|X| \geqslant M(X)\} \leqslant 2\exp\left(-\frac{c_0 M(X)^2}{K^2}\right),$$

  i.e.,

$$\exp\left(\frac{c_0 M(X)^2}{K^2}\right) \leqslant 4,$$
$$M(X)^2 \leqslant \frac{\log 4}{c_0}K^2.$$

  b. If $M(X)<0$, *then* X *falls in the left tail, satisfying*

$$\frac{1}{2} \leqslant \mathbb{P}\{X \leqslant M(X)\} \leqslant \mathbb{P}\{|X| \geqslant |M(X)|\} \leqslant 2\exp\left(-\frac{c_0|M(X)|^2}{K^2}\right),$$

  i.e.,

$$\exp\left(\frac{c_0 M(X)^2}{K^2}\right) \leqslant 4,$$
$$M(X)^2 \leqslant \frac{\log 4}{c_0}K^2.$$

  Thus,

$$|\mathbb{E}[Z]-M| = |M(X)| \leqslant C_3 K = C_3\|Z-\mathbb{E}[Z]\|_{\psi_2}. \qquad (C_3 = \sqrt{(\log 4)/c_0})$$

  Decompose $Z-M$,

$$\begin{aligned}\|Z-M\|_{\psi_2} &= \|(Z-\mathbb{E}[Z])+(\mathbb{E}[Z]-M)\|_{\psi_2}\\ &\leqslant \|Z-\mathbb{E}[Z]\|_{\psi_2}+\|\mathbb{E}[Z]-M\|_{\psi_2}\\ &\leqslant \|Z-\mathbb{E}[Z]\|_{\psi_2}+\frac{1}{\sqrt{\log 2}}|\mathbb{E}[Z]-M|,\end{aligned}$$

By the lemma, we obtain

$$\|Z - M\|_{\psi_2} \leqslant \|Z - \mathbb{E}[Z]\|_{\psi_2} + \frac{C_3}{\sqrt{\log 2}}\|Z - \mathbb{E}[Z]\|_{\psi_2} = C\|Z - \mathbb{E}[Z]\|_{\psi_2}. \qquad \left(C = 1 + \frac{C_3}{\sqrt{\log 2}}\right)$$

7 [**Equivalence of Measure Concentration and Volume Expansion**] In Section 5.1.4, we derived the spherical Lipschitz concentration inequality via Lemma L5.1.6 (*Blow-up*).

Let us reverse this logic and prove that Lipschitz concentration and volume expansion are always equivalent.

Let X be a random variable (or vector) taking values in a metric space $(\mathbb{T}, d)$ such that for any Lipschitz continuous function $f : \mathbb{T} \to \mathbb{R}$,

$$\|f(X) - \mathbb{E}[f(X)]\|_{\psi_2} \leqslant K\|f\|_{\mathrm{Lip}}. \qquad (K > 0)$$

For any subset $A \subset \mathbb{T}$, define the probability measure $\sigma(A) = \mathbb{P}\{X \in A\}$. Prove: if $\sigma(A) \geqslant 1/2$, then for any $t \geqslant 0$,

$$\sigma(A_t) \geqslant 1 - 2\exp\left(-\frac{ct^2}{K^2}\right). \qquad (c > 0)$$

Prf. • Determine the median of the distance function

Since

$$\mathbb{P}\{X \in A\} = \sigma(A) \geqslant \frac{1}{2},$$

when $X \in A$, its distance to A is 0, i.e., $f(X) = 0$, hence

$$\mathbb{P}\{f(X) \leqslant 0\} \geqslant \mathbb{P}\{X \in A\} \geqslant \frac{1}{2};$$

Meanwhile, $\mathbb{P}\{f(X) \geqslant 0\} = 1 \geqslant 1/2$, hence 0 is the median of $f(X)$.

• Construct the distance function and verify the Lipschitz property

Define the distance function from $x \in \mathbb{T}$ to A as

$$f(x) = d(x, A) = \inf_{y \in A} d(x, y),$$

then for any $x, z \in \mathbb{T}$, $y \in A$,

$$d(x, y) \leqslant d(x, z) + d(z, y),$$

Taking the infimum over $y \in A$ on both sides,

$$\inf_{y \in A} d(x, y) \leqslant d(x, z) + \inf_{y \in A} d(z, y),$$

$$f(x) \leqslant d(x, z) + f(z),$$
$$f(x) - f(z) \leqslant d(x, z);$$

Similarly, swapping x and z, we obtain $f(z) - f(x) \leqslant d(x, z)$, hence $|f(x) - f(z)| \leqslant d(x, z)$, i.e., $\|f\|_{\mathrm{Lip}} \leqslant 1$, thus

$$\|f(X) - \mathbb{E}[f(X)]\|_{\psi_2} \leqslant K\|f\|_{\mathrm{Lip}} \leqslant K.$$

By Exercise Q5.6 (*Equivalence of Mean and Median Concentration*), we obtain

$$\|f(X) - M(f(X))\|_{\psi_2} \leqslant C_1\|f(X) - \mathbb{E}[f(X)]\|_{\psi_2} \leqslant C_1 K,$$

i.e., (*Note that* $M(f(X)) = 0$)

$$\|f(X)\|_{\psi_2} \leqslant C_1 K,$$

By Proposition P2.6.1 (*Subgaussian properties*), for any $t \geqslant 0$,

$$\mathbb{P}\{f(X) \geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{K^2}\right).$$

If $X \notin A_t$, then $f(X) = \inf_{y \in A} d(X, y) \geqslant t$, hence $\{X \notin A_t\} \subset \{f(X) \geqslant t\}$, thus

$$\sigma((A_t)^c) = \mathbb{P}\{X \notin A_t\} \leqslant \mathbb{P}\{f(X) \geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{K^2}\right),$$

Taking complements on both sides,

$$\sigma(A_t) = 1 - \sigma((A_t)^c) \geqslant 1 - 2\exp\left(-\frac{ct^2}{K^2}\right).$$

8 [**Proof of the Gaussian Concentration Theorem**] Theorem T5.2.2 (*Gaussian isoperimetric inequality*) states: among all subsets with the same Gaussian measure $\gamma_n(A)$, the Gaussian measure $\gamma_n(A_t)$ of the t-neighborhood of half-spaces is the smallest.

Using this theorem, prove Theorem T5.2.3 (*Gaussian concentration*): Let $\boldsymbol{X} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$, and let $f : \mathbb{R}^n \to \mathbb{R}$ be Lipschitz continuous with respect to the Euclidean metric, then

$$\|f(\boldsymbol{X}) - \mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2} \leqslant C\|f\|_{\mathrm{Lip}}. \qquad (C > 0)$$

Prf. (*Consider the standard chain from the isoperimetric inequality through neighborhood expansion and median concentration to mean concentration.*)

- Establish the expansion lemma for Gaussian measure

  Let γ_n be the standard Gaussian measure on $\mathbb{R}^n$, i.e., $\gamma_n(A) = \mathbb{P}\{\boldsymbol{X} \in A\}$ where $\boldsymbol{X} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ and $A \subset \mathbb{R}^n$ is a measurable set satisfying $\gamma_n(A) \geqslant 1/2$.

  Let the standard half-space be $H = \{\boldsymbol{x} \in \mathbb{R}^n : x_1 \leqslant 0\}$, then $\gamma_n(H) = 1/2 \leqslant \gamma_n(A)$, hence by Theorem T5.2.2 (*Gaussian isoperimetric inequality*), for any $t > 0$,

  $$\gamma_n(A_t) \geqslant \gamma_n(H_t),$$

  where $H_t = \{\boldsymbol{x} \in \mathbb{R}^n : x_1 \leqslant t\}$.

  Since

  $$\gamma_n((H_t)^c) = \frac{1}{\sqrt{2\pi}} \int_t^\infty \mathrm{e}^{-x^2/2}\mathrm{d}x \leqslant \frac{1}{2}\mathrm{e}^{-t^2/2} \leqslant \mathrm{e}^{-t^2/2},$$

  hence

  $$\gamma_n((A_t)^c) \leqslant 1 - \gamma_n(H_t) = \gamma_n((H_t)^c) \leqslant \mathrm{e}^{-t^2/2}. \qquad (\mathrm{I})$$

- Construct the inclusion relation of level sets based on Lipschitz continuity

  Let $L = \|f\|_{\mathrm{Lip}}$, M be the median of $f(\boldsymbol{X})$, and define the sublevel set $A = \{\boldsymbol{x} \in \mathbb{R}^n : f(\boldsymbol{x}) \leqslant M\}$, then $\gamma_n(A) \geqslant 1/2$.

  By the definition of the neighborhood, for any $\boldsymbol{x} \in A_t$ and any $\eta > 0$, there exists $\boldsymbol{y} \in A$ such that $\|\boldsymbol{x} - \boldsymbol{y}\|_2 \leqslant t + \eta$, hence

  $$f(\boldsymbol{x}) = f(\boldsymbol{y}) + (f(\boldsymbol{x}) - f(\boldsymbol{y})) \leqslant f(\boldsymbol{y}) + L\|\boldsymbol{x} - \boldsymbol{y}\|_2 \leqslant M + L(t + \eta).$$

  Letting $\eta \downarrow 0$, thus

  $$A_t \subset \{\boldsymbol{x} \in \mathbb{R}^n : f(\boldsymbol{x}) \leqslant M + Lt\},$$

  i.e.,

  $$\{\boldsymbol{x} \in \mathbb{R}^n : f(\boldsymbol{x}) > M + Lt\} \subset (A_t)^c. \qquad (\mathrm{II})$$

- Derive the sub-Gaussian tail bound about the median

  Taking Gaussian measure on both sides of (II) and using the bound (I), we obtain

  $$\mathbb{P}\{f(\boldsymbol{X}) - M > Lt\} = \gamma_n(\{\boldsymbol{x} \in \mathbb{R}^n : f(\boldsymbol{x}) > M + Lt\}) \leqslant \gamma_n((A_t)^c) \leqslant \mathrm{e}^{-t^2/2}.$$

  Similarly, define the superlevel set $B = \{\boldsymbol{x} \in \mathbb{R}^n : f(\boldsymbol{x}) \geqslant M\}$ satisfying $\gamma_n(B) \geqslant 1/2$, then

  $$\mathbb{P}\{f(\boldsymbol{X}) - M < -Lt\} \leqslant \mathrm{e}^{-t^2/2}.$$

  Using the union bound, for any $t \geqslant 0$,

  $$\mathbb{P}\{|f(\boldsymbol{X}) - M| \geqslant Lt\} \leqslant \mathbb{P}\{f(\boldsymbol{X}) - M \geqslant Lt\} + \mathbb{P}\{f(\boldsymbol{X}) - M \leqslant -Lt\} \leqslant 2\mathrm{e}^{-t^2/2},$$

i.e.,

$$\mathbb{P}\{|f(\boldsymbol{X})-M|\geqslant u\}\leqslant 2\exp\left(-\frac{u^2}{2L^2}\right). \qquad (u=Lt)$$

By Proposition P2.6.1 (*Subgaussian properties*), we obtain

$$\|f(\boldsymbol{X})-M\|_{\psi_2}\leqslant C_1 L=C_1\|f\|_{\mathrm{Lip}}. \qquad (C_1>0)$$

- Convert the median bound to an expectation bound

  By Exercise Q5.6 (*Equivalence of Mean and Median Concentration*), we obtain

$$\|f(\boldsymbol{X})-\mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2}\leqslant C_2\|f(\boldsymbol{X})-M\|_{\psi_2}, \qquad (C_2>0)$$

  hence

$$\|f(\boldsymbol{X})-\mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2}\leqslant C\|f\|_{\mathrm{Lip}}. \qquad (C=C_1C_2)$$

9 [**Concentration of the Maximum**] In Proposition P2.7.6 (*Maximum of subgaussians*), we estimated the expectation of the maximum of n sub-Gaussian random variables.

Now we prove that the maximum concentrates tightly around its expectation.

(1) Let $X_1,\cdots,X_n$ i.i.d. $\sim N(0,1)$. Prove:

$$\left\|\max_{1\leqslant i\leqslant n}X_i-\mathbb{E}\left[\max_{1\leqslant i\leqslant n}X_i\right]\right\|_{\psi_2}\leqslant C. \qquad (C>0)$$

(2) More generally, let $X_1,\cdots,X_n$ be jointly normal random variables (not necessarily independent). Prove:

$$\left\|\max_{1\leqslant i\leqslant n}X_i-\mathbb{E}\left[\max_{1\leqslant i\leqslant n}X_i\right]\right\|_{\psi_2}\leqslant C\max_{1\leqslant i\leqslant n}\sqrt{\mathrm{Var}(X_i)}.$$

Prf. (1) Let $\boldsymbol{X}=(X_1,\cdots,X_n)^\top\sim N(\boldsymbol{0},\boldsymbol{I}_n)$, and define $f:\mathbb{R}^n\to\mathbb{R}$ as the maximum of the components of the vector,

$$f(\boldsymbol{x})=\max_{1\leqslant i\leqslant n}x_i.$$

Then f is Lipschitz continuous, satisfying $\|f\|_{\mathrm{Lip}}\leqslant 1$.

(*Detailed proof: for any $\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n$, suppose $f(\boldsymbol{x})$ attains its maximum at index k, i.e., $f(\boldsymbol{x})=x_k$, then*

$$f(\boldsymbol{x})-f(\boldsymbol{y})=x_k-\max_{1\leqslant i\leqslant n}y_i\leqslant x_k-y_k,$$

where

$$x_k-y_k\leqslant|x_k-y_k|\leqslant\sqrt{\sum_{i=1}^n(x_i-y_i)^2}=\|\boldsymbol{x}-\boldsymbol{y}\|_2,$$

hence $f(\boldsymbol{x})-f(\boldsymbol{y})\leqslant\|\boldsymbol{x}-\boldsymbol{y}\|_2$; by symmetry, we obtain $f(\boldsymbol{y})-f(\boldsymbol{x})\leqslant\|\boldsymbol{x}-\boldsymbol{y}\|_2$, thus

$$|f(\boldsymbol{x})-f(\boldsymbol{y})|\leqslant\|\boldsymbol{x}-\boldsymbol{y}\|_2.)$$

By Theorem T5.2.3 (*Gaussian concentration*), for $\boldsymbol{X}\sim N(\boldsymbol{0},\boldsymbol{I}_n)$,

$$\|f(\boldsymbol{X})-\mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2}\leqslant C\|f\|_{\mathrm{Lip}}\leqslant C,$$

i.e.,

$$\left\|\max_{1\leqslant i\leqslant n}X_i-\mathbb{E}\left[\max_{1\leqslant i\leqslant n}X_i\right]\right\|_{\psi_2}\leqslant C.$$

(2) Let $\boldsymbol{X}=(X_1,\cdots,X_n)^\top\sim N(\boldsymbol{\mu},\boldsymbol{\Sigma})$, then there exists $\boldsymbol{A}\in\mathbb{R}^{n\times n}$ satisfying $\boldsymbol{A}\boldsymbol{A}^\top=\boldsymbol{\Sigma}$ such that

$$\boldsymbol{X}=\boldsymbol{A}\boldsymbol{g}+\boldsymbol{\mu},$$

where $\boldsymbol{g}\sim N(\boldsymbol{0},\boldsymbol{I}_n)$, i.e.,

$$X_i=\boldsymbol{a}_i^\top\boldsymbol{g}+\mu_i,$$

where $\boldsymbol{a}_i^\top$ is the i-th row of $\boldsymbol{A}$.

Note that

$$\mathrm{Var}(X_i) = \mathrm{Var}(\boldsymbol{a}_i^\top \boldsymbol{g}) = \|\boldsymbol{a}_i\|_2^2,$$

Define the function $h : \mathbb{R}^n \to \mathbb{R}$ as

$$h(\boldsymbol{g}) = \max_{1\leqslant i\leqslant n} (\boldsymbol{a}_i^\top \boldsymbol{g} + \mu_i),$$

then $\max_i X_i = h(\boldsymbol{g})$.

For any $\boldsymbol{g}_1, \boldsymbol{g}_2 \in \mathbb{R}^n$, suppose $h(\boldsymbol{g}_1)$ attains its maximum at index k, then

$$\begin{aligned} h(\boldsymbol{g}_1) - h(\boldsymbol{g}_2) &= (\boldsymbol{a}_k^\top \boldsymbol{g}_1 + \mu_k) - \max_{1\leqslant j\leqslant n} (\boldsymbol{a}_j^\top \boldsymbol{g}_2 + \mu_j) \\ &\leqslant (\boldsymbol{a}_k^\top \boldsymbol{g}_1 + \mu_k) - (\boldsymbol{a}_k^\top \boldsymbol{g}_2 + \mu_k) \\ &= \boldsymbol{a}_k^\top (\boldsymbol{g}_1 - \boldsymbol{g}_2) \\ &\leqslant \|\boldsymbol{a}_k\|_2 \|\boldsymbol{g}_1 - \boldsymbol{g}_2\|_2, \end{aligned}$$

where

$$\|\boldsymbol{a}_k\|_2 = \sqrt{\mathrm{Var}(X_k)} \leqslant \max_{1\leqslant i\leqslant n} \sqrt{\mathrm{Var}(X_i)},$$

hence

$$h(\boldsymbol{g}_1) - h(\boldsymbol{g}_2) \leqslant \left(\max_{1\leqslant i\leqslant n} \sqrt{\mathrm{Var}(X_i)} \right) \|\boldsymbol{g}_1 - \boldsymbol{g}_2\|_2,$$

Therefore by symmetry, h is Lipschitz continuous, satisfying

$$\|h\|_{\mathrm{Lip}} \leqslant \max_{1\leqslant i\leqslant n} \sqrt{\mathrm{Var}(X_i)}.$$

By Theorem T5.2.3 (*Gaussian concentration*), there exists

$$\|h(\boldsymbol{g}) - \mathbb{E}[h(\boldsymbol{g})]\|_{\psi_2} \leqslant C\|h\|_{\mathrm{Lip}} \leqslant C \max_{1\leqslant i\leqslant n} \sqrt{\mathrm{Var}(X_i)}, \qquad (C > 0)$$

i.e.,

$$\left\| \max_{1\leqslant i\leqslant n} X_i - \mathbb{E}\left[\max_{1\leqslant i\leqslant n} X_i \right] \right\|_{\psi_2} \leqslant C \max_{1\leqslant i\leqslant n} \sqrt{\mathrm{Var}(X_i)}.$$

10 [**Concentration about the L^p Norm**] In Exercise Q5.6 (*Equivalence of Mean and Median Concentration*), we proved that the mean and median are interchangeable in concentration inequalities.

Now we prove that the L^p norm also enjoys a similar replacement property.

Let Z satisfy $\mu = \mathbb{E}[Z] \geqslant 0$. Prove: for any $p \geqslant 1$,

$$\|Z - \|Z\|_{L^p}\|_{\psi_2} \leqslant C\sqrt{p}\|Z - \mathbb{E}[Z]\|_{\psi_2}.$$

Prf. Let $X = Z - \mu$, then X is a zero-mean sub-Gaussian random variable. Denote $K = \|X\|_{\psi_2}$. If $K = 0$, then Z is a nonnegative constant almost everywhere; both sides of the inequality are 0, which is trivial; we only need to consider the case $K > 0$ below.

Using the triangle inequality, we obtain

$$\begin{aligned} \|Z - \|Z\|_{L^p}\|_{\psi_2} &= \|(Z - \mu) + (\mu - \|Z\|_{L^p})\|_{\psi_2} \\ &\leqslant \|Z - \mu\|_{\psi_2} + \|\mu - \|Z\|_{L^p}\|_{\psi_2} \\ &\leqslant K + \frac{1}{\sqrt{\log 2}} |\mu - \|Z\|_{L^p}|, \end{aligned}$$

It then suffices to estimate $|\mu - \|Z\|_{L^p}|$.

For the lower bound, note that when $p \geqslant 1$, the function $x \mapsto |x|^p$ is convex, hence by Jensen's inequality,

$$\|Z\|_{L^p} = (\mathbb{E}[|Z|^p])^{1/p} \geqslant |\mathbb{E}[Z]| = |\mu| = \mu,$$

For the upper bound, using Minkowski's inequality,

$$\|Z\|_{L^p} = \|\mu + X\|_{L^p} \leqslant \|\mu\|_{L^p} + \|X\|_{L^p} = \mu + \|X\|_{L^p}.$$

Thus,

$$|\|Z\|_{L^p} - \mu| \leqslant \|X\|_{L^p}.$$

By Proposition P2.6.1 (*Subgaussian properties*), we obtain

$$\|X\|_{L^p} = (\mathbb{E}[|X|^p])^{1/p} \leqslant C_1 K\sqrt{p},$$

i.e.,

$$|\mu - \|Z\|_{L^p}| \leqslant C_1 K\sqrt{p},$$

hence

$$\|Z - \|Z\|_{L^p}\|_{\psi_2} \leqslant K + \frac{C_1 K\sqrt{p}}{\sqrt{\log 2}} \leqslant C\sqrt{p}\|Z - \mathbb{E}[Z]\|_{\psi_2}. \qquad \left(C = 1 + \frac{C_1}{\sqrt{\log 2}}\right)$$

11 [**Measure Pushforward: from Gaussian to Uniform**] Let $\Phi(x)$ be the cumulative distribution function of $N(0,1)$. Consider the n-dimensional standard normal random vector $\boldsymbol{Z} = (Z_1, \cdots, Z_n)^\top \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Prove: under the pushforward map $\varphi(\boldsymbol{Z}) = (\Phi(Z_1), \cdots, \Phi(Z_n))^\top$, this random vector is distributed as $U([0,1]^n)$, i.e.,

$$\varphi(\boldsymbol{Z}) \sim U([0,1]^n).$$

Prf. Since $\Phi \in C(\mathbb{R})$ is strictly increasing with range $(0,1)$, its inverse $\Phi^{-1} : (0,1) \to \mathbb{R}$ is well-defined. For $u \in (0,1)$,

Let $U_i = \Phi(Z_i)$, where $Z_i \sim N(0,1)$, then

$$F_{U_i}(u) = \mathbb{P}\{U_i \leqslant u\} = \mathbb{P}\{\Phi(Z_i) \leqslant u\} = \mathbb{P}\{Z_i \leqslant \Phi^{-1}(u)\} = \Phi(\Phi^{-1}(u)) = u,$$

hence $U_i \sim U(0,1)$.

For $\boldsymbol{U} = \varphi(\boldsymbol{Z}) = (U_1, \cdots, U_n)^\top$, since $Z_1, \cdots, Z_n$ are independent, $U_1, \cdots, U_n$ are independent, thus

$$F_{\boldsymbol{U}}(\boldsymbol{u}) = \prod_{i=1}^n F_{U_i}(u_i) = \prod_{i=1}^n u_i,$$

i.e., $\varphi(\boldsymbol{Z}) \sim U([0,1]^n)$.

12 [**Concentration on the Continuous Hypercube**] Let $\boldsymbol{X} \sim U([0,1]^n)$. Theorem T5.2.10 (*Concentration on the continuous cube and ball*) states: for any Lipschitz continuous function $f : [0,1]^n \to \mathbb{R}$, the random variable $f(\boldsymbol{X})$ satisfies the sub-Gaussian concentration property. Prove this theorem following these steps:

(1) Express $\boldsymbol{X} = \varphi(\boldsymbol{Z})$ using Exercise Q5.11 (*Measure Pushforward: from Gaussian to Uniform*). Then use Theorem T5.2.3 (*Gaussian concentration*) (*Gaussian concentration*) to bound the deviations of $f(\varphi(\boldsymbol{Z}))$ by $\|f\|_{\text{Lip}}\|\varphi\|_{\text{Lip}}$.

(2) Check that $\|\varphi\|_{\text{Lip}}$ is bounded by a constant, and complete the proof of the theorem.

Proof • Construct the pushforward map φ

Let $\boldsymbol{Z} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$, and define $\varphi : \mathbb{R}^n \to [0,1]^n$ as

$$\varphi(\boldsymbol{z}) = (\Phi(z_1), \Phi(z_2), \cdots, \Phi(z_n))^\top,$$

By Exercise Q5.11 (*Measure Pushforward: from Gaussian to Uniform*), we obtain $\varphi(\boldsymbol{Z}) \sim U([0,1]^n)$, hence $f(\boldsymbol{X}) \overset{d}{=} f(\varphi(\boldsymbol{Z}))$. (*It then suffices to consider the concentration properties of* $g(\boldsymbol{Z}) = f(\varphi(\boldsymbol{Z}))$.)

Let $g = f \circ \varphi : \mathbb{R}^n \to \mathbb{R}$, then g is Lipschitz continuous, satisfying $\|g\|_{\text{Lip}} \leqslant \|f\|_{\text{Lip}}\|\varphi\|_{\text{Lip}}$, and (*Detailed proof: for any* $\boldsymbol{z}_1, \boldsymbol{z}_2 \in \mathbb{R}^n$,

$$\begin{aligned} |g(\boldsymbol{z}_1) - g(\boldsymbol{z}_2)| &= |f(\varphi(\boldsymbol{z}_1)) - f(\varphi(\boldsymbol{z}_2))| \\ &\leqslant \|f\|_{\text{Lip}}\|\varphi(\boldsymbol{z}_1) - \varphi(\boldsymbol{z}_2)\|_2 \\ &\leqslant \|f\|_{\text{Lip}}\|\varphi\|_{\text{Lip}}\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|_2. \end{aligned}$$

) Hence by Theorem T5.2.3 (*Gaussian concentration*), we obtain

$$\|g(\boldsymbol{Z}) - \mathbb{E}[g(\boldsymbol{Z})]\|_{\psi_2} \leqslant c_1\|g\|_{\text{Lip}}, \qquad (c_1 > 0)$$

i.e.,

$$\|f(\boldsymbol{X})-\mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2}\leqslant c_1\|f\|_{\mathrm{Lip}}\|\varphi\|_{\mathrm{Lip}}.$$

- Bound the Lipschitz constant of the map φ

  Since

$$\Phi'(x)=\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2},$$

  its maximum value is

$$\sup_{x\in\mathbb{R}}|\Phi'(x)|=\frac{1}{\sqrt{2\pi}},$$

  hence by the Lagrange mean value theorem, for any $x,y\in\mathbb{R}$,

$$|\Phi(x)-\Phi(y)|\leqslant\frac{1}{\sqrt{2\pi}}|x-y|,$$

  thus Φ is Lipschitz continuous, satisfying $\|\Phi\|_{\mathrm{Lip}}=1/\sqrt{2\pi}$.

  For any $\boldsymbol{z}_1,\boldsymbol{z}_2\in\mathbb{R}^n$,

$$\|\varphi(\boldsymbol{z}_1)-\varphi(\boldsymbol{z}_2)\|_2^2=\sum_{i=1}^n(\Phi(z_{1i})-\Phi(z_{2i}))^2\leqslant\sum_{i=1}^n\left(\frac{1}{\sqrt{2\pi}}|z_{1i}-z_{2i}|\right)^2=\frac{1}{2\pi}\sum_{i=1}^n(z_{1i}-z_{2i})^2=\frac{1}{2\pi}\|\boldsymbol{z}_1-\boldsymbol{z}_2\|_2^2,$$

  Taking square roots on both sides,

$$\|\varphi(\boldsymbol{z}_1)-\varphi(\boldsymbol{z}_2)\|_2\leqslant\frac{1}{\sqrt{2\pi}}\|\boldsymbol{z}_1-\boldsymbol{z}_2\|_2,$$

  hence $\|\varphi\|_{\mathrm{Lip}}\leqslant1/\sqrt{2\pi}$, and then

$$\|f(\boldsymbol{X})-\mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2}\leqslant c_1\cdot\frac{1}{\sqrt{2\pi}}\|f\|_{\mathrm{Lip}}=C\|f\|_{\mathrm{Lip}}.\qquad\left(C=\frac{c_1}{\sqrt{2\pi}}\right)$$

13 [**Concentration on the Continuous Solid Ball**] Theorem T5.2.10 (*Concentration on the continuous cube and ball*) states: if $\boldsymbol{X}\sim U(\mathbb{B}_2^n)$ and $f:\mathbb{B}_2^n\to\mathbb{R}$ is a Lipschitz continuous function, then $f(\boldsymbol{X})$ satisfies the sub-Gaussian concentration property with its sub-Gaussian norm bounded by $C\|f\|_{\mathrm{Lip}}/\sqrt{n}$.

Using a strategy similar to Exercise Q5.12 (*Concentration on the Continuous Hypercube*), prove this theorem following these steps:

(1) Define a function $\varphi:\mathbb{R}^n\to\mathbb{B}_2^n$ that pushes forward the Gaussian measure into the uniform measure on $\mathbb{B}_2^n$.

(2) Check that φ has bounded Lipschitz norm, and complete the proof of the theorem.

Prf. • Control the deviation of the composite function using the Gaussian concentration inequality

Decompose $\boldsymbol{Z}\sim N(\boldsymbol{0},\boldsymbol{I}_n)$ as $\boldsymbol{Z}=r\boldsymbol{U}$, where $\boldsymbol{U}=\bar{\boldsymbol{Z}}\sim U(\mathbb{S}^{n-1})$, $r=\|\boldsymbol{Z}\|_2$ and $r^2\sim\chi^2(n)$, and $\boldsymbol{U},r$ are independent.

Similarly, decompose $\boldsymbol{Y}\sim U(\sqrt{n}\mathbb{B}_2^n)$ as $\boldsymbol{Y}=\rho\boldsymbol{V}$, where $\boldsymbol{V}\sim U(\mathbb{S}^{n-1})$, $\rho\in[0,\sqrt{n}]$ has density proportional to the volume element ρ^{n-1}, satisfying

$$F_Y(\rho)=\left(\frac{\rho}{\sqrt{n}}\right)^n,\qquad(\rho\in[0,\sqrt{n}])$$

and $\boldsymbol{V},\rho$ are independent.

To map $\boldsymbol{Z}$ to $\boldsymbol{Y}$, let $\boldsymbol{V}=\boldsymbol{U}$ (*keeping the direction unchanged*), and only apply a monotone map to the radial length $\rho=g(r)$ satisfying $F_Y(\rho)=F_{\chi_n}(r)$, which yields

$$\left(\frac{g(r)}{\sqrt{n}}\right)^n=F_{\chi_n}(r),$$

$$g(r)=\sqrt{n}[F_{\chi_n}(r)]^{1/n},$$

Hence construct the pushforward map $\varphi : \mathbb{R}^n \to \sqrt{n}\mathbb{B}_2^n$ as

$$\varphi(\boldsymbol{z}) = g(\|\boldsymbol{z}\|_2)\frac{\boldsymbol{z}}{\|\boldsymbol{z}\|_2},$$

(*When* $\boldsymbol{z} = \boldsymbol{0}$, *we set* $\varphi(\boldsymbol{0}) = \boldsymbol{0}$) satisfying $\varphi(\boldsymbol{Z}) \sim U(\sqrt{n}\mathbb{B}_2^n)$.

- Verify the Lipschitz property

  Criterion for the Lipschitz constant of radial maps

  For the radial map $\varphi(\boldsymbol{z})$, its Jacobian matrix is symmetric and has two eigenspaces: the eigenvalue in the radial direction is $g'(r)$, and the eigenvalue in the tangential direction is $g(r)/r$.

  Note that the whole space is convex, hence the Lipschitz norm of φ equals the supremum of the operator norm of the Jacobian matrix,

$$\|\varphi\|_{\mathrm{Lip}} = \sup_{r>0} \max\left(g'(r), \frac{g(r)}{r}\right),$$

  It then suffices to prove that both $g'(r)$ and $g(r)/r$ are bounded by some absolute constant C_0. *Bound the tangential stretch.*

  For $f_{\chi_n}(t) = c_n t^{n-1}\mathrm{e}^{-t^2/2}$, where $c_n = [2^{n/2-1}\Gamma(n/2)]^{-1}$, its cumulative distribution function is

$$F_{\chi_n}(r) = c_n \int_0^r t^{n-1}\mathrm{e}^{-t^2/2}\mathrm{d}t = c_n r^n \int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x, \qquad (t = rx)$$

  Substituting this in, we obtain

$$g(r) = \sqrt{n}\left[c_n r^n \int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x\right]^{1/n} = \sqrt{n}c_n^{1/n} r\left[\int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x\right]^{1/n},$$

  and the tangential ratio is

$$\frac{g(r)}{r} = \sqrt{n}c_n^{1/n}\left[\int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x\right]^{1/n},$$

  where

$$\int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x \leqslant \int_0^1 x^{n-1}\mathrm{d}x = \frac{1}{n},$$

  Using Stirling's formula, we obtain

$$c_n^{1/n} \leqslant \frac{C_1}{\sqrt{n}}, \qquad (C_1 > 0)$$

  thus

$$\frac{g(r)}{r} \leqslant \sqrt{n}\cdot\frac{C_1}{\sqrt{n}}\cdot\left(\frac{1}{n}\right)^{1/n} \leqslant C_1.$$

  Bound the radial stretch.

  Differentiating $g(r) = \sqrt{n}[F_{\chi_n}(r)]^{1/n}$,

$$g'(r) = \sqrt{n}\cdot\frac{1}{n}[F_{\chi_n}(r)]^{\frac{1}{n}-1}f_{\chi_n}(r) = \frac{g(r)}{n}\frac{f_{\chi_n}(r)}{F_{\chi_n}(r)},$$

  where

$$\frac{f_{\chi_n}(r)}{F_{\chi_n}(r)} = \frac{c_n r^{n-1}\mathrm{e}^{-r^2/2}}{c_n r^n \int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x} = \frac{\mathrm{e}^{-r^2/2}}{r\int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x},$$

Substituting this in, we obtain

$$g'(r)=\frac{g(r)}{r}\cdot\frac{\mathrm{e}^{-r^2/2}}{n\int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x},$$

where

$$n\int_0^1 x^{n-1}\mathrm{e}^{-\frac{r^2x^2}{2}}\mathrm{d}x\geqslant \mathrm{e}^{-r^2/2}\int_0^1 \mathrm{d}(x^n)=\mathrm{e}^{-r^2/2},$$

thus

$$g'(r)\leqslant\frac{g(r)}{r}\leqslant C_1.$$

Thus, φ is Lipschitz continuous, satisfying $\|\varphi\|_{\mathrm{Lip}}\leqslant C_1$.

- Prove Theorem T5.2.10 (*Concentration on the continuous cube and ball*)

  Returning to the target space $\mathbb{T}=\mathbb{B}_2^n$.

  For φ, we obtain

$$\frac{\varphi(\boldsymbol{Z})}{\sqrt{n}}\sim U(\mathbb{B}_2^n).$$

  Let $\boldsymbol{X}=\varphi(\boldsymbol{Z})/\sqrt{n}$, and consider the composite function $h(\boldsymbol{z})=f(\boldsymbol{X})=f(\varphi(\boldsymbol{z})/\sqrt{n})$, then

$$\|h\|_{\mathrm{Lip}}\leqslant\|f\|_{\mathrm{Lip}}\cdot\frac{1}{\sqrt{n}}\cdot\|\varphi\|_{\mathrm{Lip}}\leqslant\frac{C_1}{\sqrt{n}}\|f\|_{\mathrm{Lip}},$$

  Hence by Theorem T5.2.3 (*Gaussian concentration*), we obtain

$$\|f(\boldsymbol{X})-\mathbb{E}[f(\boldsymbol{X})]\|_{\psi_2}=\|h(\boldsymbol{Z})-\mathbb{E}[h(\boldsymbol{Z})]\|_{\psi_2}\leqslant C_2\|h\|_{\mathrm{Lip}}\leqslant\frac{C_1C_2}{\sqrt{n}}\|f\|_{\mathrm{Lip}},\qquad(C_2>0)$$

  Now set $C=C_1C_2$.

14 [**Johnson-Lindenstrauss Lemma for sub-Gaussian Matrices**] Let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ whose rows $\boldsymbol{A}_1,\cdots,\boldsymbol{A}_m$ are independent zero-mean, isotropic random vectors satisfying $\max_i\|\boldsymbol{A}_i\|_{\psi_2}\leqslant K$. Define the scaled projection matrix $\boldsymbol{Q}=\boldsymbol{A}/\sqrt{m}$.

(1) Prove that Theorem T5.3.1 (*Johnson-Lindenstrauss Lemma*) still holds for $\boldsymbol{Q}$.

Let $\mathcal{X}\subset\mathbb{R}^n$ be an arbitrary set of N points with error tolerance $\varepsilon\in(0,1)$. When the projection dimension satisfies $m\geqslant C\varepsilon^{-2}\log N$, the map $\boldsymbol{Q}$ can achieve approximate isometric embedding (w.h.p.), preserving the Euclidean distance of any two points in $\mathcal{X}$, i.e., for any $\boldsymbol{x},\boldsymbol{y}\in\mathcal{X}$,

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2\leqslant\|\boldsymbol{Q}\boldsymbol{x}-\boldsymbol{Q}\boldsymbol{y}\|_2\leqslant(1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2\ (\text{w.h.p.}).$$

(2) (Binary JL projection, Binary Johnson-Lindenstrauss) Let all entries of $\boldsymbol{A}$ be i.i.d. Rademacher random variables. Prove that the general conclusion of (1) still holds. In addition, briefly discuss the practical value of using such binary random matrices for dimensionality reduction.

Prf. (1) For any pair of points $\boldsymbol{x},\boldsymbol{y}\in\mathcal{X}$, let $\overline{\boldsymbol{x}-\boldsymbol{y}}\in\mathbb{S}^{n-1}$; then

$$\|\boldsymbol{Q}\overline{\boldsymbol{x}-\boldsymbol{y}}\|_2^2=\frac{1}{m}\|\boldsymbol{A}\overline{\boldsymbol{x}-\boldsymbol{y}}\|_2^2=\frac{1}{m}\sum_{i=1}^m\langle\boldsymbol{A}_i,\overline{\boldsymbol{x}-\boldsymbol{y}}\rangle^2,$$

where $Z_i=\langle\boldsymbol{A}_i,\overline{\boldsymbol{x}-\boldsymbol{y}}\rangle$ satisfies:

1. Independence: Since $\boldsymbol{A}_i$ are independent, $Z_1,\cdots,Z_m$ are independent.
2. Unit variance: Since $\boldsymbol{A}_i$ are isotropic, $\mathbb{E}[Z_i^2]=\mathbb{E}[\overline{\boldsymbol{x}-\boldsymbol{y}}^\top\boldsymbol{A}_i\boldsymbol{A}_i^\top\overline{\boldsymbol{x}-\boldsymbol{y}}]=\overline{\boldsymbol{x}-\boldsymbol{y}}^\top\mathbb{E}[\boldsymbol{A}_i\boldsymbol{A}_i^\top]\overline{\boldsymbol{x}-\boldsymbol{y}}=\overline{\boldsymbol{x}-\boldsymbol{y}}^\top\boldsymbol{I}_n\overline{\boldsymbol{x}-\boldsymbol{y}}=\|\overline{\boldsymbol{x}-\boldsymbol{y}}\|_2^2=1$.
3. sub-Gaussian property: $\|Z_i\|_{\psi_2}=\|\langle\boldsymbol{A}_i,\overline{\boldsymbol{x}-\boldsymbol{y}}\rangle\|_{\psi_2}\leqslant\|\boldsymbol{A}_i\|_{\psi_2}\leqslant K$.

Hence Z_i^2-1 are independent zero-mean sub-Exponential random variables, satisfying $\|Z_i^2-1\|_{\psi_1}\leqslant C_1K^2$.

By Proposition P2.8.1 (*Subexponential properties*), we obtain

$$\mathbb{P}\left\{\left|\frac{1}{m}\sum_{i=1}^{m}(Z_i^2-1)\right|\geqslant\varepsilon\right\}\leqslant 2\exp\left(-c_2 m\min\left(\frac{\varepsilon^2}{K^4},\frac{\varepsilon}{K^2}\right)\right),$$

i.e.,

$$\mathbb{P}\{|\|\boldsymbol{Q}\overline{\boldsymbol{x}-\boldsymbol{y}}\|_2^2-1|\geqslant\varepsilon\}\leqslant 2\exp\left(-\frac{c_3\varepsilon^2 m}{K^4}\right),$$

Taking the union bound over all $\binom{N}{2}<N^2$ pairs of points,

$$N^2\cdot 2\exp\left(-\frac{c_3\varepsilon^2 m}{K^4}\right)\leqslant 2\exp\left(2\log N-\frac{c_3\varepsilon^2 m}{K^4}\right),$$

To make the above failure probability extremely small, set

$$\frac{c_3\varepsilon^2 m}{K^4}\geqslant C_0\log N,$$

$$m\geqslant C\frac{K^4}{\varepsilon^2}\log N.$$

Then the probability that the map $\boldsymbol{Q}$ succeeds in isometric embedding tends to 1.

(2) Let A_{ij} i.i.d. $\sim$ Rademacher, then

1. Zero mean: Since $\mathbb{E}[A_{ij}]=0$, $\mathbb{E}[\boldsymbol{A}_i]=\boldsymbol{0}$.
2. Isotropic: If $j=l$, then $\mathbb{E}[A_{ij}^2]=1$; if $j\neq l$, then $\mathbb{E}[A_{ij}A_{il}]=\mathbb{E}[A_{ij}]\mathbb{E}[A_{il}]=0$, hence $\mathbb{E}[\boldsymbol{A}_i\boldsymbol{A}_i^\top]=\boldsymbol{I}_n$.
3. sub-Gaussian: For any $\boldsymbol{v}\in\mathbb{S}^{n-1}$, $\langle\boldsymbol{A}_i,\boldsymbol{v}\rangle=\sum_{j=1}^n A_{ij}v_j$ is a linear combination of independent bounded variables, hence by Hoeffding's lemma, $\|\boldsymbol{A}_i\|_{\psi_2}\leqslant K'$.

Thus the conditions of (1) are satisfied, and the conclusion still holds.

In algorithms dealing with high-dimensional data (such as approximate database search, large-scale machine learning dimensionality reduction), using a standard Gaussian matrix $\boldsymbol{G}$ for dimensionality reduction requires computationally intensive floating-point multiplications $\boldsymbol{G}\boldsymbol{x}$, which is very time-consuming on hardware.

If a binary JL matrix is used, the projection operation $\frac{1}{\sqrt{m}}\boldsymbol{A}\boldsymbol{x}$ requires no multiplication at all, only addition and subtraction. Binary JL matrices allow the dimensionality reduction algorithm to achieve extreme memory compression (the projection matrix can be stored in binary digits), and significantly improve computational speed on CPU/FPGA.

15 [**Optimality of the Johnson-Lindenstrauss Lemma**] Prove that the dimension $m=O(\log N)$ required by Theorem T5.3.1 (*Johnson-Lindenstrauss Lemma*) is essentially optimal, even if we allow nonlinear dimensionality reduction maps. Prove this following these steps:

(1) Let $\boldsymbol{z}_1,\cdots,\boldsymbol{z}_N\in\mathbb{R}^n$ be such that $1<\|\boldsymbol{z}_i-\boldsymbol{z}_j\|_2\leqslant 2$ whenever $i\neq j$. Show that $N\leqslant 5^n$.

(2) Let $n<\frac{1}{2}\log N$. Construct vectors $\boldsymbol{x}_1,\cdots,\boldsymbol{x}_N\in\mathbb{R}^N$ for which there does not exist any map $T:\mathbb{R}^N\to\mathbb{R}^n$ satisfying $0.99\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2\leqslant\|T(\boldsymbol{x}_i)-T(\boldsymbol{x}_j)\|_2\leqslant 1.01\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2$ for all $i,j=1,\cdots,N$.

Prf. • Prove an upper bound on the cardinality of a restricted point set: $N\leqslant 5^n$

Using the volume packing method, let $\boldsymbol{z}_1,\cdots,\boldsymbol{z}_N\in\mathbb{R}^n$ satisfy, for $i\neq j$,

$$1<\|\boldsymbol{z}_i-\boldsymbol{z}_j\|_2\leqslant 2,$$

then $B(\boldsymbol{z}_i,1/2)$ are mutually disjoint.

Taking $\boldsymbol{z}_1$ as the reference center, for any $\boldsymbol{x}\in B(\boldsymbol{z}_i,1/2)$, we obtain

$$\|\boldsymbol{x}-\boldsymbol{z}_1\|_2\leqslant\|\boldsymbol{x}-\boldsymbol{z}_i\|_2+\|\boldsymbol{z}_i-\boldsymbol{z}_1\|_2<1/2+2=5/2,$$

hence all N open balls $B(\boldsymbol{z}_i,1/2)$ are entirely contained in $B(\boldsymbol{z}_1,2.5)$, satisfying

$$\sum_{i=1}^N\mathrm{Vol}\left(B\left(\boldsymbol{z}_i,\frac{1}{2}\right)\right)\leqslant\mathrm{Vol}\left(B\left(\boldsymbol{z}_1,\frac{5}{2}\right)\right),$$

Let $\mathrm{Vol}(\mathbb{B}_2^n) = V_n$, then

$$N \cdot \left(\frac{1}{2}\right)^n V_n \leqslant \left(\frac{5}{2}\right)^n V_n,$$

$$N \leqslant 5^n.$$

- Prove the insurmountability of the dimensionality reduction lower bound $O(\log N)$

  Suppose the target dimension satisfies $n < \frac{1}{2}\log N$. It suffices to construct $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N \in \mathbb{R}^N$ such that no map $T : \mathbb{R}^N \to \mathbb{R}^n$ exists satisfying

  $$0.99\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \leqslant \|T(\boldsymbol{x}_i) - T(\boldsymbol{x}_j)\|_2 \leqslant 1.01\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2.$$

  Let $\boldsymbol{x}_i = \boldsymbol{e}_i$, then

  $$\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

  Assume there exists a map $T : \mathbb{R}^N \to \mathbb{R}^n$ that preserves the distances of the above vectors with error bounds $[0.99, 1.01]$, then

  $$0.99\sqrt{2} \leqslant \|\boldsymbol{y}_i - \boldsymbol{y}_j\|_2 \leqslant 1.01\sqrt{2}.$$

  (*To use the conclusion of (1), scale the point set $\{\boldsymbol{y}_i\}$ by a constant factor.*)

  Let the scaling factor be $\rho = 0.98\sqrt{2}$, and define

  $$\boldsymbol{z}_i = \frac{1}{\rho}\boldsymbol{y}_i = \frac{1}{0.98\sqrt{2}}T(\boldsymbol{x}_i),$$

  Then the distance lower bound is

  $$\|\boldsymbol{z}_i - \boldsymbol{z}_j\|_2 \geqslant \frac{0.99\sqrt{2}}{0.98\sqrt{2}} = \frac{99}{98} > 1,$$

  and the distance upper bound is

  $$\|\boldsymbol{z}_i - \boldsymbol{z}_j\|_2 \leqslant \frac{1.01\sqrt{2}}{0.98\sqrt{2}} = \frac{101}{98} \leqslant 2,$$

  Hence $1 < \|\boldsymbol{z}_i - \boldsymbol{z}_j\|_2 \leqslant 2$, which satisfies the condition of (1). Therefore, using the conclusion of (1), we obtain

  $$N \leqslant 5^n,$$

  $$n \geqslant \frac{1}{\log 5}\log N > 0.621 \log N,$$

  which contradicts $n < \frac{1}{2}\log N$. Thus, $n \geqslant C \log N$ is optimal in order.

16 [**Matrix Taylor Series**] In this exercise, we use the spectral definition of matrix functions to derive the Taylor series expansion of matrices, namely Definition D5.4.2 (*Functions of matrices*).

Let $\boldsymbol{X} \in \mathbb{R}^{n\times n}$ be a symmetric matrix.

(1) For any polynomial $f(x) = a_0 + a_1 x + \cdots + a_p x^p$, prove:

$$f(\boldsymbol{X}) = a_0\boldsymbol{I}_n + a_1\boldsymbol{X} + \cdots + a_p\boldsymbol{X}^p.$$

(2) More generally, if $f(x) = \sum_{k=1}^{\infty} a_k(x - x_0)^k$ is a power series in a neighborhood $U(x_0)$, prove:

$$f(\boldsymbol{X}) = \sum_{k=0}^{\infty} a_k(\boldsymbol{X} - x_0\boldsymbol{I}_n)^k.$$

And explicitly state under what conditions the above matrix series converges in the Frobenius norm.

(3) Prove:

$$\mathrm{e}^{\boldsymbol{X}} = \boldsymbol{I}_n + \boldsymbol{X} + \frac{\boldsymbol{X}^2}{2!} + \cdots + \frac{\boldsymbol{X}^n}{n!} + \cdots.$$

Prf. (1) Let $f(x)=\sum_{k=0}^{p} a_k x^k$, then by the spectral definition, we obtain

$$\begin{aligned} f(\boldsymbol{X}) &= \boldsymbol{Q}\begin{pmatrix} \sum_{k=0}^{p} a_k \lambda_1^k & & \\ & \ddots & \\ & & \sum_{k=0}^{p} a_k \lambda_n^k \end{pmatrix}\boldsymbol{Q}^\top \\ &= \boldsymbol{Q}\left(\sum_{k=0}^{p} a_k \operatorname{diag}(\lambda_1^k,\cdots,\lambda_n^k)\right)\boldsymbol{Q}^\top \\ &= \sum_{k=0}^{p} a_k \boldsymbol{Q}\boldsymbol{\Lambda}^k\boldsymbol{Q}^\top = \sum_{k=0}^{p} a_k \boldsymbol{X}^k. \end{aligned}$$

(2) The above matrix series converges if and only if all eigenvalues of $\boldsymbol{X}$ lie within the interval of convergence (x_0-R, x_0+R), i.e., $\|\boldsymbol{X}-x_0\boldsymbol{I}_n\|<R$.

Define the partial sum polynomial $S_N(x)=\sum_{k=0}^{N} a_k(x-x_0)^k$, then using the conclusion of (1), we obtain

$$S_N(\boldsymbol{X})=\sum_{k=0}^{N} a_k(\boldsymbol{X}-x_0\boldsymbol{I}_n)^k=\boldsymbol{Q}\operatorname{diag}(S_N(\lambda_1),\cdots,S_N(\lambda_n))\boldsymbol{Q}^\top,$$

It then suffices to prove $S_N(\boldsymbol{X})\to f(\boldsymbol{X})$.

Consider

$$\|f(\boldsymbol{X})-S_N(\boldsymbol{X})\|_F=\left\|\boldsymbol{Q}\operatorname{diag}(f(\lambda_i)-S_N(\lambda_i))\boldsymbol{Q}^\top\right\|_F,$$

Note that orthogonal transformations do not change the Frobenius norm, i.e., $\|\boldsymbol{Q}\boldsymbol{A}\boldsymbol{Q}^\top\|_F=\|\boldsymbol{A}\|_F$, hence

$$\|f(\boldsymbol{X})-S_N(\boldsymbol{X})\|_F=\|\operatorname{diag}(f(\lambda_i)-S_N(\lambda_i))\|_F=\sqrt{\sum_{i=1}^{n}(f(\lambda_i)-S_N(\lambda_i))^2}.$$

Since $|\lambda_i-x_0|<R$, we have $S_N(\lambda_i)\to f(\lambda_i)$, thus

$$\lim_{N\to\infty}\|f(\boldsymbol{X})-S_N(\boldsymbol{X})\|_F=0.$$

Hence the above matrix series converges to $f(\boldsymbol{X})$ in the Frobenius norm.

Using the conclusion of Exercise Q4.4 (*Operator and Frobenius Norms: Relations and Properties*) (1), we have $\|\boldsymbol{A}\|\leqslant\|\boldsymbol{A}\|_F$, hence the above matrix series also converges in the operator norm.

(3) Consider the Maclaurin series of $f(x)=\mathrm{e}^x$ at $x_0=0$,

$$\mathrm{e}^x=\sum_{k=0}^{\infty}\frac{x^k}{k!}=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots,$$

Its radius of convergence is $R=\infty$, hence for any symmetric matrix $\boldsymbol{X}$, all its eigenvalues λ_i satisfy $|\lambda_i-0|<\infty$.

Using the conclusion of (2), we obtain

$$f(\boldsymbol{X})=\sum_{k=0}^{\infty}\frac{1}{k!}(\boldsymbol{X}-0\cdot\boldsymbol{I}_n)^k=\sum_{k=0}^{\infty}\frac{1}{k!}\boldsymbol{X}^k,$$

i.e.,

$$\mathrm{e}^{\boldsymbol{X}}=\boldsymbol{I}_n+\boldsymbol{X}+\frac{\boldsymbol{X}^2}{2!}+\cdots+\frac{\boldsymbol{X}^n}{n!}+\cdots.$$

17 [**Monotonicity of Matrices**] In the space of real symmetric matrices, define the Loewner partial order: if the matrix $\boldsymbol{Y}-\boldsymbol{X}$ is positive semidefinite, we write $\boldsymbol{X}\preccurlyeq\boldsymbol{Y}$.

(1) (Order-preserving property for commuting matrices) Prove: for any two symmetric matrices $\boldsymbol{X},\boldsymbol{Y}$ that commute (*i.e.,* $\boldsymbol{X}\boldsymbol{Y}=\boldsymbol{Y}\boldsymbol{X}$), if $\boldsymbol{X}\preccurlyeq\boldsymbol{Y}$, then for any monotonically nondecreasing function $f:\mathbb{R}\to\mathbb{R}$, we have $f(\boldsymbol{X})\preccurlyeq f(\boldsymbol{Y})$.

(2) (Failure for non-commuting matrices) Construct a counterexample showing that when the matrices do not commute, (1) may not hold.

Prf. (1) Since $\boldsymbol{X},\boldsymbol{Y}$ are both symmetric matrices and commute $\boldsymbol{XY}=\boldsymbol{YX}$, they can be simultaneously orthogonally diagonalized, i.e., there exists an orthogonal matrix $\boldsymbol{Q}$ satisfying $\boldsymbol{QQ}^\top=\boldsymbol{I}$ such that

$$\boldsymbol{X}=\boldsymbol{Q}\boldsymbol{\Lambda}_X\boldsymbol{Q}^\top,\boldsymbol{Y}=\boldsymbol{Q}\boldsymbol{\Lambda}_Y\boldsymbol{Q}^\top,$$

where $\boldsymbol{\Lambda}_X=\operatorname{diag}(\lambda_1^X,\cdots,\lambda_n^X)$, $\boldsymbol{\Lambda}_Y=\operatorname{diag}(\lambda_1^Y,\cdots,\lambda_n^Y)$, hence

$$\boldsymbol{Y}-\boldsymbol{X}=\boldsymbol{Q}(\boldsymbol{\Lambda}_Y-\boldsymbol{\Lambda}_X)\boldsymbol{Q}^\top\succcurlyeq\boldsymbol{O}.$$

For any $\boldsymbol{u}\in\mathbb{R}^n$, let $\boldsymbol{v}=\boldsymbol{Q}^\top\boldsymbol{u}$, then

$$\boldsymbol{u}^\top\boldsymbol{Q}(\boldsymbol{\Lambda}_Y-\boldsymbol{\Lambda}_X)\boldsymbol{Q}^\top\boldsymbol{u}=\boldsymbol{v}^\top(\boldsymbol{\Lambda}_Y-\boldsymbol{\Lambda}_X)\boldsymbol{v}=\sum_{i=1}^n v_i^2(\lambda_i^Y-\lambda_i^X)\geqslant 0,$$

and we obtain $\lambda_i^X\leqslant\lambda_i^Y$, hence

$$f(\lambda_i^X)\leqslant f(\lambda_i^Y),$$

thus

$$\boldsymbol{u}^\top(f(\boldsymbol{Y})-f(\boldsymbol{X}))\boldsymbol{u}=\boldsymbol{v}^\top(f(\boldsymbol{\Lambda}_Y)-f(\boldsymbol{\Lambda}_X))\boldsymbol{v}=\sum_{i=1}^n v_i^2(f(\lambda_i^Y)-f(\lambda_i^X))\geqslant 0,$$

i.e., $f(\boldsymbol{Y})-f(\boldsymbol{X})\succcurlyeq\boldsymbol{O}$, i.e., $f(\boldsymbol{X})\preccurlyeq f(\boldsymbol{Y})$.

(2) Let

$$\boldsymbol{X}=\begin{pmatrix}1&1\\1&1\end{pmatrix},\boldsymbol{Y}=\begin{pmatrix}2&1\\1&1\end{pmatrix},$$

then $\boldsymbol{X}\preccurlyeq\boldsymbol{Y}$, but $\boldsymbol{XY}\neq\boldsymbol{YX}$.

Let $f(x)=x^3$, then

$$f(\boldsymbol{Y})-f(\boldsymbol{X})=\boldsymbol{Y}^3-\boldsymbol{X}^3=\begin{pmatrix}13&8\\8&5\end{pmatrix}-\begin{pmatrix}4&4\\4&4\end{pmatrix}=\begin{pmatrix}9&4\\4&1\end{pmatrix},$$

Its determinant is $-7<0$, which means there exists a negative eigenvalue, i.e., it is not positive semidefinite, hence $f(\boldsymbol{X})\not\preccurlyeq f(\boldsymbol{Y})$.

18 [**Matrix Monotonicity of the Functions** $1/x$ **and** $\log x$] Let $\boldsymbol{X},\boldsymbol{Y}\in\mathbb{R}^{n\times n}$ be symmetric matrices. Assume $\boldsymbol{X}$ is invertible and satisfies the Loewner partial order relation

$$\boldsymbol{O}\preccurlyeq\boldsymbol{X}\preccurlyeq\boldsymbol{Y}.$$

(1) Prove that $\boldsymbol{Y}$ is invertible and its inverse satisfies the reversed partial order relation,

$$\boldsymbol{X}^{-1}\succcurlyeq\boldsymbol{Y}^{-1}\succcurlyeq\boldsymbol{O}.$$

(2) Let $x>0$, prove the following integral identity,

$$\log x=\int_0^\infty\Big(\frac{1}{1+t}-\frac{1}{x+t}\Big)\mathrm{d}t.$$

(3) Based on the definition of matrix functions, generalize (2) to matrices, and using the conclusion of (1), prove the monotonicity of the matrix logarithm function,

$$\log\boldsymbol{X}\preccurlyeq\log\boldsymbol{Y}.$$

Sol. (1) Prove $\boldsymbol{Y}^{-1}\succcurlyeq\boldsymbol{O}$

Since $\boldsymbol{X}$ is positive semidefinite and invertible, i.e., all its eigenvalues are strictly greater than 0, $\boldsymbol{X}$ is strictly positive definite, i.e., $\boldsymbol{X}\succ\boldsymbol{O}$.

For the inequality $\boldsymbol{X}\preccurlyeq\boldsymbol{Y}$, left- and right-multiplying both sides by a nonzero vector $\boldsymbol{v}\in\mathbb{R}^n$, we obtain

$$\boldsymbol{v}^\top\boldsymbol{Y}\boldsymbol{v}\geqslant\boldsymbol{v}^\top\boldsymbol{X}\boldsymbol{v}>0,$$

hence $\boldsymbol{Y}$ is strictly positive definite, i.e., all eigenvalues of $\boldsymbol{Y}$ are greater than 0, so $\boldsymbol{Y}$ is invertible and satisfies

$\boldsymbol{Y}^{-1} \succ \boldsymbol{O}$. Prove $\boldsymbol{X}^{-1} \succcurlyeq \boldsymbol{Y}^{-1}$

Since $\boldsymbol{X} \succ \boldsymbol{O}$, $\boldsymbol{X}^{1/2}, \boldsymbol{X}^{-1/2}$ are symmetric and positive definite.

For the inequality $\boldsymbol{X} \preccurlyeq \boldsymbol{Y}$, left- and right-multiplying both sides by $\boldsymbol{X}^{-1/2}$, we obtain

$$\boldsymbol{X}^{-1/2}\boldsymbol{X}\boldsymbol{X}^{-1/2} \preccurlyeq \boldsymbol{X}^{-1/2}\boldsymbol{Y}\boldsymbol{X}^{-1/2},$$

i.e.,

$$\boldsymbol{I}_n \preccurlyeq \boldsymbol{X}^{-1/2}\boldsymbol{Y}\boldsymbol{X}^{-1/2}.$$

Let $\boldsymbol{Z} = \boldsymbol{X}^{-1/2}\boldsymbol{Y}\boldsymbol{X}^{-1/2} \succcurlyeq \boldsymbol{I}_n$, then $\lambda_i(\boldsymbol{Z}) \geqslant 1$, hence the eigenvalues of $\boldsymbol{Z}^{-1}$ satisfy $0 < 1/\lambda_i(\boldsymbol{Z}) \leqslant 1$, i.e.,

$$\begin{aligned} \boldsymbol{Z}^{-1} &\preccurlyeq \boldsymbol{I}_n, \\ (\boldsymbol{X}^{-1/2}\boldsymbol{Y}\boldsymbol{X}^{-1/2})^{-1} &\preccurlyeq \boldsymbol{I}_n, \\ \boldsymbol{Y}^{-1} &\preccurlyeq \boldsymbol{X}^{-1}. \end{aligned}$$

(2) It is easy to see that

$$\begin{aligned} \int_0^\infty \left(\frac{1}{1+t} - \frac{1}{x+t}\right)\mathrm{d}t &= \lim_{T\to\infty} [\log(1+t) - \log(x+t)]_0^T \\ &= \lim_{T\to\infty} \left[\log\left(\frac{1+t}{x+t}\right)\right]_0^T \\ &= \lim_{T\to\infty} \log\left(\frac{1+T}{x+T}\right) - \log\left(\frac{1}{x}\right) = \log x. \end{aligned}$$

(3) Using the conclusion of (2), we obtain

$$\log \boldsymbol{Y} - \log \boldsymbol{X} = \int_0^\infty ((\boldsymbol{X} + t\boldsymbol{I}_n)^{-1} - (\boldsymbol{Y} + t\boldsymbol{I}_n)^{-1})\mathrm{d}t.$$

Since $\boldsymbol{O} \prec \boldsymbol{X} \preccurlyeq \boldsymbol{Y}$, when $t \geqslant 0$, $t\boldsymbol{I}_n \preccurlyeq t\boldsymbol{I}_n$, hence

$$\boldsymbol{O} \prec \boldsymbol{X} + t\boldsymbol{I}_n \preccurlyeq \boldsymbol{Y} + t\boldsymbol{I}_n,$$

Using the conclusion of (1), we obtain

$$(\boldsymbol{X} + t\boldsymbol{I}_n)^{-1} \succcurlyeq (\boldsymbol{Y} + t\boldsymbol{I}_n)^{-1},$$

Integrating on both sides,

$$\int_0^\infty ((\boldsymbol{X} + t\boldsymbol{I}_n)^{-1} - (\boldsymbol{Y} + t\boldsymbol{I}_n)^{-1})\mathrm{d}t \succcurlyeq \boldsymbol{O},$$

i.e., $\log \boldsymbol{X} \preccurlyeq \log \boldsymbol{Y}$.

19 [**Properties of the Matrix Exponential**] Let $\boldsymbol{X}, \boldsymbol{Y} \in \mathbb{R}^{n\times n}$ be symmetric matrices.

(1) Prove: if $\boldsymbol{X}, \boldsymbol{Y}$ commute, then

$$\mathrm{e}^{\boldsymbol{X}+\boldsymbol{Y}} = \mathrm{e}^{\boldsymbol{X}}\mathrm{e}^{\boldsymbol{Y}}.$$

(2) Construct a counterexample where $\boldsymbol{X}, \boldsymbol{Y}$ do not commute, satisfying $\mathrm{e}^{\boldsymbol{X}+\boldsymbol{Y}} \neq \mathrm{e}^{\boldsymbol{X}}\mathrm{e}^{\boldsymbol{Y}}$.

Prf. (1) By the Taylor series, we obtain

$$\mathrm{e}^{\boldsymbol{X}} = \sum_{j=0}^\infty \frac{\boldsymbol{X}^j}{j!}, \mathrm{e}^{\boldsymbol{Y}} = \sum_{k=0}^\infty \frac{\boldsymbol{Y}^k}{k!},$$

By the binomial theorem, we obtain

$$(\boldsymbol{X} + \boldsymbol{Y})^m = \sum_{j=0}^m \binom{m}{j} \boldsymbol{X}^j \boldsymbol{Y}^{m-j},$$

hence

$$\mathrm{e}^{\boldsymbol{X}+\boldsymbol{Y}} = \sum_{m=0}^\infty \frac{(\boldsymbol{X}+\boldsymbol{Y})^m}{m!} = \sum_{m=0}^\infty \sum_{j=0}^m \frac{1}{m!}\binom{m}{j}\boldsymbol{X}^j\boldsymbol{Y}^{m-j} = \sum_{m=0}^\infty \sum_{j=0}^m \frac{\boldsymbol{X}^j}{j!}\frac{\boldsymbol{Y}^{m-j}}{(m-j)!}$$

$$= \sum_{j=0}^{\infty}\sum_{k=0}^{\infty}\frac{\boldsymbol{X}^j}{j!}\frac{\boldsymbol{Y}^k}{k!} = \left(\sum_{j=0}^{\infty}\frac{\boldsymbol{X}^j}{j!}\right)\left(\sum_{k=0}^{\infty}\frac{\boldsymbol{Y}^k}{k!}\right) = \mathrm{e}^{\boldsymbol{X}}\mathrm{e}^{\boldsymbol{Y}}. \qquad (k=m-j)$$

(2) Let

$$\boldsymbol{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \boldsymbol{Y} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $\boldsymbol{XY} \neq \boldsymbol{YX}$.

For $\mathrm{e}^{\boldsymbol{X}}$, since $\boldsymbol{X}^2 = \boldsymbol{I}$,

$$\mathrm{e}^{\boldsymbol{X}} = \cosh(1)\boldsymbol{I} + \sinh(1)\boldsymbol{X} = \begin{pmatrix} \cosh 1 & \sinh 1 \\ \sinh 1 & \cosh 1 \end{pmatrix};$$

For $\mathrm{e}^{\boldsymbol{Y}}$, since $\boldsymbol{Y}$ is diagonal,

$$\mathrm{e}^{\boldsymbol{Y}} = \begin{pmatrix} \mathrm{e}^0 & 0 \\ 0 & \mathrm{e} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e} \end{pmatrix},$$

hence

$$\mathrm{e}^{\boldsymbol{X}}\mathrm{e}^{\boldsymbol{Y}} = \begin{pmatrix} \cosh 1 & \sinh 1 \\ \sinh 1 & \cosh 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e} \end{pmatrix} = \begin{pmatrix} \cosh 1 & \mathrm{e}\sinh 1 \\ \sinh 1 & \mathrm{e}\cosh 1 \end{pmatrix}.$$

However, for $\mathrm{e}^{\boldsymbol{X}+\boldsymbol{Y}}$, let $\boldsymbol{S} = \boldsymbol{X} + \boldsymbol{Y}$, then

$$\mathrm{e}^{\boldsymbol{S}} = \boldsymbol{Q}\begin{pmatrix} \mathrm{e}^{\lambda_1} & 0 \\ 0 & \mathrm{e}^{\lambda_2} \end{pmatrix}\boldsymbol{Q}^{\top},$$

where the eigenvalues of $\boldsymbol{S}$ are

$$\det(\lambda\boldsymbol{I} - \boldsymbol{S}) = \lambda^2 - \lambda - 1 = 0,$$

$$\lambda_1 = \frac{1+\sqrt{5}}{2} = \varphi,$$

$$\lambda_2 = \frac{1-\sqrt{5}}{2} = -\frac{1}{\varphi}.$$

Using the matrix exponential formula

$$\mathrm{e}^{\boldsymbol{S}} = \frac{\mathrm{e}^{\lambda_1} - \mathrm{e}^{\lambda_2}}{\lambda_1 - \lambda_2}\boldsymbol{S} + \frac{\lambda_1\mathrm{e}^{\lambda_2} - \lambda_2\mathrm{e}^{\lambda_1}}{\lambda_1 - \lambda_2}\boldsymbol{I},$$

where $\lambda_1 - \lambda_2 = \sqrt{5}$, we obtain

$$\mathrm{e}^{\boldsymbol{S}} = \frac{\mathrm{e}^{\varphi} - \mathrm{e}^{-1/\varphi}}{\sqrt{5}}\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} + \frac{\varphi\mathrm{e}^{-1/\varphi} - (-1/\varphi)\mathrm{e}^{\varphi}}{\sqrt{5}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

hence $\mathrm{e}^{\boldsymbol{S}} \neq \mathrm{e}^{\boldsymbol{X}}\mathrm{e}^{\boldsymbol{Y}}$.

20 [**Expected Operator Norm Bound for Sums of Matrices**] Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N \in \mathbb{R}^{n\times n}$ be independent symmetric matrices satisfying $\mathbb{E}[\boldsymbol{X}_i] = \boldsymbol{0}$, $\|\boldsymbol{X}_i\| \leqslant K$ (a.s.). Let the operator norm of the matrix variance be $\sigma^2 = \left\|\sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i^2]\right\|$. Using Theorem T5.4.1 (*Matrix Bernstein inequality*), prove:

$$\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\| \lesssim \sigma\sqrt{\log n} + K\log n.$$

Prf. Let $S = \left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|$. By Theorem T5.4.1 (*Matrix Bernstein inequality*), for any $t \geqslant 0$,

$$\mathbb{P}\{S \geqslant t\} \leqslant 2n\exp\left(-c\min\left(\frac{t^2}{\sigma^2}, \frac{t}{K}\right)\right), \qquad (c>0)$$

Decoupling the tail bound and using the inequality $\exp(-\min(A,B)) = \max(\mathrm{e}^{-A}, \mathrm{e}^{-B}) \leqslant \mathrm{e}^{-A} + \mathrm{e}^{-B}$, we obtain

$$\mathbb{P}\{S \geqslant t\} \leqslant 2n \exp\left(-\frac{ct^2}{\sigma^2}\right) + 2n \exp\left(-\frac{ct}{K}\right),$$

(*Both terms on the right-hand side contain* $2n$*. We need to choose a threshold such that the exponential decay terms not only cancel the* $2n$ *factor but also continue to contribute a decay factor to ensure convergence of the integral. Specifically, let*

$$2n \exp\left(-\frac{ct^2}{\sigma^2}\right) = \frac{1}{2n}, \qquad t = \sigma\sqrt{\frac{2}{c}\log(2n)},$$

$$2n \exp\left(-\frac{ct}{K}\right) = \frac{1}{2n}, \qquad t = \frac{2}{c}K\log(2n),$$

Then define the threshold as the sum of the above two terms.) Using the integral identity for the expectation of a nonnegative random variable

$$\mathbb{E}[S] = \int_0^\infty \mathbb{P}\{S \geqslant t\}\mathrm{d}t,$$

let the threshold be $u = u_1 + u_2$, where

$$u_1 = \sigma\sqrt{\frac{2}{c}\log(2n)}, \quad u_2 = \frac{2}{c}K\log(2n),$$

Splitting the integral,

$$\begin{aligned}
\mathbb{E}[S] &= \int_0^u \mathbb{P}\{S \geqslant t\}\mathrm{d}t + \int_u^\infty \mathbb{P}\{S \geqslant t\}\mathrm{d}t \\
&\leqslant u + \int_u^\infty \mathbb{P}\{S \geqslant t\}\mathrm{d}t \\
&= u + \underbrace{\int_u^\infty 2n \exp\left(-\frac{ct^2}{\sigma^2}\right)\mathrm{d}t}_{I_1} + \underbrace{\int_u^\infty 2n \exp\left(-\frac{ct}{K}\right)\mathrm{d}t}_{I_2},
\end{aligned}$$

For I_1,

$$\begin{aligned}
I_1 &\leqslant 2n \int_{u_1}^\infty \exp\left(-\frac{ct^2}{\sigma^2}\right)\mathrm{d}t \\
&= 2n \int_0^\infty \exp\left(-\frac{c(u_1^2 + x^2)}{\sigma^2}\right)\mathrm{d}x = 2n \exp\left(-\frac{cu_1^2}{\sigma^2}\right)\int_0^\infty \exp\left(-\frac{cx^2}{\sigma^2}\right)\mathrm{d}x \qquad (t = u_1 + x) \\
&= 2n \cdot \frac{1}{4n^2} \cdot \frac{\sigma}{2}\sqrt{\frac{\pi}{c}} = \frac{\sigma\sqrt{\pi}}{4n\sqrt{c}} = O(\sigma),
\end{aligned}$$

For I_2,

$$\begin{aligned}
I_2 &\leqslant 2n \int_{u_2}^\infty \exp\left(-\frac{ct}{K}\right)\mathrm{d}t \\
&= 2n\left[-\frac{K}{c}\exp\left(-\frac{ct}{K}\right)\right]_{u_2}^\infty = 2n\frac{K}{c}\exp\left(-\frac{cu_2}{K}\right) \\
&\leqslant 2n\frac{K}{c}\frac{1}{4n^2} = \frac{K}{2nc} = O(K),
\end{aligned}$$

hence

$$\mathbb{E}[S] \leqslant u_1 + u_2 + I_1 + I_2 = \sigma\sqrt{\frac{2}{c}\log(2n)} + \frac{2}{c}K\log(2n) + O(\sigma) + O(K),$$

where $\log(2n) = \log 2 + \log n \asymp 1 + \log n$, and $O(\sigma), O(K)$ can both be absorbed into the constant coefficients of the leading terms, thus

$$\mathbb{E}\left\|\sum_{i=1}^{N} \boldsymbol{X}_i\right\| \lesssim \sigma\sqrt{1+\log n} + K(1+\log n).$$

21 [**Proof of the Matrix Hoeffding Inequality**] Let $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher, and $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_N \in \mathbb{R}^{n\times n}$ be deterministic symmetric matrices. Define the random matrix sum $\boldsymbol{S} = \sum_{i=1}^{N} \varepsilon_i \boldsymbol{A}_i$. Following the proof idea of the matrix Bernstein inequality, prove Theorem T5.4.13 (*Matrix Hoeffding inequality*): for any $t \geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\} \leqslant 2n\exp\left(-\frac{t^2}{2\sigma^2}\right),$$

where $\sigma^2 = \left\|\sum_{i=1}^{N} \boldsymbol{A}_i^2\right\|$ is the operator norm of the variance of the sum matrix.

Prf. (*The standard proof idea successively applies the matrix Chernoff bound, the single-term matrix MGF estimate, Lieb's theorem decoupling, and scalar function extremum optimization.*) Matrix Chernoff bound

Since $\boldsymbol{S}$ is a symmetric matrix, its operator norm equals its maximum absolute eigenvalue, i.e., $\|\boldsymbol{S}\| = \max(\lambda_{\max}(\boldsymbol{S}), \lambda_{\max}(-\boldsymbol{S}))$. Consider the upper tail bound $\lambda_{\max}(\boldsymbol{S})$: for any $\theta > 0$, using Theorem T2.3.1 (*Chernoff inequality*) (*Chernoff inequality*), we obtain

$$\begin{aligned}\mathbb{P}\{\lambda_{\max}(\boldsymbol{S}) \geqslant t\} &= \mathbb{P}\left\{\mathrm{e}^{\theta\lambda_{\max}(\boldsymbol{S})} \geqslant \mathrm{e}^{\theta t}\right\} \leqslant \mathrm{e}^{-\theta t}\mathbb{E}\left[\mathrm{e}^{\theta\lambda_{\max}(\boldsymbol{S})}\right] \\ &\leqslant \mathrm{e}^{-\theta t}\mathbb{E}[\operatorname{tr}(\mathrm{e}^{\theta\boldsymbol{S}})]. \end{aligned} \tag{I}$$

- Use the Lieb inequality for random matrices to handle the expectation of the trace

  (*The terms inside the matrix exponential* $\mathrm{e}^{\theta\sum_i \varepsilon_i \boldsymbol{A}_i}$ *generally do not commute. Lemma L5.4.9 (Lieb inequality for random matrices) directly controls the expectation of its trace.*)

  By Lemma L5.4.9 (*Lieb inequality for random matrices*) with $\boldsymbol{H} = \boldsymbol{0}$, we obtain

$$\mathbb{E}\left[\operatorname{tr}\exp\left(\sum_{i=1}^{N}\boldsymbol{X}_i\right)\right] \leqslant \operatorname{tr}\exp\left(\sum_{i=1}^{N}\log\mathbb{E}[\mathrm{e}^{\boldsymbol{X}_i}]\right). \tag{II}$$

  (*The proof of the Golden–Thompson inequality is given in the remark.*)

  Let $\boldsymbol{X}_i = \theta\varepsilon_i\boldsymbol{A}_i$. We need to compute $\mathbb{E}[\mathrm{e}^{\theta\varepsilon_i\boldsymbol{A}_i}]$. It is easy to see that

$$\mathbb{E}[\mathrm{e}^{\theta\varepsilon_i\boldsymbol{A}_i}] = \frac{1}{2}\mathrm{e}^{\theta\boldsymbol{A}_i} + \frac{1}{2}\mathrm{e}^{-\theta\boldsymbol{A}_i} = \cosh(\theta\boldsymbol{A}_i),$$

  Note that $\cosh(x) \leqslant \mathrm{e}^{x^2/2}$, and $\boldsymbol{A}_i$ is symmetric, hence

$$\mathbb{E}[\mathrm{e}^{\theta\varepsilon_i\boldsymbol{A}_i}] \preccurlyeq \exp\left(\frac{\theta^2\boldsymbol{A}_i^2}{2}\right),$$

  By Exercise Q5.18 (*Matrix Monotonicity of the Functions* $1/x$ *and* $\log x$), $\log(\cdot)$ is operator monotone, and taking the logarithm on both sides,

$$\log\mathbb{E}[\mathrm{e}^{\theta\varepsilon_i\boldsymbol{A}_i}] \preccurlyeq \frac{\theta^2}{2}\boldsymbol{A}_i^2,$$

  Substituting this into (II), we obtain

$$\mathbb{E}[\operatorname{tr}(\mathrm{e}^{\theta\boldsymbol{S}})] \leqslant \operatorname{tr}\exp\left(\sum_{i=1}^{N}\frac{\theta^2}{2}\boldsymbol{A}_i^2\right) = \operatorname{tr}\exp\left(\frac{\theta^2}{2}\sum_{i=1}^{N}\boldsymbol{A}_i^2\right),$$

where we denote the variance matrix $\boldsymbol{V}=\sum_{i=1}^{N}\boldsymbol{A}_i^2$, satisfying $\sigma^2=\|\boldsymbol{V}\|$, then

$$\operatorname{tr}\exp\left(\frac{\theta^2}{2}\boldsymbol{V}\right)\leqslant n\cdot\lambda_{\max}\left(\exp\left(\frac{\theta^2}{2}\boldsymbol{V}\right)\right)=n\exp\left(\frac{\theta^2}{2}\|\boldsymbol{V}\|\right),$$

hence

$$\mathbb{E}[\operatorname{tr}(\mathrm{e}^{\theta\boldsymbol{S}})]\leqslant n\exp\left(\frac{\theta^2\sigma^2}{2}\right),$$

Substituting this into (I), we obtain

$$\mathbb{P}\{\lambda_{\max}(\boldsymbol{S})\geqslant t\}\leqslant n\exp\left(\underbrace{-\theta t+\frac{\theta^2\sigma^2}{2}}_{f(\theta)}\right),$$

where

$$\min_{\theta>0}f(\theta)=f\left(\frac{t}{\sigma^2}\right)=-\frac{t^2}{2\sigma^2},$$

thus

$$\mathbb{P}\{\lambda_{\max}(\boldsymbol{S})\geqslant t\}\leqslant n\exp\left(-\frac{t^2}{2\sigma^2}\right);$$

Similarly, by symmetry, we obtain

$$\mathbb{P}\{\lambda_{\max}(-\boldsymbol{S})\geqslant t\}\leqslant n\exp\left(-\frac{t^2}{2\sigma^2}\right).$$

Thus, using the union bound, we obtain

$$\mathbb{P}\{\|\boldsymbol{S}\|\geqslant t\}=\mathbb{P}\{\max(\lambda_{\max}(\boldsymbol{S}),\lambda_{\max}(-\boldsymbol{S}))\geqslant t\}\leqslant\mathbb{P}\{\lambda_{\max}(\boldsymbol{S})\geqslant t\}+\mathbb{P}\{\lambda_{\max}(-\boldsymbol{S})\geqslant t\}\leqslant 2n\exp\left(-\frac{t^2}{2\sigma^2}\right).$$

Rk. For the proof of the Golden-Thompson inequality:

Solution 1 Lieb's concavity theorem

Theorem (Corollary of Lieb's concavity theorem, Elliott H. Lieb, 1973) Let $\boldsymbol{H}$ be a Hermitian matrix, and define $F(\boldsymbol{X})$ on the set of all positive definite Hermitian matrices, then

$$F(\boldsymbol{X})=\operatorname{tr}(\exp(\boldsymbol{H}+\log\boldsymbol{X}))$$

is strictly concave.

(*In multivariable calculus, if a differentiable function $f(\boldsymbol{x})$ is concave, then it must be controlled by its tangent plane at any $\boldsymbol{y}$ (i.e., the graph of the function lies below the tangent plane),*

$$f(\boldsymbol{x})\leqslant f(\boldsymbol{y})+\langle\nabla f(\boldsymbol{y}),\boldsymbol{x}-\boldsymbol{y}\rangle.$$

In the space of real matrices, the inner product is defined as the Hilbert-Schmidt inner product $\langle\boldsymbol{M},\boldsymbol{N}\rangle=tr(\boldsymbol{M}^\top\boldsymbol{N})$, which for Hermitian matrices is equivalent to $tr(\boldsymbol{M}\boldsymbol{N})$. Hence for the concave function $F(\boldsymbol{X})$,

$$F(\boldsymbol{X})\leqslant F(\boldsymbol{Y})+tr(\nabla F(\boldsymbol{Y})\cdot(\boldsymbol{X}-\boldsymbol{Y})),$$

where $\boldsymbol{X},\boldsymbol{Y}$ are positive definite matrices.

To use the above property, we must compute $\nabla F(\boldsymbol{X})$.)

State the following directional-derivative formulas:

Directional-derivative formula 1 (Directional derivative of the trace of the exponential) Let $\boldsymbol{M},\boldsymbol{N}$ be square matrices, then

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}\operatorname{tr}(\exp(\boldsymbol{M}+t\boldsymbol{N}))=\operatorname{tr}(\exp(\boldsymbol{M})\boldsymbol{N}).$$

Directional-derivative formula 2 (Directional derivative of the trace of the logarithm) Let $\boldsymbol{C}$ be a positive definite

matrix and $\boldsymbol{V}$ a Hermitian matrix, then

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \operatorname{tr}(\log(\boldsymbol{C}+t\boldsymbol{V})) = \operatorname{tr}(\boldsymbol{C}^{-1}\boldsymbol{V}).$$

Let $F(\boldsymbol{X}) = \operatorname{tr}(\exp(\boldsymbol{A}+\log \boldsymbol{X}))$, then by Lieb's theorem, $F(\boldsymbol{X})$ is concave on the domain of positive definite matrices. Let $\boldsymbol{Y} = \exp(-\boldsymbol{A})$ be a positive definite Hermitian matrix, then

$$F(\boldsymbol{Y}) = \operatorname{tr}(\exp(\boldsymbol{A}+\log(\exp(-\boldsymbol{A})))) = \operatorname{tr}(\exp(\boldsymbol{A}-\boldsymbol{A})) = \operatorname{tr}(\boldsymbol{I}) = n.$$

The directional derivative of F at $\boldsymbol{Y}$ in the direction $\boldsymbol{V}$ is

$$\nabla_{\boldsymbol{V}} F(\boldsymbol{Y}) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \operatorname{tr}(\exp(\boldsymbol{A}+\log(\boldsymbol{Y}+t\boldsymbol{V}))).$$

Let $\boldsymbol{M}(t) = \boldsymbol{A}+\log(\boldsymbol{Y}+t\boldsymbol{V})$ satisfying $\boldsymbol{M}(0) = 0$, then by directional-derivative formula 1, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \operatorname{tr}(\exp(\boldsymbol{M}(t))) = \operatorname{tr}(\exp(\boldsymbol{M}(0))\cdot \boldsymbol{M}'(0)) = \operatorname{tr}(\boldsymbol{I}\cdot \boldsymbol{M}'(0)) = \operatorname{tr}(\boldsymbol{M}'(0)),$$

By directional-derivative formula 2, we obtain

$$\operatorname{tr}(\boldsymbol{M}'(0)) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \operatorname{tr}(\boldsymbol{A}+\log(\boldsymbol{Y}+t\boldsymbol{V})) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \operatorname{tr}(\log(\boldsymbol{Y}+t\boldsymbol{V})) = \operatorname{tr}(\boldsymbol{Y}^{-1}\boldsymbol{V}),$$

Note that $\langle \nabla F(\boldsymbol{Y}), \boldsymbol{V}\rangle = \operatorname{tr}(\nabla F(\boldsymbol{Y})\cdot \boldsymbol{V})$, hence

$$\nabla F(\boldsymbol{Y}) = \boldsymbol{Y}^{-1} = (\exp(-\boldsymbol{A}))^{-1} = \exp(\boldsymbol{A}).$$

Thus, substituting $F(\boldsymbol{Y}), \nabla F(\boldsymbol{Y})$ into the first-order concavity condition $F(\boldsymbol{X}) \leqslant F(\boldsymbol{Y}) + \operatorname{tr}(\nabla F(\boldsymbol{Y})\cdot(\boldsymbol{X}-\boldsymbol{Y}))$, we obtain

$$\begin{aligned} F(\boldsymbol{X}) &\leqslant n + \operatorname{tr}(\exp(\boldsymbol{A})\cdot(\boldsymbol{X}-\exp(-\boldsymbol{A}))) \\ &\leqslant n + \operatorname{tr}(\exp(\boldsymbol{A})\boldsymbol{X}) - \operatorname{tr}(\exp(\boldsymbol{A})\exp(-\boldsymbol{A})) \\ &= n + \operatorname{tr}(\exp(\boldsymbol{A})\boldsymbol{X}) - n = \operatorname{tr}(\exp(\boldsymbol{A})\boldsymbol{X}), \end{aligned}$$

Now letting $\boldsymbol{X} = \exp(\boldsymbol{B})$, we obtain

$$\operatorname{tr}(\exp(\boldsymbol{A}+\boldsymbol{B})) \leqslant \operatorname{tr}(\exp(\boldsymbol{A})\exp(\boldsymbol{B})).$$

Solution 2 Lie-Trotter product formula

State the following auxiliary formulas:

Auxiliary formula 1 (Lie-Trotter product formula) Let $\boldsymbol{X}, \boldsymbol{Y}$ be square matrices, then

$$\exp(\boldsymbol{X}+\boldsymbol{Y}) = \lim_{n\to\infty}\left(\exp\left(\frac{\boldsymbol{X}}{n}\right)\exp\left(\frac{\boldsymbol{Y}}{n}\right)\right)^n.$$

Auxiliary formula 2 (Corollary of the Weyl majorization inequality on the trace) Let $\boldsymbol{C} \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, \lambda_2, \cdots, \lambda_n$ and singular values $\sigma_1, \sigma_2, \cdots, \sigma_n$, then for any $p \geqslant 1$,

$$\sum_{i=1}^{n} |\lambda_i|^p \leqslant \sum_{i=1}^{n} \sigma_i^p.$$

In particular, when $p = 2m$, $\sum_{i=1}^{n} \sigma_i^{2m} = \operatorname{tr}((\boldsymbol{C}\boldsymbol{C}^*)^m)$.

Auxiliary formula 3 Let $\boldsymbol{X}, \boldsymbol{Y}$ be positive semidefinite Hermitian matrices, then for any $k \in \mathbb{Z}_0^+$,

$$\operatorname{tr}((\boldsymbol{X}\boldsymbol{Y})^{2^k}) \leqslant \operatorname{tr}(\boldsymbol{X}^{2^k}\boldsymbol{Y}^{2^k}).$$

Let $\boldsymbol{C} = \boldsymbol{X}\boldsymbol{Y}$, then $\boldsymbol{C} \sim \boldsymbol{X}^{1/2}\boldsymbol{Y}\boldsymbol{X}^{1/2}$, where $\boldsymbol{X}^{1/2}\boldsymbol{Y}\boldsymbol{X}^{1/2}$ is a positive semidefinite Hermitian matrix, hence $\boldsymbol{C}$ is positive semidefinite and all its eigenvalues satisfy $\lambda_i \geqslant 0$.

We use mathematical induction: when $m = 0$, the conclusion is trivial; assume that when $m = k-1$, the conclusion holds, i.e., for any positive semidefinite Hermitian matrices $\boldsymbol{P}, \boldsymbol{Q}$,

$$tr((\boldsymbol{P}\boldsymbol{Q})^{2^{k-1}}) \leqslant tr(\boldsymbol{P}^{2^{k-1}}\boldsymbol{Q}^{2^{k-1}});$$

Then when $m = k$, the left-hand side is

$$tr(\boldsymbol{C}^{2^k}) = \sum_{i=1}^{n} \lambda_i^{2^k} = \sum_{i=1}^{n} |\lambda_i|^{2^k},$$

By auxiliary formula 2 with $p = 2^k$, we obtain

$$tr(\boldsymbol{C}^{2^k}) = \sum_{i=1}^{n} |\lambda_i|^{2^k} \leqslant tr((\boldsymbol{C}\boldsymbol{C}^*)^{2^{k-1}}).$$

For $\boldsymbol{C}\boldsymbol{C}^*$,

$$\boldsymbol{C}\boldsymbol{C}^* = (\boldsymbol{X}\boldsymbol{Y})(\boldsymbol{X}\boldsymbol{Y})^* = \boldsymbol{X}\boldsymbol{Y}\boldsymbol{Y}^*\boldsymbol{X}^* = \boldsymbol{X}\boldsymbol{Y}^2\boldsymbol{X},$$

Taking the trace on both sides,

$$tr((\boldsymbol{C}\boldsymbol{C}^*)^{2^{k-1}}) = tr((\boldsymbol{X}\boldsymbol{Y}^2\boldsymbol{X})^{2^{k-1}}),$$

where

$$\begin{aligned} tr((\boldsymbol{X}\boldsymbol{Y}^2\boldsymbol{X})^{2^{k-1}}) &= tr(\boldsymbol{X}\boldsymbol{Y}^2\boldsymbol{X}^2\boldsymbol{Y}^2\boldsymbol{X}^2 \cdots \boldsymbol{Y}^2\boldsymbol{X}) = tr(\boldsymbol{Y}^2\boldsymbol{X}^2\boldsymbol{Y}^2\boldsymbol{X}^2 \cdots \boldsymbol{Y}^2\boldsymbol{X}^2) \\ &= tr((\boldsymbol{Y}^2\boldsymbol{X}^2)^{2^{k-1}}) = tr((\boldsymbol{X}^2\boldsymbol{Y}^2)^{2^{k-1}}), \end{aligned}$$

Substituting this, we obtain

$$tr((\boldsymbol{X}\boldsymbol{Y})^{2^k}) \leqslant tr((\boldsymbol{X}^2\boldsymbol{Y}^2)^{2^{k-1}}). \tag{III}$$

On the other hand, let $\boldsymbol{P} = \boldsymbol{X}^2, \boldsymbol{Q} = \boldsymbol{Y}^2$ be positive semidefinite Hermitian matrices, then by the induction hypothesis, we obtain

$$tr((\boldsymbol{P}\boldsymbol{Q})^{2^{k-1}}) \leqslant tr(\boldsymbol{P}^{2^{k-1}}\boldsymbol{Q}^{2^{k-1}}),$$

i.e.,

$$tr((\boldsymbol{X}^2\boldsymbol{Y}^2)^{2^{k-1}}) \leqslant tr((\boldsymbol{X}^2)^{2^{k-1}}(\boldsymbol{Y}^2)^{2^{k-1}}) = tr(\boldsymbol{X}^{2^k}\boldsymbol{Y}^{2^k}). \tag{IV}$$

Combining (III) and (IV), we obtain

$$\mathrm{tr}((\boldsymbol{X}\boldsymbol{Y})^{2^k}) \leqslant \mathrm{tr}(\boldsymbol{X}^{2^k}\boldsymbol{Y}^{2^k}).$$

By auxiliary formula 1, we obtain

$$\mathrm{tr}(\exp(\boldsymbol{A}+\boldsymbol{B})) = \mathrm{tr}\left(\lim_{n\to\infty}\left(\exp\left(\frac{\boldsymbol{A}}{n}\right)\exp\left(\frac{\boldsymbol{B}}{n}\right)\right)^n\right) = \lim_{n\to\infty}\mathrm{tr}\left(\left(\exp\left(\frac{\boldsymbol{A}}{n}\right)\exp\left(\frac{\boldsymbol{B}}{n}\right)\right)^n\right),$$

Let $n = 2^k$,

$$\mathrm{tr}(\exp(\boldsymbol{A}+\boldsymbol{B})) = \lim_{k\to\infty}\mathrm{tr}\left(\left(\exp\left(\frac{\boldsymbol{A}}{2^k}\right)\exp\left(\frac{\boldsymbol{B}}{2^k}\right)\right)^{2^k}\right).$$

For the right-hand side, let

$$\boldsymbol{X}_k = \exp\left(\frac{\boldsymbol{A}}{2^k}\right), \quad \boldsymbol{Y}_k = \exp\left(\frac{\boldsymbol{B}}{2^k}\right),$$

then $\boldsymbol{X}_k, \boldsymbol{Y}_k$ are positive definite Hermitian matrices, hence by auxiliary formula 3, we obtain

$$\mathrm{tr}((\boldsymbol{X}_k\boldsymbol{Y}_k)^{2^k}) \leqslant \mathrm{tr}(\boldsymbol{X}_k^{2^k}\boldsymbol{Y}_k^{2^k}),$$

where

$$\boldsymbol{X}_k^{2^k} = \left(\exp\left(\frac{\boldsymbol{A}}{2^k}\right)\right)^{2^k} = \exp(\boldsymbol{A}), \boldsymbol{Y}_k^{2^k} = \left(\exp\left(\frac{\boldsymbol{B}}{2^k}\right)\right)^{2^k} = \exp(\boldsymbol{B}),$$

Substituting this,

$$\mathrm{tr}\left(\left(\exp\left(\frac{\boldsymbol{A}}{2^k}\right)\exp\left(\frac{\boldsymbol{B}}{2^k}\right)\right)^{2^k}\right) \leqslant \mathrm{tr}(\exp(\boldsymbol{A})\exp(\boldsymbol{B})),$$

Taking limits on both sides,

$$\lim_{k\to\infty} \operatorname{tr}\left(\left(\exp\left(\frac{\boldsymbol{A}}{2^k}\right)\exp\left(\frac{\boldsymbol{B}}{2^k}\right)\right)^{2^k}\right) \leqslant \operatorname{tr}(\exp(\boldsymbol{A})\exp(\boldsymbol{B})),$$

i.e.,

$$\operatorname{tr}(\exp(\boldsymbol{A}+\boldsymbol{B})) \leqslant \operatorname{tr}(\exp(\boldsymbol{A})\exp(\boldsymbol{B})).$$

22 [**Derivation of the Matrix Khintchine Inequality**] Let $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher, and $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_N \in \mathbb{R}^{n\times n}$ be deterministic symmetric matrices. Let $\boldsymbol{S} = \sum_{i=1}^N \varepsilon_i \boldsymbol{A}_i$, $\sigma^2 = \left\|\sum_{i=1}^N \boldsymbol{A}_i^2\right\|$. Using Theorem T5.4.13 (*Matrix Hoeffding inequality*), prove: for any $p \geqslant 1$, the matrix Khintchine inequality holds, namely Theorem T5.4.14 (*Matrix Khintchine inequality*),

$$(\mathbb{E}\|\boldsymbol{S}\|^p)^{1/p} \leqslant C(\sqrt{p}+\sqrt{\log n})\sigma. \qquad (C>0)$$

Prf. By Theorem T5.4.13 (*Matrix Hoeffding inequality*), for any $t \geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\} \leqslant 2n\exp\left(-\frac{t^2}{2\sigma^2}\right),$$

Using the tail integral formula, we obtain

$$\begin{aligned}
\mathbb{E}[\|\boldsymbol{S}\|^p] &= \int_0^\infty pt^{p-1}\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\}\mathrm{d}t \\
&= \int_0^{t_0} pt^{p-1}\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\}\mathrm{d}t + \int_{t_0}^\infty pt^{p-1}\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\}\mathrm{d}t \\
&\leqslant \int_0^{t_0} pt^{p-1}\mathrm{d}t + \int_{t_0}^\infty pt^{p-1}\cdot 2n\exp\left(-\frac{t^2}{2\sigma^2}\right)\mathrm{d}t \\
&= \underbrace{t_0^p}_{I_1} + \underbrace{\int_{t_0}^\infty pt^{p-1}\cdot 2n\exp\left(-\frac{t^2}{2\sigma^2}\right)\mathrm{d}t}_{I_2}.
\end{aligned}$$

Let $t_0 = \sigma\sqrt{2(\log(2n)+p)}$, then for I_1,

$$I_1 = \sigma^p(2(\log(2n)+p))^{p/2} \leqslant C^p\sigma^p(\sqrt{\log n}+\sqrt{p})^p.$$

For I_2,

$$\begin{aligned}
I_2 &= 2np(\sigma\sqrt{2})^p\int_{x_0}^\infty x^{p-1}\mathrm{e}^{-x^2}\mathrm{d}x && \left(x=\frac{t}{\sigma\sqrt{2}}\right) \\
&= np(\sigma\sqrt{2})^p\cdot\frac{1}{2}\int_{x_0^2}^\infty u^{\frac{p}{2}-1}\mathrm{e}^{-u}\mathrm{d}u. && (u=x^2)
\end{aligned}$$

The above is an incomplete Gamma integral. Since $t_0 = \sigma\sqrt{2(\log(2n)+p)}$,

$$x_0^2 = \frac{t_0^2}{2\sigma^2} = \log(2n)+p.$$

Let $x = x_0^2$, and define

$$J_p(x) = \int_x^\infty u^{p/2-1}\mathrm{e}^{-u}\mathrm{d}u.$$

Since $n \geqslant 1$, $x = \log(2n)+p \geqslant p$. If $p \geqslant 2$, integration by parts gives

$$\begin{aligned}
J_p(x) &= x^{p/2-1}\mathrm{e}^{-x} + \left(\frac{p}{2}-1\right)\int_x^\infty u^{p/2-2}\mathrm{e}^{-u}\mathrm{d}u \\
&\leqslant x^{p/2-1}\mathrm{e}^{-x} + \frac{p/2-1}{x}J_p(x).
\end{aligned}$$

Since $x \geqslant p$, this gives $J_p(x) \leqslant 2x^{p/2-1}\mathrm{e}^{-x}$. If $1 \leqslant p \leqslant 2$, then $u \geqslant x$ gives $u^{p/2-1} \leqslant x^{p/2-1}$, so the same bound holds.

Substituting this bound gives

$$\begin{aligned} I_2 &\leqslant 2np(\sigma\sqrt{2})^p x^{p/2-1}\mathrm{e}^{-x} \\ &= p(\sigma\sqrt{2})^p x^{p/2-1}\mathrm{e}^{-p} \\ &\leqslant C^p\sigma^p(\log(2n)+p)^{p/2} \\ &\leqslant C^p\sigma^p(\sqrt{\log n}+\sqrt{p})^p. \end{aligned}$$

Thus,

$$\mathbb{E}[\|\boldsymbol{S}\|^p] \leqslant C^p\sigma^p(\sqrt{\log n}+\sqrt{p})^p,$$

Taking the $1/p$-th power on both sides,

$$(\mathbb{E}\|\boldsymbol{S}\|^p)^{1/p} \leqslant C\sigma(\sqrt{\log n}+\sqrt{p}),$$

i.e.,

$$\left(\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{A}_i\right\|^p\right)^{1/p} \leqslant C(\sqrt{p}+\sqrt{\log n})\left\|\sum_{i=1}^N \boldsymbol{A}_i^2\right\|^{1/2}. \qquad (C>0)$$

23 [**Matrix Bernstein Inequality for Rectangular Matrices**] Let $\boldsymbol{X}_1,\cdots,\boldsymbol{X}_N \in \mathbb{R}^{m\times n}$ be independent, satisfying $\mathbb{E}[\boldsymbol{X}_i]=\boldsymbol{0}$, $\|\boldsymbol{X}_i\|\leqslant K$. Define the variance parameter σ^2 of the matrix sum as

$$\sigma^2 = \left\|\sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i^\top \boldsymbol{X}_i]\right\| + \left\|\sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i \boldsymbol{X}_i^\top]\right\|,$$

Prove: for any $t\geqslant 0$, the operator norm of the random matrix sum $\boldsymbol{S}=\sum_{i=1}^N \boldsymbol{X}_i$ satisfies

$$\mathbb{P}\{\|\boldsymbol{S}\|\geqslant t\} \leqslant 2(m+n)\exp\left(-\frac{t^2/2}{\sigma^2+Kt/3}\right).$$

Prf. (*The Bernstein inequality is mainly applicable to symmetric matrices (whose operator norm is related to its spectral radius). The standard technique for handling rectangular matrices is to use the Hermitian dilation technique.*)

For each $\boldsymbol{X}_i$, construct its symmetrized form $\boldsymbol{H}_i \in \mathbb{R}^{(m+n)\times(m+n)}$ as

$$\boldsymbol{H}_i = \begin{pmatrix} \boldsymbol{O} & \boldsymbol{X}_i \\ \boldsymbol{X}_i^\top & \boldsymbol{O} \end{pmatrix},$$

then:

1. Symmetry: $\boldsymbol{H}_i = \boldsymbol{H}_i^\top$.
2. Zero mean:

$$\mathbb{E}[\boldsymbol{H}_i] = \begin{pmatrix} \boldsymbol{O} & \mathbb{E}[\boldsymbol{X}_i] \\ \mathbb{E}[\boldsymbol{X}_i^\top] & \boldsymbol{O} \end{pmatrix} = \boldsymbol{O}.$$

3. Bounded norm: $\|\boldsymbol{H}_i\| = \|\boldsymbol{X}_i\| \leqslant K$.
4. Independence: Since $\boldsymbol{X}_i$ are independent, $\boldsymbol{H}_i$ are independent.

Construct the sum matrix

$$\boldsymbol{S}_H = \sum_{i=1}^N \boldsymbol{H}_i = \begin{pmatrix} \boldsymbol{O} & \boldsymbol{S} \\ \boldsymbol{S}^\top & \boldsymbol{O} \end{pmatrix},$$

satisfying

$$\|\boldsymbol{S}_H\| = \left\|\begin{pmatrix} \boldsymbol{O} & \boldsymbol{S} \\ \boldsymbol{S}^\top & \boldsymbol{O} \end{pmatrix}\right\| = \|\boldsymbol{S}\|,$$

Then the variance parameter σ_H^2 of the sum matrix is

$$\sigma_H^2 = \left\|\sum_{i=1}^N \mathbb{E}[\boldsymbol{H}_i^2]\right\| = \left\|\begin{pmatrix} \sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i\boldsymbol{X}_i^\top] & \boldsymbol{O} \\ \boldsymbol{O} & \sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i^\top\boldsymbol{X}_i] \end{pmatrix}\right\| = \max\left(\left\|\sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i\boldsymbol{X}_i^\top]\right\|, \left\|\sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i^\top\boldsymbol{X}_i]\right\|\right).$$

(*Since $\sigma_H^2 \leqslant \sigma^2$, using σ^2 as the bound for the variance parameter is valid.*)

For the symmetric matrix sum $\boldsymbol{S}_H \in \mathbb{R}^{(m+n)\times(m+n)}$, by Theorem T5.4.1 (*Matrix Bernstein inequality*), we obtain

$$\mathbb{P}\{\|\boldsymbol{S}_H\| \geqslant t\} \leqslant 2(m+n)\exp\left(-\frac{t^2/2}{\sigma_H^2 + Kt/3}\right) \leqslant 2(m+n)\exp\left(-\frac{t^2/2}{\sigma^2 + Kt/3}\right),$$

hence

$$\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\} \leqslant 2(m+n)\exp\left(-\frac{t^2/2}{\sigma^2 + Kt/3}\right).$$

24 [**Matrix Khintchine Inequality for Rectangular Matrices**] Let $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher, and $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_N \in \mathbb{R}^{m\times n}$ be deterministic matrices. Let the random matrix sum be $\boldsymbol{S} = \sum_{i=1}^N \varepsilon_i \boldsymbol{A}_i$, and define the variance parameter σ^2 as

$$\sigma^2 = \max\left(\left\|\sum_{i=1}^N \boldsymbol{A}_i^\top\boldsymbol{A}_i\right\|, \left\|\sum_{i=1}^N \boldsymbol{A}_i\boldsymbol{A}_i^\top\right\|\right).$$

Let $d = m + n$. Prove: for any $p \geqslant 1$,

$$(\mathbb{E}\|\boldsymbol{S}\|^p)^{1/p} \leqslant C\sigma\sqrt{p + \log d}. \qquad (C > 0)$$

Prf. By Theorem T5.4.13 (*Matrix Hoeffding inequality*), for any $t \geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\} \leqslant 2d\exp\left(-\frac{t^2}{2\sigma^2}\right).$$

Using the tail integral formula, we obtain

$$\begin{aligned}
\mathbb{E}[\|\boldsymbol{S}\|^p] &= \int_0^\infty pt^{p-1}\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\}\mathrm{d}t \\
&= \int_0^{t_0} pt^{p-1}\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\}\mathrm{d}t + \int_{t_0}^\infty pt^{p-1}\mathbb{P}\{\|\boldsymbol{S}\| \geqslant t\}\mathrm{d}t \\
&\leqslant \int_0^{t_0} pt^{p-1}\mathrm{d}t + \int_{t_0}^\infty pt^{p-1}\cdot 2d\exp\left(-\frac{t^2}{2\sigma^2}\right)\mathrm{d}t \\
&= \underbrace{t_0^p}_{I_1} + \underbrace{\int_{t_0}^\infty pt^{p-1}\cdot 2d\exp\left(-\frac{t^2}{2\sigma^2}\right)\mathrm{d}t}_{I_2}.
\end{aligned}$$

Let $t_0 = \sigma\sqrt{2(\log(2d) + p)}$, then for I_1, using the inequality $(a+b)^{p/2} \leqslant 2^{p/2-1}(a^{p/2} + b^{p/2})$, we obtain

$$I_1 = \sigma^p(2\log(2d) + 2p)^{p/2} \leqslant C^p\sigma^p(\sqrt{p} + \sqrt{\log d})^p.$$

For I_2,

$$\begin{aligned}
I_2 &= 2dp(\sigma\sqrt{2})^p\int_{x_0}^\infty x^{p-1}\mathrm{e}^{-x^2}\mathrm{d}x && \left(x = \frac{t}{\sigma\sqrt{2}}\right) \\
&= dp(\sigma\sqrt{2})^p\int_{x_0^2}^\infty u^{\frac{p}{2}-1}\mathrm{e}^{-u}\mathrm{d}u. && (u = x^2)
\end{aligned}$$

Note that the above integral is precisely $\Gamma(p/2, x_0^2)$, where

$$t_0 = \sigma\sqrt{2(\log(2d) + p)},$$

$$x_0^2 = \frac{t_0^2}{2\sigma^2} = \log(2d) + p,$$

Substituting this in, we obtain

$$\begin{aligned} I_2 &= dp(\sigma\sqrt{2})^p \int_{x_0^2}^{\infty} u^{\frac{p}{2}-1}\mathrm{e}^{-u}\mathrm{d}u \\ &\leqslant C^p\sigma^p[\log(2d)+p]^{p/2} \\ &\leqslant C^p\sigma^p\left[\sqrt{\log d}+\sqrt{p}\right]^p. \end{aligned}$$

Thus,

$$\mathbb{E}[\|\boldsymbol{S}\|^p] \leqslant C^p\sigma^p(\sqrt{\log d}+\sqrt{p})^p,$$

Taking the $1/p$-th power on both sides,

$$(\mathbb{E}\|\boldsymbol{S}\|^p)^{1/p} \leqslant C\sigma(\sqrt{p}+\sqrt{\log d}),$$

i.e.,

$$(\mathbb{E}\|\boldsymbol{S}\|^p)^{1/p} \leqslant C\sigma\sqrt{p+\log d}. \qquad (C>0)$$

25 [**SBM Model Without Self-Loops**] In Definition D4.5.1 (*Stochastic block model*) of the SBM, $G(n,p,q)$ allows self-loops, i.e., each vertex is connected to itself with probability p.

Now modify the definition as follows:

Let $G=(V,E)$ be a random graph with n vertices, partition V equally into two communities C_1, C_2, and define the probability matrix $\boldsymbol{P}$ as

$$p_{ij} = p\cdot\mathbf{1}_{\{c_i=c_j\}} + q\cdot\mathbf{1}_{\{c_i\neq c_j\}},$$

The adjacency matrix $\boldsymbol{A}$ is

$$A_{ij} \sim \begin{cases} 0, & i=j, \\ B(p), & i\neq j, c_i=c_j,, \\ B(q), & i\neq j, c_i\neq c_j \end{cases}$$

Prove: Theorem T5.5.1 (*Spectral clustering for sparse stochastic block model*) still holds.

Prf. Let $\boldsymbol{A}$ be the adjacency matrix containing self-loops, and $\tilde{\boldsymbol{A}}$ be the modified adjacency matrix without self-loops. Then they differ only on the diagonal, satisfying

$$\tilde{\boldsymbol{A}} = \boldsymbol{A} - \mathrm{diag}(A_{11}, A_{22}, \cdots, A_{nn}),$$

where A_{ii} i.i.d. $\sim$ Bernoulli(p). Taking expectations on both sides,

$$\mathbb{E}[\tilde{\boldsymbol{A}}] = \mathbb{E}[\boldsymbol{A}] - p\boldsymbol{I}_n.$$

Let $\boldsymbol{D}=\mathbb{E}[\boldsymbol{A}]$, $\tilde{\boldsymbol{D}}=\mathbb{E}[\tilde{\boldsymbol{A}}]$, then $\tilde{\boldsymbol{D}}=\boldsymbol{D}-p\boldsymbol{I}_n$ is a scalar translation, hence $\boldsymbol{D}$ and $\tilde{\boldsymbol{D}}$ share the same eigenvectors, and the eigenvalues are shifted by $\tilde{\lambda}_i=\lambda_i-p$. The spectral gap $\tilde{\delta}=\delta\asymp n(p-q)=a-b$ remains unchanged, thus by the Davis-Kahan theorem, the proof proceeds identically to that of Theorem T5.5.1 (*Spectral clustering for sparse stochastic block model*).

Let $\boldsymbol{Z}=\boldsymbol{A}-\boldsymbol{D}$ be the noise matrix with self-loops, and $\tilde{\boldsymbol{Z}}=\tilde{\boldsymbol{A}}-\tilde{\boldsymbol{D}}$ be the noise matrix without self-loops, then

$$\begin{aligned} \tilde{\boldsymbol{Z}} &= (\boldsymbol{A}-\mathrm{diag}(\boldsymbol{A})) - (\boldsymbol{D}-p\boldsymbol{I}_n) \\ &= (\boldsymbol{A}-\boldsymbol{D}) - (\mathrm{diag}(\boldsymbol{A})-p\boldsymbol{I}_n) \\ &= \boldsymbol{Z}-\boldsymbol{\Delta}, \end{aligned}$$

where $\boldsymbol{\Delta}=\mathrm{diag}(A_{11}-p,\cdots,A_{nn}-p)$ satisfies $\|\boldsymbol{\Delta}\|\leqslant 1$ (*since* $|A_{ii}-p|\leqslant\max(p,1-p)$). Substituting this, we obtain

$$\|\tilde{\boldsymbol{Z}}\| \leqslant \|\boldsymbol{Z}\|+1 \leqslant C\sqrt{a}+1 \asymp C\sqrt{a}.$$

26 [**High-Probability Bound and Effective Rank for Covariance Estimation**] Let $\boldsymbol{X}\in\mathbb{R}^n$ be a centered sub-Gaussian

random vector with population covariance matrix $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]$, and suppose that $\|\boldsymbol{X}\|_2^2 \leqslant K^2 \operatorname{tr}(\boldsymbol{\Sigma})$ almost surely,

$$\|\langle \boldsymbol{X}, \boldsymbol{x}\rangle\|_{\psi_2} \leqslant K \|\langle \boldsymbol{X}, \boldsymbol{x}\rangle\|_{L^2}. \qquad (K \geqslant 1)$$

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_m$ be i.i.d. samples of $\boldsymbol{X}$, and define the sample covariance matrix as

$$\boldsymbol{\Sigma}_m = \frac{1}{m} \sum_{i=1}^m \boldsymbol{X}_i \boldsymbol{X}_i^\top.$$

Define the effective rank of $\boldsymbol{\Sigma}$ as

$$r = \begin{cases} \frac{\operatorname{tr}(\boldsymbol{\Sigma})}{\|\boldsymbol{\Sigma}\|}, & \|\boldsymbol{\Sigma}\| > 0, \\ 0, & \|\boldsymbol{\Sigma}\| = 0 \end{cases} \leqslant n,$$

Using Theorem T5.4.1 (*Matrix Bernstein inequality*) (*Matrix Bernstein inequality*), prove the high-probability bound of Remark R5.6.5 (*High-probability guarantees*): for any $u \geqslant 0$, with high probability,

$$\mathbb{P}\left\{ \|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant C \left(\sqrt{\frac{K^4 r(\log n + u)}{m}} + \frac{K^2 r(\log n + u)}{m} \right) \|\boldsymbol{\Sigma}\| \right\} \geqslant 1 - 2\mathrm{e}^{-u}.$$

Rk. This exercise is an important result in covariance estimation theory. It requires us to abandon the previous ε-net-based proof path (*because the ε-net can only give a bound related to the dimension n*), and instead directly use the matrix Bernstein inequality, thereby introducing the effective rank r, which is much smaller than the dimension n.

Prf. **Solution 1 Bounded matrix Bernstein inequality**

Define the centered random matrix $\boldsymbol{Z}_i = \boldsymbol{X}_i \boldsymbol{X}_i^\top - \boldsymbol{\Sigma}$, then the difference between the sample covariance matrix and the population covariance matrix can be expressed as the average of the above independent zero-mean random matrices,

$$\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma} = \frac{1}{m} \sum_{i=1}^m \boldsymbol{Z}_i.$$

To apply Theorem T5.4.1 (*Matrix Bernstein inequality*) (*Matrix Bernstein inequality*), define

$$\sigma^2 = \left\| \sum_{i=1}^m \mathbb{E}[\boldsymbol{Z}_i^2] \right\|.$$

- Estimate the variance parameter σ^2

  Expand $\mathbb{E}[\boldsymbol{Z}^2]$,

  $$\mathbb{E}[\boldsymbol{Z}^2] = \mathbb{E}[(\boldsymbol{X}\boldsymbol{X}^\top - \boldsymbol{\Sigma})^2] = \mathbb{E}\|\boldsymbol{X}\|_2^2 \boldsymbol{X}\boldsymbol{X}^\top - \boldsymbol{\Sigma}^2 \preceq \mathbb{E}\|\boldsymbol{X}\|_2^2 \boldsymbol{X}\boldsymbol{X}^\top,$$

  Hence it suffices to estimate $\sup_{\boldsymbol{v} \in \mathbb{S}^{n-1}} \boldsymbol{v}^\top \mathbb{E}\|\boldsymbol{X}\|_2^2 \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{v}$. Let $\boldsymbol{w}_1, \cdots, \boldsymbol{w}_n$ be an orthonormal basis. Then, for every $\boldsymbol{v} \in \mathbb{S}^{n-1}$,

  $$\begin{aligned} \mathbb{E}\|\boldsymbol{X}\|_2^2 \langle \boldsymbol{X}, \boldsymbol{v}\rangle^2 &= \sum_{j=1}^n \mathbb{E}[\langle \boldsymbol{X}, \boldsymbol{w}_j\rangle^2 \langle \boldsymbol{X}, \boldsymbol{v}\rangle^2] \leqslant \sum_{j=1}^n \sqrt{\mathbb{E}[\langle \boldsymbol{X}, \boldsymbol{w}_j\rangle^4] \mathbb{E}[\langle \boldsymbol{X}, \boldsymbol{v}\rangle^4]} \\ &\leqslant CK^4 \sum_{j=1}^n \langle \boldsymbol{w}_j, \boldsymbol{\Sigma}\boldsymbol{w}_j\rangle \langle \boldsymbol{v}, \boldsymbol{\Sigma}\boldsymbol{v}\rangle = CK^4 \operatorname{tr}(\boldsymbol{\Sigma}) \langle \boldsymbol{v}, \boldsymbol{\Sigma}\boldsymbol{v}\rangle \leqslant CK^4 r \|\boldsymbol{\Sigma}\|^2, \end{aligned}$$

  $$\sigma^2 = \left\| \sum_{i=1}^m \mathbb{E}[\boldsymbol{Z}_i^2] \right\| \leqslant CmK^4 r \|\boldsymbol{\Sigma}\|^2.$$

  If $\|\boldsymbol{\Sigma}\| = 0$, then $\mathbb{E}\|\boldsymbol{X}\|_2^2 = \operatorname{tr}(\boldsymbol{\Sigma}) = 0$, so $\boldsymbol{X} = \boldsymbol{0}$ almost surely. Then the conclusion is trivial; afterwards only consider the case $\|\boldsymbol{\Sigma}\| > 0$.

  Here $r \geqslant 1$. By the assumption, we obtain

  $$\|\boldsymbol{Z}_i\| \leqslant \|\boldsymbol{X}_i\|_2^2 + \|\boldsymbol{\Sigma}\| \leqslant (K^2 r + 1)\|\boldsymbol{\Sigma}\| \leqslant 2K^2 r \|\boldsymbol{\Sigma}\| = M.$$

- Apply the bounded matrix Bernstein inequality

By Theorem T5.4.1 (*Matrix Bernstein inequality*) (*Matrix Bernstein inequality*), for any $t \geqslant 0$, we obtain

$$\mathbb{P}\{\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \geqslant t\} = \mathbb{P}\left\{\left\|\sum_{i=1}^m \boldsymbol{Z}_i\right\| \geqslant mt\right\} \leqslant 2n\exp\left(-\frac{m^2t^2/2}{\sigma^2 + Mmt/3}\right) \leqslant 2n\exp\left(-c\min\left(\frac{mt^2}{K^4r\|\boldsymbol{\Sigma}\|^2}, \frac{mt}{K^2r\|\boldsymbol{\Sigma}\|}\right)\right),$$

If the above probability does not exceed $2\mathrm{e}^{-u}$, then

$$c\min\left(\frac{mt^2}{C_1K^4r\|\boldsymbol{\Sigma}\|^2}, \frac{mt}{C_5K^2r\|\boldsymbol{\Sigma}\|}\right) \geqslant \log n + u,$$

i.e.,

$$\frac{mt^2}{C_1K^4r\|\boldsymbol{\Sigma}\|^2} \geqslant \frac{\log n + u}{c},$$

$$t \geqslant \sqrt{\frac{C_1}{c}}\sqrt{\frac{K^4r(\log n + u)}{m}}\|\boldsymbol{\Sigma}\|,$$

$$\frac{mt}{2K^2r\|\boldsymbol{\Sigma}\|} \geqslant \frac{\log n + u}{c},$$

$$t \geqslant \frac{2}{c}\frac{K^2r(\log n + u)}{m}\|\boldsymbol{\Sigma}\|,$$

Therefore set

$$C = \max\left(\sqrt{\frac{C_1}{c}}, \frac{2}{c}\right), t = C\left(\sqrt{\frac{K^4r(\log n + u)}{m}} + \frac{K^2r(\log n + u)}{m}\right)\|\boldsymbol{\Sigma}\|,$$

We obtain that for any $u \geqslant 0$,

$$\mathbb{P}\left\{\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant C\left(\sqrt{\frac{K^4r(\log n + u)}{m}} + \frac{K^2r(\log n + u)}{m}\right)\|\boldsymbol{\Sigma}\|\right\} \geqslant 1 - 2\mathrm{e}^{-u}.$$

Solution 2 Khintchine inequality

- Symmetrization and matrix Khintchine inequality

  Denote the error matrix as

$$\boldsymbol{R}_m = \frac{1}{m}\sum_{i=1}^m (\boldsymbol{X}_i\boldsymbol{X}_i^\top - \boldsymbol{\Sigma}),$$

  We need to estimate its L^p norm $E_p = (\mathbb{E}\|\boldsymbol{R}_m\|^p)^{1/p}$, where $p \geqslant 1$.

  Introduce $\varepsilon_1, \cdots, \varepsilon_m$ i.i.d. $\sim$ Rademacher. Using the symmetrization inequality,

$$E_p \leqslant \frac{2}{m}\left(\mathbb{E}\left\|\sum_{i=1}^m \varepsilon_i\boldsymbol{X}_i\boldsymbol{X}_i^\top\right\|^p\right)^{1/p},$$

  Then apply Theorem T5.4.14 (*Matrix Khintchine inequality*) to ε_i,

$$E_p \leqslant \frac{C_1\sqrt{p + \log n}}{m}\left(\mathbb{E}\left\|\sum_{i=1}^m (\boldsymbol{X}_i\boldsymbol{X}_i^\top)^2\right\|^{p/2}\right)^{1/p}. \tag{I}$$

- Compute the implicit bound for E_p

  Note that $(\boldsymbol{X}_i\boldsymbol{X}_i^\top)^2 = \|\boldsymbol{X}_i\|_2^2\boldsymbol{X}_i\boldsymbol{X}_i^\top$, and $\left\|\sum_{i=1}^m \boldsymbol{X}_i\boldsymbol{X}_i^\top\right\| \leqslant m(\|\boldsymbol{R}_m\| + \|\boldsymbol{\Sigma}\|)$, hence

$$\begin{aligned}\left\|\sum_{i=1}^m \|\boldsymbol{X}_i\|_2^2\boldsymbol{X}_i\boldsymbol{X}_i^\top\right\| &\leqslant \max_{1\leqslant i\leqslant m}\|\boldsymbol{X}_i\|_2^2 \cdot \left\|\sum_{i=1}^m \boldsymbol{X}_i\boldsymbol{X}_i^\top\right\| \\ &\leqslant \max_{1\leqslant i\leqslant m}\|\boldsymbol{X}_i\|_2^2 \cdot m(\|\boldsymbol{R}_m\| + \|\boldsymbol{\Sigma}\|) \\ &= Mm(\|\boldsymbol{R}_m\| + \|\boldsymbol{\Sigma}\|), \qquad \left(M = \max_{1\leqslant i\leqslant m}\|\boldsymbol{X}_i\|_2^2\right)\end{aligned}$$

Substituting this into (I) and using Cauchy's inequality,

$$E_p \leqslant \frac{C_1\sqrt{p+\log n}}{m}\left[\mathbb{E}[M^{p/2}\cdot m^{p/2}[\|\boldsymbol{R}_m\|+\|\boldsymbol{\Sigma}\|]^{p/2}]\right]^{1/p}$$
$$\leqslant C_1\sqrt{\frac{p+\log n}{m}}\|M\|_{L^p}^{1/2}\cdot[E_p+\|\boldsymbol{\Sigma}\|]^{1/2}. \tag{II}$$

Let $\alpha = C_1\sqrt{\frac{p+\log n}{m}}\|M\|_{L^p}^{1/2}$, $x = E_p$, $c = \|\boldsymbol{\Sigma}\|$, then (II) becomes

$$x \leqslant \alpha\sqrt{x+c},$$

Solving this inequality,

$$x \leqslant \alpha^2 + \alpha\sqrt{c},$$

i.e.,

$$E_p \leqslant \alpha^2 + \alpha\sqrt{\|\boldsymbol{\Sigma}\|} = C_1^2\frac{p+\log n}{m}\|M\|_{L^p} + C_1\sqrt{\frac{p+\log n}{m}}\sqrt{\|M\|_{L^p}\|\boldsymbol{\Sigma}\|}, \tag{III}$$

It then suffices to estimate $\|M\|_{L^p}$.

- Compute the explicit bound for E_p

  By the assumption, for every i, we obtain

$$\|\boldsymbol{X}_i\|_2^2 \leqslant K^2\operatorname{tr}(\boldsymbol{\Sigma}) = K^2 r\|\boldsymbol{\Sigma}\|,$$

  Hence

$$\|M\|_{L^p} \leqslant K^2 r\|\boldsymbol{\Sigma}\|. \tag{IV}$$

  Substituting (IV) into (III),

$$E_p \leqslant C_1^2\frac{K^2 r(p+\log n)}{m}\|\boldsymbol{\Sigma}\| + C_1\sqrt{\frac{K^2 r(p+\log n)}{m}}\|\boldsymbol{\Sigma}\|$$
$$\leqslant C_5\left(\frac{K^2 r(p+\log n)}{m} + \sqrt{\frac{K^4 r(p+\log n)}{m}}\right)\|\boldsymbol{\Sigma}\|. \tag{V}$$

- Convert to a high-probability bound using Markov's inequality

  By Proposition P1.6.2 (*Markov inequality*) (*Markov's inequality*), for any $t > 0$,

$$\mathbb{P}\{\|\boldsymbol{R}_m\| \geqslant \mathrm{e}\cdot E_p\} = \mathbb{P}\{\|\boldsymbol{R}_m\|^p \geqslant \mathrm{e}^p\cdot\mathbb{E}\|\boldsymbol{R}_m\|^p\} \leqslant \mathrm{e}^{-p}.$$

  Let $p = u + \log 2$. We have

$$\mathbb{P}\left\{\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant C_6\left(\sqrt{\frac{K^4 r(u+\log n)}{m}} + \frac{K^2 r(u+\log n)}{m}\right)\|\boldsymbol{\Sigma}\|\right\} \geqslant 1 - 2\mathrm{e}^{-u}.$$

Rk. In Solution 2, by the assumption,

$$\left\|\sum_{i=1}^m \|\boldsymbol{X}_i\|_2^2\boldsymbol{X}_i\boldsymbol{X}_i^\top\right\| \leqslant \max_{1\leqslant i\leqslant m}\|\boldsymbol{X}_i\|_2^2\cdot\left\|\sum_{i=1}^m \boldsymbol{X}_i\boldsymbol{X}_i^\top\right\|$$

the maximum sample norm M satisfies $M \leqslant K^2 r\|\boldsymbol{\Sigma}\|$ almost surely, so no additional logarithmic factor is introduced.

27 [**Necessity of the Boundedness Assumption**] Prove that the boundedness assumption of Theorem T5.6.1 (*General covariance estimation*) cannot be omitted. Specifically, construct a random vector $\boldsymbol{X}$ that does not satisfy the above boundedness assumption, and show that even when the sample size m is relatively large, $\boldsymbol{\Sigma}_m$ still fails to converge to the population covariance matrix $\boldsymbol{\Sigma}$.

Sol. Let $n \geqslant 2$, and define the distribution of the random vector $\boldsymbol{X} \in \mathbb{R}^n$ as

$$\boldsymbol{X} = \begin{cases} \boldsymbol{0}, & p = 1-\varepsilon, \\ \sqrt{\dfrac{n}{\varepsilon}}\boldsymbol{e}_i, & p = \dfrac{\varepsilon}{n} \end{cases},$$

then

$$\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = (1-\varepsilon)\cdot \mathbf{0} + \sum_{i=1}^{n} \frac{\varepsilon}{n}\left(\sqrt{\frac{n}{\varepsilon}}\boldsymbol{e}_i\right)\left(\sqrt{\frac{n}{\varepsilon}}\boldsymbol{e}_i\right)^\top$$
$$= \sum_{i=1}^{n} \frac{\varepsilon}{n}\cdot\frac{n}{\varepsilon}\cdot \boldsymbol{e}_i\boldsymbol{e}_i^\top = \sum_{i=1}^{n} \boldsymbol{e}_i\boldsymbol{e}_i^\top = \boldsymbol{I}_n,$$

hence $\boldsymbol{X}$ is isotropic.

However, when

$$\boldsymbol{X} = \sqrt{\frac{n}{\varepsilon}}\boldsymbol{e}_i,$$

$\|\boldsymbol{X}\|_2 = \sqrt{\frac{n}{\varepsilon}}$. Assuming there exists $K > 0$ such that $\|\boldsymbol{X}\|_2 \leqslant K\sqrt{n}$ (a.s.), then

$$\sqrt{\frac{n}{\varepsilon}} \leqslant K\sqrt{n},$$
$$K \geqslant \frac{1}{\sqrt{\varepsilon}},$$

Letting $\varepsilon \to 0$, no such K exists, a contradiction; thus $\boldsymbol{X}$ does not satisfy the boundedness assumption.

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_m$ be i.i.d. samples of $\boldsymbol{X}$, then

$$\mathbb{P}\{\boldsymbol{X}_1 = \cdots = \boldsymbol{X}_m = \mathbf{0}\} = (1-\varepsilon)^m \sim \mathrm{e}^{-\varepsilon m},$$

Let $\varepsilon = 1/n$, $m < n$, then

$$\mathbb{P}\{\boldsymbol{\Sigma}_m = 0\} = \left(1 - \frac{1}{n}\right)^m \sim \mathrm{e}^{-m/n} \sim 1,$$

In this case $\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| = \|\mathbf{0} - \boldsymbol{I}_n\| = 1$, i.e., $\boldsymbol{\Sigma}_m \nrightarrow \boldsymbol{\Sigma}$.

28 [**Covariance Estimation and the Logarithmic Factor in the Matrix Bernstein Inequality**]

(1) Construct a probability distribution such that when $m \lesssim n\log n$, the sample covariance matrix $\boldsymbol{\Sigma}_m$ does not necessarily satisfy $\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| < \|\boldsymbol{\Sigma}\|$ (w.h.p.).

(2) For the above construction, prove that the logarithmic factor $\log n$ in the matrix Bernstein inequality is unavoidable.

Sol. (1) Construct the random vector $\boldsymbol{X} \in \mathbb{R}^n$ as

$$\mathbb{P}\{\boldsymbol{X} = \boldsymbol{e}_i\} = \frac{1}{n},$$

then the population covariance matrix is

$$\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \sum_{i=1}^{n} \frac{1}{n}\boldsymbol{e}_i\boldsymbol{e}_i^\top = \frac{1}{n}\boldsymbol{I}_n,$$

whose spectral norm is $\|\boldsymbol{\Sigma}\| = 1/n$.

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_m$ be m i.i.d. samples, then the sample covariance matrix is

$$\boldsymbol{\Sigma}_m = \frac{1}{m}\sum_{j=1}^{m} \boldsymbol{X}_j\boldsymbol{X}_j^\top,$$

Note that $\boldsymbol{\Sigma}_m$ is a diagonal matrix whose k-th diagonal entry is $d_k = \mathrm{count}(k)/m$, where $\mathrm{count}(k)$ is the number of times $\boldsymbol{e}_k$ appears.

If $\boldsymbol{e}_k$ never appears, i.e., $\mathrm{count}(k) = 0$, then

$$(\boldsymbol{\Sigma}_m)_{kk} = 0, (\boldsymbol{\Sigma})_{kk} = \frac{1}{n},$$

Hence

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \geqslant \max_{1\leqslant k\leqslant n} |(\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma})_{kk}| = \frac{1}{n} = \|\boldsymbol{\Sigma}\|.$$

By the coupon collector problem, when $m \leqslant \frac{1}{2} n \log n$, the probability that at least one vector is not sampled tends to 1. Therefore, if $m \lesssim n \log n$, then $\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \geqslant \|\boldsymbol{\Sigma}\|$ (w.h.p.).

(2) Suppose the matrix Bernstein inequality takes the form

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \lesssim \|\boldsymbol{\Sigma}\| \sqrt{\frac{n}{m}},$$

If the error $\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| < \|\boldsymbol{\Sigma}\|$, then $\sqrt{n/m} < 1$, i.e., $m > n$.

However, using the conclusion of (1), even when $n < m < n \log n$, $\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \geqslant \|\boldsymbol{\Sigma}\|$ (w.h.p.) still holds, which shows that the bound $m \asymp n$ is not valid.

29 [**Effective Rank**] Let $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$ be a symmetric positive semidefinite matrix, whose effective rank is defined as

$$r(\boldsymbol{\Sigma}) = \frac{\operatorname{tr}(\boldsymbol{\Sigma})}{\|\boldsymbol{\Sigma}\|}.$$

(1) Prove: $1 \leqslant r(\boldsymbol{\Sigma}) \leqslant \operatorname{rank}(\boldsymbol{\Sigma}) \leqslant n$.

(2) Prove that the above inequalities are sharp, i.e., there exists a full-rank matrix whose effective rank can be arbitrarily close to 1.

(3) Prove: the effective rank is a continuous function of the matrix (*in the sense of the operator norm*).

(4) Prove: if a random vector $\boldsymbol{X}$ takes values in a k-dimensional subspace of $\mathbb{R}^n$, then its covariance matrix satisfies $\operatorname{rank}(\boldsymbol{\Sigma}) \leqslant k$, and consequently $r(\boldsymbol{\Sigma}) \leqslant k$.

Prf. It is easy to see that

$$\operatorname{tr}(\boldsymbol{\Sigma}) = \sum_{i=1}^{n} \lambda_i, \|\boldsymbol{\Sigma}\| = \lambda_1, r(\boldsymbol{\Sigma}) = \frac{\sum_{i=1}^{n} \lambda_i}{\lambda_1}.$$

(1) The left and right sides are obvious; we only need to prove $r(\boldsymbol{\Sigma}) \leqslant \operatorname{rank}(\boldsymbol{\Sigma})$.

Let $\operatorname{rank}(\boldsymbol{\Sigma}) = k$, then it has k nonzero eigenvalues, i.e., $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_k > 0$, $\lambda_{k+1} = \cdots = \lambda_n = 0$, hence

$$\operatorname{tr}(\boldsymbol{\Sigma}) = \sum_{i=1}^{k} \lambda_i \leqslant \sum_{i=1}^{k} \lambda_1 = k\lambda_1,$$

Thus

$$r(\boldsymbol{\Sigma}) = \frac{\operatorname{tr}(\boldsymbol{\Sigma})}{\lambda_1} \leqslant k = \operatorname{rank}(\boldsymbol{\Sigma}).$$

(2) Consider the diagonal matrix $\boldsymbol{\Sigma}_\varepsilon = \operatorname{diag}(1, \varepsilon, \varepsilon, \cdots, \varepsilon)$, then $\operatorname{rank}(\boldsymbol{\Sigma}_\varepsilon) = n$, $\operatorname{tr}(\boldsymbol{\Sigma}_\varepsilon) = 1 + (n-1)\varepsilon$, and

$$r(\boldsymbol{\Sigma}_\varepsilon) = \frac{1 + (n-1)\varepsilon}{1} = 1 + (n-1)\varepsilon \to 1,$$

Hence the effective rank can be arbitrarily close to the lower bound 1.

(3) For the trace function $g(\boldsymbol{\Sigma}) = \operatorname{tr}(\boldsymbol{\Sigma})$, it is a linear combination of matrix entries and is clearly continuous; For the operator norm $h(\boldsymbol{\Sigma}) = \|\boldsymbol{\Sigma}\|$, by Lemma L4.1.14 (*Weyl's inequality*) (*Weyl's inequality*), the eigenvalues are Lipschitz continuous functions of the matrix entries; Therefore the effective rank function $r(\boldsymbol{\Sigma}) = g(\boldsymbol{\Sigma})/h(\boldsymbol{\Sigma})$ is continuous.

(*In contrast, the ordinary rank function exhibits a jump when a matrix becomes singular (e.g., when* $\operatorname{diag}(1, \varepsilon) \to \operatorname{diag}(1, 0)$*, the rank changes from* 2 *to* 1*), which demonstrates the stability of the effective rank.*)

(4) Let $\operatorname{supp}(\boldsymbol{X})$ be $S \subset \mathbb{R}^n$ satisfying $\dim(S) = k$, then for any $\boldsymbol{v} \in S^\perp$, $\langle \boldsymbol{X}, \boldsymbol{v} \rangle = 0$, hence

$$\boldsymbol{v}^\top \boldsymbol{\Sigma} \boldsymbol{v} = \mathbb{E}[\boldsymbol{v}^\top \boldsymbol{X} \boldsymbol{X}^\top \boldsymbol{v}] = \mathbb{E}[\langle \boldsymbol{X}, \boldsymbol{v} \rangle^2] = 0,$$

Thus $S^\perp \subset \operatorname{null}(\boldsymbol{\Sigma})$, satisfying

$$\operatorname{nullity}(\boldsymbol{\Sigma}) \geqslant \dim(S^\perp) = n - k,$$

Hence

$$\operatorname{rank}(\boldsymbol{\Sigma}) = n - \operatorname{nullity}(\boldsymbol{\Sigma}) \leqslant n - (n-k) = k,$$

Using the conclusion of (1), we obtain $r(\boldsymbol{\Sigma}) \leqslant \operatorname{rank}(\boldsymbol{\Sigma}) \leqslant k$.

30 [**Sampling from a Parseval Frame**] Let $(u_1, \cdots, u_m) \in \mathbb{R}^n$ be an equi-norm Parseval frame, i.e., for any $\boldsymbol{x} \in \mathbb{R}^n$, $\sum_{i=1}^m \langle \boldsymbol{x}, u_i \rangle^2 = \|\boldsymbol{x}\|_2^2$, where $\|u_i\|_2^2 = n/m$.

Let $1 \leqslant k \leqslant m$ and let $S \subset \{1, \cdots, m\}$ be a random subset consisting of k indices chosen uniformly without replacement from $\{1, \cdots, m\}$. Prove: when $k \gtrsim n \log n$, $\{u_i\}_{i \in S}$ forms an "approximate frame" (w.h.p.), i.e., there exists $\varepsilon \in (0, 1)$ such that for any $\boldsymbol{x} \in \mathbb{S}^{n-1}$,

$$(1-\varepsilon)\|\boldsymbol{x}\|_2^2 \leqslant \frac{m}{k} \sum_{i \in S} \langle \boldsymbol{x}, u_i \rangle^2 \leqslant (1+\varepsilon)\|\boldsymbol{x}\|_2^2.$$

Equivalently, define the sample frame operator as

$$\boldsymbol{A}_S = \frac{m}{k} \sum_{i \in S} u_i u_i^\top,$$

then $\|\boldsymbol{A}_S - \boldsymbol{I}_n\| \leqslant \varepsilon$.

Rk. This problem is a special case of Rudelson's lemma in the equi-norm setting. In 1999, Rudelson published the paper Random vectors in the isotropic position, at which time the matrix Bernstein inequality had not yet been discovered; Rudelson used the noncommutative Khintchine inequality combined with symmetrization techniques.

Prf. Let $Y_i = u_i u_i^\top$. Then $\|Y_i\| = n/m$ and

$$\sum_{i=1}^m Y_i = \boldsymbol{I}_n.$$

For a uniformly chosen fixed-size sample without replacement, the without-replacement matrix Chernoff inequality gives

$$\mathbb{P}\left\{ \left\| \frac{m}{k} \sum_{i \in S} Y_i - \boldsymbol{I}_n \right\| \geqslant \varepsilon \right\} \leqslant 2n \exp\left(-c \frac{\varepsilon^2 k}{n} \right).$$

Taking $k \geqslant Cn \log n$ and C sufficiently large makes the right-hand side smaller than any prescribed constant, so $\|\boldsymbol{A}_S - \boldsymbol{I}_n\| \leqslant \varepsilon$ with high probability.

31 [**Random Matrices with General Independent Rows**] Let $n \geqslant 2$, $K \geqslant 1$, and let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a random matrix whose row vectors $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_m \in \mathbb{R}^n$ are independent and isotropic. Prove: if $\|\boldsymbol{A}_i\|_2 \leqslant K\sqrt{n}$ and $m \geqslant CK^2 t^2 n \log n$, then for any $t \geqslant 1$, the smallest singular value $s_n(\boldsymbol{A})$ and the largest singular value $s_1(\boldsymbol{A})$ of $\boldsymbol{A}$ satisfy with high probability

$$\mathcal{E}_t = \left\{ \sqrt{m} - Kt\sqrt{n \log n} \leqslant s_n(\boldsymbol{A}) \leqslant s_1(\boldsymbol{A}) \leqslant \sqrt{m} + Kt\sqrt{n \log n} \right\},$$

$$\mathbb{P}(\mathcal{E}_t) \geqslant 1 - 2n^{-ct^2}. \qquad (c > 0)$$

Prf. Denote the sample covariance matrix

$$\boldsymbol{\Sigma}_m = \frac{1}{m} \boldsymbol{A}^\top \boldsymbol{A}.$$

Note that $s_i^2(\boldsymbol{A}) = \lambda_i(\boldsymbol{A}^\top \boldsymbol{A})$, then

$$s_i(\boldsymbol{A}) = \sqrt{m \cdot \lambda_i(\boldsymbol{\Sigma}_m)}.$$

If $\|\boldsymbol{\Sigma}_m - \boldsymbol{I}_n\| \leqslant \varepsilon$ (w.h.p.), then by perturbation theory,

$$\sqrt{m(1-\varepsilon)} \leqslant s_n(\boldsymbol{A}) \leqslant s_1(\boldsymbol{A}) \leqslant \sqrt{m(1+\varepsilon)},$$

i.e.,

$$\sqrt{m}(1-\varepsilon) \leqslant s_n(\boldsymbol{A}) \leqslant s_1(\boldsymbol{A}) \leqslant \sqrt{m}(1+\varepsilon), \qquad (\mathrm{I})$$

We only need to prove the preceding estimate for $\varepsilon \asymp Kt\sqrt{\frac{n \log n}{m}}$.

• Apply the matrix Bernstein inequality

Construct the centered random matrix $\boldsymbol{Z}_i = \boldsymbol{A}_i^\top \boldsymbol{A}_i - \boldsymbol{I}_n$, then

$$\boldsymbol{\Sigma}_m - \boldsymbol{I}_n = \frac{1}{m}\sum_{i=1}^m \boldsymbol{Z}_i.$$

For the sub-Exponential upper bound R,

$$\|\boldsymbol{Z}_i\| = \|\boldsymbol{A}_i^\top \boldsymbol{A}_i - \boldsymbol{I}_n\| \leqslant \|\boldsymbol{A}_i\|_2^2 + 1 \leqslant K^2 n + 1 \leqslant 2K^2 n,$$

Now set $R = 2K^2 n$.

For the variance parameter σ^2, since

$$\mathbb{E}[(\boldsymbol{A}_i^\top \boldsymbol{A}_i)^2] \preccurlyeq K^2 n \mathbb{E}[\boldsymbol{A}_i^\top \boldsymbol{A}_i] = K^2 n \boldsymbol{I}_n,$$

Hence

$$\sigma^2 = \left\| \sum_{i=1}^m \mathbb{E}[\boldsymbol{Z}_i^2] \right\| = m\|\mathbb{E}[(\boldsymbol{A}_i^\top \boldsymbol{A}_i - \boldsymbol{I}_n)^2]\| = m\|\mathbb{E}[(\boldsymbol{A}_i^\top \boldsymbol{A}_i)^2] - \boldsymbol{I}_n\| \leqslant mK^2 n.$$

Thus, using Theorem T5.4.1 (*Matrix Bernstein inequality*) (*Matrix Bernstein inequality*), for any $\varepsilon > 0$,

$$\mathbb{P}\{\|\boldsymbol{\Sigma}_m - \boldsymbol{I}_n\| \geqslant \varepsilon\} \leqslant 2n \exp\left(-c \min\left(\frac{m^2\varepsilon^2}{mK^2 n}, \frac{m\varepsilon}{2K^2 n}\right)\right).$$

Set $\varepsilon = C_0 K t \sqrt{\frac{n \log n}{m}}$, where C_0 is a sufficiently large absolute constant. Since $m \geqslant CK^2 t^2 n \log n$, taking C sufficiently large gives $0 < \varepsilon \leqslant 1$, and

$$\frac{m\varepsilon^2}{K^2 n} = C_0^2 t^2 \log n,$$
$$\frac{m\varepsilon}{2K^2 n} \geqslant \frac{C_0}{2} t^2 \log n.$$

Therefore, after adjusting the absolute constants, let

$$\mathcal{F}_t = \left\{ \|\boldsymbol{\Sigma}_m - \boldsymbol{I}_n\| \leqslant Kt\sqrt{\frac{n \log n}{m}} \right\},$$
$$\mathbb{P}(\mathcal{F}_t) \geqslant 1 - 2n^{-ct^2}.$$

Using $\sqrt{1-\varepsilon} \geqslant 1 - \varepsilon$ and substituting this ε into (I),

$$\sqrt{m} - Kt\sqrt{n \log n} \leqslant s_n(\boldsymbol{A}) \leqslant s_1(\boldsymbol{A}) \leqslant \sqrt{m} + Kt\sqrt{n \log n}.$$

32 [**Matrix Sketching**] Some matrices $\boldsymbol{A} \in \mathbb{R}^{N \times n}$ are too large to directly compute eigenvalues and eigenvectors. A commonly used technique is subsampling, which selects rows or columns randomly to construct a smaller matrix.

Let the row vectors of $\boldsymbol{A}$ be $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_N \in \mathbb{R}^n$, and assume that all rows have the same norm, i.e., $\|\boldsymbol{a}_j\|_2^2 = \sigma^2$. Select m indices $j_1, \cdots, j_m$ by uniform sampling (*with replacement*), and construct $\boldsymbol{B} \in \mathbb{R}^{m \times n}$ whose row vectors are

$$\boldsymbol{b}_k = \sqrt{\frac{N}{m}} \boldsymbol{a}_{j_k}.$$

Prove: when $m \geqslant Cn \log n$, with high probability,

$$\mathcal{G} = \left\{ \max_{i=1,\cdots,n} \left| s_i(\boldsymbol{A})^2 - s_i(\boldsymbol{B})^2 \right| \leqslant 0.1 s_1(\boldsymbol{A})^2 \right\},$$
$$\mathbb{P}(\mathcal{G}) \geqslant 0.9.$$

Prf. It is easy to see that

$$\boldsymbol{A}^\top \boldsymbol{A} = \sum_{j=1}^N \boldsymbol{a}_j \boldsymbol{a}_j^\top,$$

$$\boldsymbol{B}^\top \boldsymbol{B} = \sum_{k=1}^m \boldsymbol{b}_k \boldsymbol{b}_k^\top = \sum_{k=1}^m \left(\sqrt{\frac{N}{m}} \boldsymbol{a}_{j_k}\right)\left(\sqrt{\frac{N}{m}} \boldsymbol{a}_{j_k}\right)^\top = \frac{N}{m}\sum_{k=1}^m \boldsymbol{a}_{j_k} \boldsymbol{a}_{j_k}^\top,$$

where j_k are indices drawn independently and uniformly from $\{1, \cdots, N\}$, then

$$\mathbb{E}[\boldsymbol{B}^\top \boldsymbol{B}] = \frac{N}{m} \sum_{k=1}^{m} \mathbb{E}[\boldsymbol{a}_{j_k} \boldsymbol{a}_{j_k}^\top] = \frac{N}{m} \cdot m \cdot \left(\frac{1}{N} \sum_{j=1}^{N} \boldsymbol{a}_j \boldsymbol{a}_j^\top \right) = \boldsymbol{A}^\top \boldsymbol{A}.$$

Let

$$\boldsymbol{X}_k = \frac{N}{m} \boldsymbol{a}_{j_k} \boldsymbol{a}_{j_k}^\top,$$

and the centered random matrix $\boldsymbol{Z}_k = \boldsymbol{X}_k - \frac{1}{m} \boldsymbol{A}^\top \boldsymbol{A}$, then $\boldsymbol{B}^\top \boldsymbol{B} - \boldsymbol{A}^\top \boldsymbol{A} = \sum_{k=1}^{m} \boldsymbol{Z}_k$. For the sub-Exponential upper bound R, since $\|\boldsymbol{a}_j\|_2^2 = \sigma^2$,

$$\|\boldsymbol{X}_k\| = \frac{N}{m} \|\boldsymbol{a}_{j_k}\|_2^2 = \frac{N\sigma^2}{m}, \left\| \frac{1}{m} \boldsymbol{A}^\top \boldsymbol{A} \right\| \leqslant \frac{N}{m} \max_j \|\boldsymbol{a}_j\|_2^2 = \frac{N\sigma^2}{m},$$

Hence

$$\|\boldsymbol{Z}_k\| \leqslant \|\boldsymbol{X}_k\| + \left\| \frac{1}{m} \boldsymbol{A}^\top \boldsymbol{A} \right\| \leqslant \frac{2N\sigma^2}{m},$$

Now set $R = 2N\sigma^2/m$.

For the variance parameter σ_{Bern}^2, since

$$\mathbb{E}[\boldsymbol{X}_1^2] = \mathbb{E}\left[\left(\frac{N}{m} \boldsymbol{a}_{j_1} \boldsymbol{a}_{j_1}^\top \right)^2 \right] = \frac{N^2}{m^2} \mathbb{E}\|\boldsymbol{a}_{j_1}\|_2^2 \boldsymbol{a}_{j_1} \boldsymbol{a}_{j_1}^\top = \frac{N^2\sigma^2}{m^2} \cdot \frac{1}{N} \sum_{j=1}^{N} \boldsymbol{a}_j \boldsymbol{a}_j^\top = \frac{N\sigma^2}{m^2} \boldsymbol{A}^\top \boldsymbol{A},$$

Hence

$$\sigma_{\text{Bern}}^2 = \left\| \sum_{k=1}^{m} \mathbb{E}[\boldsymbol{Z}_k^2] \right\| = m \|\mathbb{E}[\boldsymbol{Z}_1^2]\| \leqslant m \cdot \frac{N\sigma^2}{m^2} \|\boldsymbol{A}^\top \boldsymbol{A}\| = \frac{N\sigma^2}{m} \|\boldsymbol{A}^\top \boldsymbol{A}\|.$$

Since $\operatorname{tr}(\boldsymbol{A}^\top \boldsymbol{A}) = N\sigma^2$ and $\boldsymbol{A}^\top \boldsymbol{A}$ has rank at most n,

$$\|\boldsymbol{A}^\top \boldsymbol{A}\| \geqslant \frac{N\sigma^2}{n}.$$

Thus, using Theorem T5.4.1 (*Matrix Bernstein inequality*) (*Matrix Bernstein inequality*), for any $\varepsilon > 0$,

$$\mathbb{P}\{\|\boldsymbol{B}^\top \boldsymbol{B} - \boldsymbol{A}^\top \boldsymbol{A}\| \geqslant \varepsilon \|\boldsymbol{A}^\top \boldsymbol{A}\|\} \leqslant 2n \exp\left(-c \min\left(\frac{\varepsilon^2 \|\boldsymbol{A}^\top \boldsymbol{A}\|^2}{\sigma_{\text{Bern}}^2}, \frac{\varepsilon \|\boldsymbol{A}^\top \boldsymbol{A}\|}{R} \right) \right),$$

Set $\varepsilon = 0.1$. For $0 < \varepsilon \leqslant 1$, this lower bound makes the minimum in the exponent at least $c\varepsilon^2 m/n$. Thus taking $m \geqslant Cn \log n$ with C sufficiently large makes the probability bound at most 0.1,

$$\mathbb{P}\{\|\boldsymbol{B}^\top \boldsymbol{B} - \boldsymbol{A}^\top \boldsymbol{A}\| \geqslant \varepsilon \|\boldsymbol{A}^\top \boldsymbol{A}\|\} \leqslant 0.1,$$

Thus by Weyl's inequality (*or perturbation theory*),

$$\max_i |s_i(\boldsymbol{B})^2 - s_i(\boldsymbol{A})^2| \leqslant \|\boldsymbol{B}^\top \boldsymbol{B} - \boldsymbol{A}^\top \boldsymbol{A}\| \leqslant 0.1 \|\boldsymbol{A}^\top \boldsymbol{A}\| = 0.1 s_1(\boldsymbol{A})^2.$$

Quadratic Forms, Symmetrization and Contraction Principle

6

Keypoints

Section 6.1 Decoupling — Q6.1 ∼ Q6.4

Theorem 6.1.1 (*Decoupling*)

Let $\boldsymbol{X}$ be a zero-mean random matrix whose entries are mutually independent, and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ satisfy $\operatorname{diag}(\boldsymbol{A}) = \boldsymbol{0}$. Then

$$\mathbb{E}[|\boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{X}|] \leqslant 4\mathbb{E}[|\boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{Y}|],$$

where $\boldsymbol{Y}$ is an independent copy of $\boldsymbol{X}$.

Remark 6.1.2 (*Diagonal-free assumption*)

The condition $\operatorname{diag}(\boldsymbol{A}) = \boldsymbol{0}$ is necessary in Theorem T6.1.1 (*Decoupling*). For example, let $F(x) = x$ and $\boldsymbol{A} = \boldsymbol{I}_n$; then the left side is $\mathbb{E}[\boldsymbol{X}^\top \boldsymbol{X}] = \sum_{i=1}^n \mathbb{E}[X_i^2] > 0$ and the right side is $4\mathbb{E}[\boldsymbol{X}^\top \boldsymbol{Y}] = 0$, so the conclusion above fails.

However, the decoupling theorem can be naturally extended to general matrices containing diagonal terms. For any $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ (*without requiring* $\operatorname{diag}(\boldsymbol{A}) = \boldsymbol{0}$),

$$\mathbb{E}\left[F\left(\sum_{i\neq j} a_{ij}X_iX_j\right)\right] \leqslant \mathbb{E}\left[F\left(4\sum_{i,j} a_{ij}X_iX_j\right)\right].$$

Section 6.2 Hanson-Wright inequality — Q6.5 ∼ Q6.11

Proposition 6.2.1 (*The norm of a subgaussian random vector*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector whose entries are mutually independent, centered K-subgaussian random variables. Then

$$\mathbb{E}\|\boldsymbol{X}\|_2 \leqslant \sqrt{n}, \|\|\boldsymbol{X}\|_2 - \sqrt{n}\|_{\psi_2} \leqslant CK^2.$$

Theorem 6.2.2 (*Hanson-Wright inequality*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector whose entries are mutually independent, centered K-subgaussian random variables. Then, for any $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ and $t \geqslant 0$,

$$\mathbb{P}\{|\boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{X}]| \geqslant t\} \leqslant 2\exp\left[-c\min\left(\frac{t^2}{K^4\|\boldsymbol{A}\|_F^2}, \frac{t}{K^2\|\boldsymbol{A}\|}\right)\right].$$

Lemma 6.2.3 (*Gaussian replacement*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a subgaussian random vector whose entries are mutually independent, centered K-subgaussian random variables. Then whenever $|\lambda| \leqslant c/(K^2\|\boldsymbol{A}\|)$,

$$\mathbb{E}[\exp(\lambda \boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{X})] \leqslant \mathbb{E}[\exp(C\lambda K^2 \boldsymbol{g}^\top \boldsymbol{A}\boldsymbol{g})],$$

where $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

Lemma 6.2.4 (*MGF of a Gaussian quadratic form*)

Let $\boldsymbol{g}, \boldsymbol{g}'$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$. Then whenever $|\lambda| \leqslant 1/2\|\boldsymbol{A}\|$,

$$\mathbb{E}[\exp(\lambda \boldsymbol{g}^\top \boldsymbol{A}\boldsymbol{g}')] \leqslant \exp(\lambda^2 \|\boldsymbol{A}\|_F^2).$$

Section 6.3 Symmetrization Q6.12 ∼ Q6.22

Lemma 6.3.1 (*Constructing symmetric distributions*)

Let X be a random variable and $\varepsilon \sim$ Rademacher. Then:

(1) $\varepsilon X \overset{d}{=} \varepsilon|X|$ is symmetric.

(2) If X is symmetric, then $\varepsilon X \overset{d}{=} \varepsilon|X| \overset{d}{=} X$.

(3) If X' is an independent copy of X, then $X - X'$ is symmetric.

Lemma 6.3.2 (*Symmetrization*)

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ be mutually independent zero-mean random vectors, and $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher. Then

$$\frac{1}{2}\mathbb{E}\left\|\sum_i \varepsilon_i \boldsymbol{X}_i\right\| \leqslant \mathbb{E}\left\|\sum_i \boldsymbol{X}_i\right\| \leqslant 2\mathbb{E}\left\|\sum_i \varepsilon_i \boldsymbol{X}_i\right\|.$$

Section 6.4 Random matrices with non-i.i.d. entries Q6.23 ∼ Q6.30

Theorem 6.4.1 (*Norm of random matrices with non-i.i.d. entries*)

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric random matrix whose entries on and above the diagonal are mutually independent zero-mean random variables, and denote by $\boldsymbol{a}_i$ the row vectors of $\boldsymbol{A}$. Then

$$\mathbb{E}\left[\max_i \|\boldsymbol{a}_i\|_2\right] \leqslant \mathbb{E}\|\boldsymbol{A}\| \leqslant C\sqrt{\log n} \cdot \mathbb{E}\left[\max_i \|\boldsymbol{a}_i\|_2\right].$$

Section 6.5 Matrix completion Q6.31 ∼ Q6.32

Theorem 6.5.1 (*Matrix completion*)

Let $\boldsymbol{X} \in \mathbb{R}^{n\times n}$ be a rank-r matrix whose coherence satisfies $\mu(\boldsymbol{X}) \leqslant \mu$. Suppose we observe the entries at m positions chosen uniformly at random from $\boldsymbol{X}$. When $m \geqslant C\mu^2 rn \log^2 n$, by nuclear norm minimization,

$$\min_{\boldsymbol{M}\in\mathbb{R}^{n\times n}} \|\boldsymbol{M}\|_* \quad \text{s.t.} \quad m_{ij} = X_{ij},$$

one can recover the original matrix $\hat{\boldsymbol{M}} = \boldsymbol{X}$ with high probability $1 - 1/n$.

Section 6.6 Contraction principle Q6.33 ∼ Q6.38

Theorem 6.6.1 (*Contraction principle*)

Let $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N$ be deterministic vectors, $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$, and $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher. Then

$$\mathbb{E}\left\|\sum_i a_i \varepsilon_i \boldsymbol{x}_i\right\| \leqslant \|\boldsymbol{a}\|_\infty \cdot \mathbb{E}\left\|\sum_i \varepsilon_i \boldsymbol{x}_i\right\|.$$

Lemma 6.6.2 (*Symmetrization with Gaussians*)

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ be mutually independent zero-mean random vectors, and $g_1, \cdots, g_N$ i.i.d. $\sim N(0,1)$ independent of $\boldsymbol{X}_i$. Then

$$\frac{c}{\sqrt{\log N}}\mathbb{E}\left\|\sum_i g_i \boldsymbol{X}_i\right\| \leqslant \mathbb{E}\left\|\sum_i \boldsymbol{X}_i\right\| \leqslant 3\mathbb{E}\left\|\sum_i g_i \boldsymbol{X}_i\right\|.$$

Exercises

1 [**Decoupling with Diagonal Terms**] As noted in Remark R6.1.2 (*Diagonal-free assumption*), the diagonal-free assumption in Theorem T6.1.1 (*Decoupling*) cannot be removed in general. However, we can include the diagonal terms on the right-hand side by relaxing the constants.

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\{X_n\}$ be i.i.d. zero-mean random variables, and F be a convex function. Prove:

$$\mathbb{E}\left[F\left(\sum_{i\neq j} a_{ij} X_i X_j\right)\right] \leqslant \mathbb{E}\left[F\left(4\sum_{i,j} a_{ij} X_i X_j'\right)\right],$$

where $\{X_n'\}$ is an independent copy of $\{X_n\}$.

Rk. Such decoupling inequalities follow a fixed routine:

(1) introduce independent Bernoulli selectors, rewriting the sum without diagonal terms as a selectorized expectation;

(2) apply Jensen's inequality to move the convex function F inside the expectation, obtaining the first upper bound;

(3) according to the values of the selectors, partition the index set into $I \sqcup I^c$, so that the random variables on the two sides of the sum are independent, thus one side can be replaced by an independent copy without changing the expectation, completing the decoupling;

(4) finally introduce a second family of independent copies to construct the auxiliary variables U_i, V_j, which are identically distributed with X_i, X_j' respectively, and the cross terms can be eliminated by taking conditional expectation, completing the selectorized expectation to the target form.

Prf. • Introduce Bernoulli selectors

Introduce random variables $\delta_1, \cdots, \delta_n$ i.i.d. $\sim B(1/2)$ independent of $\{X_n\}$, then

$$\mathbb{E}_\delta[\delta_i(1-\delta_j)] = \begin{cases} 1/4, & i \neq j, \\ 0, & i = j \end{cases},$$

hence

$$\sum_{i\neq j} a_{ij} X_i X_j = 4\mathbb{E}_\delta\left[\sum_{i,j} a_{ij} X_i \delta_i X_j (1-\delta_j)\right],$$

thus by Jensen's inequality, we obtain

$$F\left(\sum_{i\neq j} a_{ij} X_i X_j\right) = F\left(\mathbb{E}_\delta\left[4\sum_{i,j} \delta_i(1-\delta_j) a_{ij} X_i X_j\right]\right) \leqslant \mathbb{E}_\delta\left[F\left(4\sum_{i,j} \delta_i(1-\delta_j) a_{ij} X_i X_j\right)\right],$$

Taking expectations on both sides with respect to X, we obtain

$$\mathbb{E}_X\left[F\left(\sum_{i\neq j} a_{ij}X_iX_j\right)\right] \leqslant \mathbb{E}_{X,\delta}\left[F\left(4\sum_{i,j}\delta_i(1-\delta_j)a_{ij}X_iX_j\right)\right]. \tag{I}$$

- Decouple the dependence

  Let $I=\{i:\delta_i=1\}$, $I^c=\{j:\delta_j=0\}$, then Since $\delta_i(1-\delta_j)\neq 0$ holds exactly when $i\in I$, $j\in I^c$, at this point the sum on the right-hand side of (I) is

  $$\sum_{i,j}\delta_i(1-\delta_j)a_{ij}X_iX_j=\sum_{i\in I}\sum_{j\in I^c}a_{ij}X_iX_j;$$

  Since $I\cap I^c=\emptyset$, we have $\{X_i\}_{i\in I}\perp\{X_j\}_{j\in I^c}$.

  Introduce $\{X'_k\}$ as an independent copy of $\{X_k\}$, then the right-hand side of (I) is

  $$\begin{aligned}\mathbb{E}_{X,\delta}\left[F\left(4\sum_{i,j}\delta_i(1-\delta_j)a_{ij}X_iX_j\right)\right] &= \mathbb{E}_{X,\delta}\left[F\left(4\sum_{i\in I}\sum_{j\in I^c}a_{ij}X_iX_j\right)\right]\\ &= \mathbb{E}_{X,X',\delta}\left[F\left(4\sum_{i\in I}\sum_{j\in I^c}a_{ij}X_iX'_j\right)\right]\\ &= \mathbb{E}_{X,X',\delta}\left[F\left(4\sum_{i,j}a_{ij}X_i\delta_iX'_j(1-\delta_j)\right)\right],\end{aligned}$$

  Substituting this into (I), we obtain

  $$\mathbb{E}_X\left[F\left(\sum_{i\neq j}a_{ij}X_iX_j\right)\right]\leqslant\mathbb{E}_{X,X',\delta}\left[F\left(4\sum_{i,j}a_{ij}X_i\delta_iX'_j(1-\delta_j)\right)\right]. \tag{II}$$

- Complete the diagonal terms via conditional expectation

  Introduce $\{X''_k\}$ as an independent copy of $\{X_k\},\{X'_k\}$, and define auxiliary random variable sequences

  $$U_i=X_i\delta_i+X''_i(1-\delta_i),V_j=X'_j(1-\delta_j)+X''_j\delta_j,$$

  then $\{U_i\}\overset{d}{=}\{X_i\}$, $\{V'_j\}\overset{d}{=}\{X'_l\}$,

  $$U_iV_j=X_i\delta_iX'_j(1-\delta_j)+X_i\delta_iX''_j\delta_j+X''_i(1-\delta_i)X'_j(1-\delta_j)+X''_i(1-\delta_i)X''_j\delta_j,$$

  Taking conditional expectation with respect to X'' on both sides (*Using* $\mathbb{E}[X'']=0$),

  $$\mathbb{E}_{X''}[U_iV_j\mid X,X',\delta]=X_i\delta_iX'_j(1-\delta_j)+0+0+0=X_i\delta_iX'_j(1-\delta_j),$$

  Using Jensen's inequality, we obtain

  $$F\left(4\sum_{i,j}a_{ij}X_i\delta_iX'_j(1-\delta_j)\right)\leqslant\mathbb{E}_{X''}\left[F\left(4\sum_{i,j}a_{ij}U_iV_j\right)\right],$$

  Taking total expectation over X,X',δ,

  $$\mathbb{E}_{X,X',\delta}\left[F\left(4\sum_{i,j}a_{ij}X_i\delta_iX'_j(1-\delta_j)\right)\right]\leqslant\mathbb{E}_{X,X',\delta,X''}\left[F\left(4\sum_{i,j}a_{ij}U_iV_j\right)\right]. \tag{III}$$

Thus, combining (II) and (III), we obtain

$$\mathbb{E}_X\left[F\left(\sum_{i\neq j}a_{ij}X_iX_j\right)\right]\leqslant\mathbb{E}_{X,X',\delta,X''}\left[F\left(4\sum_{i,j}a_{ij}U_iV_j\right)\right],$$

i.e.,

$$\mathbb{E}\left[F\left(\sum_{i\neq j}a_{ij}X_iX_j\right)\right]\leqslant\mathbb{E}\left[F\left(4\sum_{i,j}a_{ij}X_iX'_j\right)\right].$$

2 [L^p **Norm and sub-Gaussian Decoupling**] Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\{X_n\}$ be independent zero-mean random variables, and $\{X_n'\}$ be an independent copy of $\{X_n\}$.

(1) Prove: For any $p \in [1,\infty)$,

$$\left\|\sum_{i,j:i\neq j} a_{ij}X_iX_j\right\|_{L^p} \leqslant 4\left\|\sum_{i,j} a_{ij}X_iX_j'\right\|_{L^p}.$$

(2) Prove: The above inequality also holds for the ψ_2 norm, i.e.,

$$\left\|\sum_{i,j:i\neq j} a_{ij}X_iX_j\right\|_{\psi_2} \leqslant 4\left\|\sum_{i,j} a_{ij}X_iX_j'\right\|_{\psi_2}.$$

Prf. (1) By Exercise Q6.1 (*Decoupling with Diagonal Terms*), let $F(x) = |x|^p$, then

$$\begin{aligned}\mathbb{E}\left[\left|\sum_{i\neq j} a_{ij}X_iX_j\right|^p\right] &\leqslant \mathbb{E}\left[\left|4\sum_{i,j} a_{ij}X_iX_j'\right|^p\right]\\ &= 4^p\cdot\mathbb{E}\left[\left|\sum_{i,j} a_{ij}X_iX_j'\right|^p\right],\end{aligned}$$

Taking the p-th root on both sides, we obtain

$$\left(\mathbb{E}\left[\left|\sum_{i\neq j} a_{ij}X_iX_j\right|^p\right]\right)^{1/p} \leqslant 4\cdot\left(\mathbb{E}\left[\left|\sum_{i,j} a_{ij}X_iX_j'\right|^p\right]\right)^{1/p},$$

i.e.,

$$\left\|\sum_{i\neq j} a_{ij}X_iX_j\right\|_{L^p} \leqslant 4\left\|\sum_{i,j} a_{ij}X_iX_j'\right\|_{L^p}.$$

(2) Recall that the ψ_2 norm is defined as

$$\|Z\|_{\psi_2} = \sup_{p\geqslant 1}\frac{\|Z\|_{L^p}}{\sqrt{p}},$$

Dividing both sides by $\sqrt{p}$ and then taking the supremum,

$$\sup_{p\geqslant 1}\frac{1}{\sqrt{p}}\left\|\sum_{i\neq j} a_{ij}X_iX_j\right\|_{L^p} \leqslant \sup_{p\geqslant 1}\left(4\cdot\frac{1}{\sqrt{p}}\left\|\sum_{i,j} a_{ij}X_iX_j'\right\|_{L^p}\right),$$

i.e.,

$$\left\|\sum_{i\neq j} a_{ij}X_iX_j\right\|_{\psi_2} \leqslant 4\left\|\sum_{i,j} a_{ij}X_iX_j'\right\|_{\psi_2}.$$

3 [**Vector and Space Decoupling**] Let $\{X_n\}$ be independent zero-mean random variables and $\{X_n'\}$ be an independent copy.

(1) Let $\{\boldsymbol{v}_{ij}\}_{i,j=1}^n$ be defined in a vector space V. Prove: For any convex function $F: V\to\mathbb{R}$,

$$\mathbb{E}\left[F\left(\sum_{i\neq j}\boldsymbol{v}_{ij}X_iX_j\right)\right] \leqslant \mathbb{E}\left[F\left(4\sum_{i,j}\boldsymbol{v}_{ij}X_iX_j'\right)\right].$$

(2) (Decoupling for inner product terms) Let $\{\boldsymbol{X}_i\}_{i=1}^n \subset \mathbb{R}^N$ be independent zero-mean random vectors and $\boldsymbol{A}\in\mathbb{R}^{n\times n}$. Prove: For any convex function $F:\mathbb{R}\to\mathbb{R}$,

$$\mathbb{E}\left[F\left(\sum_{i\neq j} a_{ij}\langle\boldsymbol{X}_i,\boldsymbol{X}_j\rangle\right)\right] \leqslant \mathbb{E}\left[F\left(4\sum_{i,j} a_{ij}\langle\boldsymbol{X}_i,\boldsymbol{X}_j'\rangle\right)\right].$$

(3) (Decoupling for outer product terms) Let $\{\boldsymbol{X}_i\}_{i=1}^n \subset \mathbb{R}^N$ be independent zero-mean random vectors and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$. Prove: For any convex function $F : \mathbb{R}^{N\times N} \to \mathbb{R}$,

$$\mathbb{E}\left[F\left(\sum_{i\neq j} a_{ij}\boldsymbol{X}_i\boldsymbol{X}_j^\top\right)\right] \leqslant \mathbb{E}\left[F\left(4\sum_{i,j} a_{ij}\boldsymbol{X}_i(\boldsymbol{X}_j')^\top\right)\right].$$

Prf. (1) (*It suffices to replace $a_{i,j}$ in Exercise Q6.1 (Decoupling with Diagonal Terms) with $\boldsymbol{v}_{i,j}$.*)

(2) Introduce Bernoulli selectors, then

$$\sum_{i\neq j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle = 4\mathbb{E}_\delta\left[\sum_{i,j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle\delta_i(1-\delta_j)\right].$$

Using Jensen's inequality, moving $\mathbb{E}_\delta$ outside F, we obtain

$$\mathbb{E}_X\left[F\left(\sum_{i\neq j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle\right)\right] \leqslant \mathbb{E}_{X,\delta}\left[F\left(4\sum_{i,j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle\delta_i(1-\delta_j)\right)\right]. \tag{I}$$

Define the index set $I = \{i : \delta_i = 1\}$, then

$$\sum_{i,j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle\delta_i(1-\delta_j) = \sum_{i\in I}\sum_{j\in I^c} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle.$$

Then replace $\{\boldsymbol{X}_j\}_{j\in I^c}$ with an independent copy $\{\boldsymbol{X}_j'\}_{j\in I^c}$,

$$\mathbb{E}_X\left[F\left(4\sum_{i\in I}\sum_{j\in I^c} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle\right)\right] = \mathbb{E}_{X,X'}\left[F\left(4\sum_{i\in I}\sum_{j\in I^c} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j'\rangle\right)\right],$$

Substituting this into (I), we obtain

$$\mathbb{E}_X\left[F\left(\sum_{i\neq j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle\right)\right] \leqslant \mathbb{E}_{X,X',\delta}\left[F\left(4\sum_{i,j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j'\rangle\delta_i(1-\delta_j)\right)\right]. \tag{II}$$

Construct an independent copy $\{\boldsymbol{X}_k''\}$ and define auxiliary variables

$$\boldsymbol{U}_i = \boldsymbol{X}_i\delta_i + \boldsymbol{X}_i''(1-\delta_i), \boldsymbol{V}_j = \boldsymbol{X}_j'(1-\delta_j) + \boldsymbol{X}_j''\delta_j,$$

then

$$\langle \boldsymbol{U}_i, \boldsymbol{V}_j\rangle = \langle \boldsymbol{X}_i\delta_i + \boldsymbol{X}_i''(1-\delta_i), \boldsymbol{X}_j'(1-\delta_j) + \boldsymbol{X}_j''\delta_j\rangle,$$

Taking conditional expectation with respect to X'' on both sides, we obtain

$$\mathbb{E}_{X''}[\langle \boldsymbol{U}_i, \boldsymbol{V}_j\rangle \mid \boldsymbol{X}, \boldsymbol{X}', \delta] = \langle \boldsymbol{X}_i, \boldsymbol{X}_j'\rangle\delta_i(1-\delta_j) + 0 + 0 + 0,$$

hence by Jensen's inequality, we obtain

$$F\left(4\sum_{i,j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j'\rangle\delta_i(1-\delta_j)\right) \leqslant \mathbb{E}_{X''}\left[F\left(4\sum_{i,j} a_{ij}\langle \boldsymbol{U}_i, \boldsymbol{V}_j\rangle\right)\right],$$

i.e.,

$$\mathbb{E}\left[F\left(\sum_{i\neq j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j\rangle\right)\right] \leqslant \mathbb{E}\left[F\left(4\sum_{i,j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}_j'\rangle\right)\right].$$

(3) **Solution 1 Decoupling**

(*Simply replace X_i, X_j with $\boldsymbol{X}_i, \boldsymbol{X}_j^\top$; the proof is omitted.*)

Solution 2 Variational representation

Regard the matrix space as a finite-dimensional vector space. The following representation writes the outer-product sum through a linear functional while allowing the dual variable to depend jointly on the random variables.

• An exact variational representation under expectation (*Rockafellar duality*)

By the Rockafellar integral functional interchange theorem, we obtain

$$\mathbb{E}[F(\boldsymbol{S})] = \sup_{\boldsymbol{Y}\in L^\infty} \mathbb{E}[\operatorname{tr}(\boldsymbol{Y}^\top \boldsymbol{S}) - F^*(\boldsymbol{Y})], \tag{III}$$

where

$$\begin{aligned}
\operatorname{tr}(\boldsymbol{Y}^\top \boldsymbol{S}) &= \operatorname{tr}\left(\boldsymbol{Y}^\top \sum_{i\neq j} a_{ij} \boldsymbol{X}_i \boldsymbol{X}_j^\top\right) \\
&= \sum_{i\neq j} a_{ij} \operatorname{tr}(\boldsymbol{Y}^\top \boldsymbol{X}_i \boldsymbol{X}_j^\top) \\
&= \sum_{i\neq j} a_{ij} \boldsymbol{X}_j^\top \boldsymbol{Y}^\top \boldsymbol{X}_i \\
&= \sum_{i\neq j} a_{ij} \langle \boldsymbol{Y}^\top \boldsymbol{X}_i, \boldsymbol{X}_j \rangle.
\end{aligned}$$

(*The matrix outer product problem is thus successfully reduced to a vector inner product summation form, and the subsequent routine is similar to the usual decoupling.*)

- Introducing Bernoulli selectors and interchanging via the Sion minimax theorem

  Then introduce Bernoulli selectors $\delta_1, \cdots, \delta_n$ i.i.d. $\sim B(1/2)$, at this point

$$\mathbb{E}_{\boldsymbol{X}}[\operatorname{tr}(\boldsymbol{Y}^\top \boldsymbol{S})] = \mathbb{E}_{\boldsymbol{X}}\left[4\,\mathbb{E}_\delta \sum_{i\neq j} \delta_i(1-\delta_j) a_{ij} \langle \boldsymbol{Y}^\top \boldsymbol{X}_i, \boldsymbol{X}_j \rangle\right],$$

  Substituting this into (III), we obtain

$$\mathbb{E}[F(\boldsymbol{S})] = \sup_{\boldsymbol{Y}\in L^\infty} \mathbb{E}_\delta \mathbb{E}_{\boldsymbol{X}}\left[4 \sum_{i\neq j} \delta_i(1-\delta_j) a_{ij} \langle \boldsymbol{Y}^\top \boldsymbol{X}_i, \boldsymbol{X}_j \rangle - F^*(\boldsymbol{Y})\right].$$

  Since the target functional is linear in the measure $\mathbb{P}_\delta$ (*convex*) and concave in the dual variable $\boldsymbol{Y}$ (*by the concavity of* $-F^*$), by the Sion minimax theorem, we obtain

$$\mathbb{E}[F(\boldsymbol{S})] \leqslant \mathbb{E}_\delta\left[\sup_{\boldsymbol{Y}\in L^\infty} \mathbb{E}_{\boldsymbol{X}}\left(4 \sum_{i\neq j} \delta_i(1-\delta_j) a_{ij} \langle \boldsymbol{Y}^\top \boldsymbol{X}_i, \boldsymbol{X}_j \rangle - F^*(\boldsymbol{Y})\right)\right].$$

- Replacing by independent copies via conditional independence

  Fix the selector $\boldsymbol{\delta}$; the index set decomposes into $I = \{i \mid \delta_i = 1\}$ and $I^c = \{j \mid \delta_j = 0\}$ with $I \cap I^c = \emptyset$. By the independence $\{\boldsymbol{X}_i\}_{i\in I} \perp \{\boldsymbol{X}_j\}_{j\in I^c}$, replacing $\{\boldsymbol{X}_j\}_{j\in I^c}$ by the independent copy $\{\boldsymbol{X}'_j\}_{j\in I^c}$ leaves the conditional expectation unchanged:

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{X}}\left[4 \sum_{i\neq j} \delta_i(1-\delta_j) a_{ij} \langle \boldsymbol{Y}^\top \boldsymbol{X}_i, \boldsymbol{X}_j \rangle\right] \\
&= \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left[4 \sum_{i,j} \delta_i(1-\delta_j) a_{ij} \langle \boldsymbol{Y}^\top \boldsymbol{X}_i, \boldsymbol{X}'_j \rangle\right] \\
&= \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left[\operatorname{tr}\left(\boldsymbol{Y}^\top \left(4 \sum_{i,j} \delta_i(1-\delta_j) a_{ij} \boldsymbol{X}_i (\boldsymbol{X}'_j)^\top\right)\right)\right].
\end{aligned}$$

- Reversing the Rockafellar duality and contracting by conditional expectation

  Substituting the above back into $\sup_{\boldsymbol{Y}}$ and reversing via the definition of the Legendre variational representation, we obtain

$$\sup_{\boldsymbol{Y}\in L^\infty} \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left[\operatorname{tr}\left(\boldsymbol{Y}^\top \left(4 \sum_{i,j} \delta_i(1-\delta_j) a_{ij} \boldsymbol{X}_i (\boldsymbol{X}'_j)^\top\right)\right) - F^*(\boldsymbol{Y})\right]$$

$$= \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left[F\left(4\sum_{i,j}\delta_i(1-\delta_j)a_{ij}\boldsymbol{X}_i(\boldsymbol{X}'_j)^\top\right)\right].$$

By Jensen's inequality, we obtain

$$\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left[F\left(4\sum_{i,j}\delta_i(1-\delta_j)a_{ij}\boldsymbol{X}_i(\boldsymbol{X}'_j)^\top\right)\right] \leqslant \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left[F\left(4\sum_{i,j}a_{ij}\boldsymbol{X}_i(\boldsymbol{X}'_j)^\top\right)\right],$$

Taking conditional expectation with respect to δ on both sides, we obtain

$$\mathbb{E}\left[F\left(\sum_{i\neq j}a_{ij}\boldsymbol{X}_i\boldsymbol{X}_j^\top\right)\right] \leqslant \mathbb{E}\left[F\left(4\sum_{i,j}a_{ij}\boldsymbol{X}_i(\boldsymbol{X}'_j)^\top\right)\right].$$

4 [**Decoupling for the Norm of a Random Submatrix**] Let $\boldsymbol{A}\in\mathbb{R}^{n\times n}$ satisfy $\operatorname{diag}(\boldsymbol{A})=\boldsymbol{0}$, and let $\delta_1,\cdots,\delta_n$ i.i.d. $\sim B(p)$. Let $J=\{i:\delta_i=1\}$ be the corresponding random subset of indices, $J'=\{i:\delta'_i=1\}$ be the independent copy index set of J (*where* $\{\delta'_i\}$ *is independent of* $\{\delta_i\}$). Then define $\boldsymbol{A}_{J\times J}$ as the submatrix of $\boldsymbol{A}$ whose row and column indices are both taken from J, with (i,j)-entry $a_{ij}\delta_i\delta_j$. Prove:

$$\mathbb{E}\|\boldsymbol{A}_{J,J}\| \leqslant 4\mathbb{E}\|\boldsymbol{A}_{J,J'}\|,$$

where $\boldsymbol{A}_{J,J'}$ is the submatrix defined by row indices J and column indices J', with (i,j)-entry $a_{ij}\delta_i\delta'_j$.

Prf. (*Theorem T6.1.1 (Decoupling) requires the random variables to be zero-mean. If one forcibly centers the random variables and then applies the decoupling theorem, the resulting cross terms cannot be eliminated via conditional expectation. We instead use the projection contraction property of the matrix spectral norm.*)

Let $\boldsymbol{E}_{ij}$ be the standard basis matrix (*with* 1 *at position* (i,j) *and* 0 *elsewhere*), then

$$\boldsymbol{A}_{J,J}=\sum_{i,j=1}^{n}a_{ij}\delta_i\delta_j\boldsymbol{E}_{ij}=\sum_{i\neq j}a_{ij}\delta_i\delta_j\boldsymbol{E}_{ij},\ \boldsymbol{A}_{J,J'}=\sum_{i\neq j}a_{ij}\delta_i\delta'_j\boldsymbol{E}_{ij}.$$

We then only need to prove

$$\mathbb{E}_\delta\left\|\sum_{i\neq j}a_{ij}\delta_i\delta_j\boldsymbol{E}_{ij}\right\| \leqslant 4\mathbb{E}_{\delta,\delta'}\left\|\sum_{i\neq j}a_{ij}\delta_i\delta'_j\boldsymbol{E}_{ij}\right\|.$$

- Introduce Bernoulli selectors

  Introduce $\eta_1,\cdots,\eta_n$ i.i.d. $\sim B(1/2)$ independent of $\{\varepsilon_i\}$, then

  $$\sum_{i\neq j}a_{ij}\delta_i\delta_j\boldsymbol{E}_{ij}=4\mathbb{E}_\eta\left[\sum_{i\neq j}a_{ij}\delta_i\eta_i\delta_j(1-\eta_j)\boldsymbol{E}_{ij}\right],$$

  hence by Jensen's inequality, we obtain

  $$\left\|\sum_{i\neq j}a_{ij}\delta_i\delta_j\boldsymbol{E}_{ij}\right\| \leqslant \mathbb{E}_\eta\left\|4\sum_{i\neq j}a_{ij}\delta_i\eta_i\delta_j(1-\eta_j)\boldsymbol{E}_{ij}\right\|,$$

  Taking expectations on both sides with respect to δ,

  $$\mathbb{E}_\delta\left\|\sum_{i\neq j}a_{ij}\delta_i\delta_j\boldsymbol{E}_{ij}\right\| \leqslant \mathbb{E}_{\delta,\eta}\left\|4\sum_{i\neq j}a_{ij}\delta_i\eta_i\delta_j(1-\eta_j)\boldsymbol{E}_{ij}\right\|. \tag{I}$$

- Apply Jensen's inequality and decoupling

  Let $I=\{i:\eta_i=1\}$, then Since $\eta_i(1-\eta_j)\neq 0$ holds exactly when $i\in I$, $j\in I^c$, at this point the sum on the right-hand side of (I) is

  $$\sum_{i\neq j}a_{ij}\delta_i\eta_i\delta_j(1-\eta_j)\boldsymbol{E}_{ij}=\sum_{i\in I}\sum_{j\in I^c}a_{ij}\delta_i\delta_j\boldsymbol{E}_{ij};$$

  Since $I\cap I^c=\emptyset$, we have $\{\delta_i\}_{i\in I}\perp\{\delta_j\}_{j\in I^c}$.

Introduce $\{\delta'_j\}_{j\in I^c}$ as an independent copy of $\{\delta_j\}_{j\in I^c}$, then

$$\mathbb{E}_{\delta,\eta}\left\|4\sum_{i\neq j}a_{ij}\delta_i\eta_i\delta_j(1-\eta_j)\boldsymbol{E}_{ij}\right\| = \mathbb{E}_{\delta}\left\|4\sum_{i\in I}\sum_{j\in I^c}a_{ij}\delta_i\delta_j\boldsymbol{E}_{ij}\right\|$$
$$= \mathbb{E}_{\delta,\delta'}\left\|4\sum_{i\in I}\sum_{j\in I^c}a_{ij}\delta_i\delta'_j\boldsymbol{E}_{ij}\right\|. \tag{II}$$

(*We then only need to bound the sum on the right-hand side of (II), which is merely a truncation of the target matrix* $\boldsymbol{A}_{J,J'}=\sum_{i\neq j}a_{ij}\delta_i\delta'_j\boldsymbol{E}_{ij}$.)

Define the diagonal projection matrices $\boldsymbol{P}_I=\operatorname{diag}(\eta_1,\cdots,\eta_n)$, $\boldsymbol{P}_{I^c}=\operatorname{diag}(1-\eta_1,\cdots,1-\eta_n)$, then

$$\sum_{i\in I}\sum_{j\in I^c}a_{ij}\delta_i\delta'_j\boldsymbol{E}_{ij}=\boldsymbol{P}_I\left(\sum_{i\neq j}a_{ij}\delta_i\delta'_j\boldsymbol{E}_{ij}\right)\boldsymbol{P}_{I^c}=\boldsymbol{P}_I\boldsymbol{A}_{J,J'}\boldsymbol{P}_{I^c},$$

Note that

$$\left\|\boldsymbol{P}_I\boldsymbol{A}_{J,J'}\boldsymbol{P}_{I^c}\right\|\leqslant\|\boldsymbol{P}_I\|\cdot\|\boldsymbol{A}_{J,J'}\|\cdot\|\boldsymbol{P}_{I^c}\|\leqslant\|\boldsymbol{A}_{J,J'}\|,$$

hence taking total expectation on both sides, we obtain

$$\mathbb{E}_{\delta,\delta',\eta}\left[4\left\|\sum_{i\in I}\sum_{j\in I^c}a_{ij}\delta_i\delta'_j\boldsymbol{E}_{ij}\right\|\right]\leqslant\mathbb{E}_{\delta,\delta'}[4\|\boldsymbol{A}_{J,J'}\|],$$

i.e.,

$$\mathbb{E}\|\boldsymbol{A}_{J,J}\|\leqslant 4\mathbb{E}\|\boldsymbol{A}_{J,J'}\|.$$

5 [**Failure of the Deviation Bound about the Mean**] Let $\boldsymbol{X}\in\mathbb{R}^n$ be a sub-Gaussian random vector satisfying $\|\boldsymbol{X}\|_{\psi_2}\leqslant K$. One often intuitively thinks that the tail probability bound of $\|\boldsymbol{X}\|_2$ should be centered around its mean $\mathbb{E}\|\boldsymbol{X}\|_2\approx\sqrt{n}$. Prove: The above conjecture is incorrect. Construct a counterexample such that for any $t\geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{X}\|_2\geqslant C(\sqrt{n}+Kt)\}\leqslant\mathrm{e}^{-t^2}.$$

Sol. (*The above bound implicitly assumes the "isotropic" and "mean concentrated around* $\sqrt{n}$*" hypotheses, but a general sub-Gaussian random vector does not satisfy rotational symmetry, so the concentration of its norm may deviate from this bound.*)

Let $n\geqslant 2$, take $\boldsymbol{u}\in\mathbb{S}^{n-1}$, let $B\sim\text{Bernoulli}(1/n)$, and define

$$\boldsymbol{X}=\sqrt{\log n}\,B\boldsymbol{u}.$$

For any $\boldsymbol{\theta}\in\mathbb{R}^n$, we have $|\langle\boldsymbol{X},\boldsymbol{\theta}\rangle|\leqslant\sqrt{\log n}\,\|\boldsymbol{\theta}\|_2$, and by the definition of the sub-Gaussian norm,

$$\mathbb{E}\exp\left(\frac{\langle\boldsymbol{X},\boldsymbol{\theta}\rangle^2}{\|\boldsymbol{\theta}\|_2^2}\right)\leqslant 1-\frac{1}{n}+\frac{1}{n}\exp(\log n)\leqslant 2,$$
$$\|\langle\boldsymbol{X},\boldsymbol{\theta}\rangle\|_{\psi_2}\leqslant\|\boldsymbol{\theta}\|_2.$$

Thus the sub-Gaussian norm of $\boldsymbol{X}$ is bounded by an absolute constant. On the other hand,

$$\mathbb{E}\|\boldsymbol{X}\|_2=\frac{\sqrt{\log n}}{n},\qquad\|\boldsymbol{X}\|_2\leqslant\sqrt{\log n}<\sqrt{n}.$$

Take K to be the above absolute bound and let $C\geqslant 1$. Then for every $t\geqslant 0$, the event $\{\|\boldsymbol{X}\|_2\geqslant C(\sqrt{n}+Kt)\}$ is empty, and hence

$$\mathbb{P}\{\|\boldsymbol{X}\|_2\geqslant C(\sqrt{n}+Kt)\}=0\leqslant\mathrm{e}^{-t^2},$$

This tail bound therefore does not imply that the mean norm is close to $\sqrt{n}$; the computed mean shows that the original intuition is incorrect.

6 [**Norm of a Random Matrix with sub-Gaussian Columns**]

(1) Extend the maximal inequality (Exercise Q3.13 (*Maximum Norm of a Random Vector*)) to sub-Gaussian random vectors, even without independent entries. Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N \in \mathbb{R}^n$ be independent sub-Gaussian random vectors satisfying $\|\boldsymbol{X}_i\|_{\psi_2} \leqslant K$. Prove:

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant N} \|\boldsymbol{X}_i\|_2\right] \leqslant \max_{1\leqslant i\leqslant N} \mathbb{E}\|\boldsymbol{X}_i\|_2 + CK^2\sqrt{\log N}.$$

(2) Extend the bounds on the $\|\cdot\|_1$ and $\|\cdot\|_{1\to 2}$ norms (Exercise Q4.44 (*Other $p \to q$ Operator Norms of Random Matrices*)(a),(b)) to random matrices with independent sub-Gaussian columns, even if entries are not independent. Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix with independent sub-Gaussian columns, i.e., the column vectors $\boldsymbol{A}_j \in \mathbb{R}^m$ satisfy $\|\boldsymbol{A}_j\|_{\psi_2} \leqslant K$. Prove:

$$\mathbb{E}\|\boldsymbol{A}\|_{1\to\infty} \leqslant CK\sqrt{\log m + \log n},$$
$$\mathbb{E}\|\boldsymbol{A}\|_{1\to 2} \leqslant \sqrt{m} + CK^2\sqrt{\log n}.$$

Prf. (1) Using the concentration inequality for sub-Gaussian distributions, we obtain for any $t \geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{X}_i\|_2 \geqslant \mathbb{E}\|\boldsymbol{X}_i\|_2 + t\} \leqslant \exp\left(-\frac{ct^2}{K^4}\right).$$

(*We need to bound* $\mathbb{E}[\max_{1\leqslant i\leqslant N} \|\boldsymbol{X}_i\|_2]$.)

If $N = 1$, the conclusion follows from equality. Hence assume $N \geqslant 2$, and let $m_0 = \max_{1\leqslant i\leqslant N} \mathbb{E}\|\boldsymbol{X}_i\|_2$. Taking the union bound, we obtain for any $t \geqslant 0$,

$$\mathbb{P}\left\{\max_{1\leqslant i\leqslant N} \|\boldsymbol{X}_i\|_2 \geqslant m_0 + t\right\} \leqslant \sum_{i=1}^N \mathbb{P}\{\|\boldsymbol{X}_i\|_2 \geqslant \mathbb{E}\|\boldsymbol{X}_i\|_2 + t\} \leqslant N\exp\left(-\frac{ct^2}{K^4}\right),$$

Hence

$$\begin{aligned}\mathbb{E}\left[\max_{1\leqslant i\leqslant N} \|\boldsymbol{X}_i\|_2\right] &\leqslant m_0 + \int_0^\infty \mathbb{P}\left\{\max_{1\leqslant i\leqslant N} \|\boldsymbol{X}_i\|_2 \geqslant m_0 + t\right\}\mathrm{d}t \\ &\leqslant m_0 + \int_0^\infty \min\left(1, N\exp\left(-\frac{ct^2}{K^4}\right)\right)\mathrm{d}t.\end{aligned}$$

For the integral on the right-hand side, split the integral at

$$N\exp\left(-\frac{ct^2}{K^4}\right) = 1, \quad \text{that is, } t = K^2\sqrt{\frac{\log N}{c}},$$

into two parts,

$$\int_0^\infty \min\left(1, N\exp\left(-\frac{ct^2}{K^4}\right)\right)\mathrm{d}t = \int_0^{K^2\sqrt{\frac{\log N}{c}}} 1\mathrm{d}t + \int_{K^2\sqrt{\frac{\log N}{c}}}^\infty N\exp\left(-\frac{ct^2}{K^4}\right)\mathrm{d}t \leqslant CK^2\sqrt{\log N},$$

Substituting this in, we obtain

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant N} \|\boldsymbol{X}_i\|_2\right] \leqslant \max_{1\leqslant i\leqslant N} \mathbb{E}\|\boldsymbol{X}_i\|_2 + CK^2\sqrt{\log N}.$$

(2) • $1 \to \infty$ norm bound

Using the conclusion of (1), we obtain

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant m} |A_{ij}|\right] \leqslant CK\sqrt{\log m}. \qquad (1 \leqslant j \leqslant n)$$

Hence for $\|\boldsymbol{A}\|_{1\to\infty} = \max_{i,j} |A_{ij}|$,

$$\mathbb{E}\|\boldsymbol{A}\|_{1\to\infty} = \mathbb{E}\left[\max_{1\leqslant j\leqslant n}\max_{1\leqslant i\leqslant m} |A_{ij}|\right] \leqslant CK\sqrt{\log(mn)} = CK\sqrt{\log m + \log n} \leqslant CK(\sqrt{\log m} + \sqrt{\log n}).$$

• $1 \to 2$ norm bound

Using the conclusion of (1), we obtain for $\|\boldsymbol{A}\|_{1\to 2} = \max_j \|\boldsymbol{A}_j\|_2$,

$$\mathbb{E}\|\boldsymbol{A}\|_{1\to 2} \leqslant \max_j \mathbb{E}\|\boldsymbol{A}_j\|_2 + CK^2\sqrt{\log n},$$

where $\mathbb{E}\|\boldsymbol{A}_j\|_2 \leqslant \sqrt{\mathbb{E}\|\boldsymbol{A}_j\|_2^2} = \sqrt{m}$. Substituting this in, we obtain

$$\mathbb{E}\|\boldsymbol{A}\|_{1\to 2} \leqslant \sqrt{m} + CK^2\sqrt{\log n}.$$

7 [**Gaussian Hanson-Wright Inequality**] Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric matrix. Prove an alternative proof of the Hanson-Wright inequality: directly use the MGF, and do not separate diagonal and off-diagonal terms in the proof, nor use the decoupling technique.

Prf. If $\boldsymbol{A} = \boldsymbol{O}$, then $Z = 0$ and the conclusion is trivial; afterwards only consider $\|\boldsymbol{A}\| > 0$.

We need to prove that $Z = \boldsymbol{g}^\top \boldsymbol{A}\boldsymbol{g} - \mathbb{E}[\boldsymbol{g}^\top \boldsymbol{A}\boldsymbol{g}]$ satisfies

$$\mathbb{P}\{|Z| \geqslant t\} \leqslant 2\exp\left(-c\min\left(\frac{t^2}{\|\boldsymbol{A}\|_F^2}, \frac{t}{\|\boldsymbol{A}\|}\right)\right).$$

- Spectral decomposition

  Since $\boldsymbol{A}$ is a real symmetric matrix, there exists an orthogonal matrix $\boldsymbol{Q}$ and a diagonal matrix $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \cdots, \lambda_n)$ such that $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$.

  By the rotational invariance of the Gaussian distribution, define $\boldsymbol{g}' = \boldsymbol{Q}^\top \boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ satisfying $\{g_i'\}$ are i.i.d., then

$$\boldsymbol{g}^\top \boldsymbol{A}\boldsymbol{g} = (\boldsymbol{Q}\boldsymbol{g}')^\top \boldsymbol{A}(\boldsymbol{Q}\boldsymbol{g}') = (\boldsymbol{g}')^\top(\boldsymbol{Q}^\top \boldsymbol{A}\boldsymbol{Q})\boldsymbol{g}' = (\boldsymbol{g}')^\top \boldsymbol{\Lambda}\boldsymbol{g}' = \sum_{i=1}^n \lambda_i (g_i')^2,$$

  Its mean is

$$\mathbb{E}[\boldsymbol{g}^\top \boldsymbol{A}\boldsymbol{g}] = \sum_{i=1}^n \lambda_i \mathbb{E}[(g_i')^2] = \sum_{i=1}^n \lambda_i = \operatorname{tr}(\boldsymbol{A}),$$

  Hence

$$Z = \sum_{i=1}^n \lambda_i (g_i')^2 - \sum_{i=1}^n \lambda_i = \sum_{i=1}^n \lambda_i ((g_i')^2 - 1).$$

- Compute the MGF

  By independence, we obtain

$$\begin{aligned}
\mathbb{E}[\mathrm{e}^{\theta Z}] &= \prod_{i=1}^n \mathbb{E}[\mathrm{e}^{\theta\lambda_i((g_i')^2-1)}] \\
&= \prod_{i=1}^n (\mathrm{e}^{-\theta\lambda_i}(1-2\theta\lambda_i)^{-1/2}) && (\mathbb{E}[\mathrm{e}^{\theta(g^2-1)}] = \mathrm{e}^{-\theta}(1-2\theta)^{-1/2}) \\
&= \exp\left(-\sum_{i=1}^n \left(\theta\lambda_i + \frac{1}{2}\log(1-2\theta\lambda_i)\right)\right).
\end{aligned}$$

  When $|2\theta\lambda_i| \leqslant 1/2$ holds *(i.e.,* $|\theta| \leqslant 1/(4\|\boldsymbol{A}\|)$*)*, using the Taylor expansion, we obtain

$$\log(1-2\theta\lambda_i) = -2\theta\lambda_i - \frac{(2\theta\lambda_i)^2}{2} - \sum_{k=3}^\infty \frac{(2\theta\lambda_i)^k}{k},$$

  At this point

$$\begin{aligned}
\theta\lambda_i + \frac{1}{2}\log(1-2\theta\lambda_i) &= \theta\lambda_i - \theta\lambda_i - \theta^2\lambda_i^2 - \frac{1}{2}\sum_{k=3}^\infty \frac{(2\theta\lambda_i)^k}{k} \\
&= -\theta^2\lambda_i^2 - \sum_{k=3}^\infty \frac{2^{k-1}\theta^k\lambda_i^k}{k},
\end{aligned}$$

  Substituting this in, we obtain

$$\mathbb{E}[\mathrm{e}^{\theta Z}] = \exp\left(\theta^2 \sum_{i=1}^n \lambda_i^2 + \sum_{k=3}^\infty \frac{2^{k-1}\theta^k}{k} \sum_{i=1}^n \lambda_i^k\right).$$

- Use the Chernoff bound

  Since

$$(k=2)\quad \sum_{i=1}^{n}\lambda_i^2=\|\boldsymbol{A}\|_F^2,$$

$$(k=3)\quad \sum_{i=1}^{n}|\lambda_i|^k\leqslant\|\boldsymbol{A}\|^{k-2}\|\boldsymbol{A}\|_F^2,$$

  Hence

$$\mathbb{E}[\mathrm{e}^{\theta Z}]\leqslant\exp\left(\theta^2\|\boldsymbol{A}\|_F^2\sum_{k=2}^{\infty}(2\theta\|\boldsymbol{A}\|)^{k-2}\right)\leqslant\exp\left(\frac{\theta^2\|\boldsymbol{A}\|_F^2}{1-2\theta\|\boldsymbol{A}\|}\right),$$

  thus by Theorem T2.3.1 (*Chernoff inequality*) (*Chernoff inequality*), we obtain for $0<\theta\leqslant 1/(4\|\boldsymbol{A}\|)$,

$$\mathbb{P}\{Z\geqslant t\}\leqslant\mathrm{e}^{-\theta t}\mathbb{E}[\mathrm{e}^{\theta Z}]\leqslant\exp\left(-\theta t+\frac{\theta^2\|\boldsymbol{A}\|_F^2}{1-2\theta\|\boldsymbol{A}\|}\right),$$

  Set

$$\theta=\min\left\{\frac{t}{4\|\boldsymbol{A}\|_F^2},\frac{1}{4\|\boldsymbol{A}\|}\right\}.$$

  Then $0<\theta\leqslant 1/(4\|\boldsymbol{A}\|)$ and $1-2\theta\|\boldsymbol{A}\|\geqslant 1/2$, so

$$\mathbb{P}\{Z\geqslant t\}\leqslant\exp(-\theta t+2\theta^2\|\boldsymbol{A}\|_F^2).$$

  If $t\leqslant\|\boldsymbol{A}\|_F^2/\|\boldsymbol{A}\|$, then $\theta=t/(4\|\boldsymbol{A}\|_F^2)$ and the exponent is at most $-t^2/(8\|\boldsymbol{A}\|_F^2)$; if $t>\|\boldsymbol{A}\|_F^2/\|\boldsymbol{A}\|$, then $\theta=1/(4\|\boldsymbol{A}\|)$ and the exponent is at most $-t/(8\|\boldsymbol{A}\|)$. Therefore

$$\mathbb{P}\{Z\geqslant t\}\leqslant\exp\left(-\frac{1}{8}\min\left(\frac{t^2}{\|\boldsymbol{A}\|_F^2},\frac{t}{\|\boldsymbol{A}\|}\right)\right);$$

  Applying the same argument to $-Z$,

$$\mathbb{P}\{Z\leqslant -t\}\leqslant\exp\left(-c\min\left(\frac{t^2}{\|\boldsymbol{A}\|_F^2},\frac{t}{\|\boldsymbol{A}\|}\right)\right).$$

  Thus,

$$\mathbb{P}\{|Z|\geqslant t\}=\mathbb{P}\{Z\geqslant t\}+\mathbb{P}\{Z\leqslant -t\}\leqslant 2\exp\left(-c\min\left(\frac{t^2}{\|\boldsymbol{A}\|_F^2},\frac{t}{\|\boldsymbol{A}\|}\right)\right).$$

8 [**High-Dimensional Hanson-Wright Inequality**] Let $\boldsymbol{A}\in\mathbb{R}^{n\times n}$, $\boldsymbol{X}_1,\cdots,\boldsymbol{X}_n\in\mathbb{R}^d$ be independent zero-mean random vectors whose components are independent and satisfy $\max_k\|X_{ik}\|_{\psi_2}\leqslant K$. Prove: For any $t\geqslant 0$,

$$\mathbb{P}\left\{\left|\sum_{i\neq j}a_{ij}\langle\boldsymbol{X}_i,\boldsymbol{X}_j\rangle\right|\geqslant t\right\}\leqslant 2\exp\left(-c\min\left(\frac{t^2}{K^4d\|\boldsymbol{A}\|_F^2},\frac{t}{K^2\|\boldsymbol{A}\|}\right)\right).$$

Prf. • Apply the Chernoff bound

For the target random variable $S_{\text{off}}=\sum_{i\neq j}a_{ij}\langle\boldsymbol{X}_i,\boldsymbol{X}_j\rangle$, using Theorem T2.3.1 (*Chernoff inequality*) (*Chernoff inequality*), we obtain

$$\mathbb{P}\{S_{\text{off}}\geqslant t\}\leqslant\mathrm{e}^{-\lambda t}\mathbb{E}\left[\exp\left(\lambda\sum_{i\neq j}a_{ij}\langle\boldsymbol{X}_i,\boldsymbol{X}_j\rangle\right)\right],$$

where using the conclusion of Exercise Q6.3 (*Vector and Space Decoupling*) (2), letting $F(x)=\exp(\lambda x)$, we obtain

$$\mathbb{E}\left[\exp\left(\lambda\sum_{i\neq j}a_{ij}\langle\boldsymbol{X}_i,\boldsymbol{X}_j\rangle\right)\right]\leqslant\mathbb{E}\left[\exp\left(4\lambda\sum_{i,j}a_{ij}\langle\boldsymbol{X}_i,\boldsymbol{X}_j'\rangle\right)\right],$$

Substituting this in, we obtain

$$\mathbb{P}\{S_{\text{off}} \geqslant t\} \leqslant \mathrm{e}^{-\lambda t}\mathbb{E}\left[\exp\left(4\lambda \sum_{i,j} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}'_j\rangle\right)\right]. \tag{I}$$

- Conditioning

Rearranging the sum inside the exponential,

$$\sum_{i,j=1}^{n} a_{ij}\langle \boldsymbol{X}_i, \boldsymbol{X}'_j\rangle = \sum_{i=1}^{n}\left\langle \boldsymbol{X}_i, \sum_{j=1}^{n} a_{ij}\boldsymbol{X}'_j\right\rangle = \sum_{i=1}^{n}\langle \boldsymbol{X}_i, \boldsymbol{Y}_i\rangle,$$

where $\boldsymbol{Y}_i = \sum_{j=1}^{n} a_{ij}\boldsymbol{X}'_j \in \mathbb{R}^d$.

Taking conditional expectation with respect to $\boldsymbol{X}' = (\boldsymbol{X}'_1, \cdots, \boldsymbol{X}'_n)$, we obtain

$$\mathbb{E}\left[\exp\left(4\lambda\sum_{i=1}^{n}\langle \boldsymbol{X}_i, \boldsymbol{Y}_i\rangle\right)\right] = \mathbb{E}_{\boldsymbol{X}'}\left[\mathbb{E}_{\boldsymbol{X}}\left[\exp\left(4\lambda\sum_{i=1}^{n}\langle \boldsymbol{X}_i, \boldsymbol{Y}_i\rangle\right) \,\middle|\, \boldsymbol{X}'\right]\right].$$

(*Conditional on $\boldsymbol{X}'$, $\boldsymbol{Y}_1, \cdots, \boldsymbol{Y}_n \in \mathbb{R}^d$ are deterministic constant vectors, and the inner sum $\sum_{i=1}^{n}\langle \boldsymbol{X}_i, \boldsymbol{Y}_i\rangle$ is the sum of n independent random variables.*)

Since $\boldsymbol{X}_i$ are sub-Gaussian random vectors, their projection onto the constant vector $\boldsymbol{Y}_i$, namely $\langle \boldsymbol{X}_i, \boldsymbol{Y}_i\rangle$, is a scalar sub-Gaussian random variable, satisfying

$$\|\langle \boldsymbol{X}_i, \boldsymbol{Y}_i\rangle\|_{\psi_2} \leqslant K\|\boldsymbol{Y}_i\|_2,$$

hence

$$\mathbb{E}_{\boldsymbol{X}_i}[\exp(4\lambda\langle \boldsymbol{X}_i, \boldsymbol{Y}_i\rangle) \mid \boldsymbol{X}'] \leqslant \exp(C_1\lambda^2K^2\|\boldsymbol{Y}_i\|_2^2), \qquad (C_1 > 0)$$

Summing over both sides,

$$\mathbb{E}_{\boldsymbol{X}}\left[\exp\left(4\lambda\sum_{i=1}^{n}\langle \boldsymbol{X}_i, \boldsymbol{Y}_i\rangle\right) \,\middle|\, \boldsymbol{X}'\right] \leqslant \exp\left(C_1\lambda^2K^2\sum_{i=1}^{n}\|\boldsymbol{Y}_i\|_2^2\right). \tag{II}$$

Combining (I) and (II), we obtain

$$\mathbb{P}\{S_{\text{off}} \geqslant t\} \leqslant \mathrm{e}^{-\lambda t}\mathbb{E}_{\boldsymbol{X}'}\left[\exp\left(C_1\lambda^2K^2\sum_{i=1}^{n}\|\boldsymbol{Y}_i\|_2^2\right)\right]. \tag{III}$$

- Analyze the quadratic form $\sum_{i=1}^{n}\|\boldsymbol{Y}_i\|_2^2$

By definition $\boldsymbol{Y}_i = \sum_{j=1}^{n} a_{ij}\boldsymbol{X}'_j$, let $\mathcal{X}' \in \mathbb{R}^{n\times d}$ be the matrix whose rows are $(\boldsymbol{X}'_1)^\top, \cdots, (\boldsymbol{X}'_n)^\top$, then $\boldsymbol{Y}_i^\top$ is the i-th row of $\boldsymbol{A}\mathcal{X}'$, satisfying

$$\sum_{i=1}^{n}\|\boldsymbol{Y}_i\|_2^2 = \|\boldsymbol{A}\boldsymbol{X}'\|_F^2 = \sum_{k=1}^{d}\|\boldsymbol{A}\boldsymbol{g}_k\|_2^2.$$

Let $\boldsymbol{g}_1, \cdots, \boldsymbol{g}_d \in \mathbb{R}^n$ be the column vectors of $\mathcal{X}'$, then $\boldsymbol{g}_k$ are independent, each having independent K-sub-Gaussian components, hence

$$\|\boldsymbol{A}\mathcal{X}'\|_F^2 = \sum_{k=1}^{d}\|\boldsymbol{A}\boldsymbol{g}_k\|_2^2 = \sum_{k=1}^{d}\boldsymbol{g}_k^\top(\boldsymbol{A}^\top\boldsymbol{A})\boldsymbol{g}_k,$$

thus the right-hand side of (III) is

$$\begin{aligned}\mathbb{E}_{\boldsymbol{X}'}\left[\exp\left(C_1\lambda^2K^2\sum_{i=1}^{n}\|\boldsymbol{Y}_i\|_2^2\right)\right] &= \mathbb{E}_{\boldsymbol{X}'}\left[\exp\left(C_1\lambda^2K^2\sum_{k=1}^{d}\boldsymbol{g}_k^\top(\boldsymbol{A}^\top\boldsymbol{A})\boldsymbol{g}_k\right)\right]\\ &= \prod_{k=1}^{d}\mathbb{E}_{\boldsymbol{g}_k}[\exp(C_1\lambda^2K^2\cdot\boldsymbol{g}_k^\top(\boldsymbol{A}^\top\boldsymbol{A})\boldsymbol{g}_k)].\end{aligned}$$

We state the following lemma:

Lemma (Exercise Q6.9 (*Moment Generating Function of the Squared Norm*)) Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{X} \in \mathbb{R}^n$ be a K-sub-Gaussian random vector. Then when $\alpha \leqslant c/(K^2\|\boldsymbol{A}\|^2)$,

$$\mathbb{E}[\exp(\alpha\|\boldsymbol{A}\boldsymbol{X}\|_2^2)] \leqslant \exp(CK^2\alpha\|\boldsymbol{A}\|_F^2).$$

By the lemma, let $\alpha = C_1\lambda^2K^2$. Then when

$$\lambda \leqslant \frac{c_1}{K^2\|\boldsymbol{A}\|}, \qquad \left(C_1\lambda^2K^2 \leqslant \frac{c_0}{K^2\|\boldsymbol{A}\|^2}\right)$$

holds,

$$\mathbb{E}[\exp(\alpha\|\boldsymbol{A}\boldsymbol{g}_k\|_2^2)] \leqslant \exp(C_2\alpha\|\boldsymbol{A}\|_F^2), \qquad (C_2 > 0)$$

Hence

$$\mathbb{E}_{\boldsymbol{X}'}[\exp(C_1\lambda^2K^2\|\boldsymbol{A}\boldsymbol{X}'\|_F^2)] \leqslant \exp(C_3\lambda^2K^4d\|\boldsymbol{A}\|_F^2),$$

Substituting this into (III), we obtain

$$\mathbb{P}\{S_{\text{off}} \geqslant t\} \leqslant \exp(\underbrace{-\lambda t + C_3\lambda^2K^4d\|\boldsymbol{A}\|_F^2}_{h(\lambda)}).$$

- Optimize λ (*Discussion of small and large deviations*)

  When the constraint is not considered,

$$\lambda_0 = \frac{t}{2C_3K^4d\|\boldsymbol{A}\|_F^2},$$

  We discuss the values of λ:

  a. Small deviations (sub-Gaussian bound)

  If t is small, satisfying

$$\lambda_0 \leqslant \frac{c_1}{K^2\|\boldsymbol{A}\|},$$

  substitute λ_0 into (III) to obtain

$$\mathbb{P}\{S_{\text{off}} \geqslant t\} \leqslant \exp\left(-\frac{t^2}{4C_3K^4d\|\boldsymbol{A}\|_F^2}\right).$$

  b. Large deviations (sub-Exponential bound)

  If t is large, such that

$$\lambda_0 > \frac{c_1}{K^2\|\boldsymbol{A}\|},$$

  then the minimum is attained at the boundary,

$$\begin{aligned} h\left(\frac{c_1}{K^2\|\boldsymbol{A}\|}\right) &= -\frac{c_1t}{K^2\|\boldsymbol{A}\|} + C_3\left(\frac{c_1}{K^2\|\boldsymbol{A}\|}\right)^2K^4d\|\boldsymbol{A}\|_F^2 \\ &= -\frac{c_1t}{K^2\|\boldsymbol{A}\|} + C_3c_1^2\frac{d\|\boldsymbol{A}\|_F^2}{\|\boldsymbol{A}\|^2} \\ &\leqslant -\frac{c_1t}{K^2\|\boldsymbol{A}\|} + \frac{c_1t}{2K^2\|\boldsymbol{A}\|} = -\frac{c_1}{2}\frac{t}{K^2\|\boldsymbol{A}\|}, \qquad \left(\frac{c_1}{K^2\|\boldsymbol{A}\|} < \frac{t}{2C_3K^4d\|\boldsymbol{A}\|_F^2}\right) \end{aligned}$$

  Substituting this into (III), we obtain

$$\mathbb{P}\{S_{\text{off}} \geqslant t\} \leqslant \exp\left(-c_2\frac{t}{K^2\|\boldsymbol{A}\|}\right).$$

  Thus,

$$\mathbb{P}\{S_{\text{off}} \geqslant t\} \leqslant \exp\left(-c\min\left(\frac{t^2}{K^4d\|\boldsymbol{A}\|_F^2}, \frac{t}{K^2\|\boldsymbol{A}\|}\right)\right),$$

Taking the union bound on both sides, we obtain

$$\mathbb{P}\{|S_{\text{off}}| \geqslant t\} = \mathbb{P}\{S_{\text{off}} \geqslant t\} + \mathbb{P}\{-S_{\text{off}} \geqslant t\} \leqslant 2\exp\left(-c\min\left(\frac{t^2}{K^4 d\|\boldsymbol{A}\|_F^2}, \frac{t}{K^2\|\boldsymbol{A}\|}\right)\right).$$

9 [**Moment Generating Function of the Squared Norm**] Let $\boldsymbol{B} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{X} \in \mathbb{R}^n$ be a zero-mean sub-Gaussian random vector satisfying $\|\boldsymbol{X}\|_{\psi_2} \leqslant K$. Prove: For any parameter λ satisfying $|\lambda| \leqslant c/K\|\boldsymbol{B}\|$,

$$\mathbb{E}[\exp(\lambda^2\|\boldsymbol{B}\boldsymbol{X}\|_2^2)] \leqslant \exp(CK^2\lambda^2\|\boldsymbol{B}\|_F^2). \qquad (C, c > 0)$$

Rk. **MGF version of the Hanson-Wright inequality** Let $\boldsymbol{X} \in \mathbb{R}^n$ be a zero-mean sub-Gaussian random vector satisfying $\|\boldsymbol{X}\|_{\psi_2} \leqslant K$, and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric matrix. Then for any parameter θ satisfying $|\theta| \cdot K^2\|\boldsymbol{A}\| \leqslant c$,

$$\mathbb{E}[\exp(\theta(\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}]))] \leqslant \exp(C\theta^2K^4\|\boldsymbol{A}\|_F^2).$$

First perform quadratic form decoupling. Let $\boldsymbol{X}'$ be an independent copy of $\boldsymbol{X}$. By Theorem T6.1.1 (Decoupling), we obtain for any convex function F,

$$\mathbb{E}[F(\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}])] \leqslant \mathbb{E}[F(C_1\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}')],$$

Let $F(x) = \exp(\theta x)$, we obtain

$$\mathbb{E}[\exp(\theta(\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}]))] \leqslant \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}[\exp(C_1\theta\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}')], \qquad \text{(I)}$$

This is the MGF of the quadratic form $Q = \boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}'$.

Then condition. Let $\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{X}'$, then the right-hand side of (I) is

$$\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}[\exp(C_1\theta\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}')] = \mathbb{E}_{\boldsymbol{X}'}[\mathbb{E}_{\boldsymbol{X}}[\exp(C_1\theta\langle\boldsymbol{X},\boldsymbol{Y}\rangle) \mid \boldsymbol{X}']],$$

where for fixed $\boldsymbol{X}'$, $\langle\boldsymbol{X},\boldsymbol{Y}\rangle$ is a sub-Gaussian random variable whose MGF satisfies

$$\begin{aligned}\mathbb{E}_{\boldsymbol{X}}[\exp(C_1\theta\langle\boldsymbol{X},\boldsymbol{Y}\rangle)] &\leqslant \exp(C_2(C_1\theta)^2K^2\|\boldsymbol{Y}\|_2^2) && (C_2 > 0)\\ &= \exp(C_3\theta^2K^2\|\boldsymbol{A}\boldsymbol{X}'\|_2^2), && (C_3 = C_1C_2 > 0)\end{aligned}$$

Substituting this in, we obtain

$$\mathbb{E}[\exp(\theta(\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}]))] \leqslant \mathbb{E}_{\boldsymbol{X}'}[\exp(C_3\theta^2K^2\|\boldsymbol{A}\boldsymbol{X}'\|_2^2)]. \qquad \text{(II)}$$

Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ be independent of $\boldsymbol{X}'$. Applying the Gaussian MGF for fixed $\boldsymbol{X}'$ and the sub-Gaussian MGF for fixed $\boldsymbol{g}$ gives

$$\begin{aligned}\mathbb{E}_{\boldsymbol{X}'}[\exp(C_3\theta^2K^2\|\boldsymbol{A}\boldsymbol{X}'\|_2^2)] &= \mathbb{E}_{\boldsymbol{g}}\mathbb{E}_{\boldsymbol{X}'}\left[\exp(\sqrt{2C_3}|\theta|K\langle\boldsymbol{A}^\top\boldsymbol{g},\boldsymbol{X}'\rangle) \mid \boldsymbol{g}\right]\\ &\leqslant \mathbb{E}_{\boldsymbol{g}}[\exp(C_4\theta^2K^4\|\boldsymbol{A}^\top\boldsymbol{g}\|_2^2)].\end{aligned}$$

If $C_4\theta^2K^4\|\boldsymbol{A}\|^2 \leqslant 1/4$, the Gaussian quadratic-form MGF applied to the singular values of $\boldsymbol{A}$ gives

$$\mathbb{E}_{\boldsymbol{g}}[\exp(C_4\theta^2K^4\|\boldsymbol{A}^\top\boldsymbol{g}\|_2^2)] \leqslant \exp(C_5\theta^2K^4\|\boldsymbol{A}\|_F^2).$$

Choosing a sufficiently small absolute constant c, the condition $|\theta|K^2\|\boldsymbol{A}\| \leqslant c$ guarantees this condition and proves the stated MGF bound.

Prf. Let $\boldsymbol{A} = \boldsymbol{B}^\top\boldsymbol{B}$ be a positive semidefinite symmetric matrix. Express the squared norm as a quadratic form,

$$\|\boldsymbol{B}\boldsymbol{X}\|_2^2 = (\boldsymbol{B}\boldsymbol{X})^\top(\boldsymbol{B}\boldsymbol{X}) = \boldsymbol{X}^\top(\boldsymbol{B}^\top\boldsymbol{B})\boldsymbol{X} = \boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X},$$

We need to bound the MGF of $Q = \boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}$.

On the one hand, using the MGF form of Theorem T6.2.2 (*Hanson-Wright inequality*) (*Hanson-Wright inequality*), we obtain for any parameter θ satisfying $|\theta| \cdot K^2\|\boldsymbol{A}\| \leqslant c$,

$$\mathbb{E}[\exp(\theta(\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}]))] \leqslant \exp(C\theta^2K^4\|\boldsymbol{A}\|_F^2).$$

Let $\theta = \lambda^2$. Note that $\|\boldsymbol{A}\| = \|\boldsymbol{B}^\top\boldsymbol{B}\| = \|\boldsymbol{B}\|^2$, hence the constraint condition is equivalent to

$$\lambda^2K^2\|\boldsymbol{B}\|^2 \leqslant c. \qquad \text{(III)}$$

On the other hand,

$$\mathbb{E}[\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}] = \mathbb{E}[\operatorname{tr}(\boldsymbol{A}\boldsymbol{X}\boldsymbol{X}^\top)] = \operatorname{tr}(\boldsymbol{A}\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]) \leqslant K^2\operatorname{tr}(\boldsymbol{A}) \leqslant K^2\|\boldsymbol{B}\|_F^2. \qquad \text{(IV)}$$

Thus, by the separability of the MGF, combining (III) and (IV), we obtain

$$\begin{aligned}\mathbb{E}[\exp(\lambda^2 Q)] &= \exp(\lambda^2\mathbb{E}[Q])\cdot\mathbb{E}[\exp(\lambda^2(Q-\mathbb{E}[Q]))]\\ &\leqslant \exp(\lambda^2K^2\|\boldsymbol{B}\|_F^2)\cdot\exp(C\lambda^4K^4\|\boldsymbol{A}\|_F^2),\end{aligned}$$

where the exponent of the second term is

$$\begin{aligned}C\lambda^4K^4\|\boldsymbol{A}\|_F^2 &\leqslant C\lambda^4K^4(\|\boldsymbol{B}\|^2\|\boldsymbol{B}\|_F^2)\\ &= C\cdot(\lambda^2K^2\|\boldsymbol{B}\|^2)\cdot(\lambda^2K^2\|\boldsymbol{B}\|_F^2)\\ &\leqslant C\cdot c\cdot\lambda^2K^2\|\boldsymbol{B}\|_F^2,\end{aligned}$$

Substituting this in, we obtain

$$\begin{aligned}\mathbb{E}[\exp(\lambda^2\|\boldsymbol{B}\boldsymbol{X}\|_2^2)] &\leqslant \exp(\lambda^2K^2\|\boldsymbol{B}\|_F^2)\cdot\exp(Cc^2\lambda^2K^2\|\boldsymbol{B}\|_F^2)\\ &= \exp(C'\lambda^2K^2\|\boldsymbol{B}\|_F^2). \qquad (C'=1+Cc>0)\end{aligned}$$

10 [**Norm of an Anisotropic Random Vector**] Let us generalize Proposition P6.2.1 (*The norm of a subgaussian random vector*) to anisotropic random vectors.

Let $\boldsymbol{B}\in\mathbb{R}^{m\times n}$ be a fixed matrix and $\boldsymbol{X}\in\mathbb{R}^n$ be a zero-mean sub-Gaussian random vector satisfying $\|\boldsymbol{X}\|_{\psi_2}\leqslant K$. Prove: For any $t\geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{B}\boldsymbol{X}\|_2\geqslant CK(\|\boldsymbol{B}\|_F+t\|\boldsymbol{B}\|)\}\leqslant \mathrm{e}^{-t^2}. \qquad (C>0)$$

Prf. • Reduce to a quadratic form problem

Define the positive semidefinite symmetric matrix $\boldsymbol{A}=\boldsymbol{B}^\top\boldsymbol{B}$, then $\operatorname{tr}(\boldsymbol{A})=\operatorname{tr}(\boldsymbol{B}^\top\boldsymbol{B})=\|\boldsymbol{B}\|_F^2$, $\|\boldsymbol{A}\|=\|\boldsymbol{B}\|^2$.

Let $Q=\|\boldsymbol{B}\boldsymbol{X}\|_2^2=\boldsymbol{X}^\top(\boldsymbol{B}^\top\boldsymbol{B})\boldsymbol{X}=\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}$, then

$$\mathbb{E}[Q]=\mathbb{E}[\operatorname{tr}(\boldsymbol{A}\boldsymbol{X}\boldsymbol{X}^\top)]=\operatorname{tr}(\boldsymbol{A}\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top])\leqslant K^2\operatorname{tr}(\boldsymbol{A})=K^2\|\boldsymbol{B}\|_F^2.$$

We need to control $\|\boldsymbol{B}\boldsymbol{X}\|_2\geqslant CK(\|\boldsymbol{B}\|_F+t\|\boldsymbol{B}\|)$. Squaring both sides, the goal is equivalent to proving

$$Q\geqslant C^2K^2(\|\boldsymbol{B}\|_F+t\|\boldsymbol{B}\|)^2=C^2K^2(\|\boldsymbol{B}\|_F^2+2t\|\boldsymbol{B}\|_F\|\boldsymbol{B}\|+t^2\|\boldsymbol{B}\|^2). \qquad (\mathrm{I})$$

(*We need to control the probability that* $|Q-\mathbb{E}[Q]|$ *exceeds some value* u_0.)

Let the deviation target be $u=Q-\mathbb{E}[Q]$ and the deviation threshold be $u_0=C_0K^2(t\|\boldsymbol{B}\|_F\|\boldsymbol{B}\|+t^2\|\boldsymbol{B}\|^2)$. When $Q\geqslant\mathbb{E}[Q]+u_0$ occurs,

$$Q\geqslant K^2\|\boldsymbol{B}\|_F^2+C_0K^2(t\|\boldsymbol{B}\|_F\|\boldsymbol{B}\|+t^2\|\boldsymbol{B}\|^2). \qquad (\mathrm{II})$$

(*We only need to choose* C *appropriately so that* $\mathrm{RHS}_{(II)}\leqslant\mathrm{RHS}_{(I)}$, *which reduces the problem of bounding* $\|\boldsymbol{B}\boldsymbol{X}\|_2$ *to the problem of* $Q-\mathbb{E}[Q]\geqslant u_0$.)

• Apply the Hanson-Wright inequality

By Theorem T6.2.2 (*Hanson-Wright inequality*), we obtain for $u_0\geqslant 0$,

$$\mathbb{P}\{|Q-\mathbb{E}[Q]|\geqslant u_0\}\leqslant 2\exp\left(-c\min\left(\frac{u_0^2}{K^4\|\boldsymbol{A}\|_F^2},\frac{u_0}{K^2\|\boldsymbol{A}\|}\right)\right).$$

where

$$\begin{aligned}\frac{u_0^2}{K^4\|\boldsymbol{A}\|_F^2}&=\frac{C_0^2K^4(t\|\boldsymbol{B}\|_F\|\boldsymbol{B}\|+t^2\|\boldsymbol{B}\|^2)^2}{K^4\|\boldsymbol{B}^\top\boldsymbol{B}\|_F^2}\geqslant\frac{C_0^2K^4(t\|\boldsymbol{B}\|_F\|\boldsymbol{B}\|+t^2\|\boldsymbol{B}\|^2)^2}{K^4\|\boldsymbol{B}\|^2\|\boldsymbol{B}\|_F^2}=C_0^2\left(t+t^2\frac{\|\boldsymbol{B}\|}{\|\boldsymbol{B}\|_F}\right)^2\geqslant C_0^2t^2,\\ \frac{u_0}{K^2\|\boldsymbol{A}\|}&=\frac{C_0K^2(t\|\boldsymbol{B}\|_F\|\boldsymbol{B}\|+t^2\|\boldsymbol{B}\|^2)}{K^2\|\boldsymbol{B}\|^2}=C_0\left(t\frac{\|\boldsymbol{B}\|_F}{\|\boldsymbol{B}\|}+t^2\right)\geqslant C_0t^2,\end{aligned}$$

Hence

$$\min\left(\frac{u_0^2}{K^4\|\boldsymbol{A}\|_F^2},\frac{u_0}{K^2\|\boldsymbol{A}\|}\right)\geqslant\min(C_0^2,C_0)\cdot t^2,$$

i.e.,

$$\mathbb{P}\{Q-\mathbb{E}[Q]\geqslant u_0\}\leqslant 2\exp(-c't^2), \qquad (c'=c\min(C_0^2,C_0)>0)$$

thus by adjusting absolute constants, we obtain

$$\mathbb{P}\{\|\boldsymbol{B}\boldsymbol{X}\|_2 \geqslant CK(\|\boldsymbol{B}\|_F + t\|\boldsymbol{B}\|)\} \leqslant \mathrm{e}^{-t^2}.$$

(The specific process of adjusting absolute constants:

Comparing the coefficients of $\mathrm{RHS}_{(II)} \leqslant \mathrm{RHS}_{(I)}$,

$$K^2(t\|\boldsymbol{B}\|_F\|\boldsymbol{B}\| + t^2\|\boldsymbol{B}\|^2) \leqslant C^2K^2(\|\boldsymbol{B}\|_F^2 + 2t\|\boldsymbol{B}\|_F\|\boldsymbol{B}\| + t^2\|\boldsymbol{B}\|^2),$$

we obtain)

$$\begin{cases} C^2 \geqslant 1, \\ 2C^2 \geqslant C_0, \\ C^2 \geqslant C_0 \end{cases} \Longrightarrow\ C = \sqrt{\max\left(1, \frac{C_0}{2}, C_0\right)} = \sqrt{\max(1, C_0)}.$$

11 [**High-Dimensional Hanson-Wright Inequality**] Here we give a one-sided consequence of Theorem T6.2.2 (*Hanson-Wright inequality*).

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a symmetric positive semidefinite matrix and let $\boldsymbol{X} \in \mathbb{R}^n$ have independent zero-mean sub-Gaussian components satisfying $\max_i \|X_i\|_{\psi_2} \leqslant K$. Prove: For any $s \geqslant 0$,

$$\mathbb{P}\{\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X} \geqslant CK^2(\mathrm{tr}(\boldsymbol{A}) + s\|\boldsymbol{A}\|)\} \leqslant \mathrm{e}^{-s}. \qquad (C > 0)$$

(*To compare with Theorem T6.2.2 (Hanson-Wright inequality), let* $t = CK^2s\|\boldsymbol{A}\|$ *and note that for an isotropic random vector* $\boldsymbol{X}$*, we have* $\mathrm{tr}(\boldsymbol{A}) = \mathbb{E}[\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}]$.)

Then explain that even in the case where $\boldsymbol{X}$ is isotropic, only a one-sided bound exists.

Prf. • Proof process

Similarly to Exercise Q6.10 (*Norm of an Anisotropic Random Vector*), let $Q = \boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}$, then

$$\mathbb{E}[Q] = \mathbb{E}[\mathrm{tr}(\boldsymbol{A}\boldsymbol{X}\boldsymbol{X}^\top)] = \mathrm{tr}(\boldsymbol{A}\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]) = \mathrm{tr}(\boldsymbol{A}\boldsymbol{\Sigma}) \leqslant \|\boldsymbol{\Sigma}\|\,\mathrm{tr}(\boldsymbol{A}) \leqslant K^2\,\mathrm{tr}(\boldsymbol{A}).$$

For the target threshold $T = CK^2(\mathrm{tr}(\boldsymbol{A}) + s\|\boldsymbol{A}\|)$ where $C > 1$ (*otherwise* $C - 1 \not> 0$), consider the deviation

$$u_0 = T - \mathbb{E}[Q] \geqslant (C-1)K^2\,\mathrm{tr}(\boldsymbol{A}) + CK^2s\|\boldsymbol{A}\|,$$

Let $c_1 = C - 1 > 0$, $c_2 = C > 0$, then

$$u_0 = c_1K^2\,\mathrm{tr}(\boldsymbol{A}) + c_2K^2s\|\boldsymbol{A}\| \geqslant 0.$$

Using Theorem T6.2.2 (*Hanson-Wright inequality*) (*Hanson-Wright inequality*), we obtain for $u_0 \geqslant 0$,

$$\mathbb{P}\{Q - \mathbb{E}[Q] \geqslant u_0\} \leqslant 2\exp\left(-c\min\left(\frac{u_0^2}{K^4\|\boldsymbol{A}\|_F^2}, \frac{u_0}{K^2\|\boldsymbol{A}\|}\right)\right),$$

where for the second term, using $\|\boldsymbol{A}\|_F^2 \leqslant \|\boldsymbol{A}\|\,\mathrm{tr}(\boldsymbol{A})$, we obtain

$$\frac{u_0}{K^2\|\boldsymbol{A}\|} = \frac{c_1K^2\,\mathrm{tr}(\boldsymbol{A}) + c_2K^2s\|\boldsymbol{A}\|}{K^2\|\boldsymbol{A}\|} = c_1\frac{\mathrm{tr}(\boldsymbol{A})}{\|\boldsymbol{A}\|} + c_2s \geqslant c_2s,$$

For the first term, using $(x+y)^2 \geqslant 4xy$, we obtain

$$\frac{u_0^2}{K^4\|\boldsymbol{A}\|_F^2} \geqslant \frac{(c_1K^2\,\mathrm{tr}(\boldsymbol{A}) + c_2K^2s\|\boldsymbol{A}\|)^2}{K^4\|\boldsymbol{A}\|\,\mathrm{tr}(\boldsymbol{A})} \geqslant \frac{4c_1c_2K^4s\|\boldsymbol{A}\|\,\mathrm{tr}(\boldsymbol{A})}{K^4\|\boldsymbol{A}\|\,\mathrm{tr}(\boldsymbol{A})} = 4c_1c_2s,$$

Hence

$$\mathbb{P}\{Q - \mathbb{E}[Q] \geqslant u_0\} \leqslant 2\exp(-c's), \qquad (c' = c\cdot\min(4c_1c_2, c_2) > 0)$$

thus by adjusting absolute constants, we obtain

$$\mathbb{P}\{\boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X} \geqslant CK^2(\mathrm{tr}(\boldsymbol{A}) + s\|\boldsymbol{A}\|)\} \leqslant \mathrm{e}^{-s}.$$

• Explanation of the one-sided bound

Even if $\boldsymbol{X}$ is isotropic, the above inequality only possesses a right-tail bound.

In fact, since $\boldsymbol{A}$ is a symmetric positive semidefinite matrix, we have $Q = \boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X} \geqslant 0$, which makes the random

variable hard-truncated at 0 on the left, with no probability tail extending toward $-\infty$.

Specifically, the distribution of the quadratic form $\boldsymbol{X}^\top \boldsymbol{A}\boldsymbol{X}$ exhibits a pronounced right-skewed feature: its probability density function is truncated at 0 on the left, and decays at a sub-Exponential rate on the right. Therefore, for the right-tail deviation probability $\mathbb{P}\{Q \geqslant \mathbb{E}[Q] + u\}$, we can obtain a meaningful exponential decay bound; however, for the left-tail deviation probability $\mathbb{P}\{Q \leqslant \mathbb{E}[Q] - u\}$, when $u > \mathbb{E}[Q]$, this probability is 0, and there is no need to use the Hanson-Wright inequality for a complicated asymptotic analysis.

12 [**Mean Estimation**] Recall the mean estimation problem.

Consider an unknown distribution on $\mathbb{R}^n$ with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ be independent samples drawn from this distribution, and assume that the components of every $\boldsymbol{X}_i - \boldsymbol{\mu}$ are independent. Our goal is to estimate $\boldsymbol{\mu}$ using the sample mean

$$\boldsymbol{\mu}_N = \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{X}_i.$$

Further assume that this distribution satisfies the sub-Gaussian condition, i.e., for any $\boldsymbol{u} \in \mathbb{R}^n$,

$$\|\langle \boldsymbol{X}_i - \boldsymbol{\mu}, \boldsymbol{u}\rangle\|_{\psi_2} \leqslant K\|\langle \boldsymbol{X}_i - \boldsymbol{\mu}, \boldsymbol{u}\rangle\|_{L_2}.$$

Prove: For any $\alpha \in (0,1)$,

$$\|\boldsymbol{\mu}_N - \boldsymbol{\mu}\|_2 \leqslant CK^2\left(\sqrt{\frac{\operatorname{tr}(\boldsymbol{\Sigma})}{N}} + \sqrt{\frac{\|\boldsymbol{\Sigma}\|\log(1/\alpha)}{N}}\right) \text{ (w.p. } 1-\alpha).$$

Prf. Let $Z = \|\boldsymbol{\mu}_N - \boldsymbol{\mu}\|_2$. Decompose it as

$$Z \leqslant \mathbb{E}[Z] + (Z - \mathbb{E}[Z]),$$

where $\mathbb{E}[Z]$ is deterministic and we can directly bound it from above using Jensen's inequality; $Z - \mathbb{E}[Z]$ is the deviation of the random variable from its mean, and we need to use a concentration inequality to give a high-probability upper bound.

- Estimation of the expectation part

  Let $\boldsymbol{Y}_i = \boldsymbol{X}_i - \boldsymbol{\mu}$ be independent zero-mean sub-Gaussian random vectors with mean $\boldsymbol{0}$ and covariance $\boldsymbol{\Sigma}$, then

  $$\|\boldsymbol{\mu}_N - \boldsymbol{\mu}\|_2 = \left\|\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{Y}_i\right\|_2 = \|\bar{\boldsymbol{Y}}\|_2.$$

  Using Jensen's inequality, we obtain

  $$\mathbb{E}\|\boldsymbol{\mu}_N - \boldsymbol{\mu}\|_2 \leqslant \sqrt{\mathbb{E}\|\boldsymbol{\mu}_N - \boldsymbol{\mu}\|_2^2} = \sqrt{\mathbb{E}\|\bar{\boldsymbol{Y}}\|_2^2},$$

  where

  $$\mathbb{E}\|\bar{\boldsymbol{Y}}\|_2^2 = \frac{1}{N^2}\sum_{i=1}^{N}\mathbb{E}\|\boldsymbol{Y}_i\|_2^2 = \frac{1}{N^2}\cdot N \cdot \operatorname{tr}(\boldsymbol{\Sigma}) = \frac{\operatorname{tr}(\boldsymbol{\Sigma})}{N},$$

  Substituting this in, we obtain

  $$\mathbb{E}\|\boldsymbol{\mu}_N - \boldsymbol{\mu}\|_2 \leqslant \sqrt{\frac{\operatorname{tr}(\boldsymbol{\Sigma})}{N}}.$$

- Estimation of the tail probability

  Let $\sigma_j^2 = \operatorname{Var}(Y_{1j})$ and set $\boldsymbol{D} = \operatorname{diag}(\sigma_1, \ldots, \sigma_n)$. For $\sigma_j > 0$, define

  $$Z_j = \frac{1}{\sigma_j\sqrt{N}}\sum_{i=1}^{N} Y_{ij};$$

  when $\sigma_j = 0$, set $Z_j = 0$. The nondegenerate components of $\boldsymbol{Z}$ are independent, centered, standardized, and satisfy

$\|Z_j\|_{\psi_2} \leqslant CK$. Moreover,

$$\bar{\boldsymbol{Y}} = \frac{1}{\sqrt{N}}\boldsymbol{D}\boldsymbol{Z}, \qquad \|\boldsymbol{D}\|_F^2 = \operatorname{tr}(\boldsymbol{\Sigma}), \quad \|\boldsymbol{D}\|^2 = \|\boldsymbol{\Sigma}\|.$$

Applying Theorem T3.1.1 (*Concentration of the norm*) to $\boldsymbol{D}/\sqrt{N}$ gives, for every $u \geqslant 0$,

$$\|\bar{\boldsymbol{Y}}\|_2 \leqslant CK^2\left(\sqrt{\frac{\operatorname{tr}(\boldsymbol{\Sigma})}{N}} + u\sqrt{\frac{\|\boldsymbol{\Sigma}\|}{N}}\right) \text{ (w.p. } 1 - 2\mathrm{e}^{-cu^2}).$$

Taking $u = C\sqrt{\log(2/\alpha)}$ gives the stated tail bound.

Thus,

$$Z = \|\boldsymbol{\mu}_N - \boldsymbol{\mu}\|_2 \leqslant CK^2\left(\sqrt{\frac{\operatorname{tr}(\boldsymbol{\Sigma})}{N}} + \sqrt{\frac{\|\boldsymbol{\Sigma}\|\log(1/\alpha)}{N}}\right) \text{ (w.p. } 1 - \alpha).$$

13 [**Anisotropic Concentration of the Norm**] Let's extend Theorem T3.1.1 (*Concentration of the norm*) to anisotropic distributions.

Let $\boldsymbol{B} \in \mathbb{R}^{m\times n}$ be a fixed matrix and $\boldsymbol{X} \in \mathbb{R}^n$ be a centered random vector whose components X_i are independent standardized sub-Gaussian random variables. Denote $\max_i \|X_i\|_{\psi_2} = K$. Prove:

$$\|\|\boldsymbol{B}\boldsymbol{X}\|_2 - \|\boldsymbol{B}\|_F\|_{\psi_2} \leqslant CK^2\|\boldsymbol{B}\|.$$

Prf. Introduce the notation $Z = f(\boldsymbol{X}) = \|\boldsymbol{B}\boldsymbol{X}\|_2 \geqslant 0$. If $\boldsymbol{B} = 0$, the conclusion is trivial; we only need to consider the case $E = \|\boldsymbol{B}\|_F > 0$.

Let $Q = Z^2 = \|\boldsymbol{B}\boldsymbol{X}\|_2^2 = \boldsymbol{X}^\top\boldsymbol{A}\boldsymbol{X}$, where $\boldsymbol{A} = \boldsymbol{B}^\top\boldsymbol{B}$, then

$$\mathbb{E}[Q] = \operatorname{tr}(\boldsymbol{A}) = \|\boldsymbol{B}\|_F^2 = E^2.$$

Using Theorem T6.2.2 (*Hanson-Wright inequality*) (*Hanson-Wright inequality*), we obtain for any $u \geqslant 0$,

$$\mathbb{P}\{|Q - E^2| \geqslant u\} \leqslant 2\exp\left(-c\min\left(\frac{u^2}{K^4\|\boldsymbol{A}\|_F^2}, \frac{u}{K^2\|\boldsymbol{A}\|}\right)\right),$$

Substituting $\|\boldsymbol{A}\| = \|\boldsymbol{B}\|^2$, $\|\boldsymbol{A}\|_F^2 \leqslant \|\boldsymbol{B}\|^2\|\boldsymbol{B}\|_F^2$, we obtain

$$\mathbb{P}\{|Z^2 - E^2| \geqslant u\} \leqslant 2\exp\left(-c\min\left(\frac{u^2}{K^4\|\boldsymbol{B}\|^2\|\boldsymbol{B}\|_F^2}, \frac{u}{K^2\|\boldsymbol{B}\|^2}\right)\right).$$

Lower-bound $|Z^2 - E^2|$ via t:

a. If $0 \leqslant t \leqslant E$, then

$$|Z^2 - E^2| = |Z - E|(Z + E) \geqslant t \cdot E,$$

hence $\{|Z - E| \geqslant t\} \subset \{|Z^2 - E^2| \geqslant tE\}$, satisfying

$$\begin{aligned}\mathbb{P}\{|Z - E| \geqslant t\} &\leqslant \mathbb{P}\{|Z^2 - E^2| \geqslant t\|\boldsymbol{B}\|_F\} \\ &\leqslant 2\exp\left(-c\min\left(\frac{t^2\|\boldsymbol{B}\|_F^2}{K^4\|\boldsymbol{B}\|^2\|\boldsymbol{B}\|_F^2}, \frac{t\|\boldsymbol{B}\|_F}{K^2\|\boldsymbol{B}\|^2}\right)\right) \\ &= 2\exp\left(-c\min\left(\frac{t^2}{K^4\|\boldsymbol{B}\|^2}, \frac{t\|\boldsymbol{B}\|_F}{K^2\|\boldsymbol{B}\|^2}\right)\right),\end{aligned}$$

where when $t \leqslant \|\boldsymbol{B}\|_F$,

$$\frac{t\|\boldsymbol{B}\|_F}{K^2\|\boldsymbol{B}\|^2} \geqslant \frac{t^2}{K^2\|\boldsymbol{B}\|^2},$$

hence

$$\mathbb{P}\{|Z - E| \geqslant t\} \leqslant 2\exp\left(-\frac{c't^2}{K^4\|\boldsymbol{B}\|^2}\right). \tag{I}$$

b. If $t > E$, then

$$|Z^2 - E^2| = |Z - E|(Z + E) \geqslant t \cdot t = t^2,$$

hence $\{|Z - E| \geqslant t\} \subset \{|Z^2 - E^2| \geqslant t^2\}$, satisfying

$$\begin{aligned}\mathbb{P}\{|Z - E| \geqslant t\} &\leqslant \mathbb{P}\{|Z^2 - E^2| \geqslant t^2\} \\ &\leqslant 2\exp\left(-c\min\left(\frac{t^4}{K^4\|\boldsymbol{B}\|^2\|\boldsymbol{B}\|_F^2}, \frac{t^2}{K^2\|\boldsymbol{B}\|^2}\right)\right),\end{aligned}$$

where when $t > \|\boldsymbol{B}\|_F$,

$$\frac{t^4}{K^4\|\boldsymbol{B}\|^2\|\boldsymbol{B}\|_F^2} > \frac{t^2}{K^4\|\boldsymbol{B}\|^2},$$

hence

$$\mathbb{P}\{|Z - E| \geqslant t\} \leqslant 2\exp\left(-\frac{c't^2}{K^4\|\boldsymbol{B}\|^2}\right). \tag{II}$$

Thus, combining (I) and (II), we obtain

$$\mathbb{P}\{|\|\boldsymbol{B}\boldsymbol{X}\|_2 - \|\boldsymbol{B}\|_F| \geqslant t\} \leqslant 2\exp\left(-\frac{c't^2}{K^4\|\boldsymbol{B}\|^2}\right),$$

hence by the equivalent definition of the sub-Gaussian norm, we obtain

$$\|\|\boldsymbol{B}\boldsymbol{X}\|_2 - \|\boldsymbol{B}\|_F\|_{\psi_2} \leqslant CK^2\|\boldsymbol{B}\|.$$

14 [**Distance of a Random Vector to a Subspace**] Let $E \subset \mathbb{R}^n$ satisfy $\dim(E) = d$, and let $\boldsymbol{X} \in \mathbb{R}^n$ be a centered isotropic random vector, that is, $\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \boldsymbol{I}_n$, whose components X_i are independent sub-Gaussian random variables. Denote $\max_i \|X_i\|_{\psi_2} = K$.

(1) Prove: $(\mathbb{E}[\operatorname{dist}(\boldsymbol{X}, E)^2])^{1/2} = \sqrt{n-d}$.

(2) Using Exercise Q6.13 (*Anisotropic Concentration of the Norm*), prove: For any $t \geqslant 0$,

$$\mathbb{P}\{|\operatorname{dist}(\boldsymbol{X}, E) - \sqrt{n-d}| > t\} \leqslant 2\exp\left(-\frac{ct^2}{K^4}\right).$$

Prf. (1) Let $\boldsymbol{P}$ be the orthogonal projection matrix onto the orthogonal complement $E^\perp$, then

$$\operatorname{dist}(\boldsymbol{X}, E)^2 = \|\boldsymbol{P}\boldsymbol{X}\|_2^2 = (\boldsymbol{P}\boldsymbol{X})^\top(\boldsymbol{P}\boldsymbol{X}) = \boldsymbol{X}^\top\boldsymbol{P}^\top\boldsymbol{P}\boldsymbol{X} = \boldsymbol{X}^\top\boldsymbol{P}\boldsymbol{X},$$

Hence

$$\mathbb{E}[\operatorname{dist}(\boldsymbol{X}, E)^2] = \mathbb{E}[\boldsymbol{X}^\top\boldsymbol{P}\boldsymbol{X}] = \mathbb{E}[\operatorname{tr}(\boldsymbol{X}^\top\boldsymbol{P}\boldsymbol{X})] = \operatorname{tr}(\boldsymbol{P}\mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]) = \operatorname{tr}(\boldsymbol{P}) = \dim(E^\perp) = n - d,$$

i.e., $(\mathbb{E}[\operatorname{dist}(\boldsymbol{X}, E)^2])^{1/2} = \sqrt{n-d}$.

(2) If $d = n$, then $E = \mathbb{R}^n$ and both conclusions are immediate. Hence assume $d < n$. It is easy to see that

$$\|\boldsymbol{P}\|_F = \sqrt{\operatorname{tr}(\boldsymbol{P}^\top\boldsymbol{P})} = \sqrt{\operatorname{tr}(\boldsymbol{P})} = \sqrt{n-d}, \|\boldsymbol{P}\| = 1.$$

By Exercise Q6.13 (*Anisotropic Concentration of the Norm*), let $\boldsymbol{B} = \boldsymbol{P}$. At this point $\|\boldsymbol{P}\boldsymbol{X}\|_2 = \operatorname{dist}(\boldsymbol{X}, E)$, we obtain

$$\|\operatorname{dist}(\boldsymbol{X}, E) - \sqrt{n-d}\|_{\psi_2} \leqslant CK^2 \cdot 1 = CK^2.$$

Hence $Z = \operatorname{dist}(\boldsymbol{X}, E) - \sqrt{n-d}$ is a CK^2-sub-Gaussian random variable, and by the equivalent definition of sub-Gaussian random variables, we obtain

$$\mathbb{P}\{|\operatorname{dist}(\boldsymbol{X}, E) - \sqrt{n-d}| > t\} \leqslant 2\exp\left(-\frac{c_0t^2}{(CK^2)^2}\right) = 2\exp\left(-\frac{ct^2}{K^4}\right). \qquad \left(c = \frac{c_0}{C^2}\right)$$

15 [**Random Cut Graph**] Consider a graph with E edges. Randomly partition its vertices into two groups (*each vertex is independently assigned with probability* $1/2$). Denote the cut of this graph (*i.e., the number of edges crossing the two groups*) by

cut. As we saw in Proposition P3.6.3 (0.5-*approximation algorithm for maximum cut*), $\mathbb{E}[\text{cut}] = E/2$. Prove: For any $s \geqslant 1$,

$$\mathbb{P}\left\{\left|\text{cut} - \frac{E}{2}\right| \geqslant s\sqrt{E}\right\} \leqslant 2\mathrm{e}^{-cs}.$$

Prf. Define the random vector $\boldsymbol{X} \in \mathbb{R}^n$ where $X_i \sim$ Rademacher is the group to which vertex i is assigned, then X_i are sub-Gaussian random variables.

For any $e = \{i, j\} \in \mathcal{E}$, it is cut if and only if $X_i \neq X_j$, i.e., $X_i X_j = -1$. Construct the indicator function

$$\mathbf{1}_{\{e \text{ is cut}\}} = \frac{1 - X_i X_j}{2},$$

then

$$\text{cut} = \sum_{\{i,j\} \in \mathcal{E}} \frac{1 - X_i X_j}{2} = \frac{E}{2} - \frac{1}{2} \sum_{\{i,j\} \in \mathcal{E}} X_i X_j.$$

Let $\boldsymbol{A}$ be the adjacency matrix of this graph, i.e.,

$$a_{ij} = \begin{cases} 1, \{i, j\} \in \mathcal{E}, \\ 0, \{i, j\} \notin \mathcal{E} \end{cases}$$

then

$$\boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X} = \sum_{i,j} a_{ij} X_i X_j = 2 \sum_{\{i,j\} \in \mathcal{E}} X_i X_j,$$

Hence

$$\text{cut} - \frac{E}{2} = -\frac{1}{4} \boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X},$$

(*Note that* $\mathbb{E}[\boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X}] = \operatorname{tr}(\boldsymbol{A}) = 0$, *which is consistent with the known condition* $\mathbb{E}[\text{cut}] = E/2$.)

We then only need to estimate

$$\mathbb{P}\left\{\frac{1}{4}|\boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X}| \geqslant s\sqrt{E}\right\} = \mathbb{P}\{|\boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X}| \geqslant 4s\sqrt{E}\}.$$

Using Theorem T6.2.2 (*Hanson-Wright inequality*) (*Hanson-Wright inequality*), we obtain for any $t \geqslant 0$

$$\mathbb{P}\{|\boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X} - \mathbb{E}[\boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X}]| \geqslant t\} \leqslant 2 \exp\left(-c_0 \min\left(\frac{t^2}{\|\boldsymbol{A}\|_F^2}, \frac{t}{\|\boldsymbol{A}\|}\right)\right),$$

Let $t = 4s\sqrt{E}$, then

$$\frac{t^2}{\|\boldsymbol{A}\|_F^2} = \frac{16 s^2 E}{2E} = 8s^2,$$
$$\frac{t}{\|\boldsymbol{A}\|} \geqslant \frac{t}{\|\boldsymbol{A}\|_F} = \frac{4s\sqrt{E}}{\sqrt{2E}} = 2\sqrt{2}s,$$

Hence

$$\mathbb{P}\{|\boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X} - \mathbb{E}[\boldsymbol{X}^\top \boldsymbol{A} \boldsymbol{X}]| \geqslant t\} \leqslant 2 \exp\left(-c_0 \min(8s^2, 2\sqrt{2}s)\right),$$

i.e.,

$$\mathbb{P}\left\{\left|\text{cut} - \frac{E}{2}\right| \geqslant s\sqrt{E}\right\} \leqslant 2 \exp(-c_0 \cdot 2\sqrt{2}s) = 2\mathrm{e}^{-cs}. \qquad (c = 2\sqrt{2}c_0)$$

16 [**Constructing Symmetric Distributions**] Prove Lemma L6.3.1 (*Constructing symmetric distributions*).

Let X be a random variable and $\xi \sim$ Rademacher satisfy $X \perp \xi$.

(1) $\xi X \overset{d}{=} \xi |X|$, and both are symmetric distributions.

(2) If X is symmetric, then $\xi X, \xi |X| \overset{d}{=} X$.

(3) If X' is an independent copy of X, then $X - X'$ is symmetric.

Prf. (1) • Prove symmetry

For any $A \in \mathscr{B}(\mathbb{R})$, using the law of total probability, we obtain

$$\begin{aligned}\mathbb{P}\{\xi X \in A\} &= \mathbb{P}\{\xi X \in A \mid \xi = 1\}\mathbb{P}\{\xi = 1\} + \mathbb{P}\{\xi X \in A \mid \xi = -1\}\mathbb{P}\{\xi = -1\} \\ &= \frac{1}{2}\mathbb{P}\{X \in A\} + \frac{1}{2}\mathbb{P}\{-X \in A\}, \\ \mathbb{P}\{-\xi X \in A\} &= \mathbb{P}\{-\xi X \in A \mid \xi = 1\}\mathbb{P}\{\xi = 1\} + \mathbb{P}\{-\xi X \in A \mid \xi = -1\}\mathbb{P}\{\xi = -1\} \\ &= \frac{1}{2}\mathbb{P}\{X \in A\} + \frac{1}{2}\mathbb{P}\{-X \in A\},\end{aligned}$$

Hence $\mathbb{P}\{\xi X \in A\} = \mathbb{P}\{-\xi X \in A\}$, i.e., ξX is symmetric.

Similarly, replacing X with $|X|$, we obtain

$$\mathbb{P}\{\xi|X| \in A\} = \frac{1}{2}\mathbb{P}\{|X| \in A\} + \frac{1}{2}\mathbb{P}\{-|X| \in A\} = \mathbb{P}\{-\xi|X| \in A\},$$

Hence $\xi|X|$ is symmetric.

- Prove equality in distribution

  For any $A \in \mathscr{B}(\mathbb{R})$, splitting X into its positive, negative, and zero parts gives

$$\begin{aligned}\mathbb{P}\{\xi X \in A\} &= \frac{1}{2}\mathbb{P}\{X \in A\} + \frac{1}{2}\mathbb{P}\{-X \in A\} \\ &= \frac{1}{2}\mathbb{P}\{|X| \in A\} + \frac{1}{2}\mathbb{P}\{-|X| \in A\} = \mathbb{P}\{\xi|X| \in A\}.\end{aligned}$$

  That is, $\xi X \overset{d}{=} \xi|X|$.

(2) Assume X is symmetric, then for any $A \in \mathscr{B}(\mathbb{R})$, $\mathbb{P}\{-X \in A\} = \mathbb{P}\{X \in A\}$, hence

$$\begin{aligned}\mathbb{P}\{\xi X \in A\} &= \frac{1}{2}\mathbb{P}\{X \in A\} + \frac{1}{2}\mathbb{P}\{-X \in A\} \\ &= \frac{1}{2}\mathbb{P}\{X \in A\} + \frac{1}{2}\mathbb{P}\{X \in A\} = \mathbb{P}\{X \in A\},\end{aligned}$$

i.e., $\xi X \overset{d}{=} X$, and by combining $\xi X \overset{d}{=} \xi|X|$, we obtain $\xi|X| \overset{d}{=} X$.

(3) Note that $(X, X') \overset{d}{=} (X', X)$. Construct the function $g(u, v) = u - v$, then

$$X - X' = g(X, X') \overset{d}{=} g(X', X) = X' - X = -(X - X'),$$

Hence $X - X'$ is symmetric.

17 [**Symmetrization of Bernoulli and Exponential Distributions**] Recall Lemma L6.3.1 (*Constructing symmetric distributions*) (3): if X' is an independent copy of X, then $X - X'$ is symmetric.

Now consider two typical cases and derive the distribution of $X - X'$ when X follows a Bernoulli distribution or an exponential distribution, respectively.

(1) Let $X \sim \text{Bernoulli}(p)$. Find the PMF of $X - X'$.

(2) Let $X \sim E(1)$. Prove $X - X' \sim \text{Laplace}(0, 1)$.

Prf. (1) Let $Y = X - X'$, then $\text{supp}(Y) = \{-1, 0, 1\}$.

a. When $Y = 1$,

$$\mathbb{P}\{Y = 1\} = \mathbb{P}\{X = 1, X' = 0\} = \mathbb{P}\{X = 1\}\mathbb{P}\{X' = 0\} = p(1 - p).$$

b. When $Y = -1$,

$$\mathbb{P}\{Y = -1\} = \mathbb{P}\{X = 0, X' = 1\} = \mathbb{P}\{X = 0\}\mathbb{P}\{X' = 1\} = (1 - p)p,$$

c. When $Y = 0$,

$$\mathbb{P}\{Y = 0\} = \mathbb{P}\{X = 1, X' = 1\} + \mathbb{P}\{X = 0, X' = 0\} = p \cdot p + (1 - p) \cdot (1 - p) = p^2 + (1 - p)^2.$$

Thus,

$$\mathbb{P}\{X-X'=y\}=\begin{cases}p(1-p), & y=-1,1\\ p^2+(1-p)^2, & y=0\\ 0, & \text{otherwise}\end{cases}.$$

It is easy to see that $\mathbb{P}\{Y=-1\}=\mathbb{P}\{Y=1\}$, hence $Y=X-X'$ is symmetric about 0.

(2) Let $W=-X'$, then

$$f_X(x)=\begin{cases}\mathrm{e}^{-x}, & x\geqslant 0,\\ 0, & x<0\end{cases}, f_W(w)=f_{X'}(-w)=\begin{cases}\mathrm{e}^{w}, & w\leqslant 0,\\ 0, & w>0\end{cases}.$$

Let $Y=X-X'=X+W$, then

$$f_Y(y)=\int_{-\infty}^{\infty}f_X(x)f_W(y-x)\mathrm{d}x.$$

If the density is nonzero, then

$$\begin{aligned}f_X(x)>0 &\implies x\geqslant 0,\\ f_W(y-x)>0 &\implies y-x\leqslant 0 \implies x\geqslant y,\end{aligned}$$

i.e., $x\geqslant\max(0,y)$.

a. If $y\geqslant 0$, then

$$\begin{aligned}f_Y(y)&=\int_y^{\infty}\mathrm{e}^{-x}\mathrm{e}^{y-x}\mathrm{d}x=\mathrm{e}^y\int_y^{\infty}\mathrm{e}^{-2x}\mathrm{d}x=\mathrm{e}^y\left[-\frac{1}{2}\mathrm{e}^{-2x}\right]_y^{\infty}\\ &=\mathrm{e}^y\left(0-\left(-\frac{1}{2}\mathrm{e}^{-2y}\right)\right)=\frac{1}{2}\mathrm{e}^{-y}.\end{aligned}$$

b. If $y<0$,

$$\begin{aligned}f_Y(y)&=\int_0^{\infty}\mathrm{e}^{-x}\mathrm{e}^{y-x}\mathrm{d}x=\mathrm{e}^y\int_0^{\infty}\mathrm{e}^{-2x}\mathrm{d}x=\mathrm{e}^y\left[-\frac{1}{2}\mathrm{e}^{-2x}\right]_0^{\infty}\\ &=\mathrm{e}^y\left(0-\left(-\frac{1}{2}\mathrm{e}^{0}\right)\right)=\frac{1}{2}\mathrm{e}^{y},\end{aligned}$$

Thus,

$$f_Y(y)=\frac{1}{2}\mathrm{e}^{-|y|},$$

i.e., $Y\sim\text{Laplace}(0,1)$.

18 [**Norm of a Translated Random Vector**] Let $\boldsymbol{X}$ be a symmetric random vector in some normed space and $\boldsymbol{v}$ be a fixed vector in this space. Prove:

$$\mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|\asymp\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|.$$

Prf. To prove $\mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|\asymp\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|$, it suffices to prove

$$c_1(\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|)\leqslant\mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|\leqslant c_2(\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|). \qquad (c_1,c_2>0)$$

- Prove the upper bound

  (*Symmetry is not needed*) Using the triangle inequality, we obtain

  $$\|\boldsymbol{X}+\boldsymbol{v}\|\leqslant\|\boldsymbol{X}\|+\|\boldsymbol{v}\|,$$

  Taking expectations on both sides,

  $$\mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|\leqslant\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|=\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|,$$

  At this point let $c_2=1$.

- Prove the lower bound

For $2\boldsymbol{v}$, using the triangle inequality, we obtain

$$\|2\boldsymbol{v}\| = \|\boldsymbol{v}+\boldsymbol{v}+\boldsymbol{X}-\boldsymbol{X}\| = \|(\boldsymbol{v}+\boldsymbol{X})+(\boldsymbol{v}-\boldsymbol{X})\| \leqslant \|\boldsymbol{v}+\boldsymbol{X}\|+\|\boldsymbol{v}-\boldsymbol{X}\|,$$

Taking expectations on both sides,

$$2\|\boldsymbol{v}\| \leqslant \mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|+\mathbb{E}\|\boldsymbol{v}-\boldsymbol{X}\|.$$

Since $\boldsymbol{X} \overset{d}{=} -\boldsymbol{X}$, we have $\boldsymbol{v}-\boldsymbol{X} \overset{d}{=} \boldsymbol{v}+\boldsymbol{X}$, hence $\mathbb{E}\|\boldsymbol{v}-\boldsymbol{X}\| = \mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|$. Substituting this in, we obtain

$$2\|\boldsymbol{v}\| \leqslant 2\mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\| \implies \|\boldsymbol{v}\| \leqslant \mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|. \tag{I}$$

Similarly, for $2\boldsymbol{X}$,

$$\begin{aligned}
&\|2\boldsymbol{X}\| = \|\boldsymbol{X}+\boldsymbol{X}+\boldsymbol{v}-\boldsymbol{v}\| = \|(\boldsymbol{X}+\boldsymbol{v})+(\boldsymbol{X}-\boldsymbol{v})\| \leqslant \|\boldsymbol{X}+\boldsymbol{v}\|+\|\boldsymbol{X}-\boldsymbol{v}\| \\
&\implies 2\mathbb{E}\|\boldsymbol{X}\| \leqslant \mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|+\mathbb{E}\|\boldsymbol{X}-\boldsymbol{v}\| \\
&\implies 2\mathbb{E}\|\boldsymbol{X}\| \leqslant 2\mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\| \implies \mathbb{E}\|\boldsymbol{X}\| \leqslant \mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|. \qquad \text{(II)}
\end{aligned}$$

Combining (I) and (II), we obtain

$$\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\| \leqslant \mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|+\mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\| = 2\mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\| \implies \frac{1}{2}(\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|) \leqslant \mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\|,$$

At this point $c_1 = 1/2$.

Thus,

$$\frac{1}{2}(\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|) \leqslant \mathbb{E}\|\boldsymbol{X}+\boldsymbol{v}\| \leqslant 1\cdot(\mathbb{E}\|\boldsymbol{X}\|+\|\boldsymbol{v}\|).$$

19 [**Symmetrization for Non-Zero Means**]

(1) Prove the generalized form of Lemma L6.3.2 (*Symmetrization*) for random vectors $\boldsymbol{X}_i$ that may not have zero mean:

$$\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i - \sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i]\right\| \leqslant 2\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|.$$

where $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher.

(2) Prove: There does not exist any nontrivial reverse inequality.

Prf. (1) Let $\boldsymbol{X}_1', \cdots, \boldsymbol{X}_N'$ be an independent copy of $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$. Write $\mathbb{E}[\boldsymbol{X}_i]$ as $\mathbb{E}_{\boldsymbol{X}'}[\boldsymbol{X}_i']$, then the sum of centered random variables is

$$\sum_{i=1}^N \boldsymbol{X}_i - \sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i] = \sum_{i=1}^N \boldsymbol{X}_i - \mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^N \boldsymbol{X}_i'\right] = \mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^N (\boldsymbol{X}_i - \boldsymbol{X}_i')\right],$$

Taking the norm on both sides and then taking expectation with respect to the outer variable $\boldsymbol{X}$,

$$\mathbb{E}_{\boldsymbol{X}}\left\|\sum_{i=1}^N \boldsymbol{X}_i - \sum_{i=1}^N \mathbb{E}[\boldsymbol{X}_i]\right\| = \mathbb{E}_{\boldsymbol{X}}\left\|\mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^N (\boldsymbol{X}_i - \boldsymbol{X}_i')\right]\right\|. \tag{I}$$

For the right-hand side, using Jensen's inequality, we obtain

$$\mathbb{E}_{\boldsymbol{X}}\left\|\mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^N (\boldsymbol{X}_i - \boldsymbol{X}_i')\right]\right\| \leqslant \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left\|\sum_{i=1}^N (\boldsymbol{X}_i - \boldsymbol{X}_i')\right\|,$$

where

$$\begin{aligned}
\mathbb{E}\left\|\sum_{i=1}^N (\boldsymbol{X}_i - \boldsymbol{X}_i')\right\| &= \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i(\boldsymbol{X}_i - \boldsymbol{X}_i')\right\| && (\boldsymbol{X}_i - \boldsymbol{X}_i' \overset{d}{=} \varepsilon_i(\boldsymbol{X}_i - \boldsymbol{X}_i')) \\
&\leqslant \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\| + \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i'\right\| \\
&= 2\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|,
\end{aligned}$$

Hence

$$\mathbb{E}_{\boldsymbol{X}}\left\|\mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^{N}(\boldsymbol{X}_i-\boldsymbol{X}_i')\right]\right\|\leqslant 2\mathbb{E}\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{X}_i\right\|. \tag{II}$$

Combining (I) and (II), we obtain

$$\mathbb{E}\left\|\sum_{i=1}^{N}\boldsymbol{X}_i-\sum_{i=1}^{N}\mathbb{E}[\boldsymbol{X}_i]\right\|\leqslant 2\mathbb{E}\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{X}_i\right\|.$$

(2) To prove that

$$c\cdot\mathbb{E}\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{X}_i\right\|\leqslant\mathbb{E}\left\|\sum_{i=1}^{N}\boldsymbol{X}_i-\sum_{i=1}^{N}\mathbb{E}[\boldsymbol{X}_i]\right\| \qquad (c>0)$$

does not exist, we only need to construct a counterexample.

Let $\boldsymbol{X}_i\equiv\boldsymbol{v}$ be a non-zero degenerate vector with $\|\boldsymbol{v}\|>0$, then

$$\begin{aligned}\text{RHS}&=\mathbb{E}\left\|\sum_{i=1}^{N}\boldsymbol{X}_i-\sum_{i=1}^{N}\mathbb{E}[\boldsymbol{X}_i]\right\|=\mathbb{E}\|N\boldsymbol{v}-N\boldsymbol{v}\|=0,\\ \text{LHS}&=\mathbb{E}\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{X}_i\right\|=\mathbb{E}\left\|\left(\sum_{i=1}^{N}\varepsilon_i\right)\boldsymbol{v}\right\|=\mathbb{E}\left|\sum_{i=1}^{N}\varepsilon_i\right|\cdot\|\boldsymbol{v}\|,\end{aligned}$$

where $\sum_{i=1}^{N}\varepsilon_i$ is a one-dimensional symmetric random walk, satisfying $\mathbb{E}|\sum_{i=1}^{N}\varepsilon_i|\asymp\sqrt{N}$, hence

$$\mathbb{E}\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{X}_i\right\|\asymp\sqrt{N}\|\boldsymbol{v}\|>0,$$

which is a contradiction. This shows that the lower bound of Lemma L6.3.2 (*Symmetrization*) depends on zero means; if the zero-mean restriction is ignored, only the upper bound can be preserved, while the lower bound fails.

20 [**Symmetrization with Convex Functions**] Prove the generalized form of Lemma L6.3.2 (*Symmetrization*). Let $F:\mathbb{R}^+\to\mathbb{R}$ be a monotonically increasing convex function, $\boldsymbol{X}_1,\cdots,\boldsymbol{X}_N$ be independent zero-mean random vectors, and $\varepsilon_1,\cdots,\varepsilon_N$ i.i.d. $\sim$ Rademacher. Prove: If $\|\cdot\|$ is replaced by $F(\|\cdot\|)$, the inequality of Lemma L6.3.2 (*Symmetrization*) still holds, i.e.,

$$\mathbb{E}\left[F\left(\frac{1}{2}\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{X}_i\right\|\right)\right]\leqslant\mathbb{E}\left[F\left(\left\|\sum_{i=1}^{N}\boldsymbol{X}_i\right\|\right)\right]\leqslant\mathbb{E}\left[F\left(2\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{X}_i\right\|\right)\right].$$

Prf. For any $t\in[0,1]$,

$$G(t\boldsymbol{x}+(1-t)\boldsymbol{y})=F(\|t\boldsymbol{x}+(1-t)\boldsymbol{y}\|)\leqslant F(t\|\boldsymbol{x}\|+(1-t)\|\boldsymbol{y}\|)\leqslant tF(\|\boldsymbol{x}\|)+(1-t)F(\|\boldsymbol{y}\|)=tG(\boldsymbol{x})+(1-t)G(\boldsymbol{y}),$$

hence G is convex.

- Proof of the upper bound

Consider $\sum_{i=1}^{N}\boldsymbol{X}_i$ and perform the reverse construction,

$$\sum_{i=1}^{N}\boldsymbol{X}_i=\sum_{i=1}^{N}\boldsymbol{X}_i-\mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^{N}\boldsymbol{X}_i'\right]=\mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^{N}(\boldsymbol{X}_i-\boldsymbol{X}_i')\right],$$

Using Jensen's inequality, we obtain

$$F\left(\left\|\sum_{i=1}^{N}\boldsymbol{X}_i\right\|\right)=F\left(\left\|\mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^{N}(\boldsymbol{X}_i-\boldsymbol{X}_i')\right]\right\|\right)\leqslant\mathbb{E}_{\boldsymbol{X}'}\left[F\left(\left\|\sum_{i=1}^{N}(\boldsymbol{X}_i-\boldsymbol{X}_i')\right\|\right)\right],$$

Taking expectation with respect to $\boldsymbol{X}$ on both sides,

$$\mathbb{E}_{\boldsymbol{X}}\left[F\left(\left\|\sum_{i=1}^{N}\boldsymbol{X}_i\right\|\right)\right]\leqslant\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left[F\left(\left\|\sum_{i=1}^{N}(\boldsymbol{X}_i-\boldsymbol{X}_i')\right\|\right)\right].$$

For the right-hand side, substitute $\boldsymbol{X}_i - \boldsymbol{X}_i' \overset{d}{=} \varepsilon_i(\boldsymbol{X}_i - \boldsymbol{X}_i')$, using the triangle inequality, we obtain

$$\begin{aligned}\text{RHS} &= \mathbb{E}\left[F\left(\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i - \sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i'\right\|\right)\right] \\ &\leqslant \mathbb{E}\left[F\left(\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\| + \left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i'\right\|\right)\right],\end{aligned}$$

Using the convexity inequality

$$F(a+b) = F\left(\frac{1}{2}(2a) + \frac{1}{2}(2b)\right) \leqslant \frac{1}{2}F(2a) + \frac{1}{2}F(2b),$$

we obtain

$$\begin{aligned}\text{RHS} &\leqslant \frac{1}{2}\mathbb{E}\left[F\left(2\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|\right)\right] + \frac{1}{2}\mathbb{E}\left[F\left(2\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i'\right\|\right)\right] \\ &\leqslant \mathbb{E}\left[F\left(2\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|\right)\right],\end{aligned}$$

Hence

$$\mathbb{E}\left[F\left(\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|\right)\right] \leqslant \mathbb{E}\left[F\left(2\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|\right)\right].$$

- Proof of the lower bound

  Consider $\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i$ and perform the reverse construction,

$$\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i = \sum_{i=1}^N \varepsilon_i(\boldsymbol{X}_i - \mathbb{E}_{\boldsymbol{X}'}[\boldsymbol{X}_i']) = \mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^N \varepsilon_i(\boldsymbol{X}_i - \boldsymbol{X}_i')\right],$$

  Using Jensen's inequality, we obtain

$$F\left(\frac{1}{2}\left\|\mathbb{E}_{\boldsymbol{X}'}\left[\sum_{i=1}^N \varepsilon_i(\boldsymbol{X}_i - \boldsymbol{X}_i')\right]\right\|\right) \leqslant \mathbb{E}_{\boldsymbol{X}'}\left[F\left(\frac{1}{2}\left\|\sum_{i=1}^N \varepsilon_i(\boldsymbol{X}_i - \boldsymbol{X}_i')\right\|\right)\right],$$

  Taking expectation with respect to $\boldsymbol{X}, \varepsilon$ on both sides,

$$\mathbb{E}\left[F\left(\frac{1}{2}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|\right)\right] \leqslant \mathbb{E}\left[F\left(\frac{1}{2}\left\|\sum_{i=1}^N \varepsilon_i(\boldsymbol{X}_i - \boldsymbol{X}_i')\right\|\right)\right].$$

  For the right-hand side, substitute $\varepsilon_i(\boldsymbol{X}_i - \boldsymbol{X}_i') \overset{d}{=} \boldsymbol{X}_i - \boldsymbol{X}_i'$, using the triangle inequality, we obtain

$$\begin{aligned}\text{RHS} &= \mathbb{E}\left[F\left(\frac{1}{2}\left\|\sum_{i=1}^N (\boldsymbol{X}_i - \boldsymbol{X}_i')\right\|\right)\right] \\ &\leqslant \mathbb{E}\left[F\left(\frac{1}{2}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\| + \frac{1}{2}\left\|\sum_{i=1}^N \boldsymbol{X}_i'\right\|\right)\right],\end{aligned}$$

  Using the convexity inequality

$$F(a+b) = F\left(\frac{1}{2}(2a) + \frac{1}{2}(2b)\right) \leqslant \frac{1}{2}F(2a) + \frac{1}{2}F(2b),$$

  we obtain

$$\text{RHS} \leqslant \frac{1}{2}\mathbb{E}\left[F\left(\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|\right)\right] + \frac{1}{2}\mathbb{E}\left[F\left(\left\|\sum_{i=1}^N \boldsymbol{X}_i'\right\|\right)\right] = \mathbb{E}\left[F\left(\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|\right)\right],$$

  Hence

$$\mathbb{E}\left[F\left(\frac{1}{2}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|\right)\right] \leqslant \mathbb{E}\left[F\left(\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|\right)\right].$$

21 [**Symmetrization of sub-Gaussian Sums**] Let $X_1, \cdots, X_N$ be independent zero-mean random variables. Prove: $\sum_{i=1}^N X_i$ is a sub-Gaussian random variable if and only if its symmetrized version $\sum_{i=1}^N \varepsilon_i X_i$ is a sub-Gaussian random variable, and in this case

$$c\left\|\sum_{i=1}^N \varepsilon_i X_i\right\|_{\psi_2} \leqslant \left\|\sum_{i=1}^N X_i\right\|_{\psi_2} \leqslant C\left\|\sum_{i=1}^N \varepsilon_i X_i\right\|_{\psi_2}.$$

Prf. Denote $S = \sum_{i=1}^N X_i$, $S_\varepsilon = \sum_{i=1}^N \varepsilon_i X_i$. We only need to prove

$$c\|S_\varepsilon\|_{\psi_2} \leqslant \|S\|_{\psi_2} \leqslant C\|S_\varepsilon\|_{\psi_2}.$$

(Recall the standard definition of the sub-Gaussian norm

$$\|Z\|_{\psi_2} = \inf\left\{K > 0 : \mathbb{E}\left[\exp\left(\frac{|Z|^2}{K^2}\right)\right] \leqslant 2\right\},$$

hence construct $F(x) = \exp(x^2/t^2)$.)

- Proof of the upper bound

  If S_ε is a sub-Gaussian random variable with $\|S_\varepsilon\|_{\psi_2} = K < \infty$, then

  $$\mathbb{E}\left[\exp\left(\frac{|S_\varepsilon|^2}{K^2}\right)\right] \leqslant 2.$$

  By Exercise Q6.20 (*Symmetrization with Convex Functions*), let $F(x) = \exp(x^2/t^2)$, we obtain

  $$\mathbb{E}\left[\exp\left(\frac{|S|^2}{t^2}\right)\right] \leqslant \mathbb{E}\left[\exp\left(\frac{(2|S_\varepsilon|)^2}{t^2}\right)\right] = \mathbb{E}\left[\exp\left(\frac{4|S_\varepsilon|^2}{t^2}\right)\right],$$

  Then set $\frac{4}{t^2} = \frac{1}{K^2}$, i.e., $t = 2K$, we obtain

  $$\mathbb{E}\left[\exp\left(\frac{|S|^2}{(2K)^2}\right)\right] \leqslant \mathbb{E}\left[\exp\left(\frac{|S_\varepsilon|^2}{K^2}\right)\right] \leqslant 2,$$

  Hence

  $$\|S\|_{\psi_2} \leqslant 2K = 2\|S_\varepsilon\|_{\psi_2},$$

  At this point $C = 2$.

- Proof of the lower bound

  If S is a sub-Gaussian random variable with $\|S\|_{\psi_2} = M < \infty$, then

  $$\mathbb{E}\left[\exp\left(\frac{|S|^2}{M^2}\right)\right] \leqslant 2.$$

  By Q6.20 (*Symmetrization with Convex Functions*), let $F(x) = \exp(x^2/t^2)$, we obtain

  $$\mathbb{E}\left[\exp\left(\frac{(|S_\varepsilon|/2)^2}{t^2}\right)\right] \leqslant \mathbb{E}\left[\exp\left(\frac{|S|^2}{t^2}\right)\right],$$

  Then set $t = M$, we obtain

  $$\mathbb{E}\left[\exp\left(\frac{|S_\varepsilon|^2}{(2M)^2}\right)\right] \leqslant \mathbb{E}\left[\exp\left(\frac{|S|^2}{M^2}\right)\right] \leqslant 2,$$

  Hence

  $$\|S_\varepsilon\|_{\psi_2} \leqslant 2M = 2\|S\|_{\psi_2} \implies \frac{1}{2}\|S_\varepsilon\|_{\psi_2} \leqslant \|S\|_{\psi_2},$$

  At this point $c = 1/2$.

22 [**Self-Normalized Sums**] Let $X_1, \cdots, X_N$ be independent symmetric random variables. Prove: For any $t > 0$,

$$\mathbb{P}\left\{\left|\sum_{i=1}^N X_i\right| > t\left(\sum_{i=1}^N X_i^2\right)^{1/2}\right\} \leqslant 2\mathrm{e}^{-t^2/2}.$$

Prf. Introduce $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher independent of X_i, then $X_i \overset{d}{=} \varepsilon_i |X_i|$.

Denote $A_i = |X_i|$, then

$$\text{LHS} = \mathbb{P}\left\{ \left| \sum_{i=1}^N \varepsilon_i A_i \right| > t \left(\sum_{i=1}^N (\varepsilon_i A_i)^2 \right)^{1/2} \right\} = \mathbb{P}\left\{ \left| \sum_{i=1}^N \varepsilon_i A_i \right| > t \left(\sum_{i=1}^N A_i^2 \right)^{1/2} \right\}.$$

(Since X_i may not have any moments, directly bounding the above probability is infeasible. However, since ε_i and A_i are independent, we can adopt a conditioning strategy.)

Let $\mathscr{F} = \sigma(A_1, \cdots, A_N)$. Using Theorem T2.2.1 (*Hoeffding inequality*) (*Hoeffding inequality*), we obtain for any $u > 0$,

$$\mathbb{P}\left\{ \left| \sum_{i=1}^N \varepsilon_i A_i \right| \geqslant u \,\middle|\, \mathscr{F} \right\} \leqslant 2 \exp\left(-\frac{u^2}{2 \sum_{i=1}^N A_i^2} \right),$$

If $\sum_{i=1}^N A_i^2 = 0$, the strict event is empty. Under the condition $\sum_{i=1}^N A_i^2 > 0$, substituting the threshold $u = t(\sum_{i=1}^N A_i^2)^{1/2}$ gives

$$\mathbb{P}\left\{ \left| \sum_{i=1}^N \varepsilon_i A_i \right| > t \left(\sum_{i=1}^N A_i^2 \right)^{1/2} \,\middle|\, \mathscr{F} \right\} \leqslant 2 \exp\left(-\frac{t^2}{2} \right),$$

Deconditioning on both sides,

$$\mathbb{E}_{\mathscr{F}} \left[\mathbb{P}\left\{ \left| \sum_{i=1}^N \varepsilon_i A_i \right| > t \left(\sum_{i=1}^N A_i^2 \right)^{1/2} \,\middle|\, \mathscr{F} \right\} \right] \leqslant \mathbb{E}_{\mathscr{F}}[2\mathrm{e}^{-t^2/2}].$$

That is,

$$\mathbb{P}\left\{ \left| \sum_{i=1}^N X_i \right| > t \left(\sum_{i=1}^N X_i^2 \right)^{1/2} \right\} \leqslant 2\mathrm{e}^{-t^2/2}.$$

23 [**Type p Property**] Provide a more macroscopic perspective: For a normed space X, if there exists $K > 0$ such that for any $N \in \mathbb{Z}^+$, $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N \in X$, it holds that

$$\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i \right\|^p \leqslant K^p \sum_{i=1}^N \|\boldsymbol{x}_i\|^p.$$

Then the space is said to satisfy the Type p property. In Exercise Q6.24 (*Type 2 Property*), Exercise Q6.25 (*Approximate ℓ^p-Carathéodory Theorem*), we will prove: ℓ^p space satisfies the Type p property if and only if $p \in [1, 2]$; satisfies the Type 2 property if and only if $p \in [2, \infty)$.

Let $p \in [1, 2]$.

(1) Let $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N \in \mathbb{R}^n$ be fixed vectors. Prove:

$$\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i \right\|_p^p \leqslant \sum_{i=1}^N \|\boldsymbol{x}_i\|_p^p.$$

(2) Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N \in \mathbb{R}^n$ be independent mean-zero random vectors. Prove:

$$\mathbb{E} \left\| \sum_{i=1}^N \boldsymbol{X}_i \right\|_p^p \leqslant \sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i\|_p^p.$$

(3) Give an example showing that when $p > 2$, the above conclusion does not necessarily hold.

Sol. (1) Let $\boldsymbol{x}_i = (x_{i1}, \cdots, x_{in}) \in \mathbb{R}^n$, then

$$\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i \right\|_p^p = \mathbb{E}\left[\sum_{j=1}^n \left| \sum_{i=1}^N \varepsilon_i x_{ij} \right|^p \right] = \sum_{j=1}^n \mathbb{E} \left| \sum_{i=1}^N \varepsilon_i x_{ij} \right|^p.$$

Using Khintchine's inequality, then using the inequality $(\sum a_i)^r \leqslant \sum a_i^r$ $(0 \leqslant r \leqslant 1)$, we obtain

$$\mathbb{E}\left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^p \leqslant \left(\mathbb{E}\left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^2\right)^{p/2} = \left(\sum_{i=1}^N x_{ij}^2\right)^{p/2}$$
$$\leqslant \sum_{i=1}^N (x_{ij}^2)^{p/2} = \sum_{i=1}^N |x_{ij}|^p,$$

Substituting this in, we obtain

$$\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|_p^p \leqslant \sum_{i=1}^N \|\boldsymbol{x}_i\|_p^p.$$

(2) **Solution 1**

By Exercise Q6.20 (*Symmetrization with Convex Functions*), let $F(t) = t^p$, we obtain

$$\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_p^p \leqslant \mathbb{E}\left[2^p \left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|_p^p\right] = 2^p \mathbb{E}_{\boldsymbol{X}}\left[\mathbb{E}_\varepsilon \left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|_p^p\right],$$

Using the conclusion of (1), we obtain

$$\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_p^p \leqslant 2^p \sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i\|_p^p.$$

Solution 2 Symmetrization technique

- If $\boldsymbol{X}_i$ is symmetrically distributed, the coefficient can be 1
  If $\boldsymbol{X}_i \overset{d}{=} -\boldsymbol{X}_i$, then $\boldsymbol{X}_i \overset{d}{=} \varepsilon_i \boldsymbol{X}_i$.
  Using the conclusion of (1), we obtain

$$\mathbb{E}_\varepsilon \left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|_p^p \leqslant \sum_{i=1}^N \|\boldsymbol{X}_i\|_p^p,$$

  Taking expectation with respect to $\boldsymbol{X}$ on both sides, we obtain

$$\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_p^p = \mathbb{E}_{\boldsymbol{X}}\left[\mathbb{E}_\varepsilon \left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|_p^p\right] \leqslant \mathbb{E}_{\boldsymbol{X}}\left[\sum_{i=1}^N \|\boldsymbol{X}_i\|_p^p\right] = \sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i\|_p^p.$$

- If only the zero-mean assumption holds, we can only obtain a coefficient of 2^p
  Introduce independent copies $\boldsymbol{X}_i'$, we obtain

$$\mathbb{E}_{\boldsymbol{X}}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_p^p = \mathbb{E}_{\boldsymbol{X}}\left\|\sum_{i=1}^N (\boldsymbol{X}_i - \mathbb{E}[\boldsymbol{X}_i'])\right\|_p^p \leqslant \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left\|\sum_{i=1}^N (\boldsymbol{X}_i - \boldsymbol{X}_i')\right\|_p^p.$$

  For the right-hand side, since $\boldsymbol{X}_i - \boldsymbol{X}_i'$ is symmetrically distributed,

$$\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left\|\sum_{i=1}^N (\boldsymbol{X}_i - \boldsymbol{X}_i')\right\|_p^p \leqslant \sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i - \boldsymbol{X}_i'\|_p^p,$$

  where using the inequality $(a+b)^p \leqslant 2^{p-1}(a^p + b^p)$, we obtain

$$\mathbb{E}\|\boldsymbol{X}_i - \boldsymbol{X}_i'\|_p^p \leqslant 2^{p-1}\mathbb{E}[(\|\boldsymbol{X}_i\|_p + \|\boldsymbol{X}_i'\|_p)^p] \leqslant 2^{p-1}(2\mathbb{E}\|\boldsymbol{X}_i\|_p^p) = 2^p \mathbb{E}\|\boldsymbol{X}_i\|_p^p,$$

  Substituting this in, we obtain

$$\sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i - \boldsymbol{X}_i'\|_p^p \leqslant 2^p \sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i\|_p^p,$$

Hence

$$\mathbb{E}\left\|\sum_{i=1}^{N}\boldsymbol{X}_i\right\|_p^p \leqslant 2^p\sum_{i=1}^{N}\mathbb{E}\|\boldsymbol{X}_i\|_p^p.$$

(3) Take the scalar space $X=\mathbb{R}$ and set $x_1=\cdots=x_N=1$. If the Type p inequality held, then

$$\mathbb{E}\left|\sum_{i=1}^{N}\varepsilon_i\right|^p \leqslant K^pN.$$

On the other hand, by monotonicity of L^p norms,

$$\mathbb{E}\left|\sum_{i=1}^{N}\varepsilon_i\right|^p \geqslant \left(\mathbb{E}\left|\sum_{i=1}^{N}\varepsilon_i\right|^2\right)^{p/2} = N^{p/2}.$$

Hence $N^{p/2-1}\leqslant K^p$ for every N, which is impossible as $N\to\infty$ when $p>2$.

Rk. von Bahr-Esseen's 1965 paper Inequalities for the rth Absolute Moment of a Sum of Random Variables, $1\leqslant r\leqslant 2$ points out that for general zero-mean variables, the optimal coefficient is 2; if and only if the random variables are symmetric, the optimal coefficient is 1.

- Proof of the symmetric case *(coefficient = 1)*

  When $p=2$, by orthogonality, directly expanding the square yields constant 1; when $p=1$, using the triangle inequality yields constant 1; we only need to consider the nontrivial case $p\in(1,2)$.

  - First consider the scalar case

    We make the following lemma:

    Lemma For $p\in(0,2)$,

    $$|x|^p = C_p\int_0^\infty \frac{1-\cos(tx)}{t^{p+1}}\mathrm{d}t. \qquad \left(C_p=\left(\int_0^\infty\frac{1-\cos u}{u^{p+1}}\mathrm{d}u\right)^{-1}\right)$$

    Let $Y_1,\cdots,Y_N$ be independent symmetric random variables. By the lemma, we obtain

    $$\mathbb{E}\left|\sum_{k=1}^{N}Y_k\right|^p = C_p\int_0^\infty\frac{1-\mathbb{E}\left[\cos(t\sum_{k=1}^{N}Y_k)\right]}{t^{p+1}}\mathrm{d}t.$$

    Note that

    $$\mathbb{E}\left[\cos\left(t\sum_{k=1}^{N}Y_k\right)\right] = \Re\left(\mathbb{E}\left[\exp\left(\mathrm{i}t\sum_{k=1}^{N}Y_k\right)\right]\right) = \Re\left(\prod_{k=1}^{N}\mathbb{E}[\mathrm{e}^{\mathrm{i}tY_k}]\right),$$

    Since Y_k is symmetric, the expectation of the imaginary part of the characteristic function is $\mathbb{E}[\sin(tY_k)]=0$, so

    $$\varphi_k(t)=\mathbb{E}[\cos(tY_k)],$$

    Thus

    $$1-\mathbb{E}\left[\cos\left(t\sum_{k=1}^{N}Y_k\right)\right] = 1-\prod_{k=1}^{N}\varphi_k(t),$$

    (We need to bound $1-\prod_{k=1}^{N}\varphi_k(t)$.)

    Using the inequality

    $$1-\prod_{k=1}^{N}u_k\leqslant\sum_{k=1}^{N}(1-u_k), \qquad (u_k\leqslant 1)$$

    Let $u_k=\varphi_k(t)$, we obtain

    $$\mathbb{E}\left|\sum_{k=1}^{N}Y_k\right|^p \leqslant C_p\int_0^\infty\frac{\sum_{k=1}^{N}(1-\mathbb{E}[\cos(tY_k)])}{t^{p+1}}\mathrm{d}t$$

$$= \sum_{k=1}^{N} \left(C_p \int_0^{\infty} \frac{1 - \mathbb{E}[\cos(tY_k)]}{t^{p+1}} \mathrm{d}t \right) = \sum_{k=1}^{N} \mathbb{E}|Y_k|^p.$$

- Then generalize to ℓ_p norm

  Let $\boldsymbol{X}_i = (X_{i1}, \cdots, X_{in}) \in \mathbb{R}^n$, then

$$\mathbb{E}\left\| \sum_{i=1}^{N} \boldsymbol{X}_i \right\|_p^p = \mathbb{E}\left[\sum_{j=1}^{n} \left| \sum_{i=1}^{N} X_{ij} \right|^p \right] = \sum_{j=1}^{n} \mathbb{E}\left| \sum_{i=1}^{N} X_{ij} \right|^p.$$

  For the right-hand side, using the conclusion of (1), we obtain

$$\sum_{j=1}^{n} \mathbb{E}\left| \sum_{i=1}^{N} X_{ij} \right|^p \leqslant \sum_{j=1}^{n} \left(\sum_{i=1}^{N} \mathbb{E}|X_{ij}|^p \right) = \sum_{i=1}^{N} \mathbb{E}\left[\sum_{j=1}^{n} |X_{ij}|^p \right] = \sum_{i=1}^{N} \mathbb{E}\|\boldsymbol{X}_i\|_p^p,$$

  Hence

$$\mathbb{E}\left\| \sum_{i=1}^{N} \boldsymbol{X}_i \right\|_p^p = \mathbb{E}\left[\sum_{j=1}^{n} \left| \sum_{i=1}^{N} X_{ij} \right|^p \right] = \sum_{i=1}^{N} \mathbb{E}\|\boldsymbol{X}_i\|_p^p.$$

- Proof of the general case *(coefficient = 2)*

  We make the following lemma:

  Lemma For $p \in [1, 2]$,

$$|x + y|^p \leqslant |x|^p + p|x|^{p-1}\mathrm{sgn}(x)y + 2|y|^p.$$

  Let X, Y be independent random variables with $\mathbb{E}[Y] = 0$. By the lemma, we obtain

$$|X + Y|^p \leqslant |X|^p + p|X|^{p-1}\mathrm{sgn}(X)Y + 2|Y|^p,$$

  Taking conditional expectation with respect to X on both sides,

$$\mathbb{E}[|X + Y|^p \mid X] \leqslant |X|^p + p|X|^{p-1}\mathrm{sgn}(X)\mathbb{E}[Y] + 2\mathbb{E}[|Y|^p] = |X|^p + 2\mathbb{E}[|Y|^p],$$

  Generalize this to N terms. Let $Y_1, \cdots, Y_N$ be independent zero-mean random variables, and set $X = \sum_{i=1}^{k-1} Y_i$, $Y = Y_k$. We obtain

$$\mathbb{E}\left| \sum_{i=1}^{k} Y_i \right|^p \leqslant \mathbb{E}\left| \sum_{i=1}^{k-1} Y_i \right|^p + 2\mathbb{E}|Y_k|^p,$$

  By mathematical induction, we obtain

$$\mathbb{E}\left| \sum_{i=1}^{N} Y_i \right|^p \leqslant \mathbb{E}|Y_1|^p + 2\mathbb{E}|Y_2|^p + \cdots + 2\mathbb{E}|Y_N|^p \leqslant 2\sum_{i=1}^{N} \mathbb{E}|Y_i|^p.$$

  Then generalize this to the ℓ_p norm. Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ be independent zero-mean random vectors. Then

$$\mathbb{E}\left\| \sum_{i=1}^{N} \boldsymbol{X}_i \right\|_p^p = \mathbb{E}\left[\sum_{j=1}^{n} \left| \sum_{i=1}^{N} X_{ij} \right|^p \right] = \sum_{j=1}^{n} \mathbb{E}\left| \sum_{i=1}^{N} X_{ij} \right|^p.$$

  For the right-hand side, note that the coordinate components $X_{1j}, \cdots, X_{Nj}$ are independent zero-mean random variables, hence

$$\sum_{j=1}^{n} \mathbb{E}\left| \sum_{i=1}^{N} X_{ij} \right|^p \leqslant \sum_{j=1}^{n} \left(2\sum_{i=1}^{N} \mathbb{E}|X_{ij}|^p \right) = 2\sum_{i=1}^{N} \mathbb{E}\left[\sum_{j=1}^{n} |X_{ij}|^p \right] = 2\sum_{i=1}^{N} \mathbb{E}\|\boldsymbol{X}_i\|_p^p,$$

  Substituting this in, we obtain

$$\mathbb{E}\left\| \sum_{i=1}^{N} \boldsymbol{X}_i \right\|_p^p \leqslant 2\sum_{i=1}^{N} \mathbb{E}\|\boldsymbol{X}_i\|_p^p.$$

Rk. **Burkholder-Davis-Gundy (BDG) Inequality** Define the quadratic variation as the sum of squared differences

$$[S]_N = \sum_{i=1}^N \Delta_i^2 = \sum_{i=1}^N Y_i^2,$$

Then for any $p > 0$,

$$\mathbb{E}|S_N|^p \leqslant C_p \mathbb{E}[[S]_N^{p/2}]. \qquad (C_p > 0)$$

Using the BDG inequality, we can prove: For any $p > 0$,

$$\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_p^p \lesssim \sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i\|_p^p.$$

Note that

$$\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_p^p = \mathbb{E}\left[\sum_{j=1}^n \left|\sum_{i=1}^N X_{ij}\right|^p\right] = \sum_{j=1}^n \mathbb{E}\left|\sum_{i=1}^N X_{ij}\right|^p, \qquad (\text{I})$$

Hence we only need to prove that for any dimension j,

$$\mathbb{E}\left|\sum_{i=1}^N X_{ij}\right|^p \lesssim \sum_{i=1}^N \mathbb{E}|X_{ij}|^p.$$

Let $Y_i = X_{ij}$, then $Y_1, \cdots, Y_N$ are independent zero-mean random variables.
Define the partial sum $S_k = \sum_{i=1}^k Y_i$ and the natural filtration $\mathscr{F}_k = \sigma(Y_1, \cdots, Y_k)$, then $\mathbb{E}[Y_k \mid \mathscr{F}_{k-1}] = \mathbb{E}[Y_k] = 0$, hence $\{S_k\}_{k=1}^N$ is a discrete-time martingale with martingale differences $\Delta_k = S_k - S_{k-1} = Y_k$.
Using the BDG inequality, we obtain

$$\mathbb{E}\left|\sum_{i=1}^N Y_i\right|^p \leqslant C_p \mathbb{E}\left[\left(\sum_{i=1}^N Y_i^2\right)^{p/2}\right], \qquad (\text{II})$$

Note that $f(t) = t^{p/2}$ is a concave function, satisfying subadditivity

$$\left(\sum_{i=1}^N a_i\right)^{p/2} \leqslant \sum_{i=1}^N a_i^{p/2}, \qquad (a_i \geqslant 0)$$

Let $a_i = Y_i^2 \geqslant 0$, we obtain

$$\left(\sum_{i=1}^N Y_i^2\right)^{p/2} \leqslant \sum_{i=1}^N (Y_i^2)^{p/2} = \sum_{i=1}^N |Y_i|^p, \qquad (\text{III})$$

Substituting this into (II), we obtain

$$\mathbb{E}\left|\sum_{i=1}^N Y_i\right|^p \leqslant C_p \mathbb{E}\left[\sum_{i=1}^N |Y_i|^p\right] = C_p \sum_{i=1}^N \mathbb{E}|Y_i|^p,$$

Substituting this into (I), we obtain

$$\begin{aligned}\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_p^p &= \sum_{j=1}^n \mathbb{E}\left|\sum_{i=1}^N X_{ij}\right|^p \leqslant C_p \sum_{j=1}^n \sum_{i=1}^N \mathbb{E}|X_{ij}|^p \\ &= C_p \sum_{i=1}^N \left(\sum_{j=1}^n \mathbb{E}|X_{ij}|^p\right) = C_p \sum_{i=1}^N \mathbb{E}\left[\sum_{j=1}^n |X_{ij}|^p\right] = C_p \sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i\|_p^p.\end{aligned}$$

24 [**Type 2 Property**] Let $p \in [2, \infty)$.[1]

(1) Let $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N \in \mathbb{R}^n$ be arbitrary fixed vectors. Prove:

$$\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|_p^2 \leqslant Cp \sum_{i=1}^N \|\boldsymbol{x}_i\|_p^2. \qquad (C>0)$$

(2) Generalize (1). Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N \in \mathbb{R}^n$ be independent zero-mean random vectors. Prove:

$$\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|_p^2 \leqslant Cp \sum_{i=1}^N \mathbb{E}\|\boldsymbol{X}_i\|_p^2. \qquad (C>0)$$

(3) Give an example showing that when $p<2$, the above conclusion does not necessarily hold.

Prf. (1) Let $\boldsymbol{x}_i = (x_{i1}, \cdots, x_{in}) \in \mathbb{R}^n$, then

$$\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|_p^2 = \mathbb{E}\left[\left(\sum_{j=1}^n \left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^p\right)^{2/p}\right].$$

- Apply Jensen's inequality

  Note that $2/p \leqslant 1$, the function $t \mapsto t^{2/p}$ is concave on $t \geqslant 0$, hence by Jensen's inequality, moving the expectation inside the concave function,

$$\mathbb{E}\left[\left(\sum_{j=1}^n \left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^p\right)^{2/p}\right] \leqslant \left(\mathbb{E}\left[\sum_{j=1}^n \left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^p\right]\right)^{2/p} = \left(\sum_{j=1}^n \mathbb{E}\left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^p\right)^{2/p}. \qquad (\mathrm{I})$$

- Apply Khintchine's inequality

  For the inner scalar Rademacher sum, using Khintchine's inequality, we obtain

$$\left(\mathbb{E}\left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^p\right)^{1/p} \leqslant c_0\sqrt{p}\left(\sum_{i=1}^N x_{ij}^2\right)^{1/2},$$

  Raising both sides to the pth power,

$$\mathbb{E}\left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^p \leqslant (c_0\sqrt{p})^p \left(\sum_{i=1}^N x_{ij}^2\right)^{p/2},$$

  Substituting this into (I), we obtain

$$\begin{aligned}\left(\sum_{j=1}^n \mathbb{E}\left|\sum_{i=1}^N \varepsilon_i x_{ij}\right|^p\right)^{2/p} &\leqslant \left(\sum_{j=1}^n (c_0\sqrt{p})^p \left(\sum_{i=1}^N x_{ij}^2\right)^{p/2}\right)^{2/p} \\ &=: Cp\left(\sum_{j=1}^n \left(\sum_{i=1}^N x_{ij}^2\right)^{p/2}\right)^{2/p}. \qquad (C=c_0^2)(\mathrm{II})\end{aligned}$$

- Apply Minkowski's inequality *(generalized triangle inequality)*

  Define the n-dimensional sequence $\boldsymbol{v}_i = (x_{i1}^2, x_{i2}^2, \cdots, x_{in}^2)$.

  (Note that $(\sum_{j=1}^n |v_j|^{p/2})^{2/p}$ is precisely the $L_{p/2}$ norm of $\boldsymbol{v}_i$, where $p/2 \geqslant 1$, so $L_{p/2}$ space forms a normed space.)

  Using Minkowski's inequality, we obtain

$$\left\|\sum_{i=1}^N \boldsymbol{v}_i\right\|_{p/2} \leqslant \sum_{i=1}^N \|\boldsymbol{v}_i\|_{p/2},$$

[1]The original stem called p a parameter without requiring $p \geqslant 2$. It is changed to $p \in [2, \infty)$. If $1 \leqslant p < 2$ and $\boldsymbol{x}_i = \boldsymbol{e}_i \in \ell_p^N$, the left-hand side is $N^{2/p}$ and the right-hand side is CpN; the inequality fails for sufficiently large N.

That is,

$$\left(\sum_{j=1}^{n}\left(\sum_{i=1}^{N}x_{ij}^2\right)^{p/2}\right)^{2/p}\leqslant\sum_{i=1}^{N}\left(\sum_{j=1}^{n}(x_{ij}^2)^{p/2}\right)^{2/p}=\sum_{i=1}^{N}\left(\sum_{j=1}^{n}|x_{ij}|^p\right)^{2/p}=\sum_{i=1}^{N}\|\boldsymbol{x}_i\|_p^2,$$

Substituting this into (II), we obtain

$$\mathbb{E}\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{x}_i\right\|_p^2\leqslant Cp\sum_{i=1}^{N}\|\boldsymbol{x}_i\|_p^2.$$

(2) *(Since the norm $\|\cdot\|_p$ is convex, its square $\|\cdot\|_p^2$ is also convex. We can directly apply the symmetrization lemma.)* Introduce independent copies $\boldsymbol{X}_i'$ of $\boldsymbol{X}_i$ with $\mathbb{E}[\boldsymbol{X}_i']=\boldsymbol{0}$. Using Jensen's inequality, we obtain

$$\mathbb{E}_{\boldsymbol{X}}\left\|\sum_{i=1}^{N}\boldsymbol{X}_i\right\|_p^2=\mathbb{E}_{\boldsymbol{X}}\left\|\sum_{i=1}^{N}(\boldsymbol{X}_i-\mathbb{E}[\boldsymbol{X}_i'])\right\|_p^2\leqslant\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left\|\sum_{i=1}^{N}(\boldsymbol{X}_i-\boldsymbol{X}_i')\right\|_p^2.$$

For the right-hand side, introduce $\varepsilon_1,\cdots,\varepsilon_N$ i.i.d. $\sim$ Rademacher, we obtain

$$\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}'}\left\|\sum_{i=1}^{N}(\boldsymbol{X}_i-\boldsymbol{X}_i')\right\|_p^2=\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}',\varepsilon}\left\|\sum_{i=1}^{N}\varepsilon_i(\boldsymbol{X}_i-\boldsymbol{X}_i')\right\|_p^2.$$

Condition on fixed $\boldsymbol{X},\boldsymbol{X}'$, treating $(\boldsymbol{X}_i-\boldsymbol{X}_i')$ as fixed vectors. Using the conclusion of (1), we obtain

$$\mathbb{E}_{\varepsilon}\left\|\sum_{i=1}^{N}\varepsilon_i(\boldsymbol{X}_i-\boldsymbol{X}_i')\right\|_p^2\leqslant Cp\sum_{i=1}^{N}\|\boldsymbol{X}_i-\boldsymbol{X}_i'\|_p^2,$$

where using the triangle inequality and $(a+b)^2\leqslant 2(a^2+b^2)$, we obtain

$$\mathbb{E}\|\boldsymbol{X}_i-\boldsymbol{X}_i'\|_p^2\leqslant 2\mathbb{E}\|\boldsymbol{X}_i\|_p^2+2\mathbb{E}\|\boldsymbol{X}_i'\|_p^2=4\mathbb{E}\|\boldsymbol{X}_i\|_p^2,$$

Substituting this in, we obtain

$$\mathbb{E}_{\boldsymbol{X},\boldsymbol{X}',\varepsilon}\left\|\sum_{i=1}^{N}\varepsilon_i(\boldsymbol{X}_i-\boldsymbol{X}_i')\right\|_p^2\leqslant C'p\sum_{i=1}^{N}\mathbb{E}\|\boldsymbol{X}_i\|_p^2,\qquad(C'=4C)$$

Hence

$$\mathbb{E}\left\|\sum_{i=1}^{N}\boldsymbol{X}_i\right\|_p^2\leqslant C'p\sum_{i=1}^{N}\mathbb{E}\|\boldsymbol{X}_i\|_p^2.$$

(3) Let $n=N$, $\boldsymbol{x}_i=\boldsymbol{e}_i$, then for the right-hand side,

$$\text{RHS}=Cp\sum_{i=1}^{N}\|\boldsymbol{e}_i\|_p^2=Cp\sum_{i=1}^{N}1^2=Cp\cdot N;$$

For the left-hand side, since

$$\left\|\sum_{i=1}^{N}\varepsilon_i\boldsymbol{e}_i\right\|_p=\left(\sum_{i=1}^{N}|\varepsilon_i|^p\right)^{1/p}=\left(\sum_{i=1}^{N}1\right)^{1/p}=N^{1/p},$$

Hence

$$\text{LHS}=\mathbb{E}[(N^{1/p})^2]=N^{2/p}.$$

Thus,

$$N^{\frac{2}{p}-1}\leqslant Cp,$$

When $N\to\infty$, the above inequality does not hold.

25 [**Approximate ℓ^p-Carathéodory Theorem**] State and prove the generalization of Theorem T0.1.2 (*Approximate*

Caratheodory theorem) to the ℓ^p norm, covering all $p \in [1, \infty)$.

Rk. **Approximate ℓ^p-Carathéodory Theorem** Let $\mathbb{T} \subset \mathbb{B}_p^n$, then for any $\boldsymbol{x} \in \operatorname{conv}(\mathbb{T})$, there exist $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_k \in \mathbb{T}$ *(repetition allowed)* such that:

1. When $p \in [1, 2]$,

$$\left\| \boldsymbol{x} - \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{x}_j \right\|_p \leqslant \frac{4}{k^{1-1/p}}.$$

2. When $p \in [2, \infty)$,

$$\left\| \boldsymbol{x} - \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{x}_j \right\|_p \leqslant C\sqrt{\frac{p}{k}}.$$

Prf. • Construct a probabilistic model

For $\boldsymbol{x} \in \operatorname{conv}(\mathbb{T})$, let

$$\boldsymbol{x} = \sum_{i=1}^{m} \lambda_i \boldsymbol{z}_i,$$

where $\boldsymbol{z}_i \in \mathbb{T}$, $\lambda_i \geqslant 0$, $\sum_{i=1}^m \lambda_i = 1$.

Treat the convex combination coefficients λ_i as probabilities, and define a random vector $\boldsymbol{Z}$ that takes the value $\boldsymbol{z}_i$ with probability λ_i, i.e.,

$$\mathbb{P}\{\boldsymbol{Z} = \boldsymbol{z}_i\} = \lambda_i,$$

then $\mathbb{E}[\boldsymbol{Z}] = \sum_{i=1}^m \lambda_i \boldsymbol{z}_i = \boldsymbol{x}$, $\|\boldsymbol{Z}\|_p \leqslant 1$ (a.s.).

Let $\boldsymbol{Z}_1, \cdots, \boldsymbol{Z}_k$ be independent copies of $\boldsymbol{Z}$, and define the centered random vectors

$$\boldsymbol{X}_j = \boldsymbol{Z}_j - \boldsymbol{x},$$

then $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_k$ are independent zero-mean random vectors whose norms satisfy

$$\|\boldsymbol{X}_j\|_p = \|\boldsymbol{Z}_j - \boldsymbol{x}\|_p \leqslant \|\boldsymbol{Z}_j\|_p + \|\boldsymbol{x}\|_p \leqslant 1 + 1 = 2.$$

• When $p \in [1, 2]$ *(using Type p property)*

Using the conclusion of Exercise Q6.24 (*Type 2 Property*) (2), we obtain

$$\mathbb{E}\left\| \sum_{j=1}^{k} \boldsymbol{X}_j \right\|_p^p \leqslant 2^p \sum_{j=1}^{k} \mathbb{E}\|\boldsymbol{X}_j\|_p^p \leqslant 2^p \sum_{j=1}^{k} 2^p = k4^p.$$

Hence the error of the empirical mean is

$$\mathbb{E}\left\| \boldsymbol{x} - \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{Z}_j \right\|_p^p = \mathbb{E}\left\| \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{X}_j \right\|_p^p = \frac{1}{k^p} \mathbb{E}\left\| \sum_{j=1}^{k} \boldsymbol{X}_j \right\|_p^p \leqslant \frac{k4^p}{k^p} = \frac{4^p}{k^{p-1}},$$

Thus by the probabilistic method *(if the expectation of a nonnegative random variable is at most V, then there must exist at least one sample point in the sample space such that the value of the random variable is at most V)*, there exists a concrete realization $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_k \in \mathbb{T}$ such that

$$\left\| \boldsymbol{x} - \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{x}_j \right\|_p^p \leqslant \frac{4^p}{k^{p-1}},$$

Raising both sides to the $1/p$ power,

$$\left\| \boldsymbol{x} - \frac{1}{k} \sum_{j=1}^{k} \boldsymbol{x}_j \right\|_p \leqslant \frac{4}{k^{1-1/p}}.$$

• When $p \in [2, \infty)$ *(using Type 2 property)*

Using the conclusion of Exercise Q6.25 (*Approximate ℓ^p-Carathéodory Theorem*) (2), we obtain

$$\mathbb{E}\left\|\sum_{j=1}^{k}\boldsymbol{X}_j\right\|_p^2 \leqslant C_0 p\sum_{j=1}^{k}\mathbb{E}\|\boldsymbol{X}_j\|_p^2 \leqslant C_0 p\sum_{j=1}^{k}4 = 4C_0 pk, \qquad (C_0>0)$$

Hence the error of the empirical mean is

$$\mathbb{E}\left\|\boldsymbol{x}-\frac{1}{k}\sum_{j=1}^{k}\boldsymbol{Z}_j\right\|_p^2 = \frac{1}{k^2}\mathbb{E}\left\|\sum_{j=1}^{k}\boldsymbol{X}_j\right\|_p^2 \leqslant \frac{4C_0pk}{k^2} = \frac{4C_0p}{k},$$

Thus there exists a concrete realization $\boldsymbol{x}_1,\cdots,\boldsymbol{x}_k\in\mathbb{T}$ such that

$$\left\|\boldsymbol{x}-\frac{1}{k}\sum_{j=1}^{k}\boldsymbol{x}_j\right\|_p^2 \leqslant \frac{4C_0p}{k},$$

Taking square roots on both sides,

$$\left\|\boldsymbol{x}-\frac{1}{k}\sum_{j=1}^{k}\boldsymbol{x}_j\right\|_p \leqslant C\sqrt{\frac{p}{k}}. \qquad (C=2\sqrt{C_0})$$

26 [**Marcinkiewicz-Zygmund Inequality**] Let $X_1,\cdots,X_N$ be independent mean-zero random variables, $p\in[2,\infty)$. Prove:

$$\left\|\sum_{i=1}^{N}X_i\right\|_{L^p}^2 \leqslant Cp\sum_{i=1}^{N}\|X_i\|_{L^p}^2. \qquad (C>0)$$

Prf. • Introduce symmetrization

Introduce $\varepsilon_1,\cdots,\varepsilon_N$ i.i.d. $\sim$ Rademacher. By Exercise Q6.20 (*Symmetrization with Convex Functions*), let $F(x)=x^p$, we obtain the L^p norm symmetrization principle

$$\left\|\sum_{i=1}^{N}X_i\right\|_{L^p} \leqslant 2\left\|\sum_{i=1}^{N}\varepsilon_iX_i\right\|_{L^p},$$

Squaring both sides,

$$\left\|\sum_{i=1}^{N}X_i\right\|_{L^p}^2 \leqslant 4\left\|\sum_{i=1}^{N}\varepsilon_iX_i\right\|_{L^p}^2, \qquad (\mathrm{I})$$

We only need to bound the right-hand side.

• Conditioning and Khintchine's inequality

For the right-hand side of (I), expand it,

$$\left\|\sum_{i=1}^{N}\varepsilon_iX_i\right\|_{L^p}^p = \mathbb{E}_{\boldsymbol{X}}\left[\mathbb{E}_{\varepsilon}\left|\sum_{i=1}^{N}\varepsilon_iX_i\right|^p\right],$$

Using Khintchine's inequality, we obtain

$$\left(\mathbb{E}_{\varepsilon}\left|\sum_{i=1}^{N}\varepsilon_iX_i\right|^p\right)^{1/p} \leqslant c_0\sqrt{p}\left(\sum_{i=1}^{N}X_i^2\right)^{1/2},$$

Raising both sides to the pth power and substituting,

$$\mathbb{E}_{\boldsymbol{X}}\left[\mathbb{E}_{\varepsilon}\left|\sum_{i=1}^{N}\varepsilon_iX_i\right|^p\right] \leqslant \mathbb{E}_{\boldsymbol{X}}\left[(c_0\sqrt{p})^p\left(\sum_{i=1}^{N}X_i^2\right)^{p/2}\right].$$

Raising both sides to the $2/p$ power, converting the large expectation back to the squared L^p norm form,

$$\left\|\sum_{i=1}^N \varepsilon_i X_i\right\|_{L^p}^2 \leqslant c_0^2 p \left(\mathbb{E}\left[\left(\sum_{i=1}^N X_i^2\right)^{p/2}\right]\right)^{2/p}. \tag{II}$$

- Apply Minkowski's inequality to simplify

  Note that the right-hand side of (II) is precisely the $L^{p/2}$ norm of $\sum_{i=1}^N X_i^2$, i.e.,

$$\left(\mathbb{E}\left[\left(\sum_{i=1}^N X_i^2\right)^{p/2}\right]\right)^{2/p} = \left\|\sum_{i=1}^N X_i^2\right\|_{L^{p/2}},$$

  Using Minkowski's inequality, we obtain

$$\left\|\sum_{i=1}^N X_i^2\right\|_{L^{p/2}} \leqslant \sum_{i=1}^N \|X_i^2\|_{L^{p/2}} = \sum_{i=1}^N \|X_i\|_{L^p}^2,$$

  Substituting this in,

$$\left(\mathbb{E}\left[\left(\sum_{i=1}^N X_i^2\right)^{p/2}\right]\right)^{2/p} \leqslant \sum_{i=1}^N \|X_i\|_{L^p}^2. \tag{III}$$

  Substituting (III) into (II) and then into (I), we obtain

$$\left\|\sum_{i=1}^N X_i\right\|_{L^p}^2 \leqslant 4\left\|\sum_{i=1}^N \varepsilon_i X_i\right\|_{L^p}^2 \leqslant 4\left(c_0^2 p \sum_{i=1}^N \|X_i\|_{L^p}^2\right),$$

  That is,

$$\left\|\sum_{i=1}^N X_i\right\|_{L^p}^2 \leqslant Cp \sum_{i=1}^N \|X_i\|_{L^p}^2. \qquad (C = 4c_0^2)$$

Rk. The proof logic of this problem is highly similar to Exercise Q6.25 (*Approximate ℓ^p-Carathéodory Theorem*) (2). The difference is that Exercise Q6.25 (*Approximate ℓ^p-Carathéodory Theorem*) deals with the ℓ^p norm of $\mathbb{R}^n$, while Exercise Q6.27 (*Norm Concentration under Finite Moment Conditions*) deals with the L^p norm of the probability space Ω. This shows that regardless of the space, as long as the power is at least 2 *(i.e., $p \geqslant 2$)*, the Type 2 property is naturally satisfied.

27 [**Norm Concentration under Finite Moment Conditions**] Let us prove a variant of Theorem T3.1.1 (*Concentration of the norm*) under weaker conditions that only require the existence of finite moments.

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a random vector whose entries X_i are mutually independent[2], satisfying $\mathbb{E}[X_i^2] = 1$, and there exist $p \geqslant 2, K \geqslant 0$ such that $\|X_i\|_{L^{2p}} \leqslant K$. Prove:

$$\big\|\|\boldsymbol{X}\|_2 - \sqrt{n}\big\|_{L^p} \leqslant C\sqrt{p}K^2. \qquad (C > 0)$$

Prf. First linearize the algebra,

$$|\|\boldsymbol{X}\|_2 - \sqrt{n}| = \frac{|\|\boldsymbol{X}\|_2^2 - n|}{\|\boldsymbol{X}\|_2 + \sqrt{n}} \leqslant \frac{|\|\boldsymbol{X}\|_2^2 - n|}{\sqrt{n}},$$

Taking the L^p norm on both sides,

$$\big\|\|\boldsymbol{X}\|_2 - \sqrt{n}\big\|_{L^p} \leqslant \frac{1}{\sqrt{n}}\big\|\|\boldsymbol{X}\|_2^2 - n\big\|_{L^p}. \tag{I}$$

[2]The original stem imposed the moment assumptions on the coordinates but did not require their independence. The independence condition is added. Otherwise, let every X_i equal one random variable Z with $\mathbb{P}\{Z = 0\} = \mathbb{P}\{Z = \sqrt{2}\} = 1/2$. Then $\mathbb{E}[Z^2] = 1$ and all moments are finite, but $\|\|\boldsymbol{X}\|_2 - \sqrt{n}\|_{L^p}$ is of order $\sqrt{n}$, so it cannot be controlled by a right-hand side depending only on p, K.

For the right-hand side, let $Y_i = X_i^2 - 1$ be independent zero-mean random variables, then

$$\|\boldsymbol{X}\|_2^2 - n = \sum_{i=1}^n X_i^2 - \sum_{i=1}^n \mathbb{E}[X_i^2] = \sum_{i=1}^n (X_i^2 - 1) = \sum_{i=1}^n Y_i,$$

The right-hand side of (I) is

$$\frac{1}{\sqrt{n}} \left\| \sum_{i=1}^n Y_i \right\|_{L^p}.$$

Using the Marcinkiewicz-Zygmund inequality, we obtain

$$\left\| \sum_{i=1}^n Y_i \right\|_{L^p}^2 \leqslant C_0 p \sum_{i=1}^n \|Y_i\|_{L^p}^2,$$

Taking square roots on both sides,

$$\left\| \sum_{i=1}^n Y_i \right\|_{L^p} \leqslant \sqrt{C_0 p} \left(\sum_{i=1}^n \|Y_i\|_{L^p}^2 \right)^{1/2}, \tag{II}$$

We only need to estimate $\|Y_i\|_{L^p}$.

Using Jensen's inequality, we obtain

$$1 = \mathbb{E}[X_i^2] = \|X_i\|_{L^2}^2 \leqslant \|X_i\|_{L^{2p}}^2 \leqslant K^2,$$

Solving gives $K \geqslant 1$.

Using the triangle inequality, we obtain

$$\|Y_i\|_{L^p} = \|X_i^2 - 1\|_{L^p} \leqslant \|X_i^2\|_{L^p} + \|1\|_{L^p} = \|X_i^2\|_{L^p} + 1 = \|X_i\|_{L^{2p}}^2 + 1 \leqslant K^2 + 1 \leqslant 2K^2,$$

Substituting this into (II), we obtain

$$\left\| \sum_{i=1}^n Y_i \right\|_{L^p} \leqslant \sqrt{C_0 p} \left(\sum_{i=1}^n (2K^2)^2 \right)^{1/2} = \sqrt{C_0 p} (n \cdot 4K^4)^{1/2} = \sqrt{C_0 p} \cdot 2K^2 \sqrt{n},$$

Substituting this into (I), we obtain

$$\left\| \|\boldsymbol{X}\|_2 - \sqrt{n} \right\|_{L^p} \leqslant \frac{1}{\sqrt{n}} (2\sqrt{C_0}\sqrt{p} K^2 \sqrt{n}) = C\sqrt{p} K^2. \qquad (C = 2\sqrt{C_0})$$

28 [**Norm of Rectangular Random Matrices**] Generalize Theorem T6.4.1 (*Norm of random matrices with non-i.i.d. entries*) to non-symmetric rectangular matrices.

Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a random matrix whose entries are independent and zero-mean. Prove: The expectation of its operator norm is approximately equal to the expectation of the maximum of the Euclidean norms of its row and column vectors *(up to a logarithmic factor)*, i.e.,

$$\mathbb{E}\left[\max\left\{ \max_{1 \leqslant i \leqslant m} \|\boldsymbol{A}_{i*}\|_2, \max_{1 \leqslant j \leqslant n} \|\boldsymbol{A}_{*j}\|_2 \right\} \right] \leqslant \mathbb{E}\|\boldsymbol{A}\| \leqslant C\sqrt{\log(m+n)}\, \mathbb{E}\left[\max\left\{ \max_{1 \leqslant i \leqslant m} \|\boldsymbol{A}_{i*}\|_2, \max_{1 \leqslant j \leqslant n} \|\boldsymbol{A}_{*j}\|_2 \right\} \right]. \qquad (C > 0)$$

Prf. • Proof of the lower bound

Let $\boldsymbol{e}_j \in \mathbb{R}^n$ be the jth standard basis vector, then $\boldsymbol{A}\boldsymbol{e}_j = \boldsymbol{A}_{*j}$, hence by the definition of the norm, we obtain

$$\|\boldsymbol{A}\| \geqslant \frac{\|\boldsymbol{A}\boldsymbol{e}_j\|_2}{\|\boldsymbol{e}_j\|_2} = \|\boldsymbol{A}_{*j}\|_2.$$

Similarly, consider $\boldsymbol{A}^\top$, let $\boldsymbol{u}_i \in \mathbb{R}^m$ be the ith standard basis vector, then $\boldsymbol{A}^\top \boldsymbol{u}_i = (\boldsymbol{A}_{i*})^\top$, hence

$$\|\boldsymbol{A}\| = \|\boldsymbol{A}^\top\| \geqslant \|\boldsymbol{A}^\top \boldsymbol{u}_i\|_2 = \|\boldsymbol{A}_{i*}\|_2.$$

Thus,

$$\|\boldsymbol{A}\| \geqslant \max\left\{ \max_{1 \leqslant i \leqslant m} \|\boldsymbol{A}_{i*}\|_2, \max_{1 \leqslant j \leqslant n} \|\boldsymbol{A}_{*j}\|_2 \right\},$$

Taking expectation on both sides,

$$\mathbb{E}\|\boldsymbol{A}\| \geqslant \mathbb{E}\left[\max\left\{\max_{1\leqslant i\leqslant m}\|\boldsymbol{A}_{i*}\|_2, \max_{1\leqslant j\leqslant n}\|\boldsymbol{A}_{*j}\|_2\right\}\right].$$

- Proof of the upper bound

  Construct a block matrix $\tilde{\boldsymbol{A}} \in \mathbb{R}^{(m+n)\times(m+n)}$ as

$$\tilde{\boldsymbol{A}} = \begin{pmatrix} \boldsymbol{0}_{m\times m} & \boldsymbol{A}_{m\times n} \\ \boldsymbol{A}_{n\times m}^{\top} & \boldsymbol{0}_{n\times n} \end{pmatrix},$$

  Then $\tilde{\boldsymbol{A}}$ is symmetric, its entries are independent and zero-mean, satisfying the conditions of Theorem T6.4.1 (*Norm of random matrices with non-i.i.d. entries*).

  Since

$$\tilde{\boldsymbol{A}}^2 = \begin{pmatrix} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^{\top} & \boldsymbol{0} \end{pmatrix}\begin{pmatrix} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^{\top} & \boldsymbol{0} \end{pmatrix} = \begin{pmatrix} \boldsymbol{A}\boldsymbol{A}^{\top} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{A}^{\top}\boldsymbol{A} \end{pmatrix},$$

  Hence

$$\|\tilde{\boldsymbol{A}}\|^2 = \|\tilde{\boldsymbol{A}}^2\| = \max(\|\boldsymbol{A}\boldsymbol{A}^{\top}\|, \|\boldsymbol{A}^{\top}\boldsymbol{A}\|) = \max(\|\boldsymbol{A}\|^2, \|\boldsymbol{A}\|^2) = \|\boldsymbol{A}\|^2,$$

  Solving gives $\|\tilde{\boldsymbol{A}}\| = \|\boldsymbol{A}\|$.

  Then consider $\tilde{\boldsymbol{A}}_{k*}$:

  a. When $1 \leqslant k \leqslant m$, $\tilde{\boldsymbol{A}}_{k*} = (\boldsymbol{0}, \boldsymbol{A}_{k*})$, whose Euclidean norm is $\|\boldsymbol{A}_{k*}\|_2$.

  b. When $m+1 \leqslant k \leqslant m+n$, let $j = k-m$, then $\tilde{\boldsymbol{A}}_{k*} = ((\boldsymbol{A}^{\top})_{j*}, \boldsymbol{0})$. Since the jth row of $\boldsymbol{A}^{\top}$ is precisely the jth column of $\boldsymbol{A}$, its Euclidean norm is $\|\boldsymbol{A}_{*j}\|_2$.

  Hence

$$\max_k \|\tilde{\boldsymbol{A}}_{k*}\|_2 = \max\left\{\max_{1\leqslant i\leqslant m}\|\boldsymbol{A}_{i*}\|_2, \max_{1\leqslant j\leqslant n}\|\boldsymbol{A}_{*j}\|_2\right\},$$

  Thus by Theorem T6.4.1 (*Norm of random matrices with non-i.i.d. entries*), letting $N = m+n$, we obtain

$$\mathbb{E}\|\tilde{\boldsymbol{A}}\| \leqslant C\sqrt{\log N}\cdot\mathbb{E}\left[\max_k \|\tilde{\boldsymbol{A}}_{k*}\|_2\right],$$

  That is,

$$\mathbb{E}\|\boldsymbol{A}\| \leqslant C\sqrt{\log(m+n)}\mathbb{E}\left[\max\left\{\max_{1\leqslant i\leqslant m}\|\boldsymbol{A}_{i*}\|_2, \max_{1\leqslant j\leqslant n}\|\boldsymbol{A}_{*j}\|_2\right\}\right].$$

29 [**The Logarithmic Factor Cannot Be Removed**] Prove that the logarithmic factor in Theorem T6.4.1 (*Norm of random matrices with non-i.i.d. entries*) cannot be completely removed in general. Construct a random matrix $\boldsymbol{A}$ satisfying the hypotheses of the theorem *(i.e., symmetric matrix, entries on and above the diagonal are independent and zero-mean)* such that

$$\mathbb{E}\|\boldsymbol{A}\| \geqslant c\log^{1/4}(n)\cdot\mathbb{E}\left[\max_i \|\boldsymbol{A}_{i*}\|_2\right].$$

Prf. *(We need to construct a matrix whose maximum row norm is very small (and concentrated), but whose operator norm becomes large due to the extreme value effect of multiple independent blocks.)*

Consider constructing a block diagonal matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$. Let n be a sufficiently large integer and choose the block size $m = \lfloor\sqrt{\log n}\rfloor$. Set $K = \lfloor n/m\rfloor$, use K diagonal blocks of size m, and fill the remaining coordinates with zeros; this changes no order estimate. Denote these diagonal blocks by

$$\boldsymbol{A} = \operatorname{diag}(\boldsymbol{D}_1, \boldsymbol{D}_2, \cdots, \boldsymbol{D}_K).$$

Let

$$(\boldsymbol{D}_k)_{ij} = \begin{cases} 0, & i = j, \\ \varepsilon_{ij}^{(k)}, & i < j, , \\ \varepsilon_{ji}^{(k)}, & i > j \end{cases}$$

where the variables $\varepsilon_{ij}^{(k)}$ are independent Rademacher variables across all $i < j$ and k, then $\boldsymbol{A}$ satisfies the hypotheses of

Theorem T6.4.1 (*Norm of random matrices with non-i.i.d. entries*).

- Compute the maximum row norm $\mathbb{E}[\max_i \|\boldsymbol{A}_{i*}\|_2]$

  It is easy to see that

$$\|\boldsymbol{A}_{i*}\|_2 = \sqrt{\sum_{j\neq i}(\boldsymbol{D}_k)_{ij}^2} = \sqrt{m-1}$$

  Hence

$$\mathbb{E}\left[\max_i \|\boldsymbol{A}_{i*}\|_2\right] = \sqrt{m-1} \leqslant \sqrt{m} \leqslant (\log n)^{1/4}.$$

- Compute the lower bound of the operator norm $\mathbb{E}\|\boldsymbol{A}\|$

  Let $r = m(m-1)/2$, and let $\mathcal{E}_k$ be the event that all upper-triangular off-diagonal entries of the kth block equal 1. Then

$$\mathbb{P}(\mathcal{E}_k) = 2^{-r}, \qquad \mathbb{P}\{\bigcup_{k=1}^{K} \mathcal{E}_k\} = 1-(1-2^{-r})^K.$$

  Since $m = \lfloor\sqrt{\log n}\rfloor$, we have $K2^{-r} \to \infty$, so the last probability is at least $1/2$ for sufficiently large n. On $\mathcal{E}_k$, $\boldsymbol{D}_k = 11^\top - \boldsymbol{I}$, and hence $\|\boldsymbol{D}_k\| = m-1$. Therefore

$$\mathbb{E}\|\boldsymbol{A}\| = \mathbb{E}\left[\max_{1\leqslant k\leqslant K} \|\boldsymbol{D}_k\|\right] \geqslant \frac{m-1}{2}.$$

  Thus,

$$\frac{\mathbb{E}\|\boldsymbol{A}\|}{\mathbb{E}[\max_i \|\boldsymbol{A}_{i*}\|_2]} \geqslant \frac{(m-1)/2}{\sqrt{m-1}} = \frac{1}{2}\sqrt{m-1} \gtrsim (\log n)^{1/4},$$

  That is,

$$\mathbb{E}\|\boldsymbol{A}\| \geqslant c\log^{1/4}(n)\cdot \mathbb{E}\left[\max_i \|\boldsymbol{A}_{i*}\|_2\right].$$

30 [**Row Norms of Random Matrices**] Let δ_{ij} i.i.d. $\sim B(p)$, where $i,j = 1,\cdots,n$, $np \geqslant \log n$. Prove:

$$\mathbb{E}\left[\max_{i\leqslant n}\sum_{j=1}^{n}(\delta_{ij}-p)^2\right] \leqslant Cpn. \qquad (C>0)$$

Prf. **Solution 1 Bernstein's inequality**

Let $Z_{ij} = (\delta_{ij}-p)^2$, then

$$Z_{ij} = \begin{cases}(1-p)^2, \text{ w.p. } p,\\ p^2, \qquad \text{w.p. } 1-p\end{cases},$$

Its numerical characteristics are

$$\mu = p(1-p)^2 + (1-p)p^2 = p(1-p),$$
$$\sigma^2 = [p(1-p)^4 + (1-p)p^4] - [p(1-p)]^2 = p(1-p)(1-2p)^2,$$

Hence $\sigma^2 \leqslant p$, $|Z_{ij}-\mu| \leqslant \max((1-p)^2, p^2) \leqslant 1$.

Let $S_i = \sum_{j=1}^n Z_{ij}$, then $\mathbb{E}[S_i] = np(1-p) \leqslant np$. Using Theorem T2.9.5 (*Bernstein inequality for bounded distributions*) (*Bernstein inequality for bounded distributions*), we obtain for any $t>0$,

$$\mathbb{P}\{|S_i - \mathbb{E}[S_i]| \geqslant t\} \leqslant 2\exp\left(-\frac{t^2/2}{n\sigma^2 + t/3}\right),$$

Taking the union bound on both sides,

$$\mathbb{P}\left\{\max_i |S_i - \mathbb{E}[S_i]| \geqslant t\right\} \leqslant n\cdot 2\exp\left(-\frac{t^2}{2(np+t/3)}\right).$$

By the integral definition of expectation $\mathbb{E}[Y] \leqslant y_0 + \int_{y_0}^{\infty}\mathbb{P}\{Y>t\}\mathrm{d}t$, let $Y = \max_i S_i$; note that $\max_i \mathbb{E}[S_i] = n\mu \leqslant np$,

let $t_0 = np$, then

$$\begin{aligned}\mathbb{E}\Big[\max_i S_i\Big] &\leqslant np + \int_{np}^{\infty} \mathbb{P}\Big\{\max_i S_i > np + t\Big\}\mathrm{d}t \\ &\leqslant np + \int_{np}^{\infty} \mathbb{P}\Big\{\max_i |S_i - \mathbb{E}[S_i]| > t\Big\}\mathrm{d}t \\ &\leqslant np + \int_0^{\infty} 2n\exp\Big(-\frac{t^2}{2np + 2t/3}\Big)\mathrm{d}t.\end{aligned}$$

Let $M = \max_{1\leqslant i\leqslant n}(S_i - \mathbb{E}[S_i])$, and consider the expectation of its nonnegative part,

$$\mathbb{E}[M_+] = \int_0^{\infty} \mathbb{P}\{M_+ \geqslant t\}\mathrm{d}t = \int_0^{\infty} \mathbb{P}\Big\{\max_{1\leqslant i\leqslant n}(S_i - \mathbb{E}[S_i]) \geqslant t\Big\}\mathrm{d}t,$$

Set the threshold $t_0 = 6pn$ and split the above integral,

$$\begin{aligned}\mathbb{E}[M_+] &\leqslant \int_0^{t_0} 1\mathrm{d}t + \int_{t_0}^{\infty} n\exp\Big(-\frac{t^2}{2np + 2t/3}\Big)\mathrm{d}t \\ &\leqslant \int_0^{t_0} 1\mathrm{d}t + \int_{t_0}^{\infty} n\exp\Big(-\frac{t^2}{2np + 2t/3}\Big)\mathrm{d}t \\ &\leqslant 6pn + \int_{6pn}^{\infty} n\exp\Big(-\frac{t}{3}\Big)\mathrm{d}t = 6pn + 3n\exp(-2pn),\end{aligned}$$

where

$$3n\exp(-2pn) = 3n\exp(-pn)\exp(-pn) \leqslant 3n\exp(-\log n)\exp(-pn) = 3\exp(-pn) \leqslant 3,$$

Substituting, we obtain $\mathbb{E}[M_+] \leqslant 6pn + 3$, i.e., $\mathbb{E}[M_+] \leqslant cpn$.

Using the triangle inequality, we obtain

$$\max_{1\leqslant i\leqslant n} S_i \leqslant \max_{1\leqslant i\leqslant n}(S_i - \mathbb{E}[S_i]) + \max_{1\leqslant i\leqslant n}\mathbb{E}[S_i] \leqslant M_+ + pn,$$

Taking expectation on both sides,

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n}\sum_{j=1}^{n}(\delta_{ij} - p)^2\right] \leqslant \mathbb{E}[M_+] + np \leqslant Cpn. \qquad (C = c + 1)$$

Solution 2 MGF bound

It is easy to see that

$$\begin{aligned}(\delta_{ij} - p)^2 &= \delta_{ij}^2 - 2p\delta_{ij} + p^2 \\ &= \delta_{ij} - 2p\delta_{ij} + p^2 = (1 - 2p)\delta_{ij} + p^2,\end{aligned}$$

Summing on both sides, let $S_i = \sum_{j=1}^{n}\delta_{ij} \sim B(n, p)$, we obtain

$$\begin{aligned}\sum_{j=1}^{n}(\delta_{ij} - p)^2 &= \sum_{j=1}^{n}((1 - 2p)\delta_{ij} + p^2) = (1 - 2p)\sum_{j=1}^{n}\delta_{ij} + np^2 \\ &= (1 - 2p)S_i + np^2 \leqslant S_i + np,\end{aligned}$$

Hence

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n}\sum_{j=1}^{n}(\delta_{ij} - p)^2\right] \leqslant \mathbb{E}\Big[\max_{1\leqslant i\leqslant n}(S_i + np)\Big] = \mathbb{E}\Big[\max_{1\leqslant i\leqslant n} S_i\Big] + np,$$

We only need to prove $\mathbb{E}[\max_i S_i] \leqslant C'pn$.

Using Jensen's inequality, let $F(x) = \exp(\lambda x)$ with $\lambda > 0$, we obtain

$$\exp\Big(\lambda\mathbb{E}\Big[\max_i S_i\Big]\Big) \leqslant \mathbb{E}\Big[\exp\Big(\lambda\max_i S_i\Big)\Big] = \mathbb{E}\Big[\max_i \mathrm{e}^{\lambda S_i}\Big] \leqslant \mathbb{E}\left[\sum_{i=1}^{n}\mathrm{e}^{\lambda S_i}\right] = n\cdot\mathbb{E}[\mathrm{e}^{\lambda S_1}],$$

Taking logarithms on both sides and dividing by λ, we obtain

$$\mathbb{E}\left[\max_i S_i\right] \leqslant \frac{\log n}{\lambda} + \frac{1}{\lambda}\log \mathbb{E}[\mathrm{e}^{\lambda S_1}],$$

where

$$\mathbb{E}[\mathrm{e}^{\lambda S_1}] = \prod_{j=1}^{n} \mathbb{E}[\mathrm{e}^{\lambda \delta_{1j}}] = ((1-p)+p\mathrm{e}^{\lambda})^n = (1+p(\mathrm{e}^{\lambda}-1))^n$$
$$\leqslant (\mathrm{e}^{p(\mathrm{e}^{\lambda}-1)})^n = \exp(np(\mathrm{e}^{\lambda}-1)),$$

Substituting this in, we obtain

$$\mathbb{E}\left[\max_i S_i\right] \leqslant \frac{\log n}{\lambda} + \frac{1}{\lambda}\log(\exp(np(\mathrm{e}^{\lambda}-1))) = \frac{\log n}{\lambda} + \frac{np(\mathrm{e}^{\lambda}-1)}{\lambda}$$
$$\leqslant \frac{pn}{\lambda} + \frac{np(\mathrm{e}^{\lambda}-1)}{\lambda} = \frac{\mathrm{e}^{\lambda}}{\lambda} pn.$$

Thus,

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} \sum_{j=1}^{n} (\delta_{ij}-p)^2\right] \leqslant \mathbb{E}\left[\max_{1\leqslant i\leqslant n} S_i\right] + np \leqslant \frac{\mathrm{e}^{\lambda}}{\lambda}\cdot pn + np = Cpn. \qquad \left(C = \frac{\mathrm{e}^{\lambda}}{\lambda} + 1\right)$$

31 [**Matrix Completion for Rectangular Matrices**] State and prove the generalization of Theorem T6.5.1 (*Matrix completion*) to general matrices.

Rk. **Rectangular Matrix Completion** Let $n_1, n_2 \geqslant 2$, let $\boldsymbol{A} \in \mathbb{R}^{n_1\times n_2}$ be a fixed matrix, and denote $r = \operatorname{rank}(\boldsymbol{A})$. Take $0 < p \leqslant 1$. Independently observe each entry position with probability p, let δ_{ij} i.i.d. $\sim B(p)$, and set $s = pn_1n_2$ to be the expected number of observations. Define the sparse observation matrix $\boldsymbol{Y} \in \mathbb{R}^{n_1\times n_2}$ by

$$Y_{ij} = \delta_{ij}a_{ij} = \begin{cases} a_{ij}, & \delta_{ij} = 1, \\ 0, & \delta_{ij} = 0 \end{cases}.$$

Let $\hat{\boldsymbol{A}}$ be the best rank-r approximation of the unbiased estimator matrix $p^{-1}\boldsymbol{Y}$. Denote $n = \max(n_1, n_2)$. If the expected number of observations satisfies $s \geqslant C' n\log n$, then

$$\mathbb{E}\left[\frac{1}{\sqrt{n_1n_2}}\|\hat{\boldsymbol{A}} - \boldsymbol{A}\|_F\right] \leqslant C\sqrt{\frac{rn\log n}{s}}\|\boldsymbol{A}\|_{\infty}.$$

Prf. By the Eckart-Young-Mirsky theorem, we obtain $\|\hat{\boldsymbol{A}} - p^{-1}\boldsymbol{Y}\| \leqslant \|\boldsymbol{A} - p^{-1}\boldsymbol{Y}\|$, hence

$$\|\hat{\boldsymbol{A}} - \boldsymbol{A}\| \leqslant \|\hat{\boldsymbol{A}} - p^{-1}\boldsymbol{Y}\| + \|p^{-1}\boldsymbol{Y} - \boldsymbol{A}\| \leqslant 2\|p^{-1}\boldsymbol{Y} - \boldsymbol{A}\| = \frac{2}{p}\|\boldsymbol{Y} - p\boldsymbol{A}\|,$$

Taking expectation on both sides,

$$\mathbb{E}\|\hat{\boldsymbol{A}} - \boldsymbol{A}\| \leqslant \frac{2}{p}\mathbb{E}\|\boldsymbol{Y} - p\boldsymbol{A}\|, \tag{I}$$

Let $\boldsymbol{Z} = \boldsymbol{Y} - p\boldsymbol{A}$. (*It then remains to bound* $\mathbb{E}\|\boldsymbol{Z}\|$.)

By Exercise Q6.28 (*Norm of Rectangular Random Matrices*), we obtain

$$\mathbb{E}\|\boldsymbol{Y} - p\boldsymbol{A}\| \leqslant C_1\sqrt{\log(n_1+n_2)}\left(\mathbb{E}\left[\max_{1\leqslant i\leqslant n_1}\|\boldsymbol{Z}_{i*}\|_2\right] + \mathbb{E}\left[\max_{1\leqslant j\leqslant n_2}\|\boldsymbol{Z}_{*j}\|_2\right]\right), \tag{II}$$

Then we only need to bound the row and column norms of $\boldsymbol{Z}$.

- Bounding the row and column norms

  For the row norm,

$$\|\boldsymbol{Z}_{i*}\|_2^2 = \sum_{j=1}^{n_2}(\delta_{ij}-p)^2A_{ij}^2 \leqslant \|\boldsymbol{A}\|_{\infty}^2\sum_{j=1}^{n_2}(\delta_{ij}-p)^2,$$

where by Exercise Q6.30 (*Row Norms of Random Matrices*), when $pn_2 \geqslant C' \log n_1$, we obtain

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n_1}\sum_{j=1}^{n_2}(\delta_{ij}-p)^2\right] \leqslant C_2 pn_2.$$

Substituting this in, then using Jensen's inequality $\mathbb{E}[\sqrt{W}] \leqslant \sqrt{\mathbb{E}[W]}$, we obtain

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n_1}\|\boldsymbol{Z}_{i*}\|_2\right] \leqslant \sqrt{C_2 pn_2}\|\boldsymbol{A}\|_\infty. \tag{III}$$

Similarly, for the column norm, when $pn_1 \geqslant C' \log n_2$,

$$\mathbb{E}\left[\max_{1\leqslant j\leqslant n_2}\|\boldsymbol{Z}_{*j}\|_2\right] \leqslant \sqrt{C_3 pn_1}\|\boldsymbol{A}\|_\infty. \tag{IV}$$

If (III) and (IV) both hold, then $pn \geqslant C_4 \log n$. Substituting $p = s/n_1 n_2$, we obtain $s \geqslant C' n \log n$.

Substituting (III) and (IV) into (II), we obtain

$$\mathbb{E}\|\boldsymbol{Y}-p\boldsymbol{A}\| \leqslant C_5\sqrt{\log n}(\sqrt{pn_2}+\sqrt{pn_1})\|\boldsymbol{A}\|_\infty \leqslant C_6\sqrt{pn\log n}\|\boldsymbol{A}\|_\infty,$$

Substituting this into (I), we obtain

$$\mathbb{E}\|\hat{\boldsymbol{A}}-\boldsymbol{A}\| \leqslant \frac{2}{p}C_6\sqrt{pn\log n}\|\boldsymbol{A}\|_\infty = C_7\sqrt{\frac{n\log n}{p}}\|\boldsymbol{A}\|_\infty. \tag{V}$$

- Use the low-rank assumption to transition to the Frobenius norm

  Since $\operatorname{rank}(\boldsymbol{A}), \operatorname{rank}(\hat{\boldsymbol{A}}) \leqslant r$, we have $\operatorname{rank}(\hat{\boldsymbol{A}}-\boldsymbol{A}) \leqslant 2r$, satisfying

  $$\|\hat{\boldsymbol{A}}-\boldsymbol{A}\|_F \leqslant \sqrt{2r}\|\hat{\boldsymbol{A}}-\boldsymbol{A}\|,$$

  Taking expectation on both sides, combined with (V), we obtain

  $$\mathbb{E}\|\hat{\boldsymbol{A}}-\boldsymbol{A}\|_F \leqslant C_8\sqrt{r}\sqrt{\frac{n\log n}{p}}\|\boldsymbol{A}\|_\infty = C_8\sqrt{\frac{rn\log n}{p}}\|\boldsymbol{A}\|_\infty,$$

  Dividing both sides by $\sqrt{n_1 n_2}$,

  $$\mathbb{E}\left[\frac{1}{\sqrt{n_1 n_2}}\|\hat{\boldsymbol{A}}-\boldsymbol{A}\|_F\right] \leqslant C_8\frac{1}{\sqrt{n_1 n_2}}\sqrt{\frac{rn\log n}{p}}\|\boldsymbol{A}\|_\infty = C_8\sqrt{\frac{rn\log n}{pn_1 n_2}}\|\boldsymbol{A}\|_\infty,$$

  That is,

  $$\mathbb{E}\left[\frac{1}{\sqrt{n_1 n_2}}\|\hat{\boldsymbol{A}}-\boldsymbol{A}\|_F\right] \leqslant C\sqrt{\frac{rn\log n}{s}}\|\boldsymbol{A}\|_\infty.$$

Rk. Under an independent sampling-with-replacement model, one obtains the following estimate. Let $(I_k, J_k)_{k=1}^s$ be independent and uniform over all index pairs satisfying $1 \leqslant i \leqslant n_1$ and $1 \leqslant j \leqslant n_2$. Set $\boldsymbol{Y}^{(k)} = A_{I_k J_k}\boldsymbol{e}_{I_k}\boldsymbol{e}_{J_k}^\top$ and $\boldsymbol{Z}_k = (n_1 n_2/s)\boldsymbol{Y}^{(k)} - \boldsymbol{A}/s$. Then $\boldsymbol{v}_i = \sum_{k=1}^s \varepsilon_k(\boldsymbol{Z}_k)_{i*}$ is a sub-Gaussian random vector satisfying

$$\|\boldsymbol{v}_i\|_{\psi_2} \lesssim \sqrt{\sum_{k=1}^s\|(\boldsymbol{Z}_k)_{i*}\|_2^2},$$

By the standard upper bound for the maximum of sub-Gaussian random variables, introducing the factor $\sqrt{\log n_1}$, we obtain

$$\mathbb{E}_\varepsilon\left[\max_{1\leqslant i\leqslant n_1}\|\boldsymbol{v}_i\|_2\right] \lesssim \sqrt{\log n_1}\cdot\max_{1\leqslant i\leqslant n_1}\sqrt{\sum_{k=1}^s\|(\boldsymbol{Z}_k)_{i*}\|_2^2},$$

Taking expectation with respect to $\boldsymbol{Z}$ on both sides, then using Jensen's inequality, we obtain

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n_1}\left\|\sum_{k=1}^s\varepsilon_k(\boldsymbol{Z}_k)_{i*}\right\|_2\right] \lesssim \sqrt{\log n_1}\cdot\sqrt{\mathbb{E}_{\boldsymbol{Z}}\left[\max_{1\leqslant i\leqslant n_1}\sum_{k=1}^s\|(\boldsymbol{Z}_k)_{i*}\|_2^2\right]}. \tag{I}$$

(We consider decomposing $\boldsymbol{Z}_k$ and introducing indicator variables.)

By definition, $\boldsymbol{Y}^{(k)}$ is an extremely sparse matrix containing only one nonzero entry.

Using the inequality $(a-b)^2 \leqslant 2a^2 + 2b^2$, we obtain

$$\|(\boldsymbol{Z}_k)_{i*}\|_2^2 \leqslant 2\left(\frac{n_1 n_2}{s}\right)^2 \|(\boldsymbol{Y}^{(k)})_{i*}\|_2^2 + \frac{2}{s^2}\|\boldsymbol{a}_{i*}\|_2^2.$$

Let the indicator variable $\delta_i^{(k)}$ indicate whether the kth sampling selects the ith row of the matrix, then

$$\|(\boldsymbol{Y}^{(k)})_{i*}\|_2^2 \leqslant \|\boldsymbol{A}\|_\infty^2 \cdot \delta_i^{(k)},$$

Substituting this in, then summing over the s samplings, we obtain

$$\sum_{k=1}^{s} \|(\boldsymbol{Z}_k)_{i*}\|_2^2 \leqslant 2\left(\frac{n_1 n_2}{s}\right)^2 \|\boldsymbol{A}\|_\infty^2 \sum_{k=1}^{s} \delta_i^{(k)} + \frac{2}{s}\|\boldsymbol{a}_{i*}\|_2^2.$$

Let $S_i = \sum_{k=1}^{s} \delta_i^{(k)} \sim B(s, 1/n_1)$. By the Chernoff concentration bound for binomial distributions, when $s \geqslant n_1 \log n_1$,

$$\mathbb{E}_{\boldsymbol{Z}}\left[\max_{1\leqslant i\leqslant n_1} \sum_{k=1}^{s} \delta_i^{(k)}\right] \lesssim \frac{s}{n_1},$$

Substituting this in, we obtain

$$\mathbb{E}_{\boldsymbol{Z}}\left[\max_{1\leqslant i\leqslant n_1} \sum_{k=1}^{s} \|(\boldsymbol{Z}_k)_{i*}\|_2^2\right] \lesssim \left(\frac{n_1 n_2}{s}\right)^2 \|\boldsymbol{A}\|_\infty^2 \left(\frac{s}{n_1}\right) = \frac{n_1 n_2^2}{s}\|\boldsymbol{A}\|_\infty^2. \tag{II}$$

Substituting (II) into (I), we obtain

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n_1} \left\|\sum_{k=1}^{s} \varepsilon_k (\boldsymbol{Z}_k)_{i*}\right\|_2\right] \lesssim \sqrt{\log n_1} \cdot \sqrt{\frac{n_1 n_2^2}{s}\|\boldsymbol{A}\|_\infty^2} = \frac{n_2\sqrt{n_1 \log n_1}}{\sqrt{s}}\|\boldsymbol{A}\|_\infty.$$

Similarly, when $s \geqslant n_2 \log n_2$,

$$\mathbb{E}\left[\max_{1\leqslant j\leqslant n_2} \left\|\sum_{k=1}^{s} \varepsilon_k (\boldsymbol{Z}_k)_{*j}\right\|_2\right] \lesssim \sqrt{\log n_2} \cdot \sqrt{\left(\frac{n_1 n_2}{s}\right)^2 \|\boldsymbol{A}\|_\infty^2 \left(\frac{s}{n_2}\right)} = \frac{n_1\sqrt{n_2 \log n_2}}{\sqrt{s}}\|\boldsymbol{A}\|_\infty.$$

32 [**Matrix Completion with Noisy Observations**] Generalize Theorem T6.5.1 (*Matrix completion*) to the case with noisy observations.

Specifically, let $\boldsymbol{X} \in \mathbb{R}^{n\times n}$ be a fixed matrix with $\operatorname{rank}(\boldsymbol{X}) \leqslant r$, and let the noise ν_{ij} be independent of one another and of the observation variables δ_{ij}, zero-mean sub-Gaussian random variables satisfying $\max_{i,j} \|\nu_{ij}\|_{\psi_2} \leqslant K$.

We observe not only a subset of entries of $\boldsymbol{X}$, but these entries plus noise. Define the observation matrix $\boldsymbol{Y}$ as

$$Y_{ij} = \delta_{ij}(X_{ij} + \nu_{ij}),$$

where δ_{ij} i.i.d. $\sim B(p)$ indicates whether position (i,j) is observed, and the families $\{\delta_{ij}\}$ and $\{\nu_{ij}\}$ are independent. Let $\hat{\boldsymbol{X}}$ be the best rank-r approximation of $p^{-1}\boldsymbol{Y}$, and let $m = pn^2$ be the expected number of observed entries. Prove: If $m \geqslant C' n \log n$, then

$$\mathbb{E}\left[\frac{1}{n}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F\right] \leqslant C\sqrt{\frac{rn\log n}{m}}(\|\boldsymbol{X}\|_\infty + K). \qquad (C > 0)$$

Prf. If $K = 0$, the noise vanishes almost surely and the conclusion follows from Exercise Q6.31 (*Matrix Completion for Rectangular Matrices*). Hence assume $K > 0$. Similarly to Exercise Q6.31 (*Matrix Completion for Rectangular Matrices*), by the Eckart-Young-Mirsky theorem, we obtain

$$\begin{aligned}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\| &\leqslant \|\hat{\boldsymbol{X}} - p^{-1}\boldsymbol{Y}\| + \|p^{-1}\boldsymbol{Y} - \boldsymbol{X}\| \\ &\leqslant 2\|p^{-1}\boldsymbol{Y} - \boldsymbol{X}\| = \frac{2}{p}\|\boldsymbol{Y} - p\boldsymbol{X}\|.\end{aligned}$$

For $\boldsymbol{Z} = \boldsymbol{Y} - p\boldsymbol{X}$, its (i,j)th entry is

$$Z_{ij} = \delta_{ij}(X_{ij} + \nu_{ij}) - pX_{ij} = (\delta_{ij} - p)X_{ij} + \delta_{ij}\nu_{ij},$$

Split it into $\boldsymbol{Z} = \boldsymbol{Z}^{(1)} + \boldsymbol{Z}^{(2)}$, where (signal error matrix) $Z_{ij}^{(1)} = (\delta_{ij} - p)X_{ij}$, (noise error matrix) $Z_{ij}^{(2)} = \delta_{ij}\nu_{ij}$. Using the triangle inequality, we obtain

$$\mathbb{E}\|\boldsymbol{Z}\| \leqslant \mathbb{E}\|\boldsymbol{Z}^{(1)}\| + \mathbb{E}\|\boldsymbol{Z}^{(2)}\|,$$

Then we only need to bound $\mathbb{E}\|\boldsymbol{Z}^{(1)}\|$ and $\mathbb{E}\|\boldsymbol{Z}^{(2)}\|$.

For $\boldsymbol{Z}^{(1)}$, by Theorem T6.5.1 (*Matrix completion*), we obtain

$$\mathbb{E}\|\boldsymbol{Z}^{(1)}\| \leqslant C_1\sqrt{pn\log n}\|\boldsymbol{X}\|_\infty. \tag{I}$$

For $\boldsymbol{Z}^{(2)}$, by Exercise Q6.29 (*The Logarithmic Factor Cannot Be Removed*), we obtain

$$\mathbb{E}\|\boldsymbol{Z}^{(2)}\| \leqslant C_2\sqrt{\log n}\left(\mathbb{E}\left[\max_{1\leqslant i\leqslant n}\|(\boldsymbol{Z}^{(2)})_{i*}\|_2\right] + \mathbb{E}\left[\max_{1\leqslant j\leqslant n}\|(\boldsymbol{Z}^{(2)})_{*j}\|_2\right]\right). \tag{II}$$

(By symmetry, we only need to compute the upper bound of the row norm.)

We make the following claim:

Claim Let $S_i = \|(\boldsymbol{Z}^{(2)})_{i*}\|_2^2 = \sum_{j=1}^n \delta_{ij}\nu_{ij}^2$, then

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} S_i\right] \leqslant CpnK^2.$$

Let $W_{ij} = \delta_{ij}\nu_{ij}^2$.

Since ν_{ij} is a sub-Gaussian random variable, ν_{ij}^2 is a sub-Exponential random variable, satisfying

$$\mathbb{E}[\nu_{ij}^2] \leqslant c_1K^2, \mathbb{E}[\nu_{ij}^4] \leqslant c_2K^4, \|\nu_{ij}^2\|_{\psi_1} \leqslant c_3K^2,$$

Hence the numerical characteristics of W_{ij} are

$$\mathbb{E}[W_{ij}] = \mathbb{E}[\delta_{ij}]\mathbb{E}[\nu_{ij}^2] \leqslant p\cdot c_1K^2,$$
$$\mathrm{Var}(W_{ij}) \leqslant \mathbb{E}[W_{ij}^2] = \mathbb{E}[\delta_{ij}^2]\mathbb{E}[\nu_{ij}^4] = p\cdot\mathbb{E}[\nu_{ij}^4] \leqslant c_2pK^4.$$

Moreover, since ν_{ij}^2 is sub-Exponential, $\delta_{ij}\nu_{ij}^2$ is sub-Exponential, satisfying $\|W_{ij}\|_{\psi_1} = \|\delta_{ij}\nu_{ij}^2\|_{\psi_1} \leqslant c_3K^2$.

Define the centered variable $Y_{ij} = W_{ij} - \mathbb{E}[W_{ij}]$, then $\mathbb{E}[Y_{ij}] = 0$, $v_j = \mathrm{Var}(Y_{ij}) \leqslant c_2pK^4$, $\|Y_{ij}\|_{\psi_1} \leqslant c_4K^2$.

For the fluctuation part of the ith row sum $M_i = \sum_{j=1}^n Y_{ij} = S_i - \mathbb{E}[S_i]$, using Theorem T2.9.1 (Subexponential Bernstein inequality) (sub-Exponential Bernstein inequality), we obtain for any $t \geqslant 0$,

$$\begin{aligned}\mathbb{P}\{|M_i| \geqslant t\} &\leqslant 2\exp\left(-c\min\left(\frac{t^2}{\sum_{j=1}^n v_j}, \frac{t}{\max_j\|Y_{ij}\|_{\psi_1}}\right)\right)\\ &\leqslant 2\exp\left(-c\min\left(\frac{t^2}{pnK^4}, \frac{t}{K^2}\right)\right),\end{aligned}$$

Let $M = \max_{1\leqslant i\leqslant n}|M_i|$. Using the union bound, we obtain

$$\mathbb{P}\{M \geqslant t\} \leqslant \sum_{i=1}^n \mathbb{P}\{|M_i| \geqslant t\} \leqslant 2n\exp\left(-c\min\left(\frac{t^2}{pnK^4}, \frac{t}{K^2}\right)\right).$$

To efficiently compute $\mathbb{E}[M] = \int_0^\infty \mathbb{P}\{M \geqslant t\}\mathrm{d}t$, we need to find the phase transition point t_0 such that the exponential part satisfies

$$\min\left(\frac{t^2}{pnK^4}, \frac{t}{K^2}\right) = \log n,$$

Solving gives $t \asymp \max(\sqrt{pn\log n}K^2, (\log n)K^2)$, hence let $t_0 = C_0pnK^2$, we obtain

$$\mathbb{P}\{M \geqslant t\} \leqslant \sum_{i=1}^n \mathbb{P}\{|M_i| \geqslant t\} \leqslant 2n\exp\left(-c\min\left(\frac{t^2}{pnK^4}, \frac{t}{K^2}\right)\right),$$

$$\begin{aligned}\mathbb{E}[M] &= \int_0^{t_0} \mathbb{P}\{M \geqslant t\}\mathrm{d}t + \int_{t_0}^{\infty} \mathbb{P}\{M \geqslant t\}\mathrm{d}t \\ &\leqslant t_0 + \int_{t_0}^{\infty} 2n\exp\left(-c\min\left(\frac{t^2}{pnK^4}, \frac{t}{K^2}\right)\right)\mathrm{d}t \\ &= C_0 pnK^2 + \int_{t_0}^{\infty} 2n\exp\left(-c\frac{t}{K^2}\right)\mathrm{d}t,\end{aligned}$$

where when $C_0 \geqslant 3/c$,

$$\int_{t_0}^{\infty} 2n\exp\left(-c\frac{t}{K^2}\right)\mathrm{d}t = 2n\frac{K^2}{c}\exp\left(-c\frac{t_0}{K^2}\right)$$

$$\leqslant 2n\frac{K^2}{c}\exp(-cC_0 pn) \leqslant 2n\frac{K^2}{c}\exp(-3\log n) = \frac{2K^2}{cn^2} \to 0,$$

Substituting, we obtain

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n}|S_i - \mathbb{E}[S_i]|\right] = \mathbb{E}[M] \leqslant C_0 pnK^2 + o(1) \leqslant C_1 pnK^2.$$

Thus, using the triangle inequality, we obtain

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} S_i\right] \leqslant c_1 pnK^2 + C_1 pnK^2 = (c_1 + C_1)pnK^2 = C_6 pnK^2.$$

By the claim, using Jensen's inequality $\mathbb{E}[\sqrt{S}] \leqslant \sqrt{\mathbb{E}[S]}$, *we obtain*

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant n}\|(\boldsymbol{Z}^{(2)})_{i*}\|_2\right] \leqslant \sqrt{\mathbb{E}\left[\max_{1\leqslant i\leqslant n} S_i\right]} \leqslant C_7\sqrt{pn}K,$$

Substituting this into (II), we obtain

$$\mathbb{E}\|\boldsymbol{Z}^{(2)}\| \leqslant C_8\sqrt{\log n}\cdot\sqrt{pn}K = C_8\sqrt{pn\log n}K. \tag{III}$$

Thus, combining (I) and (III), we obtain

$$\mathbb{E}\|\boldsymbol{Z}\| \leqslant \mathbb{E}\|\boldsymbol{Z}^{(1)}\| + \mathbb{E}\|\boldsymbol{Z}^{(2)}\| \leqslant C_9\sqrt{pn\log n}(\|\boldsymbol{X}\|_\infty + K),$$

Using the inequality $\|\boldsymbol{M}\|_F \leqslant \sqrt{\operatorname{rank}(\boldsymbol{M})}\|\boldsymbol{M}\|$, *we obtain*

$$\mathbb{E}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\| \leqslant \frac{2}{p}C_9\sqrt{pn\log n}(\|\boldsymbol{X}\|_\infty + K) = C_{10}\sqrt{\frac{n\log n}{p}}(\|\boldsymbol{X}\|_\infty + K),$$

By the rank bound, $\operatorname{rank}(\hat{\boldsymbol{X}} - \boldsymbol{X}) \leqslant 2r$, *so*

$$\mathbb{E}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F \leqslant \sqrt{2r}\,\mathbb{E}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\| \leqslant C_{11}\sqrt{\frac{rn\log n}{p}}(\|\boldsymbol{X}\|_\infty + K).$$

Dividing both sides by n,

$$\mathbb{E}\left[\frac{1}{n}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F\right] \leqslant \frac{1}{n}C_{11}\sqrt{\frac{rn\log n}{p}}(\|\boldsymbol{X}\|_\infty + K) = C_{11}\sqrt{\frac{rn\log n}{pn^2}}(\|\boldsymbol{X}\|_\infty + K),$$

That is,

$$\mathbb{E}\left[\frac{1}{n}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F\right] \leqslant C\sqrt{\frac{rn\log n}{m}}(\|\boldsymbol{X}\|_\infty + K),$$

The condition $pn \geqslant \log n$ *is equivalent to*

$$m \geqslant n^2\frac{\log n}{n} = n\log n.$$

33 In random matrix theory, there are very few universal results that hold without any assumptions on the matrix distribution. This problem aims to prove one such result: a variant of the matrix Bernstein inequality that applies to unbounded random matrices. (*From a broader perspective, this inequality basically says that the* L_p *spaces for* $p \geqslant 2$ *are of type 2, just like the* ℓ_p *spaces we examined in Exercise Q6.24 (Type 2 Property).*)

Let $\boldsymbol{Z}_1, \cdots, \boldsymbol{Z}_N \in \mathbb{R}^{n\times n}$ be independent positive semidefinite random matrices. Let $\boldsymbol{S} = \sum_{i=1}^N \boldsymbol{Z}_i$. Prove:

$$\mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\| \leqslant C(\sqrt{\|\mathbb{E}[\boldsymbol{S}]\| \cdot L} + L), \qquad (C > 0)$$

where $L = \log(n)\mathbb{E}[\max_{1\leqslant i\leqslant N} \|\boldsymbol{Z}_i\|]$.

Prf. • Symmetrization and matrix Khintchine inequality

Let the centered matrix $\boldsymbol{V}_i = \boldsymbol{Z}_i - \mathbb{E}[\boldsymbol{Z}_i]$. We only need to bound $\mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{V}_i\right\|$.

Introduce $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher. By the symmetrization lemma, we obtain

$$\mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\| \leqslant 2\mathbb{E}_{\boldsymbol{Z},\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{Z}_i\right\|.$$

For the right-hand side, using the matrix Khintchine inequality, we obtain

$$\mathbb{E}_\varepsilon\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{Z}_i\right\| \leqslant c_K\sqrt{\log n}\left\|\sum_{i=1}^N \boldsymbol{Z}_i^2\right\|^{1/2}, \qquad (c_K > 0)$$

Taking expectation with respect to $\boldsymbol{Z}$ on both sides, we obtain

$$\mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\| \leqslant 2c_K\sqrt{\log n}\cdot\mathbb{E}_{\boldsymbol{Z}}\left[\left\|\sum_{i=1}^N \boldsymbol{Z}_i^2\right\|^{1/2}\right]. \qquad (\mathrm{I})$$

• Use positive semidefiniteness to bound the variance term

Let $R = \max_{1\leqslant i\leqslant N}\|\boldsymbol{Z}_i\|$, then $\boldsymbol{Z}_i^2 \preccurlyeq \|\boldsymbol{Z}_i\|\boldsymbol{Z}_i \preccurlyeq R\boldsymbol{Z}_i$. Summing on both sides, we obtain

$$\sum_{i=1}^N \boldsymbol{Z}_i^2 \preccurlyeq R\sum_{i=1}^N \boldsymbol{Z}_i = R\boldsymbol{S},$$

Hence

$$\left\|\sum_{i=1}^N \boldsymbol{Z}_i^2\right\| \leqslant \|R\boldsymbol{S}\| = R\|\boldsymbol{S}\|,$$

Substituting this into (I) and using Jensen's inequality, we obtain

$$\mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\| \leqslant 2c_K\sqrt{\log n}\cdot\sqrt{\mathbb{E}[R\|\boldsymbol{S}\|]},$$

Then using Cauchy's inequality $\mathbb{E}[\sqrt{R\|\boldsymbol{S}\|}] \leqslant \sqrt{\mathbb{E}[R]}\cdot\sqrt{\mathbb{E}\|\boldsymbol{S}\|}$, let $L = \log n\cdot\mathbb{E}[R]$, we obtain

$$\mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\| \leqslant C_1\sqrt{L}\cdot\sqrt{\mathbb{E}\|\boldsymbol{S}\|}. \qquad (\mathrm{II})$$

• Solve the algebraic inequality for the expectation

Using the triangle inequality, we obtain

$$\mathbb{E}\|\boldsymbol{S}\| \leqslant \mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\| + \|\mathbb{E}[\boldsymbol{S}]\|,$$

Let $E = \mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\|$. Substituting this into (II), we obtain

$$E \leqslant C_1\sqrt{L}\sqrt{E + \|\mathbb{E}[\boldsymbol{S}]\|},$$

Squaring both sides,

$$E^2 \leqslant C_1^2 L(E + \|\mathbb{E}[\boldsymbol{S}]\|) = C_1^2 LE + C_1^2 L\|\mathbb{E}[\boldsymbol{S}]\|,$$

This is a quadratic inequality in E:

$$E^2 - (C_1^2 L)E - C_1^2 L\|\mathbb{E}[\boldsymbol{S}]\| \leqslant 0.$$

Using the inequality $\sqrt{a^2+b^2} \leqslant a + b$, we solve to obtain

$$E \leqslant \frac{C_1^2 L + \sqrt{C_1^4 L^2 + 4C_1^2 L\|\mathbb{E}[\boldsymbol{S}]\|}}{2} \leqslant \frac{C_1^2 L + C_1^2 L + 2C_1\sqrt{L\|\mathbb{E}[\boldsymbol{S}]\|}}{2} = C_1^2 L + C_1\sqrt{L\|\mathbb{E}[\boldsymbol{S}]\|},$$

That is,

$$\mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\| \leqslant C(L + \sqrt{L\|\mathbb{E}[\boldsymbol{S}]\|}). \qquad (C = \max(C_1^2, C_1))$$

34 [**Covariance Estimation under Unbounded Distributions**] Let $\boldsymbol{X} \in \mathbb{R}^n$ be a centered[3]random vector with nonzero[4]covariance matrix $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]$, and let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_m$ be independent copies of $\boldsymbol{X}$ with sample covariance matrix

$$\boldsymbol{\Sigma}_m = \frac{1}{m}\sum_{i=1}^m \boldsymbol{X}_i \boldsymbol{X}_i^\top,$$

satisfying that there exists $K \geqslant 1$ such that

$$\mathbb{E}\left[\max_{1\leqslant i\leqslant m} \|\boldsymbol{X}_i\|_2^2\right] \leqslant K^2 \mathbb{E}\|\boldsymbol{X}\|_2^2.$$

Define the effective rank as $r = \operatorname{tr}(\boldsymbol{\Sigma})/\|\boldsymbol{\Sigma}\|$. Prove:

$$\mathbb{E}\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant C\left(\sqrt{\frac{K^2 r \log n}{m}} + \frac{K^2 r \log n}{m}\right)\|\boldsymbol{\Sigma}\|. \qquad (C > 0)$$

Prf. By Exercise Q6.34 (*Covariance Estimation under Unbounded Distributions*), let $\boldsymbol{Z}_i = \boldsymbol{X}_i \boldsymbol{X}_i^\top$, $\boldsymbol{S} = \sum_{i=1}^m \boldsymbol{Z}_i = m\boldsymbol{\Sigma}_m$, we obtain

$$\mathbb{E}\|\boldsymbol{S} - \mathbb{E}[\boldsymbol{S}]\| \leqslant C_0(\sqrt{\|\mathbb{E}[\boldsymbol{S}]\| \cdot L} + L).$$

Substitute $\mathbb{E}[\boldsymbol{S}] = m\mathbb{E}[\boldsymbol{X}_1\boldsymbol{X}_1^\top] = m\boldsymbol{\Sigma}$, $\|\mathbb{E}[\boldsymbol{S}]\| = m\|\boldsymbol{\Sigma}\|$,

$$\begin{aligned} L &= \log(n)\mathbb{E}\left[\max_{1\leqslant i\leqslant m}\|\boldsymbol{Z}_i\|\right] = \log(n)\mathbb{E}\left[\max_{1\leqslant i\leqslant m}\|\boldsymbol{X}_i\|_2^2\right] \\ &\leqslant K^2\log(n)\mathbb{E}[\operatorname{tr}(\boldsymbol{X}\boldsymbol{X}^\top)] = K^2\log(n)\operatorname{tr}(\boldsymbol{\Sigma}) \leqslant K^2 r\log(n)\|\boldsymbol{\Sigma}\|, \end{aligned}$$

We obtain

$$\mathbb{E}\|m\boldsymbol{\Sigma}_m - m\boldsymbol{\Sigma}\| \leqslant C_0(\sqrt{(m\|\boldsymbol{\Sigma}\|)\cdot(K^2 r\log n\|\boldsymbol{\Sigma}\|)} + K^2 r\log(n)\|\boldsymbol{\Sigma}\|),$$

That is,

$$\mathbb{E}\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant C_0\left(\sqrt{\frac{K^2 r\log n}{m}} + \frac{K^2 r\log n}{m}\right)\|\boldsymbol{\Sigma}\|.$$

35 [**Convexity of the Expectation of Rademacher Series**] Let $\boldsymbol{z}_1, \cdots, \boldsymbol{z}_N$ be fixed vectors, $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$. Let $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher. Prove:

$$f(\boldsymbol{a}) = \mathbb{E}\left\|\sum_{i=1}^N a_i\varepsilon_i\boldsymbol{z}_i\right\|$$

is convex on $\mathbb{R}^N$.

Prf. To prove $f(\boldsymbol{a})$ is convex, we only need to show that for any $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^N$, $\lambda \in [0,1]$,

$$f(\lambda\boldsymbol{a} + (1-\lambda)\boldsymbol{b}) \leqslant \lambda f(\boldsymbol{a}) + (1-\lambda)f(\boldsymbol{b}).$$

It is easy to see that

$$\begin{aligned} f(\lambda\boldsymbol{a} + (1-\lambda)\boldsymbol{b}) &= \mathbb{E}\left\|\sum_{i=1}^N (\lambda a_i + (1-\lambda)b_i)\varepsilon_i\boldsymbol{z}_i\right\| \\ &= \mathbb{E}\left\|\sum_{i=1}^N \lambda a_i\varepsilon_i\boldsymbol{z}_i + \sum_{i=1}^N (1-\lambda)b_i\varepsilon_i\boldsymbol{z}_i\right\| \\ &\leqslant \mathbb{E}\left\|\sum_{i=1}^N \lambda a_i\varepsilon_i\boldsymbol{z}_i\right\| + \mathbb{E}\left\|\sum_{i=1}^N (1-\lambda)b_i\varepsilon_i\boldsymbol{z}_i\right\| \\ &= \lambda f(\boldsymbol{a}) + (1-\lambda)f(\boldsymbol{b}). \end{aligned}$$

[3]The original stem did not require $\boldsymbol{X}$ to be centered. This condition is added so that $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top]$ is its covariance matrix; in general, the covariance is $\mathbb{E}[(\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}])(\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}])^\top]$.

[4]The original stem did not require the covariance matrix to be nonzero. This condition is added because the effective rank is subsequently defined as $r = \operatorname{tr}(\boldsymbol{\Sigma})/\|\boldsymbol{\Sigma}\|$; if $\boldsymbol{\Sigma} = \boldsymbol{0}$, this ratio is $0/0$ and is undefined.

Rk. In fact, the norm $\|\cdot\|$ is convex, the map $\boldsymbol{a} \mapsto \sum a_i \varepsilon_i \boldsymbol{z}_i$ is linear, the composition of a convex function with a linear map remains convex, and the expectation operator $\mathbb{E}$ preserves convexity, so the above conclusion is trivial.

36 [**Contraction Principle under General Distributions**] Let $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$ be a coefficient vector, and $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_N$ be independent zero-mean random vectors. Prove:

$$\mathbb{E}\left\|\sum_{i=1}^N a_i \boldsymbol{X}_i\right\| \leqslant 4\|\boldsymbol{a}\|_\infty \cdot \mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|.$$

Prf. By the upper bound of Lemma L6.3.2 (*Symmetrization*), we obtain

$$\mathbb{E}\left\|\sum_{i=1}^N a_i \boldsymbol{X}_i\right\| \leqslant 2\mathbb{E}_{\boldsymbol{X},\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i a_i \boldsymbol{X}_i\right\|; \tag{I}$$

By the lower bound of Lemma L6.3.2 (*Symmetrization*), we obtain

$$\mathbb{E}_{\boldsymbol{X},\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\| \leqslant 2\mathbb{E}_{\boldsymbol{X}}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|. \tag{II}$$

By Theorem T6.6.1 (*Contraction principle*), we obtain

$$\mathbb{E}_{\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i a_i \boldsymbol{X}_i\right\| \leqslant \|\boldsymbol{a}\|_\infty \cdot \mathbb{E}_{\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|,$$

Taking expectation with respect to $\boldsymbol{X}$ on both sides,

$$\mathbb{E}_{\boldsymbol{X},\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i a_i \boldsymbol{X}_i\right\| \leqslant \|\boldsymbol{a}\|_\infty \cdot \mathbb{E}_{\boldsymbol{X},\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|. \tag{III}$$

Substituting (I) and (II) into (III), we obtain

$$\begin{aligned}
\mathbb{E}\left\|\sum_{i=1}^N a_i \boldsymbol{X}_i\right\| &\leqslant 2 \cdot \left(\mathbb{E}_{\boldsymbol{X},\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i a_i \boldsymbol{X}_i\right\|\right) \\
&\leqslant 2 \cdot \left(\|\boldsymbol{a}\|_\infty \cdot \mathbb{E}_{\boldsymbol{X},\varepsilon}\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{X}_i\right\|\right) \\
&\leqslant 2 \cdot \|\boldsymbol{a}\|_\infty \cdot \left(2 \cdot \mathbb{E}_{\boldsymbol{X}}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|\right) \\
&= 4\|\boldsymbol{a}\|_\infty \cdot \mathbb{E}\left\|\sum_{i=1}^N \boldsymbol{X}_i\right\|.
\end{aligned}$$

37 [**The Logarithmic Factor in Gaussian Symmetrization is Unavoidable**] Let $N \geqslant 2^5$. In Lemma L6.6.2 (*Symmetrization with Gaussians*), we proved that for any independent zero-mean random vectors $X_1, \cdots, X_N$,

$$\frac{c}{\sqrt{\log N}}\mathbb{E}\left\|\sum_{i=1}^N g_i X_i\right\| \leqslant \mathbb{E}\left\|\sum_{i=1}^N X_i\right\|, \qquad (c > 0)$$

where $g_1, \cdots, g_N \sim N(0,1)$ are independent of $X_1, \cdots, X_N$.

Construct a normed space and specific random vectors $X_1, \cdots, X_N$ in it to prove that in general, the penalty factor $\sqrt{\log N}$ on the left-hand side is optimal.

Prf. Consider the normed space ℓ_∞^N with norm $\|\boldsymbol{v}\|_\infty = \max_{1 \leqslant i \leqslant N} |v_i|$.

Let $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher, let $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_N \in \mathbb{R}^N$ be the standard orthonormal basis vectors, and define

$$\boldsymbol{X}_i = \varepsilon_i \boldsymbol{e}_i,$$

[5] The original stem did not restrict N. The condition $N \geqslant 2$ is added because the conclusion contains $1/\sqrt{\log N}$; when $N = 1$, $\log N = 0$ and this expression is undefined.

Then $\boldsymbol{X}_i$ are independent and zero-mean.

For the original expected norm,

$$\sum_{i=1}^{N} \boldsymbol{X}_i = \sum_{i=1}^{N} \varepsilon_i \boldsymbol{e}_i = (\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_N)^\top,$$

Hence

$$\left\| \sum_{i=1}^{N} \boldsymbol{X}_i \right\|_\infty = \max_{1 \leqslant i \leqslant N} |\varepsilon_i|,$$

Thus

$$\mathbb{E} \left\| \sum_{i=1}^{N} \boldsymbol{X}_i \right\|_\infty = \mathbb{E}[1] = 1. \tag{I}$$

For the Gaussian-weighted sum expected norm,

$$\sum_{i=1}^{N} g_i \boldsymbol{X}_i = \sum_{i=1}^{N} g_i \varepsilon_i \boldsymbol{e}_i = (g_1 \varepsilon_1, g_2 \varepsilon_2, \cdots, g_N \varepsilon_N)^\top,$$

Hence

$$\left\| \sum_{i=1}^{N} g_i \boldsymbol{X}_i \right\|_\infty = \max_{1 \leqslant i \leqslant N} |g_i \varepsilon_i| = \max_{1 \leqslant i \leqslant N} |g_i|,$$

Thus by the properties of the Gaussian extreme value distribution, we obtain

$$\mathbb{E} \left\| \sum_{i=1}^{N} g_i \boldsymbol{X}_i \right\|_\infty = \mathbb{E} \left[\max_{1 \leqslant i \leqslant N} |g_i| \right] \geqslant c' \sqrt{\log N}. \tag{II}$$

Combining (I) and (II), we obtain

$$\frac{c}{f(N)} \cdot (c' \sqrt{\log N}) \leqslant 1 \implies f(N) \geqslant (c \cdot c') \sqrt{\log N}.$$

38 [**Symmetrization and Contraction Principle under Convex Functions**] Let $F : \mathbb{R}^+ \to \mathbb{R}$ be an increasing convex function. Generalize Theorem T6.6.1 (*Contraction principle*) and Lemma L6.6.2 (*Symmetrization with Gaussians*) by replacing the norm $\|\cdot\|$ with $F(\|\cdot\|)$, and then prove the generalized conclusions.

Specifically, let $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N$ be fixed vectors, $\boldsymbol{a} = (a_1, \cdots, a_N) \in \mathbb{R}^N$ be a coefficient vector, $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher, $g_1, \cdots, g_N$ i.i.d. $\sim N(0,1)$. Prove:

(1) [**Generalized Contraction Principle**]

$$\mathbb{E} \left[F \left(\left\| \sum_{i=1}^{N} a_i \varepsilon_i \boldsymbol{x}_i \right\| \right) \right] \leqslant \mathbb{E} \left[F \left(\|\boldsymbol{a}\|_\infty \left\| \sum_{i=1}^{N} \varepsilon_i \boldsymbol{x}_i \right\| \right) \right].$$

(2) [**Generalized Gaussian Symmetrization (Upper Bound)**]

$$\mathbb{E} \left[F \left(\left\| \sum_{i=1}^{N} \varepsilon_i \boldsymbol{x}_i \right\| \right) \right] \leqslant \mathbb{E} \left[F \left(\sqrt{\frac{\pi}{2}} \left\| \sum_{i=1}^{N} g_i \boldsymbol{x}_i \right\| \right) \right].$$

(3) [**Generalized Gaussian Symmetrization (Lower Bound)**]

$$\mathbb{E} \left[F \left(\left\| \sum_{i=1}^{N} g_i \boldsymbol{x}_i \right\| \right) \right] \leqslant \mathbb{E}_{\boldsymbol{g}, \varepsilon} \left[F \left(\|\boldsymbol{g}\|_\infty \left\| \sum_{i=1}^{N} \varepsilon_i \boldsymbol{x}_i \right\| \right) \right].$$

Prf. (1) Let $G(\boldsymbol{v}) = F(\|\boldsymbol{v}\|)$ be convex, and define $f(\boldsymbol{a}) = \mathbb{E}_\varepsilon[G(\sum_{i=1}^{N} a_i \varepsilon_i \boldsymbol{x}_i)]$, then $f(\boldsymbol{a})$ is convex in $\boldsymbol{a}$, so the maximum of f on its domain $[-\|\boldsymbol{a}\|_\infty, \|\boldsymbol{a}\|_\infty]^N$ must be attained at a vertex.

Let $M = \|\boldsymbol{a}\|_\infty$, and let the vertex be $\boldsymbol{v} = (Ms_1, \cdots, Ms_N)$ where $s_i \in \{-1, 1\}$, then

$$f(\boldsymbol{v}) \geqslant f(\boldsymbol{a}),$$

where

$$f(\boldsymbol{v}) = \mathbb{E}_\varepsilon\left[F\left(\left\|\sum_{i=1}^N (Ms_i)\varepsilon_i \boldsymbol{x}_i\right\|\right)\right] = \mathbb{E}_\varepsilon\left[F\left(M\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|\right)\right] = \mathbb{E}_\varepsilon\left[F\left(\|\boldsymbol{a}\|_\infty\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|\right)\right],$$

Substituting this in, we obtain

$$\mathbb{E}\left[F\left(\|\boldsymbol{a}\|_\infty\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|\right)\right] \geqslant \mathbb{E}\left[F\left(\left\|\sum_{i=1}^N a_i\varepsilon_i \boldsymbol{x}_i\right\|\right)\right].$$

(2) For $g_i \sim N(0,1)$, the expectation of its absolute value is $\mathbb{E}[|g_i|] = \sqrt{2/\pi}$. Let $c = \sqrt{2/\pi}$, then $c^{-1}\mathbb{E}[|g_i|] = 1$. For the left-hand side,

$$\begin{aligned}\mathbb{E}_\varepsilon\left[F\left(\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|\right)\right] &= \mathbb{E}_\varepsilon\left[F\left(\left\|\sum_{i=1}^N \varepsilon_i\left(\frac{1}{c}\mathbb{E}_{\boldsymbol{g}}[|g_i|]\right)\boldsymbol{x}_i\right\|\right)\right] \\ &= \mathbb{E}_\varepsilon\left[F\left(\frac{1}{c}\left\|\mathbb{E}_{\boldsymbol{g}}\left[\sum_{i=1}^N \varepsilon_i|g_i|\boldsymbol{x}_i\right]\right\|\right)\right],\end{aligned}$$

Using Jensen's inequality, we obtain

$$\begin{aligned}\mathbb{E}_\varepsilon\left[F\left(\frac{1}{c}\left\|\mathbb{E}_{\boldsymbol{g}}\left[\sum_{i=1}^N \varepsilon_i|g_i|\boldsymbol{x}_i\right]\right\|\right)\right] &\leqslant \mathbb{E}_\varepsilon\left[\mathbb{E}_{\boldsymbol{g}}\left[F\left(\frac{1}{c}\left\|\sum_{i=1}^N \varepsilon_i|g_i|\boldsymbol{x}_i\right\|\right)\right]\right] \\ &= \mathbb{E}_{\varepsilon,\boldsymbol{g}}\left[F\left(\sqrt{\frac{\pi}{2}}\left\|\sum_{i=1}^N \varepsilon_i|g_i|\boldsymbol{x}_i\right\|\right)\right] \\ &= \mathbb{E}_{\boldsymbol{g}}\left[F\left(\sqrt{\frac{\pi}{2}}\left\|\sum_{i=1}^N g_i\boldsymbol{x}_i\right\|\right)\right],\end{aligned}$$

Hence

$$\mathbb{E}\left[F\left(\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|\right)\right] \leqslant \mathbb{E}\left[F\left(\sqrt{\frac{\pi}{2}}\left\|\sum_{i=1}^N g_i\boldsymbol{x}_i\right\|\right)\right].$$

(3) For the left-hand side,

$$\mathbb{E}_{\boldsymbol{g}}\left[F\left(\left\|\sum_{i=1}^N g_i \boldsymbol{x}_i\right\|\right)\right] = \mathbb{E}_{\boldsymbol{g},\varepsilon}\left[F\left(\left\|\sum_{i=1}^N |g_i|\varepsilon_i \boldsymbol{x}_i\right\|\right)\right].$$

Given $\boldsymbol{g}$, using the conclusion of (1), we obtain

$$\mathbb{E}_\varepsilon\left[F\left(\left\|\sum_{i=1}^N |g_i|\varepsilon_i \boldsymbol{x}_i\right\|\right)\middle|\boldsymbol{g}\right] \leqslant \mathbb{E}_\varepsilon\left[F\left(\max_i|g_i|\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|\right)\middle|\boldsymbol{g}\right] = \mathbb{E}_\varepsilon\left[F\left(\|\boldsymbol{g}\|_\infty\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|\right)\middle|\boldsymbol{g}\right],$$

Taking expectation with respect to $\boldsymbol{g}$ on both sides,

$$\mathbb{E}_{\boldsymbol{g},\varepsilon}\left[F\left(\left\|\sum_{i=1}^N |g_i|\varepsilon_i \boldsymbol{x}_i\right\|\right)\right] \leqslant \mathbb{E}_{\boldsymbol{g},\varepsilon}\left[F\left(\|\boldsymbol{g}\|_\infty\left\|\sum_{i=1}^N \varepsilon_i \boldsymbol{x}_i\right\|\right)\right].$$

7 Stochastic Processes

Keypoints

Section 7.1 Basic concepts and examples Q7.1 ~ Q7.5

Definition 7.1.1 (*Random process*)

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space and let $\mathbb{T}$ be an index set. If, for every $t \in \mathbb{T}$, X_t is a random variable defined on this space, then the family $(X_t : t \in \mathbb{T})$ is called a random process.

Example 7.1.3 (*Random walks*)

If $\mathbb{T} = \mathbb{N}$, a discrete-time random process $(X_n)_{n\in\mathbb{N}}$ is simply a sequence of random variables. An important example is a random walk defined as

$$X_n = \sum_{i=1}^{n} Z_i,$$

where the increments Z_i are independent, mean-zero random variables.

Definition 7.1.9 (*Gaussian process*)

A random process $(X_t)_{t\in\mathbb{T}}$ is called a Gaussian process if, for any finite subset $\mathbb{T}_0 \subset \mathbb{T}$, the random vector $(X_t)_{t\in\mathbb{T}_0}$ has normal distribution. Equivalently, every finite linear combination $\sum_{t\in\mathbb{T}_0} a_t X_t$ is a normal random variable.

Theorem 7.1.11 (*Concentration of Gaussian processes*)

Let $(X_t)_{t\in\mathbb{T}}$ be a Gaussian process with finite $\mathbb{T}$. Then

$$\| \sup_{t\in\mathbb{T}} X_t - \mathbb{E}\left[\sup_t X_t\right] \|_{\psi_2} \leqslant C \sup_{t\in\mathbb{T}} \sqrt{\mathrm{Var}(X_t)}.$$

Lemma 7.1.12 (*Gaussian random vectors*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a mean-zero Gaussian random vector. Then there exist points $\boldsymbol{t}_1, \cdots, \boldsymbol{t}_n \in \mathbb{R}^n$ such that

$$\boldsymbol{X} \overset{\text{dist}}{=} (\langle \boldsymbol{g}, \boldsymbol{t}_i \rangle)_{i=1}^n,$$

where $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

Section 7.2 Slepian, Sudakov-Fernique and Gordon Q7.6 ∼ Q7.10

Theorem 7.2.2 (*Slepian inequality*)

Let $(X_t : t \in \mathbb{T}), (Y_t : t \in \mathbb{T})$ be zero-mean Gaussian processes satisfying $\mathbb{E}[X_t^2] = \mathbb{E}[Y_t^2]$. If for every $t, s \in \mathbb{T}$, $\mathbb{E}[(X_t - X_s)^2] \leqslant \mathbb{E}[(Y_t - Y_s)^2]$, then

$$\text{For every } \tau \in \mathbb{R}, \mathbb{P}\left\{\sup_t X_t \geqslant \tau\right\} \leqslant \mathbb{P}\left\{\sup_t Y_t \geqslant \tau\right\}.$$
$$\mathbb{E}\left[\sup_t X_t\right] \leqslant \mathbb{E}\left[\sup_t Y_t\right].$$

Lemma 7.2.3 (*Gaussian integration by parts*)

Let $X \sim N(0,1)$. Then for any differentiable function $f : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}[Xf(X)] = \mathbb{E}[f'(X)],$$

assuming both expectations exist and are finite.

Lemma 7.2.4 (*Multivariate Gaussian integration by parts*)

Let $\boldsymbol{X} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma})$. Then

$$\mathbb{E}[\boldsymbol{X} f(\boldsymbol{X})] = \boldsymbol{\Sigma} \cdot \mathbb{E}[\nabla f(\boldsymbol{X})],$$
$$\mathbb{E}[X_i f(\boldsymbol{X})] = \sum_{j=1}^{n} \Sigma_{ij} \mathbb{E}\left[\frac{\partial f}{\partial x_j}(\boldsymbol{X})\right].$$

Lemma 7.2.5 (*Gaussian interpolation*)

Consider two independent Gaussian random vectors $\boldsymbol{X} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}_X)$ and $\boldsymbol{Y} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}_Y)$. Define $\boldsymbol{Z}(u) = \sqrt{u}\boldsymbol{X} + \sqrt{1-u}\boldsymbol{Y}$, $u \in [0,1]$. Then for any twice-differentiable f,

$$\frac{d}{du}\mathbb{E}[f(\boldsymbol{Z}(u))] = \frac{1}{2}\sum_{i,j}(\Sigma_{ij}^X - \Sigma_{ij}^Y)\mathbb{E}\left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{Z}(u))\right].$$

Lemma 7.2.6 (*Slepian inequality, functional form*)

Consider two mean-zero Gaussian random vectors $\boldsymbol{X}, \boldsymbol{Y}$ in $\mathbb{R}^n$ with $\mathbb{E}[X_i^2] = \mathbb{E}[Y_i^2]$ and $\mathbb{E}[(X_i - X_j)^2] \leqslant \mathbb{E}[(Y_i - Y_j)^2]$ for every i, j. For a twice-differentiable f with $\frac{\partial^2 f}{\partial x_i \partial x_j} \geqslant 0$ for $i \neq j$, we have

$$\mathbb{E}[f(\boldsymbol{X})] \geqslant \mathbb{E}[f(\boldsymbol{Y})].$$

Theorem 7.2.7 (*Slepian inequality*)

Let $\boldsymbol{X}$ and $\boldsymbol{Y}$ be Gaussian random vectors as in Lemma L7.2.6 (*Slepian inequality, functional form*). Then

$$\text{For every } \tau \geqslant 0, \mathbb{P}\left\{\max_i X_i \geqslant \tau\right\} \leqslant \mathbb{P}\left\{\max_i Y_i \geqslant \tau\right\}.$$
$$\mathbb{E}\left[\max_i X_i\right] \leqslant \mathbb{E}\left[\max_i Y_i\right].$$

Theorem 7.2.8 (*Sudakov-Fernique inequality*)

Let (X_t) and (Y_t) be zero-mean Gaussian processes. If for every $t, s \in \mathbb{T}$, $\mathbb{E}[(X_t - X_s)^2] \leqslant \mathbb{E}[(Y_t - Y_s)^2]$, then

$$\mathbb{E}\left[\sup_t X_t\right] \leqslant \mathbb{E}\left[\sup_t Y_t\right].$$

Theorem 7.2.9 (*Gordon inequality*)

Let $(X_{ut} : u \in \mathbb{U}, t \in \mathbb{T}), (Y_{ut} : u \in \mathbb{U}, t \in \mathbb{T})$ be zero-mean Gaussian processes. If, for every $u \in \mathbb{U}$ and $t, s \in \mathbb{T}$,

$$\mathbb{E}[(X_{ut} - X_{us})^2] \leqslant \mathbb{E}[(Y_{ut} - Y_{us})^2],$$

and, for every $u \neq v$ and $t, s \in \mathbb{T}$,

$$\mathbb{E}[(X_{ut} - X_{vs})^2] \geqslant \mathbb{E}[(Y_{ut} - Y_{vs})^2],$$

then

$$\mathbb{E}\left[\inf_u \sup_t X_{ut}\right] \leqslant \mathbb{E}\left[\inf_u \sup_t Y_{ut}\right].$$

Section 7.3 Sharp bounds for Gaussian matrices

Q7.11 ∼ Q7.13

Theorem 7.3.1 (*Norms of Gaussian random matrices*)

Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a standard Gaussian random matrix. Then

$$\mathbb{E}\|\boldsymbol{A}\| \leqslant \sqrt{m} + \sqrt{n}.$$

Corollary 7.3.2 (*Norms of Gaussian random matrices: tails*)

Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a standard Gaussian random matrix. Then for every $t \geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{A}\| \geqslant \sqrt{m} + \sqrt{n} + t\} \leqslant 2\exp(-ct^2).$$

Section 7.4 Sudakov inequality

Q7.14

Theorem 7.4.1 (*Sudakov inequality*)

Let $(X_t)_{t \in \mathbb{T}}$ be a mean-zero Gaussian process. Then, for any $\varepsilon \geqslant 0$,

$$\mathbb{E}\left[\sup_{t \in \mathbb{T}} X_t\right] \geqslant c\varepsilon\sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)},$$

where $d(t, s) = \|X_t - X_s\|_{L_2}$.

Corollary 7.4.2 (*Sudakov inequality in $\mathbb{R}^n$*)

Let $\mathbb{T} \subset \mathbb{R}^n$. Then, for any $\varepsilon > 0$,

$$\mathbb{E}\left[\sup_{t \in \mathbb{T}} \langle \boldsymbol{g}, t\rangle\right] \geqslant c\varepsilon\sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)},$$

where $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

Corollary 7.4.3 (*Covering numbers of polytopes*)

Let P be a polytope in $\mathbb{B}_2^n$ with N vertices. Then, for every $\varepsilon > 0$,

$$\mathcal{N}(P, \varepsilon) \leqslant N^{C/\varepsilon^2}.$$

Section 7.5 Gaussian width

Q7.15 ∼ Q7.23

Definition 7.5.1 (*Gaussian width*)

Let $\mathbb{T} \subset \mathbb{R}^n$. Its Gaussian width is defined by

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{x} \in \mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x}\rangle\right],$$

where $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

Proposition 7.5.2 (*Simple properties of Gaussian width*)

(i) $w(\mathbb{T})$ is invariant under translations. (ii) $w(\mathbb{T}) < \infty$ for any bounded $\mathbb{T}$. (iii) $w(\operatorname{conv}(\mathbb{T})) = w(\mathbb{T})$. (iv) $w(\mathbb{T} + S) = w(\mathbb{T}) + w(S)$ for independent suprema.

Definition 7.5.4 (*Spherical width*)
The spherical width is $w_S(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle\boldsymbol{\theta}, \boldsymbol{x}\rangle\right]$, where $\boldsymbol{\theta}$ is uniformly distributed on $\mathbb{S}^{n-1}$.

Lemma 7.5.5 (*Gaussian vs. spherical widths*)
For any $\mathbb{T} \subset \mathbb{R}^n$,

$$cw_S(\mathbb{T}) \leqslant \frac{1}{\sqrt{n}} w(\mathbb{T}) \leqslant Cw_S(\mathbb{T}).$$

Lemma 7.5.11 (*Almost equivalent versions of Gaussian width*)
For any set $\mathbb{T} \subset \mathbb{R}^n$, the following quantities are equivalent up to absolute constant factors: (i) $w(\mathbb{T})$; (ii) $\mathbb{E}\left[\sup_{t\in\mathbb{T}} |\langle \boldsymbol{g}, t\rangle|\right]$; (iii) $\sqrt{\mathbb{E}\left[\sup_{t\in\mathbb{T}}\langle \boldsymbol{g}, t\rangle^2\right]}$.

Definition 7.5.12 (*Effective dimension*)
For a bounded set $\mathbb{T} \subset \mathbb{R}^n$, the effective dimension is $d(\mathbb{T}) = (w(\mathbb{T})/\operatorname{rad}(\mathbb{T}))^2$, where $\operatorname{rad}(\mathbb{T}) = \sup_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{x}\|_2$.

Section 7.6 Random projections of sets Q7.24 ~ Q7.27

Theorem 7.6.1 (*Sizes of random projections of sets*)
Let $\mathbb{T} \subset \mathbb{R}^n$, and let $\boldsymbol{G} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix. Then

$$\mathbb{E}[\operatorname{rad}(\boldsymbol{G}\mathbb{T})] \leqslant \frac{w(\mathbb{T})}{\sqrt{m}} + \operatorname{rad}(\mathbb{T}),$$

$$\mathbb{E}[w(\boldsymbol{G}\mathbb{T})] \leqslant C\sqrt{m} \cdot w(\mathbb{T}) + Cm \cdot \operatorname{rad}(\mathbb{T}).$$

Exercises

1 [**Covariance and Increments**] Consider the stochastic process $(X_t)_{t\in\mathbb{T}}$.
(1) Suppose the constant zero random variable 0 belongs to this process. Express the covariance function $\Sigma(t, s) = \mathbb{E}[X_t X_s]$ in terms of the metric $d(t, s) = \|X_t - X_s\|_{L^2}$.
(2) Suppose that for every random variable X_t in this process, its opposite $-X_t$ is also included. Express the covariance function $\Sigma(t, s) = \mathbb{E}[X_t X_s]$ in terms of the metric $d(t, s) = \|X_t - X_s\|_{L^2}$.
(3) Give an example showing that, if the above assumptions are ignored, the covariance function need not be recoverable solely from the metric structure.

Prf. (1) Expand the square of the metric,

$$d(t, s)^2 = \mathbb{E}[(X_t - X_s)^2] = \mathbb{E}[X_t^2] + \mathbb{E}[X_s^2] - 2\mathbb{E}[X_t X_s],$$

solving, we obtain

$$\Sigma(t, s) = \mathbb{E}[X_t X_s] = \frac{1}{2}(\mathbb{E}[X_t^2] + \mathbb{E}[X_s^2] - d(t, s)^2). \tag{I}$$

If there exists $t_0 \in \mathbb{T}$ such that $X_{t_0} \equiv 0$, then

$$d(t, t_0)^2 = \mathbb{E}[(X_t - X_{t_0})^2] = \mathbb{E}[(X_t - 0)^2] = \mathbb{E}[X_t^2],$$

Similarly, $\mathbb{E}[X_s^2] = d(s,t_0)^2$. Substituting this into (I), we obtain

$$\Sigma(t,s) = \frac{1}{2}(d(t,t_0)^2 + d(s,t_0)^2 - d(t,s)^2),$$

which is the polarization identity.

(2) It is easy to see that

$$d(t,s)^2 = \mathbb{E}[(X_t - X_s)^2] = \mathbb{E}[X_t^2] + \mathbb{E}[X_s^2] - 2\mathbb{E}[X_t X_s]. \tag{II}$$

If there exists $s^* \in \mathbb{T}$ such that $X_{s^*} = -X_s$, then

$$\begin{aligned} d(t,s^*)^2 &= \mathbb{E}[X_t^2] + \mathbb{E}[X_s^2] + 2\mathbb{E}[X_t X_s] \\ &= \mathbb{E}[X_t^2] + \mathbb{E}[X_s^2] + 2\mathbb{E}[X_t X_s], \end{aligned} \tag{III}$$

Combining (II) and (III), we obtain

$$d(t,s^*)^2 - d(t,s)^2 = 4\mathbb{E}[X_t X_s],$$

solving, we obtain

$$\Sigma(t,s) = \frac{1}{4}(d(t,s^*)^2 - d(t,s)^2).$$

(3) (*If a global random shift is added to the entire stochastic process, the distance between variables does not change, but the covariance will change.*)

Let $\mathbb{T} = \{1,2\}$.

Define the stochastic process X as

$$X_1 = 0, X_2 = 1,$$

then $d_X(1,2)^2 = \mathbb{E}[(0-1)^2] = 1$, $\Sigma_X(1,2) = \mathbb{E}[X_1 X_2] = \mathbb{E}[0 \cdot 1] = 0$.

Next define the stochastic process Y as

$$Y_1 = X_1 + Z, Y_2 = X_2 + Z,$$

where Z satisfies $\mathbb{E}[Z] = 0$, $\mathbb{E}[Z^2] = c \neq 0$, then $d_Y(1,2)^2 = \mathbb{E}[(Y_1 - Y_2)^2] = \mathbb{E}[(Z - (1+Z))^2] = \mathbb{E}[(-1)^2] = 1$, $\Sigma_Y(1,2) = \mathbb{E}[Y_1 Y_2] = \mathbb{E}[Z(1+Z)] = \mathbb{E}[Z] + \mathbb{E}[Z^2] = 0 + c = c \neq \Sigma_X$.

Rk. The polarization identity is a bridge connecting norm and inner product. This exercise is the manifestation of the polarization identity in stochastic processes.

2 [**Symmetrization of Stochastic Processes**] Let $X_1(t), \cdots, X_N(t)$ be N mutually independent zero-mean stochastic processes defined on $\mathbb{T}$. Let $\varepsilon_1, \cdots, \varepsilon_N$ i.i.d. $\sim$ Rademacher be independent of X_i. Prove that

$$\frac{1}{2}\mathbb{E}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N \varepsilon_i X_i(t)\right|\right] \leqslant \mathbb{E}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N X_i(t)\right|\right] \leqslant 2\mathbb{E}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N \varepsilon_i X_i(t)\right|\right].$$

Prf. It is easy to see that $\Phi(Y) = \sup_{t\in\mathbb{T}}|Y(t)|$ is a convex function.

• Proof of the right-hand inequality

Introduce i.i.d. copies $X_1'(t), \cdots, X_N'(t)$, then using Jensen's inequality and the triangle inequality, we obtain

$$\begin{aligned} \mathbb{E}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N X_i(t)\right|\right] &= \mathbb{E}_X\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N (X_i(t) - \mathbb{E}_{X'}[X_i'(t)])\right|\right] \\ &\leqslant \mathbb{E}_{X,X'}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N (X_i(t) - X_i'(t))\right|\right] \\ &= \mathbb{E}_{X,X',\varepsilon}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N \varepsilon_i(X_i(t) - X_i'(t))\right|\right] \\ &\leqslant \mathbb{E}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N \varepsilon_i X_i(t)\right| + \sup_{t\in\mathbb{T}}\left|\sum_{i=1}^N \varepsilon_i X_i'(t)\right|\right] \end{aligned}$$

$$= 2\mathbb{E}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^{N}\varepsilon_i X_i(t)\right|\right].$$

- Proof of the left-hand inequality

  Similarly, introduce i.i.d. copies $X_1'(t),\cdots,X_N'(t)$, then using Jensen's inequality and the triangle inequality, we obtain

$$\begin{aligned}
\mathbb{E}_{X,\varepsilon}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^{N}\varepsilon_i X_i(t)\right|\right] &= \mathbb{E}_{X,\varepsilon}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^{N}\varepsilon_i (X_i(t)-\mathbb{E}_{X'}[X_i'(t)])\right|\right]\\
&\leqslant \mathbb{E}_{X,X',\varepsilon}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^{N}\varepsilon_i (X_i(t)-X_i'(t))\right|\right]\\
&= \mathbb{E}_{X,X'}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^{N}(X_i(t)-X_i'(t))\right|\right]\\
&\leqslant \mathbb{E}_{X,X'}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^{N}X_i(t)\right|+\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^{N}(-X_i'(t))\right|\right]\\
&= 2\mathbb{E}\left[\sup_{t\in\mathbb{T}}\left|\sum_{i=1}^{N}X_i(t)\right|\right].
\end{aligned}$$

Rk. This exercise shows that if we view stochastic processes as random vectors in the normed space $L^\infty(\mathbb{T})$, the symmetrization technique from finite-dimensional spaces remains applicable to infinite-dimensional spaces.

3 [**Talagrand's Contraction Principle**] Let $\mathbb{T}\subset\mathbb{R}^n$, and let its elements be denoted by $\boldsymbol{t}=(t_1,\cdots,t_n)$. Let $\varphi_i:\mathbb{R}\to\mathbb{R}$ be contraction maps (*i.e., satisfying* $\|\varphi_i\|_{\text{Lip}}\leqslant 1$). Let $\varepsilon_1,\cdots,\varepsilon_n$ i.i.d. $\sim$ Rademacher. Prove that

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\sum_{i=1}^{n}\varepsilon_i\varphi_i(t_i)\right]\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\sum_{i=1}^{n}\varepsilon_i t_i\right].$$

Prove this following these steps:

(1) First let $n=2$. Consider a subset $\mathbb{T}\subset\mathbb{R}^2$ and a contraction map $\varphi:\mathbb{R}\to\mathbb{R}$, and prove that

$$\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1+\varphi(t_2))+\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1-\varphi(t_2))\leqslant\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1+t_2)+\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1-t_2).$$

(2) Use induction on n to complete the proof.

Prf. - First prove the two-dimensional case

  We first prove that

$$\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1+\varphi(t_2))+\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1-\varphi(t_2))\leqslant\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1+t_2)+\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1-t_2).$$

  Introduce independent variables $\boldsymbol{s},\boldsymbol{t}\in\mathbb{T}$. Then the left-hand side is

$$\begin{aligned}
\text{LHS} &= \sup_{\boldsymbol{s}\in\mathbb{T}}(s_1+\varphi(s_2))+\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1-\varphi(t_2)) = \sup_{\boldsymbol{s},\boldsymbol{t}\in\mathbb{T}}(s_1+t_1+\varphi(s_2)-\varphi(t_2))\\
&\leqslant \sup_{\boldsymbol{s},\boldsymbol{t}\in\mathbb{T}}(s_1+t_1+|s_2-t_2|).
\end{aligned}$$

  If $s_2\geqslant t_2$, then

$$s_1+t_1+|s_2-t_2| = (s_1+s_2)+(t_1-t_2)\leqslant\sup_{\boldsymbol{u}\in\mathbb{T}}(u_1+u_2)+\sup_{\boldsymbol{v}\in\mathbb{T}}(v_1-v_2).$$

  The same bound follows after interchanging $\boldsymbol{s}$ and $\boldsymbol{t}$ when $s_2<t_2$. Thus,

$$s_1+t_1+\varphi(s_2)-\varphi(t_2)\leqslant\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1+t_2)+\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1-t_2),$$

  taking the supremum over $\boldsymbol{s},\boldsymbol{t}\in\mathbb{T}$ on both sides, we obtain

$$\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1+\varphi(t_2))+\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1-\varphi(t_2))\leqslant\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1+t_2)+\sup_{\boldsymbol{t}\in\mathbb{T}}(t_1-t_2). \tag{I}$$

- Then consider the general case

  We need to prove that

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\sum_{i=1}^{n}\varepsilon_i\varphi_i(t_i)\right]\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\sum_{i=1}^{n}\varepsilon_i t_i\right].$$

  Define the stochastic process $Y_k(\boldsymbol{t})$ as

$$Y_k(\boldsymbol{t})=\sum_{i=1}^{k}\varepsilon_i\varphi_i(t_i)+\sum_{i=k+1}^{n}\varepsilon_i t_i,$$

  then the goal is

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_n(\boldsymbol{t})\right]\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_0(\boldsymbol{t})\right],$$

  by induction, we only need to prove that for any $1\leqslant k\leqslant n$, there exists a single-step descent relation, i.e.,

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_k(\boldsymbol{t})\right]\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_{k-1}(\boldsymbol{t})\right].$$

  Let

$$f_{k-1}(\boldsymbol{t})=\sum_{i=1}^{k-1}\varepsilon_i\varphi_i(t_i)+\sum_{i=k+1}^{n}\varepsilon_i t_i,$$

  then $Y_k(\boldsymbol{t})=f_{k-1}(\boldsymbol{t})+\varepsilon_k\varphi_k(t_k)$, $Y_{k-1}(\boldsymbol{t})=f_{k-1}(\boldsymbol{t})+\varepsilon_k t_k$.

  Conditioning on all Rademacher variables except ε_k, i.e., $\varepsilon_1,\cdots,\varepsilon_{k-1},\varepsilon_{k+1},\cdots,\varepsilon_n$, and taking the conditional expectation of the supremum of $Y_k(\boldsymbol{t})$,

$$\mathbb{E}_{\varepsilon_k}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})+\varepsilon_k\varphi_k(t_k))\right]=\frac{1}{2}\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})+\varphi_k(t_k))+\frac{1}{2}\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})-\varphi_k(t_k)),$$

  For the right-hand side, using the conclusion of (I), we obtain

$$\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})+\varphi_k(t_k))+\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})-\varphi_k(t_k))\leqslant\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})+t_k)+\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})-t_k),$$

  substituting this in, we obtain

$$\begin{aligned}\mathbb{E}_{\varepsilon_k}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_k(\boldsymbol{t})\right]&\leqslant\frac{1}{2}\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})+t_k)+\frac{1}{2}\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})-t_k)\\&=\mathbb{E}_{\varepsilon_k}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}(f_{k-1}(\boldsymbol{t})+\varepsilon_k t_k)\right]\\&=\mathbb{E}_{\varepsilon_k}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_{k-1}(\boldsymbol{t})\right],\end{aligned}$$

  then taking expectation over the remaining ε,

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_k(\boldsymbol{t})\right]\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_{k-1}(\boldsymbol{t})\right].$$

  Thus,

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_n(\boldsymbol{t})\right]\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_{n-1}(\boldsymbol{t})\right]\leqslant\cdots\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_1(\boldsymbol{t})\right]\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}Y_0(\boldsymbol{t})\right],$$

  i.e.,

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\sum_{i=1}^{n}\varepsilon_i\varphi_i(t_i)\right]\leqslant\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\sum_{i=1}^{n}\varepsilon_i t_i\right].$$

4 [**General Talagrand Contraction Principle**] Generalize the Talagrand Contraction Principle from Exercise Q7.3 (*Talagrand's Contraction Principle*).

Let $\mathbb{T} \subset \mathbb{R}^n$, and let $\varphi_i : \mathbb{R} \to \mathbb{R}$ be arbitrary Lipschitz functions, and $\varepsilon_1, \cdots, \varepsilon_n$ i.i.d. $\sim$ Rademacher. State and prove the generalized Talagrand Contraction Principle inequality.

Rk. We will give two levels of generalizations.

Global bound Talagrand Contraction Principle Let $L = \max_{1 \leqslant i \leqslant n} \|\varphi_i\|_{\text{Lip}}$. Then

$$\mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i \varphi_i(t_i)\right] \leqslant L \cdot \mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i t_i\right].$$

Component-wise bound Talagrand Contraction Principle Let $L_i = \|\varphi_i\|_{\text{Lip}}$. Then

$$\mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i \varphi_i(t_i)\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i L_i t_i\right].$$

Prf. If $L = 0$, then all φ_i are constant functions, in which case

$$\mathbb{E}_{\varepsilon}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i c_i\right] = \mathbb{E}_{\varepsilon}\left[\sum_{i=1}^{n} \varepsilon_i c_i\right] = 0,$$

which is trivial.

If $L > 0$, define $\psi_i(x) = \frac{1}{L}\varphi_i(x)$, which satisfies $\|\psi_i\|_{\text{Lip}} \leqslant 1$. Then by Exercise Q7.3 (*Talagrand's Contraction Principle*), we obtain

$$\mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i \psi_i(t_i)\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i t_i\right],$$

i.e.,

$$\mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i \varphi_i(t_i)\right] \leqslant L \cdot \mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i t_i\right].$$

If $L_i = 0$, then $\varphi_i(t_i)$ is constant and its contribution to the Rademacher expectation is 0, so the conclusion is trivial; we only need to consider $L_i > 0$.

Construct a new set $\mathbb{S} \subset \mathbb{R}^n$ via a diagonal matrix linear transformation as

$$\mathbb{S} = \{\boldsymbol{s} = (L_1 t_1, L_2 t_2, \cdots, L_n t_n) : \boldsymbol{t} \in \mathbb{T}\},$$

Define $h_i(s) = \varphi_i(\frac{s}{L_i})$. Then

$$|h_i(u) - h_i(v)| = \left|\varphi_i\left(\frac{u}{L_i}\right) - \varphi_i\left(\frac{v}{L_i}\right)\right| \leqslant L_i \left|\frac{u}{L_i} - \frac{v}{L_i}\right| = |u - v|,$$

so $\|h_i\|_{\text{Lip}} \leqslant 1$. Therefore, by Exercise Q7.3 (*Talagrand's Contraction Principle*), we obtain

$$\mathbb{E}\left[\sup_{\boldsymbol{s} \in \mathbb{S}} \sum_{i=1}^{n} \varepsilon_i h_i(s_i)\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{s} \in \mathbb{S}} \sum_{i=1}^{n} \varepsilon_i s_i\right],$$

For the left-hand side, $h_i(s_i) = h_i(L_i t_i) = \varphi_i(t_i)$; for the right-hand side, $s_i = L_i t_i$; substituting these in, we obtain

$$\mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i \varphi_i(t_i)\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{t} \in \mathbb{T}} \sum_{i=1}^{n} \varepsilon_i L_i t_i\right].$$

Rk. This exercise shows that contraction maps do not change the structure of the Rademacher complexity, but merely rescale $\mathbb{T}$ along each coordinate axis.

5 [**Representing a Random Walk as a Canonical Process**] Consider the N-step Gaussian random walk of Example E7.1.3 (*Random walks*) with increments Z_i i.i.d. $\sim N(0, 1)$. Express the random walk as a canonical Gaussian process on $\mathbb{T} \subset \mathbb{R}^N$.

Prf. Let $(X_n)_{n=1}^N$ be the N-step standard Gaussian random walk. Note that

$$X_n = \sum_{i=1}^n Z_i = Z_1 \cdot 1 + Z_2 \cdot 1 + \cdots + Z_n \cdot 1 + Z_{n+1} \cdot 0 + \cdots + Z_N \cdot 0,$$

Hence define $\boldsymbol{t}_n = \sum_{i=1}^n \boldsymbol{e}_i$, i.e.,

$$\boldsymbol{t}_n = (\underbrace{1, 1, \cdots, 1}_{n}, \underbrace{0, \cdots, 0}_{N-n}),$$

which form the index set $\mathbb{T} = \{\boldsymbol{t}_1, \boldsymbol{t}_2, \cdots, \boldsymbol{t}_N\} \subset \mathbb{R}^N$ (*if one requires the inclusion of the starting point* X_0*, then add* $\boldsymbol{t}_0 = (0, \cdots, 0)$ *to* $\mathbb{T}$). Thus the standard Gaussian random walk $(X_n)_{n=1}^N$ is equivalent to the canonical Gaussian process $(\langle \boldsymbol{Z}, \boldsymbol{t} \rangle)_{\boldsymbol{t} \in \mathbb{T}}$.

6 [**Multivariate Gaussian Integration by Parts**] Prove Lemma L7.2.4 (*Multivariate Gaussian integration by parts*), also known as the multivariate Stein's lemma.

Multivariate Stein's Lemma Let $\boldsymbol{X} \sim N_n(\boldsymbol{0}, \boldsymbol{\Sigma})$. For any differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, assuming the following expectations exist and are finite, then

$$\mathbb{E}[\boldsymbol{X} f(\boldsymbol{X})] = \boldsymbol{\Sigma} \cdot \mathbb{E}[\nabla f(\boldsymbol{X})],$$

which is equivalent to the following component-wise form,

$$\mathbb{E}[X_i f(\boldsymbol{X})] = \sum_{j=1}^n \Sigma_{ij} \mathbb{E}\left[\frac{\partial f}{\partial x_j}(\boldsymbol{X})\right].$$

Prf. Since $\boldsymbol{\Sigma}$ is a positive semidefinite symmetric matrix, there exists $\boldsymbol{A} = \boldsymbol{\Sigma}^{1/2}$ such that $\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{\Sigma}$.
Let $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{g}$, where $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Then $\mathbb{E}[\boldsymbol{A}\boldsymbol{g}(\boldsymbol{A}\boldsymbol{g})^\top] = \boldsymbol{A}\mathbb{E}[\boldsymbol{g}\boldsymbol{g}^\top]\boldsymbol{A}^\top = \boldsymbol{A}\boldsymbol{I}_n\boldsymbol{A}^\top = \boldsymbol{\Sigma}$, so $\boldsymbol{X} \overset{d}{=} \boldsymbol{A}\boldsymbol{g}$, i.e.,

$$X_i = \sum_{k=1}^n A_{ik} g_k.$$

Consider the expectation of the i-th component,

$$\mathbb{E}[X_i f(\boldsymbol{X})] = \mathbb{E}\left[\left(\sum_{k=1}^n A_{ik} g_k\right) f(\boldsymbol{A}\boldsymbol{g})\right] = \sum_{k=1}^n A_{ik} \mathbb{E}[g_k f(\boldsymbol{A}\boldsymbol{g})], \tag{I}$$

Then we only need to compute $\mathbb{E}[g_k f(\boldsymbol{A}\boldsymbol{g})]$.
By the univariate Stein's lemma, we obtain

$$\mathbb{E}[g_k f(\boldsymbol{A}\boldsymbol{g})] = \mathbb{E}\left[\frac{\partial}{\partial g_k} f(\boldsymbol{A}\boldsymbol{g})\right],$$

where

$$\begin{aligned}
\frac{\partial}{\partial g_k} f(\boldsymbol{A}\boldsymbol{g}) &= \sum_{j=1}^n \frac{\partial f}{\partial x_j}(\boldsymbol{A}\boldsymbol{g}) \cdot \frac{\partial (\boldsymbol{A}\boldsymbol{g})_j}{\partial g_k} \\
&= \sum_{j=1}^n \frac{\partial f}{\partial x_j}(\boldsymbol{A}\boldsymbol{g}) \cdot \frac{\partial \boldsymbol{X}_j}{\partial g_k} \\
&= \sum_{j=1}^n \frac{\partial f}{\partial x_j}(\boldsymbol{A}\boldsymbol{g}) \cdot \frac{\partial \sum_{l=1}^n A_{jl} g_l}{\partial g_k} \\
&= \sum_{j=1}^n \frac{\partial f}{\partial x_j}(\boldsymbol{X}) \cdot A_{jk},
\end{aligned}$$

substituting this in, we obtain

$$\mathbb{E}[g_k f(\boldsymbol{A}\boldsymbol{g})] = \mathbb{E}\left[\sum_{j=1}^n A_{jk} \frac{\partial f}{\partial x_j}(\boldsymbol{X})\right], \tag{II}$$

Substituting (II) into (I), we obtain

$$\begin{aligned}\mathbb{E}[X_i f(\boldsymbol{X})] &= \sum_{k=1}^{n} A_{ik}\left(\sum_{j=1}^{n} A_{jk}\mathbb{E}\left[\frac{\partial f}{\partial x_j}(\boldsymbol{X})\right]\right)\\ &= \sum_{j=1}^{n}\left(\sum_{k=1}^{n} A_{ik}A_{jk}\right)\mathbb{E}\left[\frac{\partial f}{\partial x_j}(\boldsymbol{X})\right]\\ &= \sum_{j=1}^{n}\Sigma_{ij}\mathbb{E}\left[\frac{\partial f}{\partial x_j}(\boldsymbol{X})\right] \qquad \left(\sum_{k=1}^{n} A_{ik}A_{jk} = \Sigma_{ij}\right)\\ &= \boldsymbol{\Sigma}\cdot\mathbb{E}[\nabla f(\boldsymbol{X})].\end{aligned}$$

7 [**Derivative of the Expected Log-Partition Function**] In the proof of the Sudakov-Fernique Theorem (Theorem T7.2.8 (*Sudakov-Fernique inequality*) (*Sudakov–Fernique inequality*)), we need to prove a key step: for the interpolating Gaussian process $\boldsymbol{Z}(u) = \sqrt{u}\boldsymbol{X} + \sqrt{1-u}\boldsymbol{Y}$ (where $u \in [0,1]$), the log-partition function

$$f(\boldsymbol{x}) = \frac{1}{\beta}\log\sum_{i=1}^{n} \mathrm{e}^{\beta x_i},$$

satisfies the monotonicity

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathbb{E}[f(\boldsymbol{Z}(u))] \leqslant 0.$$

(*The premise is that* $\boldsymbol{X}, \boldsymbol{Y}$ *are mutually independent zero-mean multivariate Gaussian vectors satisfying* $\mathbb{E}[(X_i - X_j)^2] \leqslant \mathbb{E}[(Y_i - Y_j)^2]$.)

Following the steps below, prove the above conclusion:

(1) Differentiate the function f and prove that:

$$\frac{\partial f}{\partial x_i} = \frac{\mathrm{e}^{\beta x_i}}{\sum_k \mathrm{e}^{\beta x_k}} = p_i(\boldsymbol{x}),\ \frac{\partial^2 f}{\partial x_i \partial x_j} = \beta(\delta_{ij}p_i(\boldsymbol{x}) - p_i(\boldsymbol{x})p_j(\boldsymbol{x})),$$

where δ_{ij} is the Kronecker delta symbol.

(2) Apply Lemma L7.2.5 (*Gaussian interpolation*) to simplify the expression to

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathbb{E}[f(\boldsymbol{Z}(u))] = \frac{\beta}{4}\sum_{i\neq j}(\mathbb{E}[(X_i - X_j)^2] - \mathbb{E}[(Y_i - Y_j)^2])\mathbb{E}[p_i(\boldsymbol{Z}(u))p_j(\boldsymbol{Z}(u))].$$

(3) Explain that, under the given condition, the above expression is non-positive.

Prf. (1) Let the partition function be $S(\boldsymbol{x}) = \sum_{k=1}^{n} \mathrm{e}^{\beta x_k}$. Then $f(\boldsymbol{x}) = \frac{1}{\beta}\log S(\boldsymbol{x})$.

The first-order partial derivative is

$$\frac{\partial f}{\partial x_i} = \frac{1}{\beta}\cdot\frac{1}{S(\boldsymbol{x})}\cdot\frac{\partial S}{\partial x_i} = \frac{1}{\beta}\frac{1}{S(\boldsymbol{x})}(\beta\mathrm{e}^{\beta x_i}) = \frac{\mathrm{e}^{\beta x_i}}{\sum_{k=1}^{n}\mathrm{e}^{\beta x_k}} = p_i(\boldsymbol{x}),$$

Note that $p_i(\boldsymbol{x})$ satisfies $p_i(\boldsymbol{x}) > 0$, $\sum_{k=1}^{n} p_k(\boldsymbol{x}) = 1$, so it forms a probability distribution (*i.e., the Softmax function*).

For the second-order partial derivatives, we discuss two cases:

a. When $i = j$,

$$\frac{\partial p_i}{\partial x_i} = \frac{(\beta\mathrm{e}^{\beta x_i})S(\boldsymbol{x}) - \mathrm{e}^{\beta x_i}(\beta\mathrm{e}^{\beta x_i})}{[S(\boldsymbol{x})]^2} = \beta\left(\frac{\mathrm{e}^{\beta x_i}}{S(\boldsymbol{x})} - \left(\frac{\mathrm{e}^{\beta x_i}}{S(\boldsymbol{x})}\right)^2\right) = \beta(p_i - p_i^2).$$

b. When $i \neq j$,

$$\frac{\partial p_i}{\partial x_j} = \frac{0\cdot S(\boldsymbol{x}) - \mathrm{e}^{\beta x_i}(\beta\mathrm{e}^{\beta x_j})}{[S(\boldsymbol{x})]^2} = -\beta\left(\frac{\mathrm{e}^{\beta x_i}}{S(\boldsymbol{x})}\right)\left(\frac{\mathrm{e}^{\beta x_j}}{S(\boldsymbol{x})}\right) = -\beta p_i p_j.$$

Thus,

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \beta(\delta_{ij} p_i(\boldsymbol{x}) - p_i(\boldsymbol{x}) p_j(\boldsymbol{x})). \tag{I}$$

(2) By Lemma L7.2.5 (*Gaussian interpolation*), for mutually independent zero-mean Gaussian vectors $\boldsymbol{X}, \boldsymbol{Y}$, the process $\boldsymbol{Z}(u) = \sqrt{u}\boldsymbol{X} + \sqrt{1-u}\boldsymbol{Y}$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathbb{E}[f(\boldsymbol{Z}(u))] = \frac{1}{2}\sum_{i,j=1}^{n}(\mathbb{E}[X_i X_j] - \mathbb{E}[Y_i Y_j])\mathbb{E}\left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{Z}(u))\right],$$

For brevity, let $C_{ij} = \mathbb{E}[X_i X_j] - \mathbb{E}[Y_i Y_j]$, and omit the variable $\boldsymbol{Z}(u)$ inside the expectation. Using the conclusion of (1), we obtain

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathbb{E}[f] = \frac{1}{2}\sum_{i,j=1}^{n} C_{ij}\mathbb{E}[\beta(\delta_{ij}p_i - p_i p_j)] = \frac{\beta}{2}\mathbb{E}\left[\sum_{i,j=1}^{n} C_{ij}(\delta_{ij}p_i - p_i p_j)\right]. \tag{II}$$

For the inner summation, split it into diagonal ($i = j$) and off-diagonal ($i \neq j$),

$$\sum_{i,j=1}^{n} C_{ij}(\delta_{ij}p_i - p_i p_j) = \sum_{i=1}^{n} C_{ii}(p_i - p_i^2) - \sum_{i\neq j} C_{ij} p_i p_j.$$

Where

$$p_i - p_i^2 = p_i(1-p_i) = p_i \sum_{j\neq i} p_j = \sum_{j\neq i} p_i p_j,$$

substituting this in, we obtain

$$\begin{aligned}
\sum_{i,j=1}^{n} C_{ij}(\delta_{ij}p_i - p_i p_j) &= \sum_{i=1}^{n} C_{ii}\left(\sum_{j\neq i} p_i p_j\right) - \sum_{i\neq j} C_{ij} p_i p_j \\
&= \sum_{i\neq j} C_{ii} p_i p_j - \sum_{i\neq j} C_{ij} p_i p_j \\
&= \sum_{i\neq j} \frac{1}{2}(C_{ii} + C_{jj}) p_i p_j - \sum_{i\neq j} C_{ij} p_i p_j \\
&= \frac{1}{2}\sum_{i\neq j}(C_{ii} + C_{jj} - 2C_{ij}) p_i p_j,
\end{aligned}$$

where

$$\begin{aligned}
C_{ii} + C_{jj} - 2C_{ij} &= (\mathbb{E}[X_i^2] - \mathbb{E}[Y_i^2]) + (\mathbb{E}[X_j^2] - \mathbb{E}[Y_j^2]) - 2(\mathbb{E}[X_i X_j] - \mathbb{E}[Y_i Y_j]) \\
&= (\mathbb{E}[X_i^2] - 2\mathbb{E}[X_i X_j] + \mathbb{E}[X_j^2]) - (\mathbb{E}[Y_i^2] - 2\mathbb{E}[Y_i Y_j] + \mathbb{E}[Y_j^2]) \\
&= \mathbb{E}[(X_i - X_j)^2] - \mathbb{E}[(Y_i - Y_j)^2],
\end{aligned}$$

Substituting this into (II), we obtain

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}u}\mathbb{E}[f] &= \frac{\beta}{2}\mathbb{E}\left[\frac{1}{2}\sum_{i\neq j}(\mathbb{E}[(X_i - X_j)^2] - \mathbb{E}[(Y_i - Y_j)^2]) p_i p_j\right] \\
&= \frac{\beta}{4}\sum_{i\neq j}(\mathbb{E}[(X_i - X_j)^2] - \mathbb{E}[(Y_i - Y_j)^2])\mathbb{E}[p_i(\boldsymbol{Z}(u)) p_j(\boldsymbol{Z}(u))].
\end{aligned}$$

(3) By the hypothesis of the Sudakov-Fernique Theorem, the incremental second moments of the stochastic processes satisfy $\mathbb{E}[(X_i - X_j)^2] \leqslant \mathbb{E}[(Y_i - Y_j)^2]$, hence

$$\mathbb{E}[(X_i - X_j)^2] - \mathbb{E}[(Y_i - Y_j)^2] \leqslant 0.$$

Since $p_i(\boldsymbol{x}), p_j(\boldsymbol{x}) > 0$, we have

$$\mathbb{E}[p_i(\boldsymbol{Z}(u)) p_j(\boldsymbol{Z}(u))] > 0.$$

Thus,

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathbb{E}[f(\boldsymbol{Z}(u))] \leqslant 0.$$

8 [**Gaussian Contraction Inequality**] This is a Gaussian version of the Talagrand contraction principle from Exercise Q7.3 (*Talagrand's Contraction Principle*).

Let $\mathbb{T} \subset \mathbb{R}^n$, $g_1, \cdots, g_n$ i.i.d. $\sim N(0,1)$, and $\varphi_i : \mathbb{R} \to \mathbb{R}$ be contraction maps, i.e., satisfying $\|\varphi_i\|_{\mathrm{Lip}} \leqslant 1$. Using Theorem T7.2.8 (*Sudakov-Fernique inequality*) (*Sudakov–Fernique inequality*), prove that:

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} \sum_{i=1}^{n} g_i \varphi_i(t_i)\right] \leqslant \mathbb{E}\left[\sup_{t\in\mathbb{T}} \sum_{i=1}^{n} g_i t_i\right].$$

Prf. For any $\boldsymbol{t} = (t_1, \cdots, t_n) \in \mathbb{T}$, define the stochastic processes $(X_t)_{t\in\mathbb{T}}, (Y_t)_{t\in\mathbb{T}}$ as

$$X_t = \sum_{i=1}^{n} g_i \varphi_i(t_i), Y_t = \sum_{i=1}^{n} g_i t_i,$$

then $(X_t), (Y_t)$ are both zero-mean Gaussian processes.

Using Theorem T7.2.8 (*Sudakov-Fernique inequality*) (*Sudakov–Fernique inequality*), we have for any t, s,

$$\mathbb{E}[(X_t - X_s)^2] \leqslant \mathbb{E}[(Y_t - Y_s)^2],$$

hence

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \leqslant \mathbb{E}\left[\sup_{t\in\mathbb{T}} Y_t\right],$$

We only need to verify that the Sudakov-Fernique condition holds.

For $\mathbb{E}[(X_t - X_s)^2], \mathbb{E}[(Y_t - Y_s)^2]$, substituting $\mathrm{Cov}(g_i, g_j) = 0$ $(i \neq j)$, $\mathrm{Var}(g_i) = 1$, we obtain

$$\mathbb{E}[(X_t - X_s)^2] = \mathbb{E}\left[\left(\sum_{i=1}^{n} g_i(\varphi_i(t_i) - \varphi_i(s_i))\right)^2\right] = \sum_{i=1}^{n} (\varphi_i(t_i) - \varphi_i(s_i))^2,$$

$$\mathbb{E}[(Y_t - Y_s)^2] = \mathbb{E}\left[\left(\sum_{i=1}^{n} g_i(t_i - s_i)\right)^2\right] = \sum_{i=1}^{n} (t_i - s_i)^2,$$

Since φ_i are contraction maps, we have

$$(\varphi_i(t_i) - \varphi_i(s_i))^2 \leqslant (t_i - s_i)^2,$$

summing over both sides,

$$\sum_{i=1}^{n} (\varphi_i(t_i) - \varphi_i(s_i))^2 \leqslant \sum_{i=1}^{n} (t_i - s_i)^2,$$

which is precisely the Sudakov-Fernique condition,

$$\mathbb{E}[(X_t - X_s)^2] \leqslant \mathbb{E}[(Y_t - Y_s)^2].$$

Rk. Physically speaking, if the parameter space $\mathbb{T}$ is compressed via contraction maps, the Gaussian complexity of this space does not increase.

Unlike the Talagrand contraction, the Talagrand Contraction Principle applies to Rademacher random variables, while this version is specifically for Gaussian random variables. Due to the rotational invariance of the Gaussian distribution, proving the Gaussian version using the Sudakov-Fernique inequality is often more direct than proving the Rademacher version.

9 [**Gordon's Inequality**] Under the additional equal-variance assumption (*for all* u, t, $\mathbb{E}[X_{ut}^2] = \mathbb{E}[Y_{ut}^2]$), prove Theorem T7.2.9 (*Gordon inequality*).

Gordon's Inequality Let $(X_{ut}), (Y_{ut})$ be zero-mean Gaussian processes indexed by the finite pairs (u, t) with $u \in \mathbb{U}$ and

$t \in \mathbb{T}$, satisfying:

i. Within-group increments are smaller: for all $u \in \mathbb{U}$, t, s,

$$\mathbb{E}[(X_{ut} - X_{us})^2] \leqslant \mathbb{E}[(Y_{ut} - Y_{us})^2];$$

ii. Cross-group increments are larger: for all $u \neq v$ and t, s,

$$\mathbb{E}[(X_{ut} - X_{vs})^2] \geqslant \mathbb{E}[(Y_{ut} - Y_{vs})^2];$$

Then

$$\mathbb{E}\left[\min_{u\in\mathbb{U}} \max_{t\in\mathbb{T}} X_{ut}\right] \leqslant \mathbb{E}\left[\min_{u\in\mathbb{U}} \max_{t\in\mathbb{T}} Y_{ut}\right].$$

(*Just as with the Sudakov-Fernique inequality, the equal-variance assumption in Gordon's inequality can actually be dropped, but this exercise does not require proving the case without the assumption.*)

Prf. Denote the difference of incremental second moments as

$$\Delta_{ut,vs} = \mathbb{E}[(X_{ut} - X_{vs})^2] - \mathbb{E}[(Y_{ut} - Y_{vs})^2],$$

then

$$\Delta_{ut,us} \begin{cases} \leqslant 0 & u = v \\ \geqslant 0 & u \neq v \end{cases},$$

Under the equal-variance assumption, the covariance difference $C_{ut,vs} = \mathbb{E}[X_{ut}X_{vs}] - \mathbb{E}[Y_{ut}Y_{vs}]$ can be expressed as

$$C_{ut,vs} = -\frac{1}{2}\Delta_{ut,vs}.$$

- Construct a smooth approximation function

  (*To approximate* $\min_{u\in\mathbb{U}} \max_{t\in\mathbb{T}} x_{ut}$*, we need to construct a smooth approximation function for it.*)

  Introduce parameters $\beta, \gamma > 0$, and construct the nested log-partition function as

  $$f(\boldsymbol{x}) = -\frac{1}{\beta} \log \sum_{u\in\mathbb{U}} \exp\left(-\frac{\beta}{\gamma} \log \sum_{t\in\mathbb{T}} \mathrm{e}^{\gamma x_{ut}}\right),$$

  Then as $\gamma \to \infty$, the inner part tends to $\max_{t\in\mathbb{T}} x_{ut}$; then as $\beta \to \infty$, the outer part tends to $\min_{u\in\mathbb{U}} \max_{t\in\mathbb{T}} x_{ut}$.

  Then define the associated Softmax weights as

  $$\left(\textit{Within-group weights, satisfying} \sum_{t\in\mathbb{T}} p_{ut} = 1\right) \quad p_{ut} = \frac{\mathrm{e}^{\gamma x_{ut}}}{\sum_{s\in\mathbb{T}} \mathrm{e}^{\gamma x_{us}}},$$

  $$\left(\textit{Cross-group weights, satisfying} \sum_{u\in\mathbb{U}} q_u = 1\right) \quad q_u = \frac{\left(\sum_{s\in\mathbb{T}} \mathrm{e}^{\gamma x_{us}}\right)^{-\beta/\gamma}}{\sum_v \left(\sum_{s\in\mathbb{T}} \mathrm{e}^{\gamma x_{vs}}\right)^{-\beta/\gamma}},$$

  Let the global weights be $w_{ut} = q_u p_{ut}$, then $\sum_{u\in\mathbb{U},t\in\mathbb{T}} w_{ut} = 1$.

- Compute partial derivatives

  (*The process is similar to Exercise Q7.7 (Derivative of the Expected Log-Partition Function).*)

  The first-order partial derivatives are

  $$\frac{\partial f}{\partial x_{ut}} = q_u p_{ut} = w_{ut}.$$

  For the second-order partial derivatives, we discuss two cases:

  a. If $u \neq v$, i.e., cross-group differentiation,

  $$\frac{\partial^2 f}{\partial x_{ut} \partial x_{vs}} = \beta q_u p_{ut} q_v p_{vs} = \beta w_{ut} w_{vs}.$$

  b. If $u = v$, i.e., within-group differentiation,

  $$\frac{\partial^2 f}{\partial x_{ut} \partial x_{us}} = \gamma q_u (\delta_{ts} p_{ut} - p_{ut} p_{us}) - \beta q_u (1 - q_u) p_{ut} p_{us}.$$

- Apply Gaussian interpolation

  Let the interpolating process be $\boldsymbol{Z}(r) = \sqrt{r}\boldsymbol{X} + \sqrt{1-r}\boldsymbol{Y}$, where $r \in [0,1]$. Then by the Gaussian interpolation lemma, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}r}\mathbb{E}[f(\boldsymbol{Z}(r))] = \frac{1}{2}\sum_{u,t}\sum_{v,s} C_{ut,vs}\mathbb{E}\left[\frac{\partial^2 f}{\partial x_{ut}\partial x_{vs}}\right],$$

  Split the inner sum into

$$\Sigma = \Sigma_{u\neq v} + \Sigma_{u=v}.$$

  For the cross-group part $\Sigma_{u\neq v}$,

$$\Sigma_{u\neq v} = \sum_{u\neq v}\sum_{t,s}\left(-\frac{1}{2}\Delta_{ut,vs}\right)(\beta w_{ut}w_{vs}) = -\frac{\beta}{2}\sum_{u\neq v,t,s}\Delta_{ut,vs}w_{ut}w_{vs} \leqslant 0.$$

  For the within-group part $\Sigma_{u=v}$,

$$\Sigma_{u=v} = \sum_{u}\sum_{t,s} C_{ut,us}[\gamma q_u(\delta_{ts}p_{ut} - p_{ut}p_{us}) - \beta q_u(1-q_u)p_{ut}p_{us}],$$

  For the first term, substitute the symmetrization identity from Exercise Q7.7 (*Derivative of the Expected Log-Partition Function*)

$$\sum_{t,s} C_{ts}(\delta_{ts}p_t - p_t p_s) = \frac{1}{2}\sum_{t,s}(C_{tt} + C_{ss} - 2C_{ts})p_t p_s = \frac{1}{2}\sum_{t,s}\Delta_{ts}p_t p_s,$$

  For the second term, directly substitute $C_{ut,us} = -\frac{1}{2}\Delta_{ut,us}$ to obtain

$$\begin{aligned}\Sigma_{u=v} &= \gamma\sum_u q_u\left(\frac{1}{2}\sum_{t,s}\Delta_{ut,us}p_{ut}p_{us}\right) - \beta\sum_u q_u(1-q_u)\sum_{t,s}\left(-\frac{1}{2}\Delta_{ut,us}\right)p_{ut}p_{us}\\ &= \sum_u\left[\frac{\gamma}{2}q_u + \frac{\beta}{2}q_u(1-q_u)\right]\sum_{t,s}\Delta_{ut,us}p_{ut}p_{us} \leqslant 0.\end{aligned}$$

Thus,

$$\frac{\mathrm{d}}{\mathrm{d}r}\mathbb{E}[f(\boldsymbol{Z}(r))] = \frac{1}{2}\mathbb{E}[\Sigma_{u\neq v} + \Sigma_{u=v}] \leqslant 0,$$

This shows that as r increases from 0 to 1, the process smoothly transitions from $\boldsymbol{Y}$ to $\boldsymbol{X}$, and the expected smooth approximation function is monotonically decreasing. Hence

$$\mathbb{E}[f(\boldsymbol{X})] \leqslant \mathbb{E}[f(\boldsymbol{Y})],$$

Letting $\gamma, \beta \to \infty$,

$$\mathbb{E}\left[\min_{u\in\mathbb{U}}\max_{t\in\mathbb{T}} X_{ut}\right] \leqslant \mathbb{E}\left[\min_{u\in\mathbb{U}}\max_{t\in\mathbb{T}} Y_{ut}\right].$$

Rk. Complete proof without the equal-variance assumption:

Additionally define the single-point variance difference as

$$V_{ut} = \mathbb{E}[X_{ut}^2] - \mathbb{E}[Y_{ut}^2],$$

then

$$C_{ut,vs} = \frac{1}{2}(V_{ut} + V_{vs} - \Delta_{ut,vs}).$$

(*The construction of the smooth approximation function is the same as before and is omitted here.*)

Define the interpolating process $\boldsymbol{Z}(r) = \sqrt{r}\boldsymbol{X} + \sqrt{1-r}\boldsymbol{Y}$, where $r \in [0,1]$. Then by the Gaussian interpolation lemma, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}r}\mathbb{E}[f(\boldsymbol{Z}(r))] = \frac{1}{2}\sum_{u,t}\sum_{v,s} C_{ut,vs}\mathbb{E}\left[\frac{\partial^2 f}{\partial x_{ut}\partial x_{vs}}\right].$$

Split the interior of the expectation into the V-containing part S_V and the Δ-containing part S_Δ, whose expressions are

$$S_V = \frac{1}{4}\sum_{u,t}\sum_{v,s}(V_{ut}+V_{vs})\frac{\partial^2 f}{\partial x_{ut}\partial x_{vs}},$$

$$S_\Delta = -\frac{1}{4}\sum_{u\neq v}\sum_{t,s}\Delta_{ut,vs}\beta w_{ut}w_{vs} - \frac{1}{4}\sum_{u}\sum_{t,s}\Delta_{ut,us}(\gamma q_u(\delta_{ts}p_{ut}-p_{ut}p_{us})-\beta q_u(1-q_u)p_{ut}p_{us}).$$

- Compute S_V

  For S_V, note that $(u,t),(v,s)$ satisfy the symmetry

  $$\frac{\partial^2 f}{\partial x_{ut}\partial x_{vs}} = \frac{\partial^2 f}{\partial x_{vs}\partial x_{ut}},$$

  hence

  $$S_V = \frac{1}{2}\sum_{u,t}V_{ut}\left(\sum_{v,s}\frac{\partial^2 f}{\partial x_{ut}\partial x_{vs}}\right).$$

  Split the inner sum of second-order derivatives into $v\neq u, v=u$,

  $$\begin{aligned}\sum_{v,s}\frac{\partial^2 f}{\partial x_{ut}\partial x_{vs}} &= \sum_{v\neq u}\sum_{s}\frac{\partial^2 f}{\partial x_{ut}\partial x_{vs}} + \sum_{s}\frac{\partial^2 f}{\partial x_{ut}\partial x_{us}}\\ &= \sum_{v\neq u}\sum_{s\in\mathbb{T}}\beta w_{ut}w_{vs} + \sum_{s\in\mathbb{T}}(\gamma q_u(\delta_{ts}p_{ut}-p_{ut}p_{us})-\beta q_u(1-q_u)p_{ut}p_{us}),\end{aligned}$$

  where

  $$\sum_{v\neq u}\sum_{s}\beta w_{ut}q_v p_{vs} = \beta w_{ut}\sum_{v\neq u}q_v\left(\sum_s p_{vs}\right) = \beta w_{ut}\sum_{v\neq u}q_v = \beta w_{ut}(1-q_u),$$

  $$\sum_{s}\gamma q_u(\delta_{ts}p_{ut}-p_{ut}p_{us}) = \gamma q_u\left(\sum_s\delta_{ts}p_{ut}-p_{ut}\sum_s p_{us}\right) = \gamma q_u(p_{ut}\cdot 1 - p_{ut}\cdot 1) = 0,$$

  $$\sum_{s}-\beta q_u(1-q_u)p_{ut}p_{us} = -\beta q_u(1-q_u)p_{ut}\left(\sum_s p_{us}\right) = -\beta q_u p_{ut}(1-q_u) = -\beta w_{ut}(1-q_u),$$

  Substituting these in, we obtain

  $$\sum_{v,s}\frac{\partial^2 f}{\partial x_{ut}\partial x_{vs}} = \beta w_{ut}(1-q_u)+0-\beta w_{ut}(1-q_u)\equiv 0,$$

  This shows that the effect of the variance difference is canceled out by the summation of partial derivatives.

- Compute S_Δ

  Split the sum in S_Δ into $v\neq u$ and $v=u$,

  $$S_\Delta = -\frac{1}{4}\sum_{u\neq v}\sum_{t,s}\Delta_{ut,vs}\beta w_{ut}w_{vs} - \frac{1}{4}\sum_{u}\sum_{t,s}\Delta_{ut,us}(\gamma q_u(\delta_{ts}p_{ut}-p_{ut}p_{us})-\beta q_u(1-q_u)p_{ut}p_{us}).$$

  For the cross-group term ($u\neq v$), it is easy to see that

  $$-\frac{1}{4}\sum_{u\neq v}\sum_{t,s}\Delta_{ut,vs}(\beta w_{ut}w_{vs})\leqslant 0.$$

For the within-group term ($u = v$), using the identity $\sum_{t,s} \Delta_{ut,us}\delta_{ts}p_{ut} = \sum_t \Delta_{ut,ut}p_{ut} = 0$, we obtain

$$-\frac{1}{4}\sum_u \sum_{t,s} \Delta_{ut,us}(\gamma q_u(\delta_{ts}p_{ut} - p_{ut}p_{us}) - \beta q_u(1-q_u)p_{ut}p_{us}) = \sum_u \left(\frac{\gamma}{4}q_u + \frac{\beta}{4}q_u(1-q_u)\right)\sum_{t,s}\Delta_{ut,us}p_{ut}p_{us} \leqslant 0.$$

Thus, $S_\Delta \leqslant 0$.

In summary,

$$\frac{\mathrm{d}}{\mathrm{d}r}\mathbb{E}[f(\boldsymbol{Z}(r))] \leqslant 0,$$

and consequently

$$\mathbb{E}[f(\boldsymbol{X})] \leqslant \mathbb{E}[f(\boldsymbol{Y})],$$

Letting $\gamma, \beta \to \infty$,

$$\mathbb{E}\left[\min_{u\in\mathbb{U}}\max_{t\in\mathbb{T}} X_{ut}\right] \leqslant \mathbb{E}\left[\min_{u\in\mathbb{U}}\max_{t\in\mathbb{T}} Y_{ut}\right].$$

10 [**Frobenius Distance between Rank-One Matrices**] In the proof of Theorem T7.3.1 (*Norms of Gaussian random matrices*), we used the following bound.

Let $\boldsymbol{u}, \boldsymbol{w} \in \mathbb{S}^{n-1}$ and $\boldsymbol{v}, \boldsymbol{z} \in \mathbb{S}^{m-1}$. Prove that:

$$\|\boldsymbol{u}\boldsymbol{v}^\top - \boldsymbol{w}\boldsymbol{z}^\top\|_F^2 \leqslant \|\boldsymbol{u}-\boldsymbol{w}\|_2^2 + \|\boldsymbol{v}-\boldsymbol{z}\|_2^2.$$

Rk. For rank-one matrices $\boldsymbol{a}\boldsymbol{b}^\top, \boldsymbol{c}\boldsymbol{d}^\top$, their Frobenius inner product satisfies

$$\langle \boldsymbol{a}\boldsymbol{b}^\top, \boldsymbol{c}\boldsymbol{d}^\top\rangle_F = \mathrm{tr}((\boldsymbol{a}\boldsymbol{b}^\top)^\top(\boldsymbol{c}\boldsymbol{d}^\top)) = \mathrm{tr}(\boldsymbol{b}\boldsymbol{a}^\top\boldsymbol{c}\boldsymbol{d}^\top) = (\boldsymbol{a}^\top\boldsymbol{c})(\boldsymbol{b}^\top\boldsymbol{d}).$$

Prf. For the left-hand side,

$$\begin{aligned}
\mathrm{LHS} &= \|\boldsymbol{u}\boldsymbol{v}^\top - \boldsymbol{w}\boldsymbol{z}^\top\|_F^2 \\
&= \|\boldsymbol{u}\boldsymbol{v}^\top\|_F^2 + \|\boldsymbol{w}\boldsymbol{z}^\top\|_F^2 - 2\langle \boldsymbol{u}\boldsymbol{v}^\top, \boldsymbol{w}\boldsymbol{z}^\top\rangle_F \\
&= (\boldsymbol{u}^\top\boldsymbol{u})(\boldsymbol{v}^\top\boldsymbol{v}) + (\boldsymbol{w}^\top\boldsymbol{w})(\boldsymbol{z}^\top\boldsymbol{z}) - 2(\boldsymbol{u}^\top\boldsymbol{w})(\boldsymbol{v}^\top\boldsymbol{z}) \\
&= 1\cdot 1 + 1\cdot 1 - 2(\boldsymbol{u}^\top\boldsymbol{w})(\boldsymbol{v}^\top\boldsymbol{z}) = 2 - 2(\boldsymbol{u}^\top\boldsymbol{w})(\boldsymbol{v}^\top\boldsymbol{z}).
\end{aligned}$$

For the right-hand side,

$$\begin{aligned}
\mathrm{RHS} &= \|\boldsymbol{u}-\boldsymbol{w}\|_2^2 + \|\boldsymbol{v}-\boldsymbol{z}\|_2^2 \\
&= (\|\boldsymbol{u}\|_2^2 + \|\boldsymbol{w}\|_2^2 - 2\boldsymbol{u}^\top\boldsymbol{w}) + (\|\boldsymbol{v}\|_2^2 + \|\boldsymbol{z}\|_2^2 - 2\boldsymbol{v}^\top\boldsymbol{z}) \\
&= (1 + 1 - 2\boldsymbol{u}^\top\boldsymbol{w}) + (1 + 1 - 2\boldsymbol{v}^\top\boldsymbol{z}) = 4 - 2\boldsymbol{u}^\top\boldsymbol{w} - 2\boldsymbol{v}^\top\boldsymbol{z}.
\end{aligned}$$

To prove LHS $\leqslant$ RHS, we only need to prove

$$\mathrm{RHS} - \mathrm{LHS} = 2(1 - \boldsymbol{u}^\top\boldsymbol{w})(1 - \boldsymbol{v}^\top\boldsymbol{z}) \geqslant 0.$$

Since

$$|\boldsymbol{u}^\top\boldsymbol{w}| \leqslant \|\boldsymbol{u}\|_2\|\boldsymbol{w}\|_2 = 1, |\boldsymbol{v}^\top\boldsymbol{z}| \leqslant \|\boldsymbol{v}\|_2\|\boldsymbol{z}\|_2 = 1,$$

we have

$$(1 - \boldsymbol{u}^\top\boldsymbol{w})(1 - \boldsymbol{v}^\top\boldsymbol{z}) \geqslant 0.$$

11 [**Norm of GOE Random Matrices**] Apply the method of Section 7.3 to the symmetric case.

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a Gaussian random matrix whose entries on and above the main diagonal are mutually independent, satisfying

$$A_{ij} \sim \begin{cases} N(0,2), & i = j, \\ N(0,1), & i < j \end{cases}.$$

(*This is exactly the Gaussian orthogonal ensemble from Exercise Q3.19 (GOE Random Matrix).*)

(1) (*Expected bound*) Using Theorem T7.2.8 (*Sudakov-Fernique inequality*) (*Sudakov–Fernique inequality*), prove:

$$\mathbb{E}\|\boldsymbol{A}\| \leqslant 2\sqrt{n}.$$

(2) (High-probability bound) Use concentration inequalities to lift the expected bound from (1) to a high-probability bound. Prove: for any $t \geqslant 0$,

$$\mathbb{P}\{\|\boldsymbol{A}\| \geqslant 2\sqrt{n} + t\} \leqslant 2\exp(-ct^2). \qquad (c > 0)$$

Prf. (1) Since $\boldsymbol{A}$ is a real symmetric matrix, we have

$$\|\boldsymbol{A}\| = \sup_{\boldsymbol{u}\in\mathbb{S}^{n-1}} |\langle \boldsymbol{A}\boldsymbol{u}, \boldsymbol{u}\rangle|.$$

(*To remove the absolute value sign for the convenience of applying Gaussian process tools, consider introducing a sign variable.*)

Introduce a sign variable $\sigma \in \{-1, 1\}$, define the extended index set $\mathbb{T} = \{(\boldsymbol{u}, \sigma) : \boldsymbol{u} \in \mathbb{S}^{n-1}, \sigma \in \{-1, 1\}\}$, and construct the zero-mean Gaussian process $(X_{\boldsymbol{u},\sigma})$ on $\mathbb{T}$ as

$$X_{\boldsymbol{u},\sigma} = \sigma\langle \boldsymbol{A}\boldsymbol{u}, \boldsymbol{u}\rangle = \sigma \sum_{i,j=1}^{n} A_{ij}u_iu_j,$$

then $\sup_{(\boldsymbol{u},\sigma)\in\mathbb{T}} X_{\boldsymbol{u},\sigma} = \|\boldsymbol{A}\|$.

- Compute the incremental second moments of $(X_{\boldsymbol{u},\sigma})$

  Let $(\boldsymbol{u}, \sigma), (\boldsymbol{w}, \tau) \in \mathbb{T}$, and let the symmetric matrix $\boldsymbol{M} = \sigma\boldsymbol{u}\boldsymbol{u}^\top - \tau\boldsymbol{w}\boldsymbol{w}^\top$. Then the increment can be expressed as

  $$X_{\boldsymbol{u},\sigma} - X_{\boldsymbol{w},\tau} = \sum_{i,j=1}^{n} A_{ij}M_{ij},$$

  By the symmetry of $\boldsymbol{A}$, split the sum as

  $$\sum_{i,j=1}^{n} A_{ij}M_{ij} = \sum_{i=1}^{n} A_{ii}M_{ii} + 2\sum_{i<j} A_{ij}M_{ij},$$

  Compute the second moment, substituting $\mathrm{Var}(A_{ii}) = 2, \mathrm{Var}(A_{ij}) = 1$, we obtain

  $$\begin{aligned}
  \mathbb{E}[(X_{\boldsymbol{u},\sigma} - X_{\boldsymbol{w},\tau})^2] &= \sum_{i=1}^{n} \mathrm{Var}(A_{ii})M_{ii}^2 + 4\sum_{i<j} \mathrm{Var}(A_{ij})M_{ij}^2 \\
  &= 2\sum_{i=1}^{n} M_{ii}^2 + 4\sum_{i<j} M_{ij}^2 \\
  &= 2\left(\sum_{i=1}^{n} M_{ii}^2 + 2\sum_{i<j} M_{ij}^2\right) \\
  &= 2\|\boldsymbol{M}\|_F^2,
  \end{aligned}$$

  We only need to compute $2\|\boldsymbol{M}\|_F^2$.

  Computing $2\|\boldsymbol{M}\|_F^2$, substituting $\|\boldsymbol{u}\|_2 = \|\boldsymbol{w}\|_2 = 1$, $\sigma^2 = \tau^2 = 1$, we obtain

  $$\begin{aligned}
  2\|\boldsymbol{M}\|_F^2 &= 2(\sigma^2\|\boldsymbol{u}\boldsymbol{u}^\top\|_F^2 + \tau^2\|\boldsymbol{w}\boldsymbol{w}^\top\|_F^2 - 2\sigma\tau\langle \boldsymbol{u}\boldsymbol{u}^\top, \boldsymbol{w}\boldsymbol{w}^\top\rangle_F) \\
  &= 2(1 + 1 - 2\sigma\tau(\boldsymbol{u}^\top\boldsymbol{w})^2) \\
  &= 4(1 - \sigma\tau(\boldsymbol{u}^\top\boldsymbol{w})^2),
  \end{aligned}$$

  Let $y = \sigma\tau(\boldsymbol{u}^\top\boldsymbol{w}) \in [-1, 1]$. Then

  $$4(1 - y^2) = 4(1 - y)(1 + y) \leqslant 8(1 - y),$$

  hence

  $$\mathbb{E}[(X_{\boldsymbol{u},\sigma} - X_{\boldsymbol{w},\tau})^2] \leqslant 8(1 - \sigma\tau\boldsymbol{u}^\top\boldsymbol{w}) = 4\|\sigma\boldsymbol{u} - \tau\boldsymbol{w}\|_2^2.$$

- Construct a comparison process and apply Theorem T7.2.8 (*Sudakov-Fernique inequality*) (*Sudakov–Fernique inequality*)

  (*We need to construct a simpler canonical Gaussian process $(Y_{\boldsymbol{u},\sigma})$ whose increments are exactly $4\|\sigma\boldsymbol{u} - \tau\boldsymbol{w}\|_2^2$.*)

Let $\boldsymbol{g} \sim N(\boldsymbol{0}, 4\boldsymbol{I}_n)$, i.e., $\boldsymbol{g} = 2\boldsymbol{h}$, where $\boldsymbol{h} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Define the process $Y_{\boldsymbol{u},\sigma} = \langle \boldsymbol{g}, \sigma\boldsymbol{u} \rangle$. Then its incremental second moments are

$$\mathbb{E}[(Y_{\boldsymbol{u},\sigma} - Y_{\boldsymbol{w},\tau})^2] = \mathbb{E}[\langle \boldsymbol{g}, \sigma\boldsymbol{u} - \tau\boldsymbol{w} \rangle^2] = 4\|\sigma\boldsymbol{u} - \tau\boldsymbol{w}\|_2^2,$$

Hence

$$\mathbb{E}[(X_{\boldsymbol{u},\sigma} - X_{\boldsymbol{w},\tau})^2] \leqslant \mathbb{E}[(Y_{\boldsymbol{u},\sigma} - Y_{\boldsymbol{w},\tau})^2],$$

which is precisely the Sudakov-Fernique condition.

Using Theorem T7.2.8 (*Sudakov-Fernique inequality*) (*Sudakov–Fernique inequality*), we obtain

$$\begin{aligned}
\mathbb{E}\|\boldsymbol{A}\| &= \mathbb{E}\left[\sup_{(\boldsymbol{u},\sigma)\in\mathbb{T}} X_{\boldsymbol{u},\sigma}\right] \\
&\leqslant \mathbb{E}\left[\sup_{(\boldsymbol{u},\sigma)\in\mathbb{T}} Y_{\boldsymbol{u},\sigma}\right] = \mathbb{E}\left[\sup_{\boldsymbol{u}\in\mathbb{S}^{n-1},\sigma\in\{-1,1\}} \sigma\langle 2\boldsymbol{h}, \boldsymbol{u}\rangle\right] \\
&= 2\mathbb{E}\left[\sup_{\boldsymbol{u}\in\mathbb{S}^{n-1}} |\langle \boldsymbol{h}, \boldsymbol{u}\rangle|\right] \\
&= 2\mathbb{E}\|\boldsymbol{h}\|_2.
\end{aligned}$$

By Jensen's inequality, $\mathbb{E}\|\boldsymbol{h}\|_2 \leqslant (\mathbb{E}\|\boldsymbol{h}\|_2^2)^{1/2} = \sqrt{n}$, hence $\mathbb{E}\|\boldsymbol{A}\| \leqslant 2\sqrt{n}$.

(2) (*To use the Gaussian concentration inequality, we view $\boldsymbol{A}$ as a function of mutually independent standard normal random vectors.*)

A GOE matrix has $m = n(n+1)/2$ independent entries. Let $\boldsymbol{x} \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$, and explicitly parametrize $\boldsymbol{A}$ as a function of $\boldsymbol{x}$, denoted $\boldsymbol{A}(\boldsymbol{x})$. Then

$$A_{ii} = \begin{cases} \sqrt{2}x_{ii}, \text{ on the diagonal}, \\ x_{ij}, \quad i < j \end{cases}.$$

Define $f(\boldsymbol{x}) = \|\boldsymbol{A}(\boldsymbol{x})\|$. Then for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m$,

$$|f(\boldsymbol{x}) - f(\boldsymbol{y})| \leqslant \|\boldsymbol{A}(\boldsymbol{x}) - \boldsymbol{A}(\boldsymbol{y})\| \leqslant \|\boldsymbol{A}(\boldsymbol{x}) - \boldsymbol{A}(\boldsymbol{y})\|_F,$$

where

$$\begin{aligned}
\|\boldsymbol{A}(\boldsymbol{x}) - \boldsymbol{A}(\boldsymbol{y})\|_F^2 &= \sum_{i=1}^n (A_{ii}(\boldsymbol{x}) - A_{ii}(\boldsymbol{y}))^2 + 2\sum_{i<j} (A_{ij}(\boldsymbol{x}) - A_{ij}(\boldsymbol{y}))^2 \\
&= \sum_{i=1}^n (\sqrt{2}x_{ii} - \sqrt{2}y_{ii})^2 + 2\sum_{i<j} (x_{ij} - y_{ij})^2 \\
&= 2\sum_{i=1}^n (x_{ii} - y_{ii})^2 + 2\sum_{i<j} (x_{ij} - y_{ij})^2 \\
&= 2\|\boldsymbol{x} - \boldsymbol{y}\|_2^2,
\end{aligned}$$

hence $|f(\boldsymbol{x}) - f(\boldsymbol{y})| \leqslant \sqrt{2}\|\boldsymbol{x} - \boldsymbol{y}\|_2$, so $\|f\|_{\mathrm{Lip}} = \sqrt{2}$.

Using Theorem T7.1.11 (*Concentration of Gaussian processes*) (*Concentration of Gaussian processes*), we obtain

$$\mathbb{P}\{f(\boldsymbol{x}) \geqslant \mathbb{E}[f(\boldsymbol{x})] + t\} \leqslant \exp\left(-\frac{t^2}{2L^2}\right),$$

i.e.,

$$\mathbb{P}\{\|\boldsymbol{A}\| \geqslant \mathbb{E}\|\boldsymbol{A}\| + t\} \leqslant \exp\left(-\frac{t^2}{4}\right),$$

Using the conclusion of (1), we obtain

$$\mathbb{P}\{\|\boldsymbol{A}\| \geqslant 2\sqrt{n}+t\} \leqslant \mathbb{P}\{\|\boldsymbol{A}\| \geqslant \mathbb{E}\|\boldsymbol{A}\|+t\} \leqslant \exp\left(-\frac{t^2}{4}\right),$$

here let $c = 1/4$.

12 [**Norm of a Gaussian Vector**] Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Prove that

$$f(n) = \mathbb{E}\|\boldsymbol{g}\|_2 - \sqrt{n}$$

is increasing when $n \geqslant C$, where $C > 0$.

Prf. We only need to prove that there exists $C > 0$ such that when $n \geqslant C$, $f(n+1) \geqslant f(n)$, i.e.,

$$\mathbb{E}\|\boldsymbol{g}_{n+1}\|_2 - \mathbb{E}\|\boldsymbol{g}_n\|_2 \geqslant \sqrt{n+1} - \sqrt{n},$$

Let $a_n = \mathbb{E}\|\boldsymbol{g}_n\|_2$. Then the goal is

$$a_{n+1} - a_n \geqslant \sqrt{n+1} - \sqrt{n}. \tag{I}$$

- Algebraic transformation

  Since $\|\boldsymbol{g}_n\|_2^2 \sim \chi^2(n)$, we have

  $$a_n = \sqrt{2}\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}.$$

  Since

  $$\begin{aligned} a_n a_{n+1} &= \left(\sqrt{2}\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}\right)\left(\sqrt{2}\frac{\Gamma\left(\frac{n+2}{2}\right)}{\Gamma\left(\frac{n+1}{2}\right)}\right) = 2\frac{\Gamma\left(\frac{n}{2}+1\right)}{\Gamma\left(\frac{n}{2}\right)} \\ &= 2\left(\frac{n}{2}\right) = n, \qquad (\Gamma(x+1) = x\Gamma(x)) \end{aligned}$$

  Hence (I) becomes

  $$\frac{n}{a_n} - a_n \geqslant \sqrt{n+1} - \sqrt{n},$$

  which is

  $$n - a_n^2 \geqslant a_n(\sqrt{n+1} - \sqrt{n}) = \frac{a_n}{\sqrt{n+1}+\sqrt{n}}, \tag{II}$$

  where the left-hand side $n - a_n^2 = \mathbb{E}\|\boldsymbol{g}_n\|_2^2 - (\mathbb{E}\|\boldsymbol{g}_n\|_2)^2$ is precisely $\operatorname{Var}(\|\boldsymbol{g}_n\|_2)$.

- Use asymptotic expansion to compute both sides

  By Stirling's formula, as $z \to \infty$,

  $$\frac{\Gamma(z+\alpha)}{\Gamma(z)} = z^\alpha\left(1 + \frac{\alpha(\alpha-1)}{2z} + \frac{\alpha(\alpha-1)(\alpha-2)(3\alpha-1)}{24z^2} + O(z^{-3})\right).$$

  For the left-hand side of (II), let $z = n/2$, $\alpha = 1/2$. Then

  $$\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)} = \sqrt{\frac{n}{2}}\left(1 - \frac{1}{8(n/2)} + \frac{1}{128(n/2)^2} + O(n^{-3})\right) = \sqrt{\frac{n}{2}}\left(1 - \frac{1}{4n} + \frac{1}{32n^2} + O(n^{-3})\right).$$

  Squaring both sides,

  $$\begin{aligned} a_n^2 &= 2\left(\frac{n}{2}\right)\left(1 - \frac{1}{4n} + \frac{1}{32n^2} + O(n^{-3})\right)^2 \\ &= n\left(1 - \frac{1}{2n} + \left(\frac{1}{16} + \frac{2}{32}\right)\frac{1}{n^2} + O(n^{-3})\right) \\ &= n\left(1 - \frac{1}{2n} + \frac{1}{8n^2} + O(n^{-3})\right) \\ &= n - \frac{1}{2} + \frac{1}{8n} + O(n^{-2}), \end{aligned}$$

hence

$$\text{LHS} = n - a_n^2 = \frac{1}{2} - \frac{1}{8n} + O(n^{-2}).$$

For the right-hand side of (II), since

$$a_n = \sqrt{a_n^2} = \sqrt{n}\sqrt{1 - \frac{1}{2n} + O(n^{-2})} = \sqrt{n}\left(1 - \frac{1}{4n} + O(n^{-2})\right),$$
$$\sqrt{n+1} + \sqrt{n} = \sqrt{n}\left(\sqrt{1 + \frac{1}{n}} + 1\right) = \sqrt{n}\left(2 + \frac{1}{2n} + O(n^{-2})\right),$$

hence

$$\begin{aligned}
\text{RHS} &= \frac{\sqrt{n}(1 - 1/4n + O(n^{-2}))}{\sqrt{n}(2 + 1/2n + O(n^{-2}))} \\
&= \frac{1}{2}\frac{1 - 1/4n + O(n^{-2})}{1 + 1/4n + O(n^{-2})} \\
&= \frac{1}{2}\left(1 - \frac{1}{4n}\right)\left(1 - \frac{1}{4n}\right) + O(n^{-2}) \\
&= \frac{1}{2}\left(1 - \frac{1}{2n}\right) + O(n^{-2}) \\
&= \frac{1}{2} - \frac{1}{4n} + O(n^{-2}).
\end{aligned}$$

Thus,

$$\text{LHS} - \text{RHS} = \left(\frac{1}{2} - \frac{1}{8n}\right) - \left(\frac{1}{2} - \frac{1}{4n}\right) + O(n^{-2}) = \frac{1}{8n} + O(n^{-2}) \gtrsim 0.$$

13 [**Minimum Singular Value of Gaussian Matrices**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix. We will use Theorem T7.2.9 (*Gordon inequality*) to derive a precise lower bound for its minimum singular value $s_n(\boldsymbol{A})$. Assume $m \geqslant n \geqslant C$.
(1) (Expected bound) Prove:

$$\mathbb{E}[s_n(\boldsymbol{A})] \geqslant \sqrt{m} - \sqrt{n}.$$

(2) (High-probability bound) Use concentration inequalities to lift the expected bound to a high-probability bound. Prove: for any $t \geqslant 0$,

$$\mathbb{P}\{s_n(\boldsymbol{A}) \leqslant \sqrt{m} - \sqrt{n} - t\} \leqslant 2\exp(-ct^2). \qquad (c > 0)$$

Prf. (1) By definition,

$$s_n(\boldsymbol{A}) = \min_{\boldsymbol{x}\in\mathbb{S}^{n-1}} \|\boldsymbol{A}\boldsymbol{x}\|_2, \|\boldsymbol{A}\boldsymbol{x}\|_2 = \max_{\boldsymbol{y}\in\mathbb{S}^{m-1}} \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle,$$

hence

$$s_n(\boldsymbol{A}) = \min_{\boldsymbol{x}\in\mathbb{S}^{n-1}} \max_{\boldsymbol{y}\in\mathbb{S}^{m-1}} \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle.$$

Define $\mathbb{U} = \mathbb{S}^{n-1}$, $\mathbb{T} = \mathbb{S}^{m-1}$, and construct the zero-mean Gaussian process $(Y_{\boldsymbol{x}\boldsymbol{y}})$ on $\mathbb{U}\times\mathbb{T}$ as

$$Y_{\boldsymbol{x}\boldsymbol{y}} = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}x_j y_i,$$

then

$$\mathbb{E}\left[\min_{\boldsymbol{x}}\max_{\boldsymbol{y}} Y_{\boldsymbol{x}\boldsymbol{y}}\right] = \mathbb{E}[s_n(\boldsymbol{A})].$$

Then introduce mutually independent standard Gaussian vectors $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_m), \boldsymbol{h} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$, and construct the comparison process $(X_{\boldsymbol{x}\boldsymbol{y}})$ as

$$X_{\boldsymbol{x}\boldsymbol{y}} = \langle \boldsymbol{g}, \boldsymbol{y}\rangle + \langle \boldsymbol{h}, \boldsymbol{x}\rangle,$$

then

$$\begin{aligned}\mathbb{E}\left[\min_{\boldsymbol{x}\in\mathbb{U}}\max_{\boldsymbol{y}\in\mathbb{T}} X_{\boldsymbol{xy}}\right] &= \mathbb{E}\left[\min_{\boldsymbol{x}\in\mathbb{S}^{n-1}}\max_{\boldsymbol{y}\in\mathbb{S}^{m-1}}(\langle \boldsymbol{g},\boldsymbol{y}\rangle + \langle \boldsymbol{h},\boldsymbol{x}\rangle)\right]\\ &= \mathbb{E}\left[\max_{\boldsymbol{y}\in\mathbb{S}^{m-1}}\langle \boldsymbol{g},\boldsymbol{y}\rangle + \min_{\boldsymbol{x}\in\mathbb{S}^{n-1}}\langle \boldsymbol{h},\boldsymbol{x}\rangle\right]\\ &= \mathbb{E}\|\boldsymbol{g}\|_2 - \mathbb{E}\|\boldsymbol{h}\|_2\\ &= \mathbb{E}\|\boldsymbol{g}\|_2 - \mathbb{E}\|\boldsymbol{h}\|_2.\end{aligned}$$

Verify the conditions of Gordon's inequality:

$$\begin{aligned}\mathbb{E}[(Y_{\boldsymbol{xy}} - Y_{\boldsymbol{uv}})^2] &= \mathbb{E}[\langle \boldsymbol{Ax},\boldsymbol{y}\rangle - \langle \boldsymbol{Au},\boldsymbol{v}\rangle]^2 = \|\boldsymbol{yx}^\top - \boldsymbol{vu}^\top\|_F^2,\\ \mathbb{E}[(X_{\boldsymbol{xy}} - X_{\boldsymbol{uv}})^2] &= \mathbb{E}[\langle \boldsymbol{g},\boldsymbol{y}-\boldsymbol{v}\rangle + \langle \boldsymbol{h},\boldsymbol{x}-\boldsymbol{u}\rangle]^2 = \|\boldsymbol{y}-\boldsymbol{v}\|_2^2 + \|\boldsymbol{x}-\boldsymbol{u}\|_2^2,\end{aligned}$$

a. (Within-group condition) When $\boldsymbol{x}=\boldsymbol{u}$, the increment of process Y is

$$\|\boldsymbol{yx}^\top - \boldsymbol{vx}^\top\|_F^2 = \|(\boldsymbol{y}-\boldsymbol{v})\boldsymbol{x}^\top\|_F^2 = \|\boldsymbol{y}-\boldsymbol{v}\|_2^2\|\boldsymbol{x}\|_2^2 = \|\boldsymbol{y}-\boldsymbol{v}\|_2^2,$$

and the increment of process X is

$$\|\boldsymbol{y}-\boldsymbol{v}\|_2^2 + \|\boldsymbol{x}-\boldsymbol{x}\|_2^2 = \|\boldsymbol{y}-\boldsymbol{v}\|_2^2,$$

hence $\mathbb{E}[(X_{\boldsymbol{xy}} - X_{\boldsymbol{xv}})^2] \leqslant \mathbb{E}[(Y_{\boldsymbol{xy}} - Y_{\boldsymbol{xv}})^2]$.

b. (Cross-group condition) When $\boldsymbol{x}\neq\boldsymbol{u}$, to prove $\mathbb{E}[(X_{\boldsymbol{xy}} - X_{\boldsymbol{uv}})^2] \geqslant \mathbb{E}[(Y_{\boldsymbol{xy}} - Y_{\boldsymbol{uv}})^2]$, we only need to prove

$$\|\boldsymbol{x}-\boldsymbol{u}\|_2^2 + \|\boldsymbol{y}-\boldsymbol{v}\|_2^2 \geqslant \|\boldsymbol{yx}^\top - \boldsymbol{vu}^\top\|_F^2,$$

which is precisely Exercise Q7.10 (*Frobenius Distance between Rank-One Matrices*).

Thus, using Theorem T7.2.9 (*Gordon inequality*) (*Gordon inequality*), we obtain

$$\mathbb{E}\left[\min_{\boldsymbol{x}\in\mathbb{U}}\max_{\boldsymbol{y}\in\mathbb{T}} X_{\boldsymbol{xy}}\right] \leqslant \mathbb{E}\left[\min_{\boldsymbol{x}\in\mathbb{U}}\max_{\boldsymbol{y}\in\mathbb{T}} Y_{\boldsymbol{xy}}\right],$$

i.e.,

$$\mathbb{E}[s_n(\boldsymbol{A})] \geqslant \mathbb{E}\|\boldsymbol{g}\|_2 - \mathbb{E}\|\boldsymbol{h}\|_2.$$

By Exercise Q7.12 (*Norm of a Gaussian Vector*), $f(k) = \mathbb{E}\|\boldsymbol{g}_k\|_2 - \sqrt{k}$ is increasing when $k \geqslant C$, hence $f(m) \geqslant f(n)$, i.e.,

$$\mathbb{E}\|\boldsymbol{g}\|_2 - \sqrt{m} \geqslant \mathbb{E}\|\boldsymbol{h}\|_2 - \sqrt{n},$$

i.e.,

$$\mathbb{E}\|\boldsymbol{g}\|_2 - \mathbb{E}\|\boldsymbol{h}\|_2 \geqslant \sqrt{m} - \sqrt{n},$$

i.e.,

$$\mathbb{E}[s_n(\boldsymbol{A})] \geqslant \sqrt{m} - \sqrt{n}.$$

(2) (*To lift the expected result to a high-probability bound, we use the standard Gaussian concentration inequality. View* $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ *as a vector of* mn *independent standard Gaussian variables.*)

Define $F(\boldsymbol{A}) = s_n(\boldsymbol{A})$. By Weyl's inequality, we obtain

$$|s_n(\boldsymbol{A}) - s_n(\boldsymbol{B})| \leqslant \|\boldsymbol{A}-\boldsymbol{B}\|,$$

where $\|\boldsymbol{A}-\boldsymbol{B}\| \leqslant \|\boldsymbol{A}-\boldsymbol{B}\|_F$. Hence

$$|F(\boldsymbol{A}) - F(\boldsymbol{B})| \leqslant \|\boldsymbol{A}-\boldsymbol{B}\|_F,$$

so $\|F\|_{\mathrm{Lip}} = 1$.

Using Theorem T7.1.11 (*Concentration of Gaussian processes*) (*Concentration of Gaussian processes*), we obtain

$$\mathbb{P}\{F(\boldsymbol{A}) \leqslant \mathbb{E}[F(\boldsymbol{A})] - t\} \leqslant \exp\left(-\frac{t^2}{2L^2}\right),$$

i.e.,

$$\mathbb{P}\{s_n(\boldsymbol{A}) \leqslant \mathbb{E}[s_n(\boldsymbol{A})] - t\} \leqslant \exp\left(-\frac{t^2}{2}\right),$$

Using the conclusion of (1), we obtain

$$\mathbb{P}\{s_n(\boldsymbol{A}) \leqslant \sqrt{m} - \sqrt{n} - t\} \leqslant \mathbb{P}\{s_n(\boldsymbol{A}) \leqslant \mathbb{E}[s_n(\boldsymbol{A})] - t\} \leqslant \exp\left(-\frac{t^2}{2}\right).$$

14 [**Sudakov's Inequality for Non-Compact Sets**] Let $(X_t)_{t\in\mathbb{T}}$ be a zero-mean Gaussian process whose canonical metric is defined as

$$d(t,s) = \|X_t - X_s\|_{L^2} = \sqrt{\mathbb{E}[(X_t - X_s)^2]}.$$

Assume that $\mathbb{T}$ is finite or that the process is separable. Prove: If the metric space $(\mathbb{T}, d)$ is not relatively compact (*i.e., there exists* $\varepsilon > 0$ *such that the covering number* $\mathcal{N}(\mathbb{T}, d, \varepsilon) = \infty$), then

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] = \infty.$$

Rk. **Covering number** If there exists a finite set of points $\{x_1, x_2, \cdots, x_N\} \subset \mathbb{T}$ such that for any $x \in \mathbb{T}$, there exists x_i satisfying $d(x, x_i) \leqslant \varepsilon$, then it is called an ε-cover. Among all possible ε-covers, the covering number $\mathcal{N}(\mathbb{T}, d, \varepsilon)$ is defined as the cardinality of the smallest one.

Packing number If there exists a finite set of points $\{y_1, y_2, \cdots, y_M\} \subset \mathbb{T}$ such that for any $y_i \neq y_j$, $d(y_i, y_j) \geqslant \varepsilon$, then it is called an ε-packing. Among all possible ε-packings, the packing number $\mathcal{M}(\mathbb{T}, d, \varepsilon)$ is defined as the cardinality of the largest one.

It is easy to see that

$$\mathcal{N}(\mathbb{T}, d, \varepsilon) \leqslant \mathcal{M}(\mathbb{T}, d, \varepsilon).$$

Prf. Since $(\mathbb{T}, d)$ is not relatively compact, there exists $\varepsilon > 0$ such that the covering number satisfies $\mathcal{N}(\mathbb{T}, d, \varepsilon) = \infty$, and hence the packing number satisfies

$$\mathcal{M}(\mathbb{T}, d, \varepsilon) = \infty,$$

Thus for any N, there exists a finite subset $\mathbb{T}_N = \{t_1, t_2, \cdots, t_N\} \subset \mathbb{T}$ of size N such that for any $t_i \neq t_j \in \mathbb{T}_N$,

$$d(t_i, t_j) = \sqrt{\mathbb{E}[(X_{t_i} - X_{t_j})^2]} \geqslant \varepsilon,$$

which is precisely the Sudakov condition.

For $\mathbb{T}_N$, using Theorem T7.4.1 (*Sudakov inequality*) (*Sudakov inequality*), we obtain

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}_N} X_t\right] \geqslant c \cdot \varepsilon \cdot \sqrt{\log|\mathbb{T}_N|},$$

i.e.,

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \geqslant c\varepsilon\sqrt{\log N}.$$

Since $\mathbb{T}_N \subset \mathbb{T}$, we have

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \geqslant \mathbb{E}\left[\sup_{t\in\mathbb{T}_N} X_t\right],$$

Letting $N \to \infty$, we obtain

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \geqslant \lim_{N\to\infty} c\varepsilon\sqrt{\log N} = \infty.$$

15 [**Basic Properties of Gaussian Width**] Let $\mathbb{T}, \mathbb{S} \subset \mathbb{R}^n$, $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Define the Gaussian width of $\mathbb{T}$ as

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle\right].$$

Prove the seven basic properties in Property P7.5.2 (*Simple properties of Gaussian width*).

(1) Finiteness: $w(\mathbb{T})$ is finite if and only if $\mathbb{T}$ is bounded.

(2) (Rigid motion invariance) For any orthogonal matrix $\boldsymbol{U}$ and translation vector $\boldsymbol{y} \in \mathbb{R}^n$, $w(\boldsymbol{U}\mathbb{T} + \boldsymbol{y}) = w(\mathbb{T})$.

(3) (Convex hull) $w(\mathrm{conv}(\mathbb{T})) = w(\mathbb{T})$.

(4) (Minkowski sum and scaling) $w(\mathbb{T} + \mathbb{S}) = w(\mathbb{T}) + w(\mathbb{S})$; $w(a\mathbb{T}) = |a| w(\mathbb{T})$.

(5) (Symmetry)

$$w(\mathbb{T}) = \frac{1}{2} w(\mathbb{T} - \mathbb{T}) = \frac{1}{2}\mathbb{E}\left[\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x} - \boldsymbol{y}\rangle\right].$$

(6) (Width and diameter)

$$\frac{1}{\sqrt{2\pi}} \operatorname{diam}(\mathbb{T}) \leqslant w(\mathbb{T}) \leqslant \frac{\sqrt{n}}{2} \operatorname{diam}(\mathbb{T}),$$

where $\operatorname{diam}(\mathbb{T}) = \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \|\boldsymbol{x} - \boldsymbol{y}\|_2$.

(7) (Linear map) For any $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, $w(\boldsymbol{A}\mathbb{T}) \leqslant \|\boldsymbol{A}\| w(\mathbb{T})$.

Prf. (1) Suppose $\mathbb{T}$ is bounded. Then there exists a radius $R < \infty$ such that for all $\boldsymbol{t} \in \mathbb{T}$, $\|\boldsymbol{t}\|_2 \leqslant R$.

Using Cauchy's inequality, we obtain

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \|\boldsymbol{g}\|_2 \|\boldsymbol{t}\|_2\right] \leqslant R \cdot \mathbb{E}\|\boldsymbol{g}\|_2,$$

Substituting $\mathbb{E}\|\boldsymbol{g}\|_2 \leqslant \sqrt{\mathbb{E}\|\boldsymbol{g}\|_2^2} = \sqrt{n}$, we obtain $w(\mathbb{T}) \leqslant R\sqrt{n} < \infty$.

Conversely, when $\mathbb{T}$ is unbounded, there exists $\{\boldsymbol{t}_k\} \subset \mathbb{T}$ such that $\|\boldsymbol{t}_k\|_2 \to \infty$. Passing to a subsequence, we may take $\boldsymbol{t}_k / \|\boldsymbol{t}_k\|_2 \to \boldsymbol{u} \in \mathbb{S}^{n-1}$. On the event $\{\langle \boldsymbol{g}, \boldsymbol{u}\rangle > 0\}$, which has positive probability, $\langle \boldsymbol{g}, \boldsymbol{t}_k\rangle \to \infty$. Thus $\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle = \infty$ with positive probability, and consequently $w(\mathbb{T}) = \infty$, contradicting the assumption that $w(\mathbb{T})$ is finite; thus $\mathbb{T}$ is bounded.

(2) By definition,

$$\begin{aligned} w(\boldsymbol{U}\mathbb{T} + \boldsymbol{y}) &= \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{U}\boldsymbol{t} + \boldsymbol{y}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} (\langle \boldsymbol{g}, \boldsymbol{U}\boldsymbol{t}\rangle + \langle \boldsymbol{g}, \boldsymbol{y}\rangle)\right] \\ &= \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{U}^\top \boldsymbol{g}, \boldsymbol{t}\rangle\right] + \mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{y}\rangle]. \end{aligned}$$

For the second term, $\mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{y}\rangle] = \langle \mathbb{E}[\boldsymbol{g}], \boldsymbol{y}\rangle = 0$; for the first term, since $\boldsymbol{U}^\top \boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$,

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{U}^\top \boldsymbol{g}, \boldsymbol{t}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle\right] = w(\mathbb{T});$$

hence

$$w(\boldsymbol{U}\mathbb{T} + \boldsymbol{y}) = w(\mathbb{T}) + 0 = w(\mathbb{T}).$$

(3) On one hand, since $\mathbb{T} \subset \mathrm{conv}(\mathbb{T})$, we have $w(\mathbb{T}) \leqslant w(\mathrm{conv}(\mathbb{T}))$.

On the other hand, for any $\boldsymbol{x} = \sum_{i=1}^n \lambda_i \boldsymbol{t}_i \in \mathrm{conv}(\mathbb{T})$,

$$\langle \boldsymbol{g}, \boldsymbol{x}\rangle = \sum_{i=1}^n \lambda_i \langle \boldsymbol{g}, \boldsymbol{t}_i\rangle \leqslant \sum_{i=1}^n \lambda_i \sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle = \sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle,$$

hence $w(\mathrm{conv}(\mathbb{T})) \leqslant w(\mathbb{T})$.

Thus, $w(\mathrm{conv}(\mathbb{T})) = w(\mathbb{T})$.

(4) For the Minkowski sum,

$$\begin{aligned} w(\mathbb{T}+\mathbb{S}) &= \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{g}, \boldsymbol{x}+\boldsymbol{y}\rangle\right] \\ &= \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}(\langle \boldsymbol{g}, \boldsymbol{x}\rangle + \langle \boldsymbol{g}, \boldsymbol{y}\rangle)\right] \\ &= \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{x}\rangle + \sup_{\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{g}, \boldsymbol{y}\rangle\right] \\ &= \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] + \mathbb{E}\left[\sup_{\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{g}, \boldsymbol{y}\rangle\right] = w(\mathbb{T}) + w(\mathbb{S}). \end{aligned}$$

For Minkowski scaling,

$$w(a\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, a\boldsymbol{t}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle a\boldsymbol{g}, \boldsymbol{t}\rangle\right] = |a|w(\mathbb{T}).$$

(5) Using the Minkowski scaling property,

$$w(-\mathbb{T}) = |-1|w(\mathbb{T}) = w(\mathbb{T}),$$

hence

$$\frac{1}{2}w(\mathbb{T}-\mathbb{T}) = \frac{1}{2}(w(\mathbb{T}) + w(-\mathbb{T})) = \frac{1}{2}(w(\mathbb{T}) + w(\mathbb{T})) = w(\mathbb{T}),$$

where

$$\frac{1}{2}w(\mathbb{T}-\mathbb{T}) = \frac{1}{2}\mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{v}\rangle\right] = \frac{1}{2}\mathbb{E}\left[\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{x}-\boldsymbol{y}\rangle\right].$$

(6) For the lower bound, let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{T}$. Then using the conclusion of (5),

$$w(\mathbb{T}) = \frac{1}{2}\mathbb{E}\left[\sup_{\boldsymbol{u},\boldsymbol{v}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{u}-\boldsymbol{v}\rangle\right] \geqslant \frac{1}{2}\mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{x}-\boldsymbol{y}\rangle \vee \langle \boldsymbol{g}, \boldsymbol{y}-\boldsymbol{x}\rangle] = \frac{1}{2}\mathbb{E}[|\langle \boldsymbol{g}, \boldsymbol{x}-\boldsymbol{y}\rangle|],$$

where $\langle \boldsymbol{g}, \boldsymbol{x}-\boldsymbol{y}\rangle \sim N(\boldsymbol{0}, \|\boldsymbol{x}-\boldsymbol{y}\|_2^2)$, hence

$$w(\mathbb{T}) \geqslant \frac{1}{2}\sqrt{\frac{2}{\pi}}\|\boldsymbol{x}-\boldsymbol{y}\|_2 = \frac{1}{\sqrt{2\pi}}\|\boldsymbol{x}-\boldsymbol{y}\|_2,$$

taking the supremum over $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{T}$ on both sides,

$$w(\mathbb{T}) \geqslant \frac{1}{\sqrt{2\pi}}\operatorname{diam}(\mathbb{T}).$$

(The following gives a weaker upper bound:
For the upper bound, fix $\boldsymbol{y}_0 \in \mathbb{T}$ and let $\boldsymbol{t} \in \mathbb{T}$. Then $\|\boldsymbol{t}-\boldsymbol{y}_0\|_2 \leqslant \operatorname{diam}(\mathbb{T})$.
Using translation invariance, we obtain

$$\begin{aligned} w(\mathbb{T}) = w(\mathbb{T}-\boldsymbol{y}_0) &= \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{t}-\boldsymbol{y}_0\rangle\right] \\ &\leqslant \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\|\boldsymbol{g}\|_2\|\boldsymbol{t}-\boldsymbol{y}_0\|_2\right] \leqslant \mathbb{E}\|\boldsymbol{g}\|_2 \cdot \operatorname{diam}(\mathbb{T}). \end{aligned}$$

To obtain the coefficient $\sqrt{n}/2$, use symmetry.)
For the upper bound, using symmetry, we obtain

$$w(\mathbb{T}) = \frac{1}{2}w(\mathbb{T}-\mathbb{T}) = \frac{1}{2}\mathbb{E}\left[\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{x}-\boldsymbol{y}\rangle\right] \leqslant \frac{1}{2}\mathbb{E}\|\boldsymbol{g}\|_2 \sup_{\boldsymbol{x},\boldsymbol{y}}\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \frac{\sqrt{n}}{2}\operatorname{diam}(\mathbb{T}).$$

(7) By definition,

$$w(\boldsymbol{A}\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{A}\boldsymbol{t}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{A}^\top\boldsymbol{g}, \boldsymbol{t}\rangle\right],$$

where $\boldsymbol{A}^\top\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{A}^\top\boldsymbol{A})$.

Construct the comparison process $X_{\boldsymbol{t}} = \langle \boldsymbol{A}^\top\boldsymbol{g}, \boldsymbol{t}\rangle$. Its incremental second moments are

$$\mathbb{E}[(X_{\boldsymbol{t}} - X_{\boldsymbol{s}})^2] = \mathbb{E}[\langle \boldsymbol{A}^\top\boldsymbol{g}, \boldsymbol{t}-\boldsymbol{s}\rangle^2] = \mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{A}(\boldsymbol{t}-\boldsymbol{s})\rangle^2] = \|\boldsymbol{A}(\boldsymbol{t}-\boldsymbol{s})\|_2^2 \leqslant \|\boldsymbol{A}\|^2\|\boldsymbol{t}-\boldsymbol{s}\|_2^2.$$

Then construct the process $Y_{\boldsymbol{t}} = \|\boldsymbol{A}\|\langle \boldsymbol{h}, \boldsymbol{t}\rangle$, where $\boldsymbol{h} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Its incremental second moments are

$$\mathbb{E}[(Y_{\boldsymbol{t}} - Y_{\boldsymbol{s}})^2] = \|\boldsymbol{A}\|^2\mathbb{E}[\langle \boldsymbol{h}, \boldsymbol{t}-\boldsymbol{s}\rangle^2] = \|\boldsymbol{A}\|^2\|\boldsymbol{t}-\boldsymbol{s}\|_2^2,$$

hence

$$\mathbb{E}[(X_{\boldsymbol{t}} - X_{\boldsymbol{s}})^2] \leqslant \mathbb{E}[(Y_{\boldsymbol{t}} - Y_{\boldsymbol{s}})^2],$$

which is precisely the Sudakov-Fernique condition.

Using the Sudakov-Fernique inequality, we obtain

$$w(\boldsymbol{A}\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} X_{\boldsymbol{t}}\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} Y_{\boldsymbol{t}}\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \|\boldsymbol{A}\|\langle \boldsymbol{h}, \boldsymbol{t}\rangle\right] = \|\boldsymbol{A}\|w(\mathbb{T}).$$

16 [**Optimality of Gaussian Width vs. Diameter**] In Property P7.5.2 (*Simple properties of Gaussian width*) (6), it is known that for $\mathbb{T} \subset \mathbb{R}^n$,

$$\frac{1}{\sqrt{2\pi}}\operatorname{diam}(\mathbb{T}) \leqslant w(\mathbb{T}) \leqslant \frac{\sqrt{n}}{2}\operatorname{diam}(\mathbb{T}).$$

Give examples showing that both the lower and upper bounds of the above inequality can be attained (*up to an absolute constant factor*).

Prf. (*To prove the tightness of the bounds, we need to find extreme geometric objects in $\mathbb{R}^n$, where the lower bound corresponds to an extremely "narrow" one-dimensional object, and the upper bound corresponds to an object that is fully "plump" in all directions.*)

- Lower bound is attainable (*one-dimensional line segment*)

  Construct $\mathbb{T}_1$ in $\mathbb{R}^n$ as the unit line segment along the first coordinate axis, i.e.,

  $$\mathbb{T}_1 = \{t\boldsymbol{e}_1 : t \in [-1, 1]\},$$

  then $\operatorname{diam}(\mathbb{T}_1) = \|\boldsymbol{e}_1 - (-\boldsymbol{e}_1)\|_2 = \|2\boldsymbol{e}_1\|_2 = 2$.

  Let $\boldsymbol{g} = (g_1, \cdots, g_n) \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Then

  $$w(\mathbb{T}_1) = \mathbb{E}\left[\sup_{t\in[-1,1]}\langle \boldsymbol{g}, t\boldsymbol{e}_1\rangle\right] = \mathbb{E}\left[\sup_{t\in[-1,1]} tg_1\right] = \mathbb{E}[|g_1|] = \sqrt{\frac{2}{\pi}},$$

  hence

  $$\frac{1}{\sqrt{2\pi}}\operatorname{diam}(\mathbb{T}_1) = \sqrt{\frac{2}{\pi}} = w(\mathbb{T}_1).$$

- Upper bound is attainable (*Euclidean unit ball*)

  Construct $\mathbb{T}_2$ as the Euclidean unit ball in $\mathbb{R}^n$, i.e.,

  $$\mathbb{T}_2 = \mathbb{B}_2^n = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_2 \leqslant 1\},$$

  then $\operatorname{diam}(\mathbb{T}_2) = 2$.

  Let $\boldsymbol{g} = (g_1, \cdots, g_n) \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Then

  $$w(\mathbb{T}_2) = \mathbb{E}\left[\sup_{\|\boldsymbol{x}\|_2\leqslant 1}\langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] = \mathbb{E}\|\boldsymbol{g}\|_2 \gtrsim \sqrt{n},$$

hence

$$\frac{\sqrt{n}}{2}\operatorname{diam}(\mathbb{T}_2)=\sqrt{n}\asymp w(\mathbb{T}_2).$$

17 [**Gaussian Width of ℓ^p Balls**] Let $1\leqslant p\leqslant\infty$. Consider $\mathbb{B}_p^n=\{\boldsymbol{x}\in\mathbb{R}^n:\|\boldsymbol{x}\|_p\leqslant 1\}$. Prove that:

$$w(\mathbb{B}_p^n)\asymp\begin{cases}\sqrt{p'}n^{1/p'}, & p'\leqslant\log n,\\ \sqrt{\log n}, & p'\geqslant\log n\end{cases}.$$

where p' is the conjugate exponent of p.

Prf. When $n=2$, all $\ell^{p'}$ norms on $\mathbb{R}^2$ are equivalent, so both sides of the conclusion have absolute constant order; the conclusion is trivial, and afterwards only consider $n\geqslant 3$.

Let $\boldsymbol{g}\sim N(\boldsymbol{0},\boldsymbol{I}_n)$. The Gaussian width is defined as

$$w(\mathbb{B}_p^n)=\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{B}_p^n}\langle\boldsymbol{g},\boldsymbol{x}\rangle\right],$$

By the duality of norms,

$$\sup_{\|\boldsymbol{x}\|_p\leqslant 1}\langle\boldsymbol{g},\boldsymbol{x}\rangle=\|\boldsymbol{g}\|_{p'}=\left(\sum_{i=1}^n|g_i|^{p'}\right)^{1/p'},$$

hence $w(\mathbb{B}_p^n)=\mathbb{E}\|\boldsymbol{g}\|_{p'}$. Establish a global bound.

For $p'\geqslant 1$, $f(x)=x^{p'}$ is convex. Thus by Jensen's inequality,

$$\begin{aligned}\mathbb{E}\|\boldsymbol{g}\|_{p'}&\leqslant(\mathbb{E}\|\boldsymbol{g}\|_{p'}^{p'})^{1/p'}=\left[\mathbb{E}\left[\sum_{i=1}^n|g_i|^{p'}\right]\right]^{1/p'}\\&=(n\mathbb{E}[|g_1|^{p'}])^{1/p'}=n^{1/p'}(\mathbb{E}[|g_1|^{p'}])^{1/p'}=n^{1/p'}\|g_1\|_{L_{p'}}\\&\leqslant C\sqrt{p'}n^{1/p'}.\qquad(\|g\|_{L_q}\asymp\sqrt{q})\ \text{(I)}\end{aligned}$$

Then we discuss the range of p'.

a. When $p'\geqslant\log n$, for the lower bound,

$$\mathbb{E}\|\boldsymbol{g}\|_{p'}\geqslant\mathbb{E}\|\boldsymbol{g}\|_\infty\asymp\sqrt{\log n},$$

For the upper bound, using the conclusion of (I),

$$\mathbb{E}\|\boldsymbol{g}\|_{p'}\leqslant\mathbb{E}\|\boldsymbol{g}\|_{\log n}\leqslant C\sqrt{\log n}\cdot n^{1/\log n}=Ce\sqrt{\log n}\asymp\sqrt{\log n}.$$

Thus, $w(\mathbb{B}_p^n)\asymp\sqrt{\log n}$.

b. When $1\leqslant p'\leqslant\log n$, for the upper bound, using the conclusion of (I),

$$w(\mathbb{B}_p^n)\leqslant C\sqrt{p'}n^{1/p'},$$

Then we only need to estimate the lower bound.

(*Consider using the Paley-Zygmund inequality to give a rigorous lower bound.*)

Let $S=\sum_{i=1}^n|g_i|^{p'}$. Its numerical characteristics are

$$\begin{aligned}\mathbb{E}[S]&=n\mathbb{E}[|g|^{p'}],\\\mathbb{E}[S^2]&=n\mathbb{E}[|g|^{2p'}]+n(n-1)(\mathbb{E}[|g|^{p'}])^2.\end{aligned}$$

Using the Paley-Zygmund inequality, for any $\theta\in(0,1)$,

$$\mathbb{P}\{S>\theta\mathbb{E}[S]\}\geqslant(1-\theta)^2\frac{(\mathbb{E}[S])^2}{\mathbb{E}[S^2]}.$$

For the right-hand side, substituting $\mathbb{E}[|g|^q] \asymp (q/\mathrm{e})^{q/2}$, we obtain

$$\frac{\mathbb{E}[|g|^{2p'}]}{(\mathbb{E}[|g|^{p'}])^2} \asymp \frac{(2p'/\mathrm{e})^{p'}}{(p'/\mathrm{e})^{p'}} = 2^{p'},$$

hence

$$\frac{\mathbb{E}[S^2]}{(\mathbb{E}[S])^2} = \frac{n\mathbb{E}[|g|^{2p'}]}{n^2(\mathbb{E}[|g|^{p'}])^2} + \frac{n(n-1)}{n^2} \leqslant \frac{C2^{p'}}{n} + 1 \leqslant \frac{Cn^{\log 2}}{n} + 1 \to 1,$$

Thus there exists $M > 0$ such that

$$\frac{\mathbb{E}[S^2]}{(\mathbb{E}[S])^2} \leqslant M.$$

Let $\theta = 1/2$. Then

$$\mathbb{P}\left\{S > \frac{1}{2}\mathbb{E}[S]\right\} \geqslant \frac{1}{4M} > 0,$$

Using Proposition P1.6.2 (*Markov inequality*) (*Markov's inequality*),

$$\begin{aligned}\mathbb{E}\|\boldsymbol{g}\|_{p'} = \mathbb{E}[S^{1/p'}] &\geqslant \mathbb{P}\left\{S > \frac{1}{2}\mathbb{E}[S]\right\} \cdot \left(\frac{1}{2}\mathbb{E}[S]\right)^{1/p'} \\ &\geqslant \frac{1}{4M}\left(\frac{1}{2}n\mathbb{E}[|g|^{p'}]\right)^{1/p'} = \frac{1}{4M}\left(\frac{1}{2}\right)^{1/p'} n^{1/p'}\|g\|_{L_{p'}} \\ &\geqslant c\sqrt{p'}n^{1/p'}.\end{aligned}$$

Thus, $w(\mathbb{B}_p^n) \asymp \sqrt{p'}n^{1/p'}$.

18 [**Nuclear Norm**] Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ with singular values $s_1 \geqslant s_2 \geqslant \cdots \geqslant s_n \geqslant 0$. It is known that the operator norm $\|\boldsymbol{A}\|$ is the ℓ^∞ norm of the singular values (*the largest singular value*), and the Frobenius norm $\|\boldsymbol{A}\|_F$ is the ℓ^2 norm of the singular values (see Lemma L4.1.11 (*Matrix norms via singular values*)). Define the nuclear norm as the ℓ^1 norm of its singular values, i.e.,

$$\|\boldsymbol{A}\|_* = \sum_{i=1}^n s_i(\boldsymbol{A}).$$

(1) Just as the ℓ^1 and ℓ^∞ norms of vectors are dual (see Exercise Q1.19 (*Duality of ℓ_p Norms*)(b)), prove that the nuclear norm and operator norm are dual, i.e.,

$$\|\boldsymbol{A}\|_* = \max_{\|\boldsymbol{B}\|\leqslant 1} \langle \boldsymbol{A}, \boldsymbol{B}\rangle.$$

(2) Using the conclusion of (1), prove that the nuclear norm $\|\cdot\|_*$ is a legitimate norm.

Prf. Let the SVD of $\boldsymbol{A}$ be $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, where $\boldsymbol{U}, \boldsymbol{V}$ are orthogonal matrices and $\boldsymbol{\Sigma} = \mathrm{diag}(s_1, \cdots, s_n)$ is a diagonal matrix containing the nonnegative singular values.

(1) Let $\|\boldsymbol{B}\| \leqslant 1$. Then

$$\langle \boldsymbol{A}, \boldsymbol{B}\rangle = \mathrm{tr}(\boldsymbol{A}^\top\boldsymbol{B}) = \mathrm{tr}(\boldsymbol{V}\boldsymbol{\Sigma}\boldsymbol{U}^\top\boldsymbol{B}) = \mathrm{tr}(\boldsymbol{\Sigma}\boldsymbol{U}^\top\boldsymbol{B}\boldsymbol{V}).$$

Let $\boldsymbol{C} = \boldsymbol{U}^\top\boldsymbol{B}\boldsymbol{V}$. Then

$$\|\boldsymbol{C}\| = \|\boldsymbol{U}^\top\boldsymbol{B}\boldsymbol{V}\| \leqslant \|\boldsymbol{U}^\top\|\|\boldsymbol{B}\|\|\boldsymbol{V}\| = 1 \cdot \|\boldsymbol{B}\| \cdot 1 \leqslant 1,$$

substituting this in, we obtain

$$\langle \boldsymbol{A}, \boldsymbol{B}\rangle = \mathrm{tr}(\boldsymbol{\Sigma}\boldsymbol{C}) = \sum_{i=1}^n s_i c_{ii} \leqslant \sum_{i=1}^n s_i|c_{ii}| \leqslant \sum_{i=1}^n s_i = \|\boldsymbol{A}\|_*,$$

thus for any $\|\boldsymbol{B}\| \leqslant 1$, $\langle \boldsymbol{A}, \boldsymbol{B}\rangle \leqslant \|\boldsymbol{A}\|_*$.

(*Then we only need to prove the attainability of the maximum. We simply construct $\boldsymbol{B}^*$ such that $\langle \boldsymbol{A}, \boldsymbol{B}^*\rangle = \|\boldsymbol{A}\|_*$.*)

Let $\boldsymbol{C} = \boldsymbol{U}^\top \boldsymbol{B}^* \boldsymbol{V} = \boldsymbol{I}$, i.e., $\boldsymbol{B}^* = \boldsymbol{U}\boldsymbol{V}^\top$. Then $\|\boldsymbol{B}^*\| = 1 \leqslant 1$, and

$$\langle \boldsymbol{A}, \boldsymbol{B}^* \rangle = \operatorname{tr}(\boldsymbol{V}\boldsymbol{\Sigma}\boldsymbol{U}^\top \cdot \boldsymbol{U}\boldsymbol{V}^\top) = \operatorname{tr}(\boldsymbol{V}\boldsymbol{\Sigma}\boldsymbol{V}^\top) = \operatorname{tr}(\boldsymbol{\Sigma}) = \sum_{i=1}^{n} s_i = \|\boldsymbol{A}\|_*.$$

Thus,

$$\|\boldsymbol{A}\|_* = \max_{\|\boldsymbol{B}\| \leqslant 1} \langle \boldsymbol{A}, \boldsymbol{B} \rangle.$$

(2) 1. (Positive definiteness)

If $s_i \geqslant 0$, then $\|\boldsymbol{A}\|_* = \sum_{i=1}^n s_i \geqslant 0$; if $\|\boldsymbol{A}\|_* = 0$, then $s_i = 0$, which implies $\boldsymbol{A} = \boldsymbol{0}$.

(*This can also be proved via the dual formulation. If* $\boldsymbol{B} = \boldsymbol{0}$*, then* $\|\boldsymbol{A}\|_* \geqslant 0$*; if* $\max_{\|\boldsymbol{B}\| \leqslant 1} \langle \boldsymbol{A}, \boldsymbol{B} \rangle = 0$*, then letting* $\boldsymbol{B} = \boldsymbol{A}/\|\boldsymbol{A}\|$ *yields* $\boldsymbol{A} = \boldsymbol{0}$.)

2. (Absolute homogeneity)

If $c \geqslant 0$, then

$$\|c\boldsymbol{A}\|_* = \max_{\|\boldsymbol{B}\| \leqslant 1} \langle c\boldsymbol{A}, \boldsymbol{B} \rangle = \max_{\|\boldsymbol{B}\| \leqslant 1} c\langle \boldsymbol{A}, \boldsymbol{B} \rangle = c\|\boldsymbol{A}\|_*.$$

If $c < 0$, then

$$\|c\boldsymbol{A}\|_* = \max_{\|\boldsymbol{B}\| \leqslant 1} |c| \langle \boldsymbol{A}, -\boldsymbol{B} \rangle = |c| \max_{\|\boldsymbol{B}\| \leqslant 1} \langle \boldsymbol{A}, \boldsymbol{B} \rangle = |c|\|\boldsymbol{A}\|_*.$$

3. (Triangle inequality)

Let $\boldsymbol{A}, \boldsymbol{C} \in \mathbb{R}^{n \times n}$. Then

$$\begin{aligned}
\|\boldsymbol{A} + \boldsymbol{C}\|_* &= \max_{\|\boldsymbol{B}\| \leqslant 1} \langle \boldsymbol{A} + \boldsymbol{C}, \boldsymbol{B} \rangle \\
&= \max_{\|\boldsymbol{B}\| \leqslant 1} (\langle \boldsymbol{A}, \boldsymbol{B} \rangle + \langle \boldsymbol{C}, \boldsymbol{B} \rangle) \\
&\leqslant \max_{\|\boldsymbol{B}_1\| \leqslant 1} \langle \boldsymbol{A}, \boldsymbol{B}_1 \rangle + \max_{\|\boldsymbol{B}_2\| \leqslant 1} \langle \boldsymbol{C}, \boldsymbol{B}_2 \rangle \\
&= \|\boldsymbol{A}\|_* + \|\boldsymbol{C}\|_*.
\end{aligned}$$

19 [**Gaussian Width of the Operator Norm Unit Ball**] Let $\mathbb{T}$ be the operator norm unit ball in the matrix space $\mathbb{R}^{n \times n}$, i.e.,

$$\mathbb{T} = \{\boldsymbol{B} \in \mathbb{R}^{n \times n} : \|\boldsymbol{B}\| \leqslant 1\}.$$

Prove:

$$w(\mathbb{T}) \asymp n^{3/2}.$$

Prf. By the definition of Gaussian width,

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{B} \in \mathbb{T}} \langle \boldsymbol{G}, \boldsymbol{B} \rangle\right] = \mathbb{E}\left[\max_{\|\boldsymbol{B}\| \leqslant 1} \langle \boldsymbol{G}, \boldsymbol{B} \rangle\right],$$

By Exercise Q7.18 (*Nuclear Norm*), we obtain

$$w(\mathbb{T}) = \mathbb{E}\|\boldsymbol{G}\|_* = \mathbb{E}\left[\sum_{i=1}^{n} s_i(\boldsymbol{G})\right],$$

Then we only need to estimate the expectation of the sum of singular values of an $n \times n$ standard Gaussian matrix.

- Proof of the upper bound $w(\mathbb{T}) \lesssim n^{3/2}$

  For the singular value vector $(s_1, \cdots, s_n)$ and the all-ones vector $(1, \cdots, 1)$, using Cauchy's inequality,

$$\sum_{i=1}^{n} s_i \cdot 1 \leqslant \left(\sum_{i=1}^{n} s_i^2\right)^{1/2} \left(\sum_{i=1}^{n} 1^2\right)^{1/2},$$

i.e.,

$$\|\boldsymbol{G}\|_* \leqslant \sqrt{n}\|\boldsymbol{G}\|_F,$$

taking expectation on both sides,

$$\mathbb{E}\|\boldsymbol{G}\|_* \leqslant \sqrt{n}\mathbb{E}\|\boldsymbol{G}\|_F.$$

where

$$\mathbb{E}\|\boldsymbol{G}\|_F \leqslant \sqrt{\mathbb{E}\|\boldsymbol{G}\|_F^2} = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n}\mathbb{E}[G_{ij}^2]} = n.$$

Substituting this in gives $w(\mathbb{T}) \leqslant n^{3/2}$.

- Proof of the lower bound $w(\mathbb{T}) \gtrsim n^{3/2}$

Since

$$\|\boldsymbol{G}\|_F^2 = \sum_{i=1}^{n} s_i^2 \leqslant \left(\max_i s_i\right)\left(\sum_{i=1}^{n} s_i\right) = \|\boldsymbol{G}\| \cdot \|\boldsymbol{G}\|_*,$$

we have

$$\|\boldsymbol{G}\|_* \geqslant \frac{\|\boldsymbol{G}\|_F^2}{\|\boldsymbol{G}\|}. \tag{I}$$

(*We cannot simply split the expectation into numerator and denominator. To handle the operator norm in the denominator, we introduce a high-probability "benign event" $\mathcal{E}$.*)

For the denominator, by Theorem T7.3.1 (*Norms of Gaussian random matrices*),

$$\mathbb{E}\|\boldsymbol{G}\| \leqslant 2\sqrt{n}.$$

By Theorem T5.2.3 (*Gaussian concentration*),

$$\mathbb{P}\{\|\boldsymbol{G}\| \geqslant \mathbb{E}\|\boldsymbol{G}\| + t\} \leqslant \exp\left(-\frac{t^2}{2}\right),$$

Let $t = \sqrt{n}$. Then

$$\mathbb{P}\{\|\boldsymbol{G}\| \geqslant 3\sqrt{n}\} \leqslant \exp\left(-\frac{1}{2}n\right).$$

Define $\mathcal{E} = \{\|\boldsymbol{G}\| \leqslant 3\sqrt{n}\}$. Then

$$\mathbb{E}\|\boldsymbol{G}\|_* \geqslant \mathbb{E}\|\boldsymbol{G}\|_*\mathbf{1}_{\mathcal{E}} \geqslant \mathbb{E}\left[\frac{\|\boldsymbol{G}\|_F^2}{\|\boldsymbol{G}\|}\mathbf{1}_{\mathcal{E}}\right] \geqslant \frac{1}{3\sqrt{n}}\mathbb{E}\|\boldsymbol{G}\|_F^2\mathbf{1}_{\mathcal{E}}.$$

For the right-hand side,

$$\mathbb{E}\|\boldsymbol{G}\|_F^2\mathbf{1}_{\mathcal{E}} = \mathbb{E}\|\boldsymbol{G}\|_F^2 - \mathbb{E}\|\boldsymbol{G}\|_F^2\mathbf{1}_{\mathcal{E}^c} = n^2 - \mathbb{E}\|\boldsymbol{G}\|_F^2\mathbf{1}_{\mathcal{E}^c}, \tag{II}$$

where using Cauchy's inequality,

$$\mathbb{E}\|\boldsymbol{G}\|_F^2\mathbf{1}_{\mathcal{E}^c} \leqslant \sqrt{\mathbb{E}\|\boldsymbol{G}\|_F^4}\sqrt{\mathbb{P}(\mathcal{E}^c)} \leqslant \sqrt{3n^4} \cdot \sqrt{\mathrm{e}^{-n/2}} = \sqrt{3}n^2\mathrm{e}^{-n/2} \lesssim \frac{1}{2}n^2,$$

(*Since $\|\boldsymbol{G}\|_F^2 \sim \chi^2(n^2)$, its second moment is $\mathbb{E}\|\boldsymbol{G}\|_F^4 = \mathrm{Var}(\|\boldsymbol{G}\|_F^2) + (\mathbb{E}\|\boldsymbol{G}\|_F^2)^2 = 2n^2 + n^4 \leqslant 3n^4$.*)

Substituting this into (I),

$$\mathbb{E}\|\boldsymbol{G}\|_F^2\mathbf{1}_{\mathcal{E}} \geqslant n^2 - \frac{1}{2}n^2 = \frac{1}{2}n^2,$$

Substituting this into (II),

$$\mathbb{E}\|\boldsymbol{G}\|_* \geqslant \frac{1}{3\sqrt{n}} \cdot \frac{1}{2}n^2 = \frac{1}{6}n^{3/2},$$

Thus $w(\mathbb{T}) \gtrsim n^{3/2}$.

Thus, $w(\mathbb{T}) \asymp n^{3/2}$.

20 [**Gaussian Width and Gaussian Complexity**] Prove a conclusion from Lemma L7.5.11 (*Almost equivalent versions of Gaussian width*) that was left unproven.

Let $\boldsymbol{g} \sim N(\mathbf{0}, \boldsymbol{I}_n)$. Define the Gaussian width of $\mathbb{T} \subset \mathbb{R}^n$ as $w(\mathbb{T}) = \mathbb{E}\big[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{t}\rangle\big]$, and the Gaussian complexity as $\gamma(\mathbb{T}) = \mathbb{E}\big[\sup_{\boldsymbol{t}\in\mathbb{T}}|\langle \boldsymbol{g}, \boldsymbol{t}\rangle|\big]$. Prove: for any $\boldsymbol{y} \in \mathbb{T}$,

$$\gamma(\mathbb{T}) \asymp w(\mathbb{T}) + \|\boldsymbol{y}\|_2.$$

Rk. The Gaussian width $w(\mathbb{T})$ measures the intrinsic geometric size of the set and satisfies translation invariance;
The Gaussian complexity $\gamma(\mathbb{T})$ measures the maximum absolute magnitude of the vectors of the set under Gaussian projection, does not satisfy translation invariance, and its magnitude is closely related to the distance of the set from the origin.

Prf. • Proof of the upper bound $\gamma(\mathbb{T}) \lesssim w(\mathbb{T}) + \|\boldsymbol{y}\|_2$

Let $\boldsymbol{y} \in \mathbb{T}$. For any $\boldsymbol{t} \in \mathbb{T}$, decompose it as $\boldsymbol{t} = (\boldsymbol{t} - \boldsymbol{y}) + \boldsymbol{y}$. By the definition of Gaussian complexity,

$$\begin{aligned}
\gamma(\mathbb{T}) &= \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}|\langle \boldsymbol{g}, \boldsymbol{t}\rangle|\right] \\
&= \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}|\langle \boldsymbol{g}, \boldsymbol{t}-\boldsymbol{y}\rangle + \langle \boldsymbol{g}, \boldsymbol{y}\rangle|\right] \\
&\leqslant \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}(|\langle \boldsymbol{g}, \boldsymbol{t}-\boldsymbol{y}\rangle| + |\langle \boldsymbol{g}, \boldsymbol{y}\rangle|)\right] \\
&= \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}|\langle \boldsymbol{g}, \boldsymbol{t}-\boldsymbol{y}\rangle|\right] + \mathbb{E}[|\langle \boldsymbol{g}, \boldsymbol{y}\rangle|].
\end{aligned}$$

For the second term, since $\langle \boldsymbol{g}, \boldsymbol{y}\rangle \sim N(\mathbf{0}, \|\boldsymbol{y}\|_2^2)$,

$$\mathbb{E}[|\langle \boldsymbol{g}, \boldsymbol{y}\rangle|] = \sqrt{\frac{2}{\pi}}\|\boldsymbol{y}\|_2.$$

For the first term, let $\mathbb{T}' = \mathbb{T} - \boldsymbol{y}$. Then $\mathbf{0} \in \mathbb{T}'$ (*since* $\boldsymbol{y} \in \mathbb{T}$). Hence (*for a set containing the origin, the supremum of the inner product must be nonnegative*)

$$\sup_{\boldsymbol{t}'\in\mathbb{T}'}\langle \boldsymbol{g}, \boldsymbol{t}'\rangle,\ \sup_{\boldsymbol{t}'\in\mathbb{T}'}\langle -\boldsymbol{g}, \boldsymbol{t}'\rangle \geqslant 0,$$

thus

$$\begin{aligned}
\sup_{\boldsymbol{t}'\in\mathbb{T}'}|\langle \boldsymbol{g}, \boldsymbol{t}'\rangle| &= \max\left(\sup_{\boldsymbol{t}'\in\mathbb{T}'}\langle \boldsymbol{g}, \boldsymbol{t}'\rangle,\ \sup_{\boldsymbol{t}'\in\mathbb{T}'}\langle -\boldsymbol{g}, \boldsymbol{t}'\rangle\right) \\
&\leqslant \sup_{\boldsymbol{t}'\in\mathbb{T}'}\langle \boldsymbol{g}, \boldsymbol{t}'\rangle + \sup_{\boldsymbol{t}'\in\mathbb{T}'}\langle -\boldsymbol{g}, \boldsymbol{t}'\rangle,
\end{aligned}$$

taking expectation on both sides (*note that* $\boldsymbol{g} \overset{d}{=} -\boldsymbol{g}$), then by the translation invariance of Gaussian width (Q7.15 (*Basic Properties of Gaussian Width*) (2)),

$$\mathbb{E}\left[\sup_{\boldsymbol{t}'\in\mathbb{T}'}|\langle \boldsymbol{g}, \boldsymbol{t}'\rangle|\right] \leqslant w(\mathbb{T}') + w(\mathbb{T}') = 2w(\mathbb{T}') = 2w(\mathbb{T}).$$

Thus,

$$\gamma(\mathbb{T}) \leqslant 2w(\mathbb{T}) + \sqrt{\frac{2}{\pi}}\|\boldsymbol{y}\|_2 \leqslant 2(w(\mathbb{T}) + \|\boldsymbol{y}\|_2).$$

• Proof of the lower bound $w(\mathbb{T}) + \|\boldsymbol{y}\|_2 \lesssim \gamma(\mathbb{T})$

For the first term, it is easy to see that

$$\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{t}\rangle \leqslant \sup_{\boldsymbol{t}\in\mathbb{T}}|\langle \boldsymbol{g}, \boldsymbol{t}\rangle|,$$

taking expectation on both sides,

$$w(\mathbb{T}) \leqslant \gamma(\mathbb{T}).$$

For the second term, it is easy to see that

$$\sup_{\boldsymbol{t}\in\mathbb{T}} |\langle \boldsymbol{g}, \boldsymbol{t}\rangle| \geqslant |\langle \boldsymbol{g}, \boldsymbol{y}\rangle|,$$

taking expectation on both sides,

$$\gamma(\mathbb{T}) \geqslant \mathbb{E}[|\langle \boldsymbol{g}, \boldsymbol{y}\rangle|] = \sqrt{\frac{2}{\pi}}\|\boldsymbol{y}\|_2,$$

and hence

$$\|\boldsymbol{y}\|_2 \leqslant \sqrt{\frac{\pi}{2}}\gamma(\mathbb{T}).$$

Thus,

$$w(\mathbb{T}) + \|\boldsymbol{y}\|_2 \leqslant \gamma(\mathbb{T}) + \sqrt{\frac{\pi}{2}}\gamma(\mathbb{T}) = \left(1 + \sqrt{\frac{\pi}{2}}\right)\gamma(\mathbb{T}).$$

21 [**Effective Dimension and Algebraic Dimension**] Let $\mathbb{T} \subset \mathbb{R}^n$ be a nonzero finite set. In Definition D7.5.12 (*Effective dimension*) we introduced its effective dimension,

$$d(\mathbb{T}) = \left(\frac{w(\mathbb{T})}{\sup_{\boldsymbol{t}\in\mathbb{T}} \|\boldsymbol{t}\|_2}\right)^2,$$

(1) Prove that the effective dimension is always controlled by the linear algebraic dimension, i.e.,

$$d(\mathbb{T}) \leqslant \dim(\mathbb{T}).$$

(2) Prove that when $\mathbb{T} \subset \mathbb{R}^n$ is the Euclidean ball of a linear subspace, the above inequality becomes an equality (*in the asymptotic sense*).

Prf. (1) Let $\dim(\mathbb{T}) = k$. Then there exists a k-dimensional linear subspace V such that $\mathbb{T} \subset V \subset \mathbb{R}^n$.

Let $\boldsymbol{P}$ be the orthogonal projection matrix from $\mathbb{R}^n$ onto V, satisfying $\boldsymbol{P}\boldsymbol{t} = \boldsymbol{t}$ for all $\boldsymbol{t} \in \mathbb{T}$.

Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Then the Gaussian width of $\mathbb{T}$ is

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{t}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{P}\boldsymbol{t}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{P}\boldsymbol{g}, \boldsymbol{t}\rangle\right].$$

Using Cauchy's inequality,

$$\langle \boldsymbol{P}\boldsymbol{g}, \boldsymbol{t}\rangle \leqslant \|\boldsymbol{P}\boldsymbol{g}\|_2\|\boldsymbol{t}\|_2,$$

taking the supremum on both sides,

$$\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{P}\boldsymbol{g}, \boldsymbol{t}\rangle \leqslant \|\boldsymbol{P}\boldsymbol{g}\|_2\left(\sup_{\boldsymbol{t}\in\mathbb{T}}\|\boldsymbol{t}\|_2\right),$$

then taking expectation on both sides,

$$w(\mathbb{T}) \leqslant \mathbb{E}\|\boldsymbol{P}\boldsymbol{g}\|_2 \cdot \sup_{\boldsymbol{t}\in\mathbb{T}}\|\boldsymbol{t}\|_2,$$

Then we only need to compute $\mathbb{E}\|\boldsymbol{P}\boldsymbol{g}\|_2$.

Using Jensen's inequality,

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{g}\|_2 \leqslant \sqrt{\mathbb{E}\|\boldsymbol{P}\boldsymbol{g}\|_2^2},$$

where

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{g}\|_2^2 = \operatorname{tr}(\operatorname{Cov}(\boldsymbol{P}\boldsymbol{g})) = \operatorname{tr}(\boldsymbol{P}\boldsymbol{I}_n\boldsymbol{P}^\top) = \operatorname{tr}(\boldsymbol{P}) = k,$$

substituting this in,

$$w(\mathbb{T}) \leqslant \sqrt{k} \cdot \sup_{\boldsymbol{t}\in\mathbb{T}}\|\boldsymbol{t}\|_2.$$

Hence

$$d(\mathbb{T}) = \left(\frac{w(\mathbb{T})}{\sup_{\boldsymbol{t}\in\mathbb{T}} \|\boldsymbol{t}\|_2}\right)^2 \leqslant (\sqrt{k})^2 = k.$$

(2) Let $\mathbb{T} = \{\boldsymbol{t} \in V : \|\boldsymbol{t}\|_2 \leqslant R\}$. Then $\sup_{\boldsymbol{t}\in\mathbb{T}} \|\boldsymbol{t}\|_2 = R$.

The Gaussian width of $\mathbb{T}$ is

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{Pg}, \boldsymbol{t}\rangle\right],$$

where the inner product attains its maximum when the directions of $\boldsymbol{t}$ and $\boldsymbol{Pg}$ coincide, at which point the length is the maximum radius R,

$$\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{Pg}, \boldsymbol{t}\rangle = \left\langle \boldsymbol{Pg}, R\frac{\boldsymbol{Pg}}{\|\boldsymbol{Pg}\|_2} \right\rangle = R\|\boldsymbol{Pg}\|_2,$$

taking expectation on both sides,

$$w(\mathbb{T}) = R \cdot \mathbb{E}\|\boldsymbol{Pg}\|_2,$$

Hence the effective dimension $d(\mathbb{T})$ is

$$d(\mathbb{T}) = \left(\frac{w(\mathbb{T})}{\sup_{\boldsymbol{t}\in\mathbb{T}} \|\boldsymbol{t}\|_2}\right)^2 = \left(\frac{R \cdot \mathbb{E}\|\boldsymbol{Pg}\|_2}{R}\right)^2 = (\mathbb{E}\|\boldsymbol{Pg}\|_2)^2,$$

In an orthonormal coordinate system of V, $\boldsymbol{Pg}$ has distribution $N(\boldsymbol{0}, \boldsymbol{I}_k)$. By Exercise Q7.12 (*Norm of a Gaussian Vector*),

$$(\mathbb{E}\|\boldsymbol{Pg}\|_2)^2 \sim k - \frac{1}{2} + O(k^{-1}),$$

hence

$$d(\mathbb{T}) \asymp k = \dim(\mathbb{T}).$$

22 [**Effective Dimension of Finite Sets**] Let $\mathbb{T} \subset \mathbb{R}^n$ be a finite set. Prove that:

$$d(\mathbb{T}) \leqslant C \log|\mathbb{T}|. \qquad (C > 0)$$

Prf. • Extract the maximum norm

Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Then the Gaussian width of $\mathbb{T}$ is

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle\right].$$

Let $X_{\boldsymbol{t}} = \langle \boldsymbol{g}, \boldsymbol{t}\rangle$ be a zero-mean Gaussian random variable. Its variance is

$$\mathrm{Var}(X_{\boldsymbol{t}}) = \mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{t}\rangle^2] = \|\boldsymbol{t}\|_2^2,$$

Denote $R = \max_{\boldsymbol{t}\in\mathbb{T}} \|\boldsymbol{t}\|_2$, so $X_{\boldsymbol{t}} \sim N(\boldsymbol{0}, \sigma_{\boldsymbol{t}}^2)$, where $\sigma_{\boldsymbol{t}} \leqslant R$. If $R = 0$, then $\mathbb{T} = \{\boldsymbol{0}\}$, and the conclusion is trivial; afterwards only consider $R > 0$.

• Use moment generating function and union bound to estimate the expectation of the maximum

For any $\lambda > 0$,

$$\begin{aligned}
\exp(\lambda \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} X_{\boldsymbol{t}}\right]) &\leqslant \mathbb{E}\left[\exp\left(\lambda \sup_{\boldsymbol{t}\in\mathbb{T}} X_{\boldsymbol{t}}\right)\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \exp(\lambda X_{\boldsymbol{t}})\right] \\
&\leqslant \mathbb{E}\left[\sum_{\boldsymbol{t}\in\mathbb{T}} \exp(\lambda X_{\boldsymbol{t}})\right] = \sum_{\boldsymbol{t}\in\mathbb{T}} \mathbb{E}[\exp(\lambda X_{\boldsymbol{t}})] \\
&\leqslant \sum_{\boldsymbol{t}\in\mathbb{T}} \exp\left(\frac{\lambda^2 R^2}{2}\right) = |\mathbb{T}| \exp\left(\frac{\lambda^2 R^2}{2}\right),
\end{aligned}$$

Taking the logarithm on both sides and then dividing by λ gives

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} X_{\boldsymbol{t}}\right] \leqslant \frac{\log|\mathbb{T}|}{\lambda} + \frac{\lambda R^2}{2},$$

The optimal parameter on the right-hand side is $\lambda_0 = \frac{\sqrt{2\log|\mathbb{T}|}}{R}$. Substituting it gives

$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} X_{\boldsymbol{t}}\right] \leqslant \frac{\log|\mathbb{T}|}{\frac{\sqrt{2\log|\mathbb{T}|}}{R}} + \frac{\frac{\sqrt{2\log|\mathbb{T}|}}{R}\cdot R^2}{2} = \frac{R}{\sqrt{2}}\sqrt{\log|\mathbb{T}|} + \frac{R}{\sqrt{2}}\sqrt{\log|\mathbb{T}|} = \sqrt{2}R\sqrt{\log|\mathbb{T}|},$$

(*Here we assume* $|\mathbb{T}| \geqslant 2$*; if* $|\mathbb{T}| = 1$*, then* $\log|\mathbb{T}| = 0$*, in which case* $w(\mathbb{T}) = \mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{t}\rangle] = 0$*, which is trivial.*)

Substituting this in,

$$w(\mathbb{T}) \leqslant \sqrt{2}R\sqrt{\log|\mathbb{T}|},$$

hence

$$d(\mathbb{T}) = \left(\frac{w(\mathbb{T})}{\sup_{\boldsymbol{t}\in\mathbb{T}}\|\boldsymbol{t}\|_2}\right)^2 = \left(\frac{w(\mathbb{T})}{R}\right)^2 \leqslant \left(\frac{\sqrt{2}R\sqrt{\log|\mathbb{T}|}}{R}\right)^2 = 2\log|\mathbb{T}|,$$

here let $C = 2$.

23 [**Gaussian Width and Effective Dimension of Ellipsoids**] An ellipsoid is the image of $\mathbb{B}_2^n$ under the linear transformation with matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, i.e., the set $\boldsymbol{A}(\mathbb{B}_2^n)$.

Prove the following properties of this ellipsoid:

(1) (Gaussian width) The Gaussian width of $\boldsymbol{A}(\mathbb{B}_2^n)$ is asymptotically equivalent to the Frobenius norm of $\boldsymbol{A}$, i.e.,

$$w(\boldsymbol{A}(\mathbb{B}_2^n)) \asymp \|\boldsymbol{A}\|_F.$$

(2) (Effective dimension) The effective dimension of $\boldsymbol{A}(\mathbb{B}_2^n)$ is asymptotically exactly equal to the effective rank of $\boldsymbol{A}^\top\boldsymbol{A}$ (*i.e., the stable rank of* $\boldsymbol{A}$),

$$d(\boldsymbol{A}(\mathbb{B}_2^n)) \asymp r(\boldsymbol{A}^\top\boldsymbol{A}) = s(\boldsymbol{A}).$$

where the effective dimension and stable rank are respectively defined as

$$d(\mathbb{T}) \asymp \frac{w(\mathbb{T})^2}{\operatorname{diam}(\mathbb{T})^2}, s(\boldsymbol{A}) = \frac{\|\boldsymbol{A}\|_F^2}{\|\boldsymbol{A}\|^2}.$$

Rk. **Lemma** (Moment equivalence of Gaussian quadratic forms / special case of the Kahane-Khintchine inequality) Let $\boldsymbol{M} \in \mathbb{R}^{m\times n}$, $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Then

$$\mathbb{E}\|\boldsymbol{M}\boldsymbol{g}\|_2 \geqslant c\sqrt{\mathbb{E}\|\boldsymbol{M}\boldsymbol{g}\|_2^2}. \qquad (c > 0)$$

(The proof is similar to the lower bound proof in case b. of Exercise Q7.17 (Gaussian Width of ℓ^p *Balls).)*

Let $S = \|\boldsymbol{M}\boldsymbol{g}\|_2^2 = \boldsymbol{g}^\top\boldsymbol{M}^\top\boldsymbol{M}\boldsymbol{g}$*. Note that* $\boldsymbol{M}^\top\boldsymbol{M}$ *is a symmetric positive semidefinite matrix. Perform its orthogonal decomposition* $\boldsymbol{M}^\top\boldsymbol{M} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$*, where* $\boldsymbol{U}$ *is orthogonal and* $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \cdots, \lambda_n)$ *is the eigenvalue matrix. Then*

$$S = \boldsymbol{g}^\top\boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top\boldsymbol{g} = (\boldsymbol{U}^\top\boldsymbol{g})^\top\boldsymbol{\Lambda}(\boldsymbol{U}^\top\boldsymbol{g}),$$

By the rotational invariance of the Gaussian distribution, $\tilde{\boldsymbol{g}} = \boldsymbol{U}^\top\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I})$*, hence*

$$S = \sum_{i=1}^n \lambda_i \tilde{g}_i^2,$$

Thus the numerical characteristics of S *are*

$$\mathbb{E}[S] = \sum_{i=1}^n \lambda_i \mathbb{E}[\tilde{g}_i^2] = \sum_{i=1}^n \lambda_i,$$

$$\operatorname{Var}(S) = \sum_{i=1}^n \lambda_i^2 \operatorname{Var}(\tilde{g}_i^2) = 2\sum_{i=1}^n \lambda_i^2,$$

$$\mathbb{E}[S^2] = \mathrm{Var}(S) + (\mathbb{E}[S])^2 = 2\sum_{i=1}^{n}\lambda_i^2 + \left(\sum_{i=1}^{n}\lambda_i\right)^2.$$

Using the Paley-Zygmund inequality, for any $\theta \in (0,1)$,

$$\mathbb{P}\{S > \theta\mathbb{E}[S]\} \geqslant (1-\theta)^2\frac{(\mathbb{E}[S])^2}{\mathbb{E}[\mathbb{S}^2]},$$

where

$$\frac{\mathbb{E}[S^2]}{(\mathbb{E}[S])^2} = \frac{2\sum_{i=1}^{n}\lambda_i^2 + \left(\sum_{i=1}^{n}\lambda_i\right)^2}{\left(\sum_{i=1}^{n}\lambda_i\right)^2} = 2\frac{\sum_{i=1}^{n}\lambda_i^2}{\left(\sum_{i=1}^{n}\lambda_i\right)^2} + 1 \leqslant 2\cdot 1 + 1 = 3,$$

substituting this in,

$$\mathbb{P}\{S > \theta\mathbb{E}[S]\} \geqslant \frac{1}{3}(1-\theta)^2,$$

then let $\theta = 1/2$, *we obtain*

$$\mathbb{P}\left\{S > \frac{1}{2}\mathbb{E}[S]\right\} \geqslant \left(1-\frac{1}{2}\right)^2\cdot\frac{1}{3} = \frac{1}{12}.$$

Convert the above high-probability event into a lower bound for $\mathbb{E}[\sqrt{S}]$,

$$\begin{aligned}\mathbb{E}[\sqrt{S}] &\geqslant \mathbb{E}\left[\sqrt{S}\cdot\mathbf{1}_{\{S>\frac{1}{2}\mathbb{E}[S]\}}\right]\\ &\geqslant \sqrt{\frac{1}{2}\mathbb{E}[S]}\cdot\mathbb{P}\left\{S > \frac{1}{2}\mathbb{E}[S]\right\}\\ &\geqslant \sqrt{\frac{1}{2}\mathbb{E}[S]}\cdot\frac{1}{12} = \frac{1}{12\sqrt{2}}\sqrt{\mathbb{E}[S]}.\end{aligned}$$

i.e.,

$$\mathbb{E}\|\boldsymbol{M}\boldsymbol{g}\|_2 \geqslant \frac{1}{12\sqrt{2}}\sqrt{\mathbb{E}\|\boldsymbol{M}\boldsymbol{g}\|_2^2}.$$

Prf. (1) Let $\boldsymbol{g}\sim N(\boldsymbol{0},\boldsymbol{I}_m)$ be in $\mathbb{R}^m$. By the definition of Gaussian width,

$$w(\boldsymbol{A}(\mathbb{B}_2^n)) = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{B}_2^n}\langle\boldsymbol{g},\boldsymbol{A}\boldsymbol{x}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{B}_2^n}\langle\boldsymbol{A}^\top\boldsymbol{g},\boldsymbol{x}\rangle\right] = \mathbb{E}\|\boldsymbol{A}^\top\boldsymbol{g}\|_2.$$

It is easy to see that

$$\mathbb{E}\|\boldsymbol{A}^\top\boldsymbol{g}\|_2^2 = \mathrm{tr}(\boldsymbol{A}\boldsymbol{A}^\top) = \|\boldsymbol{A}\|_F^2.$$

For the upper bound, using Jensen's inequality,

$$\mathbb{E}\|\boldsymbol{A}^\top\boldsymbol{g}\|_2 \leqslant \sqrt{\mathbb{E}\|\boldsymbol{A}^\top\boldsymbol{g}\|_2^2} = \sqrt{\|\boldsymbol{A}\|_F^2} = \|\boldsymbol{A}\|_F.$$

For the lower bound, by the moment equivalence of Gaussian quadratic forms,

$$\mathbb{E}\|\boldsymbol{A}^\top\boldsymbol{g}\|_2 \geqslant c\|\boldsymbol{A}\|_F.$$

Thus,

$$w(\boldsymbol{A}(\mathbb{B}_2^n)) \asymp \|\boldsymbol{A}\|_F.$$

(2) For the numerator, using the conclusion of (1),

$$w(\boldsymbol{A}(\mathbb{B}_2^n))^2 \asymp \|\boldsymbol{A}\|_F^2.$$

For the denominator, since the ellipsoid after linear transformation is a convex body symmetric about the origin, its diameter is determined by the point farthest from the origin, i.e.,

$$\mathrm{diam}(\boldsymbol{A}(\mathbb{B}_2^n)) = 2\sup_{\boldsymbol{x}\in\mathbb{B}_2^n}\|\boldsymbol{A}\boldsymbol{x}\|_2 = 2\|\boldsymbol{A}\|.$$

Thus,

$$d(\boldsymbol{A}(\mathbb{B}_2^n)) \asymp \frac{w(\boldsymbol{A}(\mathbb{B}_2^n))^2}{\operatorname{diam}(\boldsymbol{A}(\mathbb{B}_2^n))^2} \asymp \frac{\|\boldsymbol{A}\|_F^2}{\|\boldsymbol{A}\|^2} = s(\boldsymbol{A}).$$

Then let $\boldsymbol{\Sigma} = \boldsymbol{A}^\top \boldsymbol{A}$. Its effective rank is

$$r(\boldsymbol{\Sigma}) = \frac{\operatorname{tr}(\boldsymbol{\Sigma})}{\|\boldsymbol{\Sigma}\|},$$

and $\|\boldsymbol{A}^\top \boldsymbol{A}\| = \|\boldsymbol{A}\|^2$. Hence

$$d(\boldsymbol{A}(\mathbb{B}_2^n)) \asymp s(\boldsymbol{A}) = r(\boldsymbol{A}^\top \boldsymbol{A}),$$

therefore

$$d(\boldsymbol{A}(\mathbb{B}_2^n)) \asymp r(\boldsymbol{A}^\top \boldsymbol{A}) = s(\boldsymbol{A}).$$

24 [**Modeling Random Vectors on the Sphere**] Let $1 \leqslant m \leqslant n$, let $\boldsymbol{z} \in \mathbb{S}^{m-1}$, and let $\boldsymbol{U} \sim U(O(n))$ be a random orthogonal matrix uniformly distributed on the n-dimensional orthogonal group $O(n)$ (*i.e., the Haar measure*).
Let $\boldsymbol{B} \in \mathbb{R}^{n\times m}$ consist of the first m columns of $\boldsymbol{U}$. Prove that:

$$\boldsymbol{B}\boldsymbol{z} \sim U(\mathbb{S}^{n-1}).$$

Prf. (*Consider using the dimension padding trick to directly convert this into a basic property of the Haar measure on the orthogonal group.*)

- Dimension padding

  Let the column vectors of the random orthogonal matrix $\boldsymbol{U}$ be $\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n$. Then $\boldsymbol{B} = (\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_m)$, and

  $$\boldsymbol{B}\boldsymbol{z} = z_1\boldsymbol{u}_1 + z_2\boldsymbol{u}_2 + \cdots + z_m\boldsymbol{u}_m.$$

  Construct the padded fixed vector $\tilde{\boldsymbol{z}}$ as

  $$\tilde{\boldsymbol{z}} = \begin{pmatrix} z_1 \\ \vdots \\ z_m \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{S}^{n-1},$$

  then

  $$\boldsymbol{U}\tilde{\boldsymbol{z}} = (\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n) \begin{pmatrix} z_1 \\ \vdots \\ z_m \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \sum_{i=1}^{m} z_i\boldsymbol{u}_i + \sum_{j=m+1}^{n} 0 \cdot \boldsymbol{u}_j = \boldsymbol{B}\boldsymbol{z},$$

  Then we only need to prove $\boldsymbol{x} = \boldsymbol{U}\tilde{\boldsymbol{z}} \sim U(\mathbb{S}^{n-1})$.

- Using the rotational invariance of the orthogonal group

  (*If* $\operatorname{supp}(\boldsymbol{x}) \subset \mathbb{S}^{n-1}$ *and its distribution is invariant under any orthogonal transformation (rotation or reflection), then* $\boldsymbol{x} \sim U(\mathbb{S}^{n-1})$.)

  Since

  $$\|\boldsymbol{x}\|_2^2 = \|\boldsymbol{U}\tilde{\boldsymbol{z}}\|_2^2 = (\boldsymbol{U}\tilde{\boldsymbol{z}})^\top(\boldsymbol{U}\tilde{\boldsymbol{z}}) = \tilde{\boldsymbol{z}}^\top\boldsymbol{U}^\top\boldsymbol{U}\tilde{\boldsymbol{z}} = \tilde{\boldsymbol{z}}^\top\boldsymbol{I}_n\tilde{\boldsymbol{z}} = \|\tilde{\boldsymbol{z}}\|_2^2 = 1,$$

  we have $\boldsymbol{x} \in \mathbb{S}^{n-1}$.

Let $\boldsymbol{V} \in O(n)$ be a fixed orthogonal matrix. Then $\boldsymbol{V}\boldsymbol{x} \overset{d}{=} \boldsymbol{x}$, hence

$$\boldsymbol{V}\boldsymbol{x} = \boldsymbol{V}(\boldsymbol{U}\tilde{\boldsymbol{z}}) = (\boldsymbol{V}\boldsymbol{U})\tilde{\boldsymbol{z}} \overset{d}{=} \boldsymbol{U}\tilde{\boldsymbol{z}} = \boldsymbol{x}.$$

Thus, $\boldsymbol{B}\boldsymbol{z} = \boldsymbol{x} \sim U(\mathbb{S}^{n-1})$.

Rk. This exercise introduces an efficient sampling algorithm for generating uniformly distributed random vectors on the sphere. Even if only a partial rotation transformation $\boldsymbol{B}$ in a high-dimensional space is used, applying it to a fixed unit vector of the corresponding dimension still produces a random vector that is strictly isotropic in the entire space.

25 [**Gaussian Projection of Sets**] Let $\boldsymbol{G} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix.

Prove a version of Theorem T7.6.1 (*Sizes of random projections of sets*): for a set $\mathbb{T} \subset \mathbb{R}^n$, the expected diameter satisfies the asymptotic equivalence relation of Chevet-type inequality,

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{G}\mathbb{T})] \asymp w(\mathbb{T}) + \sqrt{m}\operatorname{diam}(\mathbb{T}).$$

- Define the difference set $\mathbb{T}' = \mathbb{T} - \mathbb{T} = \{\boldsymbol{x} - \boldsymbol{y} : \boldsymbol{x}, \boldsymbol{y} \in \mathbb{T}\}$. Then the diameter after projection is

$$\operatorname{diam}(\boldsymbol{G}\mathbb{T}) = \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \|\boldsymbol{G}\boldsymbol{x} - \boldsymbol{G}\boldsymbol{y}\|_2 = \sup_{\boldsymbol{v}\in\mathbb{T}'} \|\boldsymbol{G}\boldsymbol{v}\|_2,$$

  The diameter of the difference set is the original diameter,

$$\sup_{\boldsymbol{v}\in\mathbb{T}'} \|\boldsymbol{v}\|_2 = \operatorname{diam}(\mathbb{T}) = D,$$

  The Gaussian width of the difference set is twice the original Gaussian width,

$$w(\mathbb{T}') = \mathbb{E}\left[\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x} - \boldsymbol{y}\rangle\right] = 2w(\mathbb{T}).$$

- Proof of the upper bound $\mathbb{E}[\operatorname{diam}(\boldsymbol{G}\mathbb{T})] \lesssim w(\mathbb{T}) + \sqrt{m}\operatorname{diam}(\mathbb{T})$

  For

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{G}\mathbb{T})] = \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'} \|\boldsymbol{G}\boldsymbol{v}\|_2\right] = \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\sup_{\boldsymbol{w}\in\mathbb{S}^{m-1}} \langle \boldsymbol{G}\boldsymbol{v}, \boldsymbol{w}\rangle\right],$$

  Define the target Gaussian process $X_{\boldsymbol{v},\boldsymbol{w}} = \langle \boldsymbol{G}\boldsymbol{v}, \boldsymbol{w}\rangle$, and the comparison Gaussian process $Y_{\boldsymbol{v},\boldsymbol{w}} = \langle \boldsymbol{g}, \boldsymbol{v}\rangle + D\langle \boldsymbol{h}, \boldsymbol{w}\rangle$, where $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ and $\boldsymbol{h} \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$ are independent. Then

$$\begin{aligned}
\mathbb{E}[(X_{\boldsymbol{v},\boldsymbol{w}} - X_{\boldsymbol{v}',\boldsymbol{w}'})^2] &= \mathbb{E}[(\langle \boldsymbol{G}\boldsymbol{v}, \boldsymbol{w}\rangle - \langle \boldsymbol{G}\boldsymbol{v}', \boldsymbol{w}'\rangle)^2] \\
&= \|\boldsymbol{w}\boldsymbol{v}^\top - \boldsymbol{w}'\boldsymbol{v}'^\top\|_F^2 \\
&= \|\boldsymbol{v}\|_2^2\|\boldsymbol{w}\|_2^2 + \|\boldsymbol{v}'\|_2^2\|\boldsymbol{w}'\|_2^2 - 2\langle \boldsymbol{v}, \boldsymbol{v}'\rangle\langle \boldsymbol{w}, \boldsymbol{w}'\rangle \\
&= \|\boldsymbol{v}\|_2^2 + \|\boldsymbol{v}'\|_2^2 - 2\langle \boldsymbol{v}, \boldsymbol{v}'\rangle\langle \boldsymbol{w}, \boldsymbol{w}'\rangle, \\
\mathbb{E}[(Y_{\boldsymbol{v},\boldsymbol{w}} - Y_{\boldsymbol{v}',\boldsymbol{w}'})^2] &= \mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{v} - \boldsymbol{v}'\rangle^2] + D^2\mathbb{E}[\langle \boldsymbol{h}, \boldsymbol{w} - \boldsymbol{w}'\rangle^2] \\
&= \|\boldsymbol{v} - \boldsymbol{v}'\|_2^2 + D^2\|\boldsymbol{w} - \boldsymbol{w}'\|_2^2 \\
&= \|\boldsymbol{v}\|_2^2 + \|\boldsymbol{v}'\|_2^2 - 2\langle \boldsymbol{v}, \boldsymbol{v}'\rangle + 2D^2 - 2D^2\langle \boldsymbol{w}, \boldsymbol{w}'\rangle,
\end{aligned}$$

  Hence

$$\begin{aligned}
\mathbb{E}[(Y_{\boldsymbol{v},\boldsymbol{w}} - Y_{\boldsymbol{v}',\boldsymbol{w}'})^2] - \mathbb{E}[(X_{\boldsymbol{v},\boldsymbol{w}} - X_{\boldsymbol{v}',\boldsymbol{w}'})^2] &= 2D^2 - 2D^2\langle \boldsymbol{w}, \boldsymbol{w}'\rangle - 2\langle \boldsymbol{v}, \boldsymbol{v}'\rangle + 2\langle \boldsymbol{v}, \boldsymbol{v}'\rangle\langle \boldsymbol{w}, \boldsymbol{w}'\rangle \\
&= 2(D^2 - \langle \boldsymbol{v}, \boldsymbol{v}'\rangle)(1 - \langle \boldsymbol{w}, \boldsymbol{w}'\rangle),
\end{aligned}$$

  where $\langle \boldsymbol{v}, \boldsymbol{v}'\rangle \leqslant \|\boldsymbol{v}\|_2\|\boldsymbol{v}'\|_2 \leqslant D^2$, $\langle \boldsymbol{w}, \boldsymbol{w}'\rangle \leqslant 1$. Substituting these in,

$$\mathbb{E}[(Y_{\boldsymbol{v},\boldsymbol{w}} - Y_{\boldsymbol{v}',\boldsymbol{w}'})^2] - \mathbb{E}[(X_{\boldsymbol{v},\boldsymbol{w}} - X_{\boldsymbol{v}',\boldsymbol{w}'})^2] \geqslant 0,$$

  Thus the Sudakov-Fernique condition is satisfied.

  Using Theorem T7.2.8 (*Sudakov-Fernique inequality*) (*Sudakov–Fernique inequality*),

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{G}\mathbb{T})] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}',\boldsymbol{w}\in\mathbb{S}^{m-1}} Y_{\boldsymbol{v},\boldsymbol{w}}\right]$$

$$
\begin{aligned}
&= \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{g}, \boldsymbol{v}\rangle\right] + D\cdot\mathbb{E}\left[\sup_{\boldsymbol{w}\in\mathbb{S}^{m-1}}\langle \boldsymbol{h}, \boldsymbol{w}\rangle\right] \\
&= w(\mathbb{T}') + \operatorname{diam}(\mathbb{T})\cdot\mathbb{E}\|\boldsymbol{h}\|_2 \\
&\leqslant 2w(\mathbb{T}) + \sqrt{m}\operatorname{diam}(\mathbb{T}) \\
&\lesssim w(\mathbb{T}) + \sqrt{m}\operatorname{diam}(\mathbb{T}).
\end{aligned}
$$

- Proof of the lower bound $\mathbb{E}[\operatorname{diam}(\boldsymbol{G}\mathbb{T})] \gtrsim w(\mathbb{T}) + \sqrt{m}\operatorname{diam}(\mathbb{T})$

  (*Since both terms on the right-hand side are nonnegative, we only need to separately prove that it is greater than $w(\mathbb{T})$ and $\sqrt{m}\operatorname{diam}(\mathbb{T})$.*)

  First prove that it is greater than $w(\mathbb{T})$. Let $\boldsymbol{e}_1 = (1, 0, \cdots, 0)^\top \in \mathbb{R}^m$. Then

  $$\operatorname{diam}(\boldsymbol{G}\mathbb{T}) = \sup_{\boldsymbol{v}\in\mathbb{T}'}\|\boldsymbol{G}\boldsymbol{v}\|_2 \geqslant \sup_{\boldsymbol{v}\in\mathbb{T}'}|\langle \boldsymbol{G}\boldsymbol{v}, \boldsymbol{e}_1\rangle| \geqslant \sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{G}^\top\boldsymbol{e}_1, \boldsymbol{v}\rangle = \sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{g}_1, \boldsymbol{v}\rangle,$$

  taking expectation on both sides,

  $$\mathbb{E}[\operatorname{diam}(\boldsymbol{G}\mathbb{T})] \geqslant \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{g}_1, \boldsymbol{v}\rangle\right] = w(\mathbb{T}') = 2w(\mathbb{T}) \gtrsim w(\mathbb{T}).$$

  Then prove that it is greater than $\sqrt{m}\operatorname{diam}(\mathbb{T})$. For every $\eta > 0$, choose $\boldsymbol{v}_\eta \in \mathbb{T}'$ such that $\|\boldsymbol{v}_\eta\|_2 \geqslant \operatorname{diam}(\mathbb{T}) - \eta$. Then

  $$\operatorname{diam}(\boldsymbol{G}\mathbb{T}) = \sup_{\boldsymbol{v}\in\mathbb{T}'}\|\boldsymbol{G}\boldsymbol{v}\|_2 \geqslant \|\boldsymbol{G}\boldsymbol{v}_\eta\|_2.$$

  Since the rows of $\boldsymbol{G}$ are mutually independent, the components of $\boldsymbol{G}\boldsymbol{v}_\eta \in \mathbb{R}^m$ follow $N(0, \|\boldsymbol{v}_\eta\|_2^2)$, hence

  $$\boldsymbol{G}\boldsymbol{v}_\eta \overset{d}{=} \|\boldsymbol{v}_\eta\|_2\boldsymbol{h},$$

  where $\boldsymbol{h} \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$. Taking expectation on both sides,

  $$\mathbb{E}[\operatorname{diam}(\boldsymbol{G}\mathbb{T})] \geqslant \|\boldsymbol{v}_\eta\|_2\mathbb{E}\|\boldsymbol{h}\|_2 \geqslant (\operatorname{diam}(\mathbb{T}) - \eta)\mathbb{E}\|\boldsymbol{h}\|_2.$$

  Letting $\eta \downarrow 0$ and using $\mathbb{E}\|\boldsymbol{h}\|_2 \asymp \sqrt{m}$ proves the assertion.

26 [**Random Projection of Sets: Lower Bound**] Let $\mathbb{T} \subset \mathbb{R}^n$, and let $\boldsymbol{P}$ be the orthogonal projection matrix from $\mathbb{R}^n$ onto a random m-dimensional subspace $E \sim U(G_{n,m})$, where $U(G_{n,m})$ denotes the uniform distribution on the Grassmannian $G_{n,m}$. Prove:

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] \geqslant c\left[w_s(\mathbb{T}) + \sqrt{\frac{m}{n}}\operatorname{diam}(\mathbb{T})\right].$$

$$w_s(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{\theta}, \boldsymbol{t}\rangle\right]. \qquad (\boldsymbol{\theta} \sim U(\mathbb{S}^{n-1}))$$

Rk. The Gaussian projection in the previous exercise does not preserve norms and stretches the space; the orthogonal projection in this exercise preserves norms, and the diameter after projection naturally cannot exceed the original diameter.

Prf. Define the difference set $\mathbb{T}' = \mathbb{T} - \mathbb{T}$, whose diameter $\operatorname{diam}(\mathbb{T}') = \operatorname{diam}(\mathbb{T})$. Then the diameter after projection is

$$\operatorname{diam}(\boldsymbol{P}\mathbb{T}) = \sup_{\boldsymbol{v}\in\mathbb{T}'}\|\boldsymbol{P}\boldsymbol{v}\|_2.$$

- Proof that $\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] \gtrsim \sqrt{\frac{m}{n}}\operatorname{diam}(\mathbb{T})$

  For every $\eta > 0$, choose $\boldsymbol{v}_\eta \in \mathbb{T}'$ such that $\|\boldsymbol{v}_\eta\|_2 \geqslant \operatorname{diam}(\mathbb{T}) - \eta$. Then

  $$\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] \geqslant \mathbb{E}\|\boldsymbol{P}\boldsymbol{v}_\eta\|_2,$$

  Let $\boldsymbol{h}_1, \cdots, \boldsymbol{h}_m$ be a random orthonormal basis of the random subspace E. Then

  $$\|\boldsymbol{P}\boldsymbol{v}_\eta\|_2^2 = \sum_{i=1}^m\langle \boldsymbol{v}_\eta, \boldsymbol{h}_i\rangle^2,$$

taking expectation on both sides,

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{v}_\eta\|_2^2 = \sum_{i=1}^{m} \mathbb{E}[\langle \boldsymbol{v}_\eta, \boldsymbol{h}_i\rangle^2] = m \cdot \mathbb{E}[\langle \boldsymbol{v}_\eta, \boldsymbol{h}_1\rangle^2],$$

where $\boldsymbol{h}_1 \sim U(\mathbb{S}^{n-1})$, satisfying

$$\mathbb{E}[\langle \boldsymbol{v}_\eta, \boldsymbol{h}_1\rangle^2] = \frac{\|\boldsymbol{v}_\eta\|_2^2}{n},$$

substituting this in,

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{v}_\eta\|_2^2 = \frac{m}{n}\|\boldsymbol{v}_\eta\|_2^2.$$

By the moment equivalence of Gaussian quadratic forms,

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{v}_\eta\|_2 \geqslant c_1\sqrt{\frac{m}{n}}\|\boldsymbol{v}_\eta\|_2 \geqslant c_1\sqrt{\frac{m}{n}}(\operatorname{diam}(\mathbb{T}) - \eta).$$

Letting $\eta \downarrow 0$ gives the desired lower bound.

- Proof that $\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] \gtrsim w_s(\mathbb{T})$

(*A key probabilistic equivalence: randomly drawing a unit vector from the random subspace E is equivalent to first randomly drawing a vector from the global $\mathbb{S}^{n-1}$, then projecting it onto E, and finally normalizing. This method is called the random direction of a random subspace.*)

For

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] = \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\|\boldsymbol{P}\boldsymbol{v}\|_2\right] = \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}\cap E}\langle \boldsymbol{v}, \boldsymbol{w}\rangle\right],$$

(*Here the range of* sup *is the unit sphere of the projection subspace E.*)

We only need to prove that

$$\mathbb{E}_E\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}\cap E}\langle \boldsymbol{v}, \boldsymbol{w}\rangle\right] \geqslant c_2\mathbb{E}_{\boldsymbol{\theta}}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{v}, \boldsymbol{\theta}\rangle\right].$$

Since

$$\sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}\cap E}\langle \boldsymbol{v}, \boldsymbol{w}\rangle = \|\boldsymbol{P}_E\boldsymbol{v}\|_2,$$

the proof goal is

$$\mathbb{E}_E\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\|\boldsymbol{P}_E\boldsymbol{v}\|_2\right] \geqslant \mathbb{E}_{\boldsymbol{\theta}}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{v}, \boldsymbol{\theta}\rangle\right].$$

For a fixed m-dimensional subspace E, draw a unit vector $\boldsymbol{\varphi} \sim U(\mathbb{S}^{n-1}\cap E)$. Then

$$\sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}\cap E}\langle \boldsymbol{v}, \boldsymbol{w}\rangle \geqslant \langle \boldsymbol{v}, \boldsymbol{\varphi}\rangle,$$

taking the supremum over $\boldsymbol{v} \in \mathbb{T}'$ on both sides,

$$\sup_{\boldsymbol{v}\in\mathbb{T}'}\sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}\cap E}\langle \boldsymbol{v}, \boldsymbol{w}\rangle \geqslant \sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{v}, \boldsymbol{\varphi}\rangle.$$

Keeping E fixed, take the conditional expectation over $\boldsymbol{\varphi}$,

$$\sup_{\boldsymbol{v}\in\mathbb{T}'}\sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}\cap E}\langle \boldsymbol{v}, \boldsymbol{w}\rangle \geqslant \mathbb{E}_{\boldsymbol{\varphi}|E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{v}, \boldsymbol{\varphi}\rangle\right],$$

then taking expectation over $E \sim U(G_{n,m})$ on both sides,

$$\mathbb{E}_E\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}\cap E}\langle \boldsymbol{v}, \boldsymbol{w}\rangle\right] \geqslant \mathbb{E}_E\left[\mathbb{E}_{\boldsymbol{\varphi}|E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{v}, \boldsymbol{\varphi}\rangle\right]\right]. \tag{I}$$

For the right-hand side of (I), note that it is a typical two-step random generation process:

1. First, uniformly randomly select an m-dimensional subspace E in $\mathbb{R}^n$;
2. Then, uniformly randomly select a unit direction $\boldsymbol{\varphi}$ in the subspace E;

Hence by the rotational invariance of the orthogonal group $O(n)$ action (*the high symmetry of the Haar measure*),

$$\boldsymbol{\varphi} \overset{d}{=} \boldsymbol{\theta} \sim U(\mathbb{S}^{n-1}),$$

thus

$$\mathbb{E}_E\left[\mathbb{E}_{\boldsymbol{\varphi}|E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{v},\boldsymbol{\varphi}\rangle\right]\right] = \mathbb{E}_{\boldsymbol{\theta}}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{v},\boldsymbol{\theta}\rangle\right],$$

substituting this into (I),

$$\mathbb{E}_E\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\sup_{\boldsymbol{w}\in\mathbb{S}^{n-1}\cap E}\langle \boldsymbol{v},\boldsymbol{w}\rangle\right] \geqslant \mathbb{E}_{\boldsymbol{\theta}}\left[\sup_{\boldsymbol{v}\in\mathbb{T}'}\langle \boldsymbol{v},\boldsymbol{\theta}\rangle\right].$$

Thus,

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] \geqslant \max\left(c_1\sqrt{\frac{m}{n}}\operatorname{diam}(\mathbb{T}), 2c_2 w_s(\mathbb{T})\right) \geqslant c\left(w_s(\mathbb{T}) + \sqrt{\frac{m}{n}}\operatorname{diam}(\mathbb{T})\right), \qquad \left(c = \frac{1}{2}\min(c_1, 2c_2)\right)$$

27 [**Matrix Sketching**] Suppose we have a very large $n \times k$ matrix $\boldsymbol{A}$. The following two methods can reduce its dimension without sacrificing the operator norm accuracy (*in the asymptotic sense*).

(1) (Random orthogonal projection) Let $\boldsymbol{P}$ be the orthogonal projection matrix from $\mathbb{R}^n$ onto a random m-dimensional subspace uniformly chosen from the Grassmann manifold $G_{n,m}$. Prove that:

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{A}\| \asymp \frac{1}{\sqrt{n}}\|\boldsymbol{A}\|_F + \sqrt{\frac{m}{n}}\|\boldsymbol{A}\|.$$

(2) (Gaussian random matrix projection) Let $\boldsymbol{G} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix. Prove that:

$$\mathbb{E}\|\boldsymbol{G}\boldsymbol{A}\| \asymp \|\boldsymbol{A}\|_F + \sqrt{m}\|\boldsymbol{A}\|.$$

Rk. The symbol $\asymp$ means that the values on both sides are equivalent up to a positive absolute constant factor.

Prf.

- Convert the matrix norm to a set diameter

  Let $\mathbb{T} = \boldsymbol{A}(\mathbb{B}_2^k)$ be the ellipsoid set in $\mathbb{R}^n$. Then for any matrix $\boldsymbol{M}$, its operator norm is

  $$\|\boldsymbol{M}\boldsymbol{A}\| = \sup_{\boldsymbol{x}\in\mathbb{B}_2^k}\|\boldsymbol{M}\boldsymbol{A}\boldsymbol{x}\|_2 = \sup_{\boldsymbol{t}\in\mathbb{T}}\|\boldsymbol{M}\boldsymbol{t}\|_2.$$

  Since $\mathbb{T}$ is symmetric about the origin, its radius is precisely the operator norm and its diameter is twice the radius,

  $$\operatorname{diam}(\boldsymbol{M}\mathbb{T}) = 2\|\boldsymbol{M}\boldsymbol{A}\|.$$

  (*In particular, when* $\boldsymbol{M} = \boldsymbol{I}_n$, $\operatorname{diam}(\mathbb{T}) = 2\|\boldsymbol{A}\|$.)

- Compute the Gaussian width $w(\mathbb{T})$ and spherical width $w_s(\mathbb{T})$

  Using the conclusion of Exercise Q7.23 (*Gaussian Width and Effective Dimension of Ellipsoids*) (1), for the ellipsoid $\mathbb{T} = \boldsymbol{A}(\mathbb{B}_2^k)$, its Gaussian width is asymptotically equivalent to the Frobenius norm of the generating matrix,

  $$w(\mathbb{T}) \asymp \|\boldsymbol{A}\|_F.$$

  Note that $\boldsymbol{g} \overset{d}{=} \|\boldsymbol{g}\|_2\boldsymbol{\theta}$, where $\|\boldsymbol{g}\|_2 \perp \boldsymbol{\theta} \sim U(\mathbb{S}^{n-1})$. Hence

  $$\begin{aligned} w(\mathbb{T}) &= \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g},\boldsymbol{t}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \|\boldsymbol{g}\|_2\boldsymbol{\theta},\boldsymbol{t}\rangle\right] \\ &= \mathbb{E}\|\boldsymbol{g}\|_2 \cdot \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{\theta},\boldsymbol{t}\rangle\right] = \mathbb{E}\|\boldsymbol{g}\|_2 \cdot w_s(\mathbb{T}), \end{aligned}$$

  i.e.,

  $$w_s(\mathbb{T}) = \frac{w(\mathbb{T})}{\mathbb{E}\|\boldsymbol{g}\|_2} \asymp \frac{\|\boldsymbol{A}\|_F}{\sqrt{n}}.$$

(1) By Exercise Q9.8 (*Size of Random Projections of Sets*), Exercise Q7.26 (*Random Projection of Sets: Lower Bound*),

we obtain

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] \asymp w_s(\mathbb{T}) + \sqrt{\frac{m}{n}}\operatorname{diam}(\mathbb{T}),$$

i.e.,

$$2\mathbb{E}\|\boldsymbol{P}\boldsymbol{A}\| \asymp \frac{1}{\sqrt{n}}\|\boldsymbol{A}\|_F + \sqrt{\frac{m}{n}}(2\|\boldsymbol{A}\|).$$

i.e.,

$$\mathbb{E}\|\boldsymbol{P}\boldsymbol{A}\| \asymp \frac{1}{\sqrt{n}}\|\boldsymbol{A}\|_F + \sqrt{\frac{m}{n}}\|\boldsymbol{A}\|.$$

- By Exercise Q7.25 (*Gaussian Projection of Sets*), we obtain

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{G}\mathbb{T})] \asymp w(\mathbb{T}) + \sqrt{m}\operatorname{diam}(\mathbb{T}),$$

i.e.,

$$2\mathbb{E}\|\boldsymbol{G}\boldsymbol{A}\| \asymp \|\boldsymbol{A}\|_F + \sqrt{m}(2\|\boldsymbol{A}\|).$$

i.e.,

$$\mathbb{E}\|\boldsymbol{G}\boldsymbol{A}\| \asymp \|\boldsymbol{A}\|_F + \sqrt{m}\|\boldsymbol{A}\|.$$

Rk. Comparing (1) and (2), Gaussian projection is essentially orthogonal projection magnified by a factor of approximately $\sqrt{n}$. Since $\boldsymbol{P}$ is an orthogonal projection (*norm-preserving*), its expected norm is severely compressed; whereas each row of $\boldsymbol{G}$ is an independent Gaussian vector with expected length $\sqrt{n}$, which provides a stretching effect. In practical dimensionality reduction algorithms, the two types of sketching matrices are widely adopted due to their good geometric distance-preserving properties.

8 Chaining

Keypoints

Section 8.1 Dudley inequality Q8.1 ~ Q8.8

Definition 8.1.1 (*Subgaussian increments*)

A random process $(X_t)_{t\in\mathbb{T}}$ has subgaussian increments if there exists a constant $K \geqslant 0$ such that $\|X_t - X_s\|_{\psi_2} \leqslant Kd(t,s)$ for every $t, s \in \mathbb{T}$, where d is a metric on $\mathbb{T}$.

Theorem 8.1.3 (*Dudley integral inequality*)

Let $(X_t)_{t\in\mathbb{T}}$ be a random process with subgaussian increments with respect to a metric d. Then

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \leqslant CK \int_0^{\operatorname{diam}(\mathbb{T})} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\, d\varepsilon.$$

Theorem 8.1.4 (*Discrete Dudley inequality*)

Let $(X_t)_{t\in\mathbb{T}}$ be a random process on a finite set $\mathbb{T}$ with subgaussian increments: $\|X_t - X_s\|_{\psi_2} \leqslant Kd(t,s)$. Then

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \leqslant CK \sum_{k\geqslant 0} \delta_k \sqrt{\log \mathcal{N}(\mathbb{T}, d, \delta_k)},$$

where (δ_k) is a geometrically decreasing sequence of scales starting at $\operatorname{diam}(\mathbb{T})/2$.

Remark 8.1.6 (*Dudley inequality: a high-probability bound*)

Chaining actually gives a high-probability bound: assuming $\mathbb{T}$ is finite, for every $u \geqslant 0$, the bound

$$\sup_{t,s\in\mathbb{T}} |X_t - X_s| \leqslant CK \left(\int_0^{\operatorname{diam}(\mathbb{T})} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\, d\varepsilon + u \cdot \operatorname{diam}(\mathbb{T}) \right) \text{ (w.p. } 1 - 2\exp(-u^2)).$$

holds with high probability.

Remark 8.1.7 (*Limits of Dudley integral*)

Even though the Dudley integral goes over $[0, \infty)$, we can cap it at the diameter of $\mathbb{T}$: for $\varepsilon > \operatorname{diam}(\mathbb{T})$ a single ε-ball covers $\mathbb{T}$, so $\log \mathcal{N}(\mathbb{T}, d, \varepsilon) = 0$. Thus the upper limit can be replaced by $\operatorname{diam}(\mathbb{T})$.

Theorem 8.1.8 (*Dudley inequality in* $\mathbb{R}^n$)

Let $\mathbb{T} \subset \mathbb{R}^n$. Then

$$\mathbb{E}\left[\sup_{x\in\mathbb{T}}\langle g, x\rangle\right] \leqslant C\int_0^{\operatorname{diam}(\mathbb{T})}\sqrt{\log \mathcal{N}(\mathbb{T},\varepsilon)}d\varepsilon,$$

where $g \sim N(\mathbf{0}, \boldsymbol{I}_n)$.

Section 8.2 Empirical processes Q8.9 ∼ Q8.11

Theorem 8.2.3 (*Lipschitz law of large numbers*)

Let $\mathcal{F}$ be a class of L-Lipschitz functions on $\mathbb{R}^n$, and $X_1, \cdots, X_N$ be i.i.d. random vectors. Then

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{N}\sum_{i=1}^N f(X_i) - \mathbb{E}[f(X)]\right|\right] \leqslant CL\frac{w(\mathbb{T})}{\sqrt{N}},$$

where $\mathbb{T} = \{X_1, \cdots, X_N\}$ is the empirical support.

Definition 8.2.5 (*Empirical process*)

An empirical process indexed by a function class $\mathcal{F}$ is the collection of random variables

$$\left(\frac{1}{\sqrt{N}}\sum_{i=1}^N (f(X_i) - \mathbb{E}[f(X)])\right)_{f\in\mathcal{F}},$$

where $X_1, \cdots, X_N$ are i.i.d. random variables.

Section 8.3 VC dimension Q8.12 ∼ Q8.24

Definition 8.3.1 (*VC dimension*)

Let $\mathcal{F}$ be a class of $\{0,1\}$-valued functions on a set $\mathcal{X}$. The VC dimension of $\mathcal{F}$, denoted $\operatorname{vc}(\mathcal{F})$, is the largest integer n such that there exist $x_1, \cdots, x_n \in \mathcal{X}$ that are shattered by $\mathcal{F}$, i.e., for every binary labeling of these points, there exists $f \in \mathcal{F}$ that matches it. If no such largest n exists, $\operatorname{vc}(\mathcal{F}) = \infty$.

Example 8.3.5 (*Half-spaces*)

A half-space in $\mathbb{R}^n$ is a set of the form $\{\boldsymbol{x} : \langle \boldsymbol{a}, \boldsymbol{x}\rangle \leqslant b\}$, where $\boldsymbol{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Let $\mathcal{F}$ be the class of indicator functions of all half-spaces in $\mathbb{R}^n$. Then $\operatorname{vc}(\mathcal{F}) = n + 1$.

Lemma 8.3.7 (*Pajor lemma*)

For a VC class $\mathcal{F}$ of $\{0,1\}$-valued functions, the number of distinct projections of $\mathcal{F}$ onto any n points is bounded by $\sum_{i=0}^{\operatorname{vc}(\mathcal{F})}\binom{n}{i}$.

Lemma 8.3.9 (*Sauer-Shelah Lemma*)

For a class $\mathcal{F}$ with VC dimension d, the growth function satisfies

$$\Pi_{\mathcal{F}}(n) \leqslant \sum_{i=0}^d \binom{n}{i} \leqslant \left(\frac{en}{d}\right)^d$$

for $n \geqslant d$.

Definition 8.3.10 (*Growth function*)

The growth function of a class $\mathcal{F}$ of $\{0,1\}$-valued functions is $\Pi_{\mathcal{F}}(n) = \max_{x_1,\cdots,x_n\in\mathcal{X}} |\{(f(x_1), \cdots, f(x_n)) : f \in \mathcal{F}\}|$, the maximum number of distinct labelings that $\mathcal{F}$ can produce on n points.

Proposition 8.3.11 (*VC stability*)

(i) Adding a finite number of functions to $\mathcal{F}$ does not increase VC dimension beyond adding the number of functions. (ii) Intersections, unions, and complements of VC classes are VC. (iii) If $\mathcal{F}$ is a VC class of functions, then the class of its subgraphs

is also VC.

Theorem 8.3.13 (*Covering numbers via VC dimension*)

Let $\mathcal{F}$ be a uniformly bounded class of functions with $\mathrm{vc}(\mathcal{F}) = d$. Then the covering numbers with respect to the empirical L_2 metric satisfy

$$\mathcal{N}(\mathcal{F}, \varepsilon, L_2(P_n)) \leqslant \left(\frac{C}{\varepsilon}\right)^{2d}$$

for $\varepsilon \in (0, 1)$.

Lemma 8.3.14 (*Dimension reduction*)

The empirical process indexed by a VC class can be approximated by its restriction to a random subset of the data of size proportional to the VC dimension.

Theorem 8.3.15 (*VC law of large numbers*)

Let $\mathcal{F}$ be a class of $\{0, 1\}$-valued functions with VC dimension d. Then, for any $\varepsilon \in (0, 1)$,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{N}\sum_{i=1}^{N} f(X_i) - \mathbb{E}[f(X)]\right|\right] \leqslant C\sqrt{\frac{d}{N}}.$$

Theorem 8.3.17 (*Glivenko-Cantelli Theorem*)

The empirical CDF

$$F_N(t) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\{X_i \leqslant t\}},$$

converges uniformly to the true CDF:

$$\sup_{t\in\mathbb{R}} |F_N(t) - F(t)| \to 0$$

almost surely as $N \to \infty$.

Remark 8.3.19 (*Uniform Glivenko-Cantelli classes*)

A family $\mathcal{F}$ of real-valued functions defined on a set Ω is called a uniform Glivenko-Cantelli class if for every $\varepsilon > 0$,

$$\lim_{n\to\infty}\sup_{\mu}\sup \mathbb{P}\left\{\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}f(X)\right| > \varepsilon\right\} = 0,$$

where the supremum is over all probability measures μ on Ω, and $X, X_1, \cdots, X_n$ are i.i.d. samples from μ. Theorem T8.3.15 (*VC law of large numbers*) together with Markov's inequality implies that any Boolean class with finite VC dimension is a uniform Glivenko-Cantelli class. The converse is also true (see Exercise Q8.28 (*Uniform Glivenko-Cantelli Class and Finite VC Dimension*)), so the two properties are equivalent.

Section 8.4 Statistical learning theory Q8.25 ∼ Q8.32

Definition 8.4.3 (*Empirical risk minimization*)

Given a loss function ℓ and a hypothesis class $\mathcal{F}$, the empirical risk minimizer (ERM) is

$$\hat{f} = \arg\min_{f\in\mathcal{F}} \frac{1}{N}\sum_{i=1}^{N} \ell(f(X_i), Y_i).$$

Theorem 8.4.5 (*VC generalization bound*)

Let $\mathcal{F}$ be a class of $\{0, 1\}$-valued functions with VC dimension d and let $R(f) = \mathbb{P}\{f(X) \neq Y\}$ be the true risk. Then, with

high probability, for every $f \in \mathcal{F}$,

$$R(f) \leqslant \hat{R}_N(f) + C\sqrt{\frac{d + \log(1/\delta)}{N}} \text{ (w.p. } 1-\delta),$$

where $\hat{R}_N$ is the empirical risk.

Section 8.5 Generic chaining Q8.35 ∼ Q8.38

Definition 8.5.1 (γ_2 *functional*)

For a metric space $(\mathbb{T}, d)$, $\gamma_2(\mathbb{T}, d) = \inf_{(\mathbb{T}_k)} \sup_{t\in\mathbb{T}} \sum_{k=0}^{\infty} 2^{k/2} \operatorname{diam}(\mathbb{T}_k, d)$, where the infimum is over all sequences of partitions $(\mathbb{T}_k)$ of $\mathbb{T}$ with certain admissibility properties, and $\operatorname{diam}(\mathbb{T}_k, d)$ is the diameter of the partition element containing t at scale k.

Theorem 8.5.2 (*Generic chaining bound*)

Let $(X_t)_{t\in\mathbb{T}}$ be a mean-zero random process with subgaussian increments: $\|X_t - X_s\|_{\psi_2} \leqslant Kd(t,s)$. Then

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \leqslant CK\gamma_2(\mathbb{T}, d).$$

Remark 8.5.4 (*Generic chaining: a high-probability bound*)

Theorem T8.5.2 (*Generic chaining bound*) only gives an expectation bound, but Generic chaining actually yields a high-probability bound——we have already seen this logic in Remark R8.1.6 (*Dudley inequality: a high-probability bound*). Let $\mathbb{T}$ be a finite set. Then, for any $u \geqslant 0$, the event

$$\sup_{t,s\in\mathbb{T}} |X_t - X_s| \leqslant CK(\gamma_2(\mathbb{T}, d) + u \cdot \operatorname{diam}(\mathbb{T})) \text{ (w.p. } 1 - 2\exp(-u^2)).$$

You will prove this in Exercise Q8.35 (*High-Probability Bound for Generic Chaining*).

Theorem 8.5.5 (*Talagrand majorizing measure theorem*)

For a Gaussian process (X_t) on $\mathbb{T}$ with canonical metric $d(t,s) = \|X_t - X_s\|_{L_2}$, we have

$$c\gamma_2(\mathbb{T}, d) \leqslant \mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \leqslant C\gamma_2(\mathbb{T}, d).$$

Corollary 8.5.6 (*Talagrand comparison inequality*)

Let (X_t) and (Y_t) be two Gaussian processes on $\mathbb{T}$, with $d_X(t,s) = \|X_t - X_s\|_{L_2}$ and $d_Y(t,s) = \|Y_t - Y_s\|_{L_2}$. If $d_X(t,s) \leqslant d_Y(t,s)$ for every $t, s \in \mathbb{T}$, then

$$\mathbb{E}\left[\sup_t X_t\right] \leqslant C\mathbb{E}\left[\sup_t Y_t\right].$$

Corollary 8.5.8 (*Talagrand comparison inequality: geometric form*)

Let $\mathbb{T} \subset \mathbb{R}^n$ and $\mathbb{S} \subset \mathbb{R}^n$ be such that $\mathbb{T} \subset \operatorname{conv}(\mathbb{S})$. Then

$$w(\mathbb{T}) \leqslant Cw(\mathbb{S}).$$

Section 8.6 Chevet inequality Q8.39 ∼ Q8.41

Theorem 8.6.1 (*Subgaussian Chevet inequality*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix with independent standardized K-sub-Gaussian entries. Then, for any $\mathbb{T} \subset \mathbb{S}^{n-1}$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{A}\boldsymbol{x}\|_2\right] \leqslant CK(w(\mathbb{T}) + \sqrt{m} \cdot \operatorname{rad}(\mathbb{T})).$$

Exercises

1 [**Dudley's Inequality: High-Probability Bound**] Modify the chaining proof from Section 8.1 to derive the high-probability version of Dudley's inequality stated in Remark R8.1.6 (*Dudley inequality: a high-probability bound*).
Specifically, let $(X_t)_{t\in\mathbb{T}}$ be a sub-Gaussian random process defined on a finite metric space $(\mathbb{T}, d)$ satisfying $\|X_t - X_s\|_{\psi_2} \leqslant Kd(t,s)$.
Proceed according to the following steps:
(1) First upgrade the increment bound (8.11) to the following high-probability version. Prove that for any scale k,

$$\sup_{t\in\mathbb{T}} |X_{\pi_k(t)} - X_{\pi_{k-1}(t)}| \leqslant CK\varepsilon_{k-1}\left[\sqrt{\log|\mathbb{T}_k|} + z_k\right] \text{ (w.p. } 1 - 2\exp(-z_k^2)). \qquad (C>0)$$

(2) Then choose suitable z_k, apply the union bound over all chaining levels k, and prove that for any $u \geqslant 0$,

$$\sup_{t,s\in\mathbb{T}} |X_t - X_s| \leqslant CK\left[\int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon + u\cdot\operatorname{diam}(\mathbb{T})\right] \text{ (w.p. } 1 - 2\exp(-u^2)).$$

Prf. • Chaining setup
Let $\varepsilon_k = 2^{-k}$.
Since $\mathbb{T}$ is bounded, there exists $\kappa \in \mathbb{Z}$ such that $\varepsilon_\kappa \geqslant \operatorname{diam}(\mathbb{T}) > \varepsilon_{\kappa+1}$.
For every $k \geqslant \kappa$, let $\mathbb{T}_k$ be a minimal ε_k-net of $\mathbb{T}$, so $|\mathbb{T}_k| = \mathcal{N}(\mathbb{T}, d, \varepsilon_k)$.
(*In particular, since* $\varepsilon_\kappa \geqslant \operatorname{diam}(\mathbb{T})$, *we may take* $|\mathbb{T}_\kappa| = 1$, *and denote its unique element by* t_0.)
Define the projection map $\pi_k(t)$ as the point in $\mathbb{T}_k$ closest to t, so $d(t, \pi_k(t)) \leqslant \varepsilon_k$.
Since $\mathbb{T}$ is finite, there exists a sufficiently large $K_{\max}$ such that $\varepsilon_{K_{\max}}$ is smaller than the minimum distance between any two points of $\mathbb{T}$; then $\mathbb{T}_{K_{\max}} = \mathbb{T}$ and $\pi_{K_{\max}}(t) = t$.
• High-probability bound for a single level
Fix $k > \kappa$. We need to bound the increment $\Delta_k(t) = X_{\pi_k(t)} - X_{\pi_{k-1}(t)}$.
For any $t \in \mathbb{T}$, the corresponding endpoint pair of the increment is $(\pi_k(t), \pi_{k-1}(t)) \in \mathbb{T}_k \times \mathbb{T}_{k-1}$. Hence the number of such increment pairs is at most $|\mathbb{T}_k||\mathbb{T}_{k-1}| \leqslant |\mathbb{T}_k|^2$.
The distance between the endpoints is controlled by the triangle inequality,

$$d(\pi_k(t), \pi_{k-1}(t)) \leqslant d(\pi_k(t), t) + d(t, \pi_{k-1}(t)) \leqslant \varepsilon_k + \varepsilon_{k-1} = \frac{3}{2}\varepsilon_{k-1} \leqslant 2\varepsilon_{k-1},$$

so the sub-Gaussian norm of the increment satisfies $\|\Delta_k(t)\|_{\psi_2} \leqslant K\cdot 2\varepsilon_{k-1}$. By the tail inequality for sub-Gaussian random variables, we obtain

$$\mathbb{P}\{|\Delta_k(t)| > s\} \leqslant 2\exp\left(-\frac{c_0 s^2}{(2K\varepsilon_{k-1})^2}\right), \qquad (c_0 > 0)$$

Taking the union bound on both sides,

$$\mathbb{P}\left\{\sup_{t\in\mathbb{T}} |\Delta_k(t)| > s\right\} \leqslant 2|\mathbb{T}_k|^2 \exp\left(-\frac{c_0 s^2}{4K^2\varepsilon_{k-1}^2}\right) = 2\exp\left(2\log|\mathbb{T}_k| - \frac{c_0 s^2}{4K^2\varepsilon_{k-1}^2}\right),$$

If the above failure probability equals $2\exp(-z_k^2)$, then

$$\frac{c_0 s^2}{4K^2\varepsilon_{k-1}^2} - 2\log|\mathbb{T}_k| = z_k^2 \implies s^2 = \frac{4K^2\varepsilon_{k-1}^2}{c_0}(z_k^2 + 2\log|\mathbb{T}_k|)$$
$$\implies s \leqslant \frac{2}{\sqrt{c_0}} K\varepsilon_{k-1}(z_k + \sqrt{2}\sqrt{\log|\mathbb{T}_k|}) \leqslant CK\varepsilon_{k-1}\left[\sqrt{\log|\mathbb{T}_k|} + z_k\right].$$

Using the elementary inequality $\sqrt{a^2+b^2} \leqslant a+b$ (when $a, b \geqslant 0$), we obtain

$$s \leqslant \frac{2}{\sqrt{c_0}} K\varepsilon_{k-1}(z_k + \sqrt{2}\sqrt{\log|\mathbb{T}_k|}) \leqslant CK\varepsilon_{k-1}(\sqrt{\log|\mathbb{T}_k|} + z_k). \qquad \left(C = \frac{2\sqrt{2}}{\sqrt{c_0}}\right)$$

- Choosing z_k and combining all levels

  (*Since* $X_t - X_{t_0} = \sum_{k=\kappa+1}^{K_{\max}} \Delta_k(t)$, *we want to use the union bound to ensure that the increment bounds hold simultaneously for all* $k \in (\kappa, K_{\max}]$. *Specifically, we need to choose* z_k *so that the total failure probability satisfies* $\sum_k 2\exp(-z_k^2) \leqslant 2\exp(-u^2)$.)

  For $k = \kappa+1, \kappa+2, \cdots$, construct

$$z_k = u + \sqrt{k-\kappa},$$

  then the total failure probability is

$$\begin{aligned} \sum_{k=\kappa+1}^{K_{\max}} 2\exp(-z_k^2) &\leqslant 2\sum_{j=1}^{\infty} \exp(-(u+\sqrt{j})^2) \\ &\leqslant 2\exp(-u^2)\sum_{j=1}^{\infty} \mathrm{e}^{-j} && ((u+\sqrt{j})^2 \geqslant u^2 + j) \\ &= 2\exp(-u^2)\frac{\mathrm{e}^{-1}}{1-\mathrm{e}^{-1}} < 2\exp(-u^2), \end{aligned}$$

  hence, the bounds hold for all levels (w.p. $1 - 2\exp(-u^2)$).

- Summation and conversion to an integral

  For any $t, s \in \mathbb{T}$,

$$|X_t - X_s| \leqslant |X_t - X_{t_0}| + |X_s - X_{t_0}| \leqslant 2\sup_t |X_t - X_{t_0}|,$$

  where

$$|X_t - X_{t_0}| \leqslant \sum_{k=\kappa+1}^{K_{\max}} |\Delta_k(t)| \leqslant CK \sum_{k=\kappa+1}^{K_{\max}} \varepsilon_{k-1}(\sqrt{\log|\mathbb{T}_k|} + u + \sqrt{k-\kappa}),$$

  We split the right-hand side series into three parts S_1, S_2, S_3.

  For S_1,

$$S_1 = CK \sum_{k=\kappa+1}^{K_{\max}} 2\varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon_k)} \leqslant 4CK \int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon.$$

  For S_2,

$$S_2 = CKu \sum_{k=\kappa+1}^{\infty} \varepsilon_{k-1} = CKu\varepsilon_\kappa \sum_{j=1}^{\infty} 2^{-j+1} = 2CKu\varepsilon_\kappa \leqslant 4CKu \cdot \mathrm{diam}(\mathbb{T}).$$

  For S_3,

$$\begin{aligned} S_3 &= CK \sum_{k=\kappa+1}^{\infty} \varepsilon_{k-1}\sqrt{k-\kappa} = CK\varepsilon_\kappa \sum_{j=1}^{\infty} 2^{-j+1}\sqrt{j} \\ &\leqslant c_1 CK\varepsilon_\kappa \leqslant 2c_1 CK \cdot \mathrm{diam}(\mathbb{T}). && \left(\sum_{j=1}^{\infty} 2^{-j+1}\sqrt{j} = c_1\right) \end{aligned}$$

  If $|\mathbb{T}| \geqslant 2$, then for $\varepsilon < \mathrm{diam}(\mathbb{T})/2$, the covering number satisfies $\mathcal{N} \geqslant 2$, i.e., $\log \mathcal{N} \geqslant \log 2$, hence

$$\int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant \int_0^{\mathrm{diam}(\mathbb{T})/2} \sqrt{\log 2}\mathrm{d}\varepsilon = \frac{\sqrt{\log 2}}{2}\mathrm{diam}(\mathbb{T}),$$

thus

$$S_3 \leqslant C'K \int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon.$$

Thus,

$$\sup_{t,s\in\mathbb{T}} |X_t - X_s| \leqslant C^* K \left[\int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon + u \cdot \operatorname{diam}(\mathbb{T}) \right] \text{ (w.p. } 1 - 2\exp(-u^2)). \qquad (C^* > 0)$$

2 [**High-Probability Dudley Bound for Gaussian Processes**] For general sub-Gaussian processes, obtaining a high-probability version of Dudley's inequality requires modifying the chaining proof (*as shown in Exercise Q8.1 (Dudley's Inequality: High-Probability Bound)*).

However, if we assume $(X_t)_{t\in\mathbb{T}}$ is a zero-mean Gaussian process, we can obtain the high-probability bound via a simpler route. Combining the expected version of Dudley's inequality with the concentration inequality for Gaussian processes (Theorem T7.1.11 (*Concentration of Gaussian processes*)), give an alternative proof of the high-probability bound result stated in Remark R8.1.6 (*Dudley inequality: a high-probability bound*).

Specifically, for a zero-mean Gaussian process defined on a finite metric space $(\mathbb{T}, d)$ with canonical metric $d(t,s) = \|X_t - X_s\|_{L^2}$, prove that for any $u \geqslant 0$,

$$\sup_{t,s\in\mathbb{T}} |X_t - X_s| \leqslant C \left[\int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon + u \cdot \operatorname{diam}(\mathbb{T}) \right] \text{ (w.p. } 1 - 2\exp(-u^2)). \qquad (C > 0)$$

Prf. Define the maximum increment of the Gaussian process on $\mathbb{T}$ as (*since we take two points, we may consider the product space* $\mathbb{T} \times \mathbb{T}$)

$$Z = \sup_{t,s\in\mathbb{T}} (X_t - X_s) = \sup_{t,s\in\mathbb{T}} |X_t - X_s|.$$

(*Remark: Since the Gaussian process is symmetric, the supremum without absolute value and the supremum with absolute value differ slightly in distribution, but the discussion of the supremum relative to the origin is fully applicable.*)

- Expectation bound

  Using Theorem T8.1.3 (*Dudley integral inequality*) (*Dudley integral inequality*), we obtain

$$\mathbb{E}[Z] = \mathbb{E}\left[\sup_{t,s\in\mathbb{T}} |X_t - X_s| \right] \leqslant C_1 \int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon. \qquad (\mathrm{I})$$

- Fluctuation bound

  Using the concentration inequality for Gaussian processes (Theorem T7.1.11 (*Concentration of Gaussian processes*)), we obtain

$$\|Z - \mathbb{E}[Z]\|_{\psi_2} \leqslant C_2 \sup_{t,s\in\mathbb{T}} \sqrt{\operatorname{Var}(X_t - X_s)}.$$

  where

$$\operatorname{Var}(X_t - X_s) = \mathbb{E}[(X_t - X_s)^2] = d(t,s)^2.$$

  Substituting this in,

$$\sup_{t,s\in\mathbb{T}} \sqrt{\operatorname{Var}(X_t - X_s)} = \sup_{t,s\in\mathbb{T}} d(t,s) = \operatorname{diam}(\mathbb{T}).$$

  Therefore

$$\|Z - \mathbb{E}[Z]\|_{\psi_2} \leqslant C_2 \cdot \operatorname{diam}(\mathbb{T}).$$

- Applying the sub-Gaussian tail inequality

  By the definition of the tail probability of a sub-Gaussian random variable, for any $s \geqslant 0$,

$$\mathbb{P}\{Z \geqslant \mathbb{E}[Z] + s\} \leqslant 2\exp\left(-\frac{cs^2}{\|Z - \mathbb{E}[Z]\|_{\psi_2}^2} \right) \leqslant 2\exp\left(-\frac{cs^2}{C_2^2 \cdot \operatorname{diam}(\mathbb{T})^2} \right). \qquad (c > 0)$$

If the above failure probability equals $2\exp(-u^2)$, then

$$\frac{cs^2}{C_2^2 \cdot \operatorname{diam}(\mathbb{T})^2} = u^2 \implies s = \frac{C_2}{\sqrt{c}} u \cdot \operatorname{diam}(\mathbb{T}) = C_3 u \cdot \operatorname{diam}(\mathbb{T}). \qquad \left(C_3 = \frac{C_2}{\sqrt{c}}\right)$$

Therefore

$$Z \leqslant \mathbb{E}[Z] + C_3 u \cdot \operatorname{diam}(\mathbb{T}) \text{ (w.p. } 1 - 2\exp(-u^2)).$$

Substituting this into (I), we obtain

$$\sup_{t,s\in\mathbb{T}} |X_t - X_s| \leqslant C_1 \int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon + C_3 u \cdot \operatorname{diam}(\mathbb{T}) \text{ (w.p. } 1 - 2\exp(-u^2)),$$

i.e.,

$$\sup_{t,s\in\mathbb{T}} |X_t - X_s| \leqslant C \left[\int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon + u \cdot \operatorname{diam}(\mathbb{T}) \right] \text{ (w.p. } 1 - 2\exp(-u^2)). \qquad (C = \max(C_1, C_3))$$

3 [**The Dudley Integral and the Dudley Sum Are Equivalent**] In the proof of Theorem T8.1.4 (*Discrete Dudley inequality*), we bounded the Dudley sum (*discrete form*) from above by the Dudley integral (*continuous form*). Prove that the two forms are equivalent up to absolute constant factors,

$$\sum_{k\in\mathbb{Z}} 2^{-k} \sqrt{\log \mathcal{N}(\mathbb{T}, d, 2^{-k})} \asymp \int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon.$$

Prf. • Proof that the Dudley sum $\lesssim$ Dudley integral

Write 2^{-k} as the integral $\int_{2^{-k-1}}^{2^{-k}} 2\mathrm{d}\varepsilon$.

$$\sum_{k\in\mathbb{Z}} 2^{-k} f(2^{-k}) = \sum_{k\in\mathbb{Z}} f(2^{-k}) \int_{2^{-k-1}}^{2^{-k}} 2\mathrm{d}\varepsilon = 2 \sum_{k\in\mathbb{Z}} \int_{2^{-k-1}}^{2^{-k}} f(2^{-k}) \mathrm{d}\varepsilon.$$

For any $\varepsilon \in [2^{-k-1}, 2^{-k}]$, since f is decreasing, $f(2^{-k}) \leqslant f(\varepsilon)$, hence

$$2 \sum_{k\in\mathbb{Z}} \int_{2^{-k-1}}^{2^{-k}} f(2^{-k}) \mathrm{d}\varepsilon \leqslant 2 \sum_{k\in\mathbb{Z}} \int_{2^{-k-1}}^{2^{-k}} f(\varepsilon) \mathrm{d}\varepsilon,$$

Combining the integrals over all intervals,

$$2 \sum_{k\in\mathbb{Z}} \int_{2^{-k-1}}^{2^{-k}} f(\varepsilon) \mathrm{d}\varepsilon = 2 \int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon,$$

(*Assuming ε is sufficiently large so that $\varepsilon > \operatorname{diam}(\mathbb{T})$, then $\mathcal{N} = 1$ and the integral converges.*)

i.e.,

$$2 \sum_{k\in\mathbb{Z}} \int_{2^{-k-1}}^{2^{-k}} f(2^{-k}) \mathrm{d}\varepsilon \leqslant 2 \int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon.$$

• Proof that the Dudley integral $\lesssim$ Dudley sum

Split the integral into a sum over dyadic intervals,

$$\int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon = \sum_{k\in\mathbb{Z}} \int_{2^{-k}}^{2^{-k+1}} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon.$$

For any $\varepsilon \in [2^{-k-1}, 2^{-k}]$, since f is decreasing, $f(\varepsilon) \geqslant f(2^{-k+1})$, hence

$$\begin{aligned} \int_0^\infty f(\varepsilon) \mathrm{d}\varepsilon &\geqslant \sum_{k\in\mathbb{Z}} \int_{2^{-k}}^{2^{-k+1}} f(2^{-k+1}) \mathrm{d}\varepsilon = \sum_{k\in\mathbb{Z}} f(2^{-k+1}) \cdot (2^{-k+1} - 2^{-k}) \\ &= \sum_{k\in\mathbb{Z}} 2^{-k} f(2^{-k+1}) \end{aligned}$$

$$= \sum_{j\in\mathbb{Z}} 2^{-(j+1)} f(2^{-j}) = \frac{1}{2}\sum_{j\in\mathbb{Z}} 2^{-j} f(2^{-j}), \qquad (j = k-1)$$

i.e.,

$$\int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant \frac{1}{2}\sum_{k\in\mathbb{Z}} 2^{-k}\sqrt{\log \mathcal{N}(\mathbb{T}, d, 2^{-k})}.$$

4 [**Dudley's Inequality Can Be Very Loose**] Let $\boldsymbol{e}_k$ be the standard orthonormal basis vectors of $\mathbb{R}^n$, and define $\mathbb{T} \subset \mathbb{R}^n$ as

$$\mathbb{T} = \left\{ \frac{\boldsymbol{e}_k}{\sqrt{1+\log k}} : k = 1, \cdots, n \right\}.$$

Prove the following two conflicting conclusions:

(1) There exists an absolute constant C independent of the dimension n such that the Gaussian width is bounded, i.e., $w(\mathbb{T}) \leqslant C$.

(2) As $n \to \infty$, its Dudley integral diverges, i.e.,

$$\int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \to \infty.$$

Prf. (1) Let $\boldsymbol{g} = (g_1, \cdots, g_n) \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. By the definition of Gaussian width, we obtain

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t}\rangle\right] = \mathbb{E}\left[\max_{1\leqslant k\leqslant n} \left\langle \boldsymbol{g}, \frac{\boldsymbol{e}_k}{\sqrt{1+\log k}} \right\rangle\right] = \mathbb{E}\left[\max_{1\leqslant k\leqslant n} \frac{g_k}{\sqrt{1+\log k}}\right],$$

Set $X_k = \frac{g_k}{\sqrt{1+\log k}}$. (*It remains to compute* $\mathbb{E}[\max_k X_k]$.)

Using the tail inequality for standard Gaussian variables $\mathbb{P}\{g_k > t\} \leqslant \exp(-t^2/2)$, we obtain

$$\mathbb{P}\{X_k > u\} = \mathbb{P}\{g_k > u\sqrt{1+\log k}\} \leqslant \exp\left(-\frac{u^2(1+\log k)}{2}\right) = \mathrm{e}^{-u^2/2}\cdot \mathrm{e}^{-\frac{u^2}{2}\log k} = \mathrm{e}^{-u^2/2}\frac{1}{k^{u^2/2}},$$

Taking the union bound on both sides,

$$\mathbb{P}\left\{\max_{1\leqslant k\leqslant n} X_k > u\right\} \leqslant \sum_{k=1}^n \mathbb{P}\{X_k > u\} \leqslant \mathrm{e}^{-u^2/2}\sum_{k=1}^n \frac{1}{k^{u^2/2}}.$$

For $u \geqslant 2$, the series is bounded by $\sum_{k=1}^\infty k^{-2} = \pi^2/6 < 2$, and hence $\mathbb{P}\{\max_k X_k > u\} \leqslant 2\mathrm{e}^{-u^2/2}$. Set $Y = \max_{1\leqslant k\leqslant n} X_k$. Since $Y \leqslant Y_+$, split the tail integral of Y_+ into two parts,

$$\begin{aligned}
\mathbb{E}[Y] \leqslant \mathbb{E}[Y_+] &= \int_0^\infty \mathbb{P}\{Y > u\}\mathrm{d}u \\
&\leqslant \int_0^2 1\mathrm{d}u + \int_2^\infty \mathbb{P}\left\{\max_k X_k > u\right\}\mathrm{d}u \\
&\leqslant 2 + \int_2^\infty 2\mathrm{e}^{-u^2/2}\mathrm{d}u \leqslant 2 + \int_{-\infty}^\infty \mathrm{e}^{-u^2/2}\mathrm{d}u = 2+\sqrt{2\pi},
\end{aligned}$$

Now set $C = 2 + \sqrt{2\pi}$.

(2) Let

$$\boldsymbol{t}_k = \frac{\boldsymbol{e}_k}{\sqrt{1+\log k}} \in \mathbb{T},$$

then

$$d(\boldsymbol{t}_i, \boldsymbol{t}_j)^2 = \|\boldsymbol{t}_i - \boldsymbol{t}_j\|_2^2 = \|\boldsymbol{t}_i\|_2^2 + \|\boldsymbol{t}_j\|_2^2 = \frac{1}{1+\log i} + \frac{1}{1+\log j}.$$

To establish a lower bound for the covering number, consider the subset $\mathbb{T}_m = \{\boldsymbol{t}_1, \cdots, \boldsymbol{t}_m\}$ consisting of the first m

points, where $m \leqslant n$. For any $i \neq j \leqslant m$,

$$d(\boldsymbol{t}_i, \boldsymbol{t}_j) \geqslant \sqrt{\frac{1}{1+\log m} + \frac{1}{1+\log m}} = \sqrt{\frac{2}{1+\log m}},$$

Define the critical radius

$$r_m = \frac{1}{2}\sqrt{\frac{2}{1+\log m}} = \frac{1}{\sqrt{2(1+\log m)}};$$

then $\operatorname{dist}(\mathbb{T}_m) \geqslant 2r_m$. For any $\varepsilon < r_m$, every closed ball of radius ε contains at most one point of $\mathbb{T}_m$, so

$$\mathcal{N}(\mathbb{T}, d, \varepsilon) \geqslant m,$$

hence

$$I_n = \int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant \sum_{m=2}^{n-1} \int_{r_{m+1}}^{r_m} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant \sum_{m=2}^{n-1} \sqrt{\log m} \cdot (r_m - r_{m+1}),$$

It remains to estimate the gap $r_m - r_{m+1}$. Let $h(x) = \frac{1}{\sqrt{2(1+\log x)}}$.

By the Lagrange mean value theorem, there exists $c_m \in (m, m+1)$ such that

$$h(m) - h(m+1) = -h'(c_m),$$

where

$$h'(x) = \frac{1}{\sqrt{2}}\left(-\frac{1}{2}\right)(1+\log x)^{-3/2} \cdot \frac{1}{x} = -\frac{1}{2\sqrt{2}x(1+\log x)^{3/2}},$$

Substituting this in, we obtain

$$r_m - r_{m+1} \geqslant \frac{1}{2\sqrt{2}(m+1)(1+\log(m+1))^{3/2}},$$

therefore

$$\begin{aligned} I_n &\geqslant \frac{1}{2\sqrt{2}} \sum_{m=2}^{n-1} \frac{\sqrt{\log m}}{(m+1)(1+\log(m+1))^{3/2}} \\ &\sim \frac{1}{2\sqrt{2}} \sum_{m=2}^{n-1} \frac{\sqrt{\log m}}{m(\log m)^{3/2}} = \frac{1}{2\sqrt{2}} \sum_{m=2}^{n-1} \frac{1}{m \log m} \to \infty. \end{aligned}$$

Rk. The reason for this anomalous phenomenon is that Dudley's inequality measures complexity via ε-nets. For this extremely sparse "spiky" geometry, covering it requires an exponential number of tiny balls, causing the logarithmic covering number to blow up; however, under Gaussian projection, the spikes in the vast majority of directions are attenuated when taking inner products with the Gaussian vector, so the actual Gaussian width is very small.

5 [**Dudley's and Sudakov's Inequalities Are Tight up to a Logarithmic Factor**] It is known that for $n \geqslant 2$ and a set $\mathbb{T} \subset \mathbb{R}^n$, Corollary C7.4.2 (*Sudakov inequality in* $\mathbb{R}^n$) and Theorem T8.1.8 (*Dudley inequality in* $\mathbb{R}^n$) give lower and upper bounds, respectively, for the Gaussian width $w(\mathbb{T})$,

$$s(\mathbb{T}) \lesssim w(\mathbb{T}) \lesssim d(\mathbb{T}).$$

where the single-scale eigenvalue $s(\mathbb{T})$ and the multi-scale integral $d(\mathbb{T})$ are defined respectively as

$$s(\mathbb{T}) = \sup_{\varepsilon>0} \varepsilon\sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)}, d(\mathbb{T}) = \int_0^\infty \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)}\mathrm{d}\varepsilon.$$

Prove that the above bounds are equivalent up to a logarithmic factor,

$$s(\mathbb{T}) \leqslant d(\mathbb{T}) \leqslant C\log(n)s(\mathbb{T}). \qquad (C > 0)$$

Prf. (*Remark: For $n = 1$, $\log(n) = 0$, in which case the right-hand side is understood as an absolute constant $C \cdot s(\mathbb{T})$. For notational uniformity, the derivation below assumes $n \geqslant 2$ by default.*)

• Proof of the lower bound $s(\mathbb{T}) \leqslant d(\mathbb{T})$

Since the covering number $\mathcal{N}(\mathbb{T}, x)$ is decreasing in the radius x, for any $\varepsilon > 0$,

$$d(\mathbb{T}) = \int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, x)}\mathrm{d}x \geqslant \int_0^{\varepsilon} \sqrt{\log \mathcal{N}(\mathbb{T}, x)}\mathrm{d}x$$
$$\geqslant \int_0^{\varepsilon} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)}\mathrm{d}x = \varepsilon\sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)},$$

Taking the supremum over ε on both sides,

$$s(\mathbb{T}) \leqslant d(\mathbb{T}).$$

- Establishing the relationship between $\mathrm{diam}(\mathbb{T})$ and $s(\mathbb{T})$

  Let $D = \mathrm{diam}(\mathbb{T})$. If $D = 0$, then $s(\mathbb{T}) = d(\mathbb{T}) = 0$ and the conclusion is trivial; we only need to consider the case $D > 0$.

  Set the covering radius to $\varepsilon = D/3$, then

$$\mathcal{N}\left(\mathbb{T}, \frac{D}{3}\right) \geqslant 2,$$

  Substituting this into the definition of $s(\mathbb{T})$, we obtain

$$s(\mathbb{T}) \geqslant \frac{D}{3}\sqrt{\log \mathcal{N}\left(\mathbb{T}, \frac{D}{3}\right)} \geqslant \frac{D}{3}\sqrt{\log 2},$$

  i.e.,

$$D \leqslant C_0 s(\mathbb{T}). \qquad (C_0 = 3/\sqrt{\log 2})$$

- Proof of the upper bound $d(\mathbb{T}) \leqslant C\log(n)s(\mathbb{T})$

  Note that when $\varepsilon > D$, only 1 ball of radius ε is needed to cover $\mathbb{T}$, in which case $\log \mathcal{N}(\mathbb{T}, \varepsilon) = \log 1 = 0$, so the Dudley integral is effective only on $[0, D]$,

$$d(\mathbb{T}) = \int_0^{D} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)}\mathrm{d}\varepsilon.$$

  Split it at $\varepsilon = D/n$,

$$d(\mathbb{T}) = \underbrace{\int_0^{D/n} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)}\mathrm{d}\varepsilon}_{I_{\text{small}}} + \underbrace{\int_{D/n}^{D} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)}\mathrm{d}\varepsilon}_{I_{\text{large}}}.$$

  For I_{large}, rewrite the integrand in the form of $s(\mathbb{T})$,

$$I_{\text{large}} = \int_{D/n}^{D} \frac{1}{\varepsilon}(\varepsilon\sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)})\mathrm{d}\varepsilon$$
$$\leqslant \int_{D/n}^{D} \frac{s(\mathbb{T})}{\varepsilon}\mathrm{d}\varepsilon = s(\mathbb{T})\left(\log D - \log\frac{D}{n}\right) = s(\mathbb{T})\log(n).$$

  For I_{small}, note that when $\varepsilon \leqslant D$,

$$\mathcal{N}(\mathbb{T}, \varepsilon) \leqslant \left(1 + \frac{2D}{\varepsilon}\right)^n \leqslant \left(\frac{3D}{\varepsilon}\right)^n,$$

  therefore

$$I_{\text{small}} \leqslant \int_0^{D/n} \sqrt{n\log\left(\frac{3D}{\varepsilon}\right)}\mathrm{d}\varepsilon$$
$$= \sqrt{n}\int_{\log(3n)}^{\infty} \sqrt{y}(3D\mathrm{e}^{-y})\mathrm{d}y = 3D\sqrt{n}\int_{\log(3n)}^{\infty} \sqrt{y}\mathrm{e}^{-y}\mathrm{d}y. \qquad (y = \log(\tfrac{3D}{\varepsilon}))$$

For the integral on the right, note that when $y \geqslant A$, $\sqrt{y} \leqslant y/\sqrt{A}$. Let $A = \log(3n)$, then

$$\int_A^\infty \sqrt{y}\mathrm{e}^{-y}\mathrm{d}y \leqslant \frac{1}{\sqrt{A}}\int_A^\infty y\mathrm{e}^{-y}\mathrm{d}y = \frac{1}{\sqrt{A}}(A+1)\mathrm{e}^{-A} = \frac{A+1}{A}\sqrt{A}\mathrm{e}^{-A} \leqslant 2\sqrt{A}\mathrm{e}^{-A},$$

Substituting this in,

$$\begin{aligned} I_{\text{small}} &\leqslant 3D\sqrt{n}\left(2\sqrt{\log(3n)}\cdot\frac{1}{3n}\right) = 2D\frac{\sqrt{\log(3n)}}{\sqrt{n}} \\ &\leqslant 2C_0 s(\mathbb{T})\frac{\sqrt{\log(3n)}}{\sqrt{n}} = C_1 s(\mathbb{T}). \end{aligned}$$

Thus,

$$d(\mathbb{T}) = I_{\text{large}} + I_{\text{small}} \leqslant s(\mathbb{T})\log(n) + C_1 s(\mathbb{T}) \leqslant C\log(n)s(\mathbb{T}).$$

Rk. This exercise reveals a counterintuitive but universally true fact in high dimensions: for any bounded set in Euclidean space, the single-scale Sudakov lower bound and the multi-scale-integral Dudley upper bound are equivalent up to a logarithmic factor $\log(n)$. This means that in $\mathbb{R}^n$, we often do not need to perform the complicated Dudley integral; it suffices to find the single scale that maximizes $\varepsilon\sqrt{\log \mathcal{N}}$ to characterize the Gaussian width up to logarithmic precision.

6 [**Dudley's Inequality with Refined Integration Limits**] Let $\mathbb{T} \subset \mathbb{R}^n$. According to Remark R8.1.7 (*Limits of Dudley integral*), we already know that the upper limit of the Dudley integral (8.17) can be set to $\operatorname{diam}(\mathbb{T})$. At the same time, optimizing the lower limit of the integral is also very valuable, because the integral may diverge as $\varepsilon \to 0$.

Prove that for any set $\mathbb{T} \subset \mathbb{R}^n$, its Gaussian width satisfies the following Dudley inequality with truncated integration limits,

$$w(\mathbb{T}) \leqslant C\int_a^b \sqrt{\log \mathcal{N}(\mathbb{T},\varepsilon)}\mathrm{d}\varepsilon. \qquad (C, c > 0)$$

where the upper integration limit is $b = \operatorname{diam}(\mathbb{T})$, and the lower integration limit is $a = \frac{cw(\mathbb{T})}{\sqrt{n}}$.

Prf. (*We do not continue chaining on infinitesimally small scales; instead, we consider truncated chaining, forcibly stopping when the scale drops to a certain threshold a, and bound the remaining tiny residual via the global Euclidean properties of $\mathbb{R}^n$.*)

- Constructing nets at different scales

  Let $D = \operatorname{diam}(\mathbb{T})$ and define the dyadic scale sequence $\varepsilon_k = D2^{-k}$. Let $\mathcal{N}_k$ be a minimal ε_k-net of $\mathbb{T}$, so $|\mathcal{N}_k| = \mathcal{N}(\mathbb{T}, d, \varepsilon_k)$.

  (*In particular, for $\varepsilon_0 = D$, we need only one point to cover $\mathbb{T}$, i.e., $|\mathcal{N}_0| = 1$.*)

  For any $\boldsymbol{t} \in \mathbb{T}$, define the projection map $\pi_k(\boldsymbol{t}) \in \mathcal{N}_k$ as the point in the net closest to $\boldsymbol{t}$, satisfying $\|\boldsymbol{t} - \pi_k(\boldsymbol{t})\|_2 \leqslant \varepsilon_k$.

- Performing the chaining decomposition

  Given a target truncation scale $a > 0$, there exists K such that $\varepsilon_K \leqslant a < \varepsilon_{K-1}$. Forcibly stopping the chaining expansion at the K-th step (*telescoping sum*), we obtain

  $$\boldsymbol{t} = \pi_0(\boldsymbol{t}) + \sum_{k=1}^{K}(\pi_k(\boldsymbol{t}) - \pi_{k-1}(\boldsymbol{t})) + (\boldsymbol{t} - \pi_K(\boldsymbol{t})),$$

  Let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Substituting the above decomposition into the definition of Gaussian width, we obtain

$$\begin{aligned} w(\mathbb{T}) &= \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{t}\rangle\right] \\ &\leqslant \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \pi_0(\boldsymbol{t})\rangle\right] + \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\sum_{k=1}^{K}\langle \boldsymbol{g}, \pi_k(\boldsymbol{t}) - \pi_{k-1}(\boldsymbol{t})\rangle\right] + \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{t} - \pi_K(\boldsymbol{t})\rangle\right] \\ &\leqslant \underbrace{\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \pi_0(\boldsymbol{t})\rangle\right]}_{I_0} + \underbrace{\sum_{k=1}^{K}\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \pi_k(\boldsymbol{t}) - \pi_{k-1}(\boldsymbol{t})\rangle\right]}_{I_{\text{chain}}} + \underbrace{\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{t} - \pi_K(\boldsymbol{t})\rangle\right]}_{I_{\text{rem}}}. \end{aligned}$$

1. For I_0, since $\mathcal{N}_0$ contains only one point, which we may denote by $\boldsymbol{t}_0$,
$$\mathbb{E}[\langle \boldsymbol{g}, \boldsymbol{t}_0 \rangle] = \langle \mathbb{E}[\boldsymbol{g}], \boldsymbol{t}_0 \rangle = 0,$$
hence $I_0 = 0$.
2. For I_{chain}, consider a single term
$$E_k = \mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \pi_k(\boldsymbol{t}) - \pi_{k-1}(\boldsymbol{t}) \rangle\right],$$
Let $\Delta_k = \{\boldsymbol{v}_k(\boldsymbol{t}) : \boldsymbol{t} \in \mathbb{T}\}$ be the set formed by $\boldsymbol{v}_k(\boldsymbol{t}) = \pi_k(\boldsymbol{t}) - \pi_{k-1}(\boldsymbol{t})$, satisfying $|\Delta_k| \leqslant |\mathcal{N}_k||\mathcal{N}_{k-1}| \leqslant |\mathcal{N}_k|^2$. Then
$$E_k = \mathbb{E}\left[\max_{\boldsymbol{v}\in\Delta_k} \langle \boldsymbol{g}, \boldsymbol{v} \rangle\right].$$
Since
$$X_{\boldsymbol{v}} = \langle \boldsymbol{g}, \boldsymbol{v} \rangle \sim N(0, \|\boldsymbol{v}\|_2^2),$$
where
$$\|\boldsymbol{v}\|_2 = \|\pi_k(\boldsymbol{t}) - \pi_{k-1}(\boldsymbol{t})\|_2 \leqslant \varepsilon_k + \varepsilon_{k-1} = 3\varepsilon_k,$$
by the Gaussian maximum lemma $\mathbb{E}[\max_{1\leqslant i\leqslant N} X_i] \leqslant c \cdot \sigma \cdot \sqrt{\log N}$, we obtain
$$\begin{aligned} E_k &\leqslant c \cdot (3\varepsilon_{k-1}) \cdot \sqrt{\log(|\mathcal{N}_k|^2)} \\ &=: C_1 \cdot \varepsilon_{k-1} \cdot \sqrt{\log |\mathcal{N}_k|}, \end{aligned} \qquad (C_1 = 3\sqrt{2}c)$$
Summing over both sides, we obtain
$$I_{\text{chain}} \leqslant C_1 \sum_{k=1}^{K} \varepsilon_{k-1} \sqrt{\log |\mathcal{N}_k|} = C_1 \sum_{k=1}^{K} \varepsilon_{k-1} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon_k)}.$$
By Exercise Q8.3 (*The Dudley Integral and the Dudley Sum Are Equivalent*), replacing the lower integration limit by $a \in [\varepsilon_K, 2\varepsilon_K)$ (*at the cost of at most a constant factor loss in precision; see the remark for details*), we obtain
$$I_{\text{chain}} \leqslant C_2 \int_a^D \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)} \mathrm{d}\varepsilon. \qquad (C_2 > 0)$$
3. For the remainder term I_{rem}, by Cauchy's inequality,
$$\langle \boldsymbol{g}, \boldsymbol{t} - \pi_K(\boldsymbol{t}) \rangle \leqslant \|\boldsymbol{g}\|_2 \cdot \|\boldsymbol{t} - \pi_K(\boldsymbol{t})\|_2 \leqslant \|\boldsymbol{g}\|_2 \varepsilon_K,$$
Taking the supremum and expectation on both sides,
$$\mathbb{E}\left[\sup_{\boldsymbol{t}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{t} - \pi_K(\boldsymbol{t}) \rangle\right] \leqslant \mathbb{E}\|\boldsymbol{g}\|_2 \cdot \varepsilon_K \leqslant \sqrt{\mathbb{E}\|\boldsymbol{g}\|_2^2} \cdot \varepsilon_K = \sqrt{n}\varepsilon_K.$$

Thus, for any stopping scale $a > 0$,
$$w(\mathbb{T}) \leqslant C_1 \sum_{k=1}^{K} \varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon_k)} + \sqrt{n}\varepsilon_K. \qquad \text{(I)}$$

- Choosing the optimal truncation level K

(To make the truncation error absorbed by $\frac{1}{2}w(\mathbb{T})$, we need $\sqrt{n}\varepsilon_K \leqslant \frac{1}{2}w(\mathbb{T})$, i.e., $\varepsilon_K \leqslant \frac{w(\mathbb{T})}{2\sqrt{n}}$.)

By the basic property of Gaussian width $w(\mathbb{T}) \leqslant \sqrt{n} \cdot \operatorname{diam}(\mathbb{T}) = \sqrt{n}D$, we obtain
$$\frac{w(\mathbb{T})}{2\sqrt{n}} \leqslant \frac{1}{2}D = \varepsilon_1,$$
hence there exists $K \in \mathbb{Z}^+$ such that
$$\varepsilon_K \leqslant \frac{w(\mathbb{T})}{2\sqrt{n}} < \varepsilon_{K-1},$$

Substituting this into (I), we obtain

$$w(\mathbb{T}) \leqslant C_1 \sum_{k=1}^{K} \varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon_k)} + \frac{1}{2} w(\mathbb{T}),$$

i.e.,

$$w(\mathbb{T}) \leqslant 2C_1 \sum_{k=1}^{K} \varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon_k)}.$$

- Converting the discrete sum to an integral

When $\varepsilon \in [\varepsilon_{k+1}, \varepsilon_k]$, $\mathcal{N}(\mathbb{T}, \varepsilon) \geqslant \mathcal{N}(\mathbb{T}, \varepsilon_k)$, hence

$$\int_{\varepsilon_{k+1}}^{\varepsilon_k} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)} \mathrm{d}\varepsilon \geqslant (\varepsilon_k - \varepsilon_{k+1}) \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon_k)} = \frac{1}{2} \varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon_k)},$$

Summing over both sides,

$$w(\mathbb{T}) \leqslant 4C_1 \sum_{k=1}^{K} \int_{\varepsilon_{k+1}}^{\varepsilon_k} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)} \mathrm{d}\varepsilon = 4C_1 \int_{\varepsilon_{K+1}}^{\varepsilon_1} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)} \mathrm{d}\varepsilon.$$

(*Now determine the upper and lower integration limits.*)

For the upper limit, since $\varepsilon_1 = D/2 \leqslant D = \operatorname{diam}(\mathbb{T})$, we relax the upper limit to $b = \operatorname{diam}(\mathbb{T})$.

For the lower limit, since

$$\varepsilon_{K+1} = \frac{1}{4} \varepsilon_{K-1} > \frac{1}{4} \cdot \frac{w(\mathbb{T})}{2\sqrt{n}} = \frac{1}{8} \frac{w(\mathbb{T})}{\sqrt{n}},$$

we set the lower integration limit to $a = c \frac{w(\mathbb{T})}{\sqrt{n}}$ with $c = 1/8$, so $a < \varepsilon_{K+1}$.

Thus,

$$\int_{\varepsilon_{K+1}}^{\varepsilon_1} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)} \mathrm{d}\varepsilon \leqslant \int_{a}^{b} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)} \mathrm{d}\varepsilon,$$

i.e.,

$$w(\mathbb{T}) \leqslant C \int_{a}^{b} \sqrt{\log \mathcal{N}(\mathbb{T}, \varepsilon)} \mathrm{d}\varepsilon, \qquad (C = 4C_1)$$

where $b = \operatorname{diam}(\mathbb{T})$ and $a = \frac{c w(\mathbb{T})}{\sqrt{n}}$.

7 [**Sub-Exponential Dudley Inequality**] Let $(\mathbb{T}, d)$ be a finite metric space, or let the process $(X_t)_{t \in \mathbb{T}}$ be separable and the covering numbers at the displayed scales be finite. Prove the following version of Theorem T8.1.3 (*Dudley integral inequality*): suppose the increments of a zero-mean random process $(X_t)_{t \in \mathbb{T}}$ no longer satisfy the standard sub-Gaussian distribution but instead satisfy sub-exponential tail decay. Specifically, there exists a constant $K > 0$ such that for any $t, s \in \mathbb{T}$, the sub-exponential norm $\|\cdot\|_{\psi_1}$ of the increments satisfies

$$\|X_t - X_s\|_{\psi_1} \leqslant K d(t, s).$$

Prove the sub-exponential version of Dudley's inequality:

$$\mathbb{E}\left[\sup_{t \in \mathbb{T}} X_t\right] \leqslant CK \int_{0}^{\infty} \log \mathcal{N}(\mathbb{T}, d, \varepsilon) \mathrm{d}\varepsilon. \qquad (C > 0)$$

Rk. **Sub-Exponential Maximum Lemma** Let $Z_1, \cdots, Z_N$ be zero-mean random variables (*not necessarily independent*) satisfying $\|Z_i\|_{\psi_1} \leqslant M$. Then

$$\mathbb{E}\left[\max_{1 \leqslant i \leqslant N} Z_i\right] \leqslant cM \log(N+1).$$

Prf. First consider finite $\mathbb{T}$. Since $\mathbb{T}$ is finite, there exists k_0 such that $\mathcal{N}_k = \mathbb{T}$ and hence $\pi_k(t) = t$ for $k \geqslant k_0$, so the telescoping

series below is a finite sum.

Let $D = \operatorname{diam}(\mathbb{T})$ and define the dyadic scale sequence $\varepsilon_k = D2^{-k}$. Let $\mathcal{N}_k$ be a minimal ε_k-net of $\mathbb{T}$, so $|\mathcal{N}_k| = \mathcal{N}(mathbbT, d, \varepsilon_k)$.

For $\varepsilon_0 = D$, we need only one point to cover $\mathbb{T}$, so $|\mathcal{N}_0| = 1$; denote this unique element by t_0.

For any $t \in \mathbb{T}$, define the projection map $\pi_k(t) \in \mathcal{N}_k$ as the point in the net closest to t.

Expand X_t as a telescoping series,

$$X_t = X_{t_0} + \sum_{k=1}^{\infty} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}).$$

Taking the supremum and expectation on both sides and substituting $\mathbb{E}[X_{t_0}] = 0$, we obtain

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \leqslant \sum_{k=1}^{\infty} \mathbb{E}\left[\sup_{t\in\mathbb{T}} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)})\right].$$

Consider the single-step increment $\Delta_k(t) = X_{\pi_k(t)} - X_{\pi_{k-1}(t)}$. The number of distinct endpoint pairs is at most $|\mathcal{N}_k||\mathcal{N}_{k-1}| \leqslant |\mathcal{N}_k|^2$. The distance between the corresponding two points is

$$d(\pi_k(t), \pi_{k-1}(t)) \leqslant d(\pi_k(t), t) + d(t, \pi_{k-1}(t)) \leqslant \varepsilon_k + \varepsilon_{k-1} \leqslant 2\varepsilon_{k-1},$$

hence by the sub-exponential condition,

$$\|\Delta_k(t)\|_{\psi_1} \leqslant K d(\pi_k(t), \pi_{k-1}(t)) \leqslant 2K\varepsilon_{k-1},$$

By the sub-exponential maximum lemma, we obtain

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} \Delta_k(t)\right] \leqslant c(2K\varepsilon_{k-1}) \log(|\mathcal{N}_k|^2) = 4cK\varepsilon_{k-1} \log \mathcal{N}(\mathbb{T}, d, \varepsilon_k),$$

i.e.,

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \leqslant C_1 K \sum_{k=1}^{\infty} \varepsilon_{k-1} \log \mathcal{N}(\mathbb{T}, d, \varepsilon_k). \qquad (C_1 = 4c)$$

For the right-hand side, let $f(\varepsilon) = \log \mathcal{N}(\mathbb{T}, d, \varepsilon)$, then

$$\sum_{k=1}^{\infty} \varepsilon_{k-1} f(\varepsilon_k) = 4 \sum_{k=1}^{\infty} (\varepsilon_k - \varepsilon_{k+1}) f(\varepsilon_k) \leqslant 4 \int_0^{\varepsilon_1} \log \mathcal{N}(\mathbb{T}, d, \varepsilon) \mathrm{d}\varepsilon,$$

Extending the upper integration limit to ∞,

$$\mathbb{E}\left[\sup_{t\in\mathbb{T}} X_t\right] \leqslant CK \int_0^{\infty} \log \mathcal{N}(\mathbb{T}, d, \varepsilon) \mathrm{d}\varepsilon. \qquad (C = 4C_1)$$

For a separable process, let $\mathbb{T}_m$ be increasing finite subsets containing t_0 such that $\sup_{t\in\mathbb{T}_m} X_t \uparrow \sup_{t\in\mathbb{T}} X_t$ almost surely. Apply the finite case to $\mathbb{T}_m$ and use $\mathcal{N}(\mathbb{T}_m, d, \varepsilon) \leqslant \mathcal{N}(\mathbb{T}, d, \varepsilon)$. Since $\sup_{t\in\mathbb{T}_m} X_t - X_{t_0} \geqslant 0$ increases monotonically, taking the limit gives the same conclusion.

Rk. In the standard Dudley inequality, the process is sub-Gaussian, and the expected cost of the maximum is $\sqrt{\log N}$. In the sub-exponential version of Dudley's inequality, the expected cost of the maximum increases to $\log N$.

8 [**Local Dudley Inequality**] Let $(\mathbb{T}, d)$ be a finite metric space, or let the process be separable and the displayed covering numbers finite. Let $(X_t)_{t\in\mathbb{T}}$ have sub-Gaussian increments, i.e., there exists $K > 0$ such that for any $s, t \in \mathbb{T}$, $\|X_s - X_t\|_{\psi_2} \leqslant K d(s, t)$. Given a radius $\delta > 0$, prove:

$$\mathbb{E}\left[\sup_{\substack{s,t\in\mathbb{T} \\ d(s,t)\leqslant\delta}} |X_s - X_t|\right] \leqslant CK \int_0^{\delta} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)} \mathrm{d}\varepsilon.$$

Prf. First consider finite $\mathbb{T}$. Since $\mathbb{T}$ is finite, there exists k_0 such that $\mathcal{N}_k = \mathbb{T}$ and hence $\pi_k(t) = t$ for $k \geqslant k_0$, so the telescoping series below is a finite sum.

Construct nets at different scales.

Define the scale sequence $\varepsilon_k = \delta 2^{-k}$, and let $\mathcal{N}_k \subset \mathbb{T}$ be a minimal ε_k-net of $\mathbb{T}$ in the metric d, so $|\mathcal{N}_k| = \mathcal{N}(\mathbb{T}, d, \varepsilon_k)$.

For any $t \in \mathbb{T}$, define the projection $\pi_k(t) \in \mathcal{N}_k$ as the point in $\mathcal{N}_k$ closest to t, so $d(t, \pi_k(t)) \leqslant \varepsilon_k$.

- Performing the chaining expansion

For any $s, t \in \mathbb{T}$ satisfying $d(s,t) \leqslant \delta$, expand the increment $X_s - X_t$ as a telescoping series,

$$X_s - X_t = (X_{\pi_0(s)} - X_{\pi_0(t)}) + \sum_{k=1}^{\infty}(X_{\pi_k(s)} - X_{\pi_{k-1}(s)}) - \sum_{k=1}^{\infty}(X_{\pi_k(t)} - X_{\pi_{k-1}(t)}),$$

Taking the supremum on both sides and then using the triangle inequality, we obtain

$$\sup_{d(s,t)\leqslant\delta} |X_s - X_t| \leqslant \sup_{(v,w)\in E_0} |X_v - X_w| + 2\sum_{k=1}^{\infty} \sup_{(v,w)\in E_k} |X_v - X_w|,$$

where $E_0 = \{(\pi_0(s), \pi_0(t)) : d(s,t) \leqslant \delta\}$ and $E_k = \{(\pi_k(u), \pi_{k-1}(u)) : u \in \mathbb{T}\}$.

(*Now bound using the sub-Gaussian maximum lemma.*)

For the second term, by the sub-Gaussian maximum lemma,

$$\mathbb{E}\left[\sup_{(v,w)\in E_k} |X_v - X_w|\right] \leqslant C' K d(v,w)\sqrt{\log(|E_k|)},$$

Substituting $d(v,w) \leqslant 3\varepsilon_k$ and $|E_k| \leqslant |\mathcal{N}_k|^2$, we obtain

$$\mathbb{E}\left[\sup_{(v,w)\in E_k} |X_v - X_w|\right] \leqslant C' K (3\varepsilon_k)\sqrt{\log(|\mathcal{N}_k|^2)} = 3\sqrt{2}C' K \varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon_k)}.$$

Similarly, for the first term, substituting $3\varepsilon_0 = 3\delta$, we obtain

$$\mathbb{E}\left[\sup_{(v,w)\in E_0} |X_v - X_w|\right] \leqslant 3\sqrt{2}C' K \delta \sqrt{\log \mathcal{N}(\mathbb{T}, d, \delta)}.$$

Thus,

$$\mathbb{E}\left[\sup_{d(s,t)\leqslant\delta} |X_s - X_t|\right] \leqslant C_1 K \sum_{k=0}^{\infty} \varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon_k)}. \qquad (C_1 = 6\sqrt{2}C')$$

- Converting the sum to an integral

When $\varepsilon \in [\varepsilon_{k+1}, \varepsilon_k]$, $\mathcal{N}(\mathbb{T}, d, \varepsilon) \geqslant \mathcal{N}(\mathbb{T}, d, \varepsilon_k)$, hence

$$\int_{\varepsilon_{k+1}}^{\varepsilon_k} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant (\varepsilon_k - \varepsilon_{k+1})\sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon_k)} = \frac{1}{2}\varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon_k)},$$

Summing over both sides, substituting $\varepsilon_0 = \delta, \varepsilon_\infty = 0$, we obtain

$$\int_0^{\delta} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant \frac{1}{2}\sum_{k=0}^{\infty} \varepsilon_k \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon_k)},$$

therefore

$$\mathbb{E}\left[\sup_{d(s,t)\leqslant\delta} |X_s - X_t|\right] \leqslant 2C_1 K \int_0^{\delta} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon,$$

Now set $C = 2C_1$.

For a separable process, let $\mathbb{T}_m$ be increasing finite subsets such that the random variables on the left-hand side increase almost surely to the corresponding random variable for the original process. Apply the finite case to $\mathbb{T}_m$ and use $\mathcal{N}(\mathbb{T}_m, d, \varepsilon) \leqslant \mathcal{N}(\mathbb{T}, d, \varepsilon)$. Since these random variables are nonnegative, the monotone convergence theorem gives the conclusion upon taking the limit.

9 [**Covering Numbers of Lipschitz Function Classes**] Consider the family of functions defined on $[0,1]$, taking values in

$[0,1]$, with Lipschitz constant not exceeding 1,

$$\mathcal{F} = \{f : [0,1] \to [0,1] : \|f\|_{\mathrm{Lip}} \leqslant 1\}.$$

Prove that for any $\varepsilon \in (0,1)$, the covering number of this function class in the L^∞ norm satisfies

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leqslant \mathrm{e}^{C/\varepsilon}. \qquad (C > 0)$$

Prf. • Mapping functions to discrete grid points

Choose the integer $N = \lceil 2/\varepsilon \rceil$ as the number of grid subdivisions. On the domain $[0,1]$ construct equally spaced grid points $x_k = \frac{k}{N}$. (*On the range* $[0,1]$ *construct the discrete value set* $Y_N = \{0, \frac{1}{N}, \cdots, 1\}$.)

(*For any* $f \in \mathcal{F}$*, project its true value* $f(x_k)$ *at* x_k *to the nearest discrete grid point in the range.*)

Define the sequence $\boldsymbol{v} = (v_0, v_1, \cdots, v_N)$, where $v_k \in Y_N$ satisfies $|f(x_k) - v_k| \leqslant \frac{1}{2N}$, so the true function values satisfy

$$|f(x_k) - f(x_{k-1})| \leqslant |x_k - x_{k-1}| = \frac{1}{N},$$

Therefore, by the triangle inequality,

$$\begin{aligned} |v_k - v_{k-1}| &\leqslant |v_k - f(x_k)| + |f(x_k) - f(x_{k-1})| + |f(x_{k-1}) - v_{k-1}| \\ &\leqslant \frac{1}{2N} + \frac{1}{N} + \frac{1}{2N} = \frac{2}{N}, \end{aligned}$$

Since $v_k, v_{k-1} \in Y_N$, the integer $N(v_k - v_{k-1})$ takes values in $\{-2, -1, 0, 1, 2\}$.

(*Count the total number of sequences* $\boldsymbol{v}$ *satisfying the increment constraint.*)

The starting point v_0 can be chosen arbitrarily from Y_N, giving $N+1$ possibilities; for v_k ($k \geqslant 1$), given v_{k-1}, there are at most 5 possibilities; hence the total number of all possible valid discrete sequences $\boldsymbol{v}$ is at most $(N+1)5^N$.

• Constructing an ε-net and performing error analysis

(*For any valid sequence* $\boldsymbol{v}$*, construct a piecewise affine function* $g_{\boldsymbol{v}} : [0,1] \to [0,1]$ *via linear interpolation.*)

For any $x \in [x_{k-1}, x_k]$, write it as $x = (1-t)x_{k-1} + tx_k$ ($t \in [0,1]$), and define $g_{\boldsymbol{v}}(x) = (1-t)v_{k-1} + tv_k$. Then the approximation error is

$$\begin{aligned} |f(x) - g_{\boldsymbol{v}}(x)| &= |f(x) - ((1-t)v_{k-1} + tv_k)| \\ &= |(1-t)(f(x) - v_{k-1}) + t(f(x) - v_k)| \\ &\leqslant (1-t)|f(x) - v_{k-1}| + t|f(x) - v_k|. \end{aligned}$$

where

$$\begin{aligned} |f(x) - v_{k-1}| &\leqslant |f(x) - f(x_{k-1})| + |f(x_{k-1}) - v_{k-1}| \\ &\leqslant (x - x_{k-1}) + \frac{1}{2N} \leqslant \frac{3}{2N}, \\ |f(x) - v_k| &\leqslant |f(x) - f(x_k)| + |f(x_k) - v_k| \\ &\leqslant (x_k - x) + \frac{1}{2N} \leqslant \frac{3}{2N}, \end{aligned}$$

Substituting this in,

$$|f(x) - g_{\boldsymbol{v}}(x)| \leqslant \frac{3}{2N},$$

i.e., on $[0,1]$,

$$\|f - g_{\boldsymbol{v}}\|_{L^\infty} \leqslant \frac{3}{2N},$$

Substituting $N = \lceil 2/\varepsilon \rceil \geqslant 2/\varepsilon$, we obtain

$$\|f - g_{\boldsymbol{v}}\|_{L^\infty} \leqslant \frac{3}{4}\varepsilon < \varepsilon,$$

Therefore $\mathcal{G} = \{g_{\boldsymbol{v}}\}$ forms an ε-net of $\mathcal{F}$ in the L^∞ norm, and hence

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leqslant |\mathcal{G}| \leqslant (N+1)5^N.$$

Taking logarithms on both sides,

$$\begin{aligned}\log|\mathcal{G}| &\leqslant \log(N+1) + N\log 5 \\ &\leqslant N + N\log 5 = N(1+\log 5),\end{aligned}$$

Substituting

$$N = \lceil 2/\varepsilon \rceil \leqslant \frac{2}{\varepsilon} + 1 < \frac{3}{\varepsilon},$$

we obtain

$$\log|\mathcal{G}| \leqslant \frac{3(1+\log 5)}{\varepsilon},$$

i.e.,

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leqslant \mathrm{e}^{C/\varepsilon}. \qquad (C = 3(1+\log 5))$$

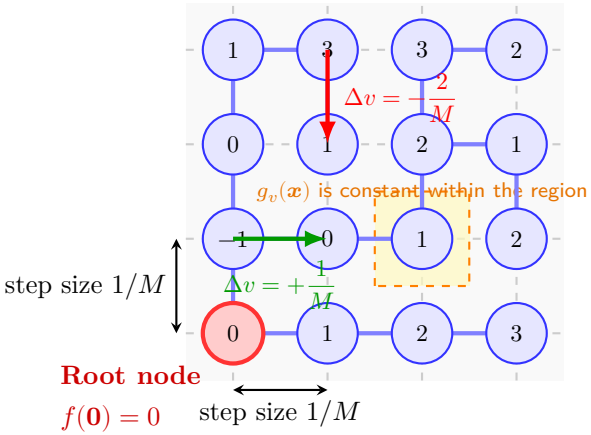
Rk. The conclusion of this exercise can be generalized:

Covering number estimate for high-dimensional Lipschitz function classes in the uniform norm Let the space dimension be $d \geqslant 1$. Define the function class $\mathcal{F}$ as

$$\mathcal{F} = \{f : [0,1]^d \to \mathbb{R} : |f(\boldsymbol{x}) - f(\boldsymbol{y})| \leqslant \|\boldsymbol{x} - \boldsymbol{y}\|_\infty, f(\mathbf{0}) = 0\}.$$

Then for any $\varepsilon \in (0,1)$, its covering number in the uniform norm $\|\cdot\|_{L^\infty}$ satisfies

$$\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leqslant C_d \varepsilon^{-d}. \qquad (C_d > 0)$$



1. Gridding (V)

Partition $[0,1]^d$ into a grid with step size $1/M$.

2. Discretization mapping (v_f)

Round the function values at the nodes $f(\boldsymbol{p})$ to $\frac{1}{M}\mathbb{Z}$.

(Note: the integers inside the circles on the left represent the numerators)

3. Increment constraint

Since f is Lipschitz, the difference of true values at adjacent grid points is $\leqslant 1/M$. Accounting for the $1/(2M)$ rounding error at each end, the discrete value increment is bounded by

$$|v_f(\boldsymbol{p}) - v_f(\boldsymbol{q})| \leqslant \frac{2}{M},$$

$\Rightarrow$ **along each edge, the state transition has only 5 possibilities,**

$$\Delta v \in \{-2, -1, 0, 1, 2\}.$$

4. Global state counting

The root node is fixed at 0, and the entire configuration is uniquely determined by the increments along the $|V| - 1$ edges of the spanning tree,

$$|\mathcal{V}| \leqslant 5^{|V|-1} \leqslant 5^{3^d \varepsilon^{-d}}.$$

Prf. (*To avoid the extra* $\log(1/\varepsilon)$ *factor arising from independently discretizing each point, we discretize the domain as a grid and exploit the spanning tree structure from graph theory. Since the function satisfies the Lipschitz property, the increments of function values at adjacent grid points along the spanning tree are constrained. By restricting the discrete states of these increments, we can count the function class effectively.*)

- Constructing the discrete grid and graph structure on the domain

Choose the integer $M = \lceil 1/\varepsilon \rceil$, and construct the equally spaced grid V on $[0,1]^d$,

$$V = \left\{ \left(\frac{i_1}{M}, \frac{i_2}{M}, \cdots, \frac{i_d}{M} \right) : i_k \in \{0, 1, \cdots, M\} \right\},$$

so the total number of grid points is

$$|V| = (M+1)^d < \left(\frac{1}{\varepsilon} + 2 \right)^d < \left(\frac{3}{\varepsilon} \right)^d = 3^d \varepsilon^{-d}.$$

View V as the vertex set of an undirected graph. Specifically, for any grid points $\boldsymbol{p}, \boldsymbol{q} \in V$, if their distance in the uniform norm satisfies $\|\boldsymbol{p} - \boldsymbol{q}\|_\infty = 1/M$, regard them as connected by an edge; then V is connected.

Choose a spanning tree of V rooted at the origin $\mathbf{0}$, denoted by T; then T has $|V| - 1$ edges.

- Projecting functions onto discrete grid states

For any $f \in \mathcal{F}$, round its true values at the grid points to the nearest integer multiple of $1/M$. Specifically, define the discrete grid function $v_f : V \to \frac{1}{M}\mathbb{Z}$ satisfying, for any $\boldsymbol{p} \in V$,

$$v_f(\boldsymbol{p}) = \frac{1}{M} \operatorname{round}(M f(\boldsymbol{p})),$$

then the projection error at the grid points is

$$|f(\boldsymbol{p}) - v_f(\boldsymbol{p})| \leqslant \frac{1}{2M},$$

(*Count the total number of valid discrete states v_f.*)

Since $f(\mathbf{0}) = 0$, we have $v_f(\mathbf{0}) = 0$.

For any $\boldsymbol{p}, \boldsymbol{q} \in V$, by the triangle inequality,

$$\begin{aligned} |v_f(\boldsymbol{p}) - v_f(\boldsymbol{q})| &\leqslant |v_f(\boldsymbol{p}) - f(\boldsymbol{p})| + |f(\boldsymbol{p}) - f(\boldsymbol{q})| + |f(\boldsymbol{q}) - v_f(\boldsymbol{q})| \\ &\leqslant \frac{1}{2M} + \|\boldsymbol{p} - \boldsymbol{q}\|_\infty + \frac{1}{2M} = \frac{1}{2M} + \frac{1}{M} + \frac{1}{2M} = \frac{2}{M}. \end{aligned}$$

Here $v_f(\boldsymbol{p}), v_f(\boldsymbol{q})$ are both integer multiples of $1/M$, so the integer $M(v_f(\boldsymbol{p}) - v_f(\boldsymbol{q}))$ takes values in $\{-2, -1, 0, 1, 2\}$.

The discrete state configuration v_f is uniquely determined by the root node (*fixed at* $\mathbf{0}$) and the increments along the $|V| - 1$ edges of the spanning tree T, so the size of the set $\mathcal{V}$ of all possible discrete configurations is bounded by

$$|\mathcal{V}| \leqslant 5^{|V|-1} \leqslant 5^{|V|} \leqslant 5^{3^d \varepsilon^{-d}}.$$

- Using discrete states to construct an ε-net

For any $v \in \mathcal{V}$, extend it to a continuous function g_v defined on $[0,1]^d$.

For any $\boldsymbol{x} \in [0,1]^d$, let $\boldsymbol{p}_{\boldsymbol{x}}$ be the grid point in V closest to $\boldsymbol{x}$, then

$$\|\boldsymbol{x} - \boldsymbol{p}_{\boldsymbol{x}}\|_\infty \leqslant \frac{1}{2M},$$

Define the piecewise constant function

$$g_v(\boldsymbol{x}) = v(\boldsymbol{p}_{\boldsymbol{x}}),$$

If the discrete configuration v_f is generated by the original function f, then for any $\boldsymbol{x} \in [0,1]^d$,

$$\begin{aligned} |f(\boldsymbol{x}) - g_{v_f}(\boldsymbol{x})| = |f(\boldsymbol{x}) - v_f(\boldsymbol{p}_{\boldsymbol{x}})| &\leqslant |f(\boldsymbol{x}) - f(\boldsymbol{p}_{\boldsymbol{x}})| + |f(\boldsymbol{p}_{\boldsymbol{x}}) - v_f(\boldsymbol{p}_{\boldsymbol{x}})| \\ &\leqslant \|\boldsymbol{x} - \boldsymbol{p}_{\boldsymbol{x}}\|_\infty + \frac{1}{2M} \leqslant \frac{1}{2M} + \frac{1}{2M} = \frac{1}{M} < \varepsilon, \end{aligned}$$

i.e., $\|f - g_{v_f}\|_{L^\infty} \leqslant \varepsilon$, so $\mathcal{G} = \{g_v : v \in \mathcal{V}\}$ forms an outer ε-net of $\mathcal{F}$, and hence

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leqslant |\mathcal{V}| \leqslant 5^{3^d \varepsilon^{-d}}.$$

Taking logarithms on both sides,

$$\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leqslant 3^d \log(5) \varepsilon^{-d}.$$

Now set $C_d = 3^d \log 5$.

10 [**High-Dimensional Lipschitz Law of Large Numbers**] Consider the family $\mathcal{F}$ of real-valued functions defined on the

d-dimensional unit cube $[0,1]^d$ ($d \geqslant 2$), satisfying the L-Lipschitz condition in the metric space $(\mathbb{R}^d, \|\cdot\|_\infty)$ and vanishing at the origin,

$$\mathcal{F} = \{f : [0,1]^d \to \mathbb{R} : \|f\|_{\mathrm{Lip}} \leqslant L, f(\mathbf{0}) = 0\}.$$

Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_n$ be i.i.d. random vectors taking values in $[0,1]^d$. Prove this theorem following these steps (*WLOG* $L = 1$):

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n f(\boldsymbol{X}_i) - \mathbb{E}[f(\boldsymbol{X}_1)]\right|\right] \leqslant C_d L \begin{cases} n^{-1/2}\log n, & d = 2, \\ n^{-1/d}, & d \geqslant 3 \end{cases}.$$

(1) Extend the covering number bound to high dimension:

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leqslant \mathrm{e}^{C/\varepsilon^d}.$$

(2) Substituting this bound directly into the Dudley integral fails because the integral diverges at $\varepsilon = 0$. To fix this, prove the following refined bound: for any $\delta \in [0,1]$,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}|X_f|\right] \leqslant \delta + \frac{C}{\sqrt{n}}\int_\delta^1 \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon)}\mathrm{d}\varepsilon.$$

Prf. • Establishing high-dimensional covering number bounds

By homogeneity, we may assume $L = 1$ without loss of generality.

For the family $\mathcal{F}$ of 1-Lipschitz functions defined on $[0,1]^d$ satisfying $f(\mathbf{0}) = 0$, by the remark of Exercise Q8.9 (*Covering Numbers of Lipschitz Function Classes*), for any $\varepsilon \in (0,1)$,

$$\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leqslant C_d \varepsilon^{-d}. \tag{I}$$

• Symmetrization and truncated Dudley integral

Let

$$Z = \sup_{f\in\mathcal{F}}|\frac{1}{n}\sum_{i=1}^n f(\boldsymbol{X}_i) - \mathbb{E}[f(\boldsymbol{X}_1)]|.$$

Introduce an i.i.d. Rademacher sequence $\sigma_i \in \{-1, 1\}$. By the symmetrization lemma, we obtain

$$\mathbb{E}[Z] \leqslant 2\mathbb{E}_{\boldsymbol{X}}\mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \sigma_i f(\boldsymbol{X}_i)\right|\right].$$

Conditional on the sample $\boldsymbol{X}_i$, the increments of the conditional empirical process

$$Y_f = \frac{1}{n}\sum_{i=1}^n \sigma_i f(\boldsymbol{X}_i),$$

satisfy the variance proxy

$$\frac{1}{n^2}\sum_{i=1}^n (f(\boldsymbol{X}_i) - g(\boldsymbol{X}_i))^2 \leqslant \frac{1}{n}\|f - g\|_{L^\infty}^2,$$

hence Y_f has sub-Gaussian increments with respect to the metric $\frac{1}{\sqrt{n}}\|\cdot\|_{L^\infty}$.

Given a truncation level $\delta \in (0,1)$, let $\pi_\delta(f)$ be the function in the δ-net (in L^∞) closest to f, and decompose Y_f as

$$Y_f = (Y_f - Y_{\pi_\delta(f)}) + Y_{\pi_\delta(f)}.$$

For the tail truncation error, by the triangle inequality,

$$\sup_{f\in\mathcal{F}}|Y_f - Y_{\pi_\delta(f)}| \leqslant \sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n |\sigma_i|\cdot|f(\boldsymbol{X}_i) - \pi_\delta(f)(\boldsymbol{X}_i)| \leqslant \|f - \pi_\delta(f)\|_{L^\infty} \leqslant \delta.$$

For the head discrete net part, by Dudley's chaining,

$$\mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}}|Y_{\pi_\delta(f)}|\right] \leqslant C'\int_{\delta/\sqrt{n}}^{1/\sqrt{n}} \sqrt{\log \mathcal{N}\left(\mathcal{F}, \frac{1}{\sqrt{n}}\|\cdot\|_{L^\infty}, u\right)}\mathrm{d}u = \frac{C'}{\sqrt{n}}\int_\delta^1 \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon)}\mathrm{d}\varepsilon.$$

Thus,

$$\mathbb{E}[Z] \leqslant 2\delta + \frac{2C'}{\sqrt{n}} \int_\delta^1 \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon)} \mathrm{d}\varepsilon. \tag{II}$$

Combining (I) and (II), we obtain

$$\mathbb{E}[Z] \leqslant 2\delta + \frac{C_1}{\sqrt{n}} \int_\delta^1 \varepsilon^{-d/2} \mathrm{d}\varepsilon. \tag{III}$$

a. When $d = 2$, $\int_\delta^1 \varepsilon^{-d/2}\mathrm{d}\varepsilon = \log(1/\delta)$, so

$$\mathbb{E}[Z] \leqslant 2\delta + \frac{C_1}{\sqrt{n}} \log(1/\delta).$$

(*To balance the two terms*) Now set $\delta = n^{-1/2}$, obtaining

$$\mathbb{E}[Z] \leqslant 2n^{-1/2} + \frac{C_1}{\sqrt{n}} \log(\sqrt{n}) \leqslant C_2 n^{-1/2} \log n.$$

b. When $d \geqslant 3$,

$$\int_\delta^1 \varepsilon^{-d/2}\mathrm{d}\varepsilon \leqslant \frac{2}{d-2}\delta^{1-d/2},$$

so

$$\mathbb{E}[Z] \leqslant 2\delta + \frac{C_3}{\sqrt{n}}\delta^{1-d/2},$$

(*To balance the two terms*) Now set $\delta = \frac{1}{\sqrt{n}}\delta^{1-d/2}$, i.e., $\delta = n^{-1/d}$, obtaining

$$\mathbb{E}[Z] \leqslant 2n^{-1/d} + \frac{C_3}{\sqrt{n}}(n^{-1/d})^{(2-d)/2} = 2n^{-1/d} + C_3 n^{-1/2} n^{-1/d+1/2} = (2 + C_3) n^{-1/d}.$$

In summary, when $L = 1$,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} \left| \frac{1}{n}\sum_{i=1}^n f(\boldsymbol{X}_i) - \mathbb{E}[f(\boldsymbol{X}_1)] \right|\right] \leqslant C_d \begin{cases} n^{-1/2}\log n, & d = 2, \\ n^{-1/d}, & d \geqslant 3 \end{cases},$$

For the general case,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} \left| \frac{1}{n}\sum_{i=1}^n f(\boldsymbol{X}_i) - \mathbb{E}[f(\boldsymbol{X}_1)] \right|\right] \leqslant C_d L \begin{cases} n^{-1/2}\log n, & d = 2, \\ n^{-1/d}, & d \geqslant 3 \end{cases}.$$

11 [**Symmetrization Lemma for Empirical Processes**] Modify the proof of Lemma L6.3.2 (*Symmetrization*) to obtain the following version of the symmetrization lemma.

Let Ω be a universe and $\mathcal{F}$ a family of real-valued functions defined on Ω. Let $X, X_1, \cdots, X_n$ be i.i.d. random variables taking values in Ω. Prove:

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} \left| \frac{1}{n}\sum_{i=1}^n f(X_i) - \mathbb{E}[f(X)] \right|\right] \leqslant 2\mathbb{E}\left[\sup_{f\in\mathcal{F}} \left| \frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i) \right|\right],$$

where $\varepsilon_1, \cdots, \varepsilon_n$ i.i.d. $\sim$ Rademacher are independent of $X_1, \cdots, X_n$.

Prf. Let $X_1', \cdots, X_n'$ be independent copies of $X_1, \cdots, X_n$. Then $\mathbb{E}[f(X)] = \mathbb{E}_{X'}[f(X_i')]$, hence

$$\frac{1}{n}\sum_{i=1}^n f(X_i) - \mathbb{E}[f(X)] = \frac{1}{n}\sum_{i=1}^n (f(X_i) - \mathbb{E}_{X'}[f(X_i')]) = \mathbb{E}_{X'}\left[\frac{1}{n}\sum_{i=1}^n (f(X_i) - f(X_i'))\right].$$

Define the convex functional $\Phi(Z) = \sup_{f\in\mathcal{F}} |Z_f|$. By Jensen's inequality, we obtain

$$\sup_{f\in\mathcal{F}} \left| \mathbb{E}_{X'}\left[\frac{1}{n}\sum_{i=1}^n (f(X_i) - f(X_i'))\right] \right| \leqslant \mathbb{E}_{X'}\left[\sup_{f\in\mathcal{F}} \left| \frac{1}{n}\sum_{i=1}^n (f(X_i) - f(X_i')) \right|\right],$$

Taking expectation over X on both sides,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}f(X_i)-\mathbb{E}[f(X)]\right|\right]\leqslant\mathbb{E}_{X,X'}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}(f(X_i)-f(X_i'))\right|\right].$$

Now introduce independent Rademacher random variables $\varepsilon_i\in\{-1,1\}$. Then $f(X_i)-f(X_i')\overset{d}{=}\varepsilon_i(f(X_i)-f(X_i'))$, therefore

$$\mathbb{E}_{X,X'}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}(f(X_i)-f(X_i'))\right|\right]=\mathbb{E}_{X,X',\varepsilon}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i(f(X_i)-f(X_i'))\right|\right],$$

where the right-hand side is

$$\begin{aligned}\mathbb{E}_{X,X',\varepsilon}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i(f(X_i)-f(X_i'))\right|\right]&\leqslant\mathbb{E}_{X,\varepsilon}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i f(X_i)\right|\right]+\mathbb{E}_{X',\varepsilon}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i f(X_i')\right|\right]\\&\leqslant 2\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i f(X_i)\right|\right],\end{aligned}$$

Substituting this in, we obtain

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}f(X_i)-\mathbb{E}[f(X)]\right|\right]\leqslant 2\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i f(X_i)\right|\right].$$

12 [**VC Dimension of the Union of Two Intervals**] On the real line $\mathbb{R}$, consider the family of sets consisting of the union of two closed intervals $\mathcal{C}=\{[a,b]\cup[c,d]:a,b,c,d\in\mathbb{R}\}$. Let $\mathcal{F}=\{\mathbf{1}_A:A\in\mathcal{C}\}$ be the family of indicator functions corresponding to $\mathcal{C}$. Prove that the Vapnik-Chervonenkis dimension of $\mathcal{F}$ is

$$\mathrm{vc}(\mathcal{F})=4.$$

Prf. Proving existence.

Choose a set of 4 strictly increasing points on $\mathbb{R}$: $\mathbb{S}=\{x_1,x_2,x_3,x_4\}$ with $x_1<x_2<x_3<x_4$. We only need to prove that for any $\mathbb{S}'\subset\mathbb{S}$, there exists $A\in\mathcal{C}$ such that $A\cap\mathbb{S}=\mathbb{S}'$.

Depending on the continuity of the points contained in $\mathbb{S}'$, we distinguish the following cases and construct the corresponding double-interval union A:

a. When $\mathbb{S}'=\emptyset$, choose $A=[x_1-2,x_1-1]\cup[x_1-2,x_1-1]$, so $A\cap\mathbb{S}=\emptyset$.

b. When $\mathbb{S}'$ consists of 1 contiguous block (*e.g.,* $\{x_1\}$, $\{x_2,x_3\}$, $\{x_1,x_2,x_3,x_4\}$), for any contiguous subset of the form $\{x_i,\cdots,x_j\}$ $(i\leqslant j)$, a single interval suffices to cover it. Choose $A=[x_i,x_j]\cup\emptyset$, so $A\cap\mathbb{S}=\mathbb{S}'$.

c. When $\mathbb{S}'$ consists of 2 separated contiguous blocks, among the set of 4 points, the following cases arise:

1. For $\{x_1,x_3\}$, take $A=[x_1,x_1]\cup[x_3,x_3]$;
2. For $\{x_1,x_4\}$, choose $A=[x_1,x_1]\cup[x_4,x_4]$;
3. For $\{x_2,x_4\}$, choose $A=[x_2,x_2]\cup[x_4,x_4]$;
4. For $\{x_1,x_2,x_4\}$, choose $A=[x_1,x_2]\cup[x_4,x_4]$;
5. For $\{x_1,x_3,x_4\}$, choose $A=[x_1,x_1]\cup[x_3,x_4]$.

Since $\mathcal{C}$ allows using 2 intervals, we can always use the first interval to cover the first block and the second interval to cover the second block.

Thus, all $2^4=16$ subsets of $\mathbb{S}$ can be picked out by $\mathcal{C}$, so $\mathbb{S}$ is shattered by $\mathcal{F}$, i.e., $\mathrm{vc}(\mathcal{F})\geqslant 4$.

Proving optimality.

We use proof by contradiction. Suppose there exists a set of 5 points $\mathbb{S}=\{x_1,x_2,x_3,x_4,x_5\}$ that can be shattered by $\mathcal{F}$. Without loss of generality, assume $x_1<x_2<x_3<x_4<x_5$. Then for any $\mathbb{S}'\subset\mathbb{S}$, there exists $A=I_1\cup I_2\in\mathcal{C}$ (*where* I_1,I_2 *are closed intervals*) such that $A\cap\mathbb{S}=\mathbb{S}'$.

Construct the specific alternating subset $\mathbb{S}'=\{x_1,x_3,x_5\}$. Since $A\cap\mathbb{S}=\mathbb{S}'$, we have $x_1\in A$, $x_2\notin A$, $x_3\in A$, $x_4\notin A$,

$x_5 \in A$.

Consider $x_1 \in A = I_1 \cup I_2$. Without loss of generality, assume $x_1 \in I_1$. Since $x_2 \notin A$, we have $x_2 \notin I_1$, so the right endpoint of I_1 is strictly less than x_2, i.e., for any $y \geqslant x_2$, $y \notin I_1$.

Consider $x_3 \in A$. By the previous deduction, $x_3 \notin I_1$. Since $x_3 \in I_1 \cup I_2$, we have $x_3 \in I_2$.

Consider $x_4 \notin A$. Since $x_4 \notin I_2$, the right endpoint of I_2 is strictly less than x_4, so for any $y \geqslant x_4$, $y \notin I_2$.

Finally consider $x_5 \in A$, i.e., $x_5 \in I_1 \cup I_2$. Since $x_5 > x_2$, $x_5 \notin I_1$; since $x_5 > x_4$, $x_5 \notin I_2$, hence $x_5 \notin I_1 \cup I_2 = A$. A contradiction.

Thus, no set of 5 points can be shattered by $\mathcal{F}$, so $\mathrm{vc}(\mathcal{F}) \leqslant 4$, i.e., $\mathrm{vc}(\mathcal{F}) = 4$.

13 [**VC Dimension of Disks in the Plane**] In the two-dimensional Euclidean space $\mathbb{R}^2$, consider the family of sets consisting of all closed disks $\mathcal{C} = \{\boldsymbol{x} \in \mathbb{R}^2 : \|\boldsymbol{x} - \boldsymbol{c}\|_2 \leqslant r, \boldsymbol{c} \in \mathbb{R}^2, r \geqslant 0\}$. Let $\mathcal{F} = \{\mathbf{1}_A : A \in \mathcal{C}\}$ be the family of indicator functions corresponding to $\mathcal{C}$. Prove that the Vapnik-Chervonenkis dimension of $\mathcal{F}$ is

$$\mathrm{vc}(\mathcal{F}) = 3.$$

Prf. Proving existence.

Choose the three vertices $\mathbb{S} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$ of an equilateral triangle. We only need to prove that all $2^3 = 8$ subsets $\mathbb{S}'$ of $\mathbb{S}$ can be picked out by some closed disk.

a. When $\mathbb{S}' = \emptyset$, choose a disk of radius 0 far away from $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3$;

b. When $\mathbb{S}'$ contains 1 point (*e.g.,* $\{\boldsymbol{x}_1\}$), choose a disk centered at $\boldsymbol{x}_1$ with a sufficiently small radius;

c. When $\mathbb{S}'$ contains 2 points (*e.g.,* $\{\boldsymbol{x}_1, \boldsymbol{x}_2\}$), choose the disk with center at the midpoint of segment $\boldsymbol{x}_1\boldsymbol{x}_2$ and radius $\frac{1}{2}\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|_2$. Since the triangle is equilateral, $\boldsymbol{x}_3$ lies strictly outside this disk;

d. When $\mathbb{S}'$ contains 3 points (*i.e.,* $\mathbb{S}$), choose the circumcircle of the triangle formed by the three points.

Thus, $\mathrm{vc}(\mathcal{F}) \geqslant 3$.

Proving optimality.

We use proof by contradiction. Suppose there exists a set of 4 points $\mathbb{S} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4\}$ that can be shattered by $\mathcal{F}$. For 4 points in the plane, their geometric-topological position can only belong to one of the following two cases; we will show that each case leads to a contradiction:

a. One point lies in the convex hull of the other three

Without loss of generality, assume $\boldsymbol{x}_4 \in \mathrm{conv}(\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3)$.

Since $\mathbb{S}$ can be shattered, there exists a disk $D \in \mathcal{C}$ that can pick out the subset $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$ alone, i.e., $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3 \in D$, $\boldsymbol{x}_4 \notin D$.

However, a closed disk D is a convex set. If $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3 \in D$, then $\mathrm{conv}(\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3) \subset D$, i.e., $\boldsymbol{x}_4 \in D$. A contradiction.

b. The 4 points form the four vertices of a convex quadrilateral

Without loss of generality, assume the four points are arranged clockwise as $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4$. The sum of the interior angles is 2π, so we may assume

$$\angle \boldsymbol{x}_1\boldsymbol{x}_2\boldsymbol{x}_3 + \angle \boldsymbol{x}_1\boldsymbol{x}_4\boldsymbol{x}_3 \geqslant \pi.$$

If the set were shattered, a closed disk D would contain $\boldsymbol{x}_1, \boldsymbol{x}_3$ and exclude $\boldsymbol{x}_2, \boldsymbol{x}_4$. We use the disk–chord lemma: if a closed disk contains the endpoints a, b of a chord and exterior points z, w lie on opposite sides of the line ab, then $\angle azb + \angle awb < \pi$. This follows from the power-of-a-point theorem and the inscribed-angle theorem: an exterior point sees the chord under a strictly smaller angle than the corresponding boundary arc, and the two boundary angles on opposite sides sum to π. Applying the lemma with $a = \boldsymbol{x}_1$, $b = \boldsymbol{x}_3$, $z = \boldsymbol{x}_2$, and $w = \boldsymbol{x}_4$ gives

$$\angle \boldsymbol{x}_1\boldsymbol{x}_2\boldsymbol{x}_3 + \angle \boldsymbol{x}_1\boldsymbol{x}_4\boldsymbol{x}_3 < \pi,$$

contradicting the preceding lower bound. Hence the convex-quadrilateral case is impossible.

Thus, no set of 4 points can be shattered by $\mathcal{F}$, so $\mathrm{vc}(\mathcal{F}) \leqslant 3$, i.e., $\mathrm{vc}(\mathcal{F}) = 3$.

14 [**VC Dimension of Axis-Aligned Rectangles**] In the two-dimensional Euclidean space $\mathbb{R}^2$, consider the family of sets consisting of all closed axis-aligned rectangles $\mathcal{C} = \{[a,b] \times [c,d] : a,b,c,d \in \mathbb{R}, a \leqslant b, c \leqslant d\}$. Let $\mathcal{F} = \{\mathbf{1}_A : A \in \mathcal{C}\}$ be the family of indicator functions corresponding to $\mathcal{C}$. Prove that the Vapnik-Chervonenkis dimension of $\mathcal{F}$ is

$$\mathrm{vc}(\mathcal{F}) = 4.$$

Prf. Proving existence.

Choose a set of 4 points in $\mathbb{R}^2$ arranged in a diamond shape: $\mathbb{S} = \{\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3, \boldsymbol{p}_4\}$ with coordinates $\boldsymbol{p}_1 = (0,1)$, $\boldsymbol{p}_2 = (1,0)$, $\boldsymbol{p}_3 = (0,-1)$, $\boldsymbol{p}_4 = (-1,0)$. We only need to prove that for any $\mathbb{S}' \subset \mathbb{S}$, there exists $R \in \mathcal{C}$ such that $R \cap \mathbb{S} = \mathbb{S}'$.

As in Exercise Q8.12 (*VC Dimension of the Union of Two Intervals*), we classify by the number of points in $\mathbb{S}'$ and construct the corresponding axis-aligned rectangle R:

a. When $\mathbb{S}'$ contains 0 points, choose $R = [2,3] \times [2,3]$, so $R \cap \mathbb{S} = \emptyset$.

b. When $\mathbb{S}'$ contains 1 point (*e.g.,* $\{\boldsymbol{p}_1\}$), choose a rectangle centered at $\boldsymbol{p}_1$ with a sufficiently small side length.

c. When $\mathbb{S}'$ contains 2 points, the following cases arise:

1. For adjacent points (*e.g.,* $\{\boldsymbol{p}_1, \boldsymbol{p}_2\}$), choose the smallest rectangle enclosing them $R = [0,1] \times [0,1]$, so $R \notin \boldsymbol{p}_3, \boldsymbol{p}_4$.
2. For opposite points (*e.g.,* $\{\boldsymbol{p}_1, \boldsymbol{p}_3\}$), choose the rectangle $R = [-0.1, 0.1] \times [-1,1]$, so $R \notin \boldsymbol{p}_2, \boldsymbol{p}_4$ (*excluding the points with x-coordinate 1 and -1*).

d. When $\mathbb{S}'$ contains 3 points (*e.g.,* $\{\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3\}$), choose the smallest axis-aligned bounding rectangle $R = [0,1] \times [-1,1]$, so $R \notin \boldsymbol{p}_4$.

e. When $\mathbb{S}'$ contains 4 points (*i.e.,* $\mathbb{S}$), choose $R = [-1,1] \times [-1,1]$.

Thus, all $2^4 = 16$ subsets of $\mathbb{S}$ can be picked out by $\mathcal{C}$, so $\mathbb{S}$ is shattered by $\mathcal{F}$, i.e., $\mathrm{vc}(\mathcal{F}) \geqslant 4$.

Proving optimality.

We use proof by contradiction. Suppose there exists a set of 5 points $\mathbb{S} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4, \boldsymbol{x}_5\}$ that can be shattered by $\mathcal{F}$.

For these 5 points, consider their coordinate extrema in the x and y axes. Let $\boldsymbol{x}_{\text{left}}, \boldsymbol{x}_{\text{right}}, \boldsymbol{x}_{\text{bottom}}, \boldsymbol{x}_{\text{top}}$ be the points with the smallest x-coordinate, largest x-coordinate, smallest y-coordinate, and largest y-coordinate, respectively. Define the subset of extreme points as $\mathbb{S}_{\text{ext}} = \{\boldsymbol{x}_{\text{left}}, \boldsymbol{x}_{\text{right}}, \boldsymbol{x}_{\text{bottom}}, \boldsymbol{x}_{\text{top}}\} \subset \mathbb{S}$.

(*Note that a point may simultaneously occupy two extreme positions, so* $|\mathbb{S}_{\text{ext}}| \leqslant 4$.)

Since $\mathbb{S}$ contains 5 points, there exists $\boldsymbol{p}_{\text{inner}}$ such that $\boldsymbol{p}_{\text{inner}} \in \mathbb{S}$ and $\boldsymbol{p}_{\text{inner}} \notin \mathbb{S}_{\text{ext}}$.

Since $\mathbb{S}$ can be shattered, there exists an axis-aligned rectangle $R = [a,b] \times [c,d] \in \mathcal{C}$ that can pick out the subset $\mathbb{S}_{\text{ext}}$ while excluding $\boldsymbol{p}_{\text{inner}}$, i.e., $\mathbb{S}_{\text{ext}} \subset R$, $\boldsymbol{p}_{\text{inner}} \notin R$.

Consider $R = [a,b] \times [c,d] \supset \mathbb{S}$. Since $\boldsymbol{x}_{\text{left}} \in R$, $a \leqslant$ the x-coordinate of $\boldsymbol{x}_{\text{left}}$; since $\boldsymbol{x}_{\text{right}} \in R$, $b \geqslant$ the x-coordinate of $\boldsymbol{x}_{\text{right}}$; since $\boldsymbol{x}_{\text{bottom}} \in R$, $c \leqslant$ the y-coordinate of $\boldsymbol{x}_{\text{bottom}}$; since $\boldsymbol{x}_{\text{top}} \in R$, $d \geqslant$ the y-coordinate of $\boldsymbol{x}_{\text{top}}$; thus the boundary of R necessarily contains the smallest axis-aligned bounding rectangle of the entire set $\mathbb{S}$.

Since the coordinates of $\boldsymbol{p}_{\text{inner}}$ are constrained by these extrema (*its x-coordinate lies between the left and right extrema, and its y-coordinate lies between the bottom and top extrema*), we have $\boldsymbol{p}_{\text{inner}} \in R$. A contradiction.

Thus, no set of 5 points can be shattered by $\mathcal{F}$, so $\mathrm{vc}(\mathcal{F}) \leqslant 4$, i.e., $\mathrm{vc}(\mathcal{F}) = 4$.

15 [**VC Dimension of Axis-Aligned Squares**] In the two-dimensional Euclidean space $\mathbb{R}^2$, consider the family of sets consisting of all closed axis-aligned squares $\mathcal{C} = \{[a, a+l] \times [b, b+l] : a,b \in \mathbb{R}, l \geqslant 0\}$. Let $\mathcal{F} = \{\mathbf{1}_A : A \in \mathcal{C}\}$ be the family of indicator functions corresponding to $\mathcal{C}$. Prove that the Vapnik-Chervonenkis dimension of $\mathcal{F}$ is

$$\mathrm{vc}(\mathcal{F}) = 3.$$

Prf. • Proving existence

Choose a set of 3 points in $\mathbb{R}^2$: $\mathbb{S} = \{\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3\}$ with coordinates $\boldsymbol{p}_1 = (1,0)$, $\boldsymbol{p}_2 = (-1,0)$, $\boldsymbol{p}_3 = (0,1)$. We only need to prove that for any $\mathbb{S}' \subset \mathbb{S}$, there exists a square $Q \in \mathcal{C}$ such that $Q \cap \mathbb{S} = \mathbb{S}'$.

Classify by the number of points in $\mathbb{S}'$ and construct the corresponding square Q:

a. When $\mathbb{S}'$ contains 0 points, choose $Q = [10, 11] \times [10, 11]$, so $Q \cap \mathbb{S} = \emptyset$.

b. When $\mathbb{S}'$ contains 1 point, the following cases arise:

1. For $\{\boldsymbol{p}_1\}$, choose $Q = [0.5, 1.5] \times [-0.5, 0.5]$;
2. For $\{\boldsymbol{p}_2\}$, choose $Q = [-1.5, -0.5] \times [-0.5, 0.5]$;
3. For $\{\boldsymbol{p}_3\}$, choose $Q = [-0.5, 0.5] \times [0.5, 1.5]$.

c. When $\mathbb{S}'$ contains 2 points, the following cases arise:

1. For $\{\boldsymbol{p}_1, \boldsymbol{p}_2\}$, choose the square $Q = [-1.1, 1.1] \times [-1.5, 0.7]$ of side length 2.2. It contains $(1, 0)$ and $(-1, 0)$, but excludes $\boldsymbol{p}_3 = (0, 1)$;
2. For $\{\boldsymbol{p}_1, \boldsymbol{p}_3\}$, choose $Q = [0, 1.5] \times [0, 1.5]$, so $Q \notin \boldsymbol{p}_2$;
3. For $\{\boldsymbol{p}_2, \boldsymbol{p}_3\}$, choose $Q = [-1.5, 0] \times [0, 1.5]$, so $Q \notin \boldsymbol{p}_1$.

d. When $\mathbb{S}'$ contains 3 points (*i.e.,* $\mathbb{S}$), choose $Q = [-1, 1] \times [-1, 1]$.

Thus, all $2^3 = 8$ subsets of $\mathbb{S}$ can be picked out by $\mathcal{C}$, so $\mathbb{S}$ is shattered by $\mathcal{F}$, i.e., $\mathrm{vc}(\mathcal{F}) \geqslant 3$.

• Proving optimality

We use proof by contradiction. Suppose there exists a set of 4 points $\mathbb{S} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4\}$ that can be shattered by $\mathcal{F}$. Consider the coordinate extrema of these 4 points. Let $x_{\min}, x_{\max}$ be the minimum and maximum x-coordinates, and $y_{\min}, y_{\max}$ the minimum and maximum y-coordinates. Define the width $W = x_{\max} - x_{\min}$ and height $H = y_{\max} - y_{\min}$. We consider two cases:

a. The 4 points do not each occupy distinct extreme positions

That is, there exists $\boldsymbol{p}_{\text{inner}} \in \mathbb{S}$ that is not the leftmost, rightmost, topmost, or bottommost point. Then $\boldsymbol{p}_{\text{inner}}$ must lie inside the smallest axis-aligned bounding rectangle determined by the other 3 points.

Since $\mathbb{S}$ can be shattered, there exists a square that can pick out the 3 points without $\boldsymbol{p}_{\text{inner}}$.

However, any square covering those 3 peripheral points necessarily covers their smallest axis-aligned bounding rectangle, and therefore inevitably contains $\boldsymbol{p}_{\text{inner}}$. A contradiction.

b. The 4 points each occupy a distinct extreme position

Let the four points be $\boldsymbol{p}_{\text{left}}, \boldsymbol{p}_{\text{right}}, \boldsymbol{p}_{\text{bottom}}, \boldsymbol{p}_{\text{top}}$. Without loss of generality, assume $W \geqslant H$ (*if $H > W$, swap the x and y axes to obtain the same conclusion*).

Consider picking out the subset $\mathbb{S}' = \{\boldsymbol{p}_{\text{left}}, \boldsymbol{p}_{\text{right}}\}$. Suppose there exists a square $Q = [a, a + l] \times [b, b + l]$ such that $Q \cap \mathbb{S} = \mathbb{S}'$.

Since $\boldsymbol{p}_{\text{left}}, \boldsymbol{p}_{\text{right}} \in Q$, we have $a \leqslant x_{\min}$ and $a + l \geqslant x_{\max}$, so the side length of the square satisfies $l \geqslant x_{\max} - x_{\min} = W$.

Since $\boldsymbol{p}_{\text{left}} \in Q$, its y-coordinate must lie within the square, i.e., $b \leqslant \boldsymbol{p}_{\text{left}}^{(y)} \leqslant b + l$.

(*To exclude $\boldsymbol{p}_{\text{top}}$, since its x-coordinate lies in $[x_{\min}, x_{\max}] \subset [a, a + l]$, its y-coordinate must fall outside $[b, b + l]$; similarly, to exclude $\boldsymbol{p}_{\text{bottom}}$, its y-coordinate must lie strictly below.*)

Since $\boldsymbol{p}_{\text{top}}^{(y)} = y_{\max} \geqslant \boldsymbol{p}_{\text{left}}^{(y)} \geqslant b$, it must lie strictly above, i.e.,

$$y_{\max} > b + l;$$

Similarly,

$$y_{\min} < b;$$

hence

$$H = y_{\max} - y_{\min} > (b + l) - b = l.$$

That is, $H > l > W$, contradicting $W \geqslant H$.

Thus, no set of 4 points can be shattered by $\mathcal{F}$, so $\mathrm{vc}(\mathcal{F}) \leqslant 3$, i.e., $\mathrm{vc}(\mathcal{F}) = 3$.

16 [**VC Dimension of Convex Polygons**] In the two-dimensional Euclidean space $\mathbb{R}^2$, consider the family $\mathcal{C}$ of all convex polygons. Let $\mathcal{F} = \{\mathbf{1}_A : A \in \mathcal{C}\}$ be the family of indicator functions corresponding to $\mathcal{C}$. Prove that the Vapnik-Chervonenkis dimension of $\mathcal{F}$ is

$$\operatorname{vc}(\mathcal{F}) = \infty.$$

Prf. For any $N \geqslant 1$, choose N distinct points on the unit circle $C = \{\boldsymbol{x} \in \mathbb{R}^2 : \|\boldsymbol{x}\|_2 = 1\}$ in $\mathbb{R}^2$ to form the set $\mathbb{S} = \{\boldsymbol{p}_1, \boldsymbol{p}_2, \cdots, \boldsymbol{p}_N\}$.

Proving that any subset can be picked out

For any subset $\mathbb{S}' \subset \mathbb{S}$, we only need to construct a convex polygon $P \in \mathcal{C}$ such that $P \cap \mathbb{S} = \mathbb{S}'$.

Classify by the number of elements in $\mathbb{S}'$:

a. When $|\mathbb{S}'| \geqslant 3$, define $P = \operatorname{conv}(\mathbb{S}')$. Since $\mathbb{S}'$ contains at least 3 discrete points on the circle, its convex hull necessarily forms a non-degenerate convex polygon whose vertices are exactly the points in $\mathbb{S}'$. Hence $P \in \mathcal{C}$ and $\mathbb{S}' \subset P$.

For any $\boldsymbol{q} \in \mathbb{S} \setminus \mathbb{S}'$, since $\boldsymbol{q}$ lies on the unit circle C, by geometric properties, all edges and interior points of the convex hull of any finite set of points on the circle lie strictly inside the circle. Since $\boldsymbol{q}$ is on the circle and is not a vertex of the convex hull P, we have $\boldsymbol{q} \notin P$, hence $P \cap \mathbb{S} = \mathbb{S}'$.

b. When $|\mathbb{S}'| \leqslant 2$, since a triangle is the simplest convex polygon (*belonging to the family* $\mathcal{C}$), we use extremely small triangles to handle boundary cases:

1. When $\mathbb{S}' = \emptyset$, construct a triangle P outside the unit circle that does not intersect the circle, so $P \cap \mathbb{S} = \emptyset$.
2. When $\mathbb{S}' = \{\boldsymbol{p}_i\}$, construct a triangle P centered at $\boldsymbol{p}_i$ with a sufficiently small side length, so $P \cap \mathbb{S} = \{\boldsymbol{p}_i\}$.
3. When $\mathbb{S}' = \{\boldsymbol{p}_i, \boldsymbol{p}_j\}$, construct a narrow triangle P containing the segment $\boldsymbol{p}_i\boldsymbol{p}_j$ and control its width to exclude the remaining $N-2$ points of $\mathbb{S}$, so $P \cap \mathbb{S} = \{\boldsymbol{p}_i, \boldsymbol{p}_j\}$.

Thus, no matter how large N is, we can always choose N points on the circle such that all 2^N of their subsets can be picked out by some convex polygon, i.e., $\mathbb{S}$ is shattered by $\mathcal{F}$, so $\operatorname{vc}(\mathcal{F}) \geqslant N$, i.e., $\operatorname{vc}(\mathcal{F}) = \infty$.

17 [**VC Dimension of Half-Spaces**] Prove the result of Example E8.3.5 (*Half-spaces*). In the n-dimensional Euclidean space $\mathbb{R}^n$, consider the family of sets consisting of all closed half-spaces $\mathcal{C} = \{\{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{w}^\top \boldsymbol{x} + b \geqslant 0\} : \boldsymbol{w} \in \mathbb{R}^n, b \in \mathbb{R}\}$. Let $\mathcal{F} = \{\mathbf{1}_A : A \in \mathcal{C}\}$ be the family of indicator functions corresponding to $\mathcal{C}$. Prove that the Vapnik-Chervonenkis dimension of $\mathcal{F}$ is

$$\operatorname{vc}(\mathcal{F}) = n + 1.$$

Prf. • Proving existence

In $\mathbb{R}^n$, choose the origin $\mathbf{0}$ and the n standard orthonormal basis vectors $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n$ to form the set of $n+1$ points $\mathbb{S} = \{\mathbf{0}, \boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n\}$. We only need to prove that for any $\mathbb{S}' \subset \mathbb{S}$, there exist $\boldsymbol{w}, b$ such that the half-space defined by $\boldsymbol{w}^\top \boldsymbol{x} + b \geqslant 0$ contains exactly $\mathbb{S}'$ and excludes $\mathbb{S} \setminus \mathbb{S}'$.

Define the target classification labels y_i as

$$y_i = \begin{cases} 1, & \boldsymbol{e}_i \in \mathbb{S}', \\ -1, & \boldsymbol{e}_i \notin \mathbb{S}' \end{cases},$$

Let the weight vector be $\boldsymbol{w} = (y_1, y_2, \cdots, y_n)^\top$, and construct the half-space parameters as

$$b = \begin{cases} 0.5, & \mathbf{0} \in \mathbb{S}', \\ -0.5, & \mathbf{0} \notin \mathbb{S}' \end{cases}.$$

Verify the correctness of this construction:

a. For the origin $\boldsymbol{x} = \mathbf{0}$, $\boldsymbol{w}^\top \mathbf{0} + b = b$. When $\mathbf{0} \in \mathbb{S}'$, $b = 0.5 > 0$ (*contained*); when $\mathbf{0} \notin \mathbb{S}'$, $b = -0.5 < 0$ (*excluded*). Hence the requirement is satisfied.

b. For the basis vector $\boldsymbol{x} = \boldsymbol{e}_i$, $\boldsymbol{w}^\top \boldsymbol{e}_i + b = y_i + b$. When $\boldsymbol{e}_i \in \mathbb{S}'$, $y_i = 1$, so regardless of whether $b = 0.5$ or -0.5,

$y_i + b \geqslant 0.5 > 0$ (*contained*); when $\boldsymbol{e}_i \notin \mathbb{S}'$, $y_i = -1$, so regardless of b, $y_i + b \leqslant -0.5 < 0$ (*excluded*). Hence the requirement is satisfied.

Thus, all 2^{n+1} subsets of $\mathbb{S}$ can be picked out by $\mathcal{C}$, so $\mathbb{S}$ is shattered by $\mathcal{F}$, i.e., $\text{vc}(\mathcal{F}) \geqslant n+1$.

- Proving optimality

We use proof by contradiction. Suppose there exists a set of $n+2$ points $V = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_{n+2}\} \subset \mathbb{R}^n$ that can be shattered by $\mathcal{F}$.

Consider the following linear system in the coefficients $\alpha_1, \alpha_2, \cdots, \alpha_{n+2}$,

$$\sum_{i=1}^{n+2} \alpha_i \boldsymbol{x}_i = \boldsymbol{0}, \quad \sum_{i=1}^{n+2} \alpha_i = 0,$$

This is a system of $n+1$ homogeneous linear equations (*the first provides n equations, the second provides* 1) with $n+2$ unknowns. Since there are more unknowns than equations, the system must have a non-zero solution $(\alpha_1, \cdots, \alpha_{n+2}) \neq \boldsymbol{0}$, so the coefficients $\alpha_1, \cdots, \alpha_{n+2}$ must contain both positive and negative values.

Partition the index set as $I = I^+ \sqcup I^-$, where

$$I^+ = \{i : \alpha_i > 0\}, \quad I^- = \{i : \alpha_i \leqslant 0\},$$

Let $A = \sum_{i \in I^+} \alpha_i > 0$, then $\sum_{i \in I^-} \alpha_i = -A$.

Construct

$$\boldsymbol{p} = \sum_{i \in I^+} \frac{\alpha_i}{A} \boldsymbol{x}_i = \sum_{i \in I^-} \frac{-\alpha_i}{A} \boldsymbol{x}_i,$$

where the left-hand side is a convex combination of $V^+ = \{\boldsymbol{x}_i : i \in I^+\}$ and the right-hand side is a convex combination of $V^- = \{\boldsymbol{x}_i : i \in I^-\}$. Hence $\boldsymbol{p} \in \text{conv}(V^+), \text{conv}(V^-)$.

(*This is Radon's theorem:* $\text{conv}(V^+) \cap \text{conv}(V^-) \neq \emptyset$.)

Since V can be shattered by $\mathcal{F}$, there exists a half-space $H = \{\boldsymbol{x} : \boldsymbol{w}^\top \boldsymbol{x} + b \geqslant 0\}$ separating V^+ and V^-, i.e., $V^+ \subset H, V^- \subset H^c$, so $\text{conv}(V^+) \subset H, \text{conv}(V^-) \subset H^c$, hence $\boldsymbol{p} \in H, \boldsymbol{p} \in H^c$, i.e., $\boldsymbol{p} \in H \cap H^c = \emptyset$. A contradiction.

Thus, no set of $n+2$ points can be shattered by $\mathcal{F}$, so $\text{vc}(\mathcal{F}) \leqslant n+1$, i.e., $\text{vc}(\mathcal{F}) = n+1$.

18 [**Relationship between VC Dimension and Algebraic Dimension**] Let Ω be a given universe and $\mathcal{F}$ a family of Boolean functions defined on Ω (*i.e., functions taking values in* $\{0,1\}$ *or* $\{-1,1\}$). View $\mathcal{F}$ as a subset of the vector space of $\mathbb{R}$-valued functions. Prove that for any finite-dimensional Boolean function family $\mathcal{F}$, its Vapnik-Chervonenkis dimension $\text{vc}(\mathcal{F})$ satisfies

$$\text{vc}(\mathcal{F}) \leqslant \dim(\mathcal{F}).$$

Prf. First suppose that the Boolean function family takes values in $\{0,1\}$. Let $\dim(\mathcal{F}) = d < \infty$. The case of $\{-1,1\}$-valued functions is identical after choosing the function with values $\text{sign}(a_i)$ in the final separating argument below.

We use proof by contradiction. Suppose $\text{vc}(\mathcal{F}) > d$. Then by the definition of the Vapnik-Chervonenkis dimension, there exists a set of at least $n = d+1$ points $\mathbb{S} = \{x_1, x_2, \cdots, x_n\} \subset \Omega$ that can be shattered by $\mathcal{F}$.

Consider the evaluation map $E : \mathcal{F} \to \mathbb{R}^n$ defined, for any $f \in \mathcal{F}$, by

$$E(f) = (f(x_1), f(x_2), \cdots, f(x_n))^\top.$$

Then $\dim(\text{span}\{E(f) : f \in \mathcal{F}\}) \leqslant d$.

Since $n = d+1 > d$, there exists a non-zero orthogonal vector, i.e., there exists a non-zero vector $\boldsymbol{a} = (a_1, a_2, \cdots, a_n)^\top \in \mathbb{R}^n$ such that for any $f \in \mathcal{F}$,

$$\boldsymbol{a}^\top E(f) = \sum_{i=1}^{n} a_i f(x_i) = 0. \tag{I}$$

Since $\boldsymbol{a} \neq \boldsymbol{0}$, partition the index set as $I = I^+ \sqcup I^-$, where

$$I^+ = \{i : a_i > 0\}, \quad I^- = \{i : a_i < 0\},$$

at least one of I^+, I^- is non-empty.

Using the property that $\mathbb{S}$ is shattered, we consider two cases:

a. When $I^+ \neq \emptyset$, since $\mathbb{S}$ is shattered, there exists $f_+ \in \mathcal{F}$ such that

$$f_+(x_i) = \begin{cases} 1, i \in I^+, \\ 0, i \notin I^+ \end{cases}$$

Substituting this into (I),

$$0 = \sum_{i=1}^{n} a_i f_+(x_i) = \sum_{i \in I^+} a_i \cdot 1 + \sum_{i \notin I^+} a_i \cdot 0 = \sum_{i \in I^+} a_i > 0,$$

a contradiction.

b. When $I^+ = \emptyset$, i.e., $I^- \neq \emptyset$, since $\mathbb{S}$ is shattered, there exists $f_- \in \mathcal{F}$ such that

$$f_-(x_i) = \begin{cases} 1, i \in I^-, \\ 0, i \notin I^- \end{cases}$$

Substituting this into (I),

$$0 = \sum_{i=1}^{n} a_i f_-(x_i) = \sum_{i \in I^-} a_i \cdot 1 = \sum_{i \in I^-} a_i < 0,$$

a contradiction.

Thus, $\mathrm{vc}(\mathcal{F}) \leqslant \dim(\mathcal{F})$.

19 [**Theorem (Tightness of Pajor's Lemma and Sauer-Shelah Lemma)**] Prove that Lemma L8.3.7 (*Pajor lemma*) and Lemma L8.3.9 (*Sauer-Shelah Lemma*) are both tight. Let $\Omega = \{1, 2, \cdots, n\}$ be a finite set containing n elements. Consider the set family $\mathcal{F} = \{A \subset \Omega : |A| \leqslant d\}$, where $0 \leqslant d \leqslant n$. Prove that the following inequalities all attain strict equality,

$$|\mathcal{F}| = |\{\Lambda \subset \Omega : \Lambda \text{ is shattered by } \mathcal{F}\}| = \sum_{k=0}^{\mathrm{vc}(\mathcal{F})} \binom{n}{k}.$$

Prf. It is easy to see that

$$|\mathcal{F}| = \sum_{k=0}^{d} \binom{n}{k}.$$

- Characterization of shattered sets

  Let $\mathcal{S} = \{\Lambda \subset \Omega : \Lambda \text{ is shattered by } \mathcal{F}\}$. Then Λ is shattered by $\mathcal{F}$ if and only if for any $\Lambda' \subset \Lambda$, there exists $A \in \mathcal{F}$ such that $A \cap \Lambda = \Lambda'$.

  a. If $|\Lambda| \leqslant d$, then for any $\Lambda' \subset \Lambda$, we have $|\Lambda'| \leqslant |\Lambda| \leqslant d$.

  Choose $A = \Lambda' \in \mathcal{F}$, satisfying $A \cap \Lambda = \Lambda'$. Hence all sets with $|\Lambda| \leqslant d$ are shattered by $\mathcal{F}$.

  b. If $|\Lambda| > d$, suppose Λ is shattered, then there exists $A \in \mathcal{F}$ such that $A \cap \Lambda = \Lambda$, i.e., $\Lambda \subset A$, so $|A| \geqslant |\Lambda| > d$, which contradicts $|A| \leqslant d$; in this case Λ cannot be shattered.

  Thus, the family of shattered sets is $\mathcal{S} = \{\Lambda \subset \Omega : |\Lambda| \leqslant d\}$, whose cardinality is $|\mathcal{S}| = \sum_{k=0}^{d} \binom{n}{k}$.

- Verification of tightness of Pajor's lemma and Sauer-Shelah lemma

  Since $|\mathcal{F}| = |\mathcal{S}|$, Pajor's lemma attains equality.

  Since every set of cardinality at most d can be shattered, and every set of cardinality greater than d cannot be shattered, we have $\mathrm{vc}(\mathcal{F}) = d$. Hence by the Sauer-Shelah lemma,

$$\sum_{k=0}^{\mathrm{vc}(\mathcal{F})} \binom{n}{k} = \sum_{k=0}^{d} \binom{n}{k} = |\mathcal{F}|,$$

  the Sauer-Shelah lemma attains equality.

Rk. For the set family $\mathcal{S}$ consisting of Hamming balls, the inequalities of both Pajor's lemma and the Sauer-Shelah lemma degenerate into strict equalities.

20 [**VC Dimension Dichotomy**] Let $\mathcal{F}$ be a family of Boolean functions defined on a universe Ω. Its growth function $\Pi_{\mathcal{F}}(n)$ is defined as the maximum number of distinct Boolean assignment patterns that $\mathcal{F}$ can induce on any n-element subset of Ω, i.e.,

$$\Pi_{\mathcal{F}}(n) = \max_{\mathbb{S}\subset\Omega, |\mathbb{S}|=n} |\{(f(x_1), f(x_2), \cdots, f(x_n)) : f \in \mathcal{F}, x_i \in \mathbb{S}\}|.$$

Prove that the growth function $\Pi_{\mathcal{F}}(n)$ can only exhibit polynomial growth or exponential growth, with no intermediate state; i.e., prove that exactly one of the following two propositions holds:

Sol. $\Pi_{\mathcal{F}}(n) = 2^n$;

Sol. There exists a polynomial $p(n)$ such that $\Pi_{\mathcal{F}}(n) \leqslant p(n)$.

Prf. For any function family $\mathcal{F}$, its VC dimension is either $\mathrm{vc}(\mathcal{F}) = \infty$ or $\mathrm{vc}(\mathcal{F}) = d$.

a. Suppose $\mathrm{vc}(\mathcal{F}) = \infty$. Then for any $n \in \mathbb{Z}^+$, the universe Ω contains a subset $\mathbb{S}$ of size at least n that can be completely shattered by $\mathcal{F}$.

By the definition of shattering, the restriction of $\mathcal{F}$ to $\mathbb{S}$ can produce all 2^n possible Boolean assignment patterns. Since $\Pi_{\mathcal{F}}(n)$ is the maximum number of patterns that can be produced by any subset of size n, and the total number of Boolean vectors of length n is at most 2^n, we have $\Pi_{\mathcal{F}}(n) = 2^n$, corresponding to exponential growth.

b. Suppose $\mathrm{vc}(\mathcal{F}) = d$. Then by the Sauer-Shelah lemma, for any $n \in \mathbb{Z}^+$, the growth function of $\mathcal{F}$ is bounded by

$$\Pi_{\mathcal{F}}(n) \leqslant \sum_{i=0}^{d} \binom{n}{i}.$$

Using the inequality $\binom{n}{i} \leqslant \frac{n^i}{i!} \leqslant n^i$, we obtain

$$\Pi_{\mathcal{F}}(n) \leqslant \sum_{i=0}^{d} n^i \leqslant \sum_{i=0}^{d} n^d = (d+1)n^d.$$

Define the polynomial $p(n) = (d+1)n^d$, which is a polynomial in n of highest degree d. Hence $\Pi_{\mathcal{F}}(n) \leqslant p(n)$, corresponding to polynomial growth.

21 [**Combinatorial Stability of VC Dimension**] In Proposition P8.3.11 (*VC stability*), we observed the stability of the VC dimension when combining two function classes pointwise via min or max.

Now extend this to any number of function classes and arbitrary Boolean combinations. Let Ω be a given universe, and let $\mathcal{F}_1, \mathcal{F}_2, \cdots, \mathcal{F}_k$ be $k \geqslant 2$ families of Boolean functions defined on Ω. Given a fixed Boolean combination function $\varphi : \{0,1\}^k \to \{0,1\}$, use φ to combine the above k function families into a new Boolean function family

$$\mathcal{F} = \{\boldsymbol{x} \mapsto \varphi(f_1(\boldsymbol{x}), \cdots, f_k(\boldsymbol{x})) : f_i \in \mathcal{F}_i, i = 1, \cdots, k\}.$$

Prove that if for every $i \in \{1, \cdots, k\}$, $\mathrm{vc}(\mathcal{F}_i) \leqslant d$, then the Vapnik-Chervonenkis dimension of the combined function family $\mathcal{F}$ satisfies

$$\mathrm{vc}(\mathcal{F}) \leqslant Cdk \log k. \qquad (C > 0)$$

Prf. If $d = 0$, each class induces only one label on every singleton, hence each class is a singleton and so is the combined class; the conclusion follows. Assume henceforth that $d \geqslant 1$. If $n \leqslant d$, the conclusion is also immediate; we only need to consider $n > d$.

- Apply the Sauer-Shelah lemma

  Let $\mathbb{S} \subset \Omega$ be a finite subset containing n elements. Define the growth function $\Pi_{\mathcal{H}}(n)$ of the function family $\mathcal{H}$ on $\mathbb{S}$ as the maximum number of distinct Boolean vectors that the restriction of $\mathcal{H}$ to $\mathbb{S}$ can produce.

  For any combined function $f = \varphi(f_1, \cdots, f_k) \in \mathcal{F}$, its output vector on $\mathbb{S}$ is completely determined by the output vectors of the k base functions $(f_1, \cdots, f_k)$ on $\mathbb{S}$. Hence the total number of distinct output patterns produced by $\mathcal{F}$ on $\mathbb{S}$ does

not exceed the product of the numbers of output patterns of the individual base function families on $\mathbb{S}$, i.e.,

$$\Pi_{\mathcal{F}}(n) \leqslant \prod_{i=1}^{k} \Pi_{\mathcal{F}_i}(n). \tag{I}$$

By the Sauer-Shelah lemma, when $n \geqslant d$,

$$\Pi_{\mathcal{F}_i}(n) \leqslant \sum_{j=0}^{d} \binom{n}{j} \leqslant \left(\frac{\mathrm{e}n}{d}\right)^{d},$$

Substituting this into (I), we obtain

$$\Pi_{\mathcal{F}}(n) \leqslant \prod_{i=1}^{k} \left(\frac{\mathrm{e}n}{d}\right)^{d} = \left(\frac{\mathrm{e}n}{d}\right)^{dk}. \tag{II}$$

- Establish a transcendental inequality for the VC dimension

  If $\mathrm{vc}(\mathcal{F}) = n$, then there exists a set of n elements completely shattered by $\mathcal{F}$, so that $\mathcal{F}$ can produce all 2^n patterns on this set, i.e., $\Pi_{\mathcal{F}}(n) = 2^n$. Substituting this into (II), we obtain

$$2^n \leqslant \left(\frac{\mathrm{e}n}{d}\right)^{dk}.$$

  Taking logarithms on both sides,

$$n \log 2 \leqslant dk \log\left(\frac{\mathrm{e}n}{d}\right).$$

  Dividing both sides by $d \log 2$, and setting $u = \frac{n}{d}$, $a = \frac{k}{\log 2}$, we obtain

$$u \leqslant a \log(\mathrm{e}u) = a(1 + \log u),$$

  Using the inequality $\log u < 2\sqrt{u}$, we obtain

$$u \leqslant a(1 + 2\sqrt{u}), \tag{III}$$

  i.e.,

$$(\sqrt{u})^2 - 2a\sqrt{u} - a \leqslant 0,$$

  Solving yields $\sqrt{u} \leqslant a + \sqrt{a^2 + a}$.

  - Solve by bootstrapping

    Since $k \geqslant 2$, we have $a = \frac{k}{\log 2} > 2$, hence $\sqrt{u} \leqslant a + \sqrt{a^2 + a} < a + \sqrt{a^2 + a^2} < 2.5a$, i.e.,

$$u \leqslant 6.25a^2. \tag{IV}$$

    Using bootstrapping, substitute the crude upper bound (IV) into (III) to obtain

$$u \leqslant a(1 + \log(6.25a^2)) = a(1 + \log 6.25 + 2 \log a).$$

    Since $a \geqslant \frac{2}{\log 2} \approx 2.88$, we have $1 + \log 6.25 \approx 2.83 \leqslant 3 \log a$. Substituting this,

$$u \leqslant a(3 \log a + 2 \log a) = 5a \log a,$$

    i.e.,

$$\frac{n}{d} \leqslant 5\left(\frac{k}{\log 2}\right) \log\left(\frac{k}{\log 2}\right).$$

    Since $k \geqslant 2$, $\log k \geqslant 0.69$, hence $\log(\frac{1}{\log 2}) \approx 0.36 \leqslant 0.6 \log k$, the right-hand side becomes

$$5\frac{k}{\log 2} \log\left(\frac{k}{\log 2}\right) \leqslant \frac{8}{\log 2} k \log k,$$

    i.e.,

$$\frac{n}{d} \leqslant \frac{8}{\log 2} k \log k,$$

Multiplying both sides by d,

$$\mathrm{vc}(\mathcal{F}) = n \leqslant Cdk\log k. \qquad \left(C = \frac{8}{\log 2}\right)$$

22 [**VC Dimension of Union**] Let $\mathcal{F}, \mathcal{G}$ be two families of Boolean functions defined on a universe Ω (*or equivalently, families of subsets*).

(1) Prove that the VC dimension of the union satisfies

$$\mathrm{vc}(\mathcal{F} \cup \mathcal{G}) \leqslant \mathrm{vc}(\mathcal{F}) + \mathrm{vc}(\mathcal{G}) + 1.$$

(2) Construct an example where the above inequality attains equality, with $\mathrm{vc}(\mathcal{F}), \mathrm{vc}(\mathcal{G}) > 0$, satisfying

$$\mathrm{vc}(\mathcal{F} \cup \mathcal{G}) = \mathrm{vc}(\mathcal{F}) + \mathrm{vc}(\mathcal{G}) + 1.$$

Prf. ***Sol.*** Let $\mathrm{vc}(\mathcal{F}) = d_1$, $\mathrm{vc}(\mathcal{G}) = d_2$. If one of them is ∞, the conclusion is trivial; we only need to consider the case $d_1, d_2 < \infty$.

Let $\Pi_{\mathcal{H}}(\mathbb{S})$ denote the number of distinct Boolean assignment patterns that the function family $\mathcal{H}$ can induce on $\mathbb{S}$; then $\Pi_{\mathcal{F}\cup\mathcal{G}}(\mathbb{S}) \leqslant \Pi_{\mathcal{F}}(\mathbb{S}) + \Pi_{\mathcal{G}}(\mathbb{S})$.

Since $|\mathbb{S}| = n = d_1 + d_2 + 2$, by the Sauer-Shelah lemma,

$$\Pi_{\mathcal{F}}(\mathbb{S}) \leqslant \sum_{i=0}^{d_1} \binom{n}{i}, \quad \Pi_{\mathcal{G}}(\mathbb{S}) \leqslant \sum_{j=0}^{d_2} \binom{n}{j}.$$

where

$$\sum_{j=0}^{d_2} \binom{n}{j} = \sum_{k=d_1+2}^{n} \binom{n}{k}, \qquad (k = n - j)$$

Hence by the binomial theorem $\sum_{m=0}^{n} \binom{n}{m} = 2^n$,

$$\Pi_{\mathcal{F}}(\mathbb{S}) + \Pi_{\mathcal{G}}(\mathbb{S}) \leqslant \sum_{i=0}^{d_1} \binom{n}{i} + \sum_{k=d_1+2}^{n} \binom{n}{k} = 2^n - \binom{n}{d_1+1} < 2^n,$$

This shows that $\mathcal{F} \cup \mathcal{G}$ cannot produce all 2^n patterns on $\mathbb{S}$, i.e., $\mathbb{S}$ cannot be shattered by $\mathcal{F} \cup \mathcal{G}$, hence

$$\mathrm{vc}(\mathcal{F} \cup \mathcal{G}) \leqslant d_1 + d_2 + 1 = \mathrm{vc}(\mathcal{F}) + \mathrm{vc}(\mathcal{G}) + 1.$$

Sol. Let the universe Ω be a finite set containing $n = d_1 + d_2 + 1$ elements. Define

$$\mathcal{F} = \{A \subset \Omega : |A| \leqslant d_1\},$$
$$\mathcal{G} = \{B \subset \Omega : |B| \geqslant d_1 + 1\}.$$

For $\mathcal{F}$, it contains all subsets of size at most d_1 and can shatter any set of size d_1; but it cannot provide the set of size $d_1 + 1$ itself, so it cannot shatter a set of size $d_1 + 1$, hence

$$\mathrm{vc}(\mathcal{F}) = d_1 > 0.$$

For $\mathcal{G}$, consider a set $\mathbb{S}$ of size d_2. For any $\mathbb{S}' \subset \mathbb{S}$, construct $B = \mathbb{S}' \cup (\Omega \setminus \mathbb{S})$, then

$$|B| = |\mathbb{S}'| + (n - d_2) = |\mathbb{S}'| + d_1 + 1 \geqslant d_1 + 1,$$

satisfying $B \in \mathcal{G}$, $B \cap \mathbb{S} = \mathbb{S}'$, so $\mathbb{S}$ is shattered.

To shatter a set $\mathbb{S}$ of size $d_2 + 1$, one would need to pick out $\emptyset$, requiring the existence of $B \in \mathcal{G}$ such that $B \cap \mathbb{S} = \emptyset$, i.e., $B \subset \Omega \setminus \mathbb{S}$, hence $|B| \leqslant n - d_2 - 1 = d_1$, which contradicts $|B| \geqslant d_1 + 1$; thus,

$$\mathrm{vc}(\mathcal{G}) = d_2 > 0.$$

For $\mathcal{F} \cup \mathcal{G}$, it contains exactly all possible subsets of Ω, i.e., $\mathcal{F} \cup \mathcal{G} = 2^{\Omega}$. Hence it can perfectly shatter Ω,

$$\mathrm{vc}(\mathcal{F} \cup \mathcal{G}) = |\Omega| = d_1 + d_2 + 1 = \mathrm{vc}(\mathcal{F}) + \mathrm{vc}(\mathcal{G}) + 1.$$

23 [**Covering Numbers at Large Scales**] Theorem T8.3.13 (*Covering numbers via VC dimension*) is stated for $\varepsilon \in (0, 1)$.

What bound holds when ε is larger?

Let Ω be a universe and μ a probability measure defined on Ω. Let $\mathcal{F}$ be a family of Boolean functions defined on Ω (*i.e., for any* $f \in \mathcal{F}, x \in \Omega$, $f(x) \in \{0,1\}$). When the allowed covering ball radius $\varepsilon \geqslant 1$, find an upper bound for the covering number $\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon)$ of $\mathcal{F}$ in the $L^2(\mu)$ norm.

Prf. Let $f, g \in \mathcal{F}$ satisfy $f(x), g(x) \in \{0,1\}$. Then the squared distance between f and g is

$$|f(x) - g(x)|^2 \leqslant 1,$$

Hence the $L^2(\mu)$ distance is

$$\|f - g\|_{L^2(\mu)}^2 = \int_\Omega |f(x) - g(x)|^2 \mathrm{d}\mu(x) \leqslant \int_\Omega 1 \mathrm{d}\mu(x) = 1.$$

If $\mathcal{F} = \emptyset$, then $\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon) = 0$. (*It then suffices to consider the case* $\mathcal{F} \neq \emptyset$.)

Let $f_0 \in \mathcal{F}$ and construct the closed ball $B(f_0, \varepsilon)$ centered at f_0 with radius $\varepsilon \geqslant 1$. When $\varepsilon \geqslant 1$,

$$\|f - f_0\|_{L^2(\mu)} \leqslant 1 \leqslant \varepsilon.$$

i.e., $f \in B(f_0, \varepsilon)$, so $\mathcal{F} \subset B(f_0, \varepsilon)$, hence

$$\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon) \leqslant 1,$$

i.e., $\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon) = 1$.

24 [**A Simpler, Weaker VC Law of Large Numbers**] Use the Sauer-Shelah lemma directly (*instead of the Pajor lemma*) to prove a weaker version of Theorem T8.3.15 (*VC law of large numbers*), with

$$C\sqrt{\frac{d}{n}\log\frac{en}{d}}$$

in the right hand side, where $d = \mathrm{vc}(\mathcal{F})$ and $1 \leqslant d \leqslant n$.

Prf. (*Symmetrization*).

Let $\Delta = \mathbb{E}\left[\sup_{f\in\mathcal{F}} |R_n(f) - R(f)|\right]$. By the symmetrization lemma,

$$\Delta \leqslant 2\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i)\right|\right],$$

where ε_i are independent Rademacher variables independent of the sample X_i.

Conditioned on the sample $X_1, \cdots, X_n$, the restrictions $\{(f(X_1), \cdots, f(X_n)) : f \in \mathcal{F}\}$ contain at most $\Pi_{\mathcal{F}}(n)$ patterns. By the Sauer-Shelah lemma, $\Pi_{\mathcal{F}}(n) \leqslant (en/d)^d$.

(*Union bound and tail integral*).

For a fixed f, since $f(X_i) \in \{0,1\}$, Hoeffding's inequality gives

$$\mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i)\right| \geqslant t\right\} \leqslant 2\mathrm{e}^{-cnt^2}.$$

Taking the union bound over all patterns,

$$\mathbb{P}\left\{\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i)\right| \geqslant t\right\} \leqslant 2(en/d)^d\mathrm{e}^{-cnt^2}.$$

Set $t_0 = \sqrt{\frac{d\log(en/d)+\log 2}{cn}}$. By the tail integral,

$$\begin{aligned}\Delta &\leqslant 2t_0 + 2\int_{t_0}^\infty \exp(-cn(t^2 - t_0^2))\mathrm{d}t \\ &\leqslant 2t_0 + \frac{1}{cnt_0} \leqslant C\sqrt{\frac{d}{n}\log\frac{en}{d}},\end{aligned}$$

which proves the weaker version of Theorem T8.3.15 (*VC law of large numbers*).

25 [**Uniform Estimation of One-Dimensional Marginal Distributions**] Let $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_m \in \mathbb{R}^n$ be i.i.d. random vectors. Prove that uniformly estimating the marginal distributions $\langle \boldsymbol{X}, \boldsymbol{u} \rangle$ along all one-dimensional projection directions $\boldsymbol{u} \in \mathbb{S}^{n-1}$ via the empirical cumulative distribution function based on m samples satisfies the following bound on the expected maximum deviation:

$$\mathbb{E}\left[\sup_{\boldsymbol{u}\in\mathbb{S}^{n-1}, t\in\mathbb{R}} \left| \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}_{\{\langle \boldsymbol{X}_i, \boldsymbol{u}\rangle \leqslant t\}} - \mathbb{P}\{\langle \boldsymbol{X}, \boldsymbol{u}\rangle \leqslant t\} \right|\right] \lesssim \sqrt{\frac{n}{m}}.$$

Prf. (Denote the supremum of the empirical process by Δ.)

Let $\boldsymbol{u} \in \mathbb{S}^{n-1}$, $t \in \mathbb{R}$. Define $f_{\boldsymbol{u},t} : \mathbb{R}^n \to \{0, 1\}$ by

$$f_{\boldsymbol{u},t}(\boldsymbol{x}) = \mathbf{1}_{\{\langle \boldsymbol{x}, \boldsymbol{u}\rangle \leqslant t\}},$$

and set $\mathcal{F} = \{f_{\boldsymbol{u},t} : \boldsymbol{u} \in \mathbb{S}^{n-1}, t \in \mathbb{R}\}$.

For $f = f_{\boldsymbol{u},t} \in \mathcal{F}$, its expectation with respect to the sample $\boldsymbol{X}$ is $\mathbb{E}[f(\boldsymbol{X})] = \mathbb{P}\{\langle \boldsymbol{X}, \boldsymbol{u}\rangle \leqslant t\}$, and the empirical average is

$$\frac{1}{m}\sum_{i=1}^{m} f(\boldsymbol{X}_i) = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}_{\{\langle \boldsymbol{X}_i, \boldsymbol{u}\rangle \leqslant t\}}.$$

Hence the goal is to prove

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} \left| \frac{1}{m}\sum_{i=1}^{m} f(\boldsymbol{X}_i) - \mathbb{E}[f(\boldsymbol{X})] \right|\right] \lesssim \sqrt{\frac{n}{m}}.$$

By Theorem T8.3.15 (*VC law of large numbers*),

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} \left| \frac{1}{m}\sum_{i=1}^{m} f(\boldsymbol{X}_i) - \mathbb{E}[f(\boldsymbol{X})] \right|\right] \leqslant C_1 \sqrt{\frac{\mathrm{vc}(\mathcal{F})}{m}},$$

where by Exercise Q8.17 (*VC Dimension of Half-Spaces*),

$$\mathrm{vc}(\mathcal{F}) = n + 1,$$

Substituting this,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} \left| \frac{1}{m}\sum_{i=1}^{m} f(\boldsymbol{X}_i) - \mathbb{E}[f(\boldsymbol{X})] \right|\right] \leqslant C_1 \sqrt{\frac{n+1}{m}} \leqslant \sqrt{2} C_1 \sqrt{\frac{n}{m}}.$$

26 [**One-Binary-Digit Quantization and Locality-Sensitive Hashing**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix. Define the map $\Phi : \mathbb{S}^{n-1} \to \{-1, 1\}^m$ by

$$\Phi(\boldsymbol{u}) = \mathrm{sgn}(\boldsymbol{A}\boldsymbol{u}),$$

where $\mathrm{sgn}(\cdot)$ is applied coordinate-wise to the vector.

(1) Prove that for any $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{S}^{n-1}$,

$$\mathbb{E}\left[\frac{1}{m} d(\Phi(\boldsymbol{u}), \Phi(\boldsymbol{v}))\right] = \frac{1}{\pi}\rho(\boldsymbol{u}, \boldsymbol{v}).$$

where $d(\cdot, \cdot)$ is the Hamming distance and $\rho(\cdot, \cdot)$ is the spherical geodesic distance.

(2) Using Theorem T8.3.15 (*VC law of large numbers*), prove that for any $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{S}^{n-1}$,

$$\sup_{\boldsymbol{u},\boldsymbol{v}\in\mathbb{S}^{n-1}} \left| \frac{1}{m} d(\Phi(\boldsymbol{u}), \Phi(\boldsymbol{v})) - \frac{1}{\pi}\rho(\boldsymbol{u}, \boldsymbol{v}) \right| \lesssim \sqrt{\frac{n}{m}} \text{ (w.p. 0.99)}.$$

Prf. ***Sol.*** Let the row vectors of $\boldsymbol{A}$ be $\boldsymbol{A}_1^\top, \cdots, \boldsymbol{A}_m^\top$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_n)$.

For the Hamming distance $d(\Phi(\boldsymbol{u}), \Phi(\boldsymbol{v}))$, it corresponds to the number of coordinates where the signs differ. Express it as a sum of indicator functions,

$$d(\Phi(\boldsymbol{u}), \Phi(\boldsymbol{v})) = \sum_{i=1}^{m} \mathbf{1}_{\{\mathrm{sgn}(\langle \boldsymbol{A}_i, \boldsymbol{u}\rangle) \neq \mathrm{sgn}(\langle \boldsymbol{A}_i, \boldsymbol{v}\rangle)\}},$$

Taking expectations on both sides,

$$\mathbb{E}\left[\frac{1}{m}d(\Phi(\boldsymbol{u}),\Phi(\boldsymbol{v}))\right]=\mathbb{P}\{\operatorname{sgn}(\langle\boldsymbol{A}_1,\boldsymbol{u}\rangle)\neq\operatorname{sgn}(\langle\boldsymbol{A}_1,\boldsymbol{v}\rangle)\}.$$

By the rotational invariance of the standard Gaussian distribution, rotate the coordinate system so that the plane spanned by $\boldsymbol{u},\boldsymbol{v}$ forms the first two coordinate axes; then the projection of $\boldsymbol{A}_1$ is a standard two-dimensional Gaussian vector.

(*The geometric meaning of having opposite signs of inner products is that the hyperplane orthogonal to* $\boldsymbol{A}_1$ *passes exactly through the angle between* $\boldsymbol{u}$ *and* $\boldsymbol{v}$.)

Since the standard Gaussian vector is isotropic, the above probability equals the ratio of the angle to the half-circle radian π, i.e.,

$$\mathbb{P}\{\operatorname{sgn}(\langle\boldsymbol{A}_1,\boldsymbol{u}\rangle)\neq\operatorname{sgn}(\langle\boldsymbol{A}_1,\boldsymbol{v}\rangle)\}=\frac{\rho(\boldsymbol{u},\boldsymbol{v})}{\pi},$$

Substituting this,

$$\mathbb{E}\left[\frac{1}{m}d(\Phi(\boldsymbol{u}),\Phi(\boldsymbol{v}))\right]=\frac{1}{\pi}\rho(\boldsymbol{u},\boldsymbol{v}).$$

Sol. Let $(\boldsymbol{u},\boldsymbol{v})\in\mathbb{S}^{n-1}\times\mathbb{S}^{n-1}$. For $\boldsymbol{x}\in\mathbb{R}^n$, define

$$f_{\boldsymbol{u},\boldsymbol{v}}(\boldsymbol{x})=\mathbf{1}_{\{\operatorname{sgn}(\langle\boldsymbol{x},\boldsymbol{u}\rangle)\neq\operatorname{sgn}(\langle\boldsymbol{x},\boldsymbol{v}\rangle)\}}.$$

Set $\mathcal{F}=\{f_{\boldsymbol{u},\boldsymbol{v}}:\boldsymbol{u},\boldsymbol{v}\in\mathbb{S}^{n-1}\}$.

Similarly to Exercise Q8.24 (*A Simpler, Weaker VC Law of Large Numbers*), for $f=f_{\boldsymbol{u},\boldsymbol{v}}\in\mathcal{F}$, its expectation with respect to $\boldsymbol{A}$ is

$$\mathbb{E}[f_{\boldsymbol{u},\boldsymbol{v}}(\boldsymbol{A})]=\frac{1}{\pi}\rho(\boldsymbol{u},\boldsymbol{v}),$$

and the empirical mean is $\frac{1}{m}d(\Phi(\boldsymbol{u}),\Phi(\boldsymbol{v}))$. The goal becomes proving

$$\underbrace{\sup_{f\in\mathcal{F}}\left|\frac{1}{m}\sum_{i=1}^{m}f(\boldsymbol{A}_i)-\mathbb{E}[f(\boldsymbol{A})]\right|}_{\Delta}\lesssim\sqrt{\frac{n}{m}}\text{ (w.p. 0.99)}.$$

By Theorem T8.3.15 (*VC law of large numbers*),

$$\mathbb{E}[\Delta]\leqslant C_1\sqrt{\frac{\operatorname{vc}(\mathcal{F})}{m}}.\tag{I}$$

Note that $f_{\boldsymbol{u},\boldsymbol{v}}(\boldsymbol{x})$ is essentially the XOR logical operation,

$$f_{\boldsymbol{u},\boldsymbol{v}}(\boldsymbol{x})=\mathbf{1}_{\{\langle\boldsymbol{x},\boldsymbol{u}\rangle\geqslant0\}}\oplus\mathbf{1}_{\{\langle\boldsymbol{x},\boldsymbol{v}\rangle\geqslant0\}},$$

(*The VC dimension of the family of unbiased half-spaces is* n*, and* $\mathcal{F}$ *is formed by combining* $k=2$ *half-space families.*) Hence by the stability theorem for VC dimension,

$$\operatorname{vc}(\mathcal{F})\leqslant C_2n.\tag{II}$$

Substituting (II) into (I),

$$\mathbb{E}[\Delta]\leqslant C_3\sqrt{\frac{n}{m}}.$$

Using Proposition P1.6.2 (*Markov inequality*) (*Markov's inequality*), for any $t>0$,

$$\mathbb{P}\{\Delta\geqslant t\}\leqslant\frac{\mathbb{E}[\Delta]}{t},$$

If the above probability does not exceed 0.01, then $t\geqslant100\mathbb{E}[\Delta]$. Take $t=100C_3\sqrt{\frac{n}{m}}$.

27 [**Minimum Singular Value of Random Matrices without Moment Assumptions**] Most results for random matrices assume something good about the distribution——such as sub-Gaussian tails (see Theorem T4.6.1 (*Two-sided bounds for sub-Gaussian matrices*)) or at least finite second moments (see Exercise Q5.31 (*Random Matrices with General Independent Rows*)).

Here we will use the power of VC theory (specifically, Theorem T8.3.15 (*VC law of large numbers*)) to show that even without any moment assumptions, a tall thin random matrix is well invertible. Let $\varepsilon, \delta > 0$. Let the random vector $\boldsymbol{X} \in \mathbb{R}^n$ satisfy the anti-concentration property, i.e., for any $\boldsymbol{u} \in \mathbb{S}^{n-1}$,

$$\mathbb{P}\{|\langle \boldsymbol{X}, \boldsymbol{u}\rangle| \geqslant \varepsilon\} \geqslant \delta.$$

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix whose rows are i.i.d. copies of $\boldsymbol{X}$.

(1) Prove that there exists $C > 0$ such that when the number of rows $m \geqslant C\delta^{-2}n$, the minimum singular value of $\boldsymbol{A}$ satisfies with high probability

$$s_n(\boldsymbol{A}) \geqslant 0.99\varepsilon\sqrt{\delta m} \text{ (w.p. 0.99)}.$$

(2) Prove that the bound in (1) is optimal. Specifically, for any positive integers $m \geqslant n$, $\varepsilon > 0$, $\delta < 1/3$, construct a specific random vector $\boldsymbol{X}$ satisfying the anti-concentration property such that

$$\mathbb{E}[s_n(\boldsymbol{A})] \leqslant 1.01\varepsilon\sqrt{\delta m}.$$

Prf. ***Sol.*** The minimum singular value of the matrix $\boldsymbol{A}$ is

$$s_n(\boldsymbol{A})^2 = \inf_{\boldsymbol{u}\in\mathbb{S}^{n-1}} \sum_{i=1}^m \langle \boldsymbol{X}_i, \boldsymbol{u}\rangle^2.$$

Introduce indicator functions for the pointwise lower bound,

$$\sum_{i=1}^m \langle \boldsymbol{X}_i, \boldsymbol{u}\rangle^2 \geqslant \varepsilon^2 \sum_{i=1}^m \mathbf{1}_{\{|\langle \boldsymbol{X}_i, \boldsymbol{u}\rangle|\geqslant\varepsilon\}}.$$

Define the parametrized family of Boolean functions $\mathcal{F} = \{f_{\boldsymbol{u}} : \boldsymbol{u} \in \mathbb{S}^{n-1}\}$, where $f_{\boldsymbol{u}}(\boldsymbol{x}) = \mathbf{1}_{\{|\langle \boldsymbol{x}, \boldsymbol{u}\rangle|\geqslant\varepsilon\}}$. Then $\mathcal{F}$ is the union of two mutually exclusive half-spaces. Hence by the stability theorem for VC dimension,

$$\operatorname{vc}(\mathcal{F}) \leqslant C_1 n. \qquad (C_1 > 0)$$

By the anti-concentration property, the true expectation is

$$\mathbb{E}[f_{\boldsymbol{u}}(\boldsymbol{X})] = \mathbb{P}\{|\langle \boldsymbol{X}, \boldsymbol{u}\rangle| \geqslant \varepsilon\} \geqslant \delta,$$

Hence by the VC law of large numbers,

$$\mathbb{E}[\Delta] \leqslant C_2\sqrt{\frac{n}{m}}.$$

Using Proposition P1.6.2 (*Markov inequality*) (*Markov's inequality*), let $t = 100\mathbb{E}[\Delta]$; then

$$\mathbb{P}\left\{\Delta \leqslant 100C_2\sqrt{\frac{n}{m}}\right\} \geqslant 0.99,$$

Hence for any $\boldsymbol{u} \in \mathbb{S}^{n-1}$,

$$\frac{1}{m}\sum_{i=1}^m f_{\boldsymbol{u}}(\boldsymbol{X}_i) \geqslant \mathbb{E}[f_{\boldsymbol{u}}(\boldsymbol{X})] - 100C_2\sqrt{\frac{n}{m}} \geqslant \delta - 100C_2\sqrt{\frac{n}{m}}.$$

The right-hand side is at least $0.99^2\delta$, i.e.,

$$m \geqslant \left(\frac{100C_2}{1-0.99^2}\right)^2 \delta^{-2}n.$$

Sol. Let $c = 1 - \frac{1}{1.01}$, so $\frac{1}{1-c} = 1.01$. Define $\boldsymbol{X} \in \mathbb{R}^n$ as a mixture distribution: with probability δ, take $\pm\frac{\varepsilon}{1-c}\boldsymbol{e}_1$; with probability $1-\delta$, draw from a multivariate Gaussian distribution $N(\boldsymbol{0}, M^2\boldsymbol{I}_{n-1})$ in the subspace $\operatorname{span}(\boldsymbol{e}_2, \cdots, \boldsymbol{e}_n)$ (*the first coordinate is always* 0*, and* M *is sufficiently large*).

Verification of the anti-concentration assumption:

a. When $|u_1| \geqslant 1 - c$, consider only the first component. Since

$$|\langle \boldsymbol{X}, \boldsymbol{u}\rangle| = \frac{\varepsilon}{1-c}|u_1| \geqslant \varepsilon,$$

the condition holds (w.p. 1). Hence the total probability is $\mathbb{P} \geqslant \delta$.

b. When $|u_1| < 1 - c$, consider only the second component. Since $\sum_{j=2}^{n} u_j^2 = 1 - u_1^2 > 2c - c^2 > c$, the conditional inner product follows a one-dimensional normal distribution $N(0, \sigma^2)$ with $\sigma^2 > M^2 c$. Choose a sufficiently large constant M so that

$$\mathbb{P}\{|N(0,\sigma^2)| \geqslant \varepsilon\} \geqslant \frac{\delta}{1-\delta}.$$

Hence the total probability is

$$\mathbb{P} \geqslant (1-\delta)\frac{\delta}{1-\delta} = \delta.$$

(*Now compute the expectation of the singular value.*)

Since $s_n(\boldsymbol{A})^2 \leqslant \|\boldsymbol{A}\boldsymbol{e}_1\|_2^2 = \sum_{i=1}^{m} X_{i1}^2$. The first coordinate X_{i1} is nonzero only in component 1, with expectation $\mathbb{E}[X_{i1}^2] = \frac{\delta\varepsilon^2}{(1-c)^2}$.

Using Jensen's inequality,

$$\mathbb{E}[s_n(\boldsymbol{A})] \leqslant \mathbb{E}\left[\sqrt{\sum_{i=1}^{m} X_{i1}^2}\right] \leqslant \sqrt{\mathbb{E}\left[\sum_{i=1}^{m} X_{i1}^2\right]} = \sqrt{m\frac{\delta\varepsilon^2}{(1-c)^2}}.$$

Hence $\mathbb{E}[s_n(\boldsymbol{A})] \leqslant 1.01\varepsilon\sqrt{\delta m}$.

28 [**Uniform Glivenko-Cantelli Class and Finite VC Dimension**] In Remark R8.3.19 (*Uniform Glivenko-Cantelli classes*) we noted that a Boolean class with finite VC dimension is a uniform Glivenko-Cantelli class.

Prove the converse: let $\mathcal{F}$ be a family of Boolean functions defined on a universe Ω. $\mathcal{F}$ is called a uniform Glivenko-Cantelli (uGC) class if for any probability measure μ on Ω, when $X_1, X_2, \cdots, X_n \overset{\text{i.i.d.}}{\sim} \mu$ with $n \to \infty$,

$$\lim_{n\to\infty} \sup_{\mu} \mathbb{E}_{X_1,\cdots,X_n\sim\mu}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}_{\mu}[f(X_1)]\right|\right] = 0.$$

Prove that if a Boolean function family $\mathcal{F}$ is a uGC class, then its Vapnik-Chervonenkis dimension must be finite, i.e., $\mathrm{vc}(\mathcal{F}) < \infty$.

Prf. • Construct a discrete measure

We use proof by contradiction. Suppose $\mathrm{vc}(\mathcal{F}) = \infty$. Then for any $n \geqslant 1$, there exists a subset $\mathbb{S}_n = \{x_1, x_2, \cdots, x_n\} \subset \Omega$ of size n that can be shattered by $\mathcal{F}$.

Define the uniform discrete probability measure μ_n on $\mathbb{S}_n$, i.e.,

$$\mu_n(\{x_j\}) = \frac{1}{n}, \qquad (j = 1, 2, \cdots, n)$$

Then draw n i.i.d. samples $X_1, X_2, \cdots, X_n$ from μ_n. Let $V = \{X_1, X_2, \cdots, X_n\}$ be the set of distinct observed points, and $U = \mathbb{S}_n \setminus V$ be the set of points never selected.

• Find a "worst-case" function

Since $\mathbb{S}_n$ is shattered by $\mathcal{F}$, for any subset of $\mathbb{S}_n$, there must exist a function in $\mathcal{F}$ that labels the points in that subset as 1 and the points outside as 0.

In particular, for the unobserved set $U \subset \mathbb{S}_n$, there must exist a specific function $f^* \in \mathcal{F}$ satisfying

$$f^*(x) = \begin{cases} 1, & x \in U, \\ 0, & x \in V. \end{cases}$$

• Compute the empirical deviation

(*Consider the performance of f^* on the samples.*)

On one hand, since $X_i \in V$, its empirical mean is

$$\frac{1}{n}\sum_{i=1}^{n} f^*(X_i) = 0.$$

On the other hand, the expectation of f^* with respect to the true measure μ_n is

$$\mathbb{E}_{\mu_n}[f^*(X_1)] = \sum_{j=1}^{n} f^*(x_j)\mu_n(\{x_j\}) = \frac{|U|}{n},$$

Hence the deviation between the measure μ_n and the function f^* is

$$\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}_{\mu_n}[f(X_1)]\right| \geqslant \left|0 - \frac{|U|}{n}\right| = \frac{|U|}{n},$$

Taking expectations on both sides,

$$\mathbb{E}_{X_1,\cdots,X_n\sim\mu_n}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}_{\mu_n}[f(X_1)]\right|\right] \geqslant \frac{\mathbb{E}[|U|]}{n}.$$

(*Now compute* $\mathbb{E}[|U|]$.)

For $x_j \in \mathbb{S}_n$, the probability that it is not selected in a single draw is $1 - \frac{1}{n}$, so

$$\mathbb{E}[|U|] = \sum_{j=1}^{n} \mathbb{P}\{x_j \notin V\} = n\left(1 - \frac{1}{n}\right)^n,$$

Substituting this,

$$\mathbb{E}_{X_1,\cdots,X_n\sim\mu_n}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}_{\mu_n}[f(X_1)]\right|\right] \geqslant \left(1 - \frac{1}{n}\right)^n,$$

Taking limits on both sides,

$$\lim_{n\to\infty}\sup_{\mu}\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}_{\mu}[f(X_1)]\right|\right] \geqslant \frac{1}{\mathrm{e}} > 0,$$

This contradicts the fact that the limit of a uGC class tends to 0; thus, $\mathrm{vc}(\mathcal{F}) < \infty$.

29 [**VC Dimension of the Squared Error Function Family**] Let $\mathcal{F}$ be a family of Boolean functions defined on a universe Ω. Given a fixed Boolean function $h : \Omega \to \{0, 1\}$, define a new family of Boolean functions $\mathcal{G}$ by

$$\mathcal{G} = \{(f - h)^2 : f \in \mathcal{F}\}.$$

Prove that the Vapnik-Chervonenkis dimension of $\mathcal{G}$ satisfies $\mathrm{vc}(\mathcal{G}) = \mathrm{vc}(\mathcal{F})$.

Rk. For $x \in \Omega$, consider the map $\phi_x : y \mapsto (y - h(x))^2$, where $y \in \{0, 1\}$.

a. When $h(x) = 0$, $\phi_x(y) = y^2 = y$, which is the identity map.

b. When $h(x) = 1$,

$$\phi_x(y) = (y-1)^2 = \begin{cases} 1, & y = 0, \\ 0, & y = 1, \end{cases}$$

which is equivalent to the logical NOT operation $\phi_x(y) = 1 - y$. Hence the map $\phi_x : \{0, 1\} \to \{0, 1\}$ is a bijection.

Prf. Consider the projection sets on finite samples.

Let $\mathbb{S} = \{x_1, x_2, \cdots, x_n\} \subset \Omega$. Define the projections of the function families $\mathcal{F}, \mathcal{G}$ on $\mathbb{S}$ as

$$P_{\mathcal{F}}(\mathbb{S}) = \{(f(x_1), f(x_2), \cdots, f(x_n)) : f \in \mathcal{F}\} \subset \{0, 1\}^n,$$
$$P_{\mathcal{G}}(\mathbb{S}) = \{((f(x_1) - h(x_1))^2, \cdots, (f(x_n) - h(x_n))^2) : f \in \mathcal{F}\} \subset \{0, 1\}^n.$$

Let the fixed Boolean vector produced by h on $\mathbb{S}$ be $\boldsymbol{b} = (h(x_1), \cdots, h(x_n))$. Define $T : \{0, 1\}^n \to \{0, 1\}^n$ by

$$T(\boldsymbol{v}) = ((v_1 - b_1)^2, \cdots, (v_n - b_n)^2).$$

(*Since T is a bijection on each coordinate, $T : \{0, 1\}^n \to \{0, 1\}^n$ is a bijection.*)

Then

$$P_{\mathcal{G}}(\mathbb{S}) = \{T(\boldsymbol{v}) : \boldsymbol{v} \in P_{\mathcal{F}}(\mathbb{S})\} = T(P_{\mathcal{F}}(\mathbb{S})).$$

Deduce the equivalence of shattering.

By the definition of shattering, $\mathbb{S}$ is shattered by $\mathcal{F}$ if and only if $|P_{\mathcal{F}}(\mathbb{S})| = 2^n$.

Note that T is a bijective operator, so

$$|P_{\mathcal{F}}(\mathbb{S})| = 2^n \iff |P_{\mathcal{G}}(\mathbb{S})| = |T(P_{\mathcal{F}}(\mathbb{S}))| = 2^n,$$

Thus $\mathbb{S}$ can be shattered by $\mathcal{F}$ if and only if $\mathbb{S}$ can be shattered by $\mathcal{G}$, hence $\mathrm{vc}(\mathcal{G}) = \mathrm{vc}(\mathcal{F})$.

30 [**VC Uniform Convergence Theorem under Random Labels**] In the learning setting of Section 8.4, we assumed that the label $T(X)$ is completely determined by X, which is rarely the case in reality. For example, it is unrealistic to think that a diagnosis $T(X) \in \{0,1\}$ is completely determined by the health data X; more commonly, the label Y is just a random variable correlated with X.

Try to extend the learning theory (up to Theorem T8.4.5 (*VC generalization bound*)) to training data of the form (X_i, Y_i), where (X_i, Y_i) are independent copies of a random pair (X, Y) taking values in $\Omega \times \{0,1\}$. Let $\mathcal{F}$ be a family of Boolean hypothesis functions defined on a feature space Ω, with Vapnik-Chervonenkis dimension $d = \mathrm{vc}(\mathcal{F}) < \infty$. Let $Z = (X, Y)$ be a random pair defined on the augmented sample space $\mathcal{Z} = \Omega \times \{0,1\}$, and let $Z_1, \cdots, Z_n = (X_1, Y_1), \cdots, (X_n, Y_n)$ be i.i.d. training samples. For any hypothesis function $f \in \mathcal{F}$, define its true risk and empirical risk respectively as

$$R(f) = \mathbb{E}_{(X,Y)}[(f(X) - Y)^2], \quad R_n(f) = \frac{1}{n}\sum_{i=1}^{n}(f(X_i) - Y_i)^2.$$

Prove that under this random label setting, the empirical risk converges uniformly to the true risk over the entire function family, i.e.,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} |R_n(f) - R(f)|\right] \leqslant C\sqrt{\frac{\mathrm{vc}(\mathcal{F})}{n}}.$$

Prf. • Construct the error function family on the augmented space

For any $f \in \mathcal{F}$, define the corresponding error function $g_f : \mathcal{Z} \to \{0,1\}$ by

$$g_f(Z) = g_f(X, Y) = (f(X) - Y)^2.$$

Since $f(X), Y \in \{0,1\}$, its squared error is equivalent to the indicator $\mathbf{1}_{\{f(X)\neq Y\}}$, so g_f is a Boolean function.

Define the new error function family $\mathcal{G} = \{g_f : f \in \mathcal{F}\}$. Then

$$R(f) = \mathbb{E}[g_f(Z)], \quad R_n(f) = \frac{1}{n}\sum_{i=1}^{n} g_f(Z_i),$$

The goal becomes proving

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} |R_n(f) - R(f)|\right] = \mathbb{E}\left[\sup_{g\in\mathcal{G}}\left|\frac{1}{n}\sum_{i=1}^{n} g(Z_i) - \mathbb{E}[g(Z)]\right|\right].$$

• Bound the VC dimension of the new function family $\mathcal{G}$

Suppose $\mathcal{G}$ can shatter a finite subset $\mathbb{S} = \{z_1, \cdots, z_m\} \subset \mathcal{Z}$, where $z_i = (x_i, y_i)$. Then the feature elements $x_1, \cdots, x_m$ of $\mathbb{S}$ must be pairwise distinct.

(*Otherwise, there exist $i \neq j$ such that $x_i = x_j$, so $y_i \neq y_j$, and for any $f \in \mathcal{F}$, the output $f(x_i) = f(x_j)$, in which case one error must be 0 and the other 1, so it is impossible to simultaneously output $(0,0)$ or $(1,1)$.*)

Let $\mathbb{S}_X = \{x_1, \cdots, x_m\} \subset \Omega$ be the projection of $\mathbb{S}$ onto Ω; then $|\mathbb{S}_X| = m$ (*since the x_i are pairwise distinct*).

Since $\mathbb{S}$ can be shattered by $\mathcal{G}$, as f ranges over $\mathcal{F}$, the vector sequence

$$((f(x_1) - y_1)^2, \cdots, (f(x_m) - y_m)^2)$$

can take all 2^m patterns in $\{0,1\}^m$.

Note that $v \mapsto (v - y_i)^2$ is a bijection, so the error vectors take all values in $\{0,1\}^m$ if and only if the underlying feature

vector sequence

$$(f(x_1), \cdots, f(x_m))$$

can also take all patterns in $\{0,1\}^m$, which means $\mathbb{S}_X$ can be shattered by the original hypothesis family $\mathcal{F}$.

Since $\mathcal{F}$ can shatter a set of size m, we have $m \leqslant \mathrm{vc}(\mathcal{F})$, i.e., $\mathrm{vc}(\mathcal{G}) \leqslant \mathrm{vc}(\mathcal{F})$.

By the optimal VC law of large numbers,

$$\mathbb{E}\left[\sup_{g\in\mathcal{G}}\left|\frac{1}{n}\sum_{i=1}^{n} g(Z_i) - \mathbb{E}[g(Z)]\right|\right] \leqslant C\sqrt{\frac{\mathrm{vc}(\mathcal{G})}{n}}, \qquad (C>0)$$

Substituting $\mathrm{vc}(\mathcal{G}) \leqslant \mathrm{vc}(\mathcal{F})$,

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} |R_n(f) - R(f)|\right] \leqslant C\sqrt{\frac{\mathrm{vc}(\mathcal{F})}{n}}.$$

31 [**Learning a Lipschitz Function**] Let $T : [0,1] \to [0,1]$ be a function with Lipschitz constant at most 1. Consider the hypothesis class

$$\mathcal{F} = \{f : [0,1] \to [0,1],\ f_{\mathrm{Lip}} \leqslant 1\},$$

and assume that $T \in \mathcal{F}$. Let $X_1, \cdots, X_n$ i.i.d. $\sim \mathrm{Unif}[0,1]$ be independent uniform samples, and observe the labels $T(X_i)$. Show that the empirical risk minimization algorithm from Section 8.4.2 has a generalization bound similar to Theorem T8.4.5 (*VC generalization bound*):

$$\mathbb{E}[R(f_n)] \leqslant R(f) + \frac{C}{\sqrt{n}},$$

where $R(f) = \mathbb{E}|f(X) - T(X)|$ ($X \sim \mathrm{Unif}[0,1]$) is the true risk.

Prf. • (*Loss function class and covering numbers*) For $f \in \mathcal{F}$, define $g_f(x) = |f(x) - T(x)|$; then $R(f) = \mathbb{E}[g_f(X)]$ and

$$R_n(f) = \frac{1}{n}\sum_{i=1}^{n} g_f(X_i).$$

Since f and T are both 1-Lipschitz, g_f has Lipschitz constant at most 2 and takes values in $[0,1]$.

(*Estimate the covering numbers of* $\mathcal{G} = \{g_f : f \in \mathcal{F}\}$.)

Fix $0 < \varepsilon \leqslant 1$. Let $\{x_j\}_{j=1}^{L}$ be an equally spaced $\varepsilon/6$-net of $[0,1]$, where $L = \lceil 6/\varepsilon \rceil + 1 \leqslant 7/\varepsilon$. For any $g \in \mathcal{G}$, choose q_j in an $\varepsilon/6$-net of $[0,1]$ so that $|g(x_j) - q_j| \leqslant \varepsilon/6$. Let h_g be the piecewise affine function with nodes (x_j, q_j). For any $x \in [x_j, x_{j+1}]$, since g is 2-Lipschitz, its difference from the segment joining $(x_j, g(x_j))$ and $(x_{j+1}, g(x_{j+1}))$ is at most $2\varepsilon/6$, and the quantized piecewise affine function differs by at most $\varepsilon/6$. Hence $\|g - h_g\|_\infty \leqslant \varepsilon/2$. Let $\mathcal{H}_\varepsilon$ be the function class obtained from arbitrary quantized configurations. For every $h \in \mathcal{H}_\varepsilon$ satisfying $B_\infty(h, \varepsilon/2) \cap \mathcal{G} \neq \emptyset$, choose $g_h \in B_\infty(h, \varepsilon/2) \cap \mathcal{G}$. Then $\{g_h\}$ is an ε-net of $\mathcal{G}$, and each q_j has at most $7/\varepsilon$ values, so

$$\mathcal{N}(\mathcal{G}, \|\cdot\|_\infty, \varepsilon) \leqslant (7/\varepsilon)^{7/\varepsilon},$$

$$\log \mathcal{N}(\mathcal{G}, \|\cdot\|_\infty, \varepsilon) \leqslant \frac{7}{\varepsilon}\log\frac{7}{\varepsilon}.$$

• (*Symmetrization and the Dudley chaining*) By the symmetrization lemma,

$$\mathbb{E}\left[\sup_{g\in\mathcal{G}} |R_n(g) - R(g)|\right] \leqslant 2\mathbb{E}\left[\sup_{g\in\mathcal{G}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i g(X_i)\right|\right].$$

Define

$$Z(g) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i g(X_i);$$

it is a sub-Gaussian process on $\mathcal{G}$, since

$$\mathbb{E}|Z(g)-Z(g')|^2=\frac{1}{n}\sum_{i=1}^{n}(g(X_i)-g'(X_i))^2\leqslant\|g-g'\|_\infty^2.$$

By Dudley's inequality (*Theorem T8.1.4 (Discrete Dudley inequality)*),

$$\begin{aligned}\mathbb{E}\left[\sup_{g\in\mathcal{G}}|Z(g)|\right]&\leqslant C\int_0^\infty\sqrt{\log\mathcal{N}(\mathcal{G},\|\cdot\|_\infty,\varepsilon)}\mathrm{d}\varepsilon\\&\leqslant C\int_0^1\varepsilon^{-1/2}\sqrt{\log(1/\varepsilon)}\mathrm{d}\varepsilon\leqslant C',\end{aligned}$$

where the last integral converges (*with* $u=\log(1/\varepsilon)$, $\int_0^\infty u^{1/2}\mathrm{e}^{-u/2}\mathrm{d}u=2^{3/2}\Gamma(3/2)<\infty$).

- (*Optimality of ERM*) Since $T\in\mathcal{F}$ and the labels are noiseless, $R_n(f_n)\leqslant R_n(T)=0$, so $R_n(f_n)=0$. Hence

$$\mathbb{E}[R(f_n)]=\mathbb{E}[R(f_n)-R_n(f_n)]\leqslant\mathbb{E}\left[\sup_{g\in\mathcal{G}}|R_n(g)-R(g)|\right]\leqslant\frac{C}{\sqrt{n}},$$

and since $R(T)=0\leqslant R(f)$, we get

$$\mathbb{E}[R(f_n)]\leqslant R(f)+\frac{C}{\sqrt{n}}.$$

32 [**Unlearnability with a Finite Sample Size**] Theorem T8.4.5 (*VC generalization bound*) implies that, in order to learn well, the number n of training points must be at least comparable to the VC dimension of the hypothesis class.

Let $\mathcal{F}$ be a family of Boolean functions defined on a universe Ω, with Vapnik-Chervonenkis dimension $d=\mathrm{vc}(\mathcal{F})\geqslant 2$. A learning algorithm $\mathcal{A}$ receives n i.i.d. samples $\mathbb{S}=\{(x_1,f^*(x_1)),\cdots,(x_n,f^*(x_n))\}$ labeled by a target function $f^*\in\mathcal{F}$ ($x_i\sim\mu$), and outputs a hypothesis function $h_{\mathbb{S}}:\Omega\to\{0,1\}$. Define the true risk as $R(h_{\mathbb{S}})=\mathbb{P}_{x\sim\mu}(h_{\mathbb{S}}(x)\neq f^*(x))$.

Prove that if the training sample size $n<\frac{d}{2}$, then there exists a probability distribution μ on Ω such that for any learning algorithm $\mathcal{A}$, there exists a "worst-case" target function $f^*\in\mathcal{F}$ satisfying the minimax lower bound

$$\inf_{\mathcal{A}}\sup_{f^*\in\mathcal{F}}\mathbb{E}_{\mathbb{S}\sim\mu^n}[R(\mathcal{A}(\mathbb{S}))]\geqslant\frac{1}{4}.$$

Prf. Since $\mathrm{vc}(\mathcal{F})=d$, there exists $C=\{c_1,c_2,\cdots,c_d\}\subset\Omega$ that can be shattered by $\mathcal{F}$, i.e., $\mathcal{F}$ can realize all 2^d possible Boolean patterns on it.

(Construct the adversarial distribution μ.)

Construct the uniform probability measure μ concentrated on the set C as

$$\mu(\{x\})=\begin{cases}1/d, & x\in C,\\ 0, & x\notin C.\end{cases}$$

- Compute the conditional expected error given the sample

  (To prove $\sup_{f^*\in\mathcal{F}}\mathbb{E}_{\mathbb{S}}[R(\mathcal{A}(\mathbb{S}))]\geqslant 1/4$, we use the probabilistic method: if the average error over a random distribution is at least $1/4$, then there exists some f^* whose error is at least $1/4$.)

  Among the 2^d functions in $\mathcal{F}$ that induce all labelings of C, draw a target function f^* uniformly at random. Then for $c_i\in C$, $f^*(c_i)$ i.i.d. $\sim$ Bernoulli$(1/2)$.

  Let the feature sequence of the training set received by the algorithm $\mathcal{A}$ be $\mathbb{S}=(x_1,\cdots,x_n)$. Let $U=\{x_1,\cdots,x_n\}\subset C$ be the set of distinct elements contained in the sample set; then $|U|\leqslant n$ (*since the total number of draws is* n).

  Compute the expected risk with respect to the random target function f^*, conditioned on a fixed feature sequence $\mathbb{S}$,

$$\mathbb{E}_{f^*}[R(h_{\mathbb{S}})\mid\mathbb{S}]=\mathbb{E}_{f^*}\left[\sum_{i=1}^{d}\mathbf{1}_{\{h_{\mathbb{S}}(c_i)\neq f^*(c_i)\}}\mu(\{c_i\})\,g|\,\mathbb{S}\right]=\frac{1}{d}\sum_{i=1}^{d}\mathbb{E}_{f^*}[\mathbf{1}_{\{h_{\mathbb{S}}(c_i)\neq f^*(c_i)\}}\mid\mathbb{S}].$$

where:

a. For observed points $c_i \in U$, since the error is nonnegative, $\mathbb{E}_{f^*}[\mathbf{1}_{\{h_{\mathbb{S}}(c_i)\neq f^*(c_i)\}} \mid \mathbb{S}] \geqslant 0$.

b. For unobserved points $c_i \notin U$, since the values of f^* on C are independent and uniform, the labels of U provide no information about c_i, so regardless of what the algorithm outputs $h_{\mathbb{S}}(c_i)$, the probability of guessing incorrectly equals $1/2$, i.e., $\mathbb{E}_{f^*}[\mathbf{1}_{\{h_{\mathbb{S}}(c_i)\neq f^*(c_i)\}} \mid \mathbb{S}] = 1/2$.

Substituting these,

$$\mathbb{E}_{f^*}[R(h_{\mathbb{S}}) \mid \mathbb{S}] \geqslant \frac{1}{d}\sum_{c_i\in U} 0 + \frac{1}{d}\sum_{c_i\notin U}\frac{1}{2} = \frac{|C\setminus U|}{2d}.$$

• Substitute the sample size condition to derive the lower bound

Since $|U| \leqslant n$, the number of unobserved points is $|C\setminus U| = d - |U| \geqslant d - n$. Substituting,

$$\mathbb{E}_{f^*}[R(h_{\mathbb{S}}) \mid \mathbb{S}] \geqslant \frac{d-n}{2d} = \frac{1}{2} - \frac{n}{2d} > \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

Taking expectation over the distribution of $\mathbb{S}$,

$$\mathbb{E}_{f^*}\mathbb{E}_{\mathbb{S}}[R(\mathcal{A}(\mathbb{S}))] > \frac{1}{4}.$$

By the averaging principle, there exists a specific target function $f_0^* \in \mathcal{F}$ such that the expected error of the algorithm exceeds $1/4$, i.e., for any algorithm $\mathcal{A}$,

$$\sup_{f^*\in\mathcal{F}} \mathbb{E}_{\mathbb{S}\sim\mu^n}[R(\mathcal{A}(\mathbb{S}))] \geqslant \frac{1}{4},$$

Taking the infimum over all algorithms $\mathcal{A}$,

$$\inf_{\mathcal{A}}\sup_{f^*\in\mathcal{F}} \mathbb{E}_{\mathbb{S}\sim\mu^n}[R(\mathcal{A}(\mathbb{S}))] \geqslant \frac{1}{4}.$$

33 [**The γ_2 Functional is Bounded by the Dudley Integral**] Let $(\mathbb{T}, d)$ be a metric space. Define a sequence of sets $(\mathbb{T}_n)_{n=0}^{\infty}$ to be an admissible sequence if it satisfies $\mathbb{T}_n \subset \mathbb{T}$, $|\mathbb{T}_0| = 1$, and for all $n \geqslant 1$, $|\mathbb{T}_n| \leqslant 2^{2^n}$. Define the Talagrand γ_2 functional on the metric space $\mathbb{T}$ as

$$\gamma_2(\mathbb{T}, d) = \inf_{(\mathbb{T}_n)}\sup_{t\in\mathbb{T}}\sum_{n=0}^{\infty} 2^{n/2} d(t, \mathbb{T}_n),$$

where the infimum is taken over all possible admissible sequences $(\mathbb{T}_n)$, and $d(t, \mathbb{T}_n) = \inf_{s\in\mathbb{T}_n} d(t, s)$. Let $\mathcal{N}(\mathbb{T}, d, \varepsilon)$ denote the covering number of the metric space. Prove that there exists an absolute constant $C > 0$ such that the γ_2 functional is bounded by the Dudley integral,

$$\gamma_2(\mathbb{T}, d) \leqslant C\int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon.$$

Prf. • Construct a specific admissible sequence

Define the cardinality threshold sequence as

$$N_0 = 1, \quad N_n = 2^{2^n}, \qquad (n \geqslant 1)$$

If the Dudley integral equals $+\infty$, the conclusion holds. Henceforth suppose that the Dudley integral is finite. Since the covering number is monotone in the radius, it is finite at every positive radius, so $\mathbb{T}$ is totally bounded. Define the critical covering radius ε_n as

$$\varepsilon_n = \inf\{\varepsilon > 0 : \mathcal{N}(\mathbb{T}, d, \varepsilon) \leqslant N_n\}, \qquad (n \geqslant 0)$$

Fix $\eta > 0$. By the definition of the infimum, for every n there exists $r_n < \varepsilon_n + \eta 2^{-n}$ such that $\mathcal{N}(\mathbb{T}, d, r_n) \leqslant N_n$. Hence there exists a set $\mathbb{T}_n \subset \mathbb{T}$ of cardinality at most N_n that forms an r_n-net of $\mathbb{T}$, i.e., for any $t \in \mathbb{T}$,

$$d(t, \mathbb{T}_n) \leqslant r_n < \varepsilon_n + \eta 2^{-n}.$$

Since $|\mathbb{T}_0| \leqslant N_0 = 1$ and $|\mathbb{T}_n| \leqslant N_n = 2^{2^n}$, $(\mathbb{T}_n)_{n=0}^{\infty}$ is a valid admissible sequence.

Substituting $(\mathbb{T}_n)$ into the definition of the γ_2 functional,

$$\gamma_2(\mathbb{T}, d) \leqslant \sup_{t\in\mathbb{T}} \sum_{n=0}^{\infty} 2^{n/2} d(t, \mathbb{T}_n) \leqslant \sum_{n=0}^{\infty} 2^{n/2}(\varepsilon_n + \eta 2^{-n})$$
$$= \sum_{n=0}^{\infty} 2^{n/2}\varepsilon_n + \eta \sum_{n=0}^{\infty} 2^{-n/2}.$$

Letting $\eta \downarrow 0$ gives

$$\gamma_2(\mathbb{T}, d) \leqslant \sum_{n=0}^{\infty} 2^{n/2}\varepsilon_n. \tag{I}$$

- Control the series using the lower bound from the Dudley integral

Split the Dudley integral into a sum of piecewise integrals over $(\varepsilon_{n+1}, \varepsilon_n]$,

$$\int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant \sum_{n=0}^{\infty} \int_{\varepsilon_{n+1}}^{\varepsilon_n} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon,$$

When $\varepsilon \in (\varepsilon_{n+1}, \varepsilon_n)$, since $\varepsilon < \varepsilon_n$, we have $\mathcal{N}(\mathbb{T}, d, \varepsilon) > N_n$.

a. When $n = 0$, $N_0 = 1$, so $\mathcal{N} \geqslant 2$. Then $\sqrt{\log \mathcal{N}} \geqslant \sqrt{\log 2} = 2^{0/2}\sqrt{\log 2}$.

b. When $n \geqslant 1$, $N_n = 2^{2^n}$, so $\sqrt{\log \mathcal{N}} > \sqrt{\log(2^{2^n})} = \sqrt{2^n \log 2} = 2^{n/2}\sqrt{\log 2}$.

Hence

$$\int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant \sqrt{\log 2} \sum_{n=0}^{\infty} (\varepsilon_n - \varepsilon_{n+1}) 2^{n/2}. \tag{II}$$

(*Consider applying Abel's summation by parts to the right-hand series.*)

For the right-hand side,

$$\sum_{n=0}^{\infty} (\varepsilon_n - \varepsilon_{n+1}) 2^{n/2} = \varepsilon_0 + \sum_{n=1}^{\infty} \varepsilon_n (2^{n/2} - 2^{(n-1)/2})$$
$$= \varepsilon_0 + \sum_{n=1}^{\infty} \varepsilon_n 2^{n/2}(1 - 1/\sqrt{2}),$$

Let $c = 1 - 1/\sqrt{2} \in (0, 1)$. Then

$$\varepsilon_0 + c\sum_{n=1}^{\infty} \varepsilon_n 2^{n/2} \geqslant c\varepsilon_0 2^{0/2} + c\sum_{n=1}^{\infty} \varepsilon_n 2^{n/2} = c\sum_{n=0}^{\infty} 2^{n/2}\varepsilon_n,$$

Substituting this into (II),

$$\int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon \geqslant c\sqrt{\log 2} \sum_{n=0}^{\infty} 2^{n/2}\varepsilon_n,$$

i.e.,

$$\sum_{n=0}^{\infty} 2^{n/2}\varepsilon_n \leqslant \frac{1}{c\sqrt{\log 2}} \int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon,$$

Substituting this into (I),

$$\gamma_2(\mathbb{T}, d) \leqslant C \int_0^{\infty} \sqrt{\log \mathcal{N}(\mathbb{T}, d, \varepsilon)}\mathrm{d}\varepsilon. \qquad \left(C = \frac{2}{c\sqrt{\log 2}}\right)$$

34 [**An Instance where the γ_2 Functional is Strictly Better than the Dudley Discrete Sum**] Sometimes the γ_2 functional can be significantly smaller than the Dudley sum. To see this, revisit the example in Exercise Q8.4 (*Dudley's Inequality Can Be Very Loose*).

Endow $\mathbb{R}^n$ with the standard Euclidean metric $d(\boldsymbol{x},\boldsymbol{y}) = \|\boldsymbol{x}-\boldsymbol{y}\|_2$. Let $\boldsymbol{e}_1,\cdots,\boldsymbol{e}_n$ be the standard orthonormal basis of $\mathbb{R}^n$. Define

$$\mathbb{T} = \{\mathbf{0}\} \cup \left\{ \frac{\boldsymbol{e}_k}{\sqrt{1+\log k}} : k = 1,\cdots,n \right\}.$$

An admissible sequence $(\mathbb{T}_m)_{m=0}^{\infty}$ satisfies $\mathbb{T}_m \subset \mathbb{T}$, $|\mathbb{T}_0| = 1, |\mathbb{T}_m| \leqslant 2^{2^m}$ $(m \geqslant 1)$.

(1) Prove that the γ_2 functional of $\mathbb{T}$ remains uniformly bounded as $n \to \infty$, i.e., construct an admissible sequence $(\mathbb{T}_m)$ such that

$$\gamma_2(\mathbb{T}, d) = \inf_{(\mathbb{T}_m)} \sup_{\boldsymbol{t}\in\mathbb{T}} \sum_{m=0}^{\infty} 2^{m/2} d(\boldsymbol{t}, \mathbb{T}_m) \leqslant C.$$

(2) Prove that if the supremum operation is placed inside the summation (*i.e., the Dudley discrete sum*), this bound diverges as the dimension $n \to \infty$,

$$\inf_{(\mathbb{T}_m)} \sum_{m=0}^{\infty} 2^{m/2} \sup_{\boldsymbol{t}\in\mathbb{T}} d(\boldsymbol{t}, \mathbb{T}_m) \to \infty.$$

Prf. ***Sol.*** Construct the cardinality threshold sequence as $N_m = 2^{2^m}$. Let $\boldsymbol{t}_k = \frac{\boldsymbol{e}_k}{\sqrt{1+\log k}}$. Define

$$\mathbb{T}_0 = \{\mathbf{0}\}, \quad \mathbb{T}_m = \{\mathbf{0}\} \cup \{\boldsymbol{t}_k : 1 \leqslant k \leqslant \min(n, N_m - 1)\}, \qquad (m \geqslant 1)$$

Then $|\mathbb{T}_m| \leqslant N_m = 2^{2^m}$, so $(\mathbb{T}_m)_{m=0}^{\infty}$ is a valid admissible sequence.

For $\boldsymbol{t}_k \in \mathbb{T}$ and level m, if $k \leqslant N_m - 1$, then $\boldsymbol{t}_k \in \mathbb{T}_m$ and $d(\boldsymbol{t}_k, \mathbb{T}_m) = 0$. If $k > N_m - 1$, then $\boldsymbol{t}_k \notin \mathbb{T}_m$, and by orthogonality, the closest point in $\mathbb{T}_m$ must be $\mathbf{0}$, so

$$d(\boldsymbol{t}_k, \mathbb{T}_m) = \|\boldsymbol{t}_k\|_2 = \frac{1}{\sqrt{1+\log k}}.$$

(*Compute the upper bound of the series sum.*)

Let $S(\boldsymbol{t}) = \sum_{m=0}^{\infty} 2^{m/2} d(\boldsymbol{t}, \mathbb{T}_m)$. If $\boldsymbol{t} = \mathbf{0}$, then $S(\mathbf{0}) = 0$. For a non-zero point $\boldsymbol{t}_k$, the series contributes only when $2^{2^m} \leqslant k$ (*i.e.,* $2^m \leqslant \frac{\log k}{\log 2}$). Let M_k be the largest integer satisfying this condition; then $2^{M_k} \leqslant \frac{\log k}{\log 2}$.

Summing the geometric series,

$$\begin{aligned} S(\boldsymbol{t}_k) &= \sum_{m=0}^{M_k} 2^{m/2} \frac{1}{\sqrt{1+\log k}} = \frac{1}{\sqrt{1+\log k}} \cdot \frac{(\sqrt{2})^{M_k+1} - 1}{\sqrt{2}-1} \\ &\leqslant \frac{\sqrt{2}}{\sqrt{2}-1} \frac{2^{M_k/2}}{\sqrt{1+\log k}} \leqslant \frac{\sqrt{2}}{(\sqrt{2}-1)\sqrt{\log 2}} \frac{\sqrt{\log k}}{\sqrt{1+\log k}} \leqslant \frac{\sqrt{2}}{(\sqrt{2}-1)\sqrt{\log 2}}, \end{aligned}$$

Hence

$$\gamma_2(\mathbb{T}, d) \leqslant \sup_{\boldsymbol{t}\in\mathbb{T}} S(\boldsymbol{t}) \leqslant \frac{\sqrt{2}}{(\sqrt{2}-1)\sqrt{\log 2}}.$$

Sol. (*Establish a lower bound for the covering radius at a single level.*)

Let $R_m = \sup_{\boldsymbol{t}\in\mathbb{T}} d(\boldsymbol{t}, \mathbb{T}_m)$. Select the first K points of $\mathbb{T}$ to form the subset $\mathbb{T}'$ $(K \leqslant n)$.

For any $\boldsymbol{t}_i, \boldsymbol{t}_j \in \mathbb{T}'$, the lower bound on their distance is

$$\|\boldsymbol{t}_i - \boldsymbol{t}_j\|_2 = \sqrt{\|\boldsymbol{t}_i\|_2^2 + \|\boldsymbol{t}_j\|_2^2} \geqslant \sqrt{2}\|\boldsymbol{t}_K\|_2 = \frac{\sqrt{2}}{\sqrt{1+\log K}}.$$

(*If $R_m < \frac{1}{\sqrt{2(1+\log K)}}$, a ball of radius R_m can cover at most 1 point, so covering $\mathbb{T}'$ requires at least K balls, i.e., $|\mathbb{T}_m| \geqslant K$.*)

If $K > |\mathbb{T}_m|$, then $R_m \geqslant \frac{1}{\sqrt{2(1+\log K)}}$. Taking $K = 2^{2^m} + 1 \leqslant n$,

$$R_m \geqslant \frac{1}{\sqrt{2(1+\log(2^{2^m}+1))}}.$$

(*Prove that the Dudley discrete sum diverges.*)

For integers m satisfying $2^{2^m}+1 \leqslant n$,

$$R_m \geqslant \frac{1}{\sqrt{2(1+(2^m+1)\log 2)}} \geqslant \frac{c}{2^{m/2}},$$

i.e., $2^{m/2}R_m \geqslant c$.

Let $M_n = \lfloor \log_2(\log_2(n-1)) \rfloor$ be the largest integer m satisfying the condition. Then the Dudley discrete sum is

$$D_n \geqslant \sum_{m=0}^{M_n} 2^{m/2}R_m \geqslant \sum_{m=0}^{M_n} c = c(M_n+1) \to \infty.$$

35 [**High-Probability Bound for Generic Chaining**] Prove the result in Remark R8.5.4 (*Generic chaining: a high-probability bound*) using chaining with "first big jumps": $t_0 \to \pi_0(t) \to \pi_1(t) \to \cdots \to t$, where u is chosen to satisfy the $u^2/2$ condition. Let $(X_t)_{t\in\mathbb{T}}$ be a random process defined on a metric space $(\mathbb{T}, d)$ with sub-Gaussian increments, i.e., for any $s, t \in \mathbb{T}$,

$$\mathbb{P}\{|X_s - X_t| \geqslant x\} \leqslant 2\exp\left(-\frac{x^2}{K^2 d(s,t)^2}\right). \qquad (K>0)$$

Let $\gamma_2(\mathbb{T}, d)$ be the Talagrand functional of $(\mathbb{T}, d)$, and $D = \operatorname{diam}(\mathbb{T})$. Choose $t_0 \in \mathbb{T}$. Prove that for any $u \geqslant 1$,

$$\sup_{t\in\mathbb{T}} |X_t - X_{t_0}| \leqslant CK(\gamma_2(\mathbb{T}, d) + uD) \text{ (w.p. } 1 - ce^{-u^2}).$$

Prf. (*We use an improved chaining method, no longer approximating level by level starting from level* 0*, but instead considering a "first leap",*

$$t_0 \to \pi_\kappa(t) \to \pi_{\kappa+1}(t) \to \cdots \to t,$$

where the starting level κ *is chosen to satisfy* $2^{\kappa/2} \asymp u$.)

- Define the admissible sequence and choose the leap level κ

  By the definition of the γ_2 functional, there exists an admissible sequence $(\mathbb{T}_n)_{n=0}^{\infty}$ (*satisfying* $|\mathbb{T}_0| = 1$, $|\mathbb{T}_n| \leqslant 2^{2^n}$) such that

  $$\sup_{t\in\mathbb{T}} \sum_{n=0}^{\infty} 2^{n/2} d(t, \mathbb{T}_n) \leqslant 2\gamma_2(\mathbb{T}, d).$$

  For $u \geqslant 1$, choose $\kappa \in \mathbb{Z}_0^+$ as the smallest integer satisfying $2^{\kappa/2} \geqslant u$; then $u \leqslant 2^{\kappa/2} < \sqrt{2}u$. Squaring both sides,

  $$u^2 \leqslant 2^\kappa < 2u^2.$$

  For any $t \in \mathbb{T}$, let $\pi_n(t)$ be the point in the net $\mathbb{T}_n$ closest to t. Expand the increment as a telescoping series (*starting from level* κ),

  $$X_t - X_{t_0} = (X_{\pi_\kappa(t)} - X_{t_0}) + \sum_{n=\kappa}^{\infty} (X_{\pi_{n+1}(t)} - X_{\pi_n(t)}).$$

  (*It then suffices to bound the probability of the first leap and the subsequent chain.*)

- Bound the probability of the "first leap"

  For any $v \in \mathbb{T}_\kappa$, where $|\mathbb{T}_\kappa| \leqslant 2^{2^\kappa}$, define the benign event

  $$\mathcal{E}_0 = \{\forall v \in \mathbb{T}_\kappa,\ |X_v - X_{t_0}| \leqslant C_0 KDu\},$$

  By the sub-Gaussian increment property, taking a union bound on both sides, the probability that the above event fails is

  $$\mathbb{P}(\mathcal{E}_0^c) \leqslant \sum_{v\in\mathbb{T}_\kappa} \mathbb{P}\{|X_v - X_{t_0}| > C_0 KDu\} \leqslant 2^{2^\kappa} \cdot 2\exp\left(-\frac{C_0^2 K^2 D^2 u^2}{K^2 D^2}\right) = 2\exp(2^\kappa \log 2 - C_0^2 u^2),$$

  where the exponential part is

  $$2\exp(2^\kappa \log 2 - C_0^2 u^2) < 2\exp(2u^2 \log 2 - C_0^2 u^2) = 2\exp(-u^2(C_0^2 - 2\log 2)),$$

Taking $C_0 = 2$ so that $C_0^2 - 2\log 2 \geqslant 1$, we obtain $\mathbb{P}(\mathcal{E}_0^c) \leqslant 2\mathrm{e}^{-u^2}$.

- Bound the probability of the subsequent chain

  For any pair of points $(s, w) \in \mathbb{T}_{n+1} \times \mathbb{T}_n$, where $|\mathbb{T}_{n+1}| \cdot |\mathbb{T}_n| \leqslant 2^{3\cdot 2^n}$, define the benign event

  $$\mathcal{E}_n = \{\forall s \in \mathbb{T}_{n+1}, \forall w \in \mathbb{T}_n,\ |X_s - X_w| \leqslant C_1 K 2^{n/2} d(s, w)\},$$

  By the sub-Gaussian increment property, taking a union bound on both sides, the probability that the above event fails is

  $$\mathbb{P}(\mathcal{E}_n^c) \leqslant 2^{3\cdot 2^n} \cdot 2\exp\left(-\frac{C_1^2 K^2 2^n d(s,w)^2}{K^2 d(s,w)^2}\right) = 2\exp(-2^n(C_1^2 - 3\log 2)).$$

  Taking $C_1 = 3$ so that $C_1^2 - 3\log 2 \geqslant 2$, we obtain $\mathbb{P}(\mathcal{E}_n^c) \leqslant 2\exp(-2\cdot 2^n)$.

  Summing the failure probabilities over all $n \geqslant \kappa$, dominated by the first term,

  $$\sum_{n=\kappa}^{\infty} \mathbb{P}(\mathcal{E}_n^c) \leqslant 2\sum_{m=0}^{\infty} \exp(-2(2^\kappa + m)) \leqslant 3\exp(-2\cdot 2^\kappa) \leqslant 3\mathrm{e}^{-2u^2} \leqslant 3\mathrm{e}^{-u^2}.$$

- Assemble the path under the benign event

  Define the benign event $\mathcal{E} = \mathcal{E}_0 \cap \bigcap_{n=\kappa}^{\infty} \mathcal{E}_n$. Then

  $$\mathbb{P}(\mathcal{E}) \geqslant 1 - \mathbb{P}(\mathcal{E}_0^c) - \sum_{n=\kappa}^{\infty} \mathbb{P}(\mathcal{E}_n^c) \geqslant 1 - 2\mathrm{e}^{-u^2} - 3\mathrm{e}^{-u^2} = 1 - 5\mathrm{e}^{-u^2}.$$

  If $\mathcal{E}$ occurs, then for any $t \in \mathbb{T}$, using the triangle inequality $d(\pi_{n+1}(t), \pi_n(t)) \leqslant d(t, \mathbb{T}_{n+1}) + d(t, \mathbb{T}_n)$,

  $$|X_t - X_{t_0}| \leqslant C_0 K D u + \sum_{n=\kappa}^{\infty} C_1 K 2^{n/2}(d(t, \mathbb{T}_{n+1}) + d(t, \mathbb{T}_n)).$$

  where

  $$\sum_{n=\kappa}^{\infty} C_1 K 2^{n/2}(d(t, \mathbb{T}_{n+1}) + d(t, \mathbb{T}_n)) \leqslant 2C_1 K \sum_{m=0}^{\infty} 2^{m/2} d(t, \mathbb{T}_m) \leqslant 4C_1 K \gamma_2(\mathbb{T}, d),$$

  Substituting this,

  $$|X_t - X_{t_0}| \leqslant K(C_0 D u + 4C_1 \gamma_2(\mathbb{T}, d)),$$

  Taking $C = \max(C_0, 4C_1)$, $c = 5$, and taking the supremum on both sides,

  $$\sup_{t\in\mathbb{T}} |X_t - X_{t_0}| \leqslant CK(\gamma_2(\mathbb{T}, d) + uD) \text{ (w.p. } 1 - c\mathrm{e}^{-u^2}).$$

36 [**Empirical Processes: General Chaining Bound**] So far, we have bounded empirical processes for only two classes of function families: Lipschitz classes (see Theorem T8.2.3 (*Lipschitz law of large numbers*)) and Boolean classes with finite VC dimension (see Theorem T8.3.15 (*VC law of large numbers*)).

Now we consider the general case. Let $\mathcal{F}$ be a family of real-valued functions defined on a universe Ω which contains the zero function, and let $X, X_1, \cdots, X_n$ be i.i.d. random variables on Ω. Let d be a metric on $\mathcal{F}$ satisfying the sub-Gaussian increment condition, i.e., for any $f, g \in \mathcal{F}$,

$$\|f(X) - g(X)\|_{\psi_2} \leqslant d(f, g),$$

(*For example, taking* $d(f,g) = C\|f - g\|_{L^\infty}$ *satisfies this condition.*) Prove that

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}[f(X)]\right|\right] \leqslant \frac{C\gamma_2(\mathcal{F}, d)}{\sqrt{n}}. \qquad (C > 0)$$

Prf. For any $f \in \mathcal{F}$, define

$$Z_f = \frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}[f(X)],$$

Then $(Z_f)_{f\in\mathcal{F}}$ is a centered random process indexed by $\mathcal{F}$. (*It then remains to bound* $\mathbb{E}\left[\sup_{f\in\mathcal{F}} |Z_f|\right]$.)

- Compute the sub-Gaussian norm of the process increments

  Consider the increment

$$Z_f - Z_g = \frac{1}{n}\sum_{i=1}^{n}(f(X_i) - g(X_i)) - \mathbb{E}[f(X) - g(X)],$$

  Let $W_i = f(X_i) - g(X_i)$. Then

$$Z_f - Z_g = \frac{1}{n}\sum_{i=1}^{n}(W_i - \mathbb{E}[W_i]).$$

  By the sub-Gaussian increment condition,

$$\|W_i\|_{\psi_2} = \|f(X) - g(X)\|_{\psi_2} \leqslant d(f,g).$$

  Hence

$$\|W_i - \mathbb{E}[W_i]\|_{\psi_2} \leqslant c_1\|W_i\|_{\psi_2} \leqslant c_1 d(f,g).$$

  Thus

$$\begin{aligned}\left\|\sum_{i=1}^{n}(W_i - \mathbb{E}[W_i])\right\|_{\psi_2}^2 &\leqslant c_2\sum_{i=1}^{n}\|W_i - \mathbb{E}[W_i]\|_{\psi_2}^2 \\ &\leqslant c_2\sum_{i=1}^{n}(c_1 d(f,g))^2 = c_1^2 c_2 n d(f,g)^2.\end{aligned}$$

  Taking square roots on both sides and then dividing by n,

$$\|Z_f - Z_g\|_{\psi_2} = \frac{1}{n}\left\|\sum_{i=1}^{n}(W_i - \mathbb{E}[W_i])\right\|_{\psi_2} \leqslant \frac{c_1\sqrt{c_2}}{\sqrt{n}}d(f,g),$$

  i.e.,

$$\|Z_f - Z_g\|_{\psi_2} \leqslant \frac{K}{\sqrt{n}}d(f,g). \qquad (K = c_1\sqrt{c_2})$$

  Thus $(Z_f)_{f\in\mathcal{F}}$ satisfies the sub-Gaussian increment condition with respect to the new metric $d'(f,g) = \frac{K}{\sqrt{n}}d(f,g)$.

- Apply the general chaining theorem

  By Talagrand's general chaining theorem (*if the centered process (Z_f) has increments satisfying $\|Z_f - Z_g\|_{\psi_2} \leqslant d'(f,g)$, then the expectation of its supremum is bounded by the γ_2 functional of the metric space $(\mathcal{F}, d')$*),

$$\begin{aligned}\mathbb{E}\left[\sup_{f\in\mathcal{F}} Z_f\right] &\leqslant C'\gamma_2(\mathcal{F}, d') \\ &= C'\gamma_2\left(\mathcal{F}, \frac{K}{\sqrt{n}}d\right) = \frac{C'K}{\sqrt{n}}\gamma_2(\mathcal{F}, d).\end{aligned}$$

  Since the zero function belongs to $\mathcal{F}$ and $Z_0 = 0$, apply the same anchored chaining estimate to the signed index set $\mathcal{F}\times\{-1,1\}$. Within one copy the increments are controlled by d', while for opposite signs,

$$\|Z_f + Z_g\|_{\psi_2} \leqslant d'(f,\mathbf{0}) + d'(g,\mathbf{0}).$$

  Thus the γ_2 functional of the two-copy metric space joined through the zero function is bounded by an absolute constant times $\gamma_2(\mathcal{F}, d')$, and

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}f(X_i) - \mathbb{E}[f(X)]\right|\right] \leqslant \frac{C\gamma_2(\mathcal{F},d)}{\sqrt{n}}. \qquad (C>0)$$

37 [**Theorem (Geometric Form of Talagrand's Comparison Inequality)**] Let us explore some useful variants of Corollary C8.5.8 (*Talagrand comparison inequality: geometric form*).

Let $(X_x)_{x\in\mathbb{T}}$ be a random process defined on $\mathbb{T}\subset\mathbb{R}^n$ (*not necessarily zero-mean*), satisfying $X_0 = 0$ (*implicitly $0\in\mathbb{T}$ or consider*

$\mathbb{T} \cup \{0\}$). Assume its increments satisfy the sub-Gaussian condition, i.e., for any $x, y \in \mathbb{T} \cup \{0\}$, $\|X_x - X_y\|_{\psi_2} \leqslant K\|x - y\|_2$.
(1) (Expectation bound) Prove that

$$\mathbb{E}\left[\sup_{x\in\mathbb{T}} |X_x|\right] \leqslant CK\gamma(\mathbb{T}),$$

where $\gamma(\mathbb{T}) = \mathbb{E}\left[\sup_{x\in\mathbb{T}} |\langle g, x\rangle|\right]$ is the Gaussian complexity of $\mathbb{T}$.
(2) (High-probability bound) Prove that for any $u \geqslant 1$,

$$\sup_{x\in\mathbb{T}} |X_x| \leqslant CK(w(\mathbb{T}) + u \cdot \operatorname{rad}(\mathbb{T})) \text{ (w.p. } 1 - 2\mathrm{e}^{-u^2}),$$

where $w(\mathbb{T})$ is the Gaussian width and $\operatorname{rad}(\mathbb{T}) = \sup_{x\in\mathbb{T}} \|x\|_2$.
(3) (Moment bound) For any $p \geqslant 1$, using the conclusions of (1) and (2), prove that

$$\left(\mathbb{E}\left[\sup_{x\in\mathbb{T}} |X_x|^p\right]\right)^{1/p} \leqslant C\sqrt{p}K\gamma(\mathbb{T}).$$

Prf. Sol. (*Since* $X_{\mathbf{0}} = 0$) Let $\mathbb{T}' = \mathbb{T} \cup \{\mathbf{0}\}$. Then

$$\sup_{\boldsymbol{x}\in\mathbb{T}} |X_{\boldsymbol{x}}| = \sup_{\boldsymbol{x}\in\mathbb{T}} |X_{\boldsymbol{x}} - X_{\mathbf{0}}| \leqslant \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}'} |X_{\boldsymbol{x}} - X_{\boldsymbol{y}}|,$$

Taking expectations on both sides,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |X_{\boldsymbol{x}}|\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}'} |X_{\boldsymbol{x}} - X_{\boldsymbol{y}}|\right].$$

By Talagrand's general chaining theorem,

$$\mathbb{E}\left[\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}'} |X_{\boldsymbol{x}} - X_{\boldsymbol{y}}|\right] \leqslant C_1 K\gamma_2(\mathbb{T}', \|\cdot\|_2); \tag{I}$$

By the majorizing measure theorem (*in Euclidean space, the* γ_2 *functional is equivalent to the Gaussian width*),

$$\gamma_2(\mathbb{T}', \|\cdot\|_2) \leqslant C_2 w(\mathbb{T}') = C_2\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}\cup\{\mathbf{0}\}} \langle \boldsymbol{g}, \boldsymbol{x}\rangle\right], \tag{II}$$

where

$$\begin{aligned}\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}\cup\{\mathbf{0}\}} \langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] &= \mathbb{E}\left[\max(0, \sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x}\rangle)\right] \\ &\leqslant \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\langle \boldsymbol{g}, \boldsymbol{x}\rangle|\right] = \gamma(\mathbb{T}),\end{aligned}$$

Substituting this into (II),

$$\gamma_2(\mathbb{T}', \|\cdot\|_2) \leqslant C_2 w(\mathbb{T}') = C_2\gamma(\mathbb{T}). \tag{III}$$

Combining (I) and (III),

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |X_{\boldsymbol{x}}|\right] \leqslant C_1 C_2 K\gamma(\mathbb{T}),$$

Taking $C = C_1 C_2$.

Sol. By the general chaining theorem, applying the high-probability form to the augmented set $\mathbb{T}'$,

$$\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}'} |X_{\boldsymbol{x}} - X_{\boldsymbol{y}}| \leqslant C_1 K(\gamma_2(\mathbb{T}') + u \cdot \operatorname{diam}(\mathbb{T}')) \text{ (w.p. } 1 - 2\mathrm{e}^{-u^2}), \tag{IV}$$

(*Since* $\mathbf{0} \in \mathbb{T}'$ *and* $X_{\mathbf{0}} = 0$, *the left-hand side above clearly dominates* $\sup_{\boldsymbol{x}\in\mathbb{T}} |X_{\boldsymbol{x}}|$.)
For the diameter term,

$$\operatorname{diam}(\mathbb{T}') \leqslant 2\sup_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{x}\|_2 = 2\operatorname{rad}(\mathbb{T}); \tag{V}$$

For the functional term,

$$\gamma_2(\mathbb{T}') \leqslant C_2 w(\mathbb{T} \cup \{\mathbf{0}\}). \tag{VI}$$

(*We need to decompose* $w(\mathbb{T} \cup \{\mathbf{0}\})$ *into the form of* $w(\mathbb{T})$.)

Let $S = \sup_{\boldsymbol{x} \in \mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x} \rangle$. Then

$$w(\mathbb{T}) = \mathbb{E}[S], \quad w(\mathbb{T} \cup \{0\}) = \mathbb{E}[\max(0, S)],$$

Using the identity $\max(0, S) = S + \max(0, -S)$ and taking expectations on both sides,

$$w(\mathbb{T} \cup \{\mathbf{0}\}) = w(\mathbb{T}) + \mathbb{E}[\max(0, -S)];$$

For any $\boldsymbol{x}_0 \in \mathbb{T}$, $-S \leqslant -\langle \boldsymbol{g}, \boldsymbol{x}_0 \rangle$, so

$$\mathbb{E}[\max(0, -S)] \leqslant \mathbb{E}[\max(0, -\langle \boldsymbol{g}, \boldsymbol{x}_0 \rangle)] = \frac{1}{2}\mathbb{E}[|\langle \boldsymbol{g}, \boldsymbol{x}_0 \rangle|] = \frac{1}{\sqrt{2\pi}}\|\boldsymbol{x}_0\|_2 \leqslant \frac{1}{\sqrt{2\pi}} \operatorname{rad}(\mathbb{T}),$$

Substituting this into (VI),

$$w(\mathbb{T} \cup \{\mathbf{0}\}) \leqslant w(\mathbb{T}) + C_3 \operatorname{rad}(\mathbb{T}). \tag{VII}$$

Substituting (V) and (VII) into (IV),

$$\sup_{\boldsymbol{x} \in \mathbb{T}} |X_{\boldsymbol{x}}| \leqslant C_1 K (C_2 (w(\mathbb{T}) + C_3 \operatorname{rad}(\mathbb{T})) + 2u \cdot \operatorname{rad}(\mathbb{T})) \leqslant C_4 K w(\mathbb{T}) + C_4 K (C_3 + 2u) \operatorname{rad}(\mathbb{T}) \text{ (w.p. } 1 - 2\mathrm{e}^{-u^2}),$$

Since $u \geqslant 1$, taking $C_5 = C_3 + 2$ gives $C_3 + 2u \leqslant C_5 u$, and then

$$\sup_{\boldsymbol{x} \in \mathbb{T}} |X_{\boldsymbol{x}}| \leqslant C_4 K w(\mathbb{T}) + C_4 C_5 K u \operatorname{rad}(\mathbb{T}) \text{ (w.p. } 1 - 2\mathrm{e}^{-u^2}),$$

Taking $C = \max(C_4, C_4 C_5)$.

Sol. Let $Z = \sup_{\boldsymbol{x} \in \mathbb{T}} |X_{\boldsymbol{x}}|$.

By the high-probability form of the general chaining theorem, for any $u \geqslant 0$,

$$\mathbb{P}\{Z > C_1 K w(\mathbb{T}) + C_1 K \operatorname{rad}(\mathbb{T}) \cdot u\} \leqslant 2\mathrm{e}^{-u^2},$$

Hence

$$Z \leqslant C_1 K w(\mathbb{T}) + C_1 K \operatorname{rad}(\mathbb{T}) \cdot \xi \text{ (a.s.)},$$

where $\xi \geqslant 0$ satisfies $\mathbb{P}\{\xi > u\} \leqslant 2\mathrm{e}^{-u^2}$.

Using Minkowski's inequality,

$$\|Z\|_{L_p} \leqslant \|C_1 K w(\mathbb{T})\|_{L_p} + \|C_1 K \operatorname{rad}(\mathbb{T}) \cdot \xi\|_{L_p} = C_1 K w(\mathbb{T}) + C_1 K \operatorname{rad}(\mathbb{T}) \cdot \|\xi\|_{L_p}. \tag{VIII}$$

(*It remains to compute* $\|\xi\|_{L_p}$.)

Using the integral formula for the expectation of a nonnegative random variable,

$$\begin{aligned} \mathbb{E}[\xi^p] &= \int_0^\infty p u^{p-1} \mathbb{P}\{\xi > u\} \mathrm{d}u \leqslant \int_0^\infty p u^{p-1} 2\mathrm{e}^{-u^2} \mathrm{d}u \qquad (t = u^2) \\ &= \int_0^\infty p t^{\frac{p-1}{2}} \mathrm{e}^{-t} t^{-\frac{1}{2}} \mathrm{d}t = p \int_0^\infty t^{\frac{p}{2}-1} \mathrm{e}^{-t} \mathrm{d}t = p \cdot \Gamma\left(\frac{p}{2}\right), \end{aligned}$$

Hence

$$\|\xi\|_{L_p} = (p \cdot \Gamma(p/2))^{1/p} \leqslant C_2 \sqrt{p},$$

Substituting this into (VIII),

$$(\mathbb{E}[Z^p])^{1/p} \leqslant C_1 K w(\mathbb{T}) + C_1 C_2 K \operatorname{rad}(\mathbb{T}) \sqrt{p}. \tag{IX}$$

(*Now bound everything uniformly using* $\gamma(\mathbb{T})$.)

For $w(\mathbb{T})$, using the conclusion of (1),

$$w(\mathbb{T}) \leqslant \gamma(\mathbb{T});$$

For $\operatorname{rad}(\mathbb{T})$, let $\boldsymbol{x} \in \mathbb{T}$. Then $\langle \boldsymbol{g}, \boldsymbol{x} \rangle \sim N(0, \|\boldsymbol{x}\|_2^2)$, and its absolute expectation is $\mathbb{E}[|\langle \boldsymbol{g}, \boldsymbol{x} \rangle|] = \sqrt{2/\pi}\|\boldsymbol{x}\|_2$. Taking

the supremum on both sides,

$$\operatorname{rad}(\mathbb{T}) \leqslant \sqrt{\frac{\pi}{2}}\gamma(\mathbb{T});$$

Substituting these into (IX),

$$(\mathbb{E}[Z^p])^{1/p} \leqslant C_1 K\gamma(\mathbb{T}) + C_1C_2\sqrt{\frac{\pi}{2}}K\sqrt{p}\gamma(\mathbb{T}),$$

i.e.,

$$\left(\mathbb{E}\left[\sup_{x\in\mathbb{T}}|X_x|^p\right]\right)^{1/p} \leqslant C\sqrt{p}K\gamma(\mathbb{T}).$$

38 [**Expected L^p Norm of a Random Vector**] In Exercise Q3.5 (*Expected ℓ^p Norm of a Random Vector*), we obtained optimal bounds on the expected L^p norm of coordinate-wise independent sub-Gaussian random vectors.
Now drop the independence assumption: let $\boldsymbol{X} \in \mathbb{R}^N$ be a sub-Gaussian random vector with sub-Gaussian norm $K = \|\boldsymbol{X}\|_{\psi_2}$, where $N \geqslant 2$. Prove that for any $p \in [1, \infty]$,

$$\mathbb{E}\|\boldsymbol{X}\|_p \leqslant \begin{cases} CK\sqrt{p}N^{1/p}, & p \leqslant \log N, \\ CK\sqrt{\log N}, & p \geqslant \log N. \end{cases} \qquad (C > 0)$$

Prf. For the sub-Gaussian random variable $X_i = \langle \boldsymbol{X}, \boldsymbol{e}_i\rangle$, its p-th absolute moment is controlled by its ψ_2 norm, i.e., for any $p \geqslant 1$,

$$(\mathbb{E}[|X_i|^p])^{1/p} \leqslant C_0K\sqrt{p} \implies \mathbb{E}[|X_i|^p] \leqslant C_0^pK^pp^{p/2}. \qquad \text{(I)}$$

- When $p \leqslant \log N$,

$$\mathbb{E}\|\boldsymbol{X}\|_p = \mathbb{E}\left[\left(\sum_{i=1}^N |X_i|^p\right)^{1/p}\right].$$

Using Jensen's inequality, let $\varphi(x) = x^{1/p}$; then

$$\begin{aligned}\mathbb{E}\left[\left(\sum_{i=1}^N |X_i|^p\right)^{1/p}\right] &\leqslant \left(\sum_{i=1}^N \mathbb{E}[|X_i|^p]\right)^{1/p} \\ &= \left(\sum_{i=1}^N \mathbb{E}[|X_i|^p]\right)^{1/p}, \end{aligned} \qquad \text{(II)}$$

(*This step does not require the independence assumption.*) Substituting (I) into (II),

$$\left(\sum_{i=1}^N \mathbb{E}[|X_i|^p]\right)^{1/p} \leqslant \left(\sum_{i=1}^N C_0^pK^pp^{p/2}\right)^{1/p} = (N\cdot C_0^pK^pp^{p/2})^{1/p},$$

i.e.,

$$\mathbb{E}\|\boldsymbol{X}\|_p \leqslant C_0K\sqrt{p}N^{1/p}.$$

- When $p \geqslant \log N$,

By the monotonicity of norms,

$$\|\boldsymbol{X}\|_p \leqslant \|\boldsymbol{X}\|_{\log N} \text{ (a.s.)},$$

Taking expectations on both sides and applying the conclusion of case a. to $q = \log N$,

$$\mathbb{E}\|\boldsymbol{X}\|_p \leqslant \mathbb{E}\|\boldsymbol{X}\|_{\log N} \leqslant C_0K\sqrt{\log N}\cdot N^{\frac{1}{\log N}},$$

i.e.,

$$\mathbb{E}\|\boldsymbol{X}\|_p \leqslant C_0K\sqrt{\log N}\cdot \mathrm{e} =: CK\sqrt{\log N}. \qquad (C = C_0\mathrm{e})$$

39 [**Gaussian Chevet Inequality**] Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix, and let $\mathbb{T} \subset \mathbb{R}^n$, $\mathbb{S} \subset \mathbb{R}^m$ be symmetric sets containing $\mathbf{0}$.

(1) Prove that Theorem T8.6.1 (*Subgaussian Chevet inequality*) holds with exact constant 1,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}} \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle\right] \leqslant w(\mathbb{T})\operatorname{rad}(\mathbb{S}) + w(\mathbb{S})\operatorname{rad}(\mathbb{T}).$$

(2) Prove that the reverse inequality also holds, i.e.,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}} \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle\right] \geqslant c[w(\mathbb{T})\operatorname{rad}(\mathbb{S}) + w(\mathbb{S})\operatorname{rad}(\mathbb{T})]. \qquad (c>0)$$

Prf. Sol. Define the target Gaussian process $(X_{\boldsymbol{x},\boldsymbol{y}})_{(\boldsymbol{x},\boldsymbol{y})\in\mathbb{T}\times\mathbb{S}}$ by

$$X_{\boldsymbol{x},\boldsymbol{y}} = \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle;$$

Introduce independent Gaussian random vectors $\boldsymbol{g} \sim N(\mathbf{0}, \boldsymbol{I}_n), \boldsymbol{h} \sim N(\mathbf{0}, \boldsymbol{I}_m)$ (*independent of* $\boldsymbol{A}$), and construct the comparison Gaussian process $(Y_{\boldsymbol{x},\boldsymbol{y}})_{(\boldsymbol{x},\boldsymbol{y})\in\mathbb{T}\times\mathbb{S}}$ by

$$Y_{\boldsymbol{x},\boldsymbol{y}} = \|\boldsymbol{y}\|_2\langle \boldsymbol{g}, \boldsymbol{x}\rangle + \|\boldsymbol{x}\|_2\langle \boldsymbol{h}, \boldsymbol{y}\rangle;$$

Then $\mathbb{E}[X_{\boldsymbol{x},\boldsymbol{y}}] = \mathbb{E}[Y_{\boldsymbol{x},\boldsymbol{y}}] = 0$.

The increment variance of X is

$$\begin{aligned}\mathbb{E}[(X_{\boldsymbol{x},\boldsymbol{y}} - X_{\boldsymbol{u},\boldsymbol{v}})^2] &= \mathbb{E}\left[\left(\sum_{i,j} A_{ij}(x_j y_i - u_j v_i)\right)^2\right] = \sum_{i,j}(x_j y_i - u_j v_i)^2 \\ &= \|\boldsymbol{x}\|_2^2\|\boldsymbol{y}\|_2^2 + \|\boldsymbol{u}\|_2^2\|\boldsymbol{v}\|_2^2 - 2\langle \boldsymbol{x}, \boldsymbol{u}\rangle\langle \boldsymbol{y}, \boldsymbol{v}\rangle,\end{aligned}$$

The increment variance of Y is

$$\begin{aligned}\mathbb{E}[(Y_{\boldsymbol{x},\boldsymbol{y}} - Y_{\boldsymbol{u},\boldsymbol{v}})^2] &= \mathbb{E}[\langle \boldsymbol{g}, \|\boldsymbol{y}\|_2\boldsymbol{x} - \|\boldsymbol{v}\|_2\boldsymbol{u}\rangle^2] + \mathbb{E}[\langle \boldsymbol{h}, \|\boldsymbol{x}\|_2\boldsymbol{y} - \|\boldsymbol{u}\|_2\boldsymbol{v}\rangle^2] \\ &= \|\|\boldsymbol{y}\|_2\boldsymbol{x} - \|\boldsymbol{v}\|_2\boldsymbol{u}\|_2^2 + \|\|\boldsymbol{x}\|_2\boldsymbol{y} - \|\boldsymbol{u}\|_2\boldsymbol{v}\|_2^2 \\ &= 2\|\boldsymbol{x}\|_2^2\|\boldsymbol{y}\|_2^2 + 2\|\boldsymbol{u}\|_2^2\|\boldsymbol{v}\|_2^2 - 2\|\boldsymbol{y}\|_2\|\boldsymbol{v}\|_2\langle \boldsymbol{x}, \boldsymbol{u}\rangle - 2\|\boldsymbol{x}\|_2\|\boldsymbol{u}\|_2\langle \boldsymbol{y}, \boldsymbol{v}\rangle,\end{aligned}$$

Their difference is

$$\begin{aligned}&\mathbb{E}[(Y_{\boldsymbol{x},\boldsymbol{y}} - Y_{\boldsymbol{u},\boldsymbol{v}})^2] - \mathbb{E}[(X_{\boldsymbol{x},\boldsymbol{y}} - X_{\boldsymbol{u},\boldsymbol{v}})^2] \\ &= (\|\boldsymbol{x}\|_2\|\boldsymbol{v}\|_2 - \|\boldsymbol{u}\|_2\|\boldsymbol{y}\|_2)^2 + 2\|\boldsymbol{x}\|_2\|\boldsymbol{u}\|_2\|\boldsymbol{y}\|_2\|\boldsymbol{v}\|_2 - 2\|\boldsymbol{y}\|_2\|\boldsymbol{v}\|_2\langle \boldsymbol{x}, \boldsymbol{u}\rangle - 2\|\boldsymbol{x}\|_2\|\boldsymbol{u}\|_2\langle \boldsymbol{y}, \boldsymbol{v}\rangle + 2\langle \boldsymbol{x}, \boldsymbol{u}\rangle\langle \boldsymbol{y}, \boldsymbol{v}\rangle \\ &= (\|\boldsymbol{x}\|_2\|\boldsymbol{v}\|_2 - \|\boldsymbol{u}\|_2\|\boldsymbol{y}\|_2)^2 + 2(\|\boldsymbol{x}\|_2\|\boldsymbol{u}\|_2 - \langle \boldsymbol{x}, \boldsymbol{u}\rangle)(\|\boldsymbol{y}\|_2\|\boldsymbol{v}\|_2 - \langle \boldsymbol{y}, \boldsymbol{v}\rangle) \geqslant 0.\end{aligned}$$

The Sudakov-Fernique condition is satisfied, hence

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}} X_{\boldsymbol{x},\boldsymbol{y}}\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}} Y_{\boldsymbol{x},\boldsymbol{y}}\right].$$

For the right-hand side,

$$\begin{aligned}\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}} Y_{\boldsymbol{x},\boldsymbol{y}}\right] &= \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}} (\|\boldsymbol{y}\|_2\langle \boldsymbol{g}, \boldsymbol{x}\rangle + \|\boldsymbol{x}\|_2\langle \boldsymbol{h}, \boldsymbol{y}\rangle)\right] \\ &\leqslant \mathbb{E}\left[\sup_{\boldsymbol{y}\in\mathbb{S}}\|\boldsymbol{y}\|_2 \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{x}\rangle + \sup_{\boldsymbol{x}\in\mathbb{T}}\|\boldsymbol{x}\|_2 \sup_{\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{h}, \boldsymbol{y}\rangle\right] \\ &= \operatorname{rad}(\mathbb{S})\cdot\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] + \operatorname{rad}(\mathbb{T})\cdot\mathbb{E}\left[\sup_{\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{h}, \boldsymbol{y}\rangle\right] \\ &= w(\mathbb{T})\operatorname{rad}(\mathbb{S}) + w(\mathbb{S})\operatorname{rad}(\mathbb{T}),\end{aligned}$$

Substituting this,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle\right]\leqslant w(\mathbb{T})\operatorname{rad}(\mathbb{S})+w(\mathbb{S})\operatorname{rad}(\mathbb{T}).$$

Sol. For any $\varepsilon>0$, there exists $\boldsymbol{y}_\varepsilon\in\mathbb{S}$ such that $\|\boldsymbol{y}_\varepsilon\|_2\geqslant\operatorname{rad}(\mathbb{S})-\varepsilon$. Restricting the double supremum to $\boldsymbol{y}_\varepsilon$ to obtain a lower bound,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle\right]\geqslant\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}_\varepsilon\rangle\right]=\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x},\boldsymbol{A}^\top\boldsymbol{y}_\varepsilon\rangle\right].$$

Note that $\boldsymbol{A}^\top\boldsymbol{y}_\varepsilon\sim N(\boldsymbol{0},\|\boldsymbol{y}_\varepsilon\|_2^2\boldsymbol{I}_n)$, so $\boldsymbol{A}^\top\boldsymbol{y}_\varepsilon\overset{d}{=}\|\boldsymbol{y}_\varepsilon\|_2\boldsymbol{g}$. Hence

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x},\boldsymbol{A}^\top\boldsymbol{y}_\varepsilon\rangle\right]=\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x},\|\boldsymbol{y}_\varepsilon\|_2\boldsymbol{g}\rangle\right]=\|\boldsymbol{y}_\varepsilon\|_2 w(\mathbb{T})\geqslant(\operatorname{rad}(\mathbb{S})-\varepsilon)w(\mathbb{T}),$$

Letting $\varepsilon\to0^+$,

$$\mathbb{E}[\sup\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle]\geqslant w(\mathbb{T})\operatorname{rad}(\mathbb{S}).\tag{I}$$

Similarly, for any $\varepsilon>0$, there exists $\boldsymbol{x}_\varepsilon\in\mathbb{T}$ such that $\|\boldsymbol{x}_\varepsilon\|_2\geqslant\operatorname{rad}(\mathbb{T})-\varepsilon$. Using symmetry $\boldsymbol{A}\boldsymbol{x}_\varepsilon\overset{d}{=}\|\boldsymbol{x}_\varepsilon\|_2\boldsymbol{h}$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle\right]\geqslant\mathbb{E}\left[\sup_{\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{A}\boldsymbol{x}_\varepsilon,\boldsymbol{y}\rangle\right]=\|\boldsymbol{x}_\varepsilon\|_2 w(\mathbb{S})\geqslant(\operatorname{rad}(\mathbb{T})-\varepsilon)w(\mathbb{S}),$$

Letting $\varepsilon\to0^+$,

$$\mathbb{E}[\sup\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle]\geqslant w(\mathbb{S})\operatorname{rad}(\mathbb{T}).\tag{II}$$

Combining (I) and (II), (*If $Z\geqslant a,b\geqslant0$, then $Z\geqslant\max(a,b)\geqslant\frac12(a+b)$*)

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle\right]\geqslant\frac12[w(\mathbb{T})\operatorname{rad}(\mathbb{S})+w(\mathbb{S})\operatorname{rad}(\mathbb{T})],$$

Taking $c=1/2$.

40 [**Chevet Inequality: A High-Probability Version**] Under the assumptions of Theorem T8.6.1 (*Subgaussian Chevet inequality*), let $\mathbb{T}\subset\mathbb{S}^{n-1}$ and $\mathbb{S}\subset\mathbb{S}^{m-1}$. Prove that for any $u\geqslant1$,

$$\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle\leqslant CK(w(\mathbb{T})+w(\mathbb{S})+u)\ (\text{w.p. } 1-2\mathrm{e}^{-u^2}).$$

Prf. • (*A sub-Gaussian process on the product space*) Normalize $K=1$. Define $X_{(\boldsymbol{x},\boldsymbol{y})}=\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle$ for $(\boldsymbol{x},\boldsymbol{y})\in\mathbb{T}\times\mathbb{S}$, and set $X_{(\boldsymbol{0},\boldsymbol{0})}=0$.

Since A_{ij} are independent, zero-mean and sub-Gaussian, the linear combination lemma for sub-Gaussian variables gives

$$\|X_{(\boldsymbol{x},\boldsymbol{y})}-X_{(\boldsymbol{x}',\boldsymbol{y}')}\|_{\psi_2}\leqslant C\sqrt{\sum_{i,j}(x_jy_i-x'_jy'_i)^2}=C\|\boldsymbol{x}\otimes\boldsymbol{y}-\boldsymbol{x}'\otimes\boldsymbol{y}'\|_F,$$

while

$$\begin{aligned}\|\boldsymbol{x}\otimes\boldsymbol{y}-\boldsymbol{x}'\otimes\boldsymbol{y}'\|_F&\leqslant\|(\boldsymbol{x}-\boldsymbol{x}')\otimes\boldsymbol{y}\|_F+\|\boldsymbol{x}'\otimes(\boldsymbol{y}-\boldsymbol{y}')\|_F\\&\leqslant\|\boldsymbol{x}-\boldsymbol{x}'\|_2\|\boldsymbol{y}\|_2+\|\boldsymbol{x}'\|_2\|\boldsymbol{y}-\boldsymbol{y}'\|_2\\&\leqslant\|\boldsymbol{x}-\boldsymbol{x}'\|_2+\|\boldsymbol{y}-\boldsymbol{y}'\|_2\leqslant\sqrt{2}\|(\boldsymbol{x},\boldsymbol{y})-(\boldsymbol{x}',\boldsymbol{y}')\|_2,\end{aligned}$$

where $\|\cdot\|_2$ in the last line is the Euclidean norm on $\mathbb{R}^{n+m}$ (*since $\boldsymbol{x},\boldsymbol{x}',\boldsymbol{y},\boldsymbol{y}'$ lie on the unit spheres*). Thus X is a sub-Gaussian process on $\mathbb{T}\times\mathbb{S}$ with respect to the Euclidean metric, with $X_{(\boldsymbol{0},\boldsymbol{0})}=0$.

• (*Applying the high-probability Talagrand comparison*) Applying part (2) (*the high-probability bound*) of Exercise Q8.37

(*Theorem (Geometric Form of Talagrand's Comparison Inequality)*) to the set $\mathbb{T}\times\mathbb{S}\subset\mathbb{R}^{n+m}$,

$$\sup_{(\boldsymbol{x},\boldsymbol{y})\in\mathbb{T}\times\mathbb{S}} X_{(\boldsymbol{x},\boldsymbol{y})} \leqslant CK(w(\mathbb{T}\times\mathbb{S})+u\cdot\operatorname{rad}(\mathbb{T}\times\mathbb{S})) \text{ (w.p. } 1-2\mathrm{e}^{-u^2}).$$

- (*Computing the geometric parameters*) Let $\boldsymbol{g}_n\sim N(\boldsymbol{0},\boldsymbol{I}_n)$ and $\boldsymbol{g}_m\sim N(\boldsymbol{0},\boldsymbol{I}_m)$ be independent. The Gaussian width of $\mathbb{T}\times\mathbb{S}$ is

$$w(\mathbb{T}\times\mathbb{S})=\mathbb{E}\left[\sup_{(\boldsymbol{x},\boldsymbol{y})\in\mathbb{T}\times\mathbb{S}}[\langle\boldsymbol{g}_n,\boldsymbol{x}\rangle+\langle\boldsymbol{g}_m,\boldsymbol{y}\rangle]\right]\leqslant w(\mathbb{T})+w(\mathbb{S}),$$

and the radius

$$\operatorname{rad}(\mathbb{T}\times\mathbb{S})=\sup_{(\boldsymbol{x},\boldsymbol{y})\in\mathbb{T}\times\mathbb{S}}\sqrt{\|\boldsymbol{x}\|_2^2+\|\boldsymbol{y}\|_2^2}=\sqrt{\operatorname{rad}(\mathbb{T})^2+\operatorname{rad}(\mathbb{S})^2}\leqslant\sqrt{2}.$$

Substituting and absorbing the constants (*absorbing* $\sqrt{2}u$ *into* Cu), we obtain

$$\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}\langle\boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle\leqslant CK(w(\mathbb{T})+w(\mathbb{S})+u) \text{ (w.p. } 1-2\mathrm{e}^{-u^2}).$$

41 [$p\to q$ **Norm of a Random Matrix**] Now we have the tools to handle all p,q——unlike before, when we only had a few special cases (see Exercise Q4.44 (*Other* $p\to q$ *Operator Norms of Random Matrices*))!

Let $m,n\geqslant 2$, and let p',q' be the Hölder conjugates of p,q respectively. Define

$$r(n,p)=\begin{cases}1, & p\in[1,2],\\ n^{\frac{1}{2}-\frac{1}{p}}, & p\in[2,\infty]\end{cases},\quad w(n,p)=\begin{cases}\sqrt{p}n^{1/p}, & p\in[1,\log n],\\ \sqrt{\log n}, & p\in[\log n,\infty]\end{cases}.$$

(1) Let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ be a random matrix whose row vectors $\boldsymbol{A}_i$ are independent zero-mean K-sub-Gaussian random vectors. Prove that for any $p,q\in[1,\infty]$,

$$\mathbb{E}\|\boldsymbol{A}\|_{p\to q}\leqslant CK[r(n,p)w(m,q)+r(m,q')w(n,p')]. \qquad (C>0)$$

(2) Prove that the bound in (1) is tight for Gaussian matrices. Specifically, let $\boldsymbol{A}$ be a standard Gaussian random matrix; then

$$\mathbb{E}\|\boldsymbol{A}\|_{p\to q}\geqslant c[r(n,p)w(m,q)+r(m,q')w(n,p')].$$

Prf. (*First compute the geometric parameters.*)

By the definition of the operator norm and the duality of the ℓ^q norm,

$$\|\boldsymbol{A}\|_{p\to q}=\sup_{\boldsymbol{x}\in\mathbb{B}_p^n}\|\boldsymbol{A}\boldsymbol{x}\|_q=\sup_{\boldsymbol{x}\in\mathbb{B}_p^n}\sup_{\boldsymbol{y}\in\mathbb{B}_{q'}^m}\langle\boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle.$$

Let $\mathbb{T}=\mathbb{B}_p^n$, $\mathbb{S}=\mathbb{B}_{q'}^m$. Compute the radius $\operatorname{rad}(\cdot)$ of the two sets, i.e., the maximum of the ℓ^2 norm. For $\mathbb{T}$: if $1\leqslant p\leqslant 2$, the maximum is 1; if $p\geqslant 2$, by Hölder's inequality $\|\boldsymbol{x}\|_2\leqslant n^{\frac{1}{2}-\frac{1}{p}}$. Hence $\operatorname{rad}(\mathbb{T})=r(n,p)$, and similarly $\operatorname{rad}(\mathbb{S})=r(m,q')$. Compute the Gaussian width $w(\cdot)$. Let $\boldsymbol{g}\sim N(\boldsymbol{0},\boldsymbol{I}_n),\boldsymbol{h}\sim N(\boldsymbol{0},\boldsymbol{I}_m)$,

$$w(\mathbb{T})=\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{B}_p^n}\langle\boldsymbol{g},\boldsymbol{x}\rangle\right]=\mathbb{E}\|\boldsymbol{g}\|_{p'},\quad w(\mathbb{S})=\mathbb{E}\left[\sup_{\boldsymbol{y}\in\mathbb{B}_{q'}^m}\langle\boldsymbol{h},\boldsymbol{y}\rangle\right]=\mathbb{E}\|\boldsymbol{h}\|_q.$$

Since the optimal order of the expected ℓ^k norm of a standard Gaussian vector is $w(n,k)$, we have $w(\mathbb{T})\asymp w(n,p'),w(\mathbb{S})\asymp w(m,q)$.

Sol. $\boldsymbol{A}$ has independent zero-mean sub-Gaussian row vectors. By Theorem T8.6.1 (*Subgaussian Chevet inequality*) (*Subgaussian Chevet inequality*),

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T},\boldsymbol{y}\in\mathbb{S}}\langle\boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle\right]\leqslant CK[w(\mathbb{T})\operatorname{rad}(\mathbb{S})+w(\mathbb{S})\operatorname{rad}(\mathbb{T})].$$

Substituting the geometric parameters, $w(\mathbb{T})\operatorname{rad}(\mathbb{S})\leqslant C_1w(n,p')r(m,q')$ and $w(\mathbb{S})\operatorname{rad}(\mathbb{T})\leqslant C_2w(m,q)r(n,p)$. Combining yields the conclusion.

Sol. For a Gaussian matrix, take $\boldsymbol{y}_0 \in \mathbb{S}$ such that $\|\boldsymbol{y}_0\|_2 = \operatorname{rad}(\mathbb{S}) = r(m, q')$.

$$\mathbb{E}\|\boldsymbol{A}\|_{p\to q} \geqslant \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}_0\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{x}, \boldsymbol{A}^\top \boldsymbol{y}_0\rangle\right].$$

The random vector $\boldsymbol{A}^\top \boldsymbol{y}_0 \sim N(\boldsymbol{0}, \|\boldsymbol{y}_0\|_2^2 \boldsymbol{I}_n)$ is equal in distribution to $\|\boldsymbol{y}_0\|_2 \boldsymbol{g}$. Hence

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{x}, \|\boldsymbol{y}_0\|_2 \boldsymbol{g}\rangle\right] = \|\boldsymbol{y}_0\|_2 w(\mathbb{T}) = r(m, q')\mathbb{E}\|\boldsymbol{g}\|_{p'}.$$

Therefore $\mathbb{E}\|\boldsymbol{A}\|_{p\to q} \geqslant c_1 r(m, q') w(n, p')$.

Similarly, take $\boldsymbol{x}_0 \in \mathbb{T}$ such that $\|\boldsymbol{x}_0\|_2 = \operatorname{rad}(\mathbb{T}) = r(n, p)$. Then

$$\mathbb{E}\|\boldsymbol{A}\|_{p\to q} \geqslant \mathbb{E}\left[\sup_{\boldsymbol{y}\in\mathbb{S}} \langle \boldsymbol{A}\boldsymbol{x}_0, \boldsymbol{y}\rangle\right] = \|\boldsymbol{x}_0\|_2 w(\mathbb{S}) = r(n, p)\mathbb{E}\|\boldsymbol{h}\|_q.$$

Therefore $\mathbb{E}\|\boldsymbol{A}\|_{p\to q} \geqslant c_2 r(n, p) w(m, q)$.

For nonnegative numbers a, b,

$$\max(a, b) \geqslant \frac{1}{2}(a + b).$$

Taking $c = \frac{1}{2}\min(c_1, c_2)$ yields the conclusion.

Matrix Deviation Inequality 9

Keypoints

Section 9.1 Matrix deviation inequality — Q9.1 ∼ Q9.5

Theorem 9.1.1 (*Matrix deviation inequality*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix whose entries are independent standardized K-sub-Gaussian random variables. Then for any $\mathbb{T} \subset \mathbb{R}^n$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2\right|\right] \leqslant CK^2 w(\mathbb{T}).$$

Theorem 9.1.2 (*Sub-Gaussian increments*)

Under the conditions of Theorem T9.1.1 (*Matrix deviation inequality*), the process $\boldsymbol{X}_{\boldsymbol{x}} = \|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2$ has sub-Gaussian increments with respect to the Euclidean metric,

$$\|\boldsymbol{X}_{\boldsymbol{x}} - \boldsymbol{X}_{\boldsymbol{y}}\|_{\psi_2} \leqslant CK^2\|\boldsymbol{x} - \boldsymbol{y}\|_2.$$

Remark 9.1.5 (*Matrix deviations of squares*)

For deviations of the quadratic process $\|\boldsymbol{A}\boldsymbol{x}\|_2^2$, by Theorem T9.1.1 (*Matrix deviation inequality*), we obtain

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{x}\|_2^2\right|\right] \leqslant CK^4\gamma(\mathbb{T})^2 + CK^2\sqrt{m}\,\mathrm{rad}(\mathbb{T})\gamma(\mathbb{T}).$$

Exercise Q9.3 (*Quadratic Matrix Deviation*) proves the conclusion above. Exercise Q9.5 (*Quadratic Empirical Process*) extends the matrix deviation inequality to empirical processes, and Exercise Q9.6 (*Deviation of Random Projections*) proves its random projection version.

Section 9.2 Random matrices, covariance estimation, and JL — Q9.6 ∼ Q9.11

Proposition 9.2.1 (*Size of random projections of sets*)

Let $\mathbb{T} \subset \mathbb{R}^n$ and $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ a sub-Gaussian matrix. Then

$$\mathbb{E}[\mathrm{rad}(\boldsymbol{A}\mathbb{T})] \leqslant (1+\varepsilon)\sqrt{m}\cdot\mathrm{rad}(\mathbb{T}) + CK^2 w(\mathbb{T}).$$

Theorem 9.2.2 (*Covariance estimation for low-dimensional distributions*)

Let $\boldsymbol{X} \in \mathbb{R}^n$ be a sub-Gaussian random vector whose distribution is supported on a d-dimensional subspace. For N independent copies, if $N \geqslant Cd/\varepsilon^2$, then the sample covariance $\boldsymbol{\Sigma}_N$ satisfies

$$\|\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}\| \leqslant \varepsilon \|\boldsymbol{\Sigma}\| \text{ (w.h.p.)}.$$

Remark 9.2.3 (*Covariance estimation: a high-probability guarantee*)

As in Remarks R4.7.3 (*High-probability bound*) and R5.6.5 (*High-probability guarantees*), the expectation bound in Theorem T9.2.2 (*Covariance estimation for low-dimensional distributions*) admits the following high-probability version: for every $u \geqslant 0$,

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant CK^4 \left(\sqrt{\frac{r+u}{m}} + \frac{r+u}{m} \right) \|\boldsymbol{\Sigma}\| \text{ (w.h.p. } 1 - 2\mathrm{e}^{-u}),$$

where r is the effective rank of $\boldsymbol{\Sigma}$. Exercise Q9.9 (*Covariance Estimation with High Probability*) proves the conclusion above.

Lemma 9.2.4 (*Additive Johnson–Lindenstrauss lemma*)

Let $\mathbb{T} \subset \mathbb{R}^n$ and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ a sub-Gaussian matrix, where $m \geqslant C\varepsilon^{-2} w(\mathbb{T})^2 / \operatorname{rad}(\mathbb{T})^2$, the map $\boldsymbol{x} \mapsto \frac{1}{\sqrt{m}} \boldsymbol{A}\boldsymbol{x}$ approximately preserves the norm: for any $\boldsymbol{x} \in \mathbb{T}$,

$$\left| \frac{1}{\sqrt{m}} \|\boldsymbol{A}\boldsymbol{x}\|_2 - \|\boldsymbol{x}\|_2 \right| \leqslant \varepsilon \operatorname{rad}(\mathbb{T}).$$

Section 9.3 Random sections: M^* bound and escape theorem — Q9.12 ∼ Q9.16

Theorem 9.3.1 (M^* *bound*)

Let $\mathbb{T} \subset \mathbb{S}^{n-1}$ and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ a standard Gaussian random matrix. Then

$$\mathbb{E}\left[\sup_{\boldsymbol{x} \in \mathbb{T}} \left| \|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m} \right| \right] \leqslant C \cdot w(\mathbb{T}).$$

Moreover, there exists $\mathbb{T}' \subset \mathbb{T}$ such that when $\boldsymbol{x} \in \mathbb{T}'$, the above deviation holds in the reverse direction (*the lower M^* bound*).

Theorem 9.3.4 (*Escape theorem*)

Let $\mathbb{T} \subset \mathbb{S}^{n-1}$ and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ a standard Gaussian random matrix. Then

$$\min_{\boldsymbol{x} \in \mathbb{T}} \|\boldsymbol{A}\boldsymbol{x}\|_2 \geqslant C^{-1}\sqrt{m} - Cw(\mathbb{T}) \text{ (w.h.p.)}.$$

In particular, when the number of measurements (*rows*) satisfies $m \geqslant C_0 w(\mathbb{T})^2$, $\mathbb{T} \cap \ker \boldsymbol{A} = \emptyset$, achieving "escape".

Section 9.4 High-dimensional linear models — Q9.17 ∼ Q9.21

Theorem 9.4.4 (*Constrained recovery*)

Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a Gaussian matrix, let $\mathbb{T} \subset \mathbb{R}^n$ be a convex symmetric set, and let $\|\cdot\|_{\mathbb{T}}$ be the norm whose unit ball is $\mathbb{T}$. For every $\boldsymbol{x} \in \mathbb{T}$, consider

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z} \in \mathbb{R}^n} \|\boldsymbol{z}\|_{\mathbb{T}} \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}.$$

If $m \geqslant Cw(\mathbb{T})^2$, then $\hat{\boldsymbol{x}} = \boldsymbol{x}$ (w.h.p.).

Remark 9.4.7 (*Unconstrained optimization*)

Imposing the exact constraints $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$ and $\boldsymbol{x} \in \mathbb{T}$ can make the feasible set empty in the presence of noise or when $\mathbb{T}$ is misspecified. Given noisy measurements $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}$, one may instead solve the unconstrained convex optimization problem

$$\min_{\boldsymbol{x} \in \mathbb{R}^n} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2 + \lambda \|\boldsymbol{x}\|_{\mathbb{T}},$$

where $\|\cdot\|_{\mathbb{T}}$ is any norm you like and $\lambda > 0$ tunes the tradeoff between fitting the observations $\boldsymbol{y}$ and keeping the solution $\boldsymbol{x}$ structured (i.e., $\|\boldsymbol{x}\|_{\mathbb{T}}$ small). Adjusting λ lets you decide which to prioritize.

Corollary 9.4.8 (*Sparse recovery*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a Gaussian matrix with $m \geqslant Cs\log(en/s)$. Then every s-sparse vector $\boldsymbol{x} \in \mathbb{R}^n$ is the unique solution of (w.h.p.)

$$\arg\min_{\boldsymbol{y}\in\mathbb{R}^n} \|\boldsymbol{y}\|_1 \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}.$$

Corollary 9.4.11 (*Low-rank matrix recovery*)

Let $\mathcal{A}$ be a linear map from $n \times n$ matrices to $\mathbb{R}^m$ with independent Gaussian entries, where $m \geqslant Crn$. Then every rank-r matrix can be recovered from $\mathcal{A}(\boldsymbol{X})$ via nuclear norm minimization.

Section 9.5 Exact sparse recovery Q9.22 ∼ Q9.33

Theorem 9.5.1 (*Exact sparse recovery*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a sub-Gaussian matrix with $m \geqslant Cs\log(en/s)$. Then every s-sparse vector $\boldsymbol{x}$ is the unique minimizer of (w.h.p.)

$$\arg\min_{\boldsymbol{y}\in\mathbb{R}^n} \|\boldsymbol{y}\|_1 \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}.$$

Lemma 9.5.2 (*Error is heavier on the support of* $\boldsymbol{x}$)

Let $\boldsymbol{x}$ be s-sparse with $\operatorname{supp}(\boldsymbol{x}) = \mathbb{S}$, and let $\boldsymbol{h}$ be the error vector of the ℓ_1 minimization. Then

$$\|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant \|\boldsymbol{h}_{\mathbb{S}}\|_1.$$

Lemma 9.5.3 (*Error is approximately sparse*)

Under the nullspace property, the error $\boldsymbol{h}$ satisfies

$$\|\boldsymbol{h}\|_2 \leqslant C\frac{\|\boldsymbol{A}\boldsymbol{h}\|_2}{\sqrt{m}},$$

which is small when $\boldsymbol{A}$ is approximately isometric on approximately sparse vectors.

Definition 9.5.5 (*Restricted isometry property*)

A matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ satisfies the restricted isometry property (RIP) of order s and constant δ_s if $(1-\delta_s)\|\boldsymbol{x}\|_2^2 \leqslant \|\boldsymbol{A}\boldsymbol{x}\|_2^2 \leqslant (1+\delta_s)\|\boldsymbol{x}\|_2^2$ for all s-sparse vectors $\boldsymbol{x}$.

Theorem 9.5.6 (*RIP implies exact recovery*)

If $\boldsymbol{A}$ satisfies the RIP of order $2s$ with constant $\delta_{2s} < 1/3$, then every s-sparse vector $\boldsymbol{x}$ is the unique minimizer of

$$\arg\min_{\boldsymbol{y}\in\mathbb{R}^n} \|\boldsymbol{y}\|_1 \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}.$$

Theorem 9.5.7 (*Random matrices satisfy RIP*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a sub-Gaussian matrix. Then $\frac{1}{\sqrt{m}}\boldsymbol{A}$ satisfies the RIP of order s and constant δ, provided $m \geqslant C\delta^{-2}s\log(en/s)$ (w.h.p.).

Section 9.6 Matrix deviation for general norms Q9.34 ∼ Q9.40

Definition 9.6.1 (*General subadditive functions*)

The functions considered in this section need not be norms; they are only required to be positively homogeneous, subadditive, and controlled by the Euclidean norm.

Theorem 9.6.3 (*General matrix deviation inequality*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix. Let $f : \mathbb{R}^m \to \mathbb{R}$ be a positively homogeneous, subadditive function, and suppose there exists $b \in \mathbb{R}$ such that $f(\boldsymbol{x}) \leqslant b\|\boldsymbol{x}\|_2$ for every $\boldsymbol{x} \in \mathbb{R}^m$. Then for any subset $\mathbb{T} \subset \mathbb{R}^n$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |f(\boldsymbol{A}\boldsymbol{x}) - \mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]|\right] \leqslant Cb\gamma(\mathbb{T}),$$

where $\gamma(\mathbb{T})$ is the Gaussian complexity of $\mathbb{T}$ (see Section 7.5.3).

Theorem 9.6.4 (*Sub-Gaussian increments*)

Under the conditions of Theorem T9.6.3 (*General matrix deviation inequality*), the random process $Z_{\boldsymbol{x}} = f(\boldsymbol{A}\boldsymbol{x}) - \mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]$ has sub-Gaussian increments:

$$\|Z_{\boldsymbol{x}} - Z_{\boldsymbol{y}}\|_{\psi_2} \leqslant Cb\|\boldsymbol{x} - \boldsymbol{y}\|_2 \quad \text{for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$$

Section 9.7 Two-sided Chevet and Dvoretzky–Milman Q9.41 ∼ Q9.43

Theorem 9.7.1 (*Two-sided Chevet inequality*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix, and let $\mathbb{T} \subset \mathbb{R}^n$ and $\mathbb{S} \subset \mathbb{R}^m$ be arbitrary sets. Then

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\sup_{\boldsymbol{y}\in\mathbb{S}} |\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{y}\rangle - w(\mathbb{S})\|\boldsymbol{x}\|_2|\right] \leqslant C\gamma(\mathbb{T})\,\mathrm{rad}(\mathbb{S}),$$

where $\gamma(\mathbb{T})$ is the Gaussian complexity of $\mathbb{T}$ (see Section 7.5.3) and $\mathrm{rad}(\mathbb{T}) = \sup_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{x}\|_2$ is the radius of $\mathbb{T}$.

Theorem 9.7.2 (*Dvoretzky–Milman theorem*)

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix, and let $\mathbb{T} \subset \mathbb{R}^n$. Then, with high probability,

$$r_-\mathbb{B}_2^m \subset \mathrm{conv}(\boldsymbol{A}\mathbb{T}) \subset r_+\mathbb{B}_2^m \text{ (w.h.p. 0.99)},$$

where $r_\pm = w(\mathbb{T}) \pm C\sqrt{m}\cdot\mathrm{rad}(\mathbb{T})$. The left inclusion holds only when $r_- \geqslant 0$; the right inclusion always holds.

Exercises

1 [**Inverse Triangle Inequality**] Prove the following geometric fact used in the proof of Theorem T9.1.2 (*Sub-Gaussian increments*). Consider vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ satisfying $1 = \|\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{y}\|_2$. Define

$$\bar{\boldsymbol{y}} = \frac{\boldsymbol{y}}{\|\boldsymbol{y}\|_2},$$

as the projection of $\boldsymbol{y}$ onto $\mathbb{S}^{n-1}$. Prove:

$$\|\boldsymbol{x} - \boldsymbol{y}\|_2 \leqslant \|\boldsymbol{x} - \bar{\boldsymbol{y}}\|_2 + \|\bar{\boldsymbol{y}} - \boldsymbol{y}\|_2 \leqslant \sqrt{2}\|\boldsymbol{x} - \boldsymbol{y}\|_2.$$

Prf. The left inequality follows from the triangle inequality for the Euclidean norm. We prove the right inequality. Let $R = \|\boldsymbol{y}\|_2 \geqslant 1$, so that $\boldsymbol{y} = R\bar{\boldsymbol{y}}$. Then

$$\begin{aligned}\langle \boldsymbol{x} - \bar{\boldsymbol{y}}, \boldsymbol{y} - \bar{\boldsymbol{y}}\rangle &= \langle \boldsymbol{x} - \bar{\boldsymbol{y}}, (R-1)\bar{\boldsymbol{y}}\rangle = (R-1)(\langle \boldsymbol{x}, \bar{\boldsymbol{y}}\rangle - \langle \bar{\boldsymbol{y}}, \bar{\boldsymbol{y}}\rangle)\\ &= (R-1)(\langle \boldsymbol{x}, \bar{\boldsymbol{y}}\rangle - 1) \leqslant 0,\end{aligned}$$

Expanding the square of the longest side $\|\boldsymbol{x}-\boldsymbol{y}\|_2^2$ by inserting $\bar{\boldsymbol{y}}$,

$$\begin{aligned}\|\boldsymbol{x}-\boldsymbol{y}\|_2^2 &= \|(\boldsymbol{x}-\bar{\boldsymbol{y}})-(\boldsymbol{y}-\bar{\boldsymbol{y}})\|_2^2 \\ &= \|\boldsymbol{x}-\bar{\boldsymbol{y}}\|_2^2+\|\boldsymbol{y}-\bar{\boldsymbol{y}}\|_2^2-2\langle \boldsymbol{x}-\bar{\boldsymbol{y}},\boldsymbol{y}-\bar{\boldsymbol{y}}\rangle \\ &\geqslant \|\boldsymbol{x}-\bar{\boldsymbol{y}}\|_2^2+\|\boldsymbol{y}-\bar{\boldsymbol{y}}\|_2^2.\end{aligned}$$

Using the algebraic inequality $(A+B)^2 \leqslant 2(A^2+B^2)$, we obtain

$$(\|\boldsymbol{x}-\bar{\boldsymbol{y}}\|_2+\|\bar{\boldsymbol{y}}-\boldsymbol{y}\|_2)^2 \leqslant 2(\|\boldsymbol{x}-\bar{\boldsymbol{y}}\|_2^2+\|\bar{\boldsymbol{y}}-\boldsymbol{y}\|_2^2) \leqslant 2\|\boldsymbol{x}-\boldsymbol{y}\|_2^2,$$

i.e.

$$\|\boldsymbol{x}-\bar{\boldsymbol{y}}\|_2+\|\bar{\boldsymbol{y}}-\boldsymbol{y}\|_2 \leqslant \sqrt{2}\|\boldsymbol{x}-\boldsymbol{y}\|_2.$$

2 [**Matrix Deviation Inequality**] Use Theorem T9.1.1 (*Matrix deviation inequality*) to derive the centered version of the matrix deviation inequality.

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix whose rows are independent, isotropic random vectors satisfying $\max_i \|\boldsymbol{A}_i\|_{\psi_2} \leqslant K$. Let $\mathbb{T} \subset \mathbb{R}^n$. The standard matrix deviation inequality states that

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2-\sqrt{m}\|\boldsymbol{x}\|_2|\right] \leqslant C_1K^2\gamma(\mathbb{T}), \qquad (C_1>0)$$

Prove the centered version of the matrix deviation inequality,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2-\mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2|\right] \leqslant CK^2\gamma(\mathbb{T}). \qquad (C>0)$$

Prf. Fix $\boldsymbol{x}\in\mathbb{T}$. Using the triangle inequality, we split the error as

$$|\|\boldsymbol{A}\boldsymbol{x}\|_2-\mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2| \leqslant |\|\boldsymbol{A}\boldsymbol{x}\|_2-\sqrt{m}\|\boldsymbol{x}\|_2|+|\sqrt{m}\|\boldsymbol{x}\|_2-\mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2|.$$

- Bounding the mean term by a supremum

  For the second term,

  $$|\sqrt{m}\|\boldsymbol{x}\|_2-\mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2| = |\mathbb{E}[\sqrt{m}\|\boldsymbol{x}\|_2-\|\boldsymbol{A}\boldsymbol{x}\|_2] \leqslant \mathbb{E}[|\sqrt{m}\|\boldsymbol{x}\|_2-\|\boldsymbol{A}\boldsymbol{x}\|_2|],$$

  and the expected absolute error is certainly no larger than the expected supremum of the absolute error over the whole set $\mathbb{T}$, namely

  $$\mathbb{E}[|\sqrt{m}\|\boldsymbol{x}\|_2-\|\boldsymbol{A}\boldsymbol{x}\|_2|] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{y}\in\mathbb{T}} |\sqrt{m}\|\boldsymbol{y}\|_2-\|\boldsymbol{A}\boldsymbol{y}\|_2|\right] = \Delta,$$

  Hence, for any $\boldsymbol{x}\in\mathbb{T}$,

  $$|\|\boldsymbol{A}\boldsymbol{x}\|_2-\mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2| \leqslant |\|\boldsymbol{A}\boldsymbol{x}\|_2-\sqrt{m}\|\boldsymbol{x}\|_2|+\Delta.$$

- Taking the supremum and the expectation

  Taking the supremum on both sides,

  $$\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2-\mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2| \leqslant \sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2-\sqrt{m}\|\boldsymbol{x}\|_2|+\Delta,$$

  and then taking the expectation on both sides,

  $$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2-\mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2|\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2-\sqrt{m}\|\boldsymbol{x}\|_2|\right]+\Delta = 2\Delta.$$

  Using Theorem T9.1.1 (*Matrix deviation inequality*) (*Matrix deviation inequality*) in its standard form $\Delta \leqslant C_1K^2\gamma(\mathbb{T})$, we obtain

  $$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2-\mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2|\right] \leqslant 2C_1K^2\gamma(\mathbb{T}).$$

  Setting $C=2C_1$

3 [**Quadratic Matrix Deviation**] Prove the claim in Remark R9.1.5 (*Matrix deviations of squares*).

Under the conditions of Theorem T9.1.1 (*Matrix deviation inequality*), let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ be a random matrix whose rows are

independent, isotropic random vectors satisfying $\max_i \|\boldsymbol{A}_i\|_{\psi_2} \leqslant K$. Let $\mathbb{T} \subset \mathbb{R}^n$. Prove that for any $\mathbb{T} \subset \mathbb{R}^n$, the quadratic form of the matrix deviation satisfies the following upper bound:

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{x}\|_2^2|\right] \leqslant CK^4\gamma(\mathbb{T})^2 + CK^2\sqrt{m}\,\mathrm{rad}(\mathbb{T})\gamma(\mathbb{T}). \qquad (C>0)$$

Prf. Let $\boldsymbol{x} \in \mathbb{T}$. Rewriting the target quantity,

$$|\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{x}\|_2^2| = |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2| \cdot (\|\boldsymbol{A}\boldsymbol{x}\|_2 + \sqrt{m}\|\boldsymbol{x}\|_2).$$

Define the linear deviation process $Z_{\boldsymbol{x}} = |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2|$ with supremum $Z = \sup_{\boldsymbol{x}\in\mathbb{T}} Z_{\boldsymbol{x}}$. By the triangle inequality,

$$\|\boldsymbol{A}\boldsymbol{x}\|_2 \leqslant \sqrt{m}\|\boldsymbol{x}\|_2 + Z_{\boldsymbol{x}},$$

Substituting this in,

$$\|\boldsymbol{A}\boldsymbol{x}\|_2 + \sqrt{m}\|\boldsymbol{x}\|_2 \leqslant Z_{\boldsymbol{x}} + 2\sqrt{m}\|\boldsymbol{x}\|_2,$$

so the quadratic deviation is bounded by

$$|\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{x}\|_2^2| \leqslant Z_{\boldsymbol{x}}(Z_{\boldsymbol{x}} + 2\sqrt{m}\|\boldsymbol{x}\|_2) = Z_{\boldsymbol{x}}^2 + 2\sqrt{m}\|\boldsymbol{x}\|_2 Z_{\boldsymbol{x}}.$$

- Taking the supremum and the expectation

  Taking the supremum on both sides,

  $$\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{x}\|_2^2| \leqslant \sup_{\boldsymbol{x}\in\mathbb{T}}(Z_{\boldsymbol{x}}^2) + 2\sqrt{m}\sup_{\boldsymbol{x}\in\mathbb{T}}(\|\boldsymbol{x}\|_2 Z_{\boldsymbol{x}}),$$

  where $\sup Z_{\boldsymbol{x}}^2 = Z^2$ and $\sup(\|\boldsymbol{x}\|_2 Z_{\boldsymbol{x}}) \leqslant (\sup\|\boldsymbol{x}\|_2)\cdot(\sup Z_{\boldsymbol{x}}) = \mathrm{rad}(\mathbb{T})\cdot Z$. Substituting,

  $$\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{x}\|_2^2| \leqslant Z^2 + 2\sqrt{m}\,\mathrm{rad}(\mathbb{T})Z,$$

  and taking expectations on both sides,

  $$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{x}\|_2^2|\right] \leqslant \mathbb{E}[Z^2] + 2\sqrt{m}\,\mathrm{rad}(\mathbb{T})\mathbb{E}[Z]. \qquad (\mathrm{I})$$

- Computing the moments and substituting

  By Theorem T9.1.1 (*Matrix deviation inequality*) (*Matrix deviation inequality*),

  $$\mathbb{E}[Z] \leqslant C_1K^2\gamma(\mathbb{T}).$$

  By the moment version of Corollary C8.5.6 (*Talagrand comparison inequality*) (*Talagrand comparison inequality*),

  $$\|Z\|_{L_p} \leqslant C_2\sqrt{p}K^2\gamma(\mathbb{T}),$$

  and taking $p = 2$,

  $$\mathbb{E}[Z^2] \leqslant (C_2\sqrt{2}K^2\gamma(\mathbb{T}))^2 = 2C_2^2K^4\gamma(\mathbb{T})^2;$$

  substituting into (I),

  $$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{x}\|_2^2|\right] \leqslant 2C_2^2K^4\gamma(\mathbb{T})^2 + 2\sqrt{m}\,\mathrm{rad}(\mathbb{T})(C_1K^2\gamma(\mathbb{T})).$$

  Setting $C = \max(2C_2^2, 2C_1)$

4 [**Anisotropic Matrix Deviation**] In Theorem T9.1.1 (*Matrix deviation inequality*), the row vectors of the random matrix are isotropic. Remove this assumption.

Let $\boldsymbol{B} \in \mathbb{R}^{m\times n}$ be a random matrix whose rows $\boldsymbol{B}_i$ are independent and mean-zero[1]and satisfy $\mathbb{E}[\boldsymbol{B}_i\boldsymbol{B}_i^\top] = \boldsymbol{\Sigma}$. Suppose that for any $\boldsymbol{x} \in \mathbb{R}^n$, the sub-Gaussian condition $\|\langle\boldsymbol{B}_i,\boldsymbol{x}\rangle\|_{\psi_2} \leqslant K\|\langle\boldsymbol{B}_i,\boldsymbol{x}\rangle\|_{L_2}$ holds. Prove that for any $\mathbb{T} \subset \mathbb{R}^n$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{B}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\|_2|\right] \leqslant CK^2\gamma(\boldsymbol{\Sigma}^{1/2}\mathbb{T}). \qquad (C>0)$$

Prf. We have

$$\begin{aligned}\|\langle\boldsymbol{B}_i,\boldsymbol{x}\rangle\|_{L_2}^2 &= \mathbb{E}[\langle\boldsymbol{B}_i,\boldsymbol{x}\rangle^2] = \mathbb{E}[\boldsymbol{x}^\top\boldsymbol{B}_i\boldsymbol{B}_i^\top\boldsymbol{x}] = \boldsymbol{x}^\top\mathbb{E}[\boldsymbol{B}_i\boldsymbol{B}_i^\top]\boldsymbol{x} = \boldsymbol{x}^\top\boldsymbol{\Sigma}\boldsymbol{x}\\ &= \boldsymbol{x}^\top\boldsymbol{\Sigma}^{1/2}\boldsymbol{\Sigma}^{1/2}\boldsymbol{x} = \|\boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\|_2^2.\end{aligned}$$

Thus the sub-Gaussian condition reads

$$\|\langle \boldsymbol{B}_i, \boldsymbol{x}\rangle\|_{\psi_2} \leqslant K\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\|_2.$$

- Whitening on the range of the covariance

  Let $R = \mathrm{im}(\boldsymbol{\Sigma})$ and let $\boldsymbol{P}_R$ be the orthogonal projection onto R. If $\boldsymbol{v} \in R^\perp = \ker(\boldsymbol{\Sigma})$, then

$$\mathbb{E}[\langle \boldsymbol{B}_i, \boldsymbol{v}\rangle^2] = \boldsymbol{v}^\top \boldsymbol{\Sigma}\boldsymbol{v} = 0.$$

  Thus $\langle \boldsymbol{B}_i, \boldsymbol{v}\rangle = 0$ almost surely. Applying this to an orthonormal basis of $R^\perp$ gives $\boldsymbol{B}_i = \boldsymbol{P}_R\boldsymbol{B}_i$ almost surely.

  Regard $S = \boldsymbol{\Sigma}^{1/2}|_R : R \to R$ as an invertible operator and define $\boldsymbol{A}_i = S^{-1}\boldsymbol{B}_i \in R$. For every $\boldsymbol{u} \in R$,

$$\mathbb{E}[\langle \boldsymbol{A}_i, \boldsymbol{u}\rangle^2] = \mathbb{E}[\langle \boldsymbol{B}_i, S^{-1}\boldsymbol{u}\rangle^2] = \|SS^{-1}\boldsymbol{u}\|_2^2 = \|\boldsymbol{u}\|_2^2,$$

  while

$$\|\langle \boldsymbol{A}_i, \boldsymbol{u}\rangle\|_{\psi_2} \leqslant K\|SS^{-1}\boldsymbol{u}\|_2 = K\|\boldsymbol{u}\|_2.$$

  Hence the $\boldsymbol{A}_i$ are independent, isotropic K-sub-Gaussian random vectors in R.

- Applying the standard matrix deviation theorem

  For every $\boldsymbol{x} \in \mathbb{T}$, the identity $\boldsymbol{B}_i = \boldsymbol{P}_R\boldsymbol{B}_i$ gives

$$\langle \boldsymbol{B}_i, \boldsymbol{x}\rangle = \langle \boldsymbol{A}_i, S\boldsymbol{P}_R\boldsymbol{x}\rangle,$$
$$S\boldsymbol{P}_R\boldsymbol{x} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{x}.$$

  Applying the standard matrix deviation theorem in R to $\mathbb{T}' = \boldsymbol{\Sigma}^{1/2}\mathbb{T}$ yields

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{B}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\|_2|\right] \leqslant CK^2\gamma(\mathbb{T}') = CK^2\gamma(\boldsymbol{\Sigma}^{1/2}\mathbb{T}).$$

5 [**Quadratic Empirical Process**] Exercise Q8.36 (*Empirical Processes: General Chaining Bound*) bounds the deviation between the empirical mean and the sample mean for any sub-Gaussian process.

Extend Theorem T9.1.1 (*Matrix deviation inequality*) to the L_2 mean. Let $\mathcal{F}$ be a star-shaped, separable[2]family of real-valued functions on a domain Ω, and let $X_1, \cdots, X_m$ be i.i.d. random variables on Ω. Consider the L_2 metric on $\mathcal{F}$ given by $d(f,g) = (\mathbb{E}[(f(X_1) - g(X_1))^2])^{1/2}$, and assume that the process increments satisfy the sub-Gaussian condition $\|f(X_1) - g(X_1)\|_{\psi_2} \leqslant Kd(f,g)$.

(1) Prove:

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\left(\frac{1}{m}\sum_{i=1}^m f(X_i)^2\right)^{1/2} - (\mathbb{E}[f(X_1)^2])^{1/2}\right|\right] \leqslant \frac{CK^2\gamma_2(\mathcal{F}, d)}{\sqrt{m}}. \qquad (C > 0)$$

(2) When $\mathbb{T} \subset \mathbb{S}^{n-1}$, this result implies the standard matrix deviation theorem; analyze why.

Prf. (1) Multiplying both sides of the target inequality by $\sqrt{m}$, define the random process $(Z_f)_{f\in\mathcal{F}}$, anchored at 0, by

$$Z_f = \left(\sum_{i=1}^m f(X_i)^2\right)^{1/2} - \sqrt{m}\|f\|_{L_2},$$

where $\|f\|_{L_2} = (\mathbb{E}[f(X_1)^2])^{1/2} = d(f,0)$.

- Sub-Gaussian increments in the equal-norm case

  Suppose $f, g \in \mathcal{F}$ satisfy $\|f\|_{L_2} = \|g\|_{L_2} = r > 0$. Then $\sqrt{m}\|f\|_{L_2} = \sqrt{m}\|g\|_{L_2}$, so the mean terms cancel,

$$Z_f - Z_g = \left(\sum_{i=1}^m f(X_i)^2\right)^{1/2} - \left(\sum_{i=1}^m g(X_i)^2\right)^{1/2}$$
$$= \frac{\sum_{i=1}^m (f(X_i)^2 - g(X_i)^2)}{\left(\sum_{i=1}^m f(X_i)^2\right)^{1/2} + \left(\sum_{i=1}^m g(X_i)^2\right)^{1/2}} = \frac{N}{D}.$$

- Analyzing the numerator N

Let $W_i = f(X_i)^2 - g(X_i)^2 = (f(X_i) - g(X_i))(f(X_i) + g(X_i))$. Then W_i are i.i.d. mean-zero sub-exponential random variables. (*Since $\|f\|_{L_2} = \|g\|_{L_2}$, we have $\mathbb{E}[W_i] = 0$; since $f(X_i) - g(X_i)$ and $f(X_i) + g(X_i)$ are sub-Gaussian, W_i is sub-exponential.*)

Since

$$\|f(X_i) - g(X_i)\|_{\psi_2} \leqslant Kd(f, g),$$
$$\|f(X_i) + g(X_i)\|_{\psi_2} \leqslant \|f(X_i)\|_{\psi_2} + \|g(X_i)\|_{\psi_2} \leqslant 2Kr,$$

the ψ_1 norm bound for products of sub-Gaussian random variables gives

$$\|W_i\|_{\psi_1} \leqslant C_0 \|f(X_i) - g(X_i)\|_{\psi_2} \cdot \|f(X_i) + g(X_i)\|_{\psi_2} \leqslant C_0 K d(f, g) \cdot 2Kr = 2C_0 K^2 r d(f, g).$$

Hence, by Theorem T2.9.1 (*Subexponential Bernstein inequality*) (*Subexponential Bernstein inequality*), for any $t \geqslant 0$,

$$\mathbb{P}\{|N| \geqslant t\} \leqslant 2\exp\left(-c\min\left(\frac{t^2}{m(2C_0K^2rd(f,g))^2}, \frac{t}{2C_0K^2rd(f,g)}\right)\right).$$

- Analyzing the denominator D

(*The numerator N is sub-exponential, whereas the ratio N/D must be controlled on the sub-Gaussian scale. We therefore distinguish two regimes.*)

Let $U_i = r^2 - f(X_i)^2$. Then U_i are mean-zero sub-exponential variables with $\|U_i\|_{\psi_1} \leqslant c_0 K^2 r^2$.

(By star-shapedness, taking $g = 0$ shows that $f(X_i)$ is sub-Gaussian; hence $f(X_i)^2$ is sub-exponential and $\|U_i\|_{\psi_1} \leqslant c_0 K^2 r^2$.)

Note that

$$\sum_{i=1}^m f(X_i)^2 = \sum_{i=1}^m (r^2 - U_i) = mr^2 - \sum_{i=1}^m U_i,$$

therefore

$$\mathbb{P}\left\{\sum_{i=1}^m f(X_i)^2 \leqslant \frac{1}{4}mr^2\right\} = \mathbb{P}\left\{\sum_{i=1}^m U_i \geqslant \frac{3}{4}mr^2\right\}.$$

Applying Theorem T2.9.1 (*Subexponential Bernstein inequality*) (*Subexponential Bernstein inequality*) to $\sum_{i=1}^m U_i$, for any $t \geqslant 0$,

$$\mathbb{P}\left\{\sum_{i=1}^m U_i \geqslant t\right\} \leqslant 2\exp\left(-c\min\left(\frac{t^2}{m\|U_i\|_{\psi_1}^2}, \frac{t}{\|U_i\|_{\psi_1}}\right)\right). \tag{I}$$

Taking $t = \frac{3}{4}mr^2$ and substituting $\|U_i\|_{\psi_1} \leqslant c_0 K^2 r^2$,

$$\mathbb{P}\left\{\sum_{i=1}^m U_i \geqslant t\right\} \leqslant 2\exp\left(-c\min\left(\frac{9}{16c_0^2}\cdot\frac{m}{K^4}, \frac{3}{4c_0}\cdot\frac{m}{K^2}\right)\right)$$
$$\leqslant 2\exp\left(-\frac{c_1 m}{K^4}\right),$$

Substituting into (I),

$$\mathbb{P}\left\{\sum_{i=1}^m f(X_i)^2 \leqslant \frac{1}{4}mr^2\right\} \leqslant 2\exp\left(-\frac{c_1 m}{K^4}\right);$$

similarly,

$$\mathbb{P}\left\{\sum_{i=1}^m g(X_i)^2 \leqslant \frac{1}{4}mr^2\right\} \leqslant 2\exp\left(-\frac{c_1' m}{K^4}\right).$$

When $D = 0$, all $f(X_i)$ and $g(X_i)$ vanish, so $N = 0$; below we use the ratio N/D on $D > 0$ and define it to be 0

on $D = 0$. Define the benign event

$$\mathcal{E} = \left\{ \sum_{i=1}^{m} f(X_i)^2 \geqslant \frac{1}{4} m r^2 \right\} \cap \left\{ \sum_{i=1}^{m} g(X_i)^2 \geqslant \frac{1}{4} m r^2 \right\}.$$

Then:

1. The bad event $\mathcal{E}^c$ has probability $\mathbb{P}(\mathcal{E}^c) \leqslant 4 \exp(-\frac{c_1 m}{K^4})$;
2. On $\mathcal{E}$, we have $D = (\sum_{i=1}^{m} f(X_i)^2)^{1/2} + (\sum_{i=1}^{m} g(X_i)^2)^{1/2} \geqslant \sqrt{m} r$.

(*Next we estimate the tail probability* $\mathbb{P}\{|Z_f - Z_g| \geqslant u\}$.)

We split into two regimes according to u:

a. When $u \leqslant 2\sqrt{m} d(f, g)$: on $\mathcal{E}$, $D \geqslant \sqrt{m} r$, so $\frac{|N|}{D} \leqslant \frac{|N|}{\sqrt{m} r}$, and

$$\mathbb{P}\left\{ \frac{|N|}{D} \geqslant u \right\} \leqslant \mathbb{P}\{|N| \geqslant u\sqrt{m} r\} + \mathbb{P}(\mathcal{E}^c).$$

For the first term, by Theorem T2.9.1 (*Subexponential Bernstein inequality*) (*Subexponential Bernstein inequality*) with $t = u\sqrt{m} r$,

$$\begin{aligned} \mathbb{P}\{|N| \geqslant u\sqrt{m} r\} &\leqslant 2 \exp\left(-c \min\left(\frac{u^2}{4 C_0^2 K^4 d(f, g)^2}, \frac{u\sqrt{m}}{2 C_0 K^2 d(f, g)} \right) \right) \\ &\asymp 2 \exp\left(-c \min\left(\frac{u^2}{K^4 d(f, g)^2}, \frac{u\sqrt{m}}{K^2 d(f, g)} \right) \right), \end{aligned}$$

where

$$\frac{u^2}{K^4 d(f, g)^2} \Big/ \frac{u\sqrt{m}}{K^2 d(f, g)} = \frac{u}{K^2 \sqrt{m} d(f, g)} \leqslant \frac{2}{K^2},$$

so the quadratic term dominates the exponent:

$$\mathbb{P}\{|N| \geqslant u\sqrt{m} r\} \leqslant 2 \exp\left(-\frac{c_2 u^2}{K^4 d(f, g)^2} \right),$$

Also, $u^2 \leqslant 4 m d(f, g)^2$, so $\mathbb{P}(\mathcal{E}^c) \leqslant 4 \exp(-\frac{c_1 u^2}{4 K^4 d(f, g)^2})$. Combining the two terms gives

$$\mathbb{P}\{|Z_f - Z_g| \geqslant u\} \leqslant C \exp\left(-\frac{c_3 u^2}{K^4 d(f, g)^2} \right).$$

b. When $u > 2\sqrt{m} d(f, g)$: let $\boldsymbol{a} = (f(X_1), \cdots, f(X_m))^\top$ and $\boldsymbol{b} = (g(X_1), \cdots, g(X_m))^\top$. Then

$$|Z_f - Z_g| = \big| \|\boldsymbol{a}\|_2 - \|\boldsymbol{b}\|_2 \big| \leqslant \|\boldsymbol{a} - \boldsymbol{b}\|_2 \leqslant \left(\sum_{i=1}^{m} (f(X_i) - g(X_i))^2 \right)^{1/2},$$

hence

$$\mathbb{P}\{|Z_f - Z_g| \geqslant u\} \leqslant \mathbb{P}\left\{ \sum_{i=1}^{m} (f(X_i) - g(X_i))^2 \geqslant u^2 \right\}.$$

Let $V_i = (f(X_i) - g(X_i))^2 - d(f, g)^2$. Then V_i are i.i.d. mean-zero sub-exponential random variables with $\|V_i\|_{\psi_1} \leqslant c_4 K^2 d(f, g)^2$. By Theorem T2.9.1 (*Subexponential Bernstein inequality*) (*Subexponential Bernstein inequality*),

$$\mathbb{P}\left\{ \sum_{i=1}^{m} V_i \geqslant t \right\} \leqslant 2 \exp\left(-c \min\left(\frac{t^2}{m K^4 d(f, g)^4}, \frac{t}{K^2 d(f, g)^2} \right) \right).$$

Taking $t = u^2 - m d(f, g)^2$. When $u > 2\sqrt{m} d(f, g)$, we have $t > \frac{3}{4} u^2$. Assume $K \geqslant 1$. Substituting,

$$\mathbb{P}\{|Z_f - Z_g| \geqslant u\} \leqslant 2 \exp\left(-c \min\left(\frac{t^2}{m K^4 d(f, g)^4}, \frac{t}{K^2 d(f, g)^2} \right) \right)$$

$$\leqslant 2\exp\left(-c\min\left(\frac{u^4}{mK^4 d(f,g)^4}, \frac{u^2}{K^2 d(f,g)^2}\right)\right)$$
$$\leqslant 2\exp\left(-\frac{c_5 u^2}{K^4 d(f,g)^2}\right),$$

Thus, for any $u \geqslant 0$,

$$\mathbb{P}\{|Z_f - Z_g| \geqslant u\} \leqslant C\exp\left(-\frac{c_6 u^2}{K^4 d(f,g)^2}\right),$$

and by the equivalent characterization of the ψ_2 norm,

$$\|Z_f - Z_g\|_{\psi_2} \leqslant C_1 K^2 d(f,g). \tag{II}$$

- Extending to arbitrary norms by radial scaling, then using a comparison inequality

For arbitrary $f, g \in \mathcal{F}$, assume without loss of generality that $0 < \|f\|_{L_2} \leqslant \|g\|_{L_2}$.

Since $\mathcal{F}$ is star-shaped, define $\bar{g} = \frac{\|f\|_{L_2}}{\|g\|_{L_2}} g \in \mathcal{F}$; then $\|\bar{g}\|_{L_2} = \|f\|_{L_2}$.

Decompose the increment as

$$Z_f - Z_g = (Z_f - Z_{\bar{g}}) + (Z_{\bar{g}} - Z_g).$$

For the first term (*the equal-norm part*), using the conclusion of (II),

$$\|Z_f - Z_{\bar{g}}\|_{\psi_2} \leqslant C_1 K^2 d(f,\bar{g}). \tag{III}$$

For the second term (*the radial-scaling part*), note that $g = \frac{\|g\|_{L_2}}{\|f\|_{L_2}}\bar{g}$, so $Z_g = \frac{\|g\|_{L_2}}{\|f\|_{L_2}} Z_{\bar{g}}$, and

$$Z_{\bar{g}} - Z_g = \left(1 - \frac{\|g\|_{L_2}}{\|f\|_{L_2}}\right) Z_{\bar{g}}.$$

(*Star-shapedness guarantees* $0 \in \mathcal{F}$*, hence* $Z_0 = 0$). Using the conclusion of (II), $\|Z_{\bar{g}}\|_{\psi_2} = \|Z_{\bar{g}} - Z_0\|_{\psi_2} \leqslant C_1 K^2 \|\bar{g}\|_{L_2}$, so

$$\|Z_{\bar{g}} - Z_g\|_{\psi_2} \leqslant C_1 K^2 \left|1 - \frac{\|g\|_{L_2}}{\|f\|_{L_2}}\right| \|\bar{g}\|_{L_2} = C_1 K^2 |\|f\|_{L_2} - \|g\|_{L_2}| = C_1 K^2 d(\bar{g}, g). \tag{IV}$$

Combining (III) and (IV),

$$\|Z_f - Z_g\|_{\psi_2} \leqslant C_1 K^2 (d(f,\bar{g}) + d(\bar{g},g)).$$

Using the inverse triangle inequality $d(f,\bar{g}) + d(\bar{g},g) \leqslant \sqrt{2} d(f,g)$ (*see Exercise Q9.1 (Inverse Triangle Inequality)*),

$$\|Z_f - Z_g\|_{\psi_2} \leqslant \sqrt{2} C_1 K^2 d(f,g) = C_2 K^2 d(f,g).$$

Since $Z_0 = 0$, both $(Z_f)_{f\in\mathcal{F}}$ and $(-Z_f)_{f\in\mathcal{F}}$ start at 0 and share the increment bound $C_2 K^2 d(f,g)$. By Corollary C8.5.6 (*Talagrand comparison inequality*) (*Talagrand comparison inequality*),

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} Z_f\right], \mathbb{E}\left[\sup_{f\in\mathcal{F}}(-Z_f)\right] \leqslant C_3 C_2 K^2 \gamma_2(\mathcal{F}, d),$$

hence

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}} |Z_f|\right] \leqslant \mathbb{E}\left[\sup_{f\in\mathcal{F}} Z_f\right] + \mathbb{E}\left[\sup_{f\in\mathcal{F}}(-Z_f)\right] \leqslant 2C_3 C_2 K^2 \gamma_2(\mathcal{F}, d),$$

and setting $C = 2C_3 C_2$

(2) Let $\boldsymbol{X} \in \mathbb{R}^n$ be an isotropic random vector with the same distribution as the rows of $\boldsymbol{A}$, so that $\boldsymbol{A}_1, \cdots, \boldsymbol{A}_m$ are i.i.d. copies of $\boldsymbol{X}$.

Let $\mathbb{T} \subset \mathbb{S}^{n-1}$ and define the family of linear functions

$$\mathcal{F} = \{f_{\boldsymbol{x}}(\boldsymbol{w}) = \langle \boldsymbol{x}, \boldsymbol{w}\rangle : \boldsymbol{x} \in \mathbb{T}\}.$$

Then

$$d(f_{\boldsymbol{x}}, f_{\boldsymbol{y}})^2 = \mathbb{E}[\langle \boldsymbol{x}-\boldsymbol{y}, \boldsymbol{X}\rangle^2] = (\boldsymbol{x}-\boldsymbol{y})^\top \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top](\boldsymbol{x}-\boldsymbol{y}) = \|\boldsymbol{x}-\boldsymbol{y}\|_2^2,$$

i.e. $d = \|\cdot\|_2$, so $\gamma_2(\mathcal{F}, d) \asymp \gamma(\mathbb{T})$.

By the assumption of the exercise,

$$\|f_{\boldsymbol{x}}(\boldsymbol{X}) - f_{\boldsymbol{y}}(\boldsymbol{X})\|_{\psi_2} = \|\langle \boldsymbol{x}-\boldsymbol{y}, \boldsymbol{X}\rangle\|_{\psi_2} \leqslant K\|\boldsymbol{x}-\boldsymbol{y}\|_2 = K d(f_{\boldsymbol{x}}, f_{\boldsymbol{y}}).$$

For any $\boldsymbol{x} \in \mathbb{S}^{n-1}$, since $\|f_{\boldsymbol{x}}\|_{L_2}^2 = \mathbb{E}[\langle \boldsymbol{x}, \boldsymbol{X}\rangle^2] = \|\boldsymbol{x}\|_2^2 = 1$ and $\sum_{i=1}^m f_{\boldsymbol{x}}(\boldsymbol{X}_i)^2 = \|\boldsymbol{A}\boldsymbol{x}\|_2^2$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left| \frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\sqrt{m}} - 1 \right|\right] \leqslant \frac{CK^2\gamma_2(\mathcal{F}, d)}{\sqrt{m}} \asymp \frac{CK^2\gamma(\mathbb{T})}{\sqrt{m}},$$

Multiplying both sides by $\sqrt{m}$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}|\right] \leqslant CK^2\gamma(\mathbb{T}).$$

6 [**Deviation of Random Projections**] Prove a random-projection version of Theorem T9.1.1 (*Matrix deviation inequality*). Let $\boldsymbol{P}$ be the orthogonal projection in $\mathbb{R}^n$ onto an m-dimensional subspace uniformly distributed in the Grassmannian $G_{n,m}$. Prove: for every set $\mathbb{T} \subset \mathbb{R}^n$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left| \|\boldsymbol{P}\boldsymbol{x}\|_2 - \sqrt{\frac{m}{n}}\|\boldsymbol{x}\|_2 \right|\right] \leqslant \frac{C\gamma(\mathbb{T})}{\sqrt{n}}. \qquad (C>0)$$

Prf. Multiplying both sides of the target inequality by $\sqrt{n}$, it suffices to prove

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\sqrt{n}\|\boldsymbol{P}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2\right|\right] \leqslant C\gamma(\mathbb{T}).$$

- Connecting random projections with Gaussian matrices

  Let $\boldsymbol{G} \in \mathbb{R}^{m\times n}$ be a standard Gaussian matrix.

  Use the polar decomposition $\boldsymbol{G} = \boldsymbol{H}\boldsymbol{V}^\top$, where $\boldsymbol{H} = (\boldsymbol{G}\boldsymbol{G}^\top)^{1/2}$ and $\boldsymbol{V} \in \mathbb{R}^{n\times m}$ satisfies $\boldsymbol{V}^\top\boldsymbol{V} = \boldsymbol{I}_m$. By the rotational invariance of the Gaussian distribution:

  1. The column space of $\boldsymbol{V}$ is uniformly distributed in $G_{n,m}$, hence $\boldsymbol{P} = \boldsymbol{V}\boldsymbol{V}^\top$.
  2. $\boldsymbol{V}$ and $\boldsymbol{H}$ are independent, and $\boldsymbol{H}^2 = \boldsymbol{G}\boldsymbol{G}^\top \sim W_m(n, \boldsymbol{I}_m)$.

  Moreover,

  for any $\boldsymbol{x} \in \mathbb{R}^n$, let $\boldsymbol{u}_{\boldsymbol{x}} = \boldsymbol{V}^\top\boldsymbol{x} \in \mathbb{R}^m$; then

  $$\|\boldsymbol{P}\boldsymbol{x}\|_2 = \|\boldsymbol{V}\boldsymbol{V}^\top\boldsymbol{x}\|_2 = \|\boldsymbol{u}_{\boldsymbol{x}}\|_2, \|\boldsymbol{G}\boldsymbol{x}\|_2 = \|\boldsymbol{H}\boldsymbol{u}_{\boldsymbol{x}}\|_2.$$

- Splitting the error by the triangle inequality

  Split the deviation term $|\sqrt{n}\|\boldsymbol{P}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2|$ as

  $$\begin{aligned} |\sqrt{n}\|\boldsymbol{P}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2| &= |\sqrt{n}\|\boldsymbol{u}_{\boldsymbol{x}}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2| \\ &\leqslant |\sqrt{n}\|\boldsymbol{u}_{\boldsymbol{x}}\|_2 - \|\boldsymbol{H}\boldsymbol{u}_{\boldsymbol{x}}\|_2| + |\|\boldsymbol{H}\boldsymbol{u}_{\boldsymbol{x}}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2| \\ &\leqslant |\sqrt{n}\|\boldsymbol{u}_{\boldsymbol{x}}\|_2 - \|\boldsymbol{H}\boldsymbol{u}_{\boldsymbol{x}}\|_2| + |\|\boldsymbol{G}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2|. \end{aligned}$$

  Taking the supremum and the expectation on both sides,

  $$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\sqrt{n}\|\boldsymbol{P}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2\right|\right] \leqslant \underbrace{\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\|\boldsymbol{H}\boldsymbol{u}_{\boldsymbol{x}}\|_2 - \sqrt{n}\|\boldsymbol{u}_{\boldsymbol{x}}\|_2\right|\right]}_{I_1} + \underbrace{\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\|\boldsymbol{G}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2\right|\right]}_{I_2}.$$

- Bounding the two error terms separately

  For I_2, by the standard matrix deviation theorem,

  $$\mathbb{E}_{\boldsymbol{G}}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\|\boldsymbol{G}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2\right|\right] \leqslant C_1\gamma(\mathbb{T}).$$

For I_1, condition on $\boldsymbol{V}$; then $\mathbb{U} = \{\boldsymbol{u}_{\boldsymbol{x}} = \boldsymbol{V}^\top \boldsymbol{x} : \boldsymbol{x} \in \mathbb{T}\} \subset \mathbb{R}^m$ is a deterministic bounded set.

Introduce a standard Gaussian matrix $\tilde{\boldsymbol{G}} \in \mathbb{R}^{n\times m}$. Since $\boldsymbol{H}^2 \overset{d}{=} \tilde{\boldsymbol{G}}^\top \tilde{\boldsymbol{G}}$, for every $\boldsymbol{u}$ we have $\|\boldsymbol{H}\boldsymbol{u}\|_2^2 \overset{d}{=} \|\tilde{\boldsymbol{G}}\boldsymbol{u}\|_2^2$. Hence

$$\mathbb{E}_{\boldsymbol{H}}\left[\sup_{\boldsymbol{u}\in\mathbb{U}} \left|\|\boldsymbol{H}\boldsymbol{u}\|_2 - \sqrt{n}\|\boldsymbol{u}\|_2\right|\right] = \mathbb{E}_{\tilde{\boldsymbol{G}}}\left[\sup_{\boldsymbol{u}\in\mathbb{U}} \left|\|\tilde{\boldsymbol{G}}\boldsymbol{u}\|_2 - \sqrt{n}\|\boldsymbol{u}\|_2\right|\right].$$

By the standard matrix deviation theorem,

$$\mathbb{E}_{\boldsymbol{H}}\left[\sup_{\boldsymbol{u}\in\mathbb{U}} \left|\|\boldsymbol{H}\boldsymbol{u}\|_2 - \sqrt{n}\|\boldsymbol{u}\|_2\right|\right] \leqslant C_2\gamma(\mathbb{U}).$$

(*It remains to bound* $\gamma(\mathbb{U})$ *in terms of* $\gamma(\mathbb{T})$.)

Let $\boldsymbol{h} \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$ satisfy $\boldsymbol{V}\boldsymbol{h} \overset{d}{=} \boldsymbol{P}\boldsymbol{g}$ ($\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$). Then

$$\gamma(\mathbb{U}) = \mathbb{E}_{\boldsymbol{h}}\left[\sup_{\boldsymbol{u}\in\mathbb{U}} \langle \boldsymbol{h}, \boldsymbol{u}\rangle\right] = \mathbb{E}_{\boldsymbol{h}}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{V}\boldsymbol{h}, \boldsymbol{x}\rangle\right] = \mathbb{E}_{\boldsymbol{h}}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{P}\boldsymbol{x}\rangle\right] = \mathbb{E}_{\boldsymbol{g}}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{P}\boldsymbol{x}\rangle\right].$$

Comparing the centered Gaussian processes $X_{\boldsymbol{x}} = \langle \boldsymbol{g}, \boldsymbol{P}\boldsymbol{x}\rangle$ and $Y_{\boldsymbol{x}} = \langle \boldsymbol{g}, \boldsymbol{x}\rangle$, since

$$\mathbb{E}[(X_{\boldsymbol{x}} - X_{\boldsymbol{y}})^2] = \|\boldsymbol{P}(\boldsymbol{x}-\boldsymbol{y})\|_2^2 \leqslant \|\boldsymbol{x}-\boldsymbol{y}\|_2^2 = \mathbb{E}[(Y_{\boldsymbol{x}} - Y_{\boldsymbol{y}})^2],$$

Theorem T7.2.8 (*Sudakov-Fernique inequality*) (*Sudakov–Fernique inequality*) gives

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} X_{\boldsymbol{x}}\right] \leqslant \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} Y_{\boldsymbol{x}}\right], \text{ hence } \gamma(\mathbb{U}) \leqslant \gamma(\mathbb{T}),$$

Removing the conditioning on $\boldsymbol{V}$,

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\|\boldsymbol{H}\boldsymbol{u}_{\boldsymbol{x}}\|_2 - \sqrt{n}\|\boldsymbol{u}_{\boldsymbol{x}}\|_2\right|\right] \leqslant C_2\gamma(\mathbb{T}).$$

Thus

$$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \left|\sqrt{n}\|\boldsymbol{P}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2\right|\right] \leqslant C_2\gamma(\mathbb{T}) + C_1\gamma(\mathbb{T}) = (C_1 + C_2)\gamma(\mathbb{T}),$$

and setting $C = C_1 + C_2$

7 [**Size of Projections: High-Probability Bound**] Give a high-probability version of Proposition P9.2.1 (*Size of random projections of sets*).

Let $\mathbb{T} \subset \mathbb{R}^n$, and let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix with independent, isotropic K-sub-Gaussian rows. Consider the scaled random linear map $\boldsymbol{Q} = \frac{1}{\sqrt{n}}\boldsymbol{A}$. Prove that for any $\varepsilon > 0$,

$$\operatorname{diam}(\boldsymbol{Q}\mathbb{T}) \leqslant (1+\varepsilon)\sqrt{\frac{m}{n}}\operatorname{diam}(\mathbb{T}) + CK^2 w_s(\mathbb{T}) \left(\text{w.h.p. } 1 - 2\exp\left(-\frac{c\varepsilon^2 m}{K^4}\right)\right).$$

Prf. • Introducing the high-probability matrix deviation inequality

By the high-probability version of the matrix deviation inequality, for any bounded set $\mathbb{S} \subset \mathbb{R}^n$ and any $u \geqslant 0$,

$$\sup_{\boldsymbol{z}\in\mathbb{S}} \left|\|\boldsymbol{A}\boldsymbol{z}\|_2 - \sqrt{m}\|\boldsymbol{z}\|_2\right| \leqslant C_1K^2\gamma(\mathbb{S}) + C_1K^2 u \cdot \operatorname{rad}(\mathbb{S}) \text{ (w.h.p. } 1 - 2\exp(-u^2)). \tag{I}$$

• Applying the inequality to the difference set $\mathbb{T} - \mathbb{T}$

Take $\mathbb{S} = \mathbb{T} - \mathbb{T}$. Its radius is

$$\operatorname{rad}(\mathbb{T} - \mathbb{T}) = \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \|\boldsymbol{x} - \boldsymbol{y}\|_2 = \operatorname{diam}(\mathbb{T}),$$

and its Gaussian complexity is

$$\gamma(\mathbb{T} - \mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x} - \boldsymbol{y}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] + \mathbb{E}\left[\sup_{\boldsymbol{y}\in\mathbb{T}} \langle \boldsymbol{g}, -\boldsymbol{y}\rangle\right] = 2w(\mathbb{T}).$$

Substituting into (I),

$$\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \left|\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2 - \sqrt{m}\|\boldsymbol{x}-\boldsymbol{y}\|_2\right| \leqslant 2C_1K^2 w(\mathbb{T}) + C_1K^2 u \cdot \operatorname{diam}(\mathbb{T}) \text{ (w.h.p. } 1 - 2\exp(-u^2)). \tag{II}$$

- Deriving the diameter of $\boldsymbol{A}\mathbb{T}$

  By the triangle inequality,

  $$\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2 \leqslant \sqrt{m}\|\boldsymbol{x}-\boldsymbol{y}\|_2 + |\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2 - \sqrt{m}\|\boldsymbol{x}-\boldsymbol{y}\|_2|,$$

  Taking the supremum over $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{T}$,

  $$\sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \|\boldsymbol{A}\boldsymbol{x}-\boldsymbol{A}\boldsymbol{y}\|_2 \leqslant \sqrt{m} \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} \|\boldsymbol{x}-\boldsymbol{y}\|_2 + \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}} |\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2 - \sqrt{m}\|\boldsymbol{x}-\boldsymbol{y}\|_2|. \tag{III}$$

  The left-hand side is exactly $\operatorname{diam}(\boldsymbol{A}\mathbb{T})$.

  Combining (II) and (III),

  $$\operatorname{diam}(\boldsymbol{A}\mathbb{T}) \leqslant \sqrt{m}\operatorname{diam}(\mathbb{T}) + 2C_1K^2w(\mathbb{T}) + C_1K^2u\cdot\operatorname{diam}(\mathbb{T}) \text{ (w.h.p. } 1-2\exp(-u^2)),$$

  i.e.

  $$\operatorname{diam}(\boldsymbol{A}\mathbb{T}) \leqslant (\sqrt{m} + C_1K^2u)\operatorname{diam}(\mathbb{T}) + 2C_1K^2w(\mathbb{T}).$$

- Scaling to the projection matrix

  Substitute $\boldsymbol{Q} = \frac{1}{\sqrt{n}}\boldsymbol{A}$; then

  $$\operatorname{diam}(\boldsymbol{Q}\mathbb{T}) = \frac{1}{\sqrt{n}}\operatorname{diam}(\boldsymbol{A}\mathbb{T}).$$

  Dividing both sides by $\sqrt{n}$,

  $$\operatorname{diam}(\boldsymbol{Q}\mathbb{T}) \leqslant \left(\sqrt{\frac{m}{n}} + \frac{C_1K^2u}{\sqrt{n}}\right)\operatorname{diam}(\mathbb{T}) + 2C_1K^2\frac{w(\mathbb{T})}{\sqrt{n}}.$$

  By the definition of the spherical width, $w_s(\mathbb{T}) \asymp \frac{w(\mathbb{T})}{\sqrt{n}}$; absorbing the comparison constants into C,

  $$\operatorname{diam}(\boldsymbol{Q}\mathbb{T}) \leqslant \left(\sqrt{\frac{m}{n}} + \frac{C_1K^2u}{\sqrt{n}}\right)\operatorname{diam}(\mathbb{T}) + CK^2w_s(\mathbb{T}).$$

  Setting $\frac{C_1K^2u}{\sqrt{n}} = \varepsilon\sqrt{\frac{m}{n}}$ gives $u = \frac{\varepsilon\sqrt{m}}{C_1K^2}$, and then

  $$1-2\exp(-u^2) \geqslant 1-2\exp\left(-\left(\frac{\varepsilon\sqrt{m}}{C_1K^2}\right)^2\right) = 1-2\exp\left(-\frac{\varepsilon^2m}{C_1^2K^4}\right),$$

  so $c = 1/C_1^2$

8 **[Size of Random Projections of Sets]** Prove a version of Proposition P9.2.1 (*Size of random projections of sets*) for the original model of $\boldsymbol{P}$ in Section 7.6.

Let $\mathbb{T} \subset \mathbb{R}^n$, and let $\boldsymbol{P}$ be the orthogonal projection matrix in $\mathbb{R}^n$ onto an m-dimensional subspace E with $E \sim U(G_{n,m})$, where $G_{n,m}$ is the Grassmannian. Prove:

$$\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] \leqslant \sqrt{\frac{m}{n}}\operatorname{diam}(\mathbb{T}) + Cw_s(\mathbb{T}). \qquad (C>0)$$

where $w_s(\mathbb{T})$ is the spherical width of the set.

(Note: unlike sub-Gaussian projections with independent coordinates, a true orthogonal projection needs no sub-Gaussian norm K^2 factor.)

Prf. • Introducing the deviation inequality for random projections

By the result of Exercise Q9.6 (*Deviation of Random Projections*), for any subset $\mathbb{S}$ of $\mathbb{R}^n$, a uniformly random orthogonal projection $\boldsymbol{P}$ satisfies the deviation bound

$$\mathbb{E}\left[\sup_{\boldsymbol{z}\in\mathbb{S}}\left|\|\boldsymbol{P}\boldsymbol{z}\|_2 - \sqrt{\frac{m}{n}}\|\boldsymbol{z}\|_2\right|\right] \leqslant \frac{C_1\gamma(\mathbb{S})}{\sqrt{n}}.$$

By the triangle inequality,

$$\|\boldsymbol{P}\boldsymbol{z}\|_2 \leqslant \sqrt{\frac{m}{n}}\|\boldsymbol{z}\|_2 + \left|\|\boldsymbol{P}\boldsymbol{z}\|_2 - \sqrt{\frac{m}{n}}\|\boldsymbol{z}\|_2\right|,$$

Taking the supremum and the expectation over $\mathbb{S}$ rewrites this in terms of the radius $\operatorname{rad}(\mathbb{S}) = \sup_{\boldsymbol{z}\in\mathbb{S}}\|\boldsymbol{z}\|_2$:

$$\mathbb{E}[\operatorname{rad}(\boldsymbol{P}\mathbb{S})] \leqslant \sqrt{\frac{m}{n}}\operatorname{rad}(\mathbb{S}) + \frac{C_1\gamma(\mathbb{S})}{\sqrt{n}}. \tag{I}$$

- Applying to the difference set $\mathbb{T}-\mathbb{T}$

  Apply (I) to $\mathbb{S} = \mathbb{T}-\mathbb{T}$. Its radius is

  $$\operatorname{rad}(\mathbb{T}-\mathbb{T}) = \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{T}}\|\boldsymbol{x}-\boldsymbol{y}\|_2 = \operatorname{diam}(\mathbb{T}).$$

  Since $\boldsymbol{P}$ is linear, $\boldsymbol{P}(\mathbb{T}-\mathbb{T}) = \boldsymbol{P}\mathbb{T}-\boldsymbol{P}\mathbb{T}$, and the radius after projection is

  $$\operatorname{rad}(\boldsymbol{P}(\mathbb{T}-\mathbb{T})) = \operatorname{diam}(\boldsymbol{P}\mathbb{T}).$$

  For the Gaussian complexity,

  $$\gamma(\mathbb{T}-\mathbb{T}) = 2w(\mathbb{T}),$$

  Substituting into (I),

  $$\mathbb{E}[\operatorname{diam}(\boldsymbol{P}\mathbb{T})] \leqslant \sqrt{\frac{m}{n}}\operatorname{diam}(\mathbb{T}) + \frac{2C_1 w(\mathbb{T})}{\sqrt{n}}.$$

  By the definition of the spherical width, $w_s(\mathbb{T}) \asymp \frac{w(\mathbb{T})}{\sqrt{n}}$; absorb the comparison constants into C.

9 [**Covariance Estimation with High Probability**] Prove the covariance-estimation guarantee in Remark R9.2.3 (*Covariance estimation: a high-probability guarantee*). Let $\boldsymbol{X}_1,\cdots,\boldsymbol{X}_m \in \mathbb{R}^n$ be i.i.d. mean-zero random vectors with covariance matrix $\boldsymbol{\Sigma} = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}^\top] \neq 0$[3]. Suppose that, for every $\boldsymbol{x}\in\mathbb{R}^n$, $\boldsymbol{X}$ satisfies the sub-Gaussian condition $\|\langle\boldsymbol{X},\boldsymbol{x}\rangle\|_{\psi_2} \leqslant K\|\langle\boldsymbol{X},\boldsymbol{x}\rangle\|_{L_2}$. If $\boldsymbol{\Sigma}=0$, the conclusion is trivial.

Define the sample covariance matrix

$$\boldsymbol{\Sigma}_m = \frac{1}{m}\sum_{i=1}^m \boldsymbol{X}_i\boldsymbol{X}_i^\top,$$

and let $r = \operatorname{tr}(\boldsymbol{\Sigma})/\|\boldsymbol{\Sigma}\|$ be the effective rank. Prove that for any $u \geqslant 0$,

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant CK^4\left(\sqrt{\frac{r+u}{m}} + \frac{r+u}{m}\right)\|\boldsymbol{\Sigma}\| \text{ (w.h.p. } 1-2\mathrm{e}^{-u}). \qquad (C>0)$$

Prf. • Whitening and reformulation

Rewrite the operator norm as the supremum of the quadratic form over $\mathbb{S}^{n-1}$,

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| = \sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}}\left|\boldsymbol{x}^\top\boldsymbol{\Sigma}_m\boldsymbol{x} - \boldsymbol{x}^\top\boldsymbol{\Sigma}\boldsymbol{x}\right| = \sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}}\left|\frac{1}{m}\sum_{i=1}^m\langle\boldsymbol{X}_i,\boldsymbol{x}\rangle^2 - \|\boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\|_2^2\right|.$$

Let $R = \operatorname{im}(\boldsymbol{\Sigma})$ and set $\boldsymbol{A}_i = \boldsymbol{\Sigma}^{\dagger/2}\boldsymbol{X}_i$. Then $\operatorname{Cov}(\boldsymbol{A}_i) = \boldsymbol{P}_R$, so the rows are isotropic on R; moreover, $\boldsymbol{X}_i \in R$ almost surely and hence $\boldsymbol{X}_i = \boldsymbol{\Sigma}^{1/2}\boldsymbol{A}_i$. Thus the matrix deviation inequality may be applied on R, and

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| = \frac{1}{m}\sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}}|\|\boldsymbol{A}\boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\|_2^2 - m\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\|_2^2|.$$

Define the ellipsoid $\mathbb{T} = \boldsymbol{\Sigma}^{1/2}\mathbb{S}^{n-1}$. Writing $\boldsymbol{z} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{x} \in \mathbb{T}$, the problem reduces to the quadratic matrix deviation of $\boldsymbol{A}$ on $\mathbb{T}$:

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| = \frac{1}{m}\sup_{\boldsymbol{z}\in\mathbb{T}}|\|\boldsymbol{A}\boldsymbol{z}\|_2^2 - m\|\boldsymbol{z}\|_2^2|.$$

- Applying the high-probability matrix deviation inequality

Define the linear deviation supremum $Z = \sup_{\boldsymbol{z}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{z}\|_2 - \sqrt{m}\|\boldsymbol{z}\|_2|$. By the high-probability version of the matrix deviation inequality, there is a constant C_1 such that for any $t \geqslant 0$,

$$Z \leqslant C_1 K^2(\gamma(\mathbb{T}) + t \cdot \mathrm{rad}(\mathbb{T})) \text{ (w.h.p. } 1 - 2\mathrm{e}^{-t^2}).$$

Using $|a^2 - b^2| \leqslant |a-b|^2 + 2b|a-b|$, the quadratic deviation is controlled by Z:

$$\sup_{\boldsymbol{z}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{z}\|_2^2 - m\|\boldsymbol{z}\|_2^2| \leqslant Z^2 + 2\sqrt{m}\,\mathrm{rad}(\mathbb{T})Z.$$

Substituting the high-probability bound,

$$\sup_{\boldsymbol{z}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{z}\|_2^2 - m\|\boldsymbol{z}\|_2^2| \leqslant [C_1K^2(\gamma(\mathbb{T}) + t\cdot\mathrm{rad}(\mathbb{T}))]^2 + 2\sqrt{m}\,\mathrm{rad}(\mathbb{T})[C_1K^2(\gamma(\mathbb{T}) + t\cdot\mathrm{rad}(\mathbb{T}))] \text{ (w.h.p. } 1 - 2\mathrm{e}^{-t^2}).$$

- Computing the geometric quantities and simplifying

The radius of $\mathbb{T}$ is $\mathrm{rad}(\mathbb{T}) = \sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} \|\boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\|_2 = \|\boldsymbol{\Sigma}\|^{1/2}$.

For the Gaussian complexity, by Jensen's inequality,

$$\gamma(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{S}^{n-1}} \langle \boldsymbol{g}, \boldsymbol{\Sigma}^{1/2}\boldsymbol{x}\rangle\right] = \mathbb{E}\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{g}\|_2 \leqslant (\mathbb{E}\|\boldsymbol{\Sigma}^{1/2}\boldsymbol{g}\|_2^2)^{1/2} = (\mathrm{tr}(\boldsymbol{\Sigma}))^{1/2}.$$

With the effective rank $r = \mathrm{tr}(\boldsymbol{\Sigma})/\|\boldsymbol{\Sigma}\|$, we have $\gamma(\mathbb{T}) \leqslant \sqrt{r\|\boldsymbol{\Sigma}\|} = \sqrt{r}\cdot\mathrm{rad}(\mathbb{T})$.

Let $v = \gamma(\mathbb{T}) + t\cdot\mathrm{rad}(\mathbb{T})$, so that $v \leqslant (\sqrt{r}+t)\|\boldsymbol{\Sigma}\|^{1/2}$. Then

$$\begin{aligned} \text{(square term)} \quad & v^2 \leqslant 2(r+t^2)\|\boldsymbol{\Sigma}\|, \\ \text{(cross term)} \quad & 2\sqrt{m}\,\mathrm{rad}(\mathbb{T})v \leqslant 2\sqrt{2}\sqrt{m}\sqrt{r+t^2}\|\boldsymbol{\Sigma}\|, \end{aligned}$$

Substituting into the quadratic deviation bound and collecting the constant C_2,

$$\sup_{\boldsymbol{z}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{z}\|_2^2 - m\|\boldsymbol{z}\|_2^2| \leqslant C_2K^4((r+t^2) + \sqrt{m}\sqrt{r+t^2})\|\boldsymbol{\Sigma}\|.$$

Dividing both sides by m yields the bound for $\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\|$:

$$\|\boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}\| \leqslant C_2K^4\left(\frac{r+t^2}{m} + \sqrt{\frac{r+t^2}{m}}\right)\|\boldsymbol{\Sigma}\| \text{ (w.h.p. } 1 - 2\mathrm{e}^{-t^2}).$$

Taking $u = t^2$, the event holds (w.h.p. $1 - 2\mathrm{e}^{-u}$), and setting $C = C_2$

10 [**From Matrix Deviation to Johnson–Lindenstrauss**] Use Theorem T9.1.1 (*Matrix deviation inequality*) to give an alternative solution to Exercise Q5.14 (*Johnson-Lindenstrauss Lemma for sub-Gaussian Matrices*).

Let $\mathcal{X} = \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N\} \subset \mathbb{R}^n$, and $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ a random matrix with independent, mean-zero, isotropic K-sub-Gaussian rows. Define the dimension reduction map $\boldsymbol{Q} = \frac{1}{\sqrt{m}}\boldsymbol{A}$. Given a tolerance parameter $\varepsilon \in (0,1)$, find a lower bound on the target dimension m such that the isometry condition

$$(1-\varepsilon)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \leqslant \|\boldsymbol{Q}\boldsymbol{x}_i - \boldsymbol{Q}\boldsymbol{x}_j\|_2 \leqslant (1+\varepsilon)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \text{ (w.h.p.)}.$$

Prf. • Constructing the difference set $\mathbb{T}$ on the unit sphere

Define $\mathbb{T}$ as the set of normalized pairwise differences of $\mathcal{X}$:

$$\mathbb{T} = \left\{\overline{\boldsymbol{x}_i - \boldsymbol{x}_j} : \boldsymbol{x}_i, \boldsymbol{x}_j \in \mathcal{X}, i \neq j\right\}.$$

Then $\mathbb{T} \subset \mathbb{S}^{n-1}$ and $|\mathbb{T}| \leqslant \frac{N(N-1)}{2} < N^2$. (*It remains to show that* $|\|\boldsymbol{Q}\boldsymbol{z}\|_2 - 1| < \varepsilon$ *for all* $\boldsymbol{z} \in \mathbb{T}$.)

Since $\mathbb{T} \subset \mathbb{S}^{n-1}$, we have $\|\boldsymbol{z}\|_2 = 1$ for all $\boldsymbol{z} \in \mathbb{T}$, hence $\mathrm{rad}(\mathbb{T}) = 1$.

For the Gaussian complexity $\gamma(\mathbb{T})$, the Gaussian maximal inequality for finite sets gives a constant C_0 such that

$$\gamma(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{z}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{z}\rangle\right] \leqslant C_0\sqrt{\log|\mathbb{T}|}\cdot\sup_{\boldsymbol{z}\in\mathbb{T}}\|\boldsymbol{z}\|_2.$$

Substituting $|\mathbb{T}| < N^2$ and $\|\boldsymbol{z}\|_2 = 1$, we obtain $\gamma(\mathbb{T}) \leqslant C_0\sqrt{\log(N^2)} = C_0\sqrt{2\log N} = C_1\sqrt{\log N}$.

- Applying the high-probability matrix deviation inequality

By the high-probability version of the matrix deviation inequality, for the sub-Gaussian matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, the set $\mathbb{T}$

and any $u \geqslant 0$,

$$\sup_{\boldsymbol{z}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{z}\|_2 - \sqrt{m}\|\boldsymbol{z}\|_2| \leqslant C_2K^2(\gamma(\mathbb{T}) + u \cdot \mathrm{rad}(\mathbb{T})) \text{ (w.h.p. } 1 - 2\mathrm{e}^{-u^2}).$$

Substituting $\|\boldsymbol{z}\|_2 = 1$, $\mathrm{rad}(\mathbb{T}) = 1$ and the upper bound on $\gamma(\mathbb{T})$, this simplifies to

$$\sup_{\boldsymbol{z}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{z}\|_2 - \sqrt{m}| \leqslant C_2K^2(C_1\sqrt{\log N} + u). \tag{I}$$

- Determining the target dimension m and quantifying the probability

  (*The JL lemma requires*

  $$C_2K^2(C_1\sqrt{\log N} + u) \leqslant \varepsilon\sqrt{m},$$

  which we split into two halves: the expectation part and the tail part should each be at most $\frac{\varepsilon}{2}\sqrt{m}$.)

  1. Controlling the expectation part (*determines the lower bound on* m)

     Requiring

     $$C_2C_1K^2\sqrt{\log N} \leqslant \frac{\varepsilon}{2}\sqrt{m}, \text{ hence } m \geqslant \frac{4C_1^2C_2^2K^4}{\varepsilon^2}\log N,$$

     and setting $C = 4C_1^2C_2^2$ gives $m \geqslant C\varepsilon^{-2}K^4\log N$.

  2. Controlling the tail probability (*determines* u)

     Requiring

     $$C_2K^2u = \frac{\varepsilon}{2}\sqrt{m}, \text{ hence } u = \frac{\varepsilon\sqrt{m}}{2C_2K^2},$$

     Substituting into the probability lower bound $1 - 2\mathrm{e}^{-u^2}$, the success probability is at least

     $$1 - 2\exp\left(-\frac{\varepsilon^2 m}{4C_2^2K^4}\right) = 1 - 2\exp\left(-\frac{c\varepsilon^2 m}{K^4}\right). \qquad \left(c = \frac{1}{4C_2^2}\right)$$

On this high-probability event, for any $\boldsymbol{z} \in \mathbb{T}$,

$$|\|\boldsymbol{A}\boldsymbol{z}\|_2 - \sqrt{m}| \leqslant \varepsilon\sqrt{m}, \text{ hence } |\|\boldsymbol{Q}\boldsymbol{z}\|_2 - 1| \leqslant \varepsilon.$$

Substituting $\boldsymbol{z} = \overline{\boldsymbol{x}_i - \boldsymbol{x}_j}$, we get

$$\boldsymbol{Q}\boldsymbol{z} = \frac{\boldsymbol{Q}\boldsymbol{x}_i - \boldsymbol{Q}\boldsymbol{x}_j}{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2};$$

multiplying both sides by $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2$ gives

$$(1-\varepsilon)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \leqslant \|\boldsymbol{Q}\boldsymbol{x}_i - \boldsymbol{Q}\boldsymbol{x}_j\|_2 \leqslant (1+\varepsilon)\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2 \text{ (w.h.p. } 1 - 2\exp(-c\varepsilon^2 m/K^4)).$$

Thus, when the target dimension satisfies $m \geqslant C\varepsilon^{-2}K^4\log N$, the isometry property holds.

11 [**Additive Johnson–Lindenstrauss Lemma**] Prove that the error of Lemma L9.2.4 (*Additive Johnson–Lindenstrauss lemma*) must be absolute rather than relative.

Prove: If the set $X = \mathrm{conv}(\mathbb{T}) \subset \mathbb{R}^n$ has nonempty interior, then for $m < n$ there is no matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ such that

$$\left|\frac{1}{\sqrt{m}}\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2 - \|\boldsymbol{x}-\boldsymbol{y}\|_2\right| \leqslant \varepsilon\|\boldsymbol{x}-\boldsymbol{y}\|_2 \qquad (0 < \varepsilon < 1)$$

holds for all $\boldsymbol{x}, \boldsymbol{y} \in X$.

Prf. **Solution 1**

(*For* $m < n$, $\dim(\ker \boldsymbol{A}) = n - \mathrm{rank}(\boldsymbol{A}) \geqslant n - m \geqslant 1$, *so a nonzero vector must be mapped to zero by* $\boldsymbol{A}$.

At the same time, note that a nonempty interior contains local displacements in every direction (X *contains an open ball, and small displacements along any direction lie in* $X - X$*); take a sufficiently small nonzero vector* $\boldsymbol{v} \in \ker \boldsymbol{A}$, *and then construct*

$$\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{v}, \boldsymbol{y} = \boldsymbol{x}_0 \in X,$$

the projected distance is 0 *while the true distance is* $\|\boldsymbol{v}\|_2 > 0$*, contradicting the positive lower bound of the relative error; thus, the relative error bound cannot hold.*)

Suppose, for contradiction, that there exist a matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ $(m < n)$ and a constant $\varepsilon \in (0,1)$ such that the relative error bound above holds and is satisfied for all $\boldsymbol{x}, \boldsymbol{y} \in X$, i.e.

$$\left|\frac{1}{\sqrt{m}}\|\boldsymbol{A}\boldsymbol{x}-\boldsymbol{y}\|_2 - \|\boldsymbol{x}-\boldsymbol{y}\|_2\right| \leqslant \varepsilon\|\boldsymbol{x}-\boldsymbol{y}\|_2. \qquad (0<\varepsilon<1)$$

The lower bound on the left-hand side is

$$\frac{1}{\sqrt{m}}\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2 \geqslant (1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2. \qquad \text{(I)}$$

Since $\operatorname{int}(X) \neq \emptyset$, by the definition of the interior there exist a point $\boldsymbol{x}_0 \in X$ and a radius $r > 0$ such that the whole open ball lies in X, i.e.

$$\boldsymbol{x}_0 + r\mathbb{B}_2^n = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}-\boldsymbol{x}_0\|_2 < r\} \subset X.$$

Consider the linear map $\boldsymbol{A} : \mathbb{R}^n \to \mathbb{R}^m$. By the rank–nullity theorem,

$$\dim(\ker \boldsymbol{A}) = n - \operatorname{rank}(\boldsymbol{A}) \geqslant n - m,$$

where $m < n$; hence $\dim(\ker \boldsymbol{A}) \geqslant 1$, so there exists a nonzero vector $\boldsymbol{v}_0 \in \ker \boldsymbol{A}$ with $\boldsymbol{A}\boldsymbol{v}_0 = \boldsymbol{0}$.

Write $\overline{\boldsymbol{v}_0} = \boldsymbol{v}_0/\|\boldsymbol{v}_0\|_2$ and let $\boldsymbol{v} = \frac{r}{2}\overline{\boldsymbol{v}_0}$. It satisfies:

1. $\|\boldsymbol{v}\|_2 = \frac{r}{2} < r$;
2. $\boldsymbol{v} \in \ker \boldsymbol{A}$, hence $\boldsymbol{A}\boldsymbol{v} = \boldsymbol{0}$;

then construct

$$\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{v}, \boldsymbol{y} = \boldsymbol{x}_0 \in X,$$

then $\|\boldsymbol{x}-\boldsymbol{x}_0\|_2 = \|\boldsymbol{v}\|_2 = \frac{r}{2} < r$, so $\boldsymbol{x} \in \boldsymbol{x}_0 + r\mathbb{B}_2^n \subset X$.

For the $\boldsymbol{x}, \boldsymbol{y}$ constructed above, since $\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y}) = \boldsymbol{A}\boldsymbol{v} = \boldsymbol{0}$, we have

$$\frac{1}{\sqrt{m}}\|\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\|_2 = 0,$$

substituting this into (I) gives

$$\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \varepsilon\|\boldsymbol{x}-\boldsymbol{y}\|_2,$$

a contradiction; thus, for $m < n$, no matrix $\boldsymbol{A}$ can realize the relative error bound on a set X with nonempty interior.

Solution 2 Normalized difference set

Let $\mathbb{T}$ be the set of normalized difference vectors, i.e.

$$\mathbb{T} = \{\overline{\boldsymbol{x}-\boldsymbol{y}} : \boldsymbol{x}, \boldsymbol{y} \in X, \boldsymbol{x} \neq \boldsymbol{y}\}.$$

If $X \supset \boldsymbol{x}_0 + r\mathbb{B}_2^n$, then for any $\boldsymbol{u} \in \mathbb{S}^{n-1}$, take $\boldsymbol{x} = \boldsymbol{x}_0 + \frac{r}{2}\boldsymbol{u} \in \boldsymbol{x}_0 + r\mathbb{B}_2^n \subset X$ and $\boldsymbol{y} = \boldsymbol{x}_0 \in X$; then

$$\overline{\boldsymbol{x}-\boldsymbol{y}} = \overline{\frac{r}{2}\boldsymbol{u}} = \boldsymbol{u},$$

so $\mathbb{T} \supset \mathbb{S}^{n-1}$, and hence $\operatorname{conv}(\mathbb{T}) \supset \mathbb{B}_2^n$ and has nonempty interior.

Since $m < n$, by the rank–nullity theorem we have $\dim(\ker \boldsymbol{A}) \geqslant 1$, so $\mathbb{T} \cap \ker \boldsymbol{A} \neq \emptyset$, i.e. there exists a unit vector $\boldsymbol{u}^* \in \mathbb{T} \cap \ker \boldsymbol{A}$ with $\|\boldsymbol{u}^*\|_2 = 1$ and $\boldsymbol{A}\boldsymbol{u}^* = \boldsymbol{0}$; hence

$$\left|\frac{1}{\sqrt{m}}\|\boldsymbol{A}\boldsymbol{u}^*\|_2 - \|\boldsymbol{u}^*\|_2\right| = |0-1| = 1,$$

contradicting $\varepsilon < 1$; thus, the relative error bound does not hold for a set X with nonempty interior.

Rk. The classical JL lemma concerns finite sets ($|X| = N$), with at most $\binom{N}{2}$ difference directions. A random projection subspace almost surely avoids these directions, so a relative error is possible.

The additive JL lemma concerns general sets (*e.g. manifolds, convex bodies, bounded infinite sets*). If a set has interior,

its difference directions cover $\mathbb{S}^{n-1}$; the dimension restriction forces the kernel to intersect this continuum, so a relative error fails and only the absolute error $\delta = \mathcal{O}(w(X)/\sqrt{m})$ is possible.

12 [M^* **Bound: Affine Sections**] Extend Theorem T9.3.1 (M^* *bound*) from sections through the origin to all affine sections. Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix. Prove: For any set $\mathbb{T} \subset \mathbb{R}^n$,

$$\mathbb{E}\left[\sup_{\boldsymbol{z}\in\mathbb{R}^n} \operatorname{diam}(\mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A}))\right] \leqslant \frac{Cw(\mathbb{T})}{\sqrt{m}}.$$

Prf. (*If $\boldsymbol{x}, \boldsymbol{y} \in \boldsymbol{z} + \ker \boldsymbol{A}$, then $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{0}$, so the difference $\boldsymbol{x} - \boldsymbol{y}$ necessarily lies in the linear subspace $\ker \boldsymbol{A}$ (independent of $\boldsymbol{z}$), hence the diameters of all sections can be controlled uniformly by the scale of the difference set $(\mathbb{T} - \mathbb{T}) \cap \ker \boldsymbol{A}$. Then the matrix deviation inequality applies: when $\boldsymbol{u} \in \ker \boldsymbol{A}$, we have $\|\boldsymbol{A}\boldsymbol{u}\|_2 = 0$, which links the Euclidean length $\|\boldsymbol{u}\|_2$ with the Gaussian width $w(\mathbb{T})$.*)

- Eliminating the affine shift $\boldsymbol{z}$, reducing the diameter to the difference set

  Fix an arbitrary $\boldsymbol{z} \in \mathbb{R}^n$. If $\mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A}) = \emptyset$ or is a singleton, its diameter is 0, which is trivial; hence we only need to consider the case $\mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A}) \neq \emptyset$.

  If $\mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A}) \neq \emptyset$, take any two points $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A})$. Since $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{T}$, we have $\boldsymbol{x} - \boldsymbol{y} \in \mathbb{T} - \mathbb{T}$; and since $\boldsymbol{x}, \boldsymbol{y} \in \boldsymbol{z} + \ker \boldsymbol{A}$, there exist $\boldsymbol{k}_1, \boldsymbol{k}_2 \in \ker \boldsymbol{A}$ such that $\boldsymbol{x} = \boldsymbol{z} + \boldsymbol{k}_1$ and $\boldsymbol{y} = \boldsymbol{z} + \boldsymbol{k}_2$, hence $\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{k}_1 - \boldsymbol{k}_2 \in \ker \boldsymbol{A}$, i.e. $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{0}$.

  Therefore, for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A})$,

  $$\boldsymbol{x} - \boldsymbol{y} \in (\mathbb{T} - \mathbb{T}) \cap \ker \boldsymbol{A}.$$

  Thus

  $$\|\boldsymbol{x} - \boldsymbol{y}\|_2 \leqslant \sup_{\boldsymbol{u}\in(\mathbb{T}-\mathbb{T})\cap\ker \boldsymbol{A}} \|\boldsymbol{u}\|_2.$$

  Taking the supremum over all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A})$ on the left, we obtain

  $$\operatorname{diam}(\mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A})) \leqslant \sup_{\boldsymbol{u}\in(\mathbb{T}-\mathbb{T})\cap\ker \boldsymbol{A}} \|\boldsymbol{u}\|_2.$$

  Note that the right-hand side does not depend on the shift vector $\boldsymbol{z}$ at all; hence, taking the supremum over all $\boldsymbol{z} \in \mathbb{R}^n$, it still holds that

  $$\sup_{\boldsymbol{z}\in\mathbb{R}^n} \operatorname{diam}(\mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A})) \leqslant \sup_{\boldsymbol{u}\in(\mathbb{T}-\mathbb{T})\cap\ker \boldsymbol{A}} \|\boldsymbol{u}\|_2. \tag{I}$$

- Constructing the deviation term via the kernel property

  For any $\boldsymbol{u} \in (\mathbb{T} - \mathbb{T}) \cap \ker \boldsymbol{A}$, since $\boldsymbol{u} \in \ker \boldsymbol{A}$, we have $\boldsymbol{A}\boldsymbol{u} = \boldsymbol{0}$ (i.e. $\|\boldsymbol{A}\boldsymbol{u}\|_2 = 0$), hence

  $$\sqrt{m}\|\boldsymbol{u}\|_2 = |\|\boldsymbol{A}\boldsymbol{u}\|_2 - \sqrt{m}\|\boldsymbol{u}\|_2|.$$

  Note that $(\mathbb{T} - \mathbb{T}) \cap \ker \boldsymbol{A} \subset \mathbb{T} - \mathbb{T}$; taking the supremum over both sides, we obtain

  $$\sup_{\boldsymbol{u}\in(\mathbb{T}-\mathbb{T})\cap\ker \boldsymbol{A}} \sqrt{m}\|\boldsymbol{u}\|_2 \leqslant \sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2|.$$

  Dividing both sides by $\sqrt{m}$ and substituting into (I), we obtain

  $$\sup_{\boldsymbol{z}\in\mathbb{R}^n} \operatorname{diam}(\mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A})) \leqslant \frac{1}{\sqrt{m}} \sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2|. \tag{II}$$

- Applying the matrix deviation inequality

  Taking expectations on both sides of (II), we obtain

  $$\mathbb{E}\left[\sup_{\boldsymbol{z}\in\mathbb{R}^n} \operatorname{diam}(\mathbb{T} \cap (\boldsymbol{z} + \ker \boldsymbol{A}))\right] \leqslant \frac{1}{\sqrt{m}} \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2|\right].$$

By Theorem T9.1.1 (*Matrix deviation inequality*), setting $\mathbb{S} = \mathbb{T} - \mathbb{T}$, we obtain

$$\mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2|\right] \leqslant C_0 w(\mathbb{T}-\mathbb{T}), \qquad (C_0 > 0)$$

i.e.,

$$\mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2|\right] \leqslant 2C_0 w(\mathbb{T}). \qquad \text{(III)}$$

Substituting (III) into the expectation inequality, we obtain

$$\mathbb{E}\left[\sup_{\boldsymbol{z}\in\mathbb{R}^n} \operatorname{diam}(\mathbb{T}\cap(\boldsymbol{z}+\ker\boldsymbol{A}))\right] \leqslant \frac{2C_0 w(\mathbb{T})}{\sqrt{m}} = \frac{Cw(\mathbb{T})}{\sqrt{m}}.$$

Rk. $\ker\boldsymbol{A}$ is a random subspace of dimension $n-m$. For every $\boldsymbol{z}$, the M^* bound compresses the diameter of $\mathbb{T}\cap(\boldsymbol{z}+\ker\boldsymbol{A})$ to $\mathcal{O}(w(\mathbb{T})/\sqrt{m})$.

In a linear inverse problem, the observation model is $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}_0$, and the consistent solutions form the affine subspace $\boldsymbol{x}_0+\ker\boldsymbol{A}$. If prior information restricts the unknown signal to a convex set $\mathbb{T}$, this theorem gives a uniform reconstruction error bound for every true signal $\boldsymbol{x}_0$.

13 [M^* **Bound: High-Probability Version**] Prove the high-probability version of the Theorem T9.3.1 (M^* *bound*). Let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ be a standard Gaussian random matrix, and let $\mathbb{T}\subset\mathbb{S}^{n-1}$. Prove: For any $u\geqslant 0$,

$$\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| \leqslant C(w(\mathbb{T})+u) \quad \text{(w.h.p. } 1-2\mathrm{e}^{-u^2}\text{)}.$$

Prf. (*Define $F(\boldsymbol{A}) = \sup_{\boldsymbol{x}\in\mathbb{T}}|\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}|$ and view $\boldsymbol{A}$ as a Gaussian random vector in $\mathbb{R}^{mn}$. The matrix deviation inequality gives $\mathbb{E}[F(\boldsymbol{A})] \leqslant C_0 w(\mathbb{T})$; the Gaussian concentration inequality upgrades this expectation bound to a high-probability bound.*)

- Defining the target functional and proving its 1-Lipschitz property

  Define the real-valued function on the matrix space $\mathbb{R}^{m\times n}$:

  $$F(\boldsymbol{A}) = \sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}|.$$

  The matrix space $\mathbb{R}^{m\times n}$ is endowed with the Frobenius norm $\|\boldsymbol{M}\|_F = \sqrt{\sum_{i,j} M_{ij}^2}$, which is equivalent to the standard Euclidean norm on $\mathbb{R}^{mn}$.

  Take any two matrices $\boldsymbol{A},\boldsymbol{B}\in\mathbb{R}^{m\times n}$:

  1. For any fixed $\boldsymbol{x}\in\mathbb{T}\subset\mathbb{S}^{n-1}$ (i.e. $\|\boldsymbol{x}\|_2 = 1$), by the reverse triangle inequality $|\|\boldsymbol{u}\|_2 - \|\boldsymbol{v}\|_2| \leqslant \|\boldsymbol{u}-\boldsymbol{v}\|_2$ for the Euclidean norm:

  $$|\|\boldsymbol{A}\boldsymbol{x}\|_2 - \|\boldsymbol{B}\boldsymbol{x}\|_2| \leqslant \|(\boldsymbol{A}-\boldsymbol{B})\boldsymbol{x}\|_2.$$

  2. By the fact that the operator norm does not exceed the Frobenius norm, $\|\boldsymbol{M}\| \leqslant \|\boldsymbol{M}\|_F$, and $\|\boldsymbol{x}\|_2 = 1$:

  $$\|(\boldsymbol{A}-\boldsymbol{B})\boldsymbol{x}\|_2 \leqslant \|\boldsymbol{A}-\boldsymbol{B}\|\|\boldsymbol{x}\|_2 = \|\boldsymbol{A}-\boldsymbol{B}\| \leqslant \|\boldsymbol{A}-\boldsymbol{B}\|_F.$$

  3. By the reverse triangle inequality $||a|-|b|| \leqslant |a-b|$ for absolute values:

  $$\big||\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| - |\|\boldsymbol{B}\boldsymbol{x}\|_2 - \sqrt{m}|\big| \leqslant |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \|\boldsymbol{B}\boldsymbol{x}\|_2| \leqslant \|\boldsymbol{A}-\boldsymbol{B}\|_F.$$

  4. Taking the supremum over all $\boldsymbol{x}\in\mathbb{T}$ and using the property $|\sup f - \sup g| \leqslant \sup|f-g|$, we obtain

  $$|F(\boldsymbol{A}) - F(\boldsymbol{B})| = \left|\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| - \sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{B}\boldsymbol{x}\|_2 - \sqrt{m}|\right| \leqslant \|\boldsymbol{A}-\boldsymbol{B}\|_F.$$

  Therefore, $F:\mathbb{R}^{m\times n}\to\mathbb{R}$ is a 1-Lipschitz function with respect to the Frobenius norm.

- Controlling the expectation of the functional

  By Theorem T9.1.1 (*Matrix deviation inequality*): for a matrix $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ whose entries satisfy A_{ij} i.i.d. $\sim N(0,1)$ and

any $\mathbb{T} \subset \mathbb{S}^{n-1}$, there exists an absolute constant $C_0 > 0$ such that

$$\mathbb{E}[F(\boldsymbol{A})] = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}|\right] \leqslant C_0 w(\mathbb{T}). \tag{I}$$

- Applying the Gaussian concentration inequality

  Flatten all entries of the matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ into the vector $\mathrm{vec}(\boldsymbol{A}) \in \mathbb{R}^{mn}$. Since A_{ij} i.i.d. $\sim N(0,1)$, $\mathrm{vec}(\boldsymbol{A}) \sim N(\boldsymbol{0}, \boldsymbol{I}_{mn})$ is a standard Gaussian random vector.

  By Theorem T5.2.3 (*Gaussian concentration*): if $f : \mathbb{R}^N \to \mathbb{R}$ is an L-Lipschitz continuous function and $\boldsymbol{G} \sim N(\boldsymbol{0}, \boldsymbol{I}_N)$, then for any $t \geqslant 0$,

$$\mathbb{P}\{|f(\boldsymbol{G}) - \mathbb{E}[f(\boldsymbol{G})]| \geqslant t\} \leqslant 2\exp\left(-\frac{t^2}{2L^2}\right).$$

  In our case $N = mn$ and $L = 1$; substituting the functional $F(\boldsymbol{A})$, for any $t \geqslant 0$ we have

$$\mathbb{P}\{|F(\boldsymbol{A}) - \mathbb{E}[F(\boldsymbol{A})]| \geqslant t\} \leqslant 2\exp\left(-\frac{t^2}{2}\right). \tag{II}$$

- Combining the expectation and concentration bounds

  In particular, from the one-sided tail bound of (II),

$$\mathbb{P}\{F(\boldsymbol{A}) \geqslant \mathbb{E}[F(\boldsymbol{A})] + t\} \leqslant 2\exp\left(-\frac{t^2}{2}\right).$$

  Setting $t = \sqrt{2}u$ (so that $\frac{t^2}{2} = u^2$), we have

$$F(\boldsymbol{A}) \leqslant \mathbb{E}[F(\boldsymbol{A})] + \sqrt{2}u \text{ (w.h.p. } 1 - 2\mathrm{e}^{-u^2}).$$

  Substituting the expectation bound $\mathbb{E}[F(\boldsymbol{A})] \leqslant C_0 w(\mathbb{T})$ from (I), we obtain

$$\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| \leqslant C_0 w(\mathbb{T}) + \sqrt{2}u.$$

  Setting the absolute constant $C = \max(C_0, \sqrt{2})$, we obtain

$$\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}| \leqslant C(w(\mathbb{T}) + u).$$

Rk. Why is this result called the "M^* bound: high-probability version"? This is the most crucial step toward the high-probability version of the M^* section diameter bound, Theorem T9.3.1 (*M^* bound*):

2. Kernel truncation: if $\boldsymbol{x} \in (K - K) \cap \ker \boldsymbol{A}$, then $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$, and the high-probability inequality above gives $\sqrt{m}\|\boldsymbol{x}\|_2 = |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \sqrt{m}\|\boldsymbol{x}\|_2| \leqslant C(w(K-K) + u \cdot \mathrm{rad}(K-K))$.

3. High-probability M^* bound: dividing both sides by $\sqrt{m}$, we prove that $\mathrm{diam}(K \cap \ker \boldsymbol{A}) \leqslant \frac{C'(w(K)+u\cdot\mathrm{diam}(K))}{\sqrt{m}}$ (w.h.p. $1 - 2\mathrm{e}^{-u^2}$).

This explains why the problem title is named "M^* Bound: High-Probability Version".

14 [**Sections of the p-Ball**] Let $\mathbb{B}_p^n$ be the unit p-ball in $\mathbb{R}^n$ and let E be a random k-dimensional subspace, where $1 \leqslant k \leqslant 0.99n$, and write $m = n - k$ for its codimension[4].

(1) Prove

$$\mathbb{E}[\mathrm{diam}(\mathbb{B}_p^n \cap E)] \lesssim_p n^{\frac{1}{2}-\frac{1}{p}} \quad \text{for all } p \in (1, \infty),$$

where $\lesssim_p$ hides a positive constant depending only on p.

(2) Check that this matches the radius of the inscribed Euclidean ball of $\mathbb{B}_p^n$ (when $p \leqslant 2$) and of the circumscribed Euclidean ball (when $p \geqslant 2$). Can you give an intuitive explanation?

Prf. (1) • connecting the random section to the M^* bound

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix, and take $m = n - k$ rows (the codimension of the section). By the rotational invariance of the normal distribution, $\ker \boldsymbol{A}$ has dimension $n - m = k$ and is uniformly distributed over the manifold $G(n,k)$ of all k-dimensional subspaces (the Haar measure). Hence $E \overset{d}{=} \ker \boldsymbol{A}$.

By the Theorem T9.3.1 (M^* *bound*), for any bounded set $\mathbb{T} \subset \mathbb{R}^n$,

$$\mathbb{E}[\operatorname{diam}(\mathbb{T} \cap \ker \boldsymbol{A})] \leqslant \frac{Cw(\mathbb{T})}{\sqrt{m}},$$

where $C > 0$ is an absolute constant and $w(\mathbb{T})$ is the Gaussian width of $\mathbb{T}$. Since the problem gives $k \leqslant 0.99n$, the codimension satisfies $m = n - k \geqslant 0.01n$, so $\sqrt{m} \geqslant 0.1\sqrt{n}$. Substituting $\mathbb{T} = \mathbb{B}_p^n$ yields

$$\mathbb{E}[\operatorname{diam}(\mathbb{B}_p^n \cap E)] \leqslant \frac{10Cw(\mathbb{B}_p^n)}{\sqrt{n}}. \tag{I}$$

- computing the Gaussian width $w(\mathbb{B}_p^n)$ of the ℓ_p-ball

  By the definition of the Gaussian width, let $\boldsymbol{g} = (g_1, \cdots, g_n) \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$:

  $$w(\mathbb{B}_p^n) = \mathbb{E}_{\boldsymbol{g}}\left[\sup_{\boldsymbol{x} \in \mathbb{B}_p^n} \langle \boldsymbol{g}, \boldsymbol{x} \rangle\right].$$

  By Hölder duality, the dual norm of the ℓ_p-norm is the ℓ_q-norm, where $\frac{1}{p} + \frac{1}{q} = 1$, i.e. $q = \frac{p}{p-1} \in (1, \infty)$:

  $$\sup_{\boldsymbol{x} \in \mathbb{B}_p^n} \langle \boldsymbol{g}, \boldsymbol{x} \rangle = \|\boldsymbol{g}\|_q = \left(\sum_{i=1}^n |g_i|^q\right)^{\frac{1}{q}}.$$

  Therefore

  $$w(\mathbb{B}_p^n) = \mathbb{E}\left[\left(\sum_{i=1}^n |g_i|^q\right)^{\frac{1}{q}}\right].$$

  By Jensen's inequality (since $q \geqslant 1$, the concave function $t \mapsto t^{1/q}$ satisfies $\mathbb{E}[X^{1/q}] \leqslant (\mathbb{E}[X])^{1/q}$):

  $$w(\mathbb{B}_p^n) \leqslant \mathbb{E}\left[\sum_{i=1}^n |g_i|^q\right]^{\frac{1}{q}} = n^{\frac{1}{q}} \mathbb{E}[|g_1|^q]^{\frac{1}{q}}.$$

  Since $g_1 \sim N(0, 1)$ is a standard scalar normal random variable, its q-th absolute moment is

  $$C_p = \mathbb{E}[|g_1|^q]^{\frac{1}{q}} = \sqrt{2}\left(\frac{\Gamma\left(\frac{q+1}{2}\right)}{\sqrt{\pi}}\right)^{\frac{1}{q}} < \infty,$$

  where C_p depends only on p. Substituting $\frac{1}{q} = 1 - \frac{1}{p}$ yields the upper bound on the Gaussian width:

  $$w(\mathbb{B}_p^n) \leqslant C_p n^{1-\frac{1}{p}}. \tag{II}$$

- combining the two bounds

  Substituting (II) into (I) yields

  $$\mathbb{E}[\operatorname{diam}(\mathbb{B}_p^n \cap E)] \leqslant \frac{10CC_p n^{1-\frac{1}{p}}}{n^{\frac{1}{2}}} = (10CC_p) n^{\frac{1}{2}-\frac{1}{p}}.$$

  Therefore

  $$\mathbb{E}[\operatorname{diam}(\mathbb{B}_p^n \cap E)] \lesssim_p n^{\frac{1}{2}-\frac{1}{p}}.$$

(2) • Verifying the inscribed and circumscribed Euclidean ball radii

Let $\mathbb{B}_2^n(r) = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_2 \leqslant r\}$. The inscribed radius r_{in} is the largest r with $\mathbb{B}_2^n(r) \subset \mathbb{B}_p^n$; the circumscribed radius R_{out} is the smallest R with $\mathbb{B}_p^n \subset \mathbb{B}_2^n(R)$. From the extremal relations between the ℓ_p-norm and the ℓ_2-norm, we obtain:

When $1 \leqslant p \leqslant 2$: the inscribed radius is $r_{\text{in}} = n^{\frac{1}{2}-\frac{1}{p}}$ (determined by the body-diagonal direction, attained at $\boldsymbol{x} = (\pm n^{-1/p}, \cdots, \pm n^{-1/p})$) and the circumscribed radius is $R_{\text{out}} = 1$ (determined by the coordinate-axis extreme points, attained at the standard basis vectors $\boldsymbol{e}_i$); the section-diameter bound is exactly of the order of the inscribed radius r_{in}.

When $p \geqslant 2$: the inscribed radius is $r_{\text{in}} = 1$ (attained at the standard basis vectors $\boldsymbol{e}_i$) and the circumscribed radius is $R_{\text{out}} = n^{\frac{1}{2}-\frac{1}{p}}$ (attained at $\boldsymbol{x} = (\pm n^{-1/p}, \cdots, \pm n^{-1/p})$); the section-diameter bound is exactly of the order of the circumscribed radius R_{out}.

- Intuition

(A) Why does the section diameter equal the circumscribed radius when $p \geqslant 2$? When $p \geqslant 2$ (e.g. the cube $[-1,1]^n$ for $p = \infty$), the Euclidean circumscribed radius of the whole set is exactly $R_{\text{out}} = n^{\frac{1}{2}-\frac{1}{p}}$. Since the subset section $\mathbb{B}_p^n \cap E \subset \mathbb{B}_p^n$, the diameter of the section cannot exceed the diameter $2R_{\text{out}} \asymp n^{\frac{1}{2}-\frac{1}{p}}$ of the whole set.

(B) Why does the section diameter shrink sharply to the inscribed radius when $p \leqslant 2$? This is the most fascinating high-dimensional phenomenon. When $p < 2$ (e.g. the octahedron, i.e. the cross-polytope $\mathbb{B}_1^n$, for $p = 1$), the circumscribed radius of the whole set is as large as $R_{\text{out}} = 1$ (the extremely thin spikes extending along the n coordinate-axis extreme points $\pm\boldsymbol{e}_i$). But the solid angle and the measure occupied by these spikes are both extremely small (close to 0). Most of the volume of $\mathbb{B}_p^n$ is actually concentrated in the central core region of radius only $r_{\text{in}} = n^{\frac{1}{2}-\frac{1}{p}}$ (which is $1/\sqrt{n}$ when $p = 1$). When we place a random k-dimensional subspace E in high-dimensional space, as long as the codimension $m \geqslant 0.01n$ occupies a macroscopic fraction, the random subspace will avoid all coordinate-axis spikes (w.h.p.) and pass straight through the central core. Hence the diameter of the random section $\mathbb{B}_p^n \cap E$ is not 1 but is "shaved down" to the inscribed-ball scale $n^{\frac{1}{2}-\frac{1}{p}}$!

Figure (A): $1 \leqslant p \leqslant 2$ (Example: $p = 1$)
Random subspace misses sharp spikes;
section diameter $\asymp r_{\text{in}} = n^{\frac{1}{2}-\frac{1}{p}}$

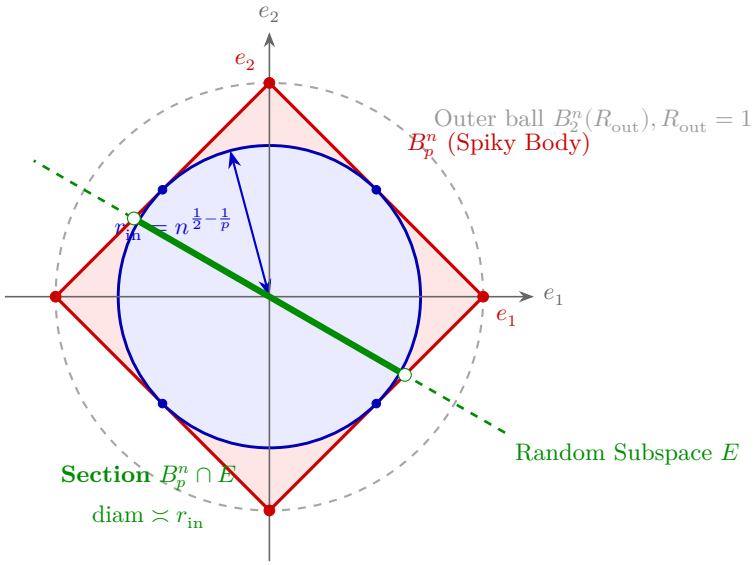
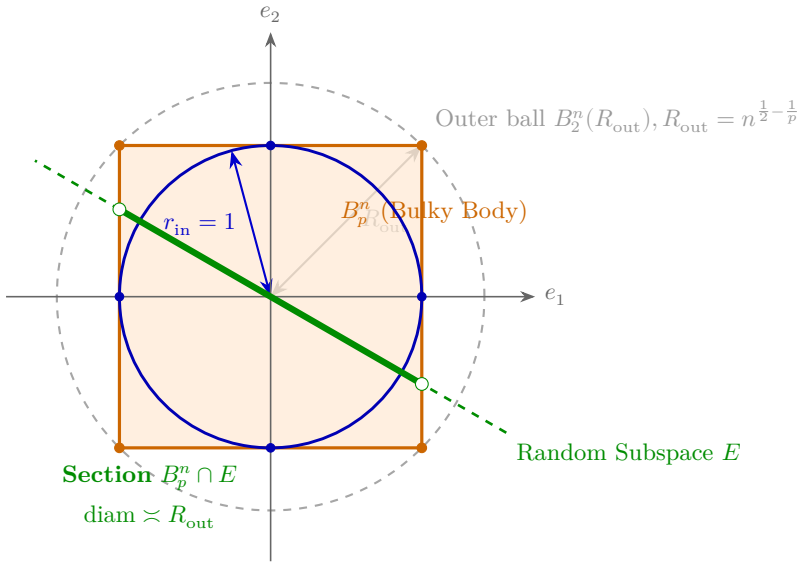


Figure (B): $p \geqslant 2$ (Example: $p = \infty$)
Bulky body;
section diameter $\asymp R_{\text{out}} = n^{\frac{1}{2}-\frac{1}{p}}$



Rk. Kashin's theorem and Dvoretzky's theorem show that a macroscopic random section of the ℓ_1-ball (*or a general ℓ_p-ball*) is close to an isotropic Euclidean ball. Its scale is controlled by the inscribed Euclidean-ball radius.

15 [**Sharpness of the Escape Theorem**] Prove that Theorem T9.3.4 (*Escape theorem*) is optimal in general for $1 \leqslant m \leqslant n$: take $\mathbb{T}$ to be the unit sphere of a subspace of $\mathbb{R}^n$.

Background: Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix, and let $\mathbb{T} \subset \mathbb{S}^{n-1}$ be a subset of the unit sphere. Then

$$\min_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{A}\boldsymbol{x}\|_2 \geqslant C^{-1}\sqrt{m} - Cw(\mathbb{T}) \text{ (w.h.p.)}.$$

In particular, when the number of measurements (rows) satisfies $m \geqslant C_0 w(\mathbb{T})^2$, the random subspace $\ker \boldsymbol{A}$ does not intersect $\mathbb{T}$ (w.h.p.) (i.e. $\mathbb{T} \cap \ker \boldsymbol{A} = \emptyset$), achieving "escape".

Hint/Method: take the test set $\mathbb{T} = V \cap \mathbb{S}^{n-1}$ to be the unit sphere of some d-dimensional linear subspace V of $\mathbb{R}^n$ $(1 \leqslant d \leqslant n)$.

(*We will show that the dimensional threshold supplied by the escape theorem has the correct order for this family.*)

Prf. • the Gaussian width $w(\mathbb{T})$ of the subspace sphere

Let $V \subset \mathbb{R}^n$ be a fixed d-dimensional linear subspace and set $\mathbb{T} = V \cap \mathbb{S}^{n-1}$. Let $\boldsymbol{g} \sim N(\mathbf{0}, \boldsymbol{I}_n)$ be a standard Gaussian vector and let $\boldsymbol{P}_V$ be the orthogonal projection onto V. By the definition of the Gaussian width,

$$w(\mathbb{T}) = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] = \mathbb{E}\left[\sup_{\boldsymbol{x}\in V, \|\boldsymbol{x}\|_2=1} \langle \boldsymbol{P}_V \boldsymbol{g}, \boldsymbol{x}\rangle\right].$$

Since the supremum of the inner products over all vectors on the unit sphere is the Euclidean norm,

$$w(\mathbb{T}) = \mathbb{E}[\|\boldsymbol{P}_V \boldsymbol{g}\|_2].$$

Since $\boldsymbol{g} \sim N(\mathbf{0}, \boldsymbol{I}_n)$ and $\dim V = d$, the projected vector $\boldsymbol{P}_V \boldsymbol{g}$ has the same distribution as a standard Gaussian vector $\boldsymbol{g}_d \sim N(\mathbf{0}, \boldsymbol{I}_d)$ in $\mathbb{R}^d$. The expected norm of a d-dimensional standard Gaussian vector satisfies

$$\mathbb{E}[\|\boldsymbol{g}_d\|_2] = \sqrt{2}\frac{\Gamma\left(\frac{d+1}{2}\right)}{\Gamma\left(\frac{d}{2}\right)} = \sqrt{d}\left(1 - O\left(\frac{1}{d}\right)\right) \asymp \sqrt{d}.$$

Hence, precisely,

$$w(\mathbb{T}) \asymp \sqrt{d}, \quad \text{i.e. } w(\mathbb{T})^2 \asymp d. \tag{I}$$

• the escape condition predicted by the escape theorem and the lower bound

By Theorem T9.3.4 (*Escape theorem*), the predicted lower bound is

$$\min_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{A}\boldsymbol{x}\|_2 \geqslant C^{-1}\sqrt{m} - Cw(\mathbb{T}) \geqslant C^{-1}\sqrt{m} - C\sqrt{d}.$$

The condition on the number of measurements that guarantees escape (i.e. $\mathbb{T} \cap \ker \boldsymbol{A} = \emptyset$) is

$$C^{-1}\sqrt{m} - Cw(\mathbb{T}) > 0 \quad \text{whenever} \quad m \geqslant C_0 w(\mathbb{T})^2 \asymp d. \tag{II}$$

• the true limit from linear algebra and random matrix theory

We now examine, from the fundamental geometric constraints of linear algebra, when $\mathbb{T} \cap \ker \boldsymbol{A} = \emptyset$ (i.e. $V \cap \ker \boldsymbol{A} = \{\mathbf{0}\}$).

When $m < d$ (a rigid obstacle to escape): by the Grassmann dimension formula for subspaces,

$$\dim(V \cap \ker \boldsymbol{A}) \geqslant \dim V + \dim \ker \boldsymbol{A} - n.$$

By the rank–nullity theorem, $\dim \ker \boldsymbol{A} = n - \operatorname{rank}(\boldsymbol{A}) \geqslant n - m$. Substituting this into the previous inequality yields

$$\dim(V \cap \ker \boldsymbol{A}) \geqslant d + (n - m) - n = d - m.$$

When $m < d$, we must have $\dim(V \cap \ker \boldsymbol{A}) \geqslant d - m \geqslant 1$: the intersection $V \cap \ker \boldsymbol{A}$ must contain at least one line through the origin, so $V \cap \ker \boldsymbol{A} \neq \{\mathbf{0}\}$ (w.h.p. 1). Hence, when $m < d$, there necessarily exists a unit vector $\boldsymbol{x} \in \mathbb{T}$ with $\boldsymbol{A}\boldsymbol{x} = \mathbf{0}$, i.e.

$$\min_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{A}\boldsymbol{x}\|_2 = 0,$$

so escape is impossible.

When $m \geqslant d$ (the smallest singular value viewpoint): choosing an orthonormal basis of V and restricting $\boldsymbol{A}$ to V yields an $m \times d$ standard Gaussian random matrix $\boldsymbol{G} \in \mathbb{R}^{m\times d}$. The target minimum is then the smallest singular value of this matrix:

$$\min_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{A}\boldsymbol{x}\|_2 = s_{\min}(\boldsymbol{G}).$$

By the classical singular value theory of Gaussian random matrices (Gordon's minimax theorem; see Exercise Q7.13 (*Minimum Singular Value of Gaussian Matrices*)),

$$s_{\min}(\boldsymbol{G}) \approx \sqrt{m} - \sqrt{d}.$$

When $m \geqslant (1+\varepsilon)d$, we have $s_{\min}(\boldsymbol{G}) \geqslant c\sqrt{m} > 0$ (w.h.p.); the true phase transition occurs exactly at $m = d$.

- Sharpness conclusion

  Comparing the two sides strictly yields the following: the Gaussian width $w(\mathbb{T}) \asymp \sqrt{d}$ corresponds precisely to the square root of the subspace dimension; the critical number of measurements $m \gtrsim w(\mathbb{T})^2 \asymp d$ agrees with the true limit $m \geqslant d$ up to the same order (the optimal order $O(d)$); the lower bound $\sqrt{m} - \sqrt{d}$ matches the singular-value spectral lower bound $s_{\min}(\boldsymbol{G}) \approx \sqrt{m} - \sqrt{d}$ in the form of the difference.

  Core conclusions: first, the quadratic dependence $m \asymp w(\mathbb{T})^2$ in the escape theorem cannot be improved (it cannot be reduced to a lower power such as the first power of $w(\mathbb{T})$); second, taking the subspace sphere as the test set proves that the escape theorem attains the optimal order (tight up to constants) of the information-theoretic and geometric-dimensional phase transition over the whole parameter range $1 \leqslant m \leqslant n$.

16 [**Putting a Sticker on a Soccer Ball**] Prove the following version of the escape theorem.

Let $\mathbb{T} \subset \mathbb{S}^{n-1}$ be an arbitrary set (think of $\mathbb{T}$ as a sticker on a soccer ball) and $X \subset \mathbb{S}^{n-1}$ a set of N points (like ink spots). Suppose

$$\sigma_{n-1}(\mathbb{T}) < \frac{1}{N},$$

where σ_{n-1} is the normalized area on the sphere. Prove that there exists a rotation $U \in O(n)$ such that

$$U\mathbb{T} \cap X = \emptyset$$

(i.e. we can place the sticker without covering any ink spot).

Prf. • the probability that a single point is covered

Fix any point $\boldsymbol{x} \in X$ of the set X. The event that $\boldsymbol{x}$ lies in the rotated sticker $U\mathbb{T}$ is equivalent to

$$\boldsymbol{x} \in U\mathbb{T} \text{ if and only if } U^{-1}\boldsymbol{x} \in \mathbb{T}.$$

Since U follows the Haar distribution on $O(n)$, its inverse $U^{-1} = U^{\top}$ also follows the Haar distribution on $O(n)$. By the rotational invariance of the spherical measure, for any fixed $\boldsymbol{x} \in \mathbb{S}^{n-1}$, the random vector $Y = U^{-1}\boldsymbol{x}$ is uniformly distributed on the unit sphere $\mathbb{S}^{n-1}$ (i.e. its probability distribution is exactly the normalized spherical measure σ_{n-1}). Hence the probability that the random sticker covers the point $\boldsymbol{x}$ is exactly the spherical measure of $\mathbb{T}$:

$$\mathbb{P}\{\boldsymbol{x} \in U\mathbb{T}\} = \mathbb{P}\{U^{-1}\boldsymbol{x} \in \mathbb{T}\} = \sigma_{n-1}(\mathbb{T}). \tag{I}$$

Solution 1: union bound

The event that the sticker intersects the set of points (i.e. covers at least one point) is

$$\{U\mathbb{T} \cap X \neq \emptyset\} = \bigcup_{\boldsymbol{x} \in X} \{\boldsymbol{x} \in U\mathbb{T}\}.$$

By the subadditivity of probability (the union bound, Boole's inequality),

$$\mathbb{P}\{U\mathbb{T} \cap X \neq \emptyset\} = \mathbb{P}\{\bigcup_{\boldsymbol{x} \in X} \{\boldsymbol{x} \in U\mathbb{T}\}\} \leqslant \sum_{\boldsymbol{x} \in X} \mathbb{P}\{\boldsymbol{x} \in U\mathbb{T}\}.$$

Substituting (I), since the set X contains N points, yields

$$\mathbb{P}\{U\mathbb{T} \cap X \neq \emptyset\} \leqslant \sum_{\boldsymbol{x} \in X} \sigma_{n-1}(\mathbb{T}) = N \cdot \sigma_{n-1}(\mathbb{T}).$$

By the assumption $\sigma_{n-1}(\mathbb{T}) < \frac{1}{N}$,

$$\mathbb{P}\{U\mathbb{T} \cap X \neq \emptyset\} < N \cdot \frac{1}{N} = 1.$$

Hence the probability of the complementary event (complete disjointness) satisfies

$$\mathbb{P}\{U\mathbb{T} \cap X = \emptyset\} = 1 - \mathbb{P}\{U\mathbb{T} \cap X \neq \emptyset\} > 1 - 1 = 0.$$

Since this event has probability strictly greater than 0, there must exist at least one orthogonal transformation $U \in O(n)$ in the sample space $O(n)$ such that $U\mathbb{T} \cap X = \emptyset$. **Solution 2: expectation method**

Define the indicator random variable $1_{\{\boldsymbol{x} \in U\mathbb{T}\}}$ and let K be the total number of ink spots covered by the rotated sticker

$U\mathbb{T}$:

$$K = |U\mathbb{T} \cap X| = \sum_{\boldsymbol{x}\in X} 1_{\{\boldsymbol{x}\in U\mathbb{T}\}}.$$

Taking expectations (using the linearity of expectation and Fubini's theorem) yields

$$\mathbb{E}[K] = \sum_{\boldsymbol{x}\in X} \mathbb{E}[1_{\{\boldsymbol{x}\in U\mathbb{T}\}}] = \sum_{\boldsymbol{x}\in X} \mathbb{P}\{\boldsymbol{x} \in U\mathbb{T}\} = N \cdot \sigma_{n-1}(\mathbb{T}) < N \cdot \frac{1}{N} = 1.$$

Since K can only take non-negative integer values ($K \in \{0, 1, \cdots, N\}$) and its expectation satisfies $\mathbb{E}[K] < 1$, the probability of $K = 0$ must be strictly positive (otherwise, if $K \geqslant 1$ always held, we would have $\mathbb{E}[K] \geqslant 1$, a contradiction):

$$\mathbb{P}\{K = 0\} > 0, \text{ hence } \mathbb{P}\{U\mathbb{T} \cap X = \emptyset\} > 0.$$

We likewise conclude that there exists $U \in O(n)$ with $U\mathbb{T} \cap X = \emptyset$.

Rk. Why is this problem "another version of the escape theorem"? It reveals a deep duality between two escape phenomena. Continuous-geometric version (Gordon's escape theorem): a continuous subspace E escapes a continuous set $\mathbb{T}$ ($E \cap \mathbb{T} = \emptyset$), with the Gaussian width $w(\mathbb{T})$ and the codimension m as the measure tools, and the criterion $m \gtrsim w(\mathbb{T})^2$. Discrete-measure version (this problem, the sticker and the ink spots): a random rotation $U\mathbb{T}$ escapes a discrete set of points X ($U\mathbb{T} \cap X = \emptyset$), with the normalized spherical area $\sigma_{n-1}(\mathbb{T})$ and the number of points N as the measure tools, and the criterion $\sigma_{n-1}(\mathbb{T}) < 1/N$. Mathematical essence: whether for a continuous subspace or a discrete pair of points, as long as the "high-dimensional capacity" (spherical measure or Gaussian width) of the target set is small enough, the isotropy of the high-dimensional rotation group and random projections guarantees enough degrees of freedom in space to separate the two completely.

17 [**Constrained Recovery: Mean Squared Error**] Prove the mean-squared error bound of Theorem T9.4.4 (*Constrained recovery*)

$$\mathbb{E}\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2^2 \leqslant \frac{CK^4 w(\mathbb{T})^2}{m}.$$

Prf. • relating the mean squared error to the deviation functional

By the statement, $\hat{\boldsymbol{x}}, \boldsymbol{x} \in \mathbb{T}$ and $\boldsymbol{A}\hat{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} = \boldsymbol{y}$. Hence the difference vector satisfies $\hat{\boldsymbol{x}} - \boldsymbol{x} \in \mathbb{T} - \mathbb{T}$, and $\boldsymbol{A}(\hat{\boldsymbol{x}} - \boldsymbol{x}) = \boldsymbol{0}$, so $\hat{\boldsymbol{x}} - \boldsymbol{x} \in (\mathbb{T} - \mathbb{T}) \cap \ker \boldsymbol{A}$.

For any $\boldsymbol{u} \in (\mathbb{T} - \mathbb{T}) \cap \ker \boldsymbol{A}$, since $\boldsymbol{A}\boldsymbol{u} = \boldsymbol{0}$ (i.e. $\|\boldsymbol{A}\boldsymbol{u}\|_2 = 0$),

$$\sqrt{m}\|\boldsymbol{u}\|_2 = \big|\|\boldsymbol{A}\boldsymbol{u}\|_2 - \sqrt{m}\|\boldsymbol{u}\|_2\big| \leqslant \sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|.$$

Therefore

$$\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2 \leqslant \frac{1}{\sqrt{m}} \sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|.$$

Squaring both sides,

$$\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2^2 \leqslant \frac{1}{m}\left(\sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|\right)^2.$$

Taking expectations on both sides yields

$$\mathbb{E}[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2^2] \leqslant \frac{1}{m}\mathbb{E}[Y^2], \tag{I}$$

where the random variable is defined as

$$Y = \sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|.$$

• controlling the first and second moments of Y

By the definition of the variance, the second moment decomposes as

$$\mathbb{E}[Y^2] = (\mathbb{E}[Y])^2 + \mathrm{Var}(Y). \tag{II}$$

Controlling the expectation $\mathbb{E}[Y]$: by Theorem T9.1.1 (*Matrix deviation inequality*),

$$\mathbb{E}[Y] = \mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}-\mathbb{T}} \left|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\right|\right] \leqslant C_1 K^2 w(\mathbb{T}-\mathbb{T}).$$

Using the difference-set property $w(\mathbb{T}-\mathbb{T}) = 2w(\mathbb{T})$ of the Gaussian width,

$$\mathbb{E}[Y] \leqslant 2C_1 K^2 w(\mathbb{T}).$$

Therefore

$$(\mathbb{E}[Y])^2 \leqslant 4C_1^2 K^4 w(\mathbb{T})^2. \tag{III}$$

Controlling the variance $\mathrm{Var}(Y)$: by the high-probability tail bound for the matrix deviation inequality (Gaussian concentration, sub-Gaussian concentration; see Exercise Q9.13 (*M^* Bound: High-Probability Version*)), the random variable Y has sub-Gaussian tail decay around its mean:

$$\mathbb{P}\{|Y-\mathbb{E}[Y]| \geqslant t\} \leqslant 2\exp\left(-\frac{ct^2}{K^4 \operatorname{rad}(\mathbb{T}-\mathbb{T})^2}\right) = 2\exp\left(-\frac{ct^2}{K^4 \operatorname{diam}(\mathbb{T})^2}\right).$$

Integrating this tail decay directly yields the variance bound (the variance property of sub-Gaussian variables):

$$\mathrm{Var}(Y) = \mathbb{E}[(Y-\mathbb{E}[Y])^2] = \int_0^\infty 2t\mathbb{P}\{|Y-\mathbb{E}[Y]| \geqslant t\}\mathrm{d}t \leqslant C_2 K^4 \operatorname{diam}(\mathbb{T})^2. \tag{IV}$$

- controlling the diameter $\operatorname{diam}(\mathbb{T})$ by the Gaussian width

  By the basic geometric property of the Gaussian width, for any two points $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{T}$, let $\boldsymbol{g} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$; then

$$w(\mathbb{T}) = \frac{1}{2}w(\mathbb{T}-\mathbb{T}) \geqslant \frac{1}{2}\mathbb{E}[|\langle \boldsymbol{g}, \boldsymbol{x}_1-\boldsymbol{x}_2\rangle|] = \frac{1}{2}\|\boldsymbol{x}_1-\boldsymbol{x}_2\|_2 \mathbb{E}[|g_1|] = \frac{1}{\sqrt{2\pi}}\|\boldsymbol{x}_1-\boldsymbol{x}_2\|_2.$$

  Taking the supremum over all $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{T}$ yields

$$\operatorname{diam}(\mathbb{T}) \leqslant \sqrt{2\pi} w(\mathbb{T}).$$

  After squaring,

$$\operatorname{diam}(\mathbb{T})^2 \leqslant 2\pi w(\mathbb{T})^2. \tag{V}$$

  Substituting (V) into (IV), the variance bound further becomes

$$\mathrm{Var}(Y) \leqslant 2\pi C_2 K^4 w(\mathbb{T})^2. \tag{VI}$$

- combining the bounds

  Substituting the expectation term (III) and the variance term (VI) into (II) yields

$$\mathbb{E}[Y^2] \leqslant 4C_1^2 K^4 w(\mathbb{T})^2 + 2\pi C_2 K^4 w(\mathbb{T})^2 =: C' K^4 w(\mathbb{T})^2,$$

  where $C' = 4C_1^2 + 2\pi C_2$ is an absolute constant independent of the dimension.

  Finally, substituting $\mathbb{E}[Y^2]$ into (I) yields

$$\mathbb{E}[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2^2] \leqslant \frac{C' K^4 w(\mathbb{T})^2}{m} =: \frac{C w(\mathbb{T})^2}{m},$$

  where $C = C' K^4$.

Rk. by the Cauchy–Schwarz inequality (Jensen's inequality), $(\mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2)^2 \leqslant \mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2^2] \leqslant \frac{Cw(\mathbb{T})^2}{m}$, hence $\mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant \frac{\sqrt{C}w(\mathbb{T})}{\sqrt{m}}$, which shows that the second-order mean squared error bound is a stronger, tighter guarantee. In compressed sensing, if $\mathbb{T} = \sqrt{s}\mathbb{B}_1^n \cap \mathbb{B}_2^n$ (the sparse-signal prior), it is known that $w(\mathbb{T})^2 \asymp s\log(n/s)$; substituting this into the conclusion of this problem directly gives the classical MSE convergence rate of sparse recovery, $\mathrm{MSE} = \mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2^2] \lesssim \frac{s\log(en/s)}{m}$.

18 [**Recovery by Optimization**] Let $\mathbb{T} = \{\boldsymbol{z} \in \mathbb{R}^n : \|\boldsymbol{z}\|_{\mathbb{T}} \leqslant 1\}$ be the unit ball of the norm $\|\cdot\|_{\mathbb{T}}$ on $\mathbb{R}^n$, let $\boldsymbol{x} \in \mathbb{T}$ be the unknown signal, and let the observations be $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} \in \mathbb{R}^m$, where $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ is a standard Gaussian matrix. Consider estimating the signal by solving the following norm-minimization convex optimization problem:

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \|\boldsymbol{z}\|_{\mathbb{T}} \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{z} = \boldsymbol{y}.$$

Prove that the error bound of Theorem T9.4.4 (*Constrained recovery*) also holds for the solution of this optimization program,

$$\mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant \frac{CK^2 w(\mathbb{T})}{\sqrt{m}}, \mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2^2 \leqslant \frac{CK^4 w(\mathbb{T})^2}{m}.$$

Prf. (*The signal $\boldsymbol{x}$ is feasible and $\|\boldsymbol{x}\|_{\mathbb{T}} \leqslant 1$. Optimality gives $\|\hat{\boldsymbol{x}}\|_{\mathbb{T}} \leqslant \|\boldsymbol{x}\|_{\mathbb{T}}$, hence $\hat{\boldsymbol{x}} \in \mathbb{T}$.*)

- reducing the error vector to the geometry of the section

  Let $\boldsymbol{x}, \hat{\boldsymbol{x}} \in \mathbb{T}$ satisfy $\boldsymbol{A}\hat{\boldsymbol{x}} = \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$. Then

  $$\hat{\boldsymbol{x}}-\boldsymbol{x} \in \mathbb{T}-\mathbb{T}, \boldsymbol{A}(\hat{\boldsymbol{x}}-\boldsymbol{x}) = \boldsymbol{0}, \text{ hence } \hat{\boldsymbol{x}}-\boldsymbol{x} \in \ker \boldsymbol{A},$$

  so $\hat{\boldsymbol{x}}-\boldsymbol{x} \in (\mathbb{T}-\mathbb{T}) \cap \ker \boldsymbol{A}$, i.e. $\boldsymbol{x}, \hat{\boldsymbol{x}} \in \mathbb{T} \cap (\boldsymbol{x}+\ker \boldsymbol{A})$. Consequently,

  $$\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant \operatorname{diam}(\mathbb{T} \cap (\boldsymbol{x}+\ker \boldsymbol{A})) \leqslant \sup_{\boldsymbol{u} \in (\mathbb{T}-\mathbb{T}) \cap \ker \boldsymbol{A}} \|\boldsymbol{u}\|_2. \tag{I}$$

- applying the matrix deviation inequality to obtain the error bound

  For any $\boldsymbol{u} \in (\mathbb{T}-\mathbb{T}) \cap \ker \boldsymbol{A}$, since $\boldsymbol{A}\boldsymbol{u} = \boldsymbol{0}$,

  $$\sqrt{m}\|\boldsymbol{u}\|_2 = \big|\|\boldsymbol{A}\boldsymbol{u}\|_2 - \sqrt{m}\|\boldsymbol{u}\|_2\big| \leqslant \sup_{\boldsymbol{v} \in \mathbb{T}-\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|.$$

  Dividing both sides by $\sqrt{m}$, substituting into (I), and then taking expectations on both sides yields

  $$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \leqslant \frac{1}{\sqrt{m}} \mathbb{E}\left[\sup_{\boldsymbol{v} \in \mathbb{T}-\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|\right].$$

  By Theorem T9.1.1 (*Matrix deviation inequality*), we obtain

  $$\mathbb{E}\left[\sup_{\boldsymbol{v} \in \mathbb{T}-\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|\right] \leqslant C_1 K^2 w(\mathbb{T}-\mathbb{T}) = 2C_1 K^2 w(\mathbb{T}).$$

  That is,

  $$\mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant \frac{CK^2 w(\mathbb{T})}{\sqrt{m}}. \qquad (C = 2C_1)$$

  Likewise, by Exercise Q9.17 (*Constrained Recovery: Mean Squared Error*), we obtain

  $$\mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2^2 \leqslant \frac{C' K^4 w(\mathbb{T})^2}{m}.$$

Rk. If the true signal $\boldsymbol{x} \neq \boldsymbol{0}$ does not lie in the unit ball, let $\boldsymbol{x}_0 = \boldsymbol{x}/\|\boldsymbol{x}\|_{\mathbb{T}} \in \mathbb{T}$, whose solution is $\hat{\boldsymbol{x}}_0 = \hat{\boldsymbol{x}}/\|\boldsymbol{x}\|_{\mathbb{T}}$. We directly generalize to the relative error bound,

$$\mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant \frac{CK^2 w(\mathbb{T})}{\sqrt{m}}\|\boldsymbol{x}\|_{\mathbb{T}}.$$

In compressed sensing basis pursuit (*Basis Pursuit*), let $\|\cdot\|_{\mathbb{T}} = \|\cdot\|_1$, $\mathbb{T} = \mathbb{B}_1^n$, the optimization problem becomes the classical basis pursuit:

$$\arg\min \|\boldsymbol{z}\|_1 \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{z} = \boldsymbol{y}.$$

If $\boldsymbol{x}$ is an s-sparse signal with $\|\boldsymbol{x}\|_1 \leqslant 1$, its Gaussian width satisfies $w(\mathbb{B}_1^n) \asymp \sqrt{\log n}$.

19 [**Constrained Optimization**] Extend Theorem T9.4.4 (*Constrained recovery*) to the noisy model:

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w},$$

where $\boldsymbol{w}$ is an unknown noise vector (*may even depend on* $\boldsymbol{A}$). Choose a vector in $\mathbb{T}$ that minimizes the residual for the observations $\boldsymbol{y}$:

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z} \in \mathbb{T}} \|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{z}\|_2.$$

Prove that the solution $\hat{\boldsymbol{x}}$ satisfies:

$$\mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \lesssim \frac{K^2 w(\mathbb{T}) + \mathbb{E}\|\boldsymbol{w}\|_2}{\sqrt{m}}.$$

Prf. • controlling $\|\boldsymbol{A}(\hat{\boldsymbol{x}}-\boldsymbol{x})\|_2$ by optimality

By the optimality of $\hat{\boldsymbol{x}}$,

$$\|\boldsymbol{y}-\boldsymbol{A}\hat{\boldsymbol{x}}\| \leqslant \|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|,$$

Substituting $\boldsymbol{y}=\boldsymbol{A}\boldsymbol{x}+\boldsymbol{w}$, we obtain

$$\|\boldsymbol{w}-\boldsymbol{A}(\hat{\boldsymbol{x}}-\boldsymbol{x})\|_2 \leqslant \|\boldsymbol{w}\|_2.$$

For the left-hand side, using the reverse triangle inequality for the Euclidean norm $\|\boldsymbol{a}-\boldsymbol{b}\|_2 \geqslant \|\boldsymbol{a}\|_2-\|\boldsymbol{b}\|_2$, we obtain

$$\|\boldsymbol{A}(\hat{\boldsymbol{x}}-\boldsymbol{x})\|_2-\|\boldsymbol{w}\|_2 \leqslant \|\boldsymbol{w}-\boldsymbol{A}(\hat{\boldsymbol{x}}-\boldsymbol{x})\|_2,$$

Substituting it in, we obtain

$$\|\boldsymbol{A}(\hat{\boldsymbol{x}}-\boldsymbol{x})\|_2 \leqslant 2\|\boldsymbol{w}\|_2. \tag{I}$$

- applying the matrix deviation inequality to establish a lower bound

Let the error vector be $\boldsymbol{h}=\hat{\boldsymbol{x}}-\boldsymbol{x}\in\mathbb{T}-\mathbb{T}$.

Using the real absolute value inequality $a-b\leqslant|a-b|$, we obtain

$$\sqrt{m}\|\boldsymbol{h}\|_2-\|\boldsymbol{A}\boldsymbol{h}\|_2 \leqslant \big|\|\boldsymbol{A}\boldsymbol{h}\|_2-\sqrt{m}\|\boldsymbol{h}\|_2\big|.$$

Taking the supremum on both sides over $\boldsymbol{h}\in\mathbb{T}-\mathbb{T}$,

$$\sqrt{m}\|\boldsymbol{h}\|_2-\|\boldsymbol{A}\boldsymbol{h}\|_2 \leqslant \sup_{\boldsymbol{u}\in\mathbb{T}-\mathbb{T}}\big|\|\boldsymbol{A}\boldsymbol{u}\|_2-\sqrt{m}\|\boldsymbol{u}\|_2\big|,$$

That is,

$$\sqrt{m}\|\boldsymbol{h}\|_2 \leqslant \|\boldsymbol{A}\boldsymbol{h}\|_2+\sup_{\boldsymbol{u}\in\mathbb{T}-\mathbb{T}}\big|\|\boldsymbol{A}\boldsymbol{u}\|_2-\sqrt{m}\|\boldsymbol{u}\|_2\big|. \tag{II}$$

Substituting (I) into (II), we obtain

$$\sqrt{m}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant 2\|\boldsymbol{w}\|_2+\sup_{\boldsymbol{u}\in\mathbb{T}-\mathbb{T}}\big|\|\boldsymbol{A}\boldsymbol{u}\|_2-\sqrt{m}\|\boldsymbol{u}\|_2\big|.$$

Dividing both sides by $\sqrt{m}$,

$$\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant \frac{2\|\boldsymbol{w}\|_2}{\sqrt{m}}+\frac{1}{\sqrt{m}}\sup_{\boldsymbol{u}\in\mathbb{T}-\mathbb{T}}\big|\|\boldsymbol{A}\boldsymbol{u}\|_2-\sqrt{m}\|\boldsymbol{u}\|_2\big|, \tag{III}$$

Then taking expectations on both sides,

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \leqslant \frac{2\mathbb{E}[\|\boldsymbol{w}\|_2]}{\sqrt{m}}+\frac{1}{\sqrt{m}}\mathbb{E}\left[\sup_{\boldsymbol{u}\in\mathbb{T}-\mathbb{T}}\big|\|\boldsymbol{A}\boldsymbol{u}\|_2-\sqrt{m}\|\boldsymbol{u}\|_2\big|\right]. \tag{IV}$$

By Theorem T9.1.1 (*Matrix deviation inequality*), we obtain

$$\mathbb{E}\left[\sup_{\boldsymbol{u}\in\mathbb{T}-\mathbb{T}}\big|\|\boldsymbol{A}\boldsymbol{u}\|_2-\sqrt{m}\|\boldsymbol{u}\|_2\big|\right] \leqslant C_0K^2w(\mathbb{T}-\mathbb{T})=2C_0K^2w(\mathbb{T}),$$

Substituting into (IV), we obtain

$$\begin{aligned}\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] &\leqslant \frac{4C_0K^2w(\mathbb{T})+2\mathbb{E}[\|\boldsymbol{w}\|_2]}{\sqrt{m}}\\ &\leqslant C\frac{K^2w(\mathbb{T})+\mathbb{E}\|\boldsymbol{w}\|_2}{\sqrt{m}}. \qquad (C=\max(4C_0,2))\end{aligned}$$

Rk. In fact, in statistical recovery theory,

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \lesssim \underbrace{\frac{K^2w(\mathbb{T})}{\sqrt{m}}}_{\text{structural undersampling error}}+\underbrace{\frac{\mathbb{E}\|\boldsymbol{w}\|_2}{\sqrt{m}}}_{\text{noise perturbation error}}.$$

When $m\gtrsim K^4w(\mathbb{T})^2$, the first term is suppressed; the second term measures the perturbation of the solution by the noise; if $\boldsymbol{w}\sim N(\boldsymbol{0},\sigma^2\boldsymbol{I}_m)$, then $\|\boldsymbol{w}\|_2\sim\sigma\sqrt{m}$ (w.h.p.), and $\|\boldsymbol{w}\|_2/\sqrt{m}\sim\sigma$; this shows that this constrained least squares algorithm achieves the optimal rate of convergence at the noise level σ.

20 [**Unconstrained Optimization**] State and prove a rigorous version of Remark R9.4.7 (*Unconstrained optimization*). For

noisy recovery, the constraint $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$ may be infeasible; construct an unconstrained recovery program by penalizing the residual. Let $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix whose row vectors are independent, isotropic K-sub-Gaussian random vectors. Consider the noisy linear model

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w},$$

where $\boldsymbol{x} \neq \boldsymbol{0}$ and $\boldsymbol{w} \neq \boldsymbol{0}$ are fixed vectors. To recover $\boldsymbol{x}$ through $\boldsymbol{y}, \boldsymbol{A}$, consider the unconstrained convex problem

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{x}\in\mathbb{R}^n} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2 + \lambda\|\boldsymbol{x}\|_{\mathbb{T}},$$

where $\|\cdot\|_{\mathbb{T}}$ is an arbitrary norm. Take the theoretical oracle parameter $\lambda = \|\boldsymbol{w}\|_2^2/\|\boldsymbol{x}\|_{\mathbb{T}}$; it depends on the unknown signal and noise and is used only to establish a benchmark bound. Prove that the solution $\hat{\boldsymbol{x}}$ satisfies

$$\mathbb{E}\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2 \lesssim \frac{K^2 w(\mathbb{T})\|\boldsymbol{x}\|_{\mathbb{T}} + \|\boldsymbol{w}\|_2}{\sqrt{m}},$$

where $\mathbb{T}$ is the unit ball of $\|\cdot\|_{\mathbb{T}}$.

Prf. • establishing the basic inequality of the optimization problem

Let the error vector be $\boldsymbol{h} = \hat{\boldsymbol{x}} - \boldsymbol{x}$; then

$$\boldsymbol{y} - \boldsymbol{A}\hat{\boldsymbol{x}} = \boldsymbol{w} - \boldsymbol{A}\boldsymbol{h}.$$

The assumptions exclude the degenerate cases $\boldsymbol{x} = \boldsymbol{0}$ and $\boldsymbol{w} = \boldsymbol{0}$.

Since $\hat{\boldsymbol{x}}$ is a global minimizer of the unconstrained objective,

$$\|\boldsymbol{y} - \boldsymbol{A}\hat{\boldsymbol{x}}\|_2^2 + \lambda\|\hat{\boldsymbol{x}}\|_{\mathbb{T}} \leqslant \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2 + \lambda\|\boldsymbol{x}\|_{\mathbb{T}}.$$

That is,

$$\|\boldsymbol{w} - \boldsymbol{A}\boldsymbol{h}\|_2^2 + \lambda\|\hat{\boldsymbol{x}}\|_{\mathbb{T}} \leqslant \|\boldsymbol{w}\|_2^2 + \lambda\|\boldsymbol{x}\|_{\mathbb{T}}. \tag{I}$$

• controlling the prior norm and the residual simultaneously

(*First control the prior norm $\|\boldsymbol{h}\|_{\mathbb{T}}$ of the error vector.*)

Using (I), we obtain

$$\lambda\|\hat{\boldsymbol{x}}\|_{\mathbb{T}} \leqslant \|\boldsymbol{w}\|_2^2 + \lambda\|\boldsymbol{x}\|_{\mathbb{T}}, \text{ hence } \|\hat{\boldsymbol{x}}\|_{\mathbb{T}} \leqslant \|\boldsymbol{x}\|_{\mathbb{T}} + \frac{\|\boldsymbol{w}\|_2^2}{\lambda}.$$

Setting $\lambda = \|\boldsymbol{w}\|_2^2/\|\boldsymbol{x}\|_{\mathbb{T}}$ gives $\|\hat{\boldsymbol{x}}\|_{\mathbb{T}} \leqslant 2\|\boldsymbol{x}\|_{\mathbb{T}}$; using the triangle inequality gives

$$\|\boldsymbol{h}\|_{\mathbb{T}} \leqslant \|\hat{\boldsymbol{x}}\|_{\mathbb{T}} + \|\boldsymbol{x}\|_{\mathbb{T}} \leqslant 3\|\boldsymbol{x}\|_{\mathbb{T}}, \tag{II}$$

Thus the error vector belongs to the rescaled unit ball $\boldsymbol{h} = \hat{\boldsymbol{x}} - \boldsymbol{x} \in 3\|\boldsymbol{x}\|_{\mathbb{T}} \cdot \mathbb{T}$.

(*Then control the prediction residual $\|\boldsymbol{A}\boldsymbol{h}\|_2$.*)

Likewise, using (I), we obtain

$$\|\boldsymbol{w} - \boldsymbol{A}\boldsymbol{h}\|_2^2 \leqslant \|\boldsymbol{w}\|_2^2 + \lambda\|\boldsymbol{x}\|_{\mathbb{T}}.$$

Substituting $\lambda = \|\boldsymbol{w}\|_2^2/\|\boldsymbol{x}\|_{\mathbb{T}}$ (*i.e.* $\lambda\|\boldsymbol{x}\|_{\mathbb{T}} = \|\boldsymbol{w}\|_2^2$) gives

$$\|\boldsymbol{w} - \boldsymbol{A}\boldsymbol{h}\|_2^2 \leqslant 2\|\boldsymbol{w}\|_2^2, \text{ hence } \|\boldsymbol{w} - \boldsymbol{A}\boldsymbol{h}\|_2 \leqslant \sqrt{2}\|\boldsymbol{w}\|_2.$$

Using the reverse triangle inequality $\|\boldsymbol{A}\boldsymbol{h}\|_2 - \|\boldsymbol{w}\|_2 \leqslant \|\boldsymbol{w} - \boldsymbol{A}\boldsymbol{h}\|_2$ gives

$$\|\boldsymbol{A}\boldsymbol{h}\|_2 \leqslant (\sqrt{2}+1)\|\boldsymbol{w}\|_2. \tag{III}$$

• applying the matrix deviation inequality

Define the rescaled set $\mathbb{S} = 3\|\boldsymbol{x}\|_{\mathbb{T}} \cdot \mathbb{T}$; then $\boldsymbol{h} \in \mathbb{S}$.

For $\boldsymbol{h} \in \mathbb{S}$, the triangle inequality gives

$$\sqrt{m}\|\boldsymbol{h}\|_2 \leqslant \|\boldsymbol{A}\boldsymbol{h}\|_2 + \sup_{\boldsymbol{u}\in\mathbb{S}} \big|\|\boldsymbol{A}\boldsymbol{u}\|_2 - \sqrt{m}\|\boldsymbol{u}\|_2\big|. \tag{IV}$$

Substituting (III) into (IV), we obtain

$$\sqrt{m}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant (\sqrt{2}+1)\|\boldsymbol{w}\|_2 + \sup_{\boldsymbol{u}\in 3\|\boldsymbol{x}\|_{\mathbb{T}}\cdot\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{u}\|_2 - \sqrt{m}\|\boldsymbol{u}\|_2\big|.$$

where

$$\sup_{\boldsymbol{u}\in c\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{u}\|_2 - \sqrt{m}\|\boldsymbol{u}\|_2\big| = c\sup_{\boldsymbol{v}\in\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|.$$

Dividing both sides by $\sqrt{m}$,

$$\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \leqslant \frac{(\sqrt{2}+1)\|\boldsymbol{w}\|_2}{\sqrt{m}} + \frac{3\|\boldsymbol{x}\|_{\mathbb{T}}}{\sqrt{m}} \sup_{\boldsymbol{v}\in\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|. \tag{V}$$

- taking expectations to obtain the final convergence bound

  Taking expectations on both sides of (V),

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \leqslant \frac{(\sqrt{2}+1)\mathbb{E}[\|\boldsymbol{w}\|_2]}{\sqrt{m}} + \frac{3\|\boldsymbol{x}\|_{\mathbb{T}}}{\sqrt{m}}\mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|\right].$$

  By Theorem T9.1.1 (*Matrix deviation inequality*), we obtain

$$\mathbb{E}\left[\sup_{\boldsymbol{v}\in\mathbb{T}} \big|\|\boldsymbol{A}\boldsymbol{v}\|_2 - \sqrt{m}\|\boldsymbol{v}\|_2\big|\right] \leqslant C_0 K^2 w(\mathbb{T}).$$

  Substituting it gives

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \leqslant \frac{3C_0K^2w(\mathbb{T})\|\boldsymbol{x}\|_{\mathbb{T}}}{\sqrt{m}} + \frac{(\sqrt{2}+1)\mathbb{E}[\|\boldsymbol{w}\|_2]}{\sqrt{m}}.$$

  Therefore,

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \lesssim \frac{K^2w(\mathbb{T})\|\boldsymbol{x}\|_{\mathbb{T}} + \|\boldsymbol{w}\|_2}{\sqrt{m}}.$$

Rk. For statistical machine learning, taking $\|\cdot\|_{\mathbb{T}} = \|\cdot\|_1$ turns this optimization problem into the classical LASSO algorithm

$$\arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \frac{1}{2}\|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{z}\|_2^2 + \lambda\|\boldsymbol{z}\|_1.$$

21 [**Lasso**] Specialize Exercise Q9.20 (*Unconstrained Optimization*) to the ℓ_1 norm and analyze the Lasso, a popular method for sparse regression. Let $\boldsymbol{x}\in\mathbb{R}^n$ be s-sparse, $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ a random matrix with independent, isotropic, K-sub-Gaussian rows. Consider the noisy linear model

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}.$$

To recover $\boldsymbol{x}$ from $\boldsymbol{y}$ and $\boldsymbol{A}$, consider the unconstrained convex problem

$$\text{minimize } \|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2^2 + \lambda\|\boldsymbol{x}\|_1 \quad \text{over } \boldsymbol{x}\in\mathbb{R}^n.$$

Assuming $\boldsymbol{x}\neq\boldsymbol{0}$ and $\boldsymbol{w}\neq\boldsymbol{0}$, choose the oracle parameter $\lambda = \|\boldsymbol{w}\|_2^2/\|\boldsymbol{x}\|_1$. Prove that the solution $\hat{\boldsymbol{x}}$ satisfies

$$\mathbb{E}\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2 \lesssim \frac{K^2\sqrt{s\log n}\|\boldsymbol{x}\|_1 + \|\boldsymbol{w}\|_2}{\sqrt{m}}.$$

Prf. - applying the universal unconstrained recovery conclusion of Exercise 20

  In Exercise Q9.20 (*Unconstrained Optimization*) we proved that for any norm $\|\cdot\|_{\mathbb{T}}$ with unit ball $\mathbb{T} = \{\boldsymbol{z}\in\mathbb{R}^n : \|\boldsymbol{z}\|_{\mathbb{T}} \leqslant 1\}$, if the tuning parameter is chosen as $\lambda = \|\boldsymbol{w}\|_2^2/\|\boldsymbol{x}\|_{\mathbb{T}}$, then the solution $\hat{\boldsymbol{x}}$ of the unconstrained program

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z}\in\mathbb{R}^n}(\|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{z}\|_2^2 + \lambda\|\boldsymbol{z}\|_{\mathbb{T}})$$

  satisfies the rigorous error bound

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \leqslant C\frac{K^2w(\mathbb{T})\|\boldsymbol{x}\|_{\mathbb{T}} + \|\boldsymbol{w}\|_2}{\sqrt{m}}, \tag{I}$$

  where $C>0$ is an absolute constant.

- specializing to the ℓ_1 norm and the unit ball $\mathbb{B}_1^n$

  In the Lasso, the penalty norm is the ℓ_1 norm $\|\cdot\|_1$, whose unit ball is the closed ℓ_1 ball

  $$\mathbb{T} = \mathbb{B}_1^n = \{\boldsymbol{z} \in \mathbb{R}^n : \|\boldsymbol{z}\|_1 \leqslant 1\},$$

  and the tuning parameter is $\lambda = \|\boldsymbol{w}\|_2^2/\|\boldsymbol{x}\|_1$. Substituting $\mathbb{T} = \mathbb{B}_1^n$ and $\|\boldsymbol{x}\|_{\mathbb{T}} = \|\boldsymbol{x}\|_1$ into (I) yields

  $$\mathbb{E}[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2] \leqslant C\frac{K^2 w(\mathbb{B}_1^n)\|\boldsymbol{x}\|_1 + \|\boldsymbol{w}\|_2}{\sqrt{m}}. \tag{II}$$

- computing the Gaussian width of the ℓ_1 unit ball

  By the definition of the Gaussian width, for a standard Gaussian random vector $\boldsymbol{g} = (g_1, g_2, \cdots, g_n) \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$,

  $$w(\mathbb{B}_1^n) = \mathbb{E}\left[\sup_{\boldsymbol{z}\in\mathbb{B}_1^n} \langle \boldsymbol{g}, \boldsymbol{z}\rangle\right].$$

  By Hölder duality, the dual norm of the ℓ_1 norm is the ℓ_∞ norm, so $\sup_{\boldsymbol{z}\in\mathbb{B}_1^n} \langle \boldsymbol{g}, \boldsymbol{z}\rangle = \|\boldsymbol{g}\|_\infty = \max_{1\leqslant i\leqslant n} |g_i|$, whence

  $$w(\mathbb{B}_1^n) = \mathbb{E}\left[\max_{1\leqslant i\leqslant n} |g_i|\right].$$

  Using the exponential moment generating function and Jensen's inequality, for any $\beta > 0$:

  $$\mathrm{e}^{\beta\mathbb{E}[\max_{1\leqslant i\leqslant n}|g_i|]} \leqslant \mathbb{E}\left[\exp\left(\beta \max_{1\leqslant i\leqslant n} |g_i|\right)\right] \leqslant \sum_{i=1}^n \mathbb{E}[\mathrm{e}^{\beta|g_i|}] \leqslant 2n\exp\left(\frac{\beta^2}{2}\right).$$

  Taking logarithms on both sides and dividing by β:

  $$\mathbb{E}\left[\max_{1\leqslant i\leqslant n} |g_i|\right] \leqslant \frac{\log(2n)}{\beta} + \frac{\beta}{2}.$$

  The optimal parameter on the right-hand side is $\beta_0 = \sqrt{2\log(2n)}$. Substituting it gives

  $$w(\mathbb{B}_1^n) = \mathbb{E}\left[\max_{1\leqslant i\leqslant n} |g_i|\right] \leqslant \sqrt{2\log(2n)} \leqslant C_0\sqrt{\log n}. \tag{III}$$

- substituting to obtain the final conclusion

  Substituting the width bound (III) into (II):

  $$\mathbb{E}[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2] \leqslant C\frac{C_0K^2\sqrt{\log n}\|\boldsymbol{x}\|_1 + \|\boldsymbol{w}\|_2}{\sqrt{m}}.$$

  Since the signal $\boldsymbol{x}$ is a nonzero s-sparse vector, its sparsity satisfies $s \geqslant 1$, so $\sqrt{\log n} \leqslant \sqrt{s\log n}$. Hence

  $$\mathbb{E}[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2] \lesssim \frac{K^2\sqrt{s\log n}\|\boldsymbol{x}\|_1 + \|\boldsymbol{w}\|_2}{\sqrt{m}}.$$

Rk. conversion from the ℓ_1 norm to the Euclidean norm —for any s-sparse vector $\boldsymbol{x}$ ($\|\boldsymbol{x}\|_0 \leqslant s$), by the Cauchy–Schwarz inequality $\|\boldsymbol{x}\|_1 = \sum_{i\in\mathrm{supp}(\boldsymbol{x})} |x_i| \leqslant \sqrt{s}(\sum_{i\in\mathrm{supp}(\boldsymbol{x})} x_i^2)^{1/2} = \sqrt{s}\|\boldsymbol{x}\|_2$, and substituting into the conclusion of this exercise yields the relative reconstruction error bound $\mathbb{E}[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2] \lesssim \frac{K^2 s\sqrt{\log n}}{\sqrt{m}}\|\boldsymbol{x}\|_2 + \|\boldsymbol{w}\|_2$ for the Euclidean norm of the true signal. This shows that to guarantee a relative estimation error bounded by a constant ε, only $m \gtrsim s^2\log n$ measurements are needed (this can be further improved to $m \gtrsim s\log(n/s)$ via tangent-cone local Gaussian width analysis).

22 [**The Sparse Recovery Problem Is Well-Posed**] Let $m \leqslant n^5$and let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ have full row rank, and give a reasonable definition of general position.

(1) Prove that if $m \geqslant 2s$, the equation $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$ has at most 1 s-sparse solution (if a solution exists).

(2) Under (1), if $\mathrm{supp}(\boldsymbol{x})$ (i.e. which components are nonzero) is known, find a scheme to compute $\boldsymbol{x}$ efficiently.

Prf. (1) Say that $\boldsymbol{A}$ is in general position if every set of m columns is linearly independent (*equivalently,* $\mathrm{spark}(\boldsymbol{A}) = m + 1$*; under* $m \leqslant n$ *and full row rank, this is also equivalent to every* $m \times m$ *minor having nonzero determinant; if the columns of* $\boldsymbol{A}$ *are sampled independently from a continuous distribution, such as a Gaussian distribution, this holds with probability* 1).

Suppose, for contradiction, that there exist distinct s-sparse solutions $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^n$, both satisfying the observation equation $\boldsymbol{A}\boldsymbol{x}_1 = \boldsymbol{A}\boldsymbol{x}_2 = \boldsymbol{y}$, with $\|\boldsymbol{x}_1\|_0, \|\boldsymbol{x}_2\|_0 \leqslant s$.

Let $\boldsymbol{h} = \boldsymbol{x}_1 - \boldsymbol{x}_2$. Then:

1. Since
$$\boldsymbol{A}\boldsymbol{h} = \boldsymbol{A}(\boldsymbol{x}_1 - \boldsymbol{x}_2) = \boldsymbol{A}\boldsymbol{x}_1 - \boldsymbol{A}\boldsymbol{x}_2 = \boldsymbol{y} - \boldsymbol{y} = \boldsymbol{0},$$
we have $\boldsymbol{h} \in \ker \boldsymbol{A}$.
2. Since
$$\operatorname{supp}(\boldsymbol{h}) = \operatorname{supp}(\boldsymbol{x}_1 - \boldsymbol{x}_2) \subset \operatorname{supp}(\boldsymbol{x}_1) \cup \operatorname{supp}(\boldsymbol{x}_2),$$
we have $\|\boldsymbol{h}\|_0 = |\operatorname{supp}(\boldsymbol{h})| \leqslant |\operatorname{supp}(\boldsymbol{x}_1)| + |\operatorname{supp}(\boldsymbol{x}_2)| \leqslant s + s = 2s \leqslant m$.

Restricting the equation $\boldsymbol{A}\boldsymbol{h} = \boldsymbol{0}$ to the set of nonzero components $\mathbb{S}_0 = \operatorname{supp}(\boldsymbol{h})$, we get
$$\boldsymbol{A}\boldsymbol{h} = \sum_{j \in \mathbb{S}_0} h_j \boldsymbol{a}_j = \boldsymbol{A}_{\mathbb{S}_0} \boldsymbol{h}_{\mathbb{S}_0} = \boldsymbol{0},$$
where $\boldsymbol{A}_{\mathbb{S}_0} \in \mathbb{R}^{m \times |\mathbb{S}_0|}$ is the submatrix of $\boldsymbol{A}$ restricted to $\mathbb{S}_0$ and $\boldsymbol{h}_{\mathbb{S}_0} \in \mathbb{R}^{|\mathbb{S}_0|}$ is the vector of nonzero components of $\boldsymbol{h}$ on $\mathbb{S}_0$.

Since $|\mathbb{S}_0| \leqslant m$ and $\boldsymbol{A}$ is in general position, the columns $\{\boldsymbol{a}_j\}_{j \in \mathbb{S}_0}$ are linearly independent, i.e. $\boldsymbol{A}_{\mathbb{S}_0}$ has full column rank; hence the homogeneous system $\boldsymbol{A}_{\mathbb{S}_0} \boldsymbol{h}_{\mathbb{S}_0} = \boldsymbol{0}$ has only the trivial solution, giving $\boldsymbol{h}_{\mathbb{S}_0} = \boldsymbol{0}$.

Note that the components of $\boldsymbol{h}$ outside $\mathbb{S}_0$ vanish; therefore $\boldsymbol{h} = \boldsymbol{0}$, i.e. $\boldsymbol{x}_1 = \boldsymbol{x}_2$. This contradicts the assumption $\boldsymbol{x}_1 \neq \boldsymbol{x}_2$; thus, when $m \geqslant 2s$, the equation $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$ has at most 1 s-sparse solution.

(2) Suppose $\mathbb{S} = \operatorname{supp}(\boldsymbol{x}) \subset \{1, 2, \cdots, n\}$, with $|\mathbb{S}| = k \leqslant s \leqslant m/2$.

Since the components outside $\mathbb{S}$ vanish, the original equation reduces to
$$\boldsymbol{y} = \boldsymbol{A}_{\mathbb{S}} \boldsymbol{x}_{\mathbb{S}},$$
where $\boldsymbol{A}_{\mathbb{S}} \in \mathbb{R}^{m \times k}$ is the column submatrix of $\boldsymbol{A}$ restricted to $\mathbb{S}$ and $\boldsymbol{x}_{\mathbb{S}} \in \mathbb{R}^k$ is the unknown nonzero vector.

Since $\boldsymbol{A}$ is in general position and $k \leqslant m$, $\boldsymbol{A}_{\mathbb{S}}$ has full column rank, so $\boldsymbol{A}_{\mathbb{S}}^\top \boldsymbol{A}_{\mathbb{S}} \in \mathbb{R}^{k \times k}$ is symmetric positive definite and invertible, and the solution has the unique closed form
$$\boldsymbol{x}_{\mathbb{S}} = (\boldsymbol{A}_{\mathbb{S}}^\top \boldsymbol{A}_{\mathbb{S}})^{-1} \boldsymbol{A}_{\mathbb{S}}^\top \boldsymbol{y} = \boldsymbol{A}_{\mathbb{S}}^\dagger \boldsymbol{y}.$$
(*Computing the inverse matrix* $(\boldsymbol{A}_{\mathbb{S}}^\top \boldsymbol{A}_{\mathbb{S}})^{-1}$ *directly is numerically ill-conditioned; here we use the QR decomposition.*)

1. Extract the submatrix $\boldsymbol{A}_{\mathbb{S}}$ and compute its economy QR decomposition $\boldsymbol{A}_{\mathbb{S}} = \boldsymbol{Q}\boldsymbol{R}$, where $\boldsymbol{Q} \in \mathbb{R}^{m \times k}$ satisfies $\boldsymbol{Q}^\top \boldsymbol{Q} = \boldsymbol{I}_k$ and $\boldsymbol{R} \in \mathbb{R}^{k \times k}$ is upper triangular and invertible.
2. Compute $\boldsymbol{c} = \boldsymbol{Q}^\top \boldsymbol{y} \in \mathbb{R}^k$ and solve the upper triangular system $\boldsymbol{R}\boldsymbol{x}_{\mathbb{S}} = \boldsymbol{c}$ by back substitution; finally reconstruct the full vector $\boldsymbol{x}$, filling in $\boldsymbol{x}_{\mathbb{S}}$ on $\mathbb{S}$ and setting the remaining entries to 0.

The QR decomposition (via Householder reflections or Gram–Schmidt orthogonalization) costs $O(mk^2)$, the orthogonal projection $\boldsymbol{c} = \boldsymbol{Q}^\top \boldsymbol{y}$ costs $O(mk)$, and back substitution costs $O(k^2)$; the total time complexity is $O(mk^2) \leqslant O(ms^2)$. Since in compressed sensing the sparsity is tiny ($s \ll n$), the computation involves only the small $m \times k$ matrix, and is extremely fast.

Rk. For (1), in information theory, $m \geqslant 2s$ is the absolute lower bound guaranteeing unambiguous recovery of the signal (conservation of degrees of freedom: each nonzero entry needs 1 measurement to determine its position and 1 to determine its value); the oracle property (Oracle Property).

For (2), the solution is called the oracle least squares estimator. If the observations contain Gaussian white noise $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}$ ($\boldsymbol{w} \sim N(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$), then
$$\mathbb{E}[\|\boldsymbol{x}_{\mathbb{S}} - \boldsymbol{x}_{\mathbb{S}}^*\|_2^2 \mid \boldsymbol{A}] = \sigma^2 \operatorname{tr}((\boldsymbol{A}_{\mathbb{S}}^\top \boldsymbol{A}_{\mathbb{S}})^{-1}).$$

If, in addition, $\sigma_{\min}(\boldsymbol{A}_{\mathbb{S}}), \sigma_{\max}(\boldsymbol{A}_{\mathbb{S}}) \asymp \sqrt{m}$, then

$$\mathbb{E}[\|\boldsymbol{x}_{\mathbb{S}} - \boldsymbol{x}_{\mathbb{S}}^*\|_2^2 \mid \boldsymbol{A}] \asymp \sigma^2 k/m.$$

Under this condition it attains the Cramér–Rao lower bound.

23 [**The "p-Norm" for $0 \leqslant p < 1$**] Equation (9.21) defines $\|\boldsymbol{x}\|_0$ as the number of nonzero components of a vector $\boldsymbol{x}$.

(1) Check that $\|\cdot\|_0$ is not a norm on $\mathbb{R}^n$.

(2) Check that if $0 < p < 1$, then $\|\cdot\|_p$ is not a norm on $\mathbb{R}^n$ (Figure 9.10 shows that the unit ball is not convex).

(3) Prove that for every $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|\boldsymbol{x}\|_0 = \lim_{p \to 0^+} \|\boldsymbol{x}\|_p^p.$$

Prf. (1) $\|\cdot\|_0$ is not a norm

Take the nonzero vector $\boldsymbol{x} = \boldsymbol{e}_1 = (1, 0, \cdots, 0)$ and the scalar $\alpha = 2$. Then $\|\alpha\boldsymbol{x}\|_0 = \|(2, 0, \cdots, 0)\|_0 = 1$, but $|\alpha| \cdot \|\boldsymbol{x}\|_0 = 2 \cdot 1 = 2$, i.e. $\|\alpha\boldsymbol{x}\|_0 = 1 \neq 2 = |\alpha| \cdot \|\boldsymbol{x}\|_0$. Absolute homogeneity fails, so $\|\cdot\|_0$ is not a norm on $\mathbb{R}^n$ (it is usually called the ℓ_0 pseudo-norm or the counting metric).

(2) $\|\cdot\|_p$ is not a norm for $0 < p < 1$

Let $\boldsymbol{x} = \boldsymbol{e}_1$ and $\boldsymbol{y} = \boldsymbol{e}_2$. Then $\|\boldsymbol{x}\|_p = \|\boldsymbol{y}\|_p = 1$, but $\|\boldsymbol{x} + \boldsymbol{y}\|_p = (1^p + 1^p)^{1/p} = 2^{1/p} > 2^1 = 2$ (since $0 < p < 1$, the exponent $1/p > 1$). This strictly violates the triangle inequality $\|\boldsymbol{x} + \boldsymbol{y}\|_p \leqslant \|\boldsymbol{x}\|_p + \|\boldsymbol{y}\|_p = 2$, so $\|\cdot\|_p$ is not a norm.

Geometric verification: the closed unit ball of any norm is necessarily convex, but when $0 < p < 1$, the unit ball $\mathbb{B}_p^n = \{\boldsymbol{z} \in \mathbb{R}^n : \|\boldsymbol{z}\|_p \leqslant 1\}$ is not convex —$\boldsymbol{e}_1, \boldsymbol{e}_2 \in \mathbb{B}_p^n$, while the midpoint $\boldsymbol{m} = \frac{1}{2}\boldsymbol{e}_1 + \frac{1}{2}\boldsymbol{e}_2$ satisfies

$$\|\boldsymbol{m}\|_p = \left(2 \cdot \left(\frac{1}{2}\right)^p\right)^{1/p} = 2^{1/p - 1} > 1,$$

i.e. the midpoint lies outside the unit ball (the ball surface caves inward, taking an astroid-like shape).

(3) The limit

For fixed $\boldsymbol{x} = (x_1, x_2, \cdots, x_n)$: if $x_i \neq 0$, then $|x_i|^p = \mathrm{e}^{p \log |x_i|} \to 1$ as $p \to 0^+$; if $x_i = 0$, then $|x_i|^p = 0$ identically. Hence the finite sum and the limit can be interchanged:

$$\|\boldsymbol{x}\|_p^p = \sum_{i=1}^n |x_i|^p \to \sum_{i=1}^n \mathbf{1}_{\{x_i \neq 0\}} = \#\{i : x_i \neq 0\} = \|\boldsymbol{x}\|_0.$$

Rk. $\|\cdot\|_0$ is the algebraic limit of the ℓ_p pseudo-norms as $p \to 0^+$.

For $p \in (0, 1)$, ℓ_p minimization approximates ℓ_0 but remains nonconvex; $p = 1$ is the minimal critical point at which the objective function is convex (*the boundary of convexity*), so replacing the ℓ_0 norm by the ℓ_1 norm (*Basis Pursuit / Lasso*) is the tightest relaxation that turns the nonconvex combinatorial problem into a high-dimensional convex program.

24 [**Approximately Sparse Recovery**] Extend Corollary C9.4.8 (*Sparse recovery*) to approximately sparse signals. Assume throughout that $n \geqslant 2$ and $1 \leqslant s \leqslant n^6$.

(1) Prove that any s-sparse signal $\boldsymbol{x}$ can be recovered from measurements $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$ by solving the optimization problem

$$\text{minimize } \|\boldsymbol{z}\|_1 \quad \text{subject to } \boldsymbol{y} = \boldsymbol{A}\boldsymbol{z},$$

whose solution $\hat{\boldsymbol{x}}$ gives the recovery error

$$\mathbb{E}\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2 \leqslant CK^2 \sqrt{\frac{s \log n}{m}} \|\boldsymbol{x}\|_2.$$

(2) Argue that a similar conclusion holds for approximately sparse signals. State and prove such a guarantee.

Prf. (1) By Exercise Q9.18 (*Recovery by Optimization*), for any norm $\|\cdot\|_{\mathbb{T}}$ on $\mathbb{R}^n$ (*with unit ball* $\mathbb{T}$), the convex program

$$\arg\min_{\boldsymbol{z} \in \mathbb{R}^n} \|\boldsymbol{z}\|_1 \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$$

has an estimator $\hat{\boldsymbol{x}}$ satisfying

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \leqslant \frac{C_1K^2w(\mathbb{T})}{\sqrt{m}}\|\boldsymbol{x}\|_{\mathbb{T}},$$

Specializing the norm to the ℓ_1 norm $\|\cdot\|_1$ with unit ball the cross-polytope $\mathbb{T}=\mathbb{B}_1^n$, we obtain

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2] \leqslant \frac{C_1K^2w(\mathbb{B}_1^n)}{\sqrt{m}}\|\boldsymbol{x}\|_1, \tag{I}$$

so it remains to compute $w(\mathbb{B}_1^n)$ and $\|\boldsymbol{x}\|_1$.

For $w(\mathbb{B}_1^n)$,

$$w(\mathbb{B}_1^n)=\mathbb{E}\left[\sup_{\boldsymbol{z}\in\mathbb{B}_1^n}\langle \boldsymbol{g},\boldsymbol{z}\rangle\right]=\mathbb{E}[\|\boldsymbol{g}\|_\infty]=\mathbb{E}\left[\max_{1\leqslant i\leqslant n}|g_i|\right]\leqslant C_0\sqrt{\log n};$$

for $\|\boldsymbol{x}\|_1$, since $\boldsymbol{x}$ is s-sparse, its number of nonzero entries satisfies $|\operatorname{supp}(\boldsymbol{x})|\leqslant s$; by the Cauchy–Schwarz inequality,

$$\|\boldsymbol{x}\|_1=\sum_{i\in\operatorname{supp}(\boldsymbol{x})}1\cdot|x_i|\leqslant\sqrt{|\operatorname{supp}(\boldsymbol{x})|}\left(\sum_{i\in\operatorname{supp}(\boldsymbol{x})}x_i^2\right)^{1/2}\leqslant\sqrt{s}\|\boldsymbol{x}\|_2;$$

substituting into (I) yields

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2]\leqslant\frac{C_2K^2\sqrt{\log n}}{\sqrt{m}}\cdot(\sqrt{s}\|\boldsymbol{x}\|_2)=CK^2\sqrt{\frac{s\log n}{m}}\|\boldsymbol{x}\|_2.$$

(2) Let $\mathbb{S}\subset\{1,2,\cdots,n\}$ be the index set of the s largest absolute components of $\boldsymbol{x}$, $\boldsymbol{x}_s=\boldsymbol{x}|_{\mathbb{S}}$ the best s-term approximation; the tail error vector is $\boldsymbol{x}-\boldsymbol{x}_s$, and the sparsification error is $\sigma_s(\boldsymbol{x})_1=\|\boldsymbol{x}-\boldsymbol{x}_s\|_1$.

We only need to prove: for every signal $\boldsymbol{x}\in\mathbb{R}^n$ (*not necessarily sparse*), the convex program

$$\arg\min_{\boldsymbol{z}\in\mathbb{R}^n}\|\boldsymbol{z}\|_{\mathbb{T}}\quad\text{s.t.}\quad\boldsymbol{A}\boldsymbol{z}=\boldsymbol{A}\boldsymbol{x}$$

has a solution $\hat{\boldsymbol{x}}$ satisfying

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2]\leqslant CK^2\left(\sqrt{\frac{s\log n}{m}}\|\boldsymbol{x}\|_2+\frac{\sqrt{\log n}}{\sqrt{m}}\|\boldsymbol{x}-\boldsymbol{x}_s\|_1\right).$$

Similarly to the conclusion of (1), for every signal $\boldsymbol{x}\in\mathbb{R}^n$ (*sparse or not*), the optimizer $\hat{\boldsymbol{x}}$ satisfies

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2]\leqslant\frac{C_2K^2\sqrt{\log n}}{\sqrt{m}}\|\boldsymbol{x}\|_1. \tag{II}$$

By the triangle inequality,

$$\|\boldsymbol{x}\|_1\leqslant\|\boldsymbol{x}_s\|_1+\|\boldsymbol{x}-\boldsymbol{x}_s\|_1;$$

for the head $\boldsymbol{x}_s$, $|\operatorname{supp}(\boldsymbol{x}_s)|=s$ ($\|\boldsymbol{x}_s\|_0\leqslant s$), the Cauchy–Schwarz inequality gives

$$\|\boldsymbol{x}_s\|_1\leqslant\sqrt{s}\|\boldsymbol{x}_s\|_2\leqslant\sqrt{s}\|\boldsymbol{x}\|_2;$$

substituting yields

$$\|\boldsymbol{x}\|_1\leqslant\sqrt{s}\|\boldsymbol{x}\|_2+\|\boldsymbol{x}-\boldsymbol{x}_s\|_1; \tag{III}$$

substituting (III) into (II) yields

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2]\leqslant\frac{C_2K^2\sqrt{\log n}}{\sqrt{m}}(\sqrt{s}\|\boldsymbol{x}\|_2+\|\boldsymbol{x}-\boldsymbol{x}_s\|_1)=CK^2\left(\sqrt{\frac{s\log n}{m}}\|\boldsymbol{x}\|_2+\frac{\sqrt{\log n}}{\sqrt{m}}\|\boldsymbol{x}-\boldsymbol{x}_s\|_1\right).$$

Rk. Observe the structure of the recovery error,

$$\mathbb{E}[\|\hat{\boldsymbol{x}}-\boldsymbol{x}\|_2]\leqslant K^2\left(\sqrt{\frac{s\log n}{m}}\|\boldsymbol{x}\|_2+\frac{\sqrt{\log n}}{\sqrt{m}}\|\boldsymbol{x}-\boldsymbol{x}_s\|_1\right),$$

the first term is the reconstruction error of the sparse head, and the second term is the model truncation error caused by

the non-sparse tail; the recovery error has the following properties:

1. Degenerate consistency: if $\boldsymbol{x}$ is exactly s-sparse, then $\|\boldsymbol{x}-\boldsymbol{x}_s\|_1=0$, degenerating to the conclusion of (1).
2. Asymptotic optimality (*Instance Optimality*): when the number of measurements is at the standard critical order $m \asymp s\log n$ of compressed sensing, the first term decays to $O(\|\boldsymbol{x}\|_2)$, and the second term becomes $\frac{\sqrt{\log n}}{\sqrt{s\log n}}\|\boldsymbol{x}-\boldsymbol{x}_s\|_1 = \frac{\|\boldsymbol{x}-\boldsymbol{x}_s\|_1}{\sqrt{s}}$, the Stechkin approximation bound of classical compressed sensing. This shows that even if the signal is perturbed non-sparsely, the recovery algorithm remains robust.

25 [**Convexification of the Set of Sparse Vectors**] Let

$$\mathbb{S}_{n,s}=\{\boldsymbol{x}\in\mathbb{R}^n : \|\boldsymbol{x}\|_0\leqslant s,\ \|\boldsymbol{x}\|_2\leqslant 1\}$$

be the set of unit s-sparse vectors,

$$\mathbb{T}_{n,s}=\sqrt{s}\mathbb{B}_1^n\cap\mathbb{B}_2^n=\{\boldsymbol{x}\in\mathbb{R}^n : \|\boldsymbol{x}\|_1\leqslant\sqrt{s},\ \|\boldsymbol{x}\|_2\leqslant 1\}$$

the truncated ℓ_1 ball. Prove:

$$\operatorname{conv}(\mathbb{S}_{n,s})\subset\mathbb{T}_{n,s}\subset 2\operatorname{conv}(\mathbb{S}_{n,s}).$$

Prf. • The first inclusion: $\operatorname{conv}(\mathbb{S}_{n,s})\subset\mathbb{T}_{n,s}$

Verify $\mathbb{S}_{n,s}\subset\mathbb{T}_{n,s}$: take $\boldsymbol{x}\in\mathbb{S}_{n,s}$; by definition $\|\boldsymbol{x}\|_2\leqslant 1$ and $\|\boldsymbol{x}\|_0=|\operatorname{supp}(\boldsymbol{x})|\leqslant s$. By the Cauchy–Schwarz inequality,

$$\|\boldsymbol{x}\|_1=\sum_{j\in\operatorname{supp}(\boldsymbol{x})}1\cdot|x_j|\leqslant\sqrt{|\operatorname{supp}(\boldsymbol{x})|}\left(\sum_{j\in\operatorname{supp}(\boldsymbol{x})}x_j^2\right)^{1/2}\leqslant\sqrt{s}\|\boldsymbol{x}\|_2\leqslant\sqrt{s},$$

so $\|\boldsymbol{x}\|_1\leqslant\sqrt{s}$ and $\|\boldsymbol{x}\|_2\leqslant 1$, i.e. $\boldsymbol{x}\in\mathbb{T}_{n,s}$; hence $\mathbb{S}_{n,s}\subset\mathbb{T}_{n,s}$. Thus

$$\operatorname{conv}(\mathbb{S}_{n,s})\subset\mathbb{T}_{n,s}.$$

• Proving $\mathbb{T}_{n,s}\subset 2\operatorname{conv}(\mathbb{S}_{n,s})$

Fix $\boldsymbol{x}\in\mathbb{T}_{n,s}$. If $\boldsymbol{x}=\boldsymbol{0}$, the conclusion is trivial; we only need to consider $\boldsymbol{x}\neq\boldsymbol{0}$.

(Partition $\operatorname{supp}(\boldsymbol{x})$ *into mutually disjoint sets* $I_1\sqcup I_2\sqcup\cdots$*, where* I_1 *indexes the* s *largest coefficients of* $\boldsymbol{x}$*,* I_2 *indexes the next* s *largest coefficients, and so on. It suffices to prove* $\sum_i\|\boldsymbol{x}_{I_i}\|_2\leqslant 2$*, where* $\boldsymbol{x}_I\in\mathbb{R}^I$ *denotes the restriction of* $\boldsymbol{x}$ *to* I*.)*

Assume without loss of generality that the components of $\boldsymbol{x}$ are arranged in nonincreasing order of absolute value:

$$|x_1|\geqslant|x_2|\geqslant\cdots\geqslant|x_n|.$$

Partition $\{1,2,\cdots,n\}=I_1\sqcup I_2\sqcup\cdots$, where $I_1=\{1,\ldots,s\}$, $I_2=\{s+1,\ldots,2s\}$, and so on. Let $\boldsymbol{x}_{I_i}\in\mathbb{R}^n$ be the truncation of $\boldsymbol{x}$ to I_i (*equal to* $\boldsymbol{x}$ *on* I_i *and zero elsewhere*); then

$$\boldsymbol{x}=\sum_{i\geqslant 1}\boldsymbol{x}_{I_i}.$$

Moreover, $\|\boldsymbol{x}_{I_i}\|_0=|I_i|\leqslant s$.

• Estimating the Euclidean norms of the blocks and summing

Write $d_i=\|\boldsymbol{x}_{I_i}\|_2$. For the first block, since $\|\boldsymbol{x}\|_2\leqslant 1$, we have $d_1\leqslant 1$. For the later blocks I_i ($i\geqslant 2$), for any $j\in I_i$ and any $l\in I_{i-1}$, the nonincreasing arrangement gives $|x_j|\leqslant|x_l|$; averaging over $l\in I_{i-1}$ ($|I_{i-1}|=s$) yields

$$|x_j|\leqslant\frac{1}{s}\sum_{l\in I_{i-1}}|x_l|=\frac{\|\boldsymbol{x}_{I_{i-1}}\|_1}{s},$$

hence

$$d_i=\|\boldsymbol{x}_{I_i}\|_2=\left(\sum_{j\in I_i}x_j^2\right)^{1/2}\leqslant\left(|I_i|\cdot\left(\frac{\|\boldsymbol{x}_{I_{i-1}}\|_1}{s}\right)^2\right)^{1/2}\leqslant\frac{\|\boldsymbol{x}_{I_{i-1}}\|_1}{\sqrt{s}}.$$

Since the blocks are disjoint, the ℓ_1 norm is additive, so

$$\sum_{i\geqslant 2}\|\boldsymbol{x}_{I_{i-1}}\|_1 \leqslant \|\boldsymbol{x}\|_1.$$

Therefore

$$\sum_{i\geqslant 1} d_i \leqslant d_1 + \frac{1}{\sqrt{s}}\sum_{i\geqslant 2}\|\boldsymbol{x}_{I_{i-1}}\|_1 \leqslant \|\boldsymbol{x}\|_2 + \frac{\|\boldsymbol{x}\|_1}{\sqrt{s}} \leqslant 1 + \frac{\sqrt{s}}{\sqrt{s}} = 2. \tag{II}$$

- Constructing the convex combination to obtain the second inclusion

  For the terms with $d_i > 0$, let $\overline{\boldsymbol{x}_{I_i}} \in \mathbb{S}_{n,s}$; then $\|\overline{\boldsymbol{x}_{I_i}}\|_0 = |I_i| \leqslant s$. Rewrite $\boldsymbol{x}$ as

$$\boldsymbol{x} = \sum_i \boldsymbol{x}_{I_i} = \sum_i d_i \overline{\boldsymbol{x}_{I_i}} = \left(\sum_i d_i\right)\sum_i \frac{d_i}{\sum_j d_j}\overline{\boldsymbol{x}_{I_i}}.$$

  Let $M = \sum_i d_i \leqslant 2$ (*using the conclusion of (II)*) and $\boldsymbol{v} = \sum_i \frac{d_i}{\sum_j d_j}\overline{\boldsymbol{x}_{I_i}}$. Then $\boldsymbol{v}$ is a convex combination of elements of $\mathbb{S}_{n,s}$, so $\boldsymbol{v} \in \operatorname{conv}(\mathbb{S}_{n,s})$. Since $\mathbf{0} \in \operatorname{conv}(\mathbb{S}_{n,s})$,

$$\boldsymbol{x} = M\boldsymbol{v} = \left(1 - \frac{M}{2}\right)\cdot \mathbf{0} + \frac{M}{2}\cdot(2\boldsymbol{v}) \in 2\operatorname{conv}(\mathbb{S}_{n,s}).$$

  Thus $\mathbb{T}_{n,s} \subset 2\operatorname{conv}(\mathbb{S}_{n,s})$.

26 [**Logarithmic Improvement in Sparse Recovery**] (1) Using Exercise Q9.25 (*Convexification of the Set of Sparse Vectors*), prove

$$w(\mathbb{T}_{n,s}) \leqslant 2w(\mathbb{S}_{n,s}) \leqslant C\sqrt{s\log(en/s)}.$$

(2) Consider the recovery program with a bounded prior: for any $\boldsymbol{x} \in \mathbb{S}_{n,s}$, let

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z}\in\mathbb{T}_{n,s}} \|\boldsymbol{z}\|_1 \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}.$$

Using the conclusion of (1), prove that the logarithmic factor in this sparse-recovery guarantee can be improved to

$$\mathbb{E}\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2 \leqslant CK^2\sqrt{\frac{s\log(en/s)}{m}},$$

showing that $m \gtrsim s\log(en/s)$ measurements suffice.

Prf. (1) • Proving $w(\mathbb{T}_{n,s}) \leqslant 2w(\mathbb{S}_{n,s})$

By Exercise Q9.25 (*Convexification of the Set of Sparse Vectors*), $\mathbb{T}_{n,s} \subset 2\operatorname{conv}(\mathbb{S}_{n,s})$. Taking Gaussian widths on both sides,

$$w(\mathbb{T}_{n,s}) \leqslant w(2\operatorname{conv}(\mathbb{S}_{n,s})) = 2w(\operatorname{conv}(\mathbb{S}_{n,s})) = 2w(\mathbb{S}_{n,s}).$$

- Proving $w(\mathbb{S}_{n,s}) \leqslant C\sqrt{s\log(en/s)}$

Let $\boldsymbol{g} = (g_1, g_2, \cdots, g_n) \sim N(\mathbf{0}, \boldsymbol{I}_n)$. By definition $w(\mathbb{S}_{n,s}) = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{S}_{n,s}}\langle \boldsymbol{g}, \boldsymbol{x}\rangle\right]$. Any $\boldsymbol{x} \in \mathbb{S}_{n,s}$ has $\operatorname{supp}(\boldsymbol{x}) \subset I \subset \{1, \cdots, n\}$ ($|I| \leqslant s$); for a fixed I, $\sup_{\boldsymbol{x}\in\mathbb{R}^I, \|\boldsymbol{x}\|_2\leqslant 1}\langle \boldsymbol{g}, \boldsymbol{x}\rangle = \|\boldsymbol{g}_I\|_2$, so taking the maximum over all subsets of size s:

$$\sup_{\boldsymbol{x}\in\mathbb{S}_{n,s}}\langle \boldsymbol{g}, \boldsymbol{x}\rangle = \max_{|I|=s}\|\boldsymbol{g}_I\|_2.$$

The number of subsets of size s is $\binom{n}{s} \leqslant (en/s)^s$. For a fixed I, $\|\boldsymbol{g}_I\|_2^2 = \sum_{i\in I} g_i^2 \sim \chi^2(s)$ has the chi-square distribution with s degrees of freedom, whose moment generating function at $0 < \lambda < 1/2$ is $\mathbb{E}[\mathrm{e}^{\lambda\|\boldsymbol{g}_I\|_2^2}] = (1 - 2\lambda)^{-s/2}$. Applying the union bound (exponential sum) to the maximum:

$$\mathbb{E}\left[\exp\left(\lambda\max_{|I|=s}\|\boldsymbol{g}_I\|_2^2\right)\right] \leqslant \sum_{|I|=s}\mathbb{E}[\mathrm{e}^{\lambda\|\boldsymbol{g}_I\|_2^2}] = \binom{n}{s}(1-2\lambda)^{-s/2} \leqslant \left(\frac{en}{s}\right)^s(1-2\lambda)^{-s/2}.$$

By Jensen's inequality $\exp(\lambda\mathbb{E}\left[\max_{|I|=s}\|\boldsymbol{g}_I\|_2^2\right]) \leqslant \mathbb{E}\left[\exp(\lambda\max_{|I|=s}\|\boldsymbol{g}_I\|_2^2)\right]$, taking logarithms and dividing by λ:

$$\mathbb{E}\left[\max_{|I|=s}\|\boldsymbol{g}_I\|_2^2\right] \leqslant \frac{s}{\lambda}\log\left(\frac{en}{s}\right) + \frac{s}{2\lambda}\log\left(\frac{1}{1-2\lambda}\right).$$

Choosing the parameter $\lambda = 1/4$ (then $1 - 2\lambda = 1/2$):

$$\mathbb{E}\left[\max_{|I|=s} \|\boldsymbol{g}_I\|_2^2\right] \leqslant 4s \log\left(\frac{en}{s}\right) + 2s \log 2 \leqslant C_0^2 s \log\left(\frac{en}{s}\right).$$

Using again Jensen's inequality ($\mathbb{E}[X] \leqslant \sqrt{\mathbb{E}[X^2]}$):

$$w(\mathbb{S}_{n,s}) = \mathbb{E}\left[\max_{|I|=s} \|\boldsymbol{g}_I\|_2\right] \leqslant \sqrt{\mathbb{E}\left[\max_{|I|=s} \|\boldsymbol{g}_I\|_2^2\right]} \leqslant C_0 \sqrt{s \log\left(\frac{en}{s}\right)}. \tag{II}$$

Combining (I) and (II) yields $w(\mathbb{T}_{n,s}) \leqslant 2w(\mathbb{S}_{n,s}) \leqslant C\sqrt{s \log(en/s)}$.

(2) • Application to sparse recovery: the cone constraint

Let $\boldsymbol{x} \in \mathbb{S}_{n,s}$ and let $\hat{\boldsymbol{x}}$ satisfy

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z} \in \mathbb{T}_{n,s}} \|\boldsymbol{z}\|_1 \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}.$$

Write $\boldsymbol{h} = \hat{\boldsymbol{x}} - \boldsymbol{x}$. By the minimality of $\hat{\boldsymbol{x}}$, $\|\hat{\boldsymbol{x}}\|_1 \leqslant \|\boldsymbol{x}\|_1$. Let $\mathbb{S} = \operatorname{supp}(\boldsymbol{x})$:

$$\|\boldsymbol{x}\|_1 = \|\boldsymbol{x}_{\mathbb{S}}\|_1 \geqslant \|\hat{\boldsymbol{x}}\|_1 = \|\boldsymbol{x}_{\mathbb{S}} + \boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \geqslant \|\boldsymbol{x}_{\mathbb{S}}\|_1 - \|\boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1,$$

so $\|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant \|\boldsymbol{h}_{\mathbb{S}}\|_1$; hence the ℓ_1 norm of the full vector satisfies

$$\|\boldsymbol{h}\|_1 = \|\boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant 2\|\boldsymbol{h}_{\mathbb{S}}\|_1 \leqslant 2\sqrt{s}\|\boldsymbol{h}_{\mathbb{S}}\|_2 \leqslant 2\sqrt{s}\|\boldsymbol{h}\|_2.$$

If $\boldsymbol{h} \neq \boldsymbol{0}$, write its normalized vector as $\bar{\boldsymbol{h}} = \boldsymbol{h}/\|\boldsymbol{h}\|_2$. Then $\|\bar{\boldsymbol{h}}\|_2 = 1$ and $\|\bar{\boldsymbol{h}}\|_1 = \|\boldsymbol{h}\|_1/\|\boldsymbol{h}\|_2 \leqslant 2\sqrt{s}$, i.e.

$$\bar{\boldsymbol{h}} \in 2(\sqrt{s}\mathbb{B}_1^n \cap \mathbb{B}_2^n) = 2\mathbb{T}_{n,s}.$$

• Application to sparse recovery: the M^* bound and the final bound

Since both $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$ belong to $\mathbb{T}_{n,s}$ and $\boldsymbol{A}\boldsymbol{h} = \boldsymbol{0}$, we have $\boldsymbol{h} \in 2\mathbb{T}_{n,s} \cap \ker \boldsymbol{A}$. By Theorem T9.3.1 ($M^*$ *bound*),

$$\mathbb{E}[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2] \leqslant \mathbb{E}[\operatorname{diam}(2\mathbb{T}_{n,s} \cap \ker \boldsymbol{A})] \leqslant \frac{C_1 K^2 w(2\mathbb{T}_{n,s})}{\sqrt{m}} = \frac{2C_1 K^2 w(\mathbb{T}_{n,s})}{\sqrt{m}}.$$

Substituting the width bound (II):

$$\mathbb{E}[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2] \leqslant \frac{2C_1 K^2 \cdot C\sqrt{s \log(en/s)}}{\sqrt{m}} = C' K^2 \sqrt{\frac{s \log(en/s)}{m}}.$$

This proves the required conclusion for $\boldsymbol{x} \in \mathbb{S}_{n,s}$.

27 [**Gaussian Width of Sparse Vectors**] Let

$$\mathbb{S}_{n,s} = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_0 \leqslant s,\ \|\boldsymbol{x}\|_2 \leqslant 1\},$$

be the set of unit s-sparse vectors, and let its truncated ℓ_1 ball (*approximate convex hull*) be

$$\mathbb{T}_{n,s} = \sqrt{s}\mathbb{B}_1^n \cap \mathbb{B}_2^n = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_1 \leqslant \sqrt{s},\ \|\boldsymbol{x}\|_2 \leqslant 1\}.$$

Prove:

$$w(\mathbb{T}_{n,s}) \asymp w(\mathbb{S}_{n,s}) \asymp \sqrt{s \log(en/s)},$$

Prf. By Exercise Q9.25 (*Convexification of the Set of Sparse Vectors*), $\operatorname{conv}(\mathbb{S}_{n,s}) \subset \mathbb{T}_{n,s} \subset 2\operatorname{conv}(\mathbb{S}_{n,s})$. Taking Gaussian widths on both sides,

$$w(\operatorname{conv}(\mathbb{S}_{n,s})) \leqslant w(\mathbb{T}_{n,s}) \leqslant w(2\operatorname{conv}(\mathbb{S}_{n,s})).$$

Simplifying with the properties above yields

$$w(\mathbb{S}_{n,s}) \leqslant w(\mathbb{T}_{n,s}) \leqslant 2w(\mathbb{S}_{n,s}),$$

i.e. $w(\mathbb{T}_{n,s}) \asymp w(\mathbb{S}_{n,s})$.

(It remains to estimate $w(\mathbb{S}_{n,s})$.)

• Proving the upper bound $w(\mathbb{S}_{n,s}) \leqslant C\sqrt{s \log(en/s)}$

Let $\boldsymbol{g} = (g_1, g_2, \cdots, g_n) \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$. Ranging over all s-sparse unit vectors:

$$w(\mathbb{S}_{n,s}) = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{S}_{n,s}} \langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] = \mathbb{E}\left[\max_{|I|=s} \|\boldsymbol{g}_I\|_2\right],$$

where $I \subset \{1, \cdots, n\}$ is an index subset of size s and $\boldsymbol{g}_I = (g_i)_{i\in I} \in \mathbb{R}^s$. The number of index subsets is $\binom{n}{s} \leqslant (en/s)^s$. For a fixed I, $\|\boldsymbol{g}_I\|_2^2 \sim \chi^2(s)$ has the chi-square distribution with s degrees of freedom, whose moment generating function at $\lambda = 1/4$ is $\mathbb{E}\left[\mathrm{e}^{\frac{1}{4}\|\boldsymbol{g}_I\|_2^2}\right] = (1-\frac{1}{2})^{-s/2} = 2^{s/2}$. By the union bound over the exponential sum:

$$\mathbb{E}\left[\exp\left(\frac{1}{4}\max_{|I|=s}\|\boldsymbol{g}_I\|_2^2\right)\right] \leqslant \sum_{|I|=s}\mathbb{E}\left[\mathrm{e}^{\frac{1}{4}\|\boldsymbol{g}_I\|_2^2}\right] \leqslant \binom{n}{s}2^{s/2} \leqslant \left(\frac{en}{s}\right)^s 2^{s/2}.$$

By Jensen's inequality, taking logarithms on both sides and multiplying by 4:

$$\mathbb{E}\left[\max_{|I|=s}\|\boldsymbol{g}_I\|_2^2\right] \leqslant 4s\log\left(\frac{en}{s}\right) + 2s\log 2 \leqslant C_1^2 s\log\left(\frac{en}{s}\right).$$

By Jensen's inequality again ($\mathbb{E}[X] \leqslant \sqrt{\mathbb{E}[X^2]}$):

$$w(\mathbb{S}_{n,s}) = \mathbb{E}\left[\max_{|I|=s}\|\boldsymbol{g}_I\|_2\right] \leqslant \sqrt{\mathbb{E}\left[\max_{|I|=s}\|\boldsymbol{g}_I\|_2^2\right]} \leqslant C_1\sqrt{s\log\left(\frac{en}{s}\right)}. \tag{II}$$

- Lower bound: $w(\mathbb{S}_{n,s}) \geqslant c\sqrt{s\log(en/s)}$

  Let $|g|_{(1)} \geqslant |g|_{(2)} \geqslant \cdots \geqslant |g|_{(n)}$ be the order statistics of the component magnitudes $|g_1|, \cdots, |g_n|$ in nonincreasing order. Then

$$\max_{|I|=s}\|\boldsymbol{g}_I\|_2 = \sqrt{\sum_{i=1}^{s}|g|_{(i)}^2} \geqslant \sqrt{s\cdot|g|_{(s)}^2} = \sqrt{s}|g|_{(s)}. \tag{III}$$

  We distinguish two ranges of s:

  a. If $s \geqslant n/4$ (*the high-dimensional dense regime*), then $\log(en/s) \leqslant \log(4\mathrm{e}) = O(1)$, i.e. $\sqrt{s\log(en/s)} \asymp \sqrt{s}$. Since $\mathbb{S}_{n,s}$ contains the unit ball $\mathbb{B}_2^s$ spanned by the first s coordinates,

$$w(\mathbb{S}_{n,s}) \geqslant \mathbb{E}[\|\boldsymbol{g}_s\|_2] = \sqrt{2}\frac{\Gamma((s+1)/2)}{\Gamma(s/2)} \geqslant c_0\sqrt{s} \geqslant c_1\sqrt{s\log(en/s)}.$$

  b. If $s < n/4$ (*the typical sparse regime*), let $p(t) = \mathbb{P}\{|g_1| \geqslant t\}$. By the intermediate value theorem, there exists $t_0 > 0$ with

$$p(t_0) = \frac{2s}{n}.$$

  By Mills' ratio,

$$t_0 \geqslant c_2\sqrt{\log(en/s)}.$$

  Define the counting random variable

$$N(t_0) = \sum_{i=1}^{n}\mathbf{1}_{\{|g_i|\geqslant t_0\}} \sim B(n, p(t_0)),$$

  whose mean is $\mathbb{E}[N(t_0)] = np(t_0) = 2s$; moreover, $\{N(t_0) \geqslant s\} = \{|g|_{(s)} \geqslant t_0\}$.

  By Theorem T2.3.1 (*Chernoff inequality*) (*Chernoff inequality*) in the binomial concentration form, with $\delta = 1/2$,

$$\mathbb{P}\{N(t_0) \geqslant s\} \geqslant 1 - \mathrm{e}^{-s/4} \geqslant 1 - \mathrm{e}^{-1/4} \geqslant \frac{1}{5}.$$

  Therefore

$$\mathbb{E}[|g|_{(s)}] \geqslant t_0\cdot\mathbb{P}\{|g|_{(s)} \geqslant t_0\} \geqslant t_0\cdot\mathbb{P}\{N(t_0)\geqslant s\} \geqslant \frac{1}{5}t_0 \geqslant c_3\sqrt{\log(en/s)}.$$

  Substituting (IV) into (III),

$$w(\mathbb{S}_{n,s}) \geqslant \sqrt{s}\mathbb{E}[|g|_{(s)}] \geqslant c\sqrt{s\log(en/s)}. \tag{V}$$

Thus

$$c\sqrt{s\log(en/s)} \leqslant w(\mathbb{S}_{n,s}) \leqslant w(\mathbb{T}_{n,s}) \leqslant 2w(\mathbb{S}_{n,s}) \leqslant C\sqrt{s\log(en/s)}.$$

Hence $w(\mathbb{T}_{n,s}) \asymp w(\mathbb{S}_{n,s}) \asymp \sqrt{s\log(en/s)}$.

28 [**The Garnaev–Gluskin Theorem**] Improve the logarithmic factor in the bound (9.3.2) on random sections of the ℓ_1 ball:

$$\mathbb{E}[\operatorname{diam}(\mathbb{B}_1^n \cap E)] \lesssim \frac{K^2\sqrt{\log(en/m)}}{\sqrt{m}},$$

where $E = \ker \boldsymbol{A}$ is the random subspace. Assume $1 \leqslant m \leqslant c_0 n$ and $K \geqslant 1$. In particular, the logarithmic factor in (9.13) can be removed.

Prf. Let the rows of $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be independent isotropic K-sub-Gaussian vectors. The implicit constant in $\lesssim$ is absolute.

- The geometry of the truncated ball and the escape event

  Choose an integer $s \in [1, n]$ and consider $\mathbb{T}_{n,s} = \sqrt{s}\mathbb{B}_1^n \cap \mathbb{B}_2^n$. Take nonzero $\boldsymbol{x} \in \mathbb{B}_1^n \cap \ker \boldsymbol{A}$. If $\|\boldsymbol{x}\|_2 > 1/\sqrt{s}$, write $\bar{\boldsymbol{x}} = \boldsymbol{x}/\|\boldsymbol{x}\|_2$. Then

  $$\boldsymbol{A}\bar{\boldsymbol{x}} = \frac{1}{\|\boldsymbol{x}\|_2}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0},$$

  and

  $$\|\bar{\boldsymbol{x}}\|_1 = \frac{\|\boldsymbol{x}\|_1}{\|\boldsymbol{x}\|_2} \leqslant \frac{1}{\|\boldsymbol{x}\|_2} < \sqrt{s},$$

  $$\|\bar{\boldsymbol{x}}\|_2 = 1.$$

  Thus $\bar{\boldsymbol{x}} \in \mathbb{T}_{n,s} \cap \mathbb{S}^{n-1} \cap \ker \boldsymbol{A}$. Consequently, if $\mathbb{T}_{n,s} \cap \mathbb{S}^{n-1} \cap \ker \boldsymbol{A} = \emptyset$, then

  $$\sup_{\boldsymbol{x}\in\mathbb{B}_1^n\cap\ker \boldsymbol{A}} \|\boldsymbol{x}\|_2 \leqslant \frac{1}{\sqrt{s}}.$$

- Choosing s by the matrix deviation inequality

  The truncated-ball Gaussian-width bound gives

  $$w(\mathbb{T}_{n,s}) \leqslant C_0\sqrt{s\log\left(\frac{en}{s}\right)}.$$

  Applying the high-probability matrix-deviation bound to $\mathbb{T}_{n,s} \cap \mathbb{S}^{n-1}$ gives

  $$\inf_{\boldsymbol{v}\in\mathbb{T}_{n,s}\cap\mathbb{S}^{n-1}} \|\boldsymbol{A}\boldsymbol{v}\|_2 \geqslant c_1\sqrt{m} - C_1K^2\sqrt{s\log\left(\frac{en}{s}\right)} \quad (\text{w.h.p. } 1-2\mathrm{e}^{-c_1 m/K^4}).$$

  Therefore it is enough that

  $$m \geqslant C_2K^4 s\log\left(\frac{en}{s}\right),$$

  $$\mathbb{T}_{n,s} \cap \mathbb{S}^{n-1} \cap \ker \boldsymbol{A} = \emptyset \ (\text{w.h.p. } 1 - 2\exp(-cm/K^4)).$$

  If $m \geqslant C_3K^4\log(en/m)$, choose

  $$s = \left\lfloor c_3\frac{m}{K^4\log(en/m)}\right\rfloor,$$

  where $c_3 > 0$ is sufficiently small. After adjusting constants, $1 \leqslant s \leqslant n$ and $s\log(en/s) \leqslant c_4 m/K^4$, so the escape event above holds. On the escape event,

  $$\operatorname{diam}(\mathbb{B}_1^n \cap \ker \boldsymbol{A}) \leqslant \frac{2}{\sqrt{s}} \leqslant C\frac{K^2\sqrt{\log(en/m)}}{\sqrt{m}}.$$

- Taking expectations and removing the logarithmic factor

Write $\mathcal{G}$ for this escape event. Since $\mathbb{B}_1^n \cap \ker \boldsymbol{A} \subset \mathbb{B}_1^n$, its diameter is at most 2. Therefore

$$\begin{aligned}\mathbb{E}[\operatorname{diam}(\mathbb{B}_1^n \cap E)] &\leqslant \frac{2}{\sqrt{s}} + 2\mathbb{P}(\mathcal{G}^c) \\ &\leqslant C\frac{K^2\sqrt{\log(en/m)}}{\sqrt{m}} + 4\exp(-cm/K^4).\end{aligned}$$

If $m < C_3K^4\log(en/m)$, the target right-hand side covers the trivial bound $\operatorname{diam}(\mathbb{B}_1^n \cap E) \leqslant 2$ after increasing the absolute constant; otherwise the second term is absorbed by the first, which proves the claim. If $m = \Theta(n)$, then $\log(en/m) = \mathcal{O}(1)$, so

$$\mathbb{E}[\operatorname{diam}(\mathbb{B}_1^n \cap E)] \lesssim \frac{1}{\sqrt{m}} \asymp \frac{1}{\sqrt{n}}.$$

Thus the logarithmic factor is removed, recovering the optimal scale in Kashin's section theorem.

29 [**Extensions of Low-Rank Matrix Recovery**] Prove the following extensions of Corollary C9.4.11 (*Low-rank matrix recovery*).

(1) Prove that the conclusion of Corollary C9.4.11 (*Low-rank matrix recovery*) also holds for the optimization program: minimize $\|\boldsymbol{X}\|_*$ subject to $y_i = \langle \boldsymbol{A}_i, \boldsymbol{X}\rangle,\ i = 1, \cdots, m$.

(2) Extend the matrix recovery result to approximately low-rank matrices.

(3) Extend the matrix recovery result to $d_1 \times d_2$ matrices.

Prf. (1) Exact low-rank recovery

Identify the matrix space $\mathbb{R}^{n\times n}$ with the n^2-dimensional Euclidean space (equipped with the Frobenius norm $\|\cdot\|_F$). By the universal convex recovery theorem of Exercise Q9.18 (*Recovery by Optimization*), for any norm unit ball $\mathbb{T}$, the convex optimizer

$$\arg\min_{\boldsymbol{Z}\in\mathbb{R}^{n\times n}} \|\boldsymbol{Z}\|_{\mathbb{T}} \quad \text{s.t.} \quad \mathcal{A}(\boldsymbol{Z}) = \mathcal{A}(\boldsymbol{X})$$

has recovery error

$$\mathbb{E}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F \leqslant \frac{C_0K^2w(\mathbb{T})}{\sqrt{m}}\|\boldsymbol{X}\|_{\mathbb{T}}. \tag{I}$$

Take the norm to be the nuclear norm $\|\cdot\|_*$ with unit ball $\mathbb{B}_* = \{\boldsymbol{Z} \in \mathbb{R}^{n\times n} : \|\boldsymbol{Z}\|_* \leqslant 1\}$. By the duality between the nuclear norm and the operator norm (the nuclear norm is the dual norm of the spectral norm), for a standard Gaussian matrix $\boldsymbol{G}$ (with i.i.d. $N(\boldsymbol{0}, 1)$ entries),

$$w(\mathbb{B}_*) = \mathbb{E}\left[\sup_{\boldsymbol{Z}\in\mathbb{B}_*} \langle \boldsymbol{G}, \boldsymbol{Z}\rangle\right] = \mathbb{E}[\|\boldsymbol{G}\|_{\mathrm{op}}] \leqslant \sqrt{n} + \sqrt{n} = 2\sqrt{n}, \tag{II}$$

where the last step is the classical bound for the operator norm of Gaussian matrices (Gordon's theorem). Let $\sigma_1 \geqslant \sigma_2 \geqslant \cdots \geqslant \sigma_n \geqslant 0$ be the singular values of $\boldsymbol{X}$. Since $\operatorname{rank}(\boldsymbol{X}) \leqslant r$, the singular value vector $\sigma(\boldsymbol{X}) \in \mathbb{R}^n$ is an r-sparse vector; by the Cauchy–Schwarz inequality:

$$\|\boldsymbol{X}\|_* = \|\sigma(\boldsymbol{X})\|_1 = \sum_{i=1}^{r}\sigma_i \leqslant \sqrt{r}\left(\sum_{i=1}^{r}\sigma_i^2\right)^{1/2} = \sqrt{r}\|\boldsymbol{X}\|_F. \tag{III}$$

Substituting (II) and (III) into (I), with $C = 2C_0$,

$$\mathbb{E}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F \leqslant \frac{2C_0K^2\sqrt{n}}{\sqrt{m}} \cdot \sqrt{r}\|\boldsymbol{X}\|_F = CK^2\sqrt{\frac{rn}{m}}\|\boldsymbol{X}\|_F. \tag{IV}$$

This is a stable relative-error bound, but it does not by itself imply exact recovery. We now prove exact recovery using the nuclear-norm descent cone.

Let $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$ be a singular value decomposition of a matrix of rank at most r, and let

$$T = \{\boldsymbol{U}\boldsymbol{M}^\top + \boldsymbol{N}\boldsymbol{V}^\top : \ \boldsymbol{M}, \boldsymbol{N} \in \mathbb{R}^{n\times r}\}$$

be the tangent space to the rank-r matrix manifold at $\boldsymbol{X}$. If a perturbation $\boldsymbol{H}$ belongs to the descent cone of the nuclear norm at $\boldsymbol{X}$, decomposability of the nuclear norm gives

$$\|\boldsymbol{P}_{T^\perp}\boldsymbol{H}\|_* \leqslant \|\boldsymbol{P}_T\boldsymbol{H}\|_*.$$

Since $\operatorname{rank}(\boldsymbol{P}_T\boldsymbol{H}) \leqslant 2r$, every unit-Frobenius-norm descent direction satisfies

$$\|\boldsymbol{H}\|_* \leqslant 2\|\boldsymbol{P}_T\boldsymbol{H}\|_* \leqslant 2\sqrt{2r}\|\boldsymbol{P}_T\boldsymbol{H}\|_F \leqslant 2\sqrt{2r}.$$

Let $\mathcal{D}(\boldsymbol{X})$ denote this descent cone. For a standard Gaussian matrix $\boldsymbol{G}$, duality between the nuclear and operator norms implies

$$w(\mathcal{D}(\boldsymbol{X}) \cap \mathbb{S}^{n^2-1}) \leqslant 2\sqrt{2r}\,\mathbb{E}\|\boldsymbol{G}\|_{\mathrm{op}} \leqslant C\sqrt{rn}. \tag{V}$$

Vectorizing matrices identifies the measurement operator with an $m \times n^2$ standard Gaussian matrix. Therefore, by Theorem T9.3.4 (*Escape theorem*) and (V), if $m \geqslant Crn$, then

$$\ker \mathcal{A} \cap \mathcal{D}(\boldsymbol{X}) = \{\boldsymbol{0}\}$$

with high probability. Thus no nonzero feasible perturbation can fail to increase the nuclear norm, so $\boldsymbol{X}$ is the unique solution of the nuclear-norm minimization problem. This proves the exact-recovery conclusion of Corollary C9.4.11 (*Low-rank matrix recovery*).

(2) Approximately low-rank recovery

Let $\boldsymbol{X}$ be an arbitrary matrix and $\boldsymbol{X}_r$ its best rank-r approximation (the SVD truncation, by the Eckart–Young–Mirsky theorem); write the nuclear norm approximation error $\|\boldsymbol{X} - \boldsymbol{X}_r\|_* = \sum_{i=r+1}^{n} \sigma_i(\boldsymbol{X})$. Substituting (II) into (I), for every matrix $\boldsymbol{X}$ we have

$$\mathbb{E}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F \leqslant \frac{CK^2\sqrt{n}}{\sqrt{m}}\|\boldsymbol{X}\|_*.$$

Decompose $\boldsymbol{X} = \boldsymbol{X}_r + (\boldsymbol{X} - \boldsymbol{X}_r)$; by the triangle inequality for the nuclear norm:

$$\|\boldsymbol{X}\|_* \leqslant \|\boldsymbol{X}_r\|_* + \|\boldsymbol{X} - \boldsymbol{X}_r\|_* \leqslant \sqrt{r}\|\boldsymbol{X}\|_F + \|\boldsymbol{X} - \boldsymbol{X}_r\|_*,$$

where the second step uses (III) and $\|\boldsymbol{X}_r\|_F \leqslant \|\boldsymbol{X}\|_F$. Substituting yields

$$\mathbb{E}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F \leqslant CK^2\left(\sqrt{\frac{rn}{m}}\|\boldsymbol{X}\|_F + \sqrt{\frac{n}{m}}\|\boldsymbol{X} - \boldsymbol{X}_r\|_*\right). \tag{VI}$$

When $m \asymp rn$, the tail error term becomes $\|\boldsymbol{X} - \boldsymbol{X}_r\|_*/\sqrt{r}$, the matrix analogue of the Stechkin optimal approximation bound in sparse recovery.

(3) Rectangular $d_1 \times d_2$ matrices

For $d_1 \times d_2$ matrices ($r \leqslant \min(d_1, d_2)$), the Gaussian width of the nuclear norm unit ball $\mathbb{B}_*$ is

$$w(\mathbb{B}_*) = \mathbb{E}[\|\boldsymbol{G}\|_{\mathrm{op}}] \leqslant \sqrt{d_1} + \sqrt{d_2}, \tag{VII}$$

Substituting into (I) and using (III) in the same way yields

$$\mathbb{E}\|\hat{\boldsymbol{X}} - \boldsymbol{X}\|_F \leqslant \frac{C_0K^2(\sqrt{d_1} + \sqrt{d_2})}{\sqrt{m}}\sqrt{r}\|\boldsymbol{X}\|_F \leqslant CK^2\sqrt{\frac{r(d_1+d_2)}{m}}\|\boldsymbol{X}\|_F, \tag{VIII}$$

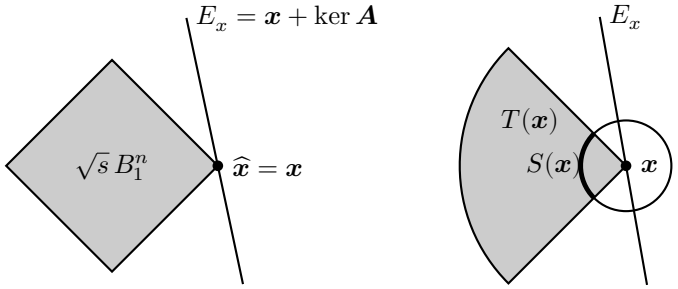
so the required number of measurements is $m \geqslant C_\varepsilon K^4 r(d_1+d_2)$, exactly of the order of the algebraic degrees of freedom $\dim(\mathcal{M}_r) = r(d_1 + d_2 - r)$ of the rank-r matrix manifold: nuclear norm convex relaxation robustly reconstructs high-dimensional rectangular low-rank matrices with a number of measurements of the same order as the intrinsic degrees of freedom.

Rk. Nuclear-norm minimization recovers low-rank matrices through convex relaxation. The number of measurements is of the order of the rank-r matrix-manifold dimension $r(d_1 + d_2 - r)$ (*rn in the square case*), far below the ambient dimension

$d_1 d_2$.

This follows the same principle as the $s\log(en/s)$ sample complexity of sparse recovery: the Gaussian width of a low-complexity convex hull determines sample complexity; here the nuclear-norm unit ball satisfies $w(\mathbb{B}_*) \leqslant \sqrt{d_1} + \sqrt{d_2}$.

30 [**Geometry of Exact Sparse Recovery**] Give a geometric interpretation of the proof of Theorem T9.5.1 (*Exact sparse recovery*). What does this proof say about the tangent cone $\mathbb{T}(\boldsymbol{x})$ and its spherical part $\mathbb{S}(\boldsymbol{x})$?



(a) Exact sparse recovery happens when the random subspace E_x is tangent to the ℓ^1 ball at the point $\boldsymbol{x}$.

(b) Tangency occurs iff E_x is disjoint from the spherical part $S(\boldsymbol{x})$ of the tangent cone $T(\boldsymbol{x})$ of the ℓ^1 ball at point $\boldsymbol{x}$.

Figure 9.7 Exact sparse recovery

Prf. • The geometry of Figure 9.7(a): strict tangency if and only if exact recovery

The feasible region of the convex optimization problem is the affine subspace $E_{\boldsymbol{x}} = \boldsymbol{x} + \ker \boldsymbol{A} = \{\boldsymbol{z} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}\}$ through $\boldsymbol{x}$. The sublevel set of the objective is the ℓ_1 ball $B = \{\boldsymbol{z} \in \mathbb{R}^n : \|\boldsymbol{z}\|_1 \leqslant \|\boldsymbol{x}\|_1\}$ of radius $\|\boldsymbol{x}\|_1$. Exact recovery $\hat{\boldsymbol{x}} = \boldsymbol{x}$ holds if and only if $\boldsymbol{x}$ is the unique minimizer in the feasible region. Geometrically, this means that the affine plane $E_{\boldsymbol{x}}$ and the ℓ_1 ball B intersect at the isolated point $\boldsymbol{x}$ and only there:

$$(\boldsymbol{x} + \ker \boldsymbol{A}) \cap \{\boldsymbol{z} \in \mathbb{R}^n : \|\boldsymbol{z}\|_1 \leqslant \|\boldsymbol{x}\|_1\} = \{\boldsymbol{x}\}.$$

Thus $E_{\boldsymbol{x}}$ is the affine feasible section through $\boldsymbol{x}$; it need not be a supporting hyperplane of the ℓ_1 ball.

• The geometry of Figure 9.7(b): tangent cone reduction if and only if escape

Consider the error vector $\boldsymbol{h} = \hat{\boldsymbol{x}} - \boldsymbol{x}$: feasibility gives $\boldsymbol{A}\hat{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$, hence $\boldsymbol{A}\boldsymbol{h} = \boldsymbol{0}$, i.e. $\boldsymbol{h} \in \ker \boldsymbol{A}$; optimality gives $\|\hat{\boldsymbol{x}}\|_1 \leqslant \|\boldsymbol{x}\|_1$, hence $\boldsymbol{x} + \boldsymbol{h} \in B$. Define the tangent cone (descent cone) of the ℓ_1 norm at $\boldsymbol{x}$ as the set of directions along which the norm does not increase:

$$\mathbb{T}(\boldsymbol{x}) = \operatorname{cone}(B - \boldsymbol{x}) = \{\boldsymbol{h} \in \mathbb{R}^n : \exists t > 0 \text{ such that } \|\boldsymbol{x} + t\boldsymbol{h}\|_1 \leqslant \|\boldsymbol{x}\|_1\}.$$

Strict tangency (i.e. there is no nonzero feasible perturbation $\boldsymbol{h} \neq \boldsymbol{0}$) is equivalent to

$$\mathbb{T}(\boldsymbol{x}) \cap \ker \boldsymbol{A} = \{\boldsymbol{0}\}.$$

Intersecting the tangent cone with the Euclidean unit sphere gives its spherical part:

$$\mathbb{S}(\boldsymbol{x}) = \mathbb{T}(\boldsymbol{x}) \cap \mathbb{S}^{n-1}.$$

Hence exact recovery is equivalent to: the random subspace $\ker \boldsymbol{A}$ completely avoids the set $\mathbb{S}(\boldsymbol{x})$, i.e. $\mathbb{S}(\boldsymbol{x}) \cap \ker \boldsymbol{A} = \emptyset$.

• What the proof of Theorem T9.5.1 (*Exact sparse recovery*) says about the tangent cone $\mathbb{T}(\boldsymbol{x})$ and $\mathbb{S}(\boldsymbol{x})$

The proof of the theorem reveals, through delicate algebraic estimates, four key geometric properties of the tangent cone $\mathbb{T}(\boldsymbol{x})$ and its spherical section $\mathbb{S}(\boldsymbol{x})$.

First, the structural property (dominance on $\operatorname{supp}(\boldsymbol{x})$): for an s-sparse signal $\boldsymbol{x}$ (with $\operatorname{supp}(\boldsymbol{x}) = \mathbb{S}$, $|\mathbb{S}| \leqslant s$), let $\boldsymbol{h} \in \mathbb{T}(\boldsymbol{x})$. There exists $t > 0$ such that

$$\|\boldsymbol{x}_{\mathbb{S}} + t\boldsymbol{h}_{\mathbb{S}}\|_1 + t\|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant \|\boldsymbol{x}_{\mathbb{S}}\|_1.$$

The reverse triangle inequality followed by division by t gives $\|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant \|\boldsymbol{h}_{\mathbb{S}}\|_1$. This shows that the tangent cone $\mathbb{T}(\boldsymbol{x})$

is not a wide cone: all its directions are confined to the narrow region where the energy off $\mathbb{S}$ is at most the energy on $\mathbb{S}$. Consequently, the ℓ_1 norm of any vector in the cone is strongly dominated by its Euclidean norm:

$$\|\boldsymbol{h}\|_1 = \|\boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant 2\|\boldsymbol{h}_{\mathbb{S}}\|_1 \leqslant 2\sqrt{s}\|\boldsymbol{h}_{\mathbb{S}}\|_2 \leqslant 2\sqrt{s}\|\boldsymbol{h}\|_2.$$

Second, the envelope property: every $\boldsymbol{u} \in \mathbb{S}(\boldsymbol{x}) = \mathbb{T}(\boldsymbol{x}) \cap \mathbb{S}^{n-1}$ satisfies $\|\boldsymbol{u}\|_2 = 1$ and $\|\boldsymbol{u}\|_1 \leqslant 2\sqrt{s}$. By the definition of the truncated ℓ_1 Euclidean ball $\mathbb{T}_{n,s} = \sqrt{s}\mathbb{B}_1^n \cap \mathbb{B}_2^n$, the proof reveals the inclusion

$$\mathbb{S}(\boldsymbol{x}) \subset 2(\sqrt{s}\mathbb{B}_1^n \cap \mathbb{B}_2^n) = 2\mathbb{T}_{n,s}.$$

Although the sphere $\mathbb{S}^{n-1}$ is an $(n-1)$-dimensional object, the projection of the tangent cone onto it, $\mathbb{S}(\boldsymbol{x})$, is completely contained in the truncated ball $2\mathbb{T}_{n,s}$, a convex body of low effective dimension.

Third, the measure property (small Gaussian width): combined with the estimates of the Gaussian width of the truncated ball in Exercises Q9.26 (*Logarithmic Improvement in Sparse Recovery*) and Q9.27 (*Gaussian Width of Sparse Vectors*), the proof gives the sharp upper bound

$$w(\mathbb{S}(\boldsymbol{x})) \leqslant 2w(\mathbb{T}_{n,s}) \leqslant C\sqrt{s\log(en/s)} \leqslant C'\sqrt{s\log n}.$$

The Gaussian width measures the average solid angle/effective cross-section of the set on the sphere. The present estimate gives only $w(\mathbb{S}(\boldsymbol{x})) \lesssim \sqrt{s\log n}$; this shows that $\mathbb{S}(\boldsymbol{x})$ occupies an exponentially small spherical measure on the high-dimensional sphere $\mathbb{S}^{n-1}$.

Fourth, the phase transition mechanism (the geometric threshold in the matrix-deviation lower bound): the high-probability matrix-deviation bound for independent isotropic K-sub-Gaussian rows guarantees that $\ker \boldsymbol{A}$ avoids the spherical set $\mathbb{S}(\boldsymbol{x})$ (i.e. $\mathbb{S}(\boldsymbol{x}) \cap \ker \boldsymbol{A} = \emptyset$) provided that the number of measurements (codimension) satisfies

$$m \geqslant CK^4 \cdot w(\mathbb{S}(\boldsymbol{x}))^2.$$

Substituting $w(\mathbb{S}(\boldsymbol{x})) \leqslant C\sqrt{s\log n}$ yields the measurement condition $m \geqslant C'K^4 s\log n$.

Geometric conclusion: as soon as the measurement dimension m reaches the squared Gaussian width $w(\mathbb{S}(\boldsymbol{x}))^2$ of the spherical part of the tangent cone, the high-dimensional random subspace $\ker \boldsymbol{A}$ passes through the large gap outside the tangent cone (w.h.p.), so that the affine plane $E_{\boldsymbol{x}}$ is strictly tangent to the ℓ_1 ball at the vertex $\boldsymbol{x}$, locking in the unique exact solution.

Rk. The affine subspace $E_{\boldsymbol{x}} = \boldsymbol{x} + \ker \boldsymbol{A}$ contains all candidate solutions consistent with the observation. The tangent cone $\mathbb{T}(\boldsymbol{x})$ collects the descent directions at $\boldsymbol{x}$ and satisfies $\|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant \|\boldsymbol{h}_{\mathbb{S}}\|_1$.

Its spherical part $\mathbb{S}(\boldsymbol{x}) = \mathbb{T}(\boldsymbol{x}) \cap \mathbb{S}^{n-1}$ is contained in $2\mathbb{T}_{n,s}$, with Gaussian width $w \lesssim \sqrt{s\log n}$. Exact recovery is equivalent to $\mathbb{S}(\boldsymbol{x}) \cap \ker \boldsymbol{A} = \emptyset$, so for $m \gtrsim s\log n$ the random kernel avoids this set with high probability.

31 [**Noisy Measurements**] Extend Theorem T9.5.1 (*Exact sparse recovery*) to noisy measurements $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}$, and modify the recovery program (9.27) accordingly.

Consider the noisy high-dimensional linear observation model $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}$, where $\boldsymbol{x} \in \mathbb{R}^n$ is an unknown s-sparse signal ($\|\boldsymbol{x}\|_0 \leqslant s$), $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ is a random measurement matrix (whose rows $\boldsymbol{A}_i$ are independent, isotropic K-sub-Gaussian random vectors), and $\boldsymbol{w} \in \mathbb{R}^m$ is unknown noise satisfying the deterministic constraint $\|\boldsymbol{w}\|_2 \leqslant \varepsilon$. Tasks: (1) modify the noiseless recovery program (9.27) accordingly to accommodate the noise model; (2) extend Theorem T9.5.1 (*Exact sparse recovery*) to the noisy case, stating and rigorously proving the corresponding recovery error bound and the high-probability matrix-deviation event used in the proof.

Prf. • Modifying the recovery program: basis pursuit denoising (BPDN)

In the noiseless case, program (9.27) uses the hard equality constraint:

$$\arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \|\boldsymbol{z}\|_1 \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{z} = \boldsymbol{y}.$$

In the noisy case $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}$, since $\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2 = \|\boldsymbol{w}\|_2 \leqslant \varepsilon$, the true value $\boldsymbol{x}$ typically no longer satisfies the hard equality constraint $\boldsymbol{A}\boldsymbol{z} = \boldsymbol{y}$.

Therefore, the most natural robustification is to relax the equality constraint to a tolerance constraint inside an ℓ_2 ball. This yields the classical basis pursuit denoising (BPDN) convex optimization program:

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \|\boldsymbol{z}\|_1 \quad \text{subject to } \|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{z}\|_2 \leqslant \varepsilon.$$

- Feasibility and residual control

Let the recovery error vector be $\boldsymbol{h} = \hat{\boldsymbol{x}} - \boldsymbol{x}$. If $\boldsymbol{h} = \boldsymbol{0}$, the error is 0 and the conclusion is trivial; we only need to consider $\boldsymbol{h} \neq \boldsymbol{0}$.

First, feasibility: since $\|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2 = \|\boldsymbol{w}\|_2 \leqslant \varepsilon$, the true value $\boldsymbol{x}$ itself satisfies the constraint $\|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2 \leqslant \varepsilon$, so $\boldsymbol{x}$ is a feasible point of the optimization problem.

Second, residual control: by the feasibility of the optimal solution $\hat{\boldsymbol{x}}$, we have $\|\boldsymbol{y}-\boldsymbol{A}\hat{\boldsymbol{x}}\|_2 \leqslant \varepsilon$. By the triangle inequality,

$$\begin{aligned}\|\boldsymbol{A}\boldsymbol{h}\|_2 &= \|\boldsymbol{A}(\hat{\boldsymbol{x}}-\boldsymbol{x})\|_2 \\ &= \|(\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}) - (\boldsymbol{y}-\boldsymbol{A}\hat{\boldsymbol{x}})\|_2 \\ &\leqslant \|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2 + \|\boldsymbol{y}-\boldsymbol{A}\hat{\boldsymbol{x}}\|_2 \leqslant 2\varepsilon. \end{aligned} \tag{I}$$

- Optimality implies that the error direction lies in the tangent cone

Since $\hat{\boldsymbol{x}}$ is a global minimizer of the objective $\|\boldsymbol{z}\|_1$ and $\boldsymbol{x}$ is feasible, we must have $\|\hat{\boldsymbol{x}}\|_1 \leqslant \|\boldsymbol{x}\|_1$.

Let $\mathbb{S} = \operatorname{supp}(\boldsymbol{x})$ ($|\mathbb{S}| \leqslant s$). Split the full vector into $\mathbb{S}$ and its complement:

$$\|\boldsymbol{x}_{\mathbb{S}}\|_1 \geqslant \|\hat{\boldsymbol{x}}\|_1 = \|\boldsymbol{x}_{\mathbb{S}} + \boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \geqslant \|\boldsymbol{x}_{\mathbb{S}}\|_1 - \|\boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1.$$

Canceling $\|\boldsymbol{x}_{\mathbb{S}}\|_1$ on both sides:

$$\|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant \|\boldsymbol{h}_{\mathbb{S}}\|_1.$$

This yields the bound on the full ℓ_1 norm of the error vector:

$$\|\boldsymbol{h}\|_1 = \|\boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant 2\|\boldsymbol{h}_{\mathbb{S}}\|_1 \leqslant 2\sqrt{s}\|\boldsymbol{h}_{\mathbb{S}}\|_2 \leqslant 2\sqrt{s}\|\boldsymbol{h}\|_2.$$

Write the normalized error vector as $\bar{\boldsymbol{h}} = \frac{\boldsymbol{h}}{\|\boldsymbol{h}\|_2}$. Then $\|\bar{\boldsymbol{h}}\|_2 = 1$ and $\|\bar{\boldsymbol{h}}\|_1 = \frac{\|\boldsymbol{h}\|_1}{\|\boldsymbol{h}\|_2} \leqslant 2\sqrt{s}$.

By the definition of the truncated ℓ_1 ball $\mathbb{T}_{n,s} = \{\boldsymbol{z} \in \mathbb{R}^n : \|\boldsymbol{z}\|_1 \leqslant \sqrt{s},\ \|\boldsymbol{z}\|_2 \leqslant 1\}$, we obtain

$$\bar{\boldsymbol{h}} \in 2\mathbb{T}_{n,s} \cap \mathbb{S}^{n-1}. \tag{II}$$

- Applying the matrix deviation inequality

Define the set $\mathbb{T} = 2\mathbb{T}_{n,s} \cap \mathbb{S}^{n-1}$. The high-probability matrix-deviation bound for independent isotropic K-sub-Gaussian rows gives

$$\min_{\boldsymbol{v}\in\mathbb{T}} \|\boldsymbol{A}\boldsymbol{v}\|_2 \geqslant C_1^{-1}\sqrt{m} - C_1 K^2 w(\mathbb{T}) \text{ (w.h.p. } 1 - 2\mathrm{e}^{-cm/K^4}). \tag{III}$$

In Exercises Q9.26 (*Logarithmic Improvement in Sparse Recovery*) and Q9.27 (*Gaussian Width of Sparse Vectors*) we have rigorously proved that the Gaussian width of this set satisfies

$$w(\mathbb{T}) = w(2\mathbb{T}_{n,s} \cap \mathbb{S}^{n-1}) \leqslant 2w(\mathbb{T}_{n,s}) \leqslant C_2\sqrt{s\log\left(\frac{\mathrm{e}n}{s}\right)}.$$

When the number of measurements satisfies

$$m \geqslant (2C_1^2C_2)^2 K^4 s\log\left(\frac{\mathrm{e}n}{s}\right) = C_0 K^4 s\log\left(\frac{\mathrm{e}n}{s}\right),$$

the second term is controlled by half of the first: $C_1K^2w(\mathbb{T}) \leqslant \frac{1}{2C_1}\sqrt{m}$. Substituting into (III), we obtain

$$\min_{\boldsymbol{v}\in\mathbb{T}} \|\boldsymbol{A}\boldsymbol{v}\|_2 \geqslant \frac{1}{2C_1}\sqrt{m} = c_0\sqrt{m}. \tag{IV}$$

- Combining the upper and lower bounds to derive the recovery error

Since the normalized vector $\bar{\boldsymbol{h}} \in \mathbb{T}$, by the isotropic lower bound (IV),

$$\|\boldsymbol{A}\boldsymbol{h}\|_2 = \|\boldsymbol{h}\|_2\|\boldsymbol{A}\bar{\boldsymbol{h}}\|_2 \geqslant c_0\sqrt{m}\|\boldsymbol{h}\|_2. \tag{V}$$

Substituting the residual upper bound $\|\boldsymbol{A}\boldsymbol{h}\|_2 \leqslant 2\varepsilon$ (i.e. (I)) into (V):

$$c_0\sqrt{m}\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2 \leqslant 2\varepsilon.$$

Dividing both sides by $c_0\sqrt{m}$ and taking $C = \frac{2}{c_0}$:

$$\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2 \leqslant \frac{C\varepsilon}{\sqrt{m}}.$$

Rk. The feasible set $\{\boldsymbol{z} : \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{z}\|_2 \leqslant \varepsilon\}$ contains both the true signal $\boldsymbol{x}$ and the BPDN solution $\hat{\boldsymbol{x}}$. The lower bound for $\boldsymbol{A}$ on $\mathbb{T}(\boldsymbol{x})$ limits the error to $\mathcal{O}(\varepsilon/\sqrt{m})$.

As $\varepsilon \to 0$, this bound tends to zero and reduces to $\hat{\boldsymbol{x}} = \boldsymbol{x}$ from Theorem T9.5.1 (*Exact sparse recovery*).

32 [**Nullspace Property**] Define the nullspace property that ensures exact recovery. A matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ satisfies the nullspace property of order s if the inequality

$$\|\boldsymbol{h}_{\mathbb{S}}\|_1 < \|\boldsymbol{h}_{\mathbb{S}^c}\|_1$$

holds for every nonzero vector $\boldsymbol{h} \in \ker \boldsymbol{A}$ and every subset $\mathbb{S} \subset \{1, \cdots, n\}$ with $|\mathbb{S}| \leqslant s$[7].

(1) Show that $\boldsymbol{A}$ satisfies the nullspace property if and only if every s-sparse vector $\boldsymbol{x} \in \mathbb{R}^n$ is the unique solution to the ℓ_1-minimization problem

$$\arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \|\boldsymbol{z}\|_1 \text{ s.t. } \boldsymbol{A}\boldsymbol{z} = \boldsymbol{y}$$

with $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$.

(2) Show that a random matrix $\boldsymbol{A}$ (with $m \geqslant Cs\log(en/s)$) satisfies the nullspace property (w.h.p.) (make this statement precise as in Theorem T9.5.7 (*Random matrices satisfy RIP*)).

Prf. (1) • NSP implies unique recovery (sufficiency)

Assume that $\boldsymbol{A}$ satisfies the nullspace property of order s. Take any s-sparse vector $\boldsymbol{x} \in \mathbb{R}^n$ (with $\text{supp}(\boldsymbol{x}) = \mathbb{S}$, $|\mathbb{S}| \leqslant s$); we show that every other feasible point $\boldsymbol{z}$ with $\boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$ and $\boldsymbol{z} \neq \boldsymbol{x}$ satisfies $\|\boldsymbol{z}\|_1 > \|\boldsymbol{x}\|_1$.

First, let the perturbation error vector be $\boldsymbol{h} = \boldsymbol{z} - \boldsymbol{x}$. Since $\boldsymbol{z} \neq \boldsymbol{x}$, we have $\boldsymbol{h} \neq \boldsymbol{0}$. By feasibility, $\boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$, hence $\boldsymbol{A}(\boldsymbol{z} - \boldsymbol{x}) = \boldsymbol{0}$, i.e. $\boldsymbol{A}\boldsymbol{h} = \boldsymbol{0}$, so $\boldsymbol{h} \in \ker \boldsymbol{A} \setminus \{\boldsymbol{0}\}$.

Second, compute the ℓ_1 norm of the feasible point $\boldsymbol{z} = \boldsymbol{x} + \boldsymbol{h}$:

$$\|\boldsymbol{z}\|_1 = \|\boldsymbol{x} + \boldsymbol{h}\|_1 = \|\boldsymbol{x}_{\mathbb{S}} + \boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{x}_{\mathbb{S}^c} + \boldsymbol{h}_{\mathbb{S}^c}\|_1.$$

Since $\mathbb{S} = \text{supp}(\boldsymbol{x})$, we have $\boldsymbol{x}_{\mathbb{S}^c} = \boldsymbol{0}$ on the complement, hence $\boldsymbol{x}_{\mathbb{S}^c} + \boldsymbol{h}_{\mathbb{S}^c} = \boldsymbol{h}_{\mathbb{S}^c}$:

$$\|\boldsymbol{z}\|_1 = \|\boldsymbol{x}_{\mathbb{S}} + \boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1.$$

Third, use the reverse triangle inequality $\|\boldsymbol{a} + \boldsymbol{b}\|_1 \geqslant \|\boldsymbol{a}\|_1 - \|\boldsymbol{b}\|_1$ on the $\mathbb{S}$ term:

$$\|\boldsymbol{x}_{\mathbb{S}} + \boldsymbol{h}_{\mathbb{S}}\|_1 \geqslant \|\boldsymbol{x}_{\mathbb{S}}\|_1 - \|\boldsymbol{h}_{\mathbb{S}}\|_1 = \|\boldsymbol{x}\|_1 - \|\boldsymbol{h}_{\mathbb{S}}\|_1.$$

Substituting back,

$$\|\boldsymbol{z}\|_1 \geqslant \|\boldsymbol{x}\|_1 - \|\boldsymbol{h}_{\mathbb{S}}\|_1 + \|\boldsymbol{h}_{\mathbb{S}^c}\|_1 = \|\boldsymbol{x}\|_1 + (\|\boldsymbol{h}_{\mathbb{S}^c}\|_1 - \|\boldsymbol{h}_{\mathbb{S}}\|_1).$$

Fourth, since $\boldsymbol{h} \in \ker \boldsymbol{A} \setminus \{\boldsymbol{0}\}$ and $|\mathbb{S}| \leqslant s$, by the definition of the nullspace property of order s we have the strict inequality $\|\boldsymbol{h}_{\mathbb{S}^c}\|_1 - \|\boldsymbol{h}_{\mathbb{S}}\|_1 > 0$. Hence $\|\boldsymbol{z}\|_1 > \|\boldsymbol{x}\|_1$.

This shows that for every feasible candidate $\boldsymbol{z} \neq \boldsymbol{x}$, the objective value is strictly larger than $\|\boldsymbol{x}\|_1$. Hence $\boldsymbol{x}$ is the unique minimizer.

• Unique recovery implies NSP (necessity)

By contradiction: assume that every s-sparse vector is the unique solution of the basis pursuit problem. Suppose $\boldsymbol{A}$ fails the nullspace property of order s; then there exist a nonzero $\boldsymbol{h} \in \ker \boldsymbol{A} \setminus \{\boldsymbol{0}\}$ and a subset $\mathbb{S}$ with $|\mathbb{S}| \leqslant s$ such that $\|\boldsymbol{h}_{\mathbb{S}}\|_1 \geqslant \|\boldsymbol{h}_{\mathbb{S}^c}\|_1$.

Define the test signal $\boldsymbol{x} = -\boldsymbol{h}_{\mathbb{S}}$. Since $\text{supp}(\boldsymbol{x}) \subset \mathbb{S}$, it is s-sparse ($\|\boldsymbol{x}\|_0 \leqslant s$). Set $\boldsymbol{z} = \boldsymbol{x} + \boldsymbol{h} = -\boldsymbol{h}_{\mathbb{S}} + \boldsymbol{h} = \boldsymbol{h}_{\mathbb{S}^c}$. Then $\boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}(\boldsymbol{x} + \boldsymbol{h}) = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{A}\boldsymbol{h} = \boldsymbol{A}\boldsymbol{x}$; moreover, $\boldsymbol{h} \neq \boldsymbol{0}$ implies $\boldsymbol{z} \neq \boldsymbol{x}$.

By the assumption, $\boldsymbol{x}$ is the unique ℓ_1-norm minimizer subject to $\boldsymbol{A}\boldsymbol{z} = \boldsymbol{A}\boldsymbol{x}$, so we must have $\|\boldsymbol{x}\|_1 < \|\boldsymbol{z}\|_1$. Computing both ℓ_1 norms: $\|\boldsymbol{x}\|_1 = \|-\boldsymbol{h}_{\mathbb{S}}\|_1 = \|\boldsymbol{h}_{\mathbb{S}}\|_1$ and $\|\boldsymbol{z}\|_1 = \|\boldsymbol{h}_{\mathbb{S}^c}\|_1$. Substituting gives

$$\|\boldsymbol{h}_{\mathbb{S}}\|_1 < \|\boldsymbol{h}_{\mathbb{S}^c}\|_1,$$

a contradiction; hence $\boldsymbol{A}$ satisfies the nullspace property of order s.

(2) Precise statement of the proposition

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a random matrix whose rows $\boldsymbol{A}_i$ are independent, isotropic K-sub-Gaussian random vectors. There exist absolute constants $C_0, c > 0$ such that if the number of measurements satisfies $m \geqslant C_0 K^4 s \log(\frac{en}{s})$, then the matrix $\boldsymbol{A}$ satisfies the nullspace property of order s (w.h.p. $1 - 2\mathrm{e}^{-cm/K^4}$).

First, define the cone of bad directions that violate the nullspace property. A vector $\boldsymbol{h} \in \mathbb{R}^n$ violates the nullspace property of order s if and only if there exists $|\mathbb{S}| \leqslant s$ with $\|\boldsymbol{h}_{\mathbb{S}}\|_1 \geqslant \|\boldsymbol{h}_{\mathbb{S}^c}\|_1$. Define the set (cone) of all bad directions:

$$\mathcal{C}_s = \{\boldsymbol{h} \in \mathbb{R}^n : \exists \mathbb{S} \subset \{1, \cdots, n\},\ |\mathbb{S}| \leqslant s,\ \text{such that } \|\boldsymbol{h}_{\mathbb{S}^c}\|_1 \leqslant \|\boldsymbol{h}_{\mathbb{S}}\|_1\}.$$

The matrix $\boldsymbol{A}$ satisfies the nullspace property if and only if there is no nonzero bad direction in its nullspace:

$$\ker \boldsymbol{A} \cap \mathcal{C}_s = \{\boldsymbol{0}\} \text{ if and only if } \ker \boldsymbol{A} \cap (\mathcal{C}_s \cap \mathbb{S}^{n-1}) = \emptyset. \tag{I}$$

Second, embed the bad spherical directions into the truncated ℓ_1 ball $\mathbb{T}_{n,s}$. Let $\boldsymbol{u} \in \mathcal{C}_s \cap \mathbb{S}^{n-1}$ be a unit vector ($\|\boldsymbol{u}\|_2 = 1$). By the definition of a bad direction, we obtain

$$\|\boldsymbol{u}\|_1 = \|\boldsymbol{u}_{\mathbb{S}}\|_1 + \|\boldsymbol{u}_{\mathbb{S}^c}\|_1 \leqslant 2\|\boldsymbol{u}_{\mathbb{S}}\|_1.$$

Using the Cauchy–Schwarz inequality (since $|\mathbb{S}| \leqslant s$):

$$\|\boldsymbol{u}\|_1 \leqslant 2\sqrt{s}\|\boldsymbol{u}_{\mathbb{S}}\|_2 \leqslant 2\sqrt{s}\|\boldsymbol{u}\|_2 = 2\sqrt{s}.$$

By the definition of the truncated ℓ_1 Euclidean ball $\mathbb{T}_{n,s} = \{\boldsymbol{z} \in \mathbb{R}^n : \|\boldsymbol{z}\|_1 \leqslant \sqrt{s},\ \|\boldsymbol{z}\|_2 \leqslant 1\}$, we immediately obtain the inclusion

$$\mathcal{C}_s \cap \mathbb{S}^{n-1} \subset 2\mathbb{T}_{n,s} \cap \mathbb{S}^{n-1}. \tag{II}$$

Third, control the Gaussian width of the bad set. In Exercises Q9.26 (*Logarithmic Improvement in Sparse Recovery*) and Q9.27 (*Gaussian Width of Sparse Vectors*) we have proved that the Gaussian width of the truncated ball satisfies

$$w(\mathcal{C}_s \cap \mathbb{S}^{n-1}) \leqslant w(2\mathbb{T}_{n,s}) = 2w(\mathbb{T}_{n,s}) \leqslant C_1\sqrt{s\log\left(\frac{en}{s}\right)}.$$

Fourth, apply the high-probability matrix-deviation bound. For the spherical set $\mathcal{T} = \mathcal{C}_s \cap \mathbb{S}^{n-1}$, if the number of measurements satisfies $m \geqslant CK^4 \cdot w(\mathcal{T})^2$, then $\ker \boldsymbol{A}$ is disjoint from $\mathcal{T}$ (i.e. $\ker \boldsymbol{A} \cap \mathcal{T} = \emptyset$) with probability at least $1 - 2\mathrm{e}^{-cm/K^4}$. Substituting the upper bound on the Gaussian width, $w(\mathcal{T})^2 \leqslant C_1^2 s\log(\frac{en}{s})$, and choosing $C_0 \geqslant CC_1^2$, whenever $m \geqslant C_0 K^4 s\log(\frac{en}{s})$, we obtain

$$\ker \boldsymbol{A} \cap (\mathcal{C}_s \cap \mathbb{S}^{n-1}) = \emptyset \text{ (w.h.p. } 1 - 2\mathrm{e}^{-cm/K^4}).$$

By the equivalence (I), this strictly guarantees that the matrix $\boldsymbol{A}$ satisfies the nullspace property of order s.

33 [**Random Projections Satisfy RIP**] Let $\boldsymbol{P}$ be the orthogonal projection in $\mathbb{R}^n$ onto a random m-dimensional subspace, uniformly distributed in the Grassmannian.

Define the scaled projection operator $\boldsymbol{Q} = \sqrt{\frac{n}{m}}\boldsymbol{P}$.

(1) Prove the RIP: for any given $\delta \in (0, 1)$, when the measurement dimension satisfies $m \geqslant C\delta^{-2} s\log(\frac{12en}{\delta s})$, the operator $\boldsymbol{Q}$ satisfies the restricted isometry property of order s and constant δ (w.h.p.), i.e. for every $\boldsymbol{x} \in \mathbb{R}^n$ with $\|\boldsymbol{x}\|_0 \leqslant s$,

$$(1-\delta)\|\boldsymbol{x}\|_2^2 \leqslant \left\|\sqrt{\frac{n}{m}}\boldsymbol{P}\boldsymbol{x}\right\|_2^2 \leqslant (1+\delta)\|\boldsymbol{x}\|_2^2.$$

(2) Exact sparse recovery: deduce from (1) a version of Theorem T9.5.1 (*Exact sparse recovery*) for random projections, namely:

if $m \geqslant Cs\log(\frac{en}{s})$, then every s-sparse vector $\boldsymbol{x} \in \mathbb{R}^n$ is the unique solution of the convex program (w.h.p.)

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \|\boldsymbol{z}\|_1 \text{ subject to } \boldsymbol{P}\boldsymbol{z} = \boldsymbol{P}\boldsymbol{x}$$

(i.e. $\hat{\boldsymbol{x}} = \boldsymbol{x}$).

Prf. (1) • Measure concentration of the single-point projection length (the JL lemma basis)

By Theorem T5.3.1 (*Johnson-Lindenstrauss Lemma*), for any fixed $\boldsymbol{x} \in \mathbb{S}^{n-1}$, since the subspace is uniformly distributed in the Grassmannian, the expectation of the squared Euclidean norm of the orthogonal projection is

$$\mathbb{E}[\|\boldsymbol{P}\boldsymbol{x}\|_2^2] = \frac{m}{n}.$$

Introduce the standard normalization operator $\boldsymbol{Q} = \sqrt{\frac{n}{m}}\boldsymbol{P}$, so that $\mathbb{E}[\|\boldsymbol{Q}\boldsymbol{x}\|_2^2] = \|\boldsymbol{x}\|_2^2 = 1$. By the measure concentration on the sphere, for any $t \in (0,1)$, we have the exponential tail decay:

$$\mathbb{P}\left|\left\|\sqrt{\frac{n}{m}}\boldsymbol{P}\boldsymbol{x}\right\|_2^2 - 1\right| \geqslant t \leqslant 2\mathrm{e}^{-c_1 m t^2}, \tag{I}$$

where $c_1 > 0$ is an absolute constant.

• Constructing an ε-net of the s-sparse unit sphere

Consider the set of all unit s-sparse vectors

$$\mathbb{S}_{n,s} = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_0 \leqslant s,\ \|\boldsymbol{x}\|_2 = 1\},$$

which is the union of $\binom{n}{s}$ unit Euclidean spheres $\mathbb{S}^{s-1}$. Take the net precision parameter $\varepsilon_0 = \frac{\delta}{4}$. For each fixed s-dimensional coordinate subspace, the size of an ε_0-net of its unit sphere is at most $(1+\frac{2}{\varepsilon_0})^s \leqslant (\frac{3}{\varepsilon_0})^s$; ranging over all $\binom{n}{s} \leqslant (\frac{en}{s})^s$ possible supp, the total capacity of an ε_0-net $\mathcal{N}$ of the whole set $\mathbb{S}_{n,s}$ satisfies

$$|\mathcal{N}| \leqslant \binom{n}{s}\left(\frac{3}{\varepsilon_0}\right)^s \leqslant \left(\frac{en}{s}\right)^s\left(\frac{12}{\delta}\right)^s = \exp\left(s\log\left(\frac{12en}{\delta s}\right)\right). \tag{II}$$

• Union bound and the continuous approximation

Apply (I) to each point of the net $\mathcal{N}$ (with $t = \delta/2$) and take the union bound:

$$\mathbb{P}\max_{\boldsymbol{z}\in\mathcal{N}}\left|\|\boldsymbol{Q}\boldsymbol{z}\|_2^2 - 1\right| \geqslant \frac{\delta}{2} \leqslant |\mathcal{N}| \cdot 2\mathrm{e}^{-c_1 m(\delta/2)^2} \leqslant 2\exp\left(s\log\left(\frac{12en}{\delta s}\right) - \frac{c_1}{4}\delta^2 m\right).$$

When the number of measurements satisfies

$$m \geqslant \frac{8}{c_1}\delta^{-2}s\log\left(\frac{12en}{\delta s}\right),$$

we have $\max_{\boldsymbol{z}\in\mathcal{N}}\left|\|\boldsymbol{Q}\boldsymbol{z}\|_2^2 - 1\right| \leqslant \frac{\delta}{2}$ (w.h.p. $1 - 2\mathrm{e}^{-c_2\delta^2 m}$). By Lemma L4.4.1 (*Estimating the operator norm on a discrete net*), for every vector $\boldsymbol{x} \in \mathbb{S}_{n,s}$ lying in the same s-dimensional subspace, we must have

$$\sup_{\boldsymbol{x}\in\mathbb{S}_{n,s}}\left|\|\boldsymbol{Q}\boldsymbol{x}\|_2^2 - 1\right| \leqslant \frac{\delta/2}{1-2\varepsilon_0} = \frac{\delta/2}{1-\delta/2} \leqslant \delta. \qquad (\delta < 1) \tag{III}$$

By positive homogeneity, for every s-sparse vector $\boldsymbol{x} \in \mathbb{R}^n$ (with arbitrary norm),

$$(1-\delta)\|\boldsymbol{x}\|_2^2 \leqslant \|\boldsymbol{Q}\boldsymbol{x}\|_2^2 \leqslant (1+\delta)\|\boldsymbol{x}\|_2^2.$$

(2) • Applying Theorem T9.5.6 (*RIP implies exact recovery*)

In part (1), take the sparsity to be $2s$ and set $\delta_{2s} = 1/4 < 1/3$. By the measurement condition of part (1),

$$m \geqslant C \cdot (1/4)^{-2}(2s)\log\left(\frac{en}{2s}\right) = C_0 s\log\left(\frac{en}{s}\right),$$

the operator $\boldsymbol{Q} = \sqrt{\frac{n}{m}}\boldsymbol{P}$ satisfies the RIP of order $2s$ and constant $\delta_{2s} \leqslant 1/4 < 1/3$ (w.h.p. $1-2\mathrm{e}^{-cm}$). By Theorem T9.5.6 (*RIP implies exact recovery*), every s-sparse vector $\boldsymbol{x}$ is the unique solution of the optimization problem

$$\arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \|\boldsymbol{z}\|_1 \text{ s.t. } \boldsymbol{Q}\boldsymbol{z} = \boldsymbol{Q}\boldsymbol{x}.$$

- Algebraic equivalence of the constraints

  Since the scalar $\sqrt{n/m} > 0$, the linear constraints are strictly equivalent:

$$\boldsymbol{Qz} = \boldsymbol{Qx} \text{ if and only if } \sqrt{\frac{n}{m}}\boldsymbol{Pz} = \sqrt{\frac{n}{m}}\boldsymbol{Px} \text{ if and only if } \boldsymbol{Pz} = \boldsymbol{Px}.$$

  Hence $\boldsymbol{x}$ is the unique exact solution of the original program $\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \|\boldsymbol{z}\|_1$ subject to $\boldsymbol{Pz} = \boldsymbol{Px}$ (i.e. $\hat{\boldsymbol{x}} = \boldsymbol{x}$).

Rk. the row space of a Gaussian matrix $\boldsymbol{A}$ is isotropically and uniformly distributed in the Grassmannian.

This exercise shows that whether one starts from the algebraic matrix sampling $\boldsymbol{Ax}$ or from the purely geometric orthogonal projection $\boldsymbol{Px}$, the restricted isometry property and the phase transition threshold $m \asymp s\log(en/s)$ of exact sparse recovery are mathematically completely consistent.

34 [**Subadditivity**] Let $f : V \to \mathbb{R}$ be a subadditive function on a vector space V. Show that for all $x, y \in V$,

$$f(x) - f(y) \leqslant f(x - y).$$

Prf. • Constructing the decomposition

Let $x, y \in V$. Since V is a vector space, the difference vector $x - y \in V$; split the vector x identically as

$$x = (x - y) + y.$$

• Applying subadditivity

Let $a = x - y \in V$ and $b = y \in V$. By the definition of subadditivity of f, we obtain

$$f(x) = f((x - y) + y) \leqslant f(x - y) + f(y).$$

• Rearranging

Subtracting $f(y)$ from both sides of the inequality yields

$$f(x) - f(y) \leqslant f(x - y).$$

Rk. If $f(-u) = f(u)$, interchange x and y and use symmetry to obtain

$$f(y) - f(x) \leqslant f(x - y),$$

hence

$$|f(x) - f(y)| \leqslant f(x - y).$$

In particular, every norm satisfies $|\|x\| - \|y\|| \leqslant \|x - y\|$. This reduces the difference between $\|\boldsymbol{Ax}\|_2$ and $\|\boldsymbol{Ay}\|_2$ to $\|\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y})\|_2$.

35 [**General Matrix Deviation for Anisotropic Distributions**] Extend Theorem T9.6.3 (*General matrix deviation inequality*) to an $m \times n$ matrix $\boldsymbol{A}$ whose row vectors are i.i.d. $N(\boldsymbol{0}, \Sigma)$ random vectors, and prove that

Let $\boldsymbol{G} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix. Let $f : \mathbb{R}^m \to \mathbb{R}$ be a positively homogeneous and subadditive function, and suppose there exists a constant $b > 0$ such that $f(v) \leqslant b\|v\|_2$ for all $v \in \mathbb{R}^m$; then for any set $\mathbb{S} \subset \mathbb{R}^n$,

$$\mathbb{E}\left[\sup_{u\in\mathbb{S}} |f(\boldsymbol{G}u) - \mathbb{E}[f(\boldsymbol{G}u)]|\right] \leqslant Cb\gamma(\mathbb{S}),$$

where $\gamma(\mathbb{S}) = w(\mathbb{S}) = \mathbb{E}_g[\sup_{u\in\mathbb{S}}\langle g, u\rangle]$ is the Gaussian width of the set (see Theorem T9.6.3 (*General matrix deviation inequality*)). Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be an anisotropic Gaussian random matrix whose rows are i.i.d. multivariate normal vectors with mean zero and covariance matrix $\Sigma \in \mathbb{R}^{n\times n}$ ($\Sigma \succeq 0$), i.e. $\boldsymbol{A}_i$ i.i.d. $\sim N(\boldsymbol{0}, \Sigma)$. Show that for any set $\mathbb{T} \subset \mathbb{R}^n$, the inequality

$$\mathbb{E}\left[\sup_{x\in\mathbb{T}} |f(\boldsymbol{Ax}) - \mathbb{E}[f(\boldsymbol{Ax})]|\right] \leqslant Cb\gamma(\Sigma^{1/2}\mathbb{T})$$

holds, where $\Sigma^{1/2}$ is the principal square root of the positive semidefinite symmetric covariance matrix Σ, and $\Sigma^{1/2}\mathbb{T} = \{\Sigma^{1/2}x : x \in \mathbb{T}\}$ is the deformed set.

Prf. • Covariance whitening decomposition

Since $\Sigma \in \mathbb{R}^{n\times n}$ is a symmetric positive semidefinite covariance matrix ($\Sigma \succeq 0$), by the spectral theorem there exists an orthogonal diagonalization $\Sigma = \boldsymbol{U}\Lambda\boldsymbol{U}^\top$. Its principal square root is defined as $\Sigma^{1/2} = \boldsymbol{U}\Lambda^{1/2}\boldsymbol{U}^\top$, which is symmetric

(i.e. $(\Sigma^{1/2})^\top = \Sigma^{1/2}$) and satisfies $\Sigma^{1/2}\Sigma^{1/2} = \Sigma$.

Let $\boldsymbol{Z} \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ be a standard isotropic Gaussian vector; the linearly transformed vector satisfies $\Sigma^{1/2}\boldsymbol{Z} \sim N(\boldsymbol{0}, \Sigma^{1/2}\boldsymbol{I}_n(\Sigma^{1/2})^\top) = N(\boldsymbol{0}, \Sigma)$. Hence each row $\boldsymbol{A}_i$ of the matrix $\boldsymbol{A}$ is distributionally equivalent to the transformation of an independent standard Gaussian vector $\boldsymbol{G}_i \sim N(\boldsymbol{0}, \boldsymbol{I}_n)$ by $\Sigma^{1/2}$. Let $\boldsymbol{G} \in \mathbb{R}^{m\times n}$ be a standard isotropic Gaussian matrix with i.i.d. $N(0,1)$ entries; by the i.i.d. property of the rows, the whole matrix satisfies the distributional identity

$$\boldsymbol{A} \overset{d}{=} \boldsymbol{G}\Sigma^{1/2}. \tag{I}$$

- Change of variables in the geometric set of the stochastic process

  For any fixed vector $\boldsymbol{x} \in \mathbb{T} \subset \mathbb{R}^n$, using (I) and the associativity of matrix multiplication,

  $$\boldsymbol{A}\boldsymbol{x} \overset{d}{=} (\boldsymbol{G}\Sigma^{1/2})\boldsymbol{x} = \boldsymbol{G}(\Sigma^{1/2}\boldsymbol{x}).$$

  Let the transformed vector be $\boldsymbol{u} = \Sigma^{1/2}\boldsymbol{x} \in \mathbb{R}^n$. As $\boldsymbol{x}$ ranges over the set $\mathbb{T}$, $\boldsymbol{u}$ ranges exactly over the deformed set $\mathbb{T}' = \Sigma^{1/2}\mathbb{T} = \{\Sigma^{1/2}\boldsymbol{x} : \boldsymbol{x} \in \mathbb{T}\} \subset \mathbb{R}^n$. Therefore, the supremum of the target stochastic process is completely equivalent in joint distribution:

  $$\sup_{\boldsymbol{x}\in\mathbb{T}} |f(\boldsymbol{A}\boldsymbol{x}) - \mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]| \overset{d}{=} \sup_{\boldsymbol{u}\in\Sigma^{1/2}\mathbb{T}} |f(\boldsymbol{G}\boldsymbol{u}) - \mathbb{E}[f(\boldsymbol{G}\boldsymbol{u})]|. \tag{II}$$

- Applying Theorem T9.6.3 (*General matrix deviation inequality*) to conclude

  Taking expectations on both sides of (II),

  $$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |f(\boldsymbol{A}\boldsymbol{x}) - \mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]|\right] = \mathbb{E}\left[\sup_{\boldsymbol{u}\in\Sigma^{1/2}\mathbb{T}} |f(\boldsymbol{G}\boldsymbol{u}) - \mathbb{E}[f(\boldsymbol{G}\boldsymbol{u})]|\right]. \tag{III}$$

  Note that here $\boldsymbol{G}$ is a standard Gaussian random matrix with i.i.d. $N(0,1)$ entries, the function f satisfies positive homogeneity, subadditivity, and $f(v) \leqslant b\|v\|_2$, and the set acted upon is $\Sigma^{1/2}\mathbb{T}$. Applying Theorem T9.6.3 (*General matrix deviation inequality*) directly to the standard Gaussian matrix $\boldsymbol{G}$ and the set $\Sigma^{1/2}\mathbb{T}$:

  $$\mathbb{E}\left[\sup_{\boldsymbol{u}\in\Sigma^{1/2}\mathbb{T}} |f(\boldsymbol{G}\boldsymbol{u}) - \mathbb{E}[f(\boldsymbol{G}\boldsymbol{u})]|\right] \leqslant Cb\gamma(\Sigma^{1/2}\mathbb{T}). \tag{IV}$$

  Substituting (IV) into (III), we have proved

  $$\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |f(\boldsymbol{A}\boldsymbol{x}) - \mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]|\right] \leqslant Cb\gamma(\Sigma^{1/2}\mathbb{T}).$$

Rk. The anisotropic case is the isotropic case applied to $\Sigma^{1/2}\mathbb{T}$. Taking $f(v) = \|v\|_2$ and $b = 1$ yields $\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} |\|\boldsymbol{A}\boldsymbol{x}\|_2 - \mathbb{E}\|\boldsymbol{A}\boldsymbol{x}\|_2|\right] \leqslant Cw(\Sigma^{1/2}\mathbb{T})$.

36 [**General Matrix Deviation: A High-Probability Bound**] Prove a high-probability version of Theorem T9.6.3 (*General matrix deviation inequality*).

Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a standard Gaussian random matrix, let $f : \mathbb{R}^m \to \mathbb{R}$ be a positively homogeneous and subadditive function, and suppose there exists a constant $b > 0$ such that $f(v) \leqslant b\|v\|_2$ for all $v \in \mathbb{R}^m$; let $\mathbb{T} \subset \mathbb{R}^n$ be an arbitrary subset, with Euclidean radius $\mathrm{rad}(\mathbb{T}) = \sup_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{x}\|_2$ and Gaussian width $\gamma(\mathbb{T}) = w(\mathbb{T}) = \mathbb{E}_g[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle g, \boldsymbol{x}\rangle]$. Show that there exists an absolute constant $C > 0$ such that for any parameter $u \geqslant 0$, the following inequality holds with high probability:

$$\sup_{\boldsymbol{x}\in\mathbb{T}} |f(\boldsymbol{A}\boldsymbol{x}) - \mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]| \leqslant Cb(\gamma(\mathbb{T}) + u\,\mathrm{rad}(\mathbb{T})) \text{ (w.h.p. } 1 - 2\mathrm{e}^{-u^2}).$$

Prf. - Using Exercise Q9.34 (*Subadditivity*) to prove that f is b-Lipschitz

  By assumption, $f : \mathbb{R}^m \to \mathbb{R}$ is subadditive and $f(z) \leqslant b\|z\|_2$. For any $v, w \in \mathbb{R}^m$: by the conclusion of Exercise Q9.34 (*Subadditivity*), $f(v) - f(w) \leqslant f(v - w)$; combined with the upper bound condition $f(v - w) \leqslant b\|v - w\|_2$, we obtain $f(v) - f(w) \leqslant b\|v - w\|_2$. Swapping the roles of v and w, similarly $f(w) - f(v) \leqslant b\|w - v\|_2 = b\|v - w\|_2$, i.e.

$f(v)-f(w)\geqslant -b\|v-w\|_2$. Combining the two inequalities,

$$|f(v)-f(w)|\leqslant b\|v-w\|_2. \tag{I}$$

This rigorously proves that f is b-Lipschitz continuous.

- Proving the Lipschitz continuity of the target functional $F(\boldsymbol{A})$

  On the matrix space $\mathbb{R}^{m\times n}$ (endowed with the Frobenius norm $\|\boldsymbol{M}\|_F=\sqrt{\sum_{ij}\boldsymbol{M}_{ij}^2}$), define the functional

  $$F(\boldsymbol{A})=\sup_{\boldsymbol{x}\in\mathbb{T}}|f(\boldsymbol{A}\boldsymbol{x})-\mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]|.$$

  Let $\boldsymbol{A},\boldsymbol{B}\in\mathbb{R}^{m\times n}$ be arbitrary two matrices.

  First, define the deterministic function $\mu(\boldsymbol{x})=\mathbb{E}_{\boldsymbol{G}}f(\boldsymbol{G}\boldsymbol{x})$, where $\boldsymbol{G}$ is a standard Gaussian matrix. For each realization of $\boldsymbol{A}$, the centered process is $f(\boldsymbol{A}\boldsymbol{x})-\mu(\boldsymbol{x})$; no comparison with an arbitrary deterministic matrix $\boldsymbol{B}$ is involved.

  Second, pointwise perturbation control: for any fixed $\boldsymbol{x}\in\mathbb{T}$, by the reverse triangle inequality for absolute values and using the conclusion of (I),

  $$||f(\boldsymbol{A}\boldsymbol{x})-\mu(\boldsymbol{x})|-|f(\boldsymbol{B}\boldsymbol{x})-\mu(\boldsymbol{x})||\leqslant|f(\boldsymbol{A}\boldsymbol{x})-f(\boldsymbol{B}\boldsymbol{x})|\leqslant b\|\boldsymbol{A}\boldsymbol{x}-\boldsymbol{B}\boldsymbol{x}\|_2=b\|(\boldsymbol{A}-\boldsymbol{B})\boldsymbol{x}\|_2.$$

  Third, using the norm compatibility and the set radius: by the relation between the operator norm and the Frobenius norm ($\|\boldsymbol{M}\|_{op}\leqslant\|\boldsymbol{M}\|_F$),

  $$\|(\boldsymbol{A}-\boldsymbol{B})\boldsymbol{x}\|_2\leqslant\|\boldsymbol{A}-\boldsymbol{B}\|_{op}\|\boldsymbol{x}\|_2\leqslant\|\boldsymbol{A}-\boldsymbol{B}\|_F\cdot\sup_{\boldsymbol{x}\in\mathbb{T}}\|\boldsymbol{x}\|_2=\operatorname{rad}(\mathbb{T})\|\boldsymbol{A}-\boldsymbol{B}\|_F.$$

  Fourth, taking the supremum: by the property $|\sup g-\sup h|\leqslant\sup|g-h|$, taking the supremum over all $\boldsymbol{x}\in\mathbb{T}$:

  $$|F(\boldsymbol{A})-F(\boldsymbol{B})|\leqslant b\operatorname{rad}(\mathbb{T})\|\boldsymbol{A}-\boldsymbol{B}\|_F. \tag{II}$$

  Hence F is an L-Lipschitz function with respect to the Frobenius norm, with Lipschitz constant $L=b\operatorname{rad}(\mathbb{T})$.

- Applying the Theorem T5.2.2 (*Gaussian isoperimetric inequality*)

  Vectorize the matrix $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ into the high-dimensional vector $\operatorname{vec}(\boldsymbol{A})\in\mathbb{R}^{mn}$. Since A_{ij} i.i.d. $\sim N(0,1)$, $\operatorname{vec}(\boldsymbol{A})\sim N(\boldsymbol{0},\boldsymbol{I}_{mn})$ is a standard Gaussian random vector.

  By the Theorem T5.2.2 (*Gaussian isoperimetric inequality*): for any L-Lipschitz function F on $\mathbb{R}^N$ and a standard normal vector $\boldsymbol{G}\sim N(\boldsymbol{0},\boldsymbol{I}_N)$, for any $t\geqslant 0$,

  $$\mathbb{P}|F(\boldsymbol{G})-\mathbb{E}[F(\boldsymbol{G})]|\geqslant t\leqslant 2\exp\left(-\frac{t^2}{2L^2}\right).$$

  Substituting $L=b\operatorname{rad}(\mathbb{T})$, the one-sided tail bound satisfies

  $$\mathbb{P}F(\boldsymbol{A})\geqslant\mathbb{E}[F(\boldsymbol{A})]+t\leqslant 2\exp\left(-\frac{t^2}{2b^2\operatorname{rad}(\mathbb{T})^2}\right). \tag{III}$$

- Substituting the expectation bound to conclude

  By Theorem T9.6.3 (*General matrix deviation inequality*), there exists an absolute constant $C_0>0$ such that

  $$\mathbb{E}[F(\boldsymbol{A})]=\mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}|f(\boldsymbol{A}\boldsymbol{x})-\mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]|\right]\leqslant C_0b\gamma(\mathbb{T}).$$

  Set $t=\sqrt{2}u\cdot b\operatorname{rad}(\mathbb{T})$; then the exponent becomes exactly

  $$\frac{t^2}{2b^2\operatorname{rad}(\mathbb{T})^2}=\frac{2u^2b^2\operatorname{rad}(\mathbb{T})^2}{2b^2\operatorname{rad}(\mathbb{T})^2}=u^2.$$

  Using the conclusion of (III), we obtain

  $$F(\boldsymbol{A})\leqslant\mathbb{E}[F(\boldsymbol{A})]+\sqrt{2}ub\operatorname{rad}(\mathbb{T})\leqslant C_0b\gamma(\mathbb{T})+\sqrt{2}ub\operatorname{rad}(\mathbb{T})\ (\text{w.h.p. } 1-2\mathrm{e}^{-u^2}).$$

  Taking the absolute constant $C=\max(C_0,\sqrt{2})$, we have proved

  $$\sup_{\boldsymbol{x}\in\mathbb{T}}|f(\boldsymbol{A}\boldsymbol{x})-\mathbb{E}[f(\boldsymbol{A}\boldsymbol{x})]|\leqslant Cb(\gamma(\mathbb{T})+u\operatorname{rad}(\mathbb{T})).$$

Rk. Taking $f(v)=\|v\|_2$, $b=1$, and $\mathbb{T}\subset\mathbb{S}^{n-1}$ recovers Exercise Q9.13 (*M^* Bound: High-Probability Version*): $\sup_{\boldsymbol{x}\in\mathbb{T}}|\|\boldsymbol{A}\boldsymbol{x}\|_2-$

$\sqrt{m}| \leqslant C(w(\mathbb{T}) + u)$.

In the general form, $b\gamma(\mathbb{T})$ controls average complexity and $ub\operatorname{rad}(\mathbb{T})$ controls the deviation scale.

37 [**Johnson–Lindenstrauss Lemma for General Norms**] The classical Johnson–Lindenstrauss lemma states that every N-point set $\boldsymbol{X} \subset \mathbb{R}^n$, where $N \geqslant 2^8$, admits a random linear embedding into $(\mathbb{R}^m, \|\cdot\|_2)$ with relative accuracy $(1 \pm \varepsilon)$, where $m \gtrsim \varepsilon^{-2} \log N$.

Using Theorem T9.6.3 (*General matrix deviation inequality*), extend the target space to a normed vector space $(\mathbb{R}^m, \|\cdot\|)$ equipped with an arbitrary norm $\|\cdot\|$; state rigorously and prove the Johnson–Lindenstrauss dimension reduction lemma for general norms.

Prf. • Constructing the set $\mathbb{T}$ of normalized difference vectors

Define the set of normalized difference vectors of all distinct pairs of points in $\boldsymbol{X}$:

$$\mathbb{T} = \{\overline{\boldsymbol{x} - \boldsymbol{y}} : \boldsymbol{x}, \boldsymbol{y} \in \boldsymbol{X},\ \boldsymbol{x} \neq \boldsymbol{y}\} \subset \mathbb{S}^{n-1}.$$

First, the cardinality of the set: $|\mathbb{T}| \leqslant \binom{N}{2} = \frac{N(N-1)}{2} < N^2$.

Second, the Gaussian complexity: by the classical bound on Gaussian maxima over finite point sets,

$$w(\mathbb{T}) = \mathbb{E}_g[\max_{\boldsymbol{u} \in \mathbb{T}} \langle g, \boldsymbol{u} \rangle] \leqslant \sqrt{2 \log |\mathbb{T}|} \leqslant \sqrt{2 \log(N^2)} = 2\sqrt{\log N}. \tag{I}$$

Third, the radius of the set: since $\mathbb{T} \subset \mathbb{S}^{n-1}$, we have $\operatorname{rad}(\mathbb{T}) = 1$.

• Verifying the conditions on the function and computing the mean

Define the function $f : \mathbb{R}^m \to \mathbb{R}$ as the target norm $f(v) = \|v\|$. Positive homogeneity: for any $\alpha \geqslant 0$, $f(\alpha v) = \|\alpha v\| = \alpha \|v\| = \alpha f(v)$; subadditivity: by the triangle inequality for norms, $f(u + v) = \|u + v\| \leqslant \|u\| + \|v\| = f(u) + f(v)$; boundedness: by the definition of b, for all $v \in \mathbb{R}^m$ we have $f(v) = \|v\| \leqslant b\|v\|_2$.

For any fixed $\boldsymbol{u} \in \mathbb{T} \subset \mathbb{S}^{n-1}$ (with $\|\boldsymbol{u}\|_2 = 1$), since $\boldsymbol{A}$ is a standard Gaussian matrix, the projection vector $\boldsymbol{A}\boldsymbol{u} \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$ has the same distribution as $\boldsymbol{g}_m$. Hence the mean function is

$$\mathbb{E}[f(\boldsymbol{A}\boldsymbol{u})] = \mathbb{E}[\|\boldsymbol{A}\boldsymbol{u}\|] = \mathbb{E}[\|\boldsymbol{g}_m\|] = h_m, \tag{II}$$

where $h_m = \mathbb{E}[\|\boldsymbol{g}_m\|]$ is the expected norm (mean width) of a standard Gaussian vector.

• Applying the high-probability version of the general matrix deviation inequality (Exercise Q9.36 (*General Matrix Deviation: A High-Probability Bound*))

By Exercise Q9.36 (*General Matrix Deviation: A High-Probability Bound*), for any parameter $t \geqslant 0$, we have

$$\sup_{\boldsymbol{u} \in \mathbb{T}} |\|\boldsymbol{A}\boldsymbol{u}\| - \mathbb{E}[\|\boldsymbol{A}\boldsymbol{u}\|]| \leqslant C_0 b(w(\mathbb{T}) + t \operatorname{rad}(\mathbb{T})) \text{ (w.h.p. } 1 - 2\mathrm{e}^{-t^2}).$$

Substituting $w(\mathbb{T}) \leqslant 2\sqrt{\log N}$ from (I), the mean h_m from (II), and $\operatorname{rad}(\mathbb{T}) = 1$:

$$\sup_{\boldsymbol{u} \in \mathbb{T}} |\|\boldsymbol{A}\boldsymbol{u}\| - h_m| \leqslant C_0 b(2\sqrt{\log N} + t).$$

Choose the parameter $t = \sqrt{\log N}$; the failure probability is then $2\mathrm{e}^{-t^2} = 2\mathrm{e}^{-\log N} = \frac{2}{N}$. Hence, we have

$$\sup_{\boldsymbol{u} \in \mathbb{T}} |\|\boldsymbol{A}\boldsymbol{u}\| - h_m| \leqslant 3C_0 b\sqrt{\log N} \left(\text{w.h.p. } 1 - \frac{2}{N}\right). \tag{III}$$

• Normalizing on both sides and recovering the pairwise distances

Divide both sides of (III) by the expected scale factor h_m:

$$\sup_{\boldsymbol{u} \in \mathbb{T}} \left| \frac{\|\boldsymbol{A}\boldsymbol{u}\|}{h_m} - 1 \right| \leqslant \frac{3C_0 b\sqrt{\log N}}{h_m}.$$

If the target dimension satisfies the condition of the statement,

$$\left(\frac{h_m}{b}\right)^2 \geqslant \frac{(3C_0)^2}{\varepsilon^2} \log N = C\varepsilon^{-2} \log N \text{ if and only if } \frac{3C_0 b\sqrt{\log N}}{h_m} \leqslant \varepsilon,$$

then we immediately obtain

$$\sup_{\boldsymbol{u}\in\mathbb{T}}\left|\left\|\frac{1}{h_m}\boldsymbol{A}\boldsymbol{u}\right\|-1\right|\leqslant\varepsilon.$$

Substituting the unit difference vector $\overline{\boldsymbol{x}-\boldsymbol{y}}\in\mathbb{T}$ and multiplying both sides by $\|\boldsymbol{x}-\boldsymbol{y}\|_2$ for all $\boldsymbol{x}\neq\boldsymbol{y}\in\boldsymbol{X}$ by absolute homogeneity of the norm:

$$\left|\left\|\frac{1}{h_m}\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\right\|-\|\boldsymbol{x}-\boldsymbol{y}\|_2\right|\leqslant\varepsilon\|\boldsymbol{x}-\boldsymbol{y}\|_2.$$

Expanding the absolute value yields

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2\leqslant\left\|\frac{1}{h_m}\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\right\|\leqslant(1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2.$$

Rk. For ℓ_2 and ℓ_1, $(h_m/b)^2\asymp m$, so $m\gtrsim\varepsilon^{-2}\log N$.
For ℓ_∞, $(h_m/b)^2\asymp\log m$, so $m\gtrsim N^{C/\varepsilon^2}$.
The critical quantity $(h_m/b)^2$ determines the dimension-reduction scale for a general norm.

38 [**Johnson–Lindenstrauss Lemma for the ℓ_1 Norm**] Specialize Exercise Q9.37 (*Johnson–Lindenstrauss Lemma for General Norms*) to the ℓ_1 norm. Let X be an N-point subset of $\mathbb{R}^n$, where $N\geqslant 2^9$, let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ be a Gaussian matrix (*entries are i.i.d.* $N(0,1)$), and let $\varepsilon\in(0,1)$. Assume that

$$m\geqslant C_\varepsilon\log N.$$

Show that the matrix (w.h.p.)

$$\boldsymbol{Q}=\sqrt{\frac{\pi}{2}}\frac{1}{m}\boldsymbol{A}$$

satisfies

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2\leqslant\|\boldsymbol{Q}\boldsymbol{x}-\boldsymbol{Q}\boldsymbol{y}\|_1\leqslant(1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2\text{ for all }\boldsymbol{x},\boldsymbol{y}\in X.$$

Prf. • Computing the Gaussian mean width h_m and the normalization operator $\boldsymbol{Q}$
Let $\boldsymbol{g}_m=(g_1,g_2,\cdots,g_m)\sim N(\boldsymbol{0},\boldsymbol{I}_m)$ be a standard Gaussian random vector.
First, compute the first absolute moment $\mathbb{E}[|g_1|]$ of a single standard normal variable:

$$\mathbb{E}[|g_1|]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}|t|\mathrm{e}^{-t^2/2}dt=\frac{2}{\sqrt{2\pi}}\int_0^{+\infty}t\mathrm{e}^{-t^2/2}dt=\sqrt{\frac{2}{\pi}}\left[-\mathrm{e}^{-t^2/2}\right]_0^{+\infty}=\sqrt{\frac{2}{\pi}}.$$

Second, by the linearity of expectation, the expected ℓ_1 norm h_m is

$$h_m=\mathbb{E}[\|\boldsymbol{g}_m\|_1]=\mathbb{E}\left[\sum_{i=1}^m|g_i|\right]=\sum_{i=1}^m\mathbb{E}[|g_i|]=m\sqrt{\frac{2}{\pi}}=\sqrt{\frac{2}{\pi}}m.\tag{I}$$

Third, by the normalization form $\boldsymbol{Q}=\frac{1}{h_m}\boldsymbol{A}$ of Exercise Q9.37 (*Johnson–Lindenstrauss Lemma for General Norms*), substituting (I):

$$\boldsymbol{Q}=\frac{1}{\sqrt{\frac{2}{\pi}}m}\boldsymbol{A}=\sqrt{\frac{\pi}{2}}\frac{1}{m}\boldsymbol{A},\tag{II}$$

which is exactly the matrix $\boldsymbol{Q}$ given in the statement.

• Computing the Euclidean proportionality constant b
By the definition of the parameter b, combined with the Cauchy–Schwarz inequality: for any $v=(v_1,\cdots,v_m)\in\mathbb{S}^{m-1}$ (i.e. $\|v\|_2=1$),

$$\|v\|_1=\sum_{i=1}^m|v_i|\leqslant\left(\sum_{i=1}^m1^2\right)^{1/2}\left(\sum_{i=1}^mv_i^2\right)^{1/2}=\sqrt{m}\|v\|_2=\sqrt{m}.$$

Taking $v = (\frac{1}{\sqrt{m}}, \cdots, \frac{1}{\sqrt{m}})$, equality holds strictly. Hence

$$b = \sup_{\|v\|_2=1} \|v\|_1 = \sqrt{m}. \tag{III}$$

- Computing the critical effective dimension and deriving the required dimension condition

  Using (I) and (III), compute the key geometric ratio:

$$\frac{h_m}{b} = \frac{\sqrt{\frac{2}{\pi}}m}{\sqrt{m}} = \sqrt{\frac{2}{\pi}}\sqrt{m}.$$

  Squaring both sides:

$$\left(\frac{h_m}{b}\right)^2 = \frac{2}{\pi}m. \tag{IV}$$

  By the general-norm JL theorem proved in Exercise Q9.37 (*Johnson–Lindenstrauss Lemma for General Norms*), $(\frac{h_m}{b})^2 \geqslant C_0\varepsilon^{-2}\log N$ is sufficient for a $(1 \pm \varepsilon)$ isometric embedding. Substituting (IV):

$$\frac{2}{\pi}m \geqslant C_0\varepsilon^{-2}\log N \text{ if and only if } m \geqslant \left(\frac{\pi}{2}C_0\varepsilon^{-2}\right)\log N = C_\varepsilon \log N. \tag{V}$$

- Deriving the final conclusion

  Under condition (V), directly applying the theorem of Exercise Q9.37 (*Johnson–Lindenstrauss Lemma for General Norms*), for all $\boldsymbol{x}, \boldsymbol{y} \in X$,

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \|\boldsymbol{Q}(\boldsymbol{x}-\boldsymbol{y})\|_1 \leqslant (1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2 \left(\text{w.h.p. } 1-\frac{2}{N}\right).$$

  By linearity $\boldsymbol{Q}(\boldsymbol{x}-\boldsymbol{y}) = \boldsymbol{Q}\boldsymbol{x} - \boldsymbol{Q}\boldsymbol{y}$, i.e.

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \|\boldsymbol{Q}\boldsymbol{x}-\boldsymbol{Q}\boldsymbol{y}\|_1 \leqslant (1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2.$$

Rk. The $\ell_2 \to \ell_1$ embedding preserves Euclidean distances with $m = \mathcal{O}(\varepsilon^{-2}\log N)$.

39 [**Embedding into** ℓ_∞] Specialize Exercise Q9.37 (*Johnson–Lindenstrauss Lemma for General Norms*) to the ℓ_∞ norm. Let X be an N-point subset of $\mathbb{R}^n$, where $N \geqslant 2^{10}$, let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ be a Gaussian matrix (*entries are i.i.d.* $N(0,1)$), and let $\varepsilon \in (0,1)$. Assume that

$$m \geqslant N^{C_\varepsilon}.$$

Show that with high probability, the matrix

$$\boldsymbol{Q} = \frac{1}{h_m}\boldsymbol{A}, \qquad h_m = \mathbb{E}\|\boldsymbol{g}_m\|_\infty \left(\text{w.h.p. } 1-\frac{2}{N}\right)$$

implements a near-isometric embedding from $(\mathbb{R}^n, \|\cdot\|_2)$ into $(\mathbb{R}^m, \|\cdot\|_\infty)$, i.e. for all $\boldsymbol{x}, \boldsymbol{y} \in X$,

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \|\boldsymbol{Q}\boldsymbol{x}-\boldsymbol{Q}\boldsymbol{y}\|_\infty \leqslant (1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2.$$

Note that in this case $m \geqslant N$, so $\boldsymbol{Q}$ is not a dimension reduction map but rather a high-dimensional near-isometric embedding.

Prf. • Computing the circumscribed Euclidean proportionality constant b

  By the relation between the ℓ_∞ norm and the ℓ_2 norm: for any $v = (v_1, v_2, \cdots, v_m) \in \mathbb{S}^{m-1}$ (*i.e.* $\sum_{i=1}^m v_i^2 = 1$),

$$\|v\|_\infty = \max_{1\leqslant i\leqslant m} |v_i| \leqslant \sqrt{\sum_{i=1}^m v_i^2} = \|v\|_2 = 1.$$

  Taking the standard basis vector $v = \boldsymbol{e}_1 = (1, 0, \cdots, 0)$, we have $\|\boldsymbol{e}_1\|_2 = 1$ and $\|\boldsymbol{e}_1\|_\infty = 1$, so equality holds. Hence

$$b = \sup_{\|v\|_2=1} \|v\|_\infty = 1. \tag{I}$$

- Computing the Gaussian mean width h_m and determining the operator $\boldsymbol{Q}$

  Let $\boldsymbol{g}_m = (g_1, g_2, \cdots, g_m) \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$ be a standard Gaussian random vector.

First, compute the expected extremum h_m:

$$h_m = \mathbb{E}[\|\boldsymbol{g}_m\|_\infty] = \mathbb{E}\Big[\max_{1\leqslant i\leqslant m} |g_i|\Big].$$

By the classical extreme value theory for the standard normal distribution, for the maximum of the absolute values of m independent standard normal variables, when $m \geqslant 2$ there exist absolute constants $c_1, c_2 > 0$ satisfying

$$c_1\sqrt{\log m} \leqslant h_m \leqslant c_2\sqrt{\log m}. \tag{II}$$

Second, determine the form of the scaled matrix $\boldsymbol{Q}$: using the conclusion of Exercise Q9.37 (*Johnson–Lindenstrauss Lemma for General Norms*), the normalization map is $\boldsymbol{Q} = \frac{1}{h_m}\boldsymbol{A}$. Using the conclusion of (II),

$$\boldsymbol{Q} = \frac{1}{h_m}\boldsymbol{A} \asymp (\log m)^{-1/2}\boldsymbol{A}, \tag{III}$$

The proportionality factor depends on m, but is bounded above and below by absolute positive constants.

- Computing the critical ratio and deriving the dimension condition $m \geqslant N^{C_\varepsilon}$

  Using the conclusions of (I) ($b = 1$) and (II) ($h_m \geqslant c_1\sqrt{\log m}$), the critical geometric ratio is

$$\left(\frac{h_m}{b}\right)^2 = \frac{h_m^2}{1^2} \geqslant c_1^2 \log m. \tag{IV}$$

  By Exercise Q9.37 (*Johnson–Lindenstrauss Lemma for General Norms*), the $(1 \pm \varepsilon)$ near-isometric embedding holds provided that $(\frac{h_m}{b})^2 \geqslant C_0\varepsilon^{-2}\log N$ (w.h.p. $1 - \frac{2}{N}$). Combining this with (IV), it suffices that

$$c_1^2 \log m \geqslant C_0\varepsilon^{-2}\log N \text{ if and only if } \log m \geqslant \left(\frac{C_0}{c_1^2}\varepsilon^{-2}\right)\log N.$$

  Define the constant $C_\varepsilon = \frac{C_0}{c_1^2}\varepsilon^{-2} = \mathcal{O}(\varepsilon^{-2})$. Taking the exponential of both sides:

$$m \geqslant \exp(C_\varepsilon \log N) = N^{C_\varepsilon}. \tag{V}$$

- Deriving the conclusion

  Under condition (V), directly applying the theorem of Exercise Q9.37 (*Johnson–Lindenstrauss Lemma for General Norms*), for all pairs $\boldsymbol{x}, \boldsymbol{y} \in X$,

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \left\|\frac{1}{h_m}\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})\right\|_\infty \leqslant (1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2 \left(\text{w.h.p. } 1-\frac{2}{N}\right).$$

  Substituting $\boldsymbol{Q} = \frac{1}{h_m}\boldsymbol{A}$, we obtain

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2 \leqslant \|\boldsymbol{Q}\boldsymbol{x}-\boldsymbol{Q}\boldsymbol{y}\|_\infty \leqslant (1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|_2.$$

Rk. For ℓ_∞^m, Gaussian width grows only as $\sqrt{\log m}$, so the Dvoretzky critical dimension is $\asymp \log m$. Embedding Euclidean geometry of dimension $\asymp \log N$ therefore requires $m \geqslant N^{C_\varepsilon}$.

40 [**Duality**] Show that for a nonempty closed, bounded set $V \subset \mathbb{R}^m$, its support function is close to the Euclidean norm if and only if its convex hull is close to the Euclidean ball. Specifically, for $r_-, r_+ \geqslant 0$, prove that

$$r_-\mathbb{B}_2^m \subset \operatorname{conv}(V) \subset r_+\mathbb{B}_2^m$$

holds if and only if

$$r_-\|\boldsymbol{y}\|_2 \leqslant \sup_{\boldsymbol{x}\in V}\langle \boldsymbol{x}, \boldsymbol{y}\rangle \leqslant r_+\|\boldsymbol{y}\|_2 \text{ for all } \boldsymbol{y} \in \mathbb{R}^m.$$

Prf. • Basic properties of support functions

Define the support function of a set $S \subset \mathbb{R}^m$ as

$$h_S(\boldsymbol{y}) = \sup_{\boldsymbol{x}\in S}\langle \boldsymbol{x}, \boldsymbol{y}\rangle. \qquad (\boldsymbol{y} \in \mathbb{R}^m)$$

This proof relies on three basic properties from convex analysis.

First, convex hull invariance: since the linear function $\langle \cdot, \boldsymbol{y}\rangle$ attains the same supremum on the set V and on its convex

hull $\operatorname{conv}(V)$, we always have

$$h_V(\boldsymbol{y}) = \sup_{\boldsymbol{x}\in V}\langle \boldsymbol{x},\boldsymbol{y}\rangle = \sup_{\boldsymbol{x}\in \operatorname{conv}(V)}\langle \boldsymbol{x},\boldsymbol{y}\rangle = h_{\operatorname{conv}(V)}(\boldsymbol{y}).$$

Second, compact convexity: since V is a compact set (*closed and bounded*), $K = \operatorname{conv}(V)$ is likewise a nonempty compact convex set in $\mathbb{R}^m$.

Third, the support function of the Euclidean ball: by the Cauchy–Schwarz inequality, the support function of the Euclidean ball $r\mathbb{B}_2^m$ of radius r is

$$h_{r\mathbb{B}_2^m}(\boldsymbol{y}) = \sup_{\|\boldsymbol{x}\|_2\leqslant r}\langle \boldsymbol{x},\boldsymbol{y}\rangle = r\|\boldsymbol{y}\|_2.$$

Therefore, the proposition to be proved is equivalent to proving, for the nonempty compact convex set $K = \operatorname{conv}(V)$,

$$r_-\mathbb{B}_2^m \subset K \subset r_+\mathbb{B}_2^m \text{ if and only if } r_-\|\boldsymbol{y}\|_2 \leqslant h_K(\boldsymbol{y}) \leqslant r_+\|\boldsymbol{y}\|_2. \qquad (\boldsymbol{y}\in\mathbb{R}^m)$$

- Outer inclusion equivalence $K \subset r_+\mathbb{B}_2^m$ if and only if $h_K(\boldsymbol{y}) \leqslant r_+\|\boldsymbol{y}\|_2$

  Necessity: assume $K \subset r_+\mathbb{B}_2^m$. For any $\boldsymbol{y}\in\mathbb{R}^m$,

  $$h_K(\boldsymbol{y}) = \sup_{\boldsymbol{x}\in K}\langle \boldsymbol{x},\boldsymbol{y}\rangle \leqslant \sup_{\boldsymbol{x}\in r_+\mathbb{B}_2^m}\langle \boldsymbol{x},\boldsymbol{y}\rangle = r_+\|\boldsymbol{y}\|_2.$$

  Sufficiency: assume that for all $\boldsymbol{y}\in\mathbb{R}^m$, we have $h_K(\boldsymbol{y}) \leqslant r_+\|\boldsymbol{y}\|_2$. The goal is to prove that every point $\boldsymbol{x}\in K$ satisfies $\|\boldsymbol{x}\|_2 \leqslant r_+$. Let $\boldsymbol{x}\in K$. If $\boldsymbol{x} = \boldsymbol{0}$, the conclusion is trivial; we only need to consider the case $\boldsymbol{x}\neq\boldsymbol{0}$, and choose the test vector $\boldsymbol{y} = \boldsymbol{x}$ in the assumption:

  $$\|\boldsymbol{x}\|_2^2 = \langle \boldsymbol{x},\boldsymbol{x}\rangle \leqslant \sup_{\boldsymbol{z}\in K}\langle \boldsymbol{z},\boldsymbol{x}\rangle = h_K(\boldsymbol{x}) \leqslant r_+\|\boldsymbol{x}\|_2.$$

  Dividing both sides by the positive number $\|\boldsymbol{x}\|_2$, we immediately get $\|\boldsymbol{x}\|_2 \leqslant r_+$. This shows that the Euclidean norm of every point of K does not exceed r_+, i.e. $K \subset r_+\mathbb{B}_2^m$.

  Thus we have proved

  $$\operatorname{conv}(V) \subset r_+\mathbb{B}_2^m \text{ if and only if } \sup_{\boldsymbol{x}\in V}\langle \boldsymbol{x},\boldsymbol{y}\rangle \leqslant r_+\|\boldsymbol{y}\|_2. \qquad (\boldsymbol{y}\in\mathbb{R}^m)(\mathrm{I})$$

- Inner inclusion equivalence $r_-\mathbb{B}_2^m \subset K$ if and only if $h_K(\boldsymbol{y}) \geqslant r_-\|\boldsymbol{y}\|_2$

  Necessity: assume $r_-\mathbb{B}_2^m \subset K$. For any $\boldsymbol{y}\in\mathbb{R}^m$,

  $$h_K(\boldsymbol{y}) = \sup_{\boldsymbol{x}\in K}\langle \boldsymbol{x},\boldsymbol{y}\rangle \geqslant \sup_{\boldsymbol{x}\in r_-\mathbb{B}_2^m}\langle \boldsymbol{x},\boldsymbol{y}\rangle = r_-\|\boldsymbol{y}\|_2.$$

  Sufficiency: assume that for all $\boldsymbol{y}\in\mathbb{R}^m$, we have $h_K(\boldsymbol{y}) \geqslant r_-\|\boldsymbol{y}\|_2$. We prove that $r_-\mathbb{B}_2^m \subset K$.

  If $r_- = 0$, then $r_-\mathbb{B}_2^m = \{\boldsymbol{0}\}$. If $\boldsymbol{0}\notin K$, by the strict separation theorem for convex sets, there exists $\boldsymbol{y}_0$ such that $h_K(\boldsymbol{y}_0) < \langle \boldsymbol{0},\boldsymbol{y}_0\rangle = 0$, contradicting $h_K(\boldsymbol{y}_0) \geqslant r_-\|\boldsymbol{y}_0\|_2 = 0$; therefore $\boldsymbol{0}\in K$.

  Now assume $r_- > 0$. By contradiction, suppose $r_-\mathbb{B}_2^m \not\subset K$; this means there exists a point $\boldsymbol{x}_0 \in r_-\mathbb{B}_2^m$ satisfying $\boldsymbol{x}_0 \notin K$. Since K is a nonempty closed convex set in $\mathbb{R}^m$ and the singleton $\{\boldsymbol{x}_0\}$ is a compact convex set not intersecting K, by the strict separating hyperplane theorem there must exist a nonzero direction vector $\boldsymbol{y}_0\in\mathbb{R}^m$ ($\boldsymbol{y}_0\neq\boldsymbol{0}$) and a real number $\alpha\in\mathbb{R}$, such that the hyperplane strictly separates K from $\boldsymbol{x}_0$:

  $$\sup_{\boldsymbol{x}\in K}\langle \boldsymbol{x},\boldsymbol{y}_0\rangle < \alpha < \langle \boldsymbol{x}_0,\boldsymbol{y}_0\rangle,$$

  i.e.

  $$h_K(\boldsymbol{y}_0) < \langle \boldsymbol{x}_0,\boldsymbol{y}_0\rangle. \qquad (\mathrm{II})$$

  On the other hand, since $\boldsymbol{x}_0 \in r_-\mathbb{B}_2^m$ (*i.e.* $\|\boldsymbol{x}_0\|_2 \leqslant r_-$), by the Cauchy–Schwarz inequality,

  $$\langle \boldsymbol{x}_0,\boldsymbol{y}_0\rangle \leqslant \|\boldsymbol{x}_0\|_2\|\boldsymbol{y}_0\|_2 \leqslant r_-\|\boldsymbol{y}_0\|_2. \qquad (\mathrm{III})$$

  Substituting (III) into (II), we get $h_K(\boldsymbol{y}_0) < r_-\|\boldsymbol{y}_0\|_2$. This directly contradicts the known assumption "for all $\boldsymbol{y}\in\mathbb{R}^m$ we have $h_K(\boldsymbol{y}) \geqslant r_-\|\boldsymbol{y}\|_2$". Hence the assumption is false, and we must have $r_-\mathbb{B}_2^m \subset K = \operatorname{conv}(V)$.

Thus we have proved

$$r_-\mathbb{B}_2^m \subset \operatorname{conv}(V) \text{ if and only if } \sup_{\boldsymbol{x}\in V}\langle \boldsymbol{x},\boldsymbol{y}\rangle \geqslant r_-\|\boldsymbol{y}\|_2. \qquad (\boldsymbol{y}\in\mathbb{R}^m)(\text{IV})$$

- Synthesizing the conclusion

  Combining (I) and (IV), the full characterization is proved:

$$r_-\mathbb{B}_2^m \subset \operatorname{conv}(V) \subset r_+\mathbb{B}_2^m \text{ if and only if } r_-\|\boldsymbol{y}\|_2 \leqslant \sup_{\boldsymbol{x}\in V}\langle \boldsymbol{x},\boldsymbol{y}\rangle \leqslant r_+\|\boldsymbol{y}\|_2. \qquad (\boldsymbol{y}\in\mathbb{R}^m)$$

Rk. The support function h_K uniquely determines a closed convex set K. Thus, if $h_K(\boldsymbol{y})$ is bounded by $(1\pm\varepsilon)\|\boldsymbol{y}\|_2$ in every direction, then K is bounded by the corresponding Euclidean balls.

41 [**High-Probability Version of the Dvoretzky–Milman Theorem**] Prove a high-probability version of the Theorem T9.7.2 (*Dvoretzky–Milman theorem*).

Let $\boldsymbol{A}\in\mathbb{R}^{m\times n}$ be a standard Gaussian random matrix, and let $\mathbb{T}\subset\mathbb{R}^n$ be a nonempty compact set. Prove that there is an absolute constant $C>0$ such that for any parameter $u\geqslant 0$, the Gaussian random projection $\operatorname{conv}(\boldsymbol{A}\mathbb{T})$ of the set $\mathbb{T}$ is with high probability tightly sandwiched between two concentric Euclidean balls, i.e.

$$r_-(u)\mathbb{B}_2^m \subset \operatorname{conv}(\boldsymbol{A}\mathbb{T}) \subset r_+(u)\mathbb{B}_2^m \text{ (w.h.p. } 1-2\mathrm{e}^{-u^2}),$$

where the inner and outer radii are respectively

$$r_\pm(u) = w(\mathbb{T}) \pm C(\sqrt{m}+u)\operatorname{rad}(\mathbb{T}).$$

(*Note: the outer inclusion always holds; the inner inclusion holds when* $r_-(u)\geqslant 0$.)

Prf. • Reduction via convex-geometric duality (Exercise Q9.40 (*Duality*))

Let $V=\boldsymbol{A}\mathbb{T}\subset\mathbb{R}^m$. Since $\mathbb{T}$ is bounded, V is bounded as well. By Exercise Q9.40 (*Duality*),

$$r_-\mathbb{B}_2^m \subset \operatorname{conv}(\boldsymbol{A}\mathbb{T}) \subset r_+\mathbb{B}_2^m \text{ if and only if } r_-\|\boldsymbol{y}\|_2 \leqslant \sup_{\boldsymbol{z}\in\boldsymbol{A}\mathbb{T}}\langle \boldsymbol{z},\boldsymbol{y}\rangle \leqslant r_+\|\boldsymbol{y}\|_2 \ (\forall \boldsymbol{y}\in\mathbb{R}^m).$$

By positive homogeneity, it suffices to prove for every $\boldsymbol{y}\in\mathbb{S}^{m-1}$ (i.e. $\|\boldsymbol{y}\|_2=1$):

$$r_-(u) \leqslant \sup_{\boldsymbol{z}\in\boldsymbol{A}\mathbb{T}}\langle \boldsymbol{z},\boldsymbol{y}\rangle \leqslant r_+(u) \ (\forall \boldsymbol{y}\in\mathbb{S}^{m-1}). \qquad (\text{I})$$

- Reformulating the projected support function through the transposed Gaussian matrix

  For any fixed $\boldsymbol{y}\in\mathbb{S}^{m-1}$, using the adjoint property of the matrix inner product,

$$\sup_{\boldsymbol{z}\in\boldsymbol{A}\mathbb{T}}\langle \boldsymbol{z},\boldsymbol{y}\rangle = \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{A}\boldsymbol{x},\boldsymbol{y}\rangle = \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x},\boldsymbol{A}^\top\boldsymbol{y}\rangle.$$

  Define the support function $f:\mathbb{R}^n\to\mathbb{R}$ of the set $\mathbb{T}$:

$$f(\boldsymbol{v}) = \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x},\boldsymbol{v}\rangle \ (\boldsymbol{v}\in\mathbb{R}^n).$$

  First, verify the geometric properties of f: positive homogeneity, for any $\alpha\geqslant 0$, $f(\alpha\boldsymbol{v})=\alpha f(\boldsymbol{v})$; subadditivity, $f(\boldsymbol{u}+\boldsymbol{v})\leqslant f(\boldsymbol{u})+f(\boldsymbol{v})$; boundedness, by the Cauchy–Schwarz inequality

$$f(\boldsymbol{v}) = \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x},\boldsymbol{v}\rangle \leqslant \left(\sup_{\boldsymbol{x}\in\mathbb{T}}\|\boldsymbol{x}\|_2\right)\|\boldsymbol{v}\|_2 = \operatorname{rad}(\mathbb{T})\|\boldsymbol{v}\|_2,$$

  so the proportionality constant is $b=\operatorname{rad}(\mathbb{T})$. Second, compute the expected mean. For any fixed $\boldsymbol{y}\in\mathbb{S}^{m-1}$ (with $\|\boldsymbol{y}\|_2=1$), since $\boldsymbol{A}$ is a standard Gaussian matrix, the transposed vector $\boldsymbol{A}^\top\boldsymbol{y}\sim N(\boldsymbol{0},\boldsymbol{I}_n)$ is a standard Gaussian random vector in $\mathbb{R}^n$. Therefore

$$\mathbb{E}[f(\boldsymbol{A}^\top\boldsymbol{y})] = \mathbb{E}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x},\boldsymbol{A}^\top\boldsymbol{y}\rangle\right] = \mathbb{E}_{\boldsymbol{g}}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x},\boldsymbol{g}\rangle\right] = w(\mathbb{T}). \qquad (\text{II})$$

  Note that this expected value is the same constant $w(\mathbb{T})$ for every $\boldsymbol{y}\in\mathbb{S}^{m-1}$.

- Applying the high-probability version of the general matrix deviation inequality (Exercise Q9.36 (*General Matrix Deviation: A High-Probability Bound*))

  Consider the transposed matrix $\boldsymbol{B}=\boldsymbol{A}^\top\in\mathbb{R}^{n\times m}$. The entries of $\boldsymbol{B}$ are i.i.d. $N(0,1)$. Apply Exercise Q9.36 (*General*

Matrix Deviation: A High-Probability Bound) to the matrix $\boldsymbol{B} \in \mathbb{R}^{n\times m}$, the functional $f(\boldsymbol{v}) = \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x}, \boldsymbol{v}\rangle$ on $\mathbb{R}^n$ (with constant $b = \operatorname{rad}(\mathbb{T})$), and the set $\mathbb{S} = \mathbb{S}^{m-1} \subset \mathbb{R}^m$ of Euclidean unit vectors. Compute two geometric characteristics of the set $\mathbb{S} = \mathbb{S}^{m-1}$: the set radius $\operatorname{rad}(\mathbb{S}) = \operatorname{rad}(\mathbb{S}^{m-1}) = 1$; the Gaussian complexity $\gamma(\mathbb{S}) = w(\mathbb{S}^{m-1}) = \mathbb{E}_{\boldsymbol{g}_m}\|\boldsymbol{g}_m\|_2 \leqslant \sqrt{m}$ (where $\boldsymbol{g}_m \sim N(\boldsymbol{0}, \boldsymbol{I}_m)$). By the high-probability bound of Exercise Q9.36 (*General Matrix Deviation: A High-Probability Bound*), for any $u \geqslant 0$, we have

$$\sup_{\boldsymbol{y}\in\mathbb{S}^{m-1}} |f(\boldsymbol{B}\boldsymbol{y}) - \mathbb{E}[f(\boldsymbol{B}\boldsymbol{y})]| \leqslant C_0 b(\gamma(\mathbb{S}) + u\operatorname{rad}(\mathbb{S})) \leqslant C_0 \operatorname{rad}(\mathbb{T})(\sqrt{m}+u) = C(\sqrt{m}+u)\operatorname{rad}(\mathbb{T}) \text{ (w.h.p. } 1-2\mathrm{e}^{-u^2}). \tag{III}$$

- Combining with duality to obtain the final geometric sandwich

  Substituting $\mathbb{E}[f(\boldsymbol{B}\boldsymbol{y})] = w(\mathbb{T})$ from (II) into (III), we have

$$\sup_{\boldsymbol{y}\in\mathbb{S}^{m-1}} \left| \sup_{\boldsymbol{z}\in\boldsymbol{A}\mathbb{T}} \langle \boldsymbol{z}, \boldsymbol{y}\rangle - w(\mathbb{T}) \right| \leqslant C(\sqrt{m}+u)\operatorname{rad}(\mathbb{T}) \text{ (w.h.p. } 1-2\mathrm{e}^{-u^2}).$$

  Expanding the absolute value, uniformly for every $\boldsymbol{y} \in \mathbb{S}^{m-1}$,

$$w(\mathbb{T}) - C(\sqrt{m}+u)\operatorname{rad}(\mathbb{T}) \leqslant \sup_{\boldsymbol{z}\in\boldsymbol{A}\mathbb{T}} \langle \boldsymbol{z}, \boldsymbol{y}\rangle \leqslant w(\mathbb{T}) + C(\sqrt{m}+u)\operatorname{rad}(\mathbb{T}).$$

  Let the inner and outer radii be $r_{\pm}(u) = w(\mathbb{T}) \pm C(\sqrt{m}+u)\operatorname{rad}(\mathbb{T})$. Applying the duality equivalence of Exercise Q9.40 (*Duality*) once more, the support-function inequality above is precisely equivalent to the spatial inclusion

$$r_-(u)\mathbb{B}_2^m \subset \operatorname{conv}(\boldsymbol{A}\mathbb{T}) \subset r_+(u)\mathbb{B}_2^m.$$

Rk. If m satisfies the Dvoretzky critical-dimension condition

$$m \leqslant c\varepsilon^2 \left(\frac{w(\mathbb{T})}{\operatorname{rad}(\mathbb{T})}\right)^2,$$

then with $u = \sqrt{m}$, the fluctuation term satisfies

$$C(\sqrt{m}+u)\operatorname{rad}(\mathbb{T}) \leqslant 2C\sqrt{m}\operatorname{rad}(\mathbb{T}) \leqslant \varepsilon w(\mathbb{T}).$$

the random m-dimensional projection $\operatorname{conv}(\boldsymbol{A}\mathbb{T})$ satisfies (w.h.p. $1-2\mathrm{e}^{-m}$)

$$(1-\varepsilon)w(\mathbb{T})\mathbb{B}_2^m \subset \operatorname{conv}(\boldsymbol{A}\mathbb{T}) \subset (1+\varepsilon)w(\mathbb{T})\mathbb{B}_2^m.$$

This is the high-probability form of the Dvoretzky–Milman theorem.

42 [**Gaussian Cloud Is Nearly Round**] Let $\boldsymbol{g}_1, \cdots, \boldsymbol{g}_n$ i.i.d. $\sim N(\boldsymbol{0}, \boldsymbol{I}_m)$. Prove that if $m \leqslant c\log n$, where $c = c_\varepsilon > 0$, then there is a constant depending only on ε such that the convex hull

$$K = \operatorname{conv}(\boldsymbol{g}_1, \boldsymbol{g}_2, \cdots, \boldsymbol{g}_n) \subset \mathbb{R}^m$$

is an almost isotropic Euclidean ball (w.h.p.) (see Figure 9.9 right), i.e. there is a radius $R \asymp \sqrt{\log n}$ satisfying

$$(1-\varepsilon)R\mathbb{B}_2^m \subset \operatorname{conv}(\boldsymbol{g}_1, \cdots, \boldsymbol{g}_n) \subset (1+\varepsilon)R\mathbb{B}_2^m.$$



Figure 9.9 A random projection of an 8-dimensional cube (left) and 10^4 Gaussian points (right) onto the plane

Prf. • Representing the Gaussian cloud as a Gaussian matrix acting on the standard basis

Let $\boldsymbol{A} = (\boldsymbol{g}_1, \cdots, \boldsymbol{g}_n) \in \mathbb{R}^{m\times n}$; then A_{ij} i.i.d. $\sim N(0,1)$.

Let $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n$ be the standard orthonormal basis of $\mathbb{R}^n$, and construct the discrete set $\mathbb{T} = \{\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n\} \subset \mathbb{R}^n$; then $\boldsymbol{A}\mathbb{T} = \{\boldsymbol{g}_1, \cdots, \boldsymbol{g}_n\}$, hence

$$\operatorname{conv}(\boldsymbol{g}_1, \cdots, \boldsymbol{g}_n) = \operatorname{conv}(\boldsymbol{A}\mathbb{T}).$$

- Computing the characteristic quantities of $\mathbb{T}$

  For $\operatorname{rad}(\mathbb{T})$,

$$\operatorname{rad}(\mathbb{T}) = \sup_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{x}\|_2 = 1.$$

  For $w(\mathbb{T})$, by the asymptotic theory of the expectation of extreme Gaussian variables (upper bound: Proposition P2.7.6 (*Maximum of subgaussians*); lower bound: Exercise Q2.38 (*Maximum of Gaussian Random Variables*)), we have

$$w(\mathbb{T}) = \mathbb{E}_{\boldsymbol{g}}\left[\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] = \mathbb{E}\left[\max_{1\leqslant i\leqslant n} \langle \boldsymbol{g}, \boldsymbol{e}_i\rangle\right] = \mathbb{E}\left[\max_{1\leqslant i\leqslant n} g_i\right] = \sqrt{2\log n}(1+o(1)),$$

  i.e. $w(\mathbb{T}) \asymp \sqrt{\log n}$.

- Applying the Dvoretzky–Milman theorem (Exercise Q9.41 (*High-Probability Version of the Dvoretzky–Milman Theorem*))

  By Exercise Q9.41 (*High-Probability Version of the Dvoretzky–Milman Theorem*), for any $u \geqslant 0$,

$$r_-(u)\mathbb{B}_2^m \subset \operatorname{conv}(\boldsymbol{A}\mathbb{T}) \subset r_+(u)\mathbb{B}_2^m \text{ (w.h.p. } 1-2\mathrm{e}^{-u^2}),$$

  where

$$r_\pm(u) = w(\mathbb{T}) \pm C_1(\sqrt{m}+u)\operatorname{rad}(\mathbb{T}) = w(\mathbb{T}) \pm C_1(\sqrt{m}+u), \tag{I}$$

  after which it suffices to control the fluctuation term $C_1(\sqrt{m}+u)$.

  Set $u = \sqrt{m}$; then the fluctuation term is $C_1(\sqrt{m}+u) = 2C_1\sqrt{m}$. If this term does not exceed $\varepsilon w(\mathbb{T})$, then

$$2C_1\sqrt{m} \leqslant \varepsilon w(\mathbb{T}), \text{ hence } 2C_1\sqrt{m} \leqslant \varepsilon c_0\sqrt{\log n}, \text{ hence } \sqrt{m} \leqslant \frac{\varepsilon c_0}{2C_1}\sqrt{\log n}, \text{ hence } m \leqslant \underbrace{\left(\frac{c_0^2}{4C_1^2}\varepsilon^2\right)}_{c_\varepsilon}\log n, \tag{II}$$

  Substituting (II) into (I), we obtain

$$r_-(u) \geqslant (1-\varepsilon)w(\mathbb{T})$$
$$r_+(u) \leqslant (1+\varepsilon)w(\mathbb{T}).$$

Thus, let the nominal radius $R = w(\mathbb{T}) \asymp \sqrt{\log n}$; we obtain

$$(1-\varepsilon)R\mathbb{B}_2^m \subset \operatorname{conv}(\boldsymbol{g}_1, \boldsymbol{g}_2, \cdots, \boldsymbol{g}_n) \subset (1+\varepsilon)R\mathbb{B}_2^m \text{ (w.h.p. } 1-2\mathrm{e}^{-m}).$$

43 [**Actual Random Projections**] Reformulate the Dvoretzky–Milman theorem for Gaussian projections as a theorem for random orthogonal projections.

Let $E \sim U(G(n,m))$ (*the random m-dimensional linear subspace of the Grassmannian $G(n,m)$ chosen uniformly with respect to the Haar measure*), and let $\boldsymbol{P} : \mathbb{R}^n \to E$ be the orthogonal projection onto E. Let $\mathbb{T} \subset \mathbb{R}^n$. Define its Euclidean circumradius and spherical mean width respectively as

$$\operatorname{rad}(\mathbb{T}) = \sup_{\boldsymbol{x}\in\mathbb{T}} \|\boldsymbol{x}\|_2, w_s(\mathbb{T}) = \mathbb{E}_{\boldsymbol{\theta}}\left(\sup_{\boldsymbol{x}\in\mathbb{T}} \langle \boldsymbol{\theta}, \boldsymbol{x}\rangle\right),$$

where $\boldsymbol{\theta} \sim U(\mathbb{S}^{n-1})$.

Prove that if $0 < \varepsilon < 1$ and the following sufficient dimension condition holds,

$$m\log\left(1 + \frac{8\operatorname{rad}(\mathbb{T})}{\varepsilon w_s(\mathbb{T})}\right) \leqslant c\varepsilon^2 n\left(\frac{w_s(\mathbb{T})}{\operatorname{rad}(\mathbb{T})}\right)^2,$$

the projected convex hull $\operatorname{conv}(\boldsymbol{P}\mathbb{T})$ approximates the Euclidean ball $B = w_s(\mathbb{T})B_2(E)$ of radius $w_s(\mathbb{T})$ in the subspace E, i.e.

$$(1-\varepsilon)B \subset \operatorname{conv}(\boldsymbol{P}\mathbb{T}) \subset (1+\varepsilon)B \text{ (w.h.p. } 1-2\mathrm{e}^{-c_0 m}).$$

Rk. Use the following lemma.

Lemma (The relation between Gaussian width and spherical width)

$$w_s(\mathbb{T}) \asymp \frac{w(\mathbb{T})}{\sqrt{n}}.$$

Likewise as in Exercise Q7.27 (Matrix Sketching), use the polar decomposition $\boldsymbol{g} = \|\boldsymbol{g}\|_2\boldsymbol{\theta}$, whose norm $\|\boldsymbol{g}\|_2$ is independent of $\boldsymbol{\theta} = \boldsymbol{g}/\|\boldsymbol{g}\|_2 \sim U(\mathbb{S}^{n-1})$. Then

$$w(\mathbb{T}) = \mathbb{E}_{\boldsymbol{g}}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{g}, \boldsymbol{x}\rangle\right] = \mathbb{E}[\|\boldsymbol{g}\|_2]\cdot \mathbb{E}_{\boldsymbol{\theta}}\left[\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{\theta}, \boldsymbol{x}\rangle\right] = \mathbb{E}[\|\boldsymbol{g}\|_2]\cdot w_s(\mathbb{T}),$$

where

$$\mathbb{E}[\|\boldsymbol{g}\|_2] = \sqrt{2}\frac{\Gamma((n+1)/2)}{\Gamma(n/2)} = \sqrt{n}\left(1 - O\left(\frac{1}{n}\right)\right) \asymp \sqrt{n},$$

and substituting this in, we obtain

$$w_s(\mathbb{T}) = \frac{w(\mathbb{T})}{\mathbb{E}\|\boldsymbol{g}\|_2} \asymp \frac{w(\mathbb{T})}{\sqrt{n}}.$$

Prf. • Applying convex-geometric duality to reformulate the problem

Denote by $\mathbb{S}(E) = E \cap \mathbb{S}^{n-1}$ the unit sphere of E, and by $B_2(E) = \{\boldsymbol{y} \in E : \|\boldsymbol{y}\|_2 \leqslant 1\}$ the corresponding closed unit Euclidean ball. Then by Exercise Q9.40 (*Duality*), letting $V = \boldsymbol{P}\mathbb{T}$,

$$(1-\varepsilon)w_s(\mathbb{T})B_2(E) \subset \operatorname{conv}(\boldsymbol{P}\mathbb{T}) \subset (1+\varepsilon)w_s(\mathbb{T})B_2(E)$$

holds if and only if for any $\boldsymbol{y} \in \mathbb{S}(E)$,

$$(1-\varepsilon)w_s(\mathbb{T}) \leqslant \sup_{\boldsymbol{z}\in\boldsymbol{P}\mathbb{T}}\langle \boldsymbol{z}, \boldsymbol{y}\rangle \leqslant (1+\varepsilon)w_s(\mathbb{T}). \tag{I}$$

For any $\boldsymbol{y} \in \mathbb{S}(E)$, since

$$\sup_{\boldsymbol{z}\in\boldsymbol{P}\mathbb{T}}\langle \boldsymbol{z}, \boldsymbol{y}\rangle = \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{P}\boldsymbol{x}, \boldsymbol{y}\rangle = \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x}, \boldsymbol{P}\boldsymbol{y}\rangle = \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x}, \boldsymbol{y}\rangle = h_{\mathbb{T}}(\boldsymbol{y}), \tag{II}$$

the support function of the projected set $\boldsymbol{P}\mathbb{T}$ in direction $\boldsymbol{y}$ is thus equivalent to the support function $h_{\mathbb{T}}(\boldsymbol{y})$ of the original set $\mathbb{T}$ in direction $\boldsymbol{y}$.

Since $E \sim U(G(n,m))$, we have $\boldsymbol{y} \sim U(\mathbb{S}^{n-1})$, and hence

$$\mathbb{E}[h_{\mathbb{T}}(\boldsymbol{y})] = \mathbb{E}_{\boldsymbol{\theta}\sim U(\mathbb{S}^{n-1})}\left(\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x}, \boldsymbol{\theta}\rangle\right) = w_s(\mathbb{T}). \tag{III}$$

It remains only to consider the support function $h_{\mathbb{T}} : \mathbb{S}^{n-1} \to \mathbb{R}$.

• Applying the spherical measure concentration inequality

For any $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{S}^{n-1}$, by the Cauchy–Schwarz inequality,

$$|h_{\mathbb{T}}(\boldsymbol{u}) - h_{\mathbb{T}}(\boldsymbol{v})| = \left|\sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x}, \boldsymbol{u}\rangle - \sup_{\boldsymbol{x}\in\mathbb{T}}\langle \boldsymbol{x}, \boldsymbol{v}\rangle\right| \leqslant \sup_{\boldsymbol{x}\in\mathbb{T}}|\langle \boldsymbol{x}, \boldsymbol{u}-\boldsymbol{v}\rangle| \leqslant \operatorname{rad}(\mathbb{T})\|\boldsymbol{u}-\boldsymbol{v}\|_2.$$

Hence $h_{\mathbb{T}}$ is an $L = \operatorname{rad}(\mathbb{T})$-Lipschitz function, and by Theorem T5.1.4 (*Isoperimetric inequality on $\mathbb{R}^n$*) (*isoperimetric inequality on $\mathbb{R}^n$*), for any $t > 0$,

$$\mathbb{P}\{|h_{\mathbb{T}}(\boldsymbol{\theta}) - w_s(\mathbb{T})| \geqslant t\} \leqslant 2\exp\left(-\frac{c_1 n t^2}{\operatorname{rad}(\mathbb{T})^2}\right). \tag{IV}$$

• Building a net on a fixed reference sphere and taking the union bound

Fix an m-dimensional subspace E_0 and an η-net $\mathcal{N}_0$ on its unit sphere, where

$$\eta = \frac{\varepsilon w_s(\mathbb{T})}{4\operatorname{rad}(\mathbb{T})}.$$

Aspect	Gaussian random projection	True orthogonal projection
Projection operator	matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ (*entries* i.i.d. $\sim N(0,1)$)	operator $\boldsymbol{P}: \mathbb{R}^n \to E$ ($E \sim \mathrm{Haar}(G(n,m))$)
Radius of the projected ball	$R = w(\mathbb{T})$ (*Gaussian width*)	$R = w_s(\mathbb{T}) \asymp \frac{w(\mathbb{T})}{\sqrt{n}}$ (*spherical mean width*)
Essence of the scaling	each row of the matrix has norm $\|\boldsymbol{A}_i\|_2 \approx \sqrt{n}$	isometric norm-preserving scaling map, dividing by the extra factor $\sqrt{n}$
Sufficient dimension condition	$m \asymp \varepsilon^2 (\frac{w(\mathbb{T})}{\mathrm{rad}(\mathbb{T})})^2$	$\underset{\lesssim \varepsilon^2 n (w_s/r)^2}{m \log(1+8r/(\varepsilon w_s))}$, where $r = \mathrm{rad}(\mathbb{T})$ and $w_s = w_s(\mathbb{T})$

Take a Haar orthogonal map $\boldsymbol{U}$ and put $E = \boldsymbol{U}E_0$ and $\mathcal{N} = \boldsymbol{U}\mathcal{N}_0$. Then $\mathcal{N}$ is an η-net on $\mathbb{S}(E)$, with

$$|\mathcal{N}| = |\mathcal{N}_0| \leqslant \left(1 + \frac{2}{\eta}\right)^m = \left(1 + \frac{8\,\mathrm{rad}(\mathbb{T})}{\varepsilon w_s(\mathbb{T})}\right)^m.$$

For every $\boldsymbol{y}_0 \in \mathcal{N}_0$, $\boldsymbol{U}\boldsymbol{y}_0 \sim U(\mathbb{S}^{n-1})$; mutual independence is unnecessary. Set $t = \varepsilon w_s(\mathbb{T})/2$. Applying (IV) and then the union bound gives

$$\begin{aligned}\mathbb{P}\left\{\max_{\boldsymbol{y}\in\mathcal{N}} |h_{\mathbb{T}}(\boldsymbol{y}) - w_s(\mathbb{T})| \geqslant \frac{\varepsilon}{2} w_s(\mathbb{T})\right\} &\leqslant |\mathcal{N}| \cdot 2\exp\left(-\frac{c_1 n (\varepsilon w_s(\mathbb{T})/2)^2}{\mathrm{rad}(\mathbb{T})^2}\right) \\ &\leqslant 2\exp\left(m\log\left(1 + \frac{8\,\mathrm{rad}(\mathbb{T})}{\varepsilon w_s(\mathbb{T})}\right) - \frac{c_1\varepsilon^2}{4} n \left(\frac{w_s(\mathbb{T})}{\mathrm{rad}(\mathbb{T})}\right)^2\right).\end{aligned}$$

By the dimension condition, choosing the absolute constant c sufficiently small makes the exponent above at most $-c_0 m$. Hence

$$\max_{\boldsymbol{y}\in\mathcal{N}} |h_{\mathbb{T}}(\boldsymbol{y}) - w_s(\mathbb{T})| \leqslant \frac{\varepsilon}{2} w_s(\mathbb{T}) \text{ (w.h.p. } 1 - 2\mathrm{e}^{-c_0 m}). \tag{V}$$

- Continuous approximation and closing the duality loop

  For any $\boldsymbol{y} \in \mathbb{S}(E)$, choose $\boldsymbol{y}_0 \in \mathcal{N}$ with $\|\boldsymbol{y} - \boldsymbol{y}_0\|_2 \leqslant \eta$. By the Lipschitz property of $h_{\mathbb{T}}$ and (V),

$$|h_{\mathbb{T}}(\boldsymbol{y}) - w_s(\mathbb{T})| \leqslant |h_{\mathbb{T}}(\boldsymbol{y}_0) - w_s(\mathbb{T})| + \mathrm{rad}(\mathbb{T})\eta \leqslant \frac{3\varepsilon}{4} w_s(\mathbb{T}) \leqslant \varepsilon w_s(\mathbb{T}),$$

  i.e.

$$(1-\varepsilon) w_s(\mathbb{T}) \leqslant h_{\mathbb{T}}(\boldsymbol{y}) = \sup_{\boldsymbol{z}\in\boldsymbol{P}\mathbb{T}} \langle \boldsymbol{z}, \boldsymbol{y}\rangle \leqslant (1+\varepsilon) w_s(\mathbb{T}),$$

  Substituting into (I), we obtain

$$(1-\varepsilon) w_s(\mathbb{T}) B_2(E) \subset \mathrm{conv}(\boldsymbol{P}\mathbb{T}) \subset (1+\varepsilon) w_s(\mathbb{T}) B_2(E).$$

Rk. The comparison between Theorem T9.7.2 (*Dvoretzky–Milman theorem*) and the true orthogonal projection is as follows: